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SUPERSEDING

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DEPARTMENT OF DEFENSE HANDBOOK

ARMY DIGITAL PUBLICATIONS DEVELOPMENT IMPLEMENTATION GUIDE

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FOREWORD

1. This military handbook is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

2. This handbook provides guidance on the implementation of Standard Generalized Markup Language (SGML) and Extensible Markup Language (XML) as it pertains to MIL–STD–2361, Department of Defense, Interface Standard, Army Digital Publications. This handbook is for guidance only. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply.

3. This document supplements Army Departmental Manuals, Directives, and Military Standards, and provides basic and fundamental information on Standard Generalized Markup Language (SGML) and Extensible Markup Language (XML) as it applies to MIL–STD–2361.

4. The use of Courier font changes in this handbook represent SGML/XML document instance fragments. See the following example for the courier font.

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VOLUME 1
GENERAL

PART I
INTRODUCTION TO MIL-HDBK-2361

1 SCOPE.

1.1 Scope. This handbook provides implementation guidance for the development of Army publications in Standard Generalized Markup Language (SGML) and Extensible Markup Language (XML) in accordance with MIL–STD–2361, Department of Defense Interface Standard, Army Digital Publications. This handbook also provides implementation guidance for the use of the Army SGML/XML Registry and Library (ASRL) (see Section 7) and other SGML/XML implementation tools. This handbook is for guidance only and cannot be cited as a requirement.

1.2 Purpose. The purpose of the handbook is to facilitate the work of publications developers and users responsible for creating or using SGML/XML publications in compliance with the requirements of MIL–STD–2361. This handbook is a guide for use by publications developers and users to allow maximum reuse and sharing of common publications source data. Army publications data prepared in accordance with MIL-HDBK-2361 will facilitate the automated storage, retrieval, interchange, and processing of technical documents from varied data sources.

1.3 Applicability. This handbook is applicable for use by the Department of the Army (DA) and its contractors, and may be used by other Services and Departments of the Government. It applies to all publications digital data and document development required for compliance with MIL-HDBK-2361. This handbook will provide knowledge and information about SGML/XML and its application, tutorials in the various ways it may be used, helpful hints and guidance regarding specific SGML/XML characteristics and other user assistance type features.

1.4 Introduction. MIL–STD–2361, Department of Defense Interface Standard, Army Digital Publications established the Standard Generalized Markup Language (SGML) and Extensible Markup Language (XML) requirements for use in Army digital publications. The military standard was developed as part of the Army Publication Content Management Program (APCMP). Army Publishing Directorate (APD) effort to re-engineer the way Army publication source data by developing, delivering, storing, exchanging, and accessing through business processes. The application of MIL–STD–2361 requirements provides the Army a capability to develop weapons systems source databases that allow reuse and sharing of common publication data. This Department of Defense Handbook, MIL-HDBK-2361, Army Digital Publications Development Implementation Guide, provides implementation guidance for MIL–STD–2361. The handbook is designed to support the requirements contained in MIL–STD–2361, and provides guidance, tutorials, and examples to aid publication developers in the publication development process. MIL-HDBK-2361 is designed to provide users a tool that is simple to use and functionally accurate to the Army publication processes. The body of MIL-HDBK-2361 contains publication development and implementation guidance information to assist the publication developer in the use and application of SGML/XML. Detailed guidance is provided in the respective appendices. The appendices include an SGML/XML Tutorial and Reference Material, applications for the respective publication types, stylesheets and an application of the FOSI as a style guide. The handbook structure is described in Chapter 4 with a brief description of each of the handbook's major sections.

1.5 Why SGML to XML. Administrative Publications and Technical Manuals changed from SGML to XML mark up due to the wide spread application of XML. XML is a subset of SGML but has the following benefits over SGML. The change from SGML to XML are described in the present edition of MIL-HDBK-2361.

a. XML is straightforwardly usable over the Internet.
b. XML is supported by a wide variety of applications.
c. XML is compatible with SGML.
d. It is easy to write applications which process XML documents.
e. XML documents are easy to create.

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f. Well-formed XML allows data to be chunked into smaller bits of information and distributed. Examples of chunked data are:
   (1) A TM can be divided into stand alone work packages.
   (2) Work packages chunked into tasks.
   (3) Tasks chunked into procedures.

2 APPLICABLE DOCUMENTS.

2.1 General. The documents listed below are not necessarily all of the documents referenced herein, but are the ones that are needed in order to fully understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the latest issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto.

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-28001 - Markup Requirements and Generic Style Specification for Electronic Printed Output and Exchange of Text

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-1840 - Automated Interchange of Technical Information.

MIL-STD-2361 - Digital Publications Development

MIL-STD-3008 - Interactive Electronic Technical Manual (IETM) Technical Data Requirements To Support The Global Combat Support System - Army (GCSS-A)

MIL-STD-40051–1 - Preparation Of Digital Technical Information For Interactive Electronic Technical Manuals (IETMS)

MIL-STD-40051–2 - Preparation Of Digital Technical Information For Page-Based Technical Manuals

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-1222 - Guide to the General Style and Format of U.S. Army Work Package Technical Manuals

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from Document Automation & Production Services (DAPS) Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094,(**Phone: 215-697-2179, URL address: https://assist.daps.dla.mil/**).)

2.2.2 Other government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless specified otherwise, the issues are those cited in the solicitation.

REGULATIONS AND PAMPHLETS

AR 25-30 - The Army Integrated Publishing Program

DA PAM 25-40 - Administrative Publications: Action Officers Guide

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(Application for copies should be addressed to U.S. Army Publications Distribution Center, 1655 Woodson Road, St. Louis, MO 63114-6181.)

DoD 5200.1-R - Information Security Program Regulation.
DODD 5230.24 - Distribution Statements On Technical Documents
DODD 5230.25 - Withholding Of Unclassified Technical Data From Public Disclosure
DODI 7930.2 - ADP Software Exchange And Release

(Copies of DoD documents are available from the Document Automation & Production Service, Building 4/D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 **Phone: 215-697-2664**.)

TRADOC Regulation 350-70 - Training Development Management, Processes, and Products Systems Approach to Training Management Processes and Products, http://www-tradoc.army.mil/tpubs/regs/r350–70/index.html

(Copies of TRADOC Pamphlet 350-70-1 and Regulation 350-70 are available from (http://www-tradoc.army.mil/tpubs/).)
(Copies of Pamphlets, Regulations, and other Government Documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

WORLD WIDE WEB CONSORTIUM

REC-xml-20060816 - Extensible Markup Language (XML) 1.0.

(Copies can be obtained from (http://www.w3.org/TR).)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ISO 8879 - Information Processing - Text and Office Systems - Standard Generalized Markup Language (SGML)
ISO 8632-2 - CGM - Character Encoding
ISO 8632-3 - CGM - Binary Encoding
ISO 8632-4 - CGM - Clear Text Encoding

(Copies ISO documents can be obtained from (http://www.iso.org).)

2.3.1 Information documents for XML. The following documents are listed for informational purposes only and do not form a part of this handbook. They may be obtained from their publishers.

a. By Charles M. Goldfarb and Paul Prescod, The XML Handbook (5th Edition), Prentice Hall PTR, December 15, 2003, ISBN: 0-13-049765-7
b. By Neil Bradley, The XML Companion (3rd Edition), Addison Wesley Professional, December 12, 2001, ISBN-10: 0201770598, ISBN-13: 9780201770599
c. By Erik T. Ray, Learning XML 2nd Edition, O'Reilly, September 2003, ISBN: 0-596-00420-6
d. By Neil Bradley, The XML Companion (2nd Edition), Addison Wesley Professional, August 7, 2002, ISBN-10: 0201770830 ISBN-13: 978-0201770834

3 DEFINITIONS.

This section contains a listing of terms, and their definitions, that are specific to publication development in SGML/XML. There is also a list defining acronyms that are widely used in the Army publication environment.

3.1 Acronyms.

AAL Additional Authorization List
ADTLP Army-Wide Doctrine and Training Literature Program

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APD	Army Publishing Directorate
AMC	Army Materiel Command
ANSI	American National Standards Institute
APPIP	Adminstrative Publication Proponent/Editor Interface Process
AR	Army Regulation
ARTEP	Army Training and Evaluation Program
ASAT	Automated System Approach to Training
ASCII	American Standard Code for Information Interchange
ASRL	Army SGML/XML Registry and Library
ATIA	Army Training Information Architecture
AVIM	Aviation Intermediate Maintenance
AVUM	Aviation Unit Maintenance
BII	Basic Issue Items
BIT	Built-in Test
BITE	Built-in Test Equipment
BOI	Basis of Issue
BOS	Battlefield Operating System
CAD	Course Administration Data
CAGEC	Commercial and Government Entity Code
CALS	Continuous Acquisition Life-Cycle Support
CCITT	Consultative Committee for International Telephone & Telegraph
CDRL	Contract Data Requirements List
CGM	Computer Graphics Metafile
CIO	Chief Information Officer
CITIS	Contractor Integrated Technical Information Service
COEI	Components of End Item
CRD	Consolidated Requirements Document
CSI	Critical Safety Items
CTA	Common Table(s) of Allowance
DA	Department of the Army
DEP	Delayed Entry Program
DISA	Defense Information Systems Agency
DMS	Data Management System
DMWR	Depot Maintenance Work Request
DoD	Department of Defense
DoDISS	Department of Defense Index of Specifications and Standards
DPD	Digital Publications Development
DS	Direct Support
DSSSL	Document Style Semantics and Specification Language
DTD	Document Type Definition
ECP	Engineering Change Proposal
EDS	Electronic Display System
EIC	End Item Code
EP	Electronic Publishing

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EPS	Electronic Publishing System
ESD	Electrostatic Discharge
ETM	Electronic Technical Manual
FDEP	Final Draft Equipment Publication
FGC	Functional Group Code
FM	Field Manual
FOSI	Formatting Output Specification Instance
FPI	Formal Public Identifier
FSCAP	Flight Safety Critical Aircraft Parts
GFI	Government Furnished Information
GS	General Support
HCI	Hardness Critical Item
HR	Hand Receipt
IAW	In Accordance With
IETM	Interactive Electronic Technical Manual
IGES	Initial Graphics Exchange Specification
ISO	International Organization for Standardization
IPSC	Information Processing Standards for Computers
JCALS	Joint Continuous Acquisition Life-Cycle Support
LMI	Logistics Management Information
LRU	Line Replacement Unit
LSA	Logistic Support Analysis
LOGSA	Logistics Support Activity
MOS	Military Occupational Specialty
MTF	Maintenance Test Flight
MTP	Mission Training Plan
MUX	Multiplex
NATO	North Atlantic Treaty Organization
NBC	Nuclear, Biological, and Chemical
NDTI	Nondestructive Testing Inspection
NHA	Next Higher Assembly
NII	Networks and Information Integration
NIIN	National Item Identification Number
NMWR	National Maintenance Work Requirement
NSN	National Stock Number
OAASA	Office of the Adminstrative Assistance to the Secretary of the Army
OASD	Office of the Assistant Secretary of Defense
ODS	Ozone Depleting Substances
OIP	Overhaul Inspection Procedure
OS	Output Specification
OTJAG	Office of the Judge Advocate General
P/N	Part Number
PA	Preparing Activity
PAM	DA Pamphlet

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PCO	Publication Control Officer
PDEP	Preliminary Draft Equipment Publication
PDL	Page Description Language
PI	Parts Information
PMCS	Preventive Maintenance Checks and Services
PMI	Phased Maintenance Inspection
PMS	Preventive Maintenance Services
POI	Program of Instruction
QA	Quality Assurance
QTY	Quantity
RDL	Reimer Digital Library
RFP	Request For Proposal
RTF	Rich Text Format
SC	Supply Catalog
SGML	Standardized Generalized Markup Language
SME	Subject Matter Expert
SMR	Source, Maintenance, and Recoverability
SRU	Shop Replacement Units
STP	Soldier's Training Publication
STRAP	System Training Plan
TAMMS	The Army Maintenance Management System
TC	Training Circulars
TM	Technical Manual
TMDE	Test Measurement & Diagnostic Equipment
TOE	Table of Organization and Equipment
TRADOC	U.S. Army Training and Doctrine Command
U/I	Unit of Issue
UOC	Usable on Code
URL	Uniform Resource Locator
VB	Visual Basic
VBA	Visual Basic for Applications
WP	Work Package
WWW	World Wide Web
WYSIWYG	What You See Is What You Get
XML	Extensible Markup Language
XSL	XML Stylesheet Language
XSL-FO	Extensible Stylesheet Language for Formatting Objects
XSLT	Extensible Stylesheet Language Transformations

3.2 Terms.

Abstract	A narrative which describes, defines, or synopsizes a Digital Publications Development SGML/XML asset.
Attribute	A member of an attribute definition list within an attribute list declaration. It declares an attribute name, specifies the form and SGML/XML -specific aspects of possible values, and specifies the action (such as providing a default value) to be taken if an attribute's value is not specified. In the

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	display under attribute (Definition) list declaration, each attribute definition is shown as: name_of_ attribute allowable_values default.
Attribute (Definition) List Declaration	A markup declaration that associates an attribute definition list with one or more element types, shown as:*<!ATTLIST name_of_associated_element(s) name_of_attribute allowable_values default>*
Attribute (of an element)	A qualifier indicating a property of an element, other than its type (which is done by a generic identifier) or its content (which is delimited by start-tags and end-tags). Attributes are only found on start-tags, and can indicate reference identifiers, confidentiality, formatting information, and so on.
Attribute (Specification) List	Markup that is a set of one or more attribute specifications, shown as: attribute=value attribute=value attribute=value. The markup is used within a Start Tag, as in: *<element_name attribute=value attribute=value attribute=value>* .
Attribute Definition	A member of an attribute definition list within an attribute list declaration. It declares an attribute name, specifies the form and SGML/XML -specific aspects of possible values, and specifies the action (such as providing a default value) to be taken if an attribute's value is not specified. In the display under ATTRIBUTE (Definition) LIST DECLARATION, each attribute definition is shown as: name_of_attribute allowable_values default.
Contract Data Requirements List	The Contract Data Requirements List (CDRL) defines the data that is to be delivered to the Government by the contractor. This data may be in hard copy, electronic, electronic mailable, or any other form specified. The specific form of delivery needs to be specified either in the SOW and/or in each individual CDRL item.
Constructs	Document type definitions (DTDs), formatting output specification instances (FOSIs), and SGML/XML tag narrative definitions.
Declaration	The SGML/XML declaration defines which characters are used in a document instance, which syntax the DTD is written in, and which SGML/XML features are used. It should accompany each SGML/XML document, although a default to the one described in the standard may be assumed.
Declaration Subset	A delimited portion of a markup declaration in which other declarations can occur.
Document Instance	The instance is the actual document text and its accompanying SGML/XML tags conforming to the specifications and restrictions set forth in the DTD and stored in an ASCII text format.
Document Type Declaration	A markup declaration that contains the formal specifications of a document type definition, shown as: *<!DOCTYPE document_type_name optional_external_identifier [optional_document_type_declaration_subset]>* . The declaration invokes a DTD in an SGML/XML document. The document instance of an SGML/XML document is always be preceded by a document type declaration.
Document Type Definition (DTD)	A DTD, or Document Type Definition, is an SGML/XML construct used to rigorously and unambiguously describe the structure and content of classes of documents in terms of SGML/XML instances (elements, attributes, entities, etc.). The DTD is occasionally but not in compliance with ISO 8879 terminology used as an abbreviation for 'document type declaration'; it is also an SGML/XML reserved word used in formal public identifiers to indicate that the identified entity is a document type declaration set, and is often used to identify such a set.

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Electronic Publication	A electronic page-based representation that provides concise, user-friendly information for instruction, repair, policy or guidance. The EP may interact with other EP or IEP information.
Element	A component of the hierarchical structure defined by a document type declaration or DTD. It is identified in a document instance by descriptive markup, usually a start-tag and end-tag, shown as: *<element_type_name attribute= "value" attribute= "value">* content of the element *</element_type_name>* .
Element Type Declaration	A markup declaration that contains the formal specification of the part of the definition of an element type that deals with the content and markup minimization, shown as: *<!ELEMENT element_type_name start_tag_minimization end_tag_minimization content_model_group_or_declared_content content_exceptions>*
Entity	A unit of information that may be referred to by a symbol in a DTD or in a document instance. Entities may be used for character strings, characters that cannot be keyed in on a keyboard, or for separate files that may or may not contain SGML/XML data.
Entity Reference	A reference that is replaced by an entity, shown as: &entity_name; or %entity_name; the ampersand is used for general entities (referenced in the document instance); the percent sign is used for parameter entities (typically referenced in the document type declaration).
Entity Set	A set of entity (and comment) declarations that are used together.
Extensible Markup Language (XML)	Extensible Markup Language, as specified in REC-xml-20060816, is a subset of SGML and requires conformance to ISO 8879. XML defines a group of data objects called XML documents and partially describes the behavior of computer programs which process them.
Electronic Technical Manual-Interface (ETM-I).	ETM-I is a prototype software interface, through which the user can electronically transfer parts request information and work order data between the ETM platform and the Unit Level Logistics System (ULLS) and the Standard Army Maintenance System (SAMS). The interface will reduce extensive data entry and eliminate transposition errors that could lead to faulty requisitions and excess parts.
Formatting Output Specification Instance (FOSI)	An instance of the Output Specification (OS) that assigns values to the style characteristics for a particular document type definition. The FOSI uses the syntax of a SGML/XML document instance and is designed to format documents for paper delivery.
Interactive Electronic Publication	A computerized screen-based representation that provides interaction with weapon system, instructor, student or technician. The IEP can provide training feedback, troubleshoot, fault isolation, and/or training instruction. The functionality is provided by communicating and interacting with selected weapon system components.
Interactive Electronic Technical Manual (IETM)	Interactive Electronic Technical Manual (IETM). A technical manual prepared in digital form and designed for interactive display to the maintenance technicians or system operator end users by means of a computer controlled Electronic Display System (EDS).
ISO 8879 Information Processing	Text and Office Systems - Standard Generalized Markup Language (SGML) and Extensible Markup Language (XML) completely specifies the SGML/XML Meta-language with regard to the grammar and syntax required for the SGML/XML language along with the features that may be optionally

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	enabled for a given SGML/XML application. In addition, ISO 8879 also specifies various procedures for processing SGML/XML notation.
Legacy data	Legacy data, for purposes of this standard, is defined as any data (paper or digital) that has not been SGML/XML -tagged in compliance with the respective functional requirement standards or specifications, this standard, and MIL-PRF-28001.
Markup	To add text to data of a document to convey information about the document.
Output file	A text presentation metafile developed through use of a page description language (PDL) is referred to as an output file.
Output Specification	An OS, or Output Specification, provides a rigorously defined set of options for the style characteristics which provide the formatting intent for interchanged SGML/XML -tagged technical publications. The OS has a mechanism for binding the style characteristics to SGML/XML elements and attributes in a document's DTD. The OS is in the form of an SGML/XML DTD. At present, the OS is intended for hard copy composition but can be applied to digital display in limited applications (e.g., non-interactive).
Parsing	A SGML/XML parser is a computer application that breaks down an SGML/XML-coded document into a series of logical elements and checks that these elements conform to the model defined in the associated document type declaration. When parsing a document, the SGML/XML parser: • Checks each new character to see if it is part of a general delimiter string that identifies the start of a piece of markup. • Checks whether or not the character is a short reference delimiter that needs to be expanded. • Checks if the character is a separator character that should be ignored. • Identifies the various markup tags, identifying any entities that need to be expanded or recalled from external sources. • Checks if identified markup tags are valid according to the declared model.
Preparing Activity	The DoD activity or the Civilian Agency responsible for the preparation, coordination, issuance, and maintenance of standardization documents.
Reuse	The use of authored publication information or publications source data in more than one type of publication product. For example, information authored initially for a TM (e.g., a maintenance work package, task, etc.) that is used verbatim, or in part, for inclusion in a training or doctrine product (e.g., Soldier's Manual, Field Manual, etc.). The intent of reuse may also be fulfilled when the information is reused in a different TM (e.g., for a different level of maintenance, different version of an equipment/weapon system, or a different equipment/weapon system altogether).
SGML/XML Constructs	SGML/XML constructs are DTDs, FOSIs, and their fragments.
SGML Declaration	A SGML Declaration is an SGML construct which specifies an SGML implementation in terms of the values of the SGML parameters, character set, concrete syntax, optional features, and capacity requirements and the SGML features used.
SGML/XML Entity	An entity whose characters are interpreted as markup or data in accordance with (IAW) ISO 8879.
SGML/XML Instance	A SGML/XML Instance or SGML/XML -tagged document is the collection of data composing a specific document that includes SGML/XML tags (SGML/XML markup) corresponding to elements, their attributes, entity references, etc. The SGML/XML markup conforms to the document's DTD.

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SGML/XML Objects	SGML/XML objects are elements, entities, attributes of elements, public identifiers, notations, and standard tagging schemes.
SGML Parser	A SGML parser is a computer program or a specialized code compiler called a "parser". A SGML parser first processes (or "parses") a SGML Declaration defining the particular SGML implementation and stores this SGML environment. Then the SGML parser can be used to process (or "parse") a DTD to determine its conformance regarding grammar and syntax to ISO 8879 and the SGML Declaration for that SGML application. The SGML parser can then be used to process an instance of a particular document to determine the conformance of the instance to both SGML grammar and syntax and the DTD.
Standard Generalized Markup Language	Standard Generalized Markup Language, as detailed in ISO 8879. SGML is a meta-language that provides a coherent and unambiguous syntax for describing the logical structure of publications in unambiguous grammar. Formalizes the markup process and frees it of system and processing dependencies.
Table of Organization and Equipment (TOE)	The Table of Organization and Equipment (TOE) is a document that prescribes the wartime mission, capabilities, organizational structure, and mission essential personnel and equipment requirements for military units. It portrays the doctrinal modernization path (MODPATH) of a unit over time from the least modernized configuration (base TOE) to the most modernized (objective TOE).
Tag or Tagging	Adding tags (descriptive markups) to document data.
Task	A sequence of user actions with a definite beginning and an end. User tasks relate to installation checkout operation, and maintenance of systems or equipment. Tasks may contain procedures and in turn steps to complete the assigned task.
Technical Publication Verification	This term refers to the parsing of the digital data stream containing a publication to assure compliance with the standard (SGML, CCITT, CGM, IGES) to which it was written. There is no intent in this term to imply the validation/verification process used to certify the content of the publication.
XML Path Language (XPATH)	XPath is a language for addressing parts of an XML document, designed to be used by XSL-T.
XML Stylesheet Language for Formatting Objects (XSL-FO)	Extensible Stylesheet Language for Formatting Objects is a pagination markup language describing a rendering vocabulary capturing the semantics of formatting information for paginated presentation. The paginated presentation may be displaying multiple separated pages on a screen, on paper or audibly.
XML Stylesheet Language Transformations (XSL-T)	Extensible Stylesheet Language Transformations is a templating markup language used to express how a processor creates a transformed result from an instance of XML information. Else the XML transformation is a process that rearranges parts of a document into a new form.
Well-formed XML Document	Compliant with REC-xml-19980210 requirements, the basic rules for writing well-formed XML documents a. Start tags have corresponding end tags b. Elements can not overlap c. XML tags are case-sensitive d. Empty elements either have an end tag or close the empty tag with "/> " e. Reserved characters (< &> " ') are replaced with corresponding character sequence (< & > " ')

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f. Each XML document has an unique root element
g. Each attribute name in an element is unique
h. Each attribute name is followed by a value indicator (=) and a quoted string

Work Package Presentation of information functionally divided into individual task packages in the logical order of work sequence. The work packages will be stand alone general information, operating, maintenance, troubleshooting, parts, and supporting information units containing all information required for directing task performance. Work packages may be given to a soldier(s) so they may have complete instructions for accomplishing a task.

4 LAYOUT, FORMAT, AND CONTENT.

4.1 Scope. MIL-HDBK-2361 has been structured into five volumes. The volumes are comprised as follows:

- Volume 1 - General
- Volume 2 - Administrative Publications.
- Volume 3 - Training and Doctrine Publications.
- Volume 4 - Interactive Electronic Technical Manual (IETM) Technical Data Requirements To Support The Global Combat Support System-Army (GCSS-A)
- Volume 5 - Technical and Equipment Publications.

Volumes 1, 2 and 5 contain parts. One part contains the layout, format, and content of each of the volumes as described in the introduction to the respective volume. The remaining part(s) contains reference-type information that can be accessed by the user as required.

5 ACQUISITION OF ARMY PUBLICATIONS INFORMATION USING MIL-STD-2361.

5.1 Purpose. The primary purpose of MIL-STD-2361 is to provide the requirements for the acquisition, development, and delivery of common publication source data that can be used, reused, and shared throughout the Army. There are many ways in which common publication source data may be used throughout the Army for technical and equipment publications, training and doctrine publications, and administrative publications. The potential uses of common publication source data are to be understood. These uses are identified and planned for in the earliest possible stages of an acquisition contract. Close coordination is to be established and maintained by organizations throughout the publications information life cycle. This section contains information describing the processes used to acquire publications information for each Army publication domain.

5.2 Publications information process by domain. Processes and procedures for acquisition of publications information vary within each of the publication domains. Contracting for Army publication development ranges from very formal for technical manuals to practically no contracting for administrative publications.

5.2.1 Technical manual (TM). Normally, TM information is obtained as part of a formal weapon or command, control, communication, computer or intelligence system (C4I) development contract. TMs are developed by contractors and delivered as the system development moves through the stages of development. Specific levels of TMs such as operator (-10), field maintenance (-23), sustainment maintenance (-40), are specified in the development statement of work (SOW).

5.2.2 Training and doctrine. Training and doctrine information is normally developed in-house by TRADOC schools and centers. In some cases, certain training information is developed by a contractor under a weapon or C4I contract. But it is rare that information such as new equipment training or individual and collective task analysis/development, when developed by the system developer, is provided to the TRADOC proponent school.

5.2.3 Administrative. Administrative publications, such as Army regulations, circulars and pamphlets, are developed and produced in-house by Department of the Army (DA) staff organizations.

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5.2.4 Publications development requirements and guidance. Requirements for TM information development and delivery are contained in MIL–STD–40051–1/–2 and MIL–STD–2361. MIL–STD–40051–1/–2 and its companion handbook, MIL-HDBK-1222, contain functional requirements for TM source information development and guidance for its application, respectively. MIL–STD–2361, and its companion handbook, MIL-HDBK-2361, contain SGML/XML requirements and procedures for applying SGML/XML to publications development, respectively. Functional requirements for development of training and doctrine information are contained in TRADOC Regulation 350-70. TRADOC Regulation 350-70 contains guidance for the training development (TD) process to be employed by TRADOC schools and centers. (See the training and doctrine segment of this handbook for discussion of the training development process). Functional requirements for administrative publications are contained in Army Regulation (AR) 25-30.

5.3 MIL-STD-2361 acquisition, development and delivery process. Under MIL–STD–2361, the acquisition of publications information should be developed and delivered using an integrated team concept. An integrated team should be comprised of functional staff from both systems development (Army Materiel Command (AMC)), and system training development (TRADOC) organizations. Members of each functional staff should have an understanding of all potential uses of the publication data requirements and the interrelationships existing between the different types of data. They should have a thorough understanding of the need for the end-to-end flow of common source data, and be able to specify these interrelationships in the development contract as shown in Figure 1 . The figure not only depicts the interrelationships of the data, it also reflects the fact that a significant amount of information required for training and doctrine product development is derived from TM source information. The interface required to achieve the end-to-end flow of information is described in the training and doctrine segment of this handbook.

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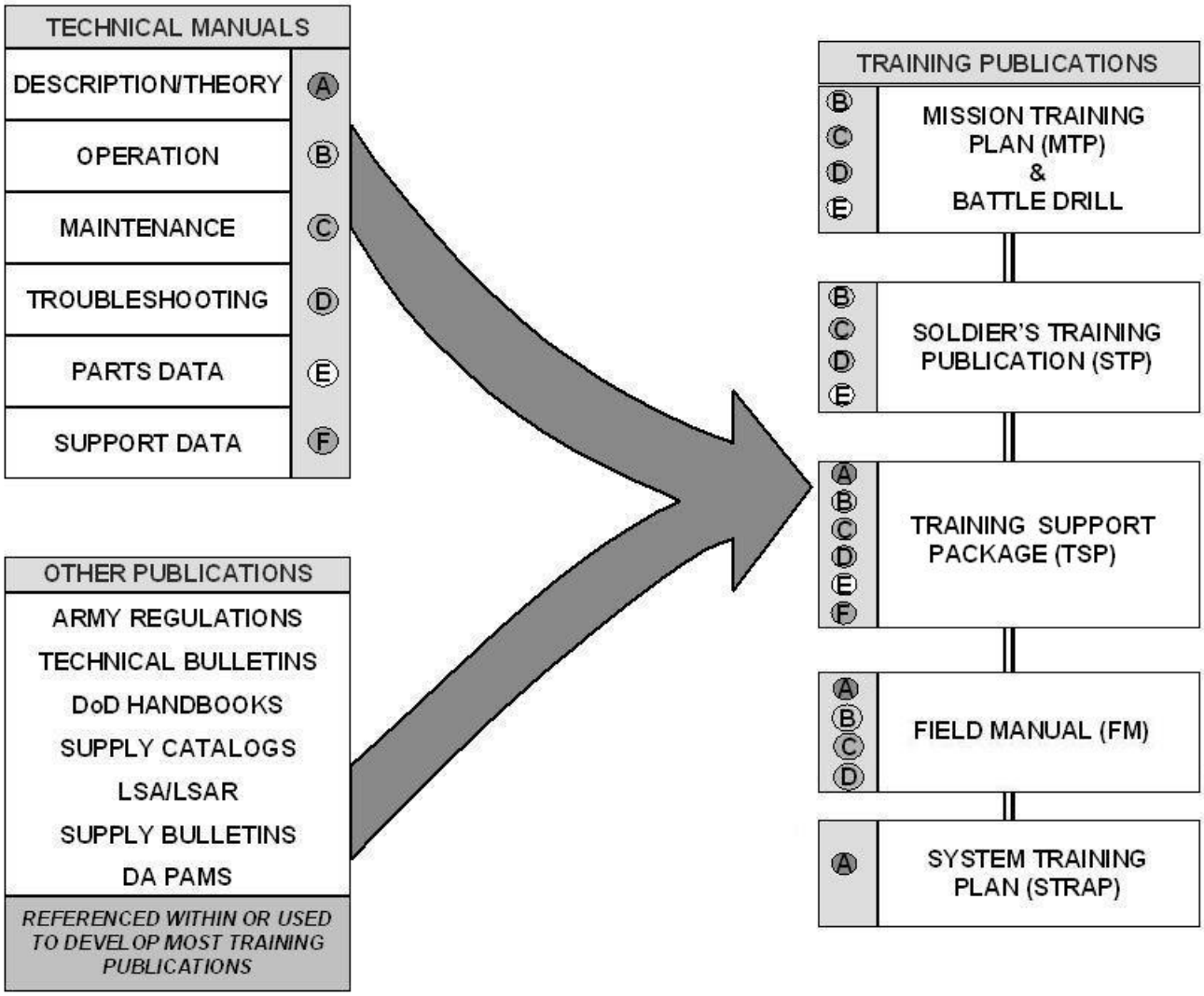

FIGURE 1. Interrelationships of Army publications data.

5.3.1 Acquisition planning. The first step in the process of acquiring common source data is the planning that take place prior to issuing the contract for publications information. An integrated team comprised of AMC and the appropriate TRADOC school(s) or center(s) personnel should perform this planning. The output of the planning process should be a data acquisition plan that clearly identifies specific data to be acquired, such as Technical Manuals, New Equipment Training (NET) Plans, literature, and training individual and collective task analysis. This plan should also contain points of contact and schedules for source data development milestones and joint in-process reviews.

5.3.2 Contract/statement of work. Guidance is to be included in the publication data development and delivery statement of work for the weapon or C4I system that clearly identifies source data to be developed and the method and destinations for delivery. MIL–STD–40051–1/–2 and MIL–STD–2361 contain requirements for development and delivery of TM and DMWR/NMWR data. MIL-STD-40051-2 contains a series of tables entitled TM and DMWR/NMWR Requirement Selection Matrices. Figure 2 contains a sample of one of the matrices. The appropriate matrix (s) should be completed by the integrated team, for use in the development of specific SOW requirements. As can be seen from the sample, the matrix contains

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specific DMWR/NMWR content requirements, as well as MIL–STD–40051–1/–2 reference paragraphs where detailed functional requirements can be found. The associated MIL–STD–2361 SGML/XML element name is also contained in the matrix. See MIL-STD-40051-2 for more information on how to use the TM and DMWR/NMWR Requirement Selection Matrices.

TABLE A-VII. DMWR/NMWR Requirements Matrix for _____.

DMWR/NMWR Content	DMWR DMWR with RPSTL	NMWR NMWR with RPSTL	MIL-STD-40051-2 Reference	Element Name
FRONT MATTER	R	R	5.2.1	<paper.frnt>
Front cover	R	R	5.2.1.1	<frntcover>
(MC) Promulgation letter			5.2.1.2	<promulgation>
Warning summary			5.2.1.3	<warnsum>
Change transmittal page			5.2.1.4	<chgsheet>
List of effective pages / work packages	R	R	5.2.1.5	<loepwp>
Title block page	R	R	5.2.1.6	<titleblk>
Table of contents	R	R	5.2.1.8	<contents>
CHAPTER 1. GENERAL INFORMATION, EQUIPMENT DESCRIPTION AND THEORY OF OPERATION	R	R	B.5.1	<gim>
GENERAL INFORMATION WORK PACKAGE	R	R	B.5.2	<ginfowp>
Scope	R	R	B.5.2.3	<scope>
Maintenance forms, records, and reports	R	R	B.5.2.4	<mfrr>
Reporting equipment improvement recommendations (EIR)	R	R	B.5.2.5	<eir>
Corrosion prevention and control (CPC)	R	R	B.5.2.7	<cpcdata>
Ozone depleting substances (ODS)			B.5.2.8	<odsdata>
Destruction of Army materiel to prevent enemy use	R	R	B.5.2.9	<destructmat>
Preparation for storage or shipment	R	R	B.5.2.10	<pssref>
Warranty information			B.5.2.11	<wrntyref>
Nomenclature cross-reference list			B.5.2.12	<nomenreflist>
List of abbreviations/acronyms	R	R	B.5.2.13	<loa>
Quality assurance (QA)			B.5.2.14	<qainfo>
Quality of material	R	R	B.5.2.15	<qualmat.info>
Safety, care, and handling			B.5.2.16	<sftyinfo>
Nuclear hardness			B.5.2.17	<hcp>
Calibration			B.5.2.18	<calref>
Engineering change proposals (ECP)	R	R	B.5.2.19	<ecp>
Modifications			B.5.2.20	<modification>
Deviations and exceptions	R	R	B.5.2.21	<deviation>
Mobilization requirements	R	R	B.5.2.22	<mobreq>
Flight safety critical aircraft parts (Aircraft only)			B.5.2.23	<fscapreq>
Cost considerations	R	R	B.5.2.24	<cost>

FIGURE 2. Example of a MIL-STD-40051-2 requirements matrix for a paper manual DMWR/NMWR.

5.3.3 Data delivery. The CDRL and underlying SOW should mandate the delivery of, or unrestricted access to, the database containing the common source data. The CDRL and SOW should specify the delivery of training and doctrine data that is not contained in TMs, such as NET Plans, training material, and task analysis summaries, to the TRADOC school(s) or center(s) requiring the data.

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6 SGML/XML REFERENCE MATERIAL.

6.1 How to keep up with SGML/XML. There are several ways to find out more about SGML/XML and to keep up to date with any SGML/XML developments, and new products. The following paragraphs describe some of the channels that users of the ASRL (see 7) can use.

6.2 SGML/XML users' group. The objectives of the SGML/XML User's Group is to promote the use of the Standard General Markup Language (SGML) or Extensible Markup Language (XML) and to provide a forum for the exchange of information. Regional chapters exist and frequent meetings are organized for developers and users of SGML/XML . For more information contact International SGML/XML Users' Group,. International SGML/XML Users' Group, Copse House, 15 Upton Close, Swindon, Wilts SN25 4UL, United Kingdom, **Phone: +44 1793 721106** or http://www.isgmlug.org.

6.3 International digital enterprise alliance (IDEAlliance) . IDEAlliance (International Digital Enterprise Alliance) is a non-profit membership organization. Its mission is to advance user-driven, cross-industry solutions for all publishing and content-related processes by developing standards, fostering business alliances, and identifying best practices. IDEAlliance has been a leader in information technology since 1966 (founded as Graphic Communications Association). For more information contact IDEAlliance, 1421 Prince St., Suite 230 Alexandria, VA 22314-2805,**Phone: 703-837-1070** or http://www.idealliance.org.

6.4 Books and magazines. For book references see the information documents (see 2.3.1) section of this handbook. The Newsletter, ***<TAG>*** is published monthly by Architag. The magazine is a reliable source of information on SGML/XML, ideas, tips literature, products and conferences. Subscriptions may be obtained from Architag International Corporation,17011 Lincoln Ave, Suite 105, Parker, CO 80134, **Phone: 1-888-302-0967** or www.architag.com

6.5 Servers. Information about SGML/XML may be obtained from "comp.text.sgml" Usenet Newsgroup on the Internet which contains an archive of messages, postings and additional information. The World Wide Web is a source for information on SGML/XML and a starting point is the SGML/XML Consortia and Users' Group at http://xml.coverpages.org/groups.html.

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PART II
ARMY SGML/XML REGISTRY AND LIBRARY (ASRL)

7 ARMY SGML/XML REGISTRY AND LIBRARY (ASRL).

7.1 Introduction. This part provides MIL-HDBK-2361 users with guidance and instruction about the U.S. Army Standard Generalized Markup Language (SGML) and Extensible Markup Language (XML) Registry and Library (ASRL), operated and maintained by Army Publishing Directorate (APD). This part is comprised of the following four major subject areas:

a. ASRL Description.
b. ASRL Concept of Operations.
c. ASRL Operations.
d. Submittal Package Delivery.

7.2 ASRL description. The ASRL is the central SGML/XML data repository and single-point source for Army-approved SGML/XML objects and constructs for departmental media developers. The ASRL contains Army-approved SGML/XML objects and constructs, such as Document Type Definitions (DTD), stylesheets, Formatting Output Specification Instances (FOSI), and SGML/XML Tag description lists authorized for use in the preparation of Army Publications. DTDs, stylesheets, FOSIs, and SGML/XML Tags officially registered with the ASRL are authorized in developing Department of the Army (DA) publications, including administrative, doctrinal and training, technical, and equipment publications, and Electronic and Interactive Electronic Technical Manuals (ETM/IETM). As part of the overall DoD Net-Centric Data Strategy, the DoD CIO established the DoD Metadata Registry (http://metadata.dod.mil) and a related metadata registration process for the collection, storage and dissemination of structural metadata information resources (schemas, data elements, attributes, document type definitions, style-sheets, data structures etc.). As OASD's Executive Agent, DISA maintains and operates the DoD Metadata Registry under the direction and oversight of OASD(NII). Registration of the Army-approved SGML/XML objects and constructs with the DoD Metadata Registry is pending. The ASRL is responsible for interfacing with the Defense Information Systems Agency (DISA) for all matters dealing with development and application of SGML/XML within the Army. Throughout this section DTDs, stylesheets, FOSIs, and tag description lists are referred to as SGML/XML objects and constructs.

7.2.1 ASRL capabilities. The ASRL may be used by publications developers to provide capabilities for the standardization and reuse of Army SGML/XML objects and constructs. In this regard, the ASRL will provide for the establishment and support of the following:

a. Infrastructures (administrative and communicative) that provide easy access to standard SGML/XML objects and constructs.
b. Processes to encourage use of standard objects and constructs by Army and defense contractor publication developers.
c. Processes to encourage timely submission of requirements not covered by existing SGML/XML objects and constructs by Army and defense contractor's publications developer.
d. Procedures for evaluation of requirements to determine whether or not they can be satisfied by approved SGML/XML objects and constructs and, if required, development of new objects and constructs.

7.2.2 Standardization and reuse. The two primary ASRL methods for SGML/XML standardization and reuse are direct reuse of existing, Army-approved SGML/XML objects and constructs; and development of new objects and constructs by APD to satisfy requirements not covered by existing SGML/XML objects and constructs.

7.2.2.1 Direct reuse. Direct reuse refers to the development of an SGML/XML document instance (which represents some class of Army publication under development) based upon Army-approved SGML/XML objects and constructs acquired from the ASRL. The publication developer will obtain the appropriate objects and constructs from the ASRL and use them to create the required SGML/XML document instance.

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7.2.2.2 New SGML/XML object and construct development. This refers to the development of new SGML/XML objects and constructs for requirements not covered by existing objects and constructs. A Government or contractor publications developer may identify a requirement(s) that supposedly is not covered by objects and constructs already included in the ASRL Library. The developer may submit a request, with full justification and rationale, to the ASRL Registry for evaluation of the requirement(s). Detailed submission procedures are contained in 7.5. Review of the ASRL Registry will determine if existing SGML/XML objects and constructs can satisfy the requirement, or have features in common with the requirement that can be reused. If the requirement evaluation determines that new SGML/XML objects and constructs are justified, they will be developed and tested by APD and included in the ASRL Library for Army-wide use.

7.3 ASRL concept of operations. This section describes the purpose, background, and overall concept of operations involving the ASRL and its related environment.

7.3.1 Purpose of the ASRL. The ASRL will contain all SGML Document Type Definitions (DTD), XML DTDs, stylesheets, Formatting Output Specification Instances (FOSI), and associated SGML/XML tags authorized for use in the preparation of Army Publications. All DTDs, stylesheets, FOSIs, and SGML/XML tags are to be officially registered with the ASRL prior to their use for developing any Department of the Army (DA) publication, including administrative, doctrinal and training, technical and equipment publications, and (ETM/IETM).

7.3.2 Background. The ASRL is part of the Army strategy to achieve the digitalization and integration of technical and business data. Employing SGML/XML facilitates the digital exchange and integration of textual data. A component of an SGML /XML application is the DTD. The DTD defines the rules that apply SGML/XML to the markup of a particular type of document, such as a technical or training manual. The development of a DTD is time consuming, but once developed, it can be reused whenever another instance of that document type is generated. Reuse saves DTD development time and document instance markup time by using previously established markup rules. The ASRL was established to facilitate the reuse of Army SGML /XML DTDs.

7.3.3 ASRL organization. Functional capabilities for the standardization and reuse of SGML/XML objects and constructs are provided by the ASRL. The Army proponent for the ASRL is the Army Publishing Directorate (APD). The ASRL is comprised of the Registry and Library. Figure 3 illustrates the organizational structure of the ASRL.

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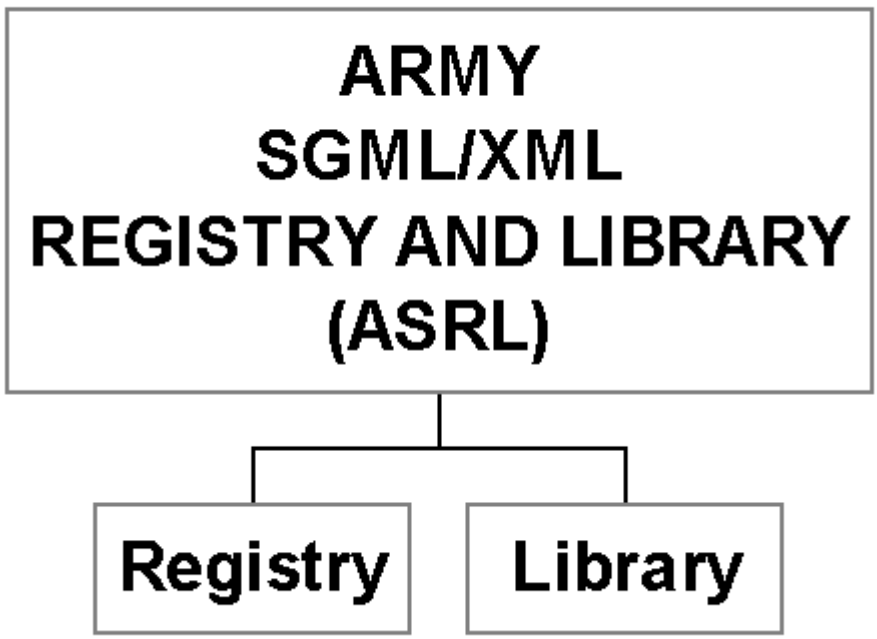

FIGURE 3. ASRL organizational structure.

7.3.4 ASRL services. The Library will provide the publications developers with a set of services that will enhance the SGML/XML publication development process. The services provided by the ASRL Library include:

a. Library services to ensure user access and utility, such as automated sign-in/check-out; virtual navigation aids; browse, search, and find utilities; and configuration control of the SGML/XML objects and constructs.
b. SGML/XML analysis capability to support ASRL user needs.
c. Administrative assistance to provide help and technical support to ASRL users needs.
d. Registry services to evaluate new SGML/XML requirements.

7.3.5 ASRL assets. ASRL assets will include, but are not limited to:

a. SGML/XML objects and constructs.
b. Army requirements and guidance documents, such as this handbook.
c. General SGML/XML reference documents or pointers.
d. ISO 8879 compliant parser list.
e. Government Furnished Information (GFI), such as stylesheets and instance processors.
f. Parser and SGML/XML system tips, examples, and lessons learned.
g. Information regarding frequently asked questions about SGML/XML, using the ASRL, etc.
h. Point-of-Contact (POC) listings.
i. ASRL access data, such as user statistics and library usage statistics.
j. SGML/XML-based Freeware.

7.3.6 Procedures for submitting existing SGML /XML objects and constructs. The ASRL registration process includes coordination with the DISA operated DoD Metadata Registry to ensure that Army SGML/XML objects and constructs are not redundant, or otherwise duplicative of those approved for use throughout the DoD. There will be an established period during which existing Army SGML/XML objects and constructs may be submitted to the ASRL Registry as candidates for incorporation into the ASRL

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Library. Such existing objects and constructs may be incorporated into the library under ”grandfather” approval. Approval will be contingent upon compliance with the appropriate requirements standards and is at the sole discretion of APD. Information and guidance regarding “grandfathering” SGML objects and constructs will be established and distributed by APD. This section of the handbook will be in effect only as long as the “grandfathering” period is in effect. This section will be deleted in future revisions. The following procedures will be followed for submission of existing SGML/XML objects and constructs to the ASRL Registry as candidates for evaluation and acceptance into the ASRL.

a. Existing SGML/XML objects and constructs (i.e., DTDs, stylesheets, FOSIs and associated tag descriptions) will be submitted to the ASRL for registration using the format and procedures described in 7.5 of this handbook.
b. Each submitted SGML/XML object and construct will be reviewed by Army Publishing Directorate (APD) to assess its compliance with guidance in this handbook, MIL-STD-2361(AC), Digital Publication Development MIL-PRF-28001, Standard Generalized Markup Language, and other specifications or standards, as required.
c. SGML/XML objects and constructs found to be not in compliance with applicable requirements standards and specifications will be returned to the submitting organization for revision.
d. SGML/XML tags complying with applicable requirements standards and specifications will be compared with other tags in the Library. Where more than one tag exists for the same information, or where joint review is required, Army Publishing Directorate (APD) will coordinate with submitting organizations to establish a common tag.
e. SGML/XML objects and constructs successfully completing the ASRL registration process will be incorporated into the library and become available for Army-wide use by publications developers.

7.3.7 Procedures for submitting new SGML/XML object and construct requirements. The ASRL Registry provides a capability for evaluating new SGML/XML requirements to determine whether or not they are covered by Army-approved SGML/XML objects and constructs. This capability is used only after a publications developer has identified a set of specific structure and content requirements that are not addressed, or only partially addressed, by approved SGML/XML objects and constructs.

a. The publications developer will submit the requirements, along with complete justification and rationale, to the ASRL Registry for evaluation. Detailed procedures and submission package format are described in 7.5.
b. The ASRL Registry will review the requirements package for completeness. Incomplete or incorrect submission packages will be coordinated with the submitting organization.
c. The requirements will be evaluated by Army Publishing Directorate (APD) to determine whether or not there are Army-approved SGML/XML objects and constructs that satisfy the submitted requirements.
d. When Army-approved SGML/XML objects and constructs are identified that satisfy the submitted requirements the objects and constructs will be provided to a submitting organization. These SGML/XML objects and constructs will be used for development or acquisition of Army publications.
e. Document structure and content requirements, or portions thereof, that cannot be satisfied using existing SGML/XML objects and constructs will be identified by the ASRL Registry evaluation process. Army Publishing Directorate (APD) will develop new SGML/XML objects and constructs for the requirements which cannot be satisfied by existing Army-approved objects and constructs.
f. The ASRL Registry evaluation and coordination process consists of technical and functional reviews by the ASRL Registry Administrator, and coordination through the ASRL Registry Registrar. This process identifies technical and functional issues associated with the new objects and constructs. This review process should take no longer than 30 days, depending on the size and complexity of the new SGML/XML objects and constructs. New SGML/ XML objects and constructs are registered and approved for use only after all technical and functional issues have been resolved.
g. Upon notification of approval by the ASRL Registry, the publications developer will use the Army-approved SGML/XML objects and constructs for the development or acquisition of Army publications. Once SGML or XML objects and constructs have been validated and approved by the ASRL, they are made available for access and Army-wide use through the ASRL Library.

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7.4 ASRL operations.

7.4.1 ASRL repository operations. The ASRL holdings, services, technical assistance, and utilities may be accessed in several different ways in order to accommodate all ASRL users, regardless of the type of equipment available to them. The primary means of access for users will be through the World Wide Web (WWW) (refer to paragraph 7.4.3 for additional information). All of the current means of access to the ASRL are provided in Table I. Additional assistance may be obtained through the ASRL/Army Publishing Directorate (APD) home page. All users should apply for a user account at the ASRL/APD home page, by contacting the ASRL Help line or mail to the following address. Once users have applied for an account, the holdings of the Army SGML/XML Registry and Library (ASRL) are available for free downloading from the Files directory on the ASRL/APD World Wide Web Site (http://www.asrl.com).

TABLE I. ASRL access.

Access	Number/Address	Comments
WWW	http://www.asrl.com	ASRL direct.
U.S. Mail	Army Publishing Directorate Attn: AAHS-PAA-PS 9351 Hall Road Building 1456 Fort Belvoir, VA 22060-5447	May request 3.5″ DOS formatted diskettes or .25" UNIX tar formatted tape.
Telephone	Commercial: (703) 428-0504	APD telephone numbers.
	DSN: 328-0504	
Electronic mail	asrl@monmouth.com	ASRL E-Mail address.
Facsimile (FAX)	(732) 578–9136	ASRL FAX number.
Support/Help Line	(800) 880-3773	ASRL help line. Toll-free. Available Monday-Friday, 8:00am - 5:00pm EST (except holidays).

7.4.2 Assistance and problem reporting. For DTD submission assistance (be prepared to state the problem or nature of assistance required), direct problem reporting (be prepared to provide a description of the problem), and customer help and technical assistance contact the ASRL administrator through any of the ASRL-designated means of access in Table I.

7.4.3 Access via the internet (WWW). The homepage for the ASRL is accessed through the Uniform Resource Locator (URL) http://www.asrl.com. The ASRL Home Page provides for a sign-up capability as well as entry to the ASRL Library. By following the links and pointers on the ASRL Home Page (see Figure 4), developers will be able to view, search, find, and download SGML/XML constructs and objects via the Internet WWW. Developers may also perform searches on key phrases. Application for a library card may be obtained on-line through the ASRL Home Page (see Figure 5). A library card is required for downloading SGML/XML constructs from the library (see Figure 6).

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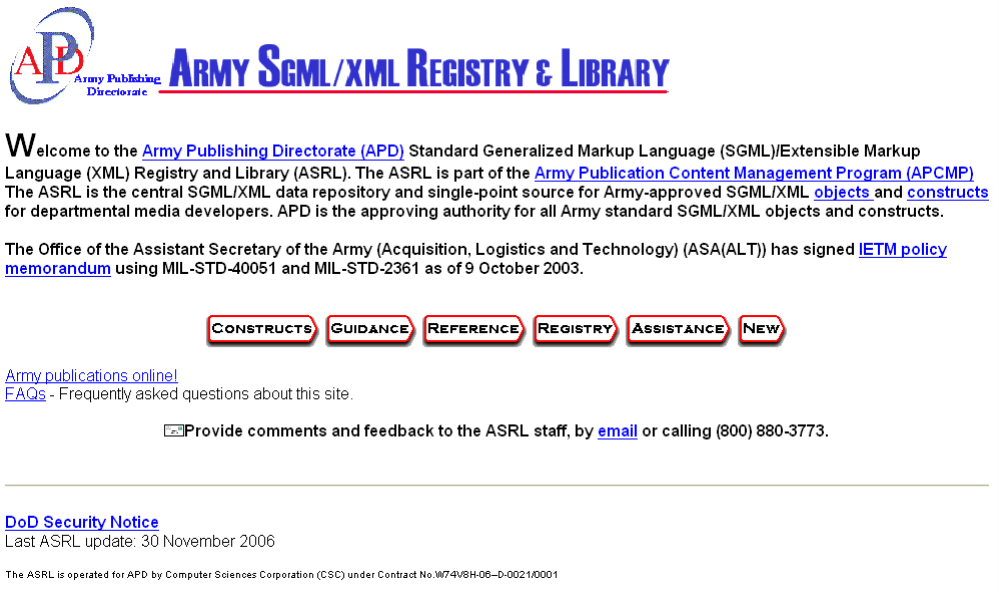

FIGURE 4. ASRL home page.

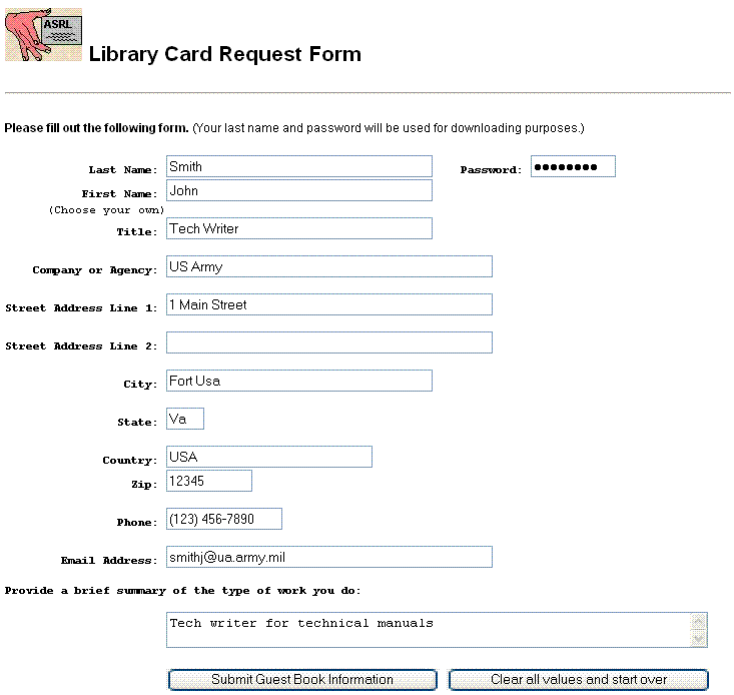

FIGURE 5. ASRL library card request.

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FIGURE 6. SGML/XML constructs.

7.5 Submittal package delivery. Submissions via E-Mail will be directed to the ASRL at asrl@monmouth.com. The submission will contain all nine parts for a variant DTD or eight parts for a new submission in one message. Each part will be delimited on a dedicated line as described in Table 1. The following paragraphs after the table describe each part of the submission package.

TABLE II. Nine parts of a submittal package.

SUBMISSION PART	PART DELIMITER
Submission Rationale	=====PART 1
DTD Information	=====PART 2
DTD Abstract	=====PART 3
DTD	=====PART 4
Parse Log	=====PART 5
Element/Attribute List	=====PART 6
Stylesheet	=====PART 7
Sample Document Instance	=====PART 8
Variant DTD Filter (if applicable)	=====PART 9

7.5.1 Rationale for submission. The submission package will contain a rationale for inclusion into the ASRL.

7.5.1.1 New requirements. A DTD submission not presently covered by registered DTDs will identify the new functional requirements and identify that current registered DTD do not duplicate these new requirements. An example for a Modification Work Order can be found under MIL-PRF-63002H.

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7.5.1.2 Variant requirements. A DTD submission previously covered by a registered DTD will identify the additional requirements and modifications to the registered DTD. The rationale will identify where the changes occurred and reason for the changes.

7.5.2 DTD information.

```
begin assetname => title
=>version => release_date =>
fpi_name => system_name
=>asset_size =>keyword
=>keyword => keyword
=> keyword => keyword
=>support_available =>producer
=>end asset
```

Table III describes the data elements, which comprise the DTD Information part of the submission package.

TABLE III. DTD information.

ELEMENT NAME	DESCRIPTION	EXAMPLE
name	This is the DTD title. It is all capital letters as per asset process requirements.	CECOM XML DTD Variant from MIL-STD-2361A SGML DTD
title	DTD Subject Matter. This format is in normal writing standard.	Provides additional management metadata and XML needs for CECOM
version	Use "-" if no version info.	–
release_date	The date the DTD was released for submission (DD_MMM_YYYY)	20 Sep 2002
alternate_name	This is the DTD's public identifier. Describes DTDs owner [USA-DOD], SGML or XML [XML], variant owner (if applicable) [CECOM from], title [MIL-STD-2361A TM SGML], date (YYYYMMDD) [20020902], and language written [EN] in.	-//USA-DOD//DTD XML Variant CECOM from MIL-STD-2361A TM SGML 20020920//EN
asset_size	The size of the DTD in Kbytes.	15 Kbytes
keyword	Text string within the asset title or asset itself that allows the user to narrow the search for specific topics. Up to five keywords can be used. Each text string can contain up to 45 characters.	
support available	Y or N	Y
producer	Personnel email address	jdoe@army.mil

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DTD information will be submitted in ASCII Format using the format template in Figure 7. Each data entry is to immediately follow the "=>" character in the template.

```
begin assetname => title
=>version => release_date =>
fpi_name => system_name
=>asset_size =>keyword
=>keyword => keyword
=> keyword => keyword
=>support_available =>producer
=>end asset
```

FIGURE 7. ASCII format for DTD information.

7.5.3 DTD abstract. Each DTD submitted will have an abstract. An abstract is a registry term for a user-provided narrative entry describing the DTD. The abstract provides a title, entity definition including formal public identifier (FPI) and file name identifier, overview of the DTD's purpose, and other information identifying the DTD.

7.5.3.1 Abstract Contents. Table IV defines the data to be included in an abstract. The abstract contains free form text, so there is no specified size provided for the different data.

TABLE IV. Abstract contents.

DATA	DESCRIPTION	EXAMPLE
Asset Name	The asset name is the first line of the abstract.	Abstract for MIL-T-38804B Supplement DTD
Entity Name, Entity Text	The entity name and entity text of the entity DTD.	% m38804bsup PUBLIC "-//USA-DOD//DTD MIL-T-38804B SUPP//EN""m38804bsup.dtd"
Description	The actual title (if applicable) and overview of the document corresponding to the asset name.	Preparation of Military Specification Time Compliance Technical Orders ----------
Legal restrictions pertaining to the asset/service	The distribution restrictions for the document and is placed at the end of the abstract.	Distribution Statement A: Approved for public release; distribution is approved for public release with unlimited distribution.

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7.5.3.2 Sample Abstract. Figure 8, displays a sample abstract.

Abstract for MIL-T-38804B Supplement DTD
% m38804bsup PUBLIC "-//USA-DOD//DTD MIL-T-38804B SUPP//EN" "m38804bsub.dtd".
The DTD describes the SGML structure and content tagging conventions for MIL-T-38804B.
The following paragraph(s) describe the requirements for this manual. This specification identifies the preparation requirements for Time Compliance Technical Orders (TCTO) manuals. TCTOs covered by this specification are used in accomplishing and providing a record of any one-time inspection, and replacement or installation of components, retrofit change or alteration to the design or construction of any aeronautical, non-aeronautical, Communication-Electronic (CE), air launched or surface launched missile, space vehicle systems or ground vehicles, their related equipment, sites, facilities, support systems and associated Aerospace Ground Equipment (AGE).
TCTOs may be used to announce each computer program change affecting weapon systems, automatic test equipment, simulator and on-board command and control systems utilizing digital computer systems.
Distribution Statement A. Approved for public release; distribution is approved for public release with unlimited distribution.

FIGURE 8. Sample abstract.

7.5.4 Sample DTD. The DTD for which registration is being requested is to be included in the submission package.

7.5.5 Parse logs. The submission package will contain parse logs (e.g., records of parsing) from at least two SGML or XML validating parsers. If a parser does not produce a report, the submitter will provide the following information:

* Title of the parser
* Date parsed
* Name of person who executed the parser
* Any other pertinent information associated with this parsing (e.g., errors, warnings).

7.5.6 Element/attribute list. The submission package will have a DTD Element/Attribute List. This list defines all the elements and attributes used in the DTD. Figure 9 provides a sample element from the element/attribute list.

7.5.6.1 Element components. Each element has the name, long name, description, and if applicable specification paragraph where requirement was derived.

7.5.6.2 Attribute components. Under each element, list the required and/or optional attributes. Each attribute has the name, description, declared value, and if applicable, default value. If parameter entities are used, indicate where the definitions are located. The declared values are described in either common terminology or as specified in the DTD. The declared value common terminology are:

- Any character - CDATA
- ID - ID
- ID Reference - IDREF
- ID Reference (one or more) - IDREFS
- Pointer - ENTITY
- List (List item value (List item name), ...) - Enumerated List

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- Name Token - NAMETOKEN
- Name Token (one or more) - NAMETOKENS

<maintwp>	Maintenance Work Packages All of the maintenance tasks required to maintain all types of equipment at all maintenance levels are contained within the maintenance work package.
Specification Paragraph:	MIL-STD-40051-4A Para. 5.3.4.9
Required Attribute(s)	
LEVEL	The maintenance level of the work package.
Declared Value :	List (DEPOT (depot level), OPERATOR (operator level), GENSUP (general support), DIRSUP (direct support), UNITLVL (unit level), and INTER (intermediate level))
WPNO	The unique number assigned to this work package by the original contracting activity. This number remains the same when the work package is reused.
Declared Value :	Any character
Optional Attribute(s)	
%WPBODYATT;	Any of the attributes in the associated attribute set may be used with this element. Refer to %wpbodyatt; for a complete description.
%SECUR;	Any of the attributes in the associated attribute set may be used with this element. Refer to %secur; for a complete description.
%TRACKING;	Any of the attributes in the associated attribute set may be used with this element. Refer to %tracking; for a complete description.
%WPRSRC-VALS;	Any of the attributes in the associated attribute set may be used with this element. Refer to %wprsrc-vals; for a complete description.

FIGURE 9. Sample element/attribute list.

7.5.7 Stylesheet. The submission package will have a stylesheet(s). Accompanying the stylesheet is a description of the stylesheet type (e.g. XSL-T, FOSI, DSSSL), identify the processing engine(s) to derive the output, information for obtaining the processing engine(s), and instructions to execute the stylesheet.

7.5.8 Sample document instance. Each submission package will have a sample document instance. The document instance will be used by the ASRL Register to verify the document instance parses against the new or variant DTD, output is produced from the stylesheet, and if applicable, the filter produces a compliant document instance to the specified Army DTD.

7.5.9 Variant DTD filter. Each variant DTD submission requires a filter to export the variant DTD source data to a specified Army DTD source data. Accompanying the filter is a description of the filter, identifying the processing engine(s) to transform the source data, information for obtaining the processing engine(s), and instructions to execute the filter.

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PART III
INTRODUCTION TO XML

8 INTRODUCTION TO XML.

8.1 Introduction. This section provides a foundation for working with, and understanding, XML. Handbook users will find this section invaluable in identifying and defining the uniqueness peculiar to XML. Users are provided information regarding "what XML is", and why its use (and "reuse") is beneficial for sharing document-based information among applications and between different computer platforms. There are references within the section that point the user to other areas of information and guidance, such as the parts for XML Tutorial (see 8.2.6) and References. There are also graphic examples of the concepts under discussion to assist in the handbook user's understanding of the material being presented.

8.1.1 XML experience. There are varying degrees of XML experience and expertise (see 8.3) required for different levels of function processes. For example, a publication author and his manager need not necessarily have the same XML background. This section addresses the levels and types of XML experience and expertise that may be needed for different implementation considerations.

8.1.2 Four primary components. The user will find easy to understand discussions regarding the four primary components of XML composition: the Document Type Definition (DTD), document instances, the XML declaration, and XML markup. Each of the components is described as to its role in the composition process. The XML markup discussion breaks markup into its essential parts (e.g., tags, elements, attributes, and entities), and describes how the DTD structure and the document instance comprised of these parts are validated through the parsing process.

8.2 What is XML. Extensible Markup Language (XML) "is the universal format for structured documents and data on the Web". Extensible Markup Language (XML) is a simplify subset of SGML and has Hyper Text Markup Language (HTML) capability to process information on the Web. It is an international, platform-neutral standard for creating and using documents and information across multiple software applications and computer platforms. XML establishes a consistent language and terminology which provides publications developer and user activities the capability to share and reuse publication information, and to preserve the organization and content of documents. The application of XML in accordance with MIL–STD–2361 and this handbook is fully compliant with the international, federal, and REC-xml-20060816 standards for XML used throughout the Government and industry (see Section 2). Refer to the 1.5 for additional information regarding XML.

8.2.1 XML requirements. MIL-HDBK-2361 establishes XML requirements which reflect the functional requirements associated with the different types of Army publications. For example, the XML requirements contained in the technical manual (TM) segment of MIL-HDBK-2361 reflect the functional requirements contained in MIL–STD–40051–1/–2. XML provides capabilities for developers and users of publication information to output the information on a variety of media (e.g., paper, CD-ROM, WWW, etc.)

8.2.2 Why use XML. XML allows developers to update and maintain critical source information over the life-cycle of weapon systems and other equipment. XML makes it easy and straightforward to use SGML on the Web and in turn provides a better means for the reuse and exchange of information among its developers and users. XML is an industry standard for sharing document-based information among applications and is compliant with open systems environments. See 1.5 for more information concerning the advantages of using XML.

8.2.3 The digital publications development (DPD) program concept. The Digital Publications Development (DPD) Program provides a seamless flow of integrated operations, diagnostics, maintenance, and training information from developer to the soldier. XML, as applied to the Digital Publications Development (DPD) Program concept and implemented by MIL–STD–2361, provides the following:

a. Description of the logical structure of documents in unambiguous syntax.
b. Assurance of automated quality control over adherence to that structure.
c. Delivery and storage of publication text in an easily maintained and updatable form.

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d. XML well form document allows data to be chunked into smaller bits of information and distributed.
e. Vendor, software, and platform independence and XML supports a wide variety of applications.

8.2.4 XML reference. XML reference information can be found in Section 2.3 of this handbook.

8.2.5 MIL–STD–2361 XML tags. The MIL–STD–2361 DTDs contain two specific types of XML tags: structural and content tags. Structural tags identify data by its place in the hierarchy of the document and by how the material is formatted on the page (i.e., primary paragraph, subparagraph, list, or table). Content tags identify material by its functional use or the type of data (i.e., maintenance task, circuit alignment, controls and indicators, or components of end item table).

8.2.6 XML tutorial. A tutorial for MIL–STD–2361 XML application and use is contained in Section 9 of this handbook. The tutorial section in this version of the handbook is intended to provide an overview of the use of XML. There is a large selection of books on XML. A few of those non government books containing XML guidance can be found in 2.3.1.

8.3 XML experience and expertise. There are varying degrees of XML experience and expertise required for different functions. For example, a publications author and his manager need not necessarily have the same XML background. This portion of the handbook will address the levels and types of XML experience and expertise that may be needed for different implementing considerations. The following considerations offered are applicable to both Government and private industry publications developers, authors, and other functions involved with the development and acquisition of publications and publication services.

8.3.1 Types of XML knowledge. This paragraph addresses the types of XML knowledge that will be required by personnel at different functional levels. The functional personnel levels and types addressed have been the primary players in the MIL–STD–2361 operational testing conducted to date. This type of information will be expanded in later revisions of the handbook, as the operational test results become more definitive.

8.3.1.1 Publications manager. A publications manager needs a good overall understanding of XML. The managers experience and depth of XML knowledge may not be as great as the publication authors or computer specialists, but it should be sufficient to evaluate their technical input and make informed decisions. These decisions may involve the evaluation and selection of an XML authoring/composition system for the organization, or contracting work to an XML conversion contractor, a publications developer, or XML consultant. XML knowledge provides the manager with tools for evaluation of technical XML input, whether it comes from within his own organization or some other source.

8.3.1.2 Publications author. The publications author will be the primary user of XML. The author will be authoring new material, developing publication revisions, and working with legacy data converted (or scheduled for conversion) to XML. The author will require the ability to read, understand, and work with XML concepts, rules, Document Type Definitions (DTDs), and stylesheets. See 8.5.1 and 8.5.4.7 for information on DTDs and stylesheets.

8.3.1.3 Publications editor. The editor will be editing new material, publication revisions, and working with legacy data converted (or scheduled for conversion) to XML. The editor will require the ability to read, understand, and work with XML concepts, rules, Document Type Definitions (DTDs), and stylesheets. See 8.5.1 and 8.5.4.7 for information on DTDs and stylesheets.

8.3.1.4 Computer specialist. computer specialist (including programmers and System Manager) will be the key technical person(s) for accomplishing the detailed technical requirements associated with computer equipment used to develop XML publications (e.g., the author/editor, composition system, database, etc.). The computer specialist XML knowledge should be sufficient to allow interpretation of DTDs in order to develop composition scripts, conversion rules, and database management requirements.

8.4 XML overview.

8.4.1 XML is a markup language. Extensible Markup Language, XML Version 1.0, as specified in REC-xml-20060816 is official recommended by the World Wide Web Consortium W3C. The Extensible Markup Language (XML) is a subset of SGML. World Wide Web Consortium's (W3C) goal "is to enable generic SGML to be served, received, and processed on the Web in the way that is now possible with

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HTML. XML has been designed for ease of implementation and for interoperability with both SGML and HTML". A MIL–STD–2361 application of XML should contain: document type declaration, 8.5.1 DTD, 8.5.2 document instance, a stylesheet or an XML Schema. The fact that the markup used in XML is standardized facilitates the exchange of tagged documents between software applications. XML markup is commonly spoken of as tagging, because tags are inserted into the text.

8.4.1.1 Well-formed XML document. Compliant with REC-xml-20060816 requirements, the basic rules for writing well-formed XML documents.

a. Start tags are required to have corresponding end tags
b. Elements can not overlap
c. Element names should start only with letters and underscores. Also, element names may contain letters, numbers, hyphens, periods, and underscores.
d. XML tags are case-sensitive
e. Empty elements are required to have either have an end tag or close the empty tag with (/>)
f. Reserved characters (< &> " ') are replaced with corresponding character sequence (< & > " ')
g. Each XML document is required to have a unique root element
h. Each attribute name in an element is unique
i. Each attribute name is followed by a value indicator (=) and a quoted string

8.4.2 XML is not a processing language. XML is not a processing language. It does not perform calculations, actions and decisions in the way that a computer program, written in a language like C++ or Java does. Instead it is a markup specification language with which you can design ways of describing text information or data information usually for storage, transmission, or processing by a program. XML supplies a structured, tagged text database upon that a software application can act. It is like having the record definition of a traditional database without the ability to sort and retrieve. Sorting and retrieving in XML are accomplished by software that “speaks” XML and can respond to Document Type Definitions (DTD).

8.4.3 Understanding XML as a text database. Understanding XML as a text database reinforces its separation of content markup and specific presentation instructions. XML is ideally suited to multiple presentation formats and media platforms. XML aims to encourage descriptive markup rather than process-oriented markup. For instance,MIL–STD–2361 uses the tag <remove> rather than a code that describes the format of this subhead; the fact that the tag will be presented as 10 point, boldface Helvetica is inferred rather than expressed directly (in MIL–STD–2361 the stylesheet provides this information).

8.5 XML composition. XML is used to define the structure and contents of a set (or class) of documents, whether those documents consist of one page or a thousand pages. This definition is accomplished and portrayed through the utility in the different parts that comprise MIL–STD–2361 XML.

8.5.1 Document type definition (DTD). A functional unit for MIL–STD–2361 XML is the DTD. MIL–STD–2361 DTDs form a generalized picture of the types of data found over a whole range of publications, rather than descriptions of the specific contents in any one publication.

a. The DTD declares which elements make up the document, the possible sequences in which they will be found, the number of occurrences in that sequence, and what lower-level elements each element may contain. In turn, lower-level elements are described in those same terms. In general, text characters form the lowest-level content of a document.
b. The DTD describes the hierarchical relationships of the elements. For instance, the maintenance information chapter in MIL–STD–2361 module contains several possible types of work packages; one type of work package, the maintenance work package, contains maintenance tasks that consist of procedures made up of steps, and so on.
c. An element can occur at different levels of the hierarchy or in different branches of the hierarchy. For example, lists can occur in Step1, Step2, or Step3 (different levels of a single branch); and in paragraphs and warnings (different branches).

8.5.2 Document instances. What we commonly think of as documents, actual pages of text and graphics, are known as document instances. In the context of a TM developed in XML in accordance with

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MIL–STD–2361, a document instance is any part of a content volume(s) with a complete structure as dictated by the relevant DTD. For instance, the minimum structure of an information chapter t includes at least one work package and any mandatory lower-level markup. The “document” described in a DTD is a generalized, virtual template. In simplified terms, a document instance consists of:

a. Document instance contains an XML declaration
b. Text marked up with XML tags.
c. Illustrations incorporated by tagged references to graphic files.
d. Every document instance either incorporates or references it's governing DTD.
e. If the document instance uses external file or text entities not defined in the DTD, those entities are to be declared in the document instance.

8.5.3 XML declaration. Every document and DTD is accompanied by an opening XML declaration. The XML declaration describes some of the most general properties of the document and telling the XML processor that it needs an XML parser to interpret this document. Each DTD in MIL–STD–2361 uses the defined XML declaration in MIL–STD–2361.

8.5.4 XML markup. XML markup includes element tag names, element attributes, and reference entities. These markup categories each have a standardized syntax defined in the ISO 8879 standard. The MIL–STD–2361 DTDs use this standard syntax.

8.5.4.1 XML tags. Tags contain two parts: start-tags and end-tags as shown in Table V. Where the end tag should occur is guided by the DTD hierarchy. It cannot be inserted until all required content of an element has been included. For instance, the end tag of a maintenance task in a work package cannot occur until at least one procedure has been inserted, insuring that a maintenance task has associated text.

EXAMPLE:

• DTD fragment:

```
<!ELEMENT crew (#PCDATA)>
```

• Document instance fragment:

```
<crew>2.6</crew>
```

TABLE V. XML Tags.

Order	Part	Description
1	<	Start Tag Open Delimiter
	</	End Tag Open Delimiter
2	crew	Generic Identifier
3	>	Tag Close Delimiter
4	/>	Empty Element Tag

8.5.4.2 Markup minimization. Compliant with REC-xml-20060816 requirements, XML start tags are to have corresponding end tags. Empty elements are to be tagged with the empty tag (/>).

• DTD fragment for corresponding end tags:

```
<!ELEMENT date (#PCDATA)>
```

• Document instance fragment for corresponding end tags:

```
<date>11 September 2001</date>
```

• DTD fragment for an empty element:

```
<!ELEMENT figno EMPTY>
 <!ATTLIST figno  idref IDREF #REQUIRED>
```

• Document instance fragment for an empty element. XML allows two methods of tagging an EMPTY element

```
<figno idref="f25"/> or
<figno idref="f25">The parser will not allow any content, like this,
between the start and end tag of an EMPTY element.</figno>
```


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8.5.4.3 Distinct categories. Elements are distinct categories of content, such as “work package,” “procedure,” “stepl,” “table,” and “tools.”

a. Elements range from the largest divisions of the document down to single words if they are significant data. An important criteria for creating an element is its usefulness for applications such as database retrieval and page composition. An element can contain other elements, such as a warning summary that contains general paragraphs, warnings, hazard symbols, etc. Elements in running text, such as “part number,” often contain only character data, which, in the DTD, is identified as “#PCDATA,” parsable character data. Elements are identified by an element name. When that name occurs in text, surrounded by characteristic syntactical markers, it is called a tag.
b. In the DTD, an element declaration includes an element name, markup minimization rules, the contents of the element, and any attributes of the element. The contents portion of the declaration is often called a “content model,” which refers to the generalized nature of this representation of all possible contents and sequencing. The content model is surrounded by parenthesis.
```
<!ELEMENT surwp (wpidinfo, wpinfo, geninfo?, %alert;, surtsk+)>
```
c. The MIL–STD–2361 elements identify generic contents rather than precise, literal contents. For instance, <remove> identifies “a removal task” rather than “removal of the Abrams M-1 rear exhaust components.” On the other hand, MIL–STD–2361 uses <remove> rather than <para0>, a structural text object from the base tag set in MIL–PRF–28001. In the base MIL–PRF–28001 DTD, <para0> could be used indiscriminately for removal, assembly, inspection, install, repair-replace, and all other types of maintenance tasks. This specific content orientation of MIL–STD–2361 provides level of granularity essential to full source data sharing and reuse.

8.5.4.4 Attributes[1]. Elements can be qualified by adding attributes. Attributes are part of the element declaration in the MIL–STD–2361 DTD and address such aspects of the data as security classifications, maintenance levels, reference IDs, and column widths of tables. An attribute has a name and an expected data type specified in the DTD and also has its status as required entry, implied value, or defaulted value defined. REC-xml-20060816 and ISO 8879 define several data types, depending on whether the value consist of numbers only, numbers and alpha characters, reference XML elements, cross-references, or an unlimited text string. Rather than a data type, the DTD can also declare a discrete list of legal values, or it can use a Boolean true-false test (any value other than “0” represents true).

```
<!ATTLIST pmcstable
    use-manhours      (yes | no)   'no'
    applicable         IDREFS                          #IMPLIED
    assocfig           IDREFS                          #IMPLIED
    changeref          IDREFS                          #IMPLIED
    comment            CDATA                           #IMPLIED
    delchlvl           CDATA                           #IMPLIED
    id                 ID                              #IMPLIED
    idref              IDREFS                          #IMPLIED
    inschlvl           CDATA                           #IMPLIED
    security           (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk           CDATA                            #IMPLIED
    tocentry           (0 | 1 | 2 | 3 | 4 | 5 )       "1"
```

8.5.4.4.1 Yes-no. A value of an attribute may be given as yes or no in which “yes” implies true and “no” implies false. The attribute **use-manhours** in the example above for the element ***<pmcstable>*** specifies whether or not the PMCS requires the man-hours entry. A “yes” value indicates that the man hours entry is required and a “no” entry indicates that it is not.

[1] This section will include examples of parameter entities. They are used by the DTD author to simplify the construction of the DTD. Parameter entities are a significant feature of XML which is why they are included in this Introduction to XML section. However, TM/IETM authors are mainly concerned with the tag or attribute names that the entity represents. Therefore, in order to familiarize the TM/IETM author with these tag and attribute names, the parameter entity notation will only be used sparingly in subsequent sections of this handbook.

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8.5.4.4.2 Value list. The attribute can contain a discrete list of legal values. The attribute **security** in the example above for the element ***<pmcstable>*** specifies the security level for the table as being unclassified ("uc"), for official use only ("fouo"), classified ("c"), secret ("s"), or top secret ("ts"). Attributes can have a default if specified. The attribute **toentry** has a default value of "1".

8.5.4.4.3 Name token (NMTOKEN). The attribute value NMTOKEN is a string of characters that begins with a letter and can contain numbers, letters and certain punctuation. See the attribute **tabstyle** in the following example for the element ***<table>***.

```
<!ATTLIST table   tabstyle NMTOKEN #IMPLIED
    %stdinfoatt;
    frame (top | bottom | topbot | all | sides | none) #IMPLIED
    colsep (0 | 1) #IMPLIED
    rowsep (0 | 1) #IMPLIED
    orient (port | land) #IMPLIED
    label CDATA #IMPLIED>
```

8.5.4.4.4 Name token list (NMTOKENS). The attribute value NMTOKENS is a sequence of one or more NMTOKENs separated by spaces. In the following example , the element ***<testblock>*** contains the attribute **valueloc** having the value NMTOKENS that contain one or more locations (usually other properties) that supply the value of the current property.

```
<!ATTLIST testblock
    type (yes | no | pass | fail | true | nottrue | value | unantic) #REQUIRED
    valueloc NMTOKENS #IMPLIED
    valuetype (boolean | string | sequence | set | real |
    integer | float | nil | input | outcome) #IMPLIED
    value CDATA #IMPLIED
    branch ID #REQUIRED
    branchlabel CDATA #IMPLIED
    branchfrom IDREFS #REQUIRED
    branchto IDREFS #REQUIRED>
```

8.5.4.4.5 Unique identifier (ID). The attribute value "ID" is a special type of attribute that gives an element a label guaranteed to be unique in the document. The instance is not allowed to have two elements having the same "ID" attribute value. It is used as a cross reference attribute with the attribute value identifier reference "IDREF" or the attribute identifier reference list "IDREFS". For example, the value "ID" of the attribute **branch** contains the unique number assigned to this test block (see 8.5.4.4.4).

8.5.4.4.6 Identifier reference (IDREF). The attribute value "IDREF" is the unique identifier reference referring to the ID of another element. The attribute **wpref** value "IDREF" is used as a cross reference attribute with the attribute value "ID" of another element. When an "IDREF" is used and there is not an "ID" with the same value as "IDREF", the parser reports an error. See the following example of the attribute value **IDREF** of the attribute **wpref** of element work package number ***<wpno>***.

```
<!ATTLIST wpno
    wpref IDREF #REQUIRED
    %secur;>
```

8.5.4.4.7 Identifier reference list (IDREFS). The attribute value "IDREFS" is a space-separated list of "IDREF" values. The attribute value of **branchfrom** and **branchto** of the element ***<testblock>***(see 8.5.4.4.4) are examples of cross reference identifiers "IDREFS" to other branches. Every "IDREF" in the list should match an "ID" value in the instance or the parser reports an error.

8.5.4.4.8 Entity name (ENTITY). Used to reference a general entity. The entity is required to be declared to identify the data being declared. See 9.3.5.3 on details using the graphic data type. The example above ***<wpno>*** (see 8.5.4.4.6) contains the entity set security **%secur;** that contains one or more attributes.

8.5.4.4.9 Entity name list (ENTITIES). Used to reference general entities. The general entity is required to be declared to identify the data being declared. See 9.3.5.3on details using the graphic data type. The

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element ***<pms-item>*** contains the entity set **%bodyatt;** that contains attributes but also contains the entity set references **%changelevel;**, **%refs;**, and **%secur;**.

a. Entity set for ***<pms-item>***

```
<!ATTLIST pms-item  %bodyatt; >
```

b. Entity set **%bodyatt;**

```
<!ENTITY % bodyatt " %changelevel;
   %refs;
   skilltrk    CDATA  #IMPLIED
   %secur;">
```

8.5.4.4.10 Character data (CDATA). The attribute value "CDATA" contains any character data. Character data can be a number, number and alpha character but can not contain elements or processing instructions. The element electronic mail (***<email>*** see 49.1.4.1.10) contains an attribute **address** that contains the value "CDATA" that specifies the address of the electronic mail. See the following example:

```
<!ATTLIST email address CDATA #REQUIRED>
```

8.5.4.5 Entities. XML allows the user to store text as entities that can then be referenced in the document instance by their entity names. Entities content may be any length (up to 300,000 characters). In MIL-HDBK-2361, for example, the entire boilerplate explanation of the maintenance allocation chart is declared as an entity and can be included in the work package by a single entity reference. Entities are also often used to reference external files, such as graphics. Indeed, users working with MIL-HDBK-2361 documents, although they do not need to be able to write DTDs (which may use parameter entities), do need to learn the syntax for declaring general entities in order to include specific illustrations within the document instance. The DTDs in MIL-HDBK-2361 also include entity declarations of standard ISO 8879 character sets. A non-keyboard character, such as "plus-or-minus" or "ohm," should be inserted into the document instance through one of the entity references in these ISO character sets. The non-keyboard character does not have to be declared in the DTD since it is an entity of ISO 8879.

a. Text entity example:

(1) In the DTD:

```
<!ENTITY surwp.surmat.chkeqp.inspect'>
<step1>
<para hcp="no" esd="no"> Inspect the equipment for damageincurred during
shipment.  If the equipment has been damaged, reportthe damage on
<extref docno="SF 361" posttext=", Transportation DiscrepancyReport"/>.
</para>
</step1>
<step1>
<para hcp="no" esd="no">Checkthe equipment against the packing slip to see
if the shipment iscomplete.  Report all discrepancies in accordance with
applicableservice instructions (e.g., for Army instructions, see
<extref docno="DAPAM 750-8"/>).
</para>
</step1>
<step1>
<para hcp="no" esd="no">Check to see whether the equipment has been modified.'
</para>
</step1>
```

(2) Referenced in the instance: **&surwp.surmat.chkeqp.inspect;**

8.5.4.6 Validation of XML markup. A distinctive characteristic of XML is the parsing process, which allows the DTD structure and the document instance markup to be tested. Unlike most proprietary markup languages, XML is self-validating through the medium of the XML parser. Refer to paragraph 27.2.4.2 for additional information on XML parsing.

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a. In order to proof document instance markup, the DTD is first parsed. Parsing creates a file that encodes the sequence, number, content model, and required tagging rules of the DTD. In short, the full structure of the DTD. Errors in the DTD are identified during this parsing process to ensure the DTD conforms to the REC-xml-20060816 and ISO 8879 standards.
b. The encoded file is used there after whenever a document instance governed by that DTD is parsed. Parsing a document instance checks that all conditions of the DTD have been met, as well as the general rules of XML. The parsing process guarantees that any XML application software receives files with the expected structure, since the operation of XML software is built on the tags context within the document.

8.5.4.7 Stylesheets. The stylesheet provides a set of formatting characteristic values used to rigorously describe composition processing functions to be performed on the elements of a text document to provide the format style required by a functional specification or standard, such as MIL–STD–40051–1/–2, AR 25-30 or TRADOC Reg 350-70. A stylesheet (i.e. FOSI, XSL-FO) delivered with the document should contain values of characteristics for every tag used in the DTD, in every context in which the tag has a unique formatting requirement, and with its attributes if they affect the formatting.

8.5.4.7.1 Generated text. Stylesheets may provide specified text that is automatically output by the stylesheet for your application during the formatting of the document. Text can be generated by an element or an attribute when applied. Generated text can be a word, title or a paragraph of text.

8.5.4.7.1.1 Generating text with elements. Elements can be used to generate text that are specified by the assigned stylesheet as a word, title or a paragraph of text. The following are examples of elements that are commonly used in MIL-STD-2361 to generate text:

• Chapter title – ***<gim>***

```
<gim>General Information, Equipment Description and Theory of
Operation....</gim>
```

• Requirement title – ***<reporting>***

```
<reporting>Reporting Errors and Recommending Improvements Statement
....</reporting>
```

• Maintenance task title – ***<disassem>***

```
<disassem>Disassembly....</dissassem>
```

• Warning – ***<warning>***

```
<warning>WARNING....</warning>
```

8.5.4.7.1.2 Generating text with attributes. Attribute values of elements can be assigned specific text to be generated by the assigned stylesheet when the attribute is applied. The following are examples of attributes that are commonly used in MIL-STD-2361 to generate text:

• Bullet list “bullet” – Specifies whether the item is preceded by a bullet character of the element ***<randlist>*** "•"

```
<randlist bullet="1"><item>Example of the attribute "bullet".<item>
<randlist>
```

• Level “level” – Specifies the level of the information module contained in the element maintlvl, in this case, "General Support".

```
<maintlvl level="gensup">
```

8.5.4.8 Extensible stylesheet language for formatting objects (XSL-FO). Extensible stylesheet language for formatting objects is a pagination markup language describing a rendering vocabulary capturing the semantics of formatting information for paginated presentation. The paginated presentation may be displaying multiple separated pages on a screen, on paper or audibly providing the format style required by a functional specification or standard, such as MIL–STD–40051–1/–2, AR 25-30 or TRADOC Reg 350-70.

8.5.4.9 Formatting output specification instance (FOSI). FOSIs can be written following MIL–STD–2361. FOSIs should follow the OS DTD included with MIL-PRF-28001.

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9 XML TUTORIAL.

9.1 Scope. This section contains tutorial-type information on MIL-HDBK-2361 XML. The information contained in this version of the handbook is intended to provide a overview of the use of XML.

9.2 Applicable documents. Refer to paragraph 2

9.3 Document type definition (DTD). DTDs describe the structure and content of a document. A DTD is comprised of document elements and their relationships elements, attributes, entities, etc.

9.3.1 Document elements. Document elements begin with document element declarations. Element declarations may be identified in the DTD by the markup declaration open (MDO) "<!" and the reserved word "ELEMENT", which is always the first word in the element declaration. The next component of the element declaration is the element name ("deflist" in the following example). The element declaration component is either declared content or a content model. Declared content is either "PCDATA" or "EMPTY." The MIL-STD-2361 application makes use of the concept of "EMPTY" elements (for example the table of contents). The content model is a model group that identifies the elements as having either mixed content or element content. " Sequence and occurrence indicators contained in the content model determine whether or not a sub-element will be in the document (see 9.3.3) and in what order the sub-elements may occur. After the content model is completed the last component is the markup declaration close (MDC) ">". An example of an element declaration is as follows:

```
<!ELEMENT deflist (title?, (term.def)+)>
```

9.3.2 Parsable character data #PCDATA. The reserved name #PCDATA is used inside the content model to indicate zero or more parsed data characters. Note no information is required to be entered to maintain a valid document. #PCDATA contains the narrative (content) text for the document, non-keyboard character general entity and text general entity, but no sub-elements are allowed. unless the current content model or higher-level permits inclusions.

9.3.3 Sequence and occurrence indicators. Sequence and occurrence indicators determine the required (mandatory) content and element groups for DTDs. Required content parameters include both the sequence and occurrence of elements within a DTD, and are identified within the DTD by standard XML codes.

9.3.3.1 Sequence indicators. Sequence indicators determine how the elements are arranged within the document. Elements which will occur in a particular order are separated by comma's (,), such as the comma after "title?" in the example above. In this example, the "title?" of the definition list ***<deflist>*** will precede term and definition ***<term.def>***. (The "?" is an occurrence indicator that indicates the title is optional). Elements that have alternative relationships (e.g., use one but not the other), are designated by vertical bars (|), such as "(xref | extref)". Elements that will be included in the document, but in no particular order are indicated by the ampersand (&) such as "(name & date)" which means the name could be followed by the date or the date could be entered first and followed by the name.

9.3.3.2 Occurrence indicators. Occurrence indicators determine the number of times an element will occur in a document. An element that may or may not occur (e.g., optional) and that, if used, will occur only once, is designated by a question mark (?). An optional element that may occur zero or more times is designated by an asterisk (*). A mandatory element that will occur at least once, or that may occur many times, is designated by a plus sign (+). An element with no occurrence indicator is a mandatory element and can occur once only.

9.3.3.3 Content model element sub-groups. Element sub-groups within a content model are enclosed in parentheses. Inside the parentheses, individual elements are governed by the same sequence and occurrence indicators as other elements, creating a sequence/occurrence model within the sub-group. The entire sub-group is governed by the sequence indicator before the open parentheses and the occurrence indicator after the close parentheses. Examples of markup for various combinations of sequence, occurrence, and element sub-groups are provided..

a. When elements are mandatory in sequence and occurrence: (name, xref) = <name> followed by an <xref>.

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b. When elements have a mandatory sequence, but the occurrence of second element is optional: (name, xref?) = <name>, which may be followed by a single optional <xref>.
c. When elements have a mandatory sequence, occurrence of third element is optional: (name, xref, extref?) = <name> followed by one <xref>, which may be followed by an optional <extref>.
d. When elements have an optional sequence and occurrence: (name? | xref?) = either one <name> or one <xref> may occur but neither has to occur.
e. When elements have an optional sequence and occurrence: (name | xref)* = either a <name> or an <xref> may occur but neither has to occur; either or both of the tags may occur many times with no specified sequence; this model could be empty of content.
f. When elements have a mandatory sequence and occurrence, but with alternative second elements: (name, (xref | extref)) = <name> followed by either one <xref> or one <extref>.
g. When elements have a mandatory sequence and minimum occurrence requirements for the alternative second element: (name, (xref | extref)+) = <name> followed by at least one <xref> or <extref>; either or both the <extref> and <xref> can occur more than once, in no particular order.
h. When elements have a mandatory sequence but occurrence is required only for the first element: (name, (xref | extref)*) = <name>, which may be followed by an <xref> or <extref>; either or both the <xref> and <extref> can occur more than once, in no particular order, but neither has to occur.
i. When elements have a mandatory sequence but are modified by having no mandatory occurrences: (name*, (xref | extref)?) = <name> may occur any number of times, but is not mandatory, followed optionally by either a single <xref> or <extref>; this model could be empty of content.

9.3.4 Attribute declaration. Attribute declarations is prefaced in the DTDs by the (MDO) "!" and the XML reserved word "ATTLIST". The next component is the element to which the attributes apply ("ginfowp" in the following example). The element is followed by the attribute name ("wpno" in the following example). The attribute name is followed by the allowable attribute values ("CDATA" in the following example). The last part of the attribute declaration is the default value or value source keyword ("#REQUIRED" in the following example). The attribute name, value and declaration may have multiple attributes to better qualify the XML element to which it is associated. The example, upon which the above descriptions are based, is as follows:

```
<!ATTLIST ginfowp wpno CDATA #REQUIRED>
```

9.3.4.1 Attribute value types and reserved XML names. The attribute value types and reserved XML names should be used in the context presented and should follow the rules contained in the definitions associated with the respective terms.

a. NMTOKEN - Will consist of either valid digits or letters.
b. NMTOKENS- Contains multiple occurrences of NMTOKEN.
c. CDATA - Free text that is not parsed internally.
d. ID - Unique identifier that may be referenced by the attribute with NAME value.
e. IDREF - Used to generate cross-references and to link elements, such as footnote text and foot note reference. The IDREF references the ID value.
f. IDREFS - Contains multiple occurrences of IDREF.
g. ENTITY - Used to reference a general entity. The general entity is required to be declared to identify the data being declared. See 9.3.5.3 on details using the graphic data type.
h. ENTITYS - Contains multiple occurrences of ENTITYS.
i. List - List of text values within parentheses usually, separated by "|" (the "or" indicator). Only one value will parse for each attribute instance and only values that are contained in the list are valid.
j. True-False (0 |1) - A value of an attribute may be given as true or false in which "0" implies "no" or "false" and "1" (or any non-zero number) implies "yes" or "true.
k. List - List of text values within parentheses usually, separated by "|" (the "or" indicator). Only one value will parse for each attribute instance and only values that are contained in the list are valid.
l. Notation - Identifies permissible notation types.
m. Name Group - Lists permissible values.

9.3.4.2 Keyword attribute defaults. The following keyword attribute defaults are used by MIL–STD–2361

a. #REQUIRED - Indicates that the value is required to be supplied in the instance.

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b. #IMPLIED - Indicates that a value is not required to be included in the instance. #IMPLIED should be used when a system is expected to resolve the current attribute value, or when it would be difficult to supply a specific value for each attribute.
   (1) In the MIL–STD–2361 application, an example of the first case is the security attribute "security". When the parent element defines the "security" attribute, the system will imply the value for the attribute to be the same as the parent attribute, unless otherwise defined.
   (2) In the MIL–STD–2361 application, an example of the second case is the attribute list of cross reference (***<xref>***). Since the reference may be to a table (attribute "tableid IDREF #IMPLIED"), or a figure (attribute "figid IDREF #IMPLIED"), but never to both. Therefore, one cannot require any of these attributes be supplied. If the user does not supply an appropriate ID, the cross-reference will not be expressed.

c. Specified default value - When a value is supplied in the keyword attribute default, the system will use the value if no value is entered for the element. In the following example, the attribute "insert" has the default value of "none".

```
<!ATTLIST null
   insert (NA | NR | dash | secure | none) "none"
    %secur;>
```

9.3.5 Entities. There are two types of entity declarations: parameter and general.

9.3.5.1 Parameter entity. Parameter entities are often used as a short cut in the DTD in order to insert common DTD declaration data. Parameter entity declarations will be prefaced by the (MDO) "<!" and the XML reserved word "ENTITY." The reserved word, ENTITY will be followed by at least one space, then a "%" followed by a space, followed by the entity name. When referenced, typically in a DTD, the entity name is preceded by the percent sign ("%") and followed by a semi-colon (";"). There is no space between the percent sign and the entity name when the entity is referenced.

9.3.5.1.1 Replacement text entities. Parameter entities can be used within the DTD to reference often used content such as:

```
<!ENTITY % titldtext "(title, (subtitle?, (%p))+)">Referenced as:
<!ELEMENT eqpdesc (%titldtext;)+>Resolved as:
<!ELEMENT eqpdesc ((title, (subtitle?, (para | specpara))+))+>
```

9.3.5.1.2 Nested entities. Parameter entities may be nested. That is, one entity may occur within another entity declaration. In the MIL–STD–2361 example, the entity ***%text;*** reference ***%pcdata;***,***%partinfo;%misc;***. In essence, ***%pcdata;***,***%partinfo;*** and ***%misc;*** have been "nested" within ***%text;***. Note however, an entity being referenced, should have been declared prior to its reference.

```
<!ENTITY % text "%pcdata; | %partinfo; | %misc; | change">
<!ENTITY % pcdata "#PCDATA | emphasis | extref | ftnote | ftnref  | indxref |
      subscrpt | supscrpt | xref">
<!ENTITY % partinfo "cageno | callout | dodac |
     dwgname | dwgno  | modelno | name | nsn | partno | sc">
<!ENTITY % misc "ctrlind | ctrlind-val | lubricant | pageloc  | symbol | torque |
     voltage | null">
```

9.3.5.1.3 External files. Parameter entities are also used to declare external files that may be referenced in the DTD. Once referenced, the content of that external file is then considered to be a part of the DTD being defined. External files may contain sets of "boilerplate" text that have been declared as general entities (see 9.3.5.2), sets of graphic entities (see 9.3.5.3), or ISO character sets (see 9.3.5.4). The entity is required to be declared then referenced in the DTD. In the following example ***%boilertxt;***, %; references the ***%boilertxt;*** entity which has given a name and location to an external file. The contents of that file "replaces" the entity reference to it. The boilerplate text entity described above contains numerous general entities that have already been defined in the MIL–STD–2361 application. Some DTDs may declare all

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general entities directly within the document rather than making use of the parameter entity reference. (These entities may be referred to as direct entities).

```
<!ENTITY % boilertext PUBLIC " -//DA-USAPA//ENTITIES MIM BoilerPlate
REV 1.0 19970301//EN"> %boilertext;
```

9.3.5.2 General entity. General entities are frequently used to enter commonly used text into the narrative. Within the DTD, general entity declarations are prefaced by the (MDO) "<!" and the XML reserved word ENTITY. The reserved word ENTITY will be followed by at least one space and the entity name. The entity name will be followed by at least one space and the text of the entity which is contained within quotation marks (typically double quotation marks are used; however single quotation marks may also be used and are required to be used if the text itself contains double quotations). When referenced in the document instance, the entity name is prefaced with an ampersand ("&") and followed by a semi-colon (";").

a. The general entity declaration in a DTD:
```
<!ENTITY siname "Driver's Night Vision Viewer">
```
b. The general entity used in a document instance:
```
<para>If your %siname; needs improvement, let us know.<para>
```
c. The resolved general entity:
```
<para>If your Driver's Night Vision Viewer needs improvement, let us
know.<para>
```

9.3.5.2.1 Replacement text entities. General entities are defined for often used text or "boilerplate" text. For example, the warranty statement may occur in numerous documents and should appear in the same wording. By defining it as a general entity in the DTD, users may reference the entity ***&pmcs.warranty;*** in the document instance without retyping and/or retagging the text.

```
<!ENTITY pmcs.warranty "<para>For equipment under manufacturer's  warranty,
hardtime oil service intervals will be followed. Intervals will  be shortened if
lubricants are known to be contaminated or if operation is  under adverse conditions
(such as longer-than-usual operating hours,  extended idling periods, extreme dust.)
<para>">
```

9.3.5.2.2 Nested entities. General entities may be nested. In the MIL–STD–2361 application, this concept is used to allow for customization for a specific document or document class. MIL–STD–2361 DTDs, provide for use of nested entities to allow local revision of boilerplate text. In the following example, the entity ***&eir.short.name;*** needs to be changed to reflect the equipment short item name.

```
<!ENTITY eir.short.name "Replace text with equipment short item name">
<!ENTITY eir.state "<para> If your &eir.short.name; needs improvement,  let
us know. Send us an EIR. You, the user, are the only one who can tell us  what you
don't like about your equipment. Let us know why you don't like  the design or
performance. Put it on an <extref docno='SF 368'  posttext=' (Product Quality
Deficiency Report)'>. Mail it to the address  specified in <extref
docno= 'DA PAM 738-750' posttext=', Functional users  Manual for the Army
Maintenance Management System (TAMMS)'>, or as specified  by the contracting
activity. We will send you a reply.">
```

9.3.5.3 Graphic entities. Graphic entities will be declared through the use of entity templates for graphics. An external file, containing graphic declarations, will be included in each DTD as in the following example:

```
<!ENTITY % graph PUBLIC "-//'ENTER OWNER'//ENTITIES 'ENTER FORMAL
PUBLIC IDENTIFIER'//EN"> %graph;
```

In the example above, the external file should contain a set of graphic entity declarations needed for a given document instance. The external file contains a template to follow when naming the graphic.

```
<!ENTITY graph PUBLIC "-//'ENTER OWNER'//ENTITIES 'ENTER FORMAL
PUBLIC IDENTIFIER'//EN" NDATA 'ENTER GRAPHIC NOTATION'>
```

9.3.5.3.1 Graphic notation identifier. The graphic notation identifier is required to be declared in a separate notation declaration located in the document file. A graphic notation states the graphical

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format type. The notation declaration example below states that Portable Network Graphics (PNG) are recognized in the document file.

```
<!NOTATION png SYSTEM "PNG">
```

9.3.5.3.2 SYSTEM vs. PUBLIC identifier. A SYSTEM identifier specifies the location of an external file in the particular system on which the document instance was developed. When the document instance is provided to another user, the document instance is required to be changed to identify the new location for the external files. A PUBLIC identifier specifies a unique name identifying the information. The formal public identifier (FPI) is mapped to the external file location. The mapping information is stored in a catalog (see Figure 10). When the document instance is provided to another user, the catalog is changed, but not the document instance. The preferred method for identifying the external files is PUBLIC.

```
PUBLIC "-//DA-APD//ENTITIES BFV Left Side 19971001//EN"
                "c:\graphics\bfvleft.cgm"
PUBLIC "-//DA-APD//ENTITIES BFV Right Side 19971001//EN"
                "c:\graphics\revised\bfvright.cgm"
```

FIGURE 10. Sample XML Catalog.

9.3.5.3.3 Graphic entity key names. Once a graphic entity is declared in a document, it will be referred to by its entity name. In a tagged document instance, the entity name will be used as the value of the graphic attribute “boardno”. The key name will point to the Public Identifier of a specific graphic file.

9.3.5.3.4 Exchanging graphic files. When tagged documents are required to be exchanged with another (external) site, the graphic files will be converted to one of the following CALS graphic formats: (i.e., Computer Graphic Metadata (CGM), Portable Network Graphic (PNG), Bit Map Picture (BMP), etc.). The appropriate graphic notation type will be added to the graphic file entity declarations. “Graph” will represent the key name used in “boardno” and “NDATA” will be used as a reserved word for coding non-XML data, such as graphic formats, as in the following example:

```
<!ENTITY graph PUBLIC "ENTER FORMAL PUBLIC IDENTIFIER" NDATA  cgm -- enter
"cgm", "png" or "bmp" here as appropriate -->
```

9.3.5.4 Unicode character set. XML recognizes Unicode as its primary character set. A major subset of Unicode is UTF-8 character set. UTF-8 is the default character encoding for XML. If the XML file does not contain an encoding statement in either the XML declaration or text declaration, the parser will assume UTF-8 text encoding.

9.3.5.4.1 Unicode two-byte value. Unicode allows for over 65,000 characters. At this point in time, Unicode is the basis for the more universal character International Standard ISO/IEC 10646. Both the ISO 10646 and Unicode encoding provide a unique number for each character defined in the applicable standard. Unicode uses a two–byte value for each character. Compared to the single byte used for the characters defined in the ASCII or ISO-8859 series character sets. ISO 10646 will use a four–byte value for each character. ISO 10646 can encode over two billion individual characters.

9.3.5.4.2 UTF-8. Though there are numerous character sets available for use, the two defined for use with the 40051 XML DTD are UTF-8 and ISO-8859-1 (Latin-1). These two character sets use the first 127 character values to define the characters that make up the ASCII (keyboard) set. UTF-8 and the extended Latin-1 set also define values for characters commonly used, but not available from the keyboard. These non-keyboard characters are often represented by character entities.

9.3.5.5 ISO character entities.

9.3.5.5.1 Non-keyboard characters. ISO character entities are normally used to insert non-keyboard characters, such as the plus/minus sign, in text. As Unicode or ISO 10646 gain support, non-keyboard characters will most likely be accessed through their character number rather than an entity name.

a. Plus or minus declared as an entity name in the ISOnum (ISO 8879:1986 Numeric and Special Graphic) entity declared in ISO 8879. %plusmn; <!ENTITY plusmn "[plusmn]">.
b. Plus or minus sign declared as through character value. %plusmn; <!ENTITY plusmn "#176;">

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At this time, all non keyboard characters should be declared using the character entity name method.

9.3.5.5.2 MIL-HDBK-2361 DTDs ISO character sets. A general entity is required to be declared for the character and the replacement text is the appropriate text or coding that allows a given system to process the non-keyboard character. If the “minus-or-plus” sign has been defined as in the example below, ± may be used in the document instance to obtain the minus-or-plus sign when the document is processed.

```
<!ENTITY plusmn SDATA "[plusmn]"--/pm B: =plus-or-minus sign-->
```

In order to use ISO character sets the ISO character set files are required to be available to your system. ISO character set files are a series of entity declarations which may be referenced with an external entity declaration. XML parsers can be used to resolve ISO entities to system-specific references to a character. The following ISO character sets are included in the MIL-HDBK-2361 DTDs, and will be used when requirements call for their use.

```
 <!ENTITY % ISOpub PUBLIC "ISO 8879-1986//ENTITIES Publishing//EN">
<!ENTITY % ISOgrk3 PUBLIC "ISO 8879-1986//ENTITIES Greek Symbols//EN">
<!ENTITY % ISOnum PUBLIC "ISO 8879-1986//ENTITIES Numeric and Special  Graphic//EN">
<!ENTITY % ISOtech PUBLIC "ISO 8879-1986//ENTITIES General Technical//EN">
```

10 INTRODUCTION TO MIL-STD-2361 XML MARKUP.

10.1 Scope. This section describes methods to markup XML documents in accordance with MIL-STD-2361 SGML constructs. Adhering to the methods defined in this section will assisted in applying MIL-STD-2361 XML constructs to both legacy and new document development.

10.2 Introduction to MIL-STD-2361 XML markup.

10.2.1 Types of tags in MIL-STD-2361. The elements used by MIL–STD–2361 fall into two types of tags: content tags and structural tags. Content tags embody the functional requirements contained in the respective functional requirements standards and specifications, such as MIL-STD-40051. Content tags have names indicating their data content, such as ***<maintsk>***, ***<install>***, or ***<mac>***. Structural tags, such as ***<para>*** (paragraphs), are based on the physical structure of the document, and are used to "chunk up" the data within the content tags. MIL-STD-2361 tags mix content identification and structural uses, such as ***<step1>***, ***<step2>***, etc.

10.2.1.1 Applying content tags. Each of MIL-STD-2361 content parts is comprised of a similar structure. The top level is an information module tag, such as ***<gim>***, ***<mim>***, ***<opim>***, ***<tim>***, ***<pim>***, or ***<sim>***. These top-level tags contain specialized sets of work package elements that are, in some cases, unique to the respective information modules, while, in other cases, common to one or more information module. For example, maintenance information chapter ***<mim>*** can contain service upon receipt work packages ***<surwp>***, preventive maintenance work packages ***<pmcswp>***, maintenance instruction work packages ***<maintwp>***, etc. The top level information chapter tag will be used for building even a single work package.

10.2.1.2 Tagging legacy data. There is a great deal of legacy data within the Army. Legacy data, for purposes of this standard, will be defined as any data (paper or digital) that has not been SGML or XML-tagged in compliance with the respective functional requirements standard or specification, MIL-STD-2361, and MIL-PRF-28001. To ensure appropriate tagging conventions and methodology are applied, the following procedures are offered as guidance for applying XML to legacy data. These procedures are oriented toward the TM requirements set forth in MIL-STD-40051, due to the maturity of that standard and its close association with MIL-STD-2361. However, all of the procedures may be applied regardless of the types of data being converted and tagged.

a. Determine the functional type(s) of publication material to be tagged (i.e., TMs, training products, etc.). Publications developed in compliance with traditional requirements documents are produced as complete books (e.g., front, body, and rear matter) in which the publication technical content is not functionally grouped. XML tagging of legacy data in compliance with this standard can be accomplished only after the publication data has been restructured into functional groupings.
b. Determine the legacy data (e.g., TM, training product, etc.) restructuring requirements for compliance with the respective functional requirements standard or specification and MIL-STD-2361. Virtually no

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TM legacy data are structured in compliance with functionally grouped requirements standards and specifications, such in MIL-STD-40051.

c. Once the restructuring requirements have been determined, an outline of the restructured document(s) should be developed. The outline will be used as a guide for restructuring the documents in compliance with the functional standard or specification and tagging the restructured document in compliance with the applicable MIL-STD-2361 DTD(s). In the case of TMs, an outline may be developed by selecting applicable MIL-STD-2361 content tags which conform to content requirements , specified in MIL-STD-40051,pertaining to the type and maintenance level of the legacy manual.

d. The appropriate top-level tag from the DTD (e.g., ***<mim>***, ***<opim>***, etc.) will be selected and applied to the legacy data. Any #REQUIRED attributes of the top-level tag can then be inserted. In particular, ***<tmno>*** and ***<imlevel>*** are necessary to construct page header, page footer, and chapter cover page.

e. Select the appropriate text for the module from your legacy data and determine the type of work package for the legacy material being tagged. Work packages will contain complete "start-to-finish" tasks for a particular component, and they may contain multiple tasks and procedures. Work packages may have starting conditions that require completing a previous work package. Starting conditions that are dependent upon completion of another work package will contain a reference to the requisite work package in the Initial Setup under the reference tag ***<ref>***. Starting conditions may be referenced in several work packages.

f. Once the work package has been determined, the appropriate work package tag may be applied and any #REQUIRED attributes of that tag inserted. In particular, ***<wpno>*** is used as source of the work package number.

g. Every work package has its own unique set of content tags. Some work package content is mandatory while other content is optional. First, determine the content-specific tags in the work package. For instance, a maintenance work package ***<maintwp>*** requires the content specific tags work package identification information ***<wpidinfo>***, work package information ***<wpinfo>***, and at least one maintenance task ***<maintsk>***. In addition, ***<maintwp>*** may include tags for warnings ***<warning>***, cautions ***<caution>***, and/or notes ***<note>*** and general information ***<geninfo>***. Apply the tags to the appropriate legacy text data and insert any #REQUIRED attributes of those tags.

h. As the writer continues to fill in the content requirements, checks of the DTD should be made for any content-oriented tags contained in the third-level tags. For instance, work package information ***<wpinfo>*** can contain eleven optional content tags. As the optional and required content tags are determined and added to the legacy data, the content requirements for the work package are being satisfied. The content tags should be applied to the legacy data to the level of the document requiring content tags.

i. There are some special content tagging considerations to be addressed when tagging legacy data. The considerations are concerned with General Information with Theory of Operation, Troubleshooting, Maintenance Allocation Chart (MAC), and Repair Parts and Special Tools List (RPSTL).

- General Information with Theory of Operation. When developing the general information with theory of operation portions of a publication chapter, several work packages may require construction, depending on the complexity of the equipment. A work package may contain a whole system theory ***<systhry>*** followed by subsystem theory ***<ssysthry>***, or it may begin immediately with subsystem theory. General information should use an introduction tag ***<intro>*** rather than a ***<systhry>*** tag. The ***<systhry>*** tag will be reserved for theory only. Complex systems may need a separate work package constructed for each subsystem. Such work packages will often include data on line replaceable units ***<lruthry>*** and shop replaceable units ***<sruthry>***.
- Maintenance Allocation Chart (MAC). The maintenance allocation charts are tagged with content tags. The functional group number ***<groupno>*** is followed by the nomenclature of the component or assembly. All of the maintenance information pertaining to that component or assembly should be contained within the qualify function tag ***<qualify>***. The qualify tag is comprised of the maintenance function ***<maintfunc>***; the level to which the maintenance function is assigned (***<maintclass>***), which contains specific maintenance level tags; reference to tools

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and equipment used in the maintenance ***<teref>***; and any additional remarks ***<remarks>***. The qualify function may contain multiple ***<maintfunc>*** entries.

• Repair Parts and Special Tool Lists (RPSTL). The DTDs containing RPSTL work packages ***<rpstlwp>*** do not currently model the list data itself. The list data should be incorporated based on the CCSS database format until revisions are made to MIL-STD-40051. Efforts are currently underway to provide requirements for content tagging the RPSTL list data. The standard RPSTL introductory material is currently included as a boilerplate text entity.

10.2.1.3 Applying structural tags. Structural tags, such as paragraph ***<para>***, contain no intelligence about the content of data, but they do flag material for specific treatment by a presentation system. For instance, tagged text can be numbered automatically by a presentation system, but untagged text cannot be automatically numbered. XML does not recognize carriage returns as processing instructions (e.g., "break line here"), but as marking record boundaries. The presentation system, in general, ignores carriage record boundaries and will not trigger paragraph returns.

10.2.1.3.1 Titles. Many elements contain a title (***<title>*** see 49.1.1.3) element content. For instance, all work packages will have a title. The title of a work package is found in work package identification information ***<wpidinfo>*** see 32.2. Most title ***<title>*** elements are mandatory and contain only character data. If a title ***<title>*** is associated with a counter, the stylesheet/FOSI will specify the appropriate automatic numbering. Titles are tagged ***<title>*** ***</title>*** for a title.

10.2.1.3.2 Paragraphs ***<para>***. Paragraphs are common structural tags, included in many content-oriented element content models. Paragraphs contain a parameter entity, %content;, which includes character data (#PCDATA) and various XML tags that may be contained in paragraph text, such as references, footnote references, index references, change notices, and emphasis tags. Paragraphs are the element through which lists, figures, and tables are included in documents developed in accordance with MIL–STD–2361

Subparagraphs. Titling paragraphs may be preceded by the usual ***<title>*** element if the content model follows the pattern "(title?, para)+". However, the ***<para>*** element uses its "parahead" attribute for the title if the DTD does not specify a ***<title>*** element before every single ***<para>***. The information will be bold and in line with the paragraphs.

a. DTD fragment for ***<systhry>***:

```
<!ELEMENT systhry (title, (specpara | para)+, (ssysthry* |
(lruthry*, sruthry*)*))>.
```

b. Sample of a XML document fragment for ***<systhry>***:

<systhry>
<title> Viewer Mount
</title>
<para parahead="Bolt Assembly, Vehicle Side"> There are three bolts needed for the plate next to the vehicle. By bolting these three bolts the viewer can be fixed to the vehicle.
</para>
<para parahead="Revolving Plate"> The viewer revolves freely due to the revolving plate that is start of the viewer mount.
</para>
</systhry>

10.2.1.3.3 Procedures. A procedure ***<proc>*** is a set of steps that comprise all or part of a task. A procedure may, but does not have to, contain a title. However, if a task, such as "SERVICING", has more than one procedure, the separate procedures will have a title. The distinction between a task name and a procedure title will be maintained and will be explicit. Task names are literals and will be appropriately inserted into the document instance by the FOSI. Procedure titles will be included as content of the ***<title>*** tag in the document instance.

10.2.1.3.4 Determination of procedure, task, or work package designation. There may be publications data where it is difficult to determine if data are procedures, tasks, or work packages. If legacy data contains

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a long or specific title for a maintenance action, it may be a procedural title. The parameter entity %maintsk; tag may have to be included above the title if not present in legacy data. On the other hand, a long or specific title may signal the beginning of a work package. If tasks like "Removal" or "Installation" are subordinate to the title, it should probably be treated as the title of a maintenance work package. The document instance developer will be required to analyze the legacy data to determine its intent for portraying either procedures, tasks, or work packages.

10.2.1.3.5 Steps . The XML structural tags for steps will be used when tagging legacy data to show the steps within a procedure. Care will be taken to ensure that the step tags are used correctly. The step tag ***<step1>*** (see 33.3) refers to primary-level step, not the first step in a procedure. The step tag ***<step2>*** refers to first-level substep, not to step number 2. Sub-steps are contained in next higher step level. Element ***<step1>*** does not end until the end of any ***<step2>*** sub-element contained within the ***<step1>*** parent element. Steps are automatically numbered by the stylesheet/FOSI. To avoid double presentation of step numbers upon printing, delete the literal step numbers from document instance. The same convention applies to indentation of sub-steps, which are indented by stylesheet/FOSI. To avoid duplication of indentation space, all tabs or spaces will be deleted from the legacy document instance.

Steps usually consist of single paragraphs, although multiple paragraphs are allowed. Good writing practice dictates precise separation into steps rather than merging several actions in one step. If a step has an associated warning, caution, or note, it will be tagged with the ***<specpara>*** tag. The ***<specpara>*** tag may replace ***<para>*** as the first element in a step. The first step will include the text for the step in a paragraph within the ***<specpara>*** tag, and the second primary-level step will include a substep. For example:

```
<step1>
<specpara>
<warning>
<warning.group>
<trim.para> Sodium peroxide can cause caustic burns from
prolonged skin contact.
</trim.para>
</warning.group>
</warning>
<para> Mix the sodium peroxide into a paste with the distilled water.
</para>
</specpara>
</step1>
<step1>
<para> Apply a thick layer of the paste over the corroded plate.
<step2>
<para> Use the toothbrush to ensure paste coverage in the threaded holes.
</para>
</step2>
</step1>
<step1>
<para> After 15 minutes, wash the plate clean of the sodium
peroxide paste with distilled water.
</para>
</step1>
```

10.2.1.3.6 Lists . Lists are usually contained within paragraphs. However, any element with a parameter entity %content; as a content model can contain a list. Warnings, cautions, and notes may also contain lists. The list tag will identify the type of list being tagged. There are three types of XML lists: random, sequential, and definition lists.

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10.2.1.3.6.1 Random lists. Random lists (***<randlists>*** see 49.1.2.2) are not numbered. Each item starts a new line of text as is regulated by the stylesheet/FOSI. If the "bullet" attribute is changed to "yes" (by entering ***<randlist bullet ="1">*** in the document instance, the items will be bulleted.

10.2.1.3.6.2 Sequential lists. The numbers on sequential lists ***<seqlist>*** see 49.1.2.1 are provided by the stylesheet/FOSI. Therefore, any numbers appearing on legacy data numbered lists should be removed during the tagging process, otherwise, the items in the lists will have duplicate item numbers. This applies to nested sequence lists also. Nested sequence list numbering is keyed to the nesting level of the list.

10.2.1.3.6.3 Definition lists. Definition lists (***<deflist>*** see 49.1.2.3) are used to for lists defining words. A definition list may have a title ***<title>***. A definition list may then have one or more term/definition element wrapper ***<term.def>*** each of which is required to be followed by a ***<term>*** and a definition ***<def>***.

10.2.1.3.6.4 Numbered lists. The numbers on numbered lists ***<seqlist>*** are provided by the stylesheet/FOSI. Therefore, any numbers appearing on legacy data numbered lists should be removed during the tagging process, otherwise, the items in the lists will have duplicate item numbers. This applies to nested sequence lists also. Nested sequence list numbering is keyed to the nesting level of the list.

10.2.1.4 Tables . There are two types of table structures allowed for use with MIL–STD–2361 DTDs: CALS table model ***<table>*** (see 42), and MIL-STD-40051 defined standard tables, in which columns and rows are inferred from content-specific tags. **All tables are numbered in accordance with MIL-STD-40051.**

10.2.1.4.1 CALS tables. The CALS table model defined in MIL-PRF-28001. The following conventions apply to CALS tables:

- May have any number and widths of columns.
- May have multi-level heads, stub columns, and spanning cells or rows.
- May be ruled and rules can be controlled locally.
- May contain graphic elements.
- May contain warnings, cautions, notes, and procedures or steps.
- May be varied by the markup in the document.

a. CALS table structure and markup. The CALS table model follows the general model:

TABLE	***<table>***
COLUMN SPECIFICATION	***<colspec>***
TABLE GROUP	***<tgroup>***
TABLE HEAD	***<thead>***
TABLE BODY	***<tbody>***

The head, body, and foot each contain:

ROWS	***<row>***
CELL ENTRY	***<entry>***

b. Column specifications ***<colspec>***. Colspec are used to define the column characteristics of a ***<table>***. Column specifications can be specified separately for head and body (the ***<colspec>*** of ***<tgroup>*** control the column specs for the body). The following colspec, used in MIL–STD–2361 is identical to the one used in MIL-PRF-28001.

```
 <!ELEMENT colspec EMPTY>
<!ATTLIST colspec
      colnum  CDATA          #IMPLIED
      colname NMTOKEN          #IMPLIED
      align  (left | right | center |justify | char) #IMPLIED
     charoff CDATA          #IMPLIED
      char  CDATA          #IMPLIED
      colwidth CDATA          #IMPLIED
      colsep (0 | 1)          #IMPLIED
       rowsep (0 | 1)          #IMPLIED>
```


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c. Table columns and spans. Columns are assigned both a column number **colnum** and a name **colname** in ***<colspec>***. The column names are referenced in a cell ***<entry>*** in **namest** and **nameend** to specify start and end of a spanning column. Spanned rows are controlled in ***<entry>*** by the attribute more rows **morerows**. The attribute more rows **morerows** is use to define the additional rows to be spanned. An example of spanning three rows :
`<entry morerows="2">.`

d. Alignment in troubleshooting tables. To indicate how material is aligned within the troubleshooting table, the second column contains, in effect, the third column. For instance, a ruled row in a ***<faultproc>*** table consists of a ***<symptom>*** (which appears in the first column) and one or more ***<malfunc>*** tags. The ***<malfunc>*** tag includes the contents the second column. It contains an ***<action>*** (in third column) to be taken in response to the ***<malfunc>***. The ***<action>*** contains a para or steps in the third column. Each ***<malfunc>*** and ***<action>*** are aligned. If a second ***<malfunc>*** tag follows, it starts a new line in the second column, separated by a line space from the material above. When another ***<malfunc>*** tag occurs, it is separated by a rule from the ***<entry>*** group above.

10.2.1.4.2 Figures . Few systems can handle composite figures in which multiple graphic files are positioned within a single figure area using the attributes of ***<graphic>***. Keep figure tagging simple. Single-page figures should contain only a ***<graphic>*** tag, not ***<subfig>*** or ***<macrograph>***. Each illustration should be a single graphic file unless made up of full-page sheets. If a figure contains several sheets, use one ***<figure>*** tag and a ***<subfig>*** for each sheet. Figures can be numbered or unnumbered using attribute "fignum". If your legacy data uses unnumbered figures, add 'fignum="0"' to the ***<figure>*** tag.

Suggestion is to perform graphic file sizing and cropping in a graphics editor. Using this suggestion will eliminate guess work and the sizing is performed in an application made specifically for the task.

10.2.1.4.3 Graphics. The ***<graphic>*** tag is used to refer to the graphic file entity and supply its size, clipping, scaling, placement, etc. The following graphic specification is used in MIL–STD–2361 and is similar to MIL-STD-28001.

```
<!ELEMENT graphic (%section508;, mapref*)>
<!ATTLIST graphic
  boardno CDATA #REQUIRED
  type NOTATION (%notation.class;) #REQUIRED
  graphsty NMTOKEN #IMPLIED
  reprowid CDATA #IMPLIED
  reprodep CDATA #IMPLIED
  hscale CDATA #IMPLIED
  vscale CDATA #IMPLIED
  scalefit (0 | 1) #IMPLIED
  hplace (left | right | center | none) #IMPLIED
  vplace (top | bottom | middle | non) #IMPLIED
  rotation CDATA #IMPLIED
  %bodyatt;
  %secur;>
```

10.2.1.4.4 Alphabetic index ***<aindx>***. The index marker reference ***<indxref>*** tag establishes a document location and index text to be referenced within the alphabetic index. Composition systems vary how they generate the data that is referenced by the ***<indxref>*** tags. A composition system may be able to generate the index entry ***<indexentry>*** to include the topic ***<topic>***, page number ***<pageno>*** and work package number ***<wpno>*** while another system needs an addition process to generate this information or the data be inserted manually. Automatic systems that need additional support to generate the work package number would use a process script such as PERL The PERL script process would review the XML instance and search for the ***<indxref>*** tag. When a ***<indxref>*** tag is found an “unique” id will be added to the tag and the attributes “ref1-4” of the ***<indxref>*** tag would be collected and stored in an array. When all the SGML instances have been processed the PERL scripts then would create an SGML instance of an Index. The array is then sorted, (alphabetical order) and each cell of the array will become an entry in the SGML index instance. The SGML index instance is imported into composition system with other SGML instances and processed

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along with the TM creating an Index. The Index entries are cross-referenced to their Work Packages. The Index would display up to a four level depth entry with its page number and work package number.

Three authoring methods are provided for generating index reference and index by the use of the index reference ***<indxref>*** tag. Methods (1) and (2) require an application to generate necessary ID and IDREF automatically. Method (3) is manually input by the author. Using any of the methods will provide the necessary information for any composition system to publish an index and furnish a mechanism to hyperlink between the index and the index reference. **Method 1** – Index reference defines the index level titles. **Method 2** – Reference the index level title generated in the index. **Method 3** – Index reference ID and index entries point to the index reference identifier.

Method 1 has the author to specify for each index reference entry the specify index level title(s). This method is the same as described in MIL-M-28001B to generate an index. MIL-STD-2361 provides a better tool for the soldier to find the indexed material, when the information is used electronically. To improve the index authoring methodology an application is required. The application will perform the following tasks: Read the document instance to obtain the index level topic and subtopics. Assign an automatic generated ID to the index reference (”). Sort the index level topics and subtopics. Generate the index using the SGML elements with the associated generated index reference IDs. The application is necessary for the composition system to generate alphabetically sorted index, the page number, and the work package number for the index reference. The author's responsibility is to enter the index level topics for each SGML element. The index reference IDs, IDREFs and index entries will be generated after completing the index reference SGML elements. The application will be applied to the document instance to prepare the document for publication

Method 2 has the author create the index entries and associate a unique identifier to each index level entry. The writer authors the document instance and applies the SGML element for index reference with attribute IDREF pointing to the index entry ID location. The method provides a consistent list of possible entries and will reduce spelling errors. The disadvantage is recalling or looking up the associated index entry ID. Again as Method 1, an application is required to provide a better tool for the soldier to find the indexed material. The application will perform the following tasks: Read the document instance to obtain the index entry reference identifier. Assign an automatic generated ID to the index reference. Assign to the associated index entry the index reference generated ID. The application is necessary for the composition system to generate the page number, the work package number for the index reference. The author's responsibility is to enter the sorted index entries with associated unique identifiers and enter the SGML element index reference with associated attribute index entry IDREF. The application will be applied to the document instance to prepare the document for publication.

Method 3 does not require a separate application applied to the document instance. The author will manually perform the following tasks to permit index generation. Generate the sorted index entries. Create the index reference with a unique identifier. Associate the index reference index to the index entry. Generate new index entries, when required. An application is not required to be applied to the document instance and the author will make the association when creating the index reference entry. However, the author is required to have a valid list of available IDs and the same index entry document instance needs to be shared or merged with other authors. The author's responsibility is to enter the sorted index entries, enter the index reference with an unique identifier for each SGML element, and associate the ID to the index entry IDREF.

```
<!ELEMENT aindx (alphaindx?, indexentry)*>
<!ATTLIST aindx  %bodyatt;   %secur;>
```

10.2.1.4.5 Warnings, cautions, and notes. Warnings, Cautions, and Notes may be added by two methods: within the ***<specpara>*** element, or explicitly by including the tags ***<warning>***, ***<caution>***, and /or ***<note>*** in the document instance. The method chosen for use will be controlled by content model of an element. If a warning or caution applies to specific step or procedure, it is required to be contained in that element. This can be accomplished through ***<specpara>***. The ***<specpara>*** element is usually invoked in grouping, i.e., (specpara | para). The ***<specpara>*** element includes a ***<para>*** element after an optional and repeating warning, caution, and or note. This ***<para>*** is mandatory in a ***<specpara>***. In steps, this ***<para>*** is the first paragraph normally contained in the step.

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10.2.1.4.6 Assigning attribute values. Almost every element has associated attributes. There are three global parameter entity sets of attributes that attach to most elements: %refs;, %bodyatts;, and %secur;. These global attributes define, respectively: IDs and ID references; text characteristics like change level or nuclear hardness critical processes; and security level. None of these attributes are required.

a. Local attributes. Many elements have additional attributes, some of which are required. For instance, “tmno” is required on the top module element and "id" is required on the footnote.
b. Cross-reference attributes. To cross-reference an element, insert an ID attribute for the element. Then refer to that ID at the cross-reference point with an ***<xref>*** or an attribute of the IDREF tag. Insert the ID as a value of the appropriate IDREF attribute. ID values are required to be unique strings, beginning with an alpha character, after which the IDs can contain both letters and numbers. To help keep ID strings unique, use the beginning alpha character to keep the IDs of work packages, tables, tasks, procedures, figures, index entries, and footnotes separate. These seven elements are the only IDs that are resolved. The FOSI specifies resolution of the IDREFs. You can use entire words or abbreviations as IDs and they are not required to contain numbers.
c. Cross reference ***<xref>*** (see 46.2.2). The cross-reference tag ***<xref>*** uses an IDREF attribute to automatically link to the ID attribute of other elements. The attribute "assocfig" (part of the %ref; attribute set) also has an IDREF value type and it is used to link text and graphics, not for cross-references. The external reference tag ***<extref>*** (see 46.2.1) is used to reference other TMs or documents outside the document instance, and the reference is inserted as a literal string in the attribute "docno."
   (1) Internal reference:

```
<!ELEMENT xref EMPTY>
<!ATTLIST xref
  taskid IDREF #IMPLIED
  wpid IDREF #IMPLIED
  stepstart IDREF #IMPLIED
  stepend IDREF #IMPLIED
  figid IDREF #IMPLIED
  itemno CDATA #IMPLIED
  itemid IDREF #IMPLIED
  callout CDATA #IMPLIED
  tableid IDREF #IMPLIED
  tslocid IDREF #IMPLIED
  pagelocid IDREF #IMPLIED
  pretext CDATA #IMPLIED
  posttext CDATA #IMPLIED
  termdefid IDREF #IMPLIED
  %secur;>
```

   (2) External reference :

```
<!ELEMENT extref EMPTY>
<!ATTLIST extref
  docno CDATA #IMPLIED
  revno CDATA #IMPLIED
  pretext CDATA #IMPLIED
  posttext CDATA #IMPLIED
  wpid CDATA #IMPLIED
  taskid CDATA #IMPLIED
  figid CDATA #IMPLIED
  tableid CDATA #IMPLIED
  partid CDATA #IMPLIED
  %secur;>
```


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d. Cross-reference resolution.

• Text. Text references will be to either a task or a titled procedure within a work package. The resolved value for a task will be its name; for a procedure it will be the title. The attribute "taskid" supplies the IDREF to the ID of a task or titled procedure. Only procedures with titles will be referenced. If a procedure does not have a title, the task containing the procedure will be referenced. The work package number will always be invoked through "wpid", if the reference location is in another work package in same information module.

• Work packages. The ***<xref>*** attribute "wpid" will always be used for cross reference resolution for work packages. The "wpid" value for text, tables, and figures located in another work package in the same information module will always be supplied. When referencing complete start-to-finish contents of another work package, only the "wpid" attribute will be used. Each work package tag will have an "id" attribute. The "wpno" attribute is not its ID. The same value will not be used for "wpno" attribute and work package "id" attribute.

• Figures and tables. Only numbered figures and tables will be referenced. The FOSI will extract the figure id ("figid") or table number ("tableid"). and use as appropriate. The cross-reference value will not include the title.

• Steps. The word “step” and the step number will be generated by the composition system when ONLY step start reference (“stepstart”) is used. When reference a sequence of steps the composition system generates the word “steps” with the first step number (using “stepstart” attribute reference) followed by an “–” and the ending step number (using “stepend attribute reference).

e. Empty tags. Empty elements contain no character data or other elements. Empty tags mark things like cross-references and index entries that will be created by the composition system. Empty tags also mark insertions of external files in non-XML notations, such as graphic files. “Empty tags” contain the information your composition system needs to resolve reference values in a series of attributes.

11 STYLESHEET/FOSI APPLICATION AS A STYLE GUIDE.

11.1 Scope. This section contains information on the application of stylesheets/FOSIs as style guides.

11.2 Using stylesheets or formatting output specification instances (FOSIs). This section describes methods for interchanging formatting requirements for technical documents whose source files are tagged according to DTDs developed in accordance with MIL–STD–2361 A DTD interprets the content and structural requirements contained in a functional specification, and the stylesheets/FOSI interprets the style and formatting requirements specified in the DTD.

11.2.1 Style. Adherence to rules described in MIL–STD–2361 allows for different receiving processing systems to unambiguously interpret the style and formatting intent of the sending system. By combining the document instance, content-tagged in accordance with the appropriate MIL–STD–2361 DTDs, with the stylesheet/FOSI, the resulting publication will preserve the information content of the original and allow similar presentation. The stylesheet/FOSI values for the style characteristics are passed to, or used by, the program that performs layout and final composition.

11.2.2 The output specification (OS). The Output Specification (OS) uses SGML/XML to define style characteristics and provide methods of linking style to elements in an SGML/XML document instance. In fact, the OS is itself a DTD. A FOSI containing specific values for the characteristics which identify the format of a document type is itself an instance of the OS DTD. In this sense, any particular FOSI is just one member of the family of possible instances for the OS DTD.

11.3 FOSIs in the ASRL. FOSIs available in the ASRL are designed to function with DTDs developed in accordance with MIL–STD–2361 The FOSIs are fully compliant with, and adhere to the rules described in, ISO 8879 and MIL-PRF-28001. The Army-approved FOSIs contained in the ASRL will facilitate the reuse of DoD SGML/XML DTDs. The overall goal is to allow for the interchange of style and formatting information between all types of publishing systems. This includes current batch and WYSIWYG systems, as well as

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future systems incorporating newer technology. This is accomplished by the interchange of style information using the semantics described, to be used as input to the formatting system, whether human or computer.

11.4 Using a FOSI. A FOSI is developed to present the formatting information of a specific document, or class of documents, based on the MIL–STD–2361 DTD with which the document instance was marked up. The FOSI is written to the Output Specification Document Type Definition (OS DTD). The OS DTD was designed to present a methodology for interchanging formatting information in a standard way. It contains many of the same features of a stylesheet and is designed to be read by both humans and machines. However, for a person to read through a FOSI, some knowledge of the SGML/XML would be beneficial as well as knowledge of the OS DTD. The process of using a FOSI to produce a document is as follows.

a. A document would be marked using the SGML/XML tagging scheme provided by the DTD written for that document type.
b. A FOSI would be prepared designating the formatting information for the document type. The FOSI provides all the necessary formatting information. Within the FOSI, this information is associated with the elements and/or attributes of the DTD written for that document. For example, the FOSI might specify that a chapter title is to be centered, bold, sans-serif, in 10 on 12 type. The document being tagged using SGML/XML tags would simply have the SGML/XML tags:
<chapter>
<title>`THIS IS THE TITLE`
<title>
<chapter>

The user does not have to alter the FOSI or provide formatting information in any manner, other than to mark up a document using the elements and attributes provided in the DTD. It should be understood, however that the FOSI has itself been written to a DTD. This is why the formatting information can be interchanged in a standard way. If formatting information needs to be changed for a given document (for example, chapter titles will now be quad left), the FOSI will have to be changed to reflect this, and parsed to make certain it is following the OS DTD. An organizational policy and procedure for such changes should be developed, implemented, and adhered to. The user should not be allowed to make arbitrary changes to the FOSI.

11.5 Stylesheets. Stylesheets define how documents are presented on screens and in print. A stylesheet provides a set of formatting characteristic values used to describe composition processing functions to be performed on the elements of a text document to provide the format style required by a functional specification or standard, such as MIL–STD–40051–1/–2, AR 25-30 or TRADOC Reg 350-70. A stylesheet (i.e. FOSI, XPath, XSL, XSL-T and XSL-FO) delivered with the document should contain values of characteristics for every tag used in the DTD, in every context in which the tag has a unique formatting requirement, and with its attributes if they affect the formatting. Stylesheets also provide generated text (see 8.5.4.7.1). Generated text is specified text that is automatically output by the stylesheet for your application during the formatting of the document.

11.5.1 XPath (XPATH). XPath is a type of stylesheet used with XML documents. XPath is fabricated to be an integral part in a host language such as XSLT. It uses a constrain, non-XML syntax for locating parts of an XML document. Xpath locator syntax uses essential functions based on the node hierach of a document and evaluate expressions to determine a location object. Its locations are used by XSLT. Addition information on XPath may be obtained from the W3C's XSL website: http://www.w3.org/Style/XSL/WhatIsXSL.html.

11.5.2 Extensible stylesheet language (XSL). XSL is a language for formulating stylesheets to be use with XML documents. The XSL stylesheet describes how the XML document is presented as a page-based document or frame-based being displayed by a web browser. The XSL stylesheet is also an XML document in itself. Addition information on XSL may be obtained from the W3C's XSL website: http://www.w3.org/TR/xsl.

11.5.3 Extensible stylesheet language transformations (XSLT). XSLT is a type of stylesheet developed to use with XML to transform a document from one XML application to be used by another application. XSLT is fabricated to be used with XPath. Its instructions are written in a document resembling a stylesheet and then it uses a transformation engine to generate a new form of the document. Addition information may be on XSLT obtained from the W3C's XSLT website: http://www.w3.org/TR/xslt.

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11.5.4 Extensible stylesheet language for formatting objects (XSL-FO). XSL-FO is a type of stylesheet developed to use with XML for formatting objects. It is a pagination markup language describing a rendering vocabulary capturing the semantics of formatting information for paginated presentation. The paginated presentation may be displaying multiple separated pages on a screen, on paper or audibly. Addition information may be on XSL obtained from the W3C's XSLT website: http://www.w3.org/TR/xsl.

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VOLUME 2
ADMINISTRATIVE PUBLICATIONS

PART I
OVERVIEW, GUIDANCE AND WORKFLOW

12 INTRODUCTION.

12.1 Overview. Volume 2 provides the implementation guidance for the development of Army administrative publications in compliance with MIL–STD–2361, AR 25–30 and DA PAM 25–40. Administrative publications are discussed in sufficient detail to allow the handbook user a comprehensive understanding of the development process and how XML is applied to the process. Volume 2 is structured with two parts, as shown in "Layout, Format and Content". Part I contains general administrative publication development guidance and workflow for both proponent authors and APD editors. Part II contains detailed guidance for the application of XML to administrative publications and related training information. There are workflow charts and other graphic illustrations to amplify the narrative discussions.

12.2 Adminstrative publication proponent/editor interface process (APPIP) to XML. The output of the APPIP application is processed into a valid XML format. The use of XML supports the capability to produce either a page document or a frame document using only one source. The XML source with the use of specific stylesheets can produce a page document or a frame document. The use of XML provides more functions with APPIP to produce a more user friendly publication. XML supports functions such as:

- Hyperlinking items in the change summary to the narrative text in the document.
- Hyperlinking the index to the narrative text in the document.
- Hyperlinking to external documents.
- Roll over function, through the use of a mouse, would display the definition of the word when the user rolls over the word.
- Publication viewing the default web browser.

12.3 Objectives. Volume 2 MIL-HDBK-2361 is designed to provide users a tool that is simple to use and functionally accurate to the Army administrative publication processes. This volume contains administrative publication development and implementation guidance information, designed to assist publication developers and editors in the use and application of MIL–STD–2361 XML. Implementation guidance is presented to realize the MIL–STD–2361 objectives to share and reuse common publication information, which is the underlying theme throughout the handbook.

12.4 Layout, format and content. Volume 2 contains information relevant to administrative publications, their development, and methods for applying and using SGML which is the currentmarkup language employed by APPIP. An outline structure of the parts and sections are provided as follows:

Volume 2 – Administrative Publications

Part I – Overview, Guidance and Workflow

12 Introduction – An overview of the Administrative Publication Part I including the intent, layout, format, and contents.

13 Administrative Publication Workflow and Processes – Identifies and describes the workflow and processes associated with the development of administrative publications.

14 Administrative Publications Proponent/Editor Interface Process (APPIP) – Provides an overview and introduction to the APPIP and manuals.

15 Proponent Authoring APPIP Workflow Process – This part provides an overview and introduction to the APPIP proponent version and manual.

16 APD Editor APPIP Workflow Process – This part provides an overview and introduction to the APPIP Editor Version and manuals.

Part II – Reference

17 Administrative Publication XML Tag Description List – The Tag Description List for Administrative Publication XML tags.

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13 ADMINISTRATIVE PUBLICATION WORKFLOW AND PROCESSES.

13.1 General administrative publication development process and workflow. The workflow diagram in Figure 11 illustrates a typical administrative publication development cycle. The process shown is a composite of the efforts by the proponent author and the APD editor. The following paragraphs are a narrative description of the administrative publication development process and flow.

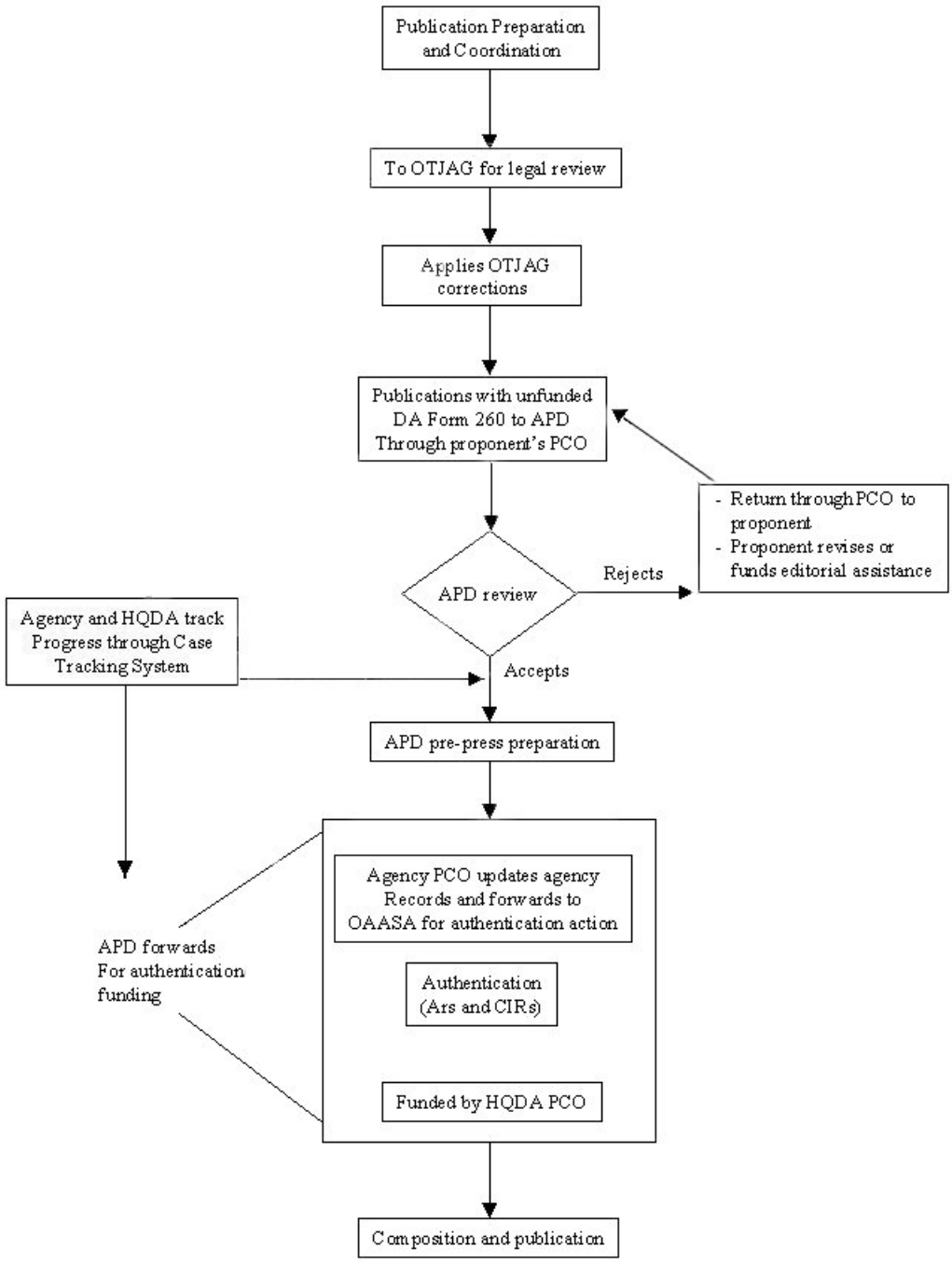

FIGURE 11. Typical administrative publication development cycle.

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13.1.1 Proponent author role in administrative publication development. The proponent author uses the APD-developed Administrative Publications Proponent/Editor Interface Process (APPIP) to develop administrative publications. APPIP allows administrative publication proponent developers to author their publications using standard Microsoft Word styles. Paragraph 15 provides a description and workflow on the APPIP proponent version. The Proponent Word Template Users Guide (obtained from the ASRL www.asrl.com) contains the detailed procedures and steps for using the APPIP proponent version.

13.1.1.1 Originating the document. The proponent author will either develop a new administrative publication, or modify an existing administrative publication, provided by APD. In either case, the author will use Microsoft Word with the integration of APPIP. The author will open Microsoft Word and follow the menus and prompts to enter the required information on the system screens. APPIP will allow the author to create a new document (or modify an existing document) by applying the Microsoft Word styles. Once a publication segment has been completed, APPIP will verify that the publication segment conforms to the requirements contained in AR 25–30, The Army Publishing Program and DA PAM 25–40, Administrative Publications: Action Officers Guide.

13.1.2 APD role in administrative publication development. APD has the responsibility to review and publish administrative publications for the Army. As part of this responsibility, APD editors will apply XML to the administrative publications. A detailed APD editor workflow is described in paragraph 16.

13.1.2.1 Review of proponent submitted document. The first step, in the APD administrative publication development, is to review the document submitted by the proponent. APD performs a review of the submitted document, including graphics, to validate its completeness, correct the application of the Microsoft Word styles, and verify compliance with the Army quality assurance policy. Once the document review is complete, APD will assign the document to an editor for XML document development.

13.1.2.2 APD review and pre-press preparation. Once the proponent author completes the creation or modification of the document, it will be forwarded to the APD editors for review, as described in DA PAM 25–40. The review will verify that the publication conforms to DA PAM 25–40 Chapter 1, Section 3 and Appendix B checklist. If the publication does not meet the prerequisites, the publication is rejected and is returned to the proponent for clarification and/or correction.

13.1.2.3 Authentication. APD forwards the corrected and legally approved document through the Publication Control Officer (PCO) of the proponent agencies to the Office of the Administrative Assistant to the Secretary of the Army (OAASA) for review and authentication.

13.1.2.4 Composition. OAASA forwards the authenticated policy document back to APD for final composition and publication.

14 ADMINISTRATIVE PUBLICATIONS PROPONENT/EDITOR INTERFACE PROCESS (APPIP).

14.1 Introduction. The Administrative Publications Proponent/Editor Interface Process (APPIP) is a user interface to provide a non-SGML application to develop and edit administrative publications. The properly formatted document will provide the necessary structures to automatically map to the administrative publication DTD and apply SGML tags. The APPIP was developed to provide a proponent version and APD editor version. This section provides an overview and introduction to the APPIP proponent version and the APPIP Editor Version.

14.2 APPIP proponent version. The APPIP proponent version provides administrative publication proponent developers a methodology for preparation of new, existing (APPIP formatted) and legacy (non-APPIP formatted) administrative publication documents using the styles available in Microsoft Word. The APPIP proponent version is based upon a series of Microsoft Word templates and Visual Basic for Applications (VBA) macros, which enable the proponent to create and modify documents without any SGML/XML knowledge or experience. By using this tool, the proponents may create documents in a standardized fashion, thus allowing the editors at APD to easily review and apply SGML tags to these documents. The following paragraphs describe the APPIP proponent version components available.

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14.2.1 APPIP editor version. The APPIP Editor Version provides the APD editors an integrated Microsoft Word with Administrative Publication templates to provide a seamless application of SGML to a document. The following paragraphs describe the template styles, users guide content and editor methodology.

14.2.1.1 Microsoft Word styles. Styles are custom formats (bolded text, underlined text, outline format, etc.) that a user creates or that are standard styles available in Microsoft Word. In Microsoft Word the styles appear in the left-hand side of your window (when view mode is Normal). The styles that are referred to in the Proponent Word Template Users Guide are user-created and are not in the standard Normal template. A special template has been designed that allows the author to apply styles either automatically when creating a new or modifying an existing (APPIP formatted) document, or manually when working with legacy (non-APPIP formatted) documents. Styles determine the formatting of the text and include special information for the system and APD editors.

14.2.1.2 Proponent Word template users guide. The Proponent Word Template Users Guide (obtained from the ASRL at www.asrl.com) is divided into four sections and seven appendices:

a. **Introduction** – provides general layout and structure for the users guide.
b. **Overview** – provides the basics and how to start APPIP.
    - APPIP Proponent Templates Installation
    - How to Install APPIP from A Disk or CD
    - How to Install APPIP from Microsoft Outlook
    - How to Install APPIP from Lotus Notes
    - Initial APPIP Start-Up
    - Utilities Button
c. **Start Using APPIP** – provides instruction on the following processes:
    - Working with APPIP Document Segments
    - Accessing Administrative Publications Segments
    - Deleting a Document Segment
    - Editing a Document Segment
    - Print Preview an Administrative Publication
    - Create a New Publication Folder
    - Renaming an “X” Administrative Publication Folder
    - Create a New Document Segment
    - Create New Front Segment
    - Create New Body Segment
    - Legacy Documents
    - General Information About Appendixes
    - Create New Appendix A Segment
    - Create New Appendix B – Z Segment
    - General Information about Glossaries
    - Create Index Segment
    - Microsoft Word Styles
d. **Style Toolbar** – describes the toolbar menus and formatting options.
    - Symbols and Fractions
    - Equations
    - Symbols
    - Inclusions
    - Inclusion Insertion
    - Address Inclusion
    - Excerpt Inclusion
    - Note Inclusion
    - List Item Inclusion
    - Add Leader
    - Proofing Format For Review

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- Increase And Decrease Indent
- Formatting and Emphasis
- Character Emphasis
- Bullets and Numbering Outline
- Table and Border Formatting
- Highlighted Text
- Working with Footnotes
- The Validation Process
- Error Marker and Error Message Display
- Validate Document Segments
- Removing the Error Markers
- Sample Error Correction

e. **Appendix A – Working With Text Styles** – This appendix contains information regarding the applying the Microsoft Word styles and occurrence each style is permit.

f. **Appendix B – Working with Headquarters Letter and Memo** – This appendix describes the creation and editing of Headquaters Letters and DA Memorandum documents.

- Body Segment Creation
- Distribution Segment Creation

g. **Appendix C – Local Command Publications** – This appendix describes the creation and editing of Local Command and Agency Publications.

- Setting Up a New Local Command
- Using the New Local Command

h. **Appendix D – Special Characters, Symbols and Fractions** – This appendix contains the list of special characters, which are available to use.

i. **Appendix E - Enable Outline Numbering** – Microsoft Word Outline Numbering requires information to be entered after the automatic numbering. When no data is entered, Microsoft Word will automatically disable the Outline Numbering. This appendix provides the procedures and steps required to enable the Outline Numbering.

j. **Appendix F - Validation Errors** – describes in detail the error messages generated as a result of errors found during validation and possible methods to correct the errors.

k. **Appendix G - APD Receiving and Sending APPIP Files** –

- APD Sending ZIP Files
- APD Sending Self-Extracting Files
- APD Receiving APPIP Documents

l. **Glossary**

m. **Index of Figures**

n. **Index of Tables**

14.2.1.3 Overview of the APPIP proponent version editing methodology. The APPIP proponent version allows administrative publication proponent developers to develop their publications using Microsoft Word styles. APD administrative publication editors can then apply SGML to the proponent-developed documents by using the APPIP Editor Version. The methodology contained in the Proponent Word Template Users Guide, combined with the APD – developed APPIP Editor Version, provides the Army the necessary tools to apply SGML to administrative publications with only minimal SGML knowledge and experience. These tools may be applied to newly created, existing (APD provided APPIP formatted) or legacy (non-APPIP formatted) documents.

14.2.1.3.1 Start using APPIP. The Start Using APPIP Section of the Proponent Word Template Users Guide contains guidance for creating and editing documents, and an explanation of the Microsoft Word styles used. The user is instructed in the use of the menus and screens regarding required information, what to do if information is not available for a required entry, etc. This section contains descriptions of the four basic templates: Outline, Table, Text Table, and Illustration and the ability to validate the styles. The

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templates will guide the author trough each body template. Each template can be selected from the menu toolbar screen. Additionally, the section provides available screen and menu examples to the user.

14.2.1.3.1.1 Outline template. The Outline Template is used to guide the author through the body outline paragraph numbering and to enter applicable title and paragraph narrative. Using input screens and pull-down menus, the outline template allows the author to:

a. Automatically apply a document number.
b. Divide a document into individual pieces (parts, chapters, sections and paragraphs) (each piece can be authored individually).
c. Assign paragraph numbering.
d. Assign paragraph titles.
e. Assign paragraph text.

14.2.1.3.1.2 Table template. The Table Template combines the Microsoft Word table editor with the required table label and title styles needed according to the APD requirements for the CALS table model. The Table Template is used to specify where the table will be located in the document body. Tables may be placed anywhere in the body section. The author will use the standard Microsoft Word table editor to create or modify rows, columns, merge cells, and enter table cell data.

14.2.1.3.1.3 Text table template. The Text Table Template is used to create and enter a text table anywhere in the body section. The user is prompted by data entry screens for each prescribed text table components (i.e. entry heading, legend, etc.). A text table is another way to list information in columns and rows in a document. A text table shows data with the column headings running vertically down the page instead of horizontally across the page. The column headings are listed with each information group down the page. Text tables are used if the information is long and best suited to be listed in this format.

14.2.1.3.1.4 Illustration template. The Illustration Template is used to create and enter a placeholder illustration with the illustration number and caption. The user is prompted by data entry screen for the illustration components. The illustrated digital files are inserted by APD editors for final review and publication.

14.2.1.3.2 Information on appendixes, glossaries and indexes. The Rear Matter Section describes how to use the Microsoft Word predefined templates. The templates will assist the author in implementing the necessary styles for the administrative publication rear. The templates will guide the author through each of the rear section pieces. The templates are selected through the rear menu option screen. The templates associated with the administrative publication rear are as follows:

- Appendix A (Reference)
- Appendix B-Z
- Glossary
- Index

14.2.1.3.2.1 Appendix A (references) section. The Appendix A Section provides the author document and form references. The manual contains detailed procedures, examples and illustrations to guide the author through the steps of creating or modifying (APPIP formatted) Appendix A or applying styles to a legacy (non-APPIP formatted) Appendix A. The Appendix A contains four sections:

- Section I – Required Publications
- Section II – Related Publications
- Section III – Prescribed Forms
- Section IV – Referenced Forms

14.2.1.3.2.2 Appendix B-Z section. The Appendix B-Z Section provides templates for the user to enter supplemental information about the document. The manual contains detailed procedures, examples, and illustrations to guide the author through the steps of creating or modifying (APPIP formatted) Appendix B-Z documents, or applying styles to a legacy (non-APPIP formatted) Appendix B-Z document. The document components are the same as described in the Start APPIP Section. Procedures for validating the documents are also provided.

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14.2.1.3.2.3 Glossary section. The Glossary Section provides the author definitions of terminology. The manual contains detailed procedures, examples and illustrations to guide the author through the steps of creating or modifying (APPIP formatted) glossary or applying styles to a legacy (non-APPIP formatted) glossary. The glossary contains three sections:

- Section I – Abbreviations
- Section II – Terms
- Section III – Special Abbreviations and Terms

14.2.1.3.2.4 Index section. The Index Section provides a keyword cross-reference to the associated paragraph number. The index for an administrative publication is entered manually by the author. The manual contains detailed procedures, examples and illustrations to guide the author through the steps of creating or modifying (APPIP formatted) index or applying styles to a legacy (non-APPIP formatted) index.

14.2.1.3.3 Validation. The APPIP proponent version includes a capability to validate that all Microsoft Word styles have been entered in the correct order by the proponent author. Only valid documents can be forwarded to APD for review and publishing. During and after each administrative publication piece has been entered, the author will execute the validation process for each piece. The style validation will verify that the titles, sections, paragraph outline numbering, tables and text tables follow the prescribed formatting rules stated in AR 25–30 and DA PAM 25–40. If any errors are encountered in the document, the system will present the error descriptions and locations to the author. The errors are corrected by the proponent author and the validation process rerun until a valid document is produced.

14.2.2 APPIP proponent version legacy. The APPIP proponent version Legacy import provides the proponent and APD editors a tool that will apply the APPIP Word template styles automatically to a standard document loaded into Microsoft Word. The import does not always provide a 100% valid document after importing. Many factors will deteriorate the import recognition capability. The following paragraphs describes the legacy import consideration, users guide content and import methodology.

14.2.2.1 Legacy import consideration. The legacy import (conversion) is used to bring previously developed non-APPIP formatted documents into the APPIP process. As in any import, if the document does not conform to the rules provided, incomplete conversion and validation errors will occur. Before executing the legacy conversion, review the document with the prescribed conversion rules. Doing this cursory review can enhance the conversion process and reduce the validation errors.

14.2.2.2 Legacy to Word conversion users guide. The Legacy to Word Conversion Users Guide (obtained from the ASRL at www.asrl.com) is divided into six sections:

a. **Introduction** – provides general layout and structure for the users guide.
b. **Overview** – provides an overview of the legacy conversion process.
c. **Invoking Legacy Conversion Process** – provides the steps to execute the legacy conversion.
d. **Front Section** – describes the requirements in the legacy document needed for best conversion results.
e. **Body Section** – describes the requirements in the legacy document needed for best conversion results.
f. **Rear Section** – describes the requirements in the legacy document needed for best conversion results for the following rear section components:
    - Appendix A – Reference
    - General Appendices
    - Glossary
    - Index

14.2.2.3 Overview of the APPIP proponent version legacy conversion methodology. The APPIP proponent version allows administrative publication proponent and APD editors to convert legacy administrative publication document into an APPIP formatted document. An important step is to review the conversion rules for each publication section. If the document conforms to these rules, the conversion process can deliver a 100% valid document. The conversion rules details are contained in the Legacy to Word Conversion Users Guide and provide the necessary requirements for converting each administrative publication section. After conversion is performed on an administrative publication section, validation is necessary to check the correct application of the APPIP Word styles.

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14.2.2.3.1 Error correction hints. Many errors can be corrected before conversion is started. Some of the most common errors are the following:

- Not having the required title paragraph(s) or correct titles in the front section.
- Not having two paragraphs for each level.
- Wrong paragraph level (i.e. subparagraph 1 level numbering to subparagraph 3 level numbering.
- After a main level paragraph the subparagraph 1 level does not starts with "a.".
- Mixing paragraph numbering (i.e. 1, 2, c, d).
- Not spelling the section headings correctly in Appendix A or Glossary.
- Not having the correct number of sections in Appendix A (four sections) or Glossary (three sections).
- Missing the obtain, cited or prescribed reference in Appendix A Section I documents and Section III forms.

14.2.2.3.2 Correct legacy or convert document. When errors occur after conversion, where should the corrections be performed? The errors can be corrected either in the legacy or the converted document depending on the severity and type of error. The following paragraphs provide cases to correct in the legacy and the converted document.

14.2.2.3.2.1 Case 1: incorrect subparagraph 1 level number. When the first subparagraph 1 level after each main paragraph does not start with “a.”, the corrective action is performed in the legacy document. The error causes the remaining subparagraphs not to be converted to the correct APPIP Word styles. After inserting the correcting subparagraph 1 level number, then both the legacy conversion and validation will be rerun.

14.2.2.3.2.2 Case 2: wrong subparagraph level. When the subparagraph is indented to the wrong level (i.e. subparagraph 1level is followed by a subparagraph 3 level), the corrective action is performed in the converted document. The editor would highlight the incorrect subparagraph(s) (i.e. all subparagraph 3s after the subparagraph 1) and press the “Decrease Indent” icon on the tool bar. The subparagraph numbering and APPIP Word style will be modified to the previous subparagraph level.

14.2.2.3.2.3 Case 3: wrong Appendix A or Glossary section title. After converting either Appendix A or Glossary and the error message “Section # Style Missing“ is displayed, the corrective action should be performed in the legacy document. When the legacy document does have the incorrect Section number and/or Section title, the conversion does not recognize that a new section has been started. Refer to the corresponding legacy conversion rules for the correct section numbering and title information. After the corrections are made, both the legacy conversion and validation will be rerun.

14.2.2.3.2.4 Case 4: missing title page statement titles. After converting the front section, the validation indicates required statements (i.e. styles) are missing. The corrective action depends on the number of missing style statements. The most common cause for the style missing error message is an incorrect statement title. Refer to the front rules for the prescribed statement title to be used. If only one or two styles are missing, the correct title and style can be applied in Word with APPIP styles. If more than two styles are missing, the incorrect titles should be corrected in the legacy document (see front conversion rules for the prescribed titles), then both the front conversion and validation will be rerun.

14.3 APPIP editor version. The APPIP Editor Version provides APD editors a methodology for preparation of new, existing (APPIP formatted) and legacy (non-APPIP formatted) administrative publication documents using the styles available in Microsoft Word. The APPIP Editor Version is based on a series of Microsoft Word templates, Visual Basic for Applications (VBA) macros, Visual Basic (VB), and an SGML parser. These processes integrated will enable the editor to modify documents without any SGML (Standard Generalized Markup Language) knowledge or experience, import non-APPIP formatted and SGML formatted administrative publications, export to SGML, edit SGML documents and produce final proofing document. By using this tool, the editors have a tool that will reduce the labor intensive application of SGML tags and reduce the processing required to produce an administrative publication.

14.3.1 SGML conversion with APPIP editor version. The APPIP Editor Version has many software facets associated to the process of applying SGML to administrative publication documents. The APPIP Editor Version has a user interface to guide the editors from reviewing the publication to applying SGML to

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providing final proofing of the publication. The APPIP Editor Version user interface provides a step-by-step flow from receiving the proponent administrative publication to final proofed publication. The interface provides the following functions (detail flow and description is provided in Section 16).

SGML Conversion	Transform the APPIP Word document to valid SGML
Preparation for SGML Editor	Validates the SGML document and prepares the SGML publication for an SGML Editor

15 PROPONENT AUTHORING APPIP WORKFLOW PROCESS.

15.1 Authoring APPIP workflow. The workflow diagram in Figure 12 illustrates a typical proponent author APPIP development workflow. The following paragraphs are a narrative description of the process and flow.

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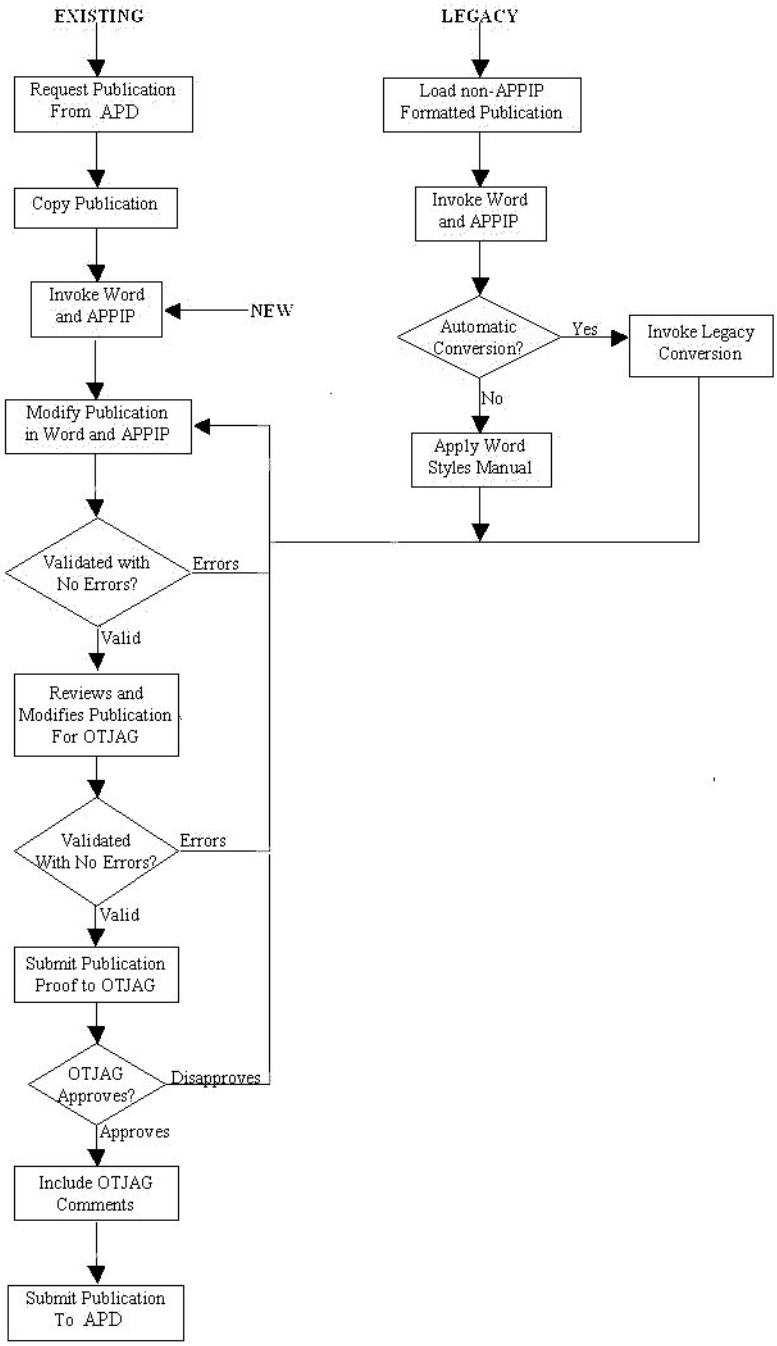

FIGURE 12. Typical proponent author APPIP development workflow.

15.2 Administrative publication format. The proponent author has two possible administrative publication formats; new and existing APPIP formatted publication or legacy non-APPIP formatted publication. The following paragraphs will describe the different formatting characteristics.

15.2.1 New publication. When a new administrative publication is required, the proponent author will create each publication segment through Microsoft Word using APPIP templates.

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15.2.2 Existing publication. An existing administrative publication is a file already containing APPIP styles. The author will obtain the existing publication from APD, which maintains the official source SGML publication. APD will provide the APPIP–formatted publication to the proponent author. The proponent will author new modifications to the existing publication through Microsoft Word with APPIP templates.

15.2.3 Legacy publication. A legacy administrative publication is a file already created with a word processor. The proponent can download the APPIP Legacy Conversion User Guide (from the ASRL at www.asrl.com), which contains the series of procedures to convert the legacy publication into APPIP. The proponent will use the following paragraphs to convert legacy to an APPIP formatted publication.

15.2.3.1 Manual review. The first process in the legacy conversion is to review the legacy publication. The proponent should perform a cursory review and correct the major formatting problems. Possible problem examples are incorrect paragraph number (using the wrong paragraph numbering), incorrect section titles (glossary, Appendix A), etc.

15.2.3.2 Automated legacy conversion. The APPIP provides a tool to convert non-APPIP formatted (legacy) publications to APPIP formatted publications. As with a conversion from any semi-structured document, the conversion is not always 100% correct. The legacy conversion process applies the APPIP styles to the publication correctly, if all the manuscript preparation rules from AR 25–30 and DA PAM 25–40 are followed.

15.2.3.3 Validate legacy conversion. After the legacy conversion process, the proponent will perform the APPIP validation on each publication segment. If no validation errors are found, the document is ready for modification. If validation errors are found, the errors are required to be corrected. The proponent will determine if the error corrections should be made in the legacy document or manual insertion. The proponent would manually insert the styles when there are minimum errors to correct. An example is if a subparagraph group is indented one paragraph level too far. The proponent would correct the error and revalidate the segment. The proponent would correct or reject the legacy publication if major formatting problems exist. After correcting the legacy publication, the publication would repeat the validation process and be ready for modification.

15.3 Administrative publication modification. The proponent can update or create a publication in APPIP and follow the series of procedures in APPIP Word Template User Guide. After modification or creation has been invoked by the proponent, each publication segment requires validation. The validation will confirm that the Microsoft Word styles are correctly structured and comply with AR 25–30 and DA PAM 25–40.

15.4 Submit administrative publication for legal review. APPIP provides a tool to automatically change from single space document to double spacing (proofing) document. Use standard Microsoft Word printing function for draft publication to provide for HQDA agencies, MACOMs and OTJAG. After preparation is completed, the proponent submits the publication through the PCO for OTJAG's comments and approval. After OTJAG's comments have been incorporated, the document is ready to have the SGML tags applied.

15.5 Submit publication to APD. After all HQDA agencies, MACOMs and OTJAGS corrections have been included and APPIP styles are validated, the publication is submitted to APD for editorial review, authentication, and publishing.

16 APD EDITOR APPIP WORKFLOW PROCESS.

16.1 Editor APPIP workflow. The workflow diagram in Figure 13 and Figure 14 illustrates a typical APD editor APPIP development workflow. The following paragraphs are a narrative description of the process and flow.

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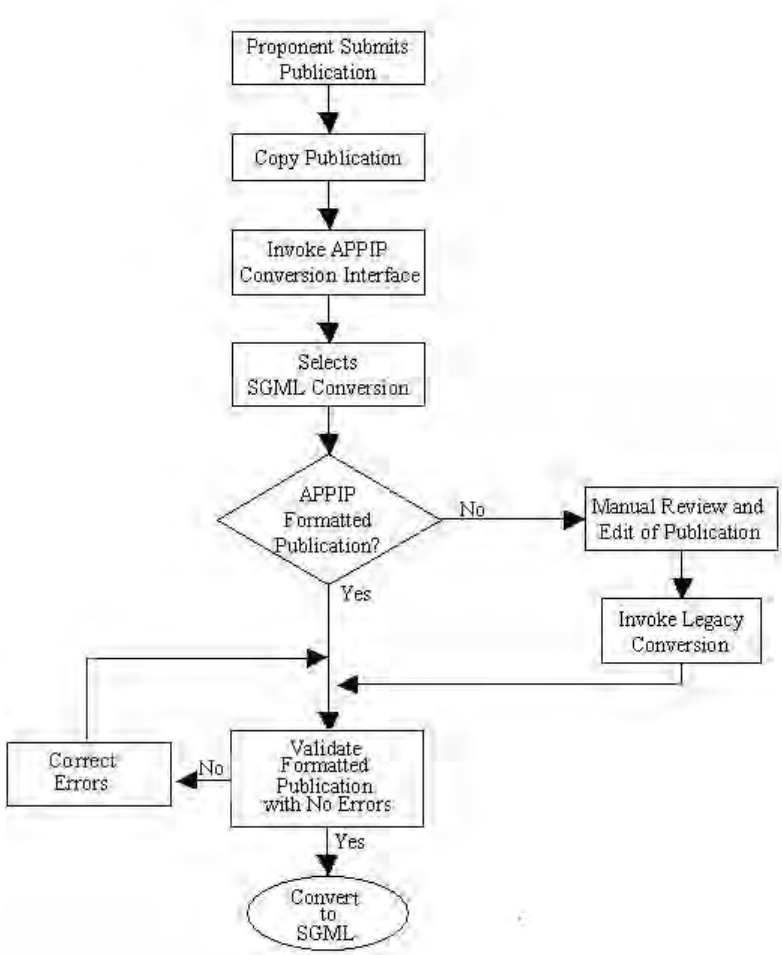

FIGURE 13. Typical APD editor APPIP development workflow (part 1).

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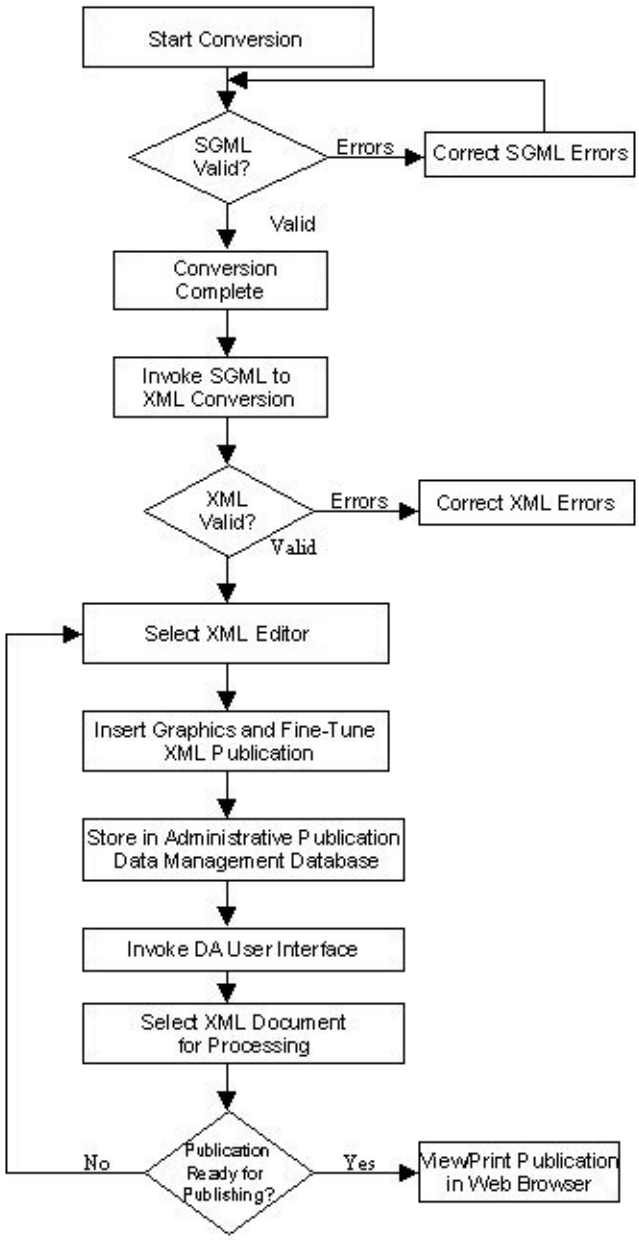

FIGURE 14. Typical APD editor APPIP development workflow (part 2).

16.2 Administrative publication receipt from the proponent. The proponent submits an electronic APPIP validated publication to APD. The editor will copy all of the APPIP components of the document into the appropriate directories in the APPIP Editor Version prescribed directory. The APD Editor's User's Guide (obtained from the ASRL, www.asrl.com, library) provides users detailed guidance on the use of the APPIP SGML Conversion processing application. The APPIP Conversion processes were designed to provide the

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APD Editors a graphical interface and a suite of tools to make the editing and publication process easier and more efficient. The system provides an interface to the following functionality:

- APD SGML Conversion – The option provides a link to the Microsoft Word templates and VBA macros to automatically SGML tag files in which the proponent authors have applied APPIP Microsoft Word styles. See paragraph 16.4 for process details.
- Preparation for SGML Editor – Validates the SGML document to ensure it adheres to the APD Document Type Definition (DTDs) and inserts the close or end tags as specified in the corresponding DTD. See paragraph 16.5 for process details.

16.3 APPIP or legacy publication. During the transition to the APPIP process, the proponents will be submitting both APPIP and legacy (see paragraph 15.2.3 for legacy publication description) publications. The APD editor will determine from the styles if the document is in APPIP format. If the document is not formatted, the APD editor follows the typical legacy conversion as described in the following paragraphs.

16.3.1 Manual review. The first process in the legacy conversion is to review the publication provided by the proponent. This review can determine if the publication has any major paragraph number problems (using the wrong paragraph numbering), incorrect section titles (glossary, Appendix A), etc. The editor will perform this cursory review and correct the major formatting problems.

16.3.2 Automated legacy conversion. The APPIP provides a tool to convert non-APPIP formatted (legacy) publications to APPIP formatted publications. The editor can obtain the APPIP Legacy Conversion User Guide (obtained from the ASRL www.asrl.com). As with any conversion from a semi-structured document, the conversion is not always 100% correct. The legacy process provides the majority of APPIP styles to narrative, if all the rules from AR 25–30 and DA PAM 25–40 are followed.

16.3.3 Validate legacy conversion. After the legacy conversion process, the editor will perform the APPIP validation on each publication segment. The APPIP process will not allow SGML tags to be applied until a publication segment is validated.

16.3.3.1 If no validation errors are found, the document is ready for preparation for OTJAG.

16.3.3.2 If validation errors are found, the document is required to be modified to correct the errors. The editor will determine if the error corrections should be made in the legacy document or manual insertion. The editor would manually insert the styles when there are minimum errors to correct. An example is if a subparagraph group is indented one paragraph level too far. The editor would correct the error and revalidate the segment. The editor would correct or reject the legacy publication if major formatting problems exist. After correcting the legacy publication, repeat the validation process and prepare the publication for OTJAG.

16.3.4 Preparation, submittal and approval for OTJAG. The editor will prepare the document for review by OTJAG. The editorial review includes grammatical review, content review, and typical editorial functions. After the editorial review, the editor will prepare the draft publication in double spaced format. The APPIP provides an option to automatically apply proofing (double spacing) to the publication for OTJAG review. After preparation is completed, the proponent submits the publication through the PCO for OTJAG's comments and approval. After OTJAG's comments have been incorporated, the document is ready to have the SGML tags applied.

16.4 Using the SGML conversion. The editor will select the SGML Conversion command from APPIP Conversion Interface menu. This invokes Microsoft Word with the integrated APPIP templates and VBA macros. The editor will open each administrative publication segment in Microsoft Word and the editor will use the Word to SGML command buttons, menus and prompt screens to convert the publication to SGML. The SGML to APPIP Conversion User Guide can be obtained from the ASRL www.asrl.com.

16.4.1 Validate before conversion. Before the SGML conversion will proceed with SGML conversion, the conversion routine will confirm the publication segment has followed the formatting rules. If validation errors are found, APPIP does not invoke the SGML conversion.

16.4.2 Applying SGML. When no validation errors are found during validation, APPIP will convert the APPIP styles to SGML tags with the appropriate attributes. The original APPIP format is not changed

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during the SGML conversion process. During the conversion process, the display screen will become very active applying the SGML tags to the publication. After the SGML conversion is completed for each segment, the publication is ready for SGML Editor preparation.

16.5 SGML editor preparation. The editor will invoke the APPIP Conversion application via the Word-SGML option, for SGML Normalization, Validation, and SGML Editor Preparation. The editor will select the "Full Publication" command button. Prompts will appear for identification of the document type, document number and the applicable document segment. The editor will enter the appropriate information and invoke the "SGML Validation and Normalization" process for that document segment. The validation application will determine that the document parses error-free, or identify any SGML errors in the document. When the document segment parses SGML error-free, then it is ready for "Full Document Processing." The application will return to the File List Menu. Once all of the associated document segments have been validated, the individual document segments will be gathered into a complete publication, normalized and validated.

16.6 Publication database. The editor will maintain the publication in the APD administrative publications data management system (DMS). The DMS will maintain configuration control for each administrative publication version.

16.7 Publish document. Once all reviews have been completed and the document is authenticated and approved for publication, APD will publish the document.

16.8 Proponent requests an administrative publication. When starting a revision the proponent will request from APD the electronic published document for modification. The Electronic Publishing System (EPS) maintains all documents in an SGML data management system (DMS). The APPIP provides an aid to convert the valid SGML document into a valid APPIP formatted document. The editor will select the SGML Conversion command from APPIP Conversion Interface menu. This invokes Microsoft Word with the integrated APPIP templates and VBA macros. The editor will open each administrative publication SGML segment in Microsoft Word. The editor selects the SGML to Word command buttons, menus and prompt screens to convert the SGML document to an APPIP formatted document. The APPIP to SGML Conversion User Guide is available from the ASRL www.asrl.com. If the proponent does not use the APPIP add-in, APD will provide the APPIP formatted administrative publication, APPIP program and the electronic APPIP Word Template User Guide. Otherwise, APD provides a validated APPIP formatted administrative publication to the proponent for revision.

16.9 SGML to XML document conversion. In order to migrate the Administrative Publications from SGML to XML, another application (Administrative Publications Document Transformation) is required. The Administrative Publications Document Transformation application will convert a valid SGML tagged document into a valid XML tagged document. The application is described in the SGML to XML Document Conversion User's Guide. The User's Guide contains the following sections:

- Introduction
- Installation
- Running the Document Conversion Application
- Error Handling

16.9.1 Introduction. This section describes and details the purpose of the Administrative Publications Document Conversion application.

16.9.2 Installation. The Installation section describes how to obtain the application software, and gives detailed instructions concerning the execution of the self-extracting “ZIP” file which will install the application to the desired location on the user's hard drive.

16.9.3 Running the document conversion application. This section provides detailed instruction on the operation of the application, along with figures illustrating the forms and dialog boxes which control the process.

16.9.4 Error handling. The SGML to XML conversion process is prone to errors. In the event that the user encounters an error during processing, notices will be displayed which will inform the user that an

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error has occurred, and provide the opportunity to review the list of errors contained in an error file. Once all of the errors have been corrected, the conversion process can be run again

16.10 XML to HTML composer application. The Composer Application is used to display and/or print Administrative Publications documents that have been tagged in XML format in a web browser.

16.10.1 Processing a XML document for viewing. The user is prompted for a path to the XML document and for the location of the graphics files that are associated with the document. Once the XML document and the graphics directory have been selected, the user clicks the Process button to begin the conversion and display process. The user will be presented with a Processing form to indicate the status of the current process.

16.10.1.1 XSL stylesheet. The XML document is opened along with the XSL stylesheet that is associated with the document. The XML document and the XSL stylesheet are processed through SAXON to be converted into HTML for web browser display. Once processing is complete, the document is displayed in the user's default web browser.

16.10.1.2 Document display in a web browser. Once the XML document has been processed, the user's default web browser will be activated and the cover page of the document will be displayed as shown in Figure 15.

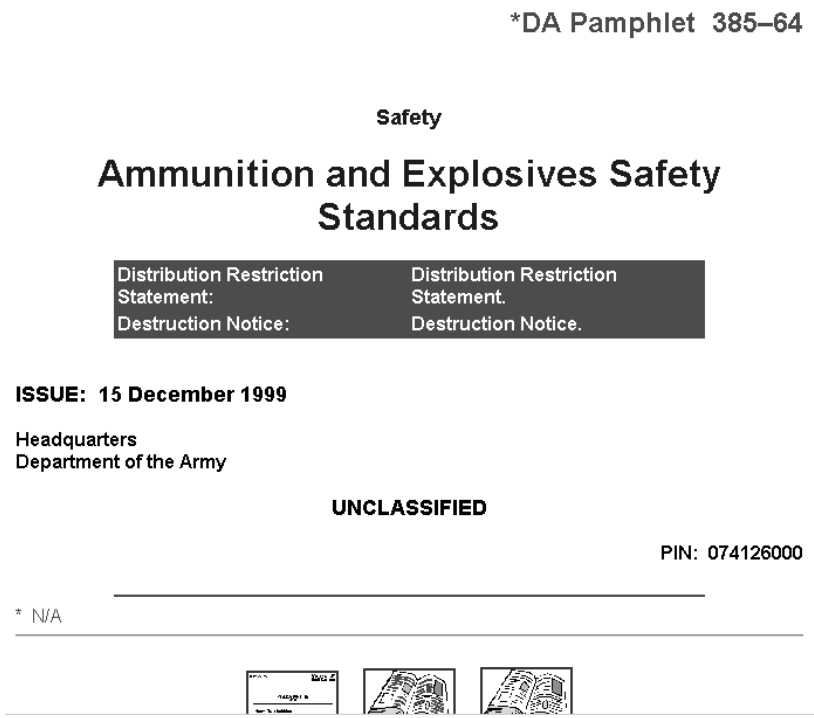

FIGURE 15. Example of a cover page displayed in a web browser. .

16.10.1.3 Navigating in a document. At the bottom of the page, there are three navigation icons which provide access to the rest of the document. From left to right, these icons represent the Title Page; the Change Summary; and the body of the document, respectively. The following steps will demonstrate how to navigate the document using the navigation icons.

Step 1 Click on the left-most icon for the Title page. The document's title page will then be displayed. (See Figure 15.)

Step 2 From either the Cover page or the Title page; click the center navigation icon to display the Change Summary page.

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Step 3 From either the Cover, Title or Change Summary pages, click on the right-most navigation icon to display the Body of the document. For the display of the body of the document, the web browser screen is divided into three frames; the Table of Contents frame, DA Form 2028, the document text frame, and a frame in the lower left which provides buttons for the DA Form 2028 for submitting comments and for printing the document.

16.10.1.4 Printing a hardcopy of a document. From the Body display of the document, the following steps will describe the document printing process.

From the Body display of the document, the following steps will describe the document printing process.

Step 1 Click the Print Publication button.

Step 2 The application will create an RTF file for printing. Once the RTF file creation process is complete, the document will be displayed in another browser window.

Step 3 Click the File menu button, and click the Print button.

Step 4 Select the proper printer and settings as desired, and click the OK button. the file will be sent to the appropriate printer.

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PART II
REFERENCE

17 ADMINISTRATIVE PUBLICATION XML TAG DESCRIPTION LIST.

17.1 Administrative XML tag description list . A copy of the Administrative Tag Description List can be obtained by downloading it from the ASRL Website,http://www.asrl.com under the link “CONSTRUCTS”.

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VOLUME 3
TRAINING AND DOCTRINE PUBLICATIONS

18 INTRODUCTION.

18.1 Overview. Volume 3 provides implementation guidance for the development of Army training and doctrine publications in compliance with MIL–STD–2361, TRADOC Regulation 350-70, and other training and doctrine requirements documents. Training and doctrine publications are discussed in sufficient detail to allow the handbook user a comprehensive understanding of their development processes and how SGML is applied to those processes. The volume contains general training and doctrine publication development guidance described in three sections. There are flow charts and other graphic illustrations to amplify the narrative discussions. An outline for each part is shown in "Layout, Format and Content" .

18.2 Objectives. Volume 3 is designed to provide users with a tool that is simple to use and functionally accurate to the Army training and doctrine publication processes. The volume contains training and doctrine publication development and implementation guidance information, designed to assist publication developers and editors in the uses of MIL–STD–2361 SGML. Implementation guidance to realize the MIL–STD–2361 objectives to share and reuse common publication information is an underlying theme throughout the handbook.

18.2.1 Layout, format and content. Volume 3 contains information relevant to training and doctrine publications, their development, and methods for applying and using SGML. An outline of each of the sections is provided. Volume 3 is structured as follows:

Volume 3 – Training and Doctrine Publications

18 –	**Introduction**	An overview of the Training and Doctrine Publication volume: including the intent, layout, format, and contents.
19 –	**Training and Doctrine Publication Workflow and Processes**	Identifies and describes the workflow and processes associated with the development of training and doctrine publications.
20 –	**Training and Doctrine Publication Consolidated Requirements Documents (CRD)**	This section provides an overview and introduction (including the intent, layout, format, and contents) to the training and doctrine publication Consolidated Requirements Documents (CRD). The CRDs for the respective training and doctrine publications can be obtained from the Army SGML/XML Registry and Library (ASRL) home page.

19 TRAINING AND DOCTRINE PUBLICATION WORKFLOW AND PROCESSES.

19.1 General training and doctrine publication development process and flow. The publication development process in use for training and doctrine products is different from the process used for developing technical and equipment (TM) publications. Technical manuals are normally developed by contractors as part of a formal weapon system contract. The development environment depends upon the digital TM used by the developing contractor. Training and doctrine publications are normally developed by government employees in functional organizations such as TRADOC centers and schools. Within TRADOC, the development environment consists of a number of organic systems, processes and procedures used to develop and deliver training and doctrine products. These systems include the Reimer Digital Library (RDL), Automated Systems Approach to Training (ASAT). and the Digital Training Management System (DTMS). The RDL provides document storage for DA and Commandant approved products. ASAT is the Army-wide system for automated training development. It is used by proponent schools for developing and producing training and

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doctrine information and products. Army Performance Improvement (API) process identifies the difference between the current performance of an organization or individual and the desired performance.

19.1.1 Proposed process flow. The Army Training Information Architecture (ATIA) was developed to support Army training by providing a blueprint to guide training investments and align other training architectures and programs. ATIA was developed by Army Training Support Center (ATSC) to replace their three current systems; namely; RDL, ASAT and DTMS. These sytems are currently used to develop and deliver training and doctrine products. Three component architectures are being developed in accordance with the Army Enterprise Architecture Guidance Document found at (www.train.army.mil/tddt/atia.htm). They are:

- Operational Architecture (ATIA-OA) – which provides a functional description of Army training organizations, activities and requirements.
- Technical Architecture (ATIA-TA) – describes the standards for Army training information and information systems.
- System Architecture (ATIA-SA) – describes the system components of Army training.

The Army Training Information Architecture (ATIA) serves to:

- Organizes the functional activities into a logical arrangement for implementation.
- Define the relationships between the activities, data entities, and standards necessary to design and implement the supporting systems.
- Define the underlying technical standards.
- Forms the basis for total system design and ensures the commonality of effort in the software segment development and design process.

19.1.2 Common source publication data. Training and Doctrine digital publications are developed using common source publication data. Once developed, the data can be reused and distributed via the Reimer Digital Library (RDL). Uses include product approval and staffing, future product development and field use. This requires a view of digital publications in which the data developed can be reduced to a common kernel (task level) of knowledge. Once this data is developed it will be the foundation for any training and doctrine publication with a set of SGML instructions to produce both approved Army publications (FM, STP, MTP, Drill, TC) and customized reports.

19.1.3 Training and doctrine product development responsibilities. Figure 16 reflects the TRADOC organizational structure and associated responsibility for training and doctrine development and delivery. TRADOC headquarters has overall responsibility for training and doctrine policy including development of TRADOC Regulation 350-70.

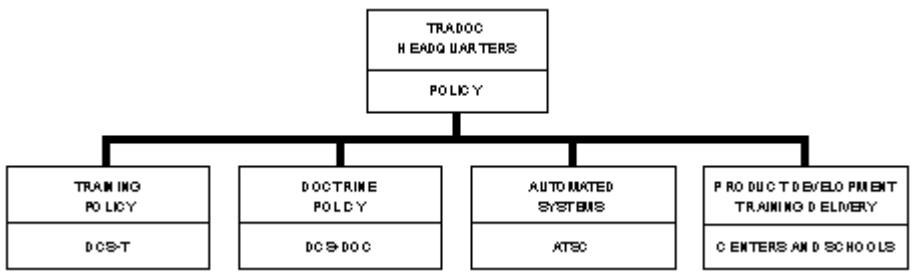

FIGURE 16. TRADOC product development responsibilities..

Responsibility for developing training and doctrine publications information is the Deputy Chief of Staff for Operations, Plans and Training (DCSOPS&T), www.tradoc.army.mil/dcsopst/index.htm. DCSOPS&T or the DCS, G-3/5/7 is responsible for developing policies, procedures and guidance for training and doctrine publication information development. DCS, G-3/5/7 synchronizes personnel developments activities of TRADOC centers and schools; that apply these policies and procedures to training and doctrine product or material development.

19.1.3.1 Proponent agency. A proponent agency is an Army organization or staff, which has been assigned primary responsibility for materiel or subject matter in its area of interest. Proponent agencies are considered

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subject matter experts in a specific subject area such as Soldier Training Publications (STP) or Field Manuals (FM). A proponent agency could be a proponent school, proponent staff agency, proponent center, etc. Proponent agencies serve as the functional expert within a subject area and prescribe format and content requirements for an assigned TRADOC publication. TRADOC systems developers (ASAT, RDL, etc.) use these format and content requirements as the basis for developing publications development and distribution systems.

19.1.3.2 Proponent school. A proponent school is a proponent agency designated by the Commanding General, TRADOC or other appropriate Army Command (ACOM) as training proponent to exercise supervisory management of all combat/training development aspects of a materiel system, functional area or task. It analyzes, designs, and implements training/training products. Basically, proponent schools prepare training publications products such as STP, MTP, etc. using formats and content requirements established by the proponent agency.

19.1.3.3 TRADOC publications information development process. Figure 17 reflects the TRADOC publication information development flow. The needs analysis is the minimum essential requirement before the development of any training product. The triggering circumstance leads to needs analysis usually starts with the receipt of a triggering circumstance. The triggering circumstance leads to needs analysis, mission analysis and ultimately to individual or collective task analysis. The culmination of this task analysis process is the publication of a specific training product. Following are examples of triggering circumstances.

- Directed by appropriate authority
- Changes in Doctrine
- Organizational Changes
- Materiel/Systems Changes
- Evaluation Finding (Changes in training methods, sites, etc.)
- Law or regulation changes (Environmental, HAZCOM, HAZMAT, etc.)

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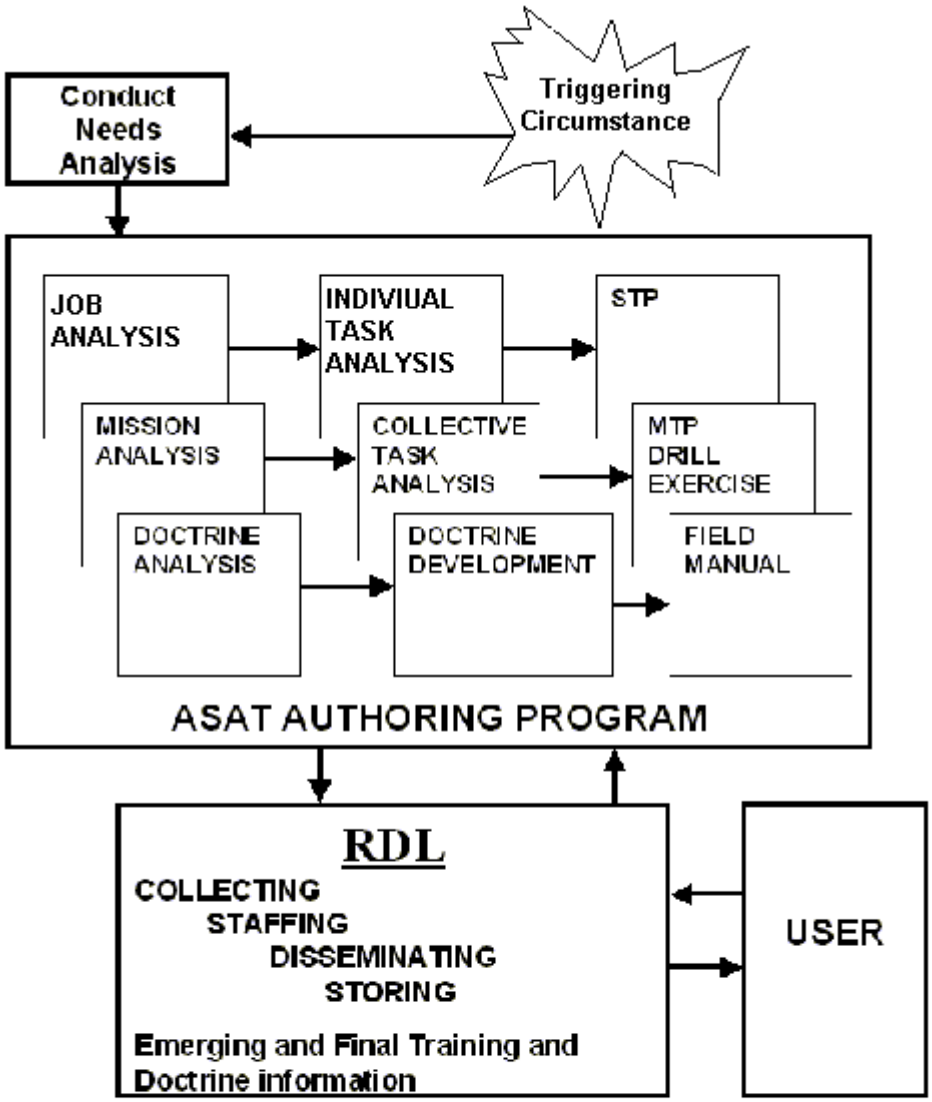

FIGURE 17. TRADOC publication development flow.

19.2 TRADOC publications information development and delivery systems. There are a number of information management systems used within TRADOC for developing, storing and distributing training and doctrine publication information and products. The following subparagraphs provide a brief description and the roles within TRADOC for these information management systems. These TRADOC information management systems and their relationship to each other are depicted in Figure 18 and are explained in paragraphs 19.2.1 through 19.3.1.2. The significance of these systems is the interface required to implement

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SGML and reuse, re-purpose, and sharing Army publication information within the Army and with Joint Services. This interface will be discussed in paragraphs 16.2.1 through 16.3.1.2.

FIGURE 18. TRADOC information management systems relationship.

19.2.1 Automated systems approach to training (ASAT). ASAT is a software application, which supports Army doctrine and training development, support, and management functions. It operates as a training information system, a tool for decision-making, and doctrine and training products production system (printed or electronic media). ASAT is a training information and doctrine management system that provides the following capabilities:

- Task creation
- Task management and training product
- Import/Export to data capability to other Army Proponent Schools
- Produces reports in Rich Text Format (RTF), which are sent to the Reimer Digital Library (RDL) and converted to HTML
- Program of Instruction

The functional modules in ASAT include:

- Collective Individual
- Combined Arms Training Strategies (CATS)
- Program of Instruction (POI) and Course Administration Data (CAD)

19.2.1.1 Collective module. The Collective Module allows training developers to create, edit, copy, delete, or review collective and drill tasks, Mission Training Plans (MTP) and Drill Books.

19.2.1.2 Individual module. The Individual Module allows training developers to create, edit, copy, delete, or review individual tasks, Soldier Training Publications (STP), Training Support Packages (TSP) and Lesson Plans, Course/Program of Instruction (POI) and Course administrations data (CAD). The POIs are produced from previously developed lesson plans and individual task data in the individual module. Report formats include production of an:

- Individual Task Summary
- Individual Task Development
- Individual Synopsis
- STP
- TSP
- Lesson Plan
- POI/CAD

19.2.1.3 Combined arms training strategies (CATS) module. The CATS Module produces Combined Arms Training Strategies to include resource information. The CATS Module within ASAT allows Training Developers to produce dynamic training strategies. Eventually, these strategies will be made available to the Unit via the Digital Training Management System (DTMS). The process of strategy development begins with the building of an Action, which consists of:

- Organization (TOE)
- Task Selection (a grouping of collective tasks)
- Exercise
- Training Aid, Device, Simulator, and Simulation (TADSS) Selection

The CATS Module allows Training/CATS developers to build actions. (DEVELOPMENT PROCESS-Build actions-Refine actions-Create Reports) Once several Actions have been developed, the Training/CATS developer can easily produce the following reports:

- Calendar
- CATS Task to Action

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- Critical Gates Summary
- Matrix Report
- Program of Instruction

19.2.2 Reimer digital library (RDL). RDL is the approved repository of official Army training and doctrine information. Access controls are included to restrict access to all non-authenticated products, and authenticated products, as appropriate.

19.2.2.1 RDL document formats. The RDL uses Hypertext Markup Language (HTML) as its primary document output format for online viewing. The RDL makes use of some advanced features of HTML, such as "frames", "Java Script", and the Secure Socket Layer (SSL). Users are required to use a browser that supports 128 bit encryption. Additionally, many documents are available in alternative formats for print on demand, downloading, or on line multimedia streaming.

19.2.2.2 Products availability from RDL. Products accessible via the RDL include:

a. Army Doctrinal and Training Literature Program (ADTLP) products includes:
   (1) Field Manuals (FM)
   (2) Soldier Training Publications (STP)
   (3) Mission Training Plans (MTP)
   (4) Drills
   (5) Training Circulars (TCs)
   (6) Combined Arms Training Strategy (CATS)

b. Army Courseware and training products includes:
   (1) Total Army Training System (TATS) Courseware
   (2) Army Correspondence Course Program (ACCP)
   (3) TRADOC Common Core Training Support Packages (TSP)
   (4) Graphic Training Aids (GTA)
   (5) Civilian Training Materials
   (6) Other Approved Courseware

19.2.3 Digital training management system (DTMS). Digital Training Management System (DTMS) is the Army's automated training management system for unit and individual training management within the Active Component, National Guard, and Reserve units. DTMS automates training management and is customized to provide the ability to plan, resource, and manage unit training at all levels IAW FM7-0 (Training the Force), FM 7-1 (Battle Focused Training), and FM 3-0 (operations). DTMS provides:

- Tools to plan, conduct, and assess training according to Army doctrine
- Training management that delivers long-range calendars, short range calendars, training schedules, and installation schools allocation/management
- A variety of tools to program managers, supervisors, human resources and administrators, and training/event coordinators that enhance the entire training/event management spectrum
- Interfaces with the Integrated Total Army Personal database (ITAPdb) and consolidates all training proponent Automated Systems Approach to Training (ASAT) databases individual and collective tasks
- The capabilities to interface with the following systems; Army Training, Tracking, and Resourcing System (ATTRS), Center for Army Lessons Learned (CALL), Web Based Total Army Authorization Document System (WEBTADDS), Army Knowledge On-line (AKO), ARFORGEN Synchronization Tool (AST), and others

19.3 Application of SGML to the training and doctrine development process. As indicated earlier, the TRADOC publication information development environment is different from the AMC development environment. TRADOC automated systems are used for authoring, storage and distribution of training and doctrine products. Each of these systems is based on database data elements to represent the product information. For example, ASAT contains 300 plus tables, each containing a number of data elements. In order to apply MIL-STD-2361 SGML to a training or doctrine product, the relationships between the MIL-STD-2361 TRADOC SGML elements and TRADOC data elements is required to be clearly identified.

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Software was developed to automatically relate an SGML element to an ASAT data element and populate the ASAT database from the SGML-tagged data for selected TRADOC products (STP, MTP, Drill and FM).

19.3.1 MIL-STD-2361 SGML element to TRADOC system data element mapping methodology. Implementation of DPD Program in TRADOC requires two types of mapping. First, mapping was conducted between MIL-STD-2361 TRADOC SGML elements for each TRADOC product and the associated ASAT data elements. The second is a mapping between a complete technical manual (TM) and the RDL data elements. This second type of mapping is required so that TM information can be applied directly to a specific TRADOC product with out re-authoring. The results of both types of mapping was approved by TRADOC and configuration managed by the APD ASRL (see Section 7). This ensures that the SGML elements and data element versions used in the mapping stay current as publication requirements reflected in the SGML elements and TRADOC data elements change.

19.3.1.1 MIL-STD-2361 SGML element to ASAT data element mapping. The first type of mapping is the establishment and implementation of the relationship of MIL-STD-2361 TRADOC SGML to specific training and doctrine products. The product-related tables and fields in ASAT for each product (STP, MTP, Drills, and FM) were identified and paired with the corresponding MIL-STD-2361 SGML element(s) within their respective TRADOC product CRD. ASAT data elements are those provided by ATSC. The mapping results are contained in the appropriate TRADOC product CRD. Table I is an example of the SGML to ASAT data element mapping.

TABLE I. MIL-STD-2361 TRADOC SGML element to ASAT data element mapping example.

SGML ELEMENT	ATTRIBUTE	ASAT TABLE	ASAT COLUMN
int.task.wp	bos	ind_task	combat_function_para_id
ind.task.wp	reference	ind_task	parent_ind_task_id
ind.task.wp	mopp	ind_task	mopp_cd
ind.task.wp	mos.code	ind_task_occupation_speciality	combat_speciality_cd
ind.task.wp	level	ind_task_occupation_speciality	skill_level_cd
ind.task.wp	subj.area	subject_area	product_id
ind.task.wp	subj.area	subject_area	record_sequence_cd
ind.task.wp	subj.area	subject_area	subject_area_cd
ind.task.wp	subj.area	subject_area	subject_area_nm
ind.task.wp	subj.area	subject_area	record_sequence_cd

19.3.1.2 MIL-STD-2361 TM elements to RDL data elements mapping. Figure 19 depicts the mapping process used to relate SGML-tagged TM information to the RDL. The RDL is the TRADOC interface with external information processing systems. Exchange of weapon system logistics (TM and other logistics analysis information) between TRADOC and AMC organizations will be through the RDL. The mapping process used for MIL-STD-2361 TRADOC SGML to ASAT data element is used for TM mapping. The TM mapping is from the TM SGML elements to the TRADOC SGML elements. Interface into both the RDL and ASAT will be through the MIL-STD-2361 SGML element to ASAT data element mapping process. The ASAT data elements will be identical to the RDL data elements once RDL development is completed.

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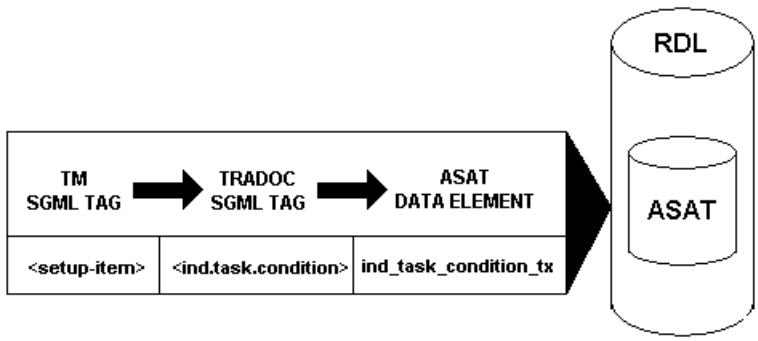

FIGURE 19. MIL-STD-2361 TM SGML element to TRADOC SGML element to RDL data element mapping.

20 TRAINING AND DOCTRINE PUBLICATION CONSOLIDATED REQUIREMENTS DOCUMENTS (CRD).

20.1 Consolidated requirements documents (CRD). CRDs were developed for the training and doctrine publication products described in the following paragraphs. The CRDs were designed by APD, in conjunction with TRADOC functional requirements experts, as a tool for applying SGML to training and doctrine publication products. The CRDs are based on concepts outlined in the SGML standards ISO 8879 and MIL-PRF-28001. An SGML tagging scheme was designed to meet the training and doctrine publication product requirements contained in TRADOC Regulation (TREG) 350-70. Training and doctrine publications are comprised of data elements. The first step in designing the generalized language of the SGML applied to training and doctrine products was to identify the relationship between the significant data elements of the respective publication products and assign each element a mnemonic name or "generic identifier." A DTD was then created using element and attribute declarations for generic identifiers. Processing information was then developed (i.e., stylesheets) to describe or specify a documents formatted output. These concepts formed the baseline for the development of a Standard Generalized Markup Language for TREG 350-70 training and doctrine publications. The CRDs provide implementation guidance and information for development of training and doctrine publications in SGML, and that conform in every way to the functional requirements contained in TREG 350-70 and other relevant TRADOC requirements documents.

20.2 Obtain consolidated requirements document (CRD). The CRDs for Soldier Training Publication (STP), Army Training and Evaluation Program (ARTEP), Training Support Package (TSP), Field Manual (FM) and System Training Plan (STRAP) are available from the ASRL http://www.asrl.com/tdconstr.htm and located in the guidance library.

20.3 Consolidated requirements document structure. CRDs include general information, a non-SGML structure outline for the training or doctrine product; applicable SGML DTDs; sample input/output text of the training and doctrine product; a detailed description of elements; the FOSI required to produce a final copy for the product; and Automated Systems Approach to Training (ASAT) data element mapping into SGML elements.

20.3.1 Non-SGML structure outline. To assist handbook users in a better understanding of the DTDs, a plain-English expression of the document structure is included in Section 2 of the CRD. Although this is not a definitive expression of the structure (which is the role of the conforming DTDs), it is intended to supplement the element descriptions and provide a functional overview of the structure as encoded in the DTDs.

20.3.2 Document type definitions (DTDs). The DTDs provided in (see Section 7) of the CRD were developed by interpreting TREG 350-70 and other pertinent requirements documents, content and structure requirements.

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20.3.3 Training and doctrine input and output samples. ASRL http://www.asrl.com/tdconstr.htm contains sample files (i.e., marked-up text as it would appear in a file prior to composition), along with a representation of the output which would be produced by this markup. The format and appearance of the printed pages are not controlled by the DTDs, but will be controlled by the FOSI or stylesheet. Samples provided in the CRD are not intended to portray the use of all the elements and/or attributes available, but only to exemplify the use of many of the elements which are particular to the respective training and doctrine products.

20.3.4 SGML element definitions. Each SGML element necessary to produce the document is described in (see Section 20.3). A discussion of the element attributes, their possible values, and the default values are also included. These descriptions can be used as an aid in interpreting the SGML elements and serve as a reference for reading the DTDs.

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VOLUME 4
INTERACTIVE ELECTRONIC TECHNICAL MANUAL (IETM) TECHNICAL DATA REQUIREMENTS TO SUPPORT THE GLOBAL COMBAT SUPPORT SYSTEM-ARMY (GCSS-A)

21 INTRODUCTION.

21.1 Overview. This Volume 4 provides the implementation guidance for the development of Interactive Electronic Technical Manual (IETM) Technical Data Requirements to Support the Global Combat Support System-Army (GCSS-A) in compliance with MIL–STD–2361 and MIL-STD-3008. GCSS-A data provided in this handbook provides non-regulatory guidance and information for the preparation of collecting and reporting operations, historical, maintenance, and ammunition data for the management and support of aviation and non-aviation weapons systems and their related systems, equipment, components/modules, including flight and mission safety parts.

21.2 Objectives. Volume 4 MIL-HDBK-2361 is designed to provide users a tool that is simple to use to develop data to populate the Interactive Electronic Technical Manual (IETM) Technical Data Requirements to Support the Global Combat Support System-Army database. This volume contains GCSS-A publication development and implementation guidance information, designed to assist publication developers and editors in the use and application of MIL-STD-3008 using XML. The data, used in conjunction with the GCSS-A, will provide the capability to support worldwide deployment, employment, and sustainment of combat forces in various scenarios and areas of operations.

21.3 Layout, format and content. Volume 4 contains information relevant to Interactive Electronic Technical Manual (IETM) Technical Data Requirements to Support the Global Combat Support System-Army, its development, and methods for applying and using XML. An outline structure of the sections are provided as follows:

- Volume 4 – Interactive Electronic Technical Manual (IETM) Technical Data Requirements to Support the Global Combat Support System-Army (GCSS-A)
  - 21 Introduction – An overview of the Global Combat Support System-Army (GCSS-A).
  - 22 GCSS-A Capabilities – Identifies and describes GCSS-A capabilities.
  - 23 GCSS-A data sources – Identifies and describes GCSS-A data source and requirements.
  - 24 GCSS-A Implementation Guidance – Identifies and describes GCSS-A data source and requirements.

22 GCSS-A CAPABILITIES.

22.1 Capabilities. The data that supports GCSS-A provides the Army a seamless, integrated, and interactive communications and Automated Information System (AIS) at all force levels of Combat Service Support (CSS). It provides users a responsive and efficient means to rapidly anticipate, allocate, and synchronize the flow of available CSS resources to equip, deploy, project, sustain, reconstitute, and re-deploy tactical forces in support of the national military strategy. GCSS-A data collected from operations, maintenance, historical, ammunition and parts requisition data will provide for the major capabilities .

a. Provide the capability to request all classes of supplies, manage prescribed load lists, bench stock, shop stock, combat spares, and reparable items.
b. Provide the capability to manage maintenance workloads and coordinate repair activities, establish maintenance priorities, and control subordinate elements such as combat repair teams (CRT) and mobile support teams (MST) and contact teams operating from the unit maintenance collection point (UMCP).
c. Provide the capability to manage maintenance workloads and coordinate repair activities, establish maintenance priorities, and control subordinate elements such as combat repair teams (CRT) and mobile support teams (MST) and contact teams operating from the unit maintenance collection point (UMCP).
d. Provide the functionality needed to report supply and maintenance financial information at all levels.
e. Provide the functionality to report maintenance status and Army Materiel Status System (AMSS) data from organizational maintenance at the unit level, shop operations of Direct Support (DS) and General

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Support (GS) maintenance units, installation level maintenance support at TDA activities, and Aviation Unit Maintenance (AVUM) and Aviation Intermediate Maintenance (AVIM).

f. Provide the functionality for the equipment dispatch procedures, maintenance and inspection worksheet procedures, recording and disposition of maintenance faults, repair parts removal and installation, equipment services performed, and all The Army Maintenance Management System (TAMMS) and TAMMS-Aviation (TAMMS-A) record management.
g. Downloading of existing weapon system historical and maintenance records and data from GCSS-A to an external device for deployment with the unit
h. Transfer of equipment, maintenance, historical, supply requests, and other information back to the parent unit and the task force command while operating in a deployed status.
i. Upload records, information, and data for system updates upon return of units.
j. Provide MST, CRT, maintenance contact team, maintainers, and technical repair personnel for all weapon systems with maintenance information at the point of repair or on site wherever maintenance is performed.
k. Provide users with an embedded capability that allows access to the ETM, ETM-I, or IETM.
l. Provide total asset visibility and readiness of sub-weapon systems and sub-components by weapons system or end item tail number, bumper number, or serial number.
m. Provide the functionality needed to dynamically query unit readiness status and trends on any weapon system, end item, subsystem or serial numbered component for any preceding period of time.
n. Provide the capability to manage phased maintenance requirements for weapon systems and end items by model and mission design series.
o. Provide the functionality and capabilities needed to monitor, control and track commercial vendor provided maintenance support.

23 GCSS-A DATA SOURCES.

23.1 Collecting data. Operational, maintenance, historical, ammunition and parts requisitioning data is collected to support aviation and non-aviation weapons systems and their related systems, equipment, components/modules, including flight and mission safety parts. Electronic Technical Manuals (ETM) and Interactive Electronic Technical Manuals (IETM) are the primary source of collecting data for GCSS-A data base. Other sources of data can be obtained by means of manual or an automated process.

23.2 Data preparation. The data will be prepared as SGML or XML in accordance with MIL-STD-3008 and using the Document Type Definition DTD MIL-STD-3008 provides. MIL-STD-3008 DTD can be obtained from the Army Publish Directorate's ASRL website "www.asrl.com".

23.3 Army sources for providing data. ETMs and IETMs are the primary source for providing data. Other data sources are Army systems such as the Army Maintenance Management System-Aviation (TAMMS-A), Unit Level Logistics System (ULLS), or Standard Army Maintenance System (SAMS) or the data can be manually provided in the interim using form templates.

23.3.1 Army maintenance management system-aviation (TAMMS-A). An Army system “TAMMS-A” provides data on aircraft, aviation-associated equipment, mission related equipment, and maintenance. The data is collected by means of forms and records administered by TAMMS-A. These forms and records forms are use d to manage maintenance, control the use, and report warranty actions and deficiencies on Army aircraft and aviation associated equipment.

23.3.2 Unit level logistics system (ULLS). ULLS automates the logistics system for unit supply, maintenance and materiel readiness management operations. It prepares supply documents, maintenance management records, materiel readiness reports and property records. GCSS-A will absorb ULLS functionality into its Maintenance Module.

23.3.3 Standard army maintenance system (SAMS). SAMS provides centralized information management to support maintenance operations at the Direct Support, General Support, and Installation Activities (both Table of Organization and Equipment (TOE) and Table of Distribution and Allowances (TDA) units). SAMS is divided into two levels: SAMS-1 which operates at the DS/GS maintenance company; and

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SAMS-2 which operates at command levels above the maintenance company. GCSS-A will absorb SAMS functionality into its Management Module.

24 GCSS-A IMPLEMENTATION GUIDANCE.

24.1 GCSS-A's DTD. GCSS-A DTD interprets the technical content and structure for the data requirements contained in MIL-STD-3008. GCSS-A's Document Type Definition (DTD) and associated tag and attribute descriptions are used for preparation and reporting of operations, historical, maintenance, and ammunition data. The data developed using this DTD will be provided to the Global Combat Support System - Army (GCSS-A) to facilitate efficient management and support of aviation and non-aviation weapons systems and their related systems, equipment, components/modules, including flight and mission safety parts.

24.1.1 Data types required. GCSS-A requires specific types of data to populate its database for aviation and non-aviation weapons systems and their related systems, equipment, components/modules, including flight and mission safety parts. GCSS-A requires data for five major functional areas and also requires cross-functional data.

24.1.1.1 Major data types. GCSS-A requires data for the five major functional areas listed :

a. Operational data
b. Maintenance data
c. Historical data
d. Ammunition data
e. Parts requisitioning data.

24.1.1.2 Cross-functional data. Cross-functional data is developed for multiple functional areas and are used to link with other specific functional data required byMIL-STD-3008. Cross-functional data requirements are applicable to both aviation and non-aviation areas.

24.1.2 Data tables. Specific data is developed and organized in the form of functional data tables. GCSS-A MIL-STD-3008 consist of sixty-five tables. These tables are structured according to the data associations which dictate the table configuration. The interrelationships and data hierarchy between tables are only established through the use of the applicable Document Type Definition. The tables support cross data requirements, aviation data requirements and non-data requirements. All the tables except Table 4 consist of the "Data Element Title" and the "Data Name". The "Data Element Title" provides the noun phrase to identify the data element with sufficient modifiers to ensure title uniqueness for a specific data element definition. The "Data Name" is defined in the DTD. Table 4 consist of “Table No”, Table Title, Equipment/Personnel Tables broken into sections consisting of Table 1, Table 2, and Table 3.

24.1.2.1 Meta tables. GCSS-A meta tables ***<meta-tables>*** consists of Table 1. Equipment type, Table 2. Equipment assignment and Table 3. Personnel data. These tables are allowed by the GCSS-A DTD's as stand-alone submissions.

24.1.2.1.1 Table 1. Equipment type. Table 1 provides standardized information for identifying aviation and non-aviation equipment, the latter including watercraft and rail equipment. This table also provides specific information on a piece of equipment, both by alpha and/or numerical designation and origin. It encompasses an equipment grouping that includes: assembly, sub-assembly, component, sub-component, end item, part, module, piece, and accessory.

24.1.2.1.2 Table 2. Equipment assignment. Table 2 provides necessary information to identify the organization that has ownership, custody and/or control of the specified aviation or non-aviation equipment.

24.1.2.1.3 Table 3. Personnel data. Table 3 provides identification and operational information on personnel who are involved in the daily operations of aviation and non-aviation equipment. The information includes both the operating crew and the maintenance personnel who provide daily support services and perform daily inspections.

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24.1.2.2 Table 4. Equipment/personnel matrix. Table 4, Figure 20 identifies Tables 5 through 65 in this standard that are linked to required identifiers contained in Tables 1 through 3. A table that has no assignment to either Table 1, Table 2 or Table 3 is a standalone table containing relevant identifying information.

TABLE 4. Equipment/personnel matrix

		Equipment/Personnel Tables		
Table No.	Table Title	Table 1	Table 2	Table 3
5	Servicing Data	X	X	X
6	Preventive Maintenance Scheduled Data	X	X	X
7	Equipment Inspection and Maintenance Data	X	X	X
8	Equipment Inspection/Modification Data	X	X	
9	Maintenance Request Register Data	X	X	
10	Maintenance Required Data	X		
11	Parts Data	X	X	
12	PLL Inventory Data		X	
13	Cost Data	X	X	X
14	Environmental Conditions Data	X	X	X
15	Operations/Maintenance Fault Data	X	X	X
16	Fault Correcting Data	X	X	
17	Related Maintenance Actions Data	X	X	X
18	Uncorrected Fault Data	X	X	
19	Not Mission Capable Data	X	X	
20	IETM Fault Result Data	X	X	
21	IETM Maintenance Reporting Data	X	X	
22	Oil Analysis Request Data			
23	Oil Analysis And Recommendation Feedback Data			
24	Product Quality Deficiency Data			
25	Ammunition Accounting Data	X	X	X
26	Ammunition Condition Data	X		X
27	Ammunition Peculiar Equipment Utilization Data			
28	General Requisition Data			
29	Technical Manual Deficiency Data			
30	IETM Deficiency Data			
31	Flight Data	X	X	X
32	Flight Manual Exceedance Data	X	X	X
33	System Status Data	X	X	
34	Armament System Data	X	X	X
35	Armament System Sighting Data	X	X	X
36	Component Data	X	X	
37	Removal Data	X	X	X
38	Repair/Overhaul/Gain Data	X	X	X
39	Installation/Loss Data	X	X	
40	Life Raft Data			
41	Helmet And Oxygen Mask/Connector Data			
42	Survival Radio/Emergency Locator Transmitter Data			
43	Survival Kit Inspection And Maintenance Data			
44	Mesh Net Survival Vest Data			
45	Life Preserver Data			
46	Oxygen Console Service Data			
47	Anti-Exposure Coveralls Data			
48	NVG Inspection And Maintenance Data			
49	Vibration Data	X	X	
50	Component Data	X	X	
51	Component/Module Recorder Data	X	X	
52	Aircraft Inventory Data	X	X	
53	Engine Turbine Wheel Data	X	X	X
54	Turbine Analysis Check Data	X	X	X
55	Engine History Recorder Operating Hours Data			
56	Meter Tracked Component Data			
57	Equipment Utilization Data	X	X	
58	Equipment Deadlined Data	X	X	
59	Armament System Data			
60	Equipment Control Data	X	X	X
61	Equipment Maintenance And Calibration Data	X	X	
62	Dry-Docking, Painting And Condition Of Vessel Bottom Data	X	X	X
63	Daily Inspection Data	X	X	X
64	Field Inspection Data	X	X	X
65	Locomotive Inspection And Repair Data	X	X	X

FIGURE 20. Example of Table 4. Equipment/personnel matrix.

24.1.2.3 Tables 5–65. Tables 5 through 65 consist of required data to support cross-functional data requirements, aviation data requirements and non-aviation data requirements.

24.1.2.4 Cross-functional data requirements. Cross-functional data requirements, located in MIL-STD-3008 have attributes that can span multiple functional areas and are used to link with other functional tables throughout MIL-STD-3008. The cross-functional data contained in the tables are inherent to the identification,

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description, origin, custody, operation, support and location of a specified weapon system or piece of equipment. Cross-functional data requirements are applicable to both aviation and non-aviation areas.

24.1.2.5 Aviation data requirements. Aviation data requirements, located in MIL-STD-3008 provide necessary information to manage operations and maintenance, control the use, and report warranty actions and deficiencies on Army aircraft and aviation-associated equipment. Aviation-associated equipment is construed to mean related mission equipment such as armament systems, electronic systems, aircraft training devices, aircraft simulators and life support equipment.

24.1.2.6 Non-aviation data requirements. Non-aviation data requirements, located in MIL-STD-3008 provide necessary information to manage operations and maintenance, to control the use, and to report warranty actions and deficiencies of Army equipment that include self-powered vehicles, towed vehicles, stationary powered equipment, watercraft and rail equipment.

24.2 Example of tagged IETM history. The following is a scenario of a soldier using an IETM to troubleshoot a piece of equipment and to complete the maintenance task on the equipment.

a. A soldier is given a work order and an IETM to troubleshoot the malfunction of a piece of equipment and to complete the maintenance task on the equipment.
b. The soldier logs on to the IETM using his personal identification number ***<pid>***.
    (1) The IETM will help identify the fault of the equipment malfunction by stepping the soldier through a series of fault diagnostics. Performing the series of fault diagnostics will provide the soldier the correct action to order the part(s) needed and to complete the maintenance task on the equipment.
    (2) The IETM handles the majority of the reporting by prompting the soldier with questions and the soldier with yes or no questions.
    (3) Parts of the IETM the soldier does need to fill out starting with his:
        (a) Table 3. Personnel Data ***<persn-data>*** - Identifies the soldier which in turn could cross-reference and identify what equipment he has worked on and what equipment he is certified to work on. The soldier enters the:
            1 Personal identification number ***<pid>***.
        (b) Table 1. Equipment type data ***<eqp-type-data>*** - Identifies the equipment by specific identification such as administration number ***<admin-no>***, part number ***<partno>*** & CAGE code number ***<cageno>*** and/or National Stock Number ***<nsn>***. The administration number is usually given with the work order. The soldier enters the:
            1 Administration number ***<admin-no>***.
            2 Make Model-Type ***<make-model-type>***.
            3 Serial number ***<serialno>***.
        (c) Table 21. IETM maintenance reporting data ***<ietm-maint-report>*** – Covers data resulting from maintenance actions performed and including parts information.
            1 Quantity of each part needed ***<qty>***.
            2 XREF attribute value of IETM Parts Data ***<ietm-parts-data>***
    (4) Parts of the IETM that is usually fill out by the soldier:
        (a) Table 1. Equipment type data ***<eqp-type-data>***.
            1 National Stock Number (NSN) ***<nsn>***
            2 Nomenclature ***<nomen>***
        (b) Table 2. Equipment assignment data ***<eqp-assign-data>***
            1 DoD Activity Address Code ***<dodaac>***
            2 Unit Identification Code ***<uic>***
            3 Name ***<name>***
            4 Address ***<address>***
            5 Home Station ***<home-station>***
            6 Zip Code ***<zip>***
        (c) Table 20. IETM fault result data ***<ietm-fault-result-data>*** - Covers fault diagnostics/troubleshooting data.
            1 ID attribute value of IETM Fault ***<ietm-fault-result id="">***

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2 Symptom ***<symptom>***
3 Description ***<desc>***
4 Fault Code ***<faultcode>***
5 Date Discovered ***<date-fault>***
6 How Recognized ***<how-recognized-code>***
7 Failure Code ***<fail-code>***
8 IETM Mission Status Data ***<ietm-mission-status-data>***
9 Status Symbol ***<status-symbol>***
10 Mission Capability ***<mission-capability>***
11 Date Completed ***<date-completed>*** (Identifies how many days or months it took to identify the equipment's malfunction.)
12 Time Completed ***<time-completed>*** (Identifies how many minutes or hours of the day it took to identify the equipment's malfunction.)

(d) Table 21. IETM maintenance reporting data ***<ietm-maint-report-data>*** - Covers data resulting from maintenance actions performed and including parts information.
1 IETM Maintenance Report ***<ietm-maint-report>***
2 IETM Parts Data ***<ietm-parts-data>***
3 Parts Ordered Information ***<parts-ordered-info>***
4 Parts Received Information ***<parts-received-info>***
5 Part Number ***<partno>***
6 Nomenclature ***<nomen>***
7 Source, Maintenance, Recoverability (SMR) ***<smr>***
8 CAGE Code ***<cageno>***
9 NIIN ***<niin>***
10 Date ***<date>***
11 Time ***<time>***
12 Maintenance Action Performed ***<maint-performed>***
13 Maintenance Action ***<maint-action>***
14 Date Performed ***<date-performed>***

(e) The soldier will pass the data through a chain of command for verification. Then the data from the IETM after verification will be sent as a data stream of information to LOGSA.

(f) The data will be disbursed to other Army data systems that manage specific required data.

c. The following is an example of a cross-functional data tagged XML instance for Table 20. IETM fault result data and Table 21. IETM maintenance reporting data:

<gcss-a>
<meta-tables>
<persn-data>
<pid>X9991
</pid>
</persn-data>
</meta-tables>
<ietm-fault-result-data>
<eqp-type-data>
<admin-no>X99
</admin-no>
<nsn>2320-99-999-9999
</nsn>
<make-model-type>X9999
</make-model-type>
<nomen>Truck, Tractor W Crn
</nomen>
<serialno>99999
</serialno>

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</eqp-type-data>
<eqp-assign-data>
*<dodaac>*XXXWRC
</dodaac>
*<uic>*XXP
</uic>
<unit-activity>
*<name>*B 18TH BN 33RD ARM DIV
</name>
*<address>*Hwy44 North
</address>
*<home-station>*Fort Sill, Oklahoma
</home-station>
*<zip>*73507
</zip>
</unit-activity>
</eqp-assign-data>
<item-fault-result>
<item-fault-data id="F991">
*<symptom>*inoperative
</symptom>
*<desc>*deterioration, break
</desc>
*<faultcode>*099
</faultcode>
*<date-fault>*2000-01-09
</date-fault>
*<how-recognized-code>*900
</how-recognized-code>
*<fail-code>*X
</fail-code>
</item-fault-data>
<ietm-mission-status-data>
<status-symbol status="status-inoperable"/>
<mission-capability capability="none"/>
</ietm-mission-status-data>
</item-fault-result>
<item-fault-result>
<item-fault-data id="F992">
*<symptom>*inoperative
</symptom>
*<desc>*leakage
</desc>
*<faultcode>*092
</faultcode>
*<date-fault>*2000-01-09
</date-fault>
<>
*<how-recognized-code>*902
</how-recognized-code>
*<fail-code>*XY
</fail-code>
</item-fault-data>
<ietm-mission-status-data>
<status-symbol status="materiel-defect"/>
<mission-capability capability="reduced"/>

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</ietm-mission-status-data>
</item-fault-result>
</item-fault-result-data>
<ietm-maint-report-data>
<eqp-type-data>
*<admin-no>*X99
</admin-no>
*<nsn>*2320-99-999-8888
</nsn>
*<make-model-type>*X9999
</make-model-type>
*<nomen>*Truck, Tractor W Crn
</nomen>
*<serialno>*99999
</serialno>
</eqp-type-data>
<eqp-assign-data>
*<dodaac>*XXXWRC
</dodaac>
*<uic>*XXF
</uic>
<unit-activity>
*<name>*B 18TH BN 33RD ARM DIV
</name>
*<address>*Hwy44 North
</address>
*<home-station>*Fort Sill, Oklahoma
</home-station>
*<zip>*73507
</zip>
</unit-activity>
</eqp-assign-data>
<ietm-main-report>
<ietm-parts-data xref="F991">
<parts-ordered-info>
*<partno>*9XXX9
</partno>
*<nomen>*Battery
</nomen>
*<smr>*XXXXX
</smr>
*<cageno>*9999X
</cageno>
*<nsn>*XXXX-99-999-9999
</nsn>
*<niin>*01-349-7676
</niin>
*<qty>*X
</qty>
*<date>*2000-01-09
</date>
*<time>*12:12:15
</time>
</parts-ordered-info>
<parts-receive-info>
*<partno>*9XXX9

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</partno>
<nomen>Battery
</nomen>
<smr>XXXXX
</smr>
<cageno>9999X
</cageno>
<nsn>XXXX-99-999-9999
</nsn>
<niin>01-349-7676
</niin>
<qty>X
</qty>
<date>2000-01-19
</date>
<time>16:25:30
</time>
</parts-received-infor>
</ietm-parts-data>
</ietm-main-report>
<ietm-main-report>
<ietm-parts-data xref="F991">
<parts-ordered-info>
<partno>XX99999-XXX
</partno>
<nomen>nut, plain, hex, 3/8-16
</nomen>
<smr>XXYYY
</smr>
<cageno>8888X
</cageno>
<nsn>XXXX-88-888-9999
</nsn>
<niin>88-888-9999
</niin>
<qty>10
</qty>
<date>2000-01-09
</date>
<time>12:12:15
</time>
</parts-ordered-info>
<parts-receive-info>
<partno>XX99999-XXX
</partno>
<nomen>nut, plain, hex, 3/8-16
</nomen>
<smr>XXYYY
</smr>
<cageno>8888X
</cageno>
<nsn>XXXX-88-888-9999
</nsn>
<niin>88-888-9999
</niin>
<qty>10

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</qty>
*<date>*2000-01-15
</date>
*<time>*14:15:30
</time>
</parts-received-infor>
</ietm-parts-data>
</ietm-main-report>
<ietm-main-report>
<maint-performed>
*<maint-action>*BaseMaint = "X99-XX"
</maint-action>
*<date-performed>*2000-01-09
</date-performed>
</maint-performed>
</ietm-main-report>
</ietm-main-report-data>
</gcss-a>

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VOLUME 5
TECHNICAL AND EQUIPMENT PUBLICATIONS

PART I
OVERVIEW

25 INTRODUCTION.

25.1 Overview. This Volume 5 provides implementation guidance for Technical Manuals (TMs), Depot Maintenance Work Requirements (DMWRs) and National Maintenance Work Requirements (NMWRs). TMs, DMWRs and NMWRs are discussed in sufficient detail to allow the handbook user a comprehensive understanding of their development processes using XML. Part I contains general TM, DMWR and NMWR development guidance. Part II contains detailed guidance for the application of XML to TMs, DMWRs and NMWRs, related training information, and an XML Tutorial for TMs, DMWRs and NMWRs. There are flow charts and other graphic illustrations to amplify the narrative discussions. The differences between the "traditional" publication development processes and development using the Digital Publications Development (DPD) Program, (see 8.2.3) concepts are discussed to illustrate the benefits, in time and effort, derived from the DPD Program. These discussions are designed to portray to handbook users sound rationale for developing publications using XML in accordance with MIL–STD–2361. The DPD Program concept of reusing information that is common to more than one type of equipment is explained in the context in which it relates to different publication types. Part I is structured with five Sections, as shown in "Layout, Format and Content", below. Part II is structured with three parts, as shown in "Layout, Format and Content".

25.2 Objectives. Volume 5 of MIL-HDBK-2361 is designed to provide users with a tool that is simple to use and functionally accurate to the Army TMs, DMWRs, and NMWRs publication processes. The volume contains TM, DMWR and NMWR publication development and implementation guidance information, designed to assist the publication developer in the use and application of XML Implementation guidance to realize the MIL–STD–2361 objectives to share and reuse common publication information is an underlying theme throughout the handbook.

25.3 Layout, format and content. Volume 5 contains information relevant to TMs, DMWRs and NMWRs and their development using XML. An outline of each of the parts is provided.

Part I – INTRODUCTION TO MIL-HDBK-2361

25 – Introduction	An overview of the TM volume: including the intent, layout, format, and contents.
26 – Workflow and Processes	Identifies and describes the workflow and processes associated with the development of TMs.
27 – Implementation Guidance	This section will contain a descriptive narrative of the functional and technical relationships between TMs, DMWRs and NMWRs, MIL-STD-40051-2 and MIL–STD–2361. The part will include explanations, examples and descriptive narrative of how the TM contents (e.g., work packages, tasks, etc.) maps (relates) to the appropriate paragraph/page of the respective standard.
28 – TM, DMWR/ NMWR Acquisition	This section will provide the handbook user with information relating to contracting for XML/publication development services.

Part IIPart II – TM Information Chapters

29 – MIL–STD–2361 XML Applications Introduction	This section provides information on the presentation of the XML elements developed for Army digital equipment manuals publications.
30 – General Information	This section provides information on general topics in developing TMs, DMWRs and NMWRs, using MIL-STD-40051-2 and MIL–STD–2361

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31 – Production	This section provides an in-depth description and use of the XML elements specifically used in the developing a complete TM or DMWR/ NMWR manual.
32 – Work Package Basics	This section provides information on the description of the elements and attributes that provide for work package metadata, identification and setup information.
33 – Tasks, Procedures, and Steps	This section provides an in-depth description on the use of tasks, procedures and steps used in MIL-STD-2361.
34 – General Information, Equipment Description And Theory Of Operation Chapter	This section provides an in-depth description and use of the XML elements specifically used in the General Information, Equipment Description with Theory of Operation Chapter (GIM).
35 – Operator Instructions Chapter	This section provides an in-depth description and use of the XML elements specifically used in the Operating Procedures Information Chapter (OPIM).
36 – Troubleshooting Chapter	This section provides an in-depth description and use of the XML elements specifically used in the Troubleshooting Information Chapter (TIM).
37 – Maintenance Instructions Chapter *<mim>*	This section provides an in-depth description and use of theXML elements specifically used in the Maintenance Information Chapter (MIM).
38 – Parts information (Rpstl)	This section provides an in-depth description and use of the XML elements specifically used in the Parts Information Chapter (PIM).
39 – Destruction of Army Material Chapter	This section provides an in-depth description and use of the XML elements specifically used in the Destruction Information Chapter (DIM).
40 – Supporting Information Chapter	This section provides an in-depth description and use of the XML elements specifically used in the Supporting Information Chapter (SIM).
41 – Alerts	This section provides an in-depth description and use of Warnings, Cautions and Notes used in MIL-STD-40051-2 and MIL–STD–2361.
42 – Tables	This section provides an in-depth description and use of tables used in MIL-STD-40051-2 and MIL–STD–2361.
43 – Standard Information	This section provides an in-depth description and use of standard information used in MIL-STD-40051-2 and MIL–STD–2361.
44 – Illustration, Graphic, and Multimedia	This section provides an in-depth description and use of illustrations, graphics and multimedia used in MIL-STD-40051-2 and MIL–STD–2361.
45 – Footnotes	This section provides an in-depth description and use of footnotes and footnote references used in MIL-STD-40051-2 and MIL–STD–2361.
46 – Linking	This section provides an in-depth description and use of linking used in MIL-STD-40051-2 and MIL–STD–2361.
47 – Filtering	This section provides an in-depth description and use of filtering used in MIL-STD-40051-2 and MIL–STD–2361.
48 – Unique IETM Functionality	This section provides an in-depth description and use of state tables, logic engines and dialog boxes used in MIL-STD-40051-2 and MIL–STD–2361.
49 – Common Structure	This section provides an in-depth description and use of the XML elements commonly used throughout TMs, DMWRs and NMWRs development.
50 – Boilerplates	This section provides an in-depth description and how to use standard verbatim text to develop specific work packages in a TM, DMWR and NMWRs.

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26 WORKFLOW AND PROCESSES.

26.1 Technical and equipment publication workflow and processes. The flow diagram in Figure 21 illustrates a typical contractor TM development cycle. The process shown is a subset of the overall logistics support development which is depicted as “Gather Source Data” in the figure. Typically, a TM develops through three iterations: Preliminary Draft Equipment Publications (PDEP); Draft Equipment Publications (DEP); and Final Draft Equipment Publications (FDEP).

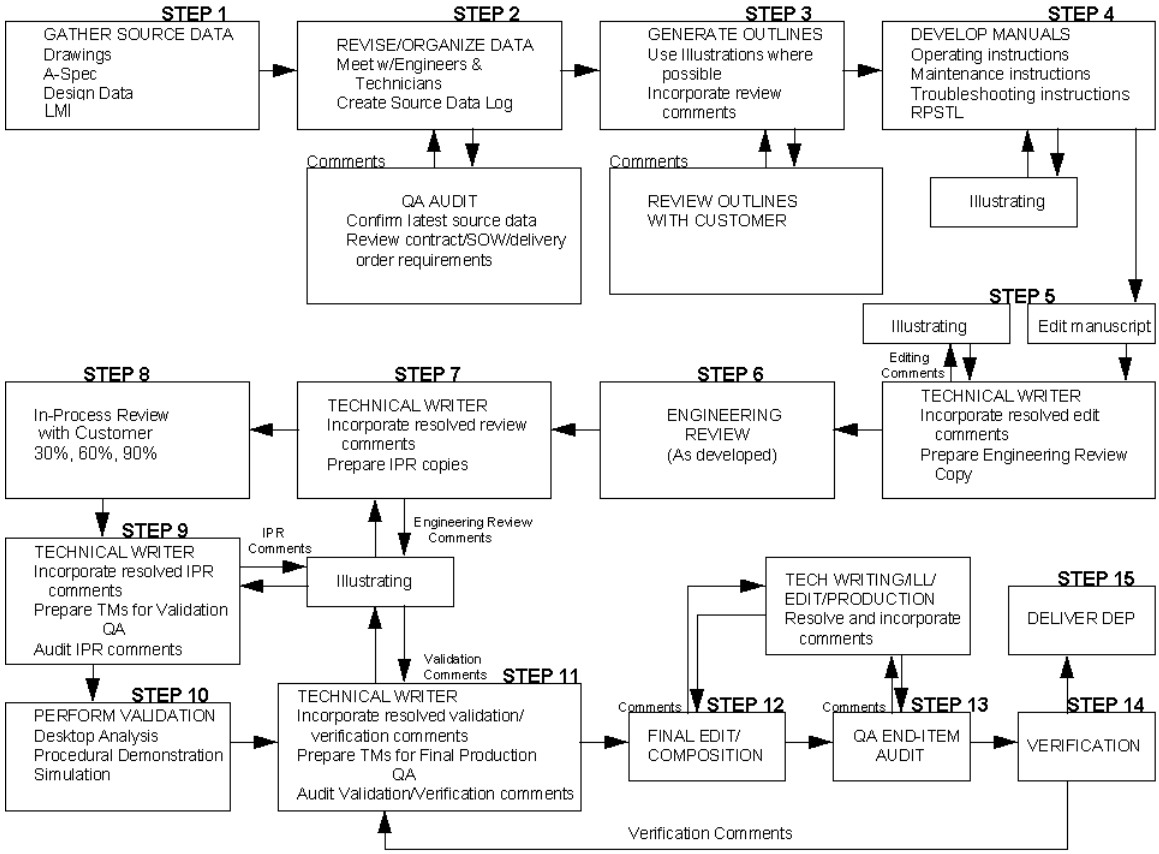

FIGURE 21. Technical manual development cycle.

26.1.1 PDEP and DEP development. The PDEP and DEP are developed during the development cycle of a weapon system and are generally used in testing the weapon system. The FDEP is developed in the production cycle of a weapon system and, when printed and distributed, is the TM used in the field. The FDEP is usually a modification of the DEP resulting from changes to the weapon system prior to the start of production.

26.1.2 Organization process of a TM. Figure 21 shows the process of how a TM is organized, developed, reviewed for accuracy during the development process, validated (usually by the writer), and verified (by the customer). Comments are constantly flowing to the database until all verification comments are resolved and included in the FDEP. The FDEP, upon delivery, may be used for printing or final digitalization.

26.2 Traditional paper TMs, DMWRs and NMWRs development: process and flow. Figure 22 shows the typical mechanics of developing paper TMs under the current specifications. The digital text file is produced in a standard word processing system and marked up for formatted output. Illustrations are prepared using CAD/CAM software, graphics illustration software, illustrators, or scanned from existing sources. The illustration files are merged into the digital file during the process of composition. Pages are composed according to the specified format for the level of maintenance for which the TM is being

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prepared. Each page created may be output on reproducible paper, film, or on a laser printer to obtain camera-ready copy. The camera-ready copy is made into offset plates and printed using conventional offset methods. The printed manual is generally distributed in loose-leaf form. The digital file of the composed TM, with its illustration files, is available for the revision process.

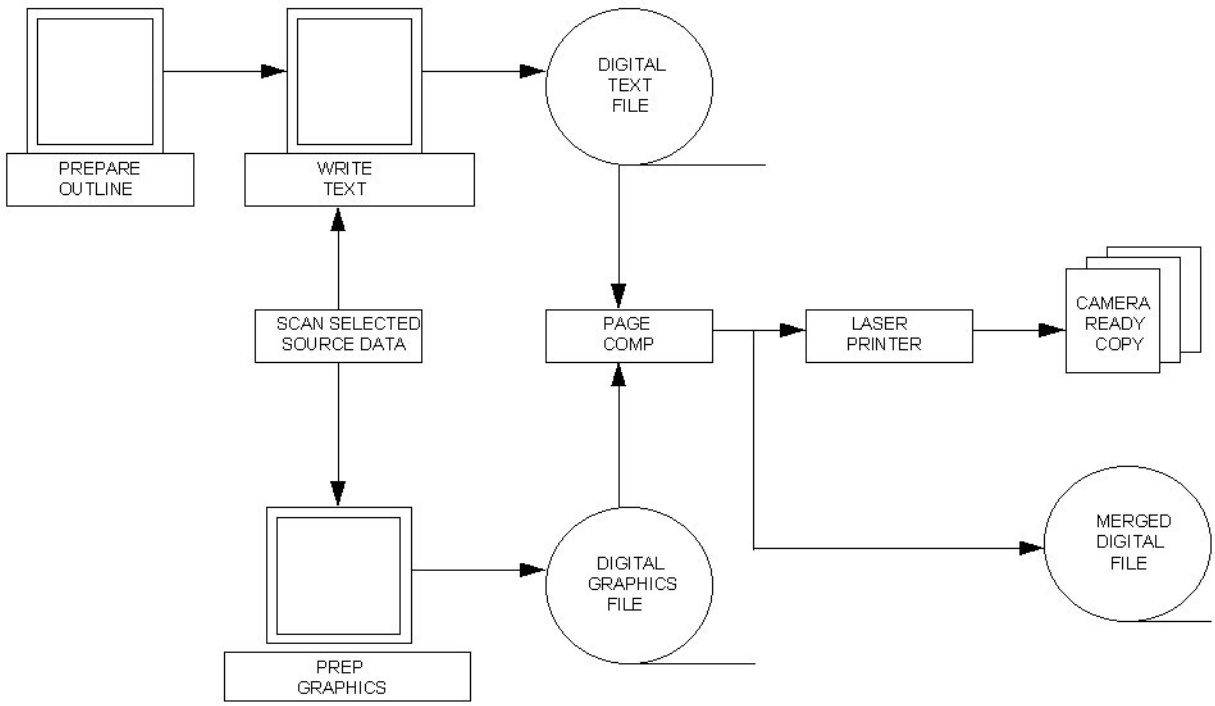

FIGURE 22. Traditional paper TM development process and flow.

26.3 MIL–STD–2361/MIL–STD–40051–1/–2 TM Development: process and flow. Figure 23 illustrates this flow of TM development under MIL–STD–2361. The gathering of source data will remain essentially unchanged at first, although improved methods of digital capture of data could flow from MIL–STD–2361 in the future.

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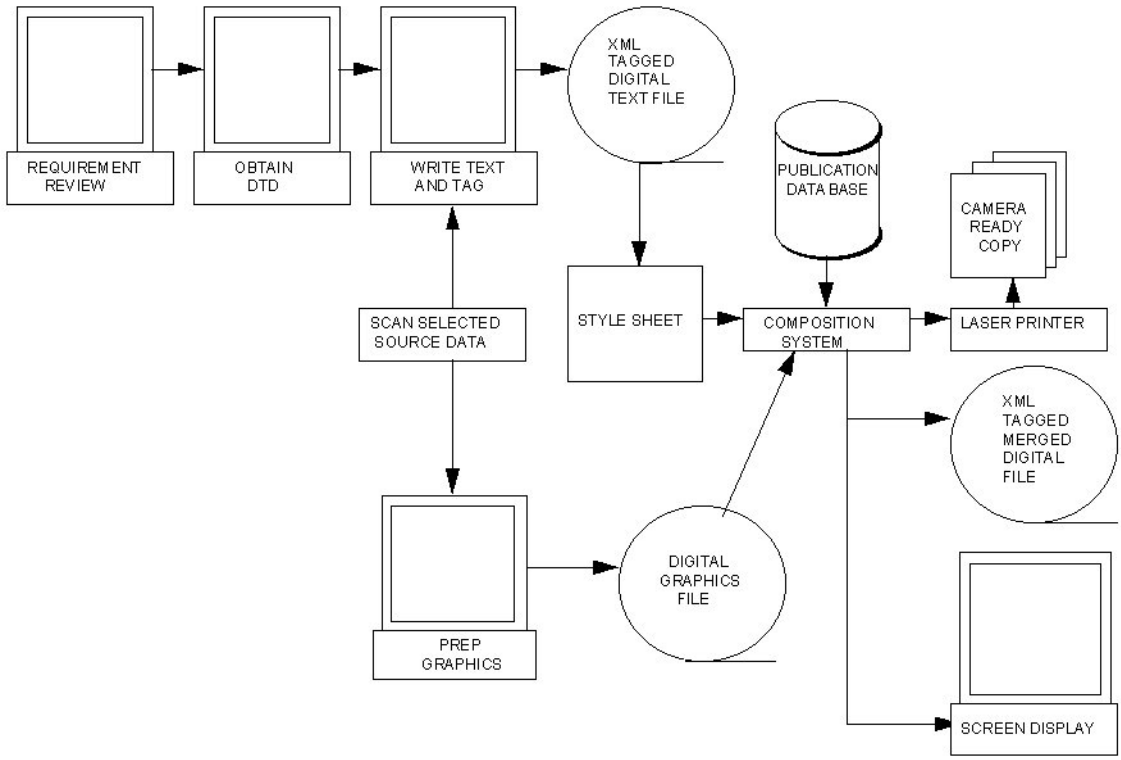

FIGURE 23. TM Development with MIL–STD–2361.

26.3.1 Requirements review. Requirements may be received as a Request for Proposal (RFP), a contract, or modification to a contract. Upon receipt of the requirements, Subject Matter Experts (SME) and managers should review the requirements to determine that they reflect the customer stated end-product (the deliverable). Areas that are unclear or apparently misstated, should be noted for discussion with the requirement proponent. The requirements review should, as a minimum, determine the following:

a. Location of documentation, such as standards or specifications, required to complete the task. The documentation may accompany the contractual document as an attachment or exhibit. Frequently, however, contractual documents specify requirements by reference and direct the TM developer to a specific address to obtain the documentation.
b. Version, identification, and availability of Document Type Definitions (DTD) and other XML objects and constructs required to develop the TMs in accordance with the Government requirements. Again, the DTDs may be provided with the contractual document. Or, they may be referenced and direction provided to contact the Administrative SGML/XML Registry and Library (refer to paragragh 7.3 for further information) to obtain the DTD and other associated XML objects and constructs.
c. Legacy and new TM development. This is an important determination for both level of effort required and cost to complete the effort. Application of XML to legacy data requires conversion from either paper or digital documents, and is normally time consuming, labor intensive, and costly.

26.3.2 XML object and construct review and setup. The DTD, and other XML objects and constructs, will be acquired, either from the contracting activity as part of the contractual document, or from the ASRL (see 7). If the DTD is not included with the contractual document, a Formal Public Identifier (FPI), location, and procedures for access should be provided for the required DTD. Figure 24 illustrates a general overview of the functional flow for the XML setup.

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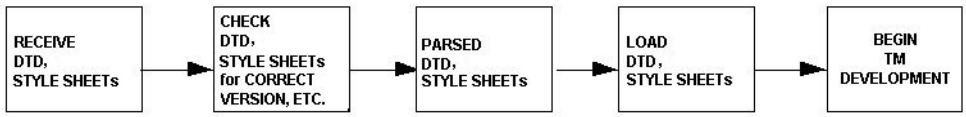

FIGURE 24. XML Setup Process.

Once the DTDs and associated XML objects and constructs (e.g., FOSIs, tag description lists, etc.) have been obtained, the following should be accomplished:

a. Check the FPI, abstract, DTD, and FOSI to ensure the correct, contractually required (e.g., version number) XML constructs have been provided. There may be more than one version of a DTD.
b. Parse the DTDs on an ISO-8879 compliant parser (e.g., SGMLS, etc.) Parsing should be done immediately after the DTDs are received. This will ensure the correctness of the DTDs and preclude TM development with a DTD containing errors. If the DTD does not parse, the provider of the DTD (contracting activity or the ASRL) should be contacted immediately.
c. Load the DTDs and, if appropriate, FOSIs on the XML platform(s) and setup for the specific XML tools (e.g., ArborText Author/Editor, Near & Far Author, and Soft Quad's Author Editor). If problems are encountered the ASRL maintains a capability for technical andXML help.
d. Determine if conversion routines will be required for the FOSI to publish on existing platforms, and ensure development of the routines is included in the development cycle schedule for the TMs.

26.3.3 Document Type Definition (DTD) as the outline. An outline of the manual, traditionally developed by the contractor as a deliverable, is not required when using the DPD concept and MIL–STD–2361. The MIL–STD–2361 DTD, provided by the Government as Government Furnished Information (GFI), can be used to develop the "outline" of the required TM in accordance with the functional requirements of MIL–STD–40051–1/–2. The DTD reflects the structure of the content volumes of MIL–STD–40051–1/–2. The main divisions of the manual are information chapters, which are comprised of work packages. Each of these divisions of the DTD have associated markup tags. In the case of TMs, the MIL–STD–2361 XML tags associated with each of the functional requirements contained in MIL–STD–40051–1/–2 have been physically embedded in the standard (see Figure 25). TM developers are able to prepare and furnish outlines by selecting applicable MIL–STD–2361 content tags which conform to content requirements specified in MIL–STD–40051–1/–2.

5.2.2.2 Description and use of controls and indicators work package <ctrlindwp>. Information shall be prepared for the description and use of all system or equipment controls and indicators. An introduction may be included in the work package. A description and use of controls and indicators <ctrlinddesc> shall be prepared for each equipment, assembly, or control panel having controls and indicators. Illustrations shall be prepared for all operator controls and indicators. For complex systems, a table <ctrlindtab> (standard information) or list may be used to explain the use of the controls and indicators. If a table is used, the controls and indicators description may be abbreviated to provide a reference to the table. For each control and indicator, the following entries shall be provided. Initial setup information is not required for this work package.

FIGURE 25. Example of embedded tags as shown in MIL–STD–40051–1/–2..

26.3.4 TM development. Upon receipt of the appropriate DTD, the writer can begin developing and writing the individual work packages. The writer can follow much the same development path as usual, EXCEPT that the material being developed will be tagged in accordance with the DTD that applies to the information chapter in which the work package appears. Most of the XML author/editor tools today have "concurrent parsing" (which can be turned on and off) that "parses as you write." In other words, it will not allow incorrect tagging. The organization to which the technical writer or editor belongs may arrange tagging to suit its own structure and work flow. Tagging may be done by the TM writers themselves, or by

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specially trained staff. In general, a subject matter expert (SME) will understand what tags are appropriate to apply. Illustrations may be incorporated by marked-up references to their placement in the manual and called into the TM for outputting the document or portions of the document.

26.3.5 Layout and style. An important difference from current TM development practices is the use of a FOSI or stylesheet, a formal method of specifying layout and style. This special XML file maps the styles to be applied to the tags in the document (see Section 11, FOSI Application as a Style Guide). The composition system interprets the FOSI or stylesheet for composition of paper or digital output.

26.3.6 Conversion of legacy data. Figure 26 shows a generalized process for converting legacy data to MIL–STD–2361 XML-tagged data. Once the decision to convert legacy data to XML is made, the legacy data should be available in electronic form (text-based, not raster or vector images or pages). If the legacy data to be converted is not available in this form, the data will be re-keyed or captured by text scanning. The digital legacy data will be restructured in accordance with MIL-STD-40051-2 requirements. The restructured data may then be tagged and reused in data resources in accordance with the MIL–STD–2361 DTD. The conversion of legacy data to MIL–STD–2361 requirements will require reallocation to other data resources in order to provide a complete implementation of the XML constructs. Remember that the MIL–STD–2361 XML content tags are included in the MIL-STD-40051-2 narrative. After tagging, the XML file will be parsed against the appropriate DTD to validate the markup, structure, and syntax. For a more detailed discussion of legacy data conversion refer to Section 8, Introduction to XML.

NOTE

Approved legacy data, previously digitized, will be restructured to MIL-STD-2361 format, tagged and parsed.

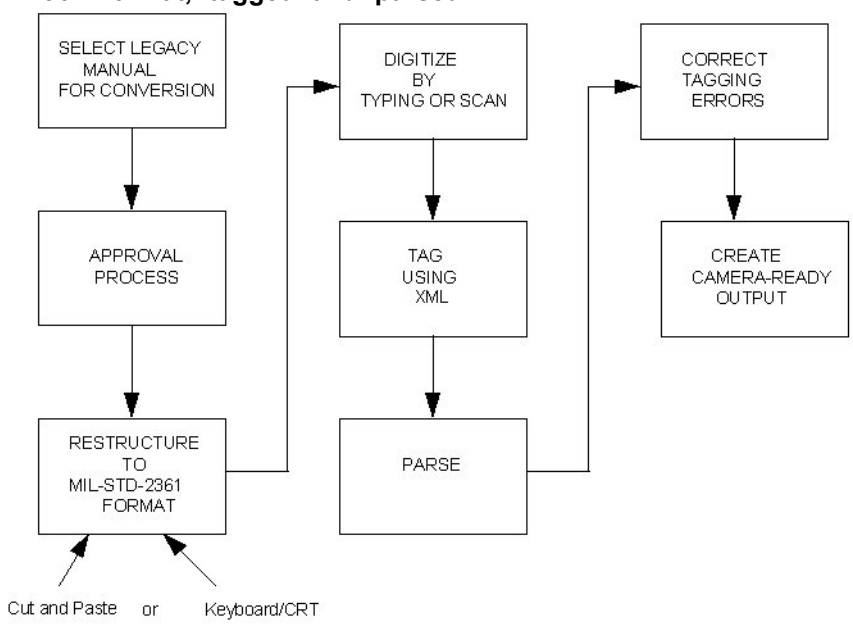

FIGURE 26. Conversion of legacy data to MIL-STD-40051 requirements.

26.3.6.1 Tagging legacy data. There is a great deal of legacy data within the Army. Legacy data, for purposes of this standard, will be defined as any data (paper or digital) that has not been SGML or XML-tagged in compliance with the respective functional requirements standard or specification, MIL–STD–2361, and MIL-PRF-28001. To ensure appropriate tagging conventions and methodology are applied, the following procedures are offered as guidance for applying XML to legacy data. These procedures are oriented toward the TM requirements set forth in MIL–STD–40051–1/–2, due to the maturity

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of that standard and its close association with MIL-HDBK-2361. However, all of the procedures may be applied regardless of the types of data being converted and tagged.

a. Determine the functional type(s) of publication material to be tagged (i.e., TMs, training products, etc.). Publications developed in compliance with traditional requirements documents are produced as complete books (e.g., front, body, and rear matter) in which the publication technical content is not functionally grouped. XML tagging of legacy data in compliance with this standard can be accomplished only after the publication data has been restructured into functional groupings.
b. Determine the legacy data (e.g., TM, training product, etc.) restructuring requirements for compliance with the respective functional requirements standard or specification and MIL-HDBK-2361MIL–STD–40051–1/–2. Virtually no TM legacy data are structured in compliance with functionally grouped requirements standards and specifications, such in MIL–STD–40051–1/–2.
c. Once the restructuring requirements have been determined, an outline of the restructured document(s) should be developed. The outline will be used as a guide for restructuring the documents in compliance with the functional standard or specification and tagging the restructured document in compliance with the applicable MIL–STD–2361 DTD(s). In the case of TMs, an outline may be developed by selecting applicable MIL–STD–2361 content tags which conform to content requirements , specified in MIL-STD-40051, pertaining to the type and maintenance level of the legacy manual.
d. The appropriate top-level tag from the DTD (e.g., ***<mim>***, ***<opim>***, etc.) will be selected and applied to the legacy data. Any #REQUIRED attributes of the top-level tag can then be inserted. In particular, ***<tmno>*** and ***<imlevel>*** are necessary to construct page header, page footer, and chapter cover page.
e. Select the appropriate text for the module from your legacy data and determine the type of work package for the legacy material being tagged. Work packages will contain complete "start-to-finish" tasks for a particular component, and they may contain multiple tasks and procedures. Work packages may have starting conditions that require completing a previous work package. Starting conditions that are dependent upon completion of another work package will contain a reference to the requisite work package in the Initial Setup under the reference tag ***<ref>***. Starting conditions may be referenced in several work packages.
f. Once the work package has been determined, the appropriate work package tag may be applied and any #REQUIRED attributes of that tag inserted. In particular, ***<wpno>*** is used as source of the work package number.
g. Every work package has its own unique set of content tags. Some work package content is mandatory while other content is optional. First, determine the content-specific tags in the work package. For instance, a maintenance work package ***<maintwp>*** requires the content specific tags work package identification information ***<wpidinfo>***, work package information ***<wpinfo>***, and at least one maintenance task ***<maintsk>***. In addition, ***<maintwp>*** may include tags for warnings ***<warning>***, cautions ***<caution>***, and/or notes ***<note>*** and general information ***<geninfo>***. Apply the tags to the appropriate legacy text data and insert any #REQUIRED attributes of those tags.
h. As the writer continues to fill in the content requirements, checks of the DTD should be made for any content-oriented tags contained in the third-level tags. For instance, work package information ***<wpinfo>*** can contain eleven optional content tags. As the optional and required content tags are determined and added to the legacy data, the content requirements for the work package are being satisfied. The content tags should be applied to the legacy data to the level of the document requiring content tags.

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i. There are some special content tagging considerations that will be addressed when tagging legacy data. The considerations are concerned with General Information with Theory of Operation, Troubleshooting, Maintenance Allocation Chart (MAC), and Repair Parts and Special Tools List (RPSTL).

- General Information with Theory of Operation. When developing the general information with theory of operation portions of a publication chapter, several work packages may require construction, depending on the complexity of the equipment. A work package may contain a whole system theory ***<systhry>*** followed by subsystem theory ***<ssysthry>***, or it may begin immediately with subsystem theory. General information should use an introduction tag ***<intro>*** rather than a ***<systhry>*** tag. The ***<systhry>*** tag will be reserved for theory only. Complex systems may need a separate work package constructed for each subsystem. Such work packages will often include data on line replaceable units ***<lruthry>*** and shop replaceable units ***<sruthry>***.
- Maintenance Allocation Chart (MAC). The maintenance allocation charts are tagged with content tags. The functional group number ***<groupno>*** is followed by the nomenclature of the component or assembly. All of the maintenance information pertaining to that component or assembly should be contained within the qualify function tag ***<qualify>***. The qualify tag is comprised of the maintenance function ***<maintfunc>***; the level to which the maintenance function is assigned (***<maintclass>***), which contains specific maintenance level tags; reference to tools and equipment used in the maintenance ***<teref>***; and any additional remarks ***<remarks>***. The qualify function may contain multiple ***<maintfunc>*** entries.
- Repair Parts and Special Tool Lists (RPSTL). The DTDs containing RPSTL work packages ***<rpstlwp>*** do not currently model the list data itself. The list data should be incorporated based on the CCSS database format until revisions are made to MIL–STD–40051–1/–2. Efforts are currently underway to provide requirements for content tagging the RPSTL list data. The standard RPSTL introductory material is currently included as a boilerplate text entity.

27 IMPLEMENTATION GUIDANCE.

27.1 MIL–STD–40051–1/–2 and MIL–STD–2361 Relationship.

27.1.1 MIL–STD–40051–1/–2, "Technical Manual Preparation". MIL–STD–40051–1/–2 establishes the technical and functional content, style, and format requirements for the preparation of paper and frame-oriented TMs, DMWRs, and NMWRs within the Department of the Army. The standard covers the development of these publications, and revisions, for operations and maintenance through depot level. The requirements contained in MIL-STD-40051-2 are divided into the following specific functional elements (parts) to enhance documentation usability in performance of weapon system/equipment and component maintenance.

a. General Information, Equipment Description and Theory of Operation Information (GIM).
b. Operator Instructions (OIPM).
c. Troubleshooting Procedures (TIM).
d. Maintenance Instructions (MIM).
e. Parts Information (PIM) and Repair Parts and Special Tools List (RPSTL).
f. Destruction Information (DIM).
g. Supporting Information (SIM).

Each of these parts provide instructions on how to develop and structure the required technical information into chapters containing individual work packages in a logical order of work sequence. These work packages are stand-alone units containing all information essential for directing task performance.

27.1.1.1 Technical content selection matrixes. MIL–STD–40051–1/–2, as show in example Table I, contains tables that list all of the technical and functional content requirements for the development of all levels of maintenance (through depot) for TMs, DMWRs and NMWR. The tables indicate which parts of MIL–STD–40051–1/–2 are applicable and lists the content requirements for each type of TM, DMWR and NMWR. The content requirements presented in the tables are in the order in which they should appear in the TM, DMWR and NMWR. The Technical Content Selection Matrixes appear in Appendix A in MIL–STD–40051–1/–2.

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TABLE I. Technical Content Selection Matrixes.

Table A-1 TM Requirements Matrix for PAGE-BASED					
TM Content	-10	-13-13&P	-14-14&P	MIL–STD–40051–1/–2Reference	Element Name
FRONT MATTER	R	R	R	5.2.1	<paper.frnt>
Front cover	R	R	R	5.2.1.1	<frntcover>
(MC) Promulgation letter				5.2.1.2	<promulgation>
Warning Summary				5.2.1.3	<warnsum>
Change transmittal page				5.2.1.4	<chgsheet>
List of effective pages /work packages	R	R	R	5.3.1.5	<loepwp>
Title block page	R	R	R	5.3.1.6	<titleblk>
Table of contents	R	R	R	5.3.1.8	<contents>
How to use this manual	R	R	R	5.3.1.9	<howtouse>
CHAPTER 1. GENERAL INFORMATION, EQUIPMENT DESCRIPTION AND THEORY OF OPERATION	R	R	R	B.5.1	<gim>
GENERAL INFORMATION WORK PACKAGE	R	R	R	B.5.2	<ginfowp>

27.1.1.1.1 Explanation of matrix column content. The columns in the Content Selection Matrixes contain specific information to assist the user in determining the appropriate functional requirements for the TM, DMWR and NMWR being developed. A brief description of each of the columns is provided.

a. Column 1 shows the name of the TM, DMWR and NMWR content part.
b. Column 2-4 contains an "R" if the content is required in the TM, DMWR and NMWR. The blank column block will be inserted with a “R”, “P”, or shading. The shaded column block is required when needed to support the equipment.
c. Column 5 contains a reference to the MIL–STD–40051–1/–2paragraph which addresses the TM, DMWR and NMWR content requirement.
d. Column 6 contains the TM, DMWR and NMWR content element name (NOTE: the element name is also the XML tag name from MIL–STD–2361).

27.1.1.1.2 Intended use on the content selection matrixes. The proponent of the TM/DMWR will determine the type(s) of TM, DMWR and NMWR required for each acquisition. Once this determination has been made, the proponent may duplicate the appropriate matrixes (i.e., the ones that contain the applicable requirements for the TM, DMWR and NMWR being developed). The proponent may indicate the type(s) of TM, DMWR and NMWR required by filling in the blank after "TM Requirements Matrix for" at the top of each matrix. The blocks that already contain an “R” are required and cannot be changed. The blocks containing “P” are prohibited for that type of TM and can not be included. For each type of TM, DMWR and NMWR selected, the proponent will indicate in the open blocks the "TM Content" desired by entering "R" for REQUIRED content; "P" for content that is PROHIBITED. The blocks that are shaded is content determined by the developer that may be changed to “R” or “P” determining what is necessary in the work package.

27.1.1.1.3 Acquisition impact. The TM, DMWR and NMWR Content Selection Matrix table(s) will become contractually binding when it is made part of a contract, statement of work, or other contractual instrument.

27.1.2 MIL-STD-2361, "Digital Publications Development". MIL–STD–2361 establishes the Extensible Markup Language (XML) requirements for use in Army digital publications. This standard is a product-oriented interface standard that addresses XML application to functional requirements set forth

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in Government functional requirements standards and specifications. This standard establishes the requirements for developing XML publications in accordance with the various Army functional requirements standards and specifications.

27.1.2.1 MIL–STD–2361 in publication development. Within MIL–STD–2361, Army publication XML requirements are separated by publication types. There are specified sections for administrative publications, training and doctrine publications, and technical and equipment publications. Each of these publication types are governed by technical and functional requirements documents, which specify the technical, functional, format, and style requirements for the respective publications. The XML requirements in MIL–STD–2361 are based on, and explicitly reflect, these technical, functional, format, and style requirements.

27.1.2.1.1 XML requirements development. This volume of MIL–STD–2361 addresses implementation guidance for the development of Army publications using the XML requirements for Army TM, DMWR and NMWR contained in MIL–STD–2361. The MIL–STD–2361 XML requirements were developed in accordance with, and directly reflect, the functional requirements contained in MIL-STD-40051-2.

27.1.2.1.1.1 Functional requirements analysis and determination. XML requirements contained in MIL–STD–2361 were developed to support Army publication proponent functional requirements. Proponent requirement documents were reviewed and analyzed with the proponent to determine the current functional publication requirements. In some cases, the functional requirements may have to be re-engineered (restructured) to accommodate the work package concept (i.e., TM, DMWR and NMWR requirements were re-engineered to become MIL-STD-40051-2. (NOTE: In no case will functional requirements be changed by ANYONE other than the proponent for the requirements.)

27.1.2.1.1.2 Determination of XML requirements. Once the current functional requirements have been identified, re-engineered (if required), and validated, existing XML DTDs and tags are reviewed to determine if the functional requirements are covered by existing XML requirements. This review will result in the identification of existing XML that may be applied to the functional requirements. Another result of the review will be the identification of functional requirements for which no existing XML requirement exists. In this case, new XML DTDs and/or tags will have to be developed.

27.1.2.1.1.3 Application of XML requirements to functional requirements. When the joint determination between the functional requirement proponent and the XML requirement proponent has been made that the XML requirements accurately reflect the functional requirements, the XML applications may be applied to the publications. In the case of TM, DMWR and NMWR, the application of XML is normally accomplished by a publication development contractor under a contract from the publication proponent.

27.1.2.2 Publication and requirement relationships. There is an explicit relationship chain between a TM, DMWR and NMWR, their functional requirements document (MIL-STD-40051-2), and the XML requirements document (MIL–STD–2361). One of the objectives of this handbook is to provide a clear understanding of these relationships to its users. This paragraph will address the relationships between the elements of the respective documents and show how the element relationships are mapped.

27.1.2.2.1 Element relationships. TM, DMWR and NMWR content is comprised of document elements as shown in Column 1 of the TM, DMWR and NMWR Content Selection Matrixes, above. These document elements have a defined relationship with both the functional requirements in MIL-STD-40051-2 and the XML requirements in MIL–STD–2361.

27.1.2.2.1.1 TM, DMWR and NMWR content to MIL–STD–40051–1/–2 Requirements. Each item in TM, DMWR and NMWR content (e.g., Front Matter, General Information Work Package, etc.) can be mapped to a functional requirement (paragraph and page) in MIL-STD-40051-2. The TM, DMWR and NMWR content is the result of it's corresponding functional requirement in MIL-STD-40051-2. The MIL-STD-40051-2 functional requirement defines the content, specifies its location in the TM, DMWR and NMWR, and defines whether or not the content is required.

27.1.2.2.1.2 TM, DMWR and NMWR to MIL–STD–40051–1/–2 to MIL–STD–2361. Both the TM, DMWR and NMWR content items and their corresponding functional requirements can be mapped to the XML requirements in MIL–STD–2361. As previously stated, the XML requirements in MIL–STD–2361 reflect

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specific functional requirements in MIL-STD-40051-2. For each TM, DMWR and NMWR content item and MIL-STD-40051-2 functional requirement there is a corresponding DTD, XML tag, and paragraph/page reference in MIL–STD–2361. Table II is an example illustrating the mapping between the TM, DMWR and NMWR, MIL–STD–40051–1/–2, and MIL–STD–2361.

TABLE II. Mapping TM, DMWR and NMWR to XML Guidance .

TM, DMWR /NMWR Content	MIL-STD-40051–2Reference	MIL-HDBK-2361DTD	MIL-HDBK-2361 Chapter XML Tag	MIL-HDBK-2361 XML Tag
	Paragraph	Paragraph		
General Information Work Package	B.-5.2	29.3.3	GIM	<ginfowp>
Troubleshooting Procedures	D.5.5.8.4.6	31.3.5.3	TIM	<tsproc>
Removal	E.5.3.5.3.6	32.3.1.5.1.12	MIM	<remove>

27.2 Element relationships. Technical Manuals (TM) developed in accordance with MIL–STD–2361 and MIL-STD-40051-2 will consist of volumes (if required by number of pages), information chapters, and work packages (WP) as indicated in Figure 27.

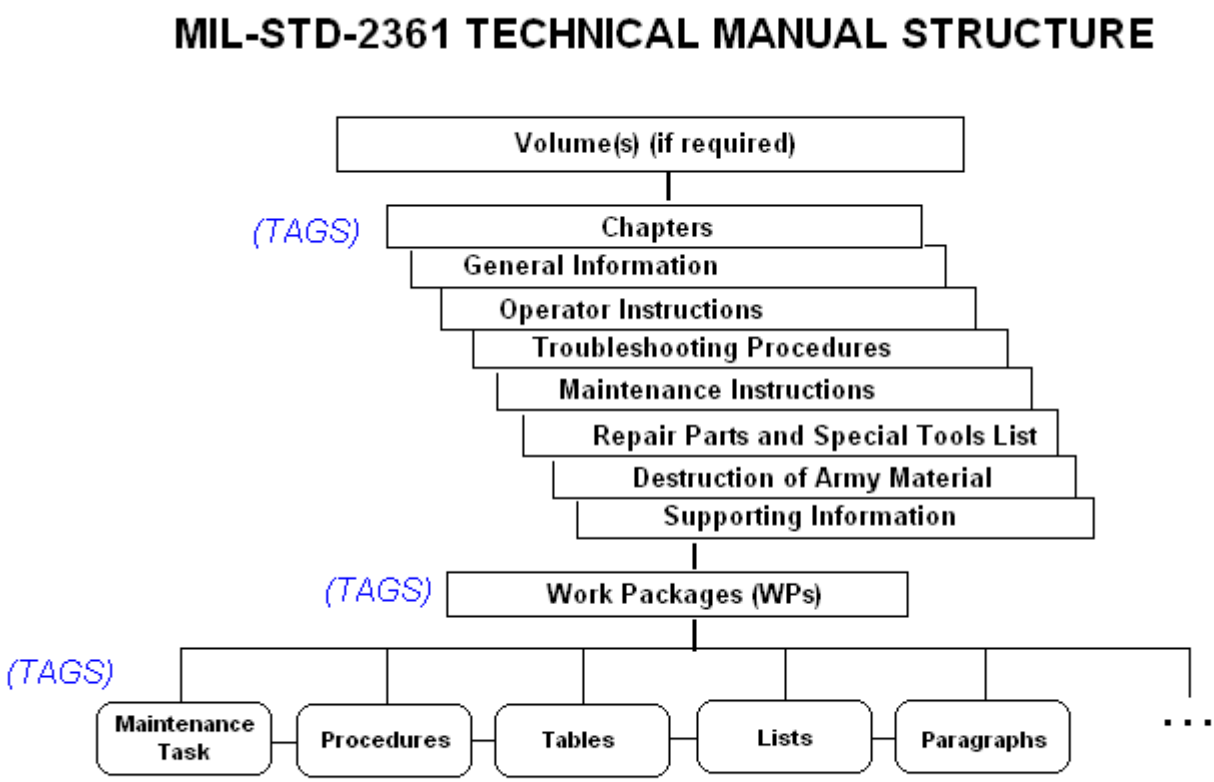

FIGURE 27. MIL–STD–2361 TM structure.

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a. Information chapter. An information chapter consists of specific functional matter required by MIL-STD-40051-2, such as introductory information with theory of operations or maintenance instructions. Each information chapter is made up of one or more work packages as shown in Figure 28. Examples of Maintenance Information Chapters (MIM) include end items (e.g., M16A1 Rifle, M109A1 Truck, or Aircraft) or system components (e.g., engine, gun carriage, landing gear, etc.). Other information chapters provide General Information (GIM), Operator Information (OPIM), Troubleshooting Information (TIM), parts or RPSTL information (PIM), Destruction Information (DIM) and Supporting Information (SIM) .

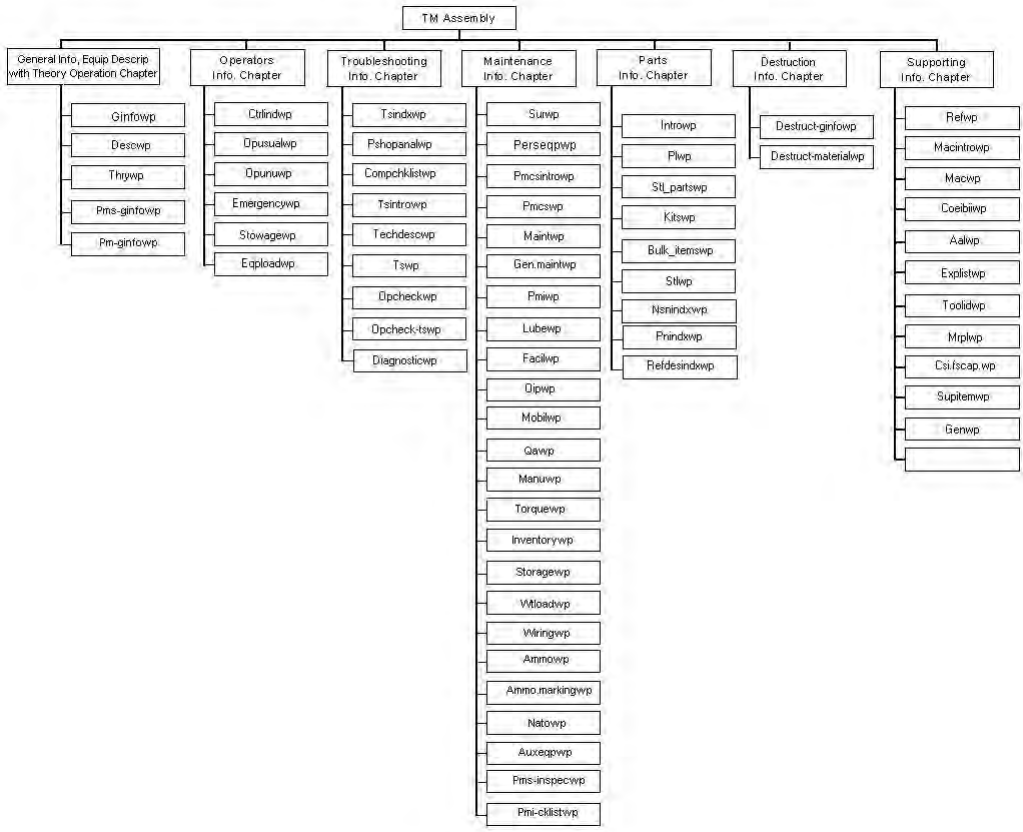

FIGURE 28. MIL–STD–2361 information chapter hierarchy.

b. Work Package. A work package (WP) consists of all data required to perform a specific function, such as service upon receipt, Preventative Maintenance Checks and Services (PMCS), or individual maintenance tasks. This structuring allows electronic access to specific pieces of information required by a technician to perform a specified task. Each work package is assigned a unique identification number for configuration control and reuse of the information contained in the WP MIL-STD-40051-2. This number does not change over the life of the work package. XML content tags allow access and use of the same work package for other weapons systems where the same function is performed. Work packages can be printed out, viewed on a computer screen, or otherwise outputted as individual documents and used separately. See Figure 29 for an example of a work package.

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TM X–XXXX–XXX 0001

FIELD MAINTENANCE

GENERAL INFORMATION WORK PACKAGE (ARMY ONLY)

SCOPE

This manual tells how to operate and maintain the hulls of the M2A3 and M3A3. TM 9-XXXX-XXX-10 tells how to operate and maintain the turret.

MAINTENANCE FORMS, RECORDS, AND REPORTS

Department of the Army forms and procedures used for equipment maintenance will be those prescribed by (as applicable) DA PAM 750-8, The Army Maintenance Management System (TAMMS) Users Manual; DA PAM 738-751, Functional Users Manual for the Army Maintenance Management Systems - Aviation (TAMMS-A); or AR 700-138, Army Logistics Readiness and Sustainability.

REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR)

If your M2A3 and M3A3 needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you don't like about your equipment. Let us know why you don't like the design or performance. If you have Internet access, the easiest and fastest way to report problems or suggestions is to go to (scroll down and choose the "Submit Quality Deficiency Report" bar). The Internet form lets you choose to submit an Equipment Improvement Recommendation (EIR), a Product Quality Deficiency Report (PQDR) or a Warranty Claim Action (WCA). You may also submit your information using a SF 368 (Product Quality Deficiency Report). You can send your SF 368 via e-mail, regular mail, or facsimile using the addresses/facsimile numbers specified in DA PAM 750-8, The Army Maintenance Management System (TAMMS) Users Manual. We will send you a reply.

HAND RECEIPT (HR) MANUALS

This manual has a companion document with a TM number followed by "-HR" (which stands for Hand Receipt). TM X-XXXX-XXX-10-HR consists of preprinted hand receipts that list end item related equipment (i.e., COEI, BII, and AAL) that must be accounted for. As an aid to property accountability, additional HR manuals may be requisitioned through normal publication channels.

DESTRUCTION OF ARMY MATERIEL TO PREVENT ENEMY USE

The following manuals tell you how and when to destroy Army materiel to prevent enemy use:

TM 750-244-2
TM 750-244-5-1
TM 750-244-6

NOMENCLATURE CROSS-REFERENCE LIST

This listing includes nomenclature cross references used in this manual.

Brush guard	Duck core rubber sheet
CVC helmet/lemn	DH 132 helmet
Dipstick	Liquid measure gage rod
Firing port weapon	M231 5.56mm submachinegun.

LIST OF ABBREVIATIONS/ACRONYMS

Many abbreviations are used in this manual. They are listed below. Learn what each one means. It will make your job easier.

A	After
Ammo	Ammunition
AP	Armor Piercing

END OF WORK PACKAGE

FIGURE 29. Work Package Sample.

27.2.1 Information grouping. Information developed in accordance with MIL-STD-40051-2 and MIL–STD–2361 is organized into chapters containing similar functional information (e.g., maintenance chapter, troubleshooting chapter, etc.), and content-tagged to support retrieval from a database. Specifically,

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the use of XML structured information, in conjunction with database technology and content-tagging, facilitates information access, sharing, reuse, management, control, and change. Database technology provides a variety of powerful XML tools and utilities, such as searching capabilities, and allowing large information repositories to be broken down and rearranged intelligently into individual documents.

27.2.2 Work package reuse. There are instances in which the same work package(s) may be used, without modification, on more than one end-item of equipment. Also, there will be instances when most of the information in a work package will apply and be used on more than one end-item of equipment, but will require minor changes to small portions of the work package information in order to tailor coverage to other end items or systems. The Digital Publications Development concept for information reuse focuses primarily on the reuse of work packages, or portions of work packages, to the maximum possible extent by either direct reuse or by filtering/overlaying information on top of existing work package data.

a. Work package reuse is defined as a single source work package with the same task/procedural intent, that may be used for TM authoring for one or more end-items.
b. Task/procedural intent is defined as tasks/procedures required to perform the same or similar function, but may or may not have the same wording. Establishing a publications source database is a mandatory element of work package reuse.

27.2.2.1 Filtering and overlaying work package information. Overlaying is defined as adding or hiding portions of original information. Filtering is defined as setting a criteria to include or exclude information. The filtered information would be associated to specific work packages by revision number. XML elements and attributes may be used to indicate variations within the modified work package text. The criteria for assigning and controlling filtering/overlaying would be the major end-item(s) NSN to which the work package is applied. This methodology could be used by authors to search for work packages by particular end-item equipment that is closely associated, or the same as, the end-item on which they are working, and to note the variations within the work package to determine the best fit or variation needed to satisfy their needs. The element ***<step1>*** and its attributes **crewmember** and **applicable** is an example of an element and its attributes that can be used to indicate variations.

27.2.2.2 Work package reuse examples. The following examples are provided as guidance for the various ways work packages may be reused. There are two types of examples: Figure 30 is for direct reuse and Figure 31 is for reuse using the technique for overlaying information.

a. Direct reuse. Direct reuse is use of the same work package on more than one end-item equipment with no modification to the work package. The direct reuse example represents an instance where one work package (WP 3 data) is used for Bradley Fighting Vehicle (BFV) A2 engine data, and is reused for two different vehicle end-items configured with the same engines (BFV A3 and Bradley Fire Support Team (BFIST) Vehicles).

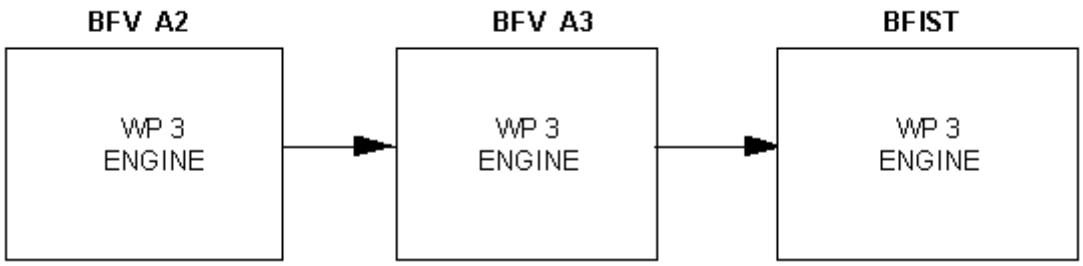

FIGURE 30. Example 1 Direct Reuse.

b. Overlay reuse. Overlay reuse starts with the original work package contained in the BFV A2 vehicle. The BFV A3 contains the same work package tasks, but requires the addition of two steps to a procedure and a change in the initial setup references to correspond to the BFV A3's system. The BFIST has the same task as the BFV A3, but requires a change of the location of a hatch from "left" to "center" of the BFIST vehicle. The remaining work package information is directly reused.

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The future prototype vehicle reuses the original configuration with some modification to the initial setup references and the removal of a step.

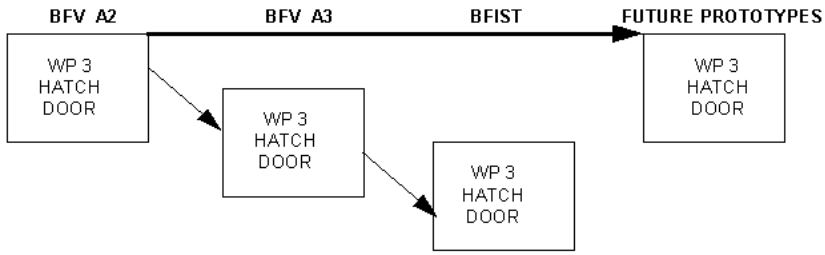

FIGURE 31. Example 2 Overlayed Reuse.

27.2.2.3 Work package initial setup. The "initial setup" is the primary guide to determine if a new work package should be created or could be filtered/overlaid by other text. The initial setup conditions will be the same, except for references, to constitute justification for reuse. For example, modifications to the initial setup which do not change the intent of the work package, but change the work package reference or applicable configuration (i.e., for another end-item) are candidates for reuse. The initial setup guide, as shown in Table III may be used to determine if a new work package should be created, or an existing work package can overlay with other text.

TABLE III. Initial Setup Guide.

Initial Setup Components	New WP	Filter	Rationale
Test Equipment	X		Changes to the test equipment would indicate changed procedures or steps are needed to use this equipment, possibly changing the intent of the work package.
Tools/Special Tools	X	X	Changes to the tools/special tools would indicate changed procedures or steps needed to use these tools, possibly changing the intent of the work package. ONLY variation is the SIM reference to the SIM work package, tool, and item number for the particular end-item.
Material/Parts	X	X	Changes to the material/parts would indicate changed procedures or steps needed to use these parts, possibly changing the intent of the work package. ONLY variation is the SIM reference to the SIM work package, part, and item number for the particular end-item.
Personnel Requirements		X	Personnel requirements may change. A change is permitted when the sub-equipment is the same, but mounted on various end-items.
Reference		X	Reference may change because of different end-items, but the intent and purpose to the reference are the same.
Equipment Condition	X	X	The equipment should be in the same condition state, otherwise the difference changes the initial purpose for the work package. ONLY variation is the referenced work package in how to put the equipment in the ready state, but should be referencing a similar type of condition.

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Table III Initial Setup Guide – Continued.

Initial Setup Components	New WP	Filter	Rationale
Special Environment	X		The special environmental conditions remain the same, otherwise how to perform the task would become varied and change the purpose of the task.
Drawing Requirement	X	X	The drawing requirements remain the same, other changes or modifications to the parts or test equipment is possible. The only exception is if the locator drawing is different, then a new work package is not required.

27.2.2.4 Additional guidance for work package reuse. Besides the initial setup guidance, the following will also govern work package reuse:

a. No variation in the work package title is allowed.
b. No tasks or procedures may be added to, or removed from the work package.
c. No information added to, or removed from, the work package can affect the initial setup new work package column.
d. No changes may be made to warnings or cautions associated with work package steps, except for references to other work packages.
e. New warnings or cautions may be associated with new steps added.
f. Warnings or cautions may be removed only when the associated step is removed.

27.2.3 XML process. The XML process follows the general outline in the following paragraphs. For a more extensive explanation of XML and its application to TM development see Section 8, Introduction to XML. Figure 32 is an overview of the XML process from the developers perspective.

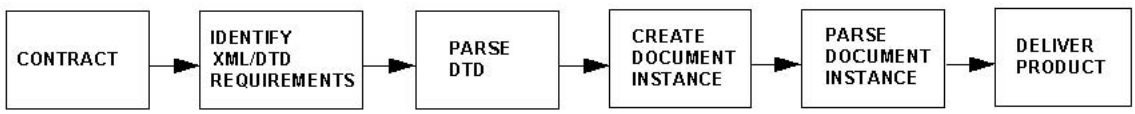

FIGURE 32. XML Process.

27.2.3.1 Applying the appropriate DTD. The first step in the process is to identify the DTD requirement(s) (called out in the contract) relating to the appropriate information chapter (e.g., MIM, TIM, etc.) of the material being authored or tagged. Once the determination is made that the correct DTD version is on-hand, it should be parsed to ensure it is valid, error-free DTD prior to use in the development process. The XML tags will have to be applied (in the case of legacy data), or the TM content information created (new publication development), within the context and parameters of the DTD and the functional requirements for technical manuals contained in MIL-STD-40051-2. The process can be accomplished by use of XML authoring software, or in a word processing system by manual insertion of tags.

27.2.3.2 Page-based vs frame-based . The same DTD can be used for both page-based and frame-based TMs, DMWRs and NMWRs. The information that is not needed for frame-based will be used to provide the detailed information for page-based, by cross-reference linking to the information needed to obtain the detail information. Different stylesheets are used for the formatting of the page-based or frame-based TMs, DMWRs and NMWRs. A page-based and frame-based TM, DMWR and NMWR have the same content coverage. However, the display and presentation of the information is different.

27.2.3.3 Creation of document instance. To create a document instance, the tags declared in the DTD should be integrated into the text of the document, whether material is being authored for the first time or legacy data is being converted. The XML markup (the tag set) takes the place of any format oriented markup. A document instance is a tagged file with a complete structure conforming to its relevant DTD.

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27.2.3.4 Validation (parse) of markup syntax. Before proceeding to output a document or information chapter, the document instance is to be tested (parsed) to validate that the markup conforms to the syntactic and structural rules of XML and the DTD. Any errors found by the parser will be corrected before proceeding further.

27.2.3.5 Creation of required output. Creation of the required output may include several forms. Requirements for the appropriate type(s) of output will be contained in the Contract Data Requirements List (CDRL) accompanying the contract vehicle. All of the following options are supported by MIL–STD–2361.

a. Using a stylesheet to direct the composition of the document in page-based format for printed output, or for the frame-based screen display in the composition software.
b. Generation of a suitable page description language file to drive printer, typesetter, or viewing software.
c. Outputting for on-screen access in a navigable database format.
d. Retrieving directly from a comprehensive XML database of all TMs for on demand printing.

27.2.4 Validation and verification process.

27.2.4.1 Validation of an XML document. Preparation of documents in an automated support environment typically consists of the following steps:

a. Downloading an approved Document Type Definition (DTD) and stylesheet or Formatting Output Specification Instance (FOSI) from the ASRL.
b. Parsing the DTDs and stylesheets or FOSIs.
c. Creating a document instance.
d. Parsing the document instance.
e. Using the approved stylesheet or FOSI and DTD to compose the composition of the document so that the produced (printed or displayed) copy corresponds to the proper format and style.

27.2.4.2 Parsing MIL–STD–2361 TM DTD. The process of validating (compiling) a DTD once it is written is known as parsing. Commonly referred to as a validating XML parser, the REC–xml–20060816 Standard defines the parser as “A program (or portion of a program or a combination of programs) that recognizes markup in XML conforming documents.” A validating XML parser will read a DTD and check the markup, and report any errors found to an error file log. It is the responsibility of the user to re-parse a DTD downloaded from the ASRL (see 7).

a. Most of the commercial XML authoring tools on the market today contain a built-in validating parser. To create XML documents which conform to a DTD downloaded from the ASRL an editor and a parser is needed.
b. The editor is used to input information and insert XML markup into the document; the parser is used to check that the markup and the way it has been used conform to the rules given in the DTD. Many commercial packages offer syntax-directed editors, which interactively ensure that any editing and markup operations conform to the rules of the DTD.
c. Once a valid XML document that conforms to a valid XML DTD has been developed, it may be wise to do some subsequent processing. For example, in order to get paper output, you will need to use a MIL–STD–2361's stylesheet/FOSI, which conforms to MIL-PRF-28001, in conjunction with a composition system to read the XML document and produce a paper output.
d. It is recommended that at least three XML parsers be used when parsing DTDs that have been downloaded from the ASRL.

27.2.4.3 Valid XML document vs. conformance to MIL-STD-40051-2. Parsing the XML document does not verify that the information or content of the TM matches the meaning for the XML element and/or MIL-STD-40051-2. The contracting activity will be responsible for verifying that the TM content is in compliance with the contract, applicable SOWs, and MIL-STD-40051-2 requirements.

27.2.4.4 Formal public identifiers for entities. Each formal public identifier (FPI) for graphics and file entities should have a file name associated with that FPI. The FPI is mapped to the file name by using an XML catalog. The format for the XML catalog is displayed:

```
PUBLIC "-//Owner//ENTITIES Public_Title Rev X.XX YYYYMMDD//EN" "(DIR)
file.name"
```


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```
PUBLIC "-//USA-DOD//DTD MIL-STD-2361 TM Assembly REV 3.13 20060621//EN"
"production.dtd
```

By making modifications as necessary in the XML catalog and changing the (DIR) to the specific directory on the target system or modifying the file name to the corresponding operating system restrictions, the correct mapping will be established. Every file entity with an FPI has an entry in the XML catalog. Verification of the FPI for entities satisfy two criteria. The first criteria is that each file identified in the XML catalog is to be included with the electronic delivery. The second criteria is to verify that each FPI in the XML document instance is defined in the XML catalog. If either criteria fails, the XML document instance is unacceptable.

27.2.4.5 System identifiers for entities. System identifiers for entities may be found after the formal public identifier in a document file. A system specifies the location of the entity on the users system. The SYSTEM entity maps the graphic “ginfowp.png” to the directory “graphics\ginfowp.png” on the user's system noting from the entity “NDATA png” that the graphic type is “png” (see 9.3.5.3.1) . An example of a PUBLIC entity containing the system identifier is found below the SYSTEM entity example.

```
<!ENTITY ginfowp.png SYSTEM “graphics\ginfowp.png” NDATA png>
<!ENTITY ginfowp PUBLIC "-//USA-DOD//ENTITIES GINFOWP DTD Hierarchy//EN"
"C:graphics\ginfowp.png">
```

27.2.4.6 ID and IDREF resolved. The document instance has an associated ID value for each IDREF in the document instance. A document that does not have an associated ID value for each ID referenced is an incomplete document and unacceptable.

The two methods to verify that all IDREF(s) are resolved are to use either an XML editor or a software application. The simplest verification method is to use an XML editor that verifies all IDREF(s) automatically. The XML editor will display the unresolved IDREF(s).

The other methodology is to develop a software application to identify all attributes with IDREF and ID, which stores the values in a separate list for each respectfully. The application sorts each list and matches the IDREF values to the ID values. The application will display the missing IDREF(s) with no resolution.

27.2.4.7 IETM - link verification. IETM link verification processes are currently being developed and will be included in the next revision of the handbook.

27.2.4.8 Published document produced by XML instance. The publications developer should verify that the published manual was produced from the XML instance. When the published manual is not produced from the XML document instance, errors may reside in the XML document instance that were corrected or updated in the published manual. To verify that a published document was composed from an XML document instance, a written certification declaring the electronic XML document instance used to compose the final delivered published manual will be included with the delivery. The certification will specify the methodology used to compose the XML document instance to the published manual.

28 TM, DMWR/ NMWR ACQUISITION.

28.1 Acquisition guidance. Technical Manual (TM) requirements proponent personnel (e.g., TM writers) develop the Portion of Procurement Data Packages (PDP) that provide the TM requirements placed on contract by procurement personnel. The TM portion of PDPs are normally compiled and included in pre-contract documents, such as solicitations and Request for Proposals (RFP). The following paragraphs address some of the different mechanisms (e.g., forms, standards, procedures, etc.) that are involved in applying TM requirements to contractual documents.

28.1.1 Contract data requirements list (CDRL). CDRLs are the primary requirements documents for conveying TM data format, context, and delivery requirements to publications developers. The CDRL is also a primary means for defining XML requirements to a publication developer. Activities involved in a weapon system development are queried, through a “Data Call”, to provide data requirements specific to their functional area. The XML aspects of the TM requirements may be provided by the activity responding to the data call, or some other functional activity (e.g., an XML specialist on a PM staff), depending on where XML functional expertise is available. One of the major results of a data call is the requirements associated

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with the development of TMs for the weapon system. Based on the TM requirements received as a result of the data call, CDRLs are developed as a means of explicitly describing the requirements. The TM CDRLs are an integral part of the PDP and each CDRL should be supported by a Statement of Work (SOW) task, describing the work effort required to develop the TMs associated with the CDRL. Following are some of the areas that may be addressed in a CDRL. These areas are not all inclusive, and are provided as guidance only.

28.1.2 Statement of work (SOW). Each CDRL will have an accompanying SOW, or be included in the contract SOW, describing the work effort and requirements required to develop the respective TMs. The SOWs are prepared by the activity that provides CDRL information in response to a data call. The SOW becomes a binding part of the contractual documentation.

28.1.3 Conversion of legacy data. The CDRL may direct the contractor to convert specified existing non-XML data (legacy data) into XML. The legacy data may consist of previously developed TMs for an existing weapon system or subsystem. The CDRL will provide the version of the DTD and stylesheet to be used for conversion, and any other applicable XML requirement information.

28.1.4 GFI/GFE source information. The CDRL will provide identification of required GFI/GFE, such as legacy TMs for conversion, that are required for conversion by the developer. If the GFI/GFE is not provided as part of the PDP, its location and procedures for acquiring it will be provided as part of the contract.

28.1.5 TM requirements and standards. TM functional requirements are developed in accordance with MIL-STD-40051-2, MIL–STD–2361, MIL-PRF-28001, ISO 8879 and REC–xml–20060816 Standard (see 2.3) to cover application of XML. The CDRL will identify these standards specifically as requirements.

28.1.6 Location of XML objects, constructs, and other information (DTDs, stylesheets/FOSIs, XML tag description lists, documentation etc.). All Army-approved XML objects and constructs are contained in the Army SGML/XML Registry and Library (ASRL). The XML objects and constructs may be provided with the contractual document, or the TM developer may be directed to obtain the required objects and constructs from the ASRL. If the ASRL is the directed source for the DTDs, stylesheets/FOSIs, tags, etc., the following information will be provided in the CDRL.

a. Formal Public Identifier (FPI) of the DTD. The FPI is the official identifying designation of a particular version of a DTD. Each DTD, and each version of a DTD, has a unique FPI.
b. Information on how to access the ASRL. The CDRL will contain, either explicitly or by reference, the procedures to follow to gain access to the ASRL through the various means available.
c. Information on downloading DTDs, stylesheets/FOSIs, tag description lists, and documentation. The CDRL will contain, either explicitly or by reference, the procedures to be followed to download the XML information needed by the TM developer.
d. Information on parsing the DTDs and stylesheets/FOSIs. The CDRL will reference the developer to the appropriate SOW, standard, etc., for detailed information on the parsing requirements for the TMs. The parsing information should contain requirements regarding parsing the digital XML tagged instance file parsed against the DTD which was provided, and requirements for submitting a parsing log record.

28.1.7 Tailoring the work packages. MIL-STD-40051-2 and MIL–STD–2361 establish the requirements for tailoring work packages for each type of TM (e.g., –10, –23, etc.). The CDRL will reference the appropriate portions of these standards for tailoring TM work packages. The appropriate SOW paragraphs will also be referenced in the CDRL.

28.1.8 Required output medium. The TM proponent will determine the output requirements and provide them to the contracting activity for inclusion in the contractual documentation (e.g., SOW, CDRL, etc.). The output requirements are included in the CDRL. Output file delivery requirements may be found in MIL–STD–2361. Output delivery requirements may include the following.

28.1.8.1 Paper. Delivery of paper products are normally camera-ready output developed from a XML document instance and stylesheet/FOSI. The CDRL will specify the appropriate requirement(s).

28.1.8.2 Electronic technical manual (ETM). ETM delivery is normally a page-oriented digital product (e.g., page turner) suitable for viewing on an electronic display. MIL–STD–2361 contains the requirements for ETMs and will be specified in the CDRL.

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28.1.8.3 Interactive electronic technical manual (IETM). The IETMs provide functionality to the soldier beyond the capability of either paper based or ETMs. In no instance will an acquisition package state a specific class of manual is being require; for example, that the contractor will deliver a class 4 IETM; but rather the acquisition package will tailor specific requirements needed to support the weapon system or equipment.

28.1.9 Delivery medium. The determination of the method in which TMs are delivered to the Government is the responsibility of the TM proponent. Determination of the appropriate method of delivery is often impacted, or directly determined, by Defense or Army policies. The method of delivery is contained in the CDRL as part of the contractual documentation. Possible, but not all inclusive, delivery mediums are provided.

a. MIL-STD-1840 - Digital Tapes.
b. Compact Disk-Read Only Memory (CD-ROM).
c. Diskettes (3 1/2" or 5 1/4").
d. Tapes - 1/4" Data Cartridge Tape.
e. Telecommunications (e.g., Internet, WWW, e-mail, etc.)
f. Paper - with one or more of the above methods.

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PART II
TM INFORMATION CHAPTERS

29 MIL–STD–2361 XML APPLICATIONS INTRODUCTION.

29.1 Scope. This section contains information on the presentation of the XML elements developed for Army digital equipment manuals publications.

29.2 Applicable documents. Refer to paragraph 2

29.3 Introduction to MIL–STD–2361 DTD model. In Chapters 34 through Chapter 43 the MIL–STD–2361 DTD content models will be displayed in the following manner:

a. The XML element is defined as used in the particular Army publication.
b. A visual representation may be given of the XML content model. The tree structure and its components are given to provide users an understanding of the relationships between the elements and the order of elements. The tree structure contains symbols defined in Figure 33.
c. A reproduction of the DTD fragment is provided for each element.
d. A description of associated attributes is provided for each element.
e. A sample document instance fragment is provided for some higher level elements. The fragmented instance shows a correct usage for the higher level element and its children.
f. When a sample document instance fragment is provided a sample output is included showing a facsimile of what is produced by a composition system.

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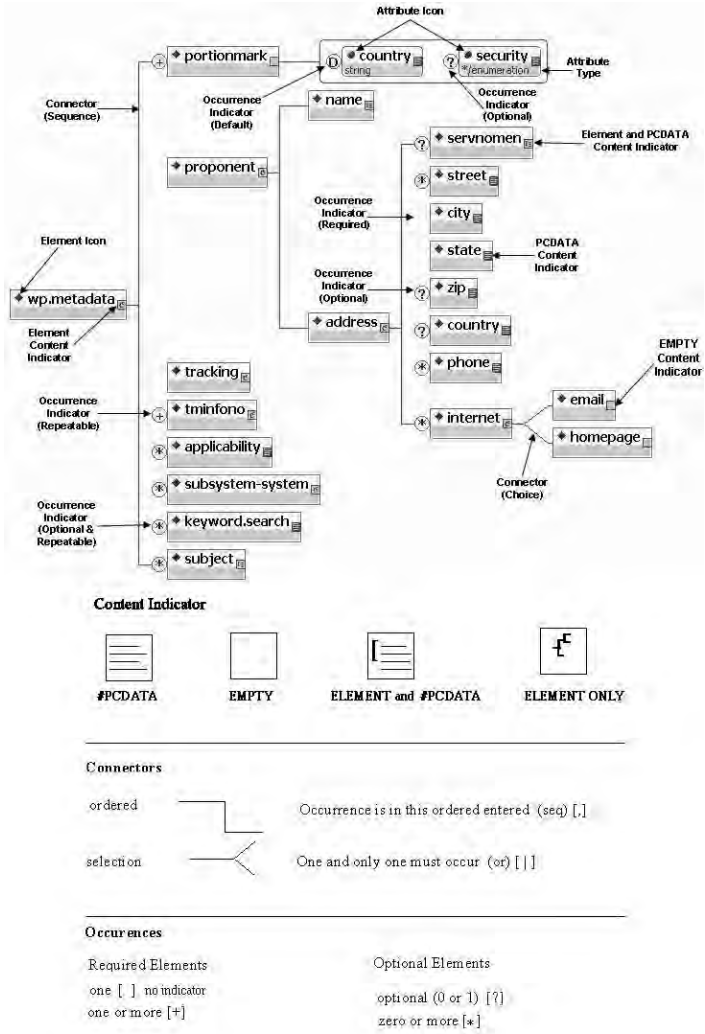

FIGURE 33. XML Tree Legend.

30 GENERAL INFORMATION.

30.1 Page-base vs. frame-base. The technical content for Interactive Electronic Technical Manuals (IETMs) as specified in MIL-STD-40051-1 and the technical content for paper, paged-based technical manuals (TMs) as specified in MIL-STD-40051-2 is in most cases, identical. Specifically, work package content, with some variations, is the same except for diagnostic work packages. Style and format differences do occur because the IETM is frame-based where as the TM is page-based. Functionality differences occur because the IETM can be interfaced with a logic engine that, based on user or other input, determines the correct sequence in which to display technical data in an IETM. There are other differences as well (e.g., links will be different). This section will address the tagging differences between IETMs and page-based TMs that result. Only elements and attributes that are pertinent to this discussion will be presented.

30.1.1 Production . Any material to be published according to the DTD in Technical Manual Production begins with this element. The available TM types are Equipment and Technical (frame-base or page-base), Ammunition, Preventive Maintenance Inspection (PMI), Phased Maintenance Services (PMS) Inspections,

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and Aviation Troubleshooting. The tagging differences between IETMs and page-based TMs start at this root level. The ***<framed.manual>*** element is for frame-based IETMS. The remaining TM types (e.g., ***<paper.manual>***, ***<ammo>***, ***<pmi>***, etc.) are essentially page-based TMs. The ***<paper.manual>*** content model will be discussed as an example of page-based markup. The markup for the remaining page-based TM types is similar. These elements contain all of the assembled technical manual contents, including the front matter, the rear matter and the work packages.

30.1.2 Manual differences. There are several differences in markup between the ***<paper.manual>*** element and the ***<framed.manual>*** element. Specifically,

a. The ***<paper.manual>*** element provides for the creation of volumes ***<volume>***, ***<vol-rear>***) and has attributes for specifying paper requirements such a page size (**fit.paper.size**).
b. Some attributes are common to both but can have different inputs. For example, the **maintlvls** attribute which specifies the maintenance level(s) authorized to use the manual does have differences between frame-based and page-based manuals (see the publication type and title in MIL-STD-40051-1 and MIL-STD-40051–2).
c. The IETM may be structured either by function or by system where as the page-based manual is structured only by function. The ***<functionhierarchy>*** and ***<systemhierarchy>*** elements in the ***<framed.manual>*** element content model provide IETMs with that capability.
d. The ***<systemhierarchy>*** element breaks down the end item work packages to the respective subsystems to the lowest defined work package subsystem item. The content model for the ***<systemhierarchy>*** element includes the ***<overallsystem>*** element, the ***<systembreakdown>*** element and the ***<systemref>*** element. Overall System Information ***<overallsystem>*** defines the work packages used overall for the equipment in a system/subsystem hierarchy manual. System breakdown ***<systembreakdown>*** defines a system/subsystem structure. Any work package related to the system is located under this element. Subsystem levels are recursively referenced down to the lowest assemble level. Some work packages are intended for all subsystems and are not identified to any one subsystem item or group. For example, the reference work package ***<systemref>*** contains the general reference work packages not specifically addressed to any system/subsystem. Page-based TMs are not structured in this manner so they will not have these elements.

30.1.3 Front matter. All front matter of a technical manual occurs before the first work package. The major difference between frame-based ***<framed.frnt>*** and page-based ***<paper.frnt>*** front matter is that a frame-based IETM contains a revision summary where as the page-based TM has a list of effective work packages , a change sheet and a title block page. The revision summary frame ***<revisionsummary>*** contains a list of work packages by title that have been revised (see 31.3.1.1.1). The list of effective pages/work packages ***<loepwp>*** lists the latest work packages in the page-based TM by work package sequence number (see 31.4.1.4). A change transmittal page ***<chgsheet>*** is prepared for each change to a page-based TM (see 31.4.1.3). The title block material ***<titleblk>*** in the TM's front matter repeats identifying information from the front cover, including the prime title (see 31.10.1.1). It also includes the reporting errors statement.

30.1.4 IETM interaction. The logic engine is a software component that determines what path to take through the IETM (see 48.2). The tags that support logic engine functionality should be ignored for page-based TMs. This includes the following:

a. Tags for determining alternative conditions. This includes the ***<precond>*** element and elements that represent a set of alternatives, i.e., those using the naming convention of ***<ELEMENTalt>***.
b. Tags that create, maintain and evaluate state information. This includes variable declarations ***<variable>***, variable references ***<variableref>***, expressions ***<expression>***, and state information manipulation ***<statemanipulation>***.
c. Tags that provide for a dialog box capability. This includes dialog ***<dialog>***, enable selection ***<enable>***, and Dialog box types (fill-in ***<fillin>***, menu ***<menu>***, binary menu ***<binarymenu>***, group dialog ***<dialog-group>***, and/or message ***<dialog-message>***.

30.1.5 Filtering. The technical manual system effectivity list ***<applic_ref_list>*** is a listing of all possible system effectivity configurations. It is included in the ***<production>*** element content model as an optional

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element. The ***<applic_ref_list>*** element consists of one or more occurrences of the ***<applic>*** element. When a work package has a system effectivity issue, the work package references or links to the system effectivity list. This level of filtering may be applied to both frame-based IETMS and page-based TMs. Filtering is not performed for page–based TMs at lower levels such as the ***<step1>*** element level. At that level, attributes that apply to filtering (e.g., the **applicable** attribute) should be ignored for page-based TMs.

30.1.6 Rear matter. Rear matter is primarily for page-based TMs. The ***<rear>*** element in the ***<paper.manual>*** content model provides for a glossary ***<glossary>***, an alphabetic index ***<aindx>***, required DA-2028 forms ***<da2028>***, a required authentication page ***<authent>***, a foldout (oversize) illustration section ***<foldsect>***, and a required back cover ***<back>***. The content model for frame-based IETMs only includes the required DA-2028 forms ***<da2028>***.

30.1.7 Linking. Both page-based TMs and frame-based IETMs have the capability to specify links to associated information. The various link methods (i.e., ***<extref>***, ***<xref>***, ***<link>*** etc.) including what is presented for page-based and frame-based display is discussed in46. Also discussed for enhanced linking ***<link>*** is when it is used for frame-based only, page-based only, or for both.

30.1.8 Frame attribute. For frame-based IETMs, the author has the capability to specify if certain elements should or should not start on a new frame. This is done by setting a value in the **frame** attribute for those elements (see 49.2.3). The **frame** attribute setting is ignored for page-based TMs.

30.1.9 Graphics. There are several tagging differences between frame-based IETMs and page-based TMs. Detailed information is provided in the graphics section (see 44.2). Examples are as follows;

a. The **application** attribute in the ***<figure>*** element identifies the figure presentation method as for page-base only (“page”), frame-base only (“frame”), or for either method (“both”).
b. The “figure size type” as specified in the **gtype** attribute in the ***<figure>*** element is for page-based TMs only. Frame–based IETMS should ignores this attribute.
c. The “new pane” as specified in the **pane** attribute in the ***<figure>*** element is for frame-based IETMs only.
d. The **alt** attribute in the ***<graphic>*** element contains a description or additional information to be displayed when a mouse-over in frame-based IETM application is used. Page–based TMs should ignore this attribute.
e. The element ***<mapref>*** provides the information to overlay hot spot reference on a graphic. The graphic hot spot is designed for use in frame-based IETM applications.

30.2 Linear vs. non-linear IETMs.

30.2.1 Description. A linear IETM contains technical data that is displayed in a sequential or document oriented manner. The sequence of the data presentation is largely predefined by the data author. It is an organization of technical data that often replicates the order of information found in a page-based document. There is generally a default “path” through the technical data. A non-linear IETM contains technical data that is not displayed in a sequential fashion. In a non-linear IETM there are high levels of interactivity between the data and the user. The order of presentation is dictated by inputs from the user, external sources or events (as in diagnostics). There are multiple paths through the data. Individual paths through the data are generally determined based on user or other input via dialog boxes. Because of this the organization of content will not follow a document or page based paradigm.

30.2.2 Functionality. The complexity factor for some functionality is different between linear and non-linear IETMs. For example, the capability to produce a fully formatted book version is not as complex for linear IETMs. In other cases, (for example filtering by model series) some functionality is not as complex to provide for non-linear IETMs. Also, a linear IETM may not be able to provide certain functions (for example, dynamic diagnostics). See the functionality matrices in MIL-STD-40051-1 for information on which functions are available and what their complexity factors are for linear and non-linear IETMs.

30.2.3 Tagging. There is no explicit tag that defines an IETM as being linear or non-linear. However, certain tags that are used to provide functionality such as dynamic diagnostics would not be found in a linear IETM. This would include the ***<precond>*** element which is evaluated by a Logic Engine

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using the expression evaluation rules. Also, any element name ending with "-alt" (for example, ***<eqpconds-setup-item-alt>***) should not be used in a linear IETM. These tags provide alternative conditions/information based upon evaluations that use preconditions.

30.3 Destruction of Army material.

30.3.1 General information. MIL–STD–40051–1/–2 require that all Army TMs contain information on how to destroy Army equipment in order to keep it from falling into enemy hands. This section discusses the various methods to present the destruction information.

30.3.1.1 Destruction information options. There are two methods of presenting destruction procedures. These are:

a. A separate destruction manual, issued as a stand alone TM (page based only). A separate manual ***<destruction_manual>*** may be developed for a stock class or item grouping at the direction of the stock class or item manager. A separate manual may also be developed for a specific weapons system, though this approach is discouraged to having the destruction procedures contained within the system TM. See 30.3.1.1.b. below.

b. As one or more work packages ***<destruct-materialwp>*** contained in a destruction procedures chapter ***<dim>***. Regardless of whether a system level destruction manual is prepared, MIL–STD–40051–1/–2 requires at least one ***<destruct-materialwp>*** be included in the basic system TM(s). This work package may reference the system destruction manual.

30.3.1.2 Information to be contained. Destruction procedures manuals and work packages are required to contain the following types of information.

a. An order of precedence for destruction of material. This order of destruction applies to all systems. This will be contained in the destruction general information work ***<destruct-ginfowp>***. A general reference to these priorities should be made in at least one of the destruction of material work packages ***<destruct-materialwp>***.
   (1) If there is any cryptographic material, it will be destroyed first. Cryptographic material consists of items, systems or information capable of allowing information to be coded or decoded. Capture of cryptographic information could give an enemy the ability to intercept and read/hear classified or sensitive information.
   (2) Any classified equipment or information will be destroyed next.
   (3) Depending on the type of destruction procedures, i.e., a generic stock class or system specific, material will be destroyed as specified in the procedures. If the entire system cannot be properly destroyed, then essential parts may be destroyed to keep the system from functioning and being rebuilt.
   (4) For units who maintain system assets, and destroy spare parts, the priority in spares destruction is the same as the essential parts listed in 30.3.1.2.a.(3) above. Destruction of the same essential components aids in preventing an enemy from reconstructing a weapons system from spare parts.

b. Destruction information and procedures.
   (1) General destruction concepts may be included. These concepts include topics such as using any self destruction capability prior to other methods, acceptable methods of destruction (i.e., burning, blowing up, smashing, etc) and potential hazards associated with various methods of destruction. The general concepts will be placed in the ***<destruct-ginfowp>*** of a stand alone destruction manual or in their own ***<destruct-materialwp>*** in a destruction chapter.
   (2) Specific destruction procedures for similar class items or specific weapons systems components. These procedures should provide sufficient detail for a soldier in a combat situation to destroy the material as quickly, safely and completely as possible. Procedures should contain warnings related to the hazards involved in destroying the material. Cautions should not be included as the intent of the procedure is to destroy the equipment. Long involved procedures should be avoided. Simple, expedient procedures are desired so the soldier can destroy the material and continue his/her escape/evasion.

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(3) Alternative procedures to destroy material when lack of time or material preclude destruction by preferred methods. Alternative methods include burying, immersing in water or physically destroying. When developing alternative procedures, consider the tools that would be available to a soldier in the field in a combat environment. For physical destruction, provide the soldier with the best locations to ensure maximum damage in a minimum of time. Alternative procedures are ideal for specifying the destruction of essential components (see 30.3.1.2.a.(4) above)

30.3.1.3 Duplication of data. To the greatest extent possible, information should not be duplicated. DOD policy is to limit duplication of information. However, given the serious nature in the destruction of material, duplication of some data is allowable. It is felt that duplication is preferable to having material captured due to lack of a procedure. Duplication should attempt to meet the following guidelines:

a. Only specific destruction procedures should be duplicated. For example if there is a stock class stand alone manual with specific instructions related to an asset on a weapons system, the weapons system may duplicate the procedures from the stock class manual. Another example would be where a weapons system has multiple destruction of material work packages based on crew duties or physical separation. Destruction procedures for common components may be duplicated for each crew member or location (e.g., a missile system where launchers and control systems are in different geographical locations).
b. Generic destruction procedures or concepts should not be duplicated.

30.4 Marine Corps requirements.

30.4.1 When to use Marine Corps variations. Both MIL-STD-40051-1 and MIL-STD-40051-2 may be used to acquire TMs that are used either exclusively by the Army, the Marine Corps, or as a multi-service manual. Both standards contain instructions on Marine Corps data requirements that either augment or, in the case of a Marine Corps only TM, replace Army requirements.

30.4.2 Marine Corp tagging. The following paragraphs provide information on how to tag TM source data IAW these Marine requirements. There are four locations in a TM where Marine Corps specific markup may be used. These locations are:

a. The technical manual number that appears in the technical manual title
b. The reporting of errors in the title block for page based TMs or in the front cover in an IETM.
c. In the equipment improvement reporting in the general information work package for both page based and IETMs.
d. In the parts list. This is required by the standard for both page based and electronic TMs.

30.4.2.1 Common tagging and text. 50.6.3 explains how to use the various boilerplate entities. Use of these entities is the recommended method to enter the common and unique text for the Marine.

30.4.3 TM number. Each TM is required to have a technical manual number. The TM number is entered in the ***<tmtitle>*** content model(see 31.3.1.1.4.1) which provides for the entry of either a single TM number, ***<tmno>***, or multiple TM numbers, ***<tminfono>***. The ***<tminfono>*** element identifies both the TM number and the branch of service that has assigned an official TM number to the manual. For single service TMs enter the technical manual number using only the ***<tmno>*** tag. For multi-service TMs, 40051 requires that each service's TM number appear. In this case, for each service the TM number and branch of service is entered using the ***<tminfono>*** tag.

30.4.3.1 Single service TM number example. Use this method of entering the TM number if there is only a single service (i.e., Marine Corps or Army) that will be using the TM. This does not preclude other services from using the TM, but indicates only that the TM was acquired primarily for the specific service.

```
<tmno>1-1520-238-10</tmno>
```

Normally, if a TM is used by another service, it will be adopted by that service and they will assign it a TM number. At that point, the TM manager of the owning service is contacted and a TM change would be issued with the cover page showing all assigned TM numbers.

30.4.3.2 Multi-service TM number example. When a TM is acquired and will be used by more than one service branch, the TM number and branch of service will be shown for each service. The order of

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appearance of the TM numbers is dictated using the following criteria. The TM number of the acquiring service is entered first. The other services TM numbers are entered and displayed in alphabetic order.

```
<tminfono>
<servbranch service="marines"/>
<tmno>12345-67/1&stago;/tmno
</tminfono>
<tminfono>
<servbranch service='army'/>
<tmno>1-1520-234-10
</tmno>
</tminfono>
```

30.4.4 Reporting of errors. Regardless of the TM type (paper or electronic) being authored, the tagging of the reporting of errors is the same. There are several options for text that may be used in the reporting of errors. These include:

a. Text for a Marine Corps only manual.
b. Text for a combined Army/Marine Corps manual.
c. Text for a multi-service manual.
d. Text for each of the above options if the manual is classified.

30.4.4.1 Reporting of errors tagging example. This example illustrates how to tag a combined Army/Marine TM. The tagging for the other options is identical, differing only in the entity selected/entered. A list of the available entities is located at 50.6.3

```
&titleblk.reporting.std.multi-service;
```

this general entity expands to the following:

```
<reporting>
<title>REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS
</title>
<para>You can help improve this manual.  If you find any mistakes
or if youknow of a way to improve the procedures, please let us
know.  Service, should be submitted as follows:
<reporting.para service="army">Mailyour letter or
<extref docno="DA Form 2028" posttext=" (RecommendedChanges to Publications
and Blank Forms)"/>, located in the backof this manual directly
to:  &amp;proponent-address.army;.  You mayalso send in
your recommended changes via electronic mail or by fax.Our
fax number is &amp;proponent-fax.army;  .  Our e-mail
address is&amp;proponent-email.army;
</reporting.para> A reply willbe furnished to you.
</para>
</reporting>
```

30.4.5 General information work package variations. Within the general information work package, requirement for Equipment Improvement Reporting **<eir>** has Marine unique requirements. As with the previous input, the required text may be entered by using the provided text entity.

```
<eir>&ginfowp.eir; </eir>
```

Where the **&ginfowp.eir;** is resolved to the following markup.

```
<eir>
<title>REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR)
</title>
<para esd="no" hcp="no">If your &amp;short.end.item.name; needsimprovement,
let us know.  Send us an EIR. You, the user, are the onlyone who can tell
us what you don&rsquo;t like about your equipment.Let us know why you
don&rsquo;t like the design or performance.  Ifyou have Internet access,
the easiest and fastest way to report problemsor suggestions is to go to
<internet show.address="yes">
```


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<homepageprotocol="https" uri="aeps.ria.army.mil/aepspublic.cfm"/>
</internet> (scroll down and choose the “Submit Quality Deficiency Report”bar). The Internet form lets you choose to submit an Equipment Improvement Recommendation (EIR), a Product Quality Deficiency Report (PQDR)or a Warranty Claim Action (WCA). You may also submit your informationusing a ***<extref docno="SF 368" posttext=" (Product Quality DeficiencyReport)"/>***. You can send your SF 368 via e-mail, regular mail, orfacsimile using the addresses/facsimile numbers specified in ***<extrefdocno="DA PAM 738-750" posttext=", Functional Users Manual for theArmy Maintenance Management System (TAMMS) "/>***. We will send you a reply.
</para>
<para esd="no" hcp="no">For Marine Corps users:Quality deficiency reports (QDR) shall be submitted on ***<extrefdocno="SF 368"/>*** in accordance with ***<extref docno="MCO 4855.10"/>***.A reply will be furnished to you.
</para>
</eir>

Note that the Marine Corps ***<para>*** is automatically included from the boilerplate entities when the marked section entity ***&usmc-tm;*** is set to 'include'. See Section 50 on using the boilerplate entities.

30.4.6 Marine Corps parts listing differences . When a TM with a RPSTL is acquired that is to be used by the Marine Corps, the following change to the ***<introwp>*** and ***<plwp>*** will be made.

a. In the ***<introwp>***, the following text will be added to the end of the "Explanation of columns in the repair parts list and special tools list work packages" For page based TMs, this will add an eighth column to the parts list. In an IETM this will add a new entry that contains the total quantity of the part used on the weapons system.
<![%usmc-tm;[<para hcp="no" esd="no">USMC QTY per Equip (<![%frame-base;[Entry]]> <![%page-base;[Column (]]>8<![%page-base;[)]]>). This <![%frame-base;[entry]]> <![%page-base;[column]]> accommodates the Marine Corps quantity per equipment requirement.<para>]]>

b. In the ***<plwp>***, there are two changes that need to be made:

(1) The first is an optional change in the Source, Maintenance, and Recoverability ***<smr>*** code. The SMR code is normally comprised of five characters found in the Joint Regulation, AR 700-82/OPNAVINST 4410.2A/MCO 4400.120, Joint Regulation Governing the Use and Application of Uniform Source. This regulation also allows a sixth character that has a service unique usage. Within the Army, this last character is used to identify demilitarization and the attribute in the ***<smr>*** element for this character is **demil**. The **demil** attribute will be used for any sixth character found in an SMR code.

<smr sourcecode="PA" maintcode="OO" recovercode="O" demil="E"/>

(2) The second change in the ***<plwp>*** required to support Marine Corps parts data is the addition of a total quantity per end item ***<qty_per_end_item>*** column. The total quantity per end item will reflect the total number assets used in a end item/weapons system.

<std.table.header>
<std.table.row>
<std.table.entry>(1)
</std.table.entry>
<std.table.entry>(2)
</std.table.entry>
<std.table.entry>(3)
</std.table.entry>
<std.table.entry>(4)
</std.table.entry>
<std.table.entry>(5)
</std.table.entry>

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<std.table.entry>(6)
</std.table.entry>
<std.table.entry>(7)
</std.table.entry>
<std.table.entry>(8)
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry>Item
</std.table.entry>
<std.table.entry>smr
</std.table.entry>
<std.table.entry>
</std.table.entry>
<std.table.entry>
</std.table.entry>
<std.table.entry>part
</std.table.entry>
<std.table.entry>description and usable on
</std.table.entry>
<std.table.entry>
</std.table.entry>
<std.table.entry>totalqty
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry>no.
</std.table.entry>
<std.table.entry>code
</std.table.entry>
<std.table.entry>nsn
</std.table.entry>
<std.table.entry>cagec
</std.table.entry>
<std.table.entry>number
</std.table.entry>
<std.table.entry>code(uoc)
</std.table.entry>
<std.table.entry>qty
</std.table.entry>
<std.table.entry>perend item
</std.table.entry>
</std.table.row>
</std.table.header>
<fncgrp>
<fnccode>01
</fnccode>
<fnctitle>top level item
</fnctitle>
<partno>a1234
</partno>
</fncgrp>
<pi.item>
<callout assocfig="fig1" label="1"/>
<uoc>all
</uoc>

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Check the source to verify that this is the current version before use.

```
<qty>1
</qty>
<smr  maintcode="da"recovercode="z"  sourcecode="pa"/>
<nsn>
<fsc>6610
</fsc>
<niin>012345678
</niin>
</nsn>
<partno>a1234&ndash;1
</partno>
<cageno>81115
</cageno>
<name>what's  in  a  name
</name>
<desc>top  level  partfor  this  example
</desc>
<qty_per_end_item>10
</qty_per_end_item>
<ui>ea
</ui>
</pi.item>
```

30.5 How to use LiveDTD. LiveDTD is an indexed HTML version of a DTD. This enables a user to readily navigate through DTD files, entity declarations, attribute lists and element content models to examine details of the document model. This section will identify:

a. how to access/download LiveDTD, and
b. features and functions provided by LiveDTD.

30.5.1 Accessing/downloading LiveDTD. LiveDTD is located at the Army SGML/XML Registry and Library (ASRL). The URL is http://www.asrl.com. LiveDTD is available for selected publications. For example, to access LiveDTD for technical manual publications enter the following:

a. Select "CONSTRUCTS" from the list of links (see Figure 34). The SGML/XML constructs page will appear.

FIGURE 34. ASRL home page list of links.

b. Select "Technical Manual (TM) Publications" from the list of links that appear in the SGML/XML constructs page (see Figure 35). The Technical Manual (TM) constructs page will appear.

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FIGURE 35. SGML constructs page TM link.

c. The Technical Manual (TM) constructs page includes selections for construct information for MIL-STD–40051–1 and MIL-STD-40051–2 version C. There are several versions of the construct data (e.g., REVISION 3.12) that are identified on this web page. Select either the latest version of the DTD or the version that applies to your system. The construct information includes selections to view or download an "Interactive TM XML DTD" (see Figure 36). This "interactive TM XML DTD" is also referred to as LiveDTD.

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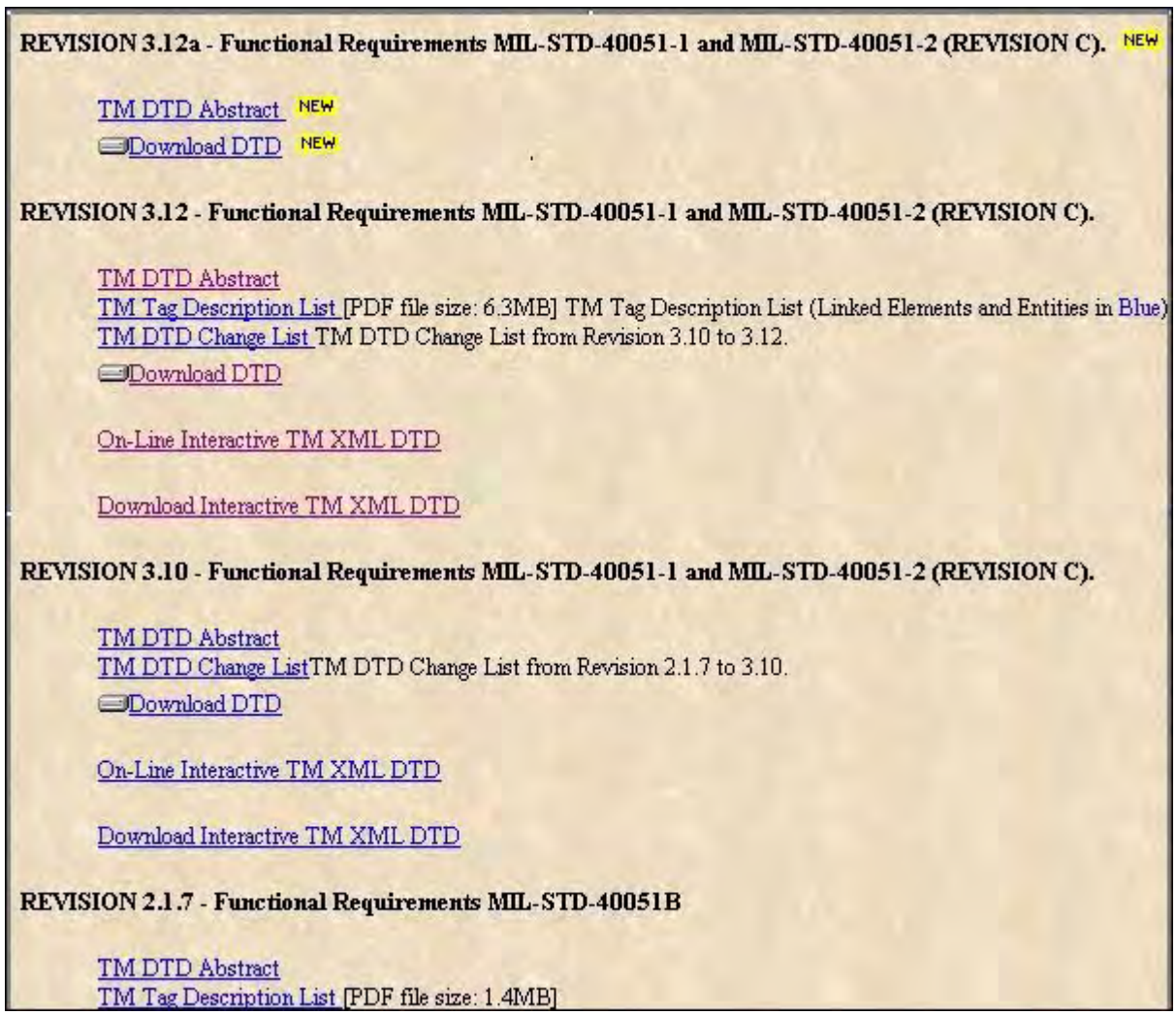

FIGURE 36. TM constructs page interactive TM XML DTD selections.

d. Selecting "On-Line Interactive TM XML DTD" will bring up a web page containing the indexed HTML from LiveDTD (see Figure 37).

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FIGURE 37. LiveDTD at startup example.

e. Selecting “Download Interactive TM XML DTD” saves a copy of LiveDTD on a local computer. The “File Download” dialog box (see Figure 38) is displayed. The file name identified in the dialog box includes the revision number. Compare this number to the revision that was intended to be downloaded to verify that the proper selection was made. The user is prompted to either open or save this file or to cancel the download. Select “cancel” if the wrong revision was selected. To download the file, select “open” or “save”.

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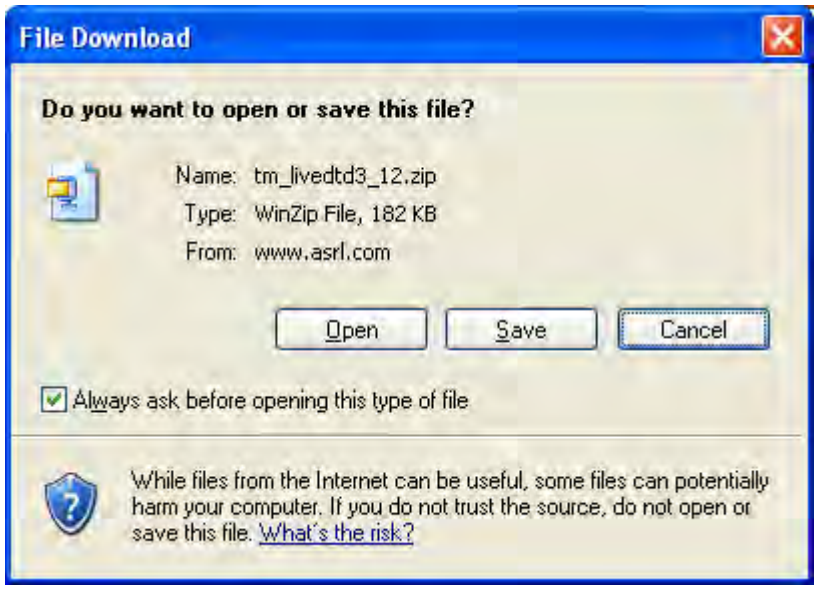

FIGURE 38. File download dialog box.

(1) Selecting "Open" will result in a ".zip" file screen being displayed. A typical example is shown (see Figure 39). Users who are accustomed to working with ".zip" files should make this selection. The ".zip" file screen shows the files that comprise LiveDTD and offers several menu choices. Selecting "extract" will result in the user being prompted to identify a folder in which to download the files. The files will be saved in a sub-folder of the selected folder named "livedtd". When the download is complete, select the "index.html" file to load LiveDTD into a browser.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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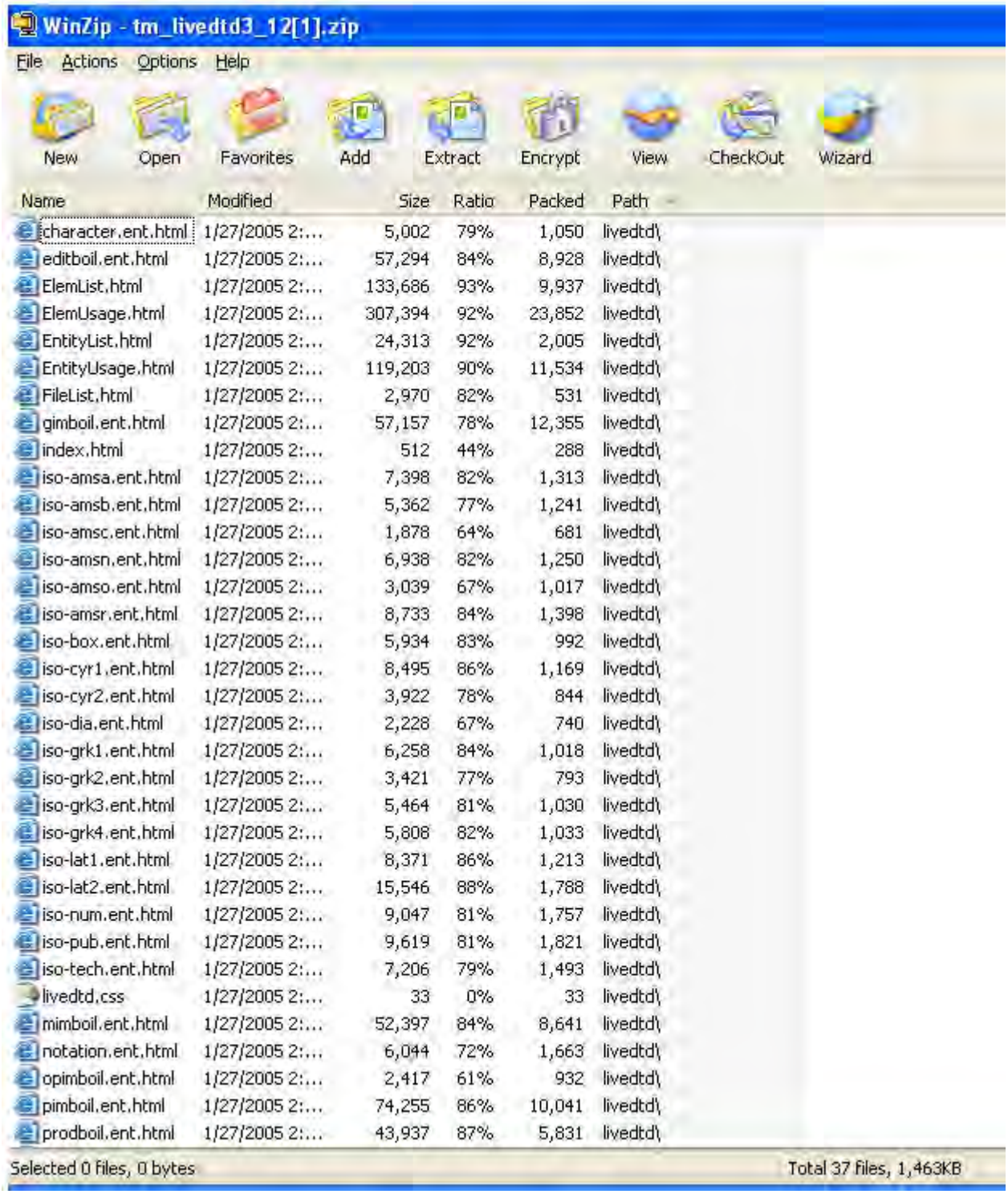

FIGURE 39. LiveDTD ZIP file example.

(2) Selecting "Save" will result in a "Save As" dialog box being displayed (see Figure 40). Select a folder in which to save LiveDTD files, then select "Save" from the "Save As" dialog box. The ".zip" file will be stored in that folder. The ".zip" file will then need to be unzipped. When the

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".zip" file is unzipped, it will place the HTML files in a sub-folder named "livedtd" that is located below the folder specified. To run LiveDTD, select the "index.html" file.

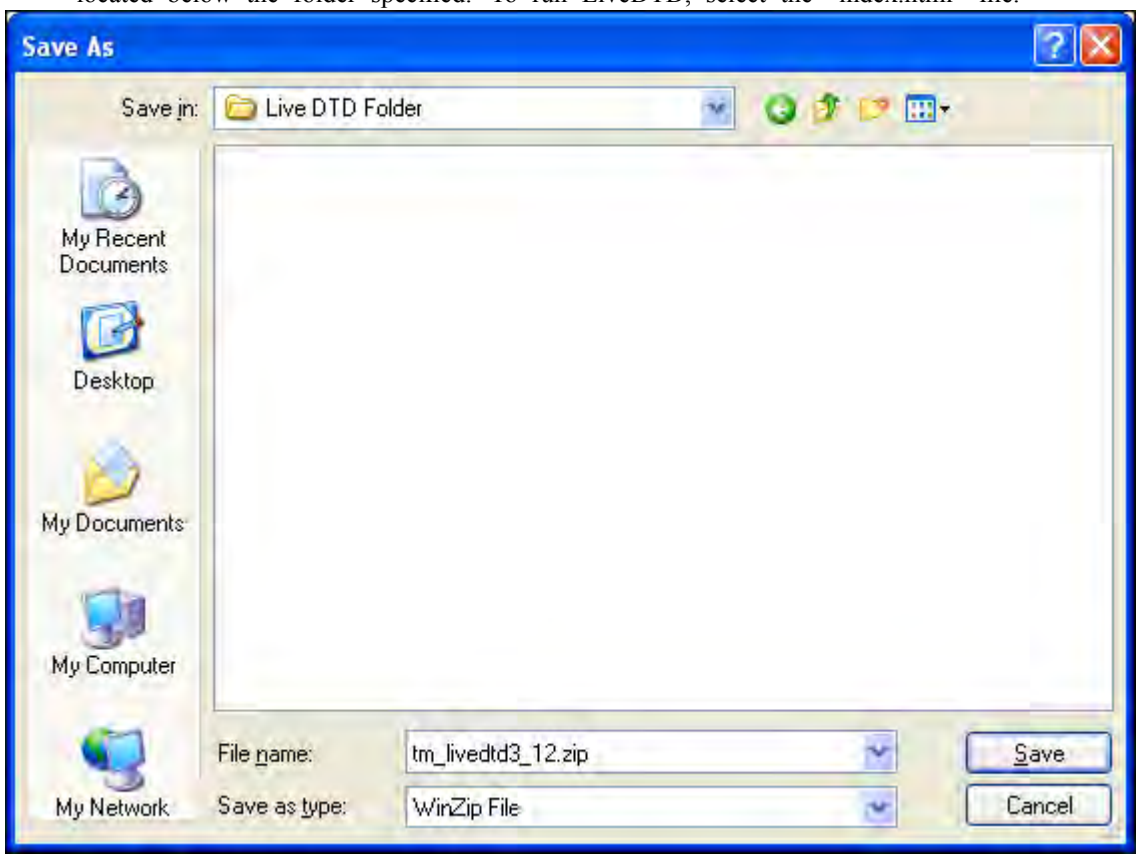

FIGURE 40. Save as dialog box.

30.5.2 Features and functions. LiveDTD presents a point and click method for examining all DTD files, element content models/usage, and parameter entity declarations/usage. As a HTML file, it will run in a browser. This permits the use of all browser functions (e.g., back). When LiveDTD is run, a window is displayed (see Figure 37) showing the DTD in the right pane. The left pane contains an index. The heading for the index pane will be the name of the DTD, in the example, this is "production". At the top of the index, under the heading "production.dtd", are three items "DTD Files", "Elements", and "% Entities". These items provide links to the major components of the LiveDTD. They will remain at the top of the left pane as the user selects components to view. Note that both panes have scroll bars. At start up, the items in the "production.dtd" list are followed by a list of elements under the heading "Elements". This view is equivalent to selecting the "Elements" link from the "production.dtd" list. A description of the features and functions provided when each of the three selections is made is as follows;

30.5.2.1 DTD files.

a. When this item is selected, an index of DTD files appears in the left pane. Files that have been previously selected are highlighted. The production DTD is shown on the right pane (see Figure 41).

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Check the source to verify that this is the current version before use.

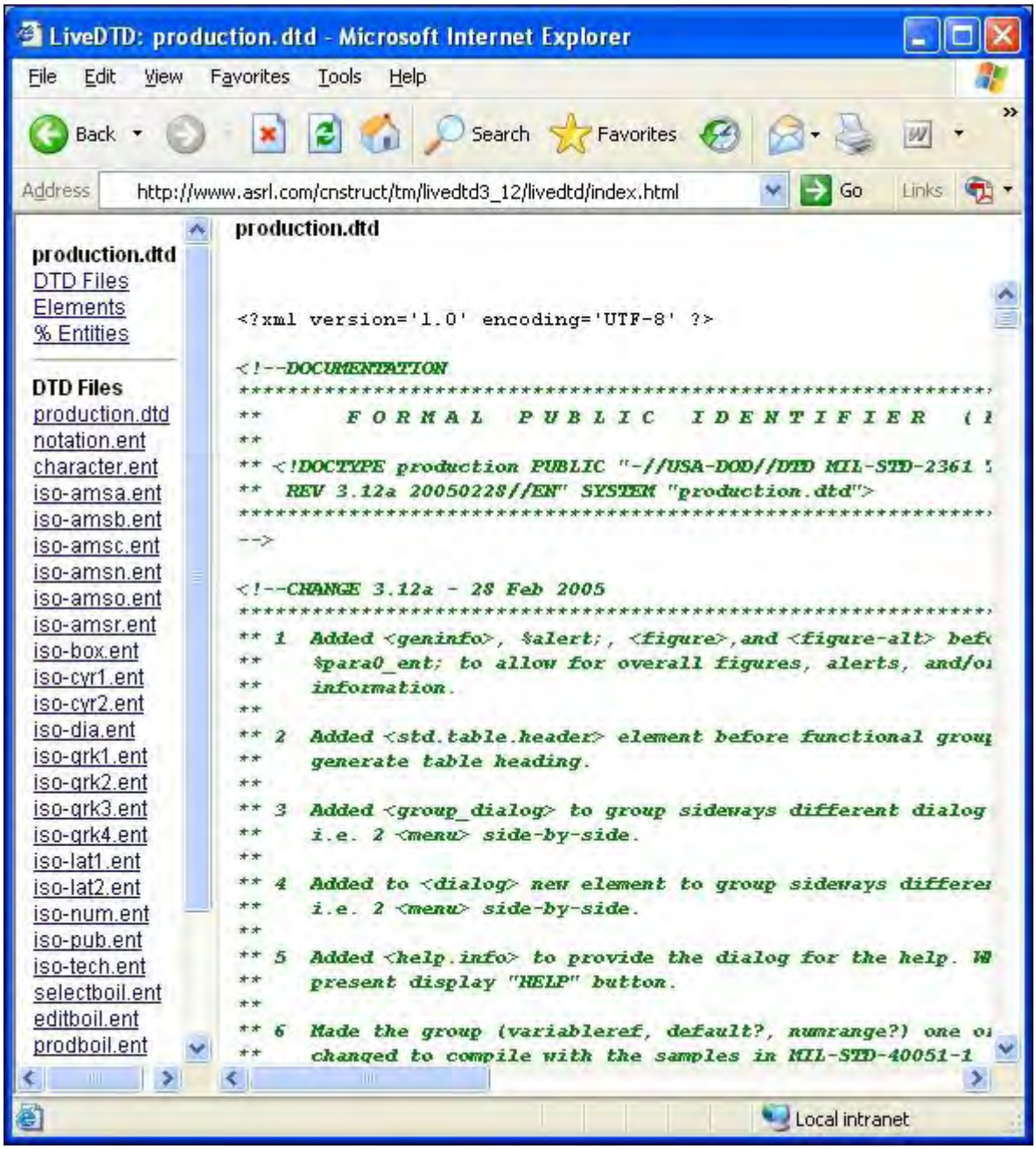

FIGURE 41. DTD files example.

b. Select any of the files under the "DTD Files" heading to view its contents in the right pane. For example, select "notation.ent" to display the contents of the notation entity file in the right pane (see Figure 42).

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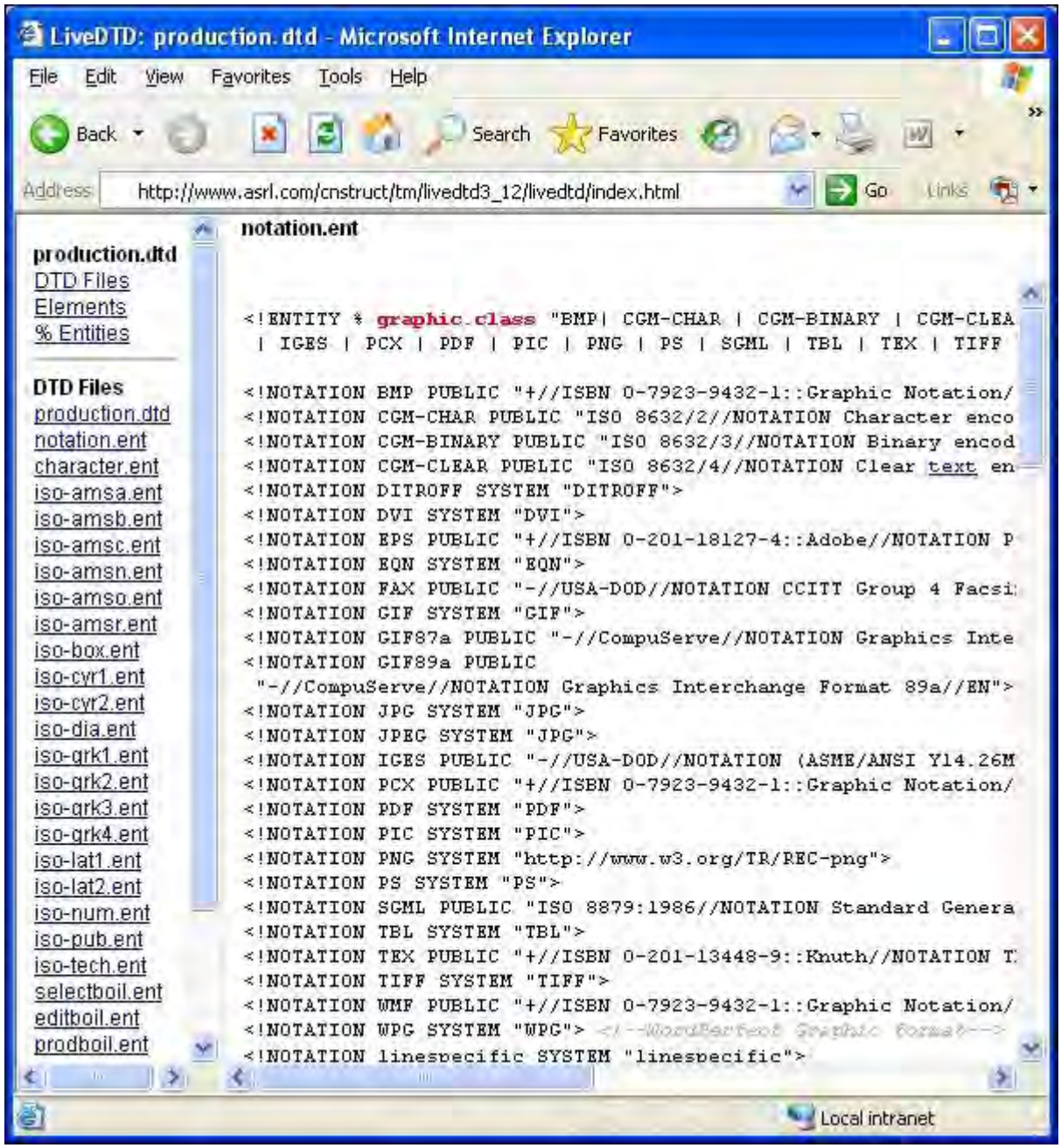

FIGURE 42. Notation file example.

c. Selecting “production.dtd” under the DTD Files heading will again display the production.dtd file in the right pane. This file has additional features/characteristics which are as follows:

(1) Identification of Elements and Entities. Scroll down the production DTD until the element declarations appear. Observe that the element and entity names are highlighted in red in their declarations. For example the declarations for element ***<memloc>*** would appear as shown below Figure 43

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```
<!ELEMENT memloc (#PCDATA)>
<!ATTLIST memloc  %bodyidatt; >
```

FIGURE 43. Highlighted element name.

(2) Links to Declarations. The element and entity names in the content models and attribute definitions are linked to their respective declarations. For example, click on the ***%bodyidatt;*** entity in the attribute list for the ***<memloc>*** element. The production.dtd file will be repositioned so that the declaration for that entity will be displayed at the top of the pane (see Figure 44).

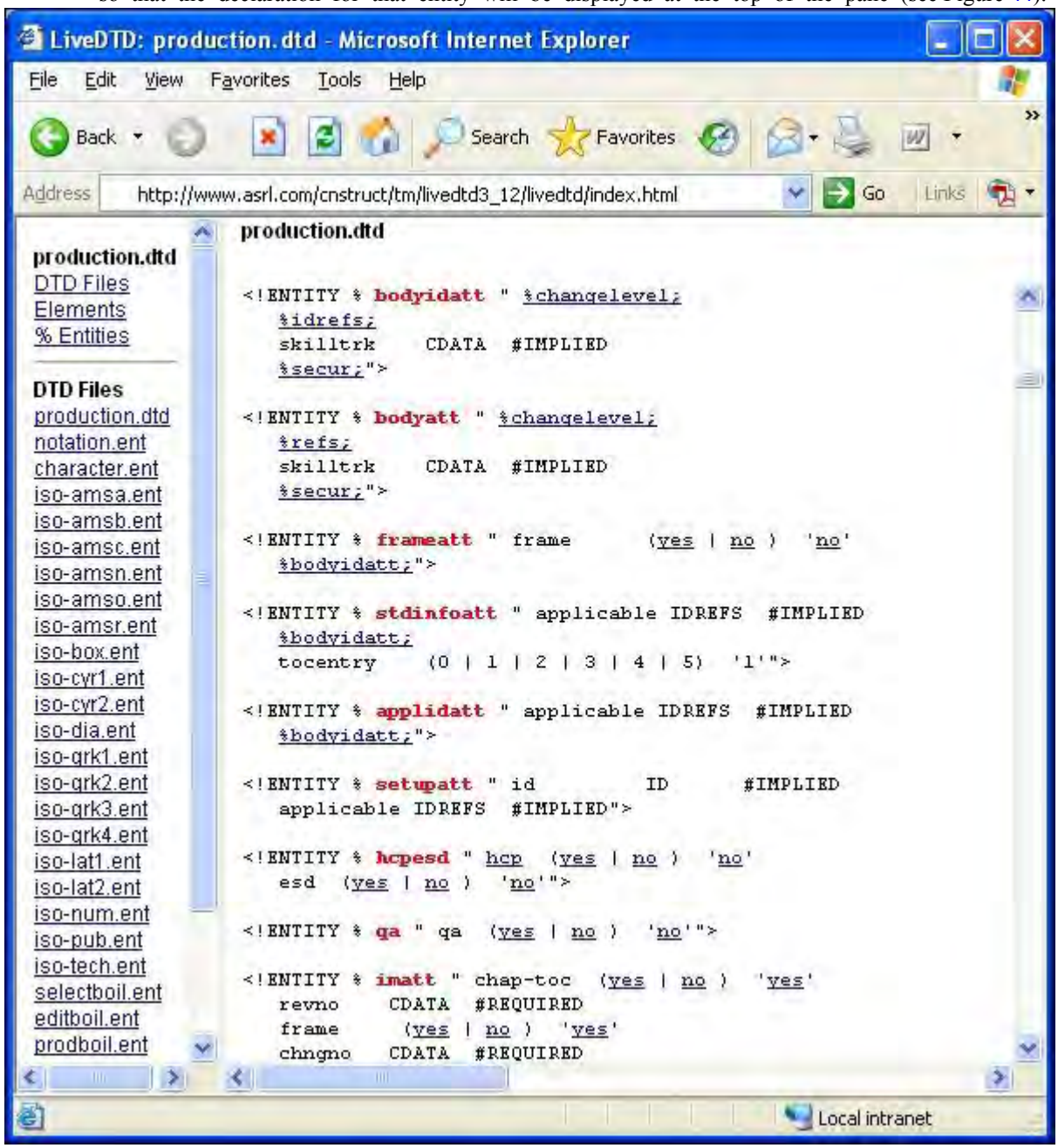

FIGURE 44. Entity declaration for %bodyidatt;.

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30.5.2.2 Elements.

a. When this item is selected, a list of elements appears in the left pane. The production DTD is shown in the right pane (see Figure 45).

FIGURE 45. List of elements example.

b. The features/characteristics which are available when the "Element" link is selected are as follows:

(1) Link from Element List to Declaration. Select any of the elements under the "Elements" heading. to view its declaration in the right pane. The production.dtd file will be repositioned so that the declaration for that element will be displayed at the top of the pane. For example, click on the element ***<action>*** in the list. The production.dtd will appear as shown (see Figure 46).

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Check the source to verify that this is the current version before use.

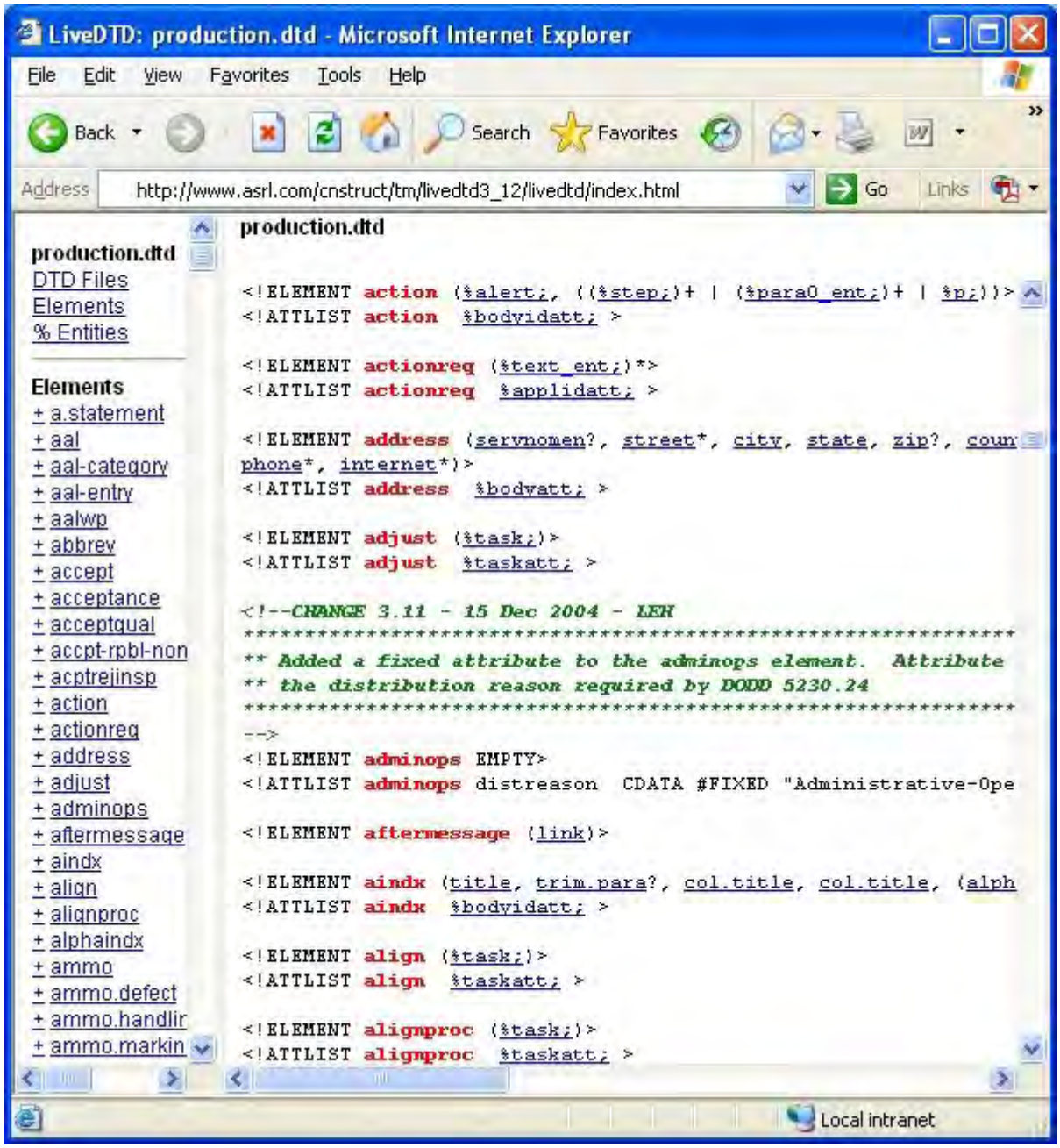

FIGURE 46. Element declaration for <action>.

(2) Link from Element List to Element Usage Table. Select the plus sign ("+") that appears to the left of any of the elements under the "Elements" heading. An element usage table will be displayed in the right pane. The selected element's usage will appear at the top of the pane. This table identifies the elements that contain a selected element in their respective content models. For example, click on the plus sign ("+") that appears to the left of the element ***<action>*** in the list. It's element usage table will be displayed (see Figure 47).

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FIGURE 47. Usage table for <action>.

The element and entity names in the usage table are linked to their respective declarations. This is the same feature as provided when "DTD Files" is selected from the production.dtd list (see 30.5.2.1.c.(2)).

(3) Identification of Elements and Entities. Element and entity names are highlighted in red in their declarations. This is the same feature as provided when "DTD Files" is selected from the production.dtd list (see 30.5.2.1.c.(1)).

(4) Links to Declarations. The element and entity names in the content models and attribute definitions are linked to their respective declarations. This is the same feature as provided when "DTD Files" is selected from the production.dtd list (see 30.5.2.1.c.(2)).

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30.5.2.3 % Entities.

a. When this item is selected, a list of entities declared in the DTD appears in the left pane. The production DTD is shown in the right pane (see Figure 48).

FIGURE 48. List of entities example.

b. The features/characteristics which are available for the "% Entities" link are the same as when "Elements" is selected (see 30.5.2.2). For example, click on the entity ***%format;*** under the "% Entities" heading to view its declaration in the right pane. The dtd file will be repositioned so that the declaration for that entity will be displayed at the top of the pane (see Figure 49).

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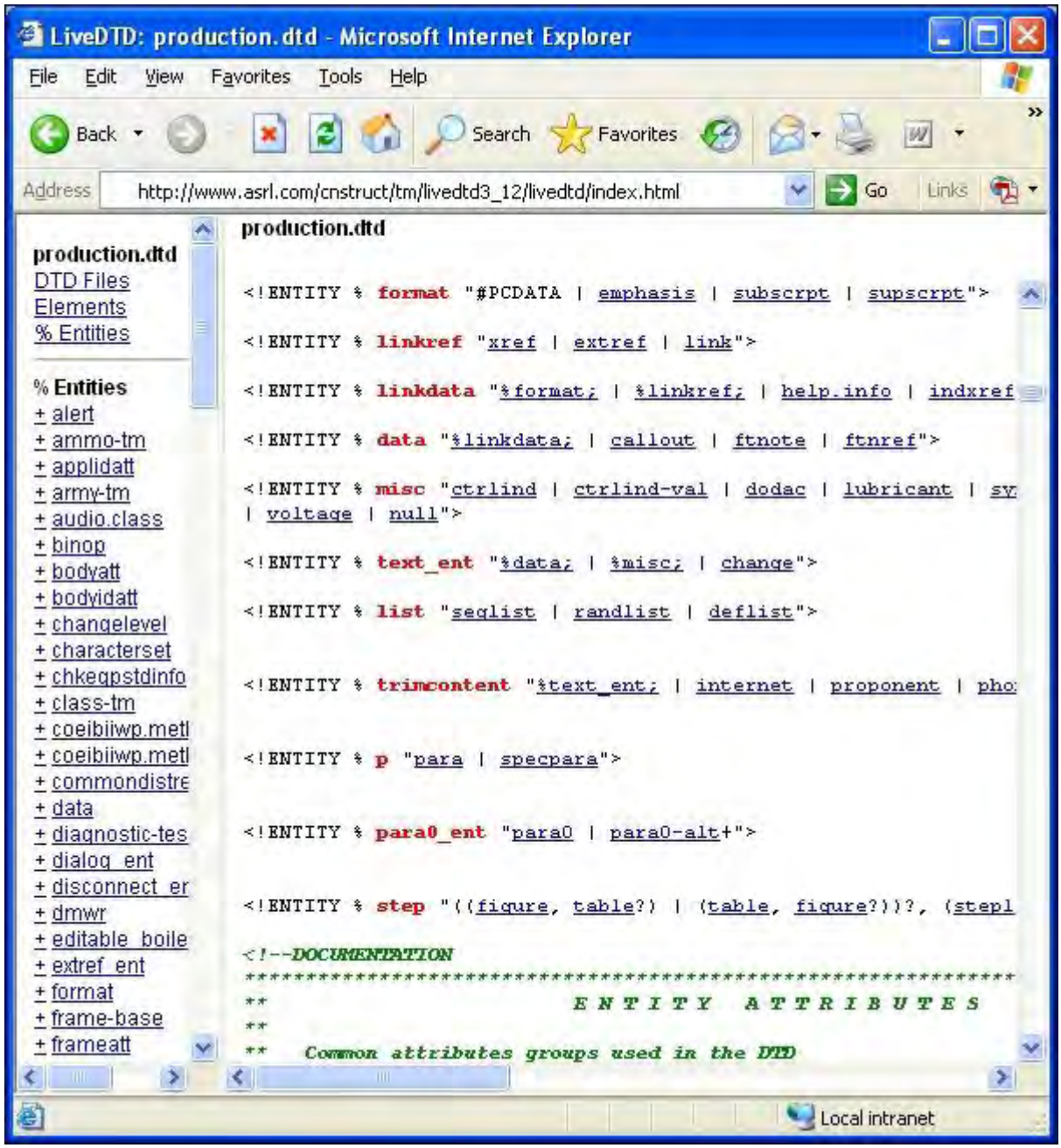

FIGURE 49. Entity declaration for %format;.

30.6 How to use Element and attribute description table.

30.6.1 Introduction. The Element and Attribute Description Table is a tool to allow users of all levels of expertise to better understand capabilities and requirements of the elements and attributes used in the DTD. A Element and Attribute Description Table may be developed for any DTD. The Tag Table is best used in conjunction with the source document that defines the requirements for the document being prepared.

30.6.2 Element and attribute description table layout. The Element and Attribute Description Table is broken into the following groups.

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30.6.2.1 Common entities. These are common constructs used at various times throughout the DTD. These common entities can provide one of three things.

TEXT ENTITY DEFINITIONS	A block of replacement text (i.e. verbatim statement) or commonly used characters that are not found on the keyboard (i.e. "1/2" character define as ***½***).
ATTRIBUTE ENTITY DEFINITIONS	A common set of attributes that is used in more than one attribute list. A common entity example is ***%bodyatt;***. The entity defines a list of attributes that are used on most work package elements found in the DTD. Rather than repeat the attributes each time it is used, the entity provides a tool to maintain consistence for the DTD developer.
ELEMENT ENTITY DEFINITIONS	A common content model that is used in more than one element in the DTD. An example is the special paragraph (group warning, caution, and/or note information with a paragraph) and paragraph. Rather than repeat "para | specpara" each time it is used, the ***%p;*** entity is used to provides a tool to maintain consistence for the DTD.

30.6.2.2 Elements list. Elements are the primary building blocks in a DTD. Elements may contain other elements, used to produce the desired document structure, character data, or a mixture of both.

30.6.3 How to read the entities lists.

30.6.3.1 Entity name components. Each entity is identified by its entity name, (i.e., ***%wpbodyatt;***) followed by its formal name. Under the formal name is a description of what the entity is designed to do.

30.6.3.2 Attribute entity components. For attribute related entities, the basic format sited in 30.6.3.1 applies and shown in Figure 50. Following the explanatory paragraph, the attributes declared in the entity are listed along with an explanation of the desired use of that attribute, a declared value and in some cases, a default value. The list is divided into two sections, REQUIRED and OPTIONAL.

%wpbodyatt; *Work Package Body Attribute Set*

Identifies a set of attributes applying to work package body elements. These attributes may be used with any element type that references this attribute set (%wpbodyatt;) in the document type declaration.

OPTIONAL ATTRIBUTE(S)

config Specifies the equipment configurations to which element applies
DECLARED VALUE: Any character

delchlvl Specifies the change level(s) at which information was deleted. An audit trail can be maintained by listing multiple change levels separated by spaces.
DECLARED VALUE: Any character

inschlvl Specifies the change level(s) at which information was inserted. An audit trail can be maintained by listing multiple change levels separated by spaces.
DECLARED VALUE: Any character

itemid Supplies an identifier of the item, such as SSSN, LRU, part number, or reference designator.
DECLARED VALUE: Name Token

FIGURE 50. Sample attribute parameter entity description.

a. Attributes may be either REQUIRED or IMPLIED (optional). Required attributes are just that, required. A value is entered or the parser will flag the omission as an error.
b. Each attribute has a value type, the declared value, that tells the parser what type of information is acceptable in the attribute entry. If you enter a invalid data type, the parser will identify this as an error.
c. In some cases, an attribute will have a pre-specified value. This is the default value and does not need to be changed unless you need to use something different. In most cases, if an attribute has a

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default value, this value is part of a list of specific choices. Each markup language application has different methods to allow these choices to be entered, but the only values that may be entered are those from the declared value type.

30.6.3.3 Element entity list. The "Element Entity Definition" list will contain the content model entity, this paragraph also contains a listing of the elements that comprise the content model as shown in Figure 51.

%list;	*Lists* Groups the lists elements. The entity contains random (<*randlist*>), sequence (<*seqlist*>), and definition (<*deflist*>)list

FIGURE 51. Sample element parameter entity description.

30.6.4 How to read the "Element/Attribute Definitions" list. The Element/Attribute Definitions comprises the majority of the Element and Attribute Description Table. The list is in alphabetical order to ease finding a specific element. An element description example is shown Figure 52.

<aal>	*Additional Authorization List* A tabular list containing standard information of all additional authorization items in the additional authorization list. These items are required to operate the equipment but are not classified as COEI or BII items. This element functions as the table element.
Specification Paragraph: 6-5.6	
OPTIONAL ATTRIBUTE(S)	
%bodyatt;	Any of the attributes in the associated attribute set may be used with this element. Refer to %bodyatt; for a complete description.
%secur;	Any of the attributes in the associated attribute set may be used with this element. Refer to %secur; for a complete description.

FIGURE 52. Sample element description.

30.6.4.1 Element definition. Each element is identified by its element name, followed by its formal name and a brief explanation on the designed use of the element. The explanation may be followed by a reference to where the requirement for this element may be found in MIL-STD-40051-1 and MIL-STD-40051-2. The reference to the standard will be found only on major component elements such as the ***<aal>*** or ***<tswp>***.

30.6.4.2 Attribute list. The element description may have a list of attributes that the element utilizes. The attribute listing will do one of two things.

a. Attributes found in the "Attribute Entity Definitions" list, as described in 30.6.3 , will be a reference to the specified parameter entity entry.
b. The remaining attributes peculiar to the element are defined individually.

30.7 Tagging information.

30.7.1 Tagging. Tagging is a method of descriptive markups by applying tags to document data. The elements used by MIL-STD-2361 fall into two types of tags: content tags (see 30.7.1.2) and structure tags (see 30.7.1.1). Information developed in accordance with MIL-STD-40051 and MIL-STD-2361 is organized into chapters containing similar functional information (e.g., maintenance chapter, troubleshooting chapter, etc.), and content-tagged to support retrieval from a database. Specifically, the use of XML structured information, in conjunction with database technology and content-tagging, facilitates information access, sharing, reuse, management, control, and change. Content tags identify the functional requirements contained in the respective functional requirements standards and specifications, such as MIL-STD-40051. Structure tags, such as ***<para>*** (paragraphs), are based on the physical structure of the document, and are used to "chunk up" the data within the content tags. MIL-STD-2361 tags mix content identification and structural uses, such as ***<step1>*** and ***<step2>***.

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30.7.1.1 Structure tagging. Structure tagging is commonly used to model and encode information according to the structure of a document. Structure tagging can be applied when the document structure can be enforced and validated. Structure tagging promotes document consistency, ease of training, ease of revision, etc. because document structures can be clearly defined and easily standardized. Structure tags, such as paragraph ***<para>***, contain no intelligence about the content of data, but they do flag material for specific treatment by a presentation system. For instance, tagged text can be numbered automatically by a presentation system, but untagged text cannot be automatically numbered. XML does not recognize carriage returns as processing instructions (e.g., "break line here"), but as marking record boundaries. The presentation system, in general, ignores carriage record boundaries and will not trigger paragraph returns. Some examples of structure tagging would be the use of generic elements listed .

a. Title ***<title>*** – A title tag (see 49.1.1.3) is a common structure element used in the content model of many elements. For instance, all work packages have a title. The title of a work package is found in the work package identification information ***<wpidinfo>*** tag. An example of a XML instance fragment with title tag used in work package identification information is shown:

```
<wpidinfo>
<maintlvl  level="operator"/>
<title>M1079 Van Preparation For Movement
</title>
</wpidinfo>
```

b. Paragraphs ***<para>*** – Paragraphs are common structure tags that are included in many content-oriented element content models. Paragraphs includes character data (#PCDATA) and various XML tags that may be contained in paragraph text, such as references, footnote references, index references, change notices, and emphasis tags. A paragraph (see 49.1.1.5) is usually the element through which lists, figures, and tables are included in documents developed in accordance with MIL–STD–2361.

```
<para>Many abbreviations are used in this manual.  They are listed
below.  Learn what each one means.  It will make your job easier.
<deflist>
<term.def>
<term>A
</term>
<def>
<para>After
</para>
</def>
</term.def>
</deflist>
</para>
```

c. Step ***<step1>*** – The XML structure tags for steps (see 33.3) will be used when to show the steps within a task or procedure. An example of an XML instance of the task remove contains the tag ***<step1>***.

```
<remove  id="m00446-9-2320-365-3">Many abbreviations are used in
this manual.  They are listed below.  Learn what each one
means.  It will make your job easier.
<title>Removal
</title>
<step1  id="m00446-9-2320-365-6">
<para>Disconnect vacuum hose from AIR FILTER RESTRICT.
</para>
</step1>
<step1  id="m00446-9-2320-365-7">
<para>Remove two screws and AIR FILTER RESTRICTION GAUGE
faceplate from instrument panel assembly.
</para>
```


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```
</step1>
</remove>
```

d. Lists – Lists are common structure tags usually contained within paragraphs. The list tag will identify the type of list being tagged. MIL–STD–2361 contain three types of lists:
   (1) Random *<randlist>* (see 49.1.2.2)
   (2) Sequential *<seqlist>* (see 49.1.2.1)
   (3) Definition lists *<deflist>* (see 49.1.2.3)

   An example of a XML instance fragment with a sequence list within a paragraph tag is below:

```
<para>AVUM activities will be staffed and equipped to perform high
frequency "On-Aircraft" maintenance tasks required to retain or
return aircraft systems to a serviceable condition.
<seqlist>
<item>Company Size Aviation Units.
</item>
<item>Less than Company Size Aviation Units.
</item>
</seqlist>
</para>
```

e. Tables *<table>* – CALS table model *<table>* (see 42), and tabular material *<tabmat>* (see 42.2) use structure tags to format general data. An example of a CALS table model *<table>* containing general information is shown:

```
<table colsep="1" frame="all" rowsep="1">
<title>General Items
</title>
<tgroup cols="3">
<tbody valign="bottom">
<row>
<entry>WP 0001
</entry>
<entry>General Information
</entry>
<entry><ginfowp>
</entry>
</row>
<row>
<entry>WP 0002
</entry>
<entry>Equipment Description And Data Work Package
</entry>
<entry><descwp>
</entry>
</row>
</tbody>
</tgroup>
</table>
```

f. Figures *<figure>* – A figure tag provides descriptive and location narrative text to a graphic (see 44.1.1).

```
<para>The Users Display is selected by depressing F2 on the HTU keyboard.
The sections within this display are explained in the following text.
<figure>
<title>HTU User's Display (Typical)
</title>
<graphic boardno="htudisplay">
```


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```
</graphic>
</figure>
</para>
```

g. Graphic **<graphic>** – The graphic tag **<graphic>** (see 44.2) is a common structure tag that is used throughout a TM to tag a graphic file entity and supply its size, clipping, scaling, placement, etc. The graphic tag is used in the XML instance example of a Reporting Errors and Recommending Improvements DA 2028-2.

```
<da2028>
<graphic boardno="da2028">
</graphic>
</da2028>
```

h. Procedure **<proc>** – The procedure tag **<proc>** (see 33.2) is a common structure tag that may contain a paragraph or a set of steps that comprise all or part of a task.

```
<proc id="m00446-9-2320-365-3">
<title>Removal
</title>
<step1 id="m00446-9-2320-365-6">
<para>Disconnect vacuum hose from AIR FILTER RESTRICT.
</para>
</step1>
<step1 id="m00446-9-2320-365-7">
<para>Remove two screws and AIR FILTER RESTRICTION GAUGE
faceplate from instrument panel assembly.
</para>
</step1>
</proc>
```

i. **<aindx>** – The alphabetic index tag **<aindx>** (see 31.4.2.2) is a common structure tag that contains reference text and its location found in a document. Below is an example of a short XML instance for an alphabetic index.

```
<aindx>
<title>ALPHABETIC INDEX
</title>
<col.title>Subject
</col.title>
<col.title>WP Sequence #
<col.title>
<alphaindx>A
</alphaindx>
<indexentry>
<title>Accessory Section
</title>
<wpno wpref="access_sect"/>
<pageno>15
</pageno>
</indexentry>
<alphaindx>B
</alphaindx>
<indexentry>
<title>Back Cover
</title>
<wpno wpref="backcover"/>
<pageno>285
```


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</pageno>
</indexentry>
</aindx>

j. Warnings, Cautions, and Notes – The warning ***<warning>*** (see 41.1.1), caution ***<caution>*** (see 41.1.2), and note ***<note>*** (see 41.1.3) tags are common structure tags that are used as alerts throughout a TM. (see 31.4.2.2). An example of a procedure XML instance fragment containing caution and note tags is shown:

```
<proc>
<title>Transmission  Fluid
</title>
<step1   id="m00446-9-2320-365-14">
<specpara>
<caution>
<trim.para>Do not overfill transmission.   The fluid level
rises as the fluid temperature increases.
</trim.para>
</caution>
<para>Check and fill transmission to proper level weekly.
Operate transmission through all operating ranges to
fill cavities and fluid passages.
</para>
</specpara>
</step1>
<step1   id="m00446-9-2320-365-14">
<specpara>
<note>
<trim.para>If oil level is above "FULL," it may be due to oil
cooler drain back.   Operate the engine for one minute, shut
down, wait one minute, then recheck oil level.
</trim.para>
</note>
<para>Check crankcase oil level daily.   Start engine and visually
check for oil leaks at drain plug and oil filter.
</para>
</specpara>
</step1>
</proc>
```

30.7.1.1.1 Complexity of tagging. Structure tagging can be complex depending what other tags are allowed to be contained within tags. For example, paragraphs are common structure tags, included in many content-oriented element content models. A paragraph contains character data (#PCDATA) and various XML tags that may be contained in paragraph text, such as references, footnote references, index references, change notices, and emphasis tags. A paragraph (see 49.1.1.5) is usually the element through which lists, figures, and tables are included in documents developed in accordance with MIL–STD–2361.

a. Content model of ***<para>***

```
<!ELEMENT para (#PCDATA | emphasis | subscrpt | supscrpt| xref | extref | link
| help.info | indxref | term | term.def | callout | ftnote | ftnref
| ctrlind | ctrlind-val | dodac | lubricant | symbol | torque | voltage | null
| change | internet | proponent | phone | instructplt |seqlist | randlist | deflist
| figure | figure-alt | inlinegraphic | table | table-alt
| verbatim | interaction | statemanipulation | statemanipulation-al)*>
```

b. The example of the paragraph ***<para>*** contains information that is tagged with footnote ***<ftnote>***, table ***<table>*** and emphasis text ***<emphasis>*** tags.

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```
<para>Table 1.  General provides information pertaining to the electrical,
operational, and environmental characteristics of the radar set
<ftnref  xrefid="AN/PPS-5XX"/>
<ftnote  id="AN/PPS-5XX">
<ftnpara>Radar  Set  AN/PPS-5XX
</ftnpara>
</ftnote>
<table>
<title>General
</title>
<tgroup  align="left"  cols="2"  colsep="1"  rowsep="1">
<colspec  colname="col1"  colwidth="2.69in"/>
<colspec  colname="col2"  colwidth="3.69in"/>
<thead  valign="bottom">
<row>
<entry  align="center">ITEM
</entry>
<entry  align="center">VALUE
</entry>
</row>
</thead>
<row>
<entry>Electrical  Characteristics
</entry>
<entry>
</entry>
</row>
<row>
<entry>Power  input
</entry>
<entry>
</entry>
</row>
<row>
<entry>Radar  Set  AN/PPS-5XX
</entry>
<entry>(18-36 volt, direct current (dc) source):  41 watts
</entry>
</row>
<row>
<entry>Operational  Characteristics
</entry>
<entry>
</entry>
</row>
<row>
<entry>Range:
</entry>
<entry>
</entry>
</row>
<row>
<entry>Maximum
</entry>
<entry>
</entry>
```


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```
</row>
<row>
<entry>Moving  personnel
</entry>
<entry>
<emphasis>10,000
</emphasis> meters  (5dBsm  Group  of  Three)
</entry>
</row>
</tbody>
</tgroup>
</table>
</para>
```

30.7.1.2 Content tagging. Content tagging is commonly used to identify document components by the nature of the information contained in the component. Content tagging can be applied when the organization of the document is dependent on the meaning of its contents. Each of MIL–STD–2361 content parts is comprised of a similar structure. The top level is an information module tag, such as general information chapter ***<gim>***, operating instructions chapter ***<opim>***, trouble shooting chapter ***<tim>***, maintenance chapter ***<mim>***, parts chapter ***<pim>***, or the supporting chapter ***<sim>***. These top-level tags contain specialized sets of work package elements that are, in some cases, unique to the respective information modules, while, in other cases, common to one or more information module. For example, the maintenance information chapter ***<mim>*** can contain service upon receipt work packages ***<surwp>***, preventive maintenance work packages ***<pmcswp>***, maintenance instruction work packages ***<maintwp>***, etc. The top level information chapter tag will be used for building even a single work package.

30.7.1.2.1 Why use content tags. Content tagging promotes information reuse and automated information management process application. The contents of content-tagged documents can be supplied by databases because database elements can be easily mapped to content tags. Similarly, databases can be populated with data from content-tagged documents. The following tags controls and indicators ***<ctrlind>*** (see 49.1.4.3), reading on – controls and indicators ***<ctrlind-val>*** (see 49.1.4.4), torque value or limit ***<torque>*** (see 49.1.4.24) are examples of content tags that are used in various work packages of MIL–STD–2361 and identifies value, such as lubricants, voltage, etc. By identifying the data by the use of these tags, the data provides information repositories to store it intelligently so it can provide reuse of the information to other types of documentation.

a. Example of how the controls and indicators ***<ctrlind>*** element is used as a content tag in the control and indicator standard table.

```
<ctrlindtab>
<title>Instrument  Panel  Controls  and  Indicators
</title>
<figure>
<title>Instrument  Panel  for  the  Controls  and  Indicators
</title>
<graphic  boardno="instr_panel">
</graphic>
</figure>
<std.table.header   cols="3">
<std.table.row>
<std.table.entry>Key
</std.table.entry>
<std.table.entry>Control/Indicator
</std.table.entry>
<std.table.entry>Function
</std.table.entry>
</std.table.row>
</std.table.header>
```


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```
<ctrlindrow>
<key>1.
</key>
<ctrlind>Radiator Fan Off Switch.
</ctrlind>
<function>When positioned to on, radiator fan off switch will illuminate
to indicate the radiator fan is disabled. Radiator fan off switch will
remain in the off position and not illuminated, unless otherwise directed.
</function>
</ctrlindrow>
<ctrlindrow>
<key>2.
</key>
<ctrlind>Lamp Test Switch.
</ctrlind>
<function>Tests the lights on high engine temperature and TRANS
OIL TEMP indicators on Lighted Indicator Display.
</function>
</ctrlindrow>
<ctrlindrow>
<key>3.
<key>
<ctrlind>Ether Start Switch.
<ctrlind>
<function>Injects ether into engine intake system to assist with
cold weather starting when switch is pressed.
<function>
<ctrlindrow>
<ctrlindtab>
```

b. Example of how the standard information Additional Authorization List (AAL)***<aal>*** (see 40.6.2) table uses the content wrapper tag Additional Authorization List entry (row) ***<aal-entry>*** to identify information that is specific to a work package. The tag ***<aal-entry>*** specifies the data in the row as part as the Additional Authorization List (AAL) table and work package. The data tag ***<aal-entry>*** contains the National Stock Number ***<nsn>*** (see 49.1.4.20), description, CAGEC and part number, usable on code ***<dcpno>*** (see 40.5.1.2.1.3.2), unit issue ***<ui>*** (see 49.1.4.20) and quantity ***<qty>***. The tag ***<nsn>*** breaks the content of the National Stock Number into the Federal Stock Classification Code ***<fsc>*** (the first four-digits) and National Item Identification Number ***<niin>*** (the last nine-digits). The tag ***<dcpno>*** contains the content tags; name ***<name>***, description ***<desc>***, part number ***<partno>*** Commercial and Government Entity Code ***<cageno>*** and useable on code ***<cageno>*** which all can be stored in a database for future references.

```
<aal>
<title>AdditionalAuthorization List (AAL)
<title>
<std.table.header cols="5" rowsep="1">
<std.table.row>
<std.table.entry colnum="1">(1)
</std.table.entry>
<std.table.entry colnum="2">(2)
</std.table.entry>
<std.table.entry colnum="3">(3)
</std.table.entry>
<std.table.entry colnum="4">(4)
</std.table.entry>
<std.table.entry colnum="5">(5)
```


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```
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry  colnum="1">NATIONAL  STOCK
</std.table.entry>
<std.table.entry  colnum="2">
</std.table.entry>
<std.table.entry  colnum="3">USABLE
</std.table.entry>
<std.table.entry  colnum="4">
</std.table.entry>
<std.table.entry  colnum="5">QTY
</std.table.entry>
</std.table.row>
<std.table.entry  colnum="1">NUMBER  (NSN)
</std.table.entry>
<std.table.entry  colnum="2">DESCRIPTION,  PART  NUMBER/(CAGEC)
</std.table.entry>
<std.table.entry  colnum="3">ON  CODE
</std.table.entry>
<std.table.entry  colnum="4">U/I
</std.table.entry>
<std.table.entry  colnum="5">RECM
</std.table.entry>
</std.table.row>
</std.table.header>
<aal-entry>
<nsn>
<fsc>6665
</fsc>
<niin>01-105-5623
</niin>
</nsn>
<dcpno>
<name>ALARM
</name>
<desc>CHEMICAL  AGENT
</desc>
<partno>8762101
</partno>
<cageno>19200
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
<aal-entry>
<nsn>
<fsc>1240
</fsc>
<niin>01-207-5787
</niin>
```


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```
</nsn>
<dcpno>
<name>BINOCULARS
</name>
<desc>MOD,CON  M22
</desc>
<partno>9370122
</partno>
<cageno>19200
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
</aal>
```

31 PRODUCTION.

31.1 Root element ***<production>***. Any material to be published according to the DTD in Technical Manual Production begins with this element. The available TM types are Equipment and Technical (frame-base or page-base), Ammunition, Preventive Maintenance Inspection (PMI), Phased Maintenance Services (PMS) Inspections, and Aviation Troubleshooting.

a. The components are:
   (1) Technical Manual System Effectivity List ***<applic_ref_list>*** (optional) (see 31.2).
   (2) Select the type of manual to develop the required TM. (required)
      (a) Frame Base Manual ***<framed.manual>*** (see 31.3).
      (b) Page Base Manual ***<paper.manual>*** (see 31.4).
      (c) Conventional and Chemical Ammunition ***<ammo>*** (see 31.5).
      (d) Phased Maintenance Inspections ***<pmi>*** (see 31.6).
      (e) Preventive Maintenance Services ***<pms>*** (see 31.7).
      (f) System-Wide Troubleshooting Aviation ***<sys-ts>*** (see 31.8).
      (g) Destruction Manual ***<destruction_manual>*** (see 31.9).

b. DTD fragment is graphically depicted.

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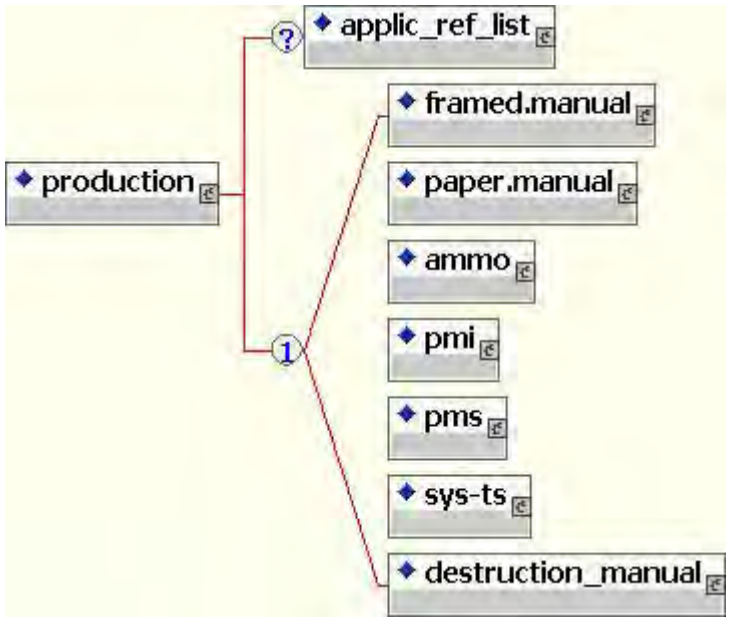

FIGURE 53. Production <production> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT production (applic_ref_list?, (framed.manual | paper.manual |
            ammo | pmi | pms | sys-ts | destruction_manual))>

<!ATTLIST production
    date          CDATA          #REQUIRED
    chnglevel     CDATA          #REQUIRED
    chngdate      CDATA          #REQUIRED>
```

d. Unique attributes:
   (1) **date** – Date of the TM. (required)
   (2) **chnglevel** – Change Level identifies the level the TM was changed to. (required)
   (3) **chngdate** – Change Date identifies the date the TM was changed to. (required)

31.2 Technical manual system effectivity list ***<applic_ref_list>***. The element ***<applic_ref_list>*** contains a listing of all possible system effectivity configurations. When a work package has a system effectivity issue, the work package reference or links to the system effectivity list.

a. The components are:
   (1) Technical Manual System Effectivity List ***<applic>*** (optional) (see 31.2.1).

b. DTD fragment is graphically depicted.

FIGURE 54. Technical manual system effectivity list <applic_ref_list> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT applic_ref_list (applic+)>
```

d. No attributes.

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31.2.1 System effectivity ***<applic>***. The element ***<applic>*** provides the qualifications to identify the effective system by NSN, part number, unique ID, etc. identifies the effective system to be displayed to in Work Package Identification Block.

a. The components are:
   (1) Name ***<name>*** (required)(see 49.1.4.19).
   (2) National Stock Number ***<nsn>*** (optional) (see 49.1.4.20).
   (3) Part Number ***<partno>*** (optional) (see 49.1.4.21).
   (4) Commercial and Government Entity Code (CAGEC) ***<cageno>*** (optional) (see 49.1.4.2).
   (5) Negate ***<not>*** (optional – zero or more) (see 31.2.1.1).
   (6) Effectivity Type Set ***<set>*** (required – one or more) (see 31.2.1.2).
   (7) Range ***<range>*** (required – one or more)(see 47.2.1.3).
   (8) Single***<single>*** (required – one or more) (see 47.2.1.3).

b. DTD fragment is graphically depicted.

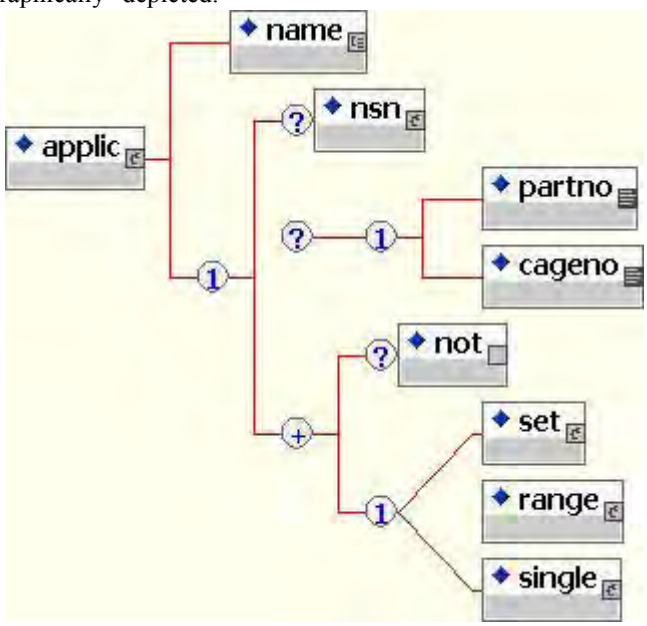

FIGURE 55. System effectivity <applic> DTD hierarchy..

c. DTD fragment:

```
<!ELEMENT applic (name, (nsn?, (partno, cageno)?,
 (not?, (set | range | single))+))>

<!ATTLIST applic
    id          ID          #REQUIRED
    abbrevcode  CDATA       #REQUIRED>
```

d. Unique attributes:
   (1) **id** – System effectivity unique identifier to reference or link. (required)
   (2) **abbrevcode** – Abbreviation code is displayed to identify the step(s) or paragraph(s) that the system has effectivity. (required)

31.2.1.1 Negate ***<not>***. The element ***<not>*** provides a return of a negative Boolean expression value. Returns 'True' if the expression is 'False', otherwise returns 'False' when the expression is 'True'.

a. The components are:
   (1) The element is EMPTY.

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b. DTD fragment:

```
<!ELEMENT not EMPTY>
```

c. No attributes.

31.2.1.2 Effectivity type set ***<set>***. The element ***<set>*** defines an effectivity filter group using the same information type.

a. The components are:
   (1) Serial Number ***<serialno>*** (required – one or more) (see 47.2.1.2.1)
   (2) Modification Work Order ***<mwo>*** (required – one or more) (see 34.1.1.17)
   (3) Usable On Code ***<uoc>*** (required – one or more) (see 38.4.2.1.3.1.3)
   (4) Model Number ***<modelno>*** (required – one or more) (see 49.1.4.18)
   (5) Software Version ***<software_version>*** (required – one or more) (see 31.2.1.2.1)
   (6) Equipment Unique Identifier ***<eqp_id>*** (required – one or more) (see 31.2.1.2.2)

b. DTD fragment is graphically depicted.

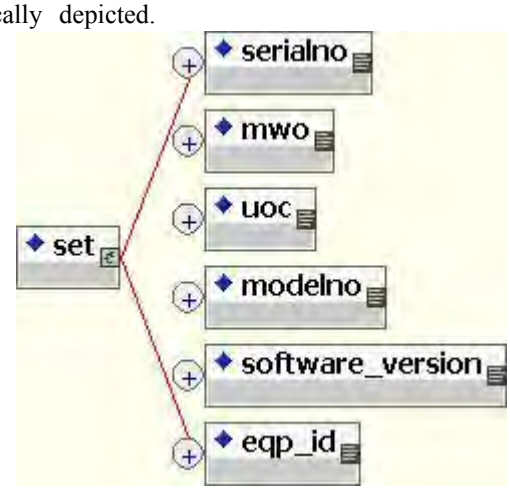

FIGURE 56. Effectivity type set <set> DTD hierarchy..

c. DTD fragment:

```
<!ELEMENT set (serialno+ | mwo+ | uoc+ | modelno+
| software_version+ | eqp_id+)>
```

d. No attributes.

31.2.1.2.1 Software version ***<software_version>***. The element ***<software_version>*** defines the software version information used to determine effectivity.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.

b. DTD fragment:

```
<!ELEMENT software_version (#PCDATA)>
```

c. No attributes.

31.2.1.2.2 Equipment unique identifier ***<eqp_id>***. The element ***<eqp_id>*** specifies the equipment unique identifier (i.e., VIN, tail number) for identification.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.

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b. DTD fragment:

```
<!ELEMENT eqp_id (#PCDATA)>
```

c. No attributes.

31.3 Frame base manual ***<framed.manual>***. The element that contains all the assembled technical manual contents, including the front and rear matter and the work packages. The frame base manual can be developed using the IETM functional hierarchy or using system/subsystem hierarchy contents. The manual is developed according to the maintenance level of the manual that is supported by the required attribute **maintlvls**.

a. The components are:
   (1) IETM Functional Hierarchy ***<functionhierarchy>*** (required). The element provides the manual to be organized as an IETM by functional grouping. (see 31.3.1)
   (2) System/Subsystem Hierarchy ***<systemhierarchy>*** (required). The element provides the manual to be organized to the system or subsystem of the end item. (see 31.3.2)

b. DTD fragment is graphically depicted.

FIGURE 57. Frame base manual <framed.manual> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT framed.manual (functionhierarchy | systemhierarchy)>
<!ATTLIST framed.manual
     maintlvls          (10 |13 |14 |24 |
                         avum-avim | dmwr | nmwr |
                         dmwr-nmwr)                    #REQUIRED
     pubno              CDATA                         #IMPLIED
     revno              CDATA                         #REQUIRED
     security           (uc | fouo | c | s | ts)      #IMPLIED>
```

d. Unique attributes:
   (1) **maintlvls** – Maintenance level. Specifies the maintenance level(s) authorized to use this manual; this attribute value is used in the stylesheet to apply the literal expression of the TM's maintenance level. Select one of the maintenance TM values (10, 13, 14, 24, avum-avim, dmwr, nmwr). (required)
   (2) **pubno** – Publication number. Specifies publication number for the technical manual. (optional)
   (3) **revno** – Revision number. Specifies revision number of the overall manual. (required)

e. Common attributes:
   (1) **security** – Security classification. (optional) (see 49.2.13)

31.3.1 IETM functional hierarchy ***<functionhierarchy>***. IETM functional hierarchy identifies the IETM manual to be organized by functional grouping. The organization of the manual is depicted by the requirements for IETM functionality and data display (look and feel) and contains data from the IETM technical content selection matrix required by a statement of work. See MIL-STD-40051-1 for more information.

a. The components are:
   (1) Frame Base Front Matter ***<framed.frnt>*** (required) The element provides the frame based front matter of a technical manual. (see 31.3.1.1).
   (2) General Information, Description Information And Theory Of Operation Chapter ***<gim>*** (required). The element provides the user with information for general requirements, descriptive data

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about the weapon system or equipment, and an explanation of how the weapon system or equipment works. (see Section 34)

(3) Operating Instructions ***<opim>*** (optional – zero or more). The element provides operating instructions for major weapon systems, and their related systems, subsystems, equipment, weapons replacement assemblies, and shop replacement assemblies. (see Section 35)

(4) Specify either a Troubleshooting Information Chapter ***<tim>*** or a Maintenance Information Chapter ***<mim>***.**At least one is required but may have both or more then one of each.**

(a) Troubleshooting Information Chapter ***<tim>***. The element provides the troubleshooting information and procedures authorized to be performed at the stated maintenance level. (see 36.4)

(b) Maintenance Information Chapter ***<mim>*** The element provides the maintenance information chapter to prepare maintenance instructions for major weapon systems, and their related systems, subsystems, equipment, weapons replacement assemblies, and shop replacement assemblies. (see Section 37)

(5) Destruction chapter ***<dim>*** (optional). The element provides destruction procedures as a chapter for a weapons system TM (see 39.1).

(6) Repair Parts and Special Tools List (RPSTL) Information Chapter ***<pim>*** (optional). The information chapter is used to prepare Parts Information or Repair Parts and Special Tools Lists (RPSTLs) for major weapon systems, and their related systems, subsystems, equipment, weapons replacement assemblies, and shop replacement assemblies (see Section 38).

(7) Supporting Information Chapter ***<sim>*** (required). The element provides the information chapter that identifies equipment supporting data(see 40).

(8) Reporting Errors and Recommending Improvements ***<da2028>*** (required – one or more). The element provides the DA Form 2028-2 for reporting errors and recommending equipment improvements. (see 31.4.2.3)

b. DTD fragment is graphically depicted.

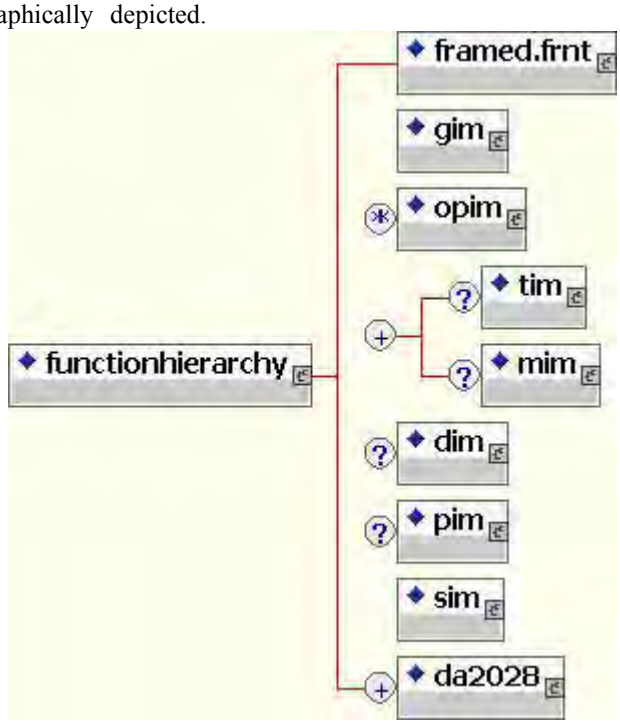

FIGURE 58. IETM functional hierarchy <functionhierarchy> DTD hierarchy.

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c. DTD fragment:

```
(framed.frnt, gim, opim*, (tim?, mim?)+, dim?,
pim?, sim, da2028+)
```

d. No attributes.

31.3.1.1 Frame base front matter ***<framed.frnt>***. Frame base front matter identifies all front matter of an IETM technical manual. Front matter occurs before the first work package.

a. The components are:
   (1) Select either: **At least one is required.**
      (a) Revision Summary ***<revisionsummary>***. The element provides a list of work packages by title that have been revised to a revised manual.(see 31.3.1.1.1)
      (b) Alternative Revision Summary ***<revisionsummary-alt>*** (see 48.2.1).
   (2) Select either: **At least one is required.**.
      (a) Front Cover ***<frntcover>*** (see 31.3.1.1.4)
      (b) Front Cover Alternative ***<frntcover-alt>*** (see 48.2.1).
   (3) Select either: **One or more may be used.**.
      (a) Promulgation Letter (USMC) ***<promulgation>***. The element provides the promulgation letter used primarily by the Marine Corp.(see 31.3.1.1.6)
      (b) Alternative Promulgation Letter ***<promulgation-alt>*** (see 48.2.1)
   (4) Select either:
      (a) Warning Summary ***<warnsum>*** (optional). The element provides the warning summary for a manual(see 31.4.1.2).
      (b) Warning Summary Alternative ***<warnsum-alt>*** (optional – zero or more)(see 48.2.1).
   (5) Select either: **At least one is required.**.
      (a) Table of Contents ***<contents>***. The element provides the table of contents for the manual(see 31.3.1.1.8).
      (b) Table of Contents Alternative ***<contents-alt>*** (see 48.2.1).
   (6) Select either: **At least one is required.**.
      (a) How to Use ***<howtouse>***. The element provides the section on how to use the manual.(see 31.3.1.1.2)
      (b) How to Use Alternative ***<howtouse-alt>*** (see 48.2.1)
b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

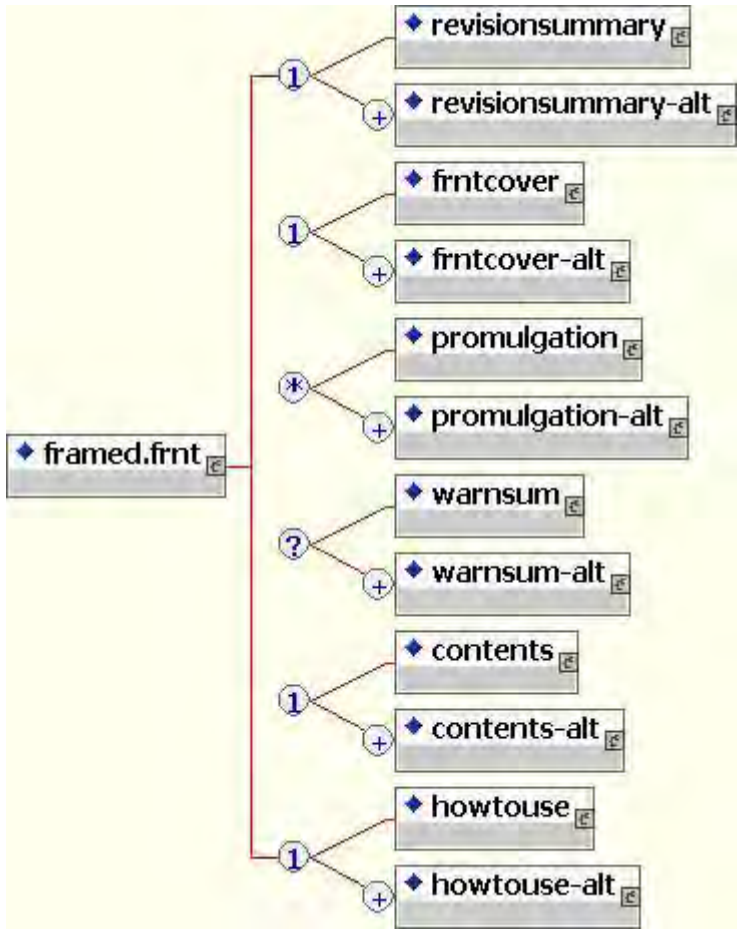

FIGURE 59. Frame base front matter <framed.frnt> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT framed.frnt ((revisionsummary | revisionsummary-alt+),
(frntcover | frntcover-alt+), (promulgation | promulgation-alt+)*,
(warnsum | warnsum-alt+)?, (contents | contents-alt+),
(howtouse | howtouse-alt+))>
```

d. No attributes.

31.3.1.1.1 Revision summary frame ***<revisionsummary>***. When a revision to an IETM is issued, a revision summary frame is displayed containing a list of work packages by title that have been revised. For each work package listed, a brief description of the major changes is provided. The revised work packages listed on the revision summary frame are linked to the work packages containing the revised information. The revision summary also indicates the work packages that have been superseded.

a. The components are:
   (1) Title ***<title>*** (required). Element provides the title for the revision summary frame of the IETM. (see 49.1.1.3)
   (2) General paragraph ***<para>*** (required). Element provides the general text explaining the function of a revision summary frame. (see 49.1.1.5)
   (3) General paragraph ***<para>***. An optional paragraph is provided if the user has a need to state further information. (see 49.1.1.5)

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(4) Standard Table Header ***<std.table.header>*** (optional). The element provides an optional heading for work package number and description.(see 42.3.1)
(5) Work package number ***<wpno>*** (required – one or more). Element provides the work package number of the revised work package in the technical manual. The element ***<wpno>*** contains the attribute **wpref** that allows linking the revision summary information to the work package containing the revised information. (see 46.2.5)
(6) Description ***<desc>*** (required – one or more). Element provides a brief description of the revised work package. (see 49.1.4.17)

b. DTD fragment is graphically depicted.

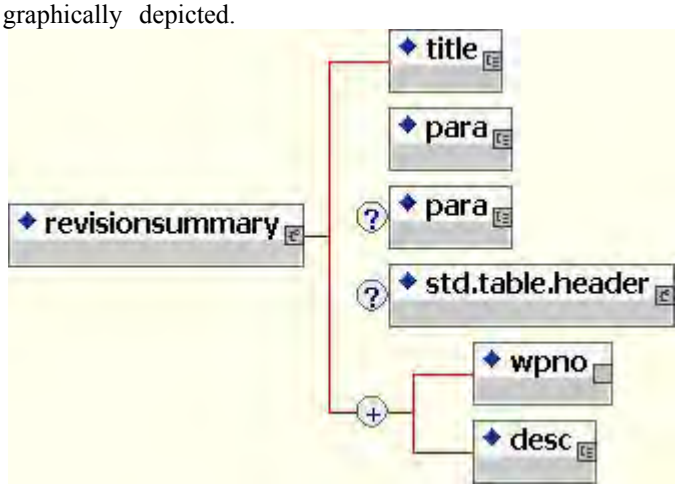

FIGURE 60. Revision summary frame <revisionsummary> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT revisionsummary (title, para, para?, std.table.header?, (wpno, desc)+)>
<!ATTLIST revisionsummary
    frame          (yes | no )       'yes'
    applicable     IDREFS            #IMPLIED>
```

Unique Attributes.
(1) **frame** – Set a frame break. (default value is "yes") (see 49.2.3)
(2) **applicable** – Set a frame break. (default value is "yes") (see 49.2.1.b.(1))

31.3.1.1.1.1 How to link revision summary to work packages. The revised work packages listed on the revision summary frame are listed by the work package title and are linked to the work packages containing the revised information. The element work package number ***<wpno>*** (see 46.2.5) required in the ***<revisionsummary>*** contains the attribute work package reference **wpref**. This attribute **wpref** references the work package required attribute **wpno** that serves as a reference target. The work package number element and it's attribute ***<wpno wpref="mxxxxx-x-xxxx-xxx">*** performs a link to the work package and it's attribute ***<maintwp wpno="mxxxxx-x-xxxx-xxx">*** in the IETM. The title of the work packages listed in the revision summary are generated by the style sheet.

31.3.1.1.1.2 Example of a revision summary frame ***<revisionsummary>*** in an IETM. The following is an example of a revision summary XML instance and the formatted display of the revision summary.

a. XML document instance fragment.

```
<revisionsummary>
<title>revision  summary
</title>
<para>The following work packages have been changed in this IETM:
</para>
```


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```
<std.table.header>
<std.table.row>
<std.table.entry  colnum="1">WP  TITLE
</std.table.entry>
<std.table.entry  colnum="2">REASON  FOR  CHANGE
</std.table.entry>
</std.table.row>
</std.table.header>
<wpno  wpref="gxxxx3-11-xxxx-xxx"/>
<desc>Addition  of  a  new  electrical  connector.
</desc>
<wpno  wpref="txxxx3-11-xxxx-xxx"/>
<desc>Additional test equipment required to perform Troubleshooting.
</desc>
</revisionsummary>
```

b. Stylesheet output.

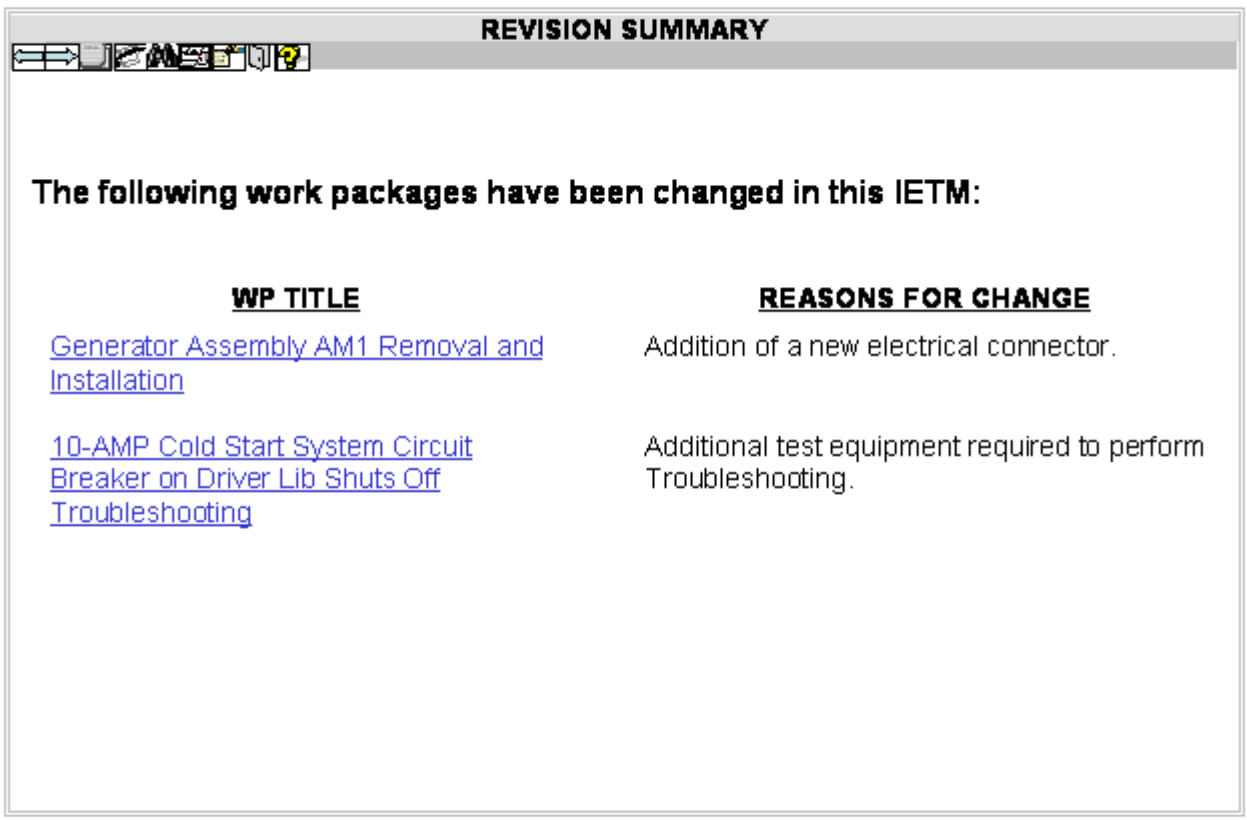

FIGURE 61. Example of a revision summary frame in an IETM.

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31.3.1.1.2 How to use ***<howtouse>***. How to use provides special or detailed information on how to read and use information and procedures in a TM. The information leads the user through the TM and explains the important features of the organization and content. When specified by the acquiring activity the "How To Use This Manual" can contain the International Standardization Agreements Statement (see 31.3.1.1.2.1). Paragraphs of text and/or procedures are used to tagged the information on "how to use" the TM.

a. The components are:
   (1) Title ***<title>*** (required). The element provides the title for the section "HOW TO USE THIS MANUAL". (see 49.1.1.3)
   (2) Select one of the following information types (optional – zero or more):
      (a) General Paragraph(s) ***<para>***. The element provides the data for the information. (see 49.1.1.5)
      (b) Paragraph Associated with an Alert ***<specpara>***. The element provides a paragraph that depicts a warning, caution, or a note. (see 49.1.1.6)
   (3) Select one of the following information types: (required – one or more)
      (a) Primary Level – Titled Paragraph ***<para0>*** (see 49.1.1.8) or/and Primary Level – Conditional – Titled Paragraph(s) ***<para0-alt>*** (see 48.2.1) that provides a title, a precondition and paragraphs of text in the description of how to use the manual.

b. DTD fragment is graphically depicted.

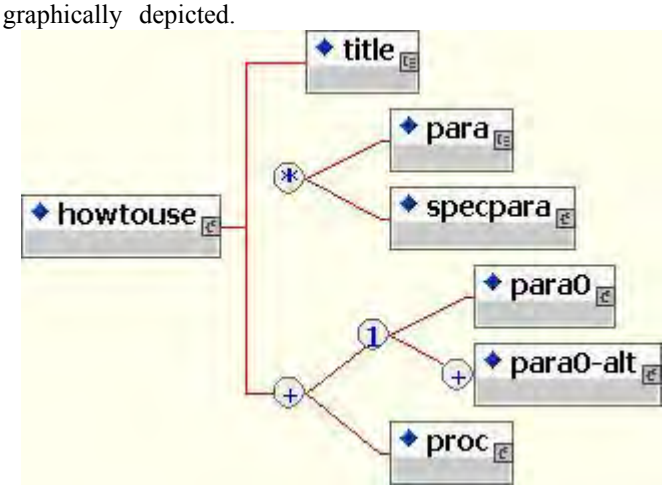

FIGURE 62. How to use <howtouse> DTD hierarchy..

c. DTD fragment:

```
<!ELEMENT howtouse (title, (para | specpara)*, ((para0 | para0-alt+)+)>
<!ATTLIST howtouse
     changeref          IDREFS                          #IMPLIED
     comment            CDATA                           #IMPLIED
     delchlvl           CDATA                           #IMPLIED
     frame              (yes | no)                      'no'
     inschlvl           CDATA                           #IMPLIED
     security           (uc | fouo | c | s | ts)        #IMPLIED>
```

d. Unique attributes:
   (1) **frame** – Set a frame break. (default value is "no") (see 49.2.3)

e. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (5) **security** – Security classification. (optional) (see 49.2.13)

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31.3.1.1.2.1 International standardization agreements. The “How To Use This Manual” can contain the International Standardization Agreements Statement when specified by the acquiring activity. The statement is different when used in a frame manual as to a paper manual. The difference is that “technical manual” is used in a paper manual and IETM is used in a frame manual. If a page-base manual is selected then “technical manual” is generated in the entity but if the frame-base manual is selected then “IETM” is generated in the International standardization agreements entity. This statement is a boiler plate entity and can be referenced using the entity ***&howtouse.intl-agree;***. The general entity also contains two nested entities ***&howtouse.intl-agree.ref;*** and ***&howtouse.intl-agree.pg;*** that need to be edited. The nested entities can be edited through the Editable Boilerplate Text file (editboil.ent). Below is the text containing the nested entities for the boiler plate entity ***&howtouse.intl-agree;***. The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

a.

<note>
<trim.para>Certain provisions of this <![%page-base;[technical manual]]><![%frame-base;[IETM]]> ***&howtouse.intl-agree.ref;*** are the subject of international standardization agreement ***&howtouse.intl-agree.pg;*** When revision or cancellation of this technical manual is proposed which will modify the international agreement concerned, the technical manual management activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement o
</trim.para>
</note>

31.3.1.1.2.2 Example of how to use. Depicted below is an example of a “HOW TO USE THIS MANUAL”. The example includes an XML document instance fragment and the stylesheet output for the element ***<howtouse>***. This example does not contain the boiler plate entity ***&howtouse.intl-agree ;***.

a. XML document instance fragment.

<howtouse>
<title>HOW TO USE THIS MANUAL
</title>
<para0>
<title>How To Use This Manual
</title>
<note>
<trim.para>This manual only covers procedures unique to the M7 Bradley FireSupport Vehicle. For any procedures that are common to both M7 and Bradley vehicles, see TM X-XXXX-XXX-XX-X series and TM X-XXXX-XXX-XX-X. Many tasks are common and can be found under a similar task name in those TMs.
</trim.para>
</note>
<para>This manual tells you how to maintain the hull on the M7 Bradley Fire Support Vehicle. Operator’s instructions are covered in TM X-XXXX-XXX-XX-X and TM X-XXXX-XXX-XX-X.
</para>
<para>Before starting any task/procedure or before applying power to the hull, make sure you have read this HOW TO USE section and the Controls and Indicators WP in TM X-XXXX-XXX-XX-X.
</para>
</para0>
<para0>
<title>WHAT’S IN THE MANUAL— FRONT TO BACK
</title>
<para>This TM supplement is divided into front and rear matter and Work Packages (WPs) for ease of use.

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</para>
*<para>*The WARNING SUMMARY section provides safety and first aid information. This section includes general warnings not found in the TM text and a list of the most important detailed warnings extracted from the WPs. All of these warnings cover hazards that could kill or injure personnel.
</para>
*<para>*The TABLE OF CONTENTS lists the WPs.
</para>
*<para>*CHAPTER 1 covers general introductory information with theory of operation. The Equipment Description WP gives a brief description of major parts and features of the hull. The Theory of Operation WP provides information that will help you understand how the hull components work.
</para>
*<para>*CHAPTER 2 covers Troubleshooting Procedures.
</para>
*<para>*CHAPTER 3 contains Unit Maintenance procedures. This includes the Preventive Maintenance Checks and Services (PMCS) WP and corrective maintenance WPs.
</para>
*<para>*CHAPTER 4 contains Direct Support Maintenance WPs. [At this time, there are no Direct Support Maintenance WPs in this document.]
</para>
*<para>*CHAPTER 5 provides supporting information for the TM. It includes the following WPs:
<randlist>
*<item>*The REFERENCES WP lists references to be used by personnel in operating and maintaining the hull. These references include technical manuals and other publications.
</item>
*<item>*The MAC WP lists maintenance functions, levels, and times assigned to each maintenance action.
</item>
*<item>*The RPSTL WP lists and authorizes spares, repair parts, and special tools required for performance of unit, direct support, and general support maintenance of the M7 hull. It authorizes the requisition, issue, and disposition of spares, repair parts, and special tools as indicated by the source, maintenance, and recoverablilty (SMR) codes.
</item>
*<item>*The TOOL IDENTIFICATION LIST WP lists all unique tools required for maintenance of the M7 hull. This list is compiled from the Initial Setup requirements of all WPs in this TM.
</item>
*<item>*The EXPENDABLE AND DURABLE ITEMS WP lists expendable supplies and materials that will be needed to maintainthe hull.
</item>
</randlist>
</para>
*<para>*For all other supporting information, see the appendices in TM X-XXXX-XXX-XX-X series and TM X-XXXX-XXX-XX-X
</para>
*<para>*The INDEX is an alphabetical listing of all the tasks in the WPs of this TM. Each entry is cross-referenced to the WP number and page number.
</para>
*<para>*The back cover includes a METRIC CONVERSION CHART that can be used to convert U.S. customary measurements to their metric

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```
equivalents.  Measurements in this manual are given in U.S.
customary unit with metric units in parentheses.
</para>
</para0>
<para0>
<title>HOW TO USE THE WORK PACKAGES
</title>
<para>See TM X-XXXX-XXX-XX-X series and TM X-XXXX-XXX-XX-X for
HOW TO USE information.  The WPs in this supplement include
the same information categories as maintenance TASKS in the TM
X-XXXX-XXX-XX-X series and TM X-XXXX-XXX-XX-X.
</para>
</para0>
</howtouse>
```

b. Stylesheet output.

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HOW TO USE THIS MANUAL

HOW TO USE THIS MANUAL

NOTE

This manual only covers procedures unique to the M7 Bradley Fire Support Vehicle. For any procedures that are common to both M7 and Bradley vehicles, see TM X-XXXX-XXX-XX-X series and TM X-XXXX-XXX-XX-X. Many tasks are common and can be found under a similar task name in those TMs.

This manual tells you how to maintain the hull on the M7 Bradley Fire Support Vehicle. Operator's instructions are covered in TM X-XXXX-XXX-XX-X and TM X-XXXX-XXX-XX-X .

Before starting any task/procedure or before applying power to the hull, make sure you have read this HOW TO USE section and the Controls and Indicators WP in TM X-XXXX-XXX-XX-X.

WHAT'S IN THE MANUAL — FRONT TO BACK

This TM supplement is divided into front and rear matter and Work Packages (WPs) for ease of use.

The WARNING SUMMARY section provides safety and first aid information. This section includes general warnings not found in the TM text and a list of the most important detailed warnings extracted from the WPs. All of these warnings cover hazards that could kill or injure personnel.

The TABLE OF CONTENTS lists the WPs.

CHAPTER 1 covers general introductory information with theory of operation. The Equipment Description WP gives a brief description of major parts and features of the hull. The Theory of Operation WP provides information that will help you understand how the hull components work.

CHAPTER 2 covers Troubleshooting Procedures.

CHAPTER 3 contains Unit Maintenance procedures. This includes the Preventive Maintenance Checks and Services (PMCS) WP and corrective maintenance WPs.

CHAPTER 4 contains Direct Support Maintenance WPs. [At this time, there are no Direct Support Maintenance WPs in this document.]

CHAPTER 5 provides supporting information for the TM. It includes the following WPs:

- The REFERENCES WP lists references to be used by personnel in operating and maintaining the hull. These references include technical manuals and other publications.
- The MAC WP lists maintenance functions, levels, and times assigned to each maintenance action.
- The RPSTL WP lists and authorizes spares, repair parts, and special tools required for performance of unit, direct support, and general support maintenance of the M7 hull. It authorizes the requisition, issue, and disposition of spares, repair parts, and special tools as indicated by the source, maintenance, and recoverablilty (SMR) codes.
- The TOOL IDENTIFICATION LIST WP lists all unique tools required for maintenance of the M7 hull. This list is compiled from the Initial Setup requirements of all WPs in this TM.
- The EXPENDABLE AND DURABLE ITEMS WP lists expendable supplies and materials that will be needed to maintain the hull.

For all other supporting information, see the appendices in TM X-XXXX-XXX-XX-X series and TM X-XXXX-XXX-XX-X.

The INDEX is an alphabetical listing of all the tasks in the WPs of this TM. Each entry is cross-referenced to the WP number and page number.

The back cover includes a METRIC CONVERSION CHART that can be used to convert U.S. customary measurements to their metric equivalents. Measurements in this manual are given in U.S. customary unit with metric units in parentheses.

HOW TO USE THE WORK PACKAGES

See TM X-XXXX-XXX-XX-X series and TM X -XXXX-XXX-XX-X for HOW TO USE information. The WPs in this supplement include the same information categories as maintenance TASKS in the TM X-XXXX-XXX-XX-X series and TM X-XXXX-XXX-XX-X.

FIGURE 63. Example of a how to use <howtouse>.

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31.3.1.1.3 How to use alternative ***<howtouse-alt>***. The element ***<howtouse-alt>*** provides configuration filtering to front matter material.

a. The components are:
   (1) How to Use ***<howtouse>*** (required) (see 31.3.1.1.2).
b. DTD fragment:

```
<!ELEMENT howtouse-alt (howtouse)>
```

31.3.1.1.4 Front cover ***<frntcover>***. The element ***<frntcover>*** is used for the front cover of a Technical Manual. The National Stock Number (NSN) ***<nsn>*** and End Item Code (EIC) ***<eic>*** is included on the front cover of equipment publications but may not be required for general equipment and other types of publications.

a. The components are:
   (1) Technical Manual Title ***<tmtitle>*** (required). (see 31.3.1.1.4.1)
   (2) Illustrations ***<graphic>*** (optional). (see 44.2)
   (3) Official Notices ***<notices>*** (required). Display the required and applicable notices on the front page.(see 31.3.1.1.4.2)
   (4) Service Nomenclature ***<servnomen>*** (required). Apply the service nomenclature of the proponent activity on the front cover.(see 31.3.1.1.4.3)
   (5) Date ***<date>*** (required). The publishing date of the technical Manual.(see 31.3.1.1.4.2)
b. DTD fragment is graphically depicted.

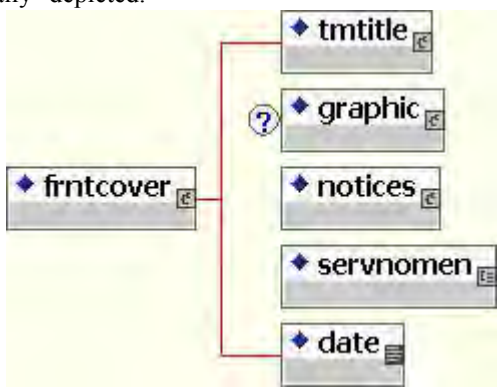

FIGURE 64. Front cover <frntcover> DTD hierarchy..

c. DTD fragment:

```
<!ELEMENT frntcover (tmtitle, graphic?, notices, servnomen, date)>
<!ATTLIST frntcover
    applicable       IDREFS                    #IMPLIED
    security        (uc | fouo |c | s | ts)    #IMPLIED

    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

d. Common elements.
   (1) **applicable** – Applicability Reference to the applicable configuration(s) specified in the WP identification information. When using an IETM that can filter information, this information is not presented when not applicable to the current configuration. Presentations that do not filter information, the information will be identified by the assigned associated abbreviation to designate applicability of the information. (optional).

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(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **inschlvl** – Insert change level (optional) (see 49.2.9).
(6) **security** – Security classification (optional) (see 49.2.13).

31.3.1.1.4.1 Technical manual title ***<tmtitle>***. The ***<tmtitle>*** element contains all the elements that identify a manual, including the manual number, primary title, and subtitle.

a. The components are:
   (1) Select either:
      (a) Technical Manual Number and Service Branch ***<tminfono>***(required – two or more) (see 31.3.1.1.4.1.1).
      (b) Technical Manual Number ***<tmno>*** (required)(see 31.3.1.1.4.1.2)
   (2) Primary Title ***<prtitle,>*** (required). Display the required and applicable notices on the front page.(see 31.3.1.1.4.1.3)
   (3) Manual Subtitle ***<stitle>*** (optional). (see 49.1.1.2)
   (4) Weapon System Title ***<weapons_system>*** (optional). (see 31.3.1.1.4.1.4)
b. DTD fragment is graphically depicted.

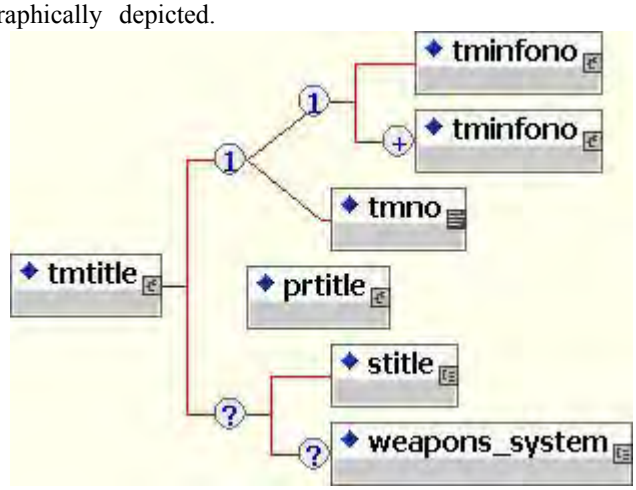

FIGURE 65. Technical manual title <tmtitle> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT tmtitle (((tminfono, tminfono+) | tmno), prtitle,
            (stitle, weapons_system?)?)>
<!ATTLIST tmtitle

    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
   security        (uc | fouo |c | s | ts)          #IMPLIED
>
```

d. Common elements.
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (5) **security** – Security classification (optional) (see 49.2.13).

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31.3.1.1.4.1.1 Technical manual number and service branch ***<tminfono>***. The element ***<tminfono>*** contains a combination of branch of service (optional) and technical manual number (required) constitutes a unique identification of the TM. If the TM is used by more than one service branch, the proponent's TM number appears first.

a. The components are:
   (1) Branch of Service ***<servbranch>*** (required)(see 31.3.1.1.4.1.1.1).
   (2) Technical Manual Number ***<tmno,>*** (required).(see 31.3.1.1.4.1.2)

b. DTD fragment is graphically depicted.

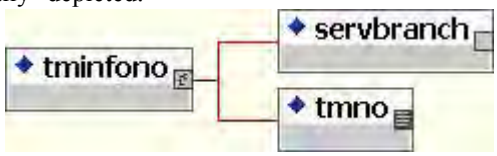

FIGURE 66. Technical manual number and service branch <tminfono> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT tminfono (servbranch, tmno)>
<!ATTLIST tminfono
    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
   security         (uc | fouo |c | s | ts)          #IMPLIED
>
```

d. Common elements.
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (5) **security** – Security classification (optional) (see 49.2.13).

31.3.1.1.4.1.1.1 Branch of service ***<servbranch>***. A branch of service that has assigned an official TM number to the manual. The branch of service is selected using the attribute **service** and is generated with an XML stylesheet.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.

b. DTD fragment:

```
<!ELEMENT servbranch EMPTY>
<!ATTLIST servbranch
    service       (army | af | navy | marines)  #REQUIRED
    qualify        CDATA         #IMPLIED
    procuring      "(yes | no)   'no'">


    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
   security         (uc | fouo |c | s | ts)          #IMPLIED
>
```

c. Unique attributes.
   (1) **service** Specifies the service branch. (required)

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(2) **qualify** Supplies any further qualification of the service, e.g., NAVAIR. (optional)
(3) **procuring** If more than one service uses the manual, specifies which branch is the procuring agency. Select either "yes or no". The default is "no".

d. Common attributes.
(1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(2) **comment** – Change information (optional) (see 49.2.9).
(3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(4) **inschlvl** – Insert change level (optional) (see 49.2.9).
(5) **security** – Security classification (optional) (see 49.2.13).

31.3.1.1.4.1.2 Technical manual number ***<tmno>***. The element ***<tmno>*** provides the number portion of the TM. It identifies the number contained in the element technical manual number and service branch.

a. The components are:
(1) The element contains narrative text #PCDATA parsable character data.

b. DTD fragment:

```
<!ELEMENT tmno (#PCDATA)>
<!ATTLIST tmno
    changeref       IDREFS                  #IMPLIED
    comment         CDATA                   #IMPLIED
    delchlvl        CDATA                   #IMPLIED
    inschlvl        CDATA                   #IMPLIED
   security        (uc | fouo |c | s | ts)        #IMPLIED>
```

c. Common attributes.
(1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(2) **comment** – Change information (optional) (see 49.2.9).
(3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(4) **inschlvl** – Insert change level (optional) (see 49.2.9).
(5) **security** – Security classification (optional) (see 49.2.13).

31.3.1.1.4.1.3 Primary title ***<prtitle>***. The element ***<prtitle>*** contains the system nomenclature with any relevant identifying numbers or qualifying subject. The primary title appears on the front cover, change sheet, and title block page of the TM.

a. The components are:
(1) The primary title may contain one or more grouping of one or more required equipment nomenclature and an optional subject.
(a) Equipment Nomenclature ***<sysnomen>***(required – one or more) (see 31.3.1.1.4.1.2).
(b) TM Subject Matter ***<subject>***(optional) (see 31.3.1.1.4.1.2).

b. DTD fragment is graphically depicted.

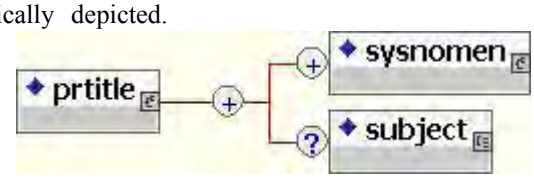

FIGURE 67. Primary title <prtitle> DTD hierarchy..

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c. DTD fragment:

```
<!ELEMENT prtitle (sysnomen+, subject?)+>
<!ATTLIST prtitle


    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
   security        (uc | fouo |c | s | ts)          #IMPLIED
>
```

d. Common attributes.

(1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(2) **comment** – Change information (optional) (see 49.2.9).
(3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(4) **inschlvl** – Insert change level (optional) (see 49.2.9).
(5) **security** – Security classification (optional) (see 49.2.13).

31.3.1.1.4.1.3.1 Equipment nomenclature ***<sysnomen>***. The element ***<sysnomen>*** contains the official end item nomenclature covered in the TM. The element consists of a name and one or more optional identifying number elements: model number, part number, NSN, and/or EIC.

a. The components are:

(1) Name ***<name>*** (see 49.1.4.19)
(2) Model Number ***<modelno>*** (see 49.1.4.18)
(3) National Stock Number ***<nsn>*** (see 49.1.4.20)
(4) Part Number ***<partno>*** (see 49.1.4.21)
(5) End-Item Code ***<eic>*** (see 31.3.1.1.4.1.3.1.1)

b. DTD fragment is graphically depicted.

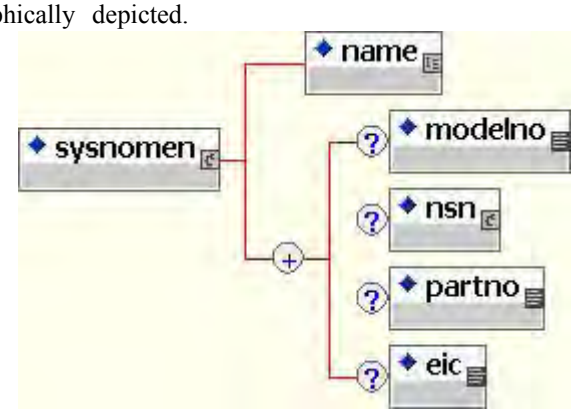

FIGURE 68. Equipment nomenclature <sysnomen> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT sysnomen (name, (modelno?, nsn?, partno?, eic?)+)>
<!ATTLIST sysnomen
    pretext          CDATA                      #IMPLIED


    changeref        IDREFS                     #IMPLIED
    comment          CDATA                      #IMPLIED
    delchlvl         CDATA                      #IMPLIED
    inschlvl         CDATA                      #IMPLIED
   security        (uc | fouo |c | s | ts)   #IMPLIED
>
```

d. Unique attributes.
   (1) **pretext** – Any text that precedes the equipment nomenclature, e.g., "FOR" or "OF." This is the only mechanism for inserting such words on the front cover.

e. Common attributes.
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (5) **security** – Security classification (optional) (see 49.2.13).

31.3.1.1.4.1.3.1.1 End-item code ***<eic>***. The element ***<eic>*** contains the assigned end-item code, a three-position alphanumeric code, of the equipment covered by the TM. When used it appears as part of the prime title on the front cover and title block page.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.(see 9.3.2)

b. DTD fragment:

```
<!ELEMENT eic (#PCDATA)>
```

c. No attributes.

31.3.1.1.4.1.4 Weapon system title ***<weapons_system>***. The element ***<weapons_system>*** specifies the weapons system component title for the TM cover page.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
   (2) Emphasis ***<emphasis>*** (optional – zero or more)(see 49.1.3.1).
   (3) Subscript ***<subscrpt>*** (optional – zero or more)(see 49.1.3.4).
   (4) Superscript ***<supscrpt>*** (optional – zero or more)(see 49.1.3.5).
   (5) Line Break ***<brk>*** (optional – zero or more)(see 49.1.3.1).

b. DTD fragment:

```
<!ELEMENT weapons_system (#PCDATA | emphasis | subscrpt | supscrpt | brk)*>
```

c. No attributes.

31.3.1.1.4.2 General notices ***<notices>***. The element ***<notices>*** contains any notices that appear on the TM front cover, change sheet, or title block page.

a. The components are:
   (1) Available ***<avail>*** (see 31.3.1.1.4.2.1).
   (2) Supersedure statement ***<super>*** (see 31.3.1.1.4.2.2).
   (3) Disclosure notice ***<disclos>*** (see 31.3.1.1.4.2.3).

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(4) Distribution statement ***<dist>*** (see 31.3.1.1.4.2.4).
(5) Export Control ***<export>*** (see 31.3.1.1.4.2.5).
(6) Destruction of the Manual ***<destr>*** (see 31.3.1.1.4.2.6).
(7) General Purpose Destruction ***<general_purpose_notices>*** (see 31.3.1.1.4.2.7).

b. DTD fragment is graphically depicted.

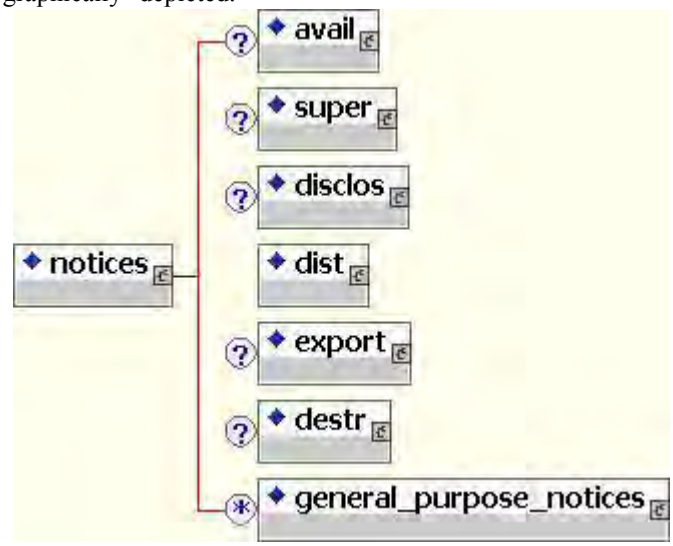

FIGURE 69. General notices <notices> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT notices (avail?, super?, disclos?, dist, export?,
 destr?, general_purpose_notices*)>
```

d. No attributes.

31.3.1.1.4.2.1 Availability statement ***<avail>***. The element ***<avail>*** is used for the standard availability notice that appears on the DMWR/NMWR front cover. This statement is a boilerplate and can be referenced using the general entity ***&notices.avail;*** (see 31.3.1.1.4.2.1.e.). **Depot only**.

a. The components are:
(1) Title ***<title>*** (optional)(see 49.1.1.3).
(2) Narrative Text ***<text>*** (required) used in two places in available (see 49.1.1.18).
(3) Proponent ***<proponent>*** (see 49.1.4.22).

b. DTD fragment is graphically depicted.

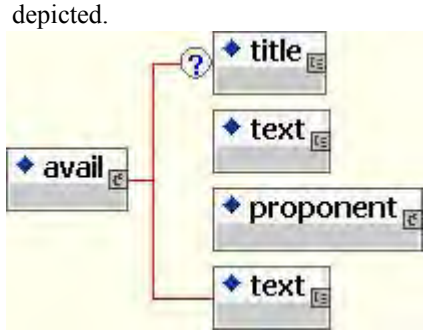

FIGURE 70. Availability statement <avail> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT avail (title?, text, proponent, text)>
```

d. No attributes.

e. XML boilerplate text for the entity ***&notices.avail;***. This entity is located in the Production Boilerplate Text file (prodboil.ent). The availability statement boilerplate contains a nested boilerplate entity ***&proponent-address.army;***. The nested entity can be edited to provide the Army's proponent address through the Editable Boilerplate Text file (editboil.ent). The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

<avail>
<text>`This publication is not available through the St. Louis Media Distribution Division. This publication is available through`
</text> ***&proponent-address.army;***
<text>.
</text>
</avail>>

31.3.1.1.4.2.2 Supersedure statement ***<super>***. The element ***<super>*** is used for the standard supersedure notice that is provided by the contracting activity when the TM, revision, or change under preparation supersedes other TMs or portions of TMs.

a. The components are:
   (1) Title ***<title>*** (optional)(see 49.1.1.3).
   (2) Paragraph Text ***<para>*** (required) used in two places in available (see 49.1.1.18).

b. DTD fragment is graphically depicted.

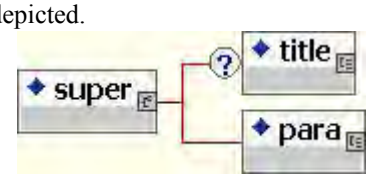

FIGURE 71. Supersedure statement <super> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT super (title?, para)>
```

d. No attributes.

31.3.1.1.4.2.3 Disclosure notice ***<disclos>***. The element ***<disclos>*** contains the standard disclosure notice. The disclosure notice is provided by the contracting activity and appears on the front cover. This statement is a boilerplate and can be referenced using the general entity ***&notices.disclos;*** (see 31.3.1.1.4.2.3.e.).

a. The components are:
   (1) Title ***<title>*** (optional) (see 49.1.1.3).
   (2) Paragraph Text ***<text>*** (required).(see 49.1.1.18).

b. DTD fragment is graphically depicted.

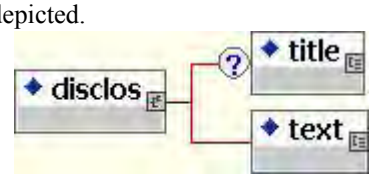

FIGURE 72. Disclosure notice<disclos> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT disclos (title?, text)>
```

d. No attributes.

e. XML boilerplate text for the entity disclosure notice ***&notices.disclos;***. This entity is located in the Production Boilerplate Text file (prodboil.ent). The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

<disclos>
<title>WARNING
</title>
</text>This information is furnished upon the condition that it will not be released to another nati on without the specific authority of the Department of the Army of the United States, that it will be used for military purposes only, that individual or corporate rights originating in the information, whether patented or not, will be respected, that the recipient will report promptly to the United States, any known or suspected compromise, and that the information will be provided substantially the same degree of security afforded it by the Department of Defense of the United States. Also, regardless of any other markings on the document, it will not be downgraded or declassified without written approval of the originating United States agency.
</text>
</disclos>

31.3.1.1.4.2.4 Distribution statement ***<dist>***. The element ***<dist>*** contains the appropriate distribution statement for the TM. The standard distribution notice is provided by the contracting activity and appears on the front cover and title block page.

a. The components are:
   (1) Select the appropriate statement for the TM according to the contracting activity. **At least one is required.**
      (a) Distribution A Statement ***<a.statement>*** (optional)(see 31.3.1.1.4.2.4.1).
      (b) Distribution B Statement ***<b.statement>*** (optional)(see 31.3.1.1.4.2.4.2).
      (c) Distribution C Statement ***<c.statement>*** (optional)(see 31.3.1.1.4.2.4.3).
      (d) Distribution D Statement ***<d.statement>*** (optional)(see 31.3.1.1.4.2.4.4).
      (e) Distribution E Statement ***<e.statement>*** (optional)(see 31.3.1.1.4.2.4.5).
      (f) Distribution F Statement ***<f.statement>*** (optional)(see 31.3.1.1.4.2.4.6).
      (g) Distribution X Statement ***<x.statement>*** (optional)(see 31.3.1.1.4.2.4.7).

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

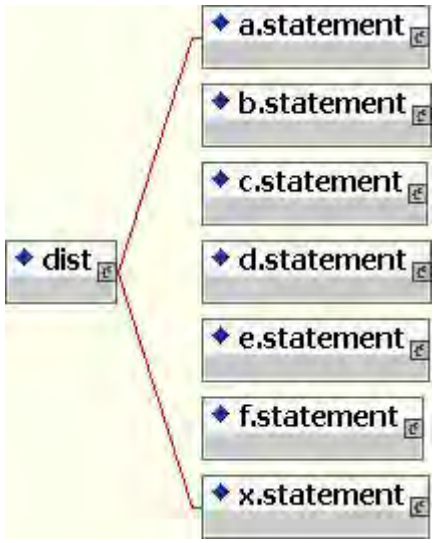

FIGURE 73. Distribution statement <dist> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT dist (a.statement | b.statement | c.statement | d.statement |
e.statement | f.statement | x.statement)>
```

d. No attributes.

31.3.1.1.4.2.4.1 Distribution A Statement ***<a.statement>***. The element ***<a.statement>*** contains the DoDD 5230.24 specified text for an unlimited distribution technical manual. This statement is a boilerplate and can be referenced using the general entity ***&notices.dist.a.statement;***(see 31.3.1.1.4.2.4.1.e.).

a. The components are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Text ***<text>*** (required) (see 49.1.1.18).

b. DTD fragment is graphically depicted.

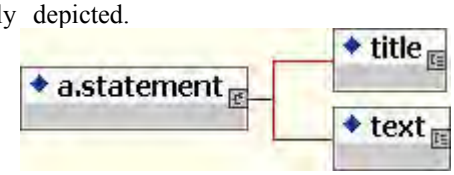

FIGURE 74. Distribution A Statement <a.statement> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT a.statement (title, text)>
```

d. No attributes.

e. XML boilerplate text for the entity Distribution A Statement entity ***&notices.dist.a.statement;***. This entity is located in the Production Boilerplate Text file (prodboil.ent). The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

<dist>
<a.statement>
<title>DISTRIBUTION STATEMENT A.
</title>
<text>Approved for public release; distribution is unlimited.

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</text>
</a.statement>
</dist>

31.3.1.1.4.2.4.2 Distribution B Statement ***<b.statement>***. The element ***<b.statement>*** contains the DODD 5230.24 specified text for an unlimited distribution technical manual. This statement is a boilerplate and can be referenced using the general entity ***&notices.dist.b.statement;***(see 31.3.1.1.4.2.4.2.e.).

a. The components are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Text ***<text>*** (required – two or more). The text element is required in four places contained in the verbatim Distribution B Statement (see 49.1.1.18).
   (3) Select one of the reasons that explains the restricted distribution for the TM. **At least one is required.**
      (a) Foreign Government Information ***<frgngvt>*** (see 31.3.1.1.4.2.4.2.1).
      (b) Software Documentation ***<softwaredoc>*** (see 31.3.1.1.4.2.4.2.2).
      (c) Critical Technology ***<crittech>*** (see 31.3.1.1.4.2.4.2.3).
      (d) Administrative or Operational Use ***<adminops>*** (see 31.3.1.1.4.2.4.2.4).
      (e) Specific Authority ***<specauth>*** (see 31.3.1.1.4.2.4.2.5).
      (f) Proprietary Information ***<proprietary>*** (see 31.3.1.1.4.2.4.2.6).
      (g) Test and Evaluation ***<testeval>*** (see 31.3.1.1.4.2.4.2.7).
      (h) Contractor Performance Evaluation ***<cntrctperform>*** (see 31.3.1.1.4.2.4.2.8).
      (i) Premature Dissemination ***<premature>*** (see 31.3.1.1.4.2.4.2.9).
   (4) Restriction Determination Date ***<reasondate>*** (see 31.3.1.1.4.2.4.2.10).
   (5) Controlling DoD Office ***<releaseagent>*** (see 31.3.1.1.4.2.4.2.11).
b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

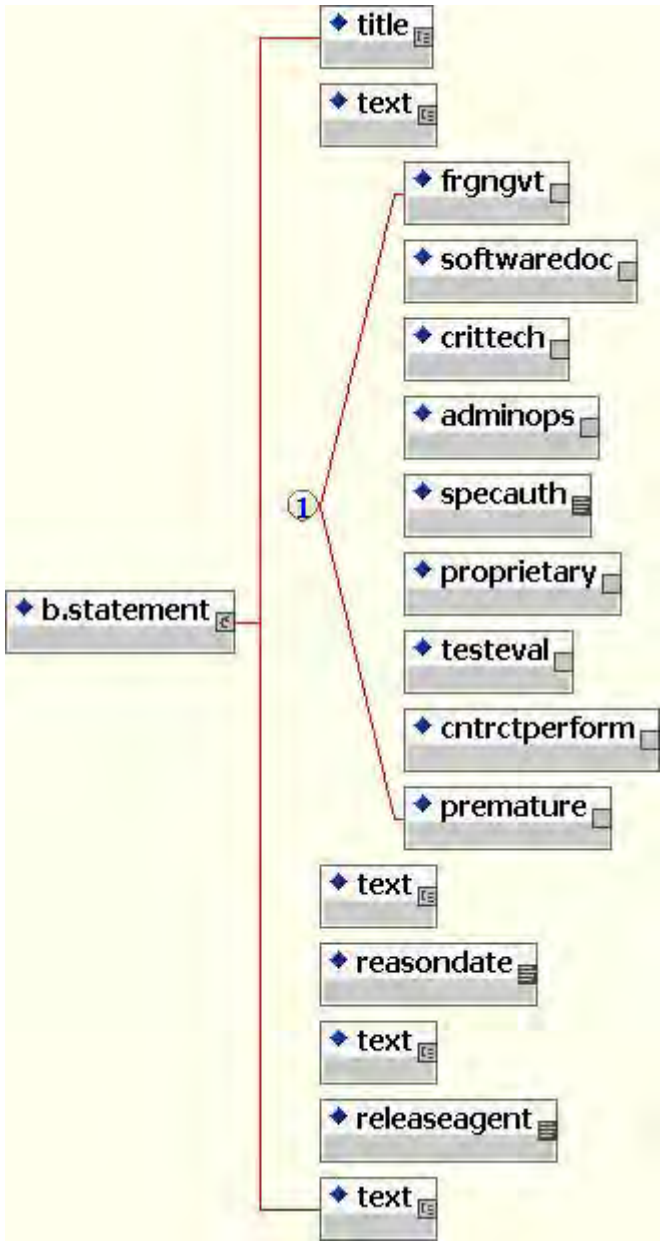

FIGURE 75. Distribution statement B<b.statement> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT b.statement (title, text, (frgngvt | softwaredoc | crittech
| adminops | specauth | proprietary | testeval | cntrctperform
| premature), text, reasondate, text, releaseagent, text)>
```

d. No attributes.

e. XML boilerplate text for the entity Distribution B Statement entity ***&notices.dist.b.statement;***. This entity is located in the Production Boilerplate Text file (prodboil.ent). Distribution B Statement boilerplate contains three nested entites ***&notices.dist.determined-reason;***, ***&notices.dist.determined-date;*** and ***&notices.dist.determined-dod-office;***. The nested entities can be edited through the Editable

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Boilerplate Text file (editboil.ent). The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

```
<dist>
<b.statement>
<title>DISTRIBUTION STATEMENT B.
</title>
<text>Distribution authorized to U.S. Government Agencies only for
</text> &notices.dist.determined-reason;
<text>, as determined on
</text> &notices.dist.determined-date;
<text>. Other requests for this document shall be referred to
</text> &notices.dist.determined-dod-office;
<text>.
</text>
</b.statement>
</dist>
```

31.3.1.1.4.2.4.2.1 Foreign Government Information ***<frgngvt>***. The element ***<frgngvt>*** provides the distribution statement restriction reason as the foreign government information and is defined as to protect and limit distribution in accordance with the desires of the foreign government that furnished the technical information. Information of this type normally is classified at the CONFIDENTIAL level or higher in accordance with DoD 5200.1-R (reference (h)). The selected distribution reason is generated through the stylesheet.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT frgngvt EMPTY>
<!ATTLIST frgngvt
    distreason          CDATA     #FIXED   "Foreign Government Information">
```

c. Unique attributes:
   (1) **distreason** – Distribution reason text for the distribution statement. (required)

31.3.1.1.4.2.4.2.2 Software Documentation ***<softwaredoc>***. The element ***<softwaredoc>*** provides the distribution statement restriction reason as the Software Documentation and is defined as relishable only in accordance with DoD Instruction 7930.2 (reference (i)). The selected distribution reason is generated through the stylesheet.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT softwaredoc EMPTY>

<!ATTLIST softwaredoc
    distreason          CDATA     #FIXED   "Software Documentation">
```

c. Unique attributes:
   (1) **distreason** – Distribution reason text for the distribution statement. (required)

31.3.1.1.4.2.4.2.3 Critical Technology ***<crittech>***. The element ***<crittech>*** provides the distribution statement restriction reason as the critical technology and is defined as to protect information and technical data that advance current technology or describe new technology in an area of significant or potentially significant military application or that relate to a specific military deficiency of a potential adversary. Information of this type may be classified or unclassified; when unclassified, it is export-controlled and subject to the provisions of DoD Directive 5230.25 (reference (c)). The selected distribution reason is generated through the stylesheet.

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a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT crittech EMPTY>
<!ATTLIST crittech
    distreason          CDATA     #FIXED   "Critical Technology">
```

c. Unique attributes:
   (1) **distreason** – Distribution reason text for the distribution statement. (required)

31.3.1.1.4.2.4.2.4 Administrative or operational use ***<adminops>***. The element ***<adminops>*** provides the distribution statement restriction reason as the Administrative or Operational Use and is defined as to protect technical or operational data or information from automatic dissemination under the International Exchange Program or by other means. This protection covers publications required solely for official use or strictly for administrative or operational purposes. This statement may be applied to manuals, pamphlets, technical orders, technical reports, and other publications containing valuable technical or operational data.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT adminops EMPTY>
<!ATTLIST adminops
    distreason          CDATA     #FIXED   "Administrative-Operational Use">
```

c. Unique attributes:
   (1) **distreason** – Distribution reason text for the distribution statement. (required)

31.3.1.1.4.2.4.2.5 Specific authority ***<specauth>***. The element ***<specauth>*** provides the distribution statement restriction reason as the Specific Authority and is defined as to protect information not specifically included in the above reasons and discussions, but which requires protection in accordance with valid documented authority such as Executive Orders, classification guidelines, DoD or DoD Component regulatory documents. The selected distribution reason is generated through the stylesheet.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT specauth EMPTY>
<!ATTLIST specauth
    distreason          CDATA     #FIXED   "Specific Authority">
```

c. Unique attributes:
   (1) **distreason** – Distribution reason text for the distribution statement. (required)

31.3.1.1.4.2.4.2.6 Proprietary information ***<proprietary>***. The element ***<proprietary>*** provides the distribution statement restriction reason as the Test and Evaluation and is defined as to protect results of test and evaluation of commercial products or military hardware when such disclosure may cause unfair advantage or disadvantage to the manufacturer of the product. The selected distribution reason is generated through the stylesheet.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.

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b. DTD fragment:

```
<!ELEMENT proprietary EMPTY>
<!ATTLIST proprietary
    distreason          CDATA           #FIXED  "Proprietary Information">

">
```

c. Unique attributes:
(1) **distreason** – Distribution reason text for the distribution statement. (required)

31.3.1.1.4.2.4.2.7 Test and evaluation ***<testeval>***. The element ***<testeval>*** provides the distribution statement restriction reason as the proprietary information and is defined as to protect information not owned by the U.S. Government and protected by a contractor's "limited rights" statement, or received with the understanding that it not be routinely transmitted outside the U.S. Government. The selected distribution reason is generated through the stylesheet.

a. The components are:
(1) The element is EMPTY and all pertinent information is entered through its attributes.

b. DTD fragment:

```
<!ELEMENT testeval EMPTY>
<!ATTLIST testeval
    distreason          CDATA           #FIXED  "Test and Evaluation">
```

c. Unique attributes:
(1) **distreason** – Distribution reason text for the distribution statement. (required)

31.3.1.1.4.2.4.2.8 Contractor performance evaluation ***<cntrctperform>***. The element ***<cntrctperform>*** provides the distribution statement restriction reason as the contractor performance evaluation and is defined as to protect information in management reviews, records of contract performance evaluation, or other advisory documents evaluating programs of contractors. The selected distribution reason is generated through the stylesheet.

a. The components are:
(1) The element is EMPTY and all pertinent information is entered through its attributes.

b. DTD fragment:

```
<!ELEMENT cntrctperform EMPTY>
<!ATTLIST cntrctperform
   distreason          CDATA     #FIXED   "Contractor Performance Evaluation">
```

c. Unique attributes:
(1) **distreason** – Distribution reason text for the distribution statement. (required)

31.3.1.1.4.2.4.2.9 Premature dissemination ***<premature>***. The element ***<premature>*** provides the distribution statement restriction reason as the premature dissemination and is defined as to protect patent able information on systems or processes in the developmental or concept stage from premature dissemination. The selected distribution reason is generated through the stylesheet.

a. The components are:
(1) The element is EMPTY and all pertinent information is entered through its attributes.

b. DTD fragment:

```
<!ELEMENT premature EMPTY>
<!ATTLIST premature
    distreason          CDATA           #FIXED  "Premature Dissemination">
```

c. Unique attributes:
(1) **distreason** – Distribution reason text for the distribution statement. (required)

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31.3.1.1.4.2.4.2.10 Restriction determination date ***<reasondate>*** . The element ***<reasondate>*** is used to specify distribution statements B-F and X restriction determination date.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT reasondate (#PCDATA)>
```

c. No attributes.

31.3.1.1.4.2.4.2.11 Controlling DoD office ***<releaseagent>***. The element ***<releaseagent>*** provides the controlling DoD office for deciding the distribution statement level and request for further distribution.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT releaseagent (#PCDATA)>
```

c. No attributes.

31.3.1.1.4.2.4.3 Distribution C statement ***<c.statement>***. The element ***<c.statement>*** contains the DODD 5230.24 specified text for C restriction distribution (Distribution authorized to U.S. Government Agencies and their contractors) technical manual. The selected distribution reason is generated through the stylesheet. The identified reason will generate text. This statement is a boilerplate and can be referenced using the general entity ***&notices.dist.c.statement;*** (see 31.3.1.1.4.2.4.3.e.).

a. The components are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Text ***<text>*** (required – two or more). The text element is required in four places contained in the verbatim Distribution B Statement(see 49.1.1.18).
   (3) Select one of the (required) reasons that explains the restricted distribution for the TM.
      (a) Foreign Government Information ***<frgngvt>*** (see 31.3.1.1.4.2.4.2.1).
      (b) Software Documentation ***<softwaredoc>*** (see 31.3.1.1.4.2.4.2.2).
      (c) Critical Technology ***<crittech>*** (see 31.3.1.1.4.2.4.2.3).
      (d) Administrative or Operational Use ***<adminops>*** (see 31.3.1.1.4.2.4.2.4).
      (e) Specific Authority ***<specauth>*** (see 31.3.1.1.4.2.4.2.5).
   (4) Restriction Determination Date ***<reasondate>*** (see 31.3.1.1.4.2.4.2.10).
   (5) Controlling DoD Office ***<releaseagent>*** (see 31.3.1.1.4.2.4.2.11).
b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

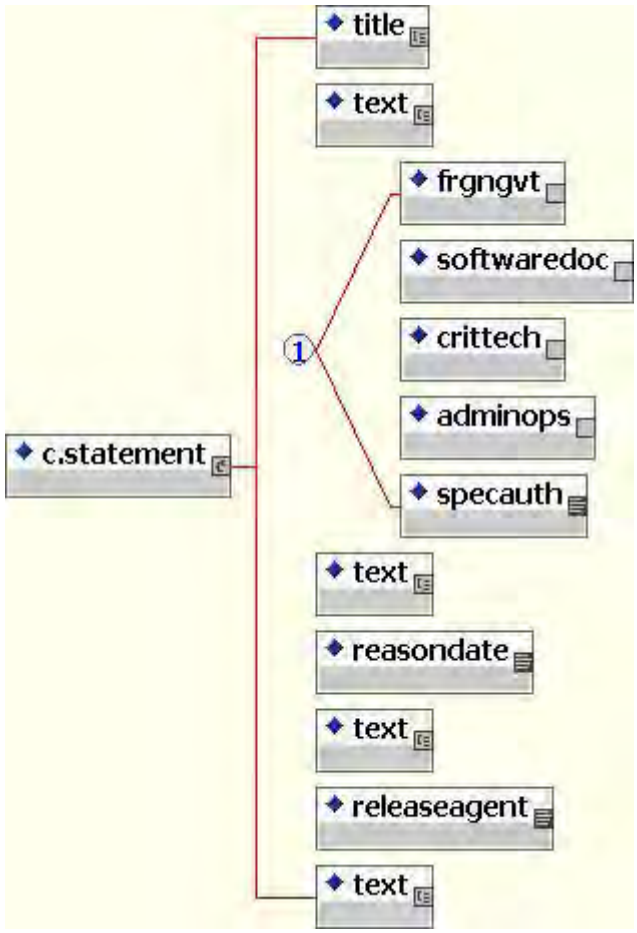

FIGURE 76. Distribution statement C <c.statement> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT c.statement (title, text, (frgngvt | softwaredoc |
crittech | adminops | specauth), text, reasondate,
text, releaseagent, text)>
```

d. No attributes.

e. XML boilerplate text for the entity Distribution C Statement entity ***&notices.dist.c.statement;***. This entity is located in the Production Boilerplate Text file (prodboil.ent). Distribution C Statement boilerplate contains three nested entites ***&notices.dist.determined-reason;***, ***&notices.dist.determined-date;*** and ***&notices.dist.determined-dod-office;***. The nested entities can be edited through the Editable Boilerplate Text file (editboil.ent). The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

<dist>
<c.statement>
<title>DISTRIBUTION STATEMENT C.
</title>
<text>Distribution authorized to U.S. Government Agencies
and their contractors only for
</text> ***&notices.dist.determined-reason;***
<text>, as determined on
</text> ***&notices.dist.determined-date;***

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<text>.  Other requests for this document shall be referred to
</text>  &notices.dist.determined-dod-office;
<text>.
</text>
</c.statement>
</dist>
```

31.3.1.1.4.2.4.4 Distribution D statement <*d.statement*>. The element <*d.statement*> contains the DODD 5230.24 specified text for D restriction distribution (Distribution authorized to the Department of Defense and U.S. DoD contractors only) technical manual. The selected distribution reason is generated through the stylesheet. The identified reason will generate text. This statement is a boilerplate and can be referenced using the general entity *&notices.dist.d.statement;* (see 31.3.1.1.4.2.4.4.e.).

a. The components are:
   (1) Title <*title*> (required) (see 49.1.1.3).
   (2) Text <*text*> (required – two or more). The text element is required in four places contained in the verbatim Distribution B Statement (see 49.1.1.18).
   (3) Select one of the (required) reasons that explains the restricted distribution for the TM.
      (a) Foreign Government Information <*frgngvt*> (see 31.3.1.1.4.2.4.2.1).
      (b) Software Documentation <*softwaredoc*> (see 31.3.1.1.4.2.4.2.2).
      (c) Critical Technology <*crittech*> (see 31.3.1.1.4.2.4.2.3).
      (d) Administrative or Operational Use <*adminops*> (see 31.3.1.1.4.2.4.2.4).
      (e) Specific Authority <*specauth*> (see 31.3.1.1.4.2.4.2.5).
   (4) Restriction Determination Date <*reasondate*> (see 31.3.1.1.4.2.4.2.10).
   (5) Controlling DoD Office <*releaseagent*> (see 31.3.1.1.4.2.4.2.11).

b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

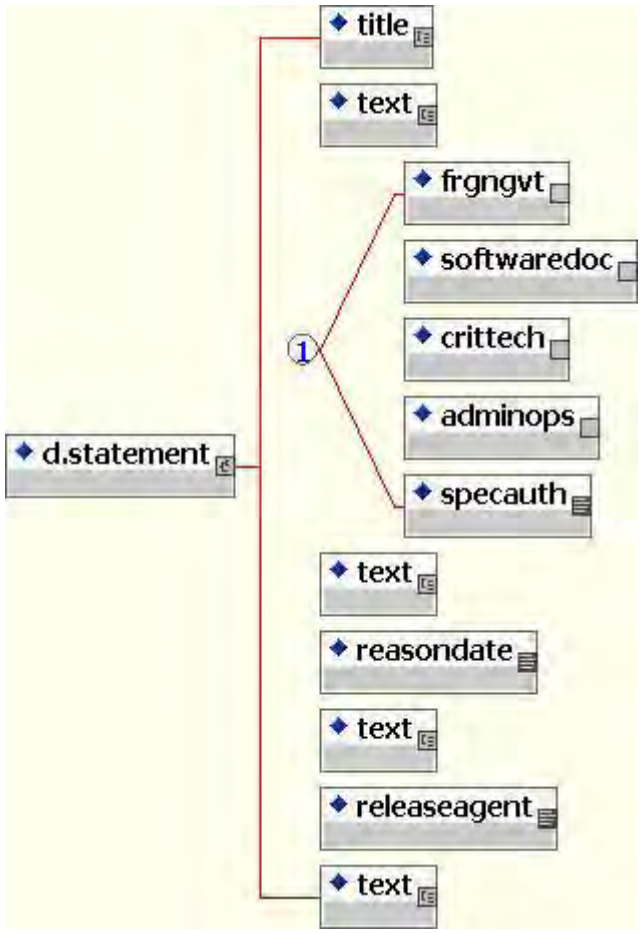

FIGURE 77. Distribution D statement <d.statement> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT d.statement (title, text, (frgngvt | softwaredoc |
crittech | adminops | specauth), text, reasondate,
text, releaseagent, text)>
```

d. No attributes.

e. XML boilerplate text for the entity Distribution D Statement entity ***&notices.dist.d.statement;*** is provided below. This entity is located in the Production Boilerplate Text file (prodboil.ent). Distribution D Statement boilerplate contains three nested entites ***&notices.dist.determined-reason;***, ***&notices.dist.determined-date;*** and ***&notices.dist.determined-dod-office;***. The nested entities can be edited through the Editable Boilerplate Text file (editboil.ent). The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

<dist>
<d.statement>
<title>DISTRIBUTION STATEMENT D.
</title>
<text>Distribution authorized to the Department of Defense
and U.S. DoD contractors only for
</text> ***&notices.dist.determined-reason;***
<text>, as determined on
</text> ***&notices.dist.determined-date;***

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<text>.  Other requests for this document shall be referred to
</text>  &notices.dist.determined-dod-office;
<text>.
</text>
</d.statement>
</dist>
```

31.3.1.1.4.2.4.5 Distribution E statement *<e.statement>*. The element *<e.statement>* contains the DODD 5230.24 specified text for E restriction distribution (Distribution authorized to DoD Components only) technical manual. The selected distribution reason is generated through the stylesheet. This statement is a boilerplate and can be referenced using the general entity *&notices.dist.e.statement;* (see 31.3.1.1.4.2.4.5.e.).

a. The components are:
   (1) Title *<title>* (required) (see 49.1.1.3).
   (2) Text *<text>* (required – two or more). The text element is required in four places contained in the verbatim Distribution B Statement.(see 49.1.1.18).
   (3) Select one of the (required) reasons that explains the restricted distribution for the TM.
      (a) Foreign Government Information *<frgngvt>* (see 31.3.1.1.4.2.4.2.1).
      (b) Software Documentation *<softwaredoc>* (see 31.3.1.1.4.2.4.2.2).
      (c) Critical Technology *<crittech>* (see 31.3.1.1.4.2.4.2.3).
      (d) Administrative or Operational Use *<adminops>* (see 31.3.1.1.4.2.4.2.4).
      (e) Specific Authority *<specauth>* (see 31.3.1.1.4.2.4.2.5).
      (f) Proprietary Information *<proprietary>* (see 31.3.1.1.4.2.4.2.6).
      (g) Test and Evaluation *<testeval>* (see 31.3.1.1.4.2.4.2.7).
      (h) Contractor Performance Evaluation *<cntrctperform>* (see 31.3.1.1.4.2.4.2.8).
      (i) Premature Dissemination *<premature>* (see 31.3.1.1.4.2.4.2.9).
   (4) Direct Military Support *<milsprt>* (see 31.3.1.1.4.2.4.5.1).
   (5) Restriction Determination Date *<reasondate>* (see 31.3.1.1.4.2.4.2.10).
   (6) Controlling DoD Office *<releaseagent>* (see 31.3.1.1.4.2.4.2.11).
b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

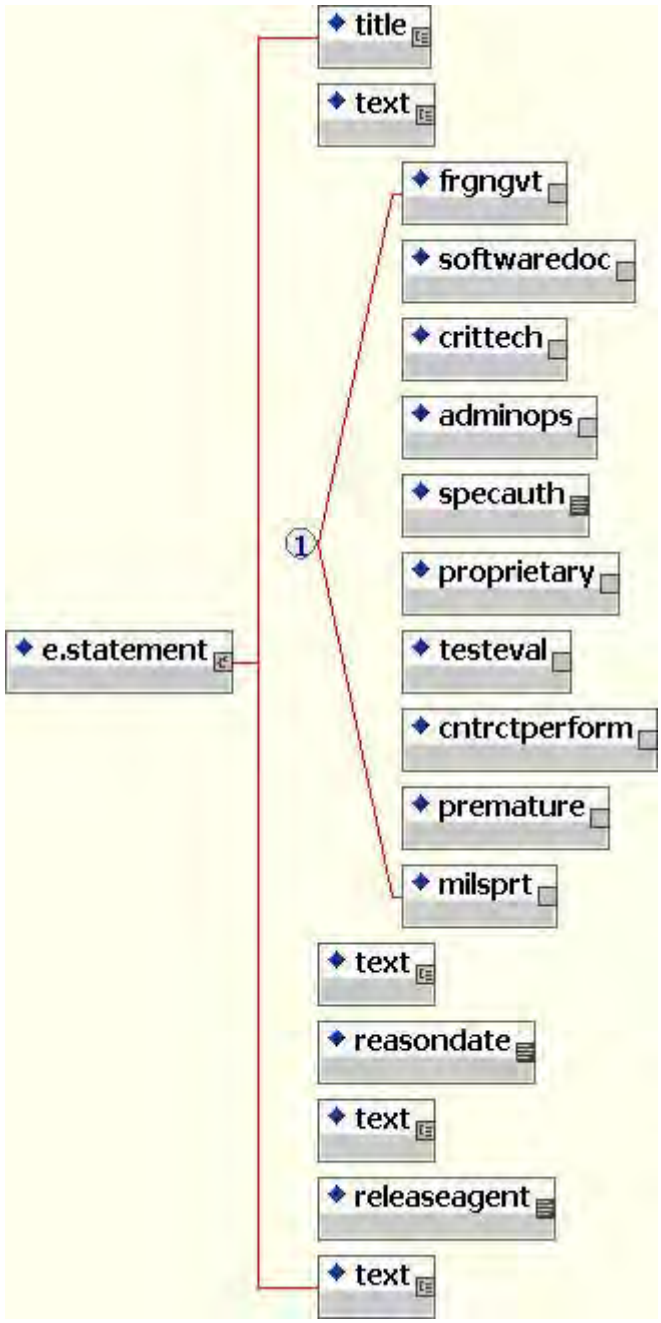

FIGURE 78. Distribution E statement <e.statement> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT e.statement (title, text, (frgngvt | softwaredoc | crittech
| adminops | specauth | proprietary | testeval | cntrctperform
| premature), text, reasondate, text, releaseagent, ext)>
```

d. No attributes.

e. XML boilerplate text for the entity Distribution E Statement entity ***&notices.dist.e.statement;*** is provided below. This entity is located in the Production Boilerplate Text file (prodboil.ent).

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Distribution E Statement boilerplate contains three nested entites ***&notices.dist.determined-reason;***, ***&notices.dist.determined-date;*** and ***&notices.dist.determined-dod-office;***. The nested entities can be edited through the Editable Boilerplate Text file (editboil.ent). The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

```
<dist>
<e.statement>
<title>DISTRIBUTION STATEMENT E.
</title>
<text>Distribution authorized to DOD Components only &no-
tices.dist.determined-reason;
<text> as determined on
</text> &notices.dist.determined-date;
<text>. Other requests for this document shall be referred to
</text> &notices.dist.determined-dod-office;
<text>.
</text>
</e.statement>
</dist>
```

31.3.1.1.4.2.4.5.1 Direct military support***<milsprt>***. The element ***<milsprt>*** provides the distribution statement restriction reason as the direct military support. The restriction reason for Direct Military Support is defined that the document contains export-controlled technical data of such military significance. The release of this type of document for purposes other than direct support of DoD-approved activities may jeopardize an important technological or operational military advantage of the United States. Designation of such data is made by competent authority in accordance with DoD Directive 5230.25 (reference (c)). The selected distribution reason is generated through the stylesheet.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT milsprt EMPTY>
<!ATTLIST milsprt
distreason     CDATA      #FIXED     "Direct Military Support">
```

c. Unique attributes:
   (1) **distreason** – Distribution reason text for the distribution statement. (required) This text is generated through the stylesheet.

31.3.1.1.4.2.4.6 Distribution F statement ***<f.statement>***. The element ***<f.statement>*** contains the DODD 5230.24 specified text for F restriction distribution (Further dissemination only as directed by controlling DoD office or higher DoD authority) technical manual. The identified reason will generate text. This statement is a boilerplate and can be referenced using the general entity ***&notices.dist.f.statement;***(see 31.3.1.1.4.2.4.6.e.).

a. The components are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Text ***<text>*** (required – two or more). The text element is required in three places contained in the verbatim Distribution F Statement (see 49.1.1.18).
   (3) Restriction Determination Date ***<reasondate>*** (see 31.3.1.1.4.2.4.2.10).
   (4) Controlling DoD Office ***<releaseagent>*** (see 31.3.1.1.4.2.4.2.11).
b. DTD fragment is graphically depicted.

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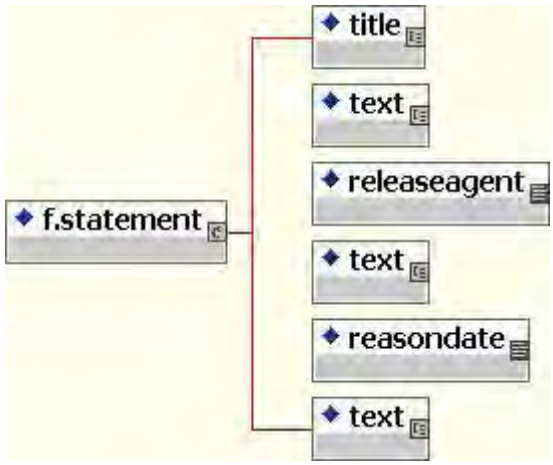

FIGURE 79. Distribution F statement <f.statement> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT f.statement (title, text, releaseagent, text, reasondate, text)>
```

d. No attributes.

e. XML boilerplate text for the entity Distribution F Statement entity ***&notices.dist.f.statement;*** is provided below. This entity is located in the Production Boilerplate Text file (prodboil.ent). Distribution F Statement boilerplate contains two nested entites,***&notices.dist.determined-dod-office;*** and ***&notices.dist.determined-date;***. The nested entities can be edited through the Editable Boilerplate Text file (editboil.ent). The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

<dist>
<f.statement>
<title>DISTRIBUTION STATEMENT F.
</title>
<text>Further dissemination only as directed by ***&notices.dist.determined-dod-office;***
<text>, as determined on
</text> ***&notices.dist.determined-date;***
<text>. or higher DOD authority.
</text>
</f.statement>
</dist>

31.3.1.1.4.2.4.7 Distribution X statement ***<x.statement>***. The element ***<x.statement>*** contains the DODD 5230.24 specified text for X restriction distribution technical manual. The selected distribution reason is generated through the stylesheet. This statement is a boilerplate and can be referenced using the general entity ***&notices.dist.x.statement;*** (see 31.3.1.1.4.2.4.7.e.).

a. The components are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Text ***<text>*** (required – two or more). The text element is required in three places contained in the verbatim Distribution F Statement (see 49.1.1.18).
   (3) Restriction Determination Date ***<reasondate>*** (see 31.3.1.1.4.2.4.2.10).
   (4) Controlling DoD Office ***<releaseagent>*** (see 31.3.1.1.4.2.4.2.11).

b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

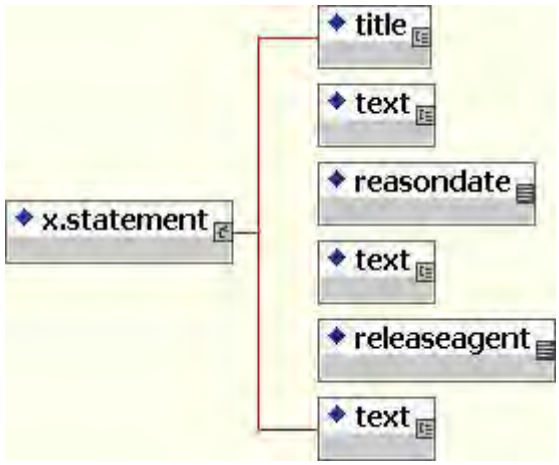

FIGURE 80. Distribution X statement <x.statement> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT x.statement (title, text, releaseagent, text, reasondate, text)>
```

d. No attributes.

e. XML boilerplate text for the entity Distribution X Statement entity ***&notices.dist.x.statement;*** is provided below. This entity is located in the Production Boilerplate Text file (prodboil.ent). Distribution X Statement boilerplate contains two nested entites ***&notices.dist.determined-date;*** and ***&notices.dist.determined-dod-office;***. The nested entities can be edited through the Editable Boilerplate Text file (editboil.ent). The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

```
<dist>
<x.statement>
<title>DISTRIBUTION STATEMENT X.
</title>
<text>Distribution authorized to U.S. Government Agencies and private
individuals or enterprises eligible to obtain export-controlled
technical data in accordance with DoD Directive 5230.25, as
determined on &notices.dist.determined-dat;
<text>Controlling DoD office is
</text> &notices.dist.determined-dod-office;
<text>.
</text>
</x.statement>
</dist>
```

31.3.1.1.4.2.5 Export control ***<export>***. The element ***<export>*** provides the export control statement warning for the pubs that have export controlled data. The statement is contained in DOD Directive 5230.24. This statement is a boilerplate and can be referenced using the general entity ***&notices.export;***. (see 31.3.1.1.4.2.5.e.)

a. The components are:
   (1) Title ***<title>*** (optional) (see 49.1.1.3).
   (2) Paragraph Text ***<text>*** (required) (see 49.1.1.18).

b. DTD fragment is graphically depicted.

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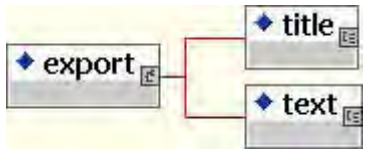

FIGURE 81. Export <export> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT export (title, text)>
```

d. No attributes.

e. XML boilerplate text for the entity disclosure notice ***&notices.export;***. This entity is located in the Production Boilerplate Text file (prodboil.ent). The use of boilerplates can reduce text entry time and errors. See Section 50 for further information on boilerplates.

```
<export>
<title>WARNING
</title>
</text>This document contains technical data whose export is restricted
by the Arms Export Control Act (Title 22, U.S.C., Sec 2751, et.
seq.)  or the Export Administration Act of 1979, as amended, Title
50, U.S.C., App.  2401 et.  seq.  Violations of these export
laws are subject to severe criminal penalties.  Disseminate in
accordance with provisions of DoD Directive 5230.25.
</text>
</export>
```

31.3.1.1.4.2.6 Destruction of the manual ***<destr>***. The element ***<destr>*** provides a standard notice concerning destruction of the manual. The destruction statement is provided by the contracting activity and appears on the front cover.

a. The components are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Paragraph Text ***<text>*** (required) (see 49.1.1.18).

b. DTD fragment is graphically depicted.

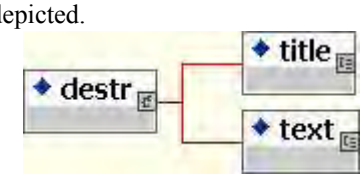

FIGURE 82. Destruction of the manual <destr> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT destr (title, text)>
```

d. No attributes.

31.3.1.1.4.2.7 General information destruction ***<general_purpose_notices>***. The element ***<general_purpose_notices>*** provides the user with information that is generic to most destruction processes.

a. The components are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Paragraph Text ***<text>*** (required) (see 49.1.1.18).

b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

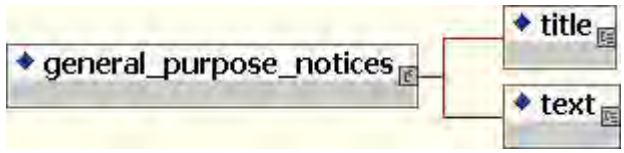

FIGURE 83. General information – destruction <general_purpose_notices> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT general_purpose_notices  (title, text)>
```

d. No attributes.

31.3.1.1.4.3 Service nomenclature ***<servnomen>***. The element ***<servnomen>*** provides the service nomenclature of the proponent activity; for most Army manuals the text is "HEADQUARTERS, DEPARTMENT OF THE ARMY."

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
   (2) Emphasis ***<emphasis>*** (optional – zero or more) (see 49.1.3.1).
   (3) Subscript ***<subscrpt>*** (optional – zero or more) (see 49.1.3.4).
   (4) Superscript ***<supscrpt>*** (optional – zero or more) (see 49.1.3.5).

b. DTD fragment:

```
<!ELEMENT servnomen (#PCDATA | emphasis | subscrpt | supscrpt)*>
```

c. No attributes.

31.3.1.1.5 Front cover alternative ***<frntcover-alt>***. The element ***<frntcover-alt>*** is used for when alternative covers are needed for a manual with multiple configurations.

a. The components are:
   (1) Front Cover ***<frntcover>*** (required). (see 31.3.1.1.4)

b. DTD fragment:

```
<!ELEMENT frntcover-alt (frntcover)>
```

31.3.1.1.6 Promulgation letter (USMC) ***<promulgation>***. The element ***<promulgation>*** contains the promulgation letter provided by the acquiring activity, used primarily by the Marine Corp (similar to authentication letter).

a. The components are:
   (1) Illustrations ***<graphic>*** (required). A graphic portraying the promulgation letter is displayed(see 44.2).

b. DTD fragment:

```
<!ELEMENT promulgation (graphic)>
<!ATTLIST promulgation
    applicable          IDREFS                  #IMPLIED
```

c. Common elements.
   (1) **applicable** – Applicability Reference to the applicable configuration(s) specified in the WP identification information. When using an IETM that can filter information, this information is not presented when not applicable to the current configuration. Presentations that do not filter information, the information will be identified by the assigned associated abbreviation to designate applicability of the information. (optional).

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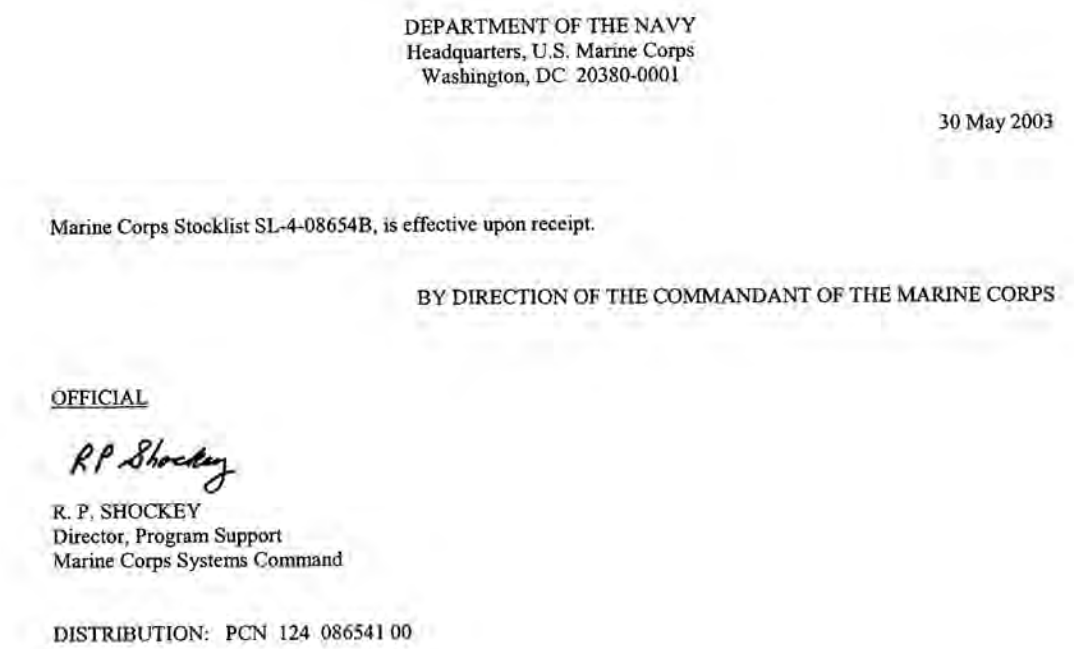
31.3.1.1.6.1 Example of a promulgation letter (USMC). An example of a promulgation letter is provided below. The promulgation letter consist of a graphic.

a. The XML document instance for a promulgation letter.

```
<promulgation>
<graphic boardno="promul.jpg">
</graphic>
</promulgation>
```

b. The XML format output for a promulgation letter using a stylesheet.



DEPARTMENT OF THE NAVY
Headquarters, U.S. Marine Corps
Washington, DC 20380-0001

30 May 2003

Marine Corps Stocklist SL-4-08654B, is effective upon receipt.

BY DIRECTION OF THE COMMANDANT OF THE MARINE CORPS

OFFICIAL

R. P. SHOCKEY
Director, Program Support
Marine Corps Systems Command

DISTRIBUTION: PCN 124 086541 00

FIGURE 84. Example of a promulgation letter (USMC)<promulgation>..

31.3.1.1.7 Alternative promulgation letter ***<promulgation-alt>***. The element ***<promulgation>*** is used when alternative promulgation letters are needed for manual with multiple configurations.

a. The components are:

(1) Promulgation Letter ***<promulgation>*** (required) (see 31.3.1.1.6).

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b. DTD fragment:

```
<!ELEMENT promulgation-alt (promulgation)>
```

31.3.1.1.8 Table of contents (TOC) ***<contents>***. The element ***<contents>*** is used for generating the TOC including an optional list of figures and/or list of tables for each work package. The TOC is automatically generated, pre-composition process populated, or manually entered. The automatically generated TOC uses an XML attribute (**tocentry**) to determine if the item is in the TOC and the indenture level. Some composition systems can generate the TOC with the assistance of a pre-composition process. The pre-composition process reads the completed TM XML instance and uses the TOC elements to populate the ***<contents>*** XML instance. While other composition systems can not handle an automated task or process such as a TOC and has to be manually created. Additional information on the process of the TOC and other optional methods (see 31.3.1.1.8.3) in developing a TOC.

a. The components (optional) are:
   (1) Title ***<title>*** (required). The element provides the title for the TOC, usually the text "TABLE OF CONTENTS" is entered (see 49.1.1.3).
   (2) Column list title ***<col.title>*** (required). The element provides the figure/table page number (column title is "Page No.") and the work package sequence number (column title is "WP Sequence No.") columns in the TOC. (see 31.4.1.4.3).
   (3) Content entry ***<contententry>*** (required – one or more) (see 31.3.1.1.8.1).

b. DTD fragment is graphically depicted.

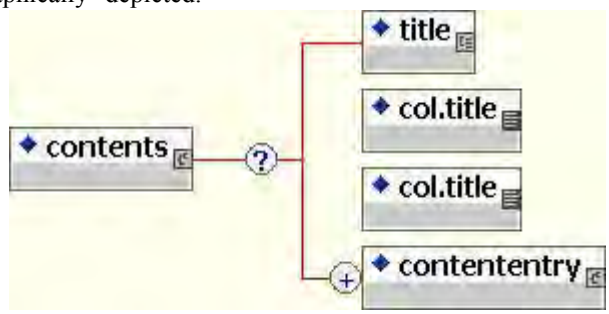

FIGURE 85. Table of content <contents> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT contents (title , col.title , col.title , contententry+)?>
```

31.3.1.1.8.1 Content entry ***<contententry>***. This element contains the necessary data to create a TOC entry and subentry either manually or through a pre-composition process. This element is recursive to allow the requested TOC levels (list of figures, list of tables, tasks, etc.) for the manual.

a. The components are:
   (1) Entry title ***<title>*** (required) (see 49.1.1.3) is the same text as used in the referenced information.
   (2) Referenced entry work package sequence or page number ***<link>*** (optional) (see 46.2.3). Note that generally warning summary, how to use, information chapters, glossary, and index do not reference page numbers.
   (3) Subentry data ***<sub-contententry>*** (optional) is used when subordinate TOC entry(s) apply. The element contains sub-content entries ***<contententry>*** for each subordinate item under the entry. When the sub-content entries contain the figure or table references the attribute **figuretable** is set to "yes".

b. DTD fragment is graphically depicted.

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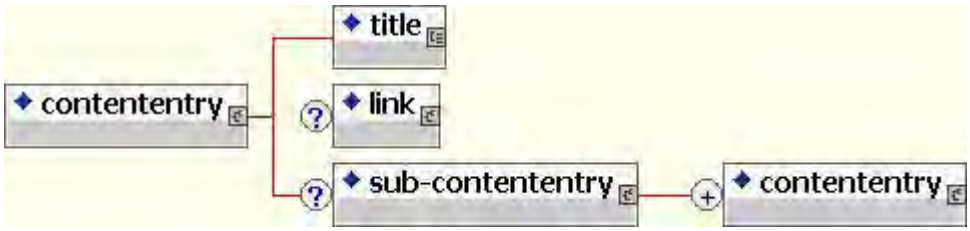

FIGURE 86. Table of content entry <contententry> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT contententry (title , link? , sub-contententry?)>
<!ATTLIST contententry
    assocfig IDREFS #IMPLIED
    id       ID     #IMPLIED
    idref    IDREFS #IMPLIED
   security          (uc | fouo |c | s | ts)           #IMPLIED
>
<!ELEMENT sub-contententry (contententry+)>
<!ATTLIST sub-contententry
    figuretable        (yes | no )                      #IMPLIED>
```

d. Unique attributes:
   (1) **figuretable** – Indicates if the entries for the subgroup are for tables and/or figures.

e. Common attributes are:
   (1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (2) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (3) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (4) **security** – Security classification (optional) (see 49.2.13).

31.3.1.1.8.2 TOC entry attribute **tocentry**. The attribute **tocentry** indicates that the data type's title and the indenture level to be used for the TOC entry. When the **tocentry** attribute is set to "0", no entry is created or used for this information. Some **tocentry** attributes have predefined indenture levels to assist the TM developer with a standard method for building the TOC. The following data types use the **tocentry** attribute (default level is in parenthesis) that are volumes ("1"), information chapters ("1"), work packages ("2"), tasks ("0"), procedures ("0"), figures ("1"), standard information tables ("1"), tables ("1"), warning summary ("1"), how to use ("1"), glossary ("1"), index ("1"), and foldout section ("1").

31.3.1.1.8.3 How to develop a TOC. A TOC is developed for each data type with the attribute **tocentry** not being "0". Titles used in the TOC are the same words as in the data type being referenced IAW MIL–STD–40051–1/–2 which states, "They shall have the exact same title and shall be listed in the same order they appear in the TM." Depending on the composition system the method used to generate the TOC may vary. The four methods to develop a TOC are:

a. Method 1 – Composition system generated TOC (not using attribute indenture level) (see 31.3.1.1.8.3.3).
b. Method 2 – Composition system generated TOC (using attribute indenture level) (see 31.3.1.1.8.3.4).
c. Method 3 – Preprocess application to generate TOC elements (see 31.3.1.1.8.3.5).
d. Method 4 – Manually enter TOC entries (see 31.3.1.1.8.3.6).

31.3.1.1.8.3.1 Linking to TOC entry to information. The TOC entry should provide a hyper link to referenced information. The work package or page reference will be active through the use of the ***<link>*** element. The TOC entry title ***<title>*** also can be linked to the target information through the attribute **idref**. A reference in the title would allow both the title and page reference to be an active link to the information.

31.3.1.1.8.3.2 Figure and table entries. Generally, figures and tables are grouped together under the work package they are related under. The purpose of listing the figures and tables under the work package is the requirement to have each figure or table to restart numbering at one for each work package.

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Grouping the figures or tables together would make the figure number or table number almost useless. When grouping figures and tables under a work package, the element ***<sub-contententry>*** attribute **figuretable** is set to "yes", which indicates to a style sheet that the page number reference is inset from the work package sequence reference column.

31.3.1.1.8.3.3 Method 1 – composition system generated TOC (not using attribute indenture level). Method 1 generates the TOC to the output file only and does not save the markup information to the source document, therefore the TOC is always current to the latest publication. The indenture level is determined by the style sheet rules, as specified by the acquiring activity. The following is the preferred hierarchy for the data types:

a. Volume (if applicable)
b. Information Chapter, Warning Summary, How to Use, Glossary, and Index
c. Work package
d. Task and Procedures (if applicable)
e. Figures and Tables (if applicable)

The style sheet stores or transverses the markup tree for the TOC marked data type titles and, if applicable, stores either the Work Package sequence number (work packages) or references the page number (tasks, procedures, figures, tables, and standard information tables).

31.3.1.1.8.3.4 Method 2 – composition system generated TOC (not using attribute indenture level). Method 2 is similar to Method 1 except the style sheet uses the **tocentry** attribute value to determine the indenture level of the data type. The TM developer needs to determine early in the development structure of the TM to avoid the need to relabel the **tocentry** later in the development cycle. Some defaulted **tocentry** attribute value will require modification (i.e., when volumes are used).

31.3.1.1.8.3.5 Method 3 – preprocess application to generate TOC elements. Method 3 uses an application, prior to applying the style sheet, to read the entire TM document instance and capture the necessary information for all data that has the **tocentry** attribute value not equal to "0". The application may use the indenture levels to determine the TOC information position or use the XML elements hierarchy (see 31.3.1.1.8.3.3 for hierarchy). Each referenced TOC entry needs an ID attribute set to provide referencing information (i.e. page number or hyper linking), if no ID value is provided the application should generate an ID value and apply the ID value to the referenced text. Generally most items in the TOC would have an ID attribute set, since all work packages require and most figures and tables have ID values. Table IV shows the mapping of data type information into TOC entry.

TABLE IV. TOC Entry Mapping.

Data Type	TOC entry ***<title>*** is derived from	TOC entry ***<link>*** attribute **label** value is derived from
Volume	Volume (***<volume>***) **label** attribute	Volume (***<volume>***) **id** attribute
Warning summary	Warning summary (***<warnsum>***) title (***<title>***)	Warning summary (***<warnsum>***) or warning summary title (***<title>***); **id** attribute
How to use	How to use (***<howtouse>***) title (***<title>***)	How to use (***<howtouse>***) or how to use title (***<title>***); **id** attribute
Glossary	Glossary (***<glossary>***) title (***<title>***)	Glossary (***<glossary>***) or glossary title (***<title>***); **id** attribute
Index	Index (***<aindx>***) title (***<title>***)	Index (***<aindx>***) or index title (***<title>***); **id** attribute

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Table IV TOC Entry Mapping – Continued.

Data Type	TOC entry ***<title>*** is derived from	TOC entry ***<link>*** attribute **label** value is derived from
Information chapter	Title page (***<titlepg>***) content ***<name>***, ***<partno>***, ***<modelno>***, and ***<nsn>***	Information chapter (***<gim>***, ***<opim>***, ***<tim>***, ***<mim>***, ***<pim>***, ***<dim>***, and ***<sim>***) attribute **id**
Work package	Work package identification information (***<wpidinfo>***) title (***<title>***)	Each work packages' (i.e., ***<maintwp>***) **wpno** attribute
Task	Task (i.e., Clean ***<clean>***) title (***<title>***)	Task (i.e., Clean ***<clean>***) or task title (***<title>***); **id** attribute
Procedure	Procedure (***<proc>***) title (***<title>***)	Procedure (***<proc>***) or procedure title (***<title>***); **id** attribute
Figure	Figure (***<figure>***) title (***<title>***)	Figure (***<figure>***) or figure title (***<title>***); **id** attribute
Table	Table (***<table>***) title (***<title>***)	Table (***<table>***) or table title (***<title>***); **id** attribute
Standard information	Standard information (i.e., PMCS ***<pmcstable>***) title (***<title>***)	Standard information (i.e., PMCS ***<pmcstable>***) or standard information title (***<title>***); **id** attribute

An example of the mapping TM data type information into TOC entry XML markup is provided below.

a. Maintenance information chapter markup example:

```
<mim chap-toc="no" id="CHAP3-X-XXXX-XXX" revno="0" frame="yes"
chngno="0" tocentry="1">
<titlepg maintlvl="field">
<name>Chapter 3
</name>
<name>Fielding Maintenance Instructions
</name>
</titlepg> . . .
<maintwp wpno="M03451–X-XXXX-XXX" tocentry="2" frame="yes" army="no"
airforce="no" navy="no" marines="no" wpseq="0029" deletewp="no">
<wpidinfo>
<maintlvl level="field"/>
<title>24-Volt Connector Receptacle
<brk/>Assembly, Repair, Reassembly
</title>
</wpidinfo> . . .
```


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```
<figure application="both" figtype="normal-page" tocentry="3" pane="no"
id="M03451–X-XXXX-XXX-fig1">
<title>24-volt connector receptacle.
</title> . . .
</figure> . . .
</maintwp> . . .
</mim>
```

b. TOC entry markup after preprocess example:

```
<contententry>
<title>Chapter 3 – Fielding Maintenance Instructions
</title>
<sub-contententry figuretable="no"> . . .
<contententry>
<title>24–Volt Connector Receptacle Assembly, Repair, Reassembly
</title>
<link xmlns:xlink="http://www.w3.org/1999/xlink" application="both"
local="M03451–X-XXXX-XXX" xlink:type="simple" linktype="goto" linkaction="prompt">
<ref.generate/>
</link>
<sub-contententry figuretable="yes"> . . .
<contententry>
<title>24-volt connector receptacle
</title>
<link xmlns:xlink="http://www.w3.org/1999/xlink" application="both"
local="M03451–X-XXXX-XXX-fig1" xlink:type="simple" linktype="goto" linkaction="prompt">
<ref.generate/>
</link>
</contententry> . . .
</sub-contententry>
</contententry>
</sub-contententry> . . .
</contententry>
```

c. Generated TOC output example:

WP Sequence No.

Page No.

...

Chapter 3 – Fielding Maintenance Instructions

...

24–Volt Connector Receptacle Assembly, Repair, Reassembly. . . . . WP 0029 . . . . . .

Figure 1. 24–volt connector receptacle . . . . . 0029–2 . . . . . .

31.3.1.1.8.3.6 Method 4 – manually enter TOC entries. Method 4 does not require a separate application applied to the document instance. The author will manually perform the following tasks to permit TOC generation. The author would generate the TOC either as each work package is created or after all work packages have been developed. It is important to note that MIL–STD–40051–1/–2 states the titles in the TOC are a exact match to the target titles. The WP sequence number and page number can be reference generated by the style sheet (this is the best method to ensure numbers are correct) or manually enter the information. The method to manually enter WP sequence numbers or page numbers, the author in using the ***<link>*** element would use the child element ***<prompt>*** instead of the element ***<ref.generate>***. The author

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would enter within the ***<prompt>*** element the WP sequence number or page number (manually entering the page number requires the author to have applied the style sheet and review the generated output). The example markup below shows how to manually enter information (using example from 31.3.1.1.8.3.5).

```
<contententry>
<title>Chapter 3 - Fielding Maintenance Instructions
</title>
<sub-contententry figuretable="no"> . . .
<contententry>
<title>24-Volt Connector Receptacle Assembly, Repair, Reassembly
</title>
<link xmlns:xlink="http://www.w3.org/1999/xlink" application="both"
local="M03451–X-XXXX-XXX" xlink:type="simple" linktype="goto" linkaction="prompt">
<prompt>WP 0029
</prompt>
</link>
<sub-contententry figuretable="yes"> . . .
<contententry>
<title>24-volt connector receptacle
</title>
<link xmlns:xlink="http://www.w3.org/1999/xlink" application="both"
local="M03451–X-XXXX-XXX-fig1" xlink:type="simple" linktype="goto" linkaction="prompt">
<prompt>0029-2
</prompt>
</link>
</contententry> . . .
</sub-contententry>
</contententry>
</sub-contententry> . . .
</contententry>
```

31.3.1.2 Table of contents alternative ***<contents-alt>***. The element ***<contents-alt>*** is used for configuration filtering of front matter material.

a. The components are:
   (1) Table of Contents ***<contents>*** (see 31.3.1.1.8)
b. DTD fragment:

```
<!ELEMENT contents-alt (contents)>
```

c. No attributes.

31.3.2 System/subsystem hierarchy ***<systemhierarchy>***. System/Subsystem hierarchy breaks down the end item work packages to the respective subsystems to the lowest defined work package subsystem item. Some work packages (i.e. general information) are intended for all subsystems and are not identified to any one subsystem item or group. The tagging of this type of IETM is grouped by system tags where the functional IETM is grouped by functional chapter tags.

a. The components are:
   (1) Revision Summary ***<revisionsummary>*** (required). The element provides a list of work packages by title that have been revised to a revised manual (see 31.3.1.1.1).
   (2) Front Cover ***<frntcover>*** (required). The element provides the data for the front cover of a TM (see 31.3.1.1.4).
   (3) Overall System Information ***<overallsystem>*** (required). The element defines the work packages used overall for the equipment in a system/subsystem hierarchy manual (see 31.3.2.1).
   (4) Work Packages ***<systembreakdown>*** (required – one or more). The element is used to define a system/subsystem structure (see 31.3.2.2).
   (5) Reference Work Packages ***<systemref>*** (required). The element provides the general reference work packages.

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(6) Parts Introductory Work Package ***<introwp>*** (required). The element provides the introductory work package (see 38.4.1).
(7) Repair Parts List Work Package ***<plwp>*** (required – one or more). The repair parts list work package contains lists and illustrations of all repair parts in accordance with the functional group code (see 38.4.2).
(8) Repair Parts For Special Tools List Work Package ***<stl_partswp>*** (optional). The element provides a work package when the special tool has repair parts that may be replaced at any maintenance level covered in the TM.(see 38.4.3).
(9) Kits Items List Work Package ***<kitswp>*** (optional – zero or more). The element provides separate work package for listing the kit items (see 38.4.4).
(10) Parts Bulk Item List Work Package ***<bulk_itemswp>*** (optional – zero or more). The element provides a work package for material that is used to make items (see 38.4.5).
(11) Special Tools List Work Package ***<stlwp>*** (optional – zero or more). The element provides a work package that contain lists and illustrations of all special tools (see 38.4.6).
(12) National Stock Number Index Work Package ***<nsnindxwp>*** (optional – zero or more). The element provides a work package that contains an index that lists the NSNs and each referenced figure number, and item number (see 38.5.1).
(13) Part Number Index Work Package ***<pnindxwp>*** (optional – zero or more). The element provides a work package that contains a part index that lists the part number, figure number, and item number for all part numbers (see 38.5.2) .
(14) Standard Information Reference Designator Index ***<refdesindxwp>*** (required). The element provides a work package that contains a cross reference of reference designators with illustration number and callout (see 38.5.3).
(15) Reporting Errors and Recommending Improvements ***<da2028>*** (required – one or more). The element provides the DA Form 2028-2 for reporting errors and recommending equipment improvements (see 34).

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

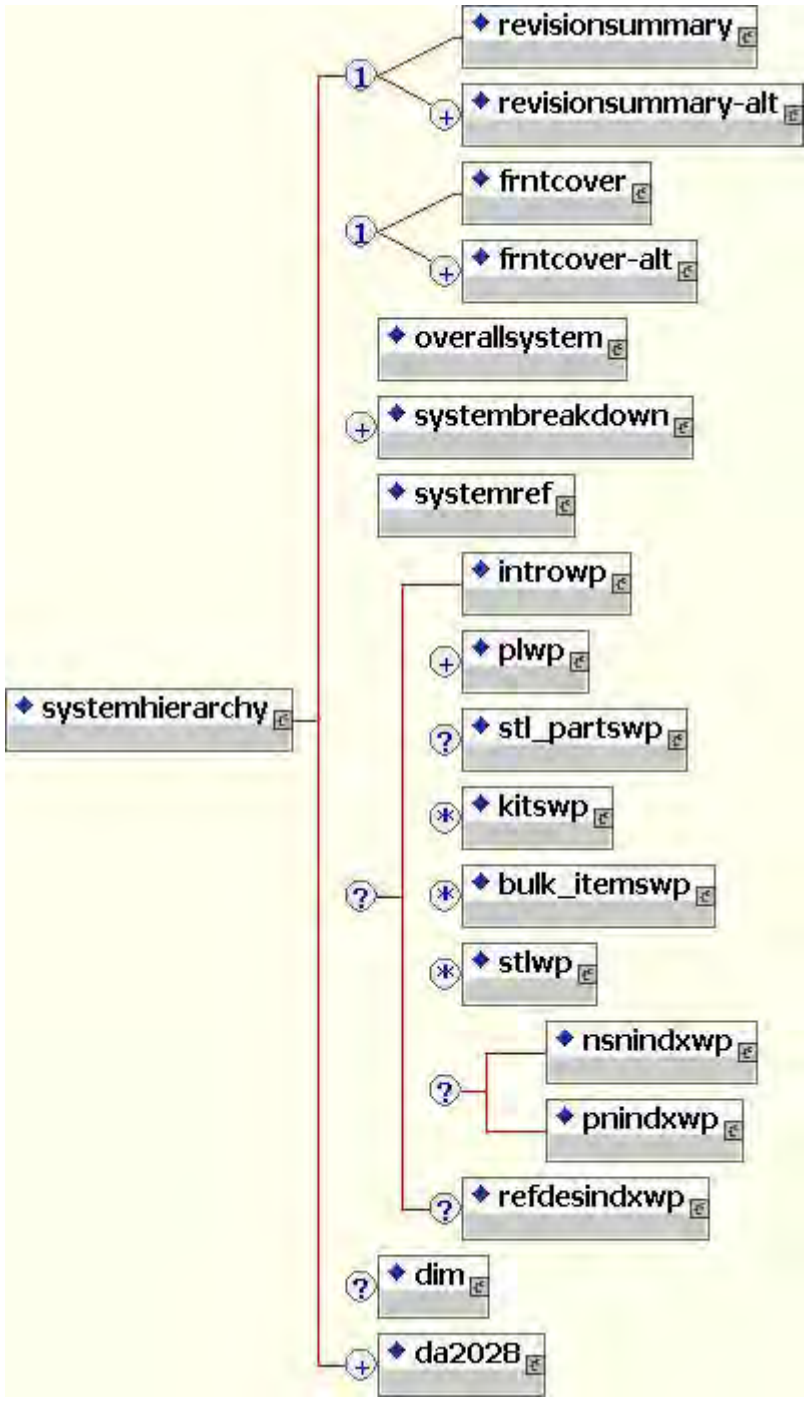

FIGURE 87. System/subsystem hierarchy <systemhierarchy> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT systemhierarchy (revisionsummary, frntcover, overallsystem,
 systembreakdown+, systemref, (introwp, plwp+, stl_partswp?, kitswp*,
bulk_itemswp*, stlwp*, (nsnindxwp, pnindxwp)?, refdesindxwp?)?, da2028+)>
```


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d. No attributes.

31.3.2.1 Overall system information ***<overallsystem>***. The element ***<overallsystem>*** is used for the overall system information that defines the work packages used overall for the equipment in a system/subsystem hierarchy manual.

a. The components are:

(1) Title ***<title>*** (required). Title for the over all system (see 49.1.1.3).

(2) How to use ***<howtouse>*** (required). Element provides the user any special information on how to read and use information and procedures (see 31.3.1.1.2).

(3) General Information Work Package ***<ginfowp>*** (required). The element provides general information, reference statements and standard statements (see 34.1.1).

(4) PMCS Introduction Work Package ***<pmcsintrowp>*** (optional). The element provides a PMCS introduction(see 37.5).

(5) PMCS Work Package ***<pmcswp>*** (optional). The element provides the required data to for a PMCS work package (see 37.6).

(6) Equipment Description and Data Work Package ***<descwp>*** (optional – zero or more). The element provides information to support the testing and troubleshooting procedures in a technical description work package (see 34.1.2)

(7) Theory of Operation Work Package ***<thrywp>*** (optional – zero or more). The element provides a functional description on how the equipment and its components function and interface (see 34.1.3).

(8) Description and Use of Controls and Indicators Work Package ***<ctrlindwp>*** (optional – zero or more). The element provides the description and use of all system and equipment controls and indicators (see 35.2).

(9) Aviation Troubleshooting Introduction Work Package ***<tsintrowp>*** (optional). The element provides general information needed to supplement the troubleshooting procedures (see 36.14).

(10) Troubleshooting Index Work Package ***<tsindxwp>*** (optional). The element provides a reference index to referenced to the applicable testing and troubleshooting WP, maintenance WP, or required corrective action(see 36.6).

b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

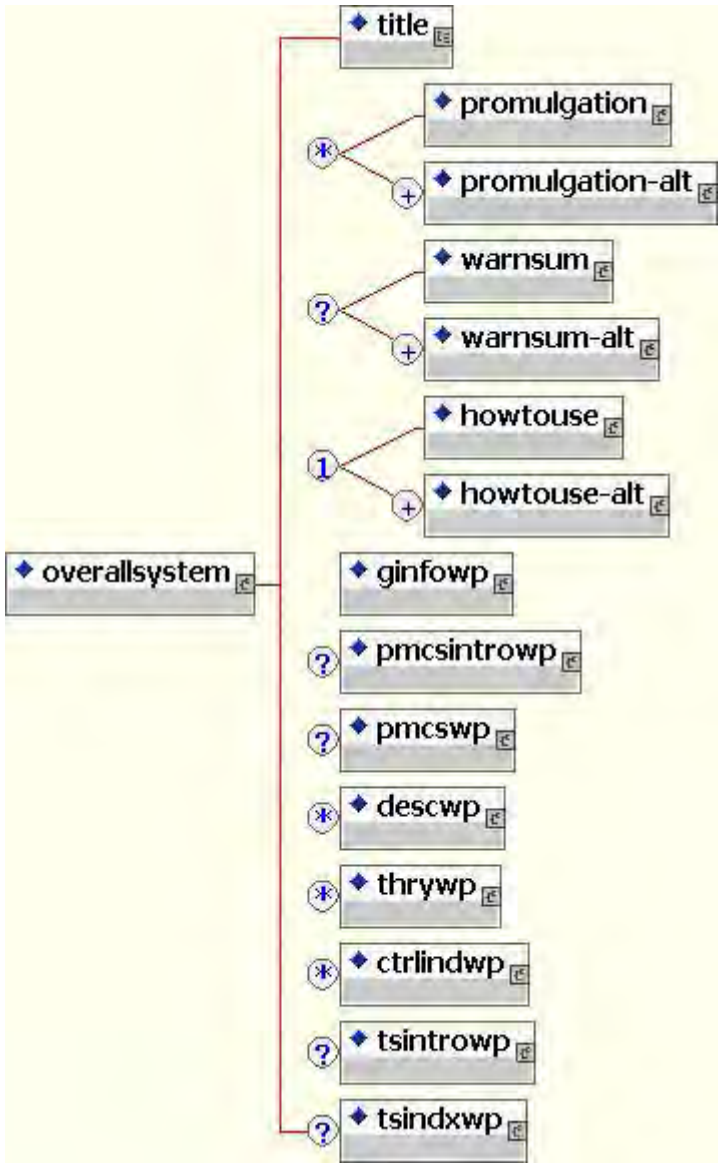

FIGURE 88. Overall system information <overallsystem> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT overallsystem (title, howtouse, ginfowp, pmcsintrowp?, pmcswp?,
 descwp*, thrywp*, ctrlindwp*, tsintrowp?, tsindxwp?)>
```

d. No attributes.

31.3.2.2 System/subsystem work packages ***<systembreakdown>***. The element ***<systembreakdown>*** is used for the system/subsystem work packages that defines a system/subsystem structure. Any work package related to the system is located under this element. Subsystem levels are recursively referenced down to the lowest assemble level.

a. The components are:

(1) System/Subsystem Nomenclature.***<systemnomen>*** (required). The element provides the nomenclature for the system/subsystem and identification numbers (see 32.3.2.4.1).

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(2) Equipment Description and Data Work Package ***<descwp>*** (required – one or more). The element provides information to support the testing and troubleshooting procedures in a technical description work package. (see 34.1.2)
(3) Theory of Operation Work Package ***<thrywp>*** (optional – zero or more). The element provides a functional description on how the equipment and its components function and interface. (see 34.1.3)
(4) Description and Use of Controls and Indicators Work Package ***<ctrlindwp>*** (optional – zero or more). The element provides the description and use of all system and equipment controls and indicators. (see 35.2)
(5) Operating Under Usual Conditions Work Package ***<opusualwp>*** (optional – zero or more). The element provides step-by-step instructions for equipment and auxiliary equipment operation in operation under usual or normal conditions modes.(see 35.3).
(6) Operating Under Unusual Conditions Work Package ***<opunuwp>*** (optional – zero or more). The element provides step-by-step instructions for equipment and auxiliary equipment operation in all under unusual condition modes. (see 35.4).
(7) Emergency Work Package ***<emergencywp>*** (optional – zero or more). The element provides emergency procedures using, but not limited to, the operating and equipment shutdown. (see 35.5).
(8) Stowage And Decal/Data Plate Guide Work Package ***<stowagewp>*** (optional – zero or more). The element provides lists and illustrates the location of all applicable COEI, BII, AAL items, decals and data plates.(see 35.6).
(9) On-Vehicle Equipment Loading Plan Work Package ***<eqploadwp>*** (optional – zero or more). The element provides a loading plan that will be prepared by the contracting activity.(see 35.7).
(10)
(a) Troubleshooting Index Work Package ***<tsindxwp>*** (optional – zero or more). The element provides a reference index to referenced to the applicable testing and troubleshooting WP, maintenance WP, or required corrective action. (see 36.6)
(b) Preshop Analysis Work Package ***<pshopanalwp>*** (optional). The element provides data used for testing or inspecting an item. (see 36.12).
(c) Component Checklist Work Package ***<compchklistwp>*** (optional). The element provides a blank form on which to list all information that is required prior to preshop analysis.(see 36.13).
(d) Aviation Troubleshooting Introduction Work Package ***<tsintrowp>*** (required). The element provides general information needed to supplement the troubleshooting procedures.(see 36.14).
(e) Technical Information and Description Work Package ***<techdescwp>*** (optional – zero or more). The element provides technical description and other supporting information about a system or subsystem/assembly/component. (see 36.15).
(f) Troubleshooting Index Work Package ***<tsindxwp>*** (optional – zero or more). The element provides a reference index to referenced to the applicable testing and troubleshooting WP, maintenance WP, or required corrective action. (see 36.6)
(11)
(a) Troubleshooting Procedures Work Package ***<tswp>*** (optional – zero or more). The element provides start-to-finish troubleshooting procedures, which result in fault isolation and rectification and ultimately either a return to readiness status or referral to a higher maintenance level. (see 36.9).
(b) Operational Checkout Work Package ***<opcheckwp>*** (optional – zero or more). The element provides operational checkout procedures that subject prescribed conditions to determine that they will function in accordance with predetermined test parameters. (see 36.8).
(c) Combined Operational Checkout and Troubleshooting Work Package ***<opcheck-tswp>*** (optional – zero or more). The element provides a combined operational checkout and troubleshooting procedures to verify proper operation to prescribed. (see 36.10).
(d) Combined Operational Checkout and Troubleshooting Work Package ***<diagnosticwp>*** (optional – zero or more). The element provides a combined operational checkout and troubleshooting procedures to verify proper operation to prescribed. (see 36.10).

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(12) Aviation Troubleshooting Introduction Work Package ***<tsintrowp>*** (required). The element provides general information needed to supplement the troubleshooting procedures.(see 36.14).
(13) Technical Information and Description Work Package ***<techdescwp>*** (optional – zero or more). The element provides technical description and other supporting information about a system or subsystem/assembly/component. (see 36.15).
(14) Service Upon Receipt Work Package ***<surwp>*** (optional – zero or more). The element provides information required for the user to ensure that the equipment will be adequately inspected, serviced and operationally tested before it is subjected to use.(see 37.3).
(15) Equipment/User Fitting Instructions Work Package ***<perseqpwp>*** (optional – zero or more). The element provides an equipment/user fitting instructions.(see 37.4).
(16) Preventive Maintenance Checks and Services Work Package ***<pmcswp>*** (optional – zero or more). The element provides the required data to perform Preventive Maintenance Checks and Services (PMCS) on the equipment.(see 37.6).
(17) Auxiliary Equipment Work Package ***<auxeqpwp>*** (optional – zero or more). The element provides maintenance instructions for peculiar support equipment. (see 37.21).
(18) Phased Maintenance InspectionsWork Package ***<pmiwp>*** (optional – zero or more). The element provides requirements for special inspections, overhaul and retirement schedule, and standards of serviceability applicable to the aircraft. **Aircraft only**. (see 37.17).
(19) Lubrication Instructions Work Package ***<lubewp>*** (optional – zero or more). The element provides the lubrication data required for equipment checks and maintenance.(see 37.9).
(20) ***<maintwp>*** (optional – zero or more). (see 37.6).
(21) ***<pmcswp>*** (optional – zero or more). (see 37.6).

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

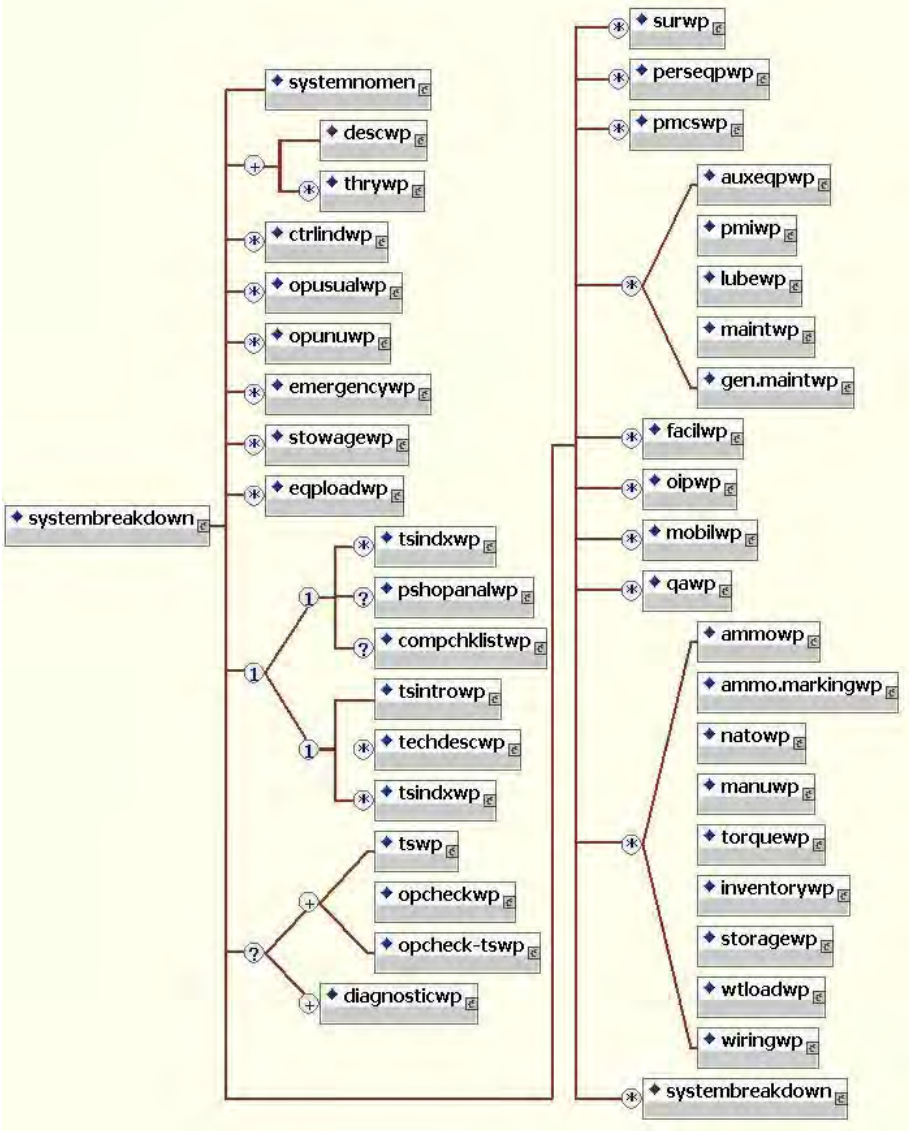

FIGURE 89. System/subsystem work packages <systembreakdown> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT systembreakdown (systemnomen, (descwp, thrywp*)+, ctrlindwp*,
opusualwp*, opunuwp*, emergencywp*, stowagewp*, eqploadwp*, ((tsindxwp*,
pshopanalwp?, compchklistwp?) | (tsintrowp, techdescwp*, tsindxwp*)),
((tswp | opcheckwp | opcheck-tswp)+ | diagnosticwp+)?, surwp*, perseqpwp*,
pmcswp*, (auxeqpwp | pmiwp | lubewp | maintwp)*, facilwp*, oipwp*,
mobilwp*, qawp*, (ammowp | ammo.markingwp | natowp | manuwp | torquewp
| inventorywp | storagewp | wtloadwp | wiringwp)*, systembreakdown*)>
```

d. No attributes.

31.3.2.2.1 System/subsystem nomenclature.***<systemnomen>***. The element defines the system/subsystem nomenclature and any identification numbers.

a. The components are:

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(1) Name ***<name>*** (required)(see 49.1.4.19).
(2) National Stock Number ***<nsn>*** (optional) (see 49.1.4.20).
(3) Part Number ***<partno>*** (optional) (see 49.1.4.21).
(4) Commercial and Government Entity Code (CAGEC) ***<cageno>*** (optional) (see 49.1.4.2).
(5) ***<modelno>*** (optional – zero or more) (see 49.1.4.18).
(6) Reference Designation***<refdes>*** (optional – zero or more) (see 38.4.2.1.3.2.10).
(7) Logistic Control Number (LCN) ***<lcn>*** (optional – zero or more) (see 31.3.2.2.1.1).
(8) Logistic Task Code ***<taskcode>*** (optional – zero or more)(see 31.3.2.2.1.2).

b. DTD fragment is graphically depicted.

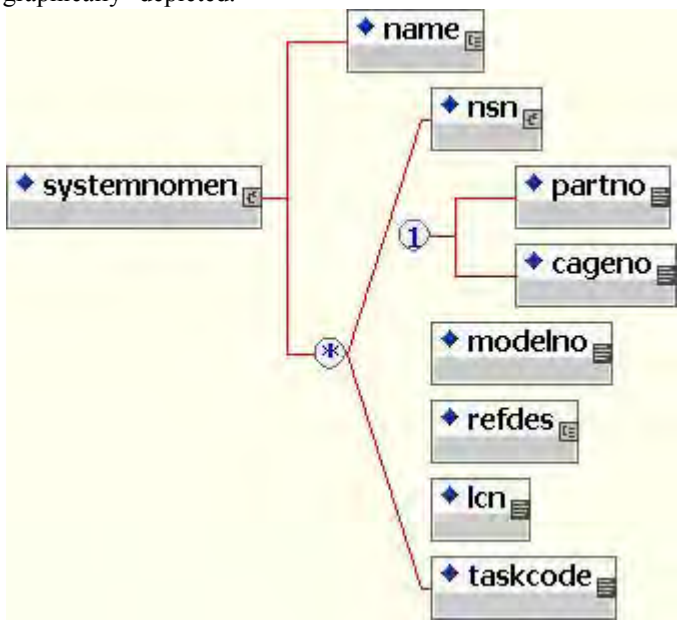

FIGURE 90. System/subsystem Nomenclature <systemnomen> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT systemnomen (name, (nsn |(partno, cageno) | modelno |
refdes | lcn | taskcode)*)>
```

d. No attributes.

31.3.2.2.1.1 Logistic control number (LCN) ***<lcn>***. The element ***<lcn>*** provides the Logistic Control Number (LCN) for additional search capability.

a. The components are:
(1) The element contains narrative text #PCDATA parsable character data.

b. DTD fragment:

```
<!ELEMENT lcn (#PCDATA)>
```

c. No attributes.

31.3.2.2.1.2 Logistic task code ***<taskcode>***. The element ***<taskcode>*** identifies information about the task generated from the Logistic Support Analysis (LSA).

a. The components are:
(1) The element contains narrative text #PCDATA parsable character data.

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b. DTD fragment:

```
<!ELEMENT taskcode (#PCDATA)>
```

c. No attributes.

31.4 Page base manual ***<paper.manual>*** . The element ***<paper.manual>*** is for all standard page-based technical manuals. Format style and requirements are prepared IAW MIL-STD-40051-2.

a. The components are:
   (1) Paper Front ***<paper.frnt>*** (required) (see 31.4.1).
   (2) General Information Chapter ***<gim>*** (required) (see 34).
   (3) Volume ***<volume>*** (optional) (see 31.10.1).
   (4) Back Matter of Volume ***<vol-rear>*** (optional) (see 31.10.2).
   (5) Operating Instructions ***<opim>*** (optional – zero or more) (see 35).
   (6) One or more is allowed but at least one of the following is required.
      (a) Maintenance Information ***<mim>*** (see 37).
      (b) Troubleshooting Information ***<tim>*** (required)(see 36.4).
   (7) Select either a Supporting Chapter or a Parts Information Chapter. **At least one is required.**
      (a) Supporting Information Chapter ***<sim>*** (see 40). If the author selects the Supporting Chapter, the following two chapters are optional.
         1 Parts Information Chapter ***<pim>*** (required)(see 38).
         2 Destruction Information Chapter ***<dim>*** (required)(see 39.1).
      (b) Parts Information Chapter ***<pim>*** (required)(see 38).
   (8) Rear ***<rear>*** (required)(see 31.4.2).

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

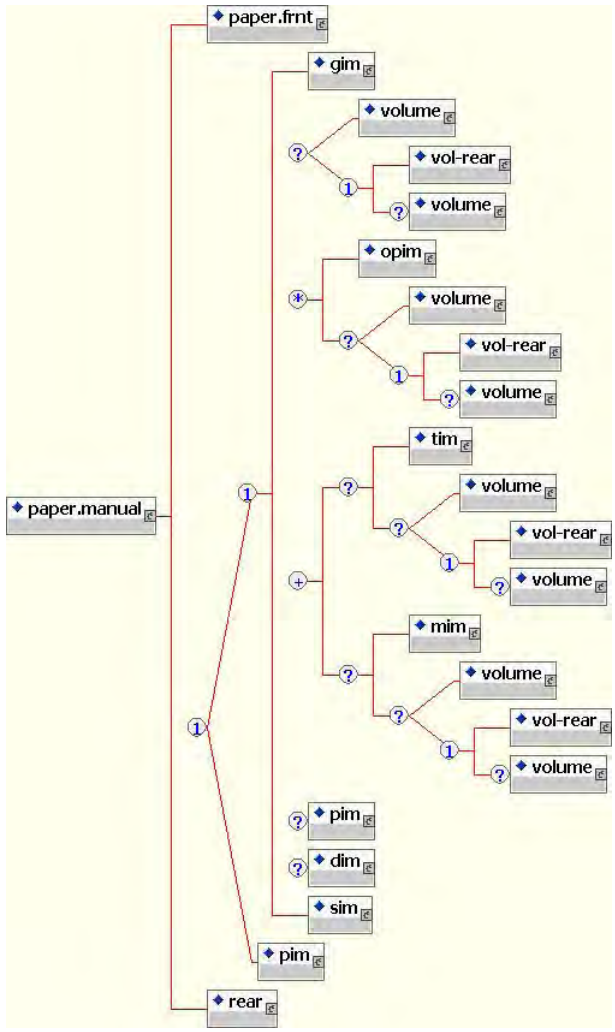

FIGURE 91. Page base manual <paper.manual> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT paper.manual (paper.frnt, ((gim, (volume | (vol-rear, volume?))?,
(opim, (volume | (vol-rear, volume?))?,)*,
((tim, (volume | (vol-rear, volume?))?,)?,
(mim, (volume | (vol-rear, volume?))?,)?)+,
pim?, dim?, sim) | pim), rear)>
<!ATTLIST paper.manual
    pubno           CDATA                   #IMPLIED
    maintitl        CDATA                   #REQUIRED
    maintlvls       (10 | 12 | 13 | 14
                    | 20 | 23 | 24 | 30
                    | 34 | 40 | avum-avim
                    | dmwr | dmwr-nmwr |
                    nmwr | NA)                #REQUIRED
    rpstl           (yes | no)                'no'
    dmwr-inclus     (parts | parts-tools)     #IMPLIED
    revno           CDATA                     #REQUIRED
    multivolume     (yes | no)                'no'
    fit.paper.size  (pocket | logbook |
                    standard | double )      'standard'
    security        (uc | fouo | c | s|ts )   #IMPLIED>
```

d. Unique attributes:
   (1) **pubno** – Publication number attribute specifies the technical manual publication number
   (2) **maintitl** – Maintenance Level Title supplies a literal version of the title for the maintenance-level. (required)
   (3) **maintlvls** – Maintenance Level specifies the maintenance level(s) authorized to use this manual; this attribute value is used in the style sheet to supply the literal expression of the TM's maintenance level. (required)
   (4) **rpstl** – Specifies whether or not the manual includes a RPSTL among its appendixes. The default value is "no".
   (5) **dmwr-inclus** – Specifies whether a DMWR/NMWR includes parts only or parts and tools.
   (6) **revno** – The overall revision number for the manual. (required)
   (7) **multivolume** – Is the manual broken into volumes. The default value is "no".
   (8) **fit.paper.size** – Fit paper size by selecting an attribute value from the list. The default value is "standard".

e. Common attributes.
   (1) **security** – Security classification (optional) (see 49.2.13).

31.4.1 Paper front matter ***<paper.frnt>***. Front matter contains all front matter of a technical manual and occurs before the first chapter of the manual. Format style and requirements are prepared for a standard page-oriented presentation. The following paragraphs describe the allowable front matter.

a. The components are:
   (1) Front cover ***<frntcover>*** (required). Element provides the data for a front cover for a TM.(see 31.3.1.1.4)
   (2) ***<promulgation>*** (optional – zero or more). Element provides the promulgation letter for a TM. (see 31.3.1.1.6)
   (3) Safety ***<warnsum>*** (optional). Element provides the explanation of the safety warning icons in the TM. (see 31.4.1.2)
   (4) Change sheet ***<chgsheet>*** (required). Element provides the change sheet required to appear in a changed document. (see 31.4.1.3)
   (5) List of effective work packages***<loepwp>*** (optional). Element provides the user a list of effective work packages that lists the latest work packages in the TM.(see 31.4.1.4)

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(6) Title block page ***<titleblk>*** (optional). Element provides the title block material from the front cover and additional data. (see 31.10.1.1)
(7) Table of contents ***<contents>*** (optional). Element provides the data to construct a table of contents. (see 31.3.1.1.8)
(8) How to use ***<howtouse>*** (optional). Element provides the user any special information on how to read and use information and procedures.(see 31.3.1.1.2)

b. DTD fragment is graphically depicted.

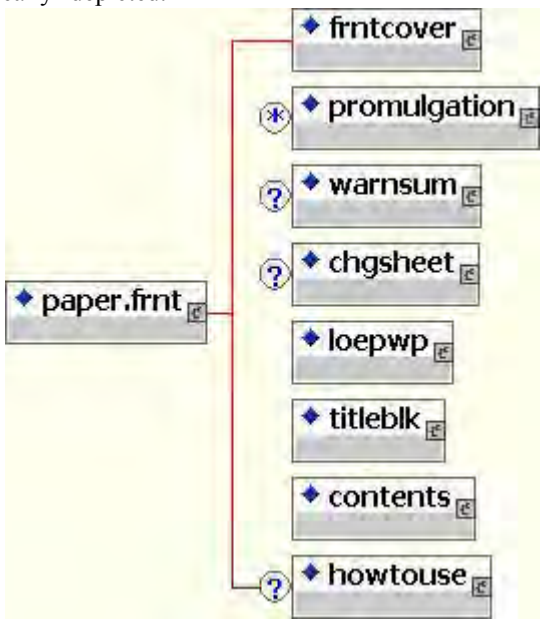

FIGURE 92. Paper front matter <frntcover> DTD hierarchy.

c. DTD fragment

```
<!ELEMENT paper.frnt (frntcover, promulgation*, warnsum?,
 chgsheet?, loepwp, titleblk, contents, howtouse?)>
```

d. No attributes:

31.4.1.1 Front cover ***<frntcover>***. The element ***<frntcover>*** is used for the front cover of a paper manual TM.

a. The components are:
(1) Technical Manual Title ***<tmtitle>*** (optional – zero or one). Element provides the elements that identify a manual.(see 31.3.1.1.4.1)
(2) Graphic ***<graphic>*** (optional – zero or one). Element provides the graphic for the front cover.(see 44.2)
(3) Official notices ***<notices>*** (required). Element provides the notices used on the front cover.(see 31.3.1.1.4.2)
(4) Service nomenclature ***<servnomen>*** (required). Element provides the service nomenclature of the proponent activity.(see 31.3.1.1.4.3)
(5) Date ***<date>*** (required). Element provides the date of the Publication for the TM.

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

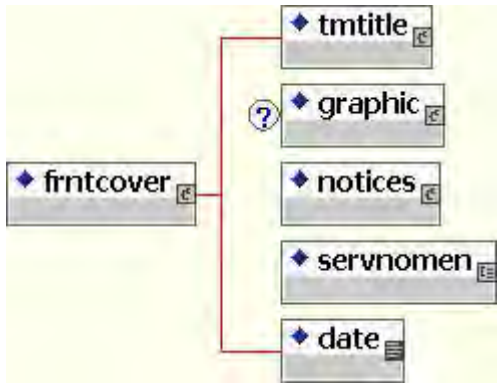

FIGURE 93. Front cover <frntcover> DTD hierarchy..

c. DTD fragment:

```
<!ELEMENT frntcover (tmtitle, graphic?, notices, servnomen, date)>
<!ATTLIST frntcover
     changeref          IDREFS                          #IMPLIED
     comment            CDATA                           #IMPLIED
     delchlvl           CDATA                           #IMPLIED
     inschlvl           CDATA                           #IMPLIED
     security           (uc | fouo | c | s | ts)        #IMPLIED>
```

d. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

31.4.1.1.1 Example of a front cover ***<frntcover>***. The following contains two examples of a XML instance for a front cover. Example (a) contains the tags and full text. Example (b) uses tags and boilerplate entities to replace standard statements contained in a front cover of a TM. Example (c) contains the stylesheet output of the front cover ***<frntcover>***.

a. Example of an XML document instance fragment containing the tags and full text for ***<frntcover>***:

```
<frntcover  security>
<tmtitle>
<tminfono>
<servbranch  service="army"/>
<tmno>TM  3-6665-339-10
</tmno>
</tminfono>
<prtitle>
<sysnomen>
<name>TECHNICAL  MANUAL
<brk/>OPERATOR'S  MANUAL  FOR  NUCLEAR-BIOLOGICAL-CHEMICAL
RECONNAISSANCE  SYSTEM  (NBCRS)
</name>
<modelno>FOX  M93A1
</modelno>
<nsn>
<fsc>  6665
</fsc>
<niin>  01-372-1303
</niin>
```


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</nsn>
</sysnomen>
</prtitle>
</tmtitle>
<notices>
<disclos>
<title>DISCLOSURE NOTICE
</title>
<text>This information is furnished upon the condition that it will not be released to another nation without the specific authority of the Department of the Army of the United States, that it will be used for military purposes only, that individual or corporate rights originating in the information, whether patented or not, will be respected, that the recipient will report promptly to the United States, any known or suspected compromise, and that the information will be provided substantially the same degree of security afforded it by the Department of Defense of the United States. Also, regardless of any other markings on the document, it will not be downgraded or declassified without written approval of the originating United States agency.
</text>
</disclos>
<dist>
<c.statement>
<title>DISTRIBUTION STATEMENT C
</title>
<text>Distribution authorized to U.S. government agencies and their contractors.
</text>
<adminops/>
<text>This publication is Administrative-Operational Use required for administrative and operational purposes, as determined on
</text>
<reasondate>22 October 1990
</reasondate>
<text>Other requests for this document must be referred to Commander,
</text>
<releaseagent>U.S. Army Chemical Research Development and Engineering Center, ATTN: SMCRR-MAT, Aberdeen Proving Ground, MD 21010-5423
</releaseagent>
<text>.
</text>
</c.statement>
</dist>
<export>
<title>WARNING
<title>
<text>This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C., Sec 2751, et. seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et. seq. Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.
</text>
</export>
<destr>
<title>DESTRUCTION NOTICE -

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</title>
<text> Destroy by any method that will prevent disclosure of
contents or reconstruction of the document.
</text>
</destr>
</notices>
<servnomen>
<emphasis emph="bold">HEADQUARTERS, DEPARTMENT OF THE ARMY
</emphasis>
</servnomen>
<date julian="1 JUNE 1996">1 JUNE 1996
</date>
<frntcover>
```

b. Example of an XML document instance fragment containing tags and the boiler plate entities to replace standard statements contained in the front cover ***<frntcover>***. The general entities disclosure notice – ***&notices.disclos;***, Distribution Statement C***&notices.dist.c.statement;***, and Export Control Notice Warning ***&notices.export;*** are used as boiler plates for standard statements.

```
<frntcover security>
<tmtitle>
<tminfono>
<servbranch service="army"/>
<tmno>TM 3-6665-339-10
</tmno>
</tminfono>
<prtitle>
<sysnomen>
<name>TECHNICAL MANUAL
<brk/>OPERATOR'S MANUAL FOR NUCLEAR-BIOLOGICAL-CHEMICAL
RECONNAISSANCE SYSTEM (NBCRS)
</name>
<modelno>FOX M93A1
</modelno>
<nsn>
<fsc> 6665
</fsc>
<niin> 01-372-1303
</niin>
</nsn>
</sysnomen>
</prtitle>
</tmtitle> &notices.disclos; &notices.dist.c.statement; &notices.export;
<destr>
<title>DESTRUCTION NOTICE -
</title>
<text> Destroy by any method that will prevent disclosure of
contents or reconstruction of the document.
</text>
</destr>
</notices>
<servnomen>
<emphasis emph="bold">HEADQUARTERS, DEPARTMENT OF THE ARMY
</emphasis>
</servnomen>
<date julian="1 JUNE 1996">1 JUNE 1996
</date>
<frntcover>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. Example of the stylesheet output for *<frntcover>*:

TM 3-6665–339–10

TECHNICAL MANUAL

OPERATOR'S MANUAL FOR NUCLEAR-BIOLOGICAL-CHEMICAL RECONNAISSANCE SYSTEM (NBCRS)

FOX M93A1 (NSN 6665-01–372–1303)

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WARNING – This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C., Sec 2751, et. seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et. seq. Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

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HEADQUARTERS, DEPARTMENT OF THE ARMY

1 JUNE 1996

FIGURE 94. Example of a front cover <frntcover>.

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Check the source to verify that this is the current version before use.

d. Example of an XML document instance fragment for a phased maintenance TM front cover<***frntcover***>:

```
<frntcover>
<tmtitle>
<tminfono>
<servbranch service="army"/>
<tmno>TM 1-1520-238-PM
</tmno>
</tminfono>
<prtitle>
<sysnomen pretext="TECHNICAL MANUAL">
<name>PHASED MAINTENANCE INSPECTION CHECKLIST
</name>
<modelno>AH-64A HELICOPTER
</modelno>
<nsn>
<fsc>6665
</fsc>
<niin>01-372-1303
</niin>
</nsn>
</sysnomen>
</prtitle>
</tmtitle>
<notices>
<a.statement>
<title>DISTRIBUTION STATEMENT A. -
</title>
<text>Approved for public release; distribution is unlimited.
</text>
</a.statement>
</dist>
</notices>
<servnomen>HEADQUARTERS, DEPARTMENT OF THE ARMY
</servnomen>
<date julian="28 FEBRUARY 2002">28 FEBRUARY 2002
</date>
</frntcover>
```

e. Example of the stylesheet output for phased maintenance TM front cover <***frntcover***>:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

*TM 1–1520–238–PM

TECHNICAL MANUAL

PHASED MAINTENANCE INSPECTION CHECKLIST FOR

ARMY
AH-64A HELICOPTER

*TM 1–1520–238–PM dated 28 February 2002 superseded TM 1–1520–238–PM dated 20 June 1994, including all changes.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

HEADQUARTERS, DEPARTMENT OF THE ARMY

28 FEBRUARY 2002

FIGURE 95. Example of a phased maintenance TM front cover front cover <frntcover>.

f. Example of an XML document instance fragment for DMWR front cover ***<frntcover>***:

```
<frntcover>
<tmtitle>
<tminfono>
<servbranch service="army"/>
<tmno>DMWR 11-5895-532-2
</tmno>
</tminfono>
<prtitle>
<sysnomen pretext="DEPOT MAINTENANCE WORK REQUIREMENT">
<name>INTERROGATOR SETS
</name>
<modelno>AN/TPX-46(V)1
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</modelno>
<nsn>
*<fsc>*5895
</fsc>
*<niin>*00-423-1693
</niin>
</nsn>
*<modelno>*AN/TPX-46(V)2
</modelno>
<nsn>
*<fsc>*5895
</fsc>
*<niin>*00-423-1694
</niin>
</nsn>
*<modelno>*AN/TPX-46(V)3
</modelno>
<nsn>
*<fsc>*5895
</fsc>
*<niin>*00-423-1696
</niin>
</nsn>
</sysnomen>
</prtitle>
</tmtitle>
<notices>
<avail>
*<title>*AVAILABILITY STATEMENT. -
</title>
*<text>*This publication is not available through the APD distribution center. This publication is available through
</text>
<proponent>
*<name>*US Army Communications-Electronics Command,
</name>
<address>
*<city>*Fort Monmouth
</city>
*<state>*NJ
</state>
*<zip>*07703-5007
</zip>
</address>
</proponent>
<text>.
</text>
</avail>
<disclos>
*<title>*DISCLOSURE NOTICE -
</title>
*<text>*This information is furnished upon the condition that it will not be released to another nation without the specific authority of the Department of the Army of the United States, that it will be used for military purposes only, that individual or corporate rights originating in the information, whether patented or not, will be respected, that

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```
the recipient will report promptly to the United States, any known
or suspected compromise, and that the information will be provided
substantially the same degree of security afforded it by the Department
of Defense of the United States.  Also, regardless of any other markings
on the document, it will not be downgraded or declassified without
written approval of the originating United States agency.
</text>
</disclos>
<dist>
<d.statement>
<title>DISTRIBUTION  STATEMENT  D.  -
</title>
<text>Distribution authorized to the Department of Defense and
U.S. DOD contractors only.   as determined.
</text>
<crittech/>
<text>This publication is critical technology, on
</text>
<reasondate>20  May  1998.
</reasondate>
<text>Other requests for this document shall be referred to
</text>
<releaseagent>AMSELL-LC-LM-LT, Fort Monmouth, NJ 07702-5007
</releaseagent>
<text>.
</text>
</d.statement>
</dist>
<destr>
<title>DESTRUCTION  NOTICE.  -
</title>
<text>Destroy by any method that will prevent disclosure of
contents or reconstruction of the document.
</text>
</destr>
</notices>
<servnomen> US ARMY COMMUNICATIONS-ELECTRONICS COMMAND, FORT MONMOUTH, NJ
</servnomen>
<date julian="30 JUNE 1998">30  JUNE  1998
</date>
</frntcover>
```

g. Example of the stylesheet output for DMWR front cover<*frntcover*>:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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DMWR 11–5895–532–2

DEPOT MAINTENANCE WORK REQUIREMENT

FOR
INTERROGATOR SETS

AN/TPX-46(V)1
(NSN 5895-00-423-1693)
AN/TPX-46(V)2
(NSN 5895-00-423-1694)
AN/TPX-46(V)3
(NSN 5895-00-423-1696)

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US ARMY COMMUNICATIONS-ELECTRONICS COMMAND, FORT MONMOUTH, NJ

30 JUNE 1998

FIGURE 96. Example of a DMWR front cover <frntcover>.

h. Example of an XML document instance fragment for National Maintenance Work Requirement (NMWR) front cover ***<frntcover>***:

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```
<frntcover>
<tmtitle>
<tminfono>
<servbranch  service="army"/>
<tmno>NMWR  X-XXXX-XXX
</tmno>
</tminfono>
<prtitle>
<sysnomen>
<name>NATIONAL  MAINTENANCE  WORK  REQUIREMENT  CONTAINING
NATIONAL  MAINTENANCE  REPAIR  STANDARDS
</name>
</sysnomen>
<sysnomen  pretext="FOR">
<name>INTERROGATOR  SETS
</name>
<modelno>AN/TPX-46(V)1
</modelno>
<nsn>
<fsc>5895
</fsc>
<niin>00-423-1693
</niin>
</nsn>
<modelno>AN/TPX-46(V)2
</modelno>
<nsn>
<fsc>5895
</fsc>
<niin>00-423-1694
</niin>
</nsn>
</sysnomen>
</prtitle>
</tmtitle>
<notices>
<avail>
<title>AVAILABILITY  STATEMENT.  -
</title>
<text>This publication is not available through the APD distribution
center.   This publication is available through
</text>
<proponent>
<name>US Army Communications-Electronics Command,
</name>
<address>
<city>Fort  Monmouth
</city>
<state>NJ
</state>
<zip>07703-5007
</zip>
</address>
</proponent>
<text>.
</text>
```


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```
</avail>
<disclos>
<title>DISCLOSURE  NOTICE  -
</title>
<text>This information is furnished upon the condition that it will not
be released to another nation without the specific authority of the
Department of the Army of the United States, that it will be used for
military purposes only, that individual or corporate rights originating
in the information, whether patented or not, will be respected, that
the recipient will report promptly to the United States, any known
or suspected compromise, and that the information will be provided
substantially the same degree of security afforded it by the Department
of Defense of the United States.  Also, regardless of any other markings
on the document, it will not be downgraded or declassified without
written approval of the originating United States agency.
</text>
</disclos>
<dist>
<d.statement>
<title>DISTRIBUTION  STATEMENT  D.  -
</title>
<text>Distribution authorized to the Department of Defense and
U.S. DOD contractors only.   as determined.
</text>
<crittech/>
<text>This publication is critical technology, on
</text>
<reasondate>20  May  1998.
</reasondate>
<text>Other requests for this document shall be referred to
</text>
<releaseagent>AMSELL-LC-LM-LT, Fort Monmouth, NJ 07702-5007
</releaseagent>
<text>.
</text>
</d.statement>
</dist>
<destr>
<title>DESTRUCTION  NOTICE.  -
</title>
<text>Destroy by any method that will prevent disclosure of
contents or reconstruction of the document.
</text>
</destr>
</notices>
<servnomen> US ARMY COMMUNICATIONS-ELECTRONICS COMMAND, FORT MONMOUTH, NJ
</servnomen>
<date julian="30 JUNE 1998">30  JUNE  1998
</date>
</frntcover>
```

i. Example of the stylesheet output for NMWR front cover ***<frntcover>***:

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NMWR X-XXXX-XXX

NATIONAL MAINTENANCE WORK REQUIREMENT
CONTAINING
NATIONAL MAINTENANCE REPAIR STANDARDS
FOR

INTERROGATOR SETS
AN/TPX-46(V)1 (NSN 5895-00-423-1693) (EIC IZA)
AN/TPX-46(V)2 (NSN 5895-00-423-1694) (EIC IZB)

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US ARMY COMMUNICATIONS-ELECTRONICS
COMMAND, FORT MONMOUTH, NJ
2 JUNE 1999

FIGURE 97. Example of a NMWR front cover <frntcover>.

31.4.1.2 Warning summary ***<warnsum>***. The element ***<warnsum>*** provides a warning summary for the TM to contain the warnings. The warning summary appears on the first right-hand page immediately after the front cover. The warning summary includes first aid data ***<first_aid>***, explanations of all general safety warning ***<safety>*** used in the manual. It also include descriptions of the general safety warnings ***<warninfo>*** and hazard icons and hazardous materials ***<hazmat>*** that have major impact throughout the manual.

a. The components are:
   (1) General paragraph ***<para>*** (required). Element provides information on what the warning summary contains. (see 49.1.1.5)
   (2) First aid data ***<first_aid>*** (required). Element provides the data for first aid for the TM. (see 31.4.1.2.1)

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(3) Safety ***<safety>*** (optional). Element provides the explanation of the safety warning icons in the TM. (see 31.4.1.2.2)
(4) General warning information ***<warninfo>*** (required). Element providesthe description of the general warnings that are included in the TM. safety. (see 31.4.1.2.3)
(5) Hazard icons and hazardous materials ***<hazmat>*** (optional). Element provides the user the different types of hazards that will be encountered in operating and maintaining the weapon system or equipment covered within the TM. (see 31.4.1.2.4)

b. DTD fragment is graphically depicted.

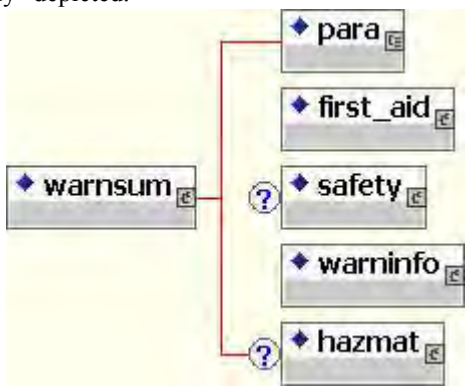

FIGURE 98. Warning summary <warnsum> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT warnsum (para, first_aid, safety?, warninfo, hazmat?)>
<!ATTLIST warnsum
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)        #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
     tocentry         (0 | 1 | 2 | 3 | 4 | 5)         '2'
```

d. Unique attributes:
(1) **tocentry** - Defines the indenture level in the TOC.

e. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.4.1.2.1 First aid ***<first_aid>***. The element ***<first_aid>*** provides first aid data in the warning summary. The first paragraph references FM 4-25.11, First Aid. Any additional first aid data not described in FM 4-25.11 is described in this section.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

a. The components are:
   (1) Title ***<title>*** (required). Title for the first aid section (see 49.1.1.3).
   (2) Figure ***<figure>*** (optional – zero or more) (see 44.1.1) or conditional figure ***<figure-alt>*** (optional – zero or more) (see .48.2.1). Provides an illustration in the description of the first aid.
   (3) Contains a grouping of
      (a) Note(s) ***<note>*** (optional – zero or more). Enable data that is important enough to highlight as a note in the description of first aid (see 41.1.3).
      (b) Paragraph(s) ***<para>*** (optional – zero or more). Provides text in the description of first aid (see 49.1.1.5).
      (c) Primary level – titled paragraph ***<para0>*** (optional – zero or more) (see 49.1.1.8) or primary level – conditional – titled paragraph ***<para0-alt>*** (optional – zero or more) (see 48.2.1). Provides a title, a precondition and paragraphs of text in the description of first aid.
   (4) OR
      (a) Primary level – titled paragraph ***<para0>*** (required – one or more) (see 49.1.1.8) or primary level – conditional – titled paragraph ***<para0-alt>*** (required – one or more) (see 48.2.1). Provides a title, precondition and paragraphs of text in the description of first aid.

b. DTD fragment is graphically depicted.

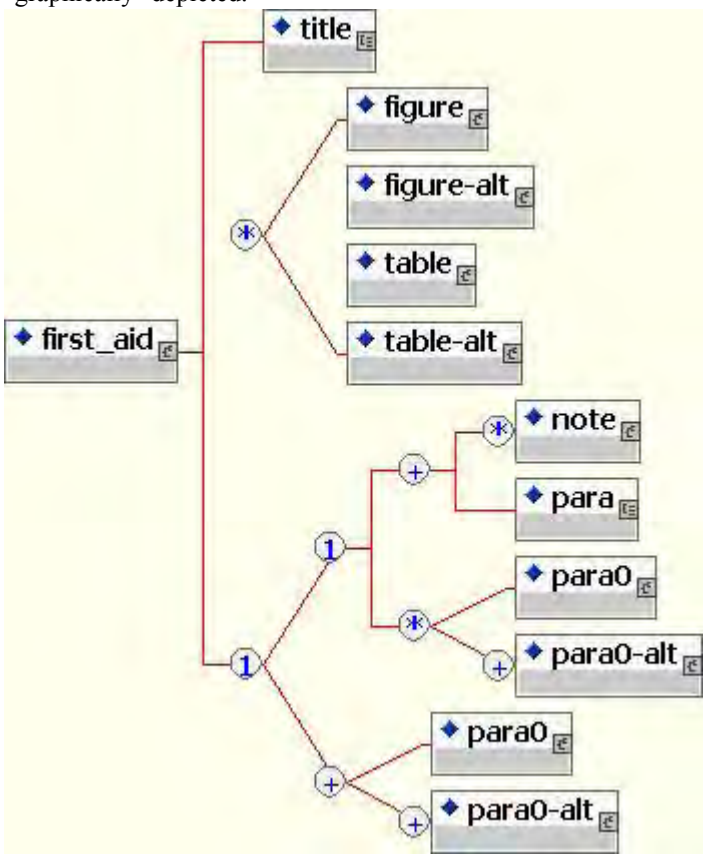

FIGURE 99. First aid <first_aid> DTD hierarchy..

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT first_aid ((title, (figure | figure-alt)*, (((note*, para)+,
     (para0 | para0-alt+)*) | (para0 | para0-alt+)+)))>
<!ATTLIST first_aid
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

e. Example of an XML fragment using "First Aid" in the Warning Summary.

<first_aid>
<title>`First Aid`
</title>
<para>`First aid can be defined as "urgent and immediate lifesaving and other measures, which can be performed for casualties by nonmedical personnel when medical personnel are not immediately available." Refer to` ***<extref docno="FM 4-25.11" pretext="FIRST AID">*** `for first aid data.`
</para>
</first_aid>

31.4.1.2.2 Safety ***<safety>***. The element ***<safety>*** provides explanations of all general safety warning icons.

a. The components are:
   (1) Title ***<title>*** (required). Title for the Safety section (see 49.1.1.3).
   (2) Safety icons ***<sfty-icons>*** (required – one or more). Provides the safety icon description for safety(see 31.4.1.2.2.1).

b. DTD fragment is graphically depicted.

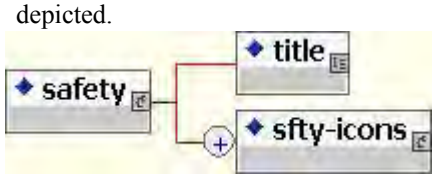

FIGURE 100. Safety <safety> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT safety (title, sfty-icons+)>
<!ATTLIST safety
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

e. Example of an XML fragment using "Safety" in the Warning Summary.

```
<safety>
<title>EXPLANATION OF SAFETY WARNING ICONS
</title>
<sfty-icons>
<symbol boardno="heavyobject_icon"/>
<sftydesc>
<title>HEAVY OBJECT
</title>
<text>- human figure stooping over heavy object shows physical
injury potential from improper lifting technique.
</text>
</sftydesc>
</sfty-icons>
</safety>
```

31.4.1.2.2.1 Safety icon section **<*sfty-icons*>**. The element **<*sfty-icons*>** provides the symbol and the safety description of the symbol.

a. The components of **<*sfty-icons*>** are:
   (1) Symbol **<*symbol*>** (required). Provides the symbol for the safety icon in the safety description. (see 44.3.1)
   (2) Safety description **<*sftydesc*>** (required). Provides the description of the safety icon in the safety description (see 31.4.1.2.2.1.1).

b. DTD fragment is graphically depicted.

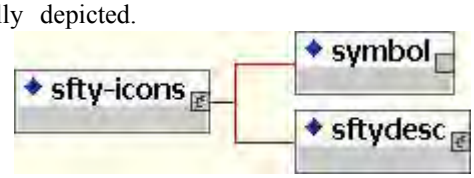

FIGURE 101. Safety icon section <sfty-icons> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT sfty-icons (symbol, sftydesc)>
<!ATTLIST sfty-icons
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo |c | s | ts)         #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (3) **comment** – Change information (optional) (see 49.2.9).
   (4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (6) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (7) **security** – Security classification (optional) (see 49.2.13).
   (8) **skilltrk** – Skill level (optional) (see 49.2.2).

31.4.1.2.2.1.1 Safety description ***<sftydesc>***. The element ***<sftydesc>*** provides a description of the hazardous condition associated with a safety icon.

a. The components are:
   (1) Title ***<title>*** (required). Title of the safety icon. (see 49.1.1.3).
   (2) Text ***<text>*** (required). Provides the description of the hazardous condition associated with of the safety icon.(see 49.1.1.18).

b. DTD fragment for :

```
<!ELEMENT sftydesc (title, text)>
<!ATTLIST sftydesc
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo |c | s | ts)         #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

c. Common attributes:
   (1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (3) **comment** – Change information (optional) (see 49.2.9).
   (4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (6) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (7) **security** – Security classification (optional) (see 49.2.13).
   (8) **skilltrk** – Skill level (optional) (see 49.2.2).

31.4.1.2.3 General warning information ***<warninfo>***. The element ***<warninfo>*** is a portion of the warning summary that contains general-purpose warnings, such as electricity or heavy objects. General warning information may contain a title, followed by at least one warning.

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a. The components are:
   (1) Title ***<title>*** (optional). Title for the general warning information section (see 49.1.1.3).
   (2) Warning(s) ***<warning>*** (required – one or more). Warning provides the icon-set and description for a general warning (see 41.1.1).
b. DTD fragment is graphically depicted.

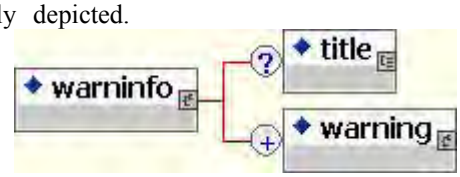

FIGURE 102. General warning information <warninfo> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT warninfo (title?, warning+)>
<!ATTLIST warninfo
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)
e. Example of an XML fragment using "General Warning Information" in the Warning Summary.

```
<warninfo>
<title>GENERAL SAFETY WARNINGS DESCRIPTION
</title>
<warning>
<warning.group>
<icon-set boardno="electrical"/>
<trim.para>Whenever possible shut off system power before
beginning work on equipment.
</trim.para>
</warning.group>
</warning>
</warninfo>
```

31.4.1.2.4 Hazard icons and hazardous materials ***<hazmat>***. The element ***<hazmat>*** provides explanations of any hazard icons in the TM and/or descriptions of hazardous materials used in performing procedures in the TM.

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Check the source to verify that this is the current version before use.

a. The components are:
   (1) Titles ***<title>*** (required). Titles for the "Explanation Of Hazardous Materials Icons" section and for the "Hazardous Materials Description" in Warning Summary (see 49.1.1.3).
   (2) Hazard icon section(s) ***<haz-icons>*** (required – one or more). Provides the hazard icons used in the warning summary (see 31.4.1.2.4.1.).
   (3) Hazardous material(s) ***<hazard>*** (required – one or more) (see 31.4.1.2.4.2).

b. DTD fragment is graphically depicted.

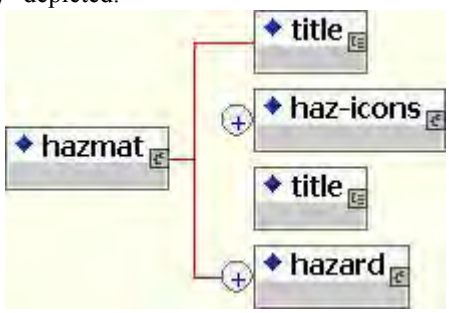

FIGURE 103. Hazard icons and hazardous materials <hazmat> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT hazmat (title, haz-icons+, title, hazard+)>
<!ATTLIST hazmat
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)        #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

e. Example of an XML fragment using “Hazard Icons And Hazardous Materials ” in the Warning Summary.

<hazmat>
<title>EXPLANATIO OF HAZARDOUS MATERIALS ICONS
</title>
<haz-icons>
<symbol boardno="chemical_icon"/>
<hazdesc>
<title>CHEMICAL
</title>

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Check the source to verify that this is the current version before use.

```
<tex> -drops of liquid on hand shows that the material will cause
burns or irritation to human skin or tissue
</text>
</hazdesc>
</haz-icons>
<title>HAZARDOUS MATERIALS DESCRIPTION
</title>
<hazard>
<hazid>DRYCLEANING SOLVENT P-D-680
</hazid>
<symbol boardno="chemical_icon"/>
<symbol boardno="vapor"/>
<par>P-D-680 solvent vapors are toxic. Avoid prolonged or repeated
breathing of vapors or solvent contact with skin. Use only with adequate
ventilation. Solvent is flammable and should not be used near open flame.
Fire extinguishers should be readily available when solvent is used.
</para>
</hazard>
</hazmat>
```

31.4.1.2.4.1 Hazard icon section <***haz-icons***>. The element <***haz-icons***> provides the hazard icons used in the TM which are defined in the warning summary.

a. The components:
   (1) Symbol <***symbol***> (required). It provides for the hazard icon.
   (2) Hazard description <***hazdesc***> (required). It provides for the description of the hazard icon (see 31.4.1.2.4.1.1).

b. DTD fragment is graphically depicted.

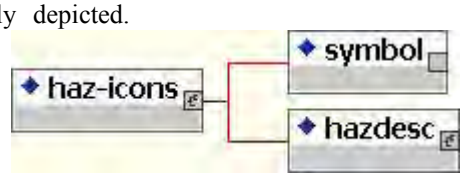

FIGURE 104. Hazard icon section <haz-icons> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT haz-icons (symbol, hazdesc)>
<!ATTLIST haz-icons
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)        #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

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(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.4.1.2.4.1.1 Hazard icon description ***<hazdesc>***. The element ***<hazdesc>*** provides a description of the hazardous condition associated with a hazard icon.

a. The components are:
(1) Title ***<title>*** (required). Title for the hazardous condition. (see 49.1.1.3)
(2) Text ***<text>*** (required). Provides the description of the hazardous condition associated with of the hazard icon.(see 49.1.1.18)

b. DTD fragment:

```
<!ELEMENT hazdesc (title, text)>
<!ATTLIST hazdesc
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)        #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

c. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.4.1.2.4.2 Hazard icon section ***<hazard>***. The element ***<hazard>*** provides the hazardous material warning appearing in the warning summary. It consists of the name or identification of the material, the associated hazard icons (usually multiple), and a description.

a. The components are:
(1) Identifying name of the hazardous material ***<hazid>*** (required). Provides the name of the hazard symbol.(see 31.4.1.2.4.2.1) is provided for the identifying name of the hazardous material.
(2) Symbol ***<symbol>*** (required – one or more). It provides for the hazard icon.
(3) Title ***<title>*** (optional). Title for the hazardous condition description (see 49.1.1.3).
(4) General paragraph(s) ***<para>*** (required – one or more) icon. Provides the description of hazardous material symbol (see 49.1.1.5).

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

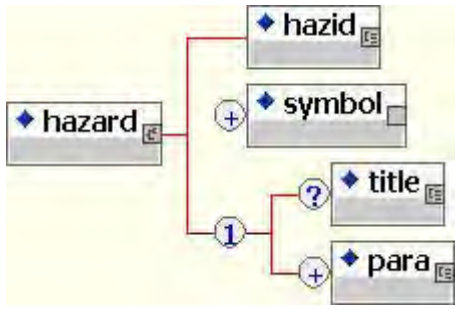

FIGURE 105. Hazard icon section <hazard> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT hazard (hazid, symbol+, (title?, para+))>
<!ATTLIST hazard
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

31.4.1.2.4.2.1 Identifying name of the hazardous material ***<hazid>***. The element ***<hazid>*** provides the name or other identification of a hazardous material.

a. The components are:
   (1) Optional zero or more of the elements listed in the DTD fragment for ***<hazid>***.
      (a) #PCDATA (see 9.3.2).
      (b) Emphasis ***<emphasis>*** (see 49.1.3.1).
      (c) Subscript ***<subscrpt>*** (see 49.1.3.4).
      (d) Superscript ***<supscrpt>*** (see 49.1.3.5).
      (e) Cross Reference ***<xref>*** (see 46.2.2).
      (f) External Reference ***<extref>*** (see 46.2.1).
      (g) Linking ***<link>*** (see 46.2.3).
      (h) Content Sensitive Help ***<help.info>*** (see 48.3.2.4).
      (i) Index Marker Reference ***<indxref>*** (see 31.4.2.2.3).
      (j) Term ***<term>*** (see 49.1.2.3.1.1).
      (k) Item Definition ***<term.def>*** (see 49.1.2.3.1).
      (l) Figure Callout ***<callout>*** (see 46.2.4.1).
      (m) Footnote ***<ftnote>*** (see 45.1.1).

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(n) Footnote Reference ***<ftnref>*** (see 45.1.2).
(o) Controls and Indicators Entry ***<ctrlind>*** (see 49.1.4.3).
(p) Reading On – Controls and Indicators ***<ctrlind-val>*** (see 49.1.4.4).
(q) Department of Defense Ammunition Code ***<dodac>*** (see 49.1.4.5).
(r) Lubricant ***<lubricant>*** (see 49.1.4.16).
(s) Symbol ***<symbol>*** (see 44.3.1).
(t) Torque Value or Limit ***<torque>*** (see 49.1.4.24).
(u) Voltage Measurement ***<voltage>*** (see 49.1.4.25).
(v) Null ***<null>*** (see 49.1.3.2).
(w) Text Change Information ***<change>*** (see 49.1.3.7).

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

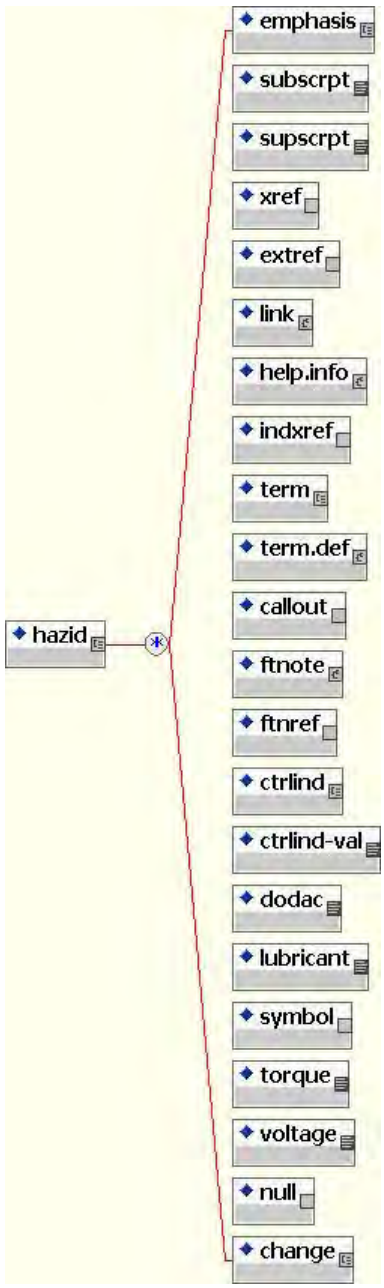

FIGURE 106. Hazardous material <hazid> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT hazid (#PCDATA | emphasis | subscrpt | supscrpt| xref | extref |
link | help.info | indxref | term | term.def | callout | ftnote |ftnref |
ctrlind | ctrlind-val | dodac | lubricant | symbol | torque |voltage |
null | change)*>
<!ATTLIST hazid
     assocfig          IDREFS                              #IMPLIED
     changeref         IDREFS                              #IMPLIED
     comment           CDATA                               #IMPLIED
     delchlvl          CDATA                               #IMPLIED
     id                ID                                  #IMPLIED
     idref             IDREFS                              #IMPLIED
     inschlvl          CDATA                               #IMPLIED
     security          (uc | fouo | c | s | ts)           #IMPLIED
     skilltrk          CDATA                               #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.4.1.2.5 Example of a ***<warnsum>***. The following is an example of a warning summary XML instance and the formatted display of the warning summary.

a. Example of an XML document instance fragment for ***<warnsum>***:

```
<warnsum>
<title>WARNING  SUMMARY
</title>
<para>This warning summary contains general safety warnings and
hazardous materials warnings that must be understood and applied
during operation and maintenance of this equipment.  Failure to
observe these precautions could result in serious injury or death to
personnel.  Also included are explanations of safety and hazardous
materials icons used within the technical manual.
</para>
<first_aid>
<title>FIRST  AID
</title>
<para>First aid can be defined as "urgent and immediate lifesaving and
other measures, which can be performed for casualties by nonmedical
personnel when medical personnel are not immediately available." Refer to
<extref  docno="FM  4-25.11"  posttext=""  pretext="FIRST  AID,  "
security="uc"/>  for  first  aid  data.
</para>
<safety>
<title>EXPLANATION  OF  SAFETY  WARNING  ICONS
</title>
<sfty-icons>
<symbol  boardno="heavyobject_icon"/>
<sftydesc>
```


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<title>HEAVY OBJECT
</title>
<text> – human figure stooping over heavy object shows physical injury potential from improper lifting technique.
</text>
</sftydesc>
</sfty-icons>
</safety>
<warninfo>
<title>GENERAL WARNING DESCRIPTION
<title>
<warning>
<warning.group>
<icon-set boardno="electricalhand_icon"/>
<trim.para>Whenever possible shut off system power before beginning work on equipment.
</trim.para>
<trim.para>Do not come in contact with electrical connectors.
<trim.para>
<trim.para>Don't be misled by low voltage. Low potentials can be dangerous.
<trim.para>
<trim.para>Do not work on electrical equipment alone. Be sure another person is nearby who can give first aid.
</warning.group>
</warning>
</warninfo>
<hazmat>
<title>EXPLANATION OF HAZARDOUS MATERIALS ICONS
</title>
<haz-icons>
<symbol boardno="chemical_icon"/>
<hazdesc>
<title>CHEMICAL
</title>
<text> - drops of liquid on hand shows that the material will cause burns or irritation to human skin or tissue.
</text>
</hazdesc>
</haz-icons>
<title>HAZARDOUS MATERIALS DESCRIPTION
</title>
<hazard>
<hazid>DRYCLEANING SOLVENT P-D-680
</hazid>
<symbol boardno="chemical_icon"/>
<symbol boardno="vapor"/>
<symbol boardno="fire"/>
<symbol boardno="eyeprotection_icon"/>
<para>P-D-680 solvent vapors are toxic. Avoid prolonged or repeated breathing of vapors or solvent contact with skin. Use only with adequate ventilation. Solvent is flammable and should not be used near open flame. Fire extinguishers should be readily available when solvent is used.
</para>
</hazard>
</hazmat>
</warnsum>

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b. Example of an formatted display of an XML document instance fragment for ***<warnsum>***

WARNING SUMMARY

This warning summary contains general safety warnings and hazardous materials warnings that must be understood and applied during operation and maintenance of this equipment. Failure to observe these precautions could result in serious injury or death to personnel. Also included are explanations of safety and hazardous materials icons used within the technical manual.

FIRST AID

First aid can be defined as "urgent and immediate lifesaving and other measures, which can be performed for casualties by nonmedical personnel when medical personnel are not immediately available". Refer to FIRST AID, FM 4-25.11 for first aid data.

EXPLANATION OF SAFETY WARNING ICONS

HEAVY OBJECT – human figure stooping over heavy object shows physical injury potential from improper lifting technique.

GENERAL WARNING DESCRIPTION

WARNING

Whenever possible shut off system power before beginning work on equipment.
Do not come in contact with electrical connectors.
Don't be misled by low voltage. Low potentials can be dangerous.
Do not work on electrical equipment alone. Be sure another person is nearby who can give first aid.

EXPLANATION OF HAZARDOUS MATERIALS ICONS

CHEMICAL – drops of liquid on hand shows that the material will cause burns or irritation to human skin or tissue.

a

FIGURE 107. Example of a Warning Summary.

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31.4.1.3 Change sheet ***<chgsheet>***. The element ***<chgsheet>*** change sheet, also know as a change transmittal page, is required in a changed document. The purpose of the change sheet is to list the reason(s) for the change to the data and to provide a table designating which pages are to be removed and which are to be inserted. Change sheet is used in the elements front matter for a page-base ***<paper.frnt>*** TM and front matter for a phased maintenance inspections/preventive maintenance services ***<pm.frnt>*** TM. The change sheet comes after the warning summary and it does not contain a page number.

a. The components are:

(1) Change number ***<chgno>*** (required). The element provides current document change level of the change sheet and title block page. The content model is #PCDATA.

(2) Title ***<title>*** (optional – zero or more) grouped with service nomenclature (see 49.1.1.3).

(3) Service Nomenclature ***<servnomen>*** (required – one or more) (see 31.3.1.1.4.3).

(4) City ***<city>*** (required). The element provides the city used in an address block of the service nomenclature (see 49.1.4.1.2).

(5) State ***<state>*** (required). The element provides the state used in an address block of the service nomenclature (see 49.1.4.1.3).

(6) Change date ***<chgdate>*** (required). The element provides the publication effective change date (see 31.4.1.3.1).

(7) Primary title ***<prtitle>*** (required). The element provides the technical manual primary title and specific information identifying system nomenclature (see 31.3.1.1.4.1.3).

(8) Subtitle ***<stitle>*** (optional). The element specifies the title of the component for the change sheet (see 49.1.1.2).

(9) Weapons System Title ***<weapons_system>*** (optional). The element specifies the title of the weapons system component (see 31.3.1.1.4.1.4).

(10) Official notices ***<notices>*** (required). The element specifies the distribution statement notice pertaining to the change sheet of the TM. (see 31.3.1.1.4.2).

(11) Reduced paragraph ***<trim.para>*** (required). The element provides text specifying the TM and the date of the TM that is being updated. (see 49.1.1.7).

(12) Change list ***<chglist>*** (required – one or more). The element provides a list of the work packages and change pages of the modified TM (see 31.4.1.3.2).

(13) Authentication page ***<authent>*** (required). The element provides the TM authentication page graphic (see 31.4.2.4).

b. DTD fragment is graphically depicted.

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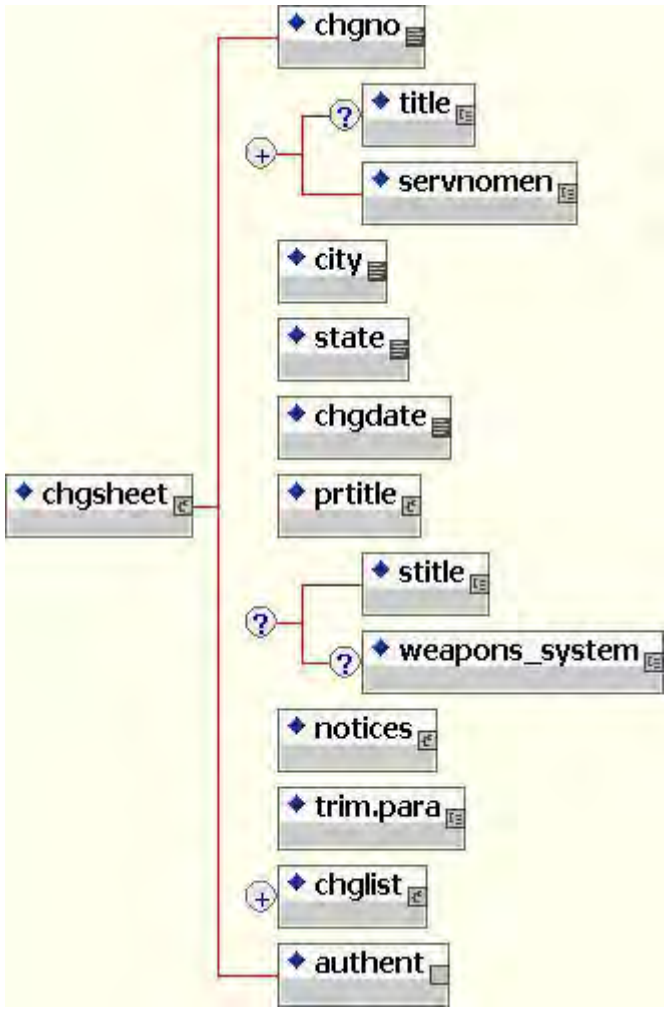

FIGURE 108. Change sheet <chgsheet> DTD hierarchy..

c. DTD fragment:

```
<!ELEMENT chgsheet (chgno, (title?, servnomen)+, city, state, chgdate,
prtitle, (stitle, weapons_system?)?, notices, trim.para, chglist+,
authent)>
<!ATTLIST chgsheet
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     date              CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)        #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Unique attributes:

(1) **date** – Effective date of the change sheet. (optional)

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e. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.4.1.3.1 Change date ***<chgdate>***. The element ***<chgdate>*** specifies the publication effective change date on a change sheet ***<chgsheet>*** and on the list of effective wp and pages ***<loepwp>***.

a. The components are:
(1) Parsable data (#PCDATA) (required). The element provides the text of the effective date. (see 9.3.2)

b.

```
<!ELEMENT chgdate (#PCDATA)>
<!ATTLIST chgdate
     julian     CDATA     #REQUIRED>
```

c. Unique attributes:
(1) **julian** – Specify the date in the following format yyyymmdd.

31.4.1.3.2 Change list ***<chglist>***. The element ***<chglist>*** specifies the pages to be removed and the current pages to be inserted. Also, change list includes the revision number of any work packages that have been deleted and if any have been added.

a. The components are:
(1) Reduced paragraph ***<trim.para>*** (required). The element provides the statement for inserting and removing pages or work packages (see 49.1.1.7).
(2) Page change list ***<chgpagelist>*** (optional). The element provides a list of the TM changed pages(see 31.4.1.3.2.1).
(3) WP change list ***<chgwplist>*** (optional). The element provides a list of the TM changed work packages(see 31.4.1.3.2.2).

b. DTD fragment:

```
<!ELEMENT chglist (trim.para, (chgpagelist | chgwplist)?)>
<!ATTLIST chglist
     label     CDATA     #IMPLIED>
```

c. Unique attributes:
(1) **label** – (optional) Output a prefix to the paragraph.

31.4.1.3.2.1 Page change list ***<chgpagelist>***. The element ***<chgpagelist>*** provides a list containing the TM changed pages. In page-based, this element functions as the table element.

a. The components are:
(1) Title ***<title>*** (required – two or more). The element is required twice to provide the titles for removing pages and inserting pages for a page change list (see 49.1.1.3).
(2) Page changes ***<chgpage>***(required – one or more). The element provides the removed and inserted page numbers (see 31.4.1.3.2.1.1).

b. DTD fragment:

```
<!ELEMENT chgpagelist (title, title, chgpage+)>
```

c. No attributes.

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31.4.1.3.2.1.1 Page changes ***<chgpage>***. The element ***<chgpage>*** specifies the removed and inserted page numbers in a change list. In page-base, the element is equivalent to a "row" element in a table.

a. The components are:
   (1) Remove page ***<removepg>***(required). The element provides the identification of the removed page (see 31.4.1.3.2.1.1.1).
   (2) Inserted page ***<insertpg>*** (required). The element provides the inserted page numbers (see 31.4.1.3.2.1.1.2).
b. DTD fragment:

```
<!ELEMENT chgpage (removepg, insertpg)>
```

c. No attributes.

31.4.1.3.2.1.1.1 Remove page ***<removepg>***. The element ***<removepg>*** identifies the page number of the removed page from the previous change front or rear matter as part of a current change. In page-base, the element is equivalent to an "entry" element in a table.

a. The components are:
   (1) Parsable data (#PCDATA) (required) (see 9.3.2).
b. DTD fragment:

```
<!ELEMENT removepg (#PCDATA)>
```

c. No attributes.

31.4.1.3.2.1.1.2 Inserted page ***<insertpg>***. The element ***<insertpg>*** identifies the page number of a page to be inserted as part of a current change in the front or rear matter. It forms a column in the change list that appears on the change sheet.

a. The components are:
   (1) Parsable data (#PCDATA). (required). The element provides the text for the page number. (see 9.3.2).
b. DTD fragment:

```
<!ELEMENT insertpg (#PCDATA)>
```

c. No attributes.

31.4.1.3.2.2 WP change list ***<chgwplist>***. The element ***<chgwplist>*** identifies the changed work packages in the TM. In page-based, this element functions as the table element.

a. The components are:
   (1) Title ***<title>*** (required). The element provides the title for the list of the changed work packages (see 49.1.1.3).
   (2) Work package number ***<wpno>*** (required – one or more). The element provides the work package sequence numbers for the work package change list (see 46.2.5).
b. DTD fragment:

```
<!ELEMENT chgwplist (title, wpno+)>
```

c. No attributes.

31.4.1.3.3 Example of a ***<chgsheet>***. The following is an example of a change sheet XML instance and the formatted display of the change sheet.

a. Example of an XML document instance fragment for ***<chgsheet>***:

```
<chgsheet>
<chgno>CHANGE  No.     1
</chgno>
```


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<servnomen>HEADQUARTERS, DEPARTMENT OF THE ARMY
</servnomen>
<city>WASHINGTON,
</city>
<state>D.C.
</state>
<chgdate julian="19930831">31 AUGUST 1993
</chgdate>
<prtitle>
<sysnomen pretext="OPERATOR AND UNIT MAINTENANCE MANUAL FOR">
<name>OPERATORS, UNIT, AND DIRECT SUPPORT MAINTENANCE MANUAL
</name>
<modelno>TEST SET RADAR AN.TON-22
</modelno>
<nsn>
<fsc>4931-
</fsc>
<niin>00-707-1229
</niin>
</nsn>
<eic>d42
</eic>
</sysnomen>
</prtitle>
<notices>
<dist>&notices.dist.a.statement;
</dist>
</notices>
<trim.para>TM X-XXX-XXXX-XX, 5 June 1987, is updated as follows:
</trim.para>
<chglist>
<trim.para>File this sheet in front of the manual for reference.
</trim.para>
</chglist>
<chglist>
<trim.para>This change is a result of new preventive maintenance checks and services procedures and new expendable/durable supplies and materials.
</trim.para>
</chglist>
<chglist>
<trim.para>Added illustrations are indicated by a vertical bar adjacent to the figure number. Changed illustrations are indicated by a miniature pointing hand adjacent to the updated area and a vertical bar adjacent to the figure number.
</trim.para>
</chglist>
<chglist>
<trim.para>Remove old pages and insert new pages asindicated below.
</trim.para>
<chgpagelist>
<title>Remove Pages
</title>
<title>Insert Pages
</title>
<chgpage>
<removepg>a through d

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```
</removepg>
<insertpg>a  through  d
</insertpg>
</chgpage>
<chgpage>
<removepg>None
</removepg>
<insertpg>e  through  g  /  (h  blank)
</insertpg>
</chgpage>
</chgpagelist>
</chglist>
<chglist>
<trim.para>Remove old pages and insert new pages as indicated below.
</trim.para>
<chgwplist>
<title>Work  package  number
</title>
<wpno   wpref="G0003-X-XXX-XXXX"/>
<wpno   wpref="O00014-X-XXX-XXXX"/>
</chgwplist>
</chglist>
<chglist>
<trim.para>Add  the  following  new  work  packages.
</trim.para>
<chgwplist>
<title>Work  package  number
</title>
<wpno   wpref="M1625.1-X-XXX-XXXX"/>
<wpno   wpref="M1700.1-X-XXX-XXXX"/>
</chgwplist>
</chglist>
<authent   boardno="xxxxxxx"   unitmeasure="in"/>
</chgsheet>
```

b. Example of an formatted display of an XML document instance fragment for ***<chgsheet>*** :

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TM X-XXXX-XXX

CHANGE
NO. CHANGE
No. 1

HEADQUARTERS, DEPARTMENT OF THE ARMY

WASHINGTON, , D.C., 31 AUGUST 1993

TECHNICAL MANUAL
FOR
OPERATORS MANUAL
TEST SET RADAR AN.TON-22
NSN 4931-00-707-1229

TM X-XXX-XXXX-XX, 5 June 1987, is updated as follows:

1. File this sheet in front of the manual for reference.
2. This change is a result of new preventive maintenance checks and services procedures and new expendable/durable supplies and materials.
3. Added illustrations are indicated by a vertical bar adjacent to the figure number. Changed illustrations are indicated by a miniature pointing hand adjacent to the updated area and a vertical bar adjacent to the figure number.
4. Remove old pages and insert new pages as indicated below.

Remove Pages	Insert Pages
a through d	a through d
None	e through g / (h blank)

5. Remove old pages and insert new pages as indicated below.

Work package number
WP
WP

6. Add the following new work packages.

Work package number
WP
WP

FIGURE 109. Example of a change sheet.

31.4.1.4 List of effective pages/work packages ***<loepwp>***. The list of effective pages/work packages (***<loepwp>***) lists the latest work packages in the TM. It is prepared along with the basic version of the TM and each subsequent change. The list of effective pages/work packages is located immediately following the warning summary (***<warnsum>***). When included in a change to a TM, it immediately follows the change transmittal sheet (***<chgsheet>***) if the warning summary is not included in the change. The examples

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shown in this section will depict a list of effective pages/work packages for a new publication, a changed publication, a revised publication, and a multi-volume, multi-service manual.

a. The components are:
   (1) Work package metadata ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Reduced content paragraph ***<trim.para>*** (optional). The ***<trim.para>*** element (see 49.1.1.7), when used in this context, provides the capability to display text before the title in the list of effective pages/work packages ((see 31.4.1.4.7 for an example).
   (3) Title ***<title>*** (required). The ***<title>*** element (see 49.1.1.3), when used in this context, contains the title text that is displayed in the list of effective pages/work packages, (i.e., **LIST OF EFFECTIVE PAGES/WORK PACKAGES**).
   (4) Note ***<note>*** (optional). The ***<note>*** element (see 41.1.3) contains the text for notes that are commonly found in a list of effective pages/work packages (e.g., Zero in the "Change No." column indicates an original page or work package.).
   (5) Issued changes list ***<issuechg>*** (required) (see 31.4.1.4.1).
   (6) Total number of changes ***<totalnumberof>*** (required – one or more) (see 31.4.1.4.2).
   (7) Column list title ***<col.title>*** (required – two) (see 31.4.1.4.3).
   (8) Change volume ***<chgvol>*** (optional) (see 31.4.1.4.4).
   (9) List of effective pages/work packages history ***<chghistory>*** (required – one or more) (see 31.4.1.4.5).

b. DTD fragment is graphically depicted.

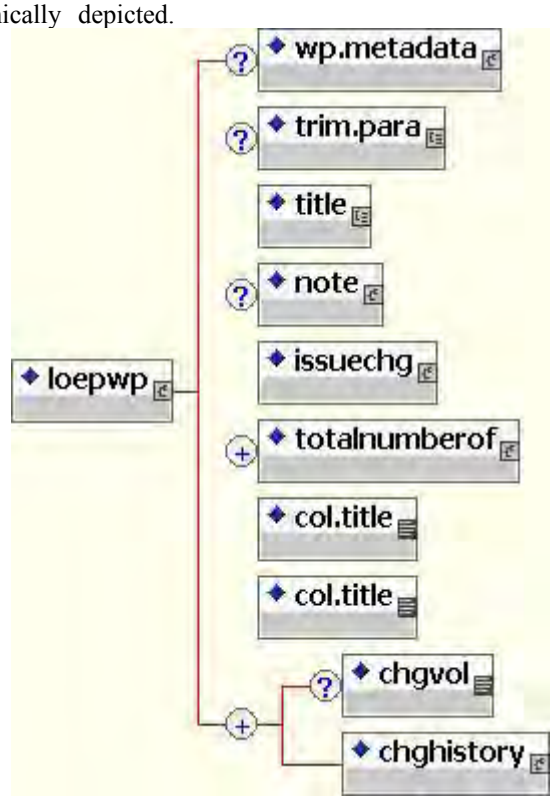

FIGURE 110. List of effective pages/work packages <loepwp> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT loepwp (wp.metadata?, trim.para?, title, note?, issuechg,
      totalnumberof+, col.title, col.title, (chgvol?, chghistory)+)>
```

d. No attributes.

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31.4.1.4.1 Issued changes list ***<issuechg>***. The element ***<issuechg>*** is required and contains the listing of issued changes following the note in the list of effective pages/work packages. If there is no note (***<note>***), the issued changes list (***<issuechg>***) follows the title ***<title>***.

a. The components are:
   (1) Reduced content paragraph ***<trim.para>*** (required). The ***<trim.para>*** element (see 49.1.1.7) contains the statement text that appears at the beginning of the issued changes list.
   (2) Issued changes list ***<issued>*** (required – one or more) (see 31.4.1.4.1.1).

b. DTD fragment is graphically depicted.

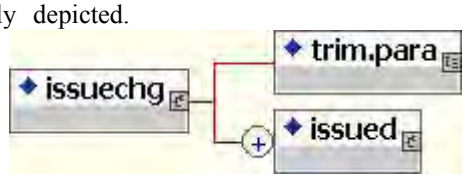

FIGURE 111. Issued changes list <issuechg> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT issuechg (trim.para, issued+)>
```

d. No attributes.

31.4.1.4.1.1 Issued changes ***<issued>***. The element ***<issued>*** provides the issued change number (***<chgno>***) and date (***<chgdate>***). Both are required.

a. The components are:
   (1) Change number ***<chgno>*** (required). The element ***<chgno>*** is the document change level of the issued change entry. The content model is #PCDATA.
   (2) Change date ***<chgdate>*** (required). The change date element ***<chgdate>***(see 31.4.1.3.1) is the publication effective change date for that change. Change date requires an attribute date **julian** which specifies the date using the julian format "yyyymmdd". The content in the element ***<chgdate>*** contains the formatted date. The content model is #PCDATA.

b. DTD fragment is graphically depicted.

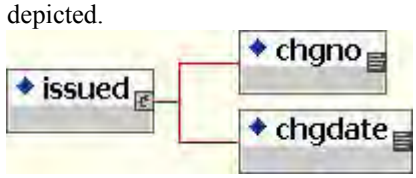

FIGURE 112. Issued changes <issuechg> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT issued (chgno, chgdate)>
```

d. No attributes.

31.4.1.4.2 Total number of changes ***<totalnumberof>***. The element ***<totalnumberof>*** contains the total number of volumes, pages, figures and/or work packages in the list of effective pages/work packages (***<loepwp>***). The element's content model starts with a required narrative (***<text>***), followed by one of the following two markups. The first markup permits the optional entry of the number of volumes (***<totnum.volumes>***), the required total number of front and rear matter pages (***<totnum.frnt-rear-pages>***) and total number of work packages (***<totnum.wps>***). The second markup permits the entry of the required total number of pages (***<totnum.pages>***), the required total number of figures (***<totnum.figures>***) and the required number of work packages (***<totnum.wps>***). In both cases text fields are included in the content models to permit the entry of textual information that will appear on the LOEP/WP.

a. The components are:

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(1) Narrative text ***<text>*** (required). The ***<text>*** element provides for the entry of textual information that will appear in the narrative for the total number of volumes, pages, figures and/or work packages (see 31.4.1.4.2.1).
(2) Total number of volumes ***<totnum.volumes>*** (optional) (see 31.4.1.4.2.2).
(3) Total number of front and rear matter pages ***<totnum.frnt-rear-pages>*** (either this element or the total number of pages (along with the total number of figures) is required) (see 31.4.1.4.2.3).
(4) Total number of work packages ***<totnum.wps>*** (required)(see 31.4.1.4.2.4).
(5) Total number of pages ***<totnum.pages>*** (either this element (along with the total number of figures) or the total number of front and rear matter pages is required) (see 31.4.1.4.2.5)
(6) Total number of figures ***<totnum.figures>*** (either this element (along with the total number of pages) or the total number of front and rear matter pages is required) (see 31.4.1.4.2.6).

b. DTD fragment is graphically depicted.

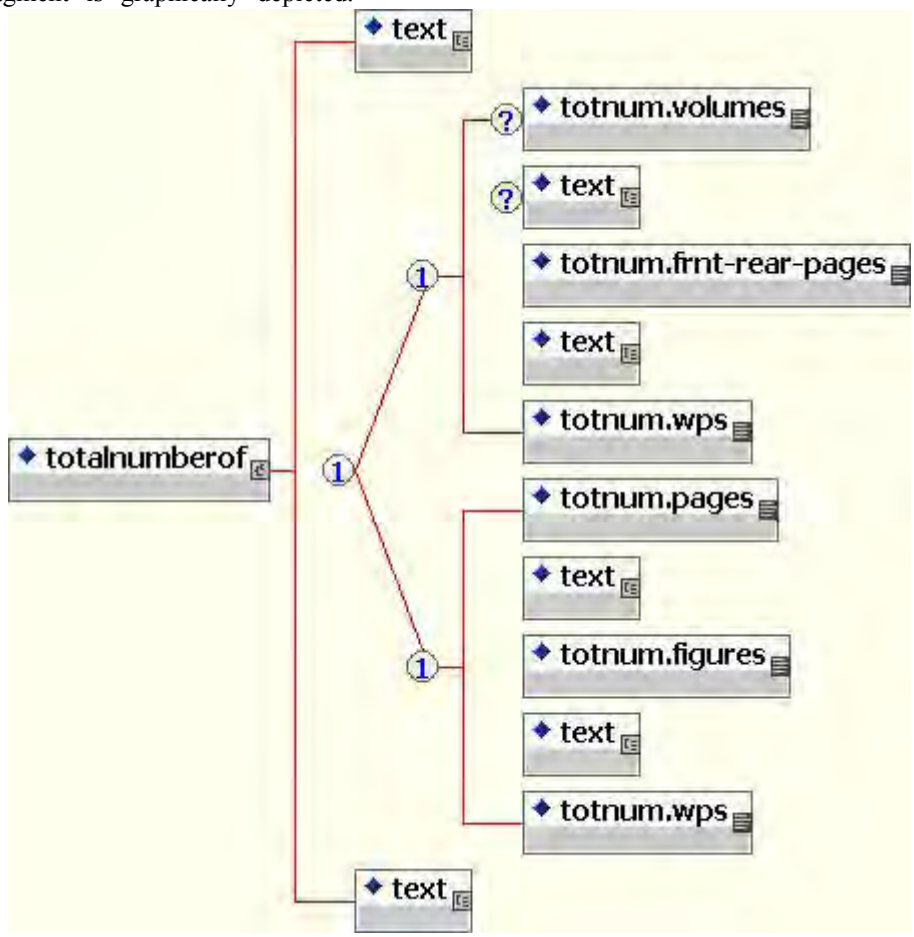

FIGURE 113. Total number of changes <totalnumberof> DTD hierarchy.

c. DTD fragment for:

```
<!ELEMENT totalnumberof (text, ((totnum.volumes?, text?,
      totnum.frnt-rear-pages, text, totnum.wps) | (totnum.pages, text,
      totnum.figures, text, totnum.wps)), text)>
```

d. No attributes.

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31.4.1.4.2.1 Narrative text ***<text>***. The ***<text>*** element appears in several places in the ***<totalnumberof>*** element's content model to permit the entry of textual information that will appear on the LOEP/WP,see 49.1.1.18. These components are optional and may be entered multiple times in any order to provide formatted textual data.

31.4.1.4.2.2 Total number of volumes ***<totnum.volumes>***. The element ***<totnum.volumes>*** specifies the total number of volumes in the TM as presented in the LOEP/WP introduction paragraph. The content model is #PCDATA.

31.4.1.4.2.3 Total number of front and rear matter pages ***<totnum.frnt-rear-pages>***. The element ***<totnum.frnt-rear-pages>*** specifies the total number of front and rear matter pages in the TM as presented in the LOEP/WP introduction paragraph. The content model is #PCDATA.

31.4.1.4.2.4 Total number of work packages ***<totnum.wps>***. The element ***<totnum.wps>*** specifies the total number of work packages in the TM as presented in the LOEP/WP introduction paragraph. The content model is #PCDATA.

31.4.1.4.2.5 Total number of pages ***<totnum.pages>***. The element ***<totnum.pages>*** specifies the total number of pages in the TM as presented in the LOEP/WP introduction paragraph. The content model is #PCDATA.

31.4.1.4.2.6 Total number of figures ***<totnum.figures>***. The element ***<totnum.figures>*** specifies the total number of figures in the TM as presented in the LOEP/WP introduction paragraph. The content model is #PCDATA.

31.4.1.4.2.7 Total number of changes example. This example shows the XML source fragment for making a total number of changes entry consisting of 23 pages of front and rear matter, and 27 work packages.

a. XML document instance fragment.

```
<totalnumberof>
<text>TOTAL NUMBER OF PAGES FOR FRONT AND REARMATTER IS
</text>
<totnum.frnt-rear-pages>23
</totnum.frnt-rear-pages>
<text> AND TOTAL NUMBER OF WORK PACKAGES IS
</text>
<totnum.wps>35
</totnum.wps>
<text>, CONSISTING OFTHE FOLLOWING:
</text>
</totalnumberof>
```

b. Stylesheet Output.

```
TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 23 AND TOTAL NUMBER OF
WORK PACKAGES IS 35, CONSISTING OF THE FOLLOWING:
```

31.4.1.4.3 Column list title ***<col.title>***. The element ***<col.title>*** specifies a list column title for the LOEP/WP. Two entries are required. The content model is #PCDATA.

31.4.1.4.4 Change volume ***<chgvol>***. Change volume is used as a volume separator in the list of effective pages/work packages. When the list of effective pages/work packages includes volumes, insert the change volume ***<chgvol>*** before the change history to add the volume number to the list. The content model is #PCDATA.

31.4.1.4.5 List of effective pages/work packages history ***<chghistory>***. All work packages and pages not included in work packages are listed in the list of effective pages/work packages except for the ones listed in MIL–STD–40051–1/–2 as being exempted pages, front and rear sections. For pages that are not included in work packages, list the page number (***<pageno>***) and its change number (***<chgno>***). For each work package, list the work package sequence number followed with the total number of pages in the work package (***<wppages>***) and its change number (***<chgno>***). The **modified** attribute indicates the change type.

a. The components are:

(1) Page number ***<pageno>*** (required for page entries– one or two) (see 31.4.1.4.5.1).

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(2) Title ***<title>*** (required for the warning summary entry) (see 49.1.1.3).
(3) Total number of pages ***<totnum.pages>*** (required for the warning summary entry) (see 31.4.1.4.5.2).
(4) Work package number ***<wpno>*** (required for work package entry) (see 31.4.1.4.5.3).
(5) Number of work package pages ***<wppages>*** (required for work package entry) (see 31.4.1.4.5.4).
(6) Change number ***<chgno>*** (required) (see 31.4.1.4.5.5).

b. DTD fragment is graphically depicted.

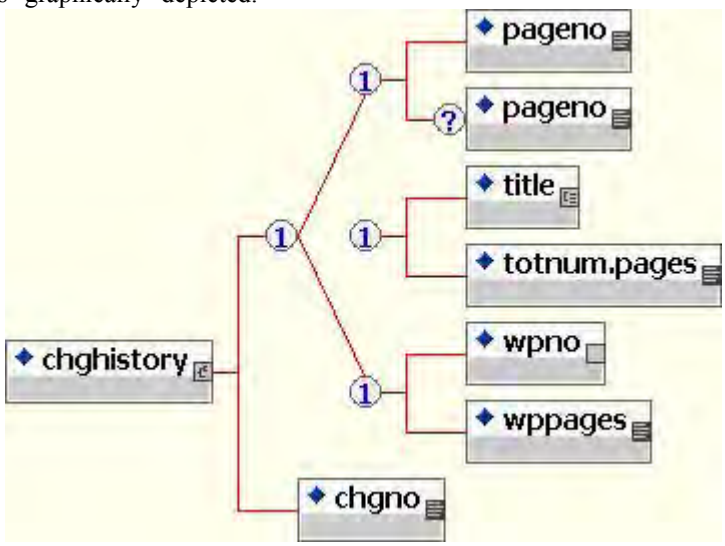

FIGURE 114. List of effective pages/work packages history <chghistory> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT chghistory (((pageno, pageno?) | (title, totnum.pages) |
      (wpno, wppages)), chgno)>
<!ATTLIST chghistory
     modified (added | deleted | blank | none | changed) "changed" >
```

d. Unique attributes:

(1) **modified** – Indicates the change type as added, deleted, blank page, no reason, and changed. If the page/work package is new to this change, set the **modified** attribute to "added". If the page/work package is deleted, set the **modified** attribute to "deleted". If a page is blank, set the **modified** attribute to "blank". Enter "none" in the **modified** attribute if there was no change to the page/ work package for that period. Enter "changed" in the **modified** attribute for a page/work package that has been changed. The stylesheet will display the appropriate entry (e.g., "Added" for a page/work package that is new to this change, "Deleted" for a page/work package that is deleted, etc.).

31.4.1.4.5.1 Page number ***<pageno>***. The element ***<pageno>*** is used to enter a manual page number in the list of effective pages/work packages. The ***<pageno>*** element may be entered twice to indicate that several pages in sequence are at the same change level (e.g., i–iii). The ***<pageno>*** element may also contain text that identifies entries for the front cover and chapter title pages. The content model is #PCDATA.

31.4.1.4.5.2 Total number of pages ***<totnum.pages>***. The element ***<totnum.pages>*** contains the number of pages in warning summary entry. The content model is #PCDATA.

31.4.1.4.5.3 Work package number ***<wpno>***. This element contains the work package identification number which is entered in the attribute **wpref**. The work package identification number (see 46.2.5) will be used to retrieve the work package sequence number that is displayed in the list of effective pages/work packages (see 32.2.2).

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31.4.1.4.5.4 Number of work package pages ***<wppages>***. Specifies the number of pages in the work package. The content model is #PCDATA.

31.4.1.4.5.5 Change number ***<chgno>***. The change level of the specified page(s)/work package(s). The content model is #PCDATA.

31.4.1.4.6 Example – List of effective pages/work packages for a new publication. When the document is a new publication, the change number is always zero. The list of effective pages/work packages will display the change number as zero for all pages and work packages listed. Note that the work package identification number is entered in the **wpref** attribute in the ***<wpno>*** element. This work package identification number is used to retrieve the work package sequence number that is displayed in the output.

a. XML document instance fragment

```
<loepwp>
<title>Listof Effective Pages/Work Packages
</title>
<note>
<trim.para esd="no" hcp="no">Zero in the&amp;amp;ldquo;Change
No.&amp;amp;rdquo;column indicates an originalpage or work package.
</trim.para>
</note>
<issuechg>
<trim.para esd="no"hcp="no">Dateof issue for the original manual is:
</trim.para>
<issued>
<chgno>Original
</chgno>
<chgdate julian="19980713">13July 1998
</chgdate>
</issued>
</issuechg>
<totalnumberof>
<text>TOTALNUMBER OF PAGESFOR FRONT AND REAR MATTER IS
</text>
<totnum.frnt-rear-pages>28
</totnum.frnt-rear-pages>
<text> AND TOTAL NUMBER OF WORK PACKAGESIS
</text>
<totnum.wps>35
</totnum.wps>
<text>, CONSISTING OFTHE FOLLOWING:
</text>
</totalnumberof>
<col.title>Page / WP No.
</col.title>
<col.title>ChangeNo.
</col.title>
<chghistory modified="none">
<pageno>Front cover
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<title>WarningSummary
</title>
<totnum.pages>4
```


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Check the source to verify that this is the current version before use.

</totnum.pages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>i
</pageno>
<pageno>iii
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="blank">
<pageno>ivblank
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp1title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="g00001-X-XxXX-XXX"/>
<wppages>4pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="g00002-X-XxXX-XXX"/>
<wppages>10pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="g00003-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="g00004-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp2title page
</pageno>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="t00001-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="t00002-X-XxXX-XXX"/>
<wppages>8pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="t00003-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp3title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00001-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00003-X-XxXX-XXX"/>
<wppages>12pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00004-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00005-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp4title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00002-X-XxXX-XXX"/>
<wppages>20pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00006-X-XxXX-XXX"/>
<wppages>30pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00007-X-XxXX-XXX"/>
<wppages>30pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00008-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00009-X-XxXX-XXX"/>
<wppages>4pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00010-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00011-X-XxXX-XXX"/>
<wppages>8pgs
</wppages>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00012-X-XxXX-XXX"/>
<wppages>12pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00013-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="m00014-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp5title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00001-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00002-X-XxXX-XXX"/>
<wppages>4pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00003-X-XxXX-XXX"/>
<wppages>6pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00004-X-XxXX-XXX"/>
<wppages>4pgs
</wppages>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00005-X-XxXX-XXX"/>
<wppages>4pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00006-X-XxXX-XXX"/>
<wppages>4pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00007-X-XxXX-XXX"/>
<wppages>4pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00008-X-XxXX-XXX"/>
<wppages>4pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00009-X-XxXX-XXX"/>
<wppages>4pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00010-X-XxXX-XXX"/>
<wppages>6pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00011-X-XxXX-XXX"/>
<wppages>6pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpnowpref="s00012-X-XxXX-XXX"/>
<wppages>2pgs

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpnowpref="s00013-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpnowpref="s00014-X-XxXX-XXX"/>
<wppages>2pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<pageno>INDEX-1
</pageno>
<pageno>INDEX-14
</pageno>
<chgno>0
</chgno>
</chghistory>
</loepwp>
```

b. Example of a stylesheet output for a list of effective pages work package for a new publication:

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MANUAL NUMBER

LIST OF EFFECTIVE PAGES/WORK PACKAGES

NOTE: Zero in the "Change No." column indicates an original page or work package.

Date of issue for the original manual is:

Original 13 July 1998

TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 28 AND TOTAL NUMBER OF WORK PACKAGES IS 35, CONSISTING OF THE FOLLOWING:

Page/WP No.	Change No.	Page/WP No.	Change No.
Front cover	0	WP 0029 (4 pgs)	0
Warning Summary (4 pgs)	0	WP 0030 (4 pgs)	0
i - iii	0	WP 0031 (6 pgs)	0
iv blank	0	WP 0032 (6 pgs)	0
Chp 1 title page	0	WP 0033 (2 pgs)	0
WP 0001 (4 pgs)	0	WP 0034 (2 pgs)	0
WP 0002 (10 pgs)	0	WP 0035 (2 pgs)	0
WP 0003 (2 pgs)	0	INDEX-1 - INDEX-14	0
WP 0004 (2 pgs)	0		
Chp 2 title page	0		
WP 0005 (2 pgs)	0		
WP 0006 (8 pgs)	0		
WP 0007 (2 pgs)	0		
Chp 3 title page	0		
WP 0008 (2 pgs)	0		
WP 0009 (12 pgs)	0		
WP 0010 (2 pgs)	0		
WP 0011 (2 pgs)	0		
Chp 4 title page	0		
WP 0012 (20 pgs)	0		
WP 0013 (30 pgs)	0		
WP 0014 (30 pgs)	0		
WP 0015 (2 pgs)	0		
WP 0016 (4 pgs)	0		
WP 0017 (2 pgs)	0		
WP 0018 (8 pgs)	0		
WP 0019 (12 pgs)	0		
WP 0020 (2 pgs)	0		
WP 0021 (2 pgs)	0		
Chp 5 title page	0		
WP 0022 (2 pgs)	0		
WP 0023 (4 pgs)	0		
WP 0024 (6 pgs)	0		
WP 0025 (4 pgs)	0		
WP 0026 (4 pgs)	0		
WP 0027 (4 pgs)	0		
WP 0028 (4 pgs)	0		

A

FIGURE 115. Example of a stylesheet output for a list of effective pages/work packages for a new publication..

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31.4.1.4.7 Example – list of effective pages/work packages for a changed publication. A list of effective pages/work packages for a changed publication starts with a ***<trim.para>*** that contains the text **"INSERT LATEST CHANGED PAGES/WORK PACKAGES. DESTROY SUPERSEDED DATA."**. This text is displayed in the list of effective pages/work packages before the title. In the issued changes list (***<issuechange>***), the change publication number is listed with the date. Where applicable, the type of modification "added" or "deleted" of a page/work package is listed to the right of entries in the list of effective pages/work packages history. The change number is displayed on the right side of the bottom of the page. The value is taken from the **chnglevel** attribute in the ***<production>*** element (see 30.1.1).

a. XML document instance fragment

```
<loepwp>
<trim.para>Insert latest changed pages/work packages.   De-
stroy superseded data.
</trim.para>
<title>List of Effective Pages/Work Packages
</title>
<note>
<trim.para esd="no" hcp="no">The portion of text affectedby the
changes is indicated by a vertical bar in the outer marginsof the
page.  Changes to illustrations are indicated by a verticalbar
adjacent to the title.  Zero in the &ldquo;Change No.&rdquo;column
indicates an original page or work package.
</trim.para>
</note>
<issuechg>
<trim.para esd="no" hcp="no">Dateof issue for the original manual
and changed pages/work packagesare:
</trim.para>
<issued>
<chgno>Original
</chgno>
<chgdate julian="19980713">13 July 1998
</chgdate>
</issued>
<issued>
<chgno>1
</chgno>
<chgdate julian="19981210">10 December 1998
</chgdate>
</issued>
<issued>
<chgno>2
</chgno>
<chgdate julian="19990302">2 March 1999
</chgdate>
</issued>
</issuechg>
<totalnumberof>
<text>TOTAL NUMBER OF PAGESFOR FRONT AND REAR MATTER IS
</text>
<totnum.frnt-rear-pages>28
</totnum.frnt-rear-pages>
<text> AND TOTAL NUMBER OF WORK PACKAGES IS
</text>
<totnum.wps>35
</totnum.wps>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<text>, CONSISTING OFTHE FOLLOWING:
</text>
</totalnumberof>
<col.title>Page / WP No.
</col.title>
<col.title>Change No.
</col.title>
<chghistory modified="none">
<pageno>Front cover
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<title>WarningSummary
</title>
<totnum.pages>4
</totnum.pages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>i
</pageno>
<pageno>iii
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="blank">
<pageno>ivblank
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp1 title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00001-X-XxXX-XXX"/>
<wppages>4 pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00002-X-XxXX-XXX"/>
<wppages>10 pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory modified="none">

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<wpno wpref="g00003-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00004-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>2
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp2 title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="t00001-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="t00002-X-XxXX-XXX"/>
<wppages>8 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="t00003-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp3 title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00001-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00003-X-XxXX-XXX"/>
<wppages>12 pgs

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00004-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00005-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory  modified="none">
<pageno>Chp4  title  page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00002-X-XxXX-XXX"/>
<wppages>20  pgs
</wppages>
<chgno>2
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00006-X-XxXX-XXX"/>
<wppages>30  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00007-X-XxXX-XXX"/>
<wppages>30  pgs
</wppages>
<chgno>2
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00008-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
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<wppages>4  pgs
</wppages>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00010-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00011-X-XxXX-XXX"/>
<wppages>8 pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00012-X-XxXX-XXX"/>
<wppages>12 pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00013-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>2
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00014-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp5 title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="s00001-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="s00002-X-XxXX-XXX"/>
<wppages>4 pgs
</wppages>
<chgno>2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00003-X-XxXX-XXX"/>
<wppages>6  pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00004-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00005-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="deleted">
<wpnowpref="s00006-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>2
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00007-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00008-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00009-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00010-X-XxXX-XXX"/>
<wppages>6  pgs
</wppages>
<chgno>2
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00011-X-XxXX-XXX"/>
<wppages>6  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00012-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>1
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00013-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="added">
<wpno  wpref="s00015-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>2
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00014-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>2
</chgno>
</chghistory>
<chghistory  modified="none">
<pageno>INDEX-1
</pageno>
<pageno>INDEX-14
</pageno>
<chgno>0
</chgno>
</chghistory>
</loepwp>
```

b. Example of a stylesheet output for a list of effective pages/work packages for a manual with changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

MANUAL NUMBER

INSERT LATEST CHANGED PAGES/WORK PACKAGES. DESTROY SUPERSEDED DATA.

LIST OF EFFECTIVE PAGES/WORK PACKAGES

NOTE: The portion of text affected by the changes is indicated by a vertical bar in the outer margins of the page. Changes to illustrations are indicated by a vertical bar adjacent to the title. Zero in the "Change No." column indicates an original page or work package.

Date of issue for the original manual and changed pages/work packages are:

Original	13 July 1998
1	10 December 1998
2	2 March 1999

TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 28 AND TOTAL NUMBER OF WORK PACKAGES IS 35, CONSISTING OF THE FOLLOWING:

Page/WP No.	Change No.	Page/WP No.	Change No.
Front cover	0	WP 0025 (4 pgs)	1
Warning Summary (4 pgs)	0	WP 0026 (4 pgs)	0
i - iii	0	WP 0027 (4 pgs) Deleted	2
iv blank	0	WP 0028 (4 pgs)	0
Chp 1 title page	0	WP 0029 (4 pgs)	0
WP 0001 (4 pgs)	1	WP 0030 (4 pgs)	1
WP 0002 (10 pgs)	1	WP 0031 (6 pgs)	2
WP 0003 (2 pgs)	0	WP 0032 (6 pgs)	0
WP 0004 (2 pgs)	2	WP 0033 (2 pgs)	1
Chp 2 title page	0	WP 0034 (2 pgs)	0
WP 0005 (2 pgs)	1	WP 0034.1 (4 pgs) Added	2
WP 0006 (8 pgs)	0	WP 0035 (2 pgs)	2
WP 0007 (2 pgs)	0	INDEX-1 - INDEX-14	0
Chp 3 title page	0		
WP 0008 (2 pgs)	0		
WP 0009 (12 pgs)	0		
WP 0010 (2 pgs)	0		
WP 0011 (2 pgs)	1		
Chp 4 title page	0		
WP 0012 (20 pgs)	2		
WP 0013 (30 pgs)	0		
WP 0014 (30 pgs)	2		
WP 0015 (2 pgs)	0		
WP 0016 (4 pgs)	0		
WP 0017 (2 pgs)	1		
WP 0018 (8 pgs)	1		
WP 0019 (12 pgs)	1		
WP 0020 (2 pgs)	2		
WP 0021 (2 pgs)	0		
Chp 5 title page	0		
WP 0022 (2 pgs)	0		
WP 0023 (4 pgs)	2		
WP 0024 (6 pgs)	1		

A Change 2

FIGURE 116. Example of a stylesheet output for a list of effective pages/work packages for a manual with changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

31.4.1.4.8 Example – list of effective pages/work packages for a revised publication. A list of effective pages/work packages for a revised publication is prepared similarly to other manuals. The note (***<note>***) and the issued changes list (***<issuechange>***) statement are different in the revised publication when compared to the note and the issued changes list statement in a new or changed publication.

a. XML document instance fragment

```
<loepwp>
<title>List of Effective Pages/Work Packages
</title>
<note>
<trim.para esd="no" hcp="no">This manual supersedes TM X-XXX-XXXdated
15 March 1998.  Zero in the &ldquo;Change No.&rdquo;
columnindicates an original page or work package.
</trim.para>
</note>
<issuechg>
<trim.para esd="no" hcp="no">Dateof issue for the revision is:
</trim.para>
<issued>
<chgno>Original
</chgno>
<chgdate julian="19980916">16September 1998
</chgdate>
</issued>
</issuechg>
<totalnumberof>
<text>TOTAL NUMBER OF PAGESFOR FRONT AND REAR MATTER IS
</text>
<totnum.frnt-rear-pages>28
</totnum.frnt-rear-pages>
<text> AND TOTAL NUMBER OF WORK PACKAGES IS
</text>
<totnum.wps>35
</totnum.wps>
<text>, CONSISTING OFTHE FOLLOWING:
</text>
</totalnumberof>
<col.title>Page / WP No.
</col.title>
<col.title>Change No.
</col.title>
<chghistory modified="none">
<pageno>Front cover
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory>
<title>Warning Summary
</title>
<totnum.pages>4
</totnum.pages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<pageno>i
</pageno>
<pageno>iii
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>ivblank
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp1 title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00001-X-XxXX-XXX"/>
<wppages>4 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00002-X-XxXX-XXX"/>
<wppages>10 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00003-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00004-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp2 title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="t00001-X-XxXX-XXX"/>
<wppages>2 pgs

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="t00002-X-XxXX-XXX"/>
<wppages>8  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="t00003-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<pageno>Chp3  title  page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00001-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00003-X-XxXX-XXX"/>
<wppages>12  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00004-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00005-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<pageno>Chp4  title  page
</pageno>
<chgno>0
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00002-X-XxXX-XXX"/>
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</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00006-X-XxXX-XXX"/>
<wppages>30  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00007-X-XxXX-XXX"/>
<wppages>30  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00008-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00009-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00010-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00011-X-XxXX-XXX"/>
<wppages>8  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00012-X-XxXX-XXX"/>
<wppages>12  pgs
</wppages>
<chgno>0
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
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<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00014-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
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</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00001-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00002-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00003-X-XxXX-XXX"/>
<wppages>6  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00004-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00005-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00006-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00007-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00008-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00009-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00010-X-XxXX-XXX"/>
<wppages>6  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00011-X-XxXX-XXX"/>
<wppages>6  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00012-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00013-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</chghistory>
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<wpno  wpref="s00014-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
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</chgno>
</chghistory>
<chghistory  modified="none">
<pageno>INDEX-1
</pageno>
<pageno>INDEX-14
</pageno>
<chgno>0
</chgno>
</chghistory>
</loepwp>
```

b. Example of a stylesheet output for a list of effective pages/work packages for a revised publication:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

MANUAL NUMBER

LIST OF EFFECTIVE PAGES/WORK PACKAGES

NOTE: This manual supersedes TM X-XXX-XXX dated 15 March 1998. Zero in the "Change No." column indicates an original page or work package.

Date of issue for the revision is:

Original 16 September 1998

TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 28 AND TOTAL NUMBER OF WORK PACKAGES IS 35, CONSISTING OF THE FOLLOWING:

Page/WP No.	Change No.	Page/WP No.	Change No.
Front cover	0	WP 0028 (4 pgs)	0
Warning Summary (4 pgs)	0	WP 0029 (4 pgs)	0
i – iii	0	WP 0030 (4 pgs)	0
iv blank	0	WP 0031 (6 pgs)	0
Chp 1 title page	0	WP 0032 (6 pgs)	0
WP 0001 (4 pgs)	0	WP 0033 (2 pgs)	0
WP 0002 (10 pgs)	0	WP 0034 (2 pgs)	0
WP 0003 (2 pgs)	0	WP 0035 (2 pgs)	0
WP 0004 (2 pgs)	0	INDEX-1 – INDEX-14	0
Chp 2 title page	0		
WP 0005 (2 pgs)	0		
WP 0006 (8 pgs)	0		
WP 0007 (2 pgs)	0		
Chp 3 title page	0		
WP 0008 (2 pgs)	0		
WP 0009 (12 pgs)	0		
WP 0010 (2 pgs)	0		
WP 0011 (2 pgs)	0		
Chp 4 title page	0		
WP 0012 (20 pgs)	0		
WP 0013 (30 pgs)	0		
WP 0014 (30 pgs)	0		
WP 0015 (2 pgs)	0		
WP 0016 (4 pgs)	0		
WP 0017 (2 pgs)	0		
WP 0018 (8 pgs)	0		
WP 0019 (12 pgs)	0		
WP 0020 (2 pgs)	0		
WP 0021 (2 pgs)	0		
Chp 5 title page	0		
WP 0022 (2 pgs)	0		
WP 0023 (4 pgs)	0		
WP 0024 (6 pgs)	0		
WP 0025 (4 pgs)	0		
WP 0026 (4 pgs)	0		
WP 0027 (4 pgs)	0		

A

FIGURE 117. Example of a stylesheet output for a list of effective pages/work packages for a revised publication..

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

31.4.1.4.9 Example – list of effective pages/work packages for a multi-volume, multi-service manual. This example shows a list of effective pages/work packages for the first volume of a multi-service TM with the Air Force being the procuring activity. The list of effective pages/work packages for the first volume covers all of the volumes. Each volume number is listed followed by the pages in that volume. Each volume, except volume 1, includes a list of effective pages/work packages for that particular volume only. For multi-service manuals, the list of effective pages/work packages contains the abbreviation of the acquiring service (e.g., USA, USN, USMC, or USAF) which is display in the lower right-hand corner of the page. To display the Air Force as acquiring service, the attribute **service** in the element (***<servbranch>***) is set to "af" and the attribute **procuring** is set to "yes". The element service branch ***<servbranch>*** is a child of TM information***<tminfono>*** which is a child of TM title ***<tmtitle>*** which is a child of front cover ***<frntcover>*** (see 31.3.1.1.4).

a. XML document instance fragment

```
<loepwp>
<title>List of Effective Pages/Work Packages
</title>
<note>
<trim.para esd="no" hcp="no">Zero in the &ldquo;Change No.&rdquo;column
indicates an original page or work package.
</trim.para>
</note>
<issuechg>
<trim.para esd="no" hcp="no">Dateof issue for the original is:
</trim.para>
<issued>
<chgno>Original
</chgno>
<chgdate julian="19900424">24April 1990
</chgdate>
</issued>
</issuechg>
<totalnumberof>
<text>Total number of Volumesis
</text>
<totnum.volumes>3
</totnum.volumes>
<text>, TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS
</text>
<totnum.frnt-rear-pages>30
</totnum.frnt-rear-pages>
<text> AND TOTAL NUMBER OF WORK PACKAGES IS
</text>
<totnum.wps>35
</totnum.wps>
<text>, CONSISTING OF THE FOLLOWING:
</text>
</totalnumberof>
<col.title>Page / WP No.
</col.title>
<col.title>Change No.
</col.title>
<chgvol>VOLUME 1
</chgvol>
<chghistory modified="none">
<pageno>Front cover
</pageno>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<chgno>0
</chgno>
</chghistory>
<chghistory>
<title>Warning Summary
</title>
<totnum.pages>4
</totnum.pages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>i
</pageno>
<pageno>iii
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>ivblank
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp1 title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00001-X-XxXX-XXX"/>
<wppages>4 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00002-X-XxXX-XXX"/>
<wppages>10 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00003-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="g00004-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp2 title page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="t00001-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="t00002-X-XxXX-XXX"/>
<wppages>8 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="t00003-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory>
<pageno>INDEX-1
</pageno>
<pageno>INDEX-2
</pageno>
<chgno>0
</chgno>
</chghistory>
<chgvol>VOLUME 2
</chgvol>
<chghistorymodified="none">
<pageno>Front cover
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>i
</pageno>
<pageno>ii
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<pageno>Chp3 title page

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00001-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00003-X-XxXX-XXX"/>
<wppages>12  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00004-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00005-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<pageno>Chp4  title  page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00002-X-XxXX-XXX"/>
<wppages>20  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00006-X-XxXX-XXX"/>
<wppages>30  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="m00007-X-XxXX-XXX"/>
<wppages>30  pgs
</wppages>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00008-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00009-X-XxXX-XXX"/>
<wppages>4 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00010-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00011-X-XxXX-XXX"/>
<wppages>8 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00012-X-XxXX-XXX"/>
<wppages>12 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00013-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="m00014-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory>
<pageno>INDEX-1
</pageno>
<pageno>INDEX-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</pageno>
<chgno>0
</chgno>
</chghistory>
<chgvol>VOLUME  3
</chgvol>
<chghistorymodified="none">
<pageno>Front  cover
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<pageno>i
</pageno>
<pageno>ii
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<pageno>Chp5  title  page
</pageno>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00001-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00002-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00003-X-XxXX-XXX"/>
<wppages>6  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00004-X-XxXX-XXX"/>
<wppages>4  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00005-X-XxXX-XXX"/>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<wppages>4 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="s00006-X-XxXX-XXX"/>
<wppages>4 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="s00007-X-XxXX-XXX"/>
<wppages>4 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="s00008-X-XxXX-XXX"/>
<wppages>4 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="s00009-X-XxXX-XXX"/>
<wppages>4 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="s00010-X-XxXX-XXX"/>
<wppages>6 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="s00011-X-XxXX-XXX"/>
<wppages>6 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="s00012-X-XxXX-XXX"/>
<wppages>2 pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory modified="none">
<wpno wpref="s00013-X-XxXX-XXX"/>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<wpno  wpref="s00014-X-XxXX-XXX"/>
<wppages>2  pgs
</wppages>
<chgno>0
</chgno>
</chghistory>
<chghistory  modified="none">
<pageno>INDEX-1
</pageno>
<pageno>INDEX-4
</pageno>
<chgno>0
</chgno>
</chghistory>
</loepwp>
```

b. Example of a stylesheet output for a list of effective pages/work packages for a multi-volume, multi-service manual:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

manual number

LIST OF EFFECTIVE PAGES/WORK PACKAGES

NOTE: Zero in the "Change No." column indicates an original page or work package.

Date of issue for the original is:

Original 24 April 1990

TOTAL NUMBER OF VOLUMES IS 3, TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 30 AND TOTAL NUMBER OF WORK PACKAGES IS 35, CONSISTING OF THE FOLLOWING:

Page/WP No.	Change No.	Page/WP No.	Change No.
VOLUME 1		i–ii	0
Front cover	0	Chp 5 title page	0
Warning Summary (4 pgs)	0	WP 0022 (2 pgs)	0
i–iii	0	WP 0023 (4 pgs)	0
iv blank	0	WP 0024 (6 pgs)	0
Chp 1 title page	0	WP 0025 (4 pgs)	0
WP 0001 (4 pgs)	0	WP 0026 (4 pgs)	0
WP 0002 (10 pgs)	0	WP 0027 (4 pgs)	0
WP 0003 (2 pgs)	0	WP 0028 (4 pgs)	0
WP 0004 (2 pgs)	0	WP 0029 (4 pgs)	0
Chp 2 title page	0	WP 0030 (4 pgs)	0
WP 0005 (2 pgs)	0	WP 0031 (6 pgs)	0
WP 0006 (8 pgs)	0	WP 0032 (6 pgs)	0
WP 0007 (2 pgs)	0	WP 0033 (2 pgs)	0
INDEX-1 – INDEX-2	0	WP 0034 (2 pgs)	0
VOLUME 2		WP 0035 (2 pgs)	0
Front cover	0	INDEX-1 – INDEX-4	0
i–ii	0		
Chp 3 title page	0		
WP 0008 (2 pgs)	0		
WP 0009 (12 pgs)	0		
WP 0010 (2 pgs)	0		
WP 0011 (2 pgs)	0		
Chp 4 title page	0		
WP 0012 (20 pgs)	0		
WP 0013 (30 pgs)	0		
WP 0014 (30 pgs)	0		
WP 0015 (2 pgs)	0		
WP 0016 (4 pgs)	0		
WP 0017 (2 pgs)	0		
WP 0018 (8 pgs)	0		
WP 0019 (12 pgs)	0		
WP 0020 (2 pgs)	0		
WP 0021 (2 pgs)	0		
INDEX-1 – INDEX-4	0		
VOLUME 3			
Front cover	0		

A USAF

FIGURE 118. Example of a stylesheet output for a list of effective pages/work packages for a multi-volume manual.

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Check the source to verify that this is the current version before use.

31.4.2 Rear matter ***<rear>***. The element ***<rear>*** is found in a page-based TM following the last work package. The rear section requires at least one reporting errors and recommending improvements DA Forms 2028–2, an authentication page, and a back cover (optional back cover graphic) and as specified a glossary, alphabetical index, and foldout section.

a. The components are:
   (1) Glossary ***<glossary>*** (optional – zero or one). The element provides the glossary (see 31.4.2.1).
   (2) Alphabetical index ***<aindx>*** (optional – zero or one). The element provides the alphabetical index (see 31.4.2.2).
   (3) Reporting errors and recommending improvements DA Forms 2028 ***<da2028>*** (required – one or more). The element provides the DA Forms 2028 (see 31.4.2.3).
   (4) Authentication page ***<authent>*** (required). The element provides the authentication page (see 31.4.2.4).
   (5) Foldout section ***<foldsect>*** (optional – zero or one). The element provides the position for the foldout section (see 31.4.2.5).
   (6) Back cover ***<back>*** (required). The element provides the back cover with optional graphic (see 31.4.2.6).

b. DTD fragment is graphically depicted.

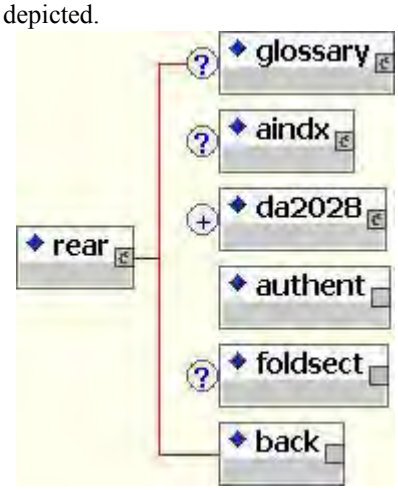

FIGURE 119. Rear matter <rear> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT rear (glossary?, aindx?, da2028+, authent, foldsect?, back)>
<!ATTLIST rear
     assocfig          IDREFS                          #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (see 49.2.6)
   (2) **id** – Unique identifier. (optional) (see 49.2.6.)
   (3) **idref** – Reference one or more identifiers. (optional) (see 49.2.6)

31.4.2.1 Glossary ***<glossary>***. The element ***<glossary>*** is used for a glossary of terms and definitions contained in a page-based TM. It contains a definition list ***<deflist>*** that contains each term ***<terms>*** and associated definition ***<def>***. The terms are listed in alphabetical order. The glossary begins on the right-hand page and immediately precedes the alphabetical index. Page numbers for a glossary begin with Glossary-1, Glossary-2, etc.

a. The components are:
   (1) Title ***<title>*** (required). The element provides the title for the glossary (see 49.1.1.3).

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(2) Reduced paragraph ***<trim.para>*** (optional – zero or one). The element provides general information on the glossary. (see 49.1.1.7).
(3) Definition list ***<deflist>*** (optional – zero or one). The element provides a list of terms and definitions used in the glossary (see 49.1.2.3).

b. DTD fragment is graphically depicted.

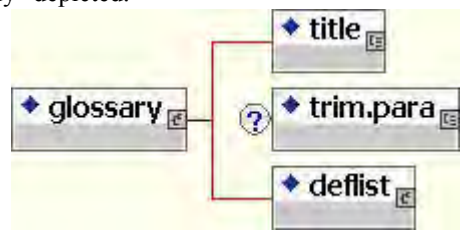

FIGURE 120. Glossary <glossary> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT glossary (title, trim.para?, deflist)>
<!ATTLIST glossary
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.4.2.1.1 How to develop a glossary. A glossary is developed by tagging uncommon words that need to be defined in the TM. The uncommon word(s), abbreviation, or term is wrapped with the element ***<term>*** and the associated definition or spelled out abbreviation is wrapped with ***<def>*** and ***<term>*** elements. Each ***<term>*** and ***<def>*** is wrapped with the element ***<term.def>*** with a required unique identifier attribute **id**. Uncommon words may be used only once or many times through out the TM, so when identifying the uncommon words in the TM that are to be tagged, also check for the suffix endings of a uncommon word like **s, ing, ed,** etc. An IETM does not contain a glossary but can contain uncommon words (see 34.1.1for information in General Information Work Package). The unique identified in ***<term.def>*** is used to link to the term when used in the TM for a hot reference (as a mouse over to provide the definition). The uncommon word is linked to the glossary by the element ***<term>*** and the attribute **idref** to the unique identifier from ***<term.def>***. An example of the mouse-over technique is by rolling over the word "Antenna Drive Unit" with the mouse which would generate a popup text box displaying “Provides support and movement control to the antenna”. See MIL-STD-40051-2 for more information on other methods using the mouse-over technique.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

31.4.2.1.2 Example of a ***<glossary>***. The following examples show how an uncommon word is tagged in a TM and how it is connected to the glossary of the TM. Example 31.4.2.1.2.c. and example 31.4.2.1.2.a. are XML instance fragments. Example 31.4.2.1.2.b. is a possible stylesheet output of a glossary in a TM.

a. Example of an XML document instance fragment for ***<glossary>***:

```
<glossary>
<title>GLOSSARY
</title>
<trim.para>The terms that are used in this TM that are
uncommon are defined as follows:
</trim.para>
<deflist>
<title.term.def>
<title>TERM
</title>
<title>DEFINITION
</title>
</title.term.def>
<term.def id="a_d_u">
<term>Antenna Drive Unit
</term>
<def>
<para>Provides support and movement control to the antenna.
</para>
</def>
</term.def>
<term.def id="feedhorn">
<term>Feedhorn
</term>
<def>
<para>Conducts echoes of RF energy from the antenna re-
flectors to the receiver transmitter
</para>
</def>
</term.def>
<term.def id="h_t_u">
<term>Handheld Terminal Unit, (HTU)
</term>
<def>
<para>Handheld Terminal Unit used by the soldier in the
field to control the radar set.
</para>
</def>
</term.def>
<term.def id="p_p_i">
<term>Planned Position Indicator
</term>
<def>
<para>Provides a graphic representation of the area of interest
drawn to scale with azimuth, range and direction displayed
on the Handheld Terminal Unit (HTU).
</para>
</def>
</term.def>
<term.def id="range_cell">
<term>Range Cell
</term>
```


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```
<def>
<para>Range window selected by user to pinpoint specific targets.
</para>
</def>
</term.def>
<term.def  idref="remote_cable">
<term>Remote  Cable
</term>
<def>
<para>The interface cable used to carry target and control
data to and from the control indicator.
</para>
</def>
</term.def>
</deflist>
</glossary>
```

b. Example of an formatted display of an XML document instance fragment for ***<glossary>***

GLOSSARY

The terms that are used in this TM that are uncommon are defined as follows:

TERM	DEFINITION
Antenna Drive Unit	Provides support and movement control to the antenna.
Feedhorn	Conducts echoes of RF energy from the antenna reflectors to the receiver transmitter
Handheld Terminal Unit, (HTU)	Handheld Terminal Unit used by the soldier in the field to control the radar set.
Planned Position Indicator	Provides a graphic representation of the area of interest drawn to scale with azimuth, range and direction displayed on the Handheld Terminal Unit (HTU).
Range Cell	Range window selected by user to pinpoint specific targets.
Remote Cable	The interface cable used to carry target and control data to and from the control indicator.

Glossary-1

FIGURE 121. Example of a Glossary Page <glossary>.

c. Example of tagging an uncommon word ***<term>*** element in a TM XML document instance fragment with reference to definition in the glossary.

```
<locdesc>
<title>LOCATION AND DESCRIPTIONS OF MAJOR COMPONENTS
</title>
<comp-item>
<title>
```


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```
<term idref="a_d_u">ANTENNA DRIVE UNIT
</term>
</title>
<para>The
<term idref="a_d_u">Antenna Drive Unit
</term> contains an electric motor, gearing, absolute elevation encoder,
optical azimuth encoder and a mounting and leveling assembly.  Two
side brackets on trunnions, one at each side of the
<term idref="a_d_u">Antenna Drive Unit
</term>, support the R-T assembly.  An elevation adjustment wheel permits
the R-T assembly to be tilted forward or backward to control the elevation
of the radar beam.  An elevation lock lever secures the R-T assembly at the
tilt angle desired.  An elevation stow lock holds the R-T upright (zero
deviation).  Locking protects the elevation mechanism during transport.
</para>
</comp-item>
</locdesc>
```

d. Example of an formatted display of an XML document instance fragment for uncommon word:

LOCATION AND DESCRIPTIONS OF MAJOR COMPONENTS

ANTENNA DRIVE UNIT

The Antenna Drive Unit contains an electric motor, gearing, absolute elevation encoder, optical azimuth encoder and a mounting and leveling assembly. Two side brackets on trunnions, one at each side of the Antenna Drive Unit, support the R-T assembly. An elevation adjustment wheel permits the R-T assembly to be tilted forward or backward to control the elevation of the radar beam. An elevation lock lever secures the R-T assembly at the tilt angle desired. An elevation stow lock holds the R-T upright (zero deviation). Locking protects the elevation mechanism during transport.

FIGURE 122. Example of an uncommon term <term> reference.

31.4.2.2 Alphabetical index ***<aindx>***. The element ***<aindx>*** is used for an alphabetical index of subjects that may be useful to the TM user. The index is automatically generated, pre-composition process populated, or manually entered. The alphabetical index is a optional task element because the composition system of applications varies how it handles specific tasks. The automatically generated index uses an XML element marker (***<indxref>***) to determine the index subject and location. Some composition systems can generate the index with the assistance of a pre-composition process. The pre-composition process reads the completed TM XML instance and uses the index reference elements to populate the ***<aindx>*** XML instance. While other composition systems can not handle an automated task or process such as an alphabetical index and has to be manually created. Additional information on the process of the alphabetical index and other optional methods (see 31.4.2.2.4) in developing an alphabetical index.

a. The components are:
   (1) Title ***<title>*** (required). The element provides the title for the alphabetical index, usually the text “INDEX” is entered (see 49.1.1.3).
   (2) Reduced paragraph ***<trim.para>*** (optional – zero or one). The element allows for an introduction to the index. Sample text is “This index is organized alphabetically by topic and subtopic. Topics and subtopics are identified by page number” (see 49.1.1.7).
   (3) Column list title ***<col.title>*** (required). The element provides the title for the subject (column title is “Subject”) and, if applicable, the page number (column title is “WP Sequence No.–Page No.”) columns in the Index list(see 31.4.1.4.3).
   (4) Category heading ***<alphaindx>*** (optional – zero or one). The element provides the a separator for each alphabetical category in an index (see 31.4.2.2.1).
   (5) Index entry ***<indexentry>*** (optional – zero or more). Index entry is used to enter the subject and page number reference (when applicable). When subentries are needed, the element

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recursively references itself to enter the subordinate subjects and page number references (when applicable) (see 31.4.2.2.2).

b. DTD fragment is graphically depicted.

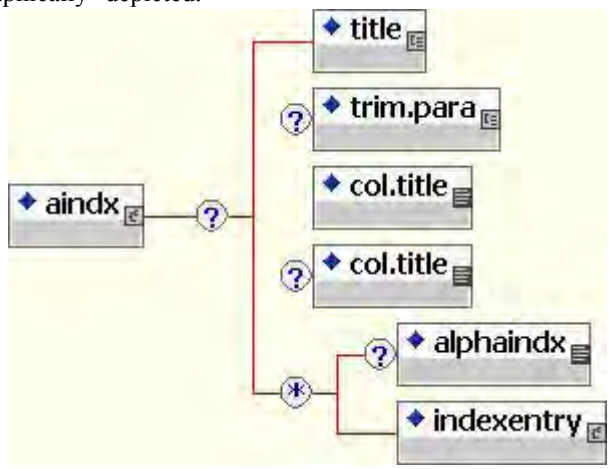

FIGURE 123. Alphabetical index <aindx> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT aindx (title, trim.para?, col.title, col.title?,
 (alphaindx?, indexentry)*)?>
<!ATTLIST aindx
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.4.2.2.1 Category heading ***<alphaindx>***. The element ***<alphaindx>*** is used for the heading of an alphabetical category in an alphabetical index. Using the category heading can make it quicker to locate the subject, since the heading is formatted differently then the index entries making easier to identify a index group.

a. The component is:

(1) The element contains narrative text #PCDATA parsable character data. The element provides the alphabetical heading (i.e., “A”).

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b. DTD fragment:

```
<!ELEMENT alphaindx (#PCDATA)>
```

c. No attributes.

31.4.2.2.2 Index entry ***<indexentry>***. This element contains the necessary data to create an index manually or through a pre-composition process. This element is recursive to allow the required index levels for the manual. The element ***<indexentry>*** establishes a cross-reference to the index location in the document and the index subject text used within the alphabetical index.

a. The components are:
   (1) Title ***<title>***. (required). The element provides the subject of the index entry (see 49.1.1.3).
   (2) Select the page number reference format (optional – zero or more) as
      (a) Work package page number reference that requires both a work package number reference ***<wpno>*** (see 46.2.5) and the associated page number***<pageno>*** (optional – zero or more). The element provides a cross-reference to the work package identification number, which the style sheet uses to generate the work package sequence number. A reference is used since the work package sequence can be changed due to a revision or change, but the work package identification number does not change. The page number is either a manually entered number or a cross-reference in the document to subject area. Note the stylesheet determines the page number during composition.
      (b) Front or rear page number ***<pageno>*** The element is used to enter the front or rear matter page number, since neither section has a work package sequence number that may change. Note the stylesheet determines the page number during composition. The content model is #PCDATA.
   (3) Index entry ***<indexentry>*** (optional – zero or more). When subordinate index entry subject(s) apply, the element is recursively referenced to enter each subordinate index entry information.

b. DTD fragment is graphically depicted.

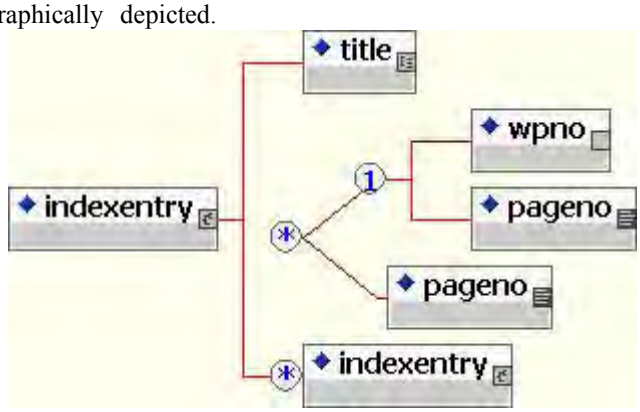

FIGURE 124. Index entry <indexentry> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT indexentry (title, ((wpno, pageno) | pageno)*, indexentry*)>
<!ATTLIST indexentry
     assocfig          IDREFS                          #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (see 49.2.6)
   (2) **id** – Unique identifier. (optional) (see 49.2.6.)
   (3) **idref** – Reference one or more identifiers. (optional) (see 49.2.6)

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(4) **security** – Security classification. (optional) (see 49.2.13)

31.4.2.2.3 Index reference marker ***<indxref>***. The element ***<indxref>*** establishes a document target (used by composition systems to determine the page number) and the index subject text to be used within the alphabetical index. The***<indxref>*** element can store up to four subject level tittles to be listed under the index entry. Note that if "**ref1**" has different tense on the word the subject information will not be grouped together, but as separate entries.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT indxref EMPTY>
<!ATTLIST indxref
     id         ID      #IMPLIED
     indxref    IDREF   #IMPLIED
     ref1       CDATA   #IMPLIED
     ref2       CDATA   #IMPLIED
     ref3       CDATA   #IMPLIED
     ref4       CDATA   #IMPLIED >
```

c. Common attributes:
   (1) **id** – Unique identifier. (optional) (see 49.2.6) The identifier can be used, during either manually or pre-composition processing, to indicate to page number to be displayed in the index.
   (2) **idref** – Reference one or more identifiers. (optional) (see 49.2.6) The cross reference can be used by the frame-based TM or IETM to provide bidirectional hot references to the index and to the index entry source (see 31.4.2.2.4)
d. Unique attributes:
   (1) **ref1** – Reference level 1(optional). Index subject text to be referenced as level one.
   (2) **ref2** – Reference level 2 (optional). Index subject text to be referenced as level two.
   (3) **ref3** – Reference level 3 (optional). Index subject text to be referenced as level three.
   (4) **ref4** – Reference level 4 (optional). Index subject text to be referenced as level four.

31.4.2.2.4 How to develop an alphabetical index. An alphabetic index can be developed by using index reference marker ***<indxref>*** element (see 31.4.2.2.3) that establishes a document location and index subject and subordinate subjects text to be referenced within the ***<aindx>*** element. A composition system may be able to generate the index entry subject and page number with no additional external applications (see 31.4.2.2.4.1), while another system needs an additional pre-composition application to generate this information (see 31.4.2.2.4.2), another the composition can use manual entered data to generate page number (see 31.4.2.2.4.3), and other composition systems requires all data inserted manually (see 31.4.2.2.4.4). These four (4) authoring methods are provided for generating index reference and index by the use of the index reference ***<indxref>*** element. Methods (1) and (2) require an application to generate necessary ID and IDREF automatically. Method (3) and (4) are manually inputted by the author. Using methods (1), (2), or (3) will provide the necessary information for any composition system to publish an index and furnish a mechanism to hyper-link between the index and the index reference. Method (4) does not provide any linking mechanism, since ***<indxref>*** element is not used.

a. Method 1 – Index reference defines the index level titles.
b. Method 2 – Reference the index level title generated in the index.
c. Method 3 – Index reference ID and index entries point to the index reference identifier.
d. Method 4 – Index subject and page number entered manually.

31.4.2.2.4.1 Method 1 – index reference defines the index level subjects. Method 1 has the author to specify for each index reference entry the specify index level title(s). This method is the same as described in MIL–PRF–28001 to generate an index. MIL–STD–2361 provides a better tool for the soldier to find the indexed material, when the information is used electronically. To improve the index authoring methodology an application is required. The application will perform the following tasks: Read the

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document instance to obtain the index level topic and subtopics. Assign an automatic generated ID to the index reference. Sort the index level topics and subtopics. Generate the index using the XML elements with the associated generated index reference IDs.

a. Read the document instance to obtain the index level topic and subtopics.
b. Assign an automatic generated ID to the index reference.
c. Sort the index level topics and subtopics.
d. Generate the index using the XML elements with the associated generated index reference IDs.

The application is necessary for the composition system to generate alphabetically sorted index, the page number, and the work package number for the index reference. The author's responsibility is to enter the index level topics for each XML element. The index reference IDs, IDREFs and index entries will be generated after completing the index reference XML elements. The application will be applied to the document instance to prepare the document for publication.

31.4.2.2.4.1.1 Method 1 – application development process. Using Method 1, the following steps can be used in developing an application. The steps provide methodology and not precise coding.

a. The process would review the XML instance and search for the ***<indxref>*** element.
b. The process would search for each ***<indxref>*** element marked in the TM.
c. When the***<indxref>*** element is found
   (1) An "unique" identifier would be added to the ***<indxref>*** element (for page number reference).
   (2) Attributes "**ref1**" through "**ref4**" would be collected and stored.
   (3) The ***<indexentry>*** element is stored for the lowest entered subject attribute entry including the index reference subject (use the attribute value), the work package identification number (if applicable), and the "unique" identifier for the page number reference. The text below is an example of the ***<indxref>*** element and the information to stored in the ***<indexentry>*** element.
      (a) Work package identification number is "**M00154–X–XXX-XXX**"
      (b) The ***<indxref>*** element is ***<indxref ref1="radio" ref2="SINCGARS"/>***
      (c) The process stores the **ref1** and **ref2** attributes for sorting.
      (d) The unique identification generated by the process could be "**M00154–X–XXX-XXX-index1**"
      (e) The ***<indxref>*** element is changed to ***<indxref id="M00154–X–XXX-XXX-index1" ref1="Radio" ref2="SINCGARS"/>***
      (f) The stored ***<indexentry>*** element would store the "**ref2**" value only as shown below

<indexentry>
<title>SINCGARS
</title>
<wpno wpref="M00154–X–XXX-XXX"/>
<pageno idref="M00154–X–XXX-XXX-index1">
</pageno>
</indexentry>

d. After locating all the ***<indxref>*** elements in the XML instance, the stored attributes ("**ref1**" through "**ref4**") are alphabetically sorted by each reference attribute level. Therefore, all "**ref1**" attribute values are sorted, then the "**ref2**" attribute values are sorted within the same "**ref1**" subject text, and so further for the four reference subject levels.
e. After the sorting is completed, each entry in the sorted list will use the stored ***<indexentry>*** data to generate the alphabetical index. If the previous subject level text (i.e., attribute "**ref1**") does not have an ***<indexentry>*** element created, an ***<indexentry>*** element is created for the stored "**ref1**" value with no page number data.
f. Example XML instance and stages in the pre-composition process:
   (1) Initial XML instance prior to pre-composition process.

<indxref ref1="radio" ref2="SINCGARS"/> SINCGARS . . .
<indxref ref1="radio" ref2="MX-6707" ref3="AT-1095/VRC Antenna"/>

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<indxref ref1="antenna" ref2="AT-1095/VRC Antenna"/> AT-1095/VRC Upper 66" Antenna Element ...
<indxref ref1="radio" ref2="MX-6707"/> MX-6707 ...
<indxref ref1="radio" ref2="MX-6707" ref3="AS-1729/VRC Antenna"/>
<indxref ref1="antenna" ref2="AS-1729/VRC Antenna"/> AS-1729/VRC Antenna

(2) XML instance after pre-composition process has inserted the unique identifiers.

<indxref id="M00154–X–XXX-XXX-index1" ref1="radio" ref2="SINCGARS"/> SINCGARS ...
<indxref id="M00154–X–XXX-XXX-index2" ref1="radio" ref2="MX-6707" ref3="AT-1095/VRC Antenna"/>
<indxref id="M00154–X–XXX-XXX-index3" ref1="antenna" ref2="AT-1095/VRC Antenna"/> AT-1095/VRC Upper 66" Antenna Element ...
<indxref id="M00154–X–XXX-XXX-index4" ref1="radio" ref2="MX-6707"/> MX-6707 ...
<indxref id="M00154–X–XXX-XXX-index5" ref1="radio" ref2="MX-6707" ref3="AS-1729/VRC Antenna"/>
<indxref id="M00154–X–XXX-XXX-index6" ref1="antenna" ref2="AS-1729/VRC Antenna"/> AS-1729/VRC Antenna

(3) The collected ***<indxref>*** data could be stored in a list (see Table V).

TABLE V. Stored ***<indxref>*** for Pre-Composition Process.

Ref1	Ref2	Ref3	Index Entry
radio	SINC-GARS		***<indexentry>*** ***<title>***SINCGARS ***</title>*** ***<wpno wpref="M00154–X–XXX-XXX"/>*** ***<pageno idref="M00154–X–XXX-XXX-index1">*** ***</pageno>*** ***</indexentry>***
radio	MX-6707	AT-1095/ VRC Antenna	***<indexentry>*** ***<title>***AT-1095/VRC Antenna ***</title>*** ***<wpno wpref="M00154–X–XXX-XXX"/>*** ***<pageno idref="M00154–X–XXX-XXX-index2">*** ***</pageno>*** ***</indexentry>***
antenna	AT-1095/ VRC Antenna		***<indexentry>*** ***<title>***AT-1095/VRC Antenna ***</title>*** ***<wpno wpref="M00154–X–XXX-XXX"/>*** ***<pageno idref="M00154–X–XXX-XXX-index3">*** ***</pageno>*** ***</indexentry>***
radio	MX-6707		***<indexentry>*** ***<title>***MX-6707 ***</title>*** ***<wpno wpref="M00154–X–XXX-XXX"/>*** ***<pageno idref="M00154–X–XXX-XXX-index4">*** ***</pageno>*** ***<indexentry>***

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Table V Stored *<indxref>* for Pre-Composition Process – Continued.

Ref1	Ref2	Ref3	Index Entry
radio	MX-6707	AS-1729/ VRC Antenna	*<indexentry>* *<title>*AS-1729/VRC Antenna *</title>* *<wpno wpref="M00154–X–XXX-XXX"/>* *<pageno idref="M00154–X–XXX-XXX-index5">* *</pageno>* *</indexentry>*
antenna	AS-1729/ VRC Antenna		*<indexentry>* *<title>*AS-1729/VRC Antenna *</title>* *<wpno wpref="M00154–X–XXX-XXX"/>* *<pageno idref="M00154–X–XXX-XXX-index6">* *</pageno>* *</indexentry>*

(4) After collecting all the *<indxref>* data, the list would be sorted in relationship to each column, (see Table VI).

TABLE VI. Sorted for Pre-Composition Process.

Ref1	Ref2	Ref3	Index Entry
antenna	AS-1729/ VRC Antenna		*<indexentry>* *<title>*AS-1729/VRC Antenna *</title>* *<wpno wpref="M00154–X–XXX-XXX"/>* *<pageno idref="M00154–X–XXX-XXX-index6">* *</pageno>* *</indexentry>*
antenna	AT-1095/ VRC Antenna		*<indexentry>* *<title>*AT-1095/VRC Antenna *</title>* *<wpno wpref="M00154–X–XXX-XXX"/>* *<pageno idref="M00154–X–XXX-XXX-index3">* *</pageno>* *</indexentry>*
radio	MX-6707		*<indexentry>* *<title>*MX-6707 *</title>* *<wpno wpref="M00154–X–XXX-XXX"/>* *<pageno idref="M00154–X–XXX-XXX-index4">* *</pageno>* *<indexentry>*

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Table VI Sorted for Pre-Composition Process – Continued.

Ref1	Ref2	Ref3	Index Entry
radio	MX-6707	AS-1729/ VRC Antenna	***<indexentry>*** ***<title>***AS-1729/VRC Antenna ***</title>*** ***<wpno wpref="M00154–X–XXX-XXX"/>*** ***<pageno idref="M00154–X–XXX-XXX-index5">*** ***</pageno>*** ***</indexentry>***
radio	MX-6707	AT-1095/ VRC Antenna	***<indexentry>*** ***<title>***AT-1095/VRC Antenna ***</title>*** ***<wpno wpref="M00154–X–XXX-XXX"/>*** ***<pageno idref="M00154–X–XXX-XXX-index2">*** ***</pageno>*** ***</indexentry>***
radio	SINCGARS		***<indexentry>*** ***<title>***SINCGARS ***</title>*** ***<wpno wpref="M00154–X–XXX-XXX"/>*** ***<pageno idref="M00154–X–XXX-XXX-index1">*** ***</pageno>*** ***</indexentry>***

(5) After sorting the list, the pre-composition process would generate the ***<aindx>*** XML instance. All "ref1" with the same data will be grouped together. Any "ref2" with the same data will be grouped together when under the same "ref1" group. The following example is the XML instance from previous index data.

```
<aindx>
<title>INDEX
</title>
<col.title>Subject
<col.title>
<col.title>WP Sequence No.&ndash;Page No.
<col.title>
<alphaindex>A
</alphaindex>
<indexentry>
<title>antenna
</title>
<indexentry>
<title>AS-1729/VRC Antenna
</title>
<wpno wpref="M00154–X–XXX-XXX"/>
<pageno idref="M00154–X–XXX-XXX-index4">
</pageno>
</indexentry>
<indexentry>
<title>AT-1095/VRC Antenna
</title>
<wpno wpref="M00154–X–XXX-XXX"/>
<pageno idref="M00154–X–XXX-XXX-index2">
```


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```
</pageno>
</indexentry>
</indexentry>
<alphaindex>R
</alphaindex>
<indexentry>
<title>radio
</title>
<indexentry>
<title>MX-6707
</title>
<wpno  wpref="M00154–X–XXX-XXX"/>
<pageno  idref="M00154–X–XXX-XXX-index3">
</pageno>
<indexentry>
<title>AS-1729/VRC  Antenna
</title>
<wpno  wpref="M00154–X–XXX-XXX"/>
<pageno  idref="M00154–X–XXX-XXX-index4">
</pageno>
</indexentry>
<indexentry>
<title>AT-1095/VRC  Antenna
</title>
<wpno  wpref="M00154–X–XXX-XXX"/>
<pageno  idref="M00154–X–XXX-XXX-index2">
</pageno>
</indexentry>
</indexentry>
<indexentry>
<title>SINCGARS
</title>
<wpno  wpref="M00154–X–XXX-XXX"/>
<pageno  idref="M00154–X–XXX-XXX-index1">
</pageno>
</indexentry>
</indexentry>
</aindx>
```


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(6) The composed instance would display the information as follows:

INDEX

Subject	WP Sequence No.–Page No.
—A—	
antenna	
AS-1729/VRC Antenna	0023–5
AT-1095/VRC Antenna	0023–4
—R—	
radio	
MX-6707	0023–4
AS-1729/VRC Antenna	0023–5
AT-1095/VRC Antenna	0023–4
SINCGARS	0023–3

FIGURE 125. Example of Method 1 Alphabetical Index Output.

31.4.2.2.4.2 Method 2 – reference the index level title generated in the index. Method 2 has the author create the index entries and associate a unique identifier to each index level entry. The writer authors the document instance and applies the XML element for index reference with attribute IDREF pointing to the index entry ID location. The method provides a consistent list of possible entries and will reduce spelling errors. The disadvantage is recalling or looking up the associated index entry ID. Again as Method 1, an application is required to provide a better tool for the soldier to find the indexed material. The application will perform the following tasks:

a. Read the document instance to obtain the index entry reference identifier.
b. Assign an automatic generated ID to the index reference.
c. Assign to the associated index entry the index reference generated ID.

The application is necessary for the composition system to generate the page number, the work package number for the index reference. The author's responsibility is to

a. Enter the sorted index entries with associated unique identifiers
b. Enter the XML element index reference with associated attribute index entry IDREF

The application will be applied to the document instance to prepare the document for publication.

31.4.2.2.4.3 Method 3 – index reference ID and index entries point to the index reference identifier. Method 3 does not require a separate application applied to the document instance. The author will manually perform the following tasks to permit index generation.

a. Generate the sorted index entries.
b. Create the index reference with a unique identifier.
c. Associate the index reference index to the index entry.
d. Generate new index entries, when required.

An application is not required to be applied to the document instance and the author will make the association when creating the index reference entry. However, the author should have a valid list of available IDs and the same index entry document instance needs to be shared or merged with other authors. The author's responsibility is to

a. Enter the sorted index entries
b. Enter the index reference with an unique identifier for each XML element
c. Associate the ID to the index entry IDREF

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31.4.2.2.4.4 Method 4 – index subject and page number entered manually. Method 4 does not require a separate application applied to the document instance. The author will manually perform the following tasks to permit index generation.

a. Generate the index entries.
b. Each index entry create the cross reference to the work package identification number (optional).
c. Each index entry enter the page number (if the work package identification number is not referenced the work package sequence number is entered).
d. Sort the index entries.
e. Generate new index entries, when required.
f. Review and update the page numbers when information is changed.

An application is not required to be applied to the document instance and the author will make the association when creating the index reference entry. However, the author is not required to identifiers for each index entry. The author's responsibility is to

a. Enter the sorted index entries
b. Identifying and maintaining the page number(s) for the indexed subject.

31.4.2.2.4.4.1 Method 4 – ***<aindx>*** XML markup . The following is an example of an XML instance and the formatted display of the alphabetical index using Method 4.

a. Example of an XML document instance fragment for Alphabetical index ***<aindx>***:

```
<aindx>
<title>INDEX
</title>
<col.title>Subject
</col.title>
<col.title>WP Sequence No.&ndash;Page No.
</col.title>
<alphaindx>A
</alphaindx>
<indexentry>
<title>Accessory Section
</title>
<indexentry>
<title>Description
</title>
<wpno wpref="G0002-X-XXX-XXXX"/>
<pageno>9
</pageno>
</indexentry>
<indexentry>
<title>Inspection
</title>
<wpno wpref="G0002-X-XXX-XXXX"/>
<pageno>3
</pageno>
</indexentry>
<indexentry>
<title>Installation
</title>
<wpno wpref="G0002-X-XXX-XXXX"/>
<pageno>10
</pageno>
</indexentry>
</indexentry>
<alphaindx>B
</alphaindx>
```


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<indexentry>
<title>Baffle and Spacer
</title>
<wpno wpref="O0045-X-XXX-XXXX"/>
<pageno>7
</pageno>
<indexentry>
<title>Installation
<title>
<wpno wpref="O0045-X-XXX-XXXX"/>
<pageno>9
</pageno>
</indexentry>
</indexentry>
<indexentry>
<title>Bearings, Anti-friction
</title>
<indexentry>
<title>Balance
</title>
<wpno wpref="O0041-X-XXX-XXXX"/>
<pageno>3
</pageno>
</indexentry>
<indexentry>
<title>Cleaning
</title>
<wpno wpref="O0041-X-XXX-XXXX"/>
<pageno>1
</pageno>
</indexentry>
</indexentry>
<alphaindx>C
</alphaindx>
<indexentry>
<title>Carbon Seals
</title>
<indexentry>
<title>Cleaning
</title>
<wpno wpref="G0008-X-XXX-XXXX"/>
<pageno>2
</pageno>
</indexentry>
<indexentry>
<title>Inspection
</title>
<wpno wpref="G0008-X-XXX-XXXX"/>
<pageno>1
</pageno>
</indexentry>
<indexentry>
<title>Replacement
</title>
<wpno wpref="G0008-X-XXX-XXXX"/>
<pageno>1

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</pageno>
</indexentry>
</indexentry>
</aindx>

b. Example of an formatted display of an XML document instance fragment for ***<aindx>***:

INDEX

Subject	WP Sequence No.–Page No.
—A—	
Accessory Section	
Description	0002–9
Inspection	0002–3
Installation	0002–10
—B—	
Baffle and Spacer	0059–7
Installation	0059–9
Bearings, Anti-friction	
Balance	0052–3
Cleaning	0052–1
—C—	
Carbon Seals	
Cleaning	0008–2
Inspection	0008–1
Replacement	0008–1

FIGURE 126. Example of Method 4 Alphabetical Index Output.

31.4.2.3 Reporting errors and recommending improvements DA Forms 2028–2 ***<da2028>***. The element ***<da2028>*** provides the form, Reporting Errors and Recommending Improvements. The DA Form 2028–2 is found in the generally only in page-based TM rear matter with one filled-out sample that includes guidelines for completing the form and three blank copies. In a frame-based TM or IETM information is generally accessed to an electronic form. Depending on the IETM viewer, many fields can be automatically filled-in by known information (i.e., TM number, WP identification, user name, etc.). The electronic form may have a direct connection to the proponent network site or stored internally to be transmitted or printed after completing the maintenance action.

a. The components are:
   (1) Choose either.
      (a) Electronic form application that is comprised of
         1 Launch the electronic form ***<link>*** (required) through a reference to web site or executes an application (see 46.2.3)
         2 TM proponent ***<proponent>*** (required) identifies the proponent name and address where to send the DA Form 2028-2 (see 49.1.4.22).
   (2) Printed DA Form 2028–2 comprised of the element ***<graphic>*** (required) that provides the paper form (see 44.2).

b. DTD fragment is graphically depicted.

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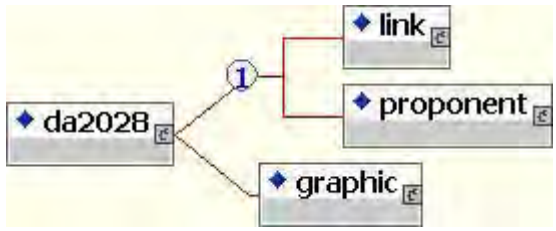

FIGURE 127. Reporting errors and recommending improvements DA Forms 2028 <da2028> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT da2028 ((link, proponent) | graphic)>
<!ATTLIST da2028
     application  (frame | page | both )  'both'>
```

d. Unique attributes:

(1) **application** – The attribute indicates if the element has application for page-based only, frame-based only, or both (default value is "both").

31.4.2.3.1 Example of a ***<da2028>***. The following are examples of a XML instance and the formatted display of the Reporting Errors and Recommending Improvements form.

a. Example of an XML document instance fragment for obtaining a page-based Reporting Errors and Recommending Improvements form ***<da2028>***:

<da2028 application=”page”>
<graphic boardno="da2028form">
</da2028>

b. Example of an XML document instance fragment for obtaining a Reporting Errors and Recommending Improvements form ***<da2028>*** for both an IETM or page-based (depending on the output media):

<da2028 application=”frame”>
<link application="frame" linkaction="prompt" linktype="return" popup="no"
xlink:href="http://www.usapa.army.mil/da2028/daform2028.asp" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
</link>
<proponent>
<name>Commander,
<brk/>US Army Communicatons-Electronics Command and Fort Monmouth,
<brk/>ATTN: AMSELLC-LEO-D-CS-CFO
</name>
<address>
<city>Fort Monmouth
</city>
<state>New Jersey
</state>
<zip>07703-5000
</zip>
</address>
</proponent>
</da2028>
<da2028 application=”page”>
<graphic boardno="da2028formexample">
</da2028>
<da2028 application=”page”>
<graphic boardno="da2028form">
</da2028>
<da2028 application=”page”>

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Check the source to verify that this is the current version before use.

<graphic boardno="da2028form">
</da2028>
<da2028 application="page">
<graphic boardno="da2028form">
</da2028>

31.4.2.4 Authentication page ***<authent>***. The element ***<authent>*** is used for positioning the authentication external graphic for page-based TM only. The authentication page specifies the document is approved for use by the U.S. Army. The external graphic is referenced from the attribute **boardno** (see 44.1 more details about graphics).

a. The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT authent EMPTY>
<!ATTLIST authent
     alt              CDATA                          #IMPLIED
     boardno          ENTITY                         #REQUIRED
     hscale           CDATA                          #IMPLIED
     reprodep         CDATA                          #IMPLIED
     reprowid         CDATA                          #IMPLIED
     scalefit         (yes | no)                     #IMPLIED
     unitmeasure      (mm | cm | px | in | pt | pi)  'in'
     vscale           CDATA                          #IMPLIED >
```

c. Common attributes:
   (1) **alt** – Additional text to assist in identifying the graphic (optional) (see 44.2.d.).
   (2) **boardno** – External graphic entity name (required) (see 44.2.d.).
   (3) **graphicunit** – Graphic unit of measurement (default value is "**in**") (see 44.2.d.).
   (4) **hscale** – Horizontal scaling factor (optional) (see 44.2.d.).
   (5) **reprowid** – Reproduction area width (optional) (see 44.2.d.).
   (6) **reprodep** – Reproduction area depth (optional) (see 44.2.d.).
   (7) **scalefit** – Scale the graphic to fit the reproduction area (optional) (see 44.2.d.).
   (8) **vscale** – Vertical scaling factor (optional) (see 44.2.d.).

31.4.2.4.1 Example of a ***<authent>***. The following is an example of an authentication page XML instance and the formatted display.

a. Example of an XML document instance fragment for ***<authent>***:
   <authent boardno ="authent_TM_X-XXX-XXXX-XX">
b. Example of an formatted display of an XML document instance fragment for ***<authent>*** :

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TM X-XXXX-XXX-XX

By Order of the Secretary of the Army:

PETER J. SCHOOMAKER

General, United States Army Chief of Staff

Official:

(Signature)

SANDRA R. RILEY

Administrative Assistant to the Secretary of the Army

0232913

DISTRIBUTION: To be distributed in accordance with the initial distribution. number XXXXXX requirements for TM X-XXXX-XXX-XX.

FIGURE 128. Example of an authentication page <authent>.

31.4.2.5 Foldout section ***<foldsect>***. The element ***<foldsect>*** is used for collecting the TM's foldout (oversize) illustrations for page-based TM only. Figures that appear in this section have been extracted from the work package (not printed with the work package) when the XML source data contains the element ***<figure figtype="fo-rear">***. The element ***<foldsect>*** designates in the TM XML instance where the foldout sheets will be placed (in the sequence order of the TM work packages). The stylesheet stores the figures which have identified the attribute **figtype** and publish them in the foldout section.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT foldsect EMPTY>
```

c. No attributes.

31.4.2.6 Back cover ***<back>***. The element ***<back>*** is used for the back cover for page-based TM only. The back cover generally contains a metric conversion chart, which is contained as a external graphic. The external graphic is referenced from the attribute **boardno** (see 44.1 more details about graphics).

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.

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b. DTD fragment:

```
<!ELEMENT back EMPTY>
<!ATTLIST back
     alt              CDATA                              #IMPLIED
     boardno          ENTITY                             #IMPLIED
     hscale           CDATA                              #IMPLIED
     reprodep         CDATA                              #IMPLIED
     reprowid         CDATA                              #IMPLIED
     scalefit         (yes | no)                        #IMPLIED
     unitmeasure      (mm | cm | px | in | pt | pi)    'in'
     vscale           CDATA                              #IMPLIED >
```

c. Unique attributes:

(1) **alt** – The attribute contains a description or additional information to be displayed when a mouse-over in frame-based/IETM application is used.

(2) **boardno** – The attribute specifies the entity name that points to the external graphic file. See 44.2.1) for how to define the graphic entity name.

(3) **hscale** – The attribute specifies the horizontal scaling factor for scaling a graphic, expressed in the form of percentages. Exclude the percent sign (%) when entering an "**hscale**" attribute value (i.e., attribute value of 10 is 10%). See 44.2.3 on further information and consideration concerning scaling.

(4) **reprodep** – The attribute specifies the reproduction area depth of the graphic. Enter a numeric value for the fixed depth, excluding the unit of measure. See 44.2.3 on further information and consideration concerning scaling.

(5) **reprowid** – The attribute specifies the reproduction area width of the graphic. Enter a numeric value for the fixed width, excluding the unit of measure. See 44.2.3 on further information and consideration concerning scaling.

(6) **scalefit** – The attribute specifies if the graphic will be scaled to fit the size of the reproduction area. Declared values are "**yes**" or "**no**". The graphic reproduction area or size is determined by the two attributes "**reprowid**" and "**reprodep**". See 44.2.3 on further information and consideration concerning scaling.

(7) **unitmeasure** – The attribute specifies the unit of measure for the value entered in "**reprowid** " or "**reprodep**" attributes. Declared values for this attribute include "mm" (millimeter), "cm" (centimeter), "px" (pixel), "in" (inch), "pt" (point), or "pi" (pica); default value is "in".

(8) **vscale** – The attribute specifies the vertical scaling factor for scaling a graphic, expressed in the form of percentages. Exclude the percent sign (%) when entering a "**vscale**" attribute value (i.e., attribute value of 10 is 10%). See 44.2.3 on further information and consideration concerning scaling.

31.5 Conventional and chemical ammunition ***<ammo>***. The element ***<ammo>*** begins the requirements for conventional and chemical ammunition manuals. This manual describes the handling, shipping, and marking for ammunition. The front matter ***<volume>*** and back matter of volume ***<vol-rear>*** is optionally for a Conventional and Chemical Ammunition Manual.

a. The components are:

(1) Paper Front ***<paper.frnt>*** (required) (see 31.4.1).

(2) General Information Chapter ***<gim>*** (required) (see 34).

(3) Volume ***<volume>*** (optional) (see 31.10.1).

(4) Back Matter of Volume ***<vol-rear>*** (optional) (see 31.10.2).

(5) Operating Instructions ***<opim>*** (optional – zero or more) (see 35).

(6) Effectivity Type Set ***<mim>*** (required – one or more)(see 37).

(7) Supporting Information Chapter ***<sim>*** (required)(see Section 40).

(8) Destruction Information Chapter ***<dim>*** (required)(see 39.1).

(9) Rear matter ***<rear>*** (required)(see 31.4.2).

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b. DTD fragment is graphically depicted.

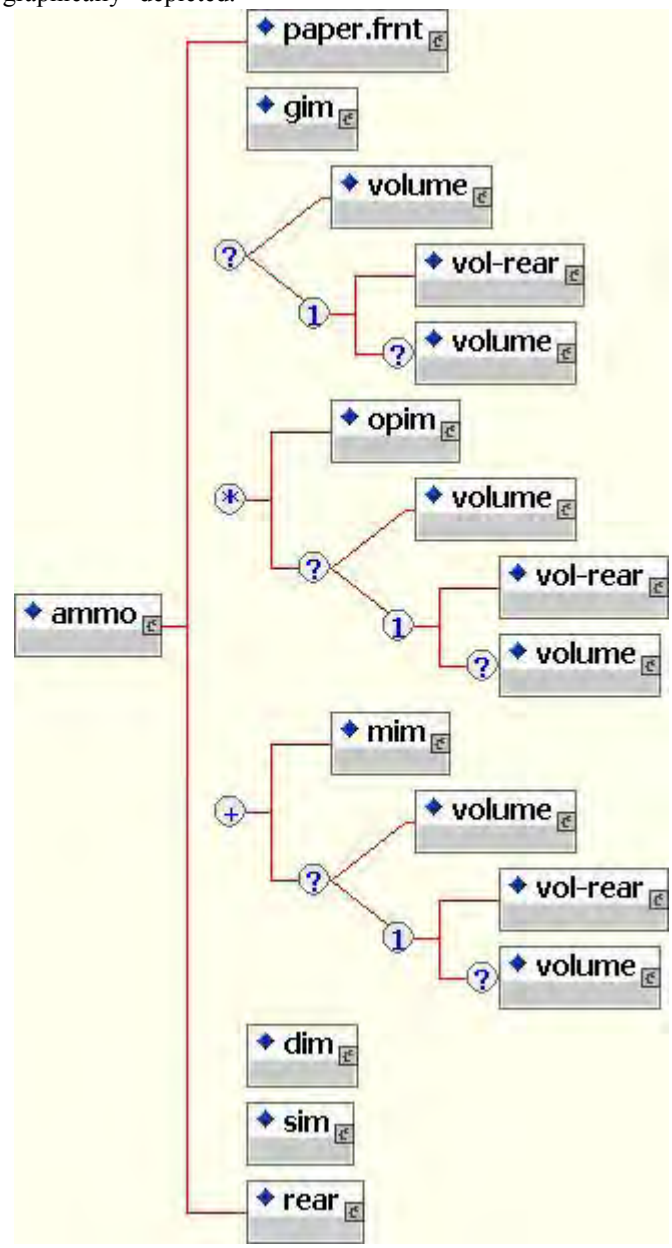

FIGURE 129. Conventional and chemical ammunition <ammo> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT ammo (paper.frnt, gim, (volume | (vol-rear, volume?))?,
(opim, (volume | (vol-rear, volume?))?)*,
(mim, (volume | (vol-rear, volume?))?)+, dim, sim, rear)>
<!ATTLIST ammo
    ammopubno           CDATA                    #IMPLIED
    maintitl            CDATA                    #REQUIRED
    maintlvls           (10 | 12 | 13 | 14 | 20 | 23
                        | 24 | 30 | 34 | 40)     #REQUIRED
    rpstl           (yes | no)    'no'
    revno           CDATA                    #REQUIRED
    multivolume          (yes | no)    'no'>
```

d. Unique attributes:

(1) **ammopubno** – Ammunition publication Number supplies the publication number for the Conventional and Chemical Ammunition Manual.
(2) **maintitl** – Maintenance Level Title supplies a literal version of the title for the maintenance-level. (required)
(3) **maintlvls** – Maintenance Level specifies the maintenance level(s) authorized to use this manual; this attribute value is used in the style sheet to supply the literal expression of the TM's maintenance level. (required)
(4) **rpstl** – Specifies whether or not the manual includes a RPSTL among its appendixes. The default value is "no".
(5) **revno** – The overall revision number for the manual. (required)
(6) **multivolume** – Is the manual broken into volumes. The default value is "no".

31.6 Phased maintenance inspections technical manual ***<pmi>***. The element ***<pmi>*** contains the Phased Maintenance Inspections (PMI) aviation manual content. It consists of PMI/Preventive Maintenance Services (PMS) front matter, Preventive Maintenance (PM) Checklist Introductory work package, the maintenance information chapter using the aviation maintenance category, and rear matter.

a. The components are:

(1) Phased Maintenance Inspections/ Preventive Maintenance Services Front Cover ***<pm.frnt>*** (required) (see 31.6.1).
(2) Phased Maintenance Checklist Introductory (required) ***<pm-ginfowp>*** (required)(see 34.1.4).
(3) Maintenance Instructions ***<mim>*** (required – one or more)(see Section 37).
(4) Phased Maintenance Inspections/ Preventive Maintenance Services Rear ***<pm.rear>*** (required)(see 31.6.2).

b. DTD fragment is graphically depicted.

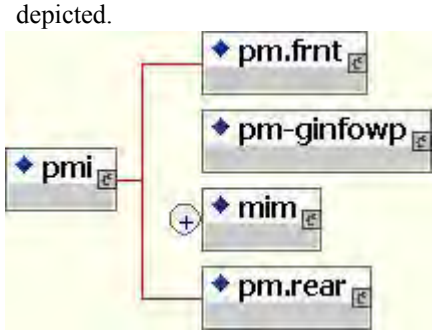

FIGURE 130. Phased maintenance inspections technical manual <pmi> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT pmi (pm.frnt, pm-ginfowp, mim+, pm.rear)>
<!ATTLIST pmi
    pubno          CDATA                              #IMPLIED
    revno          CDATA                              #REQUIRED
    maintitl       CDATA                              #REQUIRED
    maintlvls      (PMI)                              #REQUIRED
    security       (uc | fouo |c | s | ts)          #IMPLIED
>
```

d. Unique attributes:
   (1) **pubno** – Publication number. Specifies publication number for the technical manual. (optional)
   (2) **revno** – Revision number. Specifies revision number of the overall manual. (required)
   (3) **maintitl** – Maintenance Level Title supplies a literal version of the title for the maintenance-level. (required)
   (4) **maintlvls** – Maintenance level. Specifies the maintenance level(s) authorized to use this manual; this attribute value is used in the stylesheet to apply the literal expression of the TM's maintenance level. (required)

e. Common attributes:
   (1) **security** – Security classification (optional) (see 49.2.13).

31.6.1 Preventive maintenance services or phased maintenance inspections manual front matter ***<pm.frnt>***. The element ***<pm.frnt>*** contains the front matter of a preventive maintenance services or phased maintenance inspections manual. The front matter consist of a front cover, warning summary, a change sheet, list of effective pages/work packages, and title block.

a. The components are:
   (1) Front Cover ***<frntcover>*** (required) (see 31.3.1.1.4).
   (2) Warning Summary ***<warnsum>*** (optional) (see 31.4.1.2).
   (3) Change Sheet ***<chgsheet>*** (optional) (see 31.4.1.3)
   (4) List of Effective Pages and Work Package ***<loepwp>*** (required)(see 31.4.1.4).
   (5) Title Block Page ***<titleblk>*** (required) (see 31.10.1.1).

b. DTD fragment is graphically depicted.

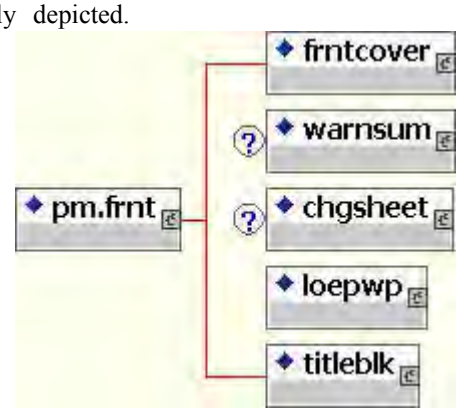

FIGURE 131. Phased maintenance inspections technical manual <pm.frnt> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT pm.frnt (frntcover, warnsum?, chgsheet?,
 loepwp, titleblk)>
```


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d. No attributes.

31.6.2 Preventive maintenance services or phased maintenance inspections manual rear matter ***<pm.rear>***. The element ***<pm.rear>*** contains the preventive maintenance services or phased maintenance inspections manual rear or back matter. It may consist of a glossary, alphabetic index, blank records, DA-2028 forms, authentication page, foldout (oversize) illustration section, and a metric conversion chart. Only the DA-2028 form, authentication page and back cover is required.

a. The components are:
   (1) Glossary ***<glossary>*** (optional – zero or one). The element provides the glossary (see 31.4.2.1).
   (2) Alphabetical index ***<aindx>*** (optional – zero or one). The element provides the alphabetical index (see 31.4.2.2).
   (3) Reporting errors and recommending improvements DA Forms 2028 ***<da2028>*** (required – one or more). The element provides the DA Forms 2028 (see 31.4.2.3).
   (4) Authentication page ***<authent>*** (required). The element provides the authentication page (see 31.4.2.4).
   (5) Foldout section ***<foldsect>*** (optional – zero or one). The element provides the position for the foldout section. (see 31.4.2.5).
   (6) Blank Aviation Record ***<blank_record>*** (optional – zero or more). The element provides a blank record for aviation (see 31.6.2.1).
   (7) Back cover ***<back>*** (required). The element provides the back cover with optional graphic (see 31.4.2.6).

b. DTD fragment is graphically depicted.

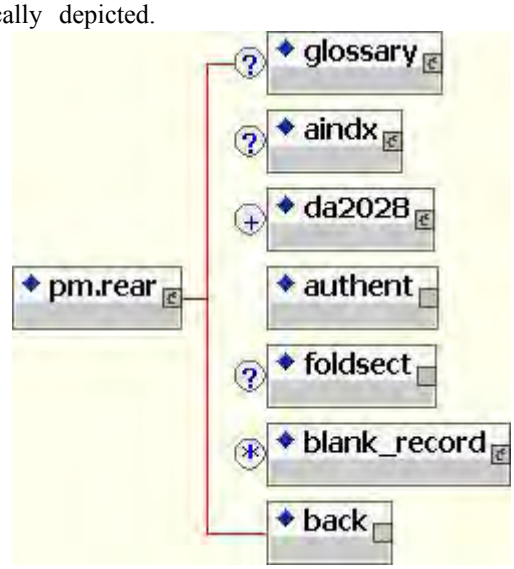

FIGURE 132. Preventive maintenance services or phased maintenance inspections manual rear matter <pm.rear> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT pm.rear (glossary?, aindx?, da2028+, authent, foldsect?,
        blank_record*, back)>
<!ATTLIST pm.rear


    assocfig IDREFS #IMPLIED
    id       ID     #IMPLIED
    idref    IDREFS #IMPLIED>
```


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d. Common attributes.
   (1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (2) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (3) **idref** – Reference identifier(s) (optional) (see 49.2.6).

31.6.2.1 Blank aviation record ***<blank_record>***. The element ***<blank_record>*** provides the capability to design a blank record for recording aviation inspection.

a. The components are:
   (1) Simple Table ***<tabmat>*** (required) (see 42.2).
b. DTD fragment:

```
<!ELEMENT blank_record (tabmat)>
```

c. No attributes.

31.7 Phased maintenance inspections technical manual ***<pms>***. The element ***<pms>*** contains the Preventive Maintenance Services (PMS) aviation manual content. It consists of Phased Maintenance Inspections (PMI) or PMS front matter, PMS general information work package, maintenance information chapter using PMS maintenance category, and rear matter.

a. The components are:
   (1) Phased Maintenance Inspections/ Preventive Maintenance Services Front Cover ***<pm.frnt>*** (required) (see 31.6.1).
   (2) Phased Maintenance Checklist Introductory (required) ***<pms-ginfowp>*** (required)(see 34.1.5).
   (3) Maintenance Instructions ***<mim>*** (required – one or more)(see Section 37).
   (4) Phased Maintenance Inspections/ Preventive Maintenance Services Rear ***<pm.rear>*** (required)(see 31.6.2).
b. DTD fragment is graphically depicted.

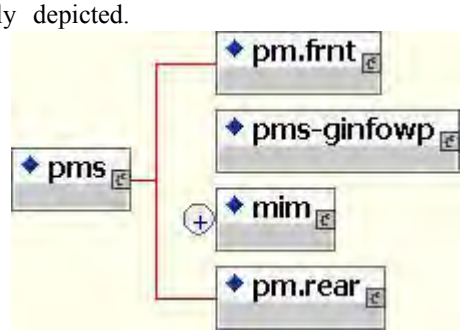

FIGURE 133. Phased maintenance inspections technical manual <pms> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT pms (pm.frnt, pms-ginfowp, mim+, pm.rear)>
<!ATTLIST pms
    pubno          CDATA              #IMPLIED
    revno          CDATA              #REQUIRED
    maintitl       CDATA              #REQUIRED
    maintlvls      (PMI)              #REQUIRED

    security        (uc | fouo |c | s | ts)          #IMPLIED
>
```

d. Unique attributes:
   (1) **pubno** – Publication number. Specifies publication number for the technical manual. (optional)
   (2) **revno** – Revision number. Specifies revision number of the overall manual. (required)

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(3) **maintitl** – Maintenance Level Title supplies a literal version of the title for the maintenance-level. (required)

(4) **maintlvls** – Maintenance level. Specifies the maintenance level(s) authorized to use this manual; this attribute value is used in the stylesheet to apply the literal expression of the TM's maintenance level. (required)

e. Common attributes:

(1) **security** – Security classification (optional) (see 49.2.13).

31.8 System-wide troubleshooting aviation – technical manual ***<sys-ts>***. The element ***<sys-ts>*** contains the contents of a system-wide troubleshooting aviation manual. It consists of front matter paged-based, troubleshooting information chapter using troubleshooting aviation category, and a rear matter.

a. The components are:

(1) Paper Front***<paper.frnt>*** (required) (see 31.4.1).

(2) Troubleshooting Instructions ***<tim>*** (required) (see 36.4).

(3) Back Matter ***<rear>*** (required) (see 31.4.2).

b. DTD fragment is graphically depicted.

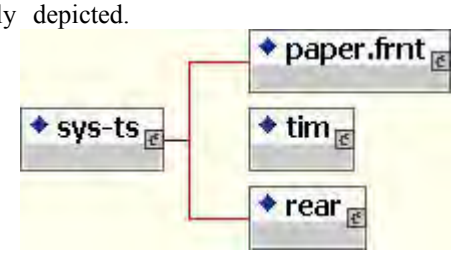

FIGURE 134. System-wide troubleshooting aviation – technical manual <sys-ts> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT sys-ts (paper.frnt, tim, rear)>

<!ATTLIST sys-ts
    pubno           CDATA
#IMPLIED
    revno           CDATA                       #REQUIRED
    multivolume     (yes | no)                  'no'
    security        (uc | fouo |c | s | ts)     #IMPLIED>
```

d. Unique attributes:

(1) **pubno** – Publication number. Specifies publication number for the technical manual. (optional)

(2) **revno** – Revision number. Specifies revision number of the overall manual. (required)

(3) **multivolume** – Multiple volumes. The manual broken into volumes.(required)

e. Common attributes:

(1) **security** – Security classification (optional) (see 49.2.13).

31.9 Destruction manual ***<destruction_manual>***. A stand alone destruction manual ***<destruction_manual>*** may be prepared when directed by a stock class, item, or program manager.

a. The components are:

(1) Front matter ***<paper.frnt>*** (required).

(2) General information work package ***<destruct_ginfowp>*** (required). This element provides the general information required for a stand alone destruction manual. MIL–STD–40051–1/–2 contains specific requirements that differ from the general information contained in a regular TM.

(3) Destruction of material work package ***<destruct-materialwp>*** (required – one or more). This work package provides detailed destruction procedures to the soldier.

(4) Rear matter ***<rear>*** (required) (see 31.4.2)

b. DTD fragment is graphically depicted.

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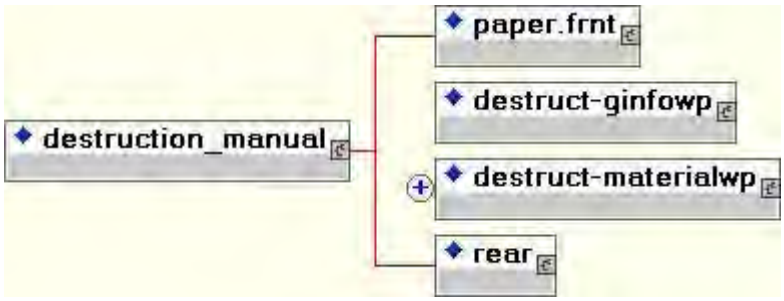

FIGURE 135. Destruction manual <destruction_manual> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT destruction_manual ((paper.frnt, destruct-ginfowp,
   destruct-materialwp+, rear) | (framed.frnt, destruct-ginfowp,
   destruct-materialwp+, da2028+))>
<!ATTLIST destruction_manual   pubno CDATA #IMPLIED
                  revno CDATA #REQUIRED
                  security  (uc | fouo | c | s | ts )  #IMPLIED >
```

d. Unique attributes:

(1) **pubno** – Publication number (optional). This allows the entry of the publication number for the TM.

(2) **revno** – Revision number (required). This contains the currnet revison number of the TM.

(3) **security** – Security classification (optional) (see 49.2.13).

31.9.1 Destruction manual front matter. The following paragraphs describe the allowable front matter. Format style and requirements are prepared for a standard page-oriented presentation. Full details are contained in 31.4.1.

a. DTD fragment is graphically depicted.

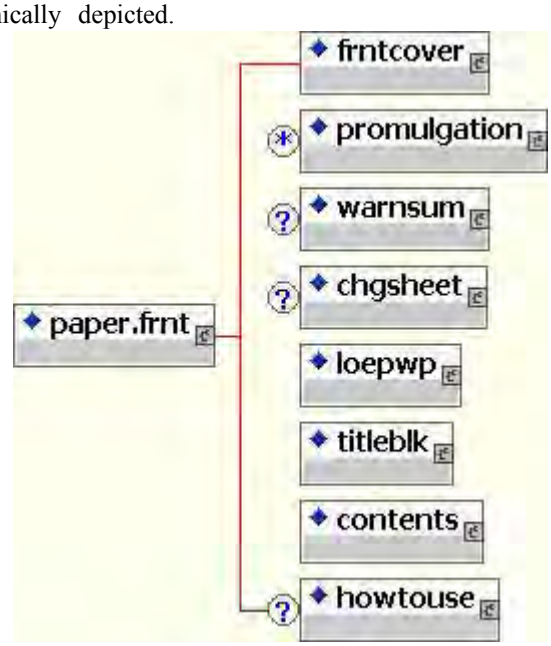

FIGURE 136. Destruction manual front matter <paper.frnt> DTD hierarchy.

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b. DTD fragment :

```
<!ELEMENT paper.frnt (frntcover, promulgation*, warnsum?, chgsheet?,
            loepwp, titleblk, contents, howtouse?)>
```

c. No attributes.

31.9.2 Destruction general information work package ***<destruct-ginfowp>***. The destruction general information work package contains those elements that present required and optional general information for a destruction manual.

a. The components are:

(1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end user (see 32.3.2).

(2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).

(3) Scope ***<scope>*** (required). The element provides a brief statement of what is covered in the technical manual and also includes the following as applicable:

(a) Type of manual.

(b) Model number(s) and equipment name(s).

(c) Purpose of equipment

(d) Special inclusions in the manual, such as drill procedures or on-vehicle loading plans. (see 49.1.4.23).

(4) Authorize to destroy ***<authorize_to_destroy>*** (required) (see 31.9.2.1).

(5) Reporting destruction ***<report_destruct>*** (required) (see 31.9.2.2).gen_dest_info

(6) General destruction information ***<general_destruct_info>*** (required) (see 31.9.2.3).

(7) Destruction of component spares ***<component_spares>*** (optional) (see 31.9.2.4).

b. DTD fragment is graphically depicted.

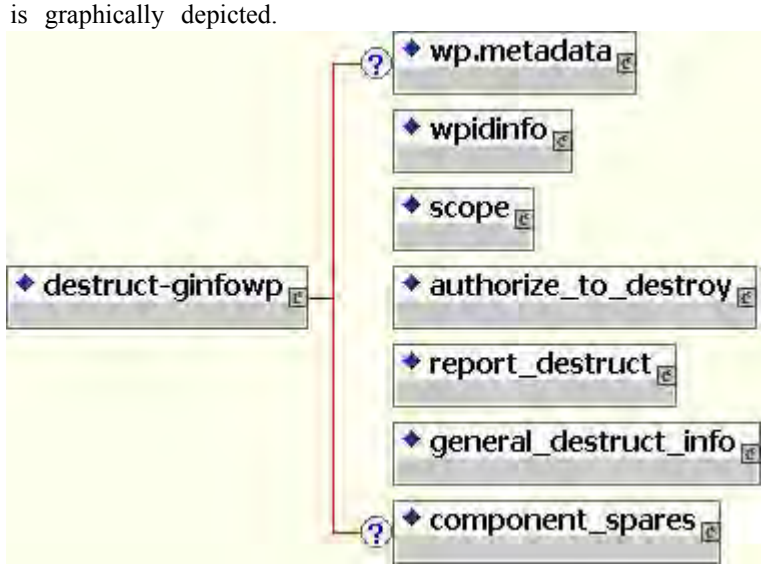

FIGURE 137. Destruction general information work package <destruct-ginfowp> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT destruct-ginfowp (wp.metadata?, wpidinfo, scope,
            authorize_to_destroy, report_destruct, general_destruct_info,
            component_spares?)>
<!ATTLIST destruct-ginfowp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)           "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
```

d. Unique attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

31.9.2.1 Authorize to destroy ***<authorize_to_destroy>***. The ***<authorize_to_destroy>*** provides a statement as to who may provide the authorization to destroy equipment. This is a boilerplate statement contained in MIL–STD–40051–1/–2. The authority to order destruction is set at division level or higher. These commanders may delegate to lower levels.

a. The components are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
b. DTD fragment is graphically depicted.

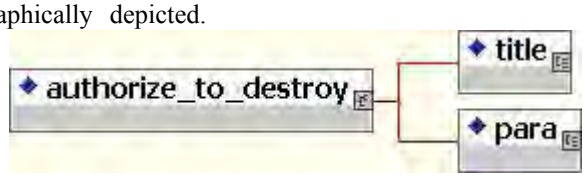

FIGURE 138. Authorize to destroy <authorize_to_destroy> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT authorize_to_destroy (title, para)>
<!ATTLIST authorize_to_destroy
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

31.9.2.2 Reporting destruction ***<report_destruct>***. The ***<report_destruct>*** provides a mechanism for units who have had to destroy equipment the instructions they need to notify upper level command.

a. The components are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).

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(2) Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).

b. DTD fragment is graphically depicted.

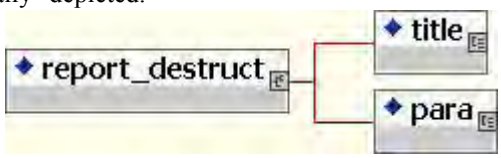

FIGURE 139. Reporting destruction <report_destruct> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT report_destruct (title, para)>
<!ATTLIST report_destruct
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.9.2.3 General destruction information ***<general_destruct_info>***. The ***<general_destruct_info>*** provides generic destruction information such as:

Information on types of destructive process such as burning, use of explosives, burying, or self destruction devices/techniques. This explanation includes the advantages and disadvantages of each process.

For complex weapons systems, the reason to perform any subordinate destruction procedures in conjunction with those for the weapons system.

Any considerations relative to physical location or weather related (wind, rain, temperature) that users should consider when destroying material.

Explanations on the priority for materiel destruction.

a. The components are:

(1) Primary Level Titled Paragraph(s) ***<para0>*** (see 49.1.1.8) or one or more Primary Level Conditional Titled Paragraph(s) ***<para0–alt>*** (required – one or more) (see 49.1.1.9).

b. DTD fragment is graphically depicted.

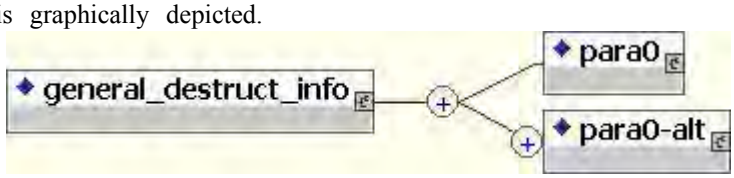

FIGURE 140. General destruction information <general_destruct_info> DTD hierarchy.

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c. DTD fragment :

```
<!ELEMENT general_destruct_info (para0 | para0-alt+)+>
<!ATTLIST general_destruct_info
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.9.2.4 Destruction of component spares ***<component_spares>***. The ***<component_spares>*** provides information on how to components not installed on the weapons system. These include supply spares or parts in the repair cycle that may be subject to capture. This section will provide a list of spare parts to be destroyed and they order they are to be destroyed. A key precept is to ensure that if a component is destroyed, that all similar components are destroyed to keep an enemy from rebuilding a weapon using captured spare parts.

a. The components are:

(1) Primary Level Titled Paragraph(s) ***<para0>*** (see 49.1.1.8) or one or more Primary Level Conditional Titled Paragraph(s) ***<para0–alt>*** (required – one or more) (see 49.1.1.9).

b. DTD fragment is graphically depicted.

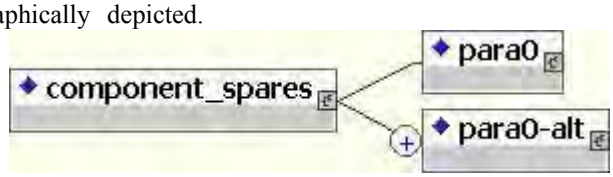

FIGURE 141. Destruction of component spares <component_spares> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT component_spares (para0 | para0-alt+)>
<!ATTLIST component_spares
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

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(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.9.3 Destruction procedures work package ***<destruct-materialwp>***. The ***<destruct-materialwp>*** provides the specific information and procedures to destroy those components or weapons system assets covered by the destruction manual.

a. The components are:
   (1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end user (see 32.3.2).
   (2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).
   (3) Work Package Initial Setup ***<wpinfo>*** (required). The element provides lists of information required by the technician so the tools, test equipment, references, parts, and other items needed to complete the tasks can be obtained (see 32.5).
   (4) Destruction of essential spares ***<essential_spares>*** (optional)(see 31.9.3.1).
   (5) Detailed destruction procedures ***<destruct_proc>*** (required – one or more) (see 31.9.3.2).

b. DTD fragment is graphically depicted.

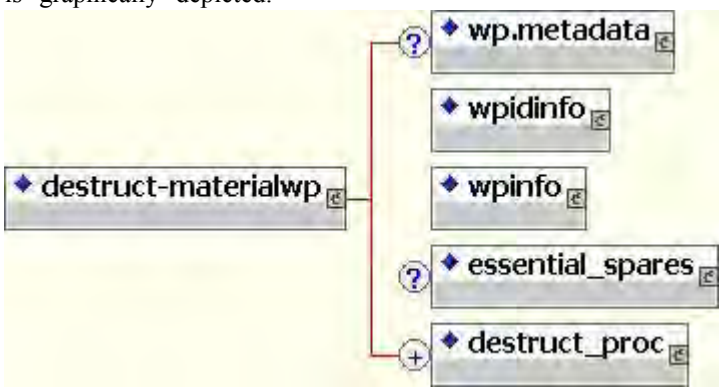

FIGURE 142. Destruction procedures <destruct-materialwp> DTD hierarchy.

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c. The DTD fragment:

```
<!ELEMENT destruct-materialwp (wp.metadata?, wpidinfo, wpinfo,
            essential_spares?, destruct_proc+)>
<!ATTLIST destruct-materialwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
```

d. Common attributes :

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).

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(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

31.9.3.1 Destruction of essential spares ***<essential_spares>***. The ***<essential_spares>*** provides a listing of specific spare parts that are essential or critical to the weapons system that should be destroyed along with the weapons system. The ***<essential_spares>*** differs from the ***<component_spares>*** in the general information. The ***<essential_spares>*** provides a listing of the spares that are to be destroyed for the weapons system.

a. The components are:
(1) Primary Level Titled Paragraph(s) ***<para0>*** (see 49.1.1.8) or one or more Primary Level Conditional Titled Paragraph(s) ***<para0–alt>*** (required – one or more) (see 49.1.1.3).

b. DTD fragment is graphically depicted.

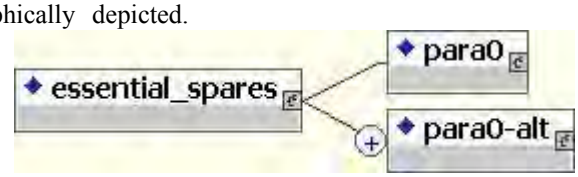

FIGURE 143. Destruction of essential spares <essential_spares> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT essential_spares (para0 | para0-alt+)>
<!ATTLIST essential_spares
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

d. Common attributes :
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.9.3.2 Detailed destruction procedures ***<destruct_proc>***. The ***<destruct_proc>*** provides detailed instructions on the destruction of a specific weapons system or stock class items. The ***<destruct_proc>*** is a single ***<proc>*** for each individual procedure or item. If there is a sequence to ensure the proper destruction, this sequence will be provided. Alternative procedures may be provided.

a. The components are:
(1) Procedure ***<proc>*** (required) (see 33.2).

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b. DTD fragment is graphically depicted.

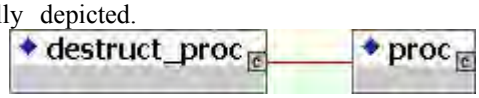

FIGURE 144. Detailed destruction procedures <destruct_proc> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT destruct_proc (proc)>
<!ATTLIST destruct_proc
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.10 Page-based volume.

31.10.1 Volume ***<volume>***. An element containing the front matter for a volume, including the front cover, warning summary, and title block. This element is used to insert the front matter only, not to indicate a containment relationship relative to surrounding TM body matter.

a. The components are:

(1) Title block page ***<titleblk>*** (required) (see 31.10.1.1).
(2) Table of contents ***<contents>*** (required) (see 31.3.1.1.8).

b. DTD fragment is graphically depicted.

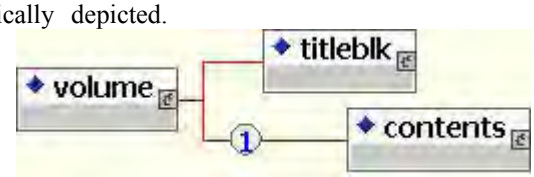

FIGURE 145. Volume <volume> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT volume (titleblk, (contents))>
<!ATTLIST volume
    tocentry          (0 | 1 | 2 | 3 | 4 | 5) "1"
    label             CDATA          #IMPLIED
    id                ID             #IMPLIED>
```

d. Unique attributes:

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(1) **tocentry** – Table of Contents Entry for the Volume defines the indenture level in the TOC. When the level is zero, no entry in the TOC is used. The default is "1".
(2) **label** – Label provides volume number for composition systems that cannot auto-generate. (optional)
(3) **id** – Identifier of the element which is assigned at origination and which remains unchanged as the document is revised or updated even though the automatically assigned enumeration or manually-assigned "labels" change (in some cases many times). The value of the "id" is used when making references to the element from other portions of the document. If no ID is given, none will be maintained and the element can then not be cross-referenced by means of an IDREF on another element or with local linking element. (optional)

31.10.1.1 Title block page ***<titleblk>***. The title block material in the TM's front matter repeats identifying information from the front cover, including the prime title; it also includes the Reporting Errors statement.

a. The components are:
(1) Change number ***<chgno>*** (required). The element is the document change level. The content model is #PCDATA.
(2) Title ***<title>*** (optional). The element provides the title for the title block page (see 49.1.1.3).
(3) Service Nomenclature ***<servnomen>*** (required) (see 31.3.1.1.4.3).
(4) City ***<city>*** (required) (see 49.1.4.1.2).
(5) State ***<state>*** (required) (see 49.1.4.1.3).
(6) Date ***<date>*** (required) .
(7) Primary Title ***<prtitle>*** (required) (see 31.3.1.1.4.1.3).
(8) Manual Subtitle ***<stitle>*** (optional) (see 49.1.1.2).
(9) Weapon System Title ***<weapons_system>*** (optional) (see 31.3.1.1.4.1.4).
(10) Reporting Errors and Recommending Improvements ***<reporting>*** (required – one or more)(see 31.4.2.3).
(11) Warning and Note Data ***<pm.warning.data>*** (optional)(see 31.10.1.1.1).
(12) Official Notices ***<notices>*** (required) (see 31.3.1.1.4.2).

b. DTD fragment:

```
<!ELEMENT titleblk (chgno?, (title?, servnomen)+,
city, state, date, prtitle,(stitle, weapons_system?)?,
reporting+, pm.warning.data?, notices)>
<!ATTLIST titleblk

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Unique attributes.
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

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(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

31.10.1.1.1 Warning and note data ***<pm.warning.data>***. The element ***<pm.warning.data>*** is used for the mandatory verbatim warning and note data contain in the phased maintenance technical manual title block.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
   (2) Emphasis ***<emphasis>*** (optional – zero or more)(see 49.1.3.1).
   (3) Subscript ***<subscrpt>*** (optional – zero or more)(see 49.1.3.4).
   (4) Superscript ***<supscrpt>*** (optional – zero or more)(see 49.1.3.5).
   (5) Line Break ***<brk>*** (optional – zero or more)(see 49.1.3.1).
b. DTD fragment:

```
<!ELEMENT weapons_system (#PCDATA | emphasis | subscrpt | supscrpt | brk)*>
```

c. No attributes.

31.10.2 Back matter of volume ***<vol-rear>***. The element ***<vol-rear>*** is used for the rear or back matter of a volume. This element is used to insert the rear matter only, not to indicate a containment relationship relative to surrounding TM body matter.

a. The components are:
   (1) Rear matter ***<rear>*** (required) (see 31.4.2).
b. DTD fragment is graphically depicted.

FIGURE 146. Back matter of volume <vol-rear> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT vol-rear (rear)>
```

d. No attributes.

31.11 Maintenance data collection (MDC) . Maintenance data collection is the ability to gather information concerning maintenance related action(s) performed to the equipment. The information collected is used later to perform analysis for possible detection of procedural issues, repeatable equipment problems, equipment status/readiness, etc. The data collection can be performed manually (filling out forms) or collected automatically by the IETM. Regardless of the method selected, the Army requires the information to be transmitted using the MIL-STD-3008, Interactive Electronic Technical Manual (IETM) Technical Data Requirements To Support The Global Combat Support System - Army (GCSS-A). An IETM can collect this information through various methods — specific program within the IETM application (application dependent), additional XML processing instructions (these are application dependent), or embedded elements in the XML sources data (these are application independent). The MIL–STD–2361 DTD provides several methods to capture this data with embedded elements in the XML source data through state table variables, external application requests, and/or mapping to MDC system elements.

31.11.1 State table variables MDC ***<mdc>*** methodology. The ***<mdc>*** element uses the state table values for collecting the maintenance data. The state variable data is not processed (sent to MDC system) until a request is indicated in the source data. A major issue with MDC is that since MIL–STD–40051–1/–2 and MIL–STD–2361 are multi-service standard documents, the data collection needs to be generic so that the various services can collect and map the information properly. The ***<mdc>*** element provides this functionality by describing the data collection system, variable/element name(s), and structural position (hierarchy).

a. The components are:

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(1) A required MDC system identifier ***<mdc.system>*** (required). Defines the MDC system that is receiving the maintenance information (i.e., MIL-STD-3008).
(2) A required MDC element structural position (hierarchy) ***<mdc.hierarchy>*** (required). The ***<mdc.hierarchy>*** consists of
(a) A required current parent level variable/element name ***<mdc.element>*** used the MDC system (required). (see 31.11.2 for an example).
(b) An optional MDC element structural position (hierarchy) ***<mdc.hierarchy>*** (optional). The recursion feature of this element allows the building a hierarchical structure to described to the external application where the state variable value is located.

b. DTD fragment is graphically depicted.

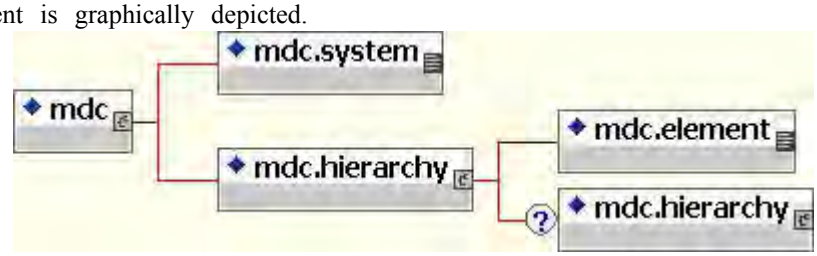

FIGURE 147. MDC <mdc> DTD hierarchy..

c. DTD fragment for ***<mdc>***, ***<mdc.system>***, ***<mdc.hierarchy>***, and ***<mdc.element>***:

```
<!ELEMENT mdc (mdc.system, mdc.hierarchy)>

<!ELEMENT mdc.system (#PCDATA)>

<!ELEMENT mdc.hierarchy (mdc.element, mdc.hierarchy?)>

<!ELEMENT mdc.element (#PCDATA)>
```

31.11.2 Example using ***<mdc>***. The example using ***<mdc>*** to capture information from a state variable and map to Army MIL-STD-3008 elements. The Army's MDC transmittal standard, MIL-STD-3008, usually requires multiple MDC variable names to input where the data belongs, but this example will only use two variables, but a complete implementation would require more variables.

a. The following is the MIL-STD-3008 DTD fragment for parts-data.

```
<!ELEMENT gcss-a (...| parts-data |...)>

<!ELEMENT parts-data (part-reqmnts?, part-order-status?, part-order-cancel?)>

<!ELEMENT part-reqmnts (date-time, part-req*, manhours-expend, manhours-cost,
                        total-parts-cost? )>

<!ELEMENT part-req (file-input-ac?, taskno?, identno?, (nsn | (partno, cageno)),
                    sfxcd?, qty-required?, qty-issued?, nmcs?, fail-code?, location?,
                    released-by-data?, part-cost?)>
```

b. The following example shows the markup for setting parts information for a part number and CAGEC.

<statemanipulation>
<variable name="used-partno" description="Part number for material used." value-type="string" scope="global">
<mdc>
<mdc.system>MIL-STD-3008A
</mdc.system>

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<mdc.hierarchy>
<mdc.element>gcss-a
</mdc.element>
<mdc.hierarchy>
<mdc.element>parts-data
</mdc.element>
<mdc.hierarchy>
<mdc.element>part-reqmnts
</mdc.element>
<mdc.hierarchy>
<mdc.element>part-req
</mdc.element>
<mdc.hierarchy>
<mdc.element>partno
</mdc.element>
</mdc.hierarchy>
</mdc.hierarchy>
</mdc.hierarchy>
</mdc.hierarchy>
</mdc.hierarchy>
</mdc>
</variable>
<expression>
<string>XF123-4512
</string>
</expression>
</statemanipulation>
<statemanipulation>
<variable name="used-cageno" description="CAGEC for material used." value-type="string" scope="global">
<mdc>
<mdc.system>MIL-STD-3008A
</mdc.system>
<mdc.hierarchy>
<mdc.element>gcss-a
</mdc.element>
<mdc.hierarchy>
<mdc.element>parts-data
</mdc.element>
<mdc.hierarchy>
<mdc.element>part-reqmnts
</mdc.element>
<mdc.hierarchy>
<mdc.element>part-req
</mdc.element>
<mdc.hierarchy>
<mdc.element>cageno
</mdc.element>
</mdc.hierarchy>
</mdc.hierarchy>
</mdc.hierarchy>
</mdc.hierarchy>
</mdc.hierarchy>
</mdc>
</variable>
<expression>

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<string>34561
</string>
</expression>
</statemanipulation>

c. The data collection information is transmitted to the MDC system through an external application. The external application will use the information in the ***<mdc>*** element to populate the logistics data base. The external application element is executed through either the ***<diagnostic>*** element or the ***<link>*** element. The following shows how to implement either options.

(1) The ***<diagnostic>*** element was developed to send parameters ***<sendparameter>*** to the application and waits until the external application is completed (see 36.11.2.2.1). Shown below is an example sending MDC data using ***<diagnostic>***.

<diagnostic application="mdc-application" testname="Sending MDC">
<desc>Sending Used parts information to the MDC system.
</desc>
<sendparameter>
<variableref="used-partno"/>
</sendparameter>
<sendparameter>
<variableref="used-cageno"/>
</sendparameter> . . .
</diagnostic>

(2) The ***<link>*** element does not have the same capability as ***<diagnostic>*** to send state table variables as parameters. The data collection application is required to know how to retrieve the state table variable from the ***<link>*** attribute "**parameter**" (see 46.2.3). Shown below is an example sending MDC data using ***<link>***.

<link xlink:href="mdc-application.exe" xlink:type="simple" linktype="return" linkaction="immediate" parameter="used-partno used-cageno ...">

d. After the external application processes the MDC data, the values contained in the variables "used-partno" and "used-cageno" will indicate the top level element (this instance the parent element is ***<gcss-a>***) and process the children until final element is reached (this instance is ***<partno>*** and ***<cageno>***) to which the values pertain. The MIL-STD-3008 markup would be similar to the following.

<gcss-a>
<parts-data>
<part-reqmnts>
<date-time> . . .
</date-time>
<part-req>
<partno>XF123-4512
</partno>
<cageno>34561
</cageno>
</part-req>
<manhours-expend> . . .
</manhours-expend>
<manhours-cost> . . .
</manhours-cost>
</part-reqmnts>
</parts-data>
</gcss-a>

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32 WORK PACKAGE BASICS.

32.1 Work package introduction. Work packages are used to logically divide TM data into functional descriptive or task-oriented information. Ideally, each work package is a independent, stand alone data unit containing only a single task and/or configuration, thus providing enhanced data exchange. It may be required to group some information or maintenance tasks in one work package and divide others into several work packages of suitable length. Division or selection of coverage will depend on various factors (see MIL-HDBK-1222). Work packages contain general information, description, theory, operating, maintenance, troubleshooting, parts, and supporting information units containing all information required for directing task performance. Work packages contain a title block, initial setups, descriptive information, operating tasks, and maintenance tasks. These data types can be further divided into paragraphs, procedural steps, tables, lists, warnings, cautions and notes, and supporting illustrations.

This section will describe the elements and attributes that provide for work package metadata, identification and setup information. Work package metadata identifies administrative and retrieval data for each work package. Work package identification information is required in every work package. The element specifies who can perform the procedure, the lowest maintenance level, the work package title, lists if there is a difference between configurations in the work package, and specifies if the work package replaces an existing work package. Work package setup information lists all of the information required by the technician to identify and obtain the tools, special tools, test equipment, source references, parts, and other items needed to complete the tasks

32.1.1 Work package organization and size.

32.1.1.1 Overview.

32.1.1.1.1 A concern of many authors and acquiring activities is how big to make a work package and what to put into a work package. Both MIL-STD-40051-1 and MIL-STD-40051-2 allow extensive leeway in what can appear. The primary limitation is in MIL-STD-40051-2 where it is recommended work packages not exceed thirty (30) pages. This limit is imposed to control the number of pages that would have to be printed if a work package is changed.

32.1.1.1.2 By definition in MIL-STD-40051-1 and MIL-STD-40051-2, a work package is a collection of data that is capable of standing alone to present a complete task or conceptual idea (theory of operation).

32.1.1.1.3 There are various, opposing, reasons for having a large or small work package. Some reasons for larger work packages (see 32.1.1.2) include:

- Paper based manuals desire to reduce print costs by producing fewer pages (see 32.1.1.2.2).
- Placing a large complex task into a single work package- (see 32.1.1.2.1).
- The standard allowing multiple tasks within a work package.

Some reasons for smaller work packages (see 32.1.1.3) include:

- Improved accessibility for IETM support.
- Less error prone in providing initial setup and follow on requirements (see 32.1.1.3.3).
- A single task is closer to the data module approach in S1000D (see 32.1.1.3.5).

32.1.1.1.4 The recommendation is to have data carried to the smallest possible level. Work packages should be a single individual task. Though it is acceptable in the DTD and allowed in the standard, placing more than one task in a work package is strongly discouraged.

32.1.1.2 Larger work packages.

32.1.1.2.1 Larger work packages are usually found in a page based manual, often containing more than one task. Large work packages are acceptable for long involved tasks that cannot be easily broken down into smaller individual tasks. When a work package is prepared to support a long task in a TM that may be used in both page and electronic presentation, the developer needs to look at potential breaks in the work package. If the work package is divided to support the 30 page limit on work packages, then it may be more advantageous to divide the task into smaller processes. If a task is long, complex,

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and cannot be split, it should be a single work package. The author should not attempt to split the task solely to support the optional page limit.

32.1.1.2.2 One of the main reasons for large work packages in page based manuals is to reduce printing costs by keeping the page count down. Each work package contain several overhead pages that add to printed page costs. Cost of printing TMs has changed in recent years. The Army no longer distributes or stocks paper manuals. All page based manuals are prepared in Adobe Acrobat's© PDF format and distributed on CD-ROM. If a hard copy of the TM is required it may be printed locally.

32.1.1.2.3 In developing some of larger work packages, more than one task may be included. Care should be taken to ensure that when more than one task is contained in a work package, that initial setup and follow on support is the same. Task with different set up and follow on procedures should not be placed in the same work package. As technical data is migrating more and more towards electronic display, page based manual support should be carefully weighed.

32.1.1.3 Smaller work packages.

32.1.1.3.1 Smaller work packages have several advantages, especially with IETM's. IETMs do not have a concern with print costs or page counts. If printing is required, then the smaller work package requires fewer local printing resources.

32.1.1.3.2 IETMs may also be able to filter data and present only what is need to the user. Smaller work packages allow easier filtering or building work packages that support a single configuration.

32.1.1.3.3 Smaller work packages allow a clearer delineation of initial setup requirements. In some larger work packages, especially those prepared for page based delivery, dissimilar tasks have been included in a single work package. Providing for and identifying the appropriate initial setup in such cases is difficult (see 32.1.1.2.3 for further guideance).

32.1.1.3.4 In an IETM, with smaller work packages, merging various processes or procedures together to perform a task is easier.

32.1.1.3.5 An final reason to implement smaller work packages is the prospect of the DOD adopting the S1000D standard. The basis for S1000D is the Data Module (DM) that is a single task.

32.1.1.4 Future concepts. As IETMs gain acceptance, there are initiatives being put forth to manage data at even lower levels than a work package. The ability to control and identify data as low as a procedural step is being discussed. Not only through a filtering function in a work package, but for reuse across multiple work packages, TMs or training material.

32.1.2 Work package number. There are two types of work package numbers; the work package identification number and the work package sequence number. These numbers are quite different in format. For example, “0027” could be a work package sequence number and “M00432-9-1425-646” could be a work package identification number. They also serve very different purposes as follows:

a. The work package identification number is for internal use only and is not seen by the IETM user. The number is unique to any other work package in the Army. The work package identification number components are an alpha designator as defined by the information type, a five digit unique identifier, and the TM number without maintenance level. The acquiring activity manages the unique five digit identifier by TM number. Since no TM number is duplicated, the management of the unique number needs to be controlled by the acquiring activity. Each information type manages its own set of unique five digit identifiers, thus each information type can start at “00000” as the first assigned number. For more information on work package identification number structure, refer toMIL-STD-40051-1.
b. The work package sequence number is used for page-based TMs (refer to MIL-STD-40051-2) to maintain a sequential ordering. It’s use is not intended for TMs developed in accordance with MIL-STD-40051-1. As speci ed in MIL-STD-40051-2, each work package is initially assigned a four digit number beginning with the number 0001. The work package sequence numbers run consecutively throughout the TM. For example, the rst work package in Chapter 2 will be assigned the number immediately following the last work package number in Chapter 1 (e.g., if 0010 is the last work

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package in Chapter 1, 0011 will be the rst work package in Chapter 2). Point numbers are assigned to work packages inserted between two work packages as a result of a change. For example, a work package going between work package sequence numbers 1010 and 1011 would be assigned the work package sequence number of 1010.1 (refer to MIL-STD-40051-2).

32.2 Work package identification.

32.2.1 Work package ID. The work package identification number is a unique number that is assigned to each work package. It is for internal use only so it is not seen by the IETM user. The work package identification number is applicable to both IETMs (refer to MIL-STD-40051-1), and page based TMs (refer to MIL-STD-40051-2). As addressed in these standards, it is assigned when preparing the document instance in accordance with the modular DTD and is not be changed throughout the life of the work package. The work package identification number may be used for database retrieval purposes and for linking (see Section 46 for more information on linking).

32.2.1.1 Assigning the work package identification number. The work package identification number consists of an alpha designation for the type of information contained in the work package, a five digit block number assigned by the acquiring activity, and the TM number less the maintenance level dash numbers. The TM number is used only to provide uniqueness and avoid duplication of a work package identification number, other than that it does not have significance. An example of how work package identification number “M00432-9-1425-646” was derived is as follows:

a. M – Identifies a WP containing maintenance instructions.
b. 00432 – Identifies the 432nd work package containing specific maintenance instructions for the M270 Armored Vehicle Mounted Rocket Launcher.
c. 9-1425-646 – Identifies the M270 Armored Vehicle Mounted Rocket Launcher TM.

This work package could have been developed for a TM such as TM 9-1425-646-20. “Maintenance Instructions Organizational for Launcher, Rocket, Armored Vehicle Mtd, M270”. If this work package were to be reused in TM 9-2320-279-20-1, “Maintenance Instructions for Organizational Maintenance M977 Series, 8 x 8 Heavy Expanded-Mobility Tactical Trucks (HEMTT)”,. the work package identification number of “M00432-9-1425-646” would not change. Using the same WP identification number from TM to TM. insures that every work package in the Army will have a single, unique work package identification number.

32.2.1.2 Identification of work package components. Components within a work package can be referenced and linked to by using a unique identifier that is based upon the work pack identification number. Making an ID unique is accomplished by using the work package identification number as a prefix to each ID (e.g, if work package identification number is “M00134-9-999-9999”, the target link ID is “M00134-9-999-9999-00001”). Since the work package identification number is not changed during the life-cycle of the work package, managing target link IDs would require managing only each individual work packages' IDs. Use caution when determining an unique ID scheme after work package identification number prefix. For example, using the step number as part of the identifier could cause problems when adding or deleting steps from a procedure. Using a simple sequencing numbering system starting from the the first target ID would avoid such problems. Any new data would be resumed from the last ID number count in the work package. For example, if the last target link ID is “M00134-9-999-9999-00152”, the next new target link is “M00134-9-999-9999-00153” (no matter where it is located in the work package).

32.2.1.3 Work package identification number **wpno** attribute example. The content model for each work package has an attribute **wpno** that contains the work package identification number. This attribute has a type of ID which permits it to serve as the target of a link (see 46 for more information on linking). The XML source depicting the use of the **wpno** attribute in both a diagnostic work package and a maintenance work package is shown below.

a. XML document instance fragment.

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```
<diagnosticwp airforce="no" army="no" deletewp="no" frame="yes"
marines="no" navy="no" tocentry="2" wpno="T00100-X-XXX-XXXX">
</diagnosticwp> ...   <maintwp airforce="no" army="no"
deletewp="no" frame="yes" marines="no" navy="no" tocentry="2"
wpno="M00432-X-XXX-XXXX"> ...  </maintwp>
```

32.2.2 Work package sequence number. The work package sequence number is used for page-based TMs (refer to MIL-STD-40051-2) to maintain a sequential ordering. It's use is not intended for TMs developed in accordance with MIL-STD-40051–1. As specified in MIL-STD-40051-2, each work package is initially assigned a four digit number beginning with the number 0001. The work package sequence numbers run consecutively throughout the TM. For example, the first work package in Chapter 2 will be assigned the number immediately following the last work package number in Chapter 1 (e.g., if 0010 is the last work package in Chapter 1, 0011 will be the first work package in Chapter 2). Point numbers are assigned to work packages inserted between two work packages as a result of a change. For example, a work package going between work package sequence numbers 1010 and 1011 would be assigned the work package sequence number of 1010.1 (refer to MIL-STD-40051-2). The **wpseq** attribute contains the work package sequence number. This attribute is of type CDATA.

32.2.2.1 Work package sequence number example. The XML source depicting the **wpseq** attribute in the markup for a Theory of Operations Work Package ***<thrywp>*** element is shown below. The work package sequence number indicates that this is the 6th work pack in the TM. Note that the work package number (see 32.2.1.1) tells us that this is the 3rd descriptive information and theory of operation type of work package that was written.

```
<thrywp airforce="no" army="no" deletewp="no" frame="no" marines="no"
navy="no" tocentry="2" wpno="g00003-9-2320-366" wpseq="0006"> ...
</thrywp>
```

32.2.2.2 Work package element in change transmittal page example. The XML source for change transmittal page is shown below. A change transmittal page is prepared for Page-based TMs when a change is issued (see MIL-STD-40051–2). The change transmittal page identifies the work packages that have been revised and the work packages that have been added to the TM. These work packages are identified by their work package identification number as entered in the **wpref** attribute in the ***<wpno>*** element. The work package sequence number is displayed on the change transmittal page under that heading of “Work Package Number”. This number is retrieved from the **wpseq** attribute that included in the content model for each work package (see 32.2.2).

a. XML document instance fragment.

```
<chglist>
<trim.para esd="no" hcp="no"> Replace the following work
packages with their revised version.
</trim.para>
<chgwplist>
<title>Work PackageNumber
</title>
<wpno wpref="G00050-X-XXX-XXXX"/>
<wpno wpref="G00038-X-XXX-XXXX"/>
</chgwplist>
</chglist>
<chglist>
<trim.para esd="no" hcp="no">Add the following new work packages
</trim.para>
<chgwplist>
<title>Work Package Number
</title>
<wpno wpref="M00455-X-XXX-XXXX"/>
<wpno wpref="M00501-X-XXX-XXXX"/>
</chgwplist>
```


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```
</chglist>  . . .
<ginfowp airforce="no" army="no" deletewp="no"frame="no" marines="no" navy="no"
tocentry="2" wpno="G00038-X-XXX-XXXX"wpseq="0042&rdquo;>  . . .
</ginfowp>  . . .
<descwp airforce="no" army="no" deletewp="no"frame="no" marines="no" navy="no"
tocentry="2" wpno="G00050-X-XXX-XXXX"wpseq="0154">  . . .
</descwp>  . . .
<surwp airforce="no" army="no" deletewp="no"frame="no" marines="no" navy="no"
tocentry="2" wpno="M00455-X-XXX-XXXX"wpseq="1625.1">. . .
</surwp>  . . .
<maintwp airforce="no" army="no" deletewp="no"frame="no" marines="no" navy="no"
tocentry="2" wpno="M00501-X-XXX-XXXX"wpseq="1700.1">  . . .
</maintwp>
```

b. Stylesheet output.

MANUAL NUMBER

CHANGE NO. 1

HEADQUARTERS, DEPARTMENT OF THE ARMY WASHINGTON, D.C., 31 AUGUST 1993

TECHNICAL MANUAL

OPERATOR'S MANUAL FOR TEST SET RADAR AN/TPM-22 NSN 4931-00–707–1229 (EIC D42)

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

TM X–XXX–XXXX–XX, 5 June 1987, is updated as follows:

1. File this sheet in front of the manual for reference.
2. This change is a result of new preventive maintenance checks and service procedures and new expendable/durable supplies and materials.
3. New or updated text is indicated by a vertical bar in the outer margin of the page.
4. Added illustrations are indicated by a vertical bar adjacent to the figure number. Changed illustrations are indicated by a miniature pointing hand adjacent to the updated area and a vertical bar adjacent to the figure number.
5. Remove old pages and insert new pages as indicated below:

Remove Pages	Insert Pages
a through d	a through d
None	e though g / h (blank)

6. Replace the following work packages with their revised version.

Work Package Number
WP 0154
WP 0042

7. Add the following new work packages.

Work Package Number
WP 1625.1
WP 1700.1

FIGURE 148. Work package element in a change transmittal page example.

32.3 Metadata. Work package metadata identifies administrative and retrieval data for each work package. This data is usually not used or seen by the end user. MIL-STD-2361 identifies the XML elements and attributes that provide the metadata for each work package. These attributes and elements are described in the following sub-paragraphs.

32.3.1 Work package metadata attributes. Each work package has optional metadata attributes. These attributes help define the work package.

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32.3.1.1 Crew member ("**crewmember**"). The attribute is used to filter work packages that have specific crew member usages. If no crew member attribute is specified, the work package has no specific crew requirements.

32.3.1.2 Functional group code ("**fgc**"). Specifies the functional group code that applies to the subject of the element. The code can assist when changes occur to higher or lower level FGCs, thus alerting the author or information custodian of possible effects.

32.3.1.3 Logistic support analysis (LSA) identifier ("**lsa-id**"). Specifies the Logistic Support Analysis (LSA) identification subject of the element in regards to the equipment covered in the TM. Providing this information allows traceability to the logistic report, if used, where the information originated and an alert when information has been updated in the logistic analysis.

32.3.1.4 Skill level ("**skilltrk**"). The skill level designation by the user or document at which the current element of information is aimed. A particular set of values common to all documents has not been created. Currently, the relevant values are set by contract.

32.3.1.5 Service specific ("**army**", "**airforce**", "**navy**", "**marines**"). The attributes "**army**", "**airforce**", "**navy**", and "**marines**", when at least one is "**yes**", identifies the work package as applicable only for the identified service(s). The attributes are used to filter information not needed for all users.

32.3.1.6 Table of contents entry ("**tocentry**"). The **tocentry** attribute is used to identify if the given element title will appear in the table of contents and for multiple level TOC's the attribute value will indicate the level the element will be placed in the TOC. The values are a list from "0" to "5". A zero indicates no TOC entry and a "5" indicates a fifth level of TOC indenture. The **tocentry** value is often defaulted based on the element using the attribute.

32.3.2 Work package metadata ***<wp.metadata>***. Work package metadata identifies information about the work package including its security classification, proponent, change history, TM usage, applicability, system/subsystem identification, keyword search list, and subject. Providing this type of information positions the Army for work packages to be a standalone document.

a. The components are:
   (1) Security portion markings ***<portionmark>*** (required – one or more). Identifies the security level of the entire work package (see 32.3.2.1).
   (2) Proponent ***<proponent>*** (required). Identifies proponent name and address for the work package (see 49.1.4.22).
   (3) Tracking Change history ***<tracking>*** (required). Tracks changes to the work package (see 32.3.2.2).
   (4) Technical Manual Number and Service Branch ***<tminfono>*** (required – one or more). Identifies the TMs using the work package (see 31.3.1.1.4.1.1).
   (5) Applicability ***<applicability>*** (optional – zero or more). The system effectivity item (see 32.3.2.3).
   (6) System/Subsystem definition ***<subsystem-system>*** (optional – zero or more). Provides a method to identify the subsystem/system hierarchy structure (see 32.3.2.4).
   (7) Keyword search list ***<keyword.search>*** (optional – zero or more). Identifies keywords to be used for searches (see 32.3.2.5).
   (8) Subject ***<subject>*** (optional – zero or more). This element contains additional qualifier information about the work package (see 32.3.2.6).

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

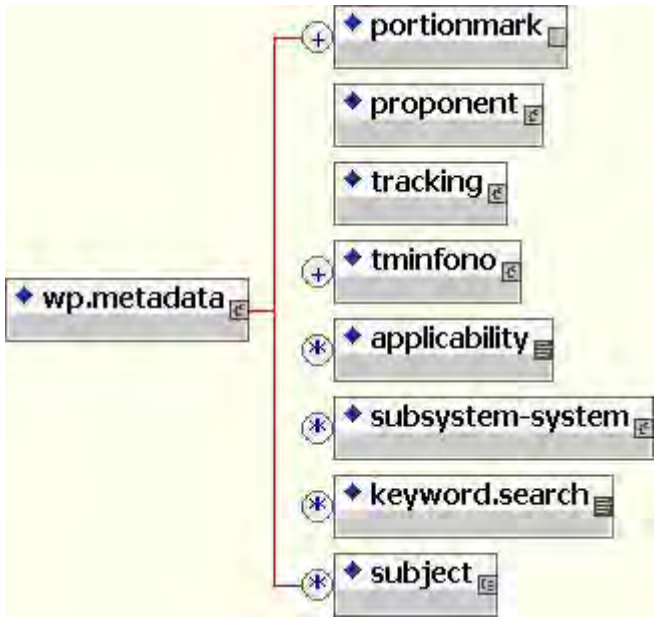

FIGURE 149. Work package metadata <wp.metadata> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT wp.metadata (portionmark+, proponent, tracking, tminfono+,
   applicability*, subsystem-system*, keyword.search*, subject*)>
```

32.3.2.1 Security portion markings ***<portionmark>***. Security portion marking identifies the security level of the work package. This security level is for the work package only and represents its highest security level. The security level of the TM(s) that contains this work package may be higher or the same. The content model for the ***<portionmark>*** element consists of the attribute **country** which identifies the security marking country of origin and the attribute **security** which identifies the associated security classification.

a. DTD fragment:

```
<!ELEMENT portionmark EMPTY>
<!ATTLIST portionmark
    country   CDATA                      "US"
    security  (uc | fouo | c | s | ts ) #IMPLIED"
>
```

b. Unique attributes:

(1) **country** – Country origin for security marking (default value is "US"). Identifies the security marking country code (derived from ISO-3166-1) of origin.

c. Common attributes:

(1) **security** – Security classification (optional) (see 49.2.13).

32.3.2.2 Tracking change history ***<tracking>***. This element maintains the work package history by identifying each modification to the work package.

a. The components are:

(1) Change history record ***<change.history>*** (required – one or more). The ***<change.history>*** element documents each change to the work package (see 32.3.2.2.1).

b. DTD fragment is graphically depicted.

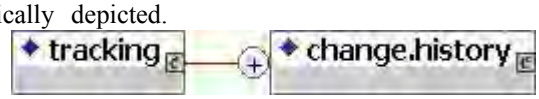

FIGURE 150. Tracking change history <tracking> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT tracking (change.history+)>
```

32.3.2.2.1 Change history record ***<change.history>***. Provides a record of each change made to the work package. This information includes the author, date of change, work package status and reason for change. All changed paragraphs, steps, and text within the work package have a reference to a change history record entry (i.e., the **id** attribute in the change history record serves as the target to be linked to from the changes in the work package).

a. The components are:
   (1) Author ***<author>*** (required). Identifies the author of the change (see 32.3.2.2.1.1).
   (2) Date ***<date>*** (required). Identifies the change date.
   (3) Work package status ***<wp.status>*** (required). Identifies the work package status (e.g., new, changed, etc) (see 32.3.2.2.1.2).
   (4) Reason ***<reason>*** (required). Identifies the reason that a change was made to the work package. This could be through an Engineering Change Proposal (ECP), a request for change form, safety issue, etc. Multiple changes at the same change level and for the same reason will reference the same change history entry. Changes made for different reasons would have different change history records, even if they are at the same change level (see 49.1.4.11).

b. DTD fragment is graphically depicted.

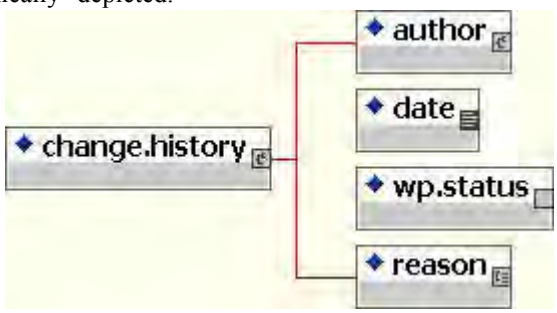

FIGURE 151. Change history record <change.history> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT change.history (author, date, wp.status, reason)>
<!ATTLIST change.history
    id      ID    #IMPLIED
>
```

d. Common attributes:
   (1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

32.3.2.2.1.1 Author ***<author>***. Identifies the change author and company/organization responsible for performing the change.

a. The components are:
   (1) Name ***<name>*** (required). Identifies the change author's name (see 49.1.4.19).
   (2) Proponent ***<proponent>*** (required). The ***<proponent>*** element, when used in this context, identifies the name and address of the organization performing the change. This can be a contractor. For more information on the content model, see 49.1.4.22.

b. DTD fragment is graphically depicted.

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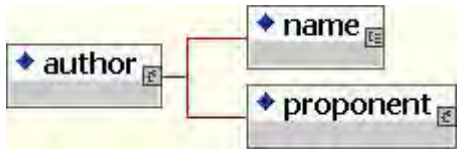

FIGURE 152. Author <author> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT author (name, proponent)>
```

32.3.2.2.1.2 Work package Status ***<wp.status>***. Specifies the change history status. This determines the work package's current usage state (i.e., Draft).

a. DTD fragment:

```
<!ELEMENT wp.status EMPTY>
<!ATTLIST wp.status
     type  (new | changed | draft | deleted | reinstated )  #REQUIRED
>
```

b. Unique attribute is **type** – Work package status type (required) that is identified by one of the following five (5) types.

(1) **new** - New work package that is approved by the acquiring activity. The creation of a work package requires an entry into the change history log. The work package status prior to the proponent approval, set to "**draft**" and after the proponent approval, set to "**new**".

(2) **changed** - Any text inserted, deleted, or modified and is approved by the acquiring activity.

(3) **deleted** - The entire work package has been removed or superseded. The information is not physically removed, but the status is set to "**deleted**".

(4) **reinstated** - A work package previously marked as "**deleted**" has been reinstated as an active work package.

(5) **draft** - The work package has not been approved by the acquiring activity and is considered work in progress. The changes made to the work package, while in draft, are not approved for release without approval from the acquiring activity. Changes are initially marked as "**draft**".

32.3.2.2.2 Tracking change history example. This example shows the XML source fragments for making a change history entry (***<change.history>***) and linking the actual change information to that change history entry. All changed paragraphs, steps, and text within the work package have a reference in the change history log that records the author and change information. In this example, only the most current change history entry is shown (i.e., 4 January 2004). Also shown is the XML source for the date of issue list (***<issuechg>***) in a TM that contains this work package. The ***<issued>*** element in the ***<issuechg>*** element content model provides the issue change number and date. The date of issue list shown contains the dates for the original TM (i.e., 31 May 2000) and three changes (i.e., changes 1–3 made on 12 June 2003, 23 February 2004, and 12 March 2005 respectively). The change level for the TM and the change level for the work package, in most instances, are not the same because work packages can be reused from manual to manual. The work package change level determination, when used in this TM, is identified from the date of issue list. The change number for the work package will be that of the TM at the first issue date after the most current work package change date. In this example, the work package would be at change number 2.

a. The XML document instance fragment containing the dates of issue list.

```
<issuechg>
<trim.para>Dates of issue of original and changed pages/work packages are:
</trim.para>
<issued>
<chgno>0
</chgno>
<chgdate julian="20000531">31 May 2000
```


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```
</chgdate>
</issued>
<issued>
<chgno>1
</chgno>
<chgdate julian="20030612">12 June 2003
</chgdate>
</issued>
<issued>
<chgno>2
</chgno>
<chgdate julian="20040223">23 February 2004
</chgdate>
</issued>
<issued>
<chgno>3
</chgno>
<chgdate julian="20050312">12 March 2005
</chgdate>
</issued>
</issuechg>
```

b. The XML document instance fragment containing the most current change history entry.

```
<change.history id="M0003-X-XXX-XXX-change3">
<author>
<name>John Doe
</name>
<proponent>
<name>Acme Inc.
</name>
<address>
<city>Anytown
</city>
<state>Anystate
</state>
</address>
</proponent>
</author>
<date julian="20040104">4 January 2004
</date>
<wp.status type="changed"/>
<reason>ECP 2034
</reason>
</change.history>
```

c. The XML document instance fragment containing linking from a changed paragraph to the change history entry.

```
<para changeref="M00003-X-XXX-XXX">. . .
</para>
```

32.3.2.3 Applicability ***<applicability>***. When a work package has variations in the narrative and/or illustration or only particular variations are applicable for the entire work package, the narrative configuration is defined using ***<applicability>*** element. The content model is #PCDATA.

32.3.2.4 System/subsystem definition ***<subsystem-system>***. The system/subsystem definition identifies the system breakdown. The system breakdown is depicted in the XML source as parent-child relationships. The first level is the current system/subsystem being referenced in the work package. Each subsequent referenced system/subsystem is at the next higher level. The top system nomenclature is not always the

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entire weapon, but a major subsystem (i.e. engine, transmission, landing gear, etc.). The system breakdown by major subsystems allows the entire subsystem to be reused in other systems without being concerned about the other system's hierarchy (i.e. LCN).

a. The components are:
   (1) Nomenclature ***<systemnomen>*** (required). This element identifies the system/subsystem nomenclature and any identification numbers. The breakdown is by nomenclature and optional identification information (i.e., NSN, CAGEC/part number, and model number) (see 32.3.2.4.1).
   (2) System/Subsystem definition ***<subsystem-system>*** (optional – zero or more). The inclusion of the ***<subsystem-system>*** element in its own content model provides for a recursion capability that permits the breakdown of the system into multiple levels.

b. DTD fragment is graphically depicted.

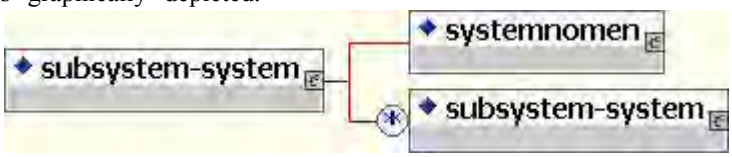

FIGURE 153. System/subsystem <subsystem-system> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT subsystem-system (systemnomen, subsystem-system*)>
```

32.3.2.4.1 System nomenclature ***<systemnomen>***. The element describes nomenclature and information to uniquely identify the system.

a. The components are:
   (1) System name ***<name>*** (required)(see 49.1.4.19).
   (2) National stock number ***<nsn>*** (optional – zero or more) (see 38.4.2.1.3.2.2).
   (3) Part number ***<partno>*** (see 49.1.4.21)/CAGE code ***<cageno>*** (see 49.1.4.2) (optional – zero or more).
   (4) Model number ***<modelno>*** (optional – zero or more) (see 49.1.4.18).
   (5) Reference designator ***<refdes>*** (optional – zero or more) (see 38.4.2.1.3.2.10).
   (6) LSA control number (LCN) ***<lcn>*** (optional – zero or more) (see 32.3.1.3) represents a topdown generation or functional breakdown of hardware and software to the standard engineering drawing structure.
   (7) LSA task code ***<taskcode>*** (optional – zero or more) uniquely identifies each operator/maintenance task associated with particular items under analysis.

b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

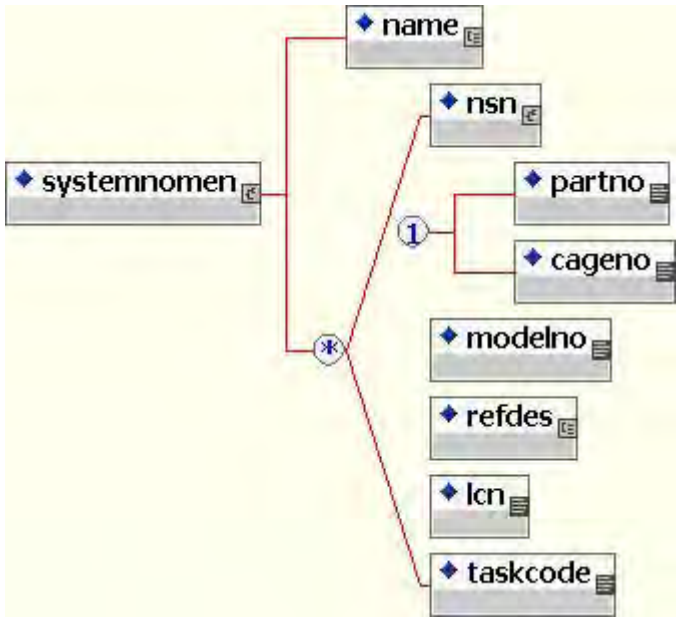

FIGURE 154. System/subsystem <subsystem-system> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT systemnomen (name, (nsn | (partno, cageno) | modelno |
   refdes | lcn | taskcode)*)>
```

32.3.2.4.2 System/subsystem example. An example is "External Lines & Hoses → Engine, Turbine → Power Plant". The XML source would be the following:

<subsystem-system>
<systemnomen>
<name>`External Lines & Hoses`
</name>
</systemnomen>
<subsystem-system>
<systemnomen>
<name>`Engine, Turbine`
</name>
</systemnomen>
<subsystem-system>
<systemnomen>
<name>`Power Plant`
</name>
</systemnomen>
</subsystem-system>
</subsystem-system>
</subsystem-system>

32.3.2.5 Keyword search ***<keyword.search>***. The IETM functionality matrix has an option to provide a keyword search capability for each work package. The element ***<keyword.search>*** permits the author to enter any number of keywords for a user to locate a certain work package. The element contains parsable character data (***%pcdata ;*** see 9.3.2). Generally, the narrative is entered using #PCDATA.

32.3.2.6 Subject ***<subject>***. This element, when used in this context, contains additional qualifier information about the work package. This includes providing major subjects and/or actions that are related to the work package, similar to a keyword search.

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32.3.2.7 Work package metadata <*wp.metadata*> example. This example shows the XML source for making a <*wp.metadatat*> entry. The sample data in this example has the following metadata:

a. Developed in U.S.A. and is unclassified.
b. Government proponent is "TACOM - PM Fighting Vehicle".
c. Work package was developed by Acme, Inc. on September 19, 2002.
d. Revised on October 19, 2003 due to ECP 3454.
e. Used in TM 9-123-123 and TM 9-321-123.
f. System hierarchy is "Machine Gun – Turret".
g. Search by keywords of "replace" or "machine gun".

```
<wp.metadata>
<portionmark country="US" security="uc"/>
<proponent>
<name>TACOM - PM Fighting Vehicles
</name>
<address>
<city>Warren
</city>
<state>MI
</state>
<zip>99999
</zip>
</address>
</proponent>
<tracking>
<change.history id="M0001-9-123-123-org">
<author>
<name>John Doe
</name>
<proponent>
<name>Acme Inc
</name>
<address>
<city>Anytown
</city>
<state>NJ
</state>
<zip>00000
</zip>
</address>
</proponent>
</author>
<date julian="20020919">19 September 2002
</date>
<wp.status type="new"/>
<reason>New work package
</reason>
</change.history>
<change.history id="M0001-9-123-123-chg1">
<author>
<name>Bruce Schanck
</name>
<proponent>
<name>Computer Sciences Corporation
</name>
<address>
```


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Check the source to verify that this is the current version before use.

```
<city>Eatontown
</city>
<state>NJ
</state>
<zip>07724
</zip>
</address>
</proponent>
</author>
<date julian="20031019">19 October 2003
</date>
<wp.status type="changed"/>
<reason>ECP 3454
</reason>
</change.history>
</tracking>
<tminfono inschlvl="1">
<servbranch procuring="yes" service="army"/>
<tmno>TM 9-123-123
</tmno>
</tminfono>
<tminfono inschlvl="1">
<servbranch procuring="yes" service="army"/>
<tmno>TM 9-321-123
</tmno>
</tminfono>
<subsystem-system>
<systemnomen>
<name>Machine Gun
</name>
</systemnomen>
<subsystem-system>
<systemnomen>
<name>Turret
</name>
</systemnomen>
</subsystem-system>
</subsystem-system>
<keyword.search>replace
</keyword.search>
<keyword.search>machine gun
</keyword.search>
</wp.metadata>
```

32.4 Work package identification information ***<wpidinfo>***. The element ***<wpidinfo>*** lists the identification information required for a work package. Work package identification information is required in every work package. The element specifies who can perform the procedure, the lowest maintenance level ***<maintlvl>***, the work package title ***<title>***, lists if there is a difference between configurations in the work package ***<config>***, and specifies if the work package replaces an existing work package ***<wpsupersede>***.

a. The components are:
   (1) Maintenance Level ***<maintlvl>*** (required) (see 32.4.2).
   (2) Title ***<title>*** (required) (see 32.4.3).
   (3) Work Package Configuration Effectivity List ***<config>*** (optional) (see 32.4.4).
   (4) Work Package Supersedure Notice ***<wpsupersede>*** (optional) (see 32.4.5).

b. DTD graphically depicted for ***<wpidinfo>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

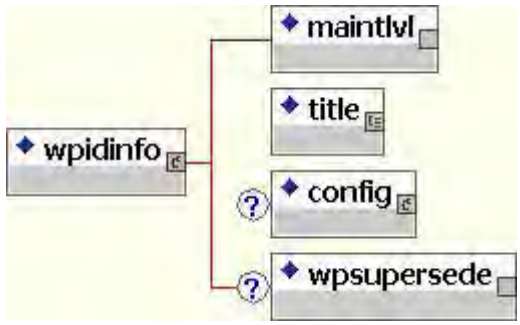

FIGURE 155. Work package identification information <wpidinfo> DTD hierarchy.

c. DTD fragment for ***<wpidinfo>***:

```
<!ELEMENT wpidinfo (maintlvl, title, config?, wpsupersede?)>
```

d. No attributes.

32.4.1 How work package identification information supports a work package. Each work package of a TM requires work package identification information. This information provides support to the work package for reuse of data and sharing by specifying the lowest maintenance level of the work package, supplying the work package title, listing the differences between configurations if there are any in the work package and by identifying if the work package is the original or if it has been modified.

32.4.2 Maintenance level ***<maintlvl>***. The element ***<maintlvl>*** provides the work package maintenance level. This is the LOWEST maintenance level allowed to perform the work specified in the work package (e.g., Depot). The element is EMPTY and the level of maintenance is entered using its attribute **level**. The maintenance level is displayed above the title of the work package. See Figure 157 for an example of how the maintenance level is displayed in the work package identification information.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. The attribute **level** contains a value that provides the choice of maintenance levels:
   (1) field – Fielding
   (2) sustain – Sustainment
   (3) depot – Depot
   (4) operator – Operator
   (5) gensup – General Support
   (6) dirsup – Direct Support
   (7) unitlvl – Aviation Unit
   (8) avum-avim – AVUM-AVIM
   (9) inter – Aviation Intermediate
   (10) tmlvls – All TM Levels
   (11) asb – Aviation support battalion
   (12) amc – Aviation maintenance company
   (13) tasmg – Theater aviation sustainment maintenance group

32.4.3 Title of the work package ***<title>***. The element ***<title>*** lists the title of the work package. Even if the work package is reused the title is not allowed to be changed. For additional information on how title is used (see 49.1.1.3).

32.4.4 Work package configuration effectivity list ***<config>***. Lists the equipment specific configurations that apply to the work package. If there are no differences between configurations, the element is not used. If certain configurations require different tasks/procedures, separate work packages are prepared.

a. Components are

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Check the source to verify that this is the current version before use.

(1) Reduced Paragraph ***<trim.para>*** (optional). (see 49.1.1.7)
(2) Work Package Configuration Effectivity List Item ***<config-setup-item>***(required – one or more). (see 32.4.4.1)

b. DTD graphically depicted for:

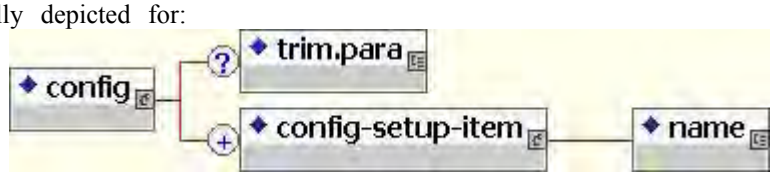

FIGURE 156. Work package configuration effectivity list <config> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT config (trim.para?, config-setup-item+)>
```

d. No attributes.

32.4.4.1 Work package configuration effectivity list item ***<config-setup-item>***. The element ***<config-setup-item>*** provides the equipment specific configurations that are effective for the work package.

a. Components are
(1) Name ***<name>*** (required). (see 49.1.4.19)

b. DTD fragment:

```
<!ELEMENT config-setup-item (name)>
<!ATTLIST config-setup-item
  applicable IDREF #REQUIRED>
```

c. Unique attributes.
(1) **applicable** - Points or links to the master effective list to determine the specific configuration.

32.4.5 Work package supersedure notice ***<wpsupersede>***. A work package supersedure notice informs the user that the work package has been replaced with a modified work package. When the supersedure notice is used, it is placed below the effectivity notice. The supersedure notice requires the work package sequence number **supersede.wpseq** of the superseded work package and the date of the superseded work package **supersede.dated** to be displayed in the notice. If the superseded WP is contained in another manual, the notice includes the publication number **from.tmno** from the technical manual from where the superseded work package is contained. When an unclassified or classified WP supersedes a classified WP a security regulation **supersed.secur** is required in the notice stating that the WP should be destroyed in accordance with applicable security regulations. The element ***<wpsupersede>*** is EMPTY and its data is derived from its attributes. The stylesheet will generate the text derived from the attributes that are used. See Figure 157 for an example of how the work package supersedure notice is displayed in the work package identification information.

a. Components are
(1) Work Package Supersedure Notice ***<wpsupersede>*** is EMPTY and all data is obtained from the attributes.

b. DTD fragment:

```
<!ELEMENT wpsupersede EMPTY>
<!ATTLIST wpsupersede
     supersede.wpseq     CDATA        #REQUIRED
     supersede.dated     CDATA        #REQUIRED
     from.tmno           CDATA        #IMPLIED
     supersed.secur      (yes | no)   'no'>
```


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Check the source to verify that this is the current version before use.

c. Unique attributes:
   (1) **supersede.wpseq** - The work package sequence number of the superseded work package.
   (2) **supersede.dated** - The date of the superseded work package.
   (3) **from.tmno** - The technical manual publication number from where the superseded work package is contained.
   (4) **supersed.secur** - Security regulations notice for a work package that supersedes a classified work package.

32.4.5.1 Page-based display. In a page-based TM the work package sequence number is displayed. The work package supersedure notice consists of one of the following three types available.

a. WP supersedes a WP in the same manual. Use the attributes **supersede.wpseq** and **supersede.dated**. See below for an example of the XML document instance fragment and the output the stylesheet will generate.
   (1) XML for a work package superseding a work package in the same manual.
```
<wpsupersede supersede.wpseq="0006" supersede.dated="January 1, 1999">
```
   (2) An example of how the work package supersedes a work package in the same manual is displayed.
```
"This WP supersedes 0006, January 1, 1999."
```

b. Superseded WP is contained in another manual, the notice includes the external publication number. Use the attributes **supersede.wpseq**, **supersede.dated**, and **from.tmno**. See below an example of the XML document instance fragment and the output the stylesheet will generate.
   (1) XML for a work package superseding a work package that is contained in another manual.
```
<wpsupersede supersede.wpseq="0006" supersede.dated="January 1, 1999"
from.tmno="TM X-XXX-XXXX-24&P">
```
   (2) An example of how the work package supersedes a work package that is contained in another manual is displayed.
```
"This WP supersedes 0006, January 1, 1999, TM X-XXX-XXXX-24&P."
```

c. Unclassified or classified WP supersedes a classified WP. Use the attributes **supersede.wpseq**, **supersede.dated**, and **supersed.secur**. See below an example of the XML document instance fragment and the output the stylesheet will generate.
   (1) XML for a work package that is unclassified or classified WP superseding a classified WP.
```
<wpsupersede supersede.wpseq="0006" supersede.dated="January 1, 1999"
supersed.secur="yes">
```
   (2) An example of how an unclassified or classified WP supersedes a classified WP is displayed.
```
"This WP supersedes (0006), (January 1, 1999), which should be destroyed
in accordance with applicable security regulations."
```

32.4.5.2 Frame-based display. A frame-based TM does not contain work package sequence numbers. Instead of displaying the work package sequence number, the title of the work package is displayed. The work package supersedure notice consists of one of the following three types available.

a. WP supersedes a WP in the same manual. Use the attributes **supersede.wpseq** and **supersede.dated**. See below for an example of the XML document instance fragment and the output the stylesheet will generate.
   (1) XML for a work package superseding a work package in the same manual.
```
<wpsupersede supersede.wpseq="Operation Under Unusual Conditions"
supersede.dated="January 1, 1999">
```
   (2) An example of how the work package supersedes a work package in the same manual is displayed.
```
"This WP supersedes Operation Under Unusual Conditions, January 1, 1999."
```

b. Superseded WP is contained in another manual, the notice includes the external publication number. Use the attributes **supersede.wpseq**, **supersede.dated**, and **from.tmno**. See below for an example of the XML document instance fragment and the output the stylesheet will generate.

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(1) XML for a work package superseding a work package that is contained in another manual.

```
<wpsupersede supersede.wpseq="Operation Under Unusual Conditions"
supersede.dated="January 1, 1999" from.tmno="TM X-XXX-XXXX-24&P">
```

(2) An example of how the work package supersedes a work package that is contained in another manual is displayed.

```
"This WP supersedes Operation Under Unusual Conditions, January 1,
1999, TM X-XXX-XXXX-24&P."
```

c. Unclassified or classified WP supersedes a classified WP. Use the attributes **supersede.wpseq**, **supersede.dated**, and **supersed.secur**. See below for an example of the XML document instance fragment and the output the stylesheet will generate.

(1) XML for a work package that is unclassified or classified WP superseding a classified WP.

```
<wpsupersede supersede.wpseq="Operation Under Unusual Conditions"
supersede.dated="January 1, 1999" supersed.secur="yes">
```

(2) An example of how an unclassified or classified WP supersedes a classified WP is displayed.

```
"This WP supersedes (Operation Under Unusual Conditions), (January 1,
1999), which should be destroyed in accordance with applicable security
regulations."
```

32.4.6 Joint use. Joint use is used by the Army for a TM in joint publications with one or more other Services. To specify a joint use TM, the tag to identify the Services is the entity attribute, service only work package of a work package. All work packages contain this attribute, service only work package. NOTE: The service only work package attributes are not part of the element work package identification ***<wpidinfo>*** but an attribute of the work package itself. See below how to specify a specific service or services for a work package.

a. An example of where and how the service attributes are located in the model content of a work package. See below the general information work package ***<ginfowp>*** model content.

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b. DTD fragment for ***<ginfowp>***:

```
<!ELEMENT ginfowp (wpidinfo, scope, reporting?, mfrr, eir,
handreceipt?,cpcdata?, odsdata?, destructmat?, pssref?, wrntyref?,
nomenreflist?, loa?, qainfo?, qual.mat.info?, sftyinfo?, hcp?,
calref?, ecp?, modification?, deviation?, mobreq?, fscapreq?, cost?,
copyrt?, supdata?)>


<!ATTLIST ginfowp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED>
```

c. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

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(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

32.4.7 Examples of work package identification. XML instance examples are provided to show the tagging for a work package identification in work packages. The style sheet output is also provided for the XML example. One example is in a page-based format and two frame-based output examples are provided. One frame-based example is a context non-filtering/login frame-based work package and the other is a filtering/login frame-based work package.

32.4.7.1 Example of a page-based work package identification. An XML instance example is provided to show the tagging for a work package identification in a work package. The style sheet output is also provided for the XML example and is in a page-based format.

a. Example of an XML document instance fragment for ***<wpidinfo>***:

```
<wpidinfo>
<maintlvl level=&ldquo;sustain&rdquo;/>
<title>ADDITIONAL AUTHORIZATION TITLE (AAL) WORK PACKAGE
</title>
<config>
<config-setup-item applicable=&ldquo;xxxx&rdquo;>
<name>Serial Numbers 12345 through 12399
</name>
</config-setup-item>
</config>
<wpsupersede supersede.wpseq=&ldquo;0006&rdquo; supersede.dated=&ldquo;January
1, 1999&rdquo; supersed.secur=&ldquo;yes&rdquo; from.tmno=&ldquo;TM
X-XXX-XXXX-24&amp;P&rdquo;>
</wpidinfo>
```

b. Example of the display for ***<wpidinfo>*** in a page-based work package.

0041

SUSTAINMENT MAINTENANCE

ADDITIONAL AUTHORIZATION TITLE (AAL) WORK PACKAGE (ARMY ONLY)

Serial Numbers 12345 through 12399

This WP supersedes 0006, dated January 1, 1999, contained in TM X-XXX-XXXX-XX, which should be destroyed in accordance with applicable security regulations.

FIGURE 157. Example of the display for <wpidinfo> in a page-based work package.

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Check the source to verify that this is the current version before use.

32.4.7.2 Examples of frame-based work package identification. An XML instance example is provided to show the tagging for a work package identification in a work package. Two frame-based output examples are also provided. One example is a context non-filtering/login frame-based work package and the other is a filtering/login frame-based work package.

a. Example of a Legacy XML document instance fragment for ***<wpidinfo>***:

<wpidinfo>
<maintlvl level=“direct”/>
<title>`Maintenance Work Package`
</title>
<config>
<config-setup-item applicable=“xxxx”>
<name>`Serial Numbers 12345 through 12399`
</name>
</config-setup-item>
</config>
<wpsupersede supersede.wpseq=“0006” supersede.date=“January 1, 1999“>
</wpidinfo>

b. Example of the display for ***<wpidinfo>*** in a context non-filtering/login frame-based work package. The title of the work package is displayed in the title bar of the outer shell. The maintenance level, list of configurations in the work package and the work package supersedure notice is displayed in the inner shell.

FIGURE 158. Example of the display for <wpidinfo> in a context non-filtering/login frame-based work package.

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Check the source to verify that this is the current version before use.

c. Example of the display for ***<wpidinfo>*** in a filtering/login frame-based work package. The title of the work package is displayed in the title bar of the outer shell. The maintenance level, list of configurations in the work package and the work package supersedure notice is not displayed in the inner shell but in the source data.

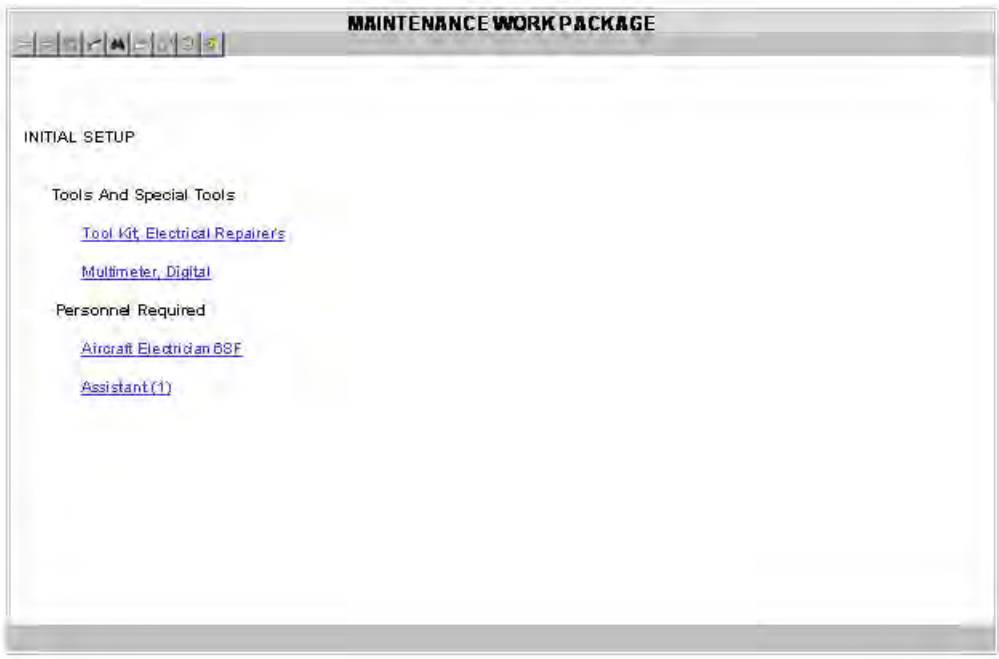

FIGURE 159. Example of the display for <wpidinfo> in a filtering/login frame-based work package.

32.5 Work package setup information ***<wpinfo>***. The element ***<wpinfo>*** lists all of the information required by the technician so the tools, special tools, test equipment, source references, parts, and other items needed to complete the tasks can be obtained and identified. Each setup category list is only referenced once and occur in the order as specified in MIL–STD–40051–1/–2. The work package setup information content model requires at least one setup category to be used. When no setup information is required, use "Not Applicable" or similar terminology to indicate to the maintainer no work package setup information is required.

a. The components are:
   (1) Test equipment list ***<testeqp>*** (optional) (see 32.5.2).
   (2) Tools and special tools list ***<tools>*** (optional) (see 32.5.3).
   (3) Expendable materials and parts required list ***<mtrlpart>*** (optional) (see 32.5.4).
   (4) Personnel required list ***<persnreq>*** (optional) (see 32.5.5).
   (5) Source reference list ***<refs>*** (optional) (see 32.5.6).
   (6) Equipment condition list ***<eqpconds>*** (optional) (see 32.5.7).
   (7) Special environmental condition list ***<specenv>*** (optional) (see 32.5.8)
   (8) Drawing required list ***<dwgreq>*** (optional) (see 32.5.9).
   (9) Estimated time to complete the task ***<time.to.comp>*** (optional) (see 32.5.10).
   (10) Not applicable ***<title>*** ***<null>*** (optional) (see 32.5.11).

b. DTD fragment is graphically depicted.

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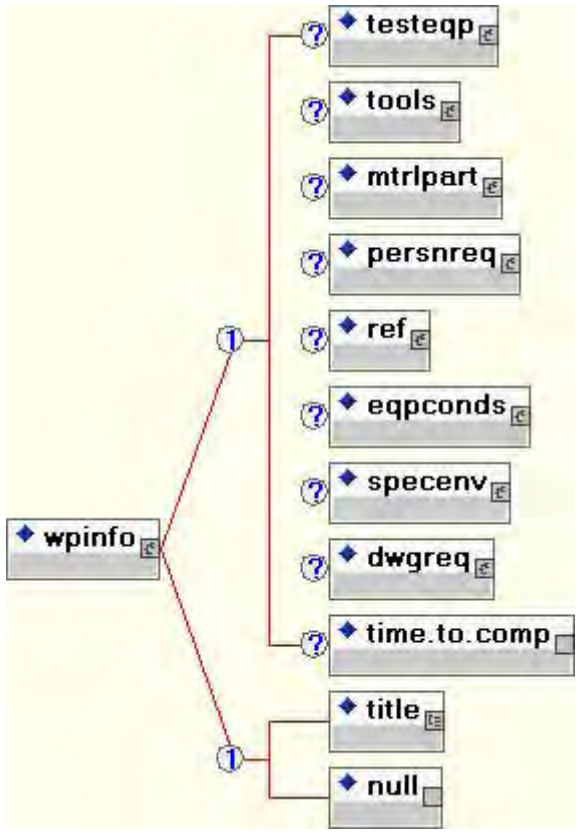

FIGURE 160. Work package initial setup graphical view.

c. DTD fragment:

```
<!ELEMENT wpinfo ((testeqp, tools?, mtrlpart?, persnreq?, ref?, eqpconds?, specenv?, dwgreq
specenv?, dwgreq?,  time.to.comp?) | (mtrlpart, persnreq?, ref?, eqpconds?,
 specenv?, dwgreq?, time.to.comp?) | (persnreq, ref?, eqpconds?, specenv?,
 dwgreq?, time.to.comp?) | (ref, eqpconds?, specenv?, dwgreq?, time.to.comp?)
 | (eqpconds, specenv?, dwgreq?, time.to.comp?) | (specenv, dwgreq?,
 time.to.comp?) | (dwgreq, time.to.comp?) | time.to.comp | (title , null))>
<!ATTLIST wpinfo
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).

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(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

32.5.1 Reference and hyper link setup item. The main purpose for initial setup is identifying the required items, by name and location, to perform the work package. MIL–STD–40051–1/–2 does not allow the NSN, part number, and CAGE code to appear in the work package (since these codes can change and a change would cause each work packages showing the code requires a change package), the identification information is obtained from the associated supporting information work packages. The page-based manual has a prescribed layout for referencing the item by work package number and item number. The frame-based manual is more flexible how the referenced item is presented by nomenclature/condition hot spot link, button link, or work package and item number hot spot link (similar to page-base look). The markup method can be accomplished by various methods and some what dependent on the IETM viewer and/or style sheet. When MIL–STD–40051–1/–2 requires an reference to the setup item either the requires the use of either the dedicated element ***<itemref>*** (see 32.5.1.1) or using referencing elements within the nomenclature/condition by using the element ***<link>***, ***<xref>***, or ***<extref>*** (see 32.5.1.2). Demonstrate the different methods the example using a tool setup item where the maintainer requires a "Handle, socket wrench, ratchet, 1/2 inch drive". The maintainer is referenced to the tools identification work package and the specify tool item information. The following paragraphs will discussion the various approaches to referencing setup items.

32.5.1.1 Dedicated setup item referencing ***<itemref>***. A common element for many of the setup items is ***<itemref>***. The element links the maintainer to the detailed information about the setup item. The element contains a cross reference within the work package/document ***<xref>***, a cross reference external to the document ***<extref>***, and enhanced cross reference to either internal or external to the work package/document ***<link>*** (see 46 for detailed information about linking). The XML referencing model uses two different methods, legacy and enhanced, to link the setup item detailed information.

a. The components are:
   (1) Internal reference ***<xref>*** (see 46.2.2) or external reference ***<extref>*** (see 46.2.1) or link ***<link>*** (required – one or more) (see 46.2.3)
b. DTD fragment is graphically depicted.

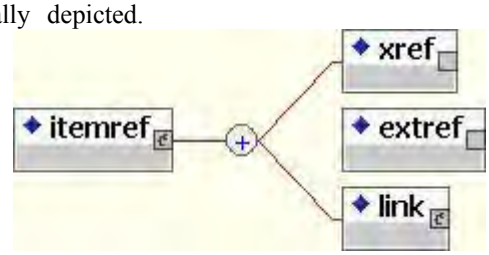

FIGURE 161. Setup item reference <itemref> DTD hierarchy.

c. DTD fragment for ***<itemref>***:

```
<!ELEMENT itemref (xref | extref | link)+>
```

32.5.1.1.1 Legacy internal referencing method ***<xref>***. The legacy referencing method uses a simple linking model of cross reference. The cross reference element ***<xref>*** points (from attribute "**wpid**") to the tools identification work package unique identifier and points (from attribute "**itemid**") to the item number unique identifier in the tool identification list.

a. The XML source code for the sample is:

```
<tools-setup-item>
<name>Handle, socket wrench, ratchet, 1/2 inch drive
</name>
<itemref>
<xref wpid="s00001-9-2350-294" itemid="s00001-9-2350-294-item3"/>
```


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</itemref>
</tools-setup-item>

b. A page-based example (assuming the tools identification work package is sequence number 0212 and the item number is "3") would produce "Handle, socket wrench, ratchet, 1/2 inch drive (WP 0212, Item 3)". The maintainer would turn to work package 0212 and lookup the item number 3 tool's specific information (i.e. NSN, part number).

c. A frame-based manual the maintainer clicks on the hot spot reference and the frame-based viewer presents the detailed tool information in a popup window, in a separate pane, or replacing the current pane. The frame-based presentation of the example could produce one of the following methods, as determined by the style sheet. Some style sheets presentation approaches are simpler to develop than other methods (order shown are from simple to complex).

(1) Textual hot spot reference after the nomenclature "Handle, socket wrench, ratchet, 1/2 inch drive See Reference"

(2) Button is the hot spot reference after the nomenclature "Handle, socket wrench, ratchet, 1/2 inch drive LINK"

(3) Work package name and item number is the hot spot reference "Handle, socket wrench, ratchet, 1/2 inch drive (Tools Identification List, Item 3)"

(4) Nomenclature is the hot spot reference "Handle, socket wrench, ratchet, 1/2 inch drive"

32.5.1.1.2 Enhanced referencing method ***<link>***. The enhanced referencing method provides greater linking functionality, more options, and defining the view for frame-based viewers (see 46 for detailed information on linking). The enhanced linking element ***<link>*** points (from attribute "**xlink:href**") to the item number unique identifier in the tool identification list. The stylesheet would dissects the attribute "**xlink:href**", therefore the attribute information before the "#" is the work package unique identifier and information following the "#" is the tool item unique identifier.

a. The XML source code for the sample has a page-based and a framed-based presentation is:

<tools-setup-item>
<name>`Handle, socket wrench, ratchet, 1/2 inch drive`
</name>
<itemref>
<link xlink:href="s00001-9-2350-294#s00001-9-2350-294-item3"
application="page"xreftype="part" xmlns:xlink="http://www.w3.org/1999/xlink"
xlink:type="simple" linkaction="prompt" linktype="return" popup="no">
<pretext> (
</pretext>
<posttext>)
</posttext>
</link>
<link xlink:href="s00001-9-2350-294#s00001-9-2350-294-item3"
application="frame"xreftype="part" xmlns:xlink="http://www.w3.org/1999/xlink"
xlink:type="simple" linkaction="prompt" linktype="return" popup="no">
<prompt>`See reference`
</prompt>
</link>
</itemref>
</tools-setup-item>

b. A page-based example (assuming the tools identification work package is sequence number 0212 and the item number is "3") would produce "Handle, socket wrench, ratchet, 1/2 inch drive (WP 0212, Item 3)". The maintainer would turn to work package 0212 and lookup the tool's specific information (i.e. NSN, part number).

c. A frame-based example would hot spot the item name "Handle, socket wrench, ratchet, 1/2 inch drive See reference".

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32.5.1.2 Nomenclature/condition contains the setup item reference. Referencing the setup item information within a nomenclature/condition is generally used by frame-based systems. The approach is to make the setup reference the nomenclature/condition text. Considerations has to be reviewed if the setup reference needs both page-based and frame-based presentations. The best and most flexible approach is using the enhanced referencing element ***<link>***. Providing for both page-based and frame-based presentation requires the need for two ***<link>*** elements. One ***<link>*** element provides the page-based presentation view and the other ***<link>*** element provides the nomenclature/condition as the hot spot reference. The page-based markup includes the nomenclature/condition in the element ***<pretext>*** and the work package and item reference information is generated by the style sheet as indicated by the element ***<ref.generate>***. The frame-based markup includes the nomenclature/condition in the element ***<prompt>***, thus making the nomenclature/condition text as the hot spot reference.

a. The XML source code for the sample has a page-based and a framed-based presentation is:

```
<tools-setup-item>
<name>
<link  xlink:href="s00001-9-2350-294#s00001-9-2350-294-item3"
application="page"xreftype="part"  xmlns:xlink="http://www.w3.org/1999/xlink"
xlink:type="simple"  linkaction="prompt"  linktype="return"  popup="no">
<pretext>Handle, socket wrench, ratchet, 1/2 inch drive (
</pretext>
<ref.generate/>
<posttext>)
</posttext>
</link>
<link  xlink:href="s00001-9-2350-294#s00001-9-2350-294-item3"
application="frame"xreftype="part"  xmlns:xlink="http://www.w3.org/1999/xlink"
xlink:type="simple"  linkaction="prompt"  linktype="return"  popup="no">
<prompt>Handle, socket wrench, ratchet, 1/2 inch drive
</prompt>
</link>
</name>
</tools-setup-item>
```

b. A page-based example (assuming the tools identification work package is sequence number 0212 and the item number is "3") would produce "Handle, socket wrench, ratchet, 1/2 inch drive (WP 0212, Item 3)".

c. A frame-based example would hot spot the item name "Handle, socket wrench, ratchet, 1/2 inch drive".

32.5.2 Test equipment list ***<testeqp>***. The element ***<testeqp>*** is the list of test equipment required to perform the tasks and procedures in the work package. Each listed test equipment items are grouped using the element ***<testeqp-setup-item>*** (required – one or more). The element ***<testeqp-setup-item>*** contains a test equipment nomenclature ***<name>*** (required) (see 49.1.4.19) and where to locate the test equipment information details (see 32.5.1).

a. The components are:

(1) Test equipment list setup ***<testeqp-setup-item>*** (required – one or more).

(a) Equipment nomenclature ***<name>*** (required) (see 32.5.1).

(b) Reference item ***<itemref>*** (optional) (see 32.5.1.1).

b. DTD fragment is graphically depicted.

FIGURE 162. Test equipment list <testeqp> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT testeqp (testeqp-setup-item+)>
<!ELEMENT testeqp-setup-item (name, itemref?)>
<!ATTLIST testeqp-setup-item
  applicable  IDREFS   #IMPLIED
   id          ID       #IMPLIED
   changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

d. Common attributes for ***<testeqp-setup-item>*** are:

(1) **applicable** – Applicablity reference (optional) (see 49.2.1.b.(1)).
(2) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(3) **inschlvl** – Insert change level (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(6) **comment** – Change information (optional) (see 49.2.9).

32.5.3 Tools and special tools list ***<tools>***. The element ***<tools>*** is the list of tools and/or special tools required to perform the tasks and procedures in the work package. Each listed tools and/or special tool is grouped using the element ***<tools-setup-item>*** (required – one or more). The element ***<tools-setup-item>*** contains a tool or special tool nomenclature ***<name>*** (required) (see 49.1.4.19) and where to locate the tool or special tool information details (see 32.5.1).

a. The components are:

(1) Tools and special tools list setup***<tools-setup-item>*** (required – one or more).
(a) Equipment nomenclature ***<name>*** (required) (see 32.5.1).
(b) Item reference information ***<itemref>*** (optional) (see 32.5.1.1).

b. DTD fragment is graphically depicted.

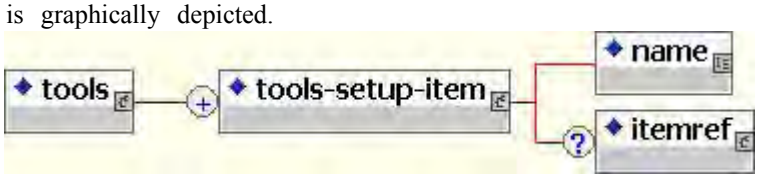

FIGURE 163. Tools and special tools list <tools> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT tools (tools-setup-item+)>
<!ELEMENT tools-setup-item (name, itemref?)>
<!ATTLIST tools-setup-item
  applicable  IDREFS   #IMPLIED
   id          ID       #IMPLIED
inschlvl       CDATA              #IMPLIED
   delchlvl       CDATA              #IMPLIED
   comment        CDATA              #IMPLIED
   changeref      IDREFS             #IMPLIED>
```

d. Common attributes for ***<tools-setup-item>*** are:

(1) **applicable** – Applicablity reference (optional) (see 49.2.1.b.(1)).
(2) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(3) **inschlvl** – Insert change level (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(6) **comment** – Change information (optional) (see 49.2.9).

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32.5.4 Materials/parts list ***<mtrlpart>***. The element ***<mtrlpart>*** is the list of expendable materials and parts required to perform the tasks and procedures in the work package. Each listed material/parts item is grouped using the element ***<mtrlpart-setup-item>*** (required – one or more). The element ***<mtrlpart-setup-item>*** contains a part nomenclature ***<name>*** (required) (see 49.1.4.19), quantity (if more than one material/part is required) ***<qty>*** (optional) (see 49.1.4.9), and where to locate the part information details (see 32.5.1).

a. The components are:
   (1) Materials/parts list setup ***<mtrlpart-setup-item>*** (required – one or more).
      (a) Equipment nomenclature ***<name>*** (required) (see 32.5.1).
      (b) Quantity ***<qty>*** (optional) (see 49.1.4.9).
      (c) Reference item ***<itemref>*** (optional) (see 32.5.1.1).

b.

FIGURE 164. Materials/parts list DTD hierarchy <mtrlpart>.

c. DTD fragment:

```
<!ELEMENT mtrlpart (mtrlpart-setup-item+)>
<!ELEMENT mtrlpart-setup-item (name, qty?, itemref?)>
<!ATTLIST mtrlpart-setup-item
  applicable  IDREFS   #IMPLIED
   id          ID       #IMPLIED
inschlvl       CDATA              #IMPLIED
   delchlvl       CDATA              #IMPLIED
   comment        CDATA              #IMPLIED
   changeref      IDREFS             #IMPLIED>
```

d. Common attributes for ***<mtrlpart-setup-item>*** are:
   (1) **applicable** – Applicablity reference (optional) (see 49.2.1.b.(1)).
   (2) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (3) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (5) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (6) **comment** – Change information (optional) (see 49.2.9).

32.5.5 Personnel required ***<persnreq>***. The element ***<persnreq>*** is the list of personnel required to perform the tasks and procedures in the work package. Each personal type is contained in the element ***<persnreq-setup-item>*** (required – one or more) that contains the type of person needed ***<name>*** (see 49.1.4.19), MOS position (if known or necessary) (***<mos>*** (optional), and quantity (if more than one person is required) ***<qty>*** (optional) (see 49.1.4.9).

a. The components are:
   (1) Personnel required setup item ***<persnreq-setup-item>*** (required – one or more).
      (a) Equipment nomenclature ***<name>*** (required) (see 32.5.1).
      (b) Military occupation specialty ***<mos>*** (optional) (see 32.5.5.1).
      (c) Quantity ***<qty>*** (optional) (see 49.1.4.9).

b. DTD graphically depicted for ***<persnreq>***:

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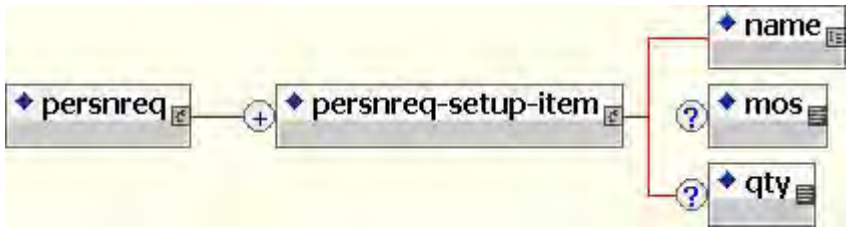

FIGURE 165. Personnel required list DTD hierarchy <persnreq>.

c. DTD fragment for ***<persnreq>***:

```
<!ELEMENT persnreq (persnreq-setup-item+)>
<!ELEMENT persnreq-setup-item (name, mos?, qty?)>
<!ATTLIST persnreq-setup-item
   id              ID                  #IMPLIED
   applicable      IDREFS              #IMPLIED
   inschlvl        CDATA               #IMPLIED
   delchlvl        CDATA               #IMPLIED
   comment         CDATA               #IMPLIED
   changeref       IDREFS              #IMPLIED
>
```

d. Common attributes for ***<persnreq-setup-item>*** are:

(1) **applicable** – Applicablity reference (optional) (see 49.2.1.b.(1)).
(2) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(3) **inschlvl** – Insert change level (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(6) **comment** – Change information (optional) (see 49.2.9).

32.5.5.1 Military occupation specialty ***<mos>***. The element ***<mos>*** contains the Military Occupation Specialty (MOS) code.

a. The components are:

(1) The element contains narrative text #PCDATA parsable character data. (required)

b. DTD fragment:

```
<!ELEMENT mos (#PCDATA)>
<!ATTLIST mos
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

c. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(6) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(7) **inschlvl** – Insert change level (optional) (see 49.2.9).
(8) **security** – Security classification (optional) (see 49.2.13).
(9) **skilltrk** – Skill level (optional) (see 49.2.2).

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32.5.6 Source reference list ***<ref>***. The element ***<ref>*** is list of all work packages and documents referenced in the work package, except for referenced work packages and documents contained in the equipment condition list. Each reference is contained in the element ***<ref-setup-item>*** (required – one or more). When the reference is a work package the element ***<link>*** (see 46.2.3) or ***<xref>*** (see 46.2.2) is used. When the reference is an external publication the element ***<link>*** (see 49.1.4.19) or ***<extref>*** (see 49.1.4.19) is used.

a. The components are:
   (1) Source reference list setup item ***<ref-setup-item>*** (required – one or more)
      (a) Internal reference ***<xref>*** (see 46.2.2) or external reference ***<extref>*** (see 46.2.1) or link ***<link>*** (see 46.2.3)

b. DTD fragment is graphically depicted.

FIGURE 166. Source reference list DTD hierarchy <ref>.

c. DTD fragment for ***<ref>***:

```
<!ELEMENT ref (ref-setup-item+)>
<!ELEMENT ref-setup-item (xref | extref | link)>
<!ATTLIST ref-setup-item
  applicable  IDREFS   #IMPLIED
   id          ID       #IMPLIED
inschlvl       CDATA              #IMPLIED
   delchlvl       CDATA              #IMPLIED
   comment        CDATA              #IMPLIED
   changeref      IDREFS             #IMPLIED>
```

d. Common attributes for ***<ref-setup-item>*** are:
   (1) **applicable** – Applicablity reference (optional) (see 49.2.1.b.(1)).
   (2) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (3) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (5) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (6) **comment** – Change information (optional) (see 49.2.9).

32.5.7 Equipment condition list ***<eqpconds>***. The element ***<eqpconds>*** is the list of equipment condition prior to beginning the tasks covered by the work package. Each listed equipment condition is grouped using the element ***<eqpconds-setup-item>*** (required – one or more). The element ***<eqpconds-setup-item>*** contains the equipment condition ***<condition>*** (required) (see 49.1.4.14) and where to locate the instructions for putting the equipment in the prescribed condition (see 32.5.1).

a. The components are:
   (1) Equipment condition setup item ***<eqpconds-setup-item>*** or Conditional Equipment Conditions Setup Item ***<eqpconds-setup-item-alt>*** (required – one or more).
      (a) Precondition ***<precond>*** (optional) (see 42.1.1.1).
      (b) System condition statement ***<condition>*** (required) (see 49.1.4.14).
      (c) Item reference information ***<itemref>*** (optional) (see 32.5.1.1).

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

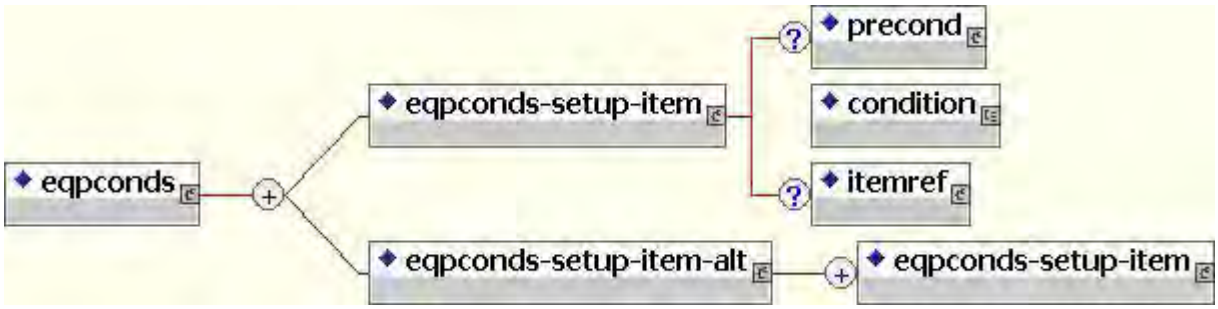

FIGURE 167. Equipment condition list DTD hierarchy <eqpconds>.

c. DTD fragment:

```
<!ELEMENT eqpconds (eqpconds-setup-item | eqpconds-setup-item-alt)+>
<!ELEMENT eqpconds-setup-item (precond?, condition, itemref?)>
<!ATTLIST eqpconds-setup-item
  applicable  IDREFS   #IMPLIED
   id          ID       #IMPLIED
inschlvl       CDATA              #IMPLIED
   delchlvl       CDATA              #IMPLIED
   comment        CDATA              #IMPLIED
   changeref      IDREFS             #IMPLIED>

<!ELEMENT eqpconds-setup-item-alt (eqpconds-setup-item+)>
```

d. Common attributes for ***<eqpconds-setup-item>*** are:
   (1) **applicable** – Applicablity reference (optional) (see 49.2.1.b.(1)).
   (2) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (3) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (5) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (6) **comment** – Change information (optional) (see 49.2.9).

32.5.7.1 Filtering equipment condition. Filtering or preconditions ***<eqpconds-setup-item-alt>*** (see 47.3 for information about using alternative or precondition requirements) on the equipment condition is permitted when using a frame-based viewer that filters information. The element ***<eqpconds-setup-item-alt>*** is used when the precondition ***<precond>*** (see 42.1.1.1) is valid the condition is displayed, otherwise when the precondition is invalid the equipment condition is not displayed to the maintainer. The preconditions are used primarily for environmental conditions (i.e. arctic, desert, marsh, etc. conditions), known completed equipment condition task, equipment model being maintained, etc. This precondition filtering is accomplished through setting of state table variable(s) during log in, information gathered from the maintainer through user interaction, and/or set state table variable during completion of a previous task.

32.5.8 Special environmental condition ***<specenv>***. The element ***<specenv>*** is special environmental condition, such as ventilation, lighting, or temperature, required to perform the procedures contained in the work package. Each special environmental condition is contained in the element ***<specenv-setup-item>*** (required – one or more) that contains the condition ***<condition>*** (required) (see 49.1.4.14) and the reason it is required ***<reason>*** (required) (see 49.1.4.11).

a. The components are:
   (1) Special environmental condition setup item ***<specenv-setup-item>*** (required – one or more).
      (a) System condition statement ***<condition>*** (required) (see 49.1.4.14).
      (b) Reason for action ***<reason>*** (required) (see 49.1.4.11).

b. DTD fragment is graphically depicted.

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FIGURE 168. Special environmental condition DTD hierarchy <specenv>.

c. DTD fragment:

```
<!ELEMENT specenv (specenv-setup-item+)>
<!ELEMENT specenv-setup-item (condition, reason)>
<!ATTLIST specenv-setup-item
  applicable  IDREFS   #IMPLIED
   id          ID       #IMPLIED
inschlvl       CDATA              #IMPLIED
   delchlvl       CDATA              #IMPLIED
   comment        CDATA              #IMPLIED
   changeref      IDREFS             #IMPLIED>
```

d. Common attributes for ***<specenv-setup-item>*** are:
   (1) **applicable** – Applicablity reference (optional) (see 49.2.1.b.(1)).
   (2) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (3) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (5) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (6) **comment** – Change information (optional) (see 49.2.9).

32.5.9 Drawing requirements list ***<dwgreq>***. The element ***<dwgreq>*** are the drawings required to perform the tasks and procedures in the work package, but which are not included in the work package (i.e. foldouts). Each listed drawing item is grouped using the element ***<dwgreq-setup-item>*** (required – one or more). The element ***<dwgreq-setup-item>*** contains a drawing title or name ***<dwgname>*** (required) (see 49.1.4.6) and the drawing number to assist in locating the information ***<dwgno>*** (required) (see 49.1.4.7).

a. The components are:
   (1) Drawing requirements list setup item ***<dwgreq-setup-item>*** (required – one or more).
      (a) Drawing name ***<dwgname>*** (required) (see 49.1.4.6).
      (b) Drawing number ***<dwgno>*** (required) (see 49.1.4.7).

b. DTD fragment is graphically depicted.

FIGURE 169. Drawing requirement list DTD hierarchy <dwgreq>.

c. DTD fragment:

```
<!ELEMENT dwgreg (dwgreq-setup-item+)>
<!ELEMENT dwgreq-setup-item (dwgname, dwgno)>
<!ATTLIST dwgreq-setup-item
  applicable  IDREFS   #IMPLIED
   id          ID       #IMPLIED
inschlvl       CDATA              #IMPLIED
   delchlvl       CDATA              #IMPLIED
   comment        CDATA              #IMPLIED
   changeref      IDREFS             #IMPLIED>
```

d. Common attributes:
   (1) **applicable** – Applicablity reference (optional) (see 49.2.1.b.(1)).

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(2) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(3) **inschlvl** – Insert change level (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(6) **comment** – Change information (optional) (see 49.2.9).

32.5.10 Estimated time to complete the task ***<time.to.comp>***. The element ***<time.to.comp>*** is the estimated time (in tenth's of a hour) to complete the task. The information for the time is entered using the attribute "**hrs**" and is required when the element is used.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment::

```
<!ELEMENT time.to.comp EMPTY>
<!ATTLIST time.to.comp
   hrs        CDATA  #REQUIRED
   applicable IDREFS #IMPLIED
>
```

c. Unique attributes:
   (1) **applicable** – Applicablity reference (optional) (see 49.2.1.b.(1)).
   (2) **hrs** - The estimated amount of hours to complete the operating task.

32.5.11 Setup not applicable ***<title><null>***. The elements ***<title>*** and ***<null>*** are used to indicate that no setup information is required. The element ***<title>*** has the text to indicate to the maintainer that no setup is required.

32.5.12 Sample work package initial setup instance. Several initial setup XML markup samples are provided below.

a. Initial setup using ***<xref>*** and ***<extref>***.

```
<wpinfo>
<persnreq>
<persnreq-setup-item>
<name>Soldier
</name>
</persnreq-setup-item>
</persnreq>
<ref>
<ref-setup-item>
<extref docno="TM 9-2350-294-10-2"/>
</ref-setup-item>
</ref>
<eqpconds>
<eqpconds-setup-item name_cond_hotspot="yes">
<condition>Engine stopped
</condition>
<itemref>
<xref wpid="O00017-9-2350-294" pretext="(" posttext=")"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item name_cond_hotspot="no">
<condition>Commander's hatch cover closed
</condition>
<itemref>
<extref docno="TM 9-2350-294-10-2" pretext=" (" posttext=")"/>
</itemref>
</eqpconds-setup-item>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<eqpconds-setup-item  name_cond_hotspot="no">
<condition>Gunner's hatch cover closed
</condition>
<itemref>
<extref docno="TM 9-2350-294-10-2" pretext=" (" posttext=")"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item  name_cond_hotspot="no">
<condition>Turret in any position except 6400 mils
</condition>
<itemref>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
```

b. Initial setup using **<link>**. The equipment condition item “Engine stopped” has two **<link>** elements page-based and frame-based.

```
<wpinfo>
<persnreq>
<persnreq-setup-item>
<name>Soldier
</name>
</persnreq-setup-item>
</persnreq>
<ref>
<ref-setup-item>
<link  xmlns:xlink="http://www.w3.org/1999/xlink"
application="both"xlink:href="TM_9-2350-294-10-2.xml"  xlink:type="simple"
xreftype="document"  linktype="return"  linkaction="prompt"  popup="no">
<prompt>TM 9-2350-294-10-2
</prompt>
</link>
</ref-setup-item>
</ref>
<eqpconds>
<eqpconds-setup-item>
<condition>
<link  xmlns:xlink="http://www.w3.org/1999/xlink"
application="page"xlink:href="O00017-9-2350-294"  xlink:type="simple"  xreftype="wp"
linktype="return"  linkaction="prompt"  popup="no">
<pretext>Engine stopped (
</pretext>
<ref.generate/>
<posttext>)
</posttext>
</link>
<link  xmlns:xlink="http://www.w3.org/1999/xlink"
application="frame"xlink:href="O00017-9-2350-294"  xlink:type="simple"  xreftype="wp"
linktype="return"  linkaction="prompt"  popup="no">
<prompt>Engine stopped
</prompt>
</link>
</condition>
</eqpconds-setup-item>
<eqpconds-setup-item>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<condition> Commander's hatch cover closed
</condition>
<itemref>
<link xmlns:xlink="http://www.w3.org/1999/xlink"
application="both"xlink:href="TM_9-2350-294-10-2.xml" xlink:type="simple"
xreftype="document" linktype="return" linkaction="prompt" popup="no">
<prompt>TM 9-2350-294-10-2
</prompt>
</link>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Gunner's hatch cover closed
</condition>
<itemref>
<link xmlns:xlink="http://www.w3.org/1999/xlink"
application="both"xlink:href="TM_9-2350-294-10-2.xml" xlink:type="simple"
xreftype="document" linktype="return" linkaction="prompt" popup="no">
<prompt>TM 9-2350-294-10-2
</prompt>
</link>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Turret in any position except 6400 mils
</condition>
<itemref>
<link xmlns:xlink="http://www.w3.org/1999/xlink"
application="both"xlink:href="TM_9-2350-294-10-2.xml" xlink:type="simple"
xreftype="document" linktype="return" linkaction="prompt" popup="no">
<prompt>TM 9-2350-294-10-2
</prompt>
</link>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
```

c. Frame-based initial setup formatted.**INITIAL SETUP:Personal Required**
Soldier

References
TM 9-2350-10-2

Equipment Condition
Engine stopped
Commander's hatch cover closed (TM 9-2350-10-2)
Gunner's hatch cover closed (TM 9-2350-10-2)

d. Page-based initial setup formatted.**INITIAL SETUP:Personal Required**
Soldier

References
TM 9-2350-10-2

Equipment Condition
Engine stopped (WP 0035)
Commander's hatch cover closed (TM 9-2350-10-2)
Gunner's hatch cover closed (TM 9-2350-10-2)

e. Initial setup using not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<wpinfo>
<title>Not  Applicable
</title>
<null  insert="none"/>
</wpinfo>
```

f. Initial setup formatted.
INITIAL SETUP: Not Applicable

33 TASKS, PROCEDURES, AND STEPS.

33.1 Tasks.

33.1.1 Task explanation. As presented in the DTD a task in both a page based and electronic TM is a collection of two or more procedures (***<proc>***), a collection of one or more steps (***<step1>*** or ***<step1–alt>***), or a single paragraph (***<para>***) that present a complete action. A task may be as simple as checking the tire pressure on a vehicle or as complex as overhauling a turbine engine. Not all tasks are related to maintenance or support. Operating procedures are also written as tasks. To support the many different levels and types of tasks, the tasks are broken down into lower levels of either specific procedures (see 33.2) or a group of procedural steps (see 33.3).

33.1.2 Task as a work package. A task may be its own work package, in which case it will have both work package identification information (***<wpidinfo>***) and initial setup (***<wpinfo>***) requirements specific to the task. Current DTD structure allows for more than one task to be included in a work package. Though the recommendation of this handbook is to make each task its own work package. 32.1.1 discusses this concept.

33.1.3 Task content model. The DTD does not contain a specific element named ***<task>***. However, there are several elements that use the same content model to develop task related data.

a. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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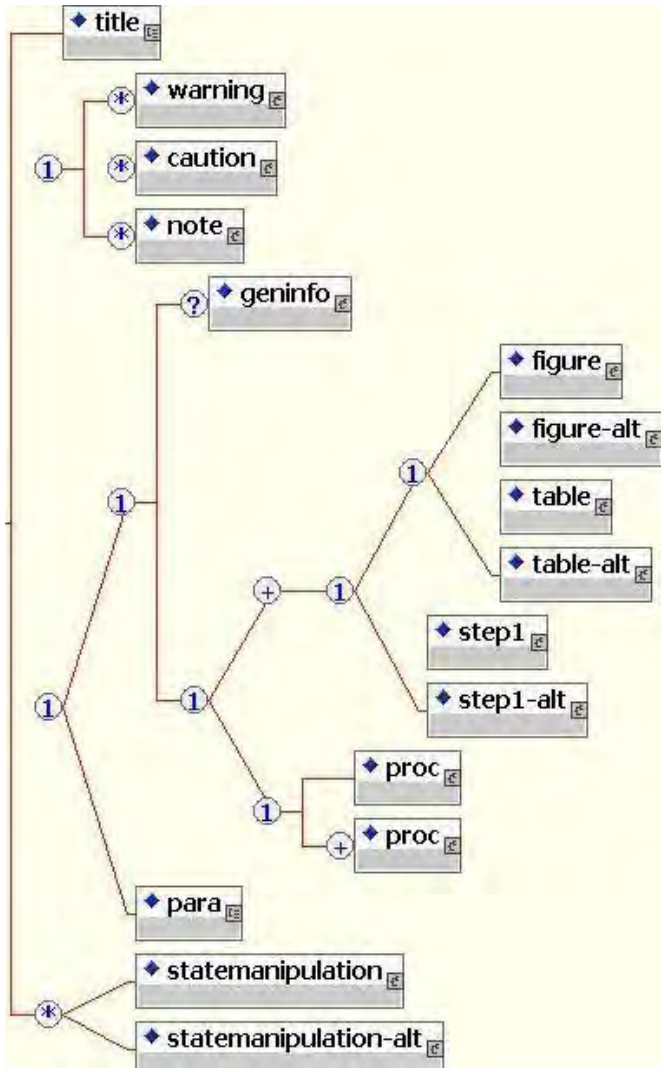

FIGURE 170. Typical task DTD hierarchy.

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b. DTD fragment:

```
(title, (warning*, caution*, note*), ((geninfo?, ((((figure | figure-alt |
table | table-alt) | step1 | step1-alt))+ | (proc, proc+))) | para),
(statemanipulation | statemanipulation-alt)*)
```

The attributes for a typical task are:

```
hcp  (yes | no )  'no'
esd  (yes | no )  'no'
frame             (yes | no )  'yes'
tocentry          (0 | 3 | 4 | 5 )  '0'
date-time-stamp   (date | time | date-time )  #IMPLIED
applicable IDREFS  #IMPLIED
 assocfig          IDREFS                          #IMPLIED
 changeref         IDREFS                          #IMPLIED
 comment           CDATA                           #IMPLIED
 delchlvl          CDATA                           #IMPLIED
 id                ID                              #IMPLIED
 idref             IDREFS                          #IMPLIED
 inschlvl          CDATA                           #IMPLIED
 security          (uc | fouo |c | s | ts)         #IMPLIED
 skilltrk          CDATA                           #IMPLIED
```

c. Common attribute specifics for a typical task element are:

(1) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(2) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(3) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(4) **tocentry** – Table of contents entry (optional). This is a list of values between zero (0) and a number between one (1) and five (5) that indicates
  (a) If the current element is to appear in the table of contents, a "0" indicates that the information will not be included in the TOC.
  (b) The level the current element will appear in an indented TOC, usually a TOC prepared for an IETM.

  In a typical task, these values are 0, 3, 4, and 5. TOC levels 1 and 2 are used for chapters and work packages.
(5) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(6) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(7) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(8) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(9) **comment** – Change information (optional) (see 49.2.9).
(10) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(11) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **security** – Security classification (optional) (see 49.2.13).
(15) **skilltrk** – Skill level (optional) (see 49.2.2).

33.1.4 Task tagging examples. When tagging a task, the model allows for three possible options.

33.1.4.1 Single paragraph task. The first option is to have a task that is a single paragraph. The single paragraph may contain a link to other information, or it may be the entire task. This option may be tagged as follows:

```
<lube>
<title>Lubricate hatch hinge pins.
```


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Check the source to verify that this is the current version before use.

```
</title>
<para>When required, apply a small amount of light weight
lubricating oil on the hatch hinge pins.
</para>
</lube>
```

33.1.4.2 Task as single procedure with multiple steps. The second option is a simple task that contains more than one step. This option may be tagged as follows:

```
<lube>
<title>Lubrication instructions.
</title>
<step1>
<para>Applylight weight lubricating oil to ..
</para>
</step1>
<step1>
<para>Apply grease to ...
</para>
</step1>
<step1>
<para>Using grease gun add grease to the following fittings:
</para>
<step2>
<para>Each wheel lubrication point.
</para>
</step2>
<step2>
<para>Steering gear joint.
</para>
</step2>
</step1>
</lube>
```

33.1.4.3 Task with multiple procedures. The third method to tag a task is one where there are two or more procedures required to complete the task or show different configurations of the same task. A task such as an engine removal may have several procedures that are to be accomplished prior to the removal of the engine. Or different models may require slightly different procedures to remove an engine. Depending on the how the task is authored, these procedures may be individual tasks that are referenced, or they may be the detailed information.

33.1.5 Three option explanation. The reason for using this three pronged approach in tagging is to be able to identify specific task functions and avoid issues in presenting the data. Previous versions of the DTD allowed an uncontrolled mixture of procedures ***<proc>***, procedural steps ***<step1>*** which are contained in the ***<proc>*** model, and paragraph ***<para>*** which are contained in the ***<step1>*** model, elements. Such intermixing can present significant problems to a presentation system, causing the output to be confusing to the end user.

33.2 Procedures. Procedures are a collection of individual steps and sub-steps that may comprise an entire task or be a subset of a task. When a single procedure is authored outside of a task model, the element ***<proc>*** is normally used. The content model of the ***<proc>*** element is very similar to that of a task. The examples below show the difference between a task and a procedure tagging.

33.2.1 Procedure using a task example. Using a multiple steps

```
<flyable>
<title>
</title>
<step1>
```


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</step1>
<step1>
</step1>
</flyable>

or using two ***<proc>***s:

<flyable>
<proc>
<title>`Procedure for model A`
</title>
<step1>
</step1>
</proc>
<proc>. . .
</proc>
<proc>
<title>`Procedurefor model B`
</title>
<step1>
</step1>
</proc>
<proc>. . .
</proc>
</flyable>

33.2.2 Procedure using ***<proc>*** example.

<flyable>
<proc>
<title>
</title>
<step1>
</step1>
</proc>
<proc>. . .
</proc>
</flyable>

33.2.3 Procedure ***<proc>*** content model information.

a. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

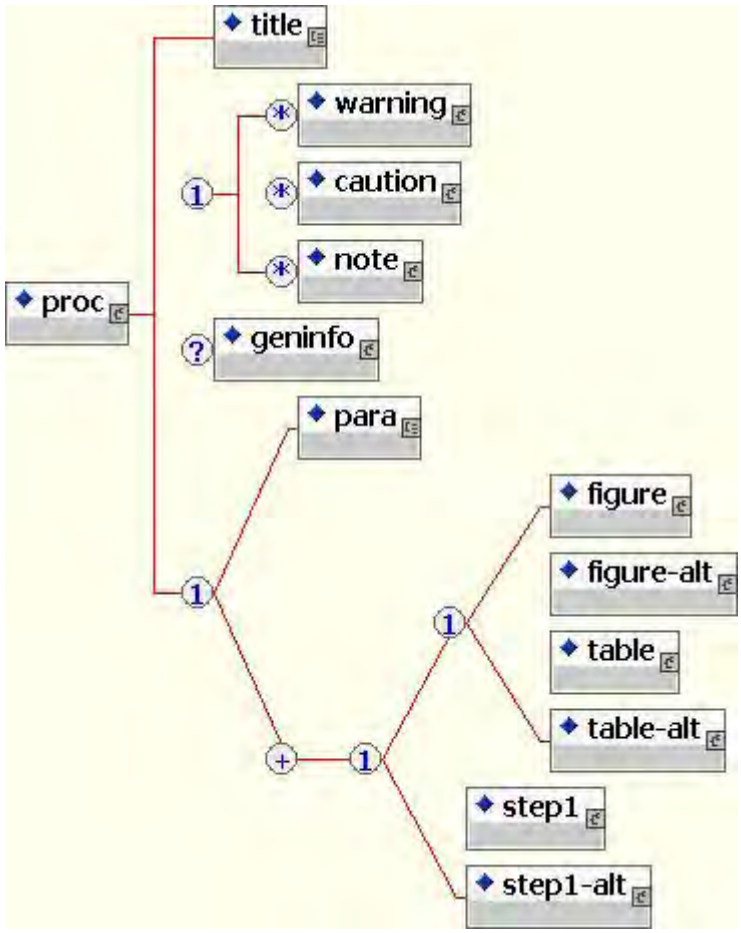

FIGURE 171. Procedure <proc> DTD hierarchy.

b. DTD fragment:

```
<!ELEMENT proc (title, %alert;, geninfo?, (para | (%step;)+))>
<!ATTLIST proc   crewmember CDATA #IMPLIED
  tocentry (0 | 3 | 4 | 5) "0"
  date-time-stamp (date | time | date-time) #IMPLIED
  frame (yes | no) "yes">
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
   esd               (yes | no)                      "no"
    hcp               (yes | no)                       "no">
```

c. Unique attributes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

(1) **crewmember** – Crewmember (optional). This attribute allows the author to identify specific functions of a task or procedure that are accomplished by different members of a crew. An example might be a gunner and loader in a gun crew.
(2) **tocentry** – Table of contents entry (optional). This is a list of values between zero (0) and a number between one (1) and five (5) that indicates
(a) If the current element is to appear in the table of contents, a "0" indicates that the information will not be included in the TOC.
(b) The level the current element will appear in an indented TOC, usually a TOC prepared for an IETM.
In a typical task, these values are 0, 3, 4, and 5. TOC levels 1 and 2 are used for chapters and work packages.
(3) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(4) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

d. Common attributes.
(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(12) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).

33.3 Steps. Steps are the key element to both tasks and procedures. They present the individual pieces of the task or procedure. The ***<step1>*** element is also used in other content models that are not typically considered a task or procedure. Exact usage of the ***<step1>*** model in this case is dependent on the element being used.

a. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

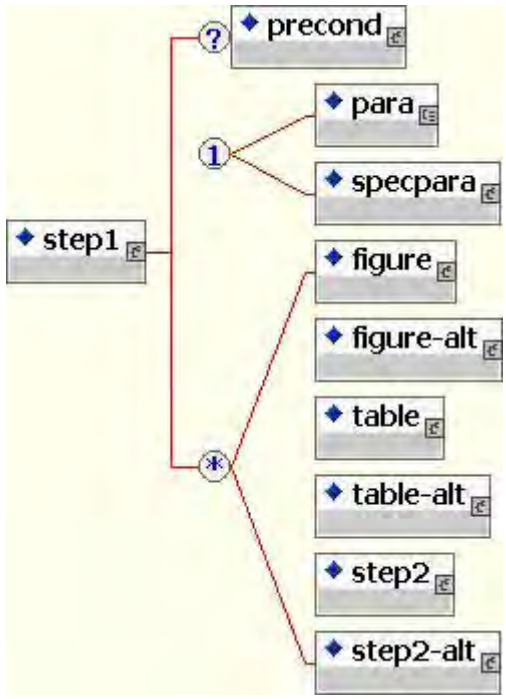

FIGURE 172. Primary-level step <step1> DTD hierarchy.

b. DTD fragment:

```
<!ELEMENT step1 (precond?, (para | specpara), (figure | figure-alt |
table | table-alt | step2 | step2-alt)*)>
<!ATTLIST step1   skilllevel (novice_expert) "novice_expert"
    label CDATA #IMPLIED
    safeflght (yes | no) #IMPLIED
    crewmember CDATA #IMPLIED
    frame (yes | no) #IMPLIED
    date-time-stamp (date | time | date-time) #IMPLIED
    warningref IDREFS #IMPLIED
    cautionref IDREFS #IMPLIED
    noteref IDREFS #IMPLIED
   esd                (yes | no)                          "no"
    hcp                (yes | no)                          "no"
    qa                 (yes | no)                          "no"
    applicable         IDREFS                              #IMPLIED
    assocfig           IDREFS                              #IMPLIED
    changeref          IDREFS                              #IMPLIED
    comment            CDATA                               #IMPLIED
    delchlvl           CDATA                               #IMPLIED
    id                 ID                                  #IMPLIED
    idref              IDREFS                              #IMPLIED
    inschlvl           CDATA                               #IMPLIED
    security           (uc | fouo |c | s | ts)            #IMPLIED
    skilltrk           CDATA                               #IMPLIED>
```

c. The attributes for ***<step1>*** are:

(1) **skilllevel** – User skill level. This is currently a single value 'novice_expert'. The intent of this attribute is to in the future allow information to be presented based on a users skill level.

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If the user is proficient, then an abbreviated set of steps could be presented. A novice would see all the steps.

(2) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(8) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(9) **inschlvl** – Insert change level (optional) (see 49.2.9).
(10) **security** – Security classification (optional) (see 49.2.13).
(11) **skilltrk** – Skill level (optional) (see 49.2.2).
(12) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(13) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(14) **label** – Label. A CDATA attribute that allows the author or system to enter various information about the ***<step1>***.
(15) **safeflght** – Safety of flight (optional). A toggle of yes or no that indicates if the ***<step1>*** is a flight safety critical step.
(16) **crewmember** – Crewmember (optional). This attribute allows the author to identify specific functions of a task or procedure that are accomplished by different members of a crew. An example might be a gunner and loader in a gun crew.
(17) **frame** – Frame (optional). A toggle of yes or no that indicates if the ***<step1>*** is to be displayed in its own frame.
(18) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(19) **warningref** – Warning reference (optional). One or more IDREF attributes that will allow the author to link the ***<step1>*** to a group of common or consolidated TM warnings.
(20) **cautionref** – Warning reference (optional). One or more IDREF attributes that will allow the author to link the ***<step1>*** to a group of common or consolidated TM cautions.
(21) **noteref** – Warning reference (optional). One or more IDREF attributes that will allow the author to link the ***<step1>*** to a group of common or consolidated TM notes.

33.3.1 Elements used in ***<step1>***. The following elements are used in ***<step1>***. They are discussed fully in their referenced paragraphs.

a. ***<precond>*** – This element is normally optional except when the ***<step1>*** is used inside a ***<step1–alt>***. See 42.1.1.1 for full details.
b. ***<para>*** – Basic paragraph of text, (see 49.1.1.5).
c. ***<specpara>*** – Paragraph that allows the inclusion of warnings, cautions, or notes. See 49.1.1.6.
d. ***<figure>*** – Allows the insertion of an illustration or other material. See 44.1.1 for full details.
e. ***<table>*** – Allows for the insertion of information in a tabular form. See Section 42 for full details.

33.3.2 Step subordination/indenture. There are six levels of step elements going from ***<step1>*** through ***<step6>***. A common concern for first time users when they see a ***<step2>*** or ***<step6>*** element is to think that a procedure or task is limited to two steps or to six steps. This is not the case. The ***<step2>*** through ***<step6>*** elements are not a sequence, but an indenture to support subordinate steps.

```
<step1>
<para>For configuration A, do the following:
</para>
<step2>
<para>Start of configuration A steps.
</para>
</step2>
</step1>
```


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```
<step1>
<para>For configurationB, do the following:
</para>
<step2>
<para>Start of configurationB steps.
</para>
</step2>
</step1>
```

33.3.3 Subordinate step content. The content models for ***<step2>*** through ***<step6>*** are identical for ***<step1>*** (see 33.3).

33.3.4 Alternate steps. The example shown in 33.3.2 may also be tagged using a step alternate element. The ***<step1–alt>*** element will be used as an example, but each of the subordinate step elements use the same approach. The alternate elements are designed to be used where the publication processing system can make decisions based on configuration or other similar data. The element used to perform this evaluation is the precondition ***<precond>*** element (see 42.1.1.1).

The example shown above could be tagged with the step-alt as follows:

```
<step1-alt>
<step1>
<precond>
<expression>
<string>configuration
</string>
<eq/>
<string>a
</string>
</expression>
</precond>
<para>Start of configuration A steps.
</para>
</step1>
<step1>
<precond>
<expression>
<string>configuration
</string>
<eq/>
<string>a
</string>
</expression>
</precond>
<para>More of configuration A steps.
</para>
</step1>
<step1>
<precond>
<expression>
<string>configuration
</string>
<ne/>
<string>a
</string>
</expression>
```


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```
</precond>
<para>Start of configuration B steps.
</para>
</step1>
<step1>
<precond>
<expression>
<string>configuration
</string>
<ne/>
<string>a
</string>
</expression>
</precond>
<para>More of configuration B steps.
</para>
</step1>
</step1-alt>
```

Note that when a step#-alt element is used, the content model is one or more of the associated step# element (see 48.2.1). When a step#-alt element is used, the optional ***<precond>*** in the parent tag is used (see Figure 172).

a. The content model for ***<step1–alt>*** and the other step alternate elements is graphically depicted below:

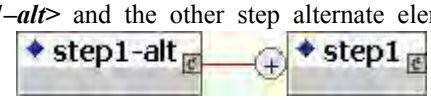

FIGURE 173. <step1–alt> content model.

b. The DTD fragment for ***<step1–alt>*** is shown below:

```
<!ELEMENT step1-alt (step1+)>
<!ATTLIST step1-alt   id ID #IMPLIED>
```

c. The attribute for ***<step1–alt>*** is a single optional **id**.

34 GENERAL INFORMATION, EQUIPMENT DESCRIPTION AND THEORY OF OPERATION CHAPTER.

34.1 General information, description information and theory of operation ***<gim>***. General information, equipment description information and theory of operation chapter is to be prepared and subdivided into individual work packages. The work packages are to provide the user with information with general requirements for the technical manual, descriptive data about the weapon system or equipment. The chapter may include a theory of operation of how the weapon system or equipment works and their related systems, subsystems, equipment, weapons replacement assemblies, and shop replacement assemblies.

a. Components are:
   (1) Chapter Title Page ***<titlepg>*** (required). The element provides the title page preceding an information chapter (see 49.1.1.1).
   (2) General Information Work Package ***<ginfowp>*** (required). The element provides general information, reference statements and standard statements that apply to the entire TM (see 34.1.1).
   (3) Equipment Description and Data Work Package ***<descwp>*** (required – one or more). The element provides the descriptive data requirements for the entire system (see 34.1.2).
   (4) Theory of Operation Work Package ***<thrywp>*** (optional – zero or more). The element provides a functional description on how the equipment and its components function and interface (see 34.1.3).

b. DTD fragment is graphically depicted.

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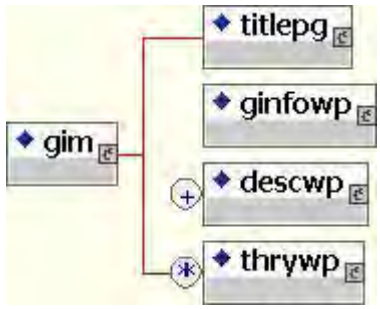

FIGURE 174. General information chapter <gim> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT gim (titlepg, ginfowp, descwp+, thrywp*)>
<!ATTLIST gim

    assocfig  IDREFS                    #IMPLIED
    changeref IDREFS                    #IMPLIED
    chap-toc  (yes | no )               "yes"
    chngno    CDATA                     #REQUIRED
    comment   CDATA                     #IMPLIED
    delchlvl  CDATA                     #IMPLIED
    frame     (yes | no )               "yes"
    idref     IDREFS                    #IMPLIED
    inschlvl  CDATA                     #IMPLIED
    revno     CDATA                     #REQUIRED
    security  (uc | fouo | c | s | ts )  #IMPLIED
    skilltrk  CDATA                     #IMPLIED
    tocentry  (0 | 1 | 2)                "1">
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **chap-toc** – Create a chapter work package table of contents (optional). Select either "**yes**" or "**no**" with a default value "**yes**".
(4) **chngno** – Change number (required) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(8) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(9) **inschlvl** – Insert change level (optional) (see 49.2.9).
(10) **revno** – Revision number (required) (see 49.2.9).
(11) **security** – Security classification (optional) (see 49.2.13).
(12) **skilltrk** – Skill level (optional) (see 49.2.2).
(13) **tocentry** – In the TM table of contents, the attribute indicates the indenture level. The possible selections are "**0**" (do not include), "**1**" (the highest level or 1st indenture level), or "**2**" (the second indentrue level) (default value is "**1**").

34.1.1 General information work package ***<ginfowp>***. All general information, reference statements and standard statements are contained within this work package. Only one general information work package is allowed.

a. The components are:

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(1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end user (see 32.3.2).
(2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).
(3) Scope ***<scope>*** (required). The element provides a brief statement of what is covered in the technical manual and also includes the following as applicable:
  (a) Type of manual.
  (b) Model number(s) and equipment name(s).
  (c) Purpose of equipment
  (d) Special inclusions in the manual, such as drill procedures or on-vehicle loading plans. (see 49.1.4.23).
(4) Maintenance Forms, Records, And Report ***<mfrr>*** (required) (see 34.1.1.1).
(5) Reporting Equipment Improvement Recommendations ***<eir>*** (required) (see 34.1.1.2).
(6) Hand Receipt (HR) Manuals Statement ***<handreceipt>*** (optional)(see 34.1.1.3).
(7) Corrosion Prevention And Control Data ***<cpcdata>*** (optional)(see 34.1.1.4).
(8) Ozone Depleting Substances ***<odsdata>*** (optional) (see 34.1.1.5).
(9) Destruction Of Army Materiel To Prevent Enemy Use ***<destructmat>*** (optional) (see 34.1.1.6).
(10) Preparation For Storage Or Shipment References ***<pssref>*** (optional)(see 34.1.1.7).
(11) Warranty Reference ***<wrntyref>*** (optional) (see 34.1.1.8).
(12) Nomenclature Cross-Reference List ***<nomenreflist>*** (optional) (see 34.1.1.9).
(13) List Of Abbreviation/Acronyms ***<loa>*** (optional) (see 34.1.1.10).
(14) Quality Assurance Information ***<qainfo>*** (optional) (see 34.1.1.11).
(15) Quality Of Material ***<qual.mat.info>*** (optional) (see 34.1.1.12).
(16) Safty, Care, And Handling (***<sftyinfo>*** (optional) (see 34.1.1.13).
(17) Nuclear Hardness ***<hcp>*** (optional) (see 34.1.1.14).
(18) Calibration Reference ***<calref>*** (optional)(see 34.1.1.15).
(19) Engineering Change Proposal ***<ecp>*** (optional)(see 34.1.1.16).
(20) Modification Work Order Statement ***<modification>*** (optional) (see 34.1.1.17).
(21) Deviations And Exceptions ***<deviation>*** (optional) (see 34.1.1.18).
(22) Mobilization Requirements ***<mobreq>*** (optional) (see 34.1.1.19).
(23) Flight Safety Critical Aircraft Parts Requirement Statement ***<fscapreq>*** (optional). (see 34.1.1.20).
(24) Cost Considerations Statement ***<cost>*** (optional) (see 34.1.1.21).
(25) Copyright Information ***<copyrt>*** (optional) (see 34.1.1.22).
(26) Supporting Information For Repair Parts, Special Tools, And Support Equipment ***<supdata>*** (optional) (see 34.1.1.23).

b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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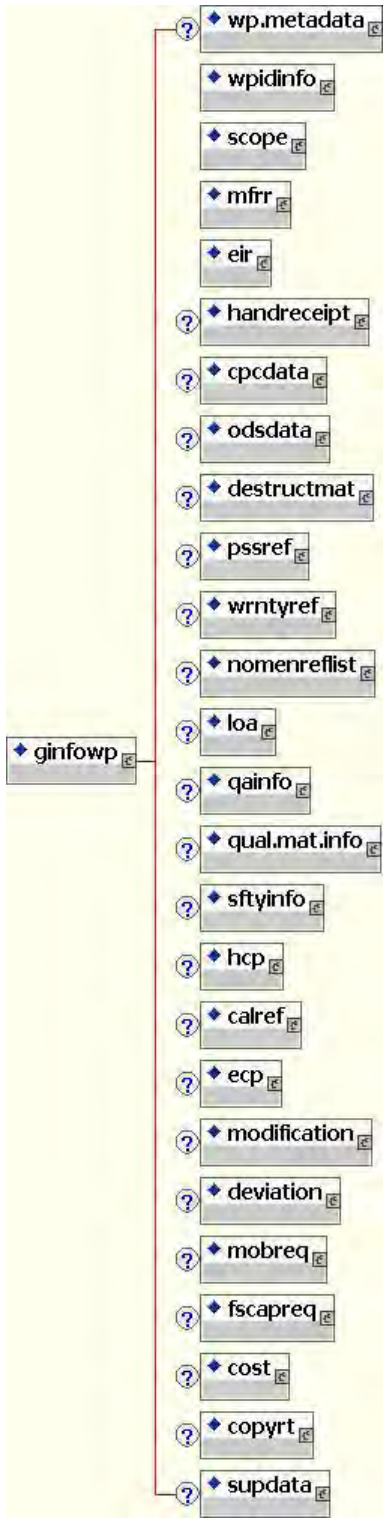

FIGURE 175. Standard general information work package <ginfowp> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT ginfowp (wp.metadata?, wpidinfo, scope, mfrr, eir,
handreceipt?, cpcdata?, odsdata?, destructmat?, pssref?,
wrntyref?, nomenreflist?, loa?, qainfo?, qual.mat.info?,
sftyinfo?, hcp?, calref?, ecp?, modification?, deviation?,
mobreq?, fscapreq?, cost?, copyrt?, supdata?)>
<!ATTLIST ginfowp
    airforce         (yes | no )               "no"
    army             (yes | no )               "no"
    assocfig         IDREFS                     #IMPLIED
    changeref        IDREFS                     #IMPLIED
    comment          CDATA                      #IMPLIED
    crewmember       CDATA                      #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                      #IMPLIED
    deletewp         (yes | no )               "no"
    fgc              CDATA                      #IMPLIED
    frame            (yes | no )               "yes"
    idref            IDREFS                     #IMPLIED
    inschlvl         CDATA                      #IMPLIED
    insertwp         CDATA                      #IMPLIED
    lsa-id           CDATA                      #IMPLIED
    marines          (yes | no )               "no"
    navy             (yes | no )               "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                      #IMPLIED
    tocentry         (2 | 3 | 4 | 5)           "2"
    wpno             ID                         #REQUIRED
    wpseq            CDATA                      #IMPLIED>
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

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(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

e. DMWR/NMWR specific tags are used only in a DMWR/NMWR General Information Work Package: quality assurance ***<qainfo>***, engineering change proposal ***<ecp>***, deviations and exceptions ***<deviation>***, modification list ***<modification>*** ***<mobreq>*** and cost considerations***<cost>***.
f. Aviation specific tags are: quality assurance ***<qainfo>***, quality of material ***<qual.mat.info>***, and flight safety critical aircraft parts ***<fscapreq>***.

34.1.1.1 Maintenance forms, records, and report statement. ***<mfrr>***. A General information package requires a ***<mfrr>*** statement. There are specific maintenance forms, records, and report statements for an Army only TM, Marines only TM, multi-service TM and an Army conventional and chemical ammunition TM. General paragraphs may be utilized or boilerplates that are accessible through the DTD per MIL–STD–40051–1/–2 and can be referenced using one of the applicable general entities below. See Section 50 for further information on boilerplates.

- Army - Maintenance forms, records and reports (MFRR) statement (***&ginfowp.mfrr-army;***).
- USMC - Maintenance forms, records and reports (MFRR) statement (***&ginfowp.mfrr-usmc;***).
- Multi-Service Maintenance forms, records and reports (MFRR) statement (***&ginfowp.mfrr-multiservice;***).
- Single-Service Maintenance forms, records and reports (MFRR) statement (***&ginfowp.mfrr-oneservice;***).

a. The components are:
(1) Title ***<title>*** (required). The element enables IAW MIL–STD–40051–1/–2 the title "MAINTENANCE FORMS, RECORDS, AND REPORTS" to be displayed for the ***<mfrr>*** statement (see 49.1.1.3).
(2) General Paragraphs ***<para>*** (required)(see 49.1.1.5) to insert the required statement **OR** one or more MFRR service specific paragraph(s) ***<mfrr.para>*** (see 34.1.1.1.1).

b. DTD fragment is graphically depicted.

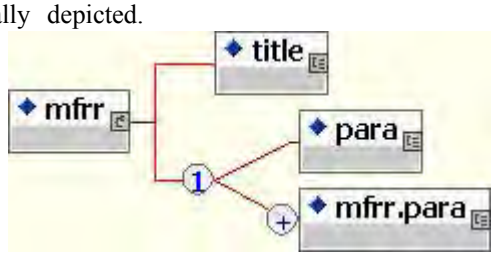

FIGURE 176. Maintenance forms, records, and report statement <mfrr> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT mfrr (title, (para | mfrr.para+))>
<!ATTLIST mfrr
     frame           (yes | no )                    'no'

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **frame** - Indicates to the IETM system the authors intended Frame break. The default value is set to 'no'.
(2) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(3) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(4) **comment** – Reason for change information. (optional) (See 49.2.9.)
(5) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(6) **id** – Unique identifier. (optional) (See 49.2.6.)
(7) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(8) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(9) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(10) **security** – Security classification. (optional) (See 49.2.13.)

34.1.1.1.1 MFRR service specific ***<mfrr.para>***. The element ***<mfrr.para>*** states by service the maintenance forms, records, and reports that are referenced. The required attribute **service** provides the correct data through the selected value of **service**. The selected value enables the correct boilerplate to be inserted if the user is using the boilerplate application. See Section 50 for further information on boilerplates.

a. The components are:

(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).

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(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).
(24) Internet address (e–mail or homepage) – ***<internet>*** (see 49.1.4.1.7).
(25) Proponent or organiztion address – ***<proponent>*** (see 49.1.4.22).
(26) Telephone number – ***<phone>*** (see 49.1.4.1.6).
(27) Instruction data plate – ***<instructplt>*** (see 49.1.4.13).

b. DTD fragment is graphically depicted.

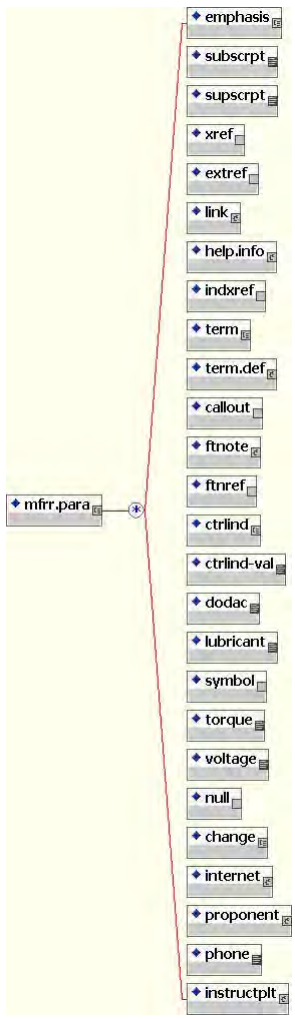

FIGURE 177. MFRR service specific <mfrr.para> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT mfrr.para (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref |
link | help.info | indxref | term | term.def | callout | ftnote | ftnref |
ctrlind | ctrlind-val | dodac | lubricant | symbol | torque | voltage | null |
change | internet | proponent | phone | instructplt)*>
<!ATTLIST mfrr.para
     service          (army | af | navy | marines )      #REQUIRED
    assocfig          IDREFS                               #IMPLIED
    changeref         IDREFS                               #IMPLIED
    comment           CDATA                                #IMPLIED
    delchlvl          CDATA                                #IMPLIED
    idref             IDREFS                               #IMPLIED
    inschlvl          CDATA                                #IMPLIED
    security          (uc | fouo |c | s | ts)            #IMPLIED
    skilltrk          CDATA                                #IMPLIED
```

d. Unique attributes:
   (1) **service** – Identifies the unique branch of service information.

e. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

34.1.1.2 Reporting equipment improvement recommendations ***<eir>***. The element ***<eir>*** provides a statement on how to report an equipment improvement recommendation. The selected value of the required attribute **service** provides the type of service branch for the TM. The selected value enables the correct boilerplate to be inserted if the user is using the boilerplate application. See Section 50 for further information on boilerplates.

a. The components are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
   (3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
   (4) Select one of the following information types:
      (a) Narrative paragraphs
         1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
         2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
      (b) Narrative paragraphs with descriptive or narrative titled text
         1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
         2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
         3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
      (c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD fragment is graphically depicted.

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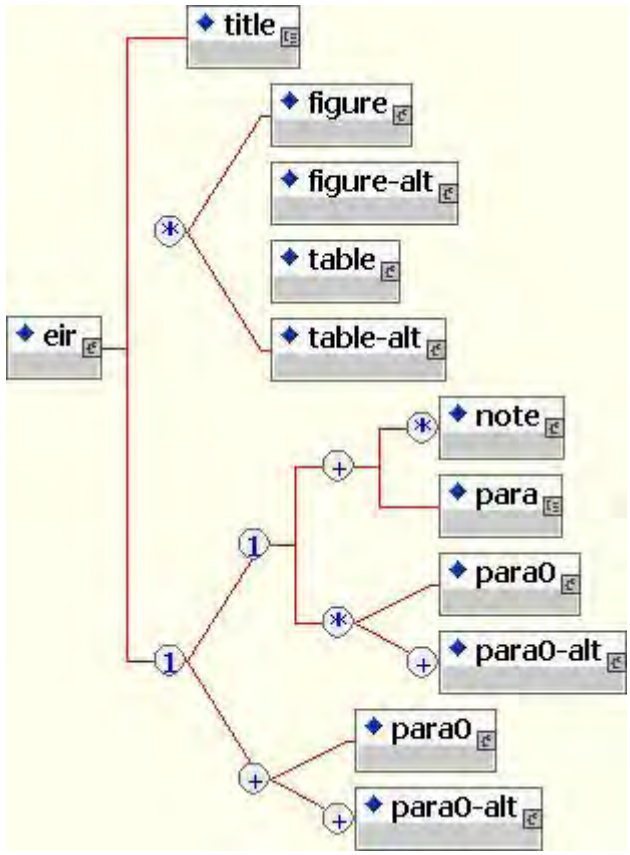

FIGURE 178. Reporting equipment improvement recommendations <eir> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT eir (title, (figure | figure-alt | table | table-alt)*,
(((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST eir
     service          (multi | army | marines )        'army'

    assocfig          IDREFS                             #IMPLIED
    changeref         IDREFS                             #IMPLIED
    comment           CDATA                              #IMPLIED
    delchlvl          CDATA                              #IMPLIED
    id                ID                                 #IMPLIED
    idref             IDREFS                             #IMPLIED
    inschlvl          CDATA                              #IMPLIED
    security          (uc | fouo |c | s | ts)           #IMPLIED
    skilltrk          CDATA                              #IMPLIED
```

d. Common attributes:

(1) **service** – Identifies the unique service information.
(2) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(3) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(4) **comment** – Reason for change information. (optional) (See 49.2.9.)
(5) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(6) **id** – Unique identifier. (optional) (See 49.2.6.)
(7) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

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(8) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(9) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(10) **security** – Security classification. (optional) (See 49.2.13.)

34.1.1.3 Hand receipt (HR) manuals statement ***<handreceipt>***. The element ***<handreceipt>*** provides the hand receipt manual statement that identifies and references the companion document. This statement is a boilerplate and can be referenced using the general entity ***&ginfowp.handreceipt;***. See Section 50 for further information on boilerplates. (Used in a operator's/unit maintenance manual.)

a. The components are:
(1) Contains paragraphs of text that may be grouped into sections or subsections.
(a) Title ***<title>*** (required) (see 49.1.1.3).
(b) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(c) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(d) Select one of the following information types:
1 Narrative paragraphs
a Note ***<note>*** (optional – zero or more) (see 41.1.3).
b Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
2 Narrative paragraphs with descriptive or narrative titled text
a Note ***<note>*** (optional – zero or more) (see 41.1.3).
b Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
c Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD fragment

```
<!ELEMENT handreceipt (title, (figure | figure-alt | table | table-alt)*,
(((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST handreceipt
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

c. Common attributes:
(1) **service** – Identifies the unique service information.
(2) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(3) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(4) **comment** – Reason for change information. (optional) (See 49.2.9.)
(5) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(6) **id** – Unique identifier. (optional) (See 49.2.6.)
(7) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(8) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(9) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(10) **security** – Security classification. (optional) (See 49.2.13.)

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34.1.1.4 Corrosion prevention and control data ***<cpcdata>***. The element ***<cpcdata>*** provides a manner in which a corrosion problem is to be reported is specified. This statement is a boilerplate and can be referenced using the general entity ***&ginfowp.cpcdata;***. See Section 50 for further information on boilerplates.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.5 Ozone depleting substances statement ***<odsdata>***. The element ***<odsdata>*** provides a listing of the ozone depleting substances that are prohibited.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.6 Destruction of materiel statement ***<destructmant>***. The element ***<destructmant>*** provides a standard paragraph references the appropriate TM(s) on destroying materiel to prevent enemy use.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.7 Preparation for storage or shipment ***<pssref>***. The element ***<pssref>*** provides references to the preparation for storage or shipment procedures, including packaging and administrative storage that are entered.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.8 Warranty reference statement ***<wrntyref>***. The element ***<wrntyref>*** provides when the TM covers equipment that is under warranty and a Warranty Technical Bulletin (WTB) is published, it is referenced. Otherwise a paragraph about the warranty is included. This statement is a boilerplate and can be referenced using the general entity ***&ginfowp.wrntyref;***. See Section 50 for further information on boilerplates.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.9 Nomenclature cross-reference list ***<nomenreflist>***. The element ***<nomenreflist>*** provides any unofficial nomenclature approved by the contracting activity that is included in the nomenclature cross-reference list.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.10 List of abbreviation/acronyms ***<loa>***. The element ***<loa>*** provides a list of all abbreviations, acronyms, signs, or symbols used in the TM.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.11 Quality assurance information statement ***<qainfo>***. The element ***<qainfo>*** provides either a reference to QA TMs or the appropriate general QA information. **(DMWR/NMWR and aviation only.)**

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.12 Quality of material statement ***<qual.mat.info>***. The element ***<qual.mat.info>*** provides statement defining the material quality requirements that are used. This statement is a boilerplate and can be referenced using the general entity ***&ginfowp.qual.mat.info;***. See Section 50 for further information on boilerplates. **(Used Field/Aviation and above only.)**.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.13 Safety, care, and handling information statement ***<sftyinfo>***. The element ***<sftyinfo>*** provides general precautions and safety regulations that are included for ammunitions TMs, equipment with radioactive parts or components, and electrical/electronic parts.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.14 Nuclear hardness statement ***<hcp>***. The element ***<hcp>*** provides when the equipment or any component has nuclear hardness survivability requirements which are to be identified. The method in which nuclear hardness (HCP) is to be identified is described in this portion of the manual. This statement is a boilerplate and can be referenced using the general entity ***&ginfowp.hcp;***. See Section 50 for further information on boilerplates.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.15 Calibration reference statement ***<calref>***. The element ***<calref>*** provides any equipment requiring calibration to be listed with a reference to the publication containing the correct calibration procedure.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.16 Engineering change proposals statement ***<ecp>***. The element ***<ecp>*** provides the method for submitting an engineering change proposal that is entered. This statement is a boilerplate and can be

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referenced using the general entity ***&ginfowp.ecp;***. See Section 50 for further information on boilerplates. The attribute **MWO** containing the value "yes or no" is provided to identify if the Modification Work Order is incorporated into the work required for the DMWR/NMWR. **(Used for DMWR/NMWR only.)**

a. See the components of ***<handreceipt>*** 34.1.1.3.a..)

b. DTD fragment:

```
<!ELEMENT ecp  (title, (figure | figure-alt | table | table-alt)*,
(((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST ecp
    mwo          (yes | no)               #IMPLIED

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

c. Common attributes:

(1) **mwo** – Identifies if Modification Work Order (MWO) are incorporated into the work required for the DMWR/NMWR.
(2) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(3) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(4) **comment** – Reason for change information. (optional) (See 49.2.9.)
(5) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(6) **id** – Unique identifier. (optional) (See 49.2.6.)
(7) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(8) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(9) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(10) **security** – Security classification. (optional) (See 49.2.13.)

34.1.1.17 Modification work order statement ***<modification>***. The element ***<modification>*** may provide a general information statement about Modification Work Orders (MWO) that are to be accomplished during overhaul. The ***<modification>*** element will list all current ECPs or MWOs that have been performed on the equipment. **(Used for DMWR/NMWR only.)**

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.18 Deviations and exceptions statement ***<deviation>***. The element ***<deviation>*** provides the method for requesting any deviations and/or exceptions to a DMWR/NMWR that is entered. This statement is a boilerplate and can be referenced using the general entity ***&ginfowp.deviation;***. See Section 50 for further information on boilerplates. **(Used for DMWR/NMWR only.)**.

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.19 Mobilization requirements statement ***<mobreq>***. The element ***<mobreq>*** provides a standard statement regarding DMWR/NMWR mobilization requirements. This statement is a boilerplate and can be referenced using the general entity ***&ginfowp.mobreq;***. See Section 50 for further information on boilerplates. **(Used for DMWR/NMWR only.)**

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.20 Flight safety critical aircraft parts requirement statement ***<fscapreq>***. The element ***<fscapreq>*** provides a standard statement that is included when defining flight safety critical aircraft parts. This statement is a boilerplate and can be referenced using the general entity ***&ginfowp.fscapreq;***. See Section 50 for further information on boilerplates. **(Used for Aircraft only.)**

a. See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

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34.1.1.21 Cost considerations statement ***<cost>***. The element ***<cost>*** provides a standard statement that is included when defining cost considerations. This statement is a boilerplate and can be referenced using the general entity ***&ginfowp.cost;***. See Section 50 for further information on boilerplates. **(Used for DMWR/NMWR only.)**.

a. (See the components, DTD fragment and attributes of ***<handreceipt>***34.1.1.3.a.

34.1.1.22 Copyright statement ***<copyrt>***. The element ***<copyrt>*** provides information on a copyright credit line that is included as the last paragraph in the general information work package for any TM that contains copyrighted data. If more than one copyrighted document is used, a statement is required for each document. The copy right credit line may be omitted if the copyright owner has allowed, in writing, the use of the copyrighted material without the statement.

a. (See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.23 Supporting information for repair parts ***<supdata>***. The element ***<supdata>*** provides a reference to the common tools and equipment; special tools, TMDE, and support equipment; and the repair parts entered within this supporting information element. This statement is a boilerplate and can be referenced using the general entity ***&ginfowp.supdata-tools;***. See Section 50 for further information on boilerplates.

a. (See the components, DTD fragment and attributes of ***<handreceipt>*** 34.1.1.3.a.

34.1.1.24 Boilerplates. MIL–STD–40051–1/–2 contains standard statements that may occur in numerous documents and should appear in the same wording. MIL–STD–2361 DTD provides boilerplates that are a type of replacement text for these standard statements. Boilerplates are defined in general entities contain in the work package or document source. By using boilerplates authors can reduce text entry time and errors. Certain entities for ***<ginfowp>*** are either page-based only, frame-based only or could be used in either one. Refer to Table for a list of general entities used in the general information work package.

TABLE VII. General information list.

Description	**Boilerplate Entity**	**Selectable Entity to Set**	**Selectable Entity to Edit**
Army - Maintenance forms, records and reports (MFRR) statement	***&ginfowp.mfrr-army;***		Not Applicable
Frame-based		***<!ENTITY % frame-base "INCLUDE"> <!ENTITY % page-base "IGNORE">***	
Page-based		***<!ENTITY % frame-base "IGNORE"> <!ENTITY % page-base "INCLUDE">***	
Ammunition TM		***<!ENTITY % ammo-tm "INCLUDE">***	
Non-Ammunition TM		***<!ENTITY % ammo-tm "IGNORE">***	

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Table VII General information list – Continued.

Description	Boilerplate Entity	Selectable Entity to Set	Selectable Entity to Edit
Multiple Service TM		*<!ENTITY % multi-tm "INCLUDE"> <!ENTITY % single-tm "IGNORE">*	
Single Service TM		*<!ENTITY % multi-tm "IGNORE"> <!ENTITY % single-tm "INCLUDE">*	
USMC - Maintenance forms, records and reports (MFRR) statement	*&ginfowp.mfrr-usmc;*	*<!ENTITY % usmc-tm " INCLUDE">*	
Multi-Service Maintenance forms, records and reports (MFRR) statement	*&ginfowp.mfrr-multiservice;*		
ARMY		*<!ENTITY % army-tm "INCLUDE">*	
Non ARMY		*<!ENTITY % army-tm "IGNORE">*	
U.S. Navy		*<!ENTITY % usn-tm "INCLUDE">*	
Non U.S. Navy		*<!ENTITY % usn-tm "IGNORE">*	
U.S. Air Force		*<!ENTITY % usaf-tm "INCLUDE">*	
Non U.S. Air Force		*<!ENTITY % usaf-tm "IGNORE">*	
U. S. Marines		*<!ENTITY % usmc-tm "INCLUDE">*	
Non U.S. Marines		*<!ENTITY % usmc-tm "IGNORE">*	

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Table VII General information list – Continued.

Description	Boilerplate Entity	Selectable Entity to Set	Selectable Entity to Edit
Single-Service Maintenance forms, records and reports (MFRR) statement	***&ginfowp.mfrr-oneservice;***	Select the settings for either Army or Marines. See above examples.	Not Applicable
Reporting equipment improvement recommendations (EIR) statement	***&ginfowp.eir;***	Select the settings for either Marines or non Marines. See above example.	Not Applicable
Hand receipt (HR) manual statement	***&ginfowp.handreceipt;***	Select the settings for either Page-based or Frame-based. See above example.	Not Applicable
Corrosion prevention and control (CPC) statement	***&ginfowp.cpcdata;***	Not Applicable	Not Applicable
Warranty information reference statement	***&ginfowp.wrntyref;***	Select the settings for either Page-based or Frame-based. See above example.	<!ENTITY ginfowp.wrntyref-time 'INSERT THE WARRANTY PERIOD MILEAGE OR TIME FRAME'>
		Not Applicable	<!ENTITY short.end.item.name "INSERT THE SHORT END ITEM NAME" >
Quality of material information statement	***&ginfowp.qual.mat.info;***	Not Applicable	<!ENTITY ginfowp.qual.mat.info-tm '<extref docno="INSERT THE TM OR IETM NUMBER"/>'>
Nuclear Hardness statement	***&ginfowp.hcp;***	Not Applicable	Not Applicable

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Table VII General information list – Continued.

Description	Boilerplate Entity	Selectable Entity to Set	Selectable Entity to Edit
Engineering Change Proposals (ECP) statement	***&ginfowp.ecp;***	Select the settings for either Page-based or Frame-based. See above example.	<!ENTITY ginfowp.ecp-address '<proponent><name>RESPONSIBLE COMMAND OR ACTIVITY PROPONENT NAME</name><address><servnomen>OPTIONAL SERVICE NOMENCLATURE</servnomen><street>0 OR MORE STREET INFORMATION </street><city>REQUIRED CITY</city><state>REQUIRED STATE<state><zip>OPTIONAL ZIP CODE</zip><country> OPTIONAL COUNTRY</country></address> </proponent>
Modification list statement	***&ginfowp.modification;***	Select the settings for either DMWR or NMWR. See below for example.	
Non Depot		***<!ENTITY % dmwr "IGNORE"> <!ENTITY % nmwr "IGNORE">***	
Depot		***<!ENTITY % dmwr "INCLUDE"> <!ENTITY % nmwr "IGNORE">***	
NMWR		***<!ENTITY % dmwr "IGNORE"> <!ENTITY % nmwr "INCLUDE">***	
Deviations and Exceptions statement	***&ginfowp.deviation;***	Select the settings for either DMWR or NMWR. See above for example.	

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Table VII General information list – Continued.

Description	Boilerplate Entity	Selectable Entity to Set	Selectable Entity to Edit
Mobilization Requirements statement	***&ginfowp.mobreq;***	Select the settings for either DMWR or NMWR. See above for example.	<!ENTITY ginfowp.mobreq-wp '<xref wpid="INSERT_THE_APPROPRIATE_WORK_PACKAGE_ID"/>'> <!ENTITY short.end.item.name "INSERT THE SHORT END ITEM NAME" >
Flight Safety Critical Aircraft Parts Statement	***&ginfowp.fscap;***	If frame-based, select frame-based turn on. See example above.	
Cost Considerations Statement	***&ginfowp.cost;***	Select the settings for either DMWR or NMWR. See above for example.	Not Applicable
Common Tools And Equipment Statement	***&ginfowp.supdata-tools;***		
Parts Information Work Package Reference	***&ginfowp.supdata-partlist.wp;***	Select the settings for either Page-based or Frame-based. See above example.	<!ENTITY ginfowp.supdata-partlist.wpref '<xref wpid="INSERT_THE_PART_LIST_WORK_PACKAGE_ID"/>'>
Parts Information TM (Page ONLY)	***&ginfowp.supdata-partlist.tm;***		<!ENTITY ginfowp.supdata-partlist.tmref '<extref docno="REPLACE WITH RPSTL TM NUMBER">'>

34.1.1.25 XML document instance fragment and output for ***<ginfowp>***. The XML instance and its stylesheet output for a General Information Work Package is provided below.

a. Example of an XML document instance fragment for selectable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % ammo-tm "IGNORE">
<!ENTITY % usmc-tm "IGNORE">
<!ENTITY % multi-tm "IGNORE">
<!ENTITY % single-tm"INCLUDE">
<!ENTITY % army-tm "INCLUDE">
<!ENTITY % usn-tm "IGNORE">
<!ENTITY % usaf-tm "IGNORE">
<!ENTITY % usmc-tm "IGNORE">
<!ENTITY % non-usmc-tm "IGNORE">
```


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```
<!ENTITY % class-tm "IGNORE">
<!ENTITY % unclass-tm "INCLUDE">
```

b. Example of an XML document instance fragment using boilerplate entities for Maintenance Forms, Records, And Reports <mfrr>, Reporting Equipment Improvement Recommendations (EIR) <eir> and Hand Receipt (HR) Manuals ***<handreceipt>*** in the XML instance for ***<ginfowp>***.

```
<ginfowp wpno="gXXXXX-11-XXXX-XXX" tocentry="2" frame="no" "army="yes"
airforce="no" navy="no" marines="no" wpseq="0001" deletewp="no">
<wpidinfo>
<maintlvl level="operator"/>
<title>GENERAL INFORMATION WORK PACKAGE
</title>
</wpidinfo>
<scope inschlvl="TM 750-244-6" frame="no">
<title>SCOPE
</title>
<para hcp="no" esd="no">This manual tells how to operate and maintain
the hulls of the M2A3 and M3A3. <extref docno="TM 9-2350-294-10-2">
tells how to operate and maintain the turret.
</para>
</scope>
<mfrr frame="no">
<mfrr.para service="army">&ginfowp.mfrr.army;
</mfrr.para>
</mfrr>
<eir service="army">&ginfowp.eir;
</eir>
<handreceipt>&ginfowp.handreceipt;
</handreceipt>
<destructmat>
<title>DESTRUCTION OF ARMY MATERIEL TO PREVENT ENEMY USE
</title>
<para>The following manuals tell you how and when to destroy
Army materiel to prevent enemy use:
<randlist bullet="no">
<item>
<extref docno="TM 750-244-2"/>
</item>
<item>
<extref docno="TM 750-244-5-1"/>
</item>
<item>
<extref docno="TM 750-244-6"/>
</item>
</randlist>
</para>
</destructmat>
<nomenreflist>
<title>Nomenclature cross-reference list
</title>
<para>This listing includes nomenclature cross refer-
ences used in this manual:
<deflist>
<term.def>
<term>Brush guard
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</term>
<def>
<para>Duck core rubber sheet
</para>
</def>
</term.def>
<term.def>
<term>CVC helmet
</term>
<def>
<para>DH 132 helmet
</para>
</def>
</term.def>
<term.def>
<term>Dipstick
</term>
<def>
<para>Liquid measure gage rod
</para>
</def>
</term.def>
<term.def>
<term>Firing port weapon
</term>
<def>
<para>M231 5.56mm sub machine gun
</para>
</def>
</term.def>
</deflist>
</para>
</nomenreflist>
<loa>
<title>List of abbreviations/acronyms
</title>
<para>Many abbreviations are used in this manual. They are listed below. Learn what each one means. It will make your job easier.
<deflist>
<term.def>
<term>A
</term>
<def>
<para>After
</para>
</def>
</term.def>
<term.def>
<term>Ammo
</term>
<def>
<para>Ammunition<
</para>
</def>
</term.def>
<term.def>

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Check the source to verify that this is the current version before use.

```
<term>AP
</term>
<def>
<para>Armor  Piercing
</para>
</def>
</term.def>
</deflist>
</para>
</loa>

</ginfowp>
```

c. Example of an XML document instance fragment for a General Information Work Package providing the text for the Maintenance Forms, Records, And Reports Statement <mfrr>, Reporting Equipment Improvement Recommendations (EIR) Statement <eir> and Hand Receipt (HR) Manuals Statement <handreceipt> in the XML instance for ***<ginfowp>***

```
<ginfowp="gXXX3-11-XXXX-XXX" tocentry="2" frame="no" "army="no" airforce="no"
navy="no "marines="no" wpseq="0001" deletewp="no">
<wpidinfo>
<maintlvl level="operator"/>
<title>GENERAL INFORMATION WORK PACKAGE
</title>
</wpidinfo>
<scope inschlvl="TM 750-244-6" frame="no">
<title>SCOPE
</title>
<para hcp="no" esd="no">This manual tells how to operate and maintain
the hulls of the M2A3 and M3A3 <extref docno="TM 9-2350-294-10-2">
tells how to operate and maintain the turret.
</para>
</scope>
<mfrr frame="no">
<mfrr.para service="army">Department of the Army forms and procedures used
for equipment maintenance will be those prescribed by (as applicable)
<extref docno="DA PAM 750-8"/>, Functional Users Manual for the
Army Maintenance Management System (TAMMS);
<extref docno="DA PAM 738-751"/>, Functional Users Manual for the Army
Maintenance Management Systems - Aviation (TAMMS-A); or
<extref docno="AR 700-138"/>, Army Logistics Readiness and Sustainability."
</mfrr.para>
</mfrr>
<eir service="army">
<title>REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR)
</title>
<para>If your (M2A3 and M3A3) needs improvement, let us know.  Send
us an EIR. You, the user, are the only one who can tell us what you
don't like about your equipment.  Let us know why you don't like the
design or performance.  If you have Internet access, the easiest and
fastest way to report problems or suggestions is to go to
<internet show.address="yes">
<homepage protocol="https" uri="aeps.ria.army.mil/aepspublic.cfm"/>
</internet> (scroll down and choose the "Submit Quality Deficiency
Report" bar).  The Internet form lets you choose to submit an
Equipment Improvement Recommendation (EIR), a Product Quality
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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Deficiency Report (PQDR) or a Warranty Claim Action (WCA). You may also submit your information using a <extref docno="SF 368" posttext="(Product Quality Deficiency Report)"/>. You can send your SF 368 via e-mail, regular mail, or facsimile using the addresses/facsimile numbers specified in <extref docno="DA PAM 750-8" posttext="(Functional Users Manual for the Army Maintenance Management System (TAMMS)"/> We will send you a reply.
</para>
</eir>
<handreceipt>
<title>HAND RECEIPT (HR)MANUALS
</title>
<para>This manual has a companion document with a TM number followed by "-HR" (which stands for Hand Receipt). TM X-XXXX-XXX-10-HR consists of preprinted hand receipts that list end item related equipment (i.e., COEI, BII, and AAL) that must be accounted for. This IETM contains As an aid to property accountability, additional HR manuals may be requisitioned through normal publication channels.
</para>
</handreceipt>
<destructmat>
<title>DESTRUCTION OF ARMY MATERIEL TO PREVENT ENEMY USE
</title>
<para>The following manuals tell you how and when to destroy Army materiel to prevent enemy use:
<randlist bullet="no">
<item>
<extref docno="TM 750-244-2"/>
</item>
<item>
<extref docno="TM 750-244-5-1"/>
</item>
<item>
<extref docno="TM 750-244-6"/>
</item>
</randlist>
</para>
</destructmat>
<nomenreflist>
<title>Nomenclature cross-reference list
</title>
<para>This listing includes nomenclature cross refer-ences used in this manual:
<deflist>
<term.def>
<term>Brush guard
</term>
<def>
<para>Duck core rubber sheet
</para>
</def>
</term.def>
<term.def>
<term>CVC helmet
</term>
<def>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<*para*>DH 132 helmet
</*para*>
</*def*>
</*term.def*>
<*term.def*>
<*term*>Dipstick
</*term*>
<*def*>
<*para*>Liquid measure gage rod
</*para*>
</*def*>
</*term.def*>
<*term.def*>
<*term*>Firing port weapon
</*term*>
<*def*>
<*para*>M231 5.56mm submachinegun
</*para*>
</*def*>
</*term.def*>
</*deflist*>
</*para*>
</*nomenreflist*>
<*loa*>
<*title*>List of abbreviations/acronyms
</*title*>
<*para*>Many abbreviations are used in this manual. They are listed below. Learn what each one means. It will make your job easier.
<*deflist*>
<*term.def*>
<*term*>A
</*term*>
<*def*>
<*para*>After
</*para*>
</*def*>
</*term.def*>
<*term.def*>
<*term*>Ammo
</*term*>
<*def*>
<*para*>Ammunition
</*para*>
</*def*>
</*term.def*>
<*term.def*>
<*term*>AP
</*term*>
<*def*>
<*para*>Armor Piercing
</*para*>
</*def*>
</*term.def*>
</*deflist*>
</*para*>
</*loa*>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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</ginfowp>

d. Example of a stylesheet output for ***<ginfowp>***:

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Check the source to verify that this is the current version before use.

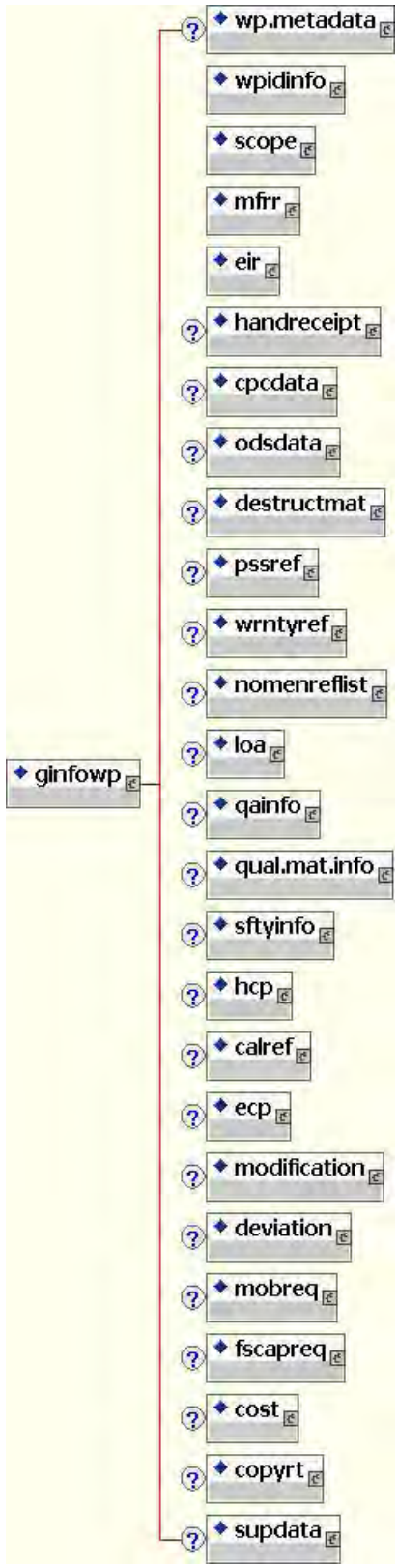

FIGURE 179. Example of a stylesheet output for a general information work package <ginfowp>..

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Check the source to verify that this is the current version before use.

34.1.2 Equipment description and data work package ***<descwp>***. The element equipment description and data work package contains the descriptive data requirements. There may be more than one equipment description and data work package in the General Information Chapter ***<gim>***.

a. The components are:
   (1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end use (see 32.3.2).
   (2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).
   (3) Equipment Characteristics, Capabilities, And Features ***<eqpinfo>*** (required – one or more). The element provides the descriptive data that contains the overall equipment description(see 34.1.2.1).
   (4) Location And Description Of Major Components ***<locdesc>*** (optional – zero or more). The element provides the descriptive data on the location and description of major components of the equipment (see 34.1.2.2).
   (5) Differences Between Models ***<eqpdiff>*** (optional) (see 34.1.2.3).
   (6) Equipment Data ***<eqpdata>*** (required). The element provides descriptive data of the equipment (see 34.1.2.4).

b. DTD fragment is graphically depicted.

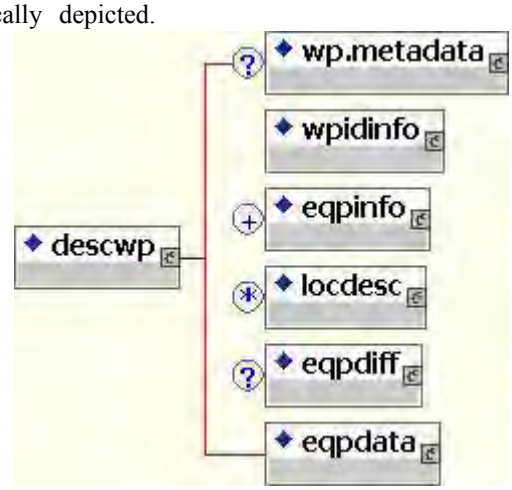

FIGURE 180. Equipment description and data work package <descwp> DTD hierarchy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT descwp (wpidinfo, eqpinfo+, locdesc*, eqpdiff?, eqpdata)>
<!ATTLIST descwp

    airforce         (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED>
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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Check the source to verify that this is the current version before use.

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

34.1.2.1 Equipment characteristics, capabilities, and features ***<eqpinfo>***. The element defines the descriptive data that contains the overall equipment description.

a. The components are:
   (1) Title ***<title>*** (required). The element provides the title for the equipment description. (see 49.1.1.3).
   (2) Equipment Description ***<eqpdesc>*** (required – one or more) (see 34.1.2.1.1).
b. DTD fragment is graphically depicted.

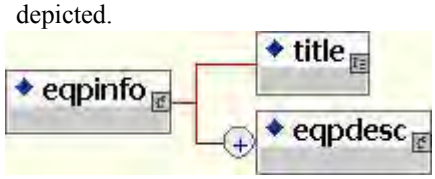

FIGURE 181. Equipment characteristics, capabilities, and features <eqpinfo> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT eqpinfo (title, eqpdesc+)>
<!ATTLIST eqpinfo

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

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34.1.2.1.1 Equipment description ***<eqpdesc>***. The element ***<eqpdesc>*** contains the general capabilities and special unique features, as well as other similar information, that will be helpful in the operation and maintenance of equipment.

a. The components of are:
   (1) Title ***<title>*** (optional) (see 49.1.1.3).
   (2) Illustration ***<figure>*** (optional – zero or more)(see 49.1.1.3).
   (3) Select one of the following information types:
      (a) Narrative paragraphs
         1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
         2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
      (b) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
   (4) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD fragment is graphically depicted.

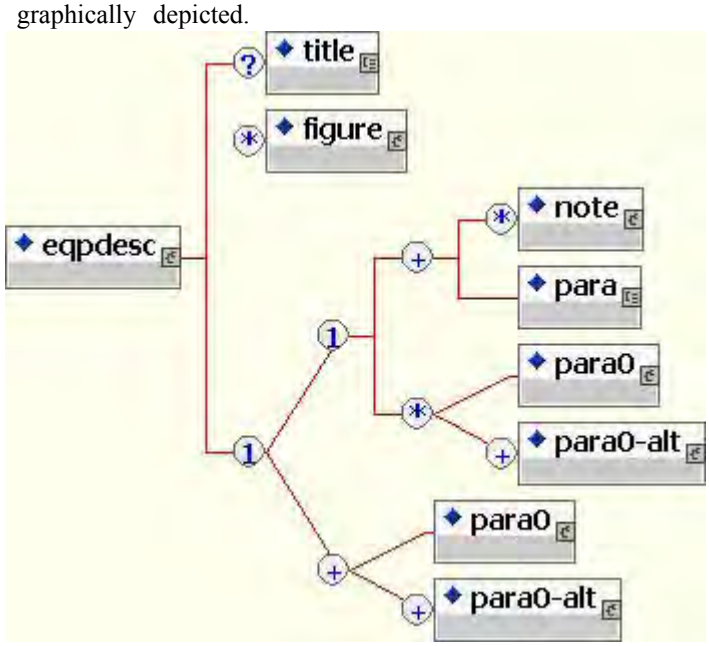

FIGURE 182. Equipment description <eqpdesc> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT eqpdesc (title?, figure*, (((note*, para)+, (para0 | para0-alt+)*)
| (para0 | para0-alt+)+))>
<!ATTLIST eqpdesc
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

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(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

34.1.2.2 Location and description of major components ***<locdesc>***. Descriptive data on the location and description of major components of the equipment.

a. The components are
   (1) Title ***<title>*** (required). Enables IAW MIL–STD–40051–1/–2 the title "Location And Description Of Major Components" to be displayed for the major components.(see 49.1.1.3)
   (2) Select one of the following information types:
      (a) Narrative paragraphs
         1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
         2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
      (b) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
   (3) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).
   (4) Component item ***<comp-item>***(required – one or more). (see 34.1.2.2.1)

b. DTD fragment is graphically depicted.

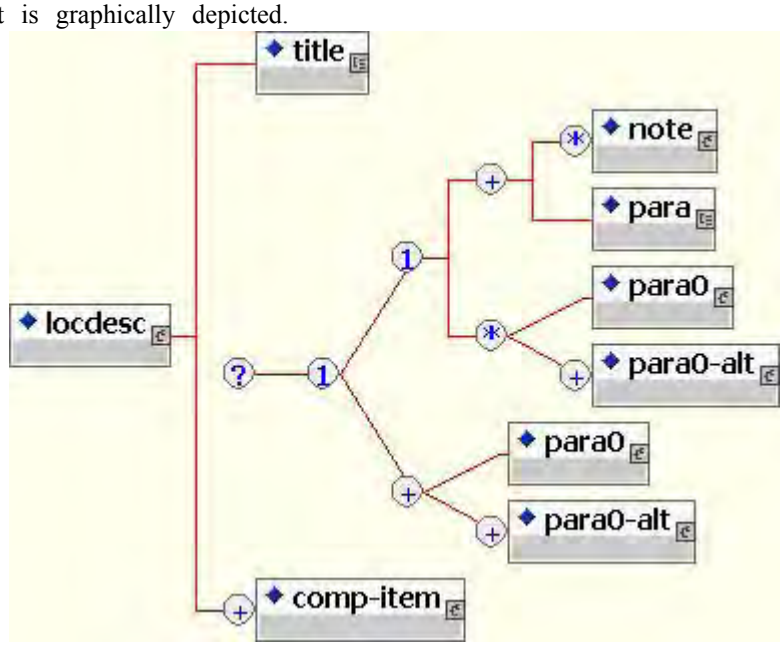

FIGURE 183. Location and description of major components <locdesc> DTD hierarchy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT locdesc (title, ((((note*, para)+, (para0 | para0-alt+)*)
| (para0 | para0-alt+)+))?, comp-item+)>
<!ATTLIST locdesc
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

34.1.2.2.1 Component item ***<comp-item>***. Component item identifies a major component of the equipment, which is covered in the location and description of equipment components.

a. The components are

(1) Title ***<title>*** (optional) (see 49.1.1.3).
(2) Illustration ***<figure>*** (optional – zero or more)(see 44.1.1).
(3) Select one of the following information types:
  (a) Narrative paragraphs
    1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
    2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
  (b) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
(4) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD fragment:

```
<!ELEMENT comp-item (title?, figure*, (((note*, para)+,
(para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST comp-item

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

34.1.2.3 Differences between models ***<eqpdiff>***. Descriptive data containing the significant differences between models or components. A title is required, "Differences Between Models", IAW MIL–STD–40051–1/–2 and the descriptive data is provided by paragraphs of text that may be grouped into systems, subsystems, or other functional or logical method of presentation.

a. The components are:
(1) Title ***<title>*** (required) (see 49.1.1.3).
(2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(4) Select one of the following information types:
(a) Narrative paragraphs
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
(b) Narrative paragraphs with descriptive or narrative titled text
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
(c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD fragment is graphically depicted.

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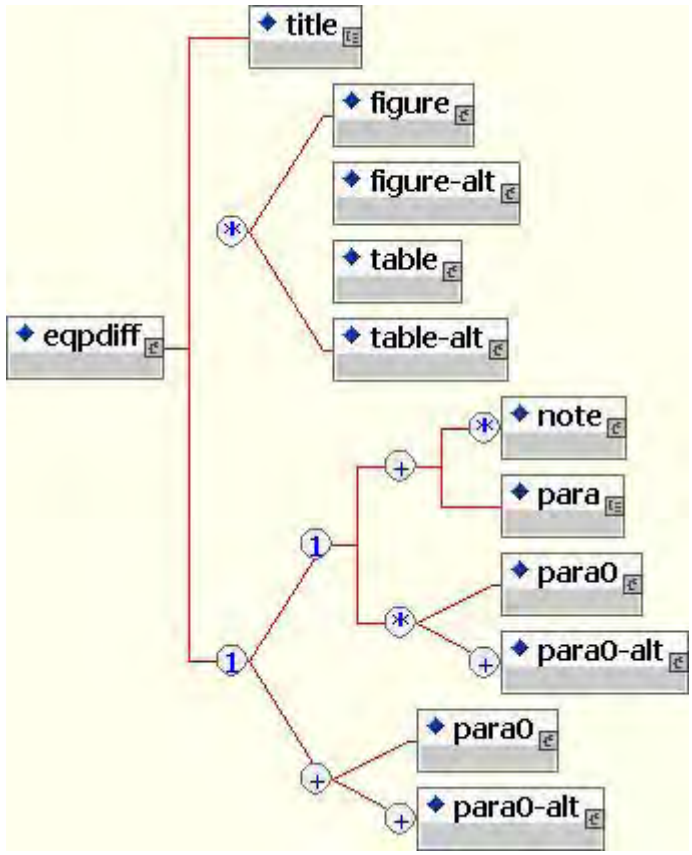

FIGURE 184. Differences between models <eqpdiff> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT eqpdiff (title, (figure | figure-alt | table | table-alt)*,
(((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST eqpdiff

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

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(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

34.1.2.4 Equipment data ***<eqpdata>***. Descriptive data, which contains a listing of the major characteristics, dimensions, capabilities and limitations, and other critical data of the equipment is defined for the equipment user.

a. The components are:
(1) Title ***<title>*** (required). Enables IAW MIL–STD–40051–1/–2 the title "Equipment Data" to be displayed for the section (see 49.1.1.3)
(2) Illustration ***<figure>*** (optional – zero or more)(see 44.1.1).
(3) Select one of the following information types:
(a) Narrative paragraphs
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
(b) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
(4) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD fragment:

```
<!ELEMENT eqpdata (title?, figure*, (((note*, para)+,
(para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST eqpdata

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

c. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

34.1.2.5 XML document instance fragments. The XML source and its stylesheet output for an Equipment Description and Data Work Package is provided below.

a. Example of an XML Document Instance Fragment for ***<descwp>***

```
<descwp wpno="gxxxx3-11-XXXX-XXX" tocentry="2" frame="no" "army="no "airforce="no
"navy="no "marines="no" wpseq="0002" deletewp="no">
<wpidinfo>
<maintlvl level="field"/>
<title>EQUIPMENT DESCRIPTION AND DATA WORK PACKAGE
</title>
```


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</wpidinfo>
<eqpinfo>
<title>EQUIPMENT CHARACTERISTICS, CAPABILITIES AND FEATURES
</title>
<eqpdesc>
<para>The radar system is a Moving Target Indicator (MTI) radar, using pulsed Doppler techniques to detect and locate moving targets, or to perform friendly fire correction/Fall-of Shot (FOS). Accuracy of target location depends on the method of surveying the radar position. Built-in-test (BIT) features enable the operator to locate faults in the Transmitter-Receiver unit and the Antenna Drive enclosure.
</para>
<para>The radar equipment presents targets as symbols on a planned position indicator (PPI) display and as audio tones in a headset. The radar equipment can be set up or taken down under blackout conditions in approximately ten minutes by two persons. Three persons can carry its components (69 pounds) with backpack.
</para>
</eqpdesc>
<eqpdesc>
<title>Searchlighting
</title>
<para>The antenna can be set to searchlight (move to either the right or the left), in response to a control actuated by the operator, and to stop moving when the control is released.
</para>
</eqpdesc>
</eqpinfo>
<locdesc>
<title>LOCATION AND DESCRIPTIONS OF MAJOR COMPONENTS
</title>
<comp-item>
<title>TRIPOD MOUNTED COMPONENTS
</title>
<figure application="both" figtype="normal-page" tocentry="1" pane="no">
<title>TRIPOD MOUNTED COMPONENTS
</title>
<graphic boardno="tripodmounted" unitmeasure="in" hscale="35" scalefit="yes" hplace="center">
</graphic>
</figure>
<para>The tripod-mounted components include most of the radar set components. The tripod and the Waveguide Feedhorn
<callout assocfig="tripodmounted" label="figure 1, item 1"/>, Antenna Reflector
<callout assocfig="tripodmounted" label="figure 1, item 2"/>, Telescope
<callout assocfig="tripodmounted" label="figure 1, item 3"/>, Receiver-Transmitter
<callout assocfig="tripodmounted" label="figure 1, item 4"/>, Antenna Drive Assembly
<callout assocfig="tripodmounted" label="figure 1, item 5"/>, Column assembly
<callout assocfig="tripodmounted" label="figure 1, item 6"/>, Tripod
<callout assocfig="tripodmounted" label="figure 1, item 7"/>, and the Mounting and Leveling assembly
<callout assocfig="tripodmounted" label="figure 1, item 8"/>.
</para>
</comp-item>
</locdesc>
<eqpdata>

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```
<title>EQUIPMENT  DATA
</title>
<para>Table 1.  General provides information pertaining to the electrical,
operational, and environmental characteristics of the radar set.
<table  tocentry="1">
<title>General
</title>
<tgroup  cols="2"  align="left">
<colspec  colname="col1"  colwidth="2.50*in">
<colspec  colname="col2"  colwidth="3.50*in">
<thead  valign="bottom">
<row>
<entry  morerows="0"  valign="top"  align="center">ITEM
</entry>
<entry  morerows="0"  align="center">VALUE
</entry>
</row>
</thead>
<tbody  valign="top">
<row>
<entry  morerows="0"  align="center">Electrical  Characteristics<
</entry>
<entry  morerows="0"  align="center">
</entry>
</row>
<row>
<entry  morerows="0"  align="center">Power  input
</entry>
<entry  morerows="0"  align="center">
</entry>
</row>
<row>
<entry  morerows="0"  align="center"> Radar  Set  AN/PPS-5XX
</entry>
<entry  morerows="0"  align="center">(18-36  volt,  direct  cur-
rent  (dc)  source):    41  watts
</entry>
</row>
<row>
<entry  morerows="0"  align="center">Operational  Characteristics
</entry>
<entry  morerows="0"  align="center">
</entry>
</row>
</para>
</eqpdata>
</descwp>
```

b. Example of a Stylesheet Output for ***<descwp>***:

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TM X-XXXX-XXX 0002

FIELD MAINTENANCE

EQUIPMENT DESCRIPTION AND DATA

EQUIPMENT CHARACTERISTICS, CAPABILITIES AND FEATURES

The radar system is a Moving Target Indicator (MTI) radar, using pulsed Doppler techniques to detect and locate moving targets, or to perform friendly fire correction/Fall-of Shot (FOS). Accuracy of target location depends on the method of surveying the radar position. Built-in-test (BIT) features enable the operator to locate faults in the Transmitter-Receiver unit and the Antenna Drive enclosure.

The radar equipment presents targets as symbols on a planned position indicator (PPI) display and as audio tones in a headset. The radar equipment can be set up or taken down under blackout conditions in approximately ten minutes by two persons. Three persons can carry its components (69 pounds) with backpack.

SEARCHLIGHTING.

The antenna can be set to searchlight (move to either the right or the left), in response to a control actuated by the operator, and to stop moving when the control is released.

LOCATION AND DESCRIPTIONS OF MAJOR COMPONENTS

TRIPOD MOUNTED COMPONENTS

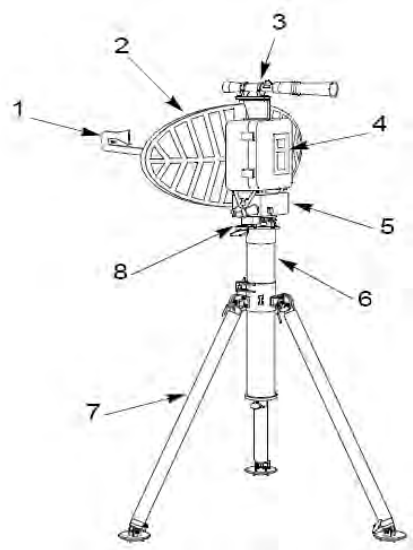

Figure 1. Tripod-Mountd Components.

The tripod-mounted components include most of the radar set components.The tripod and the Waveguide Feedhorn (Figure 1, Item 1), Antenna Reflector (Figure 1, Item 2), Telescope (Figure 1, Item 3), Receiver-Transmitter (Figure 1, Item 4), Antenna Drive Assembly (Figure 1, Item 5), Column assembly (Figure 1, Item 6) Tripod (Figure 1, Item 7), and the Mounting and Leveling assembly (Figure 1, Item 8).

0002-1

FIGURE 185. Example of a stylesheet output for an equipment description and data work package <descwp>..

34.1.3 Theory of operation work package ***<thrywp>***. The element provides a functional description on how the equipment and its components function and interface. The LMI and/or maintenance plan

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dictates the level of detail presented in this work package and is subdivided into the following content requirements. When a TM has more than one aircraft system, a theory of operation work package is to be developed for each aircraft system.

- A simple system or equipment/component can be explained in the ***<thrywp>*** using a system theory ***<systhry>***
- For more complex systems multiple ***<systhry>*** work packages may be used.
- The relative complexity of the weapon system/equipment may be that it is reasonable to first present the theory of the end item as a unit. After presented the end item as a unit then present the theory of its major system ***<systhry>***, subsystems ***<ssysthry>***, and component theory line replaceable units' theory of operation (LRU) ***<lruthry>*** and/or shop replaceable units (SRU) ***<sruthry>***.
- A TM may need to use more than one ***<thrywp>*** to explain how the equipment and its components function and interface.

a. The components of ***<thrywp>*** are

(1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end use (see 32.3.2).

(2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).

(3) Introduction ***<intro>***(optional). The element provides the introductory section contained in the work package (see 49.1.4.15).

(4) System Theory ***<systhry>*** (required – one or more). The element identifies a system's theory of operation (see 34.1.3.1).

b. Theory of operation work package DTD hierarchy ***<thrywp>***

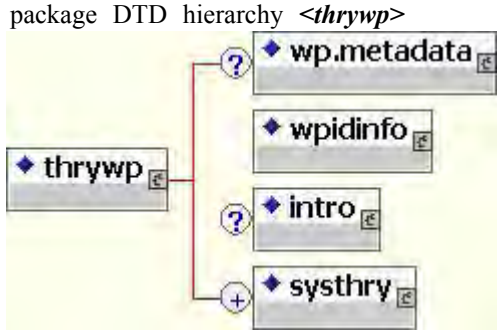

FIGURE 186. Theory of operation work package <thrywp> DTD hierarchy.

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c. DTD fragment for:

```
<!ELEMENT thrywp (wp.metadata?, wpidinfo, intro?, systhry+)>
<!ATTLIST thrywp

    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

34.1.3.1 System theory ***<systhry>***. The systems theory element ***<systhry>*** is used to identify a system's theory of operation ***<thrywp>***. Theory of operation explains how the end item and its major systems work and interface in addition to the functional effect of switches, controls, and other devices. Subordinate sections ***<ssysthry>*** on subsystem theory may be included. A simple system may only have one theory of operation work package whereas a large or complex system may contain system theory, subsystem theory, and component theory (LRU and/or SRU).

a. The components are
(1) Title ***<title>*** (required) (see 49.1.1.3).
(2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(4) Select one of the following information types:
(a) Narrative paragraphs
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
(b) Narrative paragraphs with descriptive or narrative titled text
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
(c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).
(5) Line Replaceable Unit(s) ***<lruthry>*** (optional – zero or more). The element identifies the line replaceable units in theory of operation (see 34.1.3.1.1.1).
(6) Shop Replaceable Unit(s) ***<sruthry>*** (optional – zero or more). The element identifies the shop replaceable units in theory of operation(see 34.1.3.1.1.2).
(7) Subsystem ***<ssysthry>*** (optional – zero or more) (see 34.1.3.1.1).

b. System theory DTD hierarchy ***<systhry>***

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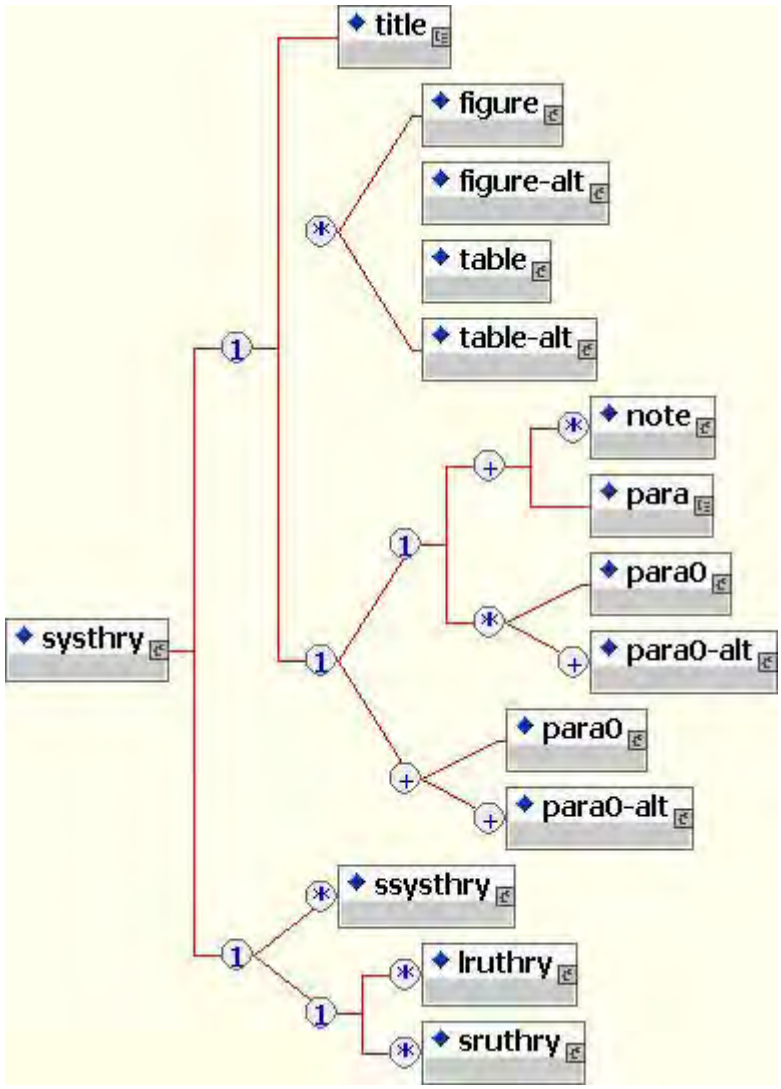

FIGURE 187. System theory <systhry> DTD hierarchy.

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c. DTD fragment for ***<systhry>***:

```
<!ELEMENT systhry (title, (figure | figure-alt | table | table-alt)*,
(((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+)),
(ssysthry* | (lruthry*, sruthry*)))>
<!ATTLIST systhry
    frame            (yes | no )                    'yes'

    assocfig          IDREFS                        #IMPLIED
    changeref         IDREFS                        #IMPLIED
    comment           CDATA                         #IMPLIED
    delchlvl          CDATA                         #IMPLIED
    id                ID                            #IMPLIED
    idref             IDREFS                        #IMPLIED
    inschlvl          CDATA                         #IMPLIED
    security          (uc | fouo |c | s | ts)      #IMPLIED
    skilltrk          CDATA                         #IMPLIED
```

d. Common attributes:

(1) **frame** – Indicates to the IETM system the authors intended Frame break. Default is 'yes'.
(2) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(3) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(4) **comment** – Reason for change information. (optional) (See 49.2.9.)
(5) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(6) **id** – Unique identifier. (optional) (See 49.2.6.)
(7) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(8) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(9) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(10) **security** – Security classification. (optional) (See 49.2.13.)

34.1.3.1.1 Subsystem ***<ssysthry>***. The element ***<ssysthry>*** is used for a complex system or multi-system equipment that may divide the theory of operation by using subsystem breakdown. This element includes theory of operation of a specific subsystem.

a. The components are:

(1) Title ***<title>*** (required) (see 49.1.1.3).
(2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(4) Select one of the following information types:
  (a) Narrative paragraphs
    1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
    2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
  (b) Narrative paragraphs with descriptive or narrative titled text
    1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
    2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
    3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
  (c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).
(5) Line Replaceable Unit(s) ***<lruthry>*** (optional – zero or more). The element identifies the line replaceable units in theory of operation (see 34.1.3.1.1.1).
(6) Shop Replaceable Unit(s) ***<sruthry>*** (optional – zero or more). The element identifies the shop replaceable units in theory of operation(see 34.1.3.1.1.2).

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b. Subsystem DTD hierarchy ***<ssysthry>***

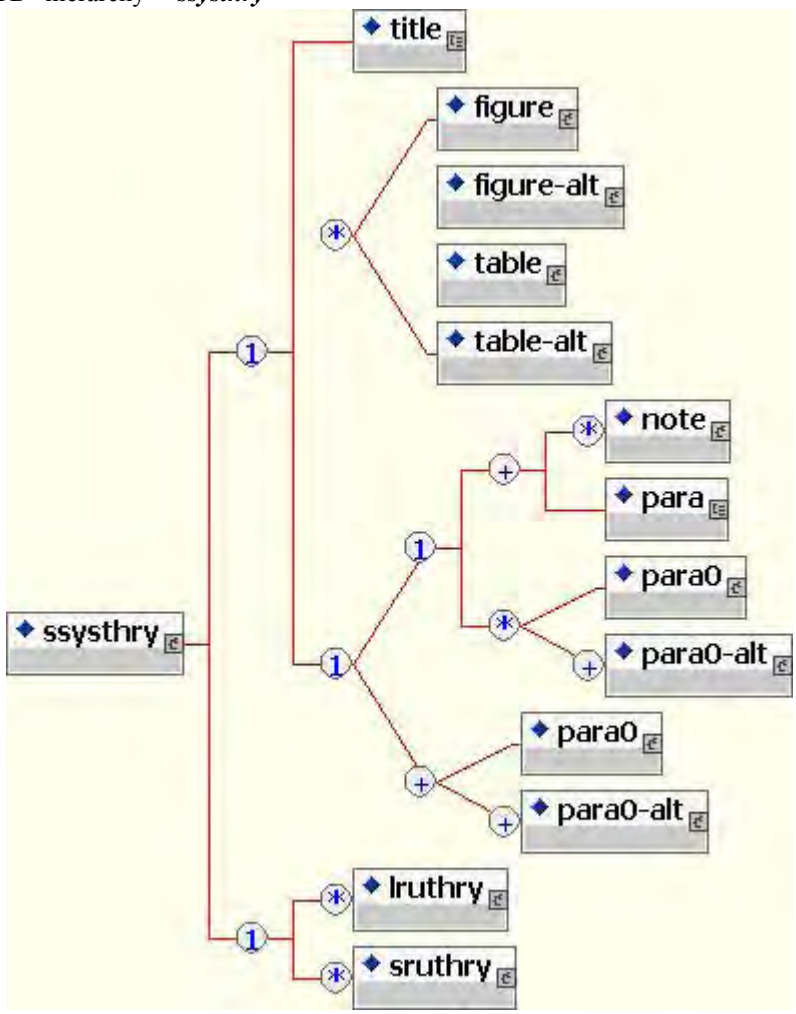

FIGURE 188. Subsystem <ssysthry> DTD hierarchy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT ssysthry ((title, (figure | figure-alt | table | table-alt)*,
(((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+)),
(lruthry*, sruthry*))>
<!ATTLIST ssysthry
    nomen           CDATA                          #IMPLIED
    nsn             CDATA                          #IMPLIED
    frame           (yes | no )   'no'
    assocfig        IDREFS                         #IMPLIED
    changeref       IDREFS                         #IMPLIED
    comment         CDATA                          #IMPLIED
    delchlvl        CDATA                          #IMPLIED
    id              ID                             #IMPLIED
    idref           IDREFS                         #IMPLIED
    inschlvl        CDATA                          #IMPLIED
    security        (uc | fouo |c | s | ts)       #IMPLIED
    skilltrk        CDATA                          #IMPLIED>
```

d. Common attributes:

(1) **nomen** – System nomenclature (optional).
(2) **nsn** – National stock number (optional).
(3) **frame** – Indicates to the IETM system the authors intended Frame break. Default is 'yes'.
(4) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(5) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(6) **comment** – Reason for change information. (optional) (See 49.2.9.)
(7) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(8) **id** – Unique identifier. (optional) (See 49.2.6.)
(9) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(10) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(11) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(12) **security** – Security classification. (optional) (See 49.2.13.)

34.1.3.1.1.1 Line replaceable unit ***<lruthry>***. The element ***<lruthry>*** identifies line replaceable units in theory of operation. A LRU is a component or unit removed at the Unit or Organizational level.

a. The components are:

(1) Title ***<title>*** (required) (see 49.1.1.3).
(2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(4) Select one of the following information types:
(a) Narrative paragraphs
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
(b) Narrative paragraphs with descriptive or narrative titled text
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
(c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. Line replaceable unit ***<lruthry>*** DTD hierarchy ***<lruthry>***.

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Check the source to verify that this is the current version before use.

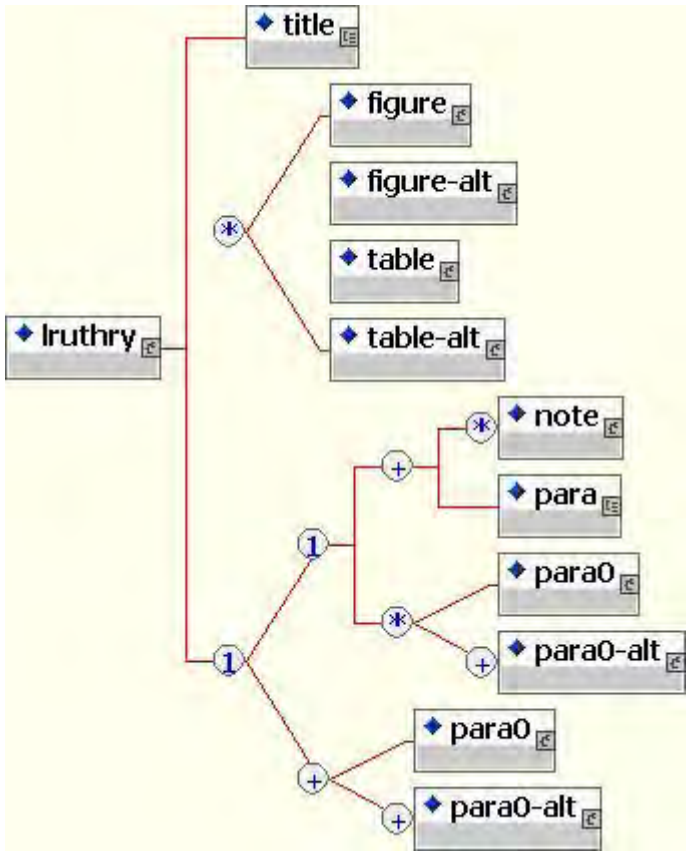

FIGURE 189. Line replaceable unit <lruthry> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT lruthry ((title, (figure | figure-alt | table | table-alt)*,
(((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST lruthry
    nomen            CDATA                          #IMPLIED
    nsn              CDATA                          #IMPLIED
    frame            (yes | no )                    'no'
    assocfig         IDREFS                         #IMPLIED
    changeref        IDREFS                         #IMPLIED
    comment          CDATA                          #IMPLIED
    delchlvl         CDATA                          #IMPLIED
    id               ID                             #IMPLIED
    idref            IDREFS                         #IMPLIED
    inschlvl         CDATA                          #IMPLIED
    security         (uc | fouo |c | s | ts)       #IMPLIED
    skilltrk         CDATA                          #IMPLIED
```

d. Common attributes:

(1) **nomen** – System nomenclature (optional).
(2) **nsn** – National stock number (optional).
(3) **frame** – Indicates to the IETM system the authors intended Frame break. Default is 'yes'.
(4) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(5) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(6) **comment** – Reason for change information. (optional) (See 49.2.9.)

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(7) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(8) **id** – Unique identifier. (optional) (See 49.2.6.)
(9) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(10) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(11) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(12) **security** – Security classification. (optional) (See 49.2.13.)

34.1.3.1.1.2 Shop replaceable unit ***<sruthry>***. The element ***<sruthry>*** identifies the shop replaceable units in theory of operation. A SRU is a component or unit that is authorized to be removed only at a repair shop.

a. The components are:
(1) Title ***<title>*** (required) (see 49.1.1.3).
(2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(4) Select one of the following information types:
(a) Narrative paragraphs
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
(b) Narrative paragraphs with descriptive or narrative titled text
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
(c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD fragment:

```
<!ELEMENT sruthry ((title, (figure | figure-alt | table | table-alt)*,
(((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+)), >
<!ATTLIST sruthry
    nomen           CDATA                       #IMPLIED
    nsn             CDATA                       #IMPLIED
    frame           (yes | no )                 'no'
    assocfig        IDREFS                       #IMPLIED
    changeref       IDREFS                       #IMPLIED
    comment         CDATA                        #IMPLIED
    delchlvl        CDATA                        #IMPLIED
    id              ID                           #IMPLIED
    idref           IDREFS                       #IMPLIED
    inschlvl        CDATA                        #IMPLIED
    security        (uc | fouo |c | s | ts)     #IMPLIED
    skilltrk        CDATA                        #IMPLIED
```

c. Common attributes:
(1) **nomen** – System nomenclature (optional).
(2) **nsn** – National stock number (optional).
(3) **frame** – Indicates to the IETM system the authors intended Frame break. Default is 'yes'.
(4) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(5) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(6) **comment** – Reason for change information. (optional) (See 49.2.9.)
(7) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(8) **id** – Unique identifier. (optional) (See 49.2.6.)
(9) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

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(10) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(11) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(12) **security** – Security classification. (optional) (See 49.2.13.)

34.1.3.2 XML document instance fragments. The XML source and its stylesheet output for a Theory of Operation Work Package ***<thrywp>*** is provided below.

a. Example of an XML Document Instance Fragment for ***<thrywp>*** containing one major system theory ***<systhry>*** that contains multiple occurrences of shop replaceable unit ***<sruthry>***.

```
<thrywp wpno="GXXXXX-X-XXXX-XXX" wpseq= 0015 summary-detail="0">
<wpidinfo>
<maintlvl level="operator"/>
<title>THEORY OF OPERATION
</title>
</wpidinfo>
<systhry>
<title>POWER ENTRY HOUSING AND SECONDARY INPUT POWER FILTER
</title>
<para>Three phase input power from an external source, i.e. diesel
powered generator or facility power, is cabled to the power entry
housing and connected to the power input connector. Inside the housing,
each phase line and the neutral line are wired to four independent surge
arresters and the primary input power fitter. The surge arresters
protect the circuits within the shelter from transients caused by
lightning, induction, switching surges and Electro Magnetic Pulse (EMP).
When a surge voltage exceeds the spark over voltage of the arrester,
the arrester becomes a short circuit and remains so until the transient
has been by-passed and the line automatically returns to normal. The
primary input power filter suppresses condition of noise caused by
Electro Magnetic Interference (EMI), into and out of the shelter over
the input power lines. A secondary input power filter is connected
in series with the primary input power fitter to ensure protection of
shelter circuits from transient voltages below the spark over voltage
of the EMP protectors coming in through the power lines.
</para>
<sruthry>
<title>POWER DISTRIBUTION ASSEMBLY (PDA)
</title>
<para>The PDA consists of a bass assembly which houses the circuit
protection and the power distribution panel which contains circuit
breakers and indicators. The circuit protection devices consist
of an over/under voltage-relay, a phase sequence relay and an over
current relay. If a condition exists in a circuit that does not
meet the parameters of any one of these devices, that fault will
cause the MAIN POWER circuit breaker to reset to the OFF position,
thereby removing power to the internal circuits of the shelter and
causes the AC POWER FAULT light to come on. When set to ON, the
MAIN POWER circuit breaker applies power to the inputs of the other
breakers and three green power indicators showing power applied for
each phase will come on. As each of the remaining circuit breakers
are set to ON, the corresponding indicators will come on providing
a visual indication of power applied to each circuit.
</para>
</sruthry>
<sruthry>
<title>ENVIRONMENTAL CONTROL UNIT (ECU)
</title>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<para> (Horizontal ECU is used only on the AN/TSM-191 (V)3 model of
the shelter.  Vertical ECU is used only on the AN/ TSM-191(V)2
and AN/TSM-191(V)4 models of the shelter.)  Once started, the air
conditioner operates automatically due to the relationship of the
components, controls and instruments.  With the model selector switch in
the OFF position, all electrical components are isolated from electrical
power except for the crankcase heater.  This device must be energized
for 30 minutes prior to operation in the cool mode.  The following
operating modes of the ECU are controlled by the mode selector switch.
</para>
</sruthry>
<sruthry>
<title>Ventilation
</title>
<para>Ventilation is provided in the VENT position by energizing
the fan motor which forces air out of the evap-orator discharge
louver.  The amount of outdoor air used for ventilation is
determined by the position of the fresh air damper.
</para>
</sruthry>
</systhry>
</thrywp>
```

b. Example of a Stylesheet Output for ***<thrywp>***:

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TM X-XXXX-XXX 0003

FIELD MAINTENANCE

THEORY OF OPERATION

POWER ENTRY HOUSING AND SECONDARY INPUT POWER FILTER

Three phase input power from an external source, i.e. diesel powered generator or facility power, is cabled to the power entry housing and connected to the power input connector. Inside the housing, each phase line and the neutral line are wired to four independent surge arresters and the primary input power fitter. The surge arresters protect the circuits within the shelter from transients caused by lightning, induction, switching surges and Electro Magnetic Pulse (EMP). When a surge voltage exceeds the spark over voltage of the arrester, the arrester becomes a short circuit and remains so until the transient has been by-passed and the line automatically returns to normal. The primary input power filter suppresses condition of noise caused by Electro Magnetic Interference (EMI), into and out of the shelter over the input power lines. A secondary input power filter is connected in series with the primary input power fitter to ensure protection of shelter circuits from transient voltages below the spark over voltage of the EMP protectors coming in through the power lines.

POWER DISTRIBUTION ASSEMBLY (PDA)

The PDA consists of a bass assembly which houses the circuit protection and the power distribution panel which contains circuit breakers and indicators. The circuit protection devices consist of an over/under voltage-relay, a phase sequence relay and an over current relay. If a condition exists in a circuit that does not meet the parameters of any one of these devices, that fault will cause the MAIN POWER circuit breaker to reset to the **OFF** position, thereby removing power to the internal circuits of the shelter and causes the AC POWER FAULT light to come on. When set to **ON**, the MAIN POWER circuit breaker applies power to the inputs of the other breakers and three green power indicators showing power applied for each phase will come on. As each of the remaining circuit breakers are set to ON, the corresponding indicators will come on providing a visual indication of power applied to each circuit.

ENVIRONMENTAL CONTROL UNIT (ECU)

(Horizontal ECU is used only on the AN/TSM-191 (V)3 model of the shelter. Vertical ECU is used only on the AN/ TSM-191(V)2 and AN/TSM-191(V)4 models of the shelter.) Once started, the air conditioner operates automatically due to the relationship of the components, controls and instruments. With the model selector switch in the OFF position, all electrical components are isolated from electrical power except for the crankcase heater. This device must be energized for 30 minutes prior to operation in the cool mode. The following operating modes of the ECU are controlled by the mode selector switch.

VENTILATION

Ventilation is provided in the VENT position by energizing the fan motor which forces air out of the evap-orator discharge louver. The amount of outdoor air used for ventilation is determined by the position of the fresh air damper

END OF WORK PACKAGE

FIGURE 190. Example of a stylesheet output for a theory of operation work package <thrywp> containing one major system theory <systhry>.

c. Example of an XML Document Instance Fragment for ***<thrywp>*** containing multiple occurrences system theory ***<systhry>***. The system theory contains multiple subsystem theory ***<ssysthry>*** that contains one or more line replaceable units' theory of operation ***<lruthry>***.

```
<thrywp wpno="gxxxx3-11-XXXX-XXX" tocentry="2" frame="no" "army="no "airforce="no
"navy="no "marines="no" wpseq="0003">
<wpidinfo>
<maintlvl level="field"/>
<title>THEORY OF OPERATION
</title>
</wpidinfo>
<intro frame="no">
<para0>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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*<title>*GENERAL
</title>
*<para>*The AN/PPS-5XX radar is a state-of-the art Pulse Doppler radar. It is a monostatic system in that it both transmits and receives pulses using the same antenna. In its most basic mode of operation, the radar transmits 512 Linear FM pulses in each 72 ms dwell. The radar scans one antenna beam width or 1.1 degrees in azimuth, during each dwell. After each pulse is transmitted, data is received and collected for one Pulse Repetition Interval (PRI) or 141 us, before the next pulse is transmitted. In this toggling manner one dwell of 512 pulses is transmitted, and the return echoes received, from targets that may have intercepted the transmitted pulses.
</para>
*<para>*Upon receiving the returned echoes, they are down converted in frequency on the RF Assembly and IF Converter circuit card. They are then sampled with A/D converters at 910KHz. The digital samples from the IF Converter are sent to the DSP/Control circuit card to a SHARC Digital Signal Processor (DSP). There Pulse Compression (Correlation), Doppler Weighting, Doppler Filtering and Detection Processing algorithms are performed. The DSP is pipe lined and works on a single dwell of data at a time. A new dwell of data, however, is being collected at the same time the previous dwell is being processed. After all DSP processing is finished, the Data Processor circuit card is triggered to read out all detections which have passed a user defined threshold in the detection processor. The Data Processor sends the detections out over the R-T to HTU cable RS-232 interface for display on the HTU's PPI.
</para>
*<para>*All radar control is performed from either the remote (HTU) or the alphanumeric Keypad/Display panel mounted on the R-T Electronics Assembly cover. The Keypad/Display, however, has no detection display capabilities.
</para>
*<para>*An audio detection capability exists in which detections can be heard on a COTS headset. This mode is available when using either the HTU or the Keypad/Display. It is intended, however, for use when only the Keypad/ Display is being used.
</para>
</para0>
</intro>
<systhry frame="no">
*<title>*RF SYSTEM
</title>
*<para>*The RF system consists of a waveguide feed and antenna, coax to waveguide transition, preselector filter and circulator. The preselector filter attenuates any out-of-band signals such as images and interference during receive, and attenuates harmonics and image signals during transmission. The circulator provides isolation between the transmitter and receiver to protect the sensitive receiver during the high energy transmit pulse. The coax to waveguide transition provides the interface between the preselector coaxial connection and the waveguide feed for the antenna. The waveguide feed collects/transmits energy as it is received/transmitted from the antenna. The antenna is a lightweight fiberglass elliptical reflector with a parabolic contour.
</para>
<ssysthry nomen="Nxxx" nsn="xxxx-xx-xxx-xxxx" frame="no">
*<title>*Antenna Positioning System

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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</title>
*<para>*The antenna drive utilizes a brush type DC servomotor that is driven by a Pulse Width Modulated (PWM) servo amplifier. An optical encoder is used to provide position feedback to the servo controller for closed loop servo control of the motor. The antenna elevation position is controlled manually by the user. An absolute encoder provides elevation position information on the PPI. A Motion Control circuit card is located in the antenna drive enclosure. It is used as the interface between control circuitry on the DSP/Control circuit card and the motion control logic in the drive box. Using the interface, the motor can be commanded to move to any position.
</para>
</ssysthry>
<ssysthry nomen="Nxxx" nsn="xxxx-xx-xxx-xxxx" frame="no">
*<title>*CABLE ASSEMBLIES
</title>
*<para>*Cable Assembly, Special Purpose, Electrical, XX-XXXXX/PPS-5XX R-T to HTU (Figure 3. Cable Assembly (Receiver-Transmitter to Handheld Terminal Unit)) is required to connect the HTU or equivalent to the Receiver Transmitter. The cable carries target and control data over a standard RS-232 interface to and from the HTU or equivalent and the Receiver-Transmitter. It is 75 feet in length and can be disconnected at both ends, coiled, and stowed for transport. The HTU and the Receiver-Transmitter may be separated an additional 75 feet, to 150 feet, using the optional Cable Assembly, Special Purpose, Electrical, XX-XXXXX/PPS-5XX extension cable.
<figure application="both" figtype="normal-page" tocentry="1" pane="no">
*<title>*Cable Assembly (Receiver-Transmitter to Handheld Terminal Unit)
</title>
<graphic boardno="Scroll.gif" unitmeasure="in">
</graphic>
</figure>
</ssysthry>
</systhry>
<systhry frame="no">
*<title>*THEORY OF OPERATION FOR THE HTU
</title>
<para0>
*<title>*Operator Control Parameters For The Handheld Terminal Unit
</title>
*<para>*The following control parameters are available to the operator from the HTU displays:
</para>
</para0>
<ssysthry nomen="Nxxx" nsn="xxxx-xx-xxx-xxxx" frame="no">
*<title>*BIT DISPLAY (Figure 48)
</title>
<para>
<figure application="both" figtype="normal-page" tocentry="1" pane="no">
*<title>*HTU Bit Display (BIT Tests)
</title>
<graphic boardno="Builtintest.gif" unitmeasure="in">
</graphic>
</figure>
</para>
</ssysthry>

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</systhry>
</thrywp>

d. Example of a stylesheet output for a theory of operation work package ***<thrywp>*** containing multiple occurrences system theory. The output is only of the first page.

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TM X-XXXX-XXX 0003

DIRECT SUPPORT MAINTENANCE

AN/PPS-5XX RADAR
THEORY OF OPERATION

GENERAL

The AN/PPS-5XX radar is a state-of-the art Pulse Doppler radar. It is a monostatic system in that it both transmits and receives pulses using the same antenna. In its most basic mode of operation, the radar transmits 512 Linear FM pulses in each 72 ms dwell. The radar scans one antenna beamwidth or 1.1 degrees in azimuth, during each dwell. After each pulse is transmitted, data is received and collected for one Pulse Repetition Interval (PRI) or 141us, before the next pulse is transmitted. In this toggling manner one dwell of 512 pulses is transmitted, and the return echoes received, from targets that may have intercepted the transmitted pulses.

Upon receiving the returned echoes, they are downconverted in frequency on the RF Assembly and IF Converter circuit card. They are then sampled with A/D converters at 910KHz. The digital samples from the IF Converter are sent to the DSP/Control circuit card to a SHARC Digital Signal Processor (DSP). There Pulse Compression (Correlation), Doppler Weighting, Doppler Filtering and Detection Processing algorithms are performed. The DSP is pipelined and works on a single dwell of data at a time. A new dwell of data, however, is being collected at the same time the previous dwell is being processed. After all DSP processing is finished, the Data Processor circuit card is triggered to read out all detections which have passed a user defined threshold in the detection processor. The Data Processor sends the detections out over the R-T to HTU cable RS-232 interface for display on the HTU's PPI.

All radar control is performed from either the remote (HTU) or the alphanumeric Keypad/Display panel mounted on the R-T Electronics Assembly cover. The Keypad/Display, however, has no detection display capabilities.

An audio detection capability exists in which detections can be heard on a COTS headset. This mode is available when using either the HTU or the Keypad/Display. It is intended, however, for use when only the Keypad/ Display is being used.

RF SYSTEM

The RF system consists of a waveguide feed and antenna, coax to waveguide transition, preselector filter and circulator. The preselector filter attenuates any out-of-band signals such as images and interference during receive, and attenuates harmonics and image signals during transmission. The circulator provides isolation between the transmitter and receiver to protect the sensitive receiver during the high energy transmit pulse. The coax to waveguide transition provides the interface between the preselector coaxial connection and the waveguide feed for the antenna. The waveguide feed collects/transmits energy as it is received/transmitted from the antenna. The antenna is a lightweight fiberglass elliptical reflector with a parabolic contour.

ANTENNA POSITIONING SYSTEM

The antenna drive utilizes a brush type DC servomotor that is driven by a Pulse Width Modulated (PWM) servo amplifier. An optical encoder is used to provide position feedback to the servo controller for closed loop servo control of the motor. The antenna elevation position is controlled manually by the user. An absolute encoder provides elevation position information on the PPI. A Motion Control circuit card is located in the antenna drive enclosure. It is used as the interface between control circuitry on the DSP/Control circuit card and the motion control logic in the drive box. Using the interface, the motor can be commanded to move to any position.

CABLE ASSEMBLY

Cable Assembly, Special Purpose, Electrical, XX-XXXXX/PPS-5XX R-T to HTU (Figure 3. Cable Assembly (Receiver-Transmitter to Handheld Terminal Unit)) is required to connect the HTU or equivalent to the Receiver Transmitter. The cable carries target and control data over a standard RS-232 interface to and from the HTU or equivalent and the Receiver-Transmitter. It is 75 feet in length and can be disconnected at both ends, coiled, and stowed for transport. The HTU and the Receiver-Transmitter may be separated an additional 75 feet, to 150 feet, using the optional Cable Assembly, Special Purpose, Electrical, XX-XXXXX/PPS-5XX extension cable.

0003-1

FIGURE 191. Example of a stylesheet output for a theory of operation work package <thrywp> containing multiple occurrences of system theory <systhry>.

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Check the source to verify that this is the current version before use.

34.1.4 General information work package (phased maintenance checklist manual only)***<pm-ginforwp>***. The element ***<pm-ginforwp>*** provides the introductory material for phased maintenance in a Phased Maintenance Checklist Manual.

a. The components are:
   (1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end user. (see 32.3.2)
   (2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package. (see 32.2)
   (3) General Information ***<geninfo>*** (required). The element provides general information, reference statements and standard statements that apply to the entire TM. (see 49.1.4.12).

b. DTD fragment for ***<pm-ginforwp>***:

```
<!ELEMENT pm-ginfowp (wp.metadata?, wpidinfo, geninfo)>
<!ATTLIST pm-ginfowp   >


   airforce          (yes | no )                "no"
   army              (yes | no )                "no"
   assocfig          IDREFS                      #IMPLIED
   changeref         IDREFS                      #IMPLIED
   comment           CDATA                       #IMPLIED
   crewmember        CDATA                       #IMPLIED
   date-time-stamp   (date | time | date-time) #IMPLIED
   delchlvl          CDATA                       #IMPLIED
   deletewp          (yes | no )                "no"
   fgc               CDATA                       #IMPLIED
   frame             (yes | no )                "yes"
   idref             IDREFS                      #IMPLIED
   inschlvl          CDATA                       #IMPLIED
   insertwp          CDATA                       #IMPLIED
   lsa-id            CDATA                       #IMPLIED
   marines           (yes | no )                "no"
   navy              (yes | no )                "no"
   security          (uc | fouo | c | s | ts ) #IMPLIED
   skilltrk          CDATA                       #IMPLIED
   tocentry          (2 | 3 | 4 | 5)            "2"
   wpno              ID                          #REQUIRED
   wpseq             CDATA                       #IMPLIED
```

c. Common attributes:
   (1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
   (2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
   (3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (5) **comment** – Change information (optional) (see 49.2.9).
   (6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
   (7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
   (8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
   (10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

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(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

34.1.4.1 Example of an XML document instance fragment and output for ***<pm-ginfowp>***. The XML instance and its stylesheet output for a General Information Work Package is provided below.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % ammo-tm "IGNORE">
<!ENTITY % usmc-tm "IGNORE">
```

b. Example of an XML document instance fragment using boilerplate entities for in the XML instance for General information work package (Phased Maintenance Checklist Manual Only) ***<pm-ginfowp>***. The boilerplates ***&pm-ginfowp.geninfo-phased;*** and ***&pm-ginfowp.geninfo-additional;*** contain verbatim text to develop this work package. The general entities contain nested entities that are editable to insert the required data for the work package. See Section 50 for further information on boilerplates.

```
<pm-ginfowp airforce="no" army="no" deletewp="no" frame="no" marines="no"
navy="no" tocentry="2" wpno="GXXXXX-X-XXXX-XXX" wpseq="0001">
<wpidinfo>
<maintlvl level="sustain"/>
</wpidinfo>
<title>Phased Maintenance Inspection Manual Only
</title> &pm-ginfowp.geninfo-phased; &pm-ginfowp.geninfo-additional;
</pm-ginfowp>
```

c. Example of a stylesheet output for ***<pm-ginfowp>***. **The example consists of only the first page of the work pagckage.**

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TM X–XXXX–XXX

SUSTAINMENT MAINTENANCE
PHASED MAINTENANCE INSPECTION MANUAL ONLY

PHASED SCHEDULE

The phased maintenance inspection checklist contains requirements for inspection of the INSERT WITH AIRCRAFT MODEL aircraft on a phased schedule having a INSERT WITH FLIGHT HOUR CYCLE hour (flight hours) cycle with INSERT WITH PHASE HOURS hour phases. Each requirement included herein is designated for accomplishment at least once, but not more than INSERT WITH NUMBER OF PHASES times during the INSERT WITH FLIGHT HOUR CYCLE hour cycle.

EXCEEDING THE PHASED SCHEDULE

The phased maintenance inspection intervals designated are the maximum and shall not be exceeded except in actual operational emergencies as explained herein. It is the Commander's responsibility to determine (on an individual aircraft basis) when inspection intervals may be exceeded. For this purpose, operational emergencies are conditions of combat, or conditions of disaster which necessitate flight to evacuate aircraft or personnel. When aircraft are operated beyond the normal inspection due time because of such emergency situations, a circled red X status symbol and an appropriate statement (to include authority) must be entered on the appropriate aircraft form as specified in DA PAM 738-751 until such time as the inspection is complete. When inspections are delayed to meet emergency requirements, Commanders will assure that the aircraft status symbol reverts to a red X and that delayed inspections are accomplished immediately upon termination of the actual emergency. When unusual local conditions (utilization, type of mission, personnel, periods of inactivity, environmental conditions, etc.) dictate, it is the prerogative and responsibility of the Maintenance Officer to increase the scope and/or frequency of maintenance or inspection as necessary to ensure safe operation (TM 1-1500-328-23).

MAINTENANCE ACTIVITIES

The inspections prescribed by this checklist will be accomplished at specified phases by Aviation Maintenance Company (AMC) activities with assistance of Aviation Support Battalion (ASB) and Depot Maintenance activities when required. The inspection of the part/component is visual unless stated otherwise.

LIMITATIONS

The checklist does not contain instructions for repair, adjustment or other means of rectifying conditions. Neither does it contain special tolerances, limits or instructions for special troubleshooting to find causes for malfunctions. Such data will be obtained from the latest issue of the aircraft INSERT WITH APPLICABLE AIRCRAFT TM series Maintenance Manuals.

CHANGEOVER TO THE PHASED MAINTENANCE SYSTEM

Changeover shall be accomplished in accordance with instructions provided in . The requirements of this TM/TB must be accomplished prior to implementation of Phase 1 inspection requirements specified in this checklist.

PRE-INSPECTION MAINTENANCE TEST FLIGHT (MTF)

A pre-inspection MTF to duplicate non-hazardous equipment problems, determine unsatisfactory conditions, determine equipment operation problems, etc., is recommended prior to start of aircraft disassembly for phased maintenance inspection. The decision to perform the pre-inspection MTF, however, shall be the responsibility of the unit Maintenance Officer.

SPECIAL INSPECTIONS, CALENDAR INSPECTIONS AND LUBRICATION REQUIREMENTS

Special inspections, calendar inspections and lubrication requirements contained in INSERT WITH APPLICABLE AIRCRAFT TM and those listed on the aircraft's DA Form 2408-18 shall be reviewed and accomplished in accordance with the "inspection due" requirements specified in those documents.

TIME BETWEEN OVERHAUL (TBO) AND RETIREMENT LIFE ITEMS CHECK

Prior to start of the applicable phased maintenance inspection, a check will be made of components and their remaining operating hours prior to removal. The latest issue of the aircraft's INSERT WITH APPLICABLE AIRCRAFT TM and DA Form 2408-16 shall be referred to for a complete listing of components and their TBO and retirement life.

USING THE PHASED INSPECTION CHECKLIST

1. A new checklist shall be used each time phased maintenance is due on the aircraft. This checklist is arranged such that it can be separated by area and distributed to the maintenance crew. For use of the checklist refer to DA PAM 738-751
   a. Space is provided on each checklist form for entering the following data:

-1

FIGURE 192. Example of a stylesheet output for a General information work package (Phased Maintenance Checklist Manual Only) <pm-ginfowp>.

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34.1.5 Preventive maintenance services inspection general information work package ***<pms-ginfowp>***. The element ***<pms-ginfowp>*** identifies a Preventive Maintenance Services TM's general information work package.

a. The components are:

(1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end user. (see 32.3.2)

(2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package. (see 32.2)

(3) Scope ***<scope>*** (required). The element provides a brief statement of what is covered in the technical manual and includes verbatim text that is available through boilerplate enitites. Also appropriate information needs to be inserted through editable entities contained in the boilerplates. See Section 50 for further information on boilerplates. .(see 49.1.4.23)

(4) Preventive Maintenance Services Inspection General Information***<pms-geninfo>*** (required). The element provides required verbatim data for the general information for the preventive maintenance services inspection general information work package (see 34.1.5.1).

b. DTD fragment is graphically depicted.

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TM X-XXXX-XXX

SUSTAINMENT MAINTENANCE
PREVENTIVE MAINTENANCE SERVICES GENERAL INFORMATION WORK PACKAGE

SCOPE

The Preventive Maintenance Services Inspection Checklist work package contains complete requirements for a (INSERT SPECIFIC INSPECTION INTERVAL(S) HERE –*&pms-ginfowp.inspect-interval;)* here for the (INSERT THE AIRCRAFT MODEL – *&pms-ginfowp.aircraft;)*. It does not contain instructions for repair, adjustment, or other means of rectifying conditions, nor does it contain instruction for troubleshooting to find causes for malfunctioning. Specific tolerances, limits, etc., can be found in the applicable maintenance manuals. Use of the alphabetical index in the applicable manuals will facilitate locating the required information.

INSPECTION REQUIREMENTS

The inspection requirements contained in this work package are stated in such a manner as to establish when certain equipment is to be inspected and what conditions are desired/undesired. Compliance with the provisions outlined herein is required in order to ensure that latent defects are discovered and corrected before malfunctioning or serious trouble results. Inspection requirements are arranged, as nearly as possible, according to the manner in which they will be performed. The requirements are divided into groups and listed under area headings in the "How To Use This Manual" portion of this manual and (INSERT FIGURE NUMBER HERE – *&pms-ginfowp.inspect-fig;*).

INSPECTION INTERVALS

The (INSERT SPECIFIC INSPECTION INTERVAL(S) –*&pms-ginfowp.inspect-interval;)* inspection will be performed every (INSERT THE AIRCRAFT HOURS –*&pms-ginfowp.inspect-hrs;*) flight hours or (INSERT THE CALENDAR DAYS – *&pms-ginfowp.inspect-days;*) days, whichever comes first. The (INSERT THE AIRCRAFT HOURS – *&pms-ginfowp.inspect-hrs;*) will not be extended except in actual operational emergencies. In no case shall the aircraft intentionally be scheduled for a flight that will cause it to exceed the (INSERT THE AIRCRAFT HOURS – *&pms-ginfowp.inspect-hrs;*) inspection due time. The (INSERT THE CALENDAR DAYS – *&pms-ginfowp.inspect-days;*) interval is a full (INSERT THE NUMBER OF WEEKS – *&pms-ginfowp.inspect-weeks;*) weeks. That is, if a (INSERT THE CALENDAR DAYS – *&pms-ginfowp.inspect-days;* day is done on Tuesday, the next (INSERT THE CALENDAR DAYS – *&pms-ginfowp.inspect-days;* day inspection will not be due until (INSERT THE DAY OF WEEK – *&pms-ginfowp.inspect-days;* INSERT THE NUMBER OF WEEKS – *&pms-ginfowp.inspect-weeks;* weeks later.

SPECIFIC NON-INSTALLED EQUIPMENT ON AIRCRAFT

This work package may contain inspection requirements applicable to specific equipment not installed on your aircraft. Those requirements should be disregarded.

DA FORMS

DA Form 2408-13-1 will be used to record all deficiencies or shortcomings discovered during the (INSERT SPECIFIC INSPECTION INTERVAL(S) –*&pms-ginfowp.inspect-interval;)*. Use DA Form 738-751DA PAM 738-751) to properly complete this form.

SPECIAL INSTRUCTIONS

WARNING

Accidental actuation of aircraft power plant or hydraulic system (INSERT AIRCRAFT EQUIPMENT(S) – *&pms-ginfowp.warning-equip;)* may cause severe injury or death. Before starting inspection, aircraft safety check must be performed, if applicable IAW INSERT THE MAINTENANCE TM NUMBER *&pms-ginfowp.warning-tm-maint;)* (and all armament must be safetied, deactivated, and cleared INSERT THE ARMAMENT TM NUMBER *&pms-ginfowp.warning-tm-arm;)*.

The (INSERT SPECIFIC INSPECTION INTERVAL(S) –*&pms-ginfowp.inspect-interval;)* will not be exceeded except in actual operational emergencies. When operational emergencies require aircraft operation beyond the normal inspection due-time, a circled red X status symbol and an appropriate statement (to include authority) must be entered in Part I, Fault Information

-1

FIGURE 193. Preventive maintenance services inspection general information work package <pms-ginfowp> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT pms-ginfowp (wp.metadata?, wpidinfo, scope, pms-geninfo)>

<!ATTLIST pms-ginfowp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time)  #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

34.1.5.1 Preventive maintenance services inspection general information ***<pms-geninfo>***. The element ***<pms-geninfo>*** identifies a Preventive Maintenance Services TM's general information work package. The general information is verbatim and some of the text requires to be edited with the appropriate information. The verbatim text is available through boilerplate enitites. Also appropriate information needs to be inserted through editable entities contained in the boilerplates. See Section 50 for further information on boilerplates.

a. The components are:
(1) Preventive Maintenance Services Inspection Paragraph ***<pms-para>*** (required – one or more)(see 34.1.5.1.1)

b. DTD fragment is graphically depicted.

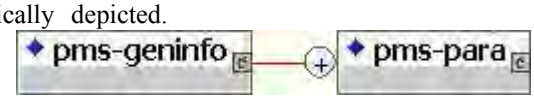

FIGURE 194. Preventive maintenance services inspection general information<pms-geninfo> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT pms-geninfo (pms-para+)>
<!ATTLIST pms-geninfo

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    frame             (yes | no)                      "no"
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:
(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **frame** – In frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**no**").
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

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34.1.5.1.1 Preventive maintenance services inspection paragraph ***<pms-para>***. The element ***<pms-geninfo>*** provides the narrative text required for the verbatim general information.

a. The components are:
   (1) Preventive Maintenance Services Inspection Paragraph ***<pms-para>*** (required – one or more)(see 34.1.5.1.1)

b. DTD fragment is graphically depicted.

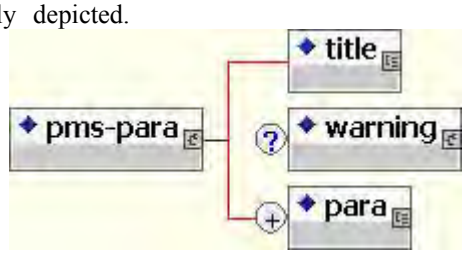

FIGURE 195. Standard general information work package <pms-ginfowp> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT pms-para (title, warning?, para+)>
```

d. No attributes.

34.1.5.2 Example of an XML document instance fragment and output for ***<pm-ginfowp>***. The XML instance and its stylesheet output for a General Information Work Package is provided below.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % ammo-tm "IGNORE">
<!ENTITY % usmc-tm "IGNORE">

b. Example of an XML document instance fragment using boilerplate entities for in the XML instance for Preventive Maintenance Services Inspection Checklist work package ***<pms-ginfowp>***. The boilerplate ***&pms-ginfowp;*** contain verbatim text to develop this work package. The general entity contains nested entities that contain editable text so the required data can be inserted for the work package. The formated example located after the XML instance displays what information should be inserted in the nested entities and provides the nested entity name in the editable boilerplate file. A list of the nested entities that are contain in the main boilerplate entity for ***&pms-ginfowp;***are below. These entities are located in the editboil.ent file of the boilerplate files. See Section 50 for further information on boilerplates.

<!ENTITY pms-ginfowp.aircraft "INSERT THE AIRCRAFT MODEL">
<!ENTITYpms-ginfowp.inspect-interval "INSERT THE INSPECTION INTERVAL">
<!ENTITYpms-ginfowp.inspect-hrs "INSERT THE AIRCRAFT HOURS">
<!ENTITY pms-ginfowp.inspect-days"INSERT THE CALENDAR DAYS">
<!ENTITY pms-ginfowp.inspect-weeks "INSERTTHE NUMBER OF WEEKS">
<!ENTITY pms-ginfowp.inspect-dayofweek "INSERTTHE DAY OF WEEK">
<!ENTITY pms-ginfowp.warning-equip "INSERT AIRCRAFTEQUIPMENT(S)">
<!ENTITY pms-ginfowp.warning-tm-maint ">
<extref docno="INSERTTHE MAINTENANCE TM NUMBER"/>
<!ENTITY pms-ginfowp.warning-tm-arm ' and all armament must be safetied,deactivated, and cleared '>
<extref docno="INSERT THE ARMAMENT TM NUMBER"/>
<!ENTITY pms-ginfowp.manhours "INSERT THE TOTAL NUMBER OF MAN-HOURS">
<!ENTITY pms-ginfowp.odc "INSERT THE ODC STATEMENT">
<!ENTITY pms-ginfowp.hazmat"INSERT THE HAZMAT STATEMENT">
<!ENTITY pms_or_pmi.eir.address "INSERTPROPONENT MAILING ADDRESS">

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```
<!ENTITY pms_or_pmi.eir.fax 'INSERT PROPONENT COMMERCIAL FAXNUMBER'>
<phone type="dsn"receive="fax"/> INSERT PROPONENT DSN FAX NUMBER or
<phone type="coml" receive="fax"/>
<!ENTITY pms_or_pmi.eir.email ">
<internet show.address="yes"/>
<email address="INSERT PROPONENT E-MAIL ADDRESS"/>
<pms-ginfowp airforce="no" army="no" deletewp="no" frame="no" marines="no"
navy="no" tocentry="2" wpno="GXXXXX-X-XXXX-XXX" wpseq="0001">
<wpidinfo>
<maintlvl level="sustain"/>
<title>Preventive Maintenance Services Inspection Checklist
</title>
</wpidinfo> &pms-ginfowp;
</pms-ginfowp>
```

c. Example of a stylesheet output for ***<pms-ginfowp>***. This formatted example of the Preventive Maintenance Services Inspection Checklist work package contains the name of the nested entity and what kind of information is required to edit the nested entity. **The example consist of only the first page of the work package.**

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TM X-XXXX-XXX

SUSTAINMENT MAINTENANCE
PREVENTIVE MAINTENANCE SERVICES GENERAL INFORMATION WORK PACKAGE

SCOPE

The Preventive Maintenance Services Inspection Checklist work package contains complete requirements for a (INSERT SPECIFIC INSPECTION INTERVAL(S) HERE –*&pms-ginfowp.inspect-interval;)* here for the (INSERT THE AIRCRAFT MODEL – *&pms-ginfowp.aircraft;)*. It does not contain instructions for repair, adjustment, or other means of rectifying conditions, nor does it contain instruction for troubleshooting to find causes for malfunctioning. Specific tolerances, limits, etc., can be found in the applicable maintenance manuals. Use of the alphabetical index in the applicable manuals will facilitate locating the required information.

INSPECTION REQUIREMENTS

The inspection requirements contained in this work package are stated in such a manner as to establish when certain equipment is to be inspected and what conditions are desired/undesired. Compliance with the provisions outlined herein is required in order to ensure that latent defects are discovered and corrected before malfunctioning or serious trouble results. Inspection requirements are arranged, as nearly as possible, according to the manner in which they will be performed. The requirements are divided into groups and listed under area headings in the "How To Use This Manual" portion of this manual and (INSERT FIGURE NUMBER HERE – *&pms-ginfowp.inspect-fig;*).

INSPECTION INTERVALS

The (INSERT SPECIFIC INSPECTION INTERVAL(S) –*&pms-ginfowp.inspect-interval;)* inspection will be performed every (INSERT THE AIRCRAFT HOURS –*&pms-ginfowp.inspect-hrs;*) flight hours or (INSERT THE CALENDAR DAYS – *&pms-ginfowp.inspect-days;*) days, whichever comes first. The (INSERT THE AIRCRAFT HOURS – *&pms-ginfowp.inspect-hrs;*) will not be extended except in actual operational emergencies. In no case shall the aircraft intentionally be scheduled for a flight that will cause it to exceed the (INSERT THE AIRCRAFT HOURS – *&pms-ginfowp.inspect-hrs;*) inspection due time. The (INSERT THE CALENDAR DAYS – *&pms-ginfowp.inspect-days;*) interval is a full (INSERT THE NUMBER OF WEEKS – *&pms-ginfowp.inspect-weeks;*) weeks. That is, if a (INSERT THE CALENDAR DAYS – *&pms-ginfowp.inspect-days;* day is done on Tuesday, the next (INSERT THE CALENDAR DAYS – *&pms-ginfowp.inspect-days;* day inspection will not be due until (INSERT THE DAY OF WEEK – *&pms-ginfowp.inspect-days;* INSERT THE NUMBER OF WEEKS – *&pms-ginfowp.inspect-weeks;* weeks later.

SPECIFIC NON-INSTALLED EQUIPMENT ON AIRCRAFT

This work package may contain inspection requirements applicable to specific equipment not installed on your aircraft. Those requirements should be disregarded.

DA FORMS

DA Form 2408-13-1 will be used to record all deficiencies or shortcomings discovered during the (INSERT SPECIFIC INSPECTION INTERVAL(S) –*&pms-ginfowp.inspect-interval;)*. Use DA Form 738-751DA PAM 738-751) to properly complete this form.

SPECIAL INSTRUCTIONS

WARNING

Accidental actuation of aircraft power plant or hydraulic system (INSERT AIRCRAFT EQUIPMENT(S) – *&pms-ginfowp.warning-equip;)* may cause severe injury or death. Before starting inspection, aircraft safety check must be performed, if applicable IAW INSERT THE MAINTENANCE TM NUMBER *&pms-ginfowp.warning-tm-maint;)* (and all armament must be safetied, deactivated, and cleared INSERT THE ARMAMENT TM NUMBER *&pms-ginfowp.warning-tm-arm;)*.

The (INSERT SPECIFIC INSPECTION INTERVAL(S) –*&pms-ginfowp.inspect-interval;)* will not be exceeded except in actual operational emergencies. When operational emergencies require aircraft operation beyond the normal inspection due-time, a circled red X status symbol and an appropriate statement (to include authority) must be entered in Part I, Fault Information

-1

FIGURE 196. Preventive maintenance services inspection checklist work package <pms-ginfowp>.

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35 OPERATOR INSTRUCTIONS CHAPTER .

35.1 Operator instructions ***<opim>***. Operator instructions chapter is to be prepared and subdivided into individual work packages that provide the operator of the weapon system/equipment with description and use of controls and indicators, operation of the weapon system/equipment under usual, unusual and emergency conditions, stowage and decal/plate guide instructions, and on-vehicle equipment loading instructions. Weapon system and equipment operator data is developed in narrative or tabular form, or by whatever method is most simple or effective to convey the specific TM application.

a. The components are:

(1) Chapter Title Page ***<titlepg>*** (required). The element provides the title page preceding an information chapter (see 49.1.1.1).

(2) Description and Use of Controls and Indicators Work Package ***<ctrlindwp>*** (required – one or more). The element provides the description and use of all system and equipment controls and indicators (see 35.2).

(3) Operation Under Usual Conditions Work Package ***<opusualwp>*** (required – one or more). The element provides the instructions to operate the weapon system/equipment and auxiliary equipment in all modes of operation (see 35.3).

(4) Operation Under Unusual Conditions Work Package ***<opunuwp>*** (required – one or more). The element provides step-by-step instructions for operation of the equipment and auxiliary equipment in all modes of operation under unusual conditions (see 35.4).

(5) Emergency Conditions Work Package ***<emergencywp>*** (optional – zero or more). The element provides instructions for temporarily adapting the equipment when a component or part of the equipment has failed under emergency conditions (see 35.5).

(6) Stowage and Decal/Data Plate Guide Work Package ***<stowagewp>*** (optional – zero or more). The element provides lists and illustrates the location of all applicable COEI, BII, AAL items, decals and data plates (see 35.6).

(7) On–Vehicle Equipment Loading Plan Work Package ***<eqploadwp>*** (optional – zero or more). The element provides a loading plan that is prepared by the contracting activity (see 35.7).

b. DTD fragment is graphically depicted.

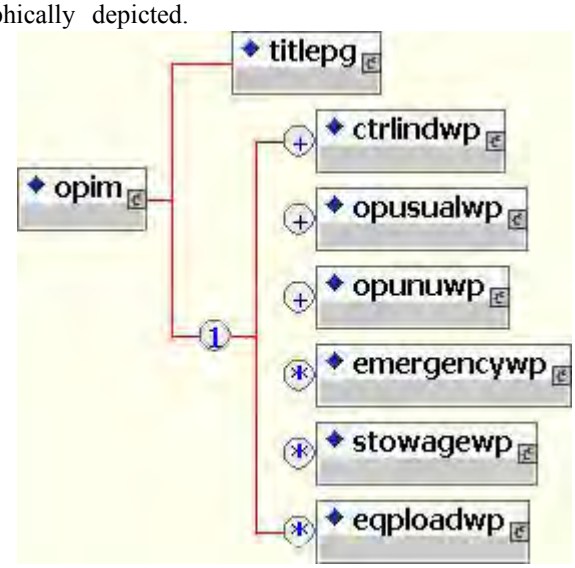

FIGURE 197. Operator instructions chapter <opim> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT opim (titlepg , (ctrlindwp+ , opusualwp+ , opunuwp+ , emergencywp*,
stowagewp* , eqploadwp*))>
<!ATTLIST opim

    assocfig  IDREFS                    #IMPLIED
    changeref IDREFS                    #IMPLIED
    chap-toc  (yes | no )              "yes"
    chngno    CDATA                     #REQUIRED
    comment   CDATA                     #IMPLIED
    delchlvl  CDATA                     #IMPLIED
    frame     (yes | no )               "yes"
    idref     IDREFS                    #IMPLIED
    inschlvl  CDATA                     #IMPLIED
    revno     CDATA                     #REQUIRED
    security  (uc | fouo | c | s | ts )  #IMPLIED
    skilltrk  CDATA                     #IMPLIED
    tocentry  (0 | 1 | 2)               "1"
>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **chap-toc** – Create a chapter work package table of contents (optional). Select either "**yes**" or "**no**" with a default value "**yes**".
(4) **chngno** – Change number (required) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(8) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(9) **inschlvl** – Insert change level (optional) (see 49.2.9).
(10) **revno** – Revision number (required) (see 49.2.9).
(11) **security** – Security classification (optional) (see 49.2.13).
(12) **skilltrk** – Skill level (optional) (see 49.2.2).
(13) **tocentry** – In the TM table of contents, the attribute indicates the indenture level. The possible selections are "**0**" (do not include), "**1**" (the highest level or 1st indenture level), or "**2**" (the second indentrue level) (default value is "**1**").

35.2 Description and use of controls and indicators work package ***<ctrlindwp>***. The description and use of controls and indicators work package provides the descriptions and use of all system or equipment controls and indicators. This work package is prepared for each equipment, assembly or control panel having controls and indicators. The controls and indicators are described using either a tabular option (see 35.2.1) or a narrative option (see 35.2.2).

a. The components are:

(1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end user (see 32.3.2).
(2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).
(3) Grouped alerts (optional – zero or more).
(a) Warnings ***<warning>***. The element provides an operation, procedure, or statement that if not performed properly may result in personal injury or death (see 41.1.1).
(b) Cautions ***<caution>***. The element provides information that identifies risk of damage to the equipment (see 41.1.2).

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(c) Notes ***<note>***. The element provides highlights to essential procedures, conditions, or statements or conveys important instructional data to the user (see 41.1.3).

(4) General or Introductory Information ***<geninfo>*** (optional). The element provides titled and subtitled paragraphs giving general or introductory information (see 49.1.4.12).

(5) Description and use of controls and indicators work package requires either a tabular form using ***<intro>*** and ***<ctrlindtab>*** or a narrative description using ***<figure>*** and ***<ctrlinddesc>***.

(a) Introduction ***<intro>*** (required). The element provides an introductory section for the Description of Controls and Indicators Tabular Form when it is used in a description and use of controls and indicators work package (see 49.1.4.15).

(b) Description of Controls and Indicators Tabular Form ***<ctrlindtab>*** (required – one or more). The element provides a description of each control and indicator in a tabular form (see 35.2.1).

(c) Illustration ***<figure>*** (required – one or more). The element provides the associated figure for each control and indicator (see 44.1.1) contained in the description of controls and indicators narrative description.

(d) Description of Controls and Indicators narrative form ***<ctrlinddesc>*** (required – one or more). The element provides paragraphs describing each control or indicator shown in the figure.

b. DTD fragment is graphically depicted.

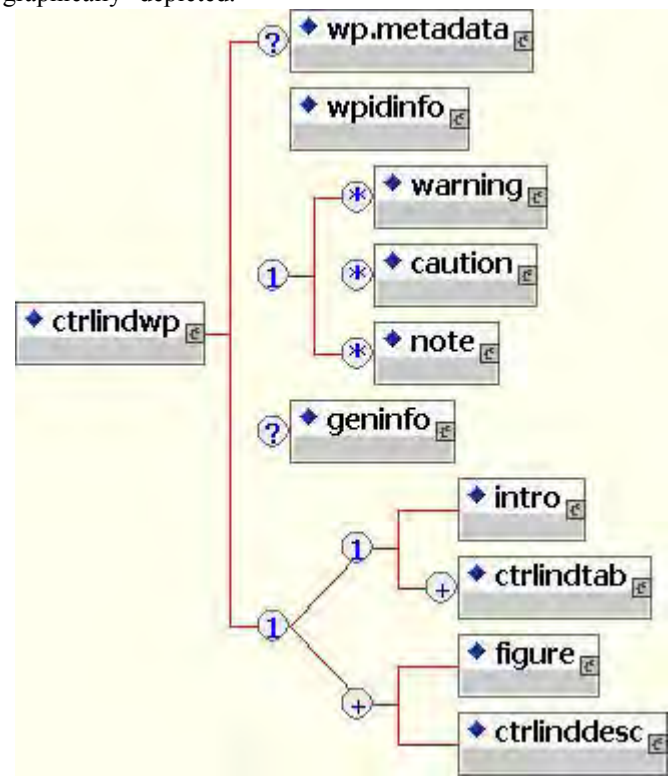

FIGURE 198. Descriptions and use of controls and indicators work package <ctrlindwp> DTD hierarchy. .

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c. DTD fragment for ***<ctrlindwp>***:

```
<!ELEMENT ctrlindwp (wp.metadata?, wpidinfo, warning*, caution*, note*,
 geninfo?,((intro , ctrlindtab+) | (figure, ctrlinddesc)+))>
<!ATTLIST ctrlindwp

    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

35.2.1 Description of controls and indicators tabular form ***<ctrlindtab>***. This work package option describes each control and indicator in a tabular format, starting with a brief introduction ***<intro>*** that identifies the basic system, area, or other breakdown, followed by one or more controls and indicators tables ***<ctrlindtab>***. The number of controls and indicators tables ***<ctrlindtab>*** or standard information required is dependent on several factors, including but not limited to system complexity, different users (crew members/stations) or configuration differences. A standard table header ***<std.table.header>*** is required for the tabular form. This table heading is accessible as boilerplate and can be referenced using the general entity ***%ctrlindwp.ctrlindtab.std-header;*** (see 50).

a. The components are:
(1) Title ***<title>*** (required). The element provides the title for the work package (see 49.1.1.3).
(2) Illustration ***<figure>*** (required). The element provides a graphic of the control or indicator that is being described (see 44.1.1). (A second figure is (optional) in the table but only after the element ***<ctrlindrow>***. If a second ***<figure>*** is used, it is followed by at least one ***<ctrlindrow>***.
(3) Standard table header ***<std.table.header>*** (required). The element provides the table header for the controls and indicators table. The text for the header can be manually inserted or the boiler plate entity ***%ctrlindwp.ctrlindtab.std-header;*** can be inserted through the MIL–STD–2361 DTD and it provides the tags and text for the header for the description of controls and indicators table (see 42.3.1).
(4) Controls and Indicators Group ***<ctrlindrow>*** (required). The element provides a table row for detailed information (see 35.2.1.1).

b. DTD fragment is graphically depicted.

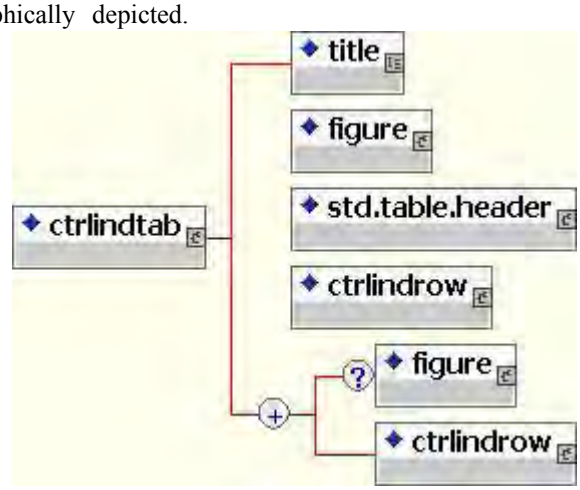

FIGURE 199. Description of controls and indicators tabular form<ctrlindtab> DTD hierarchy.

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c. DTD fragment for ***<ctrlindtab>***:

```
<!ELEMENT ctrlindtab (title, figure, std.table.header, ctrlindrow,
(figure?, ctrlindrow)+)>
<!ATTLIST ctrlindtab

    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1"
>
```

d. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

35.2.1.1 Controls and indicators group ***<ctrlindrow>***. Controls and indicators group provides detailed information. The element is similar to a "row" in a structural table.

a. The components are:

(1) Key ***<key>*** (required). The element provides the identification of a key or callout that locates a control or indicator shown on the related figure (see 35.2.1.1.1).
(2) Controls and Indicators Entry ***<ctrlind>*** (required). The element provides the name of the control or indicator(see 49.1.4.3).
(3) Note ***<note>*** (optional – zero or more). The element provides a procedure, condition, or statement that is important enough to highlight as a note (see 41.1.3).
(4) Function ***<function>*** (required). The element provides a description of the controls or indicator function.

b. DTD fragment is graphically depicted.

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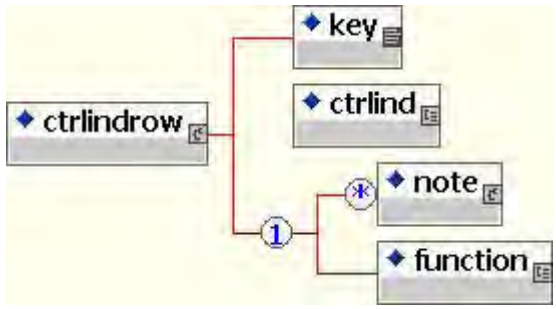

FIGURE 200. Controls and indicators group <ctrlindrow> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT ctrlindrow (key, ctrlind, (note*, function))>
<!ATTLIST ctrlindrow

    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

35.2.1.1.1 Key ***<key>***. Key provides the identification of a key or callout that locates a control or indicator shown on the related figure. The element is similar to a "cell" in a structural table and is entered in column one in the controls and indicators table.

a. The components are:

(1) Parsable data (#PCDATA) (required). The element provides the text for the TM number (see 9.3.2).

b. DTD fragment:

```
<!ELEMENT key (#PCDATA)>
```

c. No attributes.

35.2.1.1.2 Controls and indicators entry ***<ctrlind>***. Controls and indicators entry provides the name of the control or indicator and is discussed in 49.1.4.3.

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35.2.1.1.3 Function ***<function>***. Function provides a description of the function of the control or indicator. The element is similar to a "cell" in a structural table and is entered in column three in the controls and indicators table.

a. The components are:

(1) Parsable data (#PCDATA) (optional – zero or more). The element provides the text for the TM number (see 9.3.2).

(2) Emphasis ***<emphasis>*** (optional – zero or more). The element is used to emphasize the text of the entry (see 49.1.3.1).

(3) Subscript ***<subscrpt>*** (optional – zero or more). The element is used to format the text as subscript (see 49.1.3.4).

(4) Superscript ***<supscrpt>*** (optional – zero or more). The element is used to format the text as superscript (see 49.1.3.5).

(5) Cross Reference ***<xref>*** (optional – zero or more). The element is used to reference the work package sequence number, figure, table, step(s), etc. (see 46.2.2).

(6) External Reference ***<extref>*** (optional – zero or more). The element is used to reference to external document information (see 46.2.1).

(7) Linking ***<link>*** (optional – zero or more). The element provides a capability to reference internal or external targets (see 46.2.3).

(8) Content Sensitive Help ***<help.info>*** (optional – zero or more). The element is used for "help information" about the technical data used in frame-based manuals (see 48.3.2.4).

(9) Index Marker Reference ***<indxref>*** (optional – zero or more). The element establishes a document location and index text to be referenced within the alphabetical index.(see 31.4.2.2.3).

(10) Term ***<term>*** (optional – zero or more). The element provides a word, phrase, acronym, symbol, or abbreviation in a definition and legend list (see 49.1.2.3.1.1).

(11) Item Definition ***<term.def>*** (optional – zero or more). The element provides a wrapper for a term and corresponding definition (see 49.1.2.3.1).

(12) Figure Callout ***<callout>*** (optional – zero or more). The element is used for a figure reference and callout reference, number, letter, or symbol appearing in the figure (see 46.2.4.1).

(13) Footnote ***<ftnote>*** (optional – zero or more). The element contains the footnote information which is referenced in the document with footnote reference (see 45.1.1).

(14) Footnote Reference ***<ftnref>*** (optional – zero or more). The element is the reference in the text to the footnote (see 45.1.2).

(15) Controls and Indicators Entry ***<ctrlind>*** (optional – zero or more). The element provides the name of the control or indicator (see 49.1.4.3).

(16) Reading On ***<ctrlind-val>*** (optional – zero or more). – The element provides a reading from a control or indicator (see 49.1.4.4).

(17) Department of Defense Ammunition Code ***<dodac>*** (optional – zero or more). The element provides identification of an ammunition type (see 49.1.4.5).

(18) Lubricant ***<lubricant>*** (optional – zero or more). The element provides identification of a lubricant (see 49.1.4.16).

(19) Symbol ***<symbol>*** (optional – zero or more). The element provides a graphic symbol not found in standard ISO character sets that is inserted as a graphic in text (see 44.3.1).

(20) Torque Value or Limit ***<torque>*** (optional – zero or more). The element provides a torque value or limit embedded in the text or table entry (see 49.1.4.24).

(21) Voltage Measurement ***<voltage>*** (optional – zero or more). The element provides the identification of a critical voltage measurement (see 49.1.4.25).

(22) Null ***<null>*** (optional – zero or more). The element specifies that it contains no content and marking is identified by it's attribute (see 49.1.3.2).

(23) Text Change Information ***<change>*** (optional – zero or more). The element provides the scope and type of changed information that is indicated by being enclosed within start and end tag (see 49.1.3.7).

b. DTD fragment is graphically depicted.

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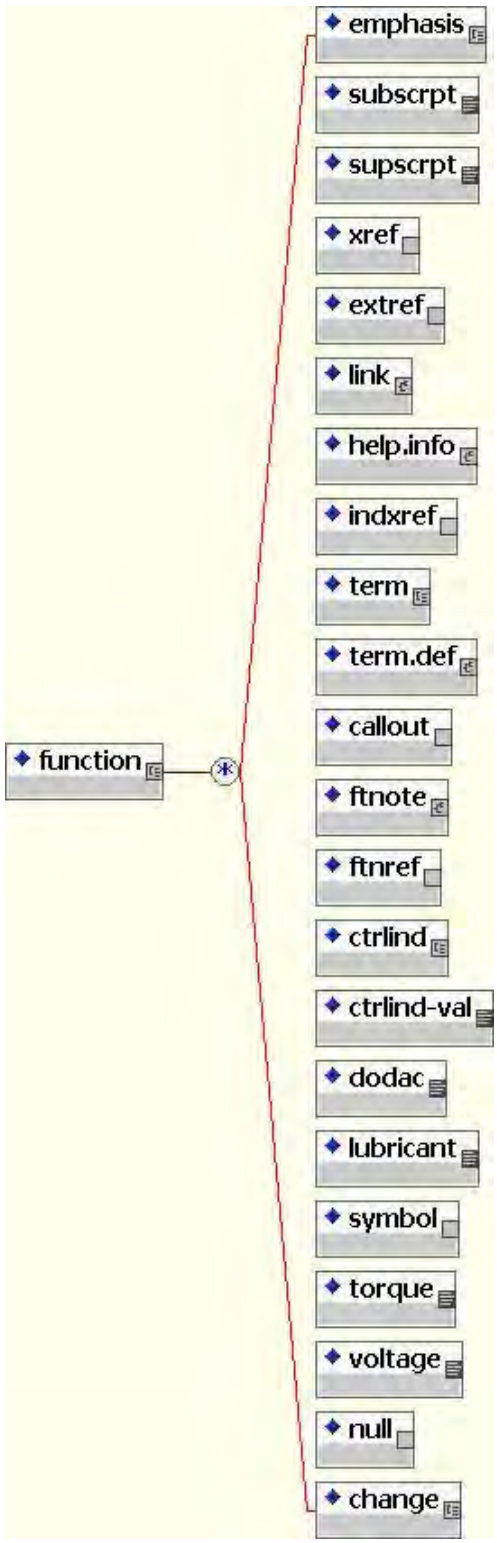

FIGURE 201. Function <function> DTD hierarchy.

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c. DTD fragment for ***<function>***:

```
<!ELEMENT function (#PCDATA | emphasis | subscrpt | supscrpt | xref |
 extref | link | help.info | indxref | term | term.def | callout |
 ftnote | ftnref | ctrlind | ctrlind-val | dodac | lubricant | symbol |
 torque | voltage | null | change)*>
<!ATTLIST function

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

35.2.2 Description of controls and indicators narrative form option ***<ctrlinddesc>***. This work package option provides a narrative approach to describe each control and indicator, starting with a figure ***<figure>*** illustrating the control or indicator being described, and followed by one or more paragraphs ***<ctrlinddesc>*** describing each control/indicator shown in the figure. More than one ***<figure>*** and ***<ctrlinddesc>*** grouping may be used to improve user understanding.

a. The components are:

(1) Title ***<title>*** (optional – zero or more). The element provides the title of the control or indicator that is being describe (see 49.1.1.3).
(2) Subtitle ***<stitle>*** (optional – zero or more). The element provides a subordinate title to the paragraph description (see 49.1.1.2).
(3) Note ***<note>*** (optional – zero or more). The element provides a procedure, condition, or statement that is important enough to highlight as a note (see 41.1.3).
(4) Reduced Paragraph ***<trim.para>*** (required – one or more). The element provides the same usage as a paragraph element, but contains a reduce content model (see 49.1.1.7).

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b. DTD fragment:

```
<!ELEMENT ctrlinddesc (title, (subtitle?, note*, trim.para+)+)>
<!ATTLIST ctrlinddesc

    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. DTD fragment is graphically depicted.

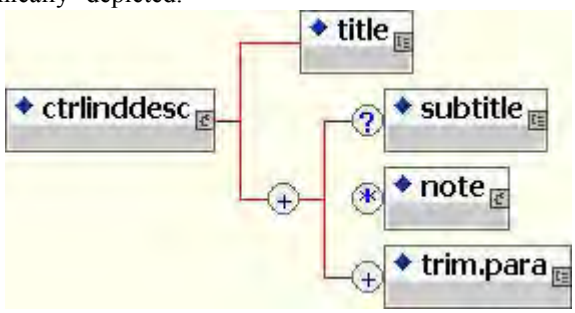

FIGURE 202. Description of controls and indicators narrative form option <ctrlinddesc> DTD hierarchy.

d. Common attributes:
   (1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
   (2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (4) **comment** – Change information (optional) (see 49.2.9).
   (5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (8) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (9) **security** – Security classification (optional) (see 49.2.13).
   (10) **skilltrk** – Skill level (optional) (see 49.2.2).

35.2.3 XML document instance fragment and output for ***<ctrlindwp>*** using the tabular option. The XML instance and its stylesheet output for a ***<ctrlindwp>*** using the tabular option is provided below.

a. Example of an XML document instance fragment for ***<ctrlindwp>*** using the tabular option:

<ctrlindwp wpno="o00001-9-2320-365">
<wpidinfo>
<maintlvl level="field"/>
<title>Description and Use of Operator's Controls and Indicators
</title>
</wpidinfo>
<intro>
<para0>
<title>Introduction

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</title>
*<para>*The following tables and illustrations provide the description and use of the controls and indicators pertaining to the instrument and auxiliary panels,center console, steering column, floor-mounted, door-mounted, seat, exterior, and M1079 van controls, and special purpose kit controls and indicators. Some switch locations on the auxiliary panel may be blank depending on the model of your vehicle controls.
</para>
</para0>
</intro>
<ctrlindtab>
*<title>*Instrument Panel Controls and Indicators
</title>
<figure application="both" figtype="normal-page" id="o00001-9-2320-365-1">
*<title>*Instrument Panel Controls and Indicators
</title>
<graphic boardno="o00001-9-2320-365_1>
</graphic>
</figure>
<std.table.header cols="3">
<std.table.row>
*<std.table.entry>*Key
</std.table.entry>
*<std.table.entry>*Control/Indicator
</std.table.entry>
*<std.table.entry>*Function
</std.table.entry>
</std.table.row>
</std.table.header>
<ctrlindrow>
*<key>*1.
</key>
*<ctrlind>*Radiator Fan Off Switch.
</ctrlind>
*<function>*When positioned to on, radiator fan off switch will illuminate to indicate the radiator fan is disabled. Radiator fan off switch will remain in the off position and not illuminated, unless otherwise directed.
</function>
</ctrlindrow>
<ctrlindrow>
*<key>*2.
</key>
*<ctrlind>*Lamp Test Switch.
</ctrlind>
*<function>*Tests the lights on high engine temperature and TRANS OIL TEMP indicators on Lighted Indicator Display.
</function>
</ctrlindrow>
<ctrlindrow>
*<key>*3.
</key>>
*<ctrlind>*Ether Start Switch.
</ctrlind>
*<function>*Injects ether into engine intake system to assist with cold weather starting when switch is pressed.
</function>

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Check the source to verify that this is the current version before use.

```
</ctrlindrow>
<ctrlindrow>
<key>4.
</key>
<ctrlind>FRONT BRAKE AIR Pressure Gage.
</ctrlind>
<function>Shows air pressure (in psi) available to operate front brakes.
Normal air pressure range is 65-120 psi (448-827 kPa).
</function>
</ctrlindrow>
<ctrlindrow>
<key>5.
</key>
<ctrlind>Lighted Indicator Display.
</ctrlind>
<function>Indicator lights to indicate operating charac-
teristics of the vehicle
<xref figid="o00001-9-2320-365-5"> shows all indicators on the
Lighted Indicator Display.
</function>
</ctrlindrow>
<key>6.
</key>
<ctrlind>Oil Press Gage.
</ctrlind>
<function>
<xref figid="o00001-9-2320-365-5"/> Gage shows engine oil pressure (in psi).
Normal oil pressure range is 15-80 psi (103-552 kPa.
</function>
</ctrlindrow>
</ctrlindtab>
</ctrlindwp>
```

b. Example of an XML document instance fragment for ***<ctrlindwp>*** using the tabular option but replacing the standard heading for the control and indicator table with the boiler plate entity ***%ctrlindwp.ctrlindtab.std-header ;***:

```
<ctrlindwp wpno="o00001-9-2320-365">
<wpidinfo>
<maintlvl level="operator"/>
<title>Description and Use of Operator's Controls and Indicators
</title>
</wpidinfo>
<intro>
<para0>
<title>Introduction
</title>
<para>The following tables and illustrations provide the description
and use of the controls and indicators pertaining to the instrument
and auxiliary panels, center console, steering column, floor-mounted,
door-mounted, seat, exterior, and M1079 van controls, and special purpose
kit controls and indicators. Some switch locations on the auxiliary
panel may be blank depending on the model of your vehicle controls.
</para>
</para0>
</intro>
<ctrlindtab>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<title>Instrument Panel Controls and Indicators
</title>
<figure application="both" figtype="normal-page" id="o00001-9-2320-365-1">
<title>Instrument Panel Controls and Indicators
</title>
<graphic boardno="o00001-9-2320-365_1"/>
</figure>%ctrlindwp.ctrlindtab.std-header;
<ctrlindrow>
<key>1.
</key>
<ctrlind>Radiator Fan Off Switch.
</ctrlind>
<function>When positioned to on, radiator fan off switch will illuminate to indicate the radiator fan is disabled. Radiator fan off switch will remain in the off position and not illuminated, unless otherwise directed.
</function>
</ctrlindrow>
<ctrlindrow>
<key>2.
</key>
<ctrlind>Lamp Test Switch.
</ctrlind>
<function>Tests the lights on high engine temperature and TRANS OIL TEMP indicators on Lighted Indicator Display.
</function>
</ctrlindrow>
<ctrlindrow>
<key>3.
</key>>
<ctrlind>Ether Start Switch.
</ctrlind>
<function>Injects ether into engine intake system to assist with cold weather starting when switch is pressed.
</function>
</ctrlindrow>
<ctrlindrow>
<key>4.
</key>
<ctrlind>FRONT BRAKE AIR Pressure Gage.
</ctrlind>
<function>Shows air pressure (in psi) available to operate front brakes. Normal air pressure range is 65-120 psi (448-827 kPa).
</function>
</ctrlindrow>
<ctrlindrow>
<key>5.
</key>
<ctrlind>Lighted Indicator Display.
</ctrlind>
<function>Indicator lights to indicate operating characteristics of the vehicle.
<xref figid="o00001-9-2320-365-5"/> shows all indicators on the Lighted Indicator Display.
</function>
</ctrlindrow>
</ctrlindtab>

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</ctrlindwp>

c. Page-based TM stylesheet output example for ***<ctrlindwp>*** using the tabular option:

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OPERATOR MAINTENANCE

DESCRIPTION AND USE OF OPERATOR'S CONTROLS AND INDICATORS

INTRODUCTION

The following tables and illustrations provide the description and use of the controls and indicators pertaining to the instrument and auxiliary panels, center console, steering column, floor-mounted, door-mounted, seat, exterior, and M1079 van controls, and special purpose kit controls and indicators. Some switch locations on the auxiliary panel may be blank depending on the model of your vehicle.controls.

Table 1. Instrument Panel Controls and Indicators

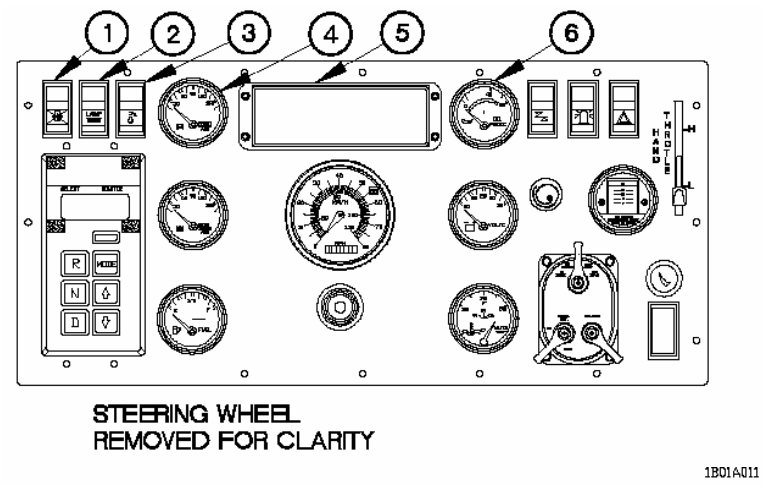

Figure 1. Instrument Panel Controls and Indicators

Key	Control/Indicator	Function
1.	Radiator Fan Off Switch.	When positioned to on, radiator fan off switch will illuminate to indicate the radiator fan is disabled. Radiator fan off switch will remain in the off position and not illuminated, unless otherwise directed.
2.	Lamp Test Switch.	Tests the lights on high engine temperature and TRANS OIL TEMP indicators on Lighted Indicator Display.
3.	Ether Start Switch.	Injects ether into engine intake system to assist with cold weather starting when switch is pressed.
4.	FRONT BRAKE AIR Pressure Gage.	Shows air pressure (in psi) available to operate front brakes. Normal air pressure range is 65-120 psi (448-827 kPa).
5.	Lighted Indicator Display.	Indicator lights to indicate operating characteristics of the vehicle. shows all indicators on the Lighted Indicator Display.

END OF WORK PACKAGE

FIGURE 203. Example of a page-based TM stylesheet output for <ctrlindwp> using the tabular option.

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Check the source to verify that this is the current version before use.

35.2.4 XML document instance fragment and output for ***<ctrlindwp>*** using the narrative option. The XML instance and its stylesheet output for a ***<ctrlindwp>*** using the narrative option is provided below.

a. Example of an XML document instance fragment for ***<ctrlindwp>*** using the narrative option:

```
<ctrlindwp wpno="o00001-9-2320-365">
<wpidinfo>
<maintlvl level="field"/>
<title>Description and Use of Operator's Controls and Indicators
</title>
</wpidinfo>
<figure id="o00001-9-2320-365-1">
<title>Instrument Panel Controls and Indicators
</title>
<graphic boardno="o00001-9-2320-365_1">
</graphic>
</figure>
<ctrlinddesc>
<title>Instrument Panel Controls and Indicators
</title>
<trim.para>When positioned to on, the RADIATOR FAN OFF SWITCH
<callout assocfig="o00001-9-2320-365-1" label="1"/> will illuminate to indicate the
radiator fan is disabled. The radiator fan off switch will remain in
the off position and not illuminated, unless otherwise directed.
</trim.para>
<trim.para>LAMP TEST SWITCH
<callout assocfig="o00001-9-2320-365-1" label="2"/> tests the lights on high engine
temperature and TRANS OIL TEMP indicators on Lighted Indicator Display.
</trim.para>
<trim.para>ETHER START SWITCH
<callout assocfig="o00001-9-2320-365-1" label="3"/> injects ether into engine intake
system to assist with cold weather starting when switch is pressed.
</trim.para>
<trim.para>FRONT BRAKE AIR PRESSURE GAGE
<callout assocfig="o00001-9-2320-365-1" label="4"/> shows air pressure
(in psi) available to operate front brakes. Normal air
pressure range is 65-120 psi (448-827 kPa).
</trim.para>
<trim.para>LIGHTED INDICATOR DISPLAY
<callout assocfig="o00001-9-2320-365-1" label="5"/> lights to indicate
operating characteristics of the vehicle.
</trim.para>
<trim.para>OIL PRESS GAGE
<callout assocfig="o00001-9-2320-365-1" label="6"/> shows engine oil pressure (in
psi). Normal oil pressure range is 15-80 psi (103-552 kPa).
</trim.para>
</ctrlinddesc>
</ctrlindwp>
```

b. Page-based TM stylesheet output example for ***<ctrlindwp>*** using the narrative option:

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OPERATOR MAINTENANCE

DESCRIPTION AND USE OF OPERATOR'S CONTROLS AND INDICATORS

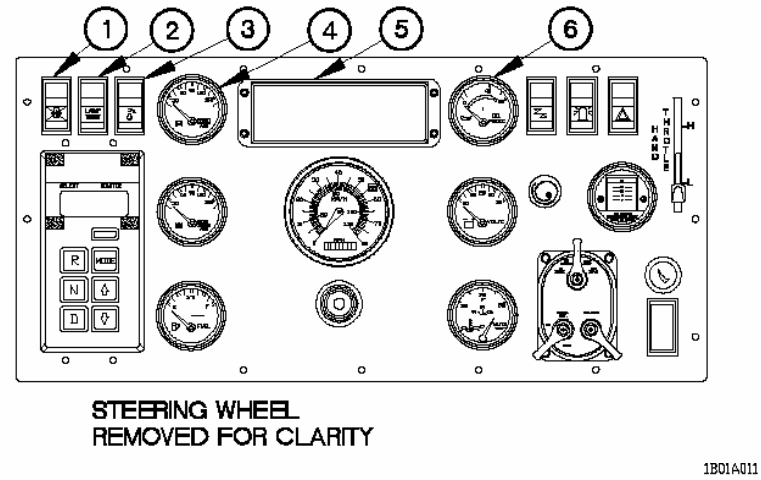

Figure 1. Instrument Panel Controls and Indicators

INSTRUMENT PANEL CONTROLS AND INDICATORS

When positioned to on, the RADIATOR FAN OFF SWITCH (Figure 1 , Item 1) will illuminate to indicate the radiator fan is disabled. The radiator fan off switch will remain in the off position and not illuminated, unless otherwise directed.

LAMP TEST SWITCH (Figure 1 , Item 2) tests the lights on high engine temperature and TRANS OIL TEMP indicators on Lighted Indicator Display

ETHER START SWITCH (Figure 1 , Item 3) injects ether into engine intake system to assist with cold weather starting when switch is pressed.

FRONT BRAKE AIR PRESSURE GAGE (Figure 1 , Item 4) shows air pressure (in psi) available to operate front brakes. Normal air pressure range is 65-120 psi (448-827 kPa).

LIGHTED INDICATOR DISPLAY (Figure 1 , Item 5) lights to indicate operating characteristics of the vehicle.

OIL PRESS GAGE (Figure 1 , Item 6) shows engine oil pressure (in psi). Normal oil pressure range is 15-80 psi (103-552 kPa).

END OF WORK PACKAGE

FIGURE 204. Example of a page-based TM stylesheet output for <ctrlindwp> using the narrative option.

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35.3 Operation under usual conditions work package ***<opusualwp>*** . The operation under usual conditions work package provides instructions to operate the weapon system/equipment and auxiliary equipment in all modes of operation, including instructions to ensure proper grounding of the equipment. Any combination of control settings that will create a hazard to personnel or cause damage to equipment are preceded by a warning or caution.

a. The components are:

(1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end use (see 32.3.2).

(2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).

(3) Work Package Initial Setup ***<wpinfo>*** (required). The element provides lists of information required by the technician so the tools, test equipment, references, parts, and other items needed to complete the tasks can be obtained (see 32.5).

(4) Grouped alerts. May have one or more of the following alerts.

(a) Warnings ***<warning>*** (optional – zero or more). The element provides an operation, procedure, or statement that if not performed properly may result in personal injury or death (see 41.1.1).

(b) Cautions ***<caution>*** (optional – zero or more). The element provides information that identifies risk of damage to the equipment (see 41.1.2).

(c) Notes ***<note>*** (optional – zero or more). The element provides highlights to essential procedures, conditions, or statements or conveys important instructional data to the user (see 41.1.3).

(5) General or Introductory Information ***<geninfo>*** (optional – zero or more). The element provides titled and subtitled paragraphs giving general or introductory information (see 49.1.4.12).

(6) Operational Tasks ***<opertsk>*** (required – one or more). The element provides operational tasks required in the operations under usual conditions work package (see 35.3.1).

b. DTD fragment is graphically depicted.

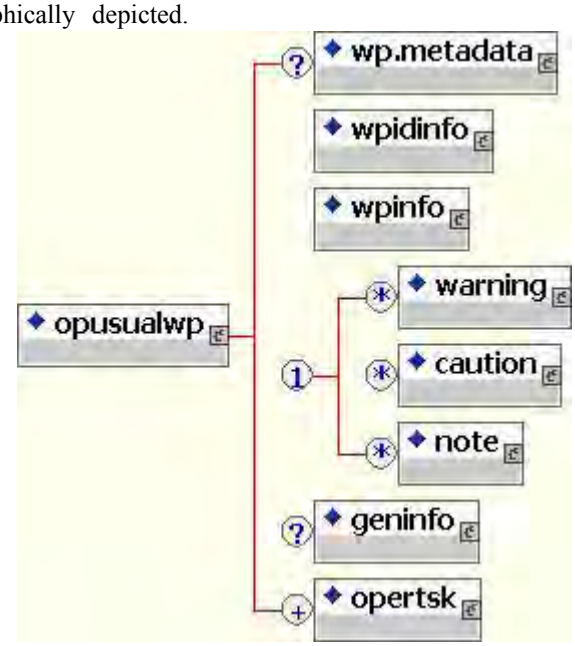

FIGURE 205. Operation under usual conditions work package <opusualwp> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT opusualwp (wp.metadata?, wpidinfo, wpinfo, warning*,
 caution*, note*, geninfo?, opertsk+)>
<!ATTLIST opusualwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

35.3.1 Operational task under usual conditions ***<opertsk>*** . The operational task under usual conditions ***<opertsk>*** element contains operator procedures on "how to" use the equipment, such as start, operate, place in standby, or shutdown the equipment. Each operation under usual condition task has only one specific task identified. If the work package requires multiple tasks, insert another ***<opertsk>***. Each operational task ***<opertsk>*** requires one of the following specific operational tasks, as defined in MIL-STD-40051-1/-2. Each specific tag has similar markup requirements for tasks (see 33.1PROCEDURES/TASKS for details on tagging.)

a. The components are:
(1) Security Measures For Electronic Data ***<secref>*** (optional – zero or more). The element provides information about the work package data and usually not used or seen by the end use (see 35.3.1.1).
(2) Sitting Requirements ***<site>*** (optional – zero or more). The element provides an operational task for site requirements that is considered prior to siting (see 35.3.1.2).
(3) Shelter Requirements ***<shelter>*** (optional – zero or more). The element provides an operational task containing the shelter requirements for equipment normally housed in a permanent or semi-permanent shelter (see 35.3.1.3).
(4) Assembly And Preparation For Use ***<prepforuse>*** (optional – zero or more). The element provides a maintenance task that is used for items that are unpacked, disassembled or removed from an assembly, subassembly or component (see 35.3.1.4).
(5) Initial Adjustments, Before Use And Self-Test ***<initial>*** (optional – zero or more). The element provides an operational task that the operator performs before putting equipment in operation (see 35.3.1.5).
(6) Operating Procedures ***<oper>*** (optional – zero or more). The element provides an operational task containing all procedures to start the equipment, operate the equipment, place the equipment in standby, or shutdown the equipment (see 35.3.1.6).
(7) Operating Auxiliary Equipment Procedures ***<operaux>*** (optional – zero or more). The element provides an operational task containing procedures to start the auxiliary equipment, operate it, place it in standby or shutdown (see 35.3.1.7).
(8) Preparation For Movement ***<prepmove>*** (optional – zero or more). The element provides an operational task containing procedures for preparing the equipment if required to move (see 35.3.1.8).

b. DTD fragment is graphically depicted.

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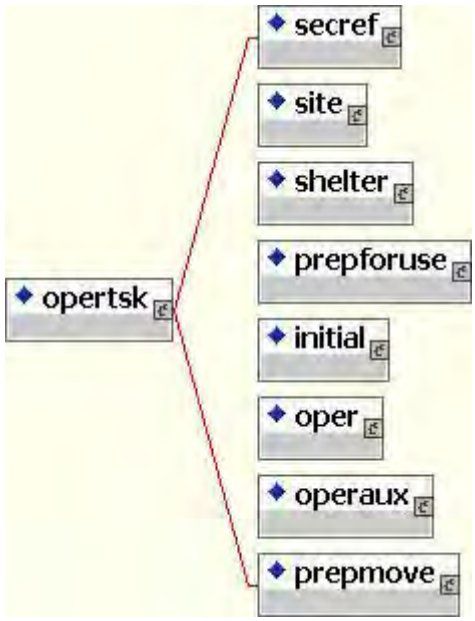

FIGURE 206. Operational task under usual conditions <opertsk> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT opertsk (secref | site | shelter | prepforuse |
 initial | oper | operaux | prepmove)>
<!ATTLIST opertsk

    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

35.3.1.1 Security measures for electronic data ***<secref>***. Security Measures for Electronic Data is an operational task. The task consists of instructions for handling, loading, scrubbing, overwriting, or unloading classified electronic data under usual or unusual conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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a. The components are: (see 33.1PROCEDURES/TASKS for details on tagging.)

35.3.1.2 Siting requirements ***<site>***. Siting requirements is an operational task for site requirements that is considered prior to siting. Includes overall site location, power sources, terrain requirements, and other similar considerations.

a. The components are: (see 33.1PROCEDURES/TASKS for details on tagging.)

35.3.1.3 Shelter requirements ***<shelter>***. Shelter Requirements is an operational task containing the shelter requirements for equipment normally housed in a permanent or semi-permanent shelter. The element includes requirements for dimensions, floor loading, layout, power or environmental conditions and other similar considerations. Shelter requirements does not apply to trucks, vans or transportable shelters.

a. The components are: (see 33.1)

35.3.1.4 Preparation for use ***<prepforuse>***. Preparation for use is a maintenance task that is used for items that are unpacked, disassembled or removed from an assembly, subassembly or component.

a. The components are: (see 33.1).

35.3.1.5 Adjustments, before use, daily checks, and self-test ***<initial>***. Adjustments, before use, daily checks, and self-test is an operational task for specification of routine checks, self-test, or adjustments that the operator performs before putting equipment in operation.

a. The components are: (see 33.1)

35.3.1.6 Operating procedures ***<oper>***. Operating procedures is an operational task containing all procedures to start the equipment, operate the equipment, place the equipment in standby, or shutdown the equipment. The element also includes the operating procedure for auxiliary equipment required to operate or support the primary equipment.

a. The components are: (see 33.1)

35.3.1.7 Operating auxiliary equipment ***<operaux>***. Operating auxiliary equipment is an operational task containing procedures to start the auxiliary equipment, operate it, place it in standby or shutdown. If procedures are in another TM, this paragraph may make reference to that TM for operating procedures. The elements external reference ***<extref>*** (see 46.2.1) and link ***<link>*** (see 46.2.3) are used for external reference.

a. The components are: (see 33.1)

35.3.1.8 Preparation for movement ***<prepmove>***. Preparation for movement is an operational task containing procedures for preparing the equipment if required to move.

a. The components are: (see 33.1)

35.3.2 XML document instance fragment and output for ***<opusualwp>*** with a single task. The XML instance and its stylesheet output for a ***<opusualwp>*** with a single task that has multiple procedures, is provided below.

a. Example of an XML document instance fragment for ***<opusualwp>***:

```
<opusualwp wpno="o00019-9-2320-365">
<wpidinfo>
<maintlvl level="field"/>
<title>M1079 Van Preparation For Movement
</title>
</wpidinfo>
<wpinfo>
<null insert="none"/>
</wpinfo>
<opertsk id="o00019-9-2320-365-3">
<prepmove id="o00019-9-2320-365-4">
<title>M1079 Weight Distibution
</title>
<proc id="o00019-9-2320-365-6">
<title>M1079 Weight Distibution
</title>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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<warning>
<warning.group>
<trim.para>Heavy objects/loads, such as tool boxes and heavy parts, must always be carried on the floor with the weight distributed as equally as possible between left and ride sides of M1079 van. Failure to comply decreases the stability of the M1079 van and will increase the likelihood of a rollover.
</trim.para>
<trim.para>Heavy cabinets must always be mounted as low as possible with the weight distributed as equally as possible between left and right sides of M1079 van. Remember to consider the weight of the items that will be stored in the cabinets. Failure to comply decreases the stability of the M1079 van and will increase the likelihood of a rollover.
</trim.para>
<trim.para>Always keep in mind, when placing items inside the M1079 van, that heavier items must always be positioned as low as possible and the weight distributed as equally as possible between left and right sides of M1079 van. Failure to comply decreases the stability of the M1079 van and will increase the likelihood of a rollover.
</trim.para>
</warning.group>
</warning>
<step1 id="o00019-9-2320-365-1">
<para>All objects that can shift during movement will be secured by using a bracing, cushioning, or tie-down method. The method used will be performed in a manner that will not cause damage to the walls or equipment.
</para>
</step1>
<step1 id="o00019-9-2320-365-2">
<para>Verify that all workbenches, lockers, cabinets, and shelves are securely attached to walls and floor.
</para>
</step1>
<step1 id="o00019-9-2320-365-3">
<para>Close van windows.
<xref taskid="o00019-9-2320-365-3" wpid="o00014-9-2320-365"/>
</para>
</step1>
<step1 id="o00019-9-2320-365-4">
<para>Disconnect van 12/24 vdc power.
<xref taskid="o00019-9-2320-365-4" wpid="o00014-9-2320-365"/>.
</para>
</step1>
<step1 id="o00019-9-2320-365-5">
<para>Disconnect van AC power
<xref taskid="o00019-9-2320-365-5" wpid="o00014-9-2320-365"/>.
</para>
</step1>
<step1 id="o00019-9-2320-365-6">
<para>Close van doors
<xref taskid="o00019-9-2320-365-6" wpid="o00014-9-2320-365"/> or
<xref taskid="o00019-9-2320-368-6" wpid="o00014-9-2320-368"/>.
</para>
</step1>
<step1 id="o00019-9-2320-365-7">
<para>Stow van ladder

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<xref taskid="o00019-9-2320-365-7" wpid="o00014-9-2320-365"/>.
</para>
</step1>
</proc>
</prepmove>
</opertsk>
</opusualwp>

b. Page-based TM stylesheet output example for ***<opusualwp>***:

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FIELD MAINTENANCE

OPERATION UNDER USUAL CONDITIONS (ARMY ONLY)

INITIAL SETUP:

NR

M1079 Van Preparation For Movement

WARNING

Heavy objects/loads, such as tool boxes and heavy parts, must always be carried on the floor with the weight distributed as equally as possible between left and ride sides of M1079 van. Failure to comply decreases the stability of the M1079 van and will increase the likelihood of a rollover.

Heavy cabinets must always be mounted as low as possible with the weight distributed as equally as possible between left and right sides of M1079 van. Remember to consider the weight of the items that will be stored in the cabinets. Failure to comply decreases the stability of the M1079 van and will increase the likelihood of a rollover.

Always keep in mind, when placing items inside the M1079 van, that heavier items must always be positioned as low as possible and the weight distributed as equally as possible between left and right sides of M1079 van. Failure to comply decreases the stability of the M1079 van and will increase the likelihood of a rollover.

1. All objects that can shift during movement will be secured by using a bracing, cushioning, or tie-down method. The method used will be performed in a manner that will not cause damage to the walls or equipment.
2. Verify that all workbenches, lockers, cabinets, and shelves are securely attached to walls and floor.
3. Close van windows (WP 0014).
4. Disconnect van 12/24 VDC power (WP 0014).
5. Disconnect van AC power (WP 0014).
6. Close van doors (WP 0014) or (WP 0017).
7. Stow van ladder (WP 0014).

END OF TASK

END OF WORK PACKAGE

FIGURE 207. Example of a page-based TM stylesheet output for <opusualwp> with one task.

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Check the source to verify that this is the current version before use.

35.3.3 XML document instance fragment and output for ***<opusualwp>*** with two tasks.

a. Example of an XML document instance fragment for ***<opusualwp>*** two operations tasks (showing skeleton tasks):

```
<opusualwp wpno="o00008-9-2320-365">
<wpidinfo>
<maintlvl level="field"/>
<title>M1079 Van 12/24 VDC Power Connection/Disconnection
<indxref ref1="M1079 Van 12/24 VDC Power" ref2="Connection"/>
<indxref ref1="Vehicle" ref2="Disconnection"/>
</title>
</wpidinfo>
<wpinfo>
<title>Not Applicable
</title>
<null insert="none>
</wpinfo>
<opertsk id="o00008-9-2320-365-36">
<oper id="o00008-9-2320-365-37">
<title>M1079 Van 12/24 VDC Power Connection
</title>
<step1 id="o00008-9-2320-365-54">
<para>. . .
</para>
</step1>
<step1 id="o00008-9-2320-365-55">
<para>. . .
</para>
</step1>
</oper>
</opertsk>
<opertsk id="o00008-9-2320-365-38">
<oper id="o00008-9-2320-365-39">
<title>M1079 Van 12/24 VDC Power Disconnection
</title>
<step1 id="o00008-9-2320-365-56">
<para>. . .
</para>
</step1>
<step1 id="o00008-9-2320-365-57">
<para>. . .
</para>
</step1>
</oper>
</opertsk>
</opusualwp>
```

b. Page-based TM stylesheet output example for ***<opusualwp>***:

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Check the source to verify that this is the current version before use.

TM X-XXXX-XXX 0009

FIELD MAINTENANCE

M1079 VAN 12/24 VDC POWER CONNECTION/DISCONNECTION
OPERATION UNDER USUAL CONDITIONS (ARMY ONLY)

INITIAL SETUP:

NR

M1079 VAN 12/24 VDC POWER CONNECTION/DISCONNECTION

M1079 Van 12/24 VDC Power Connection

1. ...
2. ...

END OF TASK

M1079 Van 12/24 VDC Power Disconnection

1. ...
2. ...

END OF TASK

END OF WORK PACKAGE

0009-1

FIGURE 208. Example of a page-based TM stylesheet output for <opusualwp> with two tasks.

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Check the source to verify that this is the current version before use.

35.4 Operation under unusual conditions work package ***<opunuwp>***. The operation under unusual conditions work package ***<opunuwp>*** provides step-by-step instructions for operation of the equipment and auxiliary equipment in all modes of operation under unusual conditions. Any combination of control settings that will create a hazard to personnel or cause damage to equipment is preceded by a warning or caution.

a. The components of ***<opunuwp>*** are:

(1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end use (see 32.3.2).

(2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).

(3) Work Package Initial Setup ***<wpinfo>*** (required). The element provides lists of information required by the technician so the tools, test equipment, references, parts, and other items needed to complete the tasks can be obtained (see 32.5).

(4) Grouped alerts. May have one or more of the following alerts.

(a) Warnings ***<warning>*** (optional – zero or more). The element provides an operation, procedure, or statement that if not performed properly may result in personal injury or death (see 41.1.1).

(b) Cautions ***<caution>*** (optional – zero or more). The element provides information that identifies risk of damage to the equipment (see 41.1.2).

(c) Notes ***<note>*** (optional – zero or more). The element provides highlights to essential procedures, conditions, or statements or conveys important instructional data to the user (see 41.1.3).

(5) General or Introductory Information ***<geninfo>*** (optional – zero or more). The element provides titled and subtitled paragraphs giving general or introductory information (see 49.1.4.12).

(6) Under Unusual Conditions Operational Tasks ***<opunutsk>*** (required – one or more). The element provides operational tasks required in the operations under unusual conditions work package (see 35.4.1).

b. DTD graphically depicted:

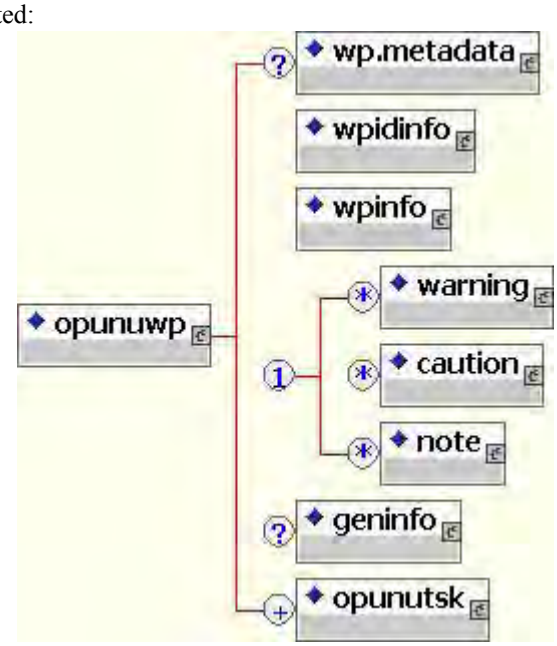

FIGURE 209. Operation under unusual conditions work package <opunuwp> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT opunuwp (wp.metadata?, wpidinfo, wpinfo, warning*,
 caution*, note*, geninfo?, opunutsk+)>

<!ATTLIST opunuwp

    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

35.4.1 Operational task under unusual conditions ***<opunutsk>*** . Operational tasks under unusual conditions contains all operational tasks required to operate the equipment under unusual conditions. Preventive or protective measures to be taken beyond the operator's capabilities are identified, in addition to instructions to ensure proper grounding of equipment. For security measures for electronic data, instructions are prepared for handing, loading, purging, overwriting, or unloading classified electronic data under unusual conditions. These instructions are to meet current security regulations as they pertain to automation. Each operational task under unusual conditions ***<opunutsk>*** requires one of the following specific operational tasks, as defined in MIL-STD-40051-1/-2. Each specific tag has similar markup requirements (see 33.1)

a. The components are:
(1) Security Measures For Electronic Data ***<secref>*** (optional – zero or more). The element provides information about the work package data and usually not used or seen by the end use (see35.3.1.1).
(2) Unusual Environment/Weather ***<unusualenv>*** (optional – zero or more). The element provides an operational task for site requirements that are considered prior to siting (see 35.4.1.1).
(3) Fording and Swimming the Equipment ***<fording>*** (optional – zero or more). The element provides an operational task containing the procedures required before, during and after fording and swimming the equipment (see 35.4.1.2).
(4) Interim Nuclear, Biological, Chemical (NBC) Decontamination ***<decon>*** (optional – zero or more). The element provides a maintenance task that is used for items that are unpacked, disassembled or removed from an assembly, subassembly or component (see 35.4.1.3).
(5) Jamming and Electronic Countermeasure Procedures ***<ecm>*** (optional – zero or more). The element provides an operational task that the operator performs before putting equipment in operation (see 35.4.1.4).
(6) Degraded operation procedures ***<degraded>*** (optional – zero or more). The element provides an operational task containing all procedures to start the equipment, operate the equipment, place the equipment in standby, or shutdown the equipment (see 35.4.1.5).

b. DTD fragment is graphically depicted.

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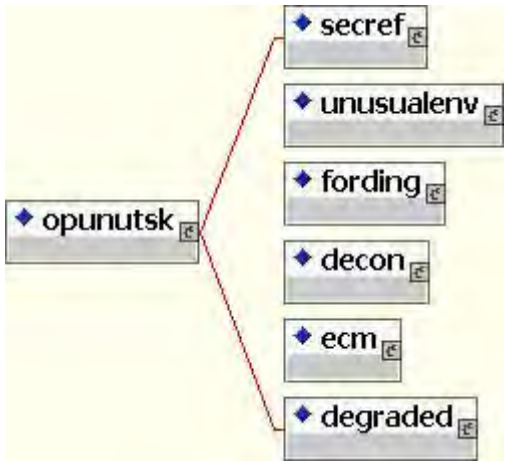

FIGURE 210. Operational task under unusual conditions <opunutsk> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT opunutsk (secref | unusualenv | fording | decon | ecm | degraded)>

<!ATTLIST opunutsk
     applicable        IDREFS                          #IMPLIED
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

35.4.1.1 Unusual environment/weather ***<unusualenv>***. Unusual environment/eeather is an unusual conditions operational task containing procedures for operating the equipment in unusual environment/weather conditions such as extreme heat or cold, sea spray, dust storm, snow, mud, or similar conditions.

a. The components are: (see 33.1)

35.4.1.2 Fording and swimming ***<fording>***. Fording and Swimming is an unusual conditions operational task containing the procedures required before, during and after fording and swimming the equipment.

a. The components are: (see 33.1)

35.4.1.3 Interim nuclear, biological, chemical (NBC) decontamination ***<decon>***. Interim nuclear, biological, chemical decontamination is an unusual conditions operational task containing procedures for interim nuclear,

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biological and chemical (NBC) decontamination; used for NBC decontamination of equipment when a normal decontamination facility is not available.

a. The components are: (see 33.1PROCEDURES/TASKS for details on tagging.)

35.4.1.4 Jamming and electronic countermeasure ***<ecm>***. Jamming and electronic countermeasure is an unusual conditions operational task containing countermeasure procedures for operation of equipment in an ECM environment through transmitted and reflected deception signals and jamming.

a. The components are: (see 33.1)

35.4.1.5 Degraded condition ***<degraded>***. Degraded condition is a task for temporarily adapting the equipment when a component or part of the equipment has failed or a power reduction or some similar condition exists and continued operation of the equipment is required.

a. The components are: (see 33.1)

35.4.2 Example of a ***<opunuwp>*** with a single task. An example of an under unusual condition work package ***<opunwp>*** containing the XML instance and its stylesheet output for is provided below. The work package includes work package initial setup ***<wpinfo>***information that references external documents and also contains a single operational task ***<unusualenv>***.

a. Example of an XML document instance fragment for ***<opunuwp>***:

```
<opunuwp wpno="o00024-9-2320-365">
<wpidinfo>
<maintlvl level="field"/>
<title>Desert Environment
</title>
</wpidinfo>
<wpinfo>
<ref>
<ref-setup-item>
<extref docno="FM 21-10"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="FM 21-11"/>
</ref-setup-item>
</ref>
</wpinfo>
<geninfo>
<title>General Information
<title/>
<para>This section provides instructions to operate the LMTV and its
auxiliary equipment under extreme and unusual conditions. Special
operating instructions are provided for these conditions.
</para>
</geninfo>
<opunutsk id="o00024-9-2320-365-8">
<unusualenv" id="o00024-9-2320-365-9">
<title>Desert Environment Operations
</title>
<proc id="o00024-9-2320-365-14">
<title>Operation In Extreme Heat
</title>
<warning>
<trim.para>When required to remain inside the vehicle during extreme
heat, occupants should follow the water intake, work/rest cycle, and
other heat stress preventive medicine measures contained in
<extref docno="FM 21-10"/> Field Hygiene and Sanitation, and
```


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Check the source to verify that this is the current version before use.

<extref docno="FM 21-11"/> First Aid for Soldiers. Failure to comply may result in serious injury or death to personnel.
</trim.para>
</warning>
<caution>
<trim.para>When operating in temperatures above 100° F (38° C), extra care must be taken to prevent overheating the engine. Watch WATER TEMP gage, STOP indicator, and engine coolant temperature indicator closely. Failure to comply may result in damage to equipment.
</trim.para>
<trim.para>Check oil levels often and keep operating strain as low as possible. Vehicle cooling and lubrication systems support each other. Failure of one system will rapidly cause failure of the other system. Failure to comply may result in damage to equipment.
</trim.para>
</caution>
<step1>
<para>Push N (Neutral) select button
<callout assocfig="o00024-9-2320-365-1" label="1"/> on WTEC II TEPSS
<callout assocfig="o00024-9-2320-365-1" label="2"/> or WTEC III TPSS
<callout assocfig="o00024-9-2320-365-1" label="2"/> while engine is running. Idle engine for approximately two minutes before engine shutdown.
<figure id="o00024-9-2320-365-1">
<title>
<title/>
<graphic boardno="o00024-9-2320-365_1" unitmeasure="in" hscale="65">
</graphic>
</figure>
</para>
</step1>
<step1 id="o00024-9-2320-365-21">
<para>Check that radiator fan off switch
<callout assocfig="o00024-9-2320-365-1" label="3"/> is in the off position and the fan off indicator
<callout assocfig="o00024-9-2320-365-1" label="4"/> is not illuminated.
</para>
</step1>
<step1 id="o00024-9-2320-365-22">
<specpara>
<caution>
<trim.para>Never operate engine for more than 30 seconds at full throttle while vehicle is not moving. Transmission oil temperature will become too hot. Failure to comply may result in damage to equipment.
</trim.para>
</caution>
<para>If the TRANS OIL TEMP indicator
<callout assocfig="o00024-9-2320-365-2" label="5"/> illuminates and WATER TEMP gage
<callout assocfig="o00024-9-2320-365-2" label="6"/> reads near 230° F (110° C), transmission oil is overheating:
<figure id="o00024-9-2320-365-2">
<title>
<title/>
<graphic boardno="o00024-9-2320-365_2">
</graphic>
</figure>
</para>

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```
</specpara>
<step2 id="o00024-9-2320-365-43">
<para>Stop vehicle.
</para>
</step2>
<step2 id="o00024-9-2320-365-44">
<para>Press the N (Neutral) select button
<callout assocfig="o00024-9-2320-365-1" label="1"/> on WTEC II TEPSS
<callout assocfig="o00024-9-2320-365-1" label="2"/> or WTEC III TPSS
<callout assocfig="o00024-9-2320-365-1" label="2"/>.
</para>
</step2>
<step2 id="o00024-9-2320-365-45">
<para>Allow engine to operate at approximately 750 rpm for three minutes.
</para>
</step2>
<step2 id="o00024-9-2320-365-46">
<para hcp="no" esd="no">Continue normal vehicle operation
when TRANS OIL TEMP indicator
<callout assocfig="o00024-9-2320-365-2" label="5"/> goes out.
</para>
</step2>
<step2 id="o00024-9-2320-365-47">
<para>Shut down engine (
<xref taskid="o00003-9-2320-365-42" wpid="o00003-9-2320-365"/>) and notify
Unit Maintenance if TRANS OIL TEMP indicator
<callout assocfig="o00024-9-2320-365-2" label="5"/> does not go out.
</para>
</step2>
</step1>
<step1 id="o00024-9-2320-365-23">
<para>Check cooling system often for the following conditions:
</para>
<step2 id="o00024-9-2320-365-48">
<para>Low coolant level in radiator overflow tank (
<xref wpid="m00033-9-2320-365" itemid="m00033-9-2320-365-117"/>).
</para>
</step2>
<step2 id="o00024-9-2320-365-49">
<para hcp="no" esd="no">Cracked or leaking radiator hoses (
<xref wpid="m00033-9-2320-365" itemid="m00033-9-2320-365-178"/>).
</para>
</step2>
<step2 id="o00024-9-2320-365-50">
<para>Radiator fins clogged with dust, leaves, or insects.
</para>
</step2>
</step1>
</proc>
</unusualenv>
</opunutsk>
</opunuwp>
```

b. Page-based TM stylesheet output example for **<opunuwp>**:

FIGURE 211. Example of a page-based TM stylesheet output for <opunuwp> with one task.

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35.4.3 Example of an ***<opunuwp>*** with multiple tasks. An example of an under unusual condition work package ***<opunwp>*** containing the XML instance and its stylesheet output for is provided below. The work package includes work package initial setup ***<wpinfo>***information that references external documents and also contains two operational tasks of ***<unusualenv>***.

a. The XML instance and its stylesheet output for a ***<opunuwp>*** with two operations tasks (showing skeleton tasks) is provided below (showing skeleton tasks):

```
<opunuwp wpno="o00024-9-2320-365">
<wpidinfo>
<maintlvl level="operator"/>
<title>Desert Environment
</title>
</wpidinfo>
<wpinfo>
<ref>
<ref-setup-item>
<extref docno="FM 21-10"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="FM 21-11"/>
</ref-setup-item>
</ref>
<wpinfo/>
<geninfo>
<title>General Information
<title/>
<para>This section provides instructions to operate the LMTV and its
auxiliary equipment under extreme and unusual conditions. Special
operating instructions are provided for these conditions.
</para>
</geninfo>
<opunutsk id="o00024-9-2320-365-8">
<unusualenv>
<title>Operations in Extreme Heat
</title>
<step1>
<specpara>
<warning>
<trim.para>
</trim.para>
</warning>
<caution>
<trim.para>
</trim.para>
</caution>
<para>. . .
</para>
</specpara>
</step1>
<step1>
<para>. . .
</para>
</step1>
</unusualenv>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</opunutsk>
<opunutsk>
<unusualenv>
<title>Operations in Sand or Mud
</title>
<step1>
<specpara>
<warning>
<trim.para>
</trim.para>
</warning>
<caution>
<trim.para>
</trim.para>
</caution>
<para>. . .
</para>
</specpara>
</step1>
<step1>
<para>. . .
</para>
</step1>
</unusualenv>
</opunutsk>
</opunuwp>

b. Page-based TM stylesheet output example for ***<opunuwp>*** with two tasks:

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FIELD MAINTENANCE

DESERT ENVIRONMENT
OPERATION UNDER USUAL CONDITIONS (ARMY ONLY)

INITIAL SETUP:

References

FM 21–10
FM 21–11

GENERAL INFORMATION

This section provides instructions to operate the LMTV and its auxiliary equipment under extreme and unusual conditions. Special operating instructions are provided for these conditions.

Desert Environment Operations

Operation In Extreme Heat

WARNING

CAUTION

1.

2.

END OF TASK

Operation in Sand or Mud

WARNING

CAUTION

1.

2.

END OF TASK

END OF WORK PACKAGE

FIGURE 212. Example of a page-based TM stylesheet output for <opunuwp> with two tasks.

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Check the source to verify that this is the current version before use.

35.5 Emergency conditions work package ***<emergencywp>***. The emergency conditions work package provides instructions for temporarily adapting the equipment when a component or part of the equipment has failed, or a power reduction or some similar condition exists and continued operation of the equipment is required.

a. The components are:
   (1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end use (see 32.3.2).
   (2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).
   (3) Work Package Initial Setup***<wpinfo>*** (optional). The element provides lists of information required by the technician so the tools, test equipment, references, parts, and other items needed to complete the tasks can be obtained (see 32.5).
   (4) Emergency Operator Task ***<emergency>*** (required – one or more). The element provides a task for the operating emergency conditions (see 35.5.1).

b. DTD fragment is graphically depicted.

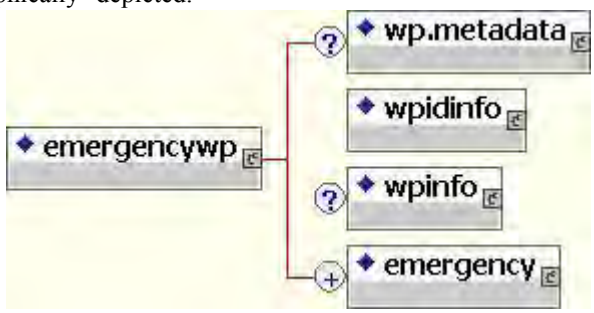

FIGURE 213. Emergency conditions work package <emergencywp> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT emergencywp (wp.metadata?, wpidinfo, wpinfo?, emergency+)>
<!ATTLIST emergencywp
    airforce         (yes | no )               "no"
    army             (yes | no )               "no"
    assocfig         IDREFS                     #IMPLIED
    changeref        IDREFS                     #IMPLIED
    comment          CDATA                      #IMPLIED
    crewmember       CDATA                      #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                      #IMPLIED
    deletewp         (yes | no )               "no"
    fgc              CDATA                      #IMPLIED
    frame            (yes | no )               "yes"
    idref            IDREFS                     #IMPLIED
    inschlvl         CDATA                      #IMPLIED
    insertwp         CDATA                      #IMPLIED
    lsa-id           CDATA                      #IMPLIED
    marines          (yes | no )               "no"
    navy             (yes | no )               "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                      #IMPLIED
    tocentry         (2 | 3 | 4 | 5)           "2"
    wpno             ID                         #REQUIRED
    wpseq            CDATA                      #IMPLIED
```

d. Common attributes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

35.5.1 Emergency operator task ***<emergency>***. Emergency operator task provide procedural instructions for operation of the equipment during emergency conditions (control failure, air failure, lube oil failure, loss of cooling water, etc.), and shutdown procedures to turn the equipment off during an emergency (fire, water, smoke, hazard to personnel, loss of coolant, normal power, etc.). Warning(s) or caution(s) are to be included to return the equipment to proper operation when the emergency is over.

a. The components are (see 33.1)

35.5.2 XML document instance fragment and output for ***<emergencywp>***. The XML instance and its stylesheet output for a ***<emergencywp>*** is provided below.

a. Example of an XML document instance fragment for ***<emergencywp>***:

```
<emergencywp  wpno="o00099-123">
<wpidinfo>
<maintlvl  level="operator"/>
<title>Highway Emergency Marker Kit Setup
</title>
</wpidinfo>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<emergency>
<title>Highway Emergency Marker Kit Setup
</title>
<proc id="o00026-9-2320-365-11">
<title>Preparing Markers For Use
</title>
<step1 id="o00026-9-2320-365-17">
<para>Position master power switch
<callout assocfig="o00026-9-2320-365-1" label="1"/> to on.
<figure id="o00026-9-2320-365-1">
<title>Preparing Markers
</title>
<graphic boardno="o00026-9-2320-365_1">
</graphic>
</figure>
</para>
</step1>
<step1 id="o00026-9-2320-365-18">
<para>Position hazard lights switch
<callout assocfig="o00026-9-2320-365-1" label="2"/> to on.
</para>
</step1>
<step1 id="o00026-9-2320-365-19">
<para>Remove emergency marker kit
<callout assocfig="o00026-9-2320-365-2" label="3"/> from TOOL KIT.
<figure id="o00026-9-2320-365-2">
<title>Preparing Markers (Cont)
</title>
<graphic boardno="o00026-9-2320-365_2">
</graphic>
</figure>
</para>
</step1>
<step1 id="o00026-9-2320-365-20">
<para>Remove three markers
<callout assocfig="o00026-9-2320-365-2" label="4"/> from emergency marker kit
<callout assocfig="o00026-9-2320-365-2" label="3"/>.
</para>
</step1>
<step1 id="o00026-9-2320-365-21">
<para>Attach two ends of marker arms
<callout assocfig="o00026-9-2320-365-2" label="5"/> with pin
<callout assocfig="o00026-9-2320-365-2" label="6"/>.
</para>
</step1>
<step1 id="o00026-9-2320-365-22">
<para>Rotate marker
<callout assocfig="o00026-9-2320-365-2" label="5"/> approximately 1/4 turn on base
<callout assocfig="o00026-9-2320-365-2" label="7"/>.
</para>
</step1>
<step1 id="o00026-9-2320-365-23">
<para>Perform
<xref stepstart="o00026-9-2320-365-20" stepend="o00026-9-2320-365-22"/>
for second and third markers.
```


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Check the source to verify that this is the current version before use.

```
</para>
</step1>
</proc>
<proc id="o00026-9-2320-365-12">
<title>Placing Markers On Undivided, Straight Highway
</title>
<step1 id="o00026-9-2320-365-24">
<para>Place one marker
<callout assocfig="o00026-9-2320-365-3" label="1"/> approximately 100 ft (30 m)
in front of vehicle with marker facing approaching traffic.
<figure id="o00026-9-2320-365-3">
<title>Placing Markers on Undivided, Straight Highway
</title>
<graphic boardno="o00026-9-2320-365_3" hscale="90">
</graphic>
</figure>
</para>
</step1>
<step1 id="o00026-9-2320-365-25">
<para>Place second marker
<callout assocfig="o00026-9-2320-365-3" label="2"/> approximately 10 ft (3
m) behind vehicle and about 5 ft (1.5 m) out from side of
vehicle with marker facing approaching traffic.
</para>
</step1>
<step1 id="o00026-9-2320-365-26">
<para>Place third marker
<callout assocfig="o00026-9-2320-365-3" label="3"/> approximately 100 ft (30
m) behind vehicle with marker facing approaching traffic.
</para>
</step1>
</proc>
</emergency>
</emergencywp>
```

b. Page-based TM stylesheet output example for ***<emergencywp>***:

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TM X-XXXX-XXX-10 0011

OPERATOR MAINTENANCE
HIGHWAY EMERGENCY MARKER KIT SETUP

HIGHWAY EMERGENCY MARKER KIT SETUP

PREPARING MARKERS FOR USE

1. Position master power switch (Figure 1, Item 1) to on.

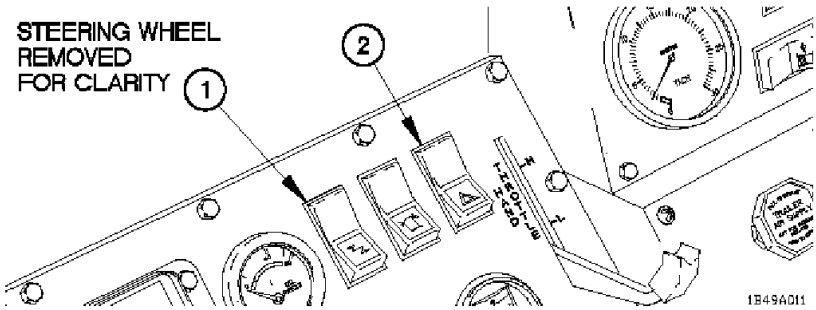

Figure 1. Preparing Markers

2. Position hazard lights switch (Figure 1, Item 2) to on.
3. Remove emergency marker kit (Figure 2, Item 3) from TOOL KIT.

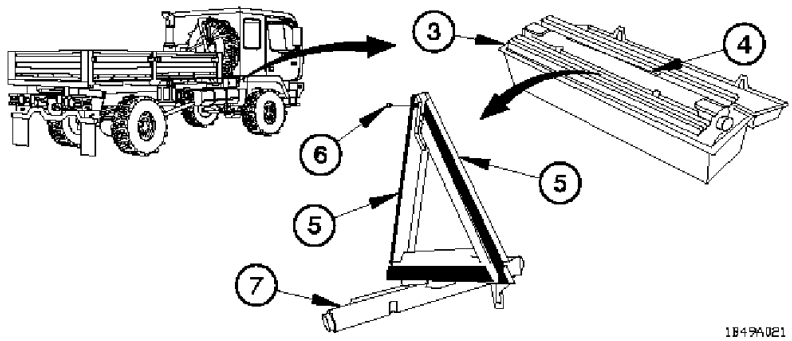

Figure 2. Preparing Markers (Cont)

4. Remove three markers (Figure 2, Item 4) from emergency marker kit (Figure 2, Item 3).
5. Attach two ends of marker arms (Figure 2, Item 5) with pin (Figure 2, Item 6).
6. Rotate marker (Figure 2, Item 5) approximately 1/4 turn on base (Figure 2, Item 7).
7. Perform Steps 4 – 6 for second and third markers.

0011-1

FIGURE 214. Example of a page-based TM stylesheet output for <emergencywp>.

35.6 Stowage and decal/data plate guide work package ***<stowagewp>***. The stowage and decal/data plate guide work package lists and illustrates the location of all applicable COEI, BII, AAL items, decals and data plates as directed by the acquiring activity.

a.

(1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end use (see 32.3.2).

(2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).

(3) Introduction ***<intro>*** (required). An introductory section explaining the purpose of the work package (see 49.1.4.15).

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Check the source to verify that this is the current version before use.

(4) Aircraft Stowage Information ***<stowinfo>*** (required – one or more). The element contains the stowage information and illustration(s) (see 35.6.1).

(5) Decal information ***<decalinfo>*** (optional – zero or more). The element provides information of decals and data plates concerning the equipment (see 35.6.2).

b. DTD graphically depicted:

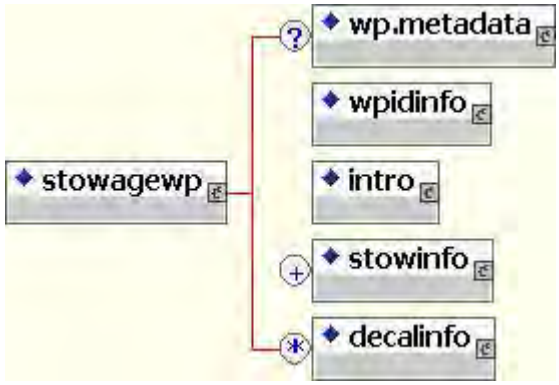

FIGURE 215. Stowage and decal/data plate guide work package <stowagewp> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT stowagewp (wp.metadata?, wpidinfo, intro, stowinfo+, decalinfo*)>
<!ATTLIST
   airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time)  #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts )  #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED>
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

35.6.1 Stowage information ***<stowinfo>***. The stowage information ***<stowinfo>*** element contains data on the location of applicable COEI, BII and AAL items. An illustration is to be included to facilitate the location of the items.

a. The components of ***<stowinfo>*** are:
(1) Introduction***<intro>*** (required). An introductory section providing the data on the location of applicable COEI, BII and AAL items. (refer to 49.1.4.15 for detail information).
(2) Figure ***<figure>*** (required – one or more). The element provides a graphic displaying the locations where items can be stored.

b. DTD graphically depicted:

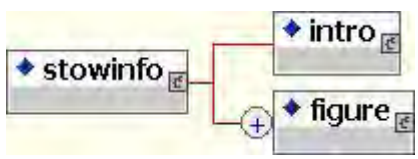

FIGURE 216. Stowage information <stowinfo> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<stowinfo>***:

```
<!ELEMENT stowinfo (intro, figure+)>
<!ATTLIST stowinfo


     applicable        IDREFS                          #IMPLIED
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED
```

35.6.2 Decal information ***<decalinfo>***. The decal information element contains data on the location of all decals and data plates in and on the equipment. Illustrations detailing the locations of the decals and data plates are to be included.

a. The components of ***<decalinfo>*** are:

(1) Title ***<title>*** (required). The element provides the title for the section on the decal information (see 49.1.1.3).

(2) Notes ***<note>***(optional – zero or more) (if note is used then the element ***<para>*** follows note). The element provides highlights to essential procedures, conditions, or statements or conveys important instructional data to the user (see 41.1.3).

(3) One of the following elements is required before the figure element:

(a) General Paragraph(s) ***<para>***. The element provides the data on the location of the decals and data plates (see 49.1.1.5).

(b) Primary Level – Titled Paragraph ***<para0>*** (see 49.1.1.8) or/and Primary Level – Conditional – Titled Paragraph(s) ***<para0-alt>*** (see 48.2.1) that provides a title, a precondition and paragraphs of text in the description of the decals and data plates located on the equipment.

(4) Figures ***<figure>*** (required – one or more). The element provides the graphic(s) to help locate where the decals and data plates are located.

b. DTD graphically depicted:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

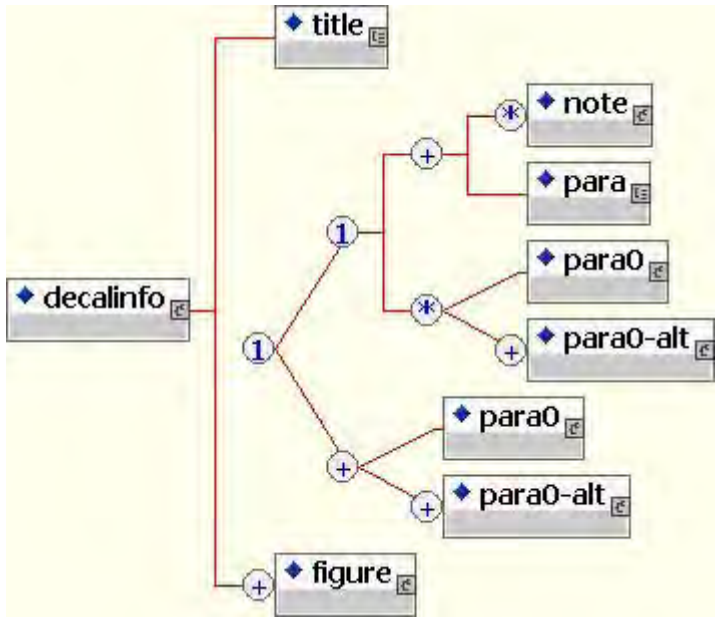

FIGURE 217. Decal information <decalinfo> DTD hierarchy.

c. DTD fragment for ***<decalinfo>***:

```
<!ELEMENT decalinfo (title, (((note*, para)+, (para0 | para0-alt+)*) |
 (para0 | para0-alt+)+), figure+)>
<!ATTLIST decalinfo


     applicable        IDREFS                          #IMPLIED
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

35.6.3 XML document instance fragment and output for ***<stowagewp>***. The XML instance and its stylesheet output for a ***<stowagewp>*** is provided below.

a. Example of an XML document instance fragment for ***<stowagewp>***:

<stowagewp wpno="o00034-9-2320-365">
<wpidinfo>
<maintlvl level="operator"/>
<title>STOWAGE LOCATION/DECAL/STENCIL GUIDE
</title>
</wpidinfo>
<intro>
<para0>
<title>Introduction
</title>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<para>This work package shows the location for stowage of equipment and material required to be carried on M1078 series vehicles and location for data plates, decals, and stencils that are required to be placed on the vehicle.
</para>
</para0>
</intro>
<stowinfo>
<intro>
<para0>
<title>Stowage Locations
</title>
<para>The equipment stowage locator is designed to help inventory items required for safe and efficient operation. The equipment locator is representative of BII and applicable AAL stowage on all M1078 series vehicles.
</para>
</para0>
</intro>
<figure id="o00034-9-2320-365-1">
<title>Stowage Locations, All Vehicles.
</title>
<graphic boardno="o00034-9-2320-365_1" hscale="80">
</graphic>
</figure>
</stowinfo>
<decalinfo>
<title>SIGN GUIDE
</title>
<para hcp="no" esd="no">The equipment sign guide details the locations of all decals and data plates on all M1078 series vehicles.
</para>
<figure id="o00034-9-2320-365-2">
<title>Decals/Stencils, All Vehicles
</title>
<subfig totalsheets="7">
<graphic boardno="o00034-9-2320-365_2" hscale="75">
</graphic>
</subfig>
<subfig totalsheets="7">
<graphic boardno="o00034-9-2320-365_3" hscale="80">
</graphic>
</subfig>
<subfig totalsheets="7">
<graphic boardno="o00034-9-2320-365_4" hscale="80">
</graphic>
</subfig>
<subfig totalsheets="7">
<graphic boardno="o00034-9-2320-365_5">
</graphic>
</subfig>
<subfig totalsheets="7">
<graphic boardno="o00034-9-2320-365_6">
</graphic>
</subfig>

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Check the source to verify that this is the current version before use.

```
<subfig totalsheets="7">
<graphic boardno="o00034-9-2320-365_7">
</graphic>
</subfig>
<subfig totalsheets="7">
<graphic boardno="o00034-9-2320-365_8">
</graphic>
</subfig>
</figure>
<figure id="o00034-9-2320-365-3">
<title>Decal/Data Plate Guide, Vehicles with 11K Self-Recovery (SRW)
</title>
<graphic boardno="o00034-9-2320-365_9">
</graphic>
</figure>
<figure id="o00034-9-2320-365-4">
<title>Decal/Data Plate Guide, Vehicles with Light Mate-
rial Handling Crane (LMHC)
</title>
<graphic boardno="o00034-9-2320-365_10">
</graphic>
</figure>
</decalinfo>
</stowagewp>
```

b. Page-based TM stylesheet output example for ***<stowagewp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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TM X-XXXX-XXX-10 0012

OPERATOR MAINTENANCE
STOWAGE LOCATION/DECAL/STENCIL GUIDE

INTRODUCTION

This work package shows the location for stowage of equipment and material required to be carried on M1078 series vehicles and location for data plates, decals, and stencils that are required to be placed on the vehicle.

STOWAGE LOCATIONS

The equipment stowage locator is designed to help inventory items required for safe and efficient operation. The equipment locator is representative of BII and applicable AAL stowage on all M1078 series vehicles.

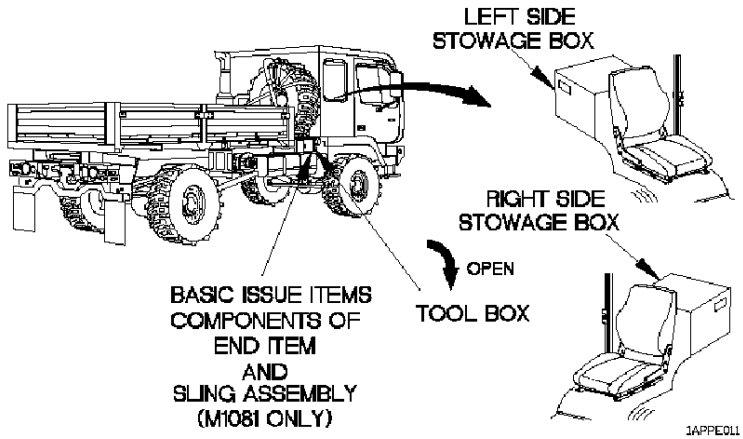

Figure 1. Stowage Locations, All Vehicles.

SIGN GUIDE

The equipment sign guide details the locations of all decals and data plates on all M1078 series vehicles.

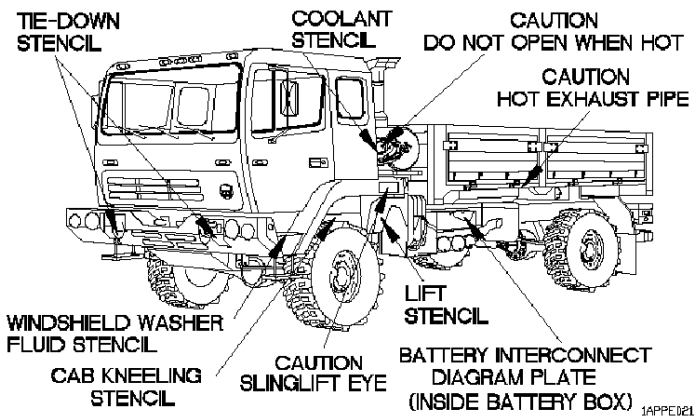

Figure 2. Decals/Stencils, All Vehicles (Sheet 1 of 7)

0012-1

FIGURE 218. Example of a page-based TM stylesheet output for <stowagewp>.

35.7 On-vehicle equipment loading plan work package ***<eqploadwp>***. The on-vehicle equipment loading plan work package is prepared when applicable to the equipment. The loading plan includes information provided by the acquiring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

a. The components of are:
   (1) Metadata ***<wp.metadata>*** (optional). The element provides information about the work package data and usually not used or seen by the end use (see 32.3.2).
   (2) Work Package Identification Information ***<wpidinfo>*** (required). The element provides the identification information required for a work package (see 32.2).
   (3) Introduction ***<intro>*** (required). An introductory section on the loading plan and identifying the equipment covered by the on-vehicle equipment loading plan work package (see 49.1.4.15).
   (4) Loading Description ***<loaddesc>*** (required – one or more). The element provides a description of equipment loading (see 35.7.1).
b. DTD graphically depicted:

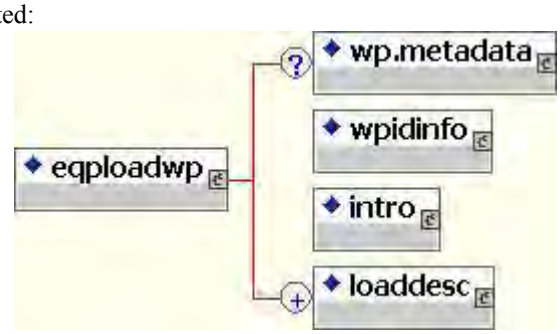

FIGURE 219. On-vehicle equipment loading plan work package <eqploadwp> DTD hierarchy.

c. DTD fragment for ***<eqploadwp>***:

```
<!ELEMENT eqploadwp (wp.metadata?, wpidinfo, intro, loaddesc+)>
<!ATTLIST eqploadwp
    airforce         (yes | no )               "no"
    army             (yes | no )               "no"
    assocfig         IDREFS                     #IMPLIED
    changeref        IDREFS                     #IMPLIED
    comment          CDATA                      #IMPLIED
    crewmember       CDATA                      #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                      #IMPLIED
    deletewp         (yes | no )               "no"
    fgc              CDATA                      #IMPLIED
    frame            (yes | no )               "yes"
    idref            IDREFS                     #IMPLIED
    inschlvl         CDATA                      #IMPLIED
    insertwp         CDATA                      #IMPLIED
    lsa-id           CDATA                      #IMPLIED
    marines          (yes | no )               "no"
    navy             (yes | no )               "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                      #IMPLIED
    tocentry         (2 | 3 | 4 | 5)           "2"
    wpno             ID                         #REQUIRED
    wpseq            CDATA                      #IMPLIED>
```

d. Attributes for ***<eqploadwp>***:
   (1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

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(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

35.7.1 Illustrated loading plan description ***<loaddesc>***. The illustrated loading plan description identifies a description of equipment loading, including illustrations of the end item with equipment locations and a standard load list table. External and internal views are to be used, if necessary. As applicable, both tactical and non-tactical situation loading configurations are to be shown. Each illustration in ***<loaddesc>*** is followed with a loading list table ***<loadlist>***.

a. The components are:
(1) Title ***<title>*** (required). The element provides the title for the loading plan description (see 49.1.1.3).
(2) Standard table header ***<std.table.header>*** (required). The element provides the table header for the loading list table (see 42.3.1).
(3) Figure ***<figure>*** (required – one or more). The element provides a graphic of the end item with equipment locations (see 44.1.1).
(4) Loading List ***<loadlist>*** (required – one or more). The element provides a standard information loading list in tabular form (see 35.7.1.1).

b. DTD graphically depicted:

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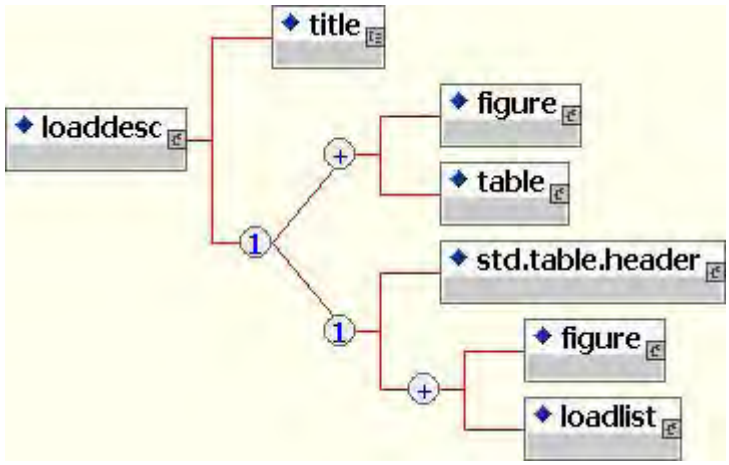

FIGURE 220. Illustrated loading plan description <eqploadwp> DTD hierarchy.

c. DTD fragment for ***<loaddesc>***

```
<!ELEMENT loaddesc (title, std.table.header, (figure, loadlist)+)>
<!ATTLIST loaddesc
    type        (tac | notac )                          #REQUIRED

    applicable          IDREFS                           #IMPLIED
    assocfig            IDREFS                           #IMPLIED
    changeref           IDREFS                           #IMPLIED
    comment             CDATA                            #IMPLIED
    delchlvl            CDATA                            #IMPLIED
    id                  ID                               #IMPLIED
    idref               IDREFS                           #IMPLIED
    inschlvl            CDATA                            #IMPLIED
    security            (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk            CDATA                            #IMPLIED>
```

d. Unique attributes:

(1) **type** – Type specifies the type of loading plan. Possible values include **tac** (tactical) and **notac** (non-tactical) (required),.

e. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

35.7.1.1 Loading list ***<loadlist>***. Loading List contains a standard information loading list that lists all applicable equipment by illustration identification number and item name. The list is on the same page or adjacent to the illustration. The element is similar to a "row" in a structural table.

a. The components are:

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(1) Call out ***<callout>*** (required – one or more). The element provides a figure reference or a callout reference, number, letter, or symbol appearing in the figure. The element is similar to a "cell" in a structural table and is entered in column (see46.2.4.1).
(2) Item ***<item>*** (required – one or more). The element provides each item in the loading list. The element is similar to a "cell" in a structural table and is entered in column (see 49.1.2.1.1).

b. DTD graphically depicted:

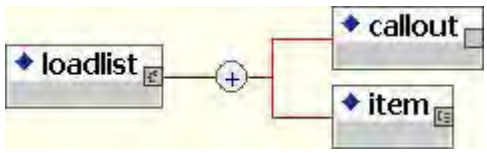

FIGURE 221. Loading list <loadlist> DTD hierarchy.

c. DTD fragment for ***<loadlist>***

```
<!ELEMENT loadlist (callout, item)+>
<!ATTLIST loadlist


     changeref          IDREFS                          #IMPLIED
     comment            CDATA                           #IMPLIED
     delchlvl           CDATA                           #IMPLIED
     inschlvl           CDATA                           #IMPLIED
     security           (uc | fouo |c | s | ts)        #IMPLIED
>
```

d. Common attributes:
(1) **comment** – Change information (optional) (see 49.2.9).
(2) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(3) **inschlvl** – Insert change level (optional) (see 49.2.9).
(4) **security** – Security classification (optional) (see 49.2.13).

35.7.2 XML document instance fragment and output for ***<eqploadwp>***. The XML instance and its stylesheet output for a ***<eqploadwp>*** is provided below.

a. Example of an XML document instance fragment for ***<eqploadwp>***:

```
<eqploadwp  wpno="O00008-9-2320-280">
<wpidinfo>
<maintlvl  level="operator"/>
<title>ON-VEHICLE  EQUIPMENT  LOADING  PLANS
</title>
</wpidinfo>
<intro>
<para0>
<title>Introduction
</title>
<para>This  work  package  shows  the  on-vehicle  equipment  loading
plan  for  TOW  carrier  (M966,  M966A1,  M1036,  M1045,  M1045A1,
M1045A2,  M1046,  M1046A1,  and  M1121).
</para>
<subpara1>
<title>TOW  CARRIER  LOAD  PLAN
</title>
<para>This is a standard load plan for the TOW carrier (M966, M966A1,
M1036, M1045, M1045A1, M1045A2, M1046, M1046A1, and M1121).  It is
designed to supplement the stowage and sign guide contained in
```


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Check the source to verify that this is the current version before use.

<xref wpid="o00034-9-2320-365" posttext=" STOWAGE AND SIGN GUIDE"/>. It includes selected items of personnel and unit equipment issued to most units within the Army or Marine Corps equipped with this vehicle. Vehicles and equipment not shown in either this document or <xref wpid="o00034-9-2320-365" posttext=" STOWAGE AND SIGN GUIDE"/> may be loaded in accordance with local command policy. Fuel cans cannot be carried in an enclosed vehicle.

<proponent>
<name>Commandant
</name>
<address>
<street>US Army Infantry School
</street>
<city>Fort Benning
</city>
<state>GA
</state>
<zip>31905
</zip>
</address>
</proponent>
</para>
</subpara1>
</intro>
<loaddesc id="O00008-9-2320-280-00001">
<title>TOW CARRIER STOWAGE PLAN
</title>
<std.table.header cols="2">
<std.table.row>
<std.table.entry cols.span="2">LOAD PLAN
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry cols.span="1">Item No.
</std.table.entry>
<std.table.entry cols.span="1">Item Description
</std.table.entry>
</std.table.row>
</std.table.header>
<figure id="O00008-9-2320-280-00002">
<title>Tow Carrier Stowage Plan
</title>
<graphic boardno="o0000213.png" hscale="50">
</graphic>
</figure>
<loadlist>
<callout assocfig="O00008-9-2320-280-00002" label="1"/>
<item>Missile Guidance Set (MGS)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="2"/>
<item>M16A1/M203 Rifle
</item>
<callout assocfig="O00008-9-2320-280-00002" label="3"/>
<item>Flashlight
</item>
<callout assocfig="O00008-9-2320-280-00002" label="4"/>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<item>Binoculars
</item>
<callout assocfig="O00008-9-2320-280-00002" label="5"/>
<item>AN/GRC-160 Radio
</item>
<callout assocfig="O00008-9-2320-280-00002" label="6"/>
<item>TSEC/KY-57
</item>
<callout assocfig="O00008-9-2320-280-00002" label="7"/>
<item>Gasoline Cook Stove (strapped to airlift bracket)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="7.1"/>
<item>Jack, Scissors (A2 only)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="8"/>
<item>M16A1/M203 Rifle
</item>
<callout assocfig="O00008-9-2320-280-00002" label="9"/>
<item>Night Sight Vehicle Power Conditioner
</item>
<callout assocfig="O00008-9-2320-280-00002" label="10"/>
<item>Adjustable Gunner's Platform
</item>
<callout assocfig="O00008-9-2320-280-00002" label="11"/>
<item>Traversing Unit
</item>
<callout assocfig="O00008-9-2320-280-00002" label="12"/>
<item>Radiac Meter
</item>
<callout assocfig="O00008-9-2320-280-00002" label="13"/>
<item>Night Vision Goggles (2)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="14"/>
<item>NBC Footwear
</item>
<callout assocfig="O00008-9-2320-280-00002" label="15"/>
<item>Battery Pack, Night Sight
</item>
<callout assocfig="O00008-9-2320-280-00002" label="16"/>
<item>NBC Overgarments (strapped to body)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="17"/>
<item>Boresight Collimator
</item>
<callout assocfig="O00008-9-2320-280-00002" label="18"/>
<item>Night Vision Sight Case
</item>
<callout assocfig="O00008-9-2320-280-00002" label="19"/>
<item>Coolant Cartridge Cases
</item>
<callout assocfig="O00008-9-2320-280-00002" label="20"/>
<item>Radio Antenna (on cargo shell)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="21"/>
<item>Combat Rations

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</item>
<callout assocfig="O00008-9-2320-280-00002" label="22"/>
<item>Optical Sight W/Cover
</item>
<callout assocfig="O00008-9-2320-280-00002" label="23"/>
<item>Night Sight Battery (under tanker roll)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="24"/>
<item>Camouflage Screen and Support System
</item>
<callout assocfig="O00008-9-2320-280-00002" label="25"/>
<item>Tanker Rolls
</item>
<callout assocfig="O00008-9-2320-280-00002" label="26"/>
<item>Tripod
</item>
<callout assocfig="O00008-9-2320-280-00002" label="27"/>
<item>Telephone
</item>
<callout assocfig="O00008-9-2320-280-00002" label="28"/>
<item>DR8 Cable Reel
</item>
<callout assocfig="O00008-9-2320-280-00002" label="29"/>
<item>Combat Rations
</item>
<callout assocfig="O00008-9-2320-280-00002" label="30"/>
<item>M13 Decon Kit
</item>
<callout assocfig="O00008-9-2320-280-00002" label="31"/>
<item>Ammunition (M16A1)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="32"/>
<item>Water Can
</item>
<callout assocfig="O00008-9-2320-280-00002" label="33"/>
<item>NBC Overgarments (strapped to body)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="34"/>
<item>MGS Battery Assemblies (2)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="35"/>
<item>NBC Footwear (2)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="36"/>
<item>Missiles (6)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="37"/>
<item>Tanker Roll (strapped to roof)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="38"/>
<item>NBC Hood Masks (4)
</item>
<callout assocfig="O00008-9-2320-280-00002" label="39"/>
<item>Launch Tube (strapped to door)
</item>
</loadlist>
```


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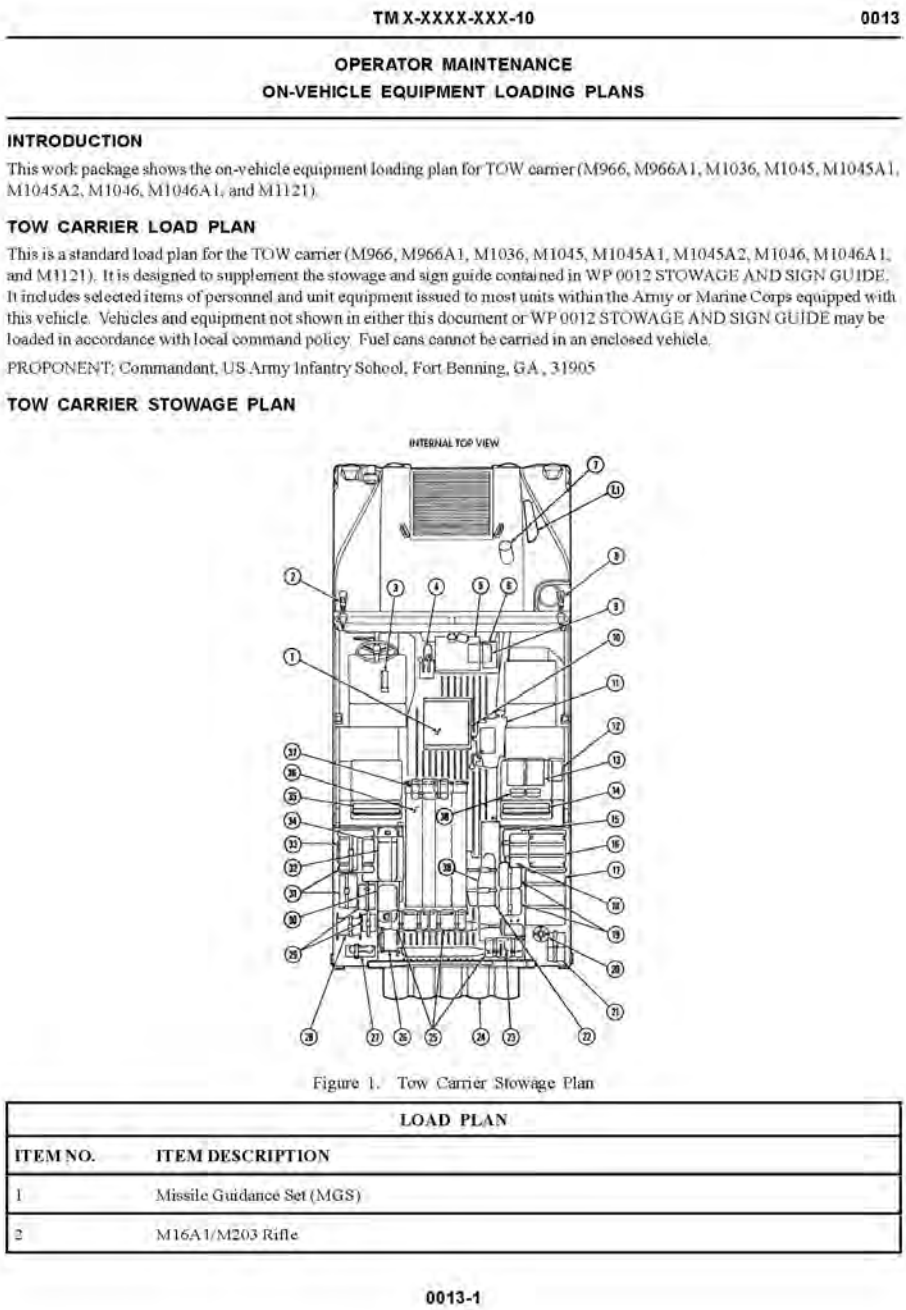
</loaddesc>
</eqploadwp>

b. Page-based TM stylesheet output example for ***<eqploadwp>***:



TM X-XXXX-XXX-10 0013

OPERATOR MAINTENANCE
ON-VEHICLE EQUIPMENT LOADING PLANS

INTRODUCTION

This work package shows the on-vehicle equipment loading plan for TOW carrier (M966, M966A1, M1036, M1045, M1045A1, M1045A2, M1046, M1046A1, and M1121).

TOW CARRIER LOAD PLAN

This is a standard load plan for the TOW carrier (M966, M966A1, M1036, M1045, M1045A1, M1045A2, M1046, M1046A1, and M1121). It is designed to supplement the stowage and sign guide contained in WP 0012 STOWAGE AND SIGN GUIDE. It includes selected items of personnel and unit equipment issued to most units within the Army or Marine Corps equipped with this vehicle. Vehicles and equipment not shown in either this document or WP 0012 STOWAGE AND SIGN GUIDE may be loaded in accordance with local command policy. Fuel cans cannot be carried in an enclosed vehicle.

PROPONENT: Commandant, US Army Infantry School, Fort Benning, GA , 31905

TOW CARRIER STOWAGE PLAN

Figure 1. Tow Carrier Stowage Plan

LOAD PLAN	
ITEM NO.	**ITEM DESCRIPTION**
1	Missile Guidance Set (MGS)
2	M16A1/M203 Rifle

0013-1

FIGURE 222. Example of a page-based TM stylesheet output for <eqploadwp>.

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36 TROUBLESHOOTING CHAPTER.

36.1 Scope. The following paragraphs give a description and use of the elements used in the MIL-STD-2361 Troubleshooting Chapter (TIM) DTD.

36.2 Applicable documents. Refer to section 2

36.3 What is troubleshooting. Troubleshooting procedures isolate faults to the part(s) authorized by the RPSTL for repair or replacement at the maintenance level addressed. Troubleshooting procedures begin with testing observed problems, a fault symptom or a malfunction and result in a diagnosis to a single fault/failure. Troubleshooting refers to the specific maintenance or repair tasks to correct the fault. Troubleshooting data is test and fault isolation oriented. Troubleshooting instructions include detailed inspection and troubleshooting information. Instructions include or reference to functional descriptions of subsystems being diagnosed to aid the operator/technician. The method used for identifying system equipment test points, including the requirements and methods of determining defects through visual inspection is explained.

36.4 Troubleshooting chapter components and categories ***<tim>***. The troubleshooting chapter has a title page (***<titlepage>***) (in frame-based this information is ignored) followed by a troubleshooting category. The troubleshooting category to select is determined from the context matrix requirements from MIL–STD–40051–1/–2.

a. The components are:
   (1) Title page***<titlepg>*** (required) (see 49.1.1.1).
   (2) One of the following maintenance categories, in accordance with MIL–STD–40051–1/–2, is required:
      (a) Troubleshooting master index category (Page-based only) ***<masterindexcategory>*** (see 36.5.1).
      (b) Standard troubleshooting category ***<troublecategory>*** (see 36.5.2).
      (c) DMWR/NMWR troubleshooting category ***<troubledmwrnwmrcategory>*** (see 36.5.3).
      (d) Aviation troubleshooting manual category ***<troubleaviationcategory>*** (see 36.5.4).

b. DTD fragment is graphically depicted for ***<tim>***:

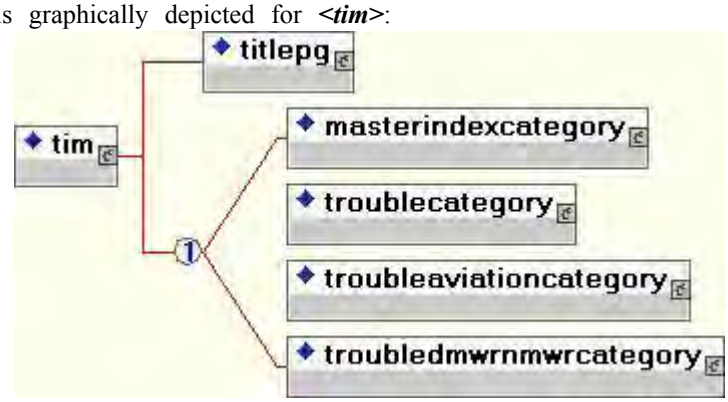

FIGURE 223. Troubleshooting instructions chapter DTD hierarchy <tim>.

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c. DTD fragment:

```
<!ELEMENT tim (titlepg, (masterindexcategory | troublecategory |
   troubleaviationcategory | troubledmwrnmwrcategory))>
<!ATTLIST tim
    assocfig  IDREFS                    #IMPLIED
    changeref IDREFS                    #IMPLIED
    chap-toc  (yes | no )               "yes"
    chngno    CDATA                     #REQUIRED
    comment   CDATA                     #IMPLIED
    delchlvl  CDATA                     #IMPLIED
    frame     (yes | no )               "yes"
    idref     IDREFS                    #IMPLIED
    inschlvl  CDATA                     #IMPLIED
    revno     CDATA                     #REQUIRED
    security  (uc | fouo | c | s | ts )  #IMPLIED
    skilltrk  CDATA                     #IMPLIED
    tocentry  (0 | 1 | 2)               "1"
>
```

d. Common attributes for ***<tim>***.

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **chap-toc** – Create a chapter work package table of contents (optional). Select either "**yes**" or "**no**" with a default value "**yes**".
(4) **chngno** – Change number (required) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(8) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(9) **inschlvl** – Insert change level (optional) (see 49.2.9).
(10) **revno** – Revision number (required) (see 49.2.9).
(11) **security** – Security classification (optional) (see 49.2.13).
(12) **skilltrk** – Skill level (optional) (see 49.2.2).
(13) **tocentry** – In the TM table of contents, the attribute indicates the indenture level. The possible selections are "**0**" (do not include), "**1**" (the highest level or 1st indenture level), or "**2**" (the second indentrue level) (default value is "**1**").

36.5 Troubleshooting categories. Depending on the type and complexity of the weapon system/equipment, the IETM or TM will contain one or more of the following troubleshooting category chapters and work packages (within the selected categories). In accordance with MIL–STD–40051–1/–2, the content matrices prescribes the optional and/or required troubleshooting categories and work packages for each technical manual type. The acquiring activity has the responsibility in selecting the troubleshooting category and the work packages within each category.

36.5.1 Master malfunction/symptom troubleshooting index category (page-based only) ***<masterindexcategory>***. As specified in MIL-STD-40051–2, one troubleshooting malfunction/symptom index work package is prepared to index all the troubleshooting work packages by malfunction and/or symptom for the system/equipment. Other troubleshooting information chapters are needed to provide the details for the operational checkout and/or troubleshooting procedures. Therefore, generally no other troubleshooting indices are need, but are not excluded from additional troubleshooting chapters.

a. The work package for ***<masterindexcategory>*** is troubleshooting index work package ***<tsindxwp>*** (required) (see 36.6).

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b. DTD fragment is graphically depicted for Master malfunction/symptom troubleshooting index category ***<masterindexcategory>***:

FIGURE 224. Master malfunction/symptom troubleshooting index category DTD hierarchy <masterindexcategory>.

c. DTD fragment for ***<masterindexcategory>***:

```
<!ELEMENT masterindexcategory (tsindxwp)>
```

36.5.2 Standard troubleshooting category ***<troublecategory>***. This category is generally used for most Technical Manuals. A critical consideration is the requirement or need for maintaining the system state information (see 48.1) and thus will be developing a frame-based manual only.

a. The work packages for ***<troublecategory>*** are:
   (1) Troubleshooting index work package ***<tsindxwp>*** (optional – zero or more) (see 36.6).
   (2) The troubleshooting method for the entire chapter using either
      (a) Interactive and intrusive diagnostic work packages (frame-based only) ***<diagnosticwp>*** (required – one or more) (see 36.11)
      (b) Standard troubleshooting work packages that include at least one of the following
         1 Operational checkout work package ***<opcheckwp>*** (see 36.8).
         2 Troubleshooting work package ***<tswp>*** (see 36.9).
         3 Combined operational checkout/troubleshooting work package ***<opcheck-tswp>*** (see 36.10).

b. DTD fragment is graphically depicted for Standard Troubleshooting Category ***<troublecategory>***:

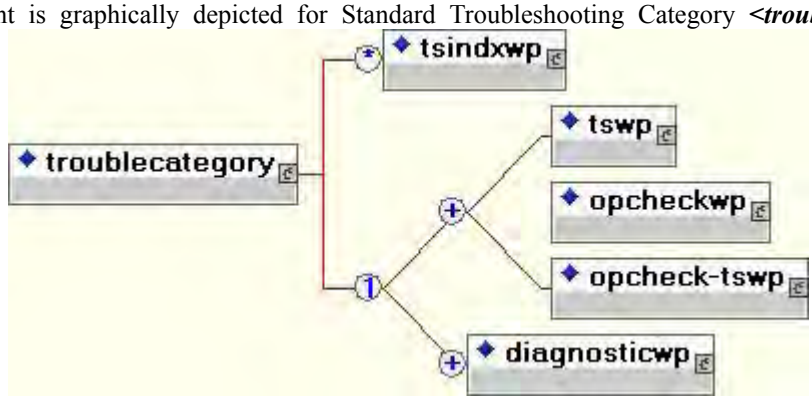

FIGURE 225. Standard troubleshooting category DTD hierarchy <troublecategory>.

c. DTD fragment for ***<troublecategory>***:

```
<!ELEMENT troublecategory (tsindxwp*, ((tswp | opcheckwp | opcheck-tswp)+ |
   diagnosticwp+))>
```

36.5.3 DMWR/NMWR troubleshooting category ***<troubledmwrnwmrcategory>***. This category is only applicable tor a DMWR or NMWR that is indicated in the manual number prefix with either DMWR or NMWR. The category clearly identifies to the developer the specify work packages needed for DMWRs or NMWRs. This category includes the same work packages in the standard troubleshooting category (refer to 36.5.2) (these work packages are only developed as specified by the acquiring activity), but has additionally the preshop analysis work package and component checklist work packages. When the DMWR or NMWR does not have a requirement in the troubleshooting chapter to use the preshop analysis work package, the developer will use the standard troubleshooting category.

a. The work packages for ***<troubledmwrnwmrcategory>*** are:
   (1) Troubleshooting index work package ***<tsindxwp>*** (optional – zero or more) (see 36.6).

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(2) Preshop analysis work package ***<pshopanalwp>*** (required) (see 36.12).
(3) Component checklist work package ***<compchklistwp>*** (optional) (see 36.13)
(4) The troubleshooting method for the entire chapter using either
(a) Interactive and intrusive diagnostic work packages (frame-based only) ***<diagnosticwp>*** (optional – zero or more) (see 36.11).
(b) Standard troubleshooting work packages that include the following (optional – zero or more):
1 Operational checkout work package ***<opcheckwp>*** (see 36.8).
2 Troubleshooting work package ***<tswp>*** (see 36.9).
3 Combined operational checkout/troubleshooting work package ***<opcheck-tswp>*** (see 36.10).

b. DTD fragment is graphically depicted for DMWR/NMWR Troubleshooting Category ***<troubledmwrnwmrcategory>***:

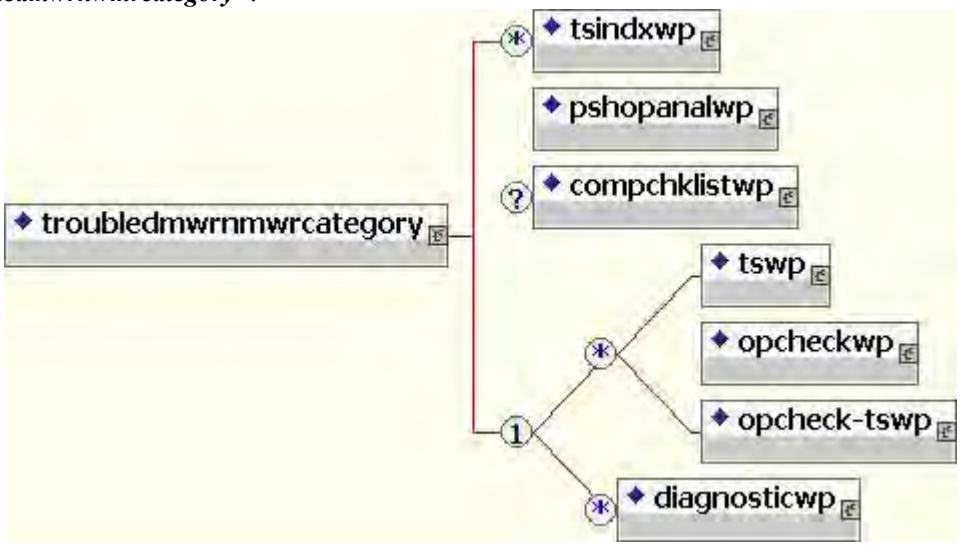

FIGURE 226. DMWR/NMWR troubleshooting category DTD hierarchy <troubledmwrnwmrcategory>.

c. DTD fragment for ***<troubledmwrnwmrcategory>***:

```
<!ELEMENT troubledmwrnwmrcategory (tsindxwp*, (pshopanalwp,
   compchklistwp?), ((tswp | opcheckwp | opcheck-tswp)* | diagnosticwp*))>
```

36.5.4 Aviation manual troubleshooting category ***<troubleaviationcategory>***. This category is similar to the standard troubleshooting category, but allows the Aviation Troubleshooting Technical Manuals to include requirements for an introduction work package and a system information work package (equipment description, theory of operation, and/or controls/indicators).

This category is only applicable for an Aviation Troubleshooting TM that is indicated in the TM number with a –T. The category identifies to the developer the specific work packages allowed for Aviation Troubleshooting TMs. This category includes the same work packages in the standard troubleshooting category (refer to 36.5.2), but has additionally the troubleshooting introduction work package and technical description work package (equipment description, theory of operation, and/or controls/indicators).

a. The work packages for ***<troubleaviationcategory>*** are:
(1) Troubleshooting introduction work package ***<tsintrowp>*** (optional). (see 36.14).
(2) Aviation troubleshooting technical description work package ***<techdescwp>*** (optional – zero or more) (see 36.15).
(3) Troubleshooting index work package ***<tsindxwp>*** (optional – zero or more) (see 36.6).
(4) The troubleshooting method for the entire chapter using either
(a) Interactive and intrusive diagnostic work packages (frame-based only) ***<diagnosticwp>*** (required – one or more) (see 36.11)

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(b) Standard troubleshooting work packages that include at least one of the following:
1 Operational checkout work package ***<opcheckwp>*** (see 36.8).
2 Troubleshooting work package ***<tswp>*** (see 36.9).
3 Combined operational checkout/troubleshooting work package ***<opcheck-tswp>*** (see 36.10).

b. DTD fragment is graphically depicted for Aviation Troubleshooting Category ***<troubleaviationcategory>***:

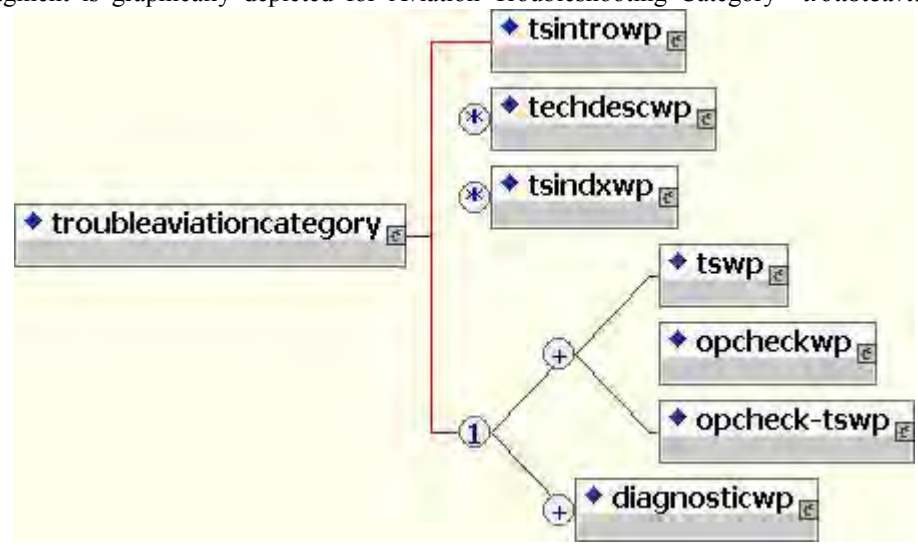

FIGURE 227. Aviation troubleshooting category DTD hierarchy <troubleaviationcategory>.

c. DTD fragment for ***<troubleaviationcategory>***:

```
<!ELEMENT troubleaviationcategory (tsintrowp?, techdescwp*, tsindxwp*,
   ((tswp | opcheckwp | opcheck-tswp)+ | diagnosticwp+))>
```

36.6 Troubleshooting index work package ***<tsindxwp>***. The troubleshooting index provides a list of system components, associated possible symptoms/malfunctions or fault codes/message words and the reference to the test or corrective action for the fault. The index is most commonly used in paper to determine the test or corrective action to next perform. The same index file can be used by IETM viewers to automatically determine the next test or corrective action by using the intrusive test equipment results (if connected to the test equipment) or user dialog interface. Generally, in the IETM or paper, this is where fault isolation starts. Sometimes the results from the diagnostic testing produces multiple fault codes, which certain combinations of the test results may indicate a different test or corrective action to perform then each result separately. The index allows for an index entry to have multiple fault codes/message words or symptoms/malfunctions grouped together to provide these differences.

a. The components of the troubleshooting index work package (***<tsindxwp>***) are:
(1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
(2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
(3) General information ***<geninfo>*** (optional) (see 49.1.4.12).
(4) Troubleshooting index (required) is by one of the following methods:
(a) System/subsystem ***<tsindx.system>*** (see 36.6.1).
(b) Symptom/malfunction ***<tsindx.symptom>*** (see 36.6.2).
(c) Fault code/message word ***<tsindx.messageword>*** (see 36.6.3).

b. DTD fragment is graphically depicted for Troubleshooting Index Work Package ***<tsindxwp>***.

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Check the source to verify that this is the current version before use.

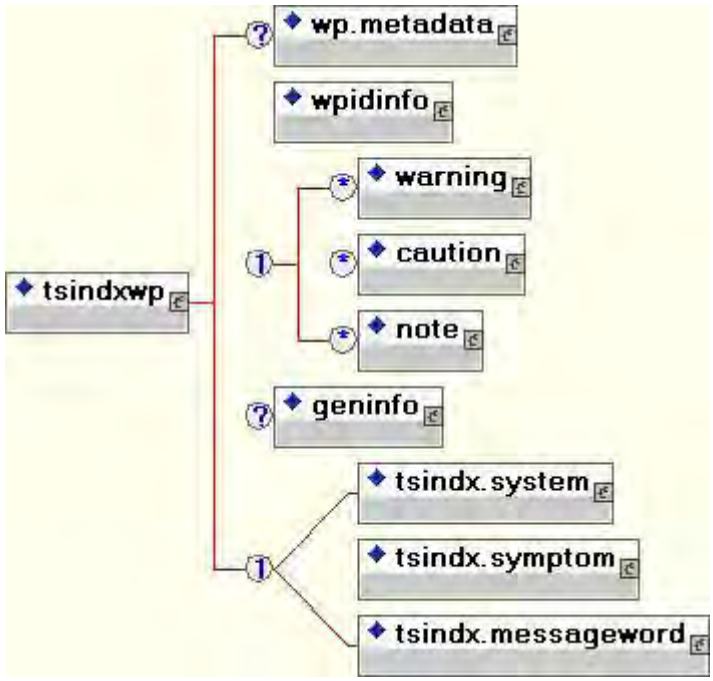

FIGURE 228. Troubleshooting index work package DTD hierarchy <tsindxwp>.

c. DTD fragment for (***<tsindxwp>***).

```
<!ELEMENT tsindxwp (wp.metadata?, wpidinfo,
   geninfo?, (tsindx.system | tsindx.symptom | tsindx.messageword))>
<!ATTLIST tsindxwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Attributes for ***<tsindxwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

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(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

36.6.1 Troubleshooting system/subsystem index. The index may group subsystems into categories as defined by the author. The index entries have the system name followed by either a reference to the testing/maintenance work package or a corrective action statement (if the action is relatively short).

a. The components of troubleshooting system/subsystem index (***<tsindx.system>***) are:
(1) Index table title ***<title>*** (required) (see 49.1.1.3).
(2) Standard table column headers ***<std.table.header>***(optional) (see 42.3.1). The predefined general entity ***&tsindx.system.header;*** is used for the troubleshooting index table header (see 50 for boilerplate entity).
(3) The system/subsystem is organized either by:
(a) System/subsystem categories ***<tsindx.system-category>*** (required – one or more). The components are:
1 System/subsystem title ***<title>*** (required) (see 49.1.1.3).
2 System/subsystem troubleshooting index entry ***<tsindx.system-entry>***(optional – zero or more)
3 Further breakdown system/subsystem category ***<tsindx.system-category>*** (optional – zero or more). This is a recursive element that allows the necessary system/subsystem levels.
(b) System/subsystem troubleshooting index entry ***<tsindx.system-entry>***.(required – one or more). The element is similar to a "row" in a structural table. The components are:

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1 System nomenclature ***<name>*** (required) (see 49.1.4.19). The element is similar to a "cell" in a structural table and is entered in column one.
2 The corrective action (The element is similar to a "cell" in a structural table and is entered in column two.) is either brief step(s) or references.
a Corrective action step(s) ***<action>*** (required) (see 36.6.1.1), generally states two or three steps or "Call maintenance supervisor."
b Reference to the troubleshooting work package to fault isolation or the maintenance work package for corrective action ***<link>/<xref> <extref>*** (required) (see 46).

b. DTD fragment is graphically depicted for

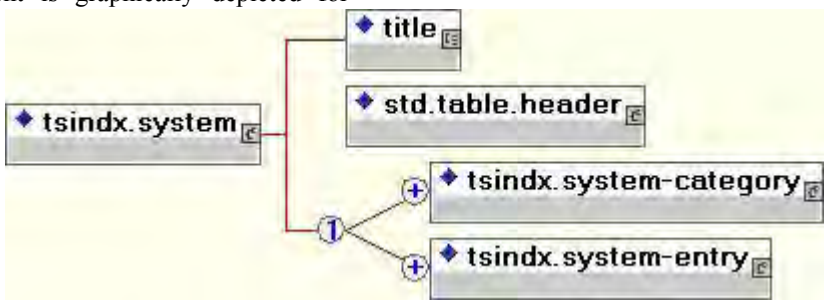

FIGURE 229. Troubleshooting system/subsystem index DTD hierarchy <tsindx.system> and children.

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Check the source to verify that this is the current version before use.

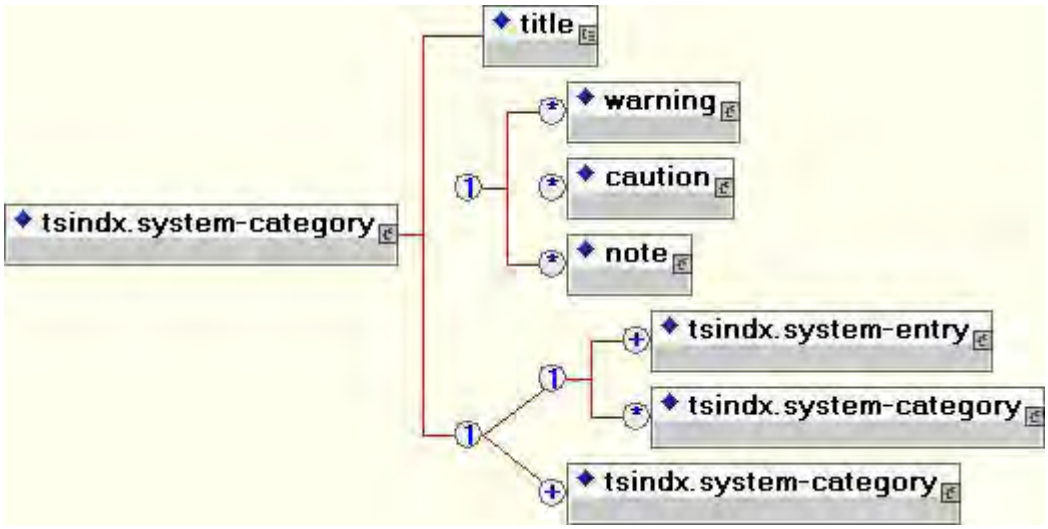

FIGURE 229. Troubleshooting system/subsystem index DTD hierarchy <tsindx.system> and children - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

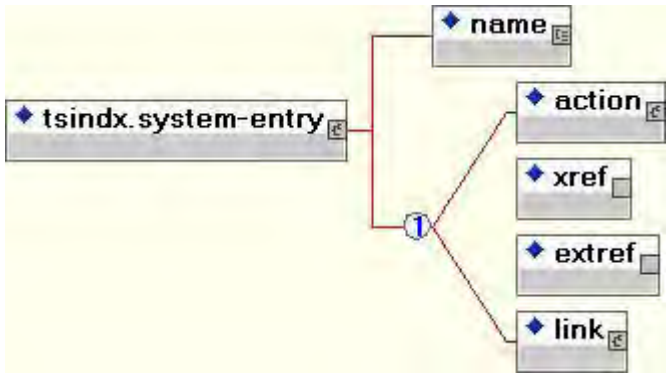

FIGURE 229. Troubleshooting system/subsystem index DTD hierarchy <tsindx.system> and children - Continued.

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c. DTD fragment for ***<tsindx.system>***, ***<tsindx.system-category>***, and ***<tsindx.system-entry>***.

```
<!ELEMENT tsindx.system (title, std.table.header?,
   (tsindx.system-category+ | tsindx.system-entry+))>
<!ATTLIST tsindx.system
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1"
>


<!ELEMENT tsindx.system-category (title,
   ((tsindx.system-entry+, tsindx.system-category*) |
   tsindx.system-category+))>
<!ATTLIST tsindx.system-category
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>


<!ELEMENT tsindx.system-entry (name, (action | link | xref | extref))>
<!ATTLIST tsindx.system-entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<tsindx.system>***, ***<tsindx.system-category>***, and ***<tsindx.system-entry>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).

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(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

36.6.1.1 Corrective action ***<action>***. The element ***<action>***provides the corrective action steps or reference to fix an error, problem, defect, or symptom.

a. The components are:
(1) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(2) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(3) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(4) At least of the following is required:
(a) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).
(b) Table ***<table>*** (see 42) and/or conditional table ***<table-alt>*** (see 48.2.1).
(c) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1-alt>*** (see 48.2.1).
(d) Paragraph ***<para>*** (see 49.1.1.5) or paragraph with alerts ***<specpara>*** (see 49.1.1.6) (required).

b. DTD fragment graphically is:

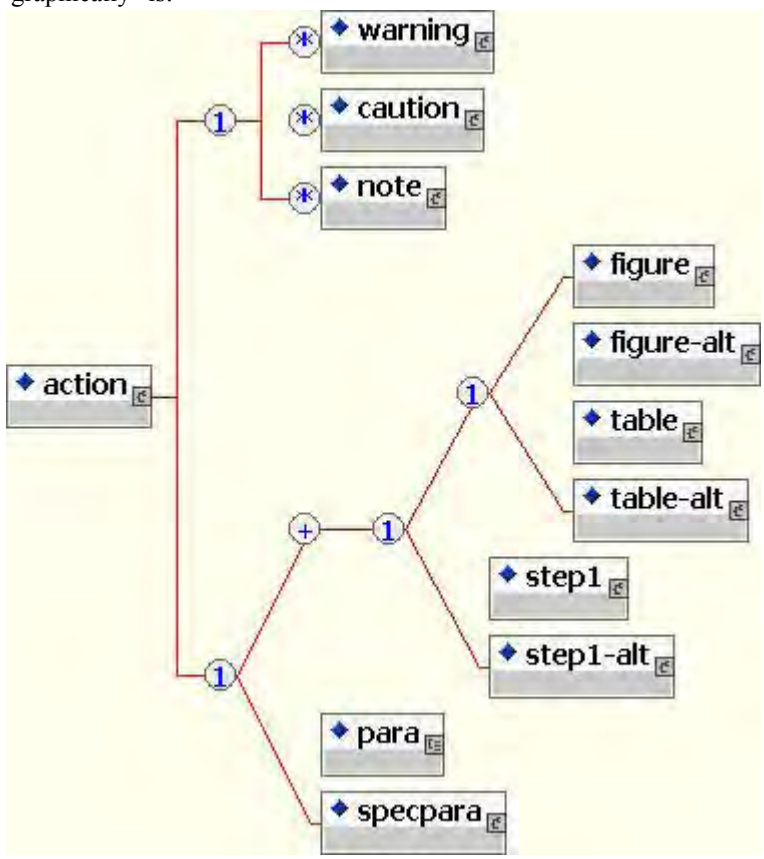

FIGURE 230. Corrective action <action>.

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c. DTD fragment for ***<action>*** is:

```
<!ELEMENT action  ((warning*, caution*, note*),
 ((((figure | figure-alt | table | table-alt) | step1 |
 step1-alt))+ | para | specpara))>
<!ATTLIST action

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Unique attributes::

(1) **label** – Specifies the column heading if the presentation format is tabular (required).

(2) **fault code** – Specifies the fault code associated with the malfunction .

e. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

(3) **comment** – Reason for change information. (optional) (See 49.2.9.)

(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)

(5) **id** – Unique identifier. (optional) (See 49.2.6.)

(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)

(9) **security** – Security classification. (optional) (See 49.2.13.)

36.6.2 Troubleshooting symptom/malfunction index. This index may group symptoms into categories as defined by the author. The index entries have a symptom or multiple symptoms (if the symptoms together determines the next action) followed by either a reference to the testing/maintenance work package or a corrective action statement (if the action is relatively short).

a. The components of troubleshooting symptom/malfunction index (***<tsindx.symptom>***) are:

(1) Index table title ***<title>*** (required) (see 49.1.1.3).

(2) Standard table column headers ***<std.table.header>*** (optional) (see 42.3.1). The predefined general entity ***&tsindx.symptom.header;*** is used for the troubleshooting index table header (see 50 for boilerplate entity).

(3) The symptoms/malfunctions are organized either by:

(a) Symptom/malfunction category ***<tsindx.symptom-category>*** (required – one or more). The components are:

1 Symptom/malfunction category title ***<title>*** (required) (see 49.1.1.3).

2 Symptom/malfunction troubleshooting index entry ***<tsindx.symptom-entry>*** (optional – zero or more)

(b) Symptom/malfunction troubleshooting index entry ***<tsindx.symptom-entry>*** (required – one or more). The element is similar to a "row" in a structural table. The components are:

1 Symptom/malfunction ***<malfunc>*** (required) (see 36.6.2.1). The element is similar to a “cell” in a structural table and is entered in column one.

2 The corrective action (The element is similar to a “cell” in a structural table and is entered in column two.) is either brief step(s) or references.

a Corrective action step(s) ***<action>*** (required) (see 36.6.1.1), generally states two or three steps or “Call maintenance supervisor.”

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b Reference to the troubleshooting work package to fault isolation or the maintenance work package for corrective action ***<link>/<xref> <extref>*** (required) (see section 46).

b. DTD fragment is graphically depicted for

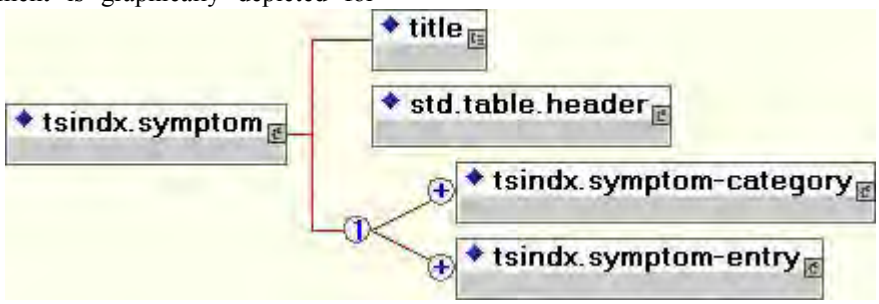

FIGURE 231. Troubleshooting symptom/malfunction index DTD hierarchy <tsindx.symptom> and children.

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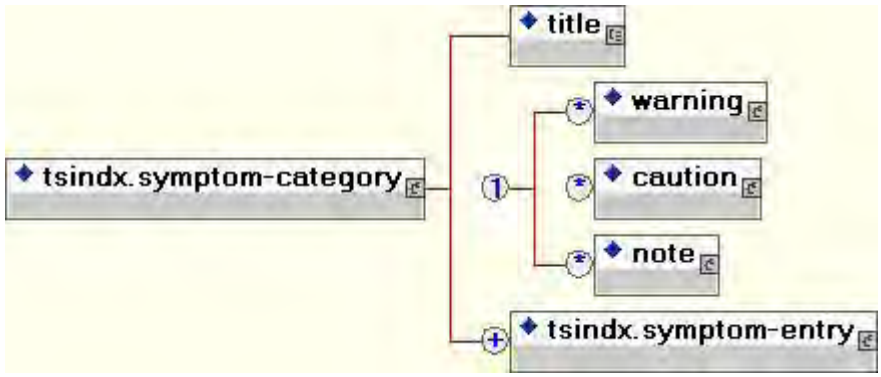

FIGURE 231. Troubleshooting symptom/malfunction index DTD hierarchy <tsindx.symptom> and children - Continued.

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Check the source to verify that this is the current version before use.

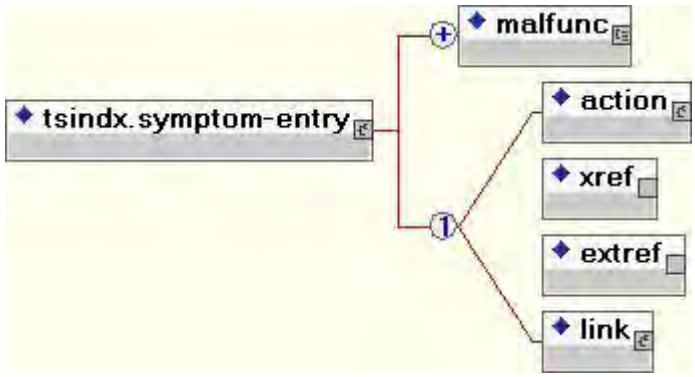

FIGURE 231. Troubleshooting symptom/malfunction index DTD hierarchy <tsindx.symptom> and children - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment for (***<tsindx.symptom-category>***), (***<tsindx.symptom-category>***), and (***<tsindx.symptom-entry>***).

```
<!ELEMENT tsindx.symptom (title, std.table.header?,
    (tsindx.symptom-category+ | tsindx.symptom-entry+))>
<!ATTLIST tsindx.symptom
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1"
>


<!ELEMENT tsindx.symptom-category (title, (warning*, caution*, note*),
     tsindx.symptom-entry+)>
<!ATTLIST tsindx.symptom-category
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>


<!ELEMENT tsindx.symptom-entry (malfunc+,(action | link | xref | extref))>
<!ATTLIST tsindx.symptom-entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. DTD attributes for ***<tsindx.symptom>***, ***<tsindx.symptom-category>***, and ***<tsindx.symptom-entry>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).

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(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

36.6.2.1 Symptom/malfunction ***<malfunc>***. The element specifies the probable malfunction or symptom.
a. The components of ***<malfunc>*** are:
(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).
b. DTD fragment graphically is:

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Check the source to verify that this is the current version before use.

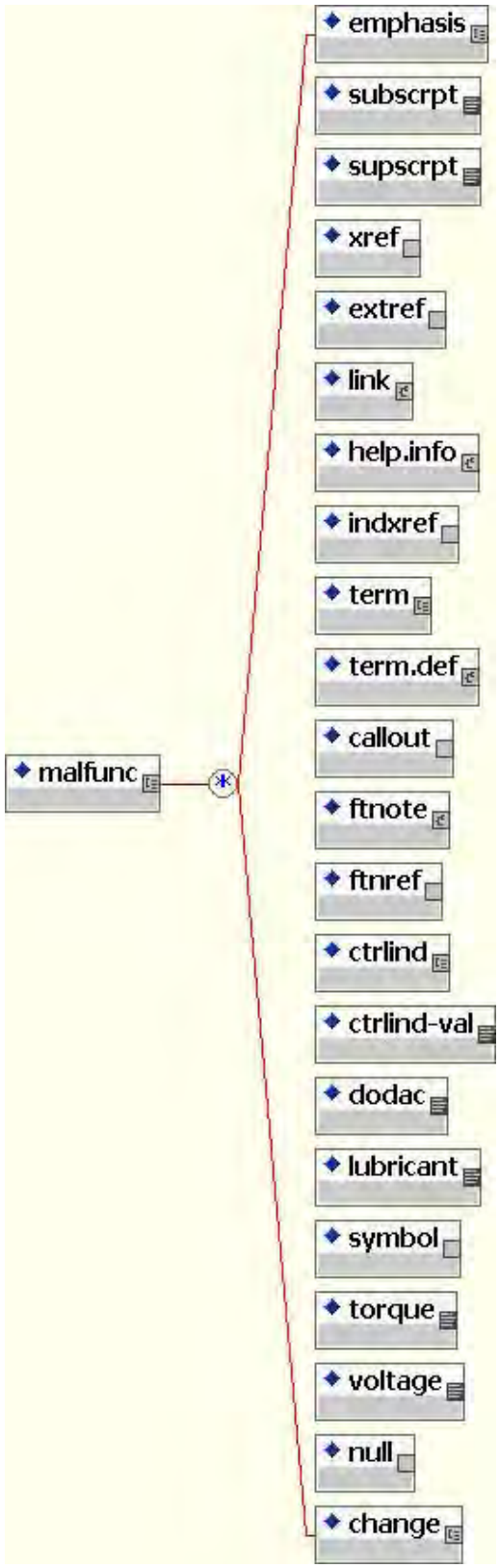

FIGURE 232. Malfunction DTD hierarchy <malfunc>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<malfunc>*** is:

```
<!ELEMENT malfunc (#PCDATA | emphasis | subscrpt | supscrpt | xref |
   extref | link | help.info | indxref | term | term.def | callout |
   ftnote | ftnref | ctrlind | ctrlind-val | dodac | lubricant |
   symbol | torque | voltage | null | change)*>
<!ATTLIST malfunc
    label             (symptom | malfunction)          #REQUIRED
    faultcode         CDATA                             #IMPLIED
    assocfig          IDREFS                            #IMPLIED
    changeref         IDREFS                            #IMPLIED
    comment           CDATA                             #IMPLIED
    delchlvl          CDATA                             #IMPLIED
    id                ID                                #IMPLIED
    idref             IDREFS                            #IMPLIED
    inschlvl          CDATA                             #IMPLIED
    security          (uc | fouo |c | s | ts)          #IMPLIED
    skilltrk          CDATA                             #IMPLIED
>
```

d. Attributes for ***<malfunc>*** are:

(1) **label** – Specifies the column heading if the presentation format is tabular (required).
(2) **fault code** – Specifies the fault code associated with the malfunction (optional).
(3) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(4) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(5) **comment** – Reason for change information. (optional) (See 49.2.9.)
(6) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(7) **id** – Unique identifier. (optional) (See 49.2.6.)
(8) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(9) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(10) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(11) **security** – Security classification. (optional) (See 49.2.13.)

36.6.3 Troubleshooting fault code/message word index. The index may group fault codes/message words into categories as defined by the author. The index entries have the fault code/message word or multiple fault codes/message words (if the fault codes/message words together determine the next action) followed by either a reference to the testing/maintenance work package or a corrective action statement (if the action is relatively short).

a. The components of troubleshooting fault code/message word index (***<tsindx.messageword>***) are:

(1) Index table title ***<title>*** (required) (see 49.1.1.3).
(2) Standard table column headers ***<std.table.header>*** (optional) (see 42.3.1). The predefined general entity ***&tsindx.messageword.header;*** is used for the troubleshooting index table header (see 50 for boilerplate entity).
(3) The symptoms/malfunctions ares organized either by:
  (a) Fault code/message word category ***<tsindx.messageword-category>*** (required – one or more) and the components are:
    1 Fault code/message word category title ***<title>*** (required) (see 49.1.1.3).
    2 Fault code/message word troubleshooting index entry ***<tsindx.messageword-entry>***(optional – zero or more)
  (b) Fault code/message word troubleshooting index entry ***<tsindx.messageword-entry>*** (required – one or more). The element is similar to a "row" in a structural table. The components are:
    1 Fault code/message word ***<messageword>*** (required) (see 36.6.2.1). The element is similar to a “cell” in a structural table and is entered in column one.
    2 The corrective action (The element is similar to a “cell” in a structural table and is entered in column two.) is either brief step(s) or references.

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a Corrective action step(s) ***<action>*** (required) (see 36.6.1.1), generally states two or three steps or "Call maintenance supervisor."

b Reference to the troubleshooting work package to fault isolation or the maintenance work package for corrective action ***<link>/<xref> <extref>*** (required) (see 46).

b. DTD fragment is graphically depicted for

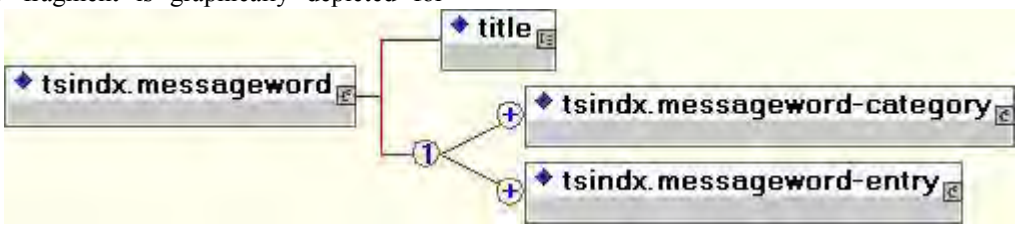

FIGURE 233. Troubleshooting fault code/message word index DTD hierarchy <tsindx.messageword> and children.

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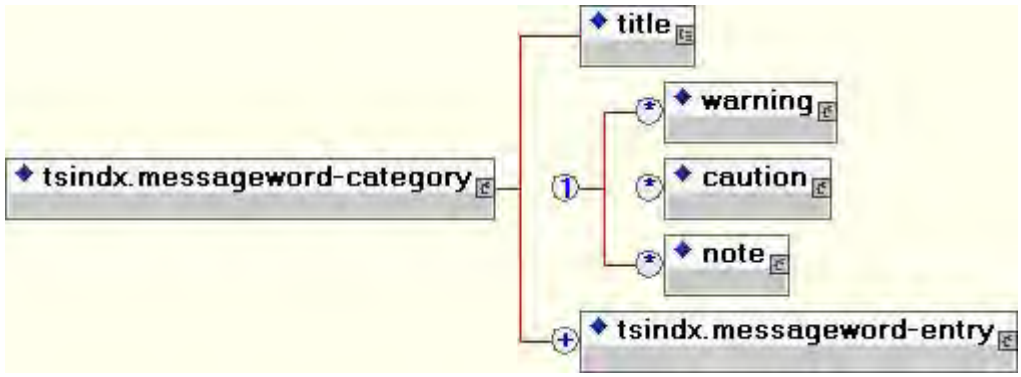

FIGURE 233. Troubleshooting fault code/message word index DTD hierarchy <tsindx.messageword> and children - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

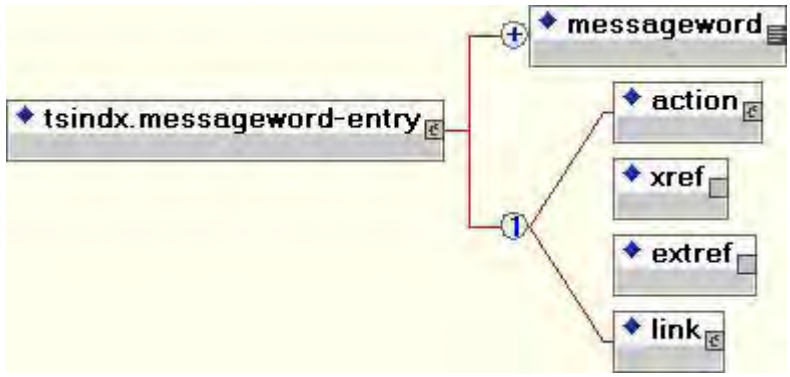

FIGURE 233. Troubleshooting fault code/message word index DTD hierarchy <tsindx.messageword> and children - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment for ***<tsindx.messageword-category>***, ***<tsindx.messageword-category>***, and ***<tsindx.messageword-entry>***.

```
<!ELEMENT tsindx.messageword (title, std.table.header,
     (tsindx.messageword-category+ | tsindx.messageword-entry+))>
<!ATTLIST tsindx.messageword
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1"
>


<!ELEMENT tsindx.messageword-category (title, (warning*, caution*, note*),
     tsindx.messageword-entry+)>
<!ATTLIST tsindx.messageword-category
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>

<!ELEMENT tsindx.messageword-entry (messageword+, (action | link |
   xref | extref))>
<!ATTLIST tsindx.messageword-entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. DTD attributes for ***<tsindx.messageword>***, ***<tsindx.messageword-category>***, and ***<tsindx.messageword-entry>***.

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).

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(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

36.6.4 XML document instance fragment and output for ***<tsindxwp>***. The XML instance and its stylesheet output for a ***<tsindxwp>*** is provided below.

a. Sample XML instance.

```
<tsindxwp wpno="T10002-X-XXXX-XXX" tocentry="2" frame="yes" army="no"
airforce="no" navy="no" marines="no" wpseq="0231" deletewp="no">
<wpidinfo>
<maintlvl level="operator"/>
<title>Troubleshooting Symptom Index
</title>
</wpidinfo>
<geninfo frame="no">
<title>General
</title>
<para0 hcp="no" esd="no">
<title>Troubleshooting Symptom Index
</title>
<para hcp="no" esd="no">The Troubleshooting Symptom Index lists common
malfunctions found during operation or crew servicing of the M2A3
and M3A3.  The Troubleshooting Symptom Index is divided into
sections.  Each section covers malfunctions common to the different
systems of the vehicle (engine, transmission, suspension).
</para>
</para0>
<para0 hcp="no" esd="no">
<title>Troubleshooting Table
</title>
<para hcp="no" esd="no">The TROUBLESHOOTING TABLE has three divisions:
MALFUNCTION, TEST OR INSPECTION, and CORRECTIVE ACTION. The MALFUNCTIONS,
or systems, are numbered in sequence through the TROUBLESHOOTING TABLE.
The MALFUNCTION is what will bring you to the TROUBLESHOOTING TABLE. The
MALFUNCTION is what will bring you to the TROUBLESHOOTING TABLE.
</para>
<para hcp="no" esd="no">TEST OR INSPECTION is a step you take to isolate the
MALFUNCTION. Each TEST OR INSPECTION has a CORRECTIVE ACTION. These are
"if" statements which tell you what to do when the MALFUNCTION is not fixed.
</para>
<para hcp="no" esd="no">After you have identified the malfunction
describing your problem (symptom) from the symptom index, turn to
the appropriate troubleshooting table.  Follow each step through
the tests or inspections until a fault is identified.  Perform the
corrective action listed to correct the fault.  If the fault cannot
be identified or corrected, notify unit maintenance be writing a
<extref docno="DA Form 2404"/> describing the symptom.
</para>
<para hcp="no" esd="no">The operator's manual cannot list all possible
malfunctions, nor all the tests and inspections required for corrective
```


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actions. If a malfunction is not listed, or is not corrected by the listed corrective action, notify your supervisor.
</para>
</para0>
</geninfo>
<tsindx.symptom tocentry="1">
<title>Troubleshooting Symptom Index
</title>
<std.table.header cols="2">
<std.table.row>
<std.table.entry cols.span="1">Symptom/Malfunction
</std.table.entry>
<std.table.entry cols.span="1">Troubleshooting WP Sequence
</std.table.entry>
</std.table.row>
</std.table.header>
<tsindx.symptom-category>
<title>ENGINE
</title>
<tsindx.symptom-entry>
<malfunc label="symptom">
<ctrlind>ENGINE OIL PRESS
</ctrlind> gauge is in red zone
</malfunc>
<xref wpid="T00003-X-XXXX-XXX"/>
</tsindx.symptom-entry>
<tsindx.symptom-entry>
<malfunc label="symptom">
<ctrlind>ENGINE COOLANT TEMP
</ctrlind> gauge is in red zone
</malfunc>
<xref wpid="T00004-X-XXXX-XXX"/>
</tsindx.symptom-entry>
<tsindx.symptom-entry>
<malfunc label="symptom">
<ctrlind>COOLANT HI TEMP
</ctrlind> indicator light flashes
</malfunc>
<xref wpid="T00005-X-XXXX-XXX"/>
</tsindx.symptom-entry>
<tsindx.symptom-entry>
<malfunc label="symptom">
<ctrlind>AIR CLEANER CLOGGED
</ctrlind> indicator light flashes
</malfunc>
<xref wpid="T00006-X-XXXX-XXX"/>
</tsindx.symptom-entry>
<tsindx.symptom-entry>
<malfunc label="symptom">
<ctrlind>FUEL FILTER CLOGGED
</ctrlind> indicator light flashes
</malfunc>
<xref wpid="T00007-X-XXXX-XXX"/>
</tsindx.symptom-entry>
<tsindx.symptom-entry>
<malfunc label="symptom">Excessive smoke, loss of power

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</malfunc>
<xref wpid="T00008-X-XXXX-XXX"/>
</tsindx.symptom-entry>
</tsindx.symptom-category>
<tsindx.symptom-category>
<title>TRANSMISSION
</title>
<tsindx.symptom-entry>
<malfunc label="symptom">
<ctrlind>TRANS OIL PRESS
</ctrlind> indicator light flashes
</malfunc>
<xref wpid="T00009-X-XXXX-XXX"/>
</tsindx.symptom-entry>
<tsindx.symptom-entry>
<malfunc label="symptom">
<ctrlind>TRANS OIL TEMP
</ctrlind> indicator light flashes
</malfunc>
<xref wpid="T00010-X-XXXX-XXX"/>
</tsindx.symptom-entry>
</tsindx.symptom-category>
<tsindx.symptom-category>
<title>BILGE PUMPS
</title>
<tsindx.symptom-entry>
<malfunc label="symptom">Bilge pumps discharge little or no water.
</malfunc>
<xref wpid="T00011-X-XXXX-XXX"/>
</tsindx.symptom-entry>
</tsindx.symptom-category>
<tsindx.symptom-category>
<title>SUSPENSION
</title>
<tsindx.symptom-entry>
<malfunc label="symptom">Vehicle pulls to one side on level ground
</malfunc>
<xref wpid="T00012-X-XXXX-XXX"/>
</tsindx.symptom-entry>
<tsindx.symptom-entry>
<malfunc label="symptom">Track thumps while driving
</malfunc>
<xref wpid="T00013-X-XXXX-XXX"/>
</tsindx.symptom-entry>
</tsindx.symptom-category>
<tsindx.symptom-category>
<title>RAMP
</title>
<tsindx.symptom-entry>
<malfunc label="symptom">Ramp will not lower
</malfunc>
<xref wpid="T00014-X-XXXX-XXX"/>
</tsindx.symptom-entry>
</tsindx.symptom-category>
</tsindx.symptom>
</tsindxwp>

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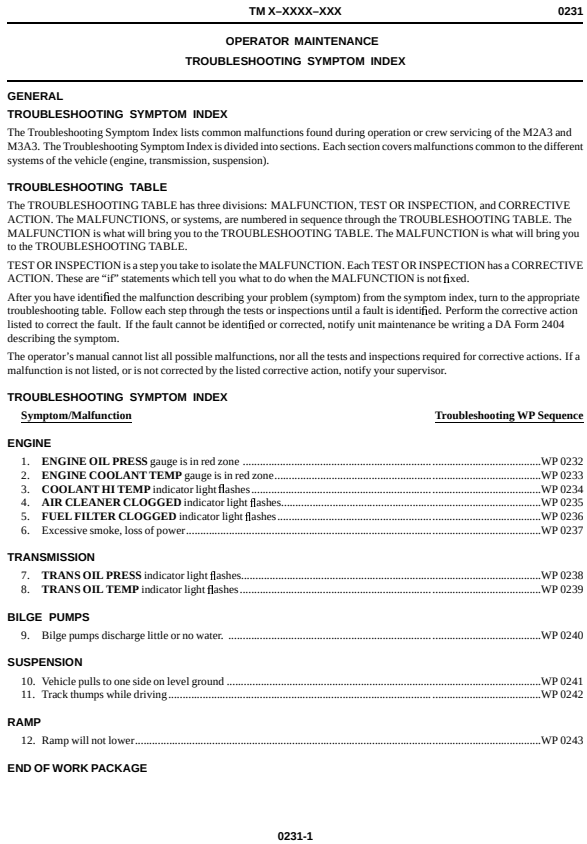
b. XML sample page-layout.



TM X–XXXX–XXX 0231

OPERATOR MAINTENANCE
TROUBLESHOOTING SYMPTOM INDEX

GENERAL

TROUBLESHOOTING SYMPTOM INDEX

The Troubleshooting Symptom Index lists common malfunctions found during operation or crew servicing of the M2A3 and M3A3. The Troubleshooting Symptom Index is divided into sections. Each section covers malfunctions common to the different systems of the vehicle (engine, transmission, suspension).

TROUBLESHOOTING TABLE

The TROUBLESHOOTING TABLE has three divisions: MALFUNCTION, TEST OR INSPECTION, and CORRECTIVE ACTION. The MALFUNCTIONS, or systems, are numbered in sequence through the TROUBLESHOOTING TABLE. The MALFUNCTION is what will bring you to the TROUBLESHOOTING TABLE. The MALFUNCTION is what will bring you to the TROUBLESHOOTING TABLE.

TEST OR INSPECTION is a step you take to isolate the MALFUNCTION. Each TEST OR INSPECTION has a CORRECTIVE ACTION. These are "if" statements which tell you what to do when the MALFUNCTION is not fixed.

After you have identified the malfunction describing your problem (symptom) from the symptom index, turn to the appropriate troubleshooting table. Follow each step through the tests or inspections until a fault is identified. Perform the corrective action listed to correct the fault. If the fault cannot be identified or corrected, notify unit maintenance be writing a DA Form 2404 describing the symptom.

The operator's manual cannot list all possible malfunctions, nor all the tests and inspections required for corrective actions. If a malfunction is not listed, or is not corrected by the listed corrective action, notify your supervisor.

TROUBLESHOOTING SYMPTOM INDEX

Symptom/Malfunction	Troubleshooting WP Sequence
ENGINE	
1. **ENGINE OIL PRESS** gauge is in red zone	WP 0232
2. **ENGINE COOLANT TEMP** gauge is in red zone	WP 0233
3. **COOLANT HI TEMP** indicator light flashes	WP 0234
4. **AIR CLEANER CLOGGED** indicator light flashes	WP 0235
5. **FUEL FILTER CLOGGED** indicator light flashes	WP 0236
6. Excessive smoke, loss of power	WP 0237
TRANSMISSION	
7. **TRANS OIL PRESS** indicator light flashes	WP 0238
8. **TRANS OIL TEMP** indicator light flashes	WP 0239
BILGE PUMPS	
9. Bilge pumps discharge little or no water.	WP 0240
SUSPENSION	
10. Vehicle pulls to one side on level ground	WP 0241
11. Track thumps while driving	WP 0242
RAMP	
12. Ramp will not lower	WP 0243

END OF WORK PACKAGE

0231-1

FIGURE 234. Troubleshooting index work package (<tsindxwp>) Page-layout.

36.7 Troubleshooting system description and support information. The elements listed below contains support information used throughout the testing in the troubleshooting, operational checkout, and combined troubleshooting/operational checkout work packages. The support information is generally referenced throughout the operational checkout and troubleshooting work packages. The use of ID/IDREF provides a mechanism to link the support information with the fault isolation steps.

a. The components are:

(1) Single procedure

(a) Procedural steps title ***<title>*** (required) (see 49.1.1.3).

(b) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).

(c) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).

(d) Note – ***<note>*** (optional – zero or more) (see 41.1.3).

(e) General information ***<geninfo>*** (optional) (see 49.1.4.12).

(f) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).

(g) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).

(h) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1-alt>*** (see 48.2.1) (required – one or more).

(2) Multiple procedure ***<proc>*** (required – two or more) (see 33.2).

(3) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

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b. DTD fragment is graphically depicted as:

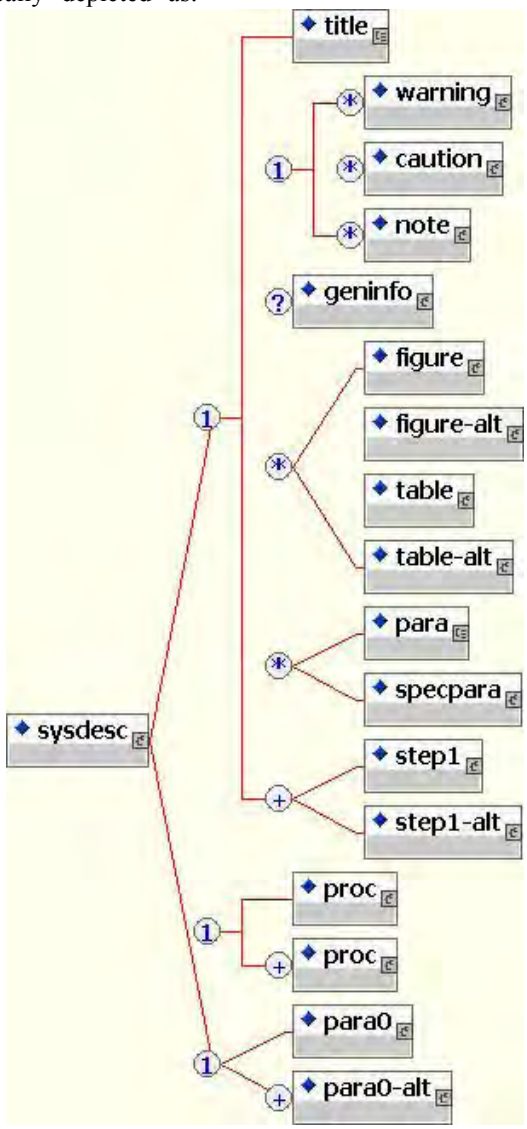

FIGURE 235. Troubleshooting system description and support information DTD hierarchy.

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Check the source to verify that this is the current version before use.

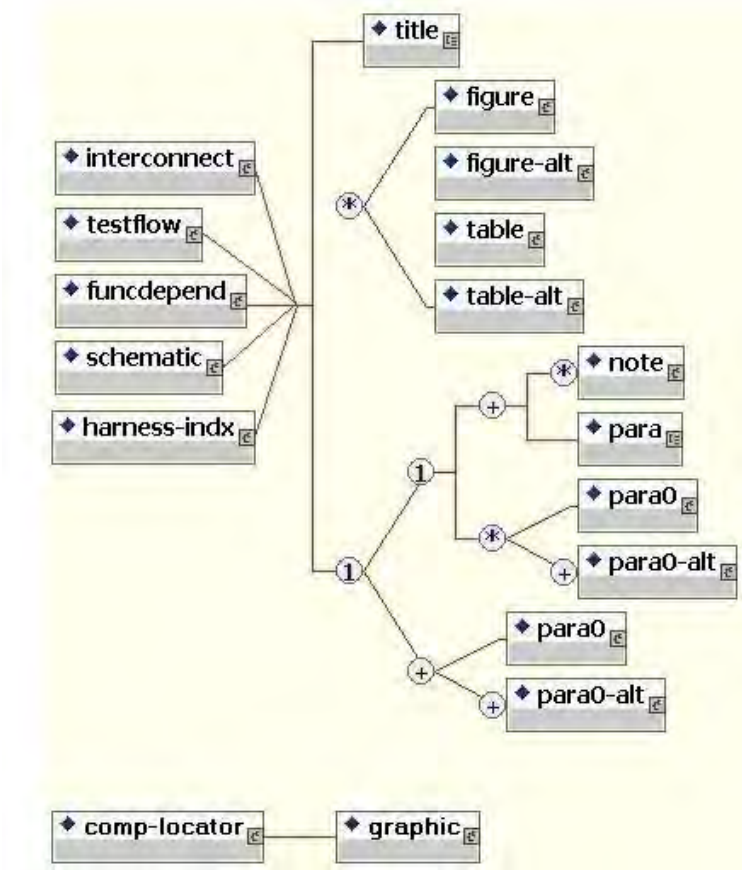

FIGURE 235. Troubleshooting system description and support information DTD hierarchy - Continued.

c. Common attributes are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.7.1 System description ***<sysdesc>***. The element is used for description of the system/subsystem under test provided as supporting technical information; contained either as optional introductory section of a troubleshooting work package.

36.7.2 Interconnection ***<interconnect>***. The element contains diagrams or other means of presenting the electrical and electronic connections between components of the system under test.

36.7.3 Test flow ***<testflow>***. The element describes the troubleshooting testing flow.

36.7.4 Functional dependencies ***<funcdepend>***. The element describes the functional dependencies of components that make up the system under test.

36.7.5 Schematic drawings ***<schematic>***. The element is used for schematic drawings included as supporting technical information during a troubleshooting procedure.

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36.7.6 Component locator ***<comp-locator>***. The element contains a figure to assist in the components under test location.

36.7.7 Harness index ***<harness-indx>***. The element is a special index of electrical wiring harnesses, needed due to the extensive interrelated circuitry.

36.8 Operational checkout work package ***<opcheckwp>***. Operational checkout procedures subjects the equipment to prescribed conditions to determine if it will function with the predetermined test parameters. The operational checkout procedure ***<opcheckproc>*** includes an index of test set message words, description, and reference to corrective action or further diagnosis (generated from a test set or BIT/BITE readings) (see 36.8.3), a reference index of test set or BIT/BITE fault codes and related follow-on actions (generated from a test set or BIT/BITE readings) (see 36.8.4), or operational testing procedures that include test steps, indication and correct actions (either fix fault or additional testing procedures) (see 36.8.1). The work package is used either after a maintenance task is completed and is used to determine if the system is operating correctly and all faults have been identified or assist in the fault deduction through operational checkout steps. When trying to identify the malfunction the first work package to reference is the troubleshooting index to direct user to the most likely task to determine and correct the fault.

a. The components of operational checkout work package ***<opcheckwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Introduction ***<intro>*** (optional) (see 49.1.4.15) explaining how the operational checkout procedures are to be used to perform testing and how they relate to the associated troubleshooting work packages.
   (5) General procedures and precautions ***<proc>*** (optional – zero or more) (see 33.2) that are performed prior to checkout and precautions that are taken during the performance of the checkout procedure.
   (6) System description and support information (optional) (see 36.7).
   (7) Test set hookup ***<hookup>*** (optional) (see 33.1 how to develop a task). If test set hookup is used test set disconnection is required.
   (8) Operational checkout procedure ***<opcheckproc>*** (required) uses only one of the following methods per work packages.
      (a) Message word reference index ***<messageindx>*** (required) (see 36.8.3).
      (b) Fault code reference report ***<faultreport>*** (required) (see 36.8.4).
      (c) Operational checkout steps ***<opcheck>*** (required) (see 36.8.1).
   (9) Test set disconnection ***<disconnect>*** (optional) (see 36.8.2 (see 33.1 how to develop a task). If test set hookup is used test set disconnection is required.

b. DTD fragment graphically is:

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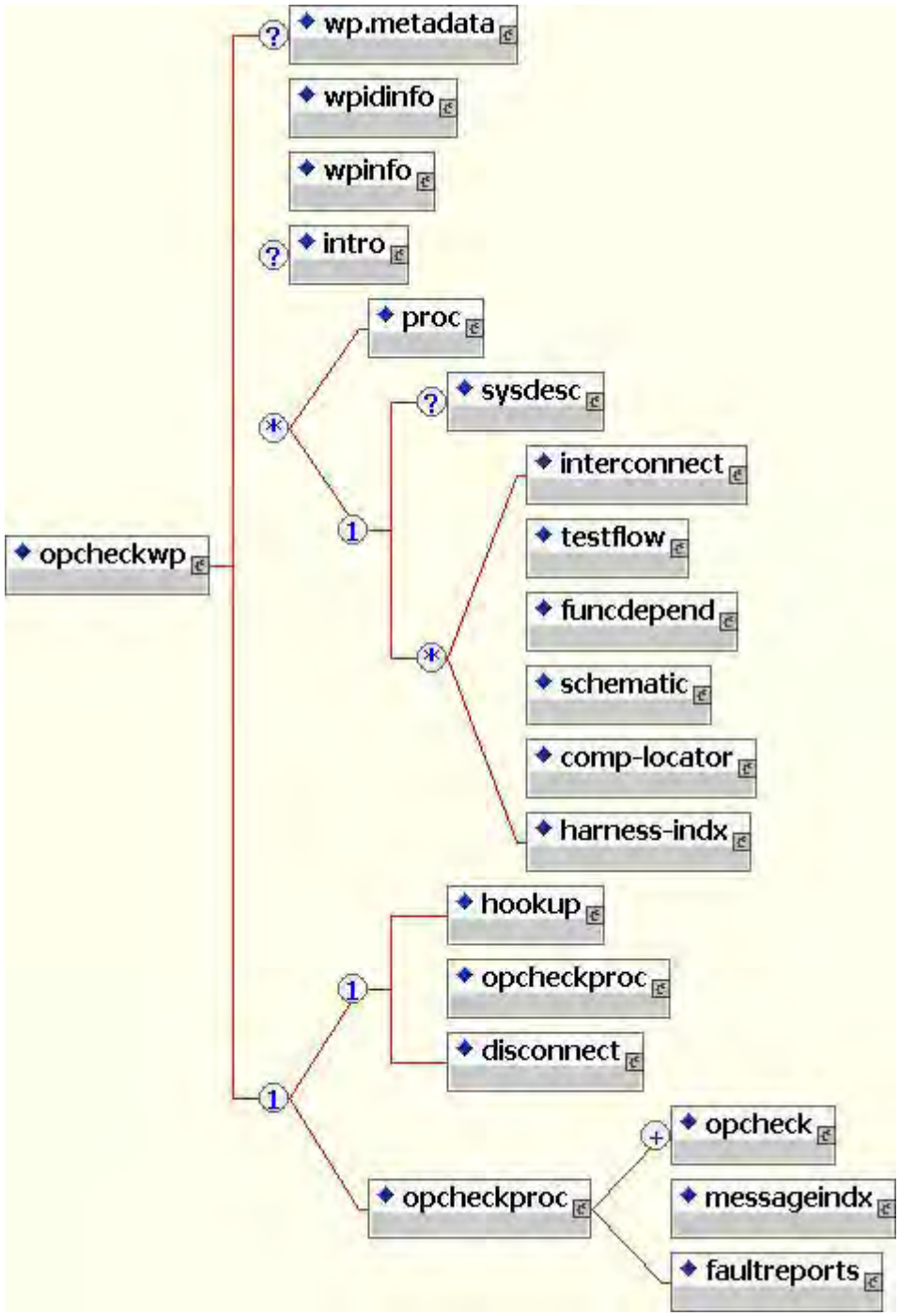

FIGURE 236. Operational checkout work package DTD hierarchy <opcheckwp>.

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c. DTD fragment is:

```
<!ELEMENT opcheckwp (wp.metadata?, wpidinfo, wpinfo, intro?,
     (proc | (sysdesc?,(interconnect | testflow | funcdepend | schematic |
     comp-locator | harness-indx)*))*,
     ((hookup, opcheckproc, disconnect) | opcheckproc))>
<!ATTLIST opcheckwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
<!ELEMENT opcheckproc (opcheck+ | messageindx | faultreports)>
<!ATTLIST opcheckproc
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<opcheckcwp>*** and ***<opcheckproc>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

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(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

36.8.1 Operational checkout testing ***<opcheck>***. The element contains an ordered set of operational checkout test procedures to obtain results that will determine if system is in either operational condition or requires corrective action. Each operational test procedure contains checkout instructions to perform that lead to an indication/condition. When a normal indication/conditional is obtained, the next operational checkout step is performed. When an abnormal condition/indication is observed then corrective action (or reference to detailed troubleshooting or maintenance work packages) is performed. The operational checkout test table can be represented in a narrative format or as tabular. When shown in tabular format, the element is similar to a "row" in a structural table.

In a frame-based viewer, the information can be presented similar to page-based or have an interactive capability. Figure 241, Figure 242 and Figure 243 show sample interactive frames how the operational checkout procedure could be presented and processed. The operational checkout instructions and indication/condition is presented to the maintainer (see Figure 241). If the indication/condition is abnormal (selected the "NO" button), then the corrective action is presented (see Figure 242). If the indication/condition is normal (selected the "YES" button), then the next checkout test is shown (see Figure 243). The frame format and interaction can vary dependent on the software and stylesheet, this is only a possible method.

a. The components of operational checkout ***<opcheck>*** are:
   (1) Overall title ***<title>*** (optional) (see 49.1.1.3) for the checkout procedures.
   (2) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (3) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (4) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (5) Operation checkout procedure ***<testproc>*** (required – one or more) that contains:
      (a) Table header ***<std.table.header>*** (optional) (see 42.3.1) that contain column headers when the information is shown tabular.
      (b) Operation checkout step ***<checkstep>*** (required) (see 36.8.1.1).
b. DTD fragment graphically is:

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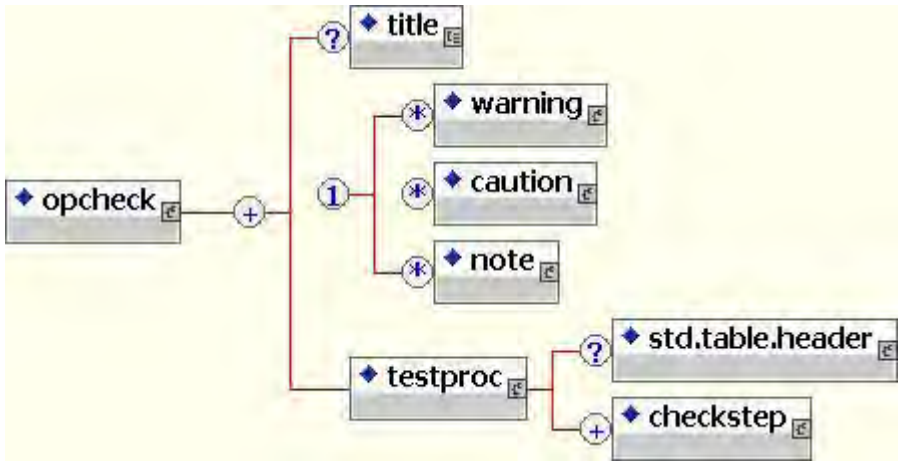

FIGURE 237. Operational check testing DTD hierarchy <opcheck>.

c. DTD fragment for ***<opcheck>***:

```
<!ELEMENT opcheck (title?, (warning*, caution*, note*), testproc)+>
<!ATTLIST opcheck
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
<!ELEMENT testproc (std.table.header?, checkstep+)>
<!ATTLIST testproc
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<opcheck>*** and ***<testproc>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.8.1.1 Operational checkout step ***<checkstep>***. The element contains a series of steps and substeps that leads to an indication or condition which concludes with a corrective action or reference to detailed troubleshooting or maintenance work packages. When a normal indication is obtained, the next operational

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checkout step is performed until either all checkout steps are completed or an abnormal condition or indication is observed. When the test procedure results in an abnormal indication or condition, a symptom or malfunction or a series of symptoms/malfunctions is provided with associated possible corrective action. The element is similar to a "row" in a structural table when shown tabular in page-based.

a. The components of operational checkout step ***<checkstep>*** are:
   (1) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).
   (2) Table ***<table>*** (see section 42) and/or conditional illustration ***<table-alt>*** (see 48.2.1).(optional – zero or more).
   (3) Checkout step level 1 ***<step1>*** (see 33.3) and/or conditional checkout step level 1 ***<step1-alt>*** (see 48.2.1) (required – one or more).
   (4) Indication or condition ***<indication>*** (required) (see 36.8.1.1.1).
   (5) When abnormal indication or condition exists then user is directed to
      (a) Malfunction or series malfunctions ***<malfunc>*** (see 36.6.2.1) with each having an associated corrective action ***<action>*** (see 36.6.1.1) or work package reference ***<xref>*** (see 46.2.2) for detailed instructions.
      (b) Corrective action ***<action>*** (see 36.6.1.1).
      (c) Work package reference ***<xref>*** (see 46.2.2) for detailed corrective action.

b. DTD fragment graphically is:

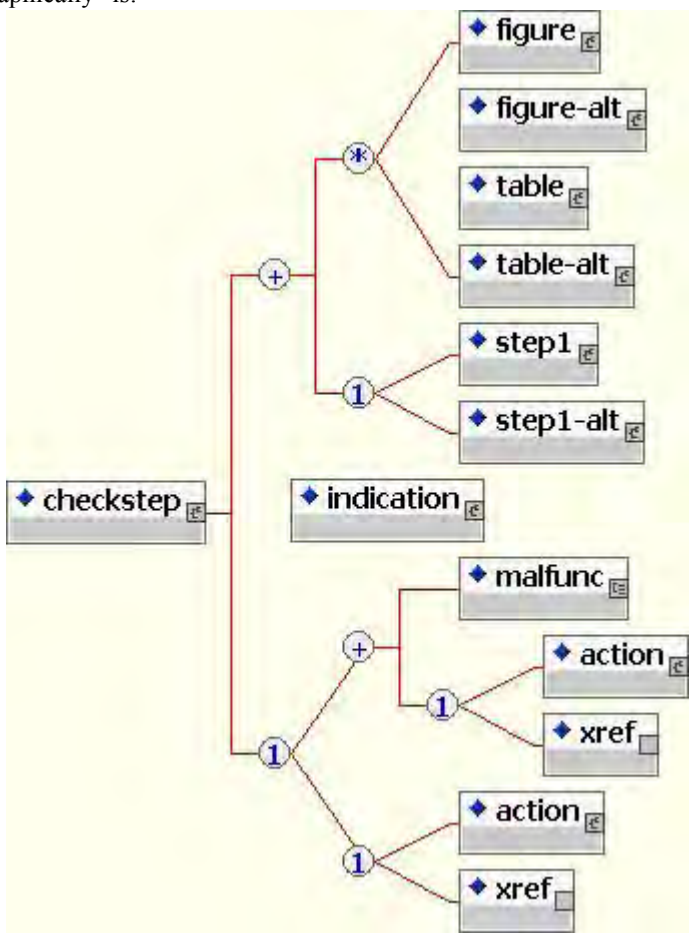

FIGURE 238. Operational checkout step DTD hierarchy <checkstep>.

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c. DTD fragment for ***<checkstep>***:

```
<!ELEMENT checkstep (((figure | figure-alt | table | table-alt)*, (step1 |
   step1-alt))+, indication, ((malfunc, (action | xref))+ |
   (action | xref)))>
<!ATTLIST checkstep
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<checkstep>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.8.1.1.1 Indication or condition ***<indication>***. The element contains an indication or condition (usually stated in the form of a question) that would be queried after the test.

a. The components of ***<indication>***, which contains one of the following information types, are:

(1) Narrative text describing the indication or condition (required – one or more)
(a) Title ***<title>*** (optional) (see 49.1.1.3).
(b) Paragraph ***<para>*** (see 49.1.1.5) or paragraph with alerts ***<specpara>*** (see 49.1.1.6) (required).
(2) Ordered/number list ***<seqlist>*** (required) (see 49.1.2.1).
(3) Unordered/bullet list ***<randlist>*** (required) (see 49.1.2.2).
(4) Definition list ***<deflist>*** (required) (see 49.1.2.3).
(5) Instructions (required) with component of
(a) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).
(b) Table ***<table>*** (see section 42) and/or conditional illustration ***<table-alt>*** (see 48.2.1).(optional – zero or more).
(c) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1-alt>*** (see 48.2.1) (required – one or more).

b. DTD fragment graphically is:

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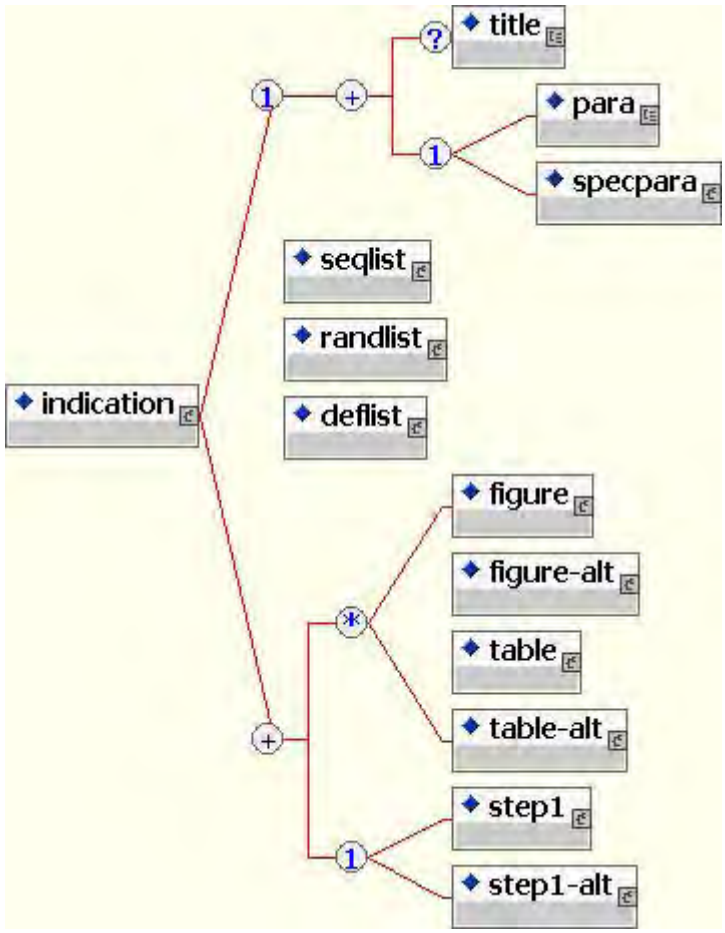

FIGURE 239. Test indication or condition DTD hierarchy <indication>..

c. DTD fragment for ***<indication>***:

```
<!ELEMENT indication (((title?, (para | specpara))+) | seqlist | randlist |
   deflist | ((figure | figure-alt | table | table-alt)*, (step1 |
   step1-alt))+)>
<!ATTLIST indication
    status            (normal | abnormal)              #IMPLIED
    assocfig          IDREFS                            #IMPLIED
    changeref         IDREFS                            #IMPLIED
    comment           CDATA                             #IMPLIED
    delchlvl          CDATA                             #IMPLIED
    id                ID                                #IMPLIED
    idref             IDREFS                            #IMPLIED
    inschlvl          CDATA                             #IMPLIED
    security          (uc | fouo |c | s | ts)          #IMPLIED
    skilltrk          CDATA                             #IMPLIED
```

d. Unique attributes for ***<indication>*** is **status** (optional) – Specifies whether the current indication element is a normal or abnormal (out-of- range) indication.

e. Common attributes for ***<indication>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

(3) **comment** – Reason for change information. (optional) (See 49.2.9.)

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(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.8.1.2 XML document instance fragment and output for ***<opcheckwp>***/***<opcheck>***. The XML instance and its stylesheet output for a ***<opcheck>*** is provided below.

a. Sample XML instance.

```
<opcheckwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="t00003-X-XXX-XXX" wpseq="0145">
<wpidinfo>
<maintlvl level="sustain"/>
<title>ARRESTING GEAR SYSTEM OPERATIONAL CHECKOUT
</title>
</wpidinfo>
<wpinfo>
<persnreq>
<persnreq-setup-item>
<name>Maintainer
</name>
<qty>2
</qty>
</persnreq-setup-item>
</persnreq>
<ref>
<ref-setup-item>
<xref wpid="o00050-X-XXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="t00010-X-XXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="t00001-X-XXX-XXX"/>
</ref-setup-item>
</ref>
<eqpconds>
<eqpconds-setup-item>
<condition>Door 103 is Installed
</condition>
<itemref>
<xref wpid="o00061-X-XXX-XXX"/>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
<opcheckproc>
<opcheck>
<testproc>
<checkstep>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Make sure door 103 is installed
<xref posttext=")" pretext="(" wpid="o00061-X-XXX-XXX"/>.
</para>
</step1>
```


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Check the source to verify that this is the current version before use.

```
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Make sure arresting
<ctrlind idref="o00042-X-XXX-XXXhook">HOOK
</ctrlind> manual control lever is set to
<ctrlind-val>up
</ctrlind-val>.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Read, record and reset nose wheelwell DDI
<xref posttext=")" pretext="(" wpid="o00050-X-XXX-XXX"/>.
</para>
</step1>
<indication>
<para esd="no" hcp="no">No maintenance code exists.
</para>
</indication>
<action>
<para esd="no" hcp="no">Perform troubleshooting
<xref posttext=")" pretext="(" taskid="t00010-X-XXX-XXXmc916"/>.
</para>
</action>
</checkstep>
<checkstep>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If arresting hook is not up, manually
raise and latch arresting hook.
</para>
</step1>
<indication>
<para esd="no" hcp="no">Arresting hook latches in up position.
</para>
</indication>
<action>
<para esd="no" hcp="no">Do arresting hook push-pull control assembly
rigging or replace push-pull control assembly
<xref posttext="0218)" pretext="(" wpid="t00001-X-XXX-XXX"/>.
</para>
</action>
</checkstep>
</testproc>
</opcheck>
</opcheckproc>
</opcheckwp>
```

b. XML sample page-layout.

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TM X–XXXX–XXX 0145

SUSTAINMENT MAINTENANCE

ARRESTING GEAR SYSTEM OPERATIONAL CHECKOUT

INITIAL SETUP:

Personnel Required
Maintainer (2)

References
WP 0040
WP 0187

References (cont.)
WP 0218

Equipment Condition
Door 103 is Installed (WP 0032)

STEP

1. Make sure door 103 is installed (WP 0032).
2. Make sure arresting **HOOK** manual control lever is set to **up**.
3. Read, record and reset nose wheelwell DDI (WP 0040).

CONDITION/INDICATION

No maintenance code exists.

CORRECTIVE ACTION

Perform troubleshooting (WP 0187, Maintenance Code 916).

STEP

4. If arresting hook is not up, manually raise and latch arresting hook.

CONDITION/INDICATION

Arresting hook latches in up position.

CORRECTIVE ACTION

Do arresting hook push-pull control assembly rigging or replace push-pull control assembly (WP 0218).

END OF WORK PACKAGE

0145-1

FIGURE 240. Operational check work package operational check testing <opcheckwp>/<opcheck> Page-layout.

c. XML sample frame-layout.

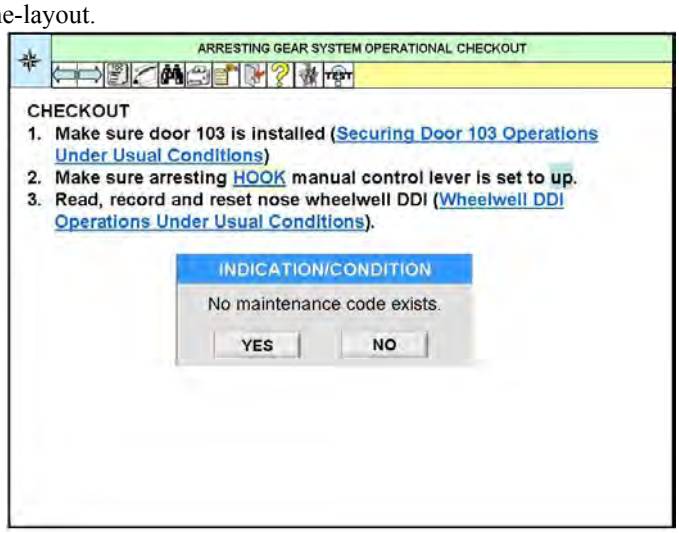

FIGURE 241. Operational check work package operational check testing <opcheckwp>/<opcheck> frame-layout.

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Check the source to verify that this is the current version before use.

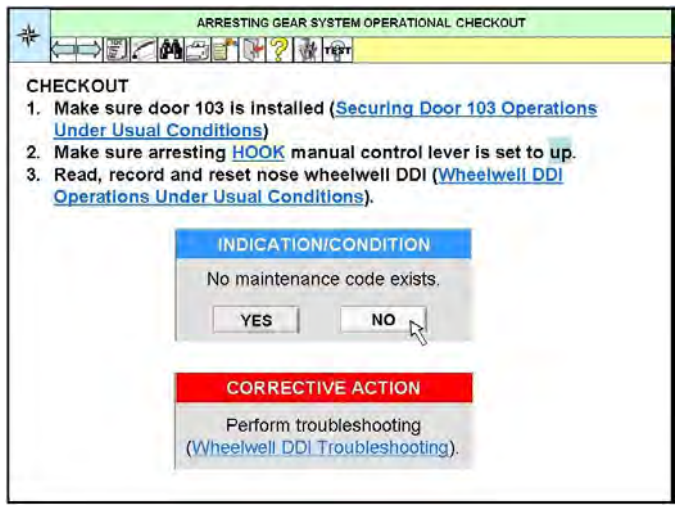

FIGURE 242. Operational check work package operational check testing <opcheckwp>/<opcheck> frame-layout – abnormal condition.

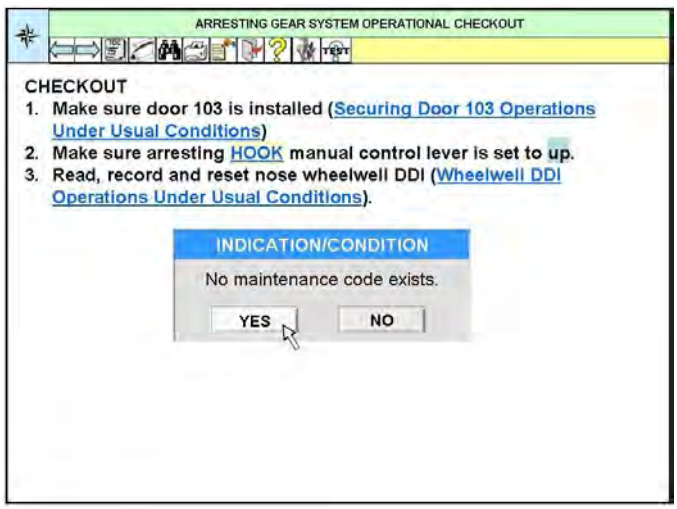

FIGURE 243. Operational check work package operational check testing <opcheckwp>/<opcheck> frame-layout – normal condition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

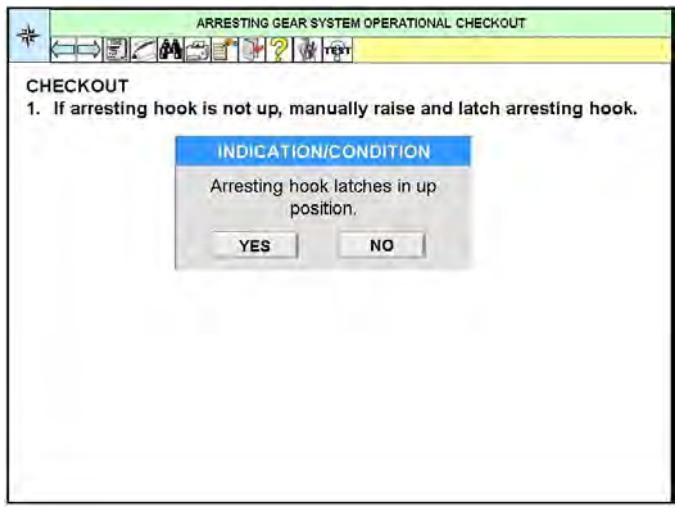

FIGURE 243. Operational check work package operational check testing <opcheckwp>/<opcheck> frame-layout – normal condition - Continued.

36.8.2 Disconnection procedures ***<disconnect>***. The element ***<disconnect>*** is used for the test set disconnection procedure.

a. The components are:

(1) Precondition check ***<precond>*** (optional) (see 42.1.1.1).
(2) Test title ***<title>*** (optional) (see 49.1.1.3).
(3) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(4) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(5) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(6) General information ***<geninfo>*** (optional) (see 49.1.4.12).
(7) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).
(8) Table ***<table>*** (see section 42) and/or conditional illustration ***<table-alt>*** (see 48.2.1).
(9) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1–alt>*** (see 48.2.1)
(10) General procedures and precautions ***<proc>*** (see 33.2).
(11) Paragraph ***<para>*** (see 49.1.1.5).
(12) Initializing state variable ***<statemanipulation>***/***<statemanipulation-alt>*** (see 48.2.3).

b. DTD fragment graphically is:

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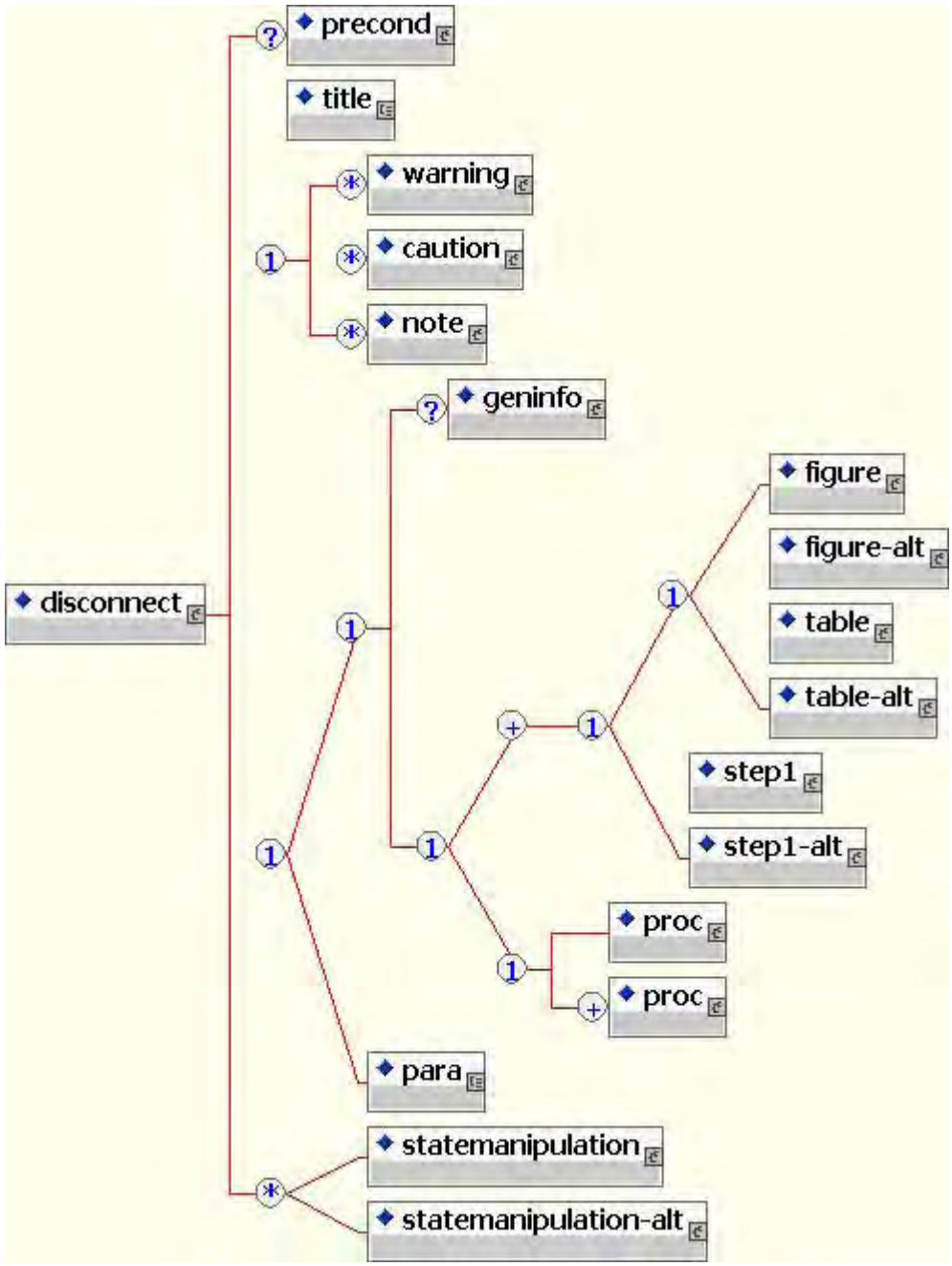

FIGURE 244. Disconnection procedures <disconnect>.

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c. DTD fragment for:

```
<!ELEMENT disconnect (precond?, title,(warning*, caution*, note*),
((geninfo?, ((((figure, table?) | (table, figure?))?,
 (step1 | step1-alt))+ | (proc, proc+))) | para),
 (statemanipulation | statemanipulation-alt)*>
<!ATTLIST disconnect
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    date-time-stamp   (date | time | date-time)       #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    esd               (yes | no)                      "no"
    frame             (yes | no)                      "yes"
    hcp               (yes | no)                      "no"
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 3 | 4 | 5)                 "0"
>
```

d. Common attributes are:

(1) **applicable** – Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(8) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(9) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(10) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(11) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(12) **inschlvl** – Insert change level (optional) (see 49.2.9).
(13) **security** – Security classification (optional) (see 49.2.13).
(14) **skilltrk** – Skill level (optional) (see 49.2.2).
(15) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**0**").

36.8.2.1 Message index entry ***<messageitem>***. The element contains a message word or bit-code word with a brief description and corrective action when message is appears. The element is similar to a "row" in a structural table.

a. The components of message index entry ***<messageitem>*** are:

(1) Message word or bit-code word ***<messageword>*** (required). The element is similar to a "cell" in a structural table and is entered in column one.
(2) Description ***<para>*** (required – one or more) (see 49.1.1.5). The element is similar to a "cell" in a structural table and is entered in column two.
(3) When message word or bit-code word exists, then the user is directed to corrective action. The element used is similar to a "cell" in a structural table and is entered in column three (required).
(a) Brief corrective action ***<action>*** (see 36.6.1.1).

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(b) Work package reference ***<xref>*** (see 46.2.2)/***<link>*** (see 46.2.3) for detailed corrective action.

b. DTD fragment graphically is:

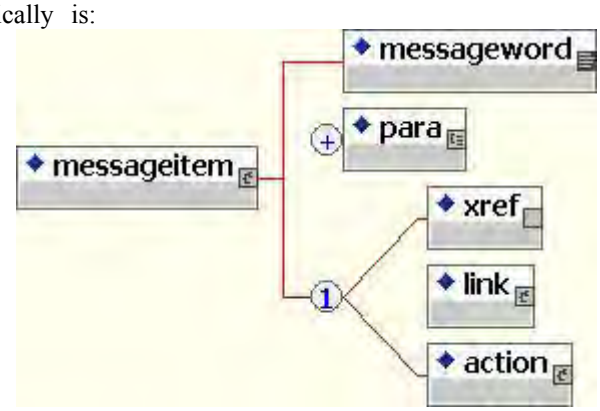

FIGURE 245. Message index entry DTD hierarchy <messageitem>..

c. DTD fragment for ***<messageitem>***:

```
<!ELEMENT messageitem (messageword, para+, (xref | link | action))>
<!ATTLIST messageitem
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
<!ELEMENT messageword (#PCDATA)>
<!ATTLIST messageword
    id                ID                              #REQUIRED
    security          (uc | fouo |c | s | ts)         #IMPLIED>
```

d. Common attributes for ***<messageword>*** are

(1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

(2) **security** – Security classification (optional) (see 49.2.13).

e. Common attributes for ***<messageitem>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

(3) **comment** – Reason for change information. (optional) (See 49.2.9.)

(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)

(5) **id** – Unique identifier. (optional) (See 49.2.6.)

(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)

(9) **security** – Security classification. (optional) (See 49.2.13.)

36.8.2.2 XML document instance fragment and output for ***<opcheckwp>***/***<messageindx>***. The XML instance and its stylesheet output for a ***<messageindx>*** is provided below.

a. Sample XML instance.

<opcheckwp airforce="no" army="no" deletewp="no" frame="yes" marines="no" navy="no" tocentry="2" wpno="t00008-X-XXX-XXX" wpseq="0132">
<wpidinfo>
<maintlvl level="sustain"/>

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<title>MESSAGE WORD INDEX
</title>
</wpidinfo>
<wpinfo>
<title>NOT APPLICABLE
</title>
<null/>
</wpinfo>
<opcheckproc>
<messageindx>
<title>Message Word Index
</title>
<std.table.header cols="3">
<std.table.row>
<std.table.entry colnum="1" cols.span="0">MESSAGE WORD NUMBER
</std.table.entry>
<std.table.entry colnum="2" cols.span="0">ITEM
</std.table.entry>
<std.table.entry colnum="3" cols.span="0">CORRECTIVE ACTION
</std.table.entry>
</std.table.row>
</std.table.header>
<messageitem>
<messageword>001
</messageword>
<para>Relay Assembly 2A27 (RY A)
</para>
<xref wpid="m0201-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-003">003
</messageword>
<para> –13°Elevation Switch 2A30 (TRL)
</para>
<xref wpid="m0202-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-005">005
</messageword>
<para>AP Low Ammo Sensor 2A21 (APL)
</para>
<xref wpid="m0203-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-006">006
</messageword>
<para> –3°Elevation Switch 2A29 (GRL)
</para>
<xref wpid="m0204-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-007">007
</messageword>
<para>Gunner's Position Indicator 2A9 (GPI)
</para>
<xref wpid="m0205-X-XXXX-XXX"/>

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Check the source to verify that this is the current version before use.

```
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-009">009
</messageword>
<para>EP Weapon 2A 15 (GUN)
</para>
<xref wpid="m0206-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-010">010
</messageword>
<para>Weapon Control Box 2A4 (WCB)
</para>
<xref wpid="m0207-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-011">011
</messageword>
<para>Slip Ring Assembly 2A7 (SLR)
</para>
<xref wpid="m0208-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-012">012
</messageword>
<para>Azimuth Encoder 2A2 (AZE)
</para>
<xref wpid="m0209-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-013">013
</messageword>
<para>Turret Control Box 2A3 (TCS)
</para>
<xref wpid="m0210-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-014">014
</messageword>
<para>7.62 Firing Solenoid 2A14 (MGS)
</para>
<xref wpid="m0212-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-015">015
</messageword>
<para>HE LO Ammo Sensor 2A22 (HEL)
</para>
<xref wpid="m0213-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-018">018
</messageword>
<para>7.62 LO Ammo Sensor .2A20 (MGL)
</para>
<xref wpid="m0214-X-XXXX-XXX"/>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-019">019
</messageword>
<para>Turret Distribution Box 2A 1 (TDB)
</para>
<xref wpid="m0215-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-020">020
</messageword>
<para>Turret Batteries 2A36/2A37 (TRB)
</para>
<xref wpid="m0216-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-022">022
</messageword>
<para>Turret Rotor Fan 2A 10 (TRF)
</para>
<xref wpid="m0217-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-029">029
</messageword>
<para>Traverse Drive Assembly 2A201 (TDA)
</para>
<xref wpid="m0218-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-030">030
</messageword>
<para>Gun Elevation Drive Assembly 2A202 (GED)
</para>
<xref wpid="m0219-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-031">031
</messageword>
<para>TOW Elevation Drive Assembly 2A203 (TED)
</para>
<xref wpid="m0220-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-032">032
</messageword>
<para>TOW Lift Actuator .Assembly 2A204 (TLM)
</para>
<xref wpid="m0221-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-033">033
</messageword>
<para>Electronic Control Assembly 2A200 (ECA)
</para>
<xref wpid="m0222-X-XXXX-XXX"/>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-034">034
</messageword>
<para>Electronic Control Assembly 2A200 (ECA)
</para>
<xref wpid="m0223-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-035">035
</messageword>
<para>Electronic Control Assembly 2A200 (ECA)
</para>
<xref wpid="m0224-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-040">040
</messageword>
<para>Defogger Fan 2A38 (FND)
</para>
<xref wpid="m0225-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-041">041
</messageword>
<para>Defogger Switch 2A40 (DFS)
</para>
<xref wpid="m0226-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-046">046
</messageword>
<para>CIB Connection and Disconnection
</para>
<xref wpid="m0227-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-047">047
</messageword>
<para>CIB Connection in Turret
</para>
<xref wpid="m0211-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-048">048
</messageword>
<para>Driver Hatch Open Switch 1S7 (DOS)
</para>
<xref wpid="m0228-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-049">049
</messageword>
<para>Driver Hatch Popped Switch 1 S9 (DPS)
</para>
<xref wpid="m0229-X-XXXX-XXX"/>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-052">052
</messageword>
<para>Vehicle Distribution Box 1 A 1 (VDB)
</para>
<xref wpid="m0230-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-053">053
</messageword>
<para>Missile Test Container (MTC)
</para>
<xref wpid="m0231-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-054">054
</messageword>
<para>Cable to Cable 2W10P3/2W12J1
</para>
<xref wpid="m0232-X-XXXX-XXX"/>
</messageitem> . . .
</messageindx>
</opcheckproc>
</opcheckwp>
```

b. XML sample page-layout.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

TM X–XXXX–XXX 0132

SUSTAINMENT MAINTENANCE
MESSAGE WORD INDEX

INITIAL SETUP:

NOT APPLICABLE

MESSAGE WORD INDEX

MESSAGE WORD NUMBER	ITEM	CORRECTIVE ACTION
001	Relay Assembly 2A27 (RY A)	WP 0201
003	-13°Elevation Switch 2A30 (TRL)	WP 0202
005	AP Low Ammo Sensor 2A21 (APL)	WP 0203
006	–3°Elevation Switch 2A29 (GRL)	WP 0204
007	Gunner's Position Indicator 2A9 (GPI)	WP 0205
009	EP Weapon 2A 15 (GUN)	WP 0206
010	Weapon Control Box 2A4 (WCB)	WP 0207
011	Slip Ring Assembly 2A7 (SLR)	WP 0208
012	Azimuth Encoder 2A2 (AZE)	WP 0209
013	Turret Control Box 2A3 (TCS)	WP 0210
014	7.62 Firing Solenoid 2A14 (MGS)	WP 0212
015	HE LO Ammo Sensor 2A22 (HEL)	WP 0213
018	7.62 LO Ammo Sensor .2A20 (MGL)	WP 0214
019	Turret Distribution Box 2A 1 (TDB)	WP 0215
020	Turret Batteries 2A36/2A37 (TRB)	WP 0216
022	Turret Rotor Fan 2A 10 (TRF)	WP 0217
029	Traverse Drive Assembly 2A201 (TDA)	WP 0218
030	Gun Elevation Drive Assembly 2A202 (GED)	WP 0219
031	TOW Elevation Drive Assembly 2A203 (TED)	WP 0220
032	TOW Lift Actuator .Assembly 2A204 (TLM)	WP 0221
033	Electronic Control Assembly 2A200 (ECA)	WP 0222
034	Electronic Control Assembly 2A200 (ECA)	WP 0223
035	Electronic Control Assembly 2A200 (ECA)	WP 0224
040	Defogger Fan 2A38 (FND)	WP 0225
041	Defogger Switch 2A40 (DFS)	WP 0226
046	CIB Connection and Disconnection	WP 0227
047	CIB Connection in Turret	WP 0211
048	Driver Hatch Open Switch 1S7 (DOS)	WP 0228
049	Driver Hatch Popped Switch 1 S9 (DPS)	WP 0229
052	Vehicle Distribution Box 1 A 1 (VDB)	WP 0230
053	Missile Test Container (MTC)	WP 0231
054	Cable to Cable 2W10P3/2W12J1	WP 0232

0132-1

FIGURE 246. Operational check work package message index <opcheckwp>/<messageindx> page-layout.

36.8.3 Test set message word index***<messageindx>***. The element is an index of automated/semi-automated test set messages or bit-code words with message word description. When used in an IETM, the index can be used to automatically lookup, in the XML source code, the generated test set message or bit-code word and display the associated corrective action, if the IETM is integrated or connected to the test set equipment or sensors. The test set message word index can be represented in a narrative format or as tabular. When in tabular layout the element is similar to a "table" in a structural table.

a. The components of test set message word index ***<messageindx>*** are:
   (1) Index title ***<title>*** (required) (see 49.1.1.3).
   (2) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (3) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (4) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (5) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (6) The index is shown either as:
      (a) Table ***<table>*** (required) (see section 42).
      (b) Test set message word index (content specific) with column header ***<std.table.header>*** (optional) (see 42.3.1) and test set message word entry ***<messageitem>*** (required – one or more) (see 36.8.2.1).

b. DTD fragment graphically is:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

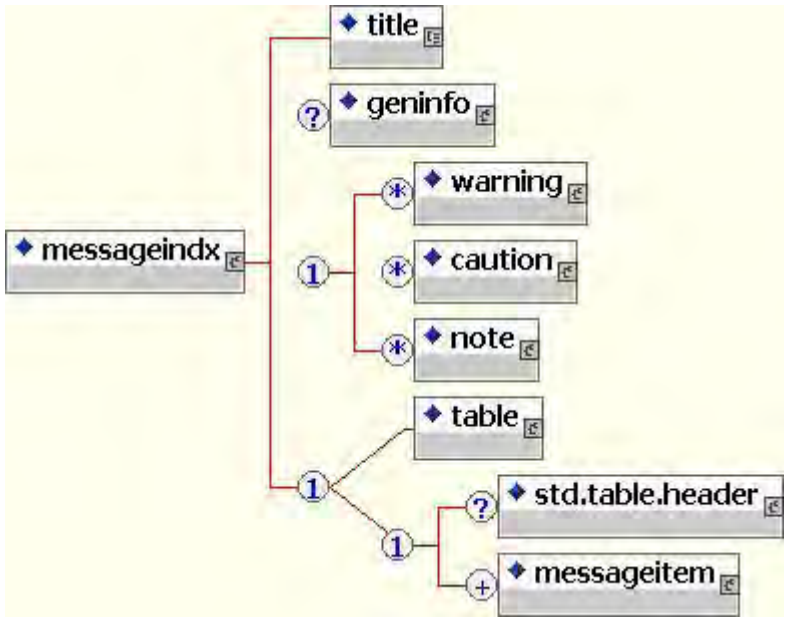

FIGURE 247. Test set message word index DTD hierarchy <messageindx>..

c. DTD fragment for ***<messageindx>***:

```
<!ELEMENT messageindx (title, geninfo?, (warning*, caution*, note*),
   std.table.header?, messageitem+)>
<!ATTLIST messageindx
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<messageindx>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.8.3.1 Message index entry ***<messageitem>***. The element contains a message word or bit-code word with a brief description and corrective action when message is appears. The element is similar to a "row" in a structural table. This element is fully discussed in 36.8.2.1

36.8.3.2 XML document instance fragment and output for ***<opcheckwp>***/***<messageindx>***. The XML instance and its stylesheet output for a ***<messageindx>*** is provided below.

a. Sample XML instance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<opcheckwp airforce="no" army="no" deletewp="no" frame="yes" marines="no" navy="no" tocentry="2" wpno="t00008-X-XXX-XXX" wpseq="0132">
<wpidinfo>
<maintlvl level="sustain"/>
<title>MESSAGE WORD INDEX
</title>
</wpidinfo>
<wpinfo>
<title>NOT APPLICABLE
</title>
<null/>
</wpinfo>
<opcheckproc>
<messageindx>
<title>Message Word Index
</title>
<std.table.header cols="3">
<std.table.row>
<std.table.entry colnum="1" cols.span="0">MESSAGE WORD NUMBER
</std.table.entry>
<std.table.entry colnum="2" cols.span="0">ITEM
</std.table.entry>
<std.table.entry colnum="3" cols.span="0">CORRECTIVE ACTION
</std.table.entry>
</std.table.row>
</std.table.header>
<messageitem>
<messageword>001
</messageword>
<para>Relay Assembly 2A27 (RY A)
</para>
<xref wpid="m0201-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-003">003
</messageword>
<para> –13°Elevation Switch 2A30 (TRL)
</para>
<xref wpid="m0202-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-005">005
</messageword>
<para>AP Low Ammo Sensor 2A21 (APL)
</para>
<xref wpid="m0203-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-006">006
</messageword>
<para> –3°Elevation Switch 2A29 (GRL)
</para>
<xref wpid="m0204-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-007">007

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</messageword>
<para>Gunner's Position Indicator 2A9 (GPI)
</para>
<xref wpid="m0205-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-009">009
</messageword>
<para>EP Weapon 2A 15 (GUN)
</para>
<xref wpid="m0206-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-010">010
</messageword>
<para>Weapon Control Box 2A4 (WCB)
</para>
<xref wpid="m0207-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-011">011
</messageword>
<para>Slip Ring Assembly 2A7 (SLR)
</para>
<xref wpid="m0208-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-012">012
</messageword>
<para>Azimuth Encoder 2A2 (AZE)
</para>
<xref wpid="m0209-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-013">013
</messageword>
<para>Turret Control Box 2A3 (TCS)
</para>
<xref wpid="m0210-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-014">014
</messageword>
<para>7.62 Firing Solenoid 2A14 (MGS)
</para>
<xref wpid="m0212-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-015">015
</messageword>
<para>HE LO Ammo Sensor 2A22 (HEL)
</para>
<xref wpid="m0213-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-018">018

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</messageword>
<para>7.62 LO Ammo Sensor .2A20 (MGL)
</para>
<xref wpid="m0214-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-019">019
</messageword>
<para>Turret Distribution Box 2A 1 (TDB)
</para>
<xref wpid="m0215-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-020">020
</messageword>
<para>Turret Batteries 2A36/2A37 (TRB)
</para>
<xref wpid="m0216-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-022">022
</messageword>
<para>Turret Rotor Fan 2A 10 (TRF)
</para>
<xref wpid="m0217-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-029">029
</messageword>
<para>Traverse Drive Assembly 2A201 (TDA)
</para>
<xref wpid="m0218-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-030">030
</messageword>
<para>Gun Elevation Drive Assembly 2A202 (GED)
</para>
<xref wpid="m0219-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-031">031
</messageword>
<para>TOW Elevation Drive Assembly 2A203 (TED)
</para>
<xref wpid="m0220-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-032">032
</messageword>
<para>TOW Lift Actuator .Assembly 2A204 (TLM)
</para>
<xref wpid="m0221-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-033">033
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</messageword>
<para>Electronic Control Assembly 2A200 (ECA)
</para>
<xref wpid="m0222-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-034">034
</messageword>
<para>Electronic Control Assembly 2A200 (ECA)
</para>
<xref wpid="m0223-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-035">035
</messageword>
<para>Electronic Control Assembly 2A200 (ECA)
</para>
<xref wpid="m0224-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-040">040
</messageword>
<para>Defogger Fan 2A38 (FND)
</para>
<xref wpid="m0225-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-041">041
</messageword>
<para>Defogger Switch 2A40 (DFS)
</para>
<xref wpid="m0226-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-046">046
</messageword>
<para>CIB Connection and Disconnection
</para>
<xref wpid="m0227-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-047">047
</messageword>
<para>CIB Connection in Turret
</para>
<xref wpid="m0211-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-048">048
</messageword>
<para>Driver Hatch Open Switch 1S7 (DOS)
</para>
<xref wpid="m0228-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-049">049
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</messageword>
<para>Driver Hatch Popped Switch 1 S9 (DPS)
</para>
<xref wpid="m0229-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-052">052
</messageword>
<para>Vehicle Distribution Box 1 A 1 (VDB)
</para>
<xref wpid="m0230-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-053">053
</messageword>
<para>Missile Test Container (MTC)
</para>
<xref wpid="m0231-X-XXXX-XXX"/>
</messageitem>
<messageitem>
<messageword id="m0201-X-XXXX-XXX-m-054">054
</messageword>
<para>Cable to Cable 2W10P3/2W12J1
</para>
<xref wpid="m0232-X-XXXX-XXX"/>
</messageitem> . . .
</messageindx>
</opcheckproc>
</opcheckwp>
```

b. XML sample page-layout.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

TM X–XXXX–XXX 0132

SUSTAINMENT MAINTENANCE
MESSAGE WORD INDEX

INITIAL SETUP:

NOT APPLICABLE

MESSAGE WORD INDEX

MESSAGE WORD NUMBER	ITEM	CORRECTIVE ACTION
001	Relay Assembly 2A27 (RY A)	WP 0201
003	-13°Elevation Switch 2A30 (TRL)	WP 0202
005	AP Low Ammo Sensor 2A21 (APL)	WP 0203
006	–3°Elevation Switch 2A29 (GRL)	WP 0204
007	Gunner's Position Indicator 2A9 (GPI)	WP 0205
009	EP Weapon 2A 15 (GUN)	WP 0206
010	Weapon Control Box 2A4 (WCB)	WP 0207
011	Slip Ring Assembly 2A7 (SLR)	WP 0208
012	Azimuth Encoder 2A2 (AZE)	WP 0209
013	Turret Control Box 2A3 (TCS)	WP 0210
014	7.62 Firing Solenoid 2A14 (MGS)	WP 0212
015	HE LO Ammo Sensor 2A22 (HEL)	WP 0213
018	7.62 LO Ammo Sensor .2A20 (MGL)	WP 0214
019	Turret Distribution Box 2A 1 (TDB)	WP 0215
020	Turret Batteries 2A36/2A37 (TRB)	WP 0216
022	Turret Rotor Fan 2A 10 (TRF)	WP 0217
029	Traverse Drive Assembly 2A201 (TDA)	WP 0218
030	Gun Elevation Drive Assembly 2A202 (GED)	WP 0219
031	TOW Elevation Drive Assembly 2A203 (TED)	WP 0220
032	TOW Lift Actuator .Assembly 2A204 (TLM)	WP 0221
033	Electronic Control Assembly 2A200 (ECA)	WP 0222
034	Electronic Control Assembly 2A200 (ECA)	WP 0223
035	Electronic Control Assembly 2A200 (ECA)	WP 0224
040	Defogger Fan 2A38 (FND)	WP 0225
041	Defogger Switch 2A40 (DFS)	WP 0226
046	CIB Connection and Disconnection	WP 0227
047	CIB Connection in Turret	WP 0211
048	Driver Hatch Open Switch 1S7 (DOS)	WP 0228
049	Driver Hatch Popped Switch 1 S9 (DPS)	WP 0229
052	Vehicle Distribution Box 1 A 1 (VDB)	WP 0230
053	Missile Test Container (MTC)	WP 0231
054	Cable to Cable 2W10P3/2W12J1	WP 0232

0132-1

FIGURE 248. Operational check work package message index <opcheckwp>/<messageindx> page-layout.

36.8.4 Fault code reference index ***<faultreports>***. The element is used for a troubleshooting reference table contained in a operational checkout work package. These fault code reference indexes are generated by automated and/or built-in diagnostics. When used in an IETM, the index can be used to automatically lookup, in the XML source code, the generated test set message or bit-code word and display the associated corrective action, if the IETM is integrated or connected to the test set equipment or sensors. The test set message word index can be represented in a narrative format or as tabular. When in tabular layout the element is similar to a "table" in a structural table.

a. The components of fault code reference index ***<faultreports>*** are:
   (1) Index title ***<title>*** (optional) (see 49.1.1.3).
   (2) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (3) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (4) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (5) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (6) Index is shown either as:
      (a) Table ***<table>*** (required) (see 42).
      (b) Fault code reference index (content specific) with column header ***<std.table.header>*** (optional) (see 42.3.1) and fault code reference entry ***<faultcode>*** (required – one or more) (see 36.8.4.1).

b. DTD fragment graphically is:

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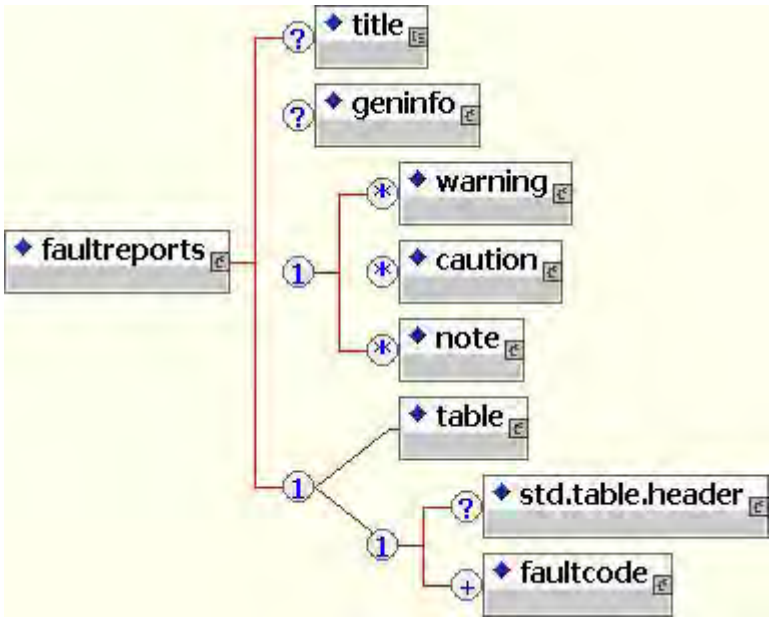

FIGURE 249. Fault code reference index DTD hierarchy <faultreports>..

c. DTD fragment for ***<faultreports>***:

```
<!ELEMENT faultreports (title?, geninfo?, (warning*, caution*, note*),
   (table | (std.table.header?, faultcode+))>
<!ATTLIST faultreports
    indxcols          (2 | 3)                          #REQUIRED
    reftype           (wp | page | tsloc)              "wp"
    assocfig          IDREFS                           #IMPLIED
    changeref         IDREFS                           #IMPLIED
    comment           CDATA                            #IMPLIED
    delchlvl          CDATA                            #IMPLIED
    id                ID                               #IMPLIED
    idref             IDREFS                           #IMPLIED
    inschlvl          CDATA                            #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                            #IMPLIED
>
```

d. Unique attributes for ***<faultreports>*** are:

(1) **indxcols** - Specifies number of columns in the index; when follow-on procedures are used the attribute value is "3", otherwise the attribute value is "2". Generally when use as an index of message words will have three columns, an index of fault reports from built-in diagnostics may have only two.

(2) **reftype** - Specifies reference type format to be used in the tables third column for the composition system. If no value is entered for the attribute the default is "WP".

(a) "wp" - Defines to use the work package sequence number to indicate where the corrective action is located.

(b) "page" - Defines to use page number to indicate where the corrective action is located.

(c) "tsloc"- Defines to use both the work package sequence number and page number to indicate where the corrective action is located.

e. Common attributes for ***<faultreports>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

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(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.8.4.1 Fault code reference entry ***<faultcode>***. The element identifies a fault code set that includes the fault code with appropriate corrective action (either as steps or reference to detail instructions) and, when required, follow-on testing after corrective actions. The element represents the row in a page-based or group in a frame-based in the fault code reference index.

a. The components of Fault code reference entry ***<faultcode>*** are:
   (1) Fault code ***<messageword>*** (required). The element is similar to a "cell" in a structural table and is entered in column one.
   (2) When a fault code reference exists, then the user is directed to the corrective action. The element used is similar to a "cell" in a structural table and is entered in column two.
      (a) Brief corrective action ***<action>*** (see 36.6.1.1).
      (b) Work package reference ***<xref>*** (see 46.2.2)/***<link>*** (see 46.2.3) for detailed corrective action.
   (3) Follow-on procedure ***<follow-on>*** (optional) (see 36.8.4.2).

b. DTD fragment graphically is:

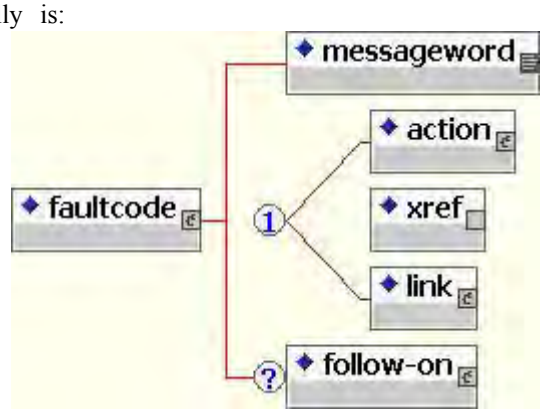

FIGURE 250. Fault code reference DTD hierarchy <faultcode>.

c. DTD fragment for ***<faultcode>***:

```
<!ELEMENT faultcode (messageword, (action | xref | link), follow-on?)>
<!ATTLIST faultcode
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>

<!ELEMENT messageword (#PCDATA)>
<!ATTLIST messageword
    id                ID                              #REQUIRED
    security          (uc | fouo |c | s | ts)         #IMPLIED>
```


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d. Common attributes for ***<faultcode>*** are:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

36.8.4.2 Follow-on procedure ***<follow-on>***. The element specifies any additional follow-on operational testing procedures to be performed after the corrective action has be completed.

a. The components of follow-on procedure ***<follow-on>*** is on of the following:
   (1) Single follow-on step ***<para>*** (required) (see 49.1.1.5).
   (2) Follow-on step (required – one or more) and the components are:
      (a) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).
      (b) Table ***<table>*** (see 42) and/or conditional illustration ***<table-alt>*** (see 48.2.1).(optional – zero or more).
      (c) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1>*** (see 48.2.1) (required).
   (3) Reference to detail follow-on work package ***<link>*** (see 46.2.3)/***<xref>*** (see 46.2.2) (required).

b. DTD fragment graphically is:

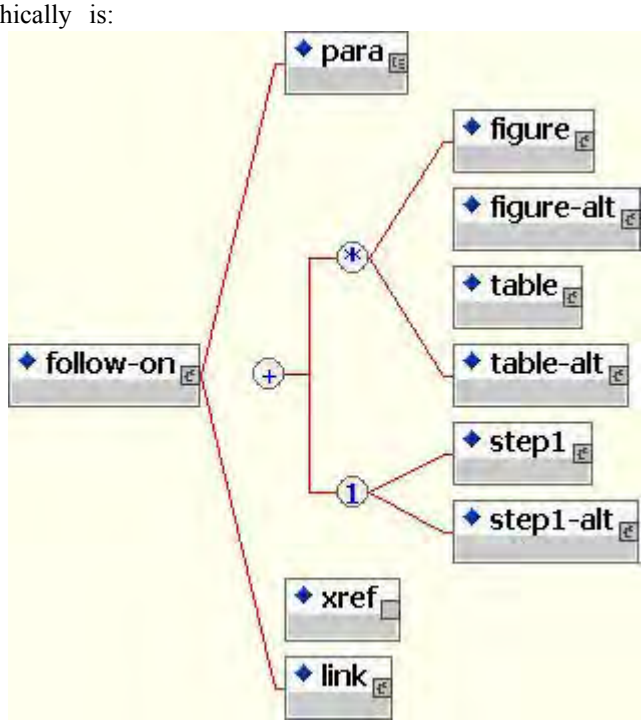

FIGURE 251. Follow-on procedure DTD hierarchy <follow-on>.

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c. DTD fragment for ***<follow-on>***:

```
<!ELEMENT follow-on (para | ((figure | figure-alt | table | table-alt)*,
   (step1 | step1-alt))+ | xref | link)>
<!ATTLIST follow-on
    id          ID                          #REQUIRED
    security    (uc | fouo |c | s | ts)     #IMPLIED>
```

d. Common attributes for ***<follow-on>*** are:

(1) **id** – Specifies unique identifier (target) to reference (required)(see 49.2.6).

(2) **security** – Security classification (optional) (see 49.2.13).

36.8.4.3 XML document instance fragment and output for ***<opcheckwp>***/***<faultreports>***. The XML instance and its stylesheet output for a ***<faultreports>*** is provided below.

a. Sample XML instance.

```
<opcheckwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="t00029-X-XXX-XXX" wpseq="0120">
<wpidinfo>
<maintlvl level="sustain"/>
<title>Fault Message Index
</title>
</wpidinfo>
<wpinfo>
<title>NOT APPLICABLE
</title>
<null/>
</wpinfo>
<opcheckproc>
<faultreports indxcols="2">
<title>Fault Message Index
</title>
<std.table.header cols="2">
<std.table.row>
<std.table.entry colnum="1" cols.span="0">FAULT MESSAGE
</std.table.entry>
<std.table.entry colnum="2" cols.span="0">CORRECTIVE ACTION
</std.table.entry>
</std.table.row>
</std.table.header>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-ABS">FAULTY ABS
</messageword>
<xref taskid="m00312-X-XXXX-XXX-task1" wpid="m00312-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-ANP">FAULTY ANP
</messageword>
<xref taskid="m00336-X-XXXX-XXX-task1" wpid="m00336-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-APL">FAULTY APL
</messageword>
<xref taskid="m00333-X-XXXX-XXX-task1" wpid="m00333-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-AZE">FAULTY AZE
</messageword>
<xref taskid="m00337-X-XXXX-XXX-task1" wpid="m00337-X-XXXX-XXX"/>
```


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Check the source to verify that this is the current version before use.

```
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-BATTERY-CHARGE">FAULTY BATTERY CHARGE
</messageword>
<xref wpid="m00211-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-BATTERY-CHARGING-SYS">FAULTY
BATTERY/ CHARGING SYS
</messageword>
<action>
<para>Notify hull maintenance of Faulty Batteries or Charging System.
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-BATTERY-CHG-SYS-SHORT">FAULTY
BATTERY/ CHG SYS SHORT
</messageword>
<action>
<para>Notify hull maintenance of short in Batteries or Charging System.
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CA1201">FAULTY CA1201
</messageword>
<action>
<para>See
<extref docno="TM X-XXXX-YYY-14&P"/>
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CA1202">FAULTY CA1202
</messageword>
<action>
<para>See
<extref docno="TM X-XXXX-YYY-14&P"/>
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CARGO-HATCH-SYSTEM">FAULTY
CARGO HATCH SYSTEM
</messageword>
<xref wpid="m00213-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CDL">FAULTY CDL
</messageword>
<xref taskid="m00341-X-XXXX-XXX-task1" wpid="m00341-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CEL">FAULTY CEL
</messageword>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<xref taskid="m00342-X-XXXX-XXX-task1" wpid="m00342-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CFOS">FAULTY CFOS
</messageword>
<action>
<para>Notify hull maintenance of Faulty/Misadjusted Cargo Hatch Full Open Switch.
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CHARGE-SYS-OVERVOLTAGE">FAULTY CHARGE SYS OVER-VOLTAGE
</messageword>
<action>
<para>Notify hull maintenance of Charging System Overcharging.
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CHS">FAULTY CHS
</messageword>
<xref taskid="m00347-X-XXXX-XXX-task1" wpid="m00347-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-COS">FAULTY COS
</messageword>
<action>
<para>Notify hull maintenance of Faulty/Misadjusted Cargo Hatch Open Switch.
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CPI">FAULTY CPI
</messageword>
<xref taskid="m00348-X-XXXX-XXX-task1" wpid="m00348-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CPS">FAULTY CPS
</messageword>
<action>
<para>Notify hull maintenance of Faulty/Misadjusted Cargo Hatch Popped Switch.
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-CX-1201">FAULTY CX 1201
</messageword>
<action>
<para>See
<extref docno="TM X-XXXX-YYY-14&P"/>
</para>
</action>

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Check the source to verify that this is the current version before use.

```
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-DCGE-Q1-THRU-99">FAULTY DCGE-01 THRU 99
</messageword>
<action>
<para>Record fault message and notify supervisor for
T2SS-SE testing of Faulty DCGE .
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-DOS">FAULTY DOS
</messageword>
<action>
<para>Notify hull maintenance of Faulty/Misadjusted
Driver's Hatch Open Switch.
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-DPS">FAULTY DPS
</messageword>
<action>
<para>Notify hull maintenance of Faulty/Misadjusted
Driver's Hatch Popped Switch.
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-DRIVE-PWR-DISTR">FAULTY DRIVE PWR DISTR
</messageword>
<xref wpid="m00217-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-DRIVER-HATCH-SYSTEM">FAULTY
DRIVER HATCH SYSTEM
</messageword>
<xref wpid="m00219-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-ECA">FAULTY ECA
</messageword>
<xref taskid="m00401-X-XXXX-XXX-task1" wpid="m00401-X-XXXX-XXX"/>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-ECA-AND-2W305">FAULTY ECA AND 2W305
</messageword>
<action>
<para>
<xref taskid="m00401-X-XXXX-XXX-task1" wpid="m00401-X-XXXX-XXX"/> and
<xref taskid="m00309-X-XXXX-XXX-task1" wpid="m00309-X-XXXX-XXX"/>.
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-ECA-AND-GEDGBX">FAULTY ECA AND GED GBX
```


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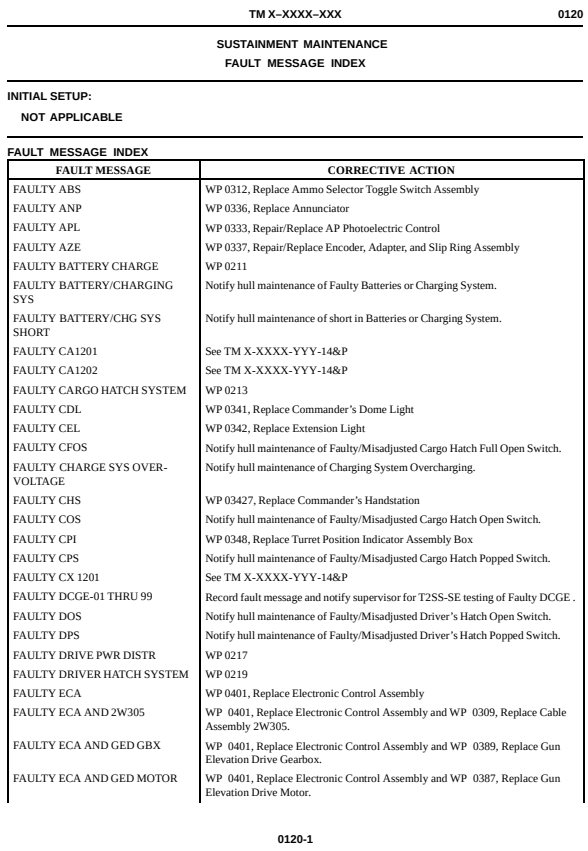
```
</messageword>
<action>
<para>
<xref taskid="m00401-X-XXXX-XXX-task1" wpid="m00401-X-XXXX-XXX"/> and
<xref taskid="m00389-X-XXXX-XXX-task1" wpid="m00389-X-XXXX-XXX"/>.
</para>
</action>
</faultcode>
<faultcode>
<messageword id="t0029-X-XXXX-XXX-f-ECA-AND-GED-MOTOR">FAULTY
ECA AND GED MOTOR
</messageword>
<action>
<para>
<xref taskid="m00401-X-XXXX-XXX-task1" wpid="m00401-X-XXXX-XXX"/> and
<xref taskid="m00387-X-XXXX-XXX-task1" wpid="m00387-X-XXXX-XXX"/>.
</para>
</action>
</faultcode> . . .
</faultreports>
</opcheckproc>
</opcheckwp>
```

b. XML sample page-layout.



TM X–XXXX–XXX 0120

SUSTAINMENT MAINTENANCE
FAULT MESSAGE INDEX

INITIAL SETUP:
NOT APPLICABLE

FAULT MESSAGE INDEX

FAULT MESSAGE	CORRECTIVE ACTION
FAULTY ABS	WP 0312, Replace Ammo Selector Toggle Switch Assembly
FAULTY ANP	WP 0336, Replace Annunciator
FAULTY APL	WP 0333, Repair/Replace AP Photoelectric Control
FAULTY AZE	WP 0337, Repair/Replace Encoder, Adapter, and Slip Ring Assembly
FAULTY BATTERY CHARGE	WP 0211
FAULTY BATTERY/CHARGING SYS	Notify hull maintenance of Faulty Batteries or Charging System.
FAULTY BATTERY/CHG SYS SHORT	Notify hull maintenance of short in Batteries or Charging System.
FAULTY CA1201	See TM X-XXXX-YYY-14&P
FAULTY CA1202	See TM X-XXXX-YYY-14&P
FAULTY CARGO HATCH SYSTEM	WP 0213
FAULTY CDL	WP 0341, Replace Commander's Dome Light
FAULTY CEL	WP 0342, Replace Extension Light
FAULTY CFOS	Notify hull maintenance of Faulty/Misadjusted Cargo Hatch Full Open Switch.
FAULTY CHARGE SYS OVER-VOLTAGE	Notify hull maintenance of Charging System Overcharging.
FAULTY CHS	WP 03427, Replace Commander's Handstation
FAULTY COS	Notify hull maintenance of Faulty/Misadjusted Cargo Hatch Open Switch.
FAULTY CPI	WP 0348, Replace Turret Position Indicator Assembly Box
FAULTY CPS	Notify hull maintenance of Faulty/Misadjusted Cargo Hatch Popped Switch.
FAULTY CX 1201	See TM X-XXXX-YYY-14&P
FAULTY DCGE-01 THRU 99	Record fault message and notify supervisor for T2SS-SE testing of Faulty DCGE .
FAULTY DOS	Notify hull maintenance of Faulty/Misadjusted Driver's Hatch Open Switch.
FAULTY DPS	Notify hull maintenance of Faulty/Misadjusted Driver's Hatch Popped Switch.
FAULTY DRIVE PWR DISTR	WP 0217
FAULTY DRIVER HATCH SYSTEM	WP 0219
FAULTY ECA	WP 0401, Replace Electronic Control Assembly
FAULTY ECA AND 2W305	WP 0401, Replace Electronic Control Assembly and WP 0309, Replace Cable Assembly 2W305.
FAULTY ECA AND GED GBX	WP 0401, Replace Electronic Control Assembly and WP 0389, Replace Gun Elevation Drive Gearbox.
FAULTY ECA AND GED MOTOR	WP 0401, Replace Electronic Control Assembly and WP 0387, Replace Gun Elevation Drive Motor.

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FIGURE 252. Operational check work package fault reports <opcheckwp>/<faultreports> page-layout.

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36.9 Troubleshooting work package ***<tswp>***. The element contains start-to-finish troubleshooting procedures, which result in fault isolation and rectification and ultimately either a return to readiness status or referral to a higher maintenance level. Troubleshooting procedures can be presented in tabular or narrative format; or in diagrammatic flow tree. In electronic presentations a ***<tswp>*** may be made up of simple sequential nodes or be traversed as filtered nodes. The work package contains three troubleshooting procedure ***<tsproc>*** methods for diagnosing the fault symptom through logic (tree) (see 36.9.1), fault (see 36.9.2), or multiplex read code data (see 36.9.3).

a. The components of ***<tswp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Introduction ***<intro>*** (optional) (see 49.1.4.15) explaining how the troubleshooting procedures are to be used to perform testing and how they relate to the associated troubleshooting work packages.
   (5) General procedures and precautions ***<proc>*** (optional – zero or more) (see 33.2) that are performed prior to troubleshooting and precautions that are taken during the performance of the troubleshooting procedure.
   (6) System description and support information (optional) (see 36.7).
   (7) Test set hookup ***<hookup>*** (optional) (see 33.1 how to develop a task). If test set hookup is used test set disconnection is required.
   (8) Troubleshooting fault isolation procedure ***<tsproc>*** (required) uses only one of the following fault symptom diagnosing methods per work packages.
      (a) Logic procedure ***<logicproc>*** (required – one or more) (see 36.9.1).
      (b) Fault procedure ***<faultproc>*** (required – one or more) (see 36.9.2).
      (c) Multiplex read code data ***<muxproc>*** (required) (see 36.9.3).
   (9) Test set disconnection ***<disconnect>*** (optional) (see 33.1 how to develop a task). If test set hookup is used test set disconnection is required.

b. DTD fragment graphically is:

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Check the source to verify that this is the current version before use.

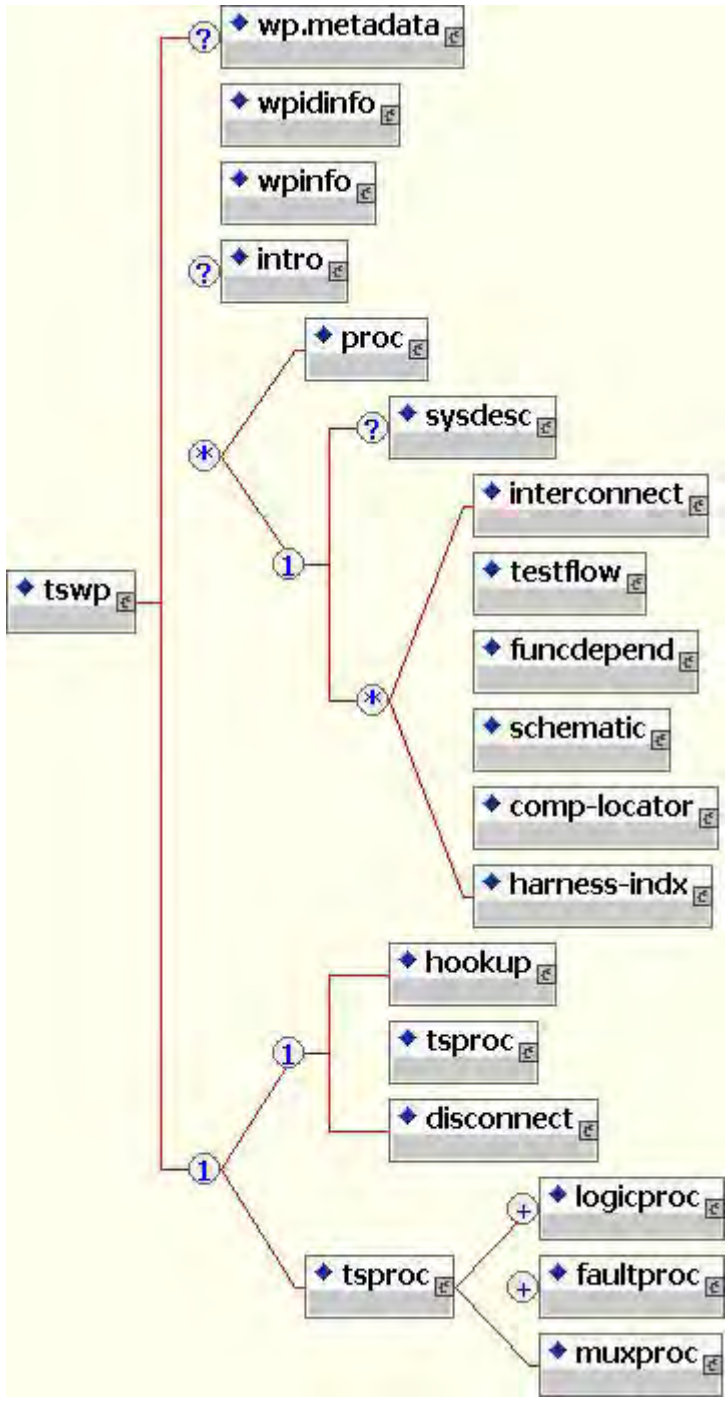

FIGURE 253. Troubleshooting work package DTD hierarchy <tswp>..

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Check the source to verify that this is the current version before use.

c. DTD fragments are:

```
<!ELEMENT tswp (wp.metadata?, wpidinfo, wpinfo, intro?, (proc | (sysdesc?,
    (interconnect | testflow | funcdepend | schematic | comp-locator |
    harness-indx)*))*, ((hookup, tsproc, disconnect) | tsproc))>
<!ATTLIST tswp
   airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
<!ELEMENT tsproc (logicproc+ | faultproc+ | muxproc)>
<!ATTLIST tsproc
    assocfig          IDREFS                            #IMPLIED
    changeref         IDREFS                            #IMPLIED
    comment           CDATA                             #IMPLIED
    delchlvl          CDATA                             #IMPLIED
    idref             IDREFS                            #IMPLIED
    inschlvl          CDATA                             #IMPLIED
    security          (uc | fouo |c | s | ts)          #IMPLIED
    skilltrk          CDATA                             #IMPLIED
>
```

d. Common attributes are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

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(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.
(23) **id** – Specifies unique identifier (target) to reference (optional)(see 49.2.6).

36.9.1 Logic procedure ***<logicproc>***. The element is used to isolate a fault through a series of tests. Each test has an indication or condition (results observed during the testing) and the maintainer selects the corresponding observed answer to determine the next action to isolate the fault. Generally, this method is referenced as a logic tree. The logic tree has various path (branches) which the maintainer will traverse to determine or isolate the fault. Each path or branch has a test, a question (what did the maintainer observe) and corresponding answers to the question. Each answer leads the maintainer to the next test or the fault was isolate. Figure 254 is a generic graphical example showing the logic procedure. Figure 262 is the sample XML markup (see 36.9.1.4.a.) showing the logical flow in frame-based output.

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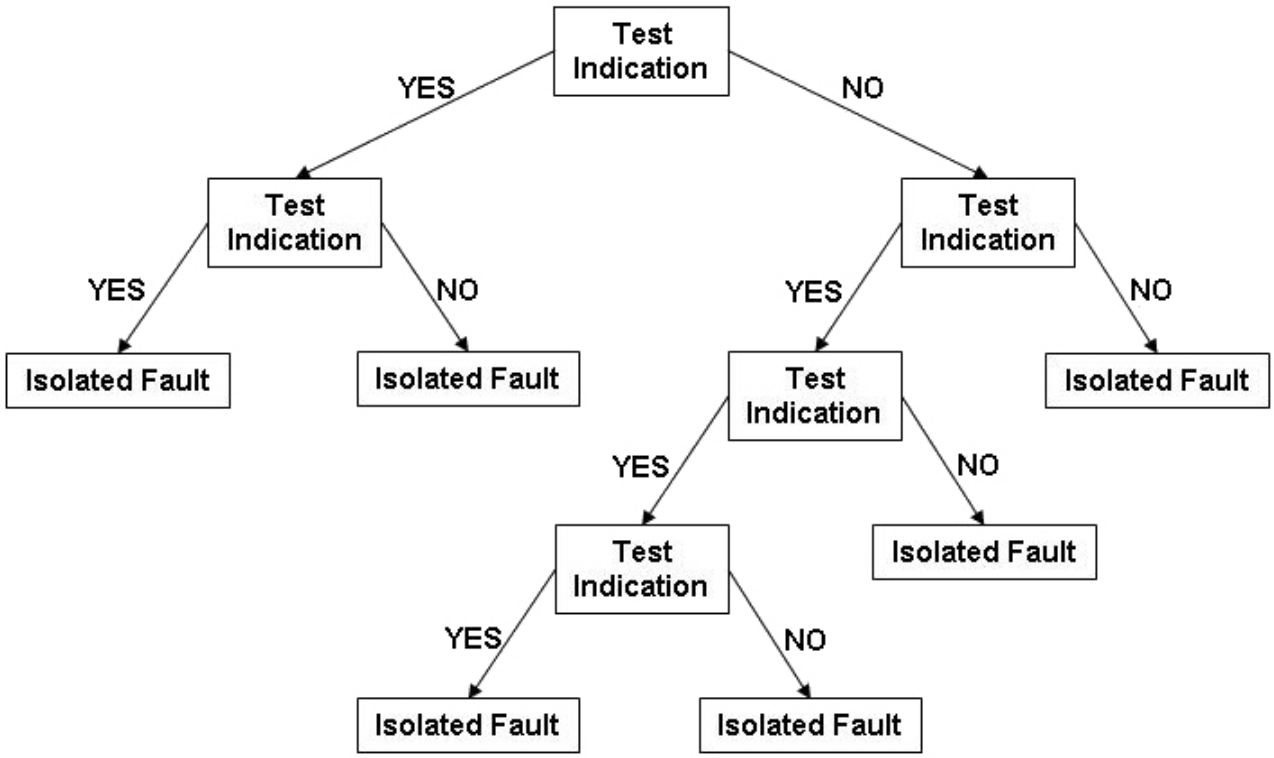

FIGURE 254. Example logic tree.

MORE INFORMATION.

The attribute **branchfrom** is used to manage where the test or end block was derived from through ID references to the block ID attribute (**branch** or **origin**). The attribute **branchto** is used to manage the possible test or end block(s) that can be called from the current block through ID references to the block ID attribute (**branch** or **origin**). The capability may be used by a viewer to traverse the logic tree in reverse or knowing the next possible test or end blocks.

a. The components of ***<logicproc>*** are:
   (1) Procedure title ***<title>*** (required) (see 49.1.1.3).
   (2) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (3) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (4) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (5) Test block origin ***<origin>*** (required) (see 36.9.1.1).
   (6) Further testing includes one or more of the following types:
      (a) Test block ***<testblock>*** (see 36.9.1.1).
      (b) End block ***<endblock>*** (see 36.9.1.2).
      (c) Test/end block reference ***<branchref>*** (see 36.9.1.3).

b. DTD fragment graphically is:

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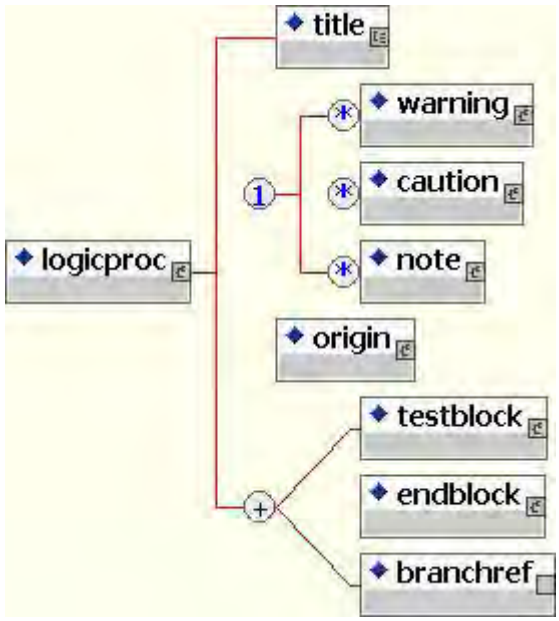

FIGURE 255. Logic procedure DTD hierarchy <logicproc>.

c. DTD fragment for ***<logicproc>***:

```
<!ELEMENT logicproc (title, warning*, caution*, note*, origin, (testblock |
   endblock | branchref)+)>
<!ATTLIST logicproc
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<logicproc>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.9.1.1 Test block ***<origin>/<testblock>***. The test block contains the prescribed test(s) ***<test>*** followed by indication or condition (usually in the form of question) ***<indication>***, and possible answers ***<answer>***.

a. The components of ***<origin>/<testblock>*** are:

(1) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(2) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(3) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(4) Test procedure ***<test>*** (required – one or more) (see 36.9.1.1.1).

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(5) Indication or condition ***<indication>*** (required) (see 36.8.1.1.1).

(6) Response answer to indication or condition***<answer>*** (required – two or more) (see 36.9.1.1.2).

b. DTD fragment graphically is:

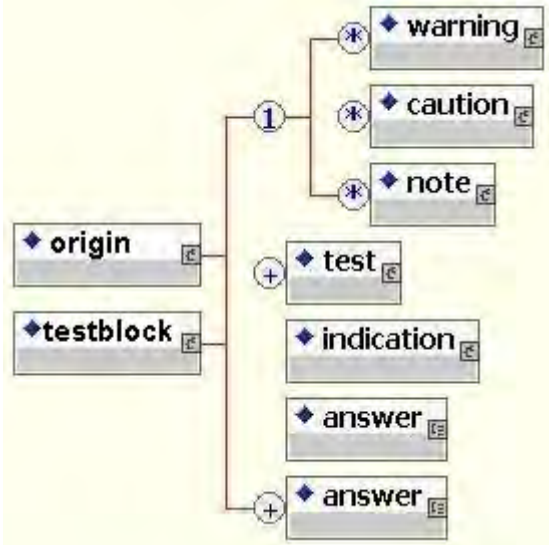

FIGURE 256. Test block DTD hierarchy <origin>/<testblock>.

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c. DTD fragments are:

```
<!ELEMENT origin (warning*, caution*, note*, test+, indication,
   answer, answer+)>
<!ATTLIST origin
   branchto    IDREFS              #REQUIRED
   branchlabel CDATA               #IMPLIED
   origin      ID                  #REQUIRED
   type        (yes | no |
                pass | fail |
                true | nottrue |
                value |
                unantic)           #IMPLIED
   value       CDATA               #IMPLIED
   valueloc    NMTOKENS            #IMPLIED
   valuetype   (boolean | string |
                sequence | set |
                real | integer |
                float | nil |
                input | outcome)    #IMPLIED>
<!ELEMENT testblock (warning*, caution*, note*, test+, indication,
   answer, answer+)>
<!ATTLIST testblock
   branch      ID                  #REQUIRED
   branchfrom  IDREFS              #REQUIRED
   branchlabel CDATA               #IMPLIED
   branchto    IDREFS              #REQUIRED
   type        (yes | no |
                pass | fail |
                true | nottrue |
                value |
                unantic)           #REQUIRED
   value       CDATA               #IMPLIED
   valueloc    NMTOKENS            #IMPLIED
   valuetype   (boolean | string |
                sequence | set |
                real | integer |
                float | nil |
                input | outcome)    #IMPLIED>
```

d. Unique attributes for ***<origin>***:

(1) **branchlabel**(optional) – Supplies an explicit reference to this branch and used in the test block presentation.

(2) **branchto**(required) – References test/end block(s) (branches) to the next test or actions after test results.

(3) **origin**(required) – Specifies the test block unique identifier.

(4) **type**(optional) - Specifies the type of branch logic current element. This value may be displayed in either paper or electronic display.

(a) “yes” – Applies to a positive answer and composition system will display ”YES".

(b) “no” – Applies to a negative answer and composition system will display ”NO".

(c) “pass” – Applies to a positive answer and composition system will display ”PASS".

(d) “fail” – Applies to a negative answer and composition system will display ”FAIL".

(e) “true” – Applies to a positive answer and composition system will display ”TRUE".

(f) “nottrue” – Applies to a negative answer and composition system will display ”NOT TRUE".

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(g) “value” – Applies to a response value from **valueloc** attribute and the composition system will display the from the **valuetype** attribute.
(h) “unantic” – Applies to a unanticipated result.

(5) **value**(optional) – Supplies an alphanumeric or numeric value if attribute **type** is "value".
(6) **valueloc**(optional) – Supplies location of value if contained in another element, such as ***<input>***.
(7) **valuetype**(optional) – Specifies the type of value if attribute **type** is "value."
(a) “boolean” - Applies to a boolean value from **valueloc** and the composition system will display either “true” or “false”.
(b) “string” - Applies to a character string from **valueloc** and the composition system will display the string.
(c) “sequence” - Applies to an ordered sequence values from **valueloc** and the composition system will display the sequence value.
(d) “set” - Applies to an unordered sequence values from **valueloc** and the composition system will display the set values.
(e) “real” - Applies to a real number from **valueloc** and the composition system will display the number.
(f) “integer” - Applies to an integer number from **valueloc** and the composition system will display the number.
(g) “float” - Applies to a floating point number from **valueloc** and the composition system will display the number.
(h) “nil” - Applies to null value from **valueloc** and the composition system will display the “nil”.
(i) “input” - Applies to a inserted query value from **valueloc** and the composition system will display the inputted value.
(j) “outcome” - Applies to a query outcome from a test from **valueloc** and the composition system will display the value.

e. Unique attributes for ***<testblock>***:
(1) **branch**(required) – Specifies the test block unique identifier.
(2) **branchfrom**(required) – References the test block(s) (branches), which are from results from parent test blocks.
(3) **branchlabel**(optional) – Supplies an explicit reference to this branch and used in the test block presentation.
(4) **branchto**(required) – References test/end block(s) (branches) to the next test or actions after test results.
(5) **type**(required) - Specifies the type of branch logic current element. This value may be displayed in either paper or electronic display.
(a) “yes” – Applies to a positive answer and composition system will display ”YES".
(b) “no” – Applies to a negative answer and composition system will display ”NO".
(c) “pass” – Applies to a positive answer and composition system will display ”PASS".
(d) “fail” – Applies to a negative answer and composition system will display ”FAIL".
(e) “true” – Applies to a positive answer and composition system will display ”TRUE".
(f) “nottrue” – Applies to a negative answer and composition system will display ”NOT TRUE".
(g) “value” – Applies to a response value from **valueloc** attribute and the composition system will display the from the **valuetype** attribute.
(h) “unantic” – Applies to a unanticipated result.
(6) **value**(optional) – Supplies an alphanumeric or numeric value if attribute **type** is "value".
(7) **valueloc**(optional) – Supplies location of value if contained in another element, such as ***<input>***.
(8) **valuetype**(optional) – Specifies the type of value if attribute **type** is "value."
(a) “boolean” - Applies to a boolean value from **valueloc** and the composition system will display either “true” or “false”.
(b) “string” - Applies to a character string from **valueloc** and the composition system will display the string.

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(c) “sequence” - Applies to an ordered sequence values from **valueloc** and the composition system will display the sequence value.
(d) “set” - Applies to an unordered sequence values from **valueloc** and the composition system will display the set values.
(e) “real” - Applies to a real number from **valueloc** and the composition system will display the number.
(f) “integer” - Applies to an integer number from **valueloc** and the composition system will display the number.
(g) “float” - Applies to a floating point number from **valueloc** and the composition system will display the number.
(h) “nil” - Applies to null value from **valueloc** and the composition system will display the “nil”.
(i) “input” - Applies to a inserted query value from **valueloc** and the composition system will display the inputted value.
(j) “outcome” - Applies to a query outcome from a test from **valueloc** and the composition system will display the value.

36.9.1.1.1 Test ***<test>***. The element describes the testing steps that leads to an indication or condition to assist in determining the actual fault or next symptom to check.

a. The components of ***<test>*** are:
(1) Test title ***<title>*** (optional) (see 49.1.1.3).
(2) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(3) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(4) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(5) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).
(6) Table ***<table>*** (see section 42) and/or conditional illustration ***<table-alt>*** (see 48.2.1).(optional – zero or more).
(7) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1–alt>*** (see 48.2.1) (required – one or more).

b. DTD fragment graphically is:

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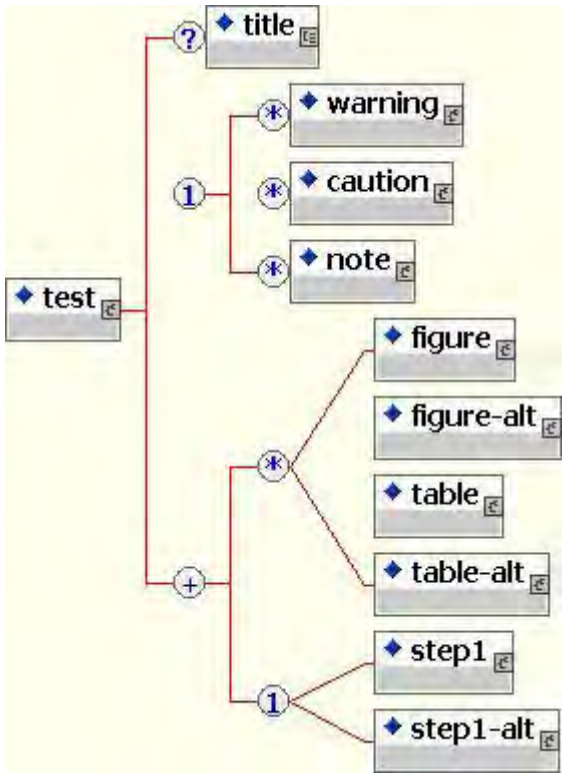

FIGURE 257. Test DTD hierarchy <test>.

c. DTD fragment for ***<test>***:

```
<!ELEMENT test (title?, (warning*, caution*, note*), ((figure |
   figure-alt | table | table-alt)*, (step1 | step1-alt))+)>
<!ATTLIST test
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<test>***:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.9.1.1.2 Answer ***<answer>***. The element specifies explicit actions, keyed to values such as "yes," "nottrue," or "value" contained in the **answerval**. The answer provides (from the indication or condition) the

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resulting value that leads to another fault isolation test block ***<testblock>*** or a malfunction/correct action block ***<endblock>***. The content contains information about the type of result (i.e., YES) and has a reference to the next test block step, next troubleshooting work package, or the determined malfunction (usually through ***<xref>*** or ***<link>*** elements).

a. The components of ***<answer>*** are:

(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).

b. DTD fragment graphically is:

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Check the source to verify that this is the current version before use.

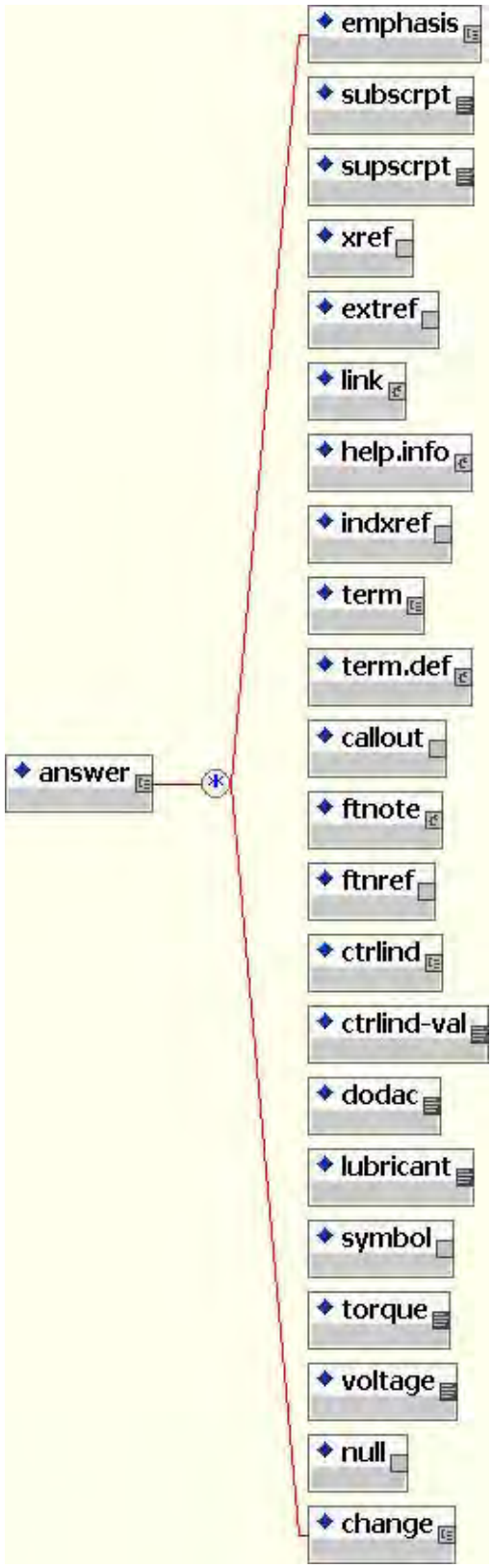

FIGURE 258. Test answer DTD hierarchy <answer>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<answer>***:

```
<!ELEMENT answer (#PCDATA | emphasis | subscrpt | supscrpt | xref |
   extref | link | help.info | indxref | term | term.def | callout |
   ftnote | ftnref | ctrlind | ctrlind-val | dodac | lubricant |
   symbol | torque | voltage | null | change)*>
<!ATTLIST answer
    answerval          (yes | no |
                        pass | fail |
                        true | nottrue |
                        value | unantic)              #IMPLIED
    value              CDATA                          #IMPLIED
    assocfig           IDREFS                         #IMPLIED
    changeref          IDREFS                         #IMPLIED
    comment            CDATA                          #IMPLIED
    delchlvl           CDATA                          #IMPLIED
    id                 ID                             #IMPLIED
    idref              IDREFS                         #IMPLIED
    inschlvl           CDATA                          #IMPLIED
    security           (uc | fouo |c | s | ts)       #IMPLIED
    skilltrk           CDATA                          #IMPLIED
>
```

d. Unique attributes for ***<answer>***:

(1) **answerval** – Specifies the logical value associated with the current element. This value may be displayed in either paper or electronic display.
   (a) “yes” - Applies to a positive answer and composition system will display ”YES".
   (b) “no” - Applies to a negative answer and composition system will display ”NO".
   (c) “pass” - Applies to a positive answer and composition system will display ”PASS".
   (d) “fail” - Applies to a negative answer and composition system will display ”FAIL".
   (e) “true” - Applies to a positive answer and composition system will display ”TRUE".
   (f) “nottrue” - Applies to a negative answer and composition system will display ”NOT TRUE".
   (g) “value” - Applies to a response value from **value** attribute.
   (h) “unantic” - Applies to an unanticipated response.

(2) **value** – Supplies an alphanumeric or numeric value if attribute **answerval** is set to "value".

e. Common attributes for ***<answer>***:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.9.1.2 End block ***<endblock>***. The element concludes a logic path or branch in a logical procedure. Based on the indicators or conditions from the previous test block procedure, the malfunction is identified and a corrective action is presented. The corrective action may contain brief instructions to correct the action, a reference to a detailed maintenance work package, or a reference for further fault isolation.

a. The components of ***<endblock>*** are:

(1) Identified malfunction (***<malfunc>*** (required) (see 36.6.2.1).
(2) Corrective action ***<action>*** (required) (see 36.6.1.1).

b. DTD fragment graphically is:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

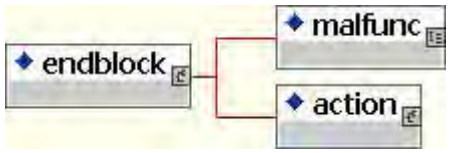

FIGURE 259. End block DTD hierarchy <endblock>.

c. DTD fragment for ***<endblock>***:

```
<!ELEMENT endblock (malfunc, action)>
<!ATTLIST endblock
   type        (yes | no |
                pass | fail |
                true | nottrue)  #REQUIRED
   branch      ID                #REQUIRED
   branchlabel CDATA             #IMPLIED
   branchfrom  IDREFS            #REQUIRED>
```

d. Unique attributes for ***<endblock>*** are:

(1) **branch** (required) – Specifies the end block unique identifier.

(2) **branchfrom** (required) – References the test block(s) (branches), which are from results from parent test blocks.

(3) **branchlabel** (optional) – Supplies an explicit reference to this branch and used in the end block presentation.

(4) **type** (required) – Specifies the type of branch logic current element. This value may be displayed in either paper or electronic display.

(a) “yes” – Applies to a positive answer and composition system will display ”YES".

(b) “no” – Applies to a negative answer and composition system will display ”NO".

(c) “pass” – Applies to a positive answer and composition system will display ”PASS".

(d) “fail” – Applies to a negative answer and composition system will display ”FAIL".

(e) “true” – Applies to a positive answer and composition system will display ”TRUE".

(f) “nottrue” – Applies to a negative answer and composition system will display ”NOT TRUE".

36.9.1.3 Test/end block reference ***<branchref>***. The element is used primarily with page-based manuals to provide a block or page reference to next corresponding test/end block. Presenting the logic procedure in narrative form on single page is not feasible, the reference provides a method to point to the next logical block of information. A good practice is to always use the same YES/NO branch for the reference. Using the same branch answer the user will have a consistent method to know when to branch and assists in developing the style sheet with graphical YES/NO symbols. In frame-based manuals the display of the element is not shown, but the link references are still used to traverse the logic flow tree. The element is EMPTY and all pertinent information is entered through its attributes.

a. DTD fragment for ***<branchref>***:

```
<!ELEMENT branchref EMPTY>
<!ATTLIST branchref
   branch      ID      #REQUIRED
   branchfrom  IDREFS  #REQUIRED
   branchlabel CDATA   #IMPLIED
   branchto    IDREFS  #REQUIRED
   textblockid IDREF   #REQUIRED>
```

b. Unique attributes for ***<branchref>*** are:

(1) **branch** (required) – Specifies the unique identifier.

(2) **branchfrom** (required) – References the test block(s) (branches), which are from results from parent test blocks.

(3) **branchlabel** (optional) – Supplies an explicit reference to this branch and used in the reference block presentation.

(4) **branchto** (required) – References the test/end block (branch) to the next test or action.

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(5) **textblockid** (required) – References the test/end block (branch) identifier on another page.

36.9.1.4 XML document instance fragment and output for ***<logicproc>***. The XML instance and its stylesheet output for a ***<logicproc>*** is provided below.

a. Sample XML instance.

```
<tswp airforce="no" army="no" deletewp="no" frame="yes" marines="no" navy="no"
tocentry="2" wpno="T0012-X-XXXX-XXX" wpseq="0151">
<wpidinfo>
<maintlvl level="field"/>
<title>Front Turn Light(s) Do Not Work
</title>
</wpidinfo>
<wpinfo>
<tools>
<tools-setup-item>
<name>General mechanic's tool kit: automotive
</name>
<itemref>
<xref itemid="s0004-X-XXXX-XXX-47" wpid="s0004-X-XXXX-XXX"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Multimeter
</name>
<itemref>
<xref itemid="s0004-X-XXXX-XXX-77" wpid="s0004-X-XXXX-XXX"/>
</itemref>
</tools-setup-item>
</tools>
<persnreq>
<persnreq-setup-item>
<name>ITV/IFV/CFV Sys Mech
</name>
<mos>63T10
</mos>
</persnreq-setup-item>
<persnreq-setup-item>
<name>Helper
</name>
<mos>H
</mos>
</persnreq-setup-item>
</persnreq>
<ref>
<ref-setup-item>
<extref docno="TM X-XXXX-XXX-10-1"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="T0013-X-XXXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="M0013-X-XXXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="M0025-X-XXXX-XXX"/>
</ref-setup-item>
```


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<ref-setup-item>
<xref wpid="M0035-X-XXXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="M0049-X-XXXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="M0183-X-XXXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="M0229-X-XXXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="M0032-X-XXXX-XXX"/>
</ref-setup-item>
</ref>
<eqpconds>
<eqpconds-setup-item>
<condition>Engine stopped
</condition>
<itemref>
<xref wpid="O0032-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>FIRE SUPPRESSION switch in MANUAL
</condition>
<itemref>
<xref wpid="O0034-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
<tsproc>
<logicproc>
<title>Troubleshooting Procedure
</title>
<note>
<trim.para>Left and right light circuits are identical. Designators for left side are given with designators for right side in parenthesis, when appropriate.
</trim.para>
</note>
<origin branchto="T0012-X-XXXX-XXX-block2 T0012-X-XXXX-XXX-block11" origin="T0012-X-XXXX-XXX-block1">
<test>
<figure id="T0012-X-XXXX-XXX-fig1">
<title>Front composite light
</title>
<graphic boardno="T0012-X-XXXX-XXX-fig1">
</graphic>
</figure>
<step1 id="T0012-X-XXXX-XXX-block1-step">
<para>Remove appropriate front composite light lens
<xref posttext=")" pretext=" (" wpid="M0032-X-XXXX-XXX"/>.
</para>

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```
</step1>
<step1>
<para>Remove turn signal lamp
<callout assocfig="T0012-X-XXXX-XXX-fig1" label="1"/> from
<callout assocfig="T0012-X-XXXX-XXX-fig1" label="2"/>.
</para>
</step1>
<step1 crewmember="H">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-master_power">MASTER POWER
</ctrlind> to
<ctrlind-val>ON
</ctrlind-val>
</para>
</step1>
<step1 crewmember="H">
<para>Move turn signal switch to appropriate position.
</para>
</step1>
<step1>
<para>Measure voltage between socket
<callout assocfig="T0012-X-XXXX-XXX-fig1" label="2"/> and ground.
</para>
</step1>
</test>
<indication>
<para>Does multimeter read less than 17 volts?
</para>
</indication>
<answer answerval="no">
<xref stepstart="T0012-X-XXXX-XXX-block11-step"/>
</answer>
<answer answerval="yes">
<xref stepstart="T0012-X-XXXX-XXX-block2-step"/>
</answer>
</origin>
<testblock branch="T0012-X-XXXX-XXX-block2" branchfrom="T0012-X-XXXX-XXX-block1"
branchto="T0012-X-XXXX-XXX-block4" type="yes">
<test>
<step1 id="T0012-X-XXXX-XXX-block2-step">
<para>Check rear lights
</para>
</step1>
</test>
<indication>
<para>Do rear stop/turn lights work?
</para>
</indication>
<answer answerval="no">Go to: Service stop light(s) malfunction
<xref wpid="M0013-X-XXXX-XXX"/>.
</answer>
<answer answerval="yes">
<xref stepstart="T0012-X-XXXX-XXX-block3-step"/>
</answer>
</testblock>
```


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```
<testblock branch="T0012-X-XXXX-XXX-block4" branchfrom="T0012-X-XXXX-XXX-block2"
branchto="T0012-X-XXXX-XXX-block5 T0012-X-XXXX-XXX-block6" type="yes">
<test>
<figure id="T0012-X-XXXX-XXX-fig2">
<title>Plug 1W13P7.
</title>
<graphic boardno="T0012-X-XXXX-XXX-fig2">
</graphic>
</figure>
<step1 crewmember="H" id="T0012-X-XXXX-XXX-block3-step">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-master_power">MASTER POWER
</ctrlind> to
<ctrlind-val>OFF
</ctrlind-val>.
</para>
</step1>
<step1>
<para>Remove plug 1W13P7 (1W13P12)
<callout assocfig="T0012-X-XXXX-XXX-fig2" label="1"/> from turn signal plug P3
<callout assocfig="T0012-X-XXXX-XXX-fig2" label="2"/>.
</para>
</step1>
<step1 crewmember="H">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-master_power">MASTER POWER
</ctrlind> to
<ctrlind-val>ON
</ctrlind-val>.
</para>
</step1>
<step1 crewmember="H">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-turn_indicator">TURN INDICATOR
</ctrlind> switch to appropriate position.
</para>
</step1>
</test>
<indication>
<para>Does multimeter read less than 17 volts?
</para>
</indication>
<answer answerval="no">
<xref stepstart="T0012-X-XXXX-XXX-block5-malfunc"/>
</answer>
<answer answerval="yes">
<xref stepstart="T0012-X-XXXX-XXX-block6-step"/>
</answer>
</testblock>
<endblock branch="T0012-X-XXXX-XXX-block5"
branchfrom="T0012-X-XXXX-XXX-block4" type="no">
<malfunc id="T0012-X-XXXX-XXX-block5-malfunc" label="symptom"> Faulty
front composite light assembly.
</malfunc>
<action>
<step1 crewmember="H">
```


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Check the source to verify that this is the current version before use.

```
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-turn_indicator">TURN INDICATOR
</ctrlind> switch to center position.
</para>
</step1>
<step1 crewmember="H">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-master_power">MASTER POWER
</ctrlind> to
<ctrlind-val>OFF
</ctrlind-val>.
</para>
</step1>
<step1>
<para>Replace front composite light assembly
<xref wpid="M0035-X-XXXX-XXX"/>.
</para>
</step1>
<step1>
<para>Verify no faults found.
</para>
</step1>
</action>
</endblock>
<testblock branch="T0012-X-XXXX-XXX-block6" branchfrom="T0012-X-XXXX-XXX-block4"
branchto="T0012-X-XXXX-XXX-block7 T0012-X-XXXX-XXX-block8" type="yes">
<test>
<figure id="T0012-X-XXXX-XXX-fig3">
<title>Jack 1A1J9.
</title>
<graphic boardno="T0012-X-XXXX-XXX-fig3">
</graphic>
</figure>
<step1 crewmember="H" id="T0012-X-XXXX-XXX-block6-step">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-master_power">MASTER POWER
</ctrlind> to
<ctrlind-val>OFF
</ctrlind-val>.
</para>
</step1>
<step1>
<para>Install front composite light door
<xref wpid="M0035-X-XXXX-XXX"/>.
</para>
</step1>
<step1>
<para>Install plug 1W13P7 (1W1P12) on turn signal on plug P3.
</para>
</step1>
<step1>
<para>Remove plug 1W13P1
<callout assocfig="T0012-X-XXXX-XXX-fig3" label="1"/> from jack 1A1J9
<callout assocfig="T0012-X-XXXX-XXX-fig3" label="2"/>.
</para>
</step1>
```


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Check the source to verify that this is the current version before use.

<step1 crewmember="H">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-master_power">MASTER POWER
</ctrlind> to
<ctrlind-val>ON
</ctrlind-val>.
</para>
</step1>
<step1 crewmember="H">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-turn_indicator">TURN INDICATOR
</ctrlind> switch to appropriate position.
</para>
</step1>
<step1>
<para>Measure voltage at jack 1A1J9
<callout assocfig="T0012-X-XXXX-XXX-fig3" label="2"/> between pin A (left)
<callout assocfig="T0012-X-XXXX-XXX-fig3" label="3"/> and ground
and between pin B (right)
<callout assocfig="T0012-X-XXXX-XXX-fig3" label="4"/> and ground.
</para>
</step1>
</test>
<indication>
<para>Does multimeter read less than 17 volts?
</para>
</indication>
<answer answerval="no">
<xref stepstart="T0012-X-XXXX-XXX-block7-malfunc"/>
</answer>
<answer answerval="yes">
<xref stepstart="T0012-X-XXXX-XXX-block8-step"/>
</answer>
</testblock>
<endblock branch="T0012-X-XXXX-XXX-block7"
branchfrom="T0012-X-XXXX-XXX-block6" type="no">
<malfunc id="T0012-X-XXXX-XXX-block7-malfunc" label="symptom">Faulty
front composite light assembly.
</malfunc>
<action>
<step1 crewmember="H">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-turn_indicator">TURN INDICATOR
</ctrlind> switch to center position.
</para>
</step1>
<step1 crewmember="H">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-master_power"> MASTER POWER
</ctrlind> to
<ctrlind-val>OFF
</ctrlind-val>.
</para>
</step1>
<step1>
<para>Replace/replace wiring harness 1W13

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```
<xref wpid="M0049-X-XXXX-XXX"/>.
</para>
</step1>
<step1>
<para>Verify no faults found.
</para>
</step1>
</action>
</endblock>
<testblock branch="T0012-X-XXXX-XXX-block8" branchfrom="T0012-X-XXXX-XXX-block6"
branchto="T0012-X-XXXX-XXX-block9 T0012-X-XXXX-XXX-block10" type="yes">
<test>
<figure id="T0012-X-XXXX-XXX-fig4">
<title>Plugs 1W10P1 and 1W10P2.
</title>
<graphic boardno="T0012-X-XXXX-XXX-fig4">
</graphic>
</figure>
<step1 id="T0012-X-XXXX-XXX-block8-step">
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-master_power">MASTER POWER
</ctrlind> to
<ctrlind-val>OFF
</ctrlind-val>.
</para>
</step1>
<step1>
<para>Install plug 1W13P1 on jack 1A1J9.
</para>
</step1>
<step1>
<para>Remove plug 1W10P1
<callout assocfig="T0012-X-XXXX-XXX-fig4" label="1"/> from jack 1A1J8
<callout assocfig="T0012-X-XXXX-XXX-fig4" label="2"/>.
</para>
</step1>
<step1>
<para>Remove plug 1W10P2
<callout assocfig="T0012-X-XXXX-XXX-fig4" label="3"/> from turn signal switch 1A4S13
<callout assocfig="T0012-X-XXXX-XXX-fig4" label="4"/>.
</para>
</step1>
<step1>
<para>Measure resistance at plug 1W10P1
<callout assocfig="T0012-X-XXXX-XXX-fig4" label="1"/> between pin AS
<callout assocfig="T0012-X-XXXX-XXX-fig4" label="5"/> and pin J
<callout assocfig="T0012-X-XXXX-XXX-fig4" label="6"/> and at plug 1W10P2
<callout assocfig="T0012-X-XXXX-XXX-fig4" label="3"/> between pin A
<callout assocfig="T0012-X-XXXX-XXX-fig4" label="7"/> and pin B
<callout assocfig="T0012-X-XXXX-XXX-fig4" label="8"/>.
</para>
</step1>
</test>
<indication>
<para>Does multimeter read 0 ohms?
</para>
```


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```
</indication>
<answer answerval="no">
<xref stepstart="T0012-X-XXXX-XXX-block9-malfunc"/>
</answer>
<answer answerval="yes">
<xref stepstart="T0012-X-XXXX-XXX-block10-malfunc"/>
</answer>
</testblock>
<endblock branch="T0012-X-XXXX-XXX-block9"
branchfrom="T0012-X-XXXX-XXX-block8" type="no">
<malfunc id="T0012-X-XXXX-XXX-block9-malfunc" label="malfunction">Faulty
wiring harness.
</malfunc>
<action>
<step1>
<para>Repair/replace wiring harness 1W10
<xref wpid="M0025-X-XXXX-XXX"/>.
</para>
</step1>
<step1>
<para>Verify no faults found.
</para>
</step1>
</action>
</endblock>
<endblock branch="T0012-X-XXXX-XXX-block10"
branchfrom="T0012-X-XXXX-XXX-block8" type="yes">
<malfunc id="T0012-X-XXXX-XXX-block10-malfunc" label="malfunction">Faulty
vehicle electrical distribution box.
</malfunc>
<action>
<step1>
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-master_power">MASTER POWER
</ctrlind> to
<ctrlind-val>OFF
</ctrlind-val>.
</para>
</step1>
<step1>
<para>Replace vehicle electrical distribution box
<xref wpid="M0013-X-XXXX-XXX"/>.
</para>
</step1>
<step1>
<para>Verify no faults found.
</para>
</step1>
</action>
</endblock>
<testblock branch="T0012-X-XXXX-XXX-block11" branchfrom="T0012-X-XXXX-XXX-block1"
branchto="T0012-X-XXXX-XXX-block12" type="no">
<test>
<figure id="T0012-X-XXXX-XXX-fig5">
<title>Composite light assembly.
</title>
```


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```
<graphic boardno="T0012-X-XXXX-XXX-fig5">
</graphic>
</figure>
<step1 crewmember="H" id="T0012-X-XXXX-XXX-block11-step">
<para>Move turn signal switch to OFF.
</para>
</step1>
<step1>
<para>Move
<ctrlind idref="O0003-X-XXXX-XXX-master_power">MASTER POWER
</ctrlind> to
<ctrlind-val>OFF
</ctrlind-val>.
</para>
</step1>
<step1>
<para>Mesaure resistance between socket body
<callout assocfig="T0012-X-XXXX-XXX-fig5" label="1"/> and ground.
</para>
</step1>
</test>
<indication>
<para>Does multimeter read 0 ohms?
</para>
</indication>
<answer answerval="no">
<xref stepstart="T0012-X-XXXX-XXX-block12-malfunc"/>
</answer>
<answer answerval="yes">
<xref stepstart="T0012-X-XXXX-XXX-block13-malfunc"/>
</answer>
</testblock>
<endblock branch="T0012-X-XXXX-XXX-block12"
branchfrom="T0012-X-XXXX-XXX-block11" type="no">
<malfunc id="T0012-X-XXXX-XXX-block12-malfunc" label="malfunction">Faulty
wiring harness 1W52.
</malfunc>
<action>
<step1>
<para>Repair/replace wiring harness 1W52
<xref wpid="M0183-X-XXXX-XXX"/>.
</para>
</step1>
</action>
</endblock>
<endblock branch="T0012-X-XXXX-XXX-block13"
branchfrom="T0012-X-XXXX-XXX-block11" type="yes">
<malfunc id="T0012-X-XXXX-XXX-block13-malfunc" label="malfunction">Faulty
front composite light assembly lamp.
</malfunc>
<action>
<step1>
<para>Repair/replace front composite light assembly lamp
<xref wpid="M0229-X-XXXX-XXX"/>.
</para>
</step1>
```


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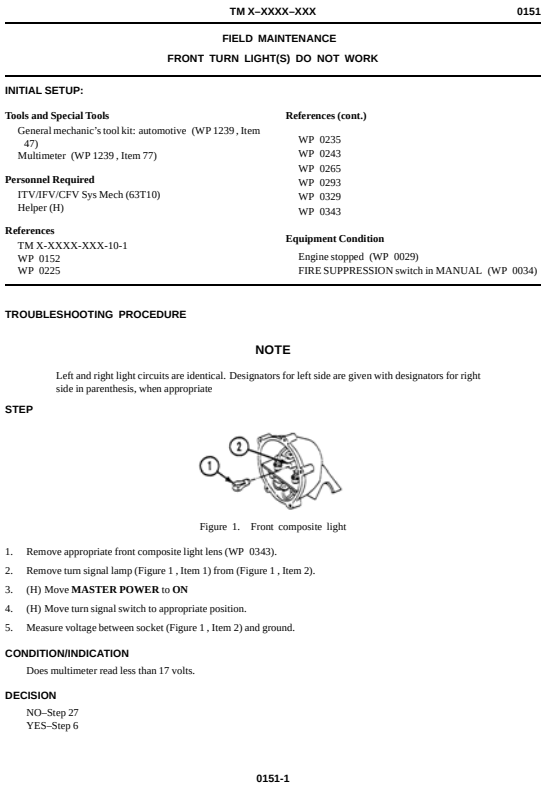
</action>
</endblock>
</logicproc>
</tsproc>
</tswp>

b. XML sample page-layout.



TM X–XXXX–XXX 0151

FIELD MAINTENANCE

FRONT TURN LIGHT(S) DO NOT WORK

INITIAL SETUP:

Tools and Special Tools
General mechanic's tool kit: automotive (WP 1239 , Item 47)
Multimeter (WP 1239 , Item 77)

Personnel Required
ITV/IFV/CFV Sys Mech (63T10)
Helper (H)

References
TM X-XXXX-XXX-10-1
WP 0152
WP 0225

References (cont.)
WP 0235
WP 0243
WP 0265
WP 0293
WP 0329
WP 0343

Equipment Condition
Engine stopped (WP 0029)
FIRE SUPPRESSION switch in MANUAL (WP 0034)

TROUBLESHOOTING PROCEDURE

NOTE

Left and right light circuits are identical. Designators for left side are given with designators for right side in parenthesis, when appropriate

STEP

Figure 1. Front composite light

1. Remove appropriate front composite light lens (WP 0343).
2. Remove turn signal lamp (Figure 1 , Item 1) from (Figure 1 , Item 2).
3. (H) Move **MASTER POWER** to **ON**
4. (H) Move turn signal switch to appropriate position.
5. Measure voltage between socket (Figure 1 , Item 2) and ground.

CONDITION/INDICATION
Does multimeter read less than 17 volts.

DECISION
NO–Step 27
YES–Step 6

0151-1

FIGURE 260. Troubleshooting work package logic procedure <logicproc> page-layout.

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STEP

6. Check rear lights

CONDITION/INDICATION

Do rear stop/turn lights work?

DECISION

NO–Go to: Service stop light(s) malfunction WP 0152.
YES–Step 7

STEP

Figure 2. Plug 1W13P7.

7. (H) Move **MASTER POWER** to **OFF**.
8. Remove plug 1W13P7 (1W13P12) (Figure 2 , Item 1) from turn signal plug P3 (Figure 2 , Item 2).
9. (H) Move **MASTER POWER** to **ON**.
10. (H) Move **TURN INDICATOR** switch to appropriate position.

CONDITION/INDICATION

Does multimeter read less than 17 volts?

DECISION

NO–Step 11
YES–Step 12

MALFUNCTION

11. Faulty front composite light assembly.

ACTION

STEP 1. (H) Move **TURN INDICATOR** switch to center position.

STEP 2. (H) Move **MASTER POWER** to **OFF**.

STEP 3. Replace front composite light assembly WP 0243.

STEP 4. Verify no faults found.

FIGURE 260. Troubleshooting work package logic procedure <logicproc> page-layout - Continued.

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Check the source to verify that this is the current version before use.

STEP

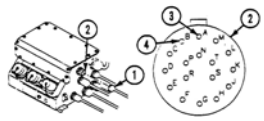

Figure 3. Jack 1A1J9.

12. (H) Move **MASTER POWER** to **OFF**.
13. Install front composite light door WP 0243.
14. Install plug 1W13P7 (1W1P12) on turn signal on plug P3.
15. Remove plug 1W13P1 (Figure 3 , Item 1) from jack 1A1J9 (Figure 3 , Item 2).
16. (H) Move **MASTER POWER** to **ON**.
17. (H) Move **TURN INDICATOR** switch to appropriate position.
18. Measure voltage at jack 1A1J9 (Figure 3 , Item 2) between pin A (left) (Figure 3 , Item 3) and ground and between pin B (right) (Figure 3 , Item 4) and ground.

CONDITION/INDICATION

Does multimeter read less than 17 volts?

DECISION

NO–Step 19
YES–Step 20

MALFUNCTION

19. Faulty front composite light assembly.

ACTION

STEP 1. (H) Move **TURN INDICATOR** switch to center position.

STEP 2. (H) Move **MASTER POWER** to **OFF**.

STEP 3. Replace/replace wiring harness 1W13 WP 0265.

STEP 4. Verify no faults found.

FIGURE 260. Troubleshooting work package logic procedure <logicproc> page-layout - Continued.

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STEP

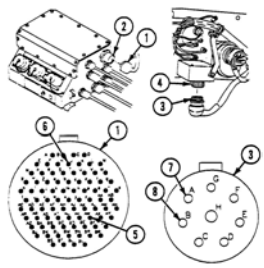

Figure 4. Plugs 1W10P1 and 1W10P2.

20. Move **MASTER POWER** to **OFF**.
21. Install plug 1W13P1 on jack 1A1J9.
22. Remove plug 1W10P1 (Figure 4 , Item 1) from jack 1A1J8 (Figure 4 , Item 2).
23. Remove plug 1W10P2 (Figure 4 , Item 3) from turn signal switch 1A4S13 (Figure 4 , Item 4).
24. Measure resistance at plug 1W10P1 (Figure 4 , Item 1) between pin AS (Figure 4 , Item 5) and pin J (Figure 4 , Item 6) and at plug 1W10P2 (Figure 4 , Item 3) between pin A (Figure 4 , Item 7) and pin B (Figure 4 , Item 8).

CONDITION/INDICATION

Does multimeter read 0 ohms?

DECISION

NO–Step 25
YES–Step 26

MALFUNCTION

25. Faulty wiring harness.

ACTION

STEP 1. Repair/replace wiring harness 1W10 WP 0235.

STEP 2. Verify no faults found.

MALFUNCTION

26. Faulty vehicle electrical distribution box.

ACTION

STEP 1. Move **MASTER POWER** to **OFF**.

STEP 2. Replace vehicle electrical distribution box WP 0225

STEP 3. Verify no faults found.

FIGURE 260. Troubleshooting work package logic procedure <logicproc> page-layout - Continued.

c. XML sample frame-layout.

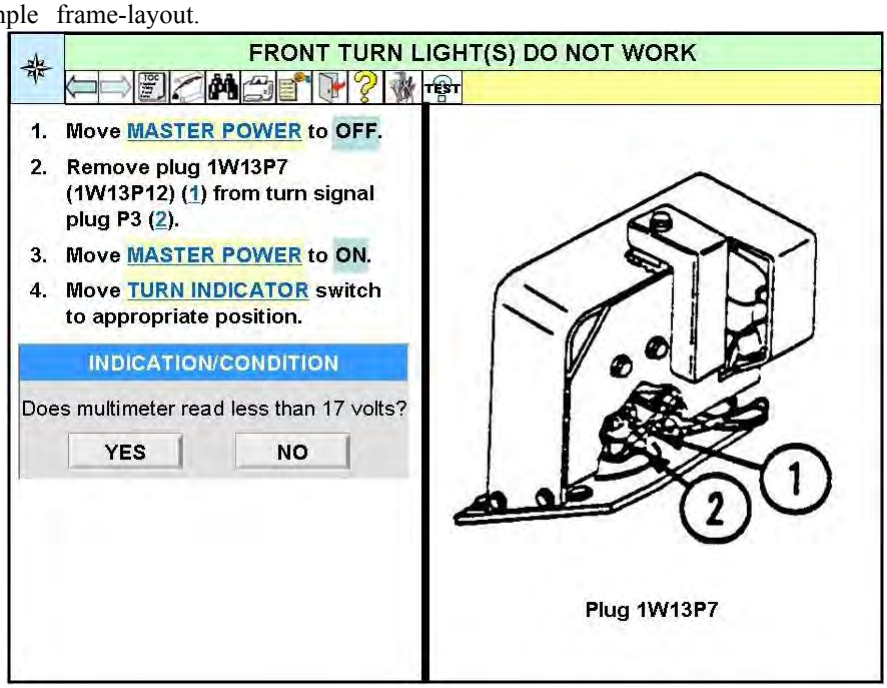

FIGURE 261. Troubleshooting work package logic procedure <logicproc> frame-layout.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

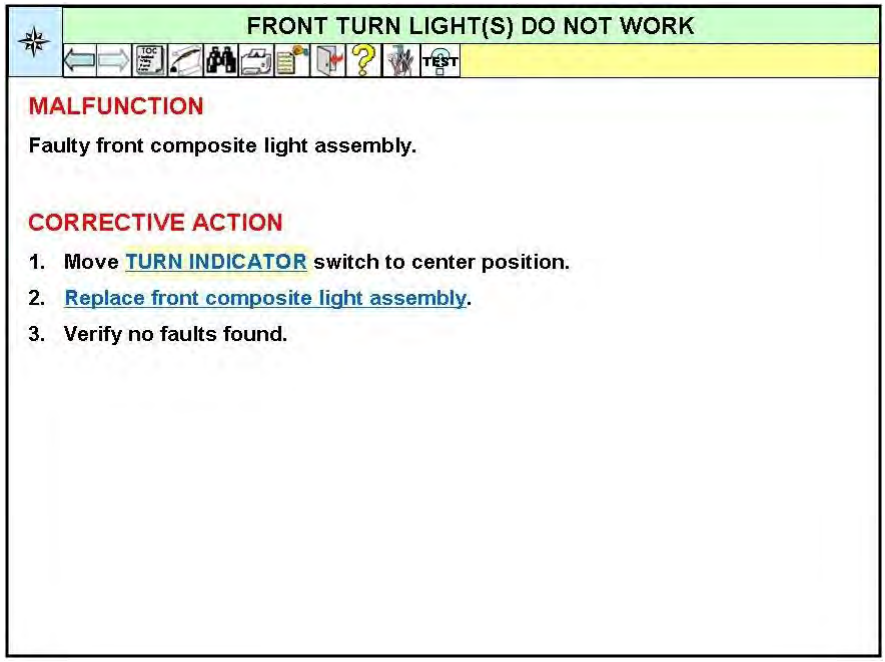

FIGURE 261. Troubleshooting work package logic procedure <logicproc> frame-layout - Continued.

d. XML sample logic flow in frame-layout.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

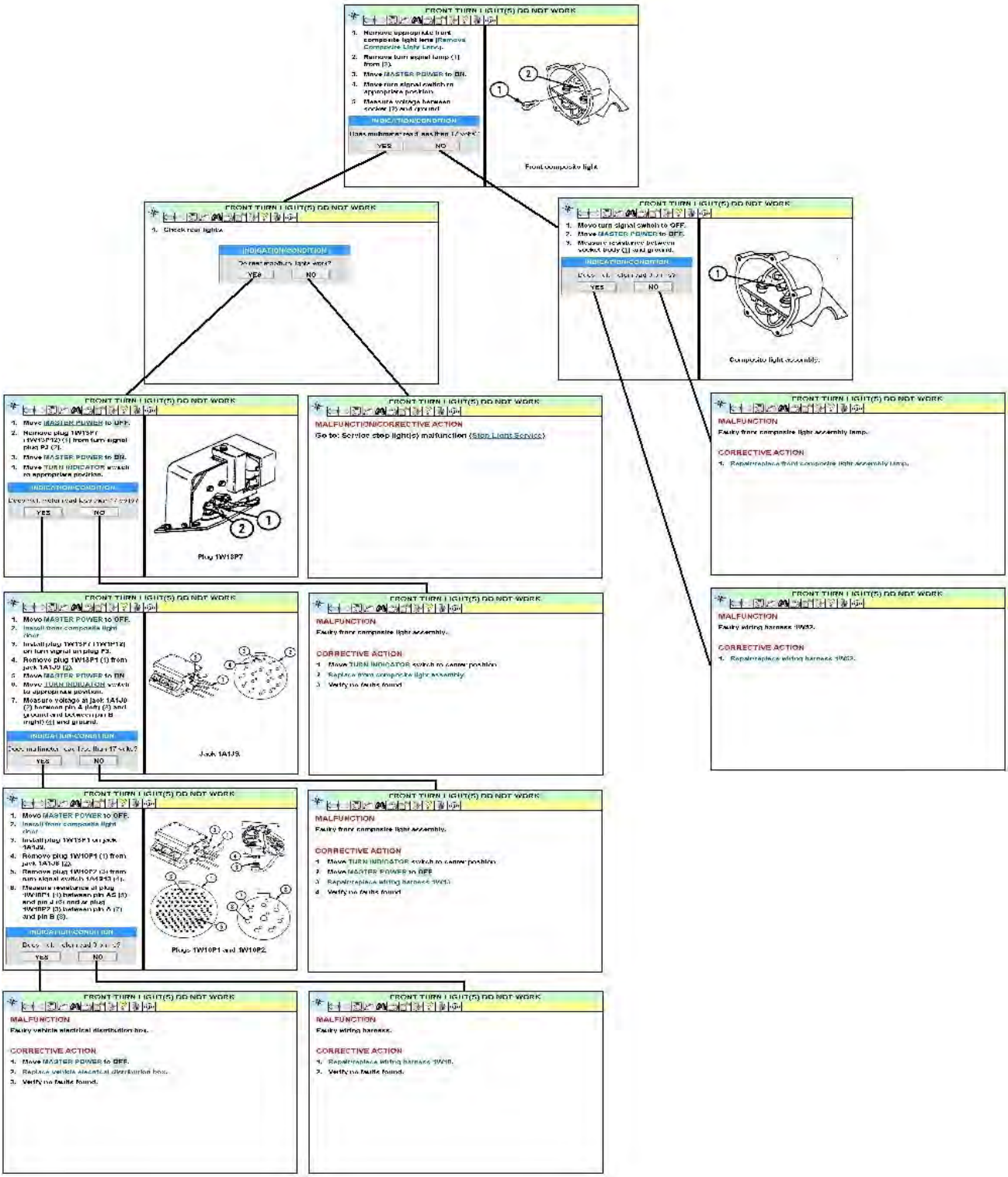

FIGURE 262. Troubleshooting logic functional flow – frame-layout.

36.9.2 Fault procedure ***<faultproc>***. The element is used for troubleshooting procedures consisting of an all inclusive series of specific fault symptoms for the system/equipment being troubleshot.

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When in a frame–based viewer, the content specific elements identifies the information and allows some information to be contracted or expanded depending on the symptom and malfunctions identified. Figure 266 provides an example where the known symptom is engine oil pressure is in the red zone and the malfunctions checked has led to the coolant temperature gauge is in the red zone. Knowing the malfunction, showing the corrective action for engine oil pressure is low is not necessary.

a. The components of ***<faultproc>*** are:
   (1) Procedure title ***<title>*** (required) (see 49.1.1.3).
   (2) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (3) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (4) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (5) Table header ***<std.table.header>*** (optional) (see 42.3.1). When the style sheet generates the fault procedure in table format, the element provides the column headers.
   (6) Each fault symptom (required – one or more) has the following components:
      (a) Symptom description ***<symptom>*** (see 36.9.2.1).
      (b) Malfunctions ***<malfunc>*** (required – one or more) (see 36.6.2.1).
      (c) Each malfunction has associated corrective action (required) as either steps ***<action>*** (see 36.6.1.1) or reference to maintenance work package/further detail troubleshooting work package ***<xref>*** (see 46.2.2)/***<link>*** (see 46.2.3).

b. DTD fragment graphically is:

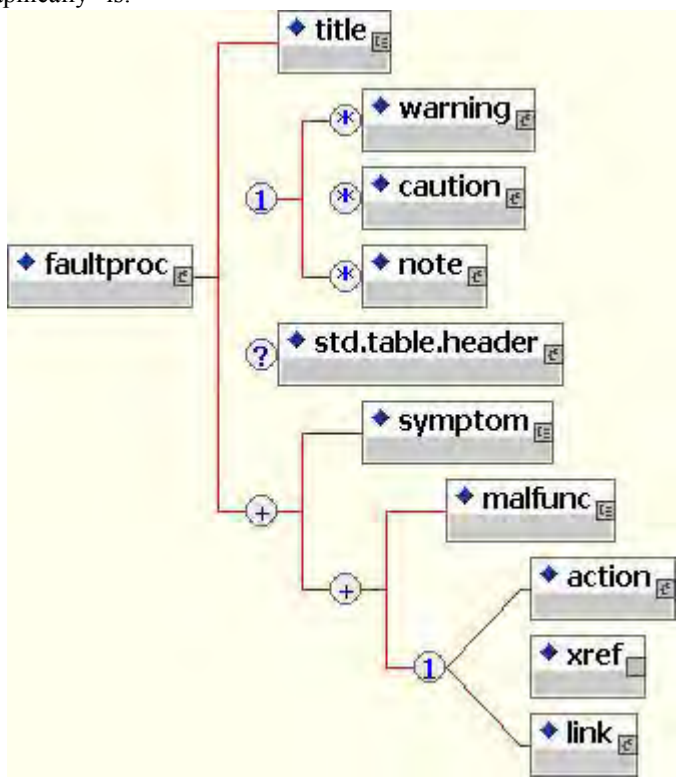

FIGURE 263. Fault procedure DTD hierarchy <faultproc>.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment for Fault Procedure ***<faultproc>***:

```
<!ELEMENT faultproc (title, (warning*, caution*, note*),
 std.table.header?, (symptom, (malfunc, (action | xref))+)+)>
<!ATTLIST faultproc
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<faultproc>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.9.2.1 Symptom ***<symptom>***. The element specifies the fault symptom for the system/equipment being diagnosed.

a. The components of ***<symptom>*** are:

(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).

b. DTD fragment graphically is:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

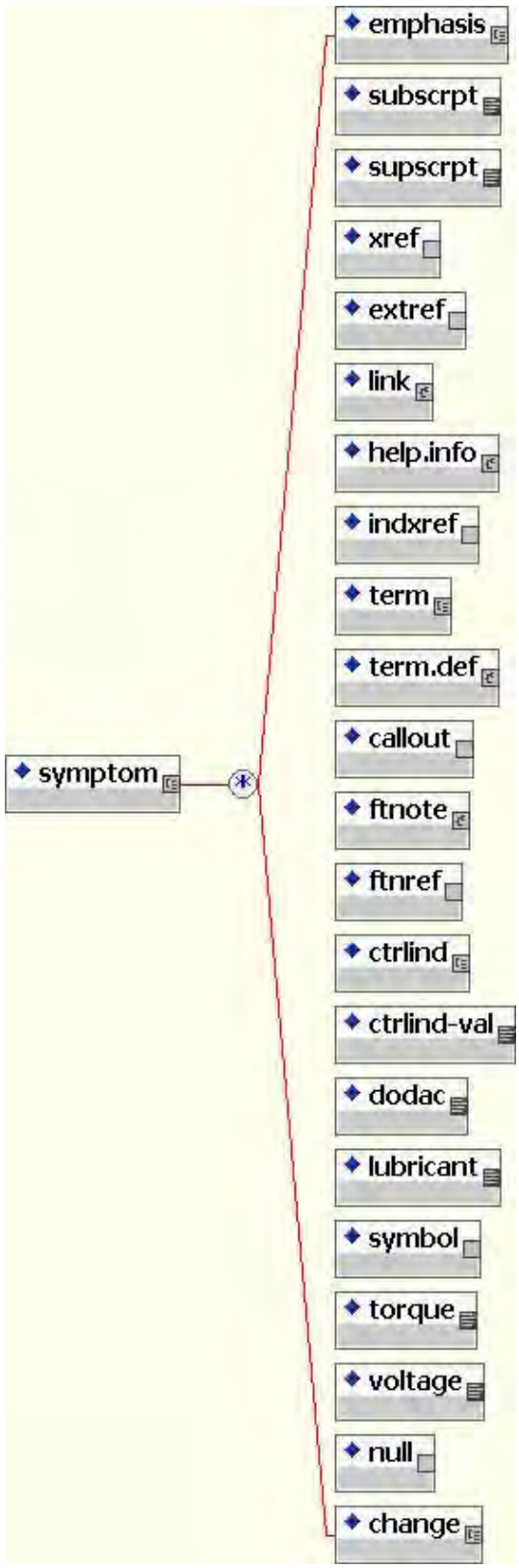

FIGURE 264. Symptom description DTD hierarchy <symptom>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<symptom>*** is:

```
<!ELEMENT symptom (#PCDATA | emphasis | subscrpt | supscrpt | xref |
   extref | link | help.info | indxref | term | term.def | callout |
   ftnote | ftnref | ctrlind | ctrlind-val | dodac | lubricant |
   symbol | torque | voltage | null | change)*>
<!ATTLIST symptom
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<symptom>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.9.2.2 XML document instance fragment and output for ***<tswp>***/***<faultproc>***. The XML instance and its stylesheet output for a ***<faultproc>*** is provided below.

a. Sample XML instance.

<tswp airforce="no" army="no" deletewp="no" frame="yes" insertwp="" marines="no" navy="no" tocentry="2" wpno="T00003-X-XXXX-XXX" wpseq="0251">
<wpidinfo>
<maintlvl level="operator"/>
<title>Engine Troubleshooting Procedures
</title>
</wpidinfo>
<wpinfo>
<persnreq>
<persnreq-setup-item>
<name>Mechanic
</name>
<qty>3
</qty>
</persnreq-setup-item>
</persnreq>
<ref>
<ref-setup-item>
<xref wpid="o00039-X-XXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="o00040-X-XXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="o00041-X-XXX-XXX"/>

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Check the source to verify that this is the current version before use.

```
</ref-setup-item>
<ref-setup-item>
<xref  wpid="o00047-X-XXX-XXX"/>
</ref-setup-item>
</ref>
</wpinfo>
<tsproc>
<faultproc>
<title>Engine  oil  pressure
</title>
<symptom>Engine oil press gauge is in red zone.
</symptom>
<malfunc  id="T00004-X-XXXX-XXX-01-01"  label="symptom">
<ctrlind  idref="o00042-X-XXXX-XXXengine_oil_low_press">ENGINE OIL LOW PRESS
</ctrlind>  indicator  light  is  flashing.
</malfunc>
<action>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">If  indicator  light  is  flashing,
stop  engine  immediately
<xref  posttext=")"  pretext="("  wpid="o00039-X-XXX-XXX"/>.
</para>
</step1>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<specpara  esd="no"  hcp="no">
<warning  haz-abbrev="no">
<trim.para  esd="no"  hcp="no">Hot  power  unit  can  burn  you.   Use
care  when  working  near  power  unit.
</trim.para>
</warning>
<para  esd="no"  hcp="no">Check  engine  oil  level
<xref  posttext=")"  pretext="("  wpid="o00047-X-XXX-XXX"/>.
</para>
</specpara>
</step1>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Check  bilge  for  oil.
</para>
<step2  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">If  oil  is  present,  check  engine  hoses,
clamps,  and  fittings  for  leaks.
</para>
</step2>
</step1>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Start  engine
<xref  posttext=")"  pretext="("  wpid="o00039-X-XXX-XXX"/>.
</para>
</step1>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Check  if
<ctrlind  idref="o00042-X-XXXX-XXXengine_oil_press">ENGINE OIL PRESS
</ctrlind>  gauge  is  still  in  the  red  zone.
</para>
<step2  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Stop  engine  immediately
```


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<xref posttext=")" pretext="(" wpid="o00039-X-XXX-XXX"/>. Notify unit maintenance.
</para>
</step2>
</step1>
</action>
<malfunc id="T00004-X-XXXX-XXX-01-02" label="symptom">ENGINE COOLANT TEMPERATURE GAUGE IS IN RED ZONE.
</malfunc>
<action>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check if
<ctrlind idref="o00042-X-XXXX-XXXcoolant_low_level">COOLANT LOW LEVEL
</ctrlind> indicator light is flashing.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If
<ctrlind idref="o00042-X-XXXX-XXXcoolant_low_level">COOLANT LOW LEVEL
</ctrlind> indicator light is off.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Press
<ctrlind idref="o00042-X-XXXX-XXXtest_sensor_button">TEST SENSOR BUTTON
</ctrlind>.
</para>
<step3 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If
<ctrlind idref="o00042-X-XXXX-XXXcoolant_low_level">COOLANT LOW LEVEL
</ctrlind> indicator does not flash, notify unit maintenance.
</para>
</step3>
<step3 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If
<ctrlind idref="o00042-X-XXXX-XXXcoolant_low_level">COOLANT LOW LEVEL
</ctrlind> indicator light is flashing,
</para>
<step4 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Stop engine immediately
<xref posttext=")" pretext="(" wpid="o00039-X-XXX-XXX"/>.
</para>
</step4>
<step4 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<specpara esd="no" hcp="no">
<warning haz-abbrev="no">
<trim.para esd="no" hcp="no">Hot coolant can burn you. Do not remove radiator cap until coolant temperature gauge reads in bottom one-quarter of green zone. Turn cap slowly to release pressure.
</trim.para>
</warning>
<para esd="no" hcp="no">Check coolant level
<xref posttext=")" pretext="(" wpid="o00040-X-XXX-XXX"/>.
</para>
</specpara>
</step4>
</step3>
</step2>

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```
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Open power unit access door
<xref posttext=")" pretext="(" wpid="o00041-X-XXX-XXX"/>.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check bilge for coolant.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If coolant is found, check hoses,
clamps, and fittings for coolant leaks.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If leaks are found, notify unit maintenance.
</para>
</step2>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check coolant system for malfunction.
</para>
</step1>
</action>
</faultproc>
</tsproc>
</tswp>
```

b. XML sample page-layout.

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OPERATOR MAINTENANCE
ENGINE TROUBLESHOOTING PROCEDURES

INITIAL SETUP:

Personnel Required
Mechanic (3)

References
WP 0039

References (cont.)
WP 0040
WP 0041
WP 0047

TROUBLESHOOTING PROCEDURE
ENGINE OIL PRESSURE

SYMPTOM

Engine oil press gauge is in red zone.

MALFUNCTION

ENGINE OIL LOW PRESS indicator light is flashing.

CORRECTIVE ACTION

STEP 1. If indicator light is flashing, stop engine immediately (WP 0039).

WARNING

Hot power unit can burn you. Use care when working near power unit.

STEP 2. Check engine oil level (WP 0047).
STEP 3. Check bilge for oil.
a. If oil is present, check engine hoses, clamps, and fittings for leaks.
STEP 4. Start engine (WP 0039).
STEP 5. Check if **ENGINE OIL PRESS** gauge is still in the red zone.
a. Stop engine immediately (WP 0039). Notify unit maintenance.

MALFUNCTION

ENGINE COOLANT TEMPERATURE GAUGE IS IN RED ZONE.

CORRECTIVE ACTION

STEP 1. Check if **COOLANT LOW LEVEL** indicator light is flashing.
a. If **COOLANT LOW LEVEL** indicator light is off.
b. Press **TEST SENSOR BUTTON**.
(1) If **COOLANT LOW LEVEL** indicator does not flash, notify unit maintenance.
(2) If **COOLANT LOW LEVEL** indicator light is flashing,
(a) Stop engine immediately (WP 0039).

WARNING

Hot coolant can burn you. Do not remove radiator cap until coolant temperature gauge reads in bottom one-quarter of green zone. Turn cap slowly to release pressure.

(b) Check coolant level(WP 0040).
STEP 2. Open power unit access door (WP 0041).

FIGURE 265. Troubleshooting work package fault procedure <faultproc> page-layout.

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TM X–XXXX–XXX 0251

CORRECTIVE ACTION –CONTINUED

STEP 3. Check bilge for coolant.

a. If coolant is found, check hoses, clamps, and fittings for coolant leaks.

b. If leaks are found, notify unit maintenance.

STEP 4. Check coolant system for malfunction.

END OF WORK PACKAGE

0251-2

FIGURE 265. Troubleshooting work package fault procedure <faultproc> page-layout - Continued.

c. XML sample frame-layout.

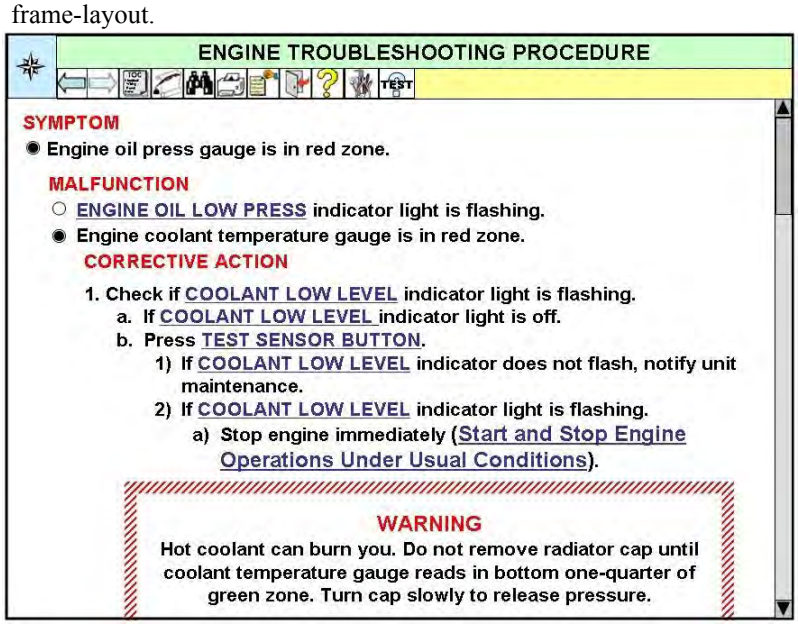

FIGURE 266. Troubleshooting work package fault procedure <faultproc> frame-layout.

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36.9.3 Multiplex read code ***<muxproc>***. The element displays computer generated multiplex (MUX) read code data which are listed in troubleshooting sequence order by signal name ***<signame>***. The method can be presented either tabular or narrative format.

a. The components of ***<muxproc>*** are:
   (1) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (2) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (3) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (4) The MUX read codes are either by
      (a) Symptom/malfunction ***<symptom>*** (optional) (see 36.9.2.1) with associated signal name MUX read codes ***<signal-item>*** (required – one or more) (see 36.9.3.1).
      (b) Signal name MUX read codes ***<signal-item>*** (required – one or more) (see 36.9.3.1).

b. DTD fragment graphically is:

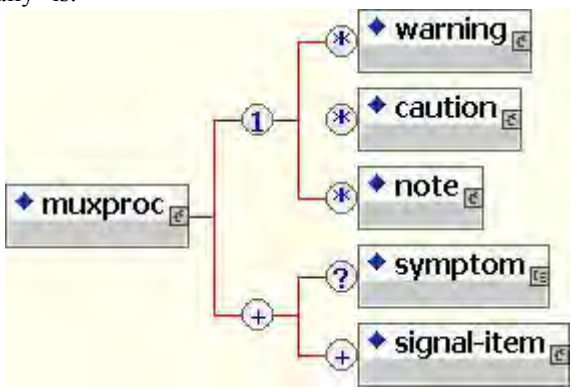

FIGURE 267. Multiplex read code procedure DTD hierarchy <muxproc>..

c. DTD fragment for ***<muxproc>***:

```
<!ELEMENT muxproc ((warning*, caution*, note*), (symptom?, signal-item+)+)>
<!ATTLIST muxproc
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<muxproc>*** are:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

36.9.3.1 Signal item ***<signal-item>***. The element contains all information required to test a particular signal, component part, process, or data item during pass/fail fault isolation. The element contains a

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signal name ***<signame>***, data items ***<dataitem>***, pass/fail check remarks ***<ckremarks>***, a pass/fail criteria ***<criteria>*** followed by a second pass/fail criteria.

a. The components are:

(1) Signal name ***<signame>*** (required) is the name of the signal item being analyzed.

(2) MUX read codes ***<dataitem>*** (required) components are:

(a) Memory location address ***<memloc>*** (required)

(b) Memory data bit(s) ***<memdata>*** (required)

(c) Conditions ***<condition>*** (required), when no conditions exist enter "none".

(d) Signal function description ***<sigfunc>*** (required)

(3) Remarks ***<remarks>*** (required) concerning about the signal item or MUX read code. May be used to describe where signal is derived from.

(4) Pass criteria ***<criteria>*** (required) (see 36.9.3.1.1).

(5) Fail criteria ***<criteria>*** (required) (see 36.9.3.1.1).

b. DTD fragment graphically is:

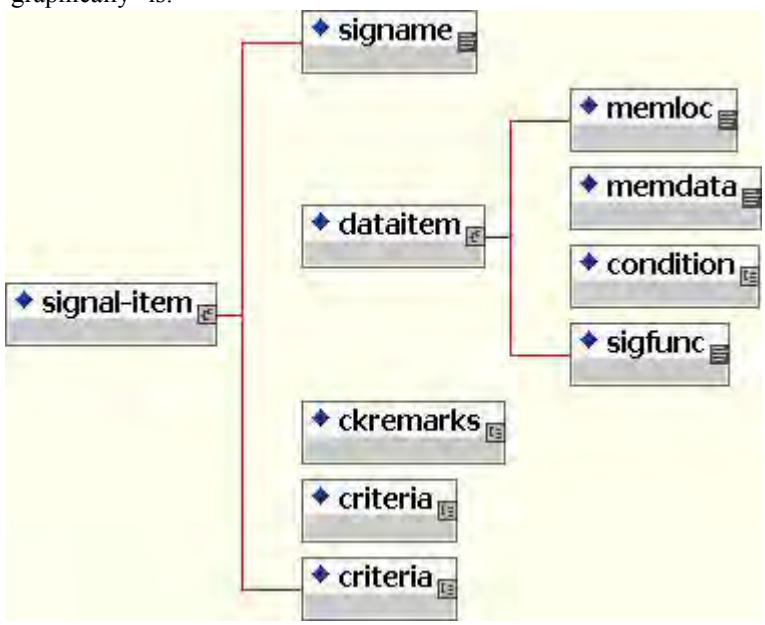

FIGURE 268. Multiplex read code procedure DTD hierarchy <muxproc>..

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c. DTD fragment for ***<signal-item>***:

```
<!ELEMENT signal-item (signame, dataitem, ckremarks, criteria, criteria)>
<!ATTLIST signal-item
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
   type              (part| signal | process | other) #IMPLIED>


<!ELEMENT dataitem (memloc, memdata, condition, sigfunc)>
<!ATTLIST dataitem
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Unique attributes for ***<signal-item>*** is:

(1) **type** (optional) – Defines the type of signal being analyzed.
   (a) "part" - Test applies to a component part.
   (b) "signal" - Test applies to a particular signal.
   (c) "process" - Test applies to a process.
   (d) "other" - Test applies to a data item.

e. Common attributes for ***<signal-item>*** and ***<dataitem>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

36.9.3.1.1 Pass/fail criteria ***<criteria>***. The element specifies the criteria for a functioning signal item with the next step in diagnosing the problem or signal item failed and the corrective action. When a pass criteria is being described a reference is made to the next signal item to check for a fault.

a. The components of ***<muxproc>*** are:

(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).

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(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).

b. DTD fragment graphically is:

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Check the source to verify that this is the current version before use.

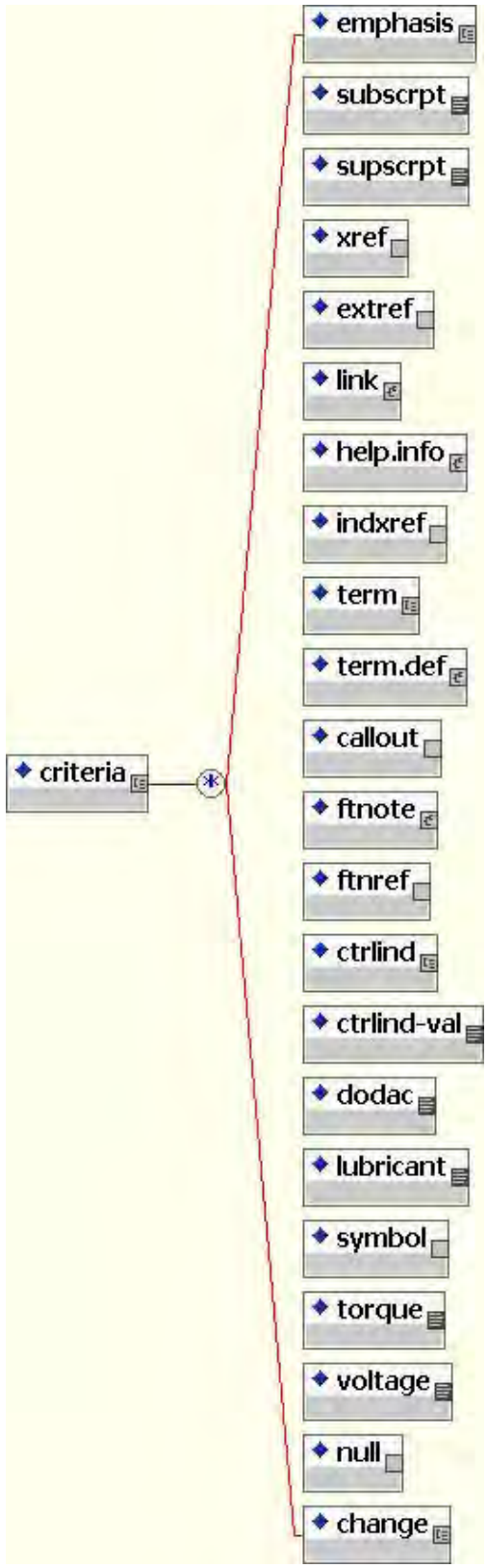

FIGURE 269. Multiplex read code pass/fail criteria DTD hierarchy <criteria>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<criteria>***:

```
<!ELEMENT criteria (#PCDATA | emphasis | subscrpt | supscrpt | xref |
   extref | link | help.info | indxref | term | term.def | callout |
   ftnote | ftnref | ctrlind | ctrlind-val | dodac | lubricant |
   symbol | torque | voltage | null | change)*>
<!ATTLIST criteria
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
   type              (pass | fail)                   #REQUIRED>
```

d. Unique attributes for ***<criteria>*** is:

(1) **type** - Specifies the type of action resulting from this criteria element.

(a) "pass" - The identifies the content as a pass criteria.

(b) "fail" - The identifies the content as a fail criteria.

e. Common attributes for ***<criteria>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

(3) **comment** – Reason for change information. (optional) (See 49.2.9.)

(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)

(5) **id** – Unique identifier. (optional) (See 49.2.6.)

(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)

(9) **security** – Security classification. (optional) (See 49.2.13.)

36.9.3.2 XML document instance fragment and output for ***<tswp>/<muxproc>***. The XML instance and its stylesheet output for a ***<tswp>/<muxproc>*** is provided below.

a. Example of an XML document instance fragment for ***<tswp>/<faultproc>***:

```
<tswp airforce="no" army="no" deletewp="no" frame="yes" marines="no" navy="no"
tocentry="2" wpno="t00006-X-XXX-XXX" wpseq="0327">
<wpidinfo>
<maintlvl level="sustain"/>
<title>Multiplexor signal Testing
</title>
</wpidinfo>
<wpinfo>
<title>NOT APPLICABLE
</title>
<null insert="none"/>
</wpinfo>
<tsproc>
<muxproc> . . .
<signal-item id="t00006-X-XXX-XXXstep23">
<signame>DASEC STATUS WORD DC ANALOG OUTPUT BIT
</signame>
<dataitem>
<memloc>002150
</memloc>
<memdata>15 (BINARY)
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</memdata>
<condition>(None)
</condition>
<sigfunc>Indicates status of DC analog circuits.
</sigfunc>
</dataitem>
<ckremarks>From DASEC to FCC.
</ckremarks>
<criteria type="pass">If second digit displayed on HOD is 3 or 7
<xref pretext=", go to " stepstart="t00006-X-XXX-XXXstep24"/>.
</criteria>
<criteria type="fail">Location of fault: replace DASEC
<extref docno="TM 1-1520-238-23" posttext=" Series)" pretext="("/>.
</criteria>
</signal-item>
<signal-item id="t00006-X-XXX-XXXstep24">
<signame>DASEC STATUS WORD AD/DA BIT
</signame>
<dataitem>
<memloc>002150
</memloc>
<memdata>13 (BINARY)
</memdata>
<condition>(None)
</condition>
<sigfunc>Indicates status of analog-to-digital and dig-
ital-to-analog circuits.
</sigfunc>
</dataitem>
<ckremarks>From DASEC to FCC.
</ckremarks>
<criteria type="pass">If third digit displayed on HOD is 1, 3, 5, or 7
<xref pretext=", go to " stepstart="t00006-X-XXX-XXXstep25"/>.
</criteria>
<criteria type="fail">Location of fault: replace DASEC
<extref docno="TM 1-1520-238-23" posttext=" Series)" pretext="("/>.
</criteria>
</signal-item>
<signal-item id="t00006-X-XXX-XXXstep25">
<signame>DASEC STATUS WORD FD/LS TEST
</signame>
<dataitem>
<memloc>002150
</memloc>
<memdata>12 (BINARY)
</memdata>
<condition>(None)
</condition>
<sigfunc>Indicates FD/LS ground test is being run.
</sigfunc>
</dataitem>
<ckremarks>From DASEC to FCC.
</ckremarks>
<criteria type="pass">If third digit displayed on HOD is 1 or 5
<xref pretext=", go to " stepstart="t00006-X-XXX-XXXstep26"/>.
</criteria>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<criteria type="fail">Location of fault:  replace DASEC
<extref docno="TM 1-1520-238-23" posttext=" Series)" pretext="("/>.
</criteria>
</signal-item>
<signal-item id="t00006-X-XXX-XXXstep26">
<signame>DASEC STATUS WORD ASE BIT
</signame>
<dataitem>
<memloc>002150
</memloc>
<memdata>11  (BINARY)
</memdata>
<condition>(None)
</condition>
<sigfunc>Indicates last FD/LS test ASE bit status.
</sigfunc>
</dataitem>
<ckremarks>DASEC to FCC.
</ckremarks>
<criteria type="pass">If third digit displayed on HOD is 1
<xref pretext=", go to " stepstart="t00006-X-XXX-XXXstep27"/>.
</criteria>
<criteria type="fail">Location of fault:  replace DASEC
<extref docno="TM 1-1520-238-23" posttext=" Series)" pretext="("/>.
</criteria>
</signal-item>
<signal-item id="t00006-X-XXX-XXXstep27">
<signame>DASEC STATUS WORD VD TEST
</signame>
<dataitem>
<memloc>002150
</memloc>
<memdata>10  (BINARY)
</memdata>
<condition>(None)
</condition>
<sigfunc>Indicates last FD/LS test VD bit status.
</sigfunc>
</dataitem>
<ckremarks>DASEC to FCC.
</ckremarks>
<criteria type="pass">If third digit displayed on HOD is 3 or 4
<xref pretext=", go to " stepstart="t00006-X-XXX-XXXstep28"/>.
</criteria>
<criteria type="fail">Location of fault:  replace DASEC
<extref docno="TM 1-1520-238-23" posttext=" Series)" pretext="("/>.
</criteria>
</signal-item>  . . .
</muxproc>
</tsproc>
</tswp>
```

b. Page-based TM stylesheet output example for *<tswp>*/*<muxproc>*:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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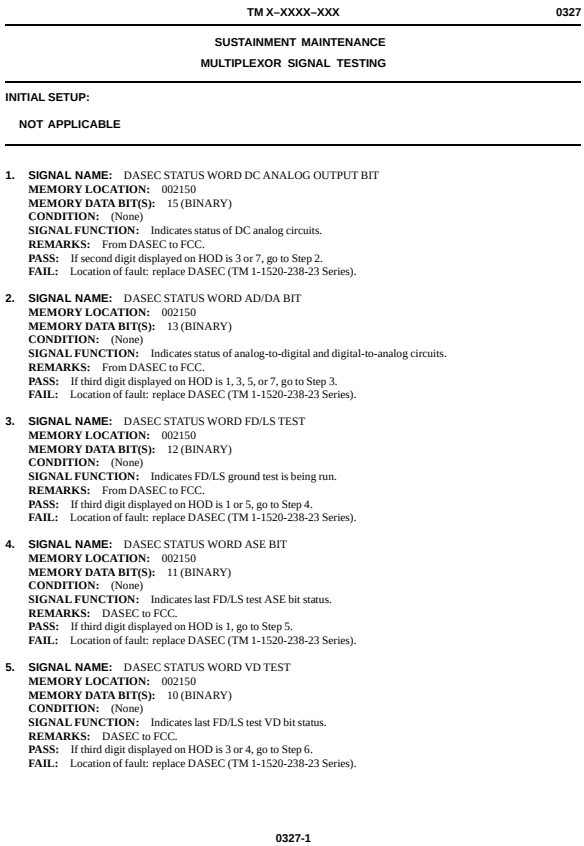

TM X–XXXX–XXX 0327

SUSTAINMENT MAINTENANCE

MULTIPLEXOR SIGNAL TESTING

INITIAL SETUP:

NOT APPLICABLE

1. **SIGNAL NAME:** DASEC STATUS WORD DC ANALOG OUTPUT BIT
MEMORY LOCATION: 002150
MEMORY DATA BIT(S): 15 (BINARY)
CONDITION: (None)
SIGNAL FUNCTION: Indicates status of DC analog circuits.
REMARKS: From DASEC to FCC.
PASS: If second digit displayed on HOD is 3 or 7, go to Step 2.
FAIL: Location of fault: replace DASEC (TM 1-1520-238-23 Series).

2. **SIGNAL NAME:** DASEC STATUS WORD AD/DA BIT
MEMORY LOCATION: 002150
MEMORY DATA BIT(S): 13 (BINARY)
CONDITION: (None)
SIGNAL FUNCTION: Indicates status of analog-to-digital and digital-to-analog circuits.
REMARKS: From DASEC to FCC.
PASS: If third digit displayed on HOD is 1, 3, 5, or 7, go to Step 3.
FAIL: Location of fault: replace DASEC (TM 1-1520-238-23 Series).

3. **SIGNAL NAME:** DASEC STATUS WORD FD/LS TEST
MEMORY LOCATION: 002150
MEMORY DATA BIT(S): 12 (BINARY)
CONDITION: (None)
SIGNAL FUNCTION: Indicates FD/LS ground test is being run.
REMARKS: From DASEC to FCC.
PASS: If third digit displayed on HOD is 1 or 5, go to Step 4.
FAIL: Location of fault: replace DASEC (TM 1-1520-238-23 Series).

4. **SIGNAL NAME:** DASEC STATUS WORD ASE BIT
MEMORY LOCATION: 002150
MEMORY DATA BIT(S): 11 (BINARY)
CONDITION: (None)
SIGNAL FUNCTION: Indicates last FD/LS test ASE bit status.
REMARKS: DASEC to FCC.
PASS: If third digit displayed on HOD is 1, go to Step 5.
FAIL: Location of fault: replace DASEC (TM 1-1520-238-23 Series).

5. **SIGNAL NAME:** DASEC STATUS WORD VD TEST
MEMORY LOCATION: 002150
MEMORY DATA BIT(S): 10 (BINARY)
CONDITION: (None)
SIGNAL FUNCTION: Indicates last FD/LS test VD bit status.
REMARKS: DASEC to FCC.
PASS: If third digit displayed on HOD is 3 or 4, go to Step 6.
FAIL: Location of fault: replace DASEC (TM 1-1520-238-23 Series).

0327-1

FIGURE 270. Example of a page-based TM stylesheet output for <tswp>/<muxproc>.

36.10 Operational checkout and troubleshooting work package ***<opcheck-tswp>***. The element is used to combine both operational checkout with troubleshooting procedures. Generally the author develops operational checkout steps and when a criteria fails the user is performs the fault isolation for the faulty component. MORE INFORMATION.

a. The components of operational checkout and troubleshooting work package ***<opcheck-tswp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Introduction ***<intro>*** (optional) (see 49.1.4.15) explaining how the operational checkout procedures are to be used to perform testing and how they relate to the associated troubleshooting work packages.
   (5) General procedures and precautions ***<proc>*** (optional – zero or more) (see 33.2) that are performed prior to checkout and precautions that are taken during the performance of the checkout procedure.
   (6) System description and support information (optional) (see 36.7).
   (7) Test set hookup ***<hookup>*** (optional) (see 33.1 how to develop a task). If test set hookup is used test set disconnection is required.
   (8) Select a method for the work package.
      (a) Operational checkout procedure ***<opcheckproc>*** (required) (see 36.8.a.(8)) followed by troubleshooting procedure ***<tsproc>*** (required) (see 36.9.a.(8)).
      (b) Combined operational checkout and troubleshooting procedure ***<opcheck-tsproc>*** (required) (see 36.8.1 for similar markup as <opcheck>).

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(9) Test set disconnection ***<disconnect>*** (optional) (see 33.1 how to develop a task). If test set hookup is used test set disconnection is required.

b. DTD fragment graphically is:

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Check the source to verify that this is the current version before use.

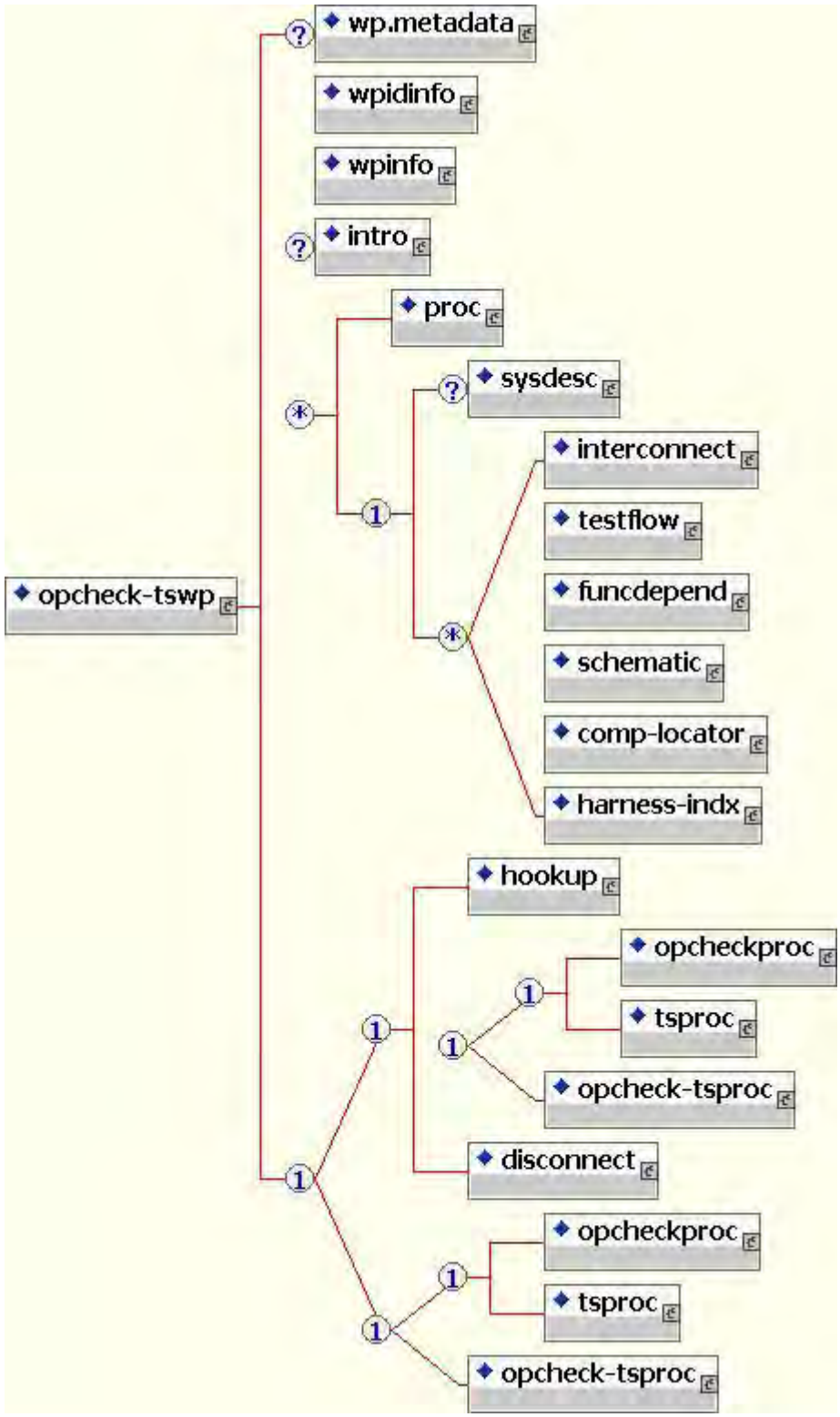

FIGURE 271. Operational checkout and troubleshooting work package DTD hierarchy <opcheck-tswp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment is:

```
<!ELEMENT opcheck-tswp (wp.metadata?, wpidinfo, wpinfo, intro?, (proc,
     (sysdesc?, (interconnect | testflow | funcdepend | schematic |
     comp-locator | harness-indx)*))*,
     ((hookup, ((opcheckproc, tsproc) | opcheck-tsproc), disconnect) |
     ((opcheckproc, tsproc) | opcheck-tsproc)))>
<!ATTLIST opcheck-tswp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>

<!ELEMENT opcheck-tsproc (title?, (warning*, caution*, note*), testproc)+>
<!ATTLIST opcheck-tsproc
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<opcheck-tswp>*** and ***<opcheck-tsproc>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

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(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

36.10.1 XML document instance fragment and output for ***<opcheck-tswp>***. The XML instance and its stylesheet output for a ***<opcheck-tswp>*** is provided below.

a. Example of an XML document instance fragment for ***<opcheck-tswp>***:

```
<opcheck-tswp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="t00008-X-XXX-XXX" wpseq="0481">
<wpidinfo>
<maintlvl level="sustain"/>
<title>Computer Processor
<brk/>Operational Checkout And Troubleshooting
</title>
</wpidinfo>
<wpinfo>
<testeqp>
<testeqp-setup-item>
<name>Test set
</name>
<itemref>
<link alt="Test set information" application="frame" linkaction="prompt"
linktype="return" local="s00654-XX-XXX-XXXitem4" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="part">
</link>
<link application="page" linkaction="prompt" linktype="undefined"
local="s00654-XX-XXX-XXXitem4" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="part">
<ref.generate/>
</link>
</itemref>
</testeqp-setup-item>
</testeqp>
```


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```
<ref>
<ref-setup-item>
<link application="both" linkaction="prompt" linktype="return" local="m00005-XX-XXX-XXX"
popup="no" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="part">
<ref.generate/>
</link>
</ref-setup-item>
</ref>
</wpinfo>
<opcheck-tsproc>
<title>Computer Processor &ndash; Test Procedure
</title>
<testproc>
<checkstep>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Remove computer processor top cover
<link application="both" linkaction="prompt" linktype="return" local="m00005-XX-XXX-XXX"
popup="no" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="part">
<pretext> (
</pretext>
<ref.generate/>
<posttext>)
</posttext>
</link>.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Apply power to test set and place test set
<ctrlind idref="o00042-X-XXX-XXXpower">POWER
</ctrlind> switch to
<ctrlind-val>ON
</ctrlind-val> position.
</para>
</step1>
<indication>
<para esd="no" hcp="no">Test set power indicator is illuminated.
</para>
</indication>
<malfunc label="malfunction">If power indicator does not light.
</malfunc>
<action>
<para esd="no" hcp="no">Check power source for 28 VDC.
</para>
</action>
</checkstep>
<checkstep>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Place
<ctrlind idref="o00042-X-XXX-XXXuut_power"> UUT POWER
</ctrlind> switch in
<ctrlind-val>CP
</ctrlind-val> position.
</para>
</step1>
<indication>
<para esd="no" hcp="no">
```


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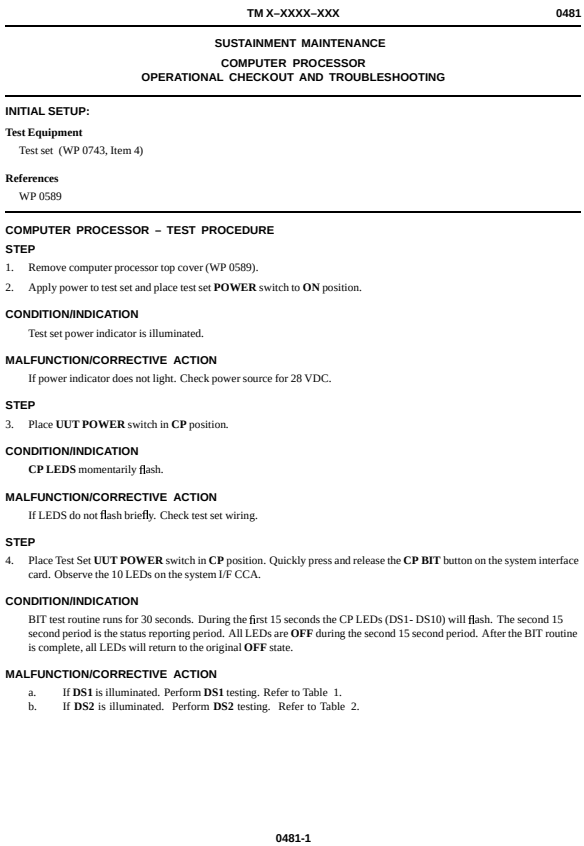
Check the source to verify that this is the current version before use.

```
<ctrlind idref="o00042-X-XXX-XXXcp_leds">CP LEDS
</ctrlind> momentarily flash.
</para>
</indication>
<malfunc label="malfunction">If LEDS do not flash briefly.
</malfunc>
<action>
<para esd="no" hcp="no"> Check test set wiring.
</para>
</action>
</checkstep>
<checkstep>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Place Test Set
<ctrlind idref="o00042-X-XXX-XXXuut_power">UUT POWER
</ctrlind> switch in
<ctrlind-val>CP
</ctrlind-val> position. Quickly press and release the
<ctrlind idref="o00042-X-XXX-XXXcp_bit">CP BIT
</ctrlind> button on the system interface card. Observe
the 10 LEDs on the system I/F CCA.
</para>
</step1>
<indication>
<para esd="no" hcp="no">BIT test routine runs for 30 seconds. During the
first 15 seconds the CP LEDs (DS1- DS10) will flash. The second 15
second period is the status reporting period. All LEDs are
<ctrlind-val>OFF
</ctrlind-val> during the second 15 second period. After the BIT
routine is complete, all LEDs will return to the original
<ctrlind-val>OFF
</ctrlind-val> state.
</para>
</indication>
<malfunc label="malfunction">If
<ctrlind idref="o00042-X-XXX-XXXds1">DS1
</ctrlind> is illuminated.
</malfunc>
<action>
<para esd="no" hcp="no"> Perform
<ctrlind idref="o00042-X-XXX-XXXds1">DS1
</ctrlind> testing.
<xref posttext="." pretext="Refer to " tableid="t00008-X-XXX-XXXtable2"/>
</para>
</action>
<malfunc label="malfunction">If
<ctrlind idref="o00042-X-XXX-XXXds2">DS2
</ctrlind> is illuminated.
</malfunc>
<action>
<para esd="no" hcp="no"> Perform
<ctrlind idref="o00042-X-XXX-XXXds2">DS2
</ctrlind> testing.
<xref posttext="." pretext="Refer to " tableid="t00008-X-XXX-XXXtable3"/>
</para>
</action>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</checkstep> . . .
</testproc>
</opcheck-tsproc>
</opcheck-tswp>

b. Page-based TM stylesheet output example for ***<opcheck-tswp>***:



TM X–XXXX–XXX 0481

SUSTAINMENT MAINTENANCE
COMPUTER PROCESSOR
OPERATIONAL CHECKOUT AND TROUBLESHOOTING

INITIAL SETUP:

Test Equipment
Test set (WP 0743, Item 4)

References
WP 0589

COMPUTER PROCESSOR – TEST PROCEDURE

STEP
1. Remove computer processor top cover (WP 0589).
2. Apply power to test set and place test set **POWER** switch to **ON** position.

CONDITION/INDICATION
Test set power indicator is illuminated.

MALFUNCTION/CORRECTIVE ACTION
If power indicator does not light. Check power source for 28 VDC.

STEP
3. Place **UUT POWER** switch in **CP** position.

CONDITION/INDICATION
CP LEDS momentarily flash.

MALFUNCTION/CORRECTIVE ACTION
If LEDS do not flash briefly. Check test set wiring.

STEP
4. Place Test Set **UUT POWER** switch in **CP** position. Quickly press and release the **CP BIT** button on the system interface card. Observe the 10 LEDs on the system I/F CCA.

CONDITION/INDICATION
BIT test routine runs for 30 seconds. During the first 15 seconds the CP LEDs (DS1- DS10) will flash. The second 15 second period is the status reporting period. All LEDs are **OFF** during the second 15 second period. After the BIT routine is complete, all LEDs will return to the original **OFF** state.

MALFUNCTION/CORRECTIVE ACTION
a. If **DS1** is illuminated. Perform **DS1** testing. Refer to Table 1.
b. If **DS2** is illuminated. Perform **DS2** testing. Refer to Table 2.

0481-1

FIGURE 272. Example of a page-based TM stylesheet output for <opcheck-tswp>.

36.11 Diagnostic work package ***<diagnosticwp>***. The diagnostic work package is used for frame-based presentation only. The work package is used to interactively diagnosis the fault or symptom and direct the maintainer to the corrective action(s). The work package is broken down into six (6) areas initial setup, introductory information, test procedure, obtaining test results, evaluating test results, and link to next action (either corrective or next test). The major work package concepts are discussed in diagnostic with state variables (see 36.11.2) and without state variables (see 36.11.1).

a. The components of intrusive diagnostic work package ***<diagnosticwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (8) System description and support information (optional) (see 36.7).
   (9) Diagnostic testing with state variables and components are:

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(a) Initializing state variable ***<statemanipulation>***/***<statemanipulation-alt>*** (optional – zero or more) (see 48.2.3).

(b) Diagnostic testing with state variables ***<testwithstate>***/***<testwithstate-alt>*** (required – one or more) (see 36.11.2).

(10) Diagnostic testing (without state variables) ***<testwithoutstate>*** (required – one or more) (see 36.11.1).

b. DTD fragment graphically is:

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Check the source to verify that this is the current version before use.

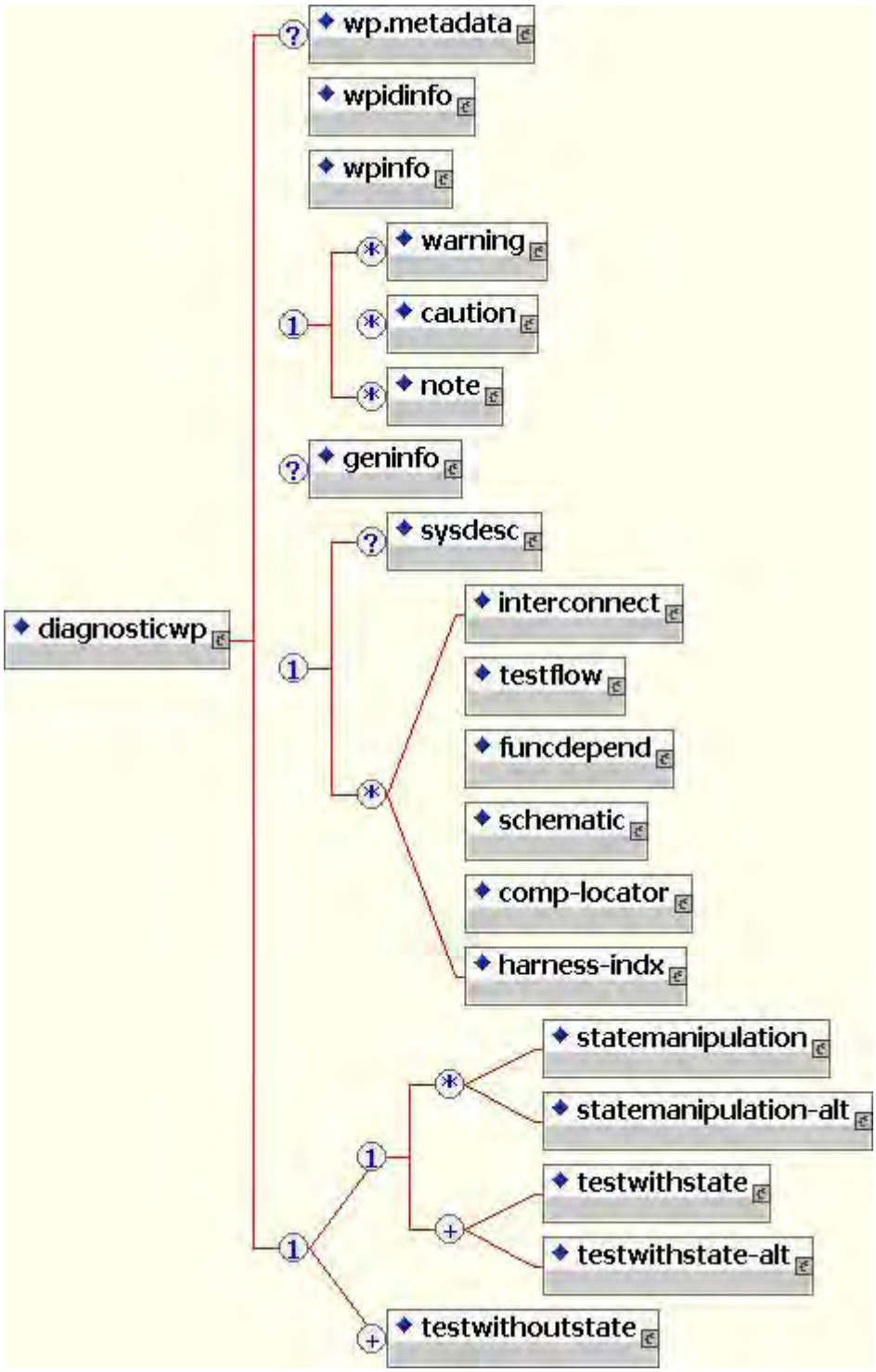

FIGURE 273. Intrusive diagnostic work package DTD hierarchy <diagnosticwp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment is:

```
<!ELEMENT diagnosticwp (wp.metadata?, wpidinfo, wpinfo, (warning* |
     caution* | note*), geninfo?, (sysdesc?, (interconnect | testflow |
     funcdepend | schematic | comp-locator | harness-indx)*),
     (((%statemanipulation_ent;)*, (testwithstate | testwithstate-alt)+) |
     testwithoutstate+))>
<!ATTLIST diagnosticwp
   airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time)  #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes for ***<diagnosticwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

36.11.1 Diagnostic testing (without state variables) ***<testwithoutstate>***. The testing (without state variables) is used for frame-based viewers that do not use a logic engine (see 48.2). The test is broken into five (5) areas: introductory information (reason for test, necessary schematics, component locator, etc.), test set hookup, test procedure, observed test results, and test result with the next action (identified fault(s), short narrative actions, completed testing, and/or link to corrective action or next test). The only method the maintainer's observed result can be processed (link to the corrective action, brief corrective action instructions, or link to next test or observed test result) by the viewer is through a dialog box. The dialog box will have predefined options, which when the option is selected, links the viewer to the next test or a corrective action. Examples of predefined responses are Yes/No; or Pressure is: <15, 15–50, or> 50.

Generally, the information layout is the test title, reason for test, introductory information, testing steps, and concluding with a observed test result entered through a dialog box with the selected menu item indicating the next action to perform. Each test can have multiple observed test results. When multiple observed test results are implemented the first observed test result is presented. After the first observed test result is entered, the remaining observed test results (in this test) requires a reference to the next observed test result menu dialog box. An example is an observed test result dialog box is prompted either YES or NO. If the maintainer selects the menu option "YES", the associated resulting action is a reference to the next observed test result dialog box for "YES". If the maintainer selects the menu option "NO", the associated resulting action is a reference to the next observed test result dialog box for "NO". Thus using multiple observed test results is similar to a logic tree.

a. The components of diagnostic testing (without state variables) ***<testwithoutstate>*** are:
   (1) Test title ***<title>*** (optional) (see 49.1.1.3).
   (2) Reason for test ***<reasonfortest>*** (optional) (see 36.11.1.1).
   (3) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (4) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (5) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (6) System description and support information (optional) (see 36.7).
   (7) Test set hookup ***<hookup>*** (optional – zero or more) (see 33.1 how to develop a task). If test set hookup is used test set disconnection is required.
   (8) Test setup steps (optional) and components are either
      (a) Procedure ***<proc>*** (required) (see 33.2).
      (b) Procedural step ***<step1>*** (see 33.3)/***<step1-alt>*** (see 48.2.1) (required – one or more) with associated figure ***<figure>***/***<figure-alt>*** (optional – zero or more) (see 44.1.1) and/or associated table ***<table>***/***<table-alt>*** (optional – zero or more) (see 42).
   (9) Observed test results (required – one or more) and components are:
      (a) Binary option dialog box ***<simple>*** (required) (see 36.11.1.2).
      (b) Multiple option (user defined) dialog box ***<multioption>*** (required) (see 36.11.1.3).

b. DTD fragment graphically is:

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Check the source to verify that this is the current version before use.

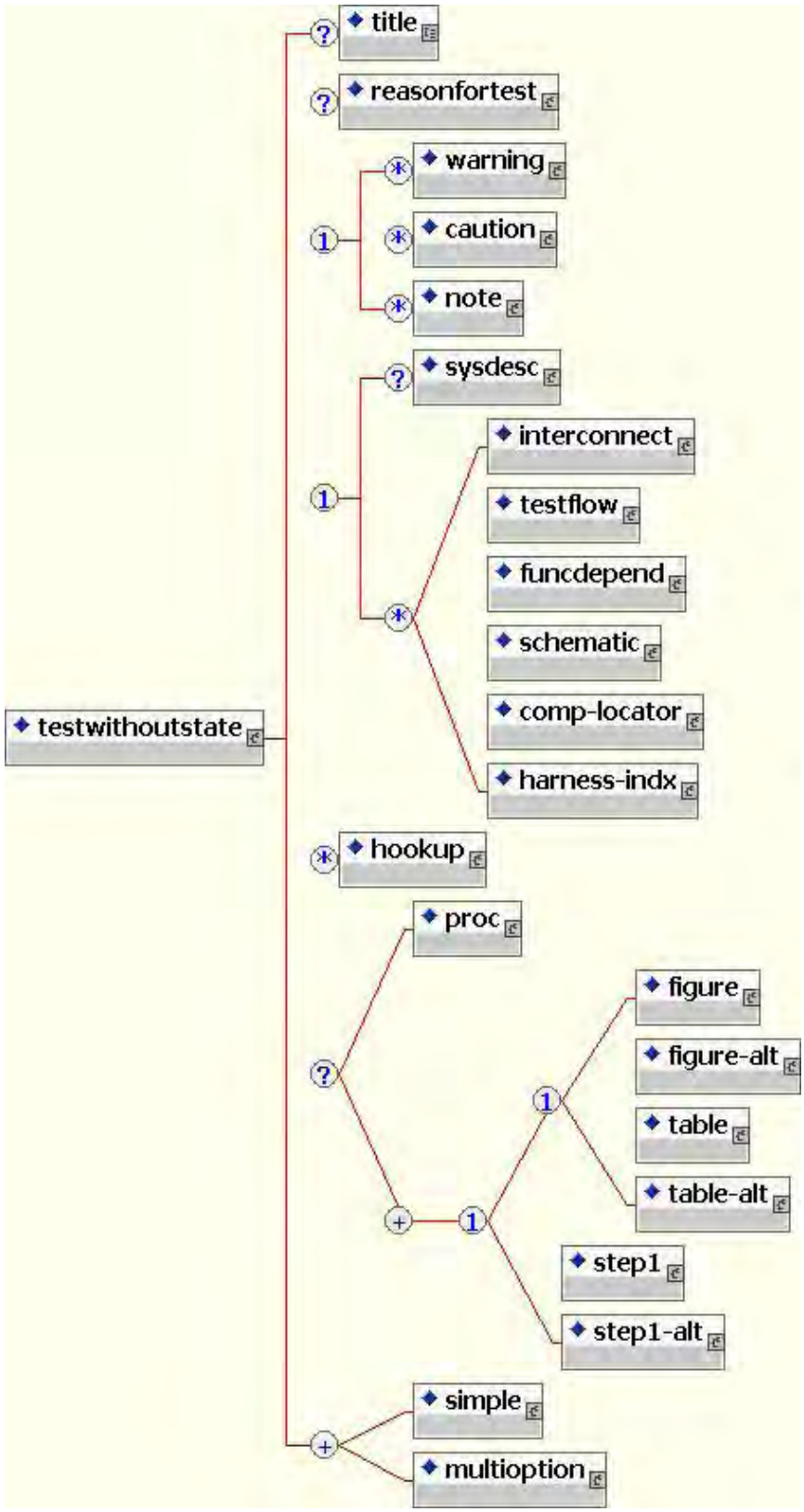

FIGURE 274. Testing (without state variables) DTD hierarchy <testwithoutstate>.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment is:

```
<!ELEMENT testwithoutstate (title?, reasonfortest?, (warning* | caution* | note*),
   (sysdesc?, (interconnect | testflow | funcdepend | schematic | comp-locator |
   harness-indx)*, hookup*, (proc | (figure | figure-alt | table | table-alt |
   step1 | step1-alt)+)?, (simple | multioption)+))>
<!ATTLIST testwithoutstate
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<testwithoutstate>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

36.11.1.1 Reason for testing ***<reasonfortest>***. The element provides a brief explanation for the testing purpose.

a. The components of reason for testing ***<reasonfortest>*** are:

(1) Title ***<title>*** (required) (see 49.1.1.3).
(2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(4) Select one of the following information types:
  (a) Narrative paragraphs
    1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
    2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
  (b) Narrative paragraphs with descriptive or narrative titled text
    1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
    2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
    3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
  (c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD fragment graphically is:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

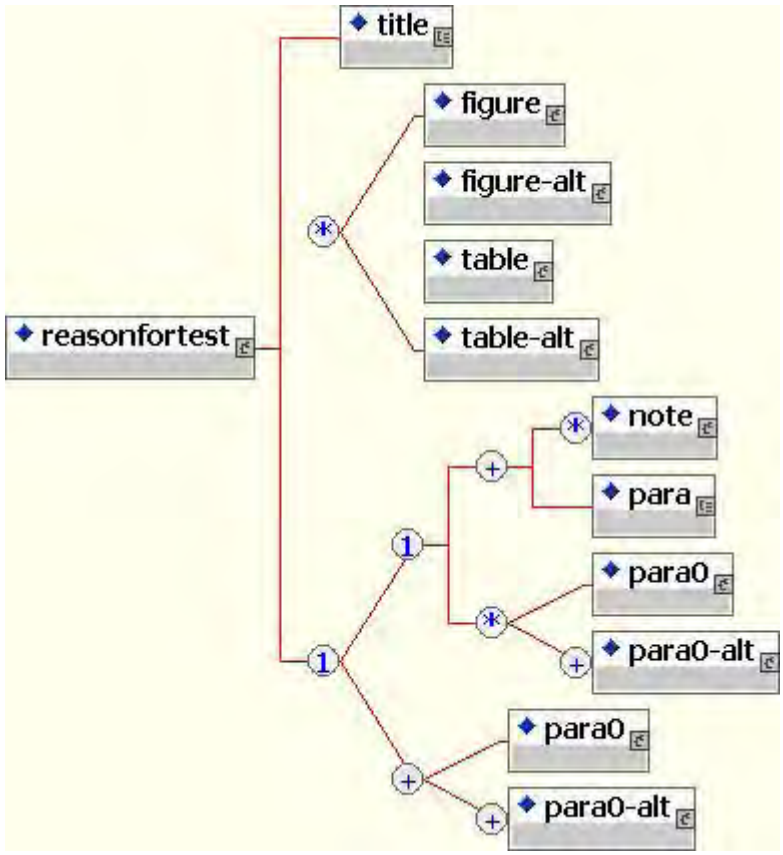

FIGURE 275. Reason for testing DTD hierarchy <reasonfortest>..

c. DTD fragment is:

```
<!ELEMENT reasonfortest (title, (figure | figure-alt | table | table-alt)*,
   (((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST reasonfortest
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<reasonfortest>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

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Check the source to verify that this is the current version before use.

36.11.1.2 Observed test result with binary options ***<simple>***. The observed test result with binary options (presented as a dialog box with menu options) provides a simple YES/NO menu options to the prompt, which when selected provides the next action. The dialog box generally contains a question/prompt with only two menu option responses as positive and negative. The menu option label is determined by the **answer** attribute value.

a. The components of observed test result with binary options ***<simple>*** are:
   (1) Dialog title ***<title>*** (optional) (see 49.1.1.3).
   (2) Question/prompt ***<prompt>*** (required) (see 48.3.1.2).
   (3) Yes (positive) response ***<yes>*** (required) and when selected the prescribed action ***<resultwithoutstate>*** (required) (see 36.11.1.4) is performed.
   (4) No (negative) response ***<no>*** (required) and when selected the prescribed action ***<resultwithoutstate>*** (required) (see 36.11.1.4) is performed.

b. DTD fragment graphically is:

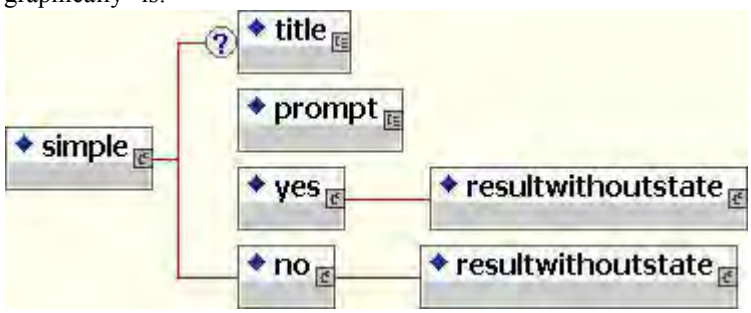

FIGURE 276. Observed test result with binary options DTD hierarchy <simple>.

c. DTD fragment is:

```
<!ELEMENT simple (title?, prompt, yes, no)>
<!ATTLIST simple
   id     ID                                #IMPLIED
   answer (yesno | truefalse | passfail) "yesno">


<!ELEMENT yes (resultwithoutstate)>

<!ELEMENT no (resultwithoutstate)>
```

d. Unique attribute for ***<simple>*** is **answer** – Specifies the binary buttons labels (optional). The possible values are "**yesno**" for YES and NO buttons; "**truefalse**" for TRUE and FALSE buttons; and "**passfail**" for PASS and FAIL buttons. The default value is "**yesno**".

e. Common attributes for ***<simple>*** is **id** – Specifies unique identifier (target) to reference (optional)(see 49.2.6).

36.11.1.3 Observed test result with predefined options ***<multioption>***. The observed test result with predefined options (presented as a dialog box with menu options) provides an author specified menu option labels, which when selected provides the next action. The dialog box generally contains a question/prompt with two or more predefined (by the author) response options.

a. The components of observed test result with predefined options ***<multioption>*** are:
   (1) Dialog title ***<title>*** (optional) (see 49.1.1.3).
   (2) Question/prompt ***<prompt>*** (required) (see 48.3.1.2).
   (3) Predefined menu option response ***<option>*** (required – two or more) and when selected the prescribed action ***<resultwithoutstate>*** (required) (see 36.11.1.4) is performed.

b. DTD fragment graphically is:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

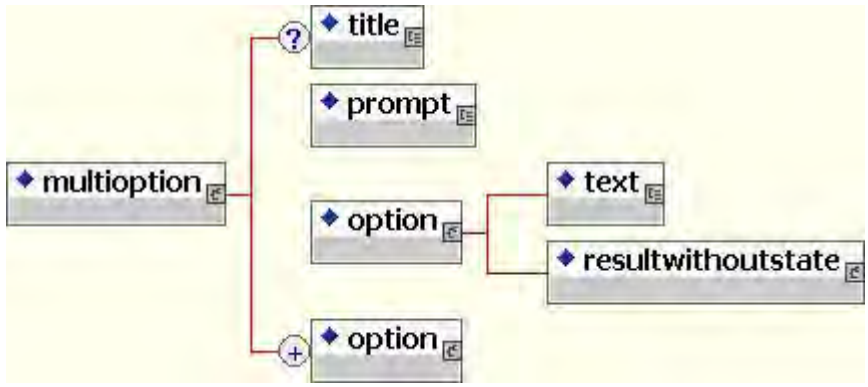

FIGURE 277. Observed test result with predefined options DTD hierarchy <multioption>.

c. DTD fragment is:

```
<!ELEMENT multioption (title?, prompt, option, option+)>
<!ATTLIST multioption
   id ID #IMPLIED>


<!ELEMENT option (text, resultwithoutstate)>
```

d. Common attributes:

(1) **id** – Specifies unique identifier (target) to reference (optional)(see 49.2.6).

36.11.1.4 Test results (without state variables) ***<resultwithoutstate>***. The element contains the next action resulting from the response from the observed test (either a dialog box option as binary or predefined). When the resulting action specifies further testing a link is provided to another diagnostic work package ***<diagnosticwp>***, another test (without state variable) ***<testwithoutstate>*** (within the same work package), or observed test (either binary or predefined option). The link generally does not require a prompt (unless specified), but seamlessly presents the next testing compontent. The author does not infer the following test is processed, but always provides a link. Using a link provides a consistent method for identifying the next test.

When the resulting action specifies a corrective action either a narrative or link is provided. If the corrective action is short and simple the step(s) (***<step1>*** or ***<para>***) are developed. An example is "Call maintenance supervisor."

When the corrective action has different initial setup requirements and/or is complex, a separate and referenced (***<link>***) maintenance work package is developed. In either situation, the author needs to define the identified fault code(s) (***<fault>***) and indicate that testing is completed (***<completed_test>***).

a. The components of test results (without state variables) ***<resultwithoutstate>*** are:

(1) Fault code ***<fault>*** (optional – zero or more) (see 48.1.3.5).

(2) Disconnect test set ***<disconnect>***/***<disconnect-alt>*** (optional – zero or more) (see 36.8.2).

(3) Action (required) component is either:

(a) Link for further testing ***<link>*** (required) (see 46.2.3).

(b) Isolated the fault ***<para>*** (required) (see 49.1.1.5) which is optionally followed by either

1 Link to the corrective action maintenance work package ***<link>*** (optional) (see 46.2.3).

2 Test completion action ***<completed_test>*** (optional) (see 36.11.1.5).

b. DTD fragment graphically is:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

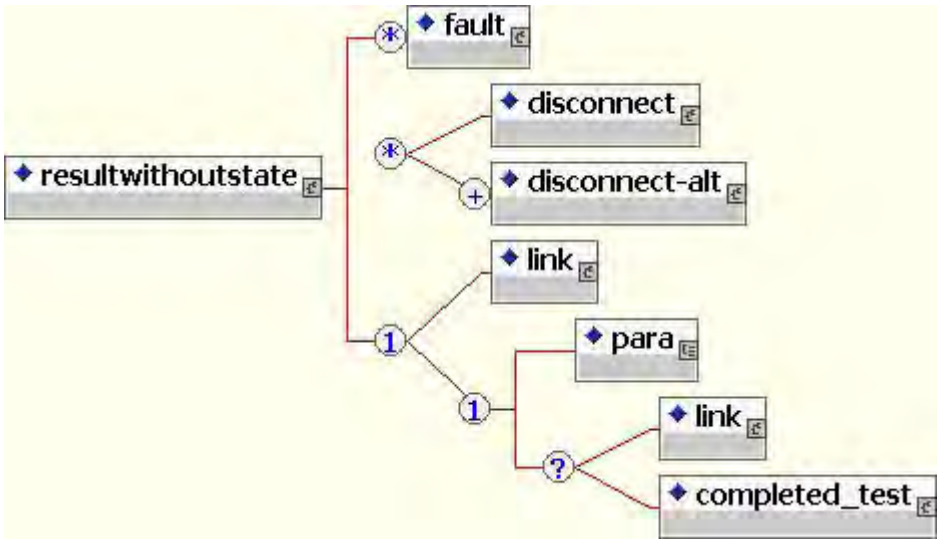

FIGURE 278. Test result (without state variables) DTD hierarchy <resultwithoutstate>. .

c. DTD fragment is:

```
<!ELEMENT resultwithoutstate (fault*, (disconnect | disconnect-alt+)*,
   (link | (para, (link | completed_test)?)))>
```

36.11.1.5 Completed test ***<completed_test>***. Indicates the test is completed and provides, if applicable, correct actions or statement to notify the supervisor.

a. The components of completed test ***<completed_test>*** is procedural step ***<step1>*** (see 33.3)/***<step1-alt>*** (see 48.2.1) (optional – zero or more) with associated figure ***<figure>***/***<figure-alt>*** (optional – zero or more) (see 44.1.1) and/or associated table ***<table>***/***<table-alt>*** (optional – zero or more) (see 42).

b. DTD fragment graphically is:

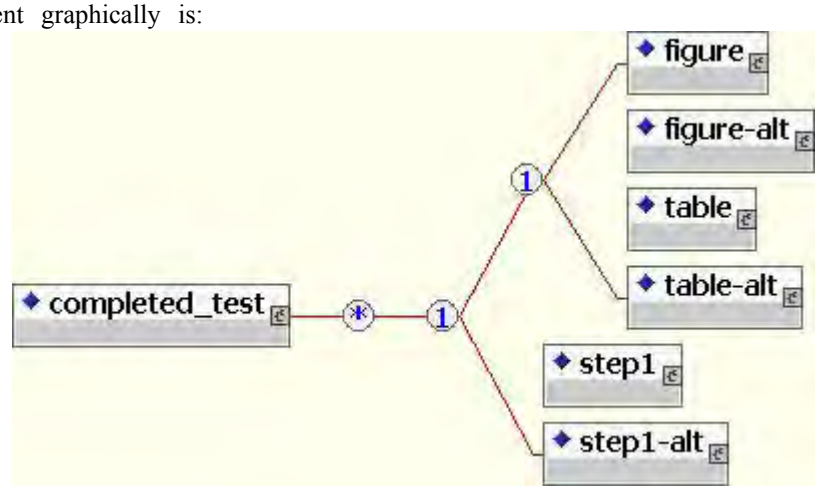

FIGURE 279. Completed test DTD hierarchy <completed_test>.

c. DTD fragment is:

```
<!ELEMENT completed_test (firgure | figure-alt | table | table-alt |
   step1 |step1-alt)*>
```

36.11.1.6 XML document instance fragment and output for ***<testwithoutstate>***. The XML instance and its stylesheet output for a ***<testwithoutstate>*** is provided below.

a. Example of an XML document instance fragment for ***<testwithoutstate>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<diagnosticwp airforce="no" army="no" deletewp="no" frame="yes" insertwp="no" marines="no" navy="no" tocentry="2" wpno="T00001-XX-XXXX-XXX">
<wpidinfo>
<maintlvl level="field"/>
<title>ENGINE CRANKS SLOWLY
</title>
</wpidinfo>
<wpinfo>
<testeqp>
<testeqp-setup-item>
<name>1553B Test Equipment
</name>
<itemref>
<link application="frame" linkaction="prompt" linktype="goto" popup="no" xlink:href="S00032-XX-XXXX-XXX#testset1553b" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink"/>
</itemref>
</testeqp-setup-item>
</testeqp>
<ref>
<ref-setup-item>
<extref docno="TM 55-1520-240-23"/>
</ref-setup-item>
</ref>
</wpinfo>
<testwithoutstate id="T00001-XX-XXXX-XXX-test1">
<step1>
<para>Have pilot perform ground run at 225 RPM, flat pitch.
</para>
</step1>
<step1>
<para>Measure ground track on both rotors
<extref docno="TM 55-1520-240-23" pretext="(" posttext=")"/>.
</para>
</step1>
<simple answer="yesno">
<prompt>Were blades within ⅜ inch of each other?
</prompt>
<yes>
<resultwithoutstate>
<link application="frame" linkaction="immediate" linktype="goto" local="T00001-XX-XXXX-XXX-test2" popup="no" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<link application="frame" linkaction="immediate" linktype="goto" local="T00001-XX-XXXX-XXX-test7" popup="no" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test"/>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate id="T00001-XX-XXXX-XXX-test2">
<step1>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<para>Have pilot establish stable hover tune VIBREX to 225 RPM.
</para>
</step1>
<step1>
<para>Record forward rotor vertical and lateral and AFT rotor vertical
and lateral vibrations in IPS and clock angles.
</para>
</step1>
<step1>
<para>Record hover track on both rotors.
</para>
</step1>
<simple answer="yesno">
<prompt>Were all measured vibration level within ranges of table 2-1.3?
</prompt>
<yes>
<resultwithoutstate>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-XX-XXXX-XXX-test9" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-XX-XXXX-XXX-test3" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test"/>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate id="T00001-XX-XXXX-XXX-test3">
<step1>
<para>Land helicopter.
</para>
</step1>
<step1>
<para>Shut down engines check recorded hover blade track.
</para>
</step1>
<simple answer="yesno" id="T00001-XX-XXXX-XXX-test3">
<prompt>Were blades within &frac12; inch of each other?
</prompt>
<yes>
<resultwithoutstate>
<para esd="no" hcp="no">Use pitch link adjustment = 0 marks for
input to PC Pitch Link-Lateral Program.
</para>
<link application="frame" linkaction="prompt" linktype="goto"
local="T00001-XX-XXXX-XXX-test4" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Continue
</prompt>
</link>
</resultwithoutstate>
</yes>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<no>
<resultwithoutstate>
<para esd="no" hcp="no">Calculate pitch link adjustment to bring blades within ½ inch of each other. Use calculated adjustment as input to PC Pitch Link-Lateral Program.
</para>
<link application="frame" linkaction="prompt" linktype="goto" local="T00001-XX-XXXX-XXX-test4" popup="no" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Continue
</prompt>
</link>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate id="T00001-XX-XXXX-XXX-test4">
<step1>
<para>Using PC Pitch Link-Lateral Program, determine VIBEX tip weight correction to compensate for any pitch link adjustment.
</para>
</step1>
<step1>
<para>Adjust tip weights as necessary
<extref docno="TM 55-1520-240-23" pretext="(" posttext=")"/>.
</para>
</step1>
<step1>
<para>Have pilot establish stable hover.
</para>
</step1>
<step1>
<para>Measure forward and AFT rotor vibrations.
</para>
</step1>
<simple answer="yesno">
<prompt>Are all vibration levels within ranges of table 2-1.3?
</prompt>
<yes>
<resultwithoutstate>
<link application="frame" linkaction="immediate" linktype="goto" local="T00001-XX-XXXX-XXX-test9" popup="no" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<link application="frame" linkaction="immediate" linktype="goto" local="T00001-XX-XXXX-XXX-test5" popup="no" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate id="T00001-XX-XXXX-XXX-test5">
<simple answer="yesno">

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<prompt>Are recorded vertical vibration levels within ranges of table 2-1.3?
</prompt>
<yes>
<resultwithoutstate>
<para esd="no" hcp="no">Adjust VIBEX blade tip weights
<extref docno="TM 55-1520-240-23" pretext="(" posttext=")"/> to correct
lateral vibration using latest hover input data in PC Pitch
Link-Lateral Program. Make necessary adjustments.
</para>
<link application="frame" linkaction="prompt" linktype="goto"
local="T00001-XX-XXXX-XXX-test2" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Perform this task again beginning with stable hover.
</prompt>
</link>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-XX-XXXX-XXX-test6" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate id="T00001-XX-XXXX-XXX-test6">
<step1>
<para>Check pitch link bearings.
</para>
</step1>
<simple answer="yesno">
<prompt>Are any pitch link bearings worn?
</prompt>
<yes>
<resultwithoutstate>
<para esd="no" hcp="no">Replace pitch links found to have worn bearings.
</para>
<link application="frame" linkaction="prompt" linktype="goto"
local="T00001-XX-XXXX-XXX-test2" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Perform this task again from the beginning.
</prompt>
</link>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<para esd="no" hcp="no">Inspect rotor head assy.
<link application="frame" linkaction="prompt" linktype="goto" popup="no" xlink:href="TM
55-1520-240-23" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink">
<pretext>Reference
</pretext>
<prompt>TM 55-1520-240-23
</prompt>
</link>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</para>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate  id="T00001-XX-XXXX-XXX-test7">
<step1>
<para>Shut  down  engines.
</para>
</step1>
<step1>
<para>Adjust pitch links to bring blades within &frac38; inch of each other.
</para>
</step1>
<step1>
<para>Perform  ground  run  again.
</para>
</step1>
<step1>
<para>Check  ground  track.
</para>
</step1>
<simple  answer="yesno">
<prompt>Are blades within &frac38; inch of each other?
</prompt>
<yes>
<resultwithoutstate>
<link  application="frame"  linkaction="immediate"  linktype="goto"
local="T00001-XX-XXXX-XXX-test2"  popup="no"  xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<link  application="frame"  linkaction="immediate"  linktype="goto"
local="T00001-XX-XXXX-XXX-test6"  popup="no"  xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink"/>
<completed_test/>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate  id="T00001-XX-XXXX-XXX-test9">
<step1>
<para>Have pilot established 130 knot cruise profile.
</para>
</step1>
<step1>
<para>Record FWD vertical and lateral and AFT vertical and lateral
vibrations in IPS and clock angle record in-flight track on both rotors.
</para>
</step1>
<simple  answer="yesno">
<prompt>Are vibration levels with ranges of Table 2-1.3?
</prompt>
<yes>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<resultwithoutstate>
<para esd="no" hcp="no">1/Rev vibration levels are satisfactory.
</para>
<completed_test/>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-XX-XXXX-XXX-test10" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate id="T00001-XX-XXXX-XXX-test10">
<step1>
<para>Land helicopter shut down engines.
</para>
</step1>
<step1>
<para>Check 130 knot blade track.
</para>
</step1>
<simple answer="yesno">
<prompt>Were blades within &frac12; inch of each other?
</prompt>
<yes>
<resultwithoutstate>
<para esd="no" hcp="no">Use "0" marks pitch link adjustment as
input to Pitch Link-Lateral Program.
</para>
<link application="frame" linkaction="prompt" linktype="goto"
local="T00001-XX-XXXX-XXX-test11" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Continue
</prompt>
</link>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<para esd="no" hcp="no">Calculate pitch link adjustment to bring blades
within &frac12; inch of each other at 130 knots.  Use calculated
adjustment as input to PC Link-Lateral Program.
</para>
<link application="frame" linkaction="prompt" linktype="goto"
local="T00001-XX-XXXX-XXX-test11" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Continue
</prompt>
</link>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
```


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Check the source to verify that this is the current version before use.

```
<testwithoutstate id="T00001-XX-XXXX-XXX-test11">
<step1>
<para>Using the pitch link-lateral program.
</para>
</step1>
<step1>
<para>Determine VIBEX tip weight correction to compensate for
any pitch link adjustment and the recorded lateral imbalance
at 130 knots adjust VIBEX tip weights
<extref docno="TM 55-1520-240-23" pretext="(" posttext=")"/>.
</para>
</step1>
<step1>
<para>Have pilot establish 130 knot cruise profile record
blade track and measure vibration.
</para>
</step1>
<simple answer="yesno">
<prompt>Are vertical and lateral vibrations within range of table 2-1.3?
</prompt>
<yes>
<resultwithoutstate>
<para esd="no" hcp="no">1/Rev vibration levels are satisfactory.
</para>
<completed_test/>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-XX-XXXX-XXX-test12" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate id="T00001-XX-XXXX-XXX-test12">
<simple answer="yesno">
<prompt>Are recorded vertical vibration levels within range of table 2-1.3?
</prompt>
<yes>
<resultwithoutstate>
<para esd="no" hcp="no">Adjust VIBEX blade tip weights (TM 55-1520-240-23)
to correct lateral vibration.  Using latest 130 knot VIB input data in
PC Pitch Link-Lateral Program.  Make necessary adjustments.
</para>
<link application="frame" linkaction="prompt" linktype="goto"
local="T00001-XX-XXXX-XXX-test9" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
<prompt>Perform this task again beginning at 130 knot cruise.
</prompt>
</link>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-XX-XXXX-XXX-test13" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test"/>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate id="T00001-XX-XXXX-XXX-test13">
<step1>
<para>Land helicopter and shut down engines.
</para>
</step1>
<step1>
<para>Observe recorded blade track at 130 knots.
</para>
</step1>
<simple answer="yesno">
<prompt>Did any blade climb or dive?
</prompt>
<yes>
<resultwithoutstate>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-XX-XXXX-XXX-test14" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<para esd="no" hcp="no">Inspect rotor head assy.
<link application="frame" linkaction="prompt" linktype="goto"
popup="no" xlink:href="TM 55-152-240-23" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="document">
<pretext>Reference
</pretext>
<prompt>TM 55-152-240-23
</prompt>
</link>.
</para>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
<testwithoutstate id="T00001-XX-XXXX-XXX-test14">
<simple answer="yesno">
<prompt>Is trim tab set correctly as stenciled?
</prompt>
<yes>
<resultwithoutstate>
<para>
<link linkaction="prompt" linktype="goto" local="M00032-XX-XXXX-XXX"
popup="no" xlink:type="simple" xreftype="wp">
<prompt>Replace blade
</prompt>
</link>
</para>
<completed_test/>
```


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Check the source to verify that this is the current version before use.

```
</resultwithoutstate>
</yes>
<no>
<resultwithoutstate>
<para>Reset trim tab with tab bending tool.
</para>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-XX-XXXX-XXX-test9" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
<prompt>Perform this task again beginning at 130 knot cruise.
</prompt>
</link>
</resultwithoutstate>
</no>
</simple>
</testwithoutstate>
</diagnosticwp>
```

b. Frame-based TM stylesheet output example for *<testwithoutstate>*:

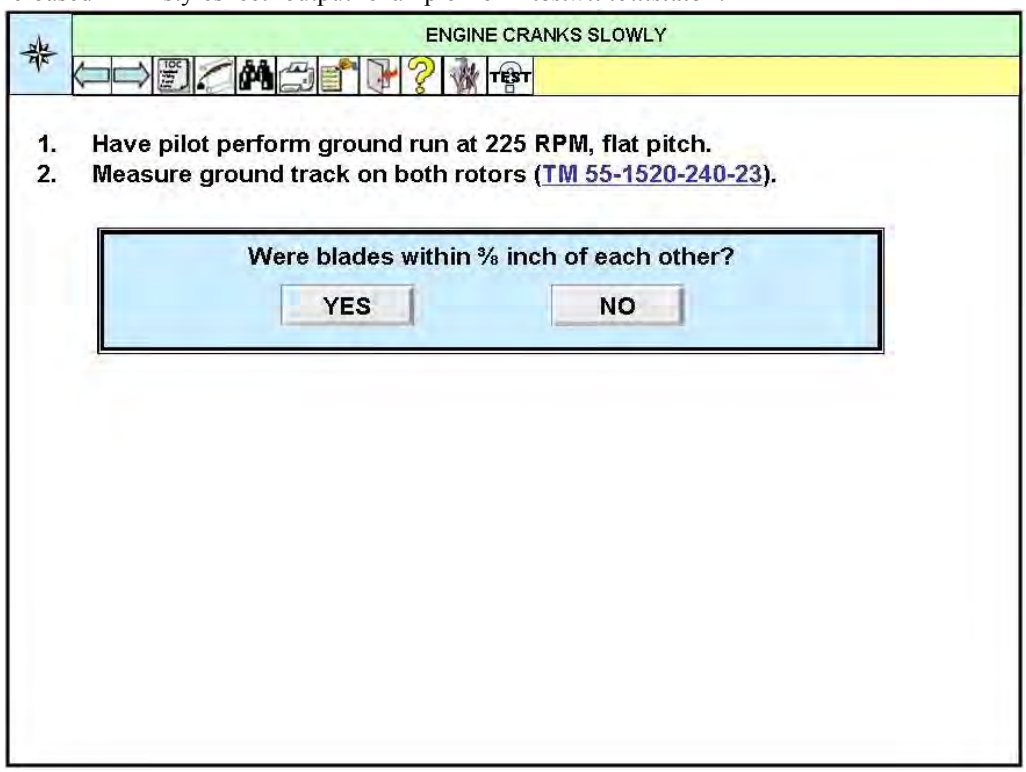

FIGURE 280. XML sample simple dialog frame-layout.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

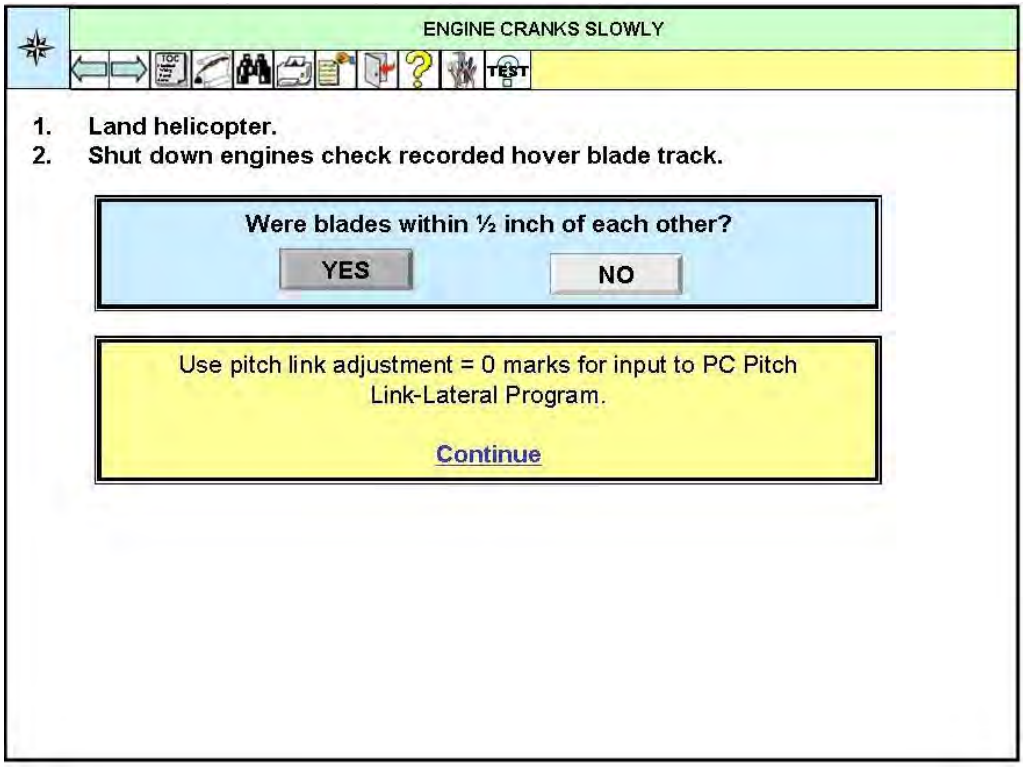

FIGURE 281. XML sample simple dialog with link prompt frame-layout.

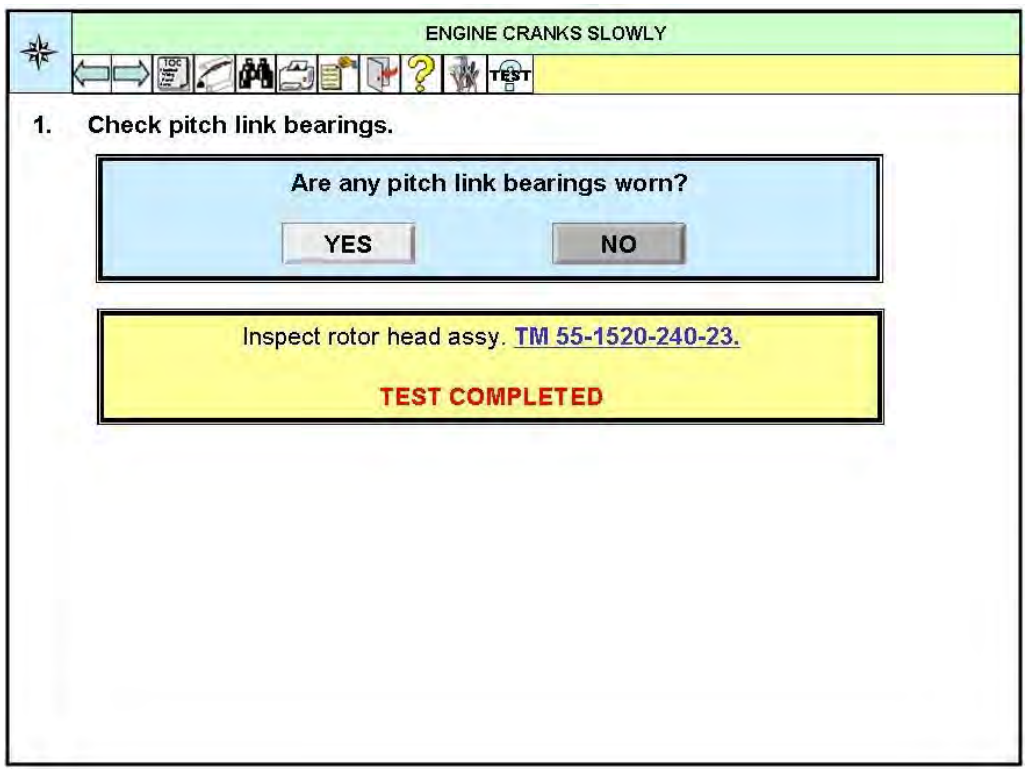

FIGURE 282. XML sample simple dialog with test completed frame-layout.

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Check the source to verify that this is the current version before use.

c. Figure 283 illustrates the XML fragment diagnostic flow. The figure shows how diagnostic prompts can be reused and how each diagnostic prompt is linked to the next diagnostic prompt (determined from answer selected [***YES*** or ***NO***]) until diagnostics testing is completed.

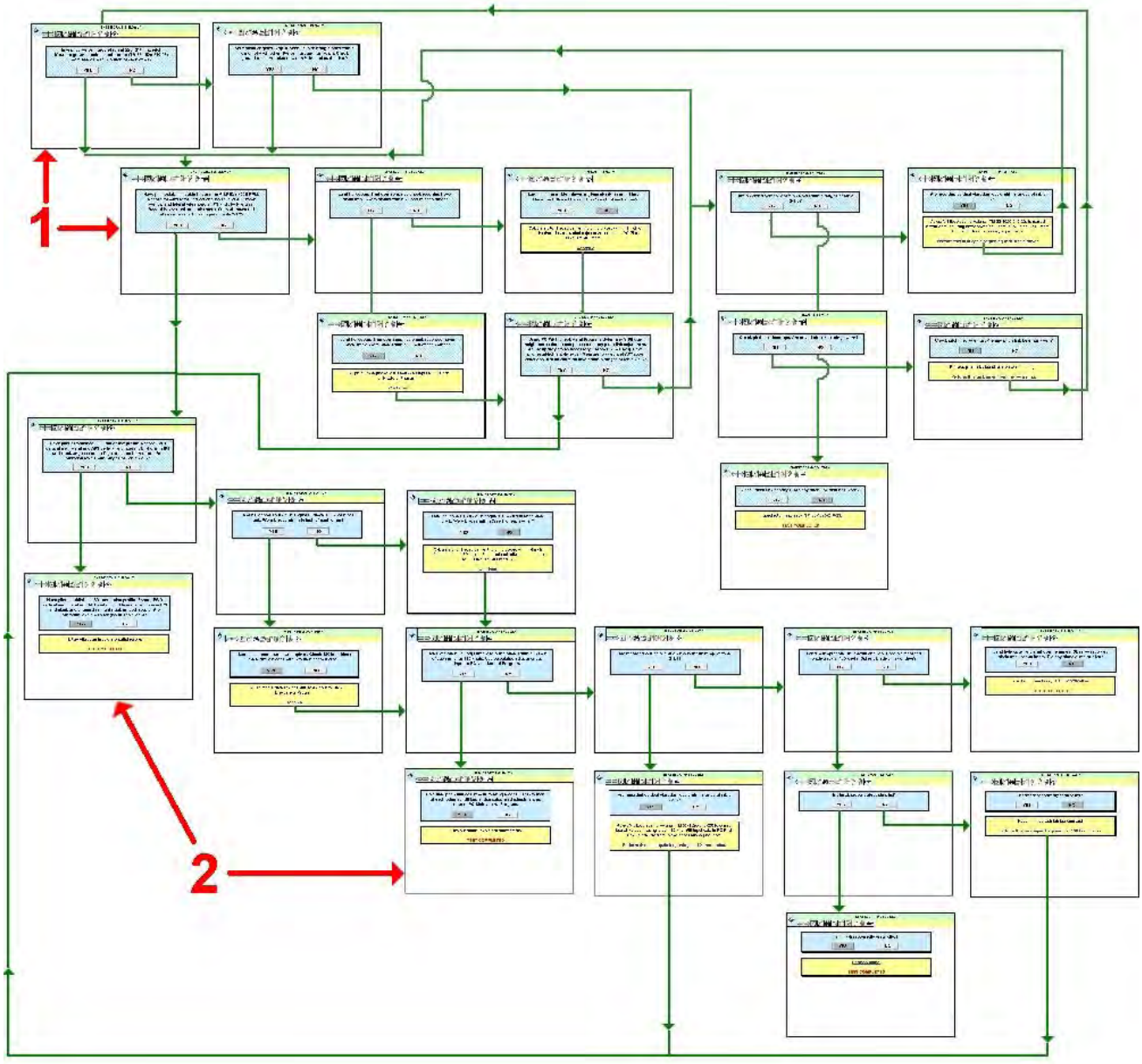

FIGURE 283. Example XML fragment diagnostic flow.

36.11.2 Diagnostic testing with state variables ***<testwithstate>***. The testing (with state variables) is used for frame-based viewers that require a logic engine (see 48.2). The test is broken into six (6) areas: introductory information (reason for test, necessary schematics, component locator, etc.), test set hookup, test procedure, observed test or BIT/BITE/automated test equipment (ATE) output results, test results evaluation leading to next action (identified fault(s), short narrative actions, completed testing, and/or link to corrective action or next test), and any follow-on maintenance task. A key concept in the diagnostic work package is after each

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evaluated test result, the next action (either another test or corrective action) is through a link. The assumption should never be the next action following the evaluation is performed, but a link to that action needs to occur (even if the next test does follow the test).

Depending on the diagnostic testing capabilities either ATE can be connected to the IETM and/or observed results are used to set the state table variables, which are then evaluated for the next action. When obtaining information from the user, this method has more robust dialog boxes (i.e., fill-in dialog and variable within dialog prompts) available to obtain the test result information. The dialog box may have predefined options or defined depending on the situation (variable usage), which when the option is selected, links the viewer to the next test or a corrective action. When obtaining information from ATE, the diagnostic resulting information is rendered to the IETM with little or no maintainer intervention. Generally, the information layout is the test title, reason for test, introductory information, testing steps, result from observed or ATE responses, evaluating results, and linking to the next action.

a. The components of diagnostic testing (without state variables) ***<testwithstate>*** are:
   (1) Precondition check ***<precond>*** (optional) (see 42.1.1.1).
   (2) Test title ***<title>*** (optional) (see 49.1.1.3).
   (3) Reason for test ***<reasonfortest>*** (optional) (see 36.11.1.1).
   (4) Setting (manipulating) state table variables ***<statemanipulation>***/***<statemanipulation-alt>*** (optional – zero or more) (see 48.2.3).
   (5) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (6) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (7) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (8) System description and support information (optional) (see 36.7).
   (9) Test set hookup ***<hookup>*** (optional – zero or more) (see 33.1 how to develop a task). If test set hookup is used test set disconnection is required.
   (10) Conduct diagnostic testing including one or more of the following components:
      (a) Test setup steps (optional) and components are either
         1 Procedure ***<proc>*** (required) (see 33.2).
         2 Procedural step ***<step1>*** (see 33.3)/***<step1–alt>*** (see 48.2.1) (required – one or more) with associated figure ***<figure>***/***<figure-alt>*** (optional – zero or more) (see 44.1.1) and/or associated table ***<table>***/***<table-alt>*** (optional – zero or more) (see 42).
      (b) Interaction with user to obtain diagnostic testing information ***<interaction>*** (see 49.1.4.10).
      (c) Assistance in finding a component's location ***<comp-locator>*** (see 36.7.6).
      (d) Initializing the BIT/BITE or integrated test equipment ***<diagnostic_initial>***/***<diagnostic_initial-alt>*** (see 36.11.2.1).
      (e) Perform BIT/BITE or integrated test equipment diagnostic testing(s) ***<diagnostic_group>*** (see 36.11.2.2).
      (f) Evaluate diagnostic test result ***<evaluate>*** (see 36.11.2.3) to determine next action.
   (11) Follow-on maintenance task ***<followon.maintsk>*** (optional) are the instructions for a maintenance condition which is accomplished sometime following the completion of a specific task to clean up or undo actions performed during the task (see 33.1 for task development).

b. DTD fragment graphically is:

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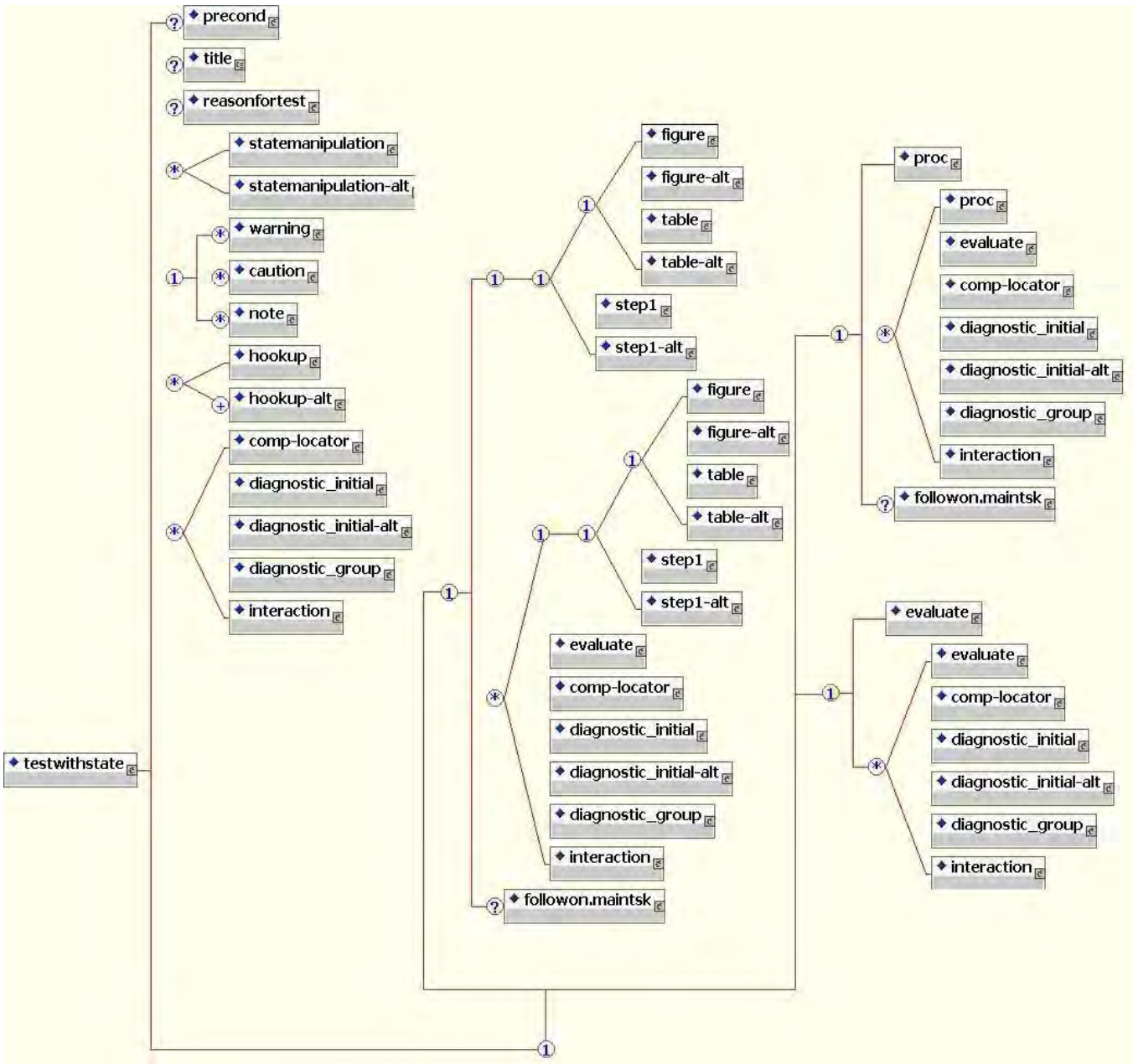

FIGURE 284. Testing (with state variables) DTD hierarchy <testwithstate>.

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c. DTD fragment is:

```
<!ELEMENT testwithstate (precond? , title? , reasonfortest? ,
   (statemanipulation | statemanipulation-alt)* , (warning* | caution* | note*) ,
   (hookup | hookup-alt+)* , (comp-locator | diagnostic_initial |
   diagnostic_initial-alt | diagnostic_group | interaction)* ,
   (((((figure | figure-alt | table | table-alt) | step1 | step1-alt)) ,
   ((((figure | figure-alt | table | table-alt) | step1 | step1-alt)) |
   evaluate | comp-locator | diagnostic_initial | diagnostic_initial-alt |
   diagnostic_group | interaction)* , followon.maintsk?) |
   (proc , (proc | evaluate | comp-locator | diagnostic_initial |
  diagnostic_initial-alt | diagnostic_group | interaction)* , followon.maintsk?) |
   (evaluate , (evaluate | comp-locator | diagnostic_initial |
   diagnostic_initial-alt | diagnostic_group | interaction)*)))>
<!ATTLIST testwithstate
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<testwithstate>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

36.11.2.1 Automatic test equipment (ATE) initialization ***<diagnostic_initial>***. The element starts the initial calibration or sets up the ATE through an external software application. The initialization will prepare the ATE for testing the equipment. If the ATE has already been initialized the action is filtered out with a precondition check (***<precond>***) by previously setting state variable(s). If some additional maintainer intervention is required before executing the external initialization software, the IETM viewer will prompt (using ***<message>***) the maintainer on the required actions (i.e., turn power on, connect cable, etc.). Depending on the external initialization software one or more parameters (***<sendparameter>***) are required to be sent to correctly set the ATE. The ATE can return parameter(s) (***<receiveparameter>***) to the logic engine that indicate if the ATE is ready or not. These stored state variables can be evaluated for readiness (this would require an evaluation ***<evaluate>*** to be performed) to determine ATE readiness followed by the appropriate actions.

a. The components of ATE initialization ***<diagnostic_initial>*** are:

(1) Precondition ***<precond>*** (optional) (see 42.1.1.1).
(2) Preparatory user message ***<message>*** (optional) (see 48.4).
(3) Input parameter(s) to ATE ***<sendparameter>*** (optional – zero or more) (see 36.11.2.1.1).
(4) Output result parameter(s) from ATE ***<receiveparameter>*** (optional – zero or more) (see 36.11.2.1.2).

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(5) Setting (manipulating) state table variables ***<statemanipulation>/<statemanipulation-alt>*** (optional – zero or more) (see 48.2.3).

b. DTD fragment graphically is:

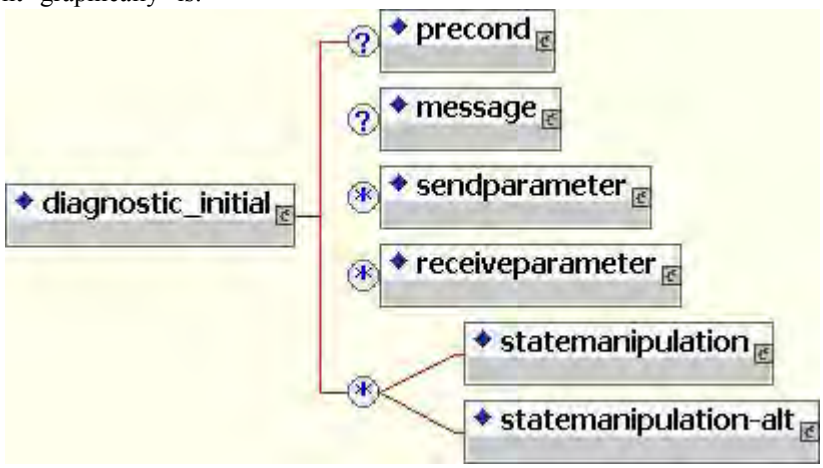

FIGURE 285. ATE initialization DTD hierarchy <diagnostic_initial>..

c. DTD fragment is:

```
<!ELEMENT diagnostic_initial (precond? , message? , sendparameter* ,
   receiveparameter* , (statemanipulation | statemanipulation-alt)*)>

<!ATTLIST diagnostic_initial
   action              (prompt | immediate )            #REQUIRED
   application         ENTITY                            #IMPLIED
   mfr                 CDATA                             #IMPLIED
   model               CDATA                             #IMPLIED
   protocol            CDATA                             #IMPLIED
    applicable         IDREFS                            #IMPLIED
    assocfig           IDREFS                            #IMPLIED
    changeref          IDREFS                            #IMPLIED
    comment            CDATA                             #IMPLIED
    delchlvl           CDATA                             #IMPLIED
    id                 ID                                #IMPLIED
    idref              IDREFS                            #IMPLIED
    inschlvl           CDATA                             #IMPLIED
    security           (uc | fouo |c | s | ts)          #IMPLIED
    skilltrk           CDATA                             #IMPLIED
>
```

d. Unique attributes for ***<diagnostic_initial>*** are:

(1) **action** Indicate if the ATE initialization is performed immediately or after message prompt is completed (required). Select either "**immediate**" or "**prompt**".

(2) **application** References to the external software application (using the reference to the ENTITY definition) to interface the IETM with the ATE (optional).

(3) **mfr** The ATE manufacturer name or code (optional).

(4) **model** The ATE model number (optional).

(5) **protocol** The interface or protocol to communicate with the ATE (optional).

e. Common attributes for ***<diagnostic_initial>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).

(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

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(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

36.11.2.1.1 Automatic test equipment (ATE) input parameters ***<sendparameter>***. The element provides a method to pass a parameter value to the ATE or external software. Each parameter value requires a separate ***<sendparameter>***. The input value can be either static (fixed valued) (***<string>***) or dynamic (state variable) (***<variableref>***). The information can be inputted by referenced order or name pair (thus order may not be important). The name pair requires each send parameter use the external software parameter name (***<name>***) and value (either ***<variableref>*** or ***<string>***). An example using a name pair is an ATE needing the number of cylinders (parameter name is "CYL" and value is 12) the XML markup is ***<sendparameter>*** ***<name>***CYL***</name>*** ***<string>***12***</string>*** ***</sendparameter>***. The format of the information is dependent on the interface between the IETM viewer and the ATE (commonly referred as middle-ware).

a. The components of ATE input parameters ***<sendparameter>*** are:
   (1) Parameter name ***<name>*** (optional) (see 49.1.4.19).
   (2) ATE input value is either:
      (a) Dynamic value (state variable) ***<variableref>*** (required) (see 48.1.2).
      (b) Static value ***<string>*** (required) (see 48.1.3.2).

b. DTD fragment graphically is:

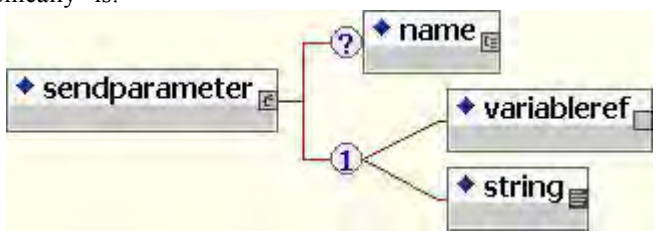

FIGURE 286. ATE input parameters DTD hierarchy <sendparameter>..

c. DTD fragment is:

```
<!ELEMENT sendparameter (name? , (variableref | string))>
<!ATTLIST sendparameter
   mode              CDATA              #IMPLIED>
>
```

36.11.2.1.2 Automatic test equipment (ATE) output parameter ***<receiveparameter>***. The element provides a method to receive a parameter value back from the ATE or external software. Each receiving parameter value requires a separate ***<receiveparameter>***. The referenced state variable name will store the received values (within the logic engine) in the order provided. The ATE test results value is generally used to determine the next action to perform during diagnostics (determined using the ***<evaluate>*** element).

a. The components of ATE output parameter ***<receiveparameter>*** is the state variable reference ***<variableref>*** (required) (see 48.1.2) to store the returned value.

b. DTD fragment graphically is:

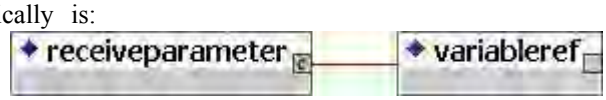

FIGURE 287. ATE output parameter DTD hierarchy <receiveparameter>..

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c. DTD fragment is:

```
<!ELEMENT receiveparameter (variableref)>
```

36.11.2.2 Grouped automatic test equipment (ATE) testing ***<diagnostic_group>***. The element provides the information to execute one or more external software ATE application(s). The external application generally returns diagnostic test results (***<receiveparameter>***) to state variables, which are used to evaluate the weapon system condition. If some maintainer intervention is required before executing the external software ATE application, then the IETM will prompt (using ***<message>***) the maintainer on the required actions (i.e., turn power on, connect cable, etc.). The ***<diagnostic_group>*** allows multiple external diagnostic tests (***<diagnostic>***) to be performed (i.e., each cylinder) and defines common input parameters that are set and reused for each diagnostic test within the group.

a. The components of grouped ATE testing ***<diagnostic_group>*** are:
   (1) Preparatory user message ***<message>*** (optional) (see 48.4).
   (2) Common input parameter(s) to ATE ***<sendparameter>*** (optional – zero or more) (see 36.11.2.1.1).
   (3) Execute ATE diagnostic(s) ***<diagnostic>*** (required – one or more) (see 36.11.2.2.1).

b. DTD fragment graphically is:

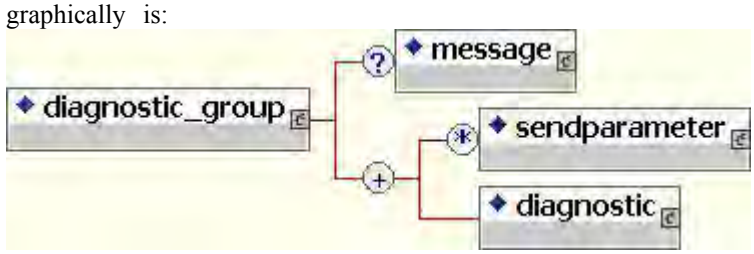

FIGURE 288. Grouped ATE testing DTD hierarchy <diagnostic_group>..

c. DTD fragment is:

```
<!ELEMENT diagnostic_group (message? , (sendparameter* , diagnostic)+)>
<!ATTLIST diagnostic_group
   action                  (prompt | immediate )          #REQUIRED >
```

d. Unique attribute for ***<diagnostic_group>*** is **action** which Indicates if the ATE initialization is performed immediately or after message prompt is completed (required). Select either "**immediate**" or "**prompt**".

Below is an example of setting common input parameters for engine cylinder testing. Each engine cylinder needs to be checked to determine if the correct compression ratio is being maintained. The common input parameters are set with the minimum and maximum compression ratio for each engine cylinder. The unique information is the engine cylinder to be tested and the storage of each cylinder test result.

<!ENTITY CylinderRatio PUBLIC "-//Engine Company//ENTITY Cylinder Compression Ration Diagnostic Test Build 4712//EN" "cylratio.exe"> . . .
<diagnostic-group action="immediate">
<sendparameter>
<name>minratio
</name>
<string>10.0
</string>
</sendparameter>
<sendparameter>
<name>maxratio
</name>
<string>16.0
</string>
</sendparameter>
<diagnostic testname="Cylinder_Compression_Ratio" application="CylinderRatio">
<desc>Check a single engine cylinder for compression ratio

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```
</desc>
<sendparameter>
<name>CYL
</name>
<string>1
</string>
</sendparameter>
<receiveparameter>
<variableref name="Check Cylinder 1"/>
</receiveparameter>
</diagnostic>  . . .
<diagnostic testname="Cylinder_Compression_Ratio" application="CylinderRatio">
<desc>Check a single engine cylinder for compression ratio
</desc>
<sendparameter>
<name>CYL
</name>
<string>16
</string>
</sendparameter>
<receiveparameter>
<variableref name="Check Cylinder 16"/>
</receiveparameter>
</diagnostic>
</diagnostic-group>
```

36.11.2.2.1 Automatic test equipment (ATE) testing ***<diagnostic>***. The element is used to execute ATE software and receive test results to be evaluated for next action. Each test has a brief description of the test to be performed and is used for documentation and/or shown to the maintainer for information. The testing requires an interface (middle-ware) between the IETM viewer and the ATE. Any additional parameter(s) may be sent to the testing equipment (these are in addition to the parameters defined in ***<diagnostic_group>***) to prepare the ATE. After the ATE has completed the diagnostic testing, the resulting values are passed to state variable(s) (in sequential order received) for evaluation. The external software application for testing is defined using the attributes **testname**, **application**, **protocol**, **mfr**, and **model** (see attribute definition for usage). The only required information is the test name that can be used by the middle-ware to determine how to execute the diagnostic test and information that is returned to the IETM logic engine.

a. The components of ATE testing ***<diagnostic>*** are:
   (1) ATE diagnostic test description for documentation ***<desc>*** (required) (see 49.1.4.17).
   (2) Input parameter(s) to ATE ***<sendparameter>*** (optional – zero or more) (see 36.11.2.1.1).
   (3) Output result parameter(s) from ATE ***<receiveparameter>*** (optional – zero or more) (see 36.11.2.1.2).

b. DTD fragment graphically is:

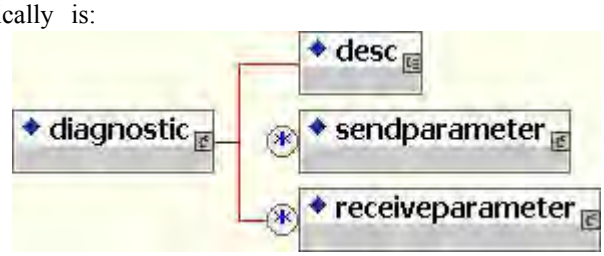

FIGURE 289. ATE testing DTD hierarchy <diagnostic>.

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c. DTD fragment is:

```
<!ELEMENT diagnostic (desc , sendparameter* , receiveparameter*)>
<!ATTLIST diagnostic
    application       ENTITY                          #IMPLIED
    mfr               CDATA                           #IMPLIED
    model             CDATA                           #IMPLIED
    protocol          CDATA                           #IMPLIED
    testname          CDATA                           #REQUIRED
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

d. Unique attributes for ***<diagnostic>*** are:
   (1) **application** – References to the external software application (using the reference to the ENTITY definition) to interface the IETM with the ATE (optional).
   (2) **mfr** – The ATE manufacturer name or code (optional).
   (3) **model** – The ATE model number (optional).
   (4) **protocol** – The interface or protocol to communicate with the ATE (optional).
   (5) **testname** – The ATE test name that can be used for identification only or by the IETM middle-ware to determine the test execute.

e. Common attributes for ***<diagnostic>*** are:
   (1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
   (2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (4) **comment** – Change information (optional) (see 49.2.9).
   (5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (8) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (9) **security** – Security classification (optional) (see 49.2.13).
   (10) **skilltrk** – Skill level (optional) (see 49.2.2).

36.11.2.3 Evaluate test result state variable(s) ***<evaluate>***. The element provides the capability to evaluate the system condition through user interaction or returned test conditions from an ATE, which ultimately determines the next test or maintenance action that is performed (accomplished through a link reference). The evaluation has three methods, IF-THEN-ELSE, loop FOR-NEXT, and loop WHILE. The evaluation or testing to be performed will determine the method. The first method is IF-THEN_ELSE, which is when an state variable(s) has been assigned a value and is evaluated against the predefined expression. If the expression returns TRUE the prescribed action is performed, otherwise the ELSE action is performed. The second method is loop FOR-NEXT, which is when a number of iterations are performed on a set of actions or tests. The final method is loop WHILE, which is when the a set of actions or tests are continuously performed while an expression evaluates to TRUE and will only exit when the expression returns FALSE.

a. The components of evaluating test result state variables ***<evaluate>*** are (one is required):
   (1) Evaluate state variable information to determine the next action. ***<if>*** (see 36.11.2.3.1).
   (2) Execute a loop action(s) a specified number of iterations ***<loopfor>*** (see 36.11.2.3.2).
   (3) Execute a loop action(s) while evaluated expression is true ***<loop>*** (see 36.11.2.3.3).

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b. DTD fragment graphically is:

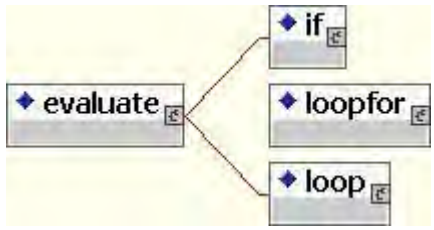

FIGURE 290. Evaluate test result state variables DTD hierarchy <evaluate>.

c. DTD fragment is:

```
<!ELEMENT evaluate (if | loopfor | loop)>
<!ATTLIST evaluate
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<evaluate>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

36.11.2.3.1 Evaluate IF condition ***<if>***. The IF condition evaluates the user dialog response or ATE test result state variables in a predefined expression using the logic engine. When an expression evaluates to TRUE, the THEN action is performed. When an expression evaluates to FALSE, each ELSE-IF (if applicable) is evaluated until a TRUE condition is located or, when all ELSE-IF evaluated expressions are FALSE, the ELSE action is performed. When no ELSE action is specified (not recommended), the IETM viewer executes next command after the IF condition. The usage of ELSE-IF is similar to a CASE statement in a programming language.

a. The components of evaluate IF condition ***<if>*** are:

(1) Conditional expression that is evaluated as TRUE or FALSE to determine next action ***<expression>*** (required) (see 48.2.2).
(2) Actions after a conditional expression is TRUE ***<then>*** (required) (see 36.11.2.3.1.1).
(3) Nested IF condition when previous conditional expression(s) is FALSE ***<elseif>*** (optional – zero or more) (see 36.11.2.3.1.2).
(4) Actions after all conditional expressions are FALSE ***<else>*** (optional) (see 36.11.2.3.1.3).

b. DTD fragment graphically is:

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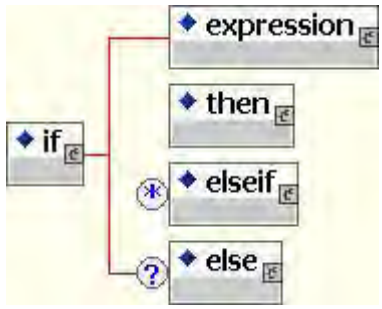

FIGURE 291. Evaluate IF condition DTD hierarchy <if>.

c. DTD fragment is:

```
<!ELEMENT if (expression , then , elseif* , else?)>
<!ATTLIST if
    id                          ID              #IMPLIED >
```

d. Common attribute for ***<if>*** is **id** – Specifies unique identifier (target) to reference (optional)(see 49.2.6).

An XML markup (with comments) example using the IF condition is shown below.

<!– EXECUTES THE ATE TEST FOR Cylinder Compression Ratio FOR AN ENGINE AND CAPTURES THE RESULTS IN THE STATE VARIABLE Test Result. –>
<diagnostic-group action="immediate">
<diagnostic testname="Cylinder_Compression_Ratio">
<desc>Check a single engine cylinder for compression ratio
</desc>
<receiveparameter>
<variableref name="Test Result"/>
</receiveparameter>
</diagnostic>
</diagnostic-group>
<if>
<!– CHECKS IF THE ATE TEST RESULT RETURNS ENGINE CYLINDER COMPRESSION RATIO IS OK. –>
<expression>
<variableref name="Test Result/>
<eq>
<string>OK
</string>
</expression>
<!– ENGINE CYLINDER COMPRESSION RATIO IS OK, THEN LINK TO TEST NODE WITH ID test0043 FOR NEXT POSSIBLE FAULT. –>
<then>
<resultwithstate>
<link application="frame" linkaction="immediate" linktype="goto" popup="no" xlink:href="test0043" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithstate>
</then>
<elseif>
<!– CHECKS IF THE ATE TEST RESULT RETURNS ENGINE CYLINDER COMPRESSION RATIO IS POSSIBLE ERROR. –>
<expression>
<variableref name="Test Result/>
<eq>
<string>POSSIBLE
</string>

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```
</expression>
<!– ENGINE CYLINDER COMPRESSION RATIO IS POSSIBLE ERROR, THEN LINK TO
TEST NODE WITH ID test0024 FOR MORE DETAILED CYLINDER TESTING. –>
<then>
<resultwithstate>
<link application="frame" linkaction="immediate" linktype="goto" popup="no"
xlink:href="test0024" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink"/>
</resultwithstate>
</then>
</elseif>
<!– CHECKS IF THE ATE TEST RESULT RETURNS ENGINE
CYLINDER COMPRESSION RATIO IS BAD. –>
<elseif>
<expression>
<variableref name=”Test Result/>
<eq>
<string>BAD
</string>
</expression>
<!– ENGINE CYLINDER COMPRESSION RATIO IS BAD, THEN LINK TO MAINTENANCE
WORK PACKAGE M00187-XX-XXXX-XXX TO CORRECT CYLINDER COMPRESSION
FAULT AND THEN RETEST THE CYLINDER COMPRESSION RATIO. –>
<then>
<resultwithstate>
<link application="frame" linkaction="immediate" linktype="return"
popup="no" xlink:href="M00187-XX-XXXX-XXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<returnlink>
<link application="frame" linkaction="immediate" linktype="goto" popup="no"
xlink:href="test0040" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink"/>
</returnlink>
</link>
</resultwithstate>
</then>
</elseif>
<!– UNEXPECTED ATE TEST RESULT RETURNS ENGINE CYLINDER COMPRESSION
RATIO, THEN NOTIFY SUPERVISOR OF PROBLEM. –>
<else>
<resultwithstate>
<interaction>
<message popup=”yes” button_title=”OK”>
<messageline>Report to supervisor invalid engine cylinder
compression ratio ATE test result of
<variableref name=”Test Result/>.
</messageline>
</message>
</interaction>
<completed_test/>
</resultwithstate>
</else>
</if>
```

36.11.2.3.1.1 Evaluated THEN condition ***<then>***. The element contains the actions after a evaluation has satisfied a condition (either the IF or ELSE-IF evaluated to "TRUE" or a loop meets the terminating condition). Generally the action contains the possible fault information, any applicable disconnection procedures, reference to another test node, reference to the corrective action and/or interact with the user to obtain additional

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information. Additional evaluation can be done to either reduce the expression complexity or after obtaining additional test result information (from the ***<resultwithstate>***) continue with further nested evaluations.

a. The components of evaluated THEN condition ***<then>*** are either :
   (1) Action to perform after IF evaluation ***<resultwithstate>*** (required) (see 36.11.2.3.1.1.1). Following the action, an additional evaluation test result state variables ***<evaluate>*** (optional) (see 36.11.2.3).
   (2) Additional evaluation test results state variables ***<evaluate>*** (required) (see 36.11.2.3).

b. DTD fragment graphically is:

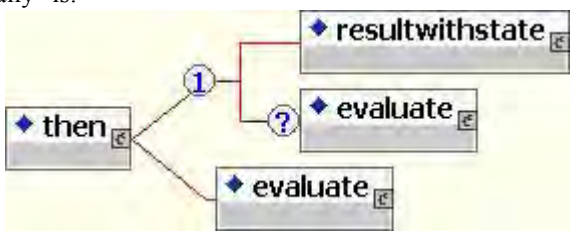

FIGURE 292. Evaluated THEN condition DTD hierarchy <then>.

c. DTD fragment is:

```
<!ELEMENT then ((resultwithstate , evaluate?) | evaluate)>
```

36.11.2.3.1.1.1 Test results (with state variables) ***<resultwithstate>***. The element contains the next actions resulting from the evaluated test response. When the resulting action specifies further testing, a link is provided to another diagnostic work package ***<diagnosticwp>***, another test (with state variable) ***<testwithstate>*** within the same work package, or conduct another test (either observed or from ATE). The link generally does not require a prompt (unless specified), but seamlessly presents the next testing component. The author does not infer the next test following is processed, but always provides a link. Using a link provides a consistent method for identifying the next action since the testing is dynamic.

When the resulting action specifies a corrective action either a narrative or link is provided. If the corrective action is short and simple the step(s) (***<step1>*** or ***<para>***) are developed. An example is "Call maintenance supervisor."

When the corrective action has different initial setup requirements and/or is complex, a separate and referenced (***<link>***) maintenance work package is developed. In either situation, the author needs to define the identified fault code(s) (***<fault>***) and indicate that testing is completed (***<completed_test>***).

a. The components of test results (with state variables) ***<resultwithstate>*** are:
   (1) Fault code ***<fault>*** (optional – zero or more) (see 48.1.3.5).
   (2) Disconnect test set ***<disconnect>***/***<disconnect-alt>*** (optional – zero or more) (see 36.8.2).
   (3) Action (required) component is either:
      (a) Setting (manipulating) state table variables ***<statemanipulation>***/***<statemanipulation-alt>*** (see 48.2.3).
      (b) Interaction with user to obtain diagnostic testing information ***<interaction>*** (see 49.1.4.10).
      (c) Link for further testing or corrective action maintenance work package ***<link>*** (see 46.2.3).
      (d) Test result information ***<para>*** (see 49.1.1.5).
   (4) Test completion action ***<completed_test>*** (optional) (see 36.11.1.5).

b. DTD fragment graphically is:

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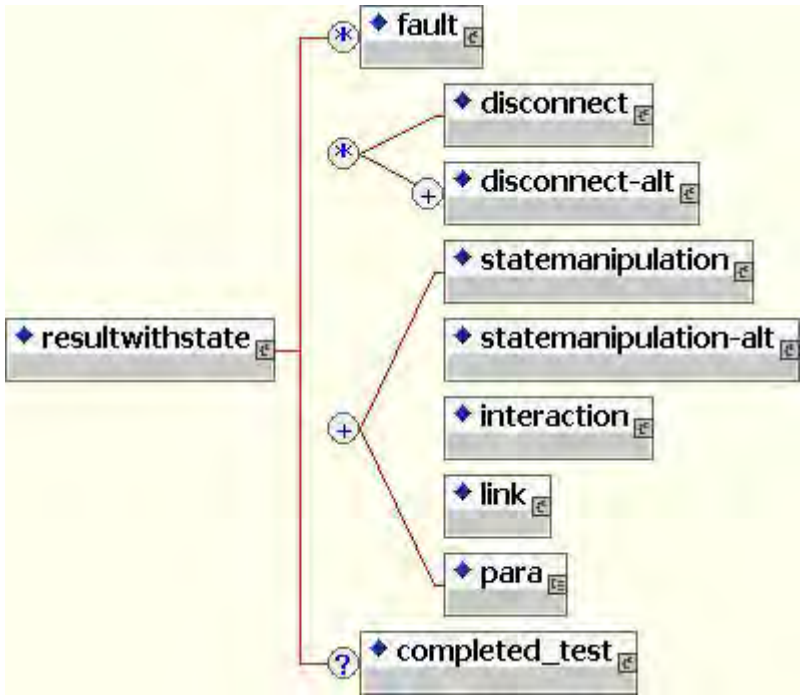

FIGURE 293. Test results (with state variables) DTD hierarchy <resultwithstate>.

c. DTD fragment is:

```
<!ELEMENT resultwithstate (fault* , (disconnect | disconnect-alt+)* ,
   (statemanipulation | statemanipulation-alt | interaction | link | para)+ ,
   completed_test?)>
```

36.11.2.3.1.2 Evaluated ELSE-IF condition ***<elseif>***. If the THEN or previous ELSE-IF condition(s) have failed, then the ELSE-IF condition is checked. If the ELSE-IF condition returns a TRUE result, the THEN condition is performed, otherwise the next ELSE-IF or ELSE is processed. The ELSE-IF is used when multiple conditions are possible for a state variable (i.e., error code).

a. The components of evaluated ELSE-IF condition ***<elseif>*** are:
   (1) Conditional expression that is evaluated as TRUE or FALSE to determine next action ***<expression>*** (required) (see 48.2.2).
   (2) Actions after a conditional expression is TRUE ***<then>*** (required) (see 36.11.2.3.1.1).

b. DTD fragment graphically is:

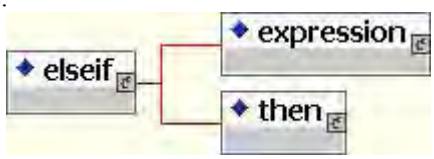

FIGURE 294. Evaluated ELSE-IF condition DTD hierarchy <elseif>..

c. DTD fragment is:

```
<!ELEMENT elseif (expression , then)>
```

An example XML markup is shown below in using the ELSE-IF element.

<if>
<expression>
<variableref name="Error Code"/>

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```
<eq>
<string>0001
</string>
</expression>
<then>
<resultwithstate>. . .
</resultwithstate>
</then>
<elseif>
<expression>
<variableref name="Error Code"/>
<eq>
<string>0002
</string>
</expression>
<then>
<resultwithstate>. . .
</resultwithstate>
</then>
</elseif> . . .
<elseif>
<expression>
<variableref name="Error Code"/>
<eq>
<string>0870
</string>
</expression>
<then>
<resultwithstate>. . .
</resultwithstate>
</then>
</elseif>
<else>
<resultwithstate>. . .
</resultwithstate>
</else>
</if>
```

36.11.2.3.1.3 Evaluated ELSE condition ***<else>***. When all IF and ELSE-IF condition evaluations return FALSE, then the ELSE action is performed. The content model is same as ***<then>*** (see36.11.2.3.1.1).

36.11.2.3.2 Evaluate loop FOR-NEXT condition ***<loopfor>***. The loop FOR-NEXT provides a method to run diagnostic tests for specified number of iterations. The author specifies the number of iterations by setting the start counter (***<initialcount>***), setting the counter incremental expression (***<increment>***), and setting the counter ending condition or expression (***<endcondition>***). The actions to be performed during the looping is defined in ***<loopaction>***. After the looping has terminated the concluding action (***<then>***) is performed, which specifies the the next action(s) to perform (generally linking to next testing or evaluation to perform).

The start, increment, and end counter generally are preset (fixed) values, but these values can be set dynamically with state variables. These dymanic values can be entered by the user through a dialog box or dependent on applicable equipment configuration (using ***<statemanipulation-alt>***).

a. The components of evaluate Loop FOR-NEXT condition ***<loopfor>*** are:
   (1) Setting (manipulating) state table variables ***<statemanipulation>***/***<statemanipulation-alt>*** (optional – zero or more) (see 48.2.3).
   (2) Initializing the FOR-NEXT state table variable counter ***<initialcount>*** (required). Sets the FOR-NEXT counter using ***<statemanipulation>***/***<statemanipulation-alt>*** (required) (see 48.2.3).

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(3) The FOR-NEXT counter incremental expression ***<increment>*** (required). The incremental amount is defined through ***<expression>*** (required) (see 48.2.2).
(4) The FOR-NEXT counter termination expression ***<endcondition>*** (required) (see 36.11.2.3.2.1).
(5) Actions to perform within the loop ***<loopaction>*** (required) (see 36.11.2.3.2.2).
(6) Follow-on action after FOR-NEXT loop completion ***<then>*** (required) (see 36.11.2.3.1.1).

b. DTD fragment graphically is:

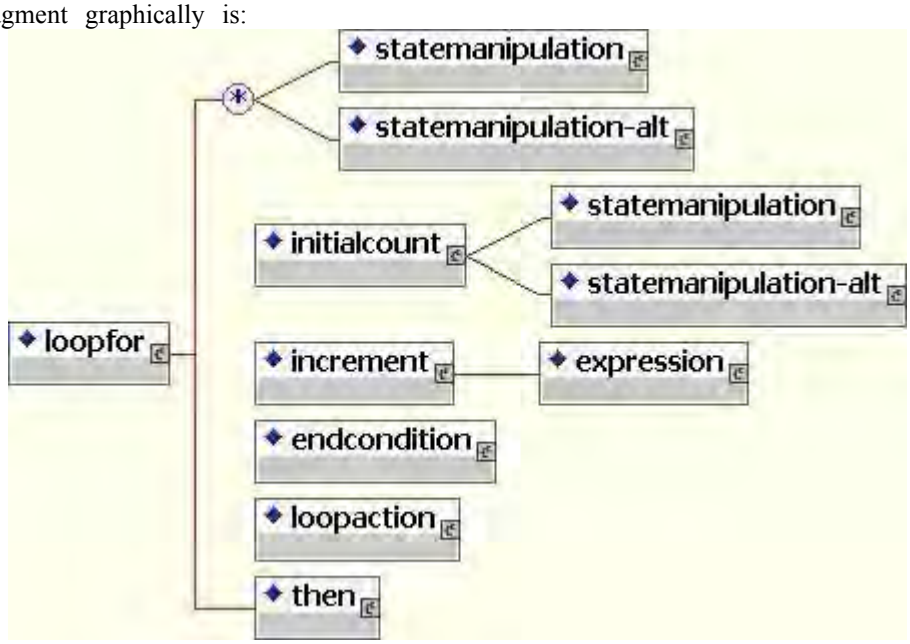

FIGURE 295. Evaluate loop FOR-NEXT condition DTD hierarchy <loopfor>..

c. DTD fragment is:

```
<!ELEMENT loopfor ((statemanipulation | statemanipulation-alt)* , initialcount ,
   increment , endcondition , loopaction , then)>
<!ATTLIST loopfor
    id                    ID          #IMPLIED >


<!ELEMENT initialcount (statemanipulation | statemanipulation-alt+)>

<!ELEMENT increment (expression)>
```

d. Common attribute for ***<loopfor>*** is **id** – Specifies unique identifier (target) to reference (optional)(see 49.2.6).

The example below shows a FOR-NEXT loop starting at 2, ending at 8, and increments by 2. In a programming language it may be expressed as `For ctr = 2 To 8 Step 2... Next ctr`.

<loopfor>
<initialcount>
<statemanipulation>
<variable config="no" description="Loop Counter" name="ctr" value-type="integer">
<initialize>
<expression>
<integer>2
</integer>
</expression>
</initialize>
</variable>

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Check the source to verify that this is the current version before use.

```
</statemanipulation>
</initialcount>
<increment>
<expression>
<variableref  name="ctr"/>
<plus/>
<integer>2
</integer>
</expression>
</increment>
<endcondition>
<expression>
<variableref  name="ctr"/>
<le/>
<integer>8
</integer>
</expression>
</endcondition>
<loopaction>. . .
</loopaction>
<then>. . .
</then>
</loopfor>
```

The example below shows a FOR-NEXT loop starting at 2, ending at 6 (Model A) or 8 (Model B) depending on applicable equipment configuration, and increments by 2. In a programming language it may be expressed as `For ctr = 2 To MaxCtr Step 2... Next ctr`.

```
<loopfor>
<statemanipulation-alt>
<statemanipulation  applicable="model-a">
<variableref  name="MaxCtr"/>
<expression>
<integer>6
</integer>
</expression>
</statemanipulation>
<statemanipulation  applicable="model-b">
<variableref  name="MaxCtr"/>
<expression>
<integer>8
</integer>
</expression>
</statemanipulation>
</statemanipulation-alt>
<statemanipulation-alt>
<variableref  name=”MaxCtr”/>
<initialize>
<expression>
<integer>2
</integer>
</expression>
</initialize>
</variable>
</statemanipulation>
<initialcount>
<statemanipulation>
<variable  config="no"  description="Loop  Counter"  name="ctr"  value-type="integer">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<initialize>
<expression>
<integer>2
</integer>
</expression>
</initialize>
</variable>
</statemanipulation>
</initialcount>
<increment>
<expression>
<variableref name="ctr"/>
<plus/>
<integer>2
</integer>
</expression>
</increment>
<endcondition>
<expression>
<variableref name="ctr"/>
<le/>
<variableref name="MaxCtr"/>
</expression>
</endcondition>
<loopaction>. . .
</loopaction>
<then>. . .
</then>
</loopfor>

36.11.2.3.2.1 Loop termination condition ***<endcondition>***. The element provides the termination condition for the loops (FOR-NEXT and WHILE). The looping action continues as long as the evaluated termination condition is TRUE. Generally, the FOR-NEXT loop termination condition is expressed as a state variable that is less than or equal (<=) to the maximum counter amount. Generally, the WHILE loop termination condition is dependent on the information obtained during the looping action (i.e., adjusting the tire pressure while not within normal range).

a. The components of Loop termination condition ***<endcondition>*** is defined through evaluated ***<expression>*** (required) (see 48.2.2).
b. DTD fragment graphically is:

FIGURE 296. Loop termination condition DTD hierarchy <endcondition>..

c. DTD fragment is:

```
<!ELEMENT endcondition (expression)>
```

36.11.2.3.2.2 Actions to perform during looping ***<loopaction>***. The element defines what actions occur while within a loop. The actions can be procedural step(s) to performed followed by ATE testing or user dialog box.

a. The components of actions to perform during looping ***<loopaction>*** are:
   (1) Procedure step(s) (optional) includes one of the following:
      (a) Procedure ***<proc>*** (required – two or more) (see 33.2).
      (b) Procedural step ***<step1>*** (see 33.3)/***<step1–alt>*** (see 48.2.1) (required – two or more) with associated figure ***<figure>***/***<figure-alt>*** (optional – zero or more) (see 44.1.1) and/or associated table ***<table>***/***<table-alt>*** (optional – zero or more) (see 42).

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(c) Single step or paragraph text ***<para>*** (required) (see 49.1.1.5).
(d) Single step with an alert ***<specpara>*** (required) (see 49.1.1.6).

(2) State variable manipulation (required – one or more) includes the following:
(a) Setting (manipulating) state table variables ***<statemanipulation>***/***<statemanipulation-alt>*** (see 48.2.3).
(b) Perform BIT/BITE or integrated test equipment diagnostic testing(s) ***<diagnostic_group>*** (see 36.11.2.2).
(c) Dialog box to obtain user input ***<dialog>***/***<dialog-alt>*** (see 48.3.2).

b. DTD fragment graphically is:

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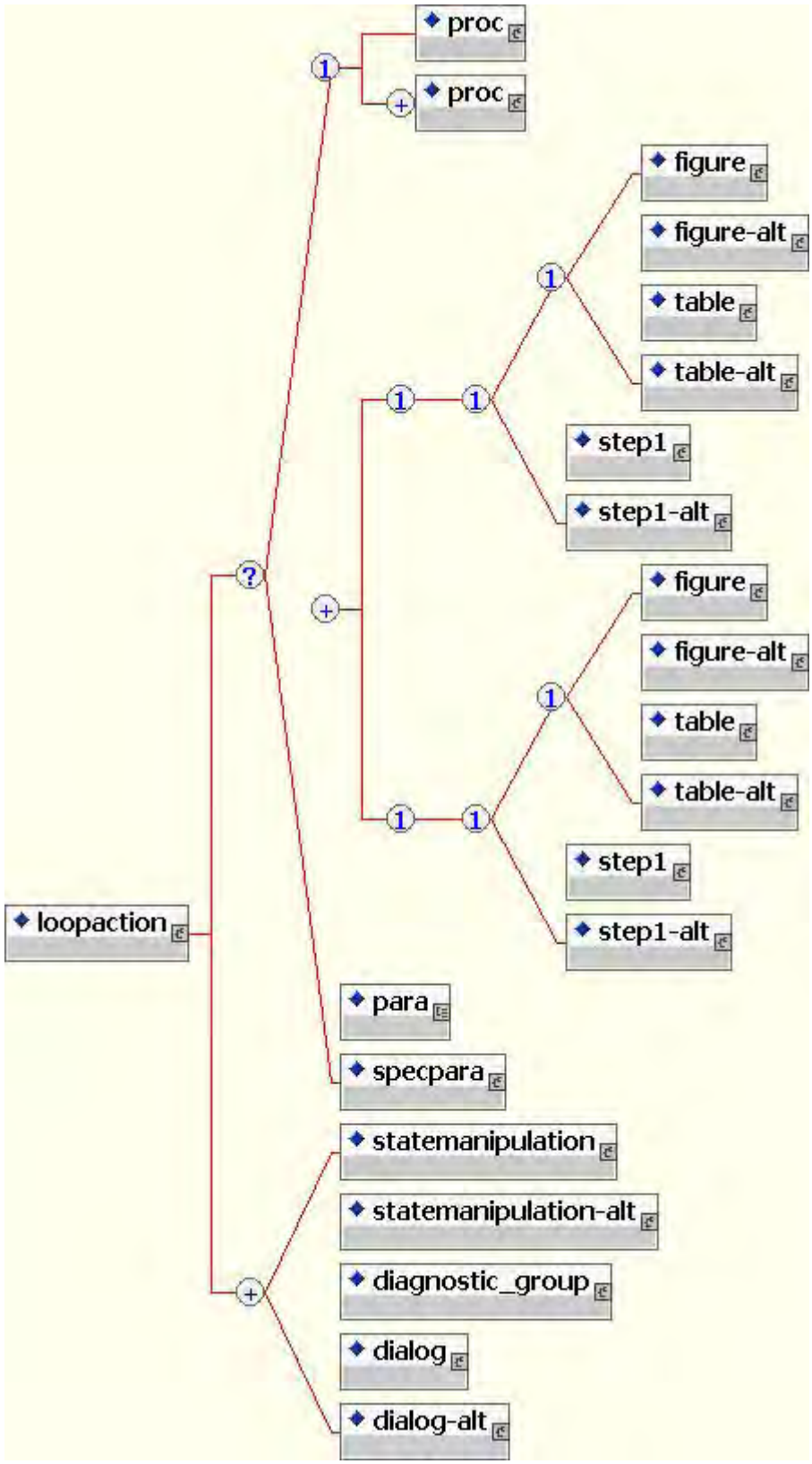

FIGURE 297. Actions to perform during looping DTD hierarchy <loopaction>.

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c. DTD fragment is:

```
<!ELEMENT loopaction (((proc , proc+) |
   ((((figure | figure-alt | table | table-alt) | step1 | step1-alt)) ,
   (((figure | figure-alt | table | table-alt) | step1 | step1-alt)))+ |
   para | specpara)? , (statemanipulation | diagnostic_group | dialog |
   dialog-alt)+)>
```

36.11.2.3.3 Evaluate loop condition ***<loop>***. The element is used to perform the actions while the evaluated loop termination expression is TRUE. The looping is used when adjustments or calibration readings are being performed on the equipment and are within range before continuing. An example is tire pressure or compression ratio that falls within a specified range. The ATE test result state variables or user dialog box responses are evaluated for the termination condition. Some caution needs to be taken to avoid an infinite looping. A good practice is to have maximum number of iterations or a dialog box option with an exit the loop condition even if the conditions is not achieved.

a. The components of evaluate Loop condition ***<loop>*** are:
   (1) Setting (manipulating) state table variables ***<statemanipulation>***/***<statemanipulation-alt>*** (optional – zero or more) (see 48.2.3).
   (2) Check for loop termination before performing any actions ***OR***
      (a) The loop termination expression ***<endcondition>*** (required) (see 36.11.2.3.2.1).
      (b) Actions to perform within the loop ***<loopaction>*** (required) (see 36.11.2.3.2.2).
   (3) Perform action(s) at least once before checking for loop termination.
      (a) Actions to perform within the loop ***<loopaction>*** (required) (see 36.11.2.3.2.2).
      (b) The loop termination expression ***<endcondition>*** (required) (see 36.11.2.3.2.1).
   (4) Actions after loop termination ***<then>*** (required) (see 36.11.2.3.1.1).

b. DTD fragment graphically is:

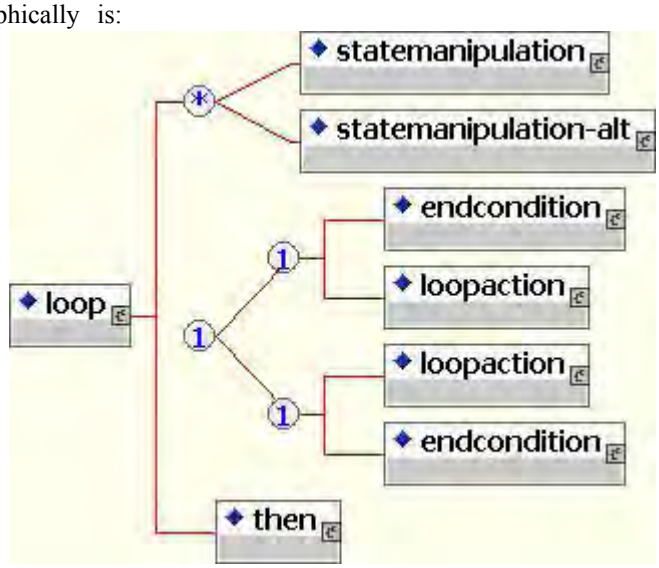

FIGURE 298. Evaluate loop condition DTD hierarchy <loop>.

c. DTD fragment is:

```
<!ELEMENT loop ((statemanipulation | statemanipulation-alt)* ,
   ((endcondition , loopaction) | (loopaction , endcondition)) , then)>
<!ATTLIST loop
    id                      ID          #IMPLIED >
```

d. Common attribute for ***<loop>*** is **id** – Specifies unique identifier (target) to reference (optional)(see 49.2.6).

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The first example below shows a loop for checking tire pressure with a dialog box option to terminate the loop without achieving proper air pressure.

```
<loop>
<loopaction>
<dialog cancel_button="no" reset_button="no">
<menu flow="list" mandatory="yes" select="single" type="radio">
<prompt>Tire air pressure is between:
</prompt>
<choice>
<text>0&ndash;40psi
</text>
<statemanipulation>
<variableref name="Tire Pressure"/>
<expression>
<string>Low
</string>
</expression>
</statemanipulation>
</choice>
<choice>
<text>40 &ndash; 60 psi
</text>
<statemanipulation>
<variableref name="Tire Pressure"/>
<expression>
<string>OK
</string>
</expression>
</statemanipulation>
</choice>
<choice>
<text>60+psi
</text>
<statemanipulation>
<variableref name="Tire Pressure"/>
<expression>
<string>High
</string>
</expression>
</statemanipulation>
</choice>
<choice>
<text>No tire gauge
</text>
<statemanipulation>
<variableref name="Tire Pressure"/>
<expression>
<string>NoGauge
</string>
</expression>
</statemanipulation>
</choice>
</menu>
</dialog>
</loopaction>
<endcondition>
```


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```
<expression>
<expression>
<variableref name="Tire Pressure"/>
<eq/>
<string>OK
</string>
</expression>
<or/>
<expression>
<variableref name="Tire Pressure"/>
<eq/>
<string>NoGauge
</string>
</expression>
</expression>
</endcondition>
<then>
<evaluate>
<if>
<expression>
<variableref name="Tire Pressure"/>
<eq/>
<string>OK
</string>
</expression>
<then>
<resultwithstate>
<link application="frame" linkaction="prompt" linktype="goto" xreftype="test"
local="T00343-X-XXX-XXX" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink">
<title>Tire pressure is correct
</title>
<prompt>Check wheel alignment.
</prompt>
</link>
</resultwithstate>
</then>
<else>
<resultwithstate>
<interaction>
<message button_title="OK">
<title>Problem
</title>
<messageline>Please obtain tire gauge from supervisor.
</messageline>
<messageline>Restart testing.
</messageline>
</message>
</interaction>
<completed_test/>
</resultwithstate>
</else>
</if>
</evaluate>
</then>
</loop>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

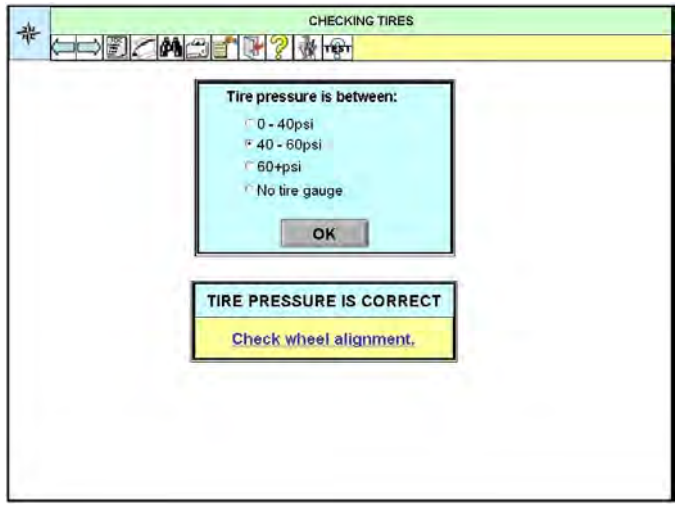

FIGURE 299. Sample loop with dialog box exit.

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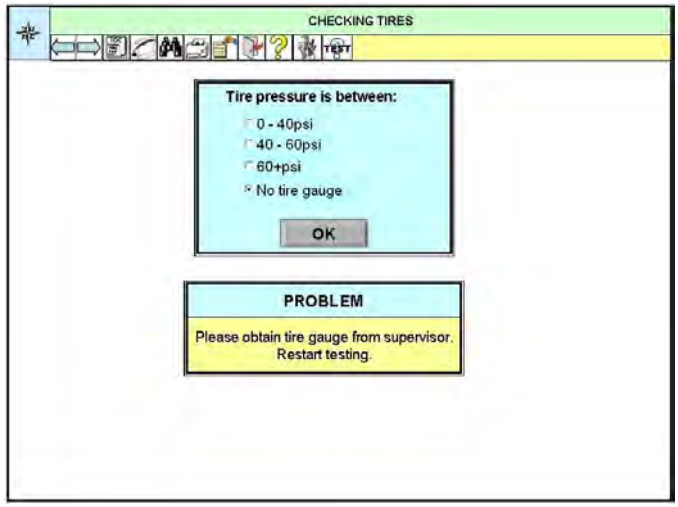

FIGURE 299. Sample loop with dialog box exit - Continued.

The second example below shows a loop for check compression ratio (through ATE). The maximum number of iterations are set to ten, after which the loop will be terminated without the correct compression ratio.

```
<loop>
<statemanipulation>
<variableref name="Counter"/>
<expression>
<integer>1
</integer>
</expression>
</statemanipulation>
<statemanipulation>
<variableref name="Max_Iteration"/>
<expression>
<integer>10
</integer>
</expression>
</statemanipulation>
<loopaction>
<statemanipulation>
<variableref name="Counter"/>
<expression>
<variableref name="Counter"/>
<plus/>
<integer>1
</integer>
</expression>
</statemanipulation>
<diagnostic_group action="immediate">
<diagnostic application="CompressionRatio" testname="Cylinder Compression Ratio">
<desc>Check the cylinder compression ratio and automati-
cally adjusts during each test.
</desc>
<receiveparameter>
<variableref name="Test_Result"/>
</receiveparameter>
</diagnostic>
</diagnostic_group>
```


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```
</loopaction>
<endcondition>
<expression>
<expression>
<variableref  name="Test_Result"/>
<ne/>
<string>PASS
</string>
</expression>
<and/>
<expression>
<variableref  name="Counter"/>
<le/>
<variableref  name="Max  Iteration"/>
</expression>
</expression>
</endcondition>
<then>
<evaluate>
<if>
<expression>
<variableref  name="Test_Result"/>
<eq/>
<string>PASS
</string>
</expression>
<then>
<resultwithstate>
<link  application="frame"  linkaction="prompt"  linktype="goto"  xreftype="test"
local="T00543-X-XXX-XXX"  xlink:type="simple"  xmlns:xlink="http://www.w3.org/1999/xlink">
<title>Cylinder  Compression  Ratio  is  Correct
</title>
<prompt>Check  fuel  filter.
</prompt>
</link>
</resultwithstate>
</then>
<else>
<resultwithstate>
<interaction>
<message  button_title="OK">
<title>Problem
</title>
<messageline>Please  contact  supervisor.
</messageline>
</message>
</interaction>
<completed_test/>
</resultwithstate>
</else>
</if>
</evaluate>
</then>
</loop>
```

The sample XML markup shown in a programming language.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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```
Counter = 1 Max_Iteration = 10 DO Counter = Counter + 1 Test_Result
= Cylinder_Compression_Ratio_Test() LOOP WHILE Test_Result <>
"PASS" AND Counter <= Max_Iteration IF Test_Result = "PASS" THEN
MSGBOX TEXT:="Check fuel filter.", TITLE:="Cylinder Compression
Ratio is Correct" GOTO T00543-X-XXX-XXX ELSE MSGBOX TEXT:="Please
contact supervisor.", TITLE:="Problem" EXIT ENDIF
```

36.11.3 XML document instance fragment and output for ***<diagnosticwp>***. The XML instance and its stylesheet output for a ***<diagnosticwp>*** is provided below.

a. Example of an XML document instance fragment for ***<diagnosticwp>***:

(1) Example XML markup in relationship to Figure 300.

```
<diagnosticwp airforce="no" army="no" deletewp="no" frame="yes" insertwp="no"
marines="no" navy="no" tocentry="2" wpno="T00001-X-XXX-XXXX">
<wpidinfo>
<maintlvl level="field"/>
<title>ENGINE COMPARTMENT FAN WILL NOT OPERATE WHEN AUX-
ILIARY POWER UNIT IS OPERATING
</title>
</wpidinfo>
<wpinfo>
<tools>
<tools-setup-item>
<name>General Mechanic's Tool Kit:  Automotive
</name>
<itemref>
<link application="frame" linkaction="prompt" linktype="return" popup="no"
xlink:href="S00034-X-XXX-XXXX#LTL-001-ICV-B00-5B2" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="part">
</link>
</itemref>
</tools-setup-item>
</tools>
<ref>
<ref-setup-item>
<link application="frame" linkaction="prompt" linktype="return"
popup="no" xlink:href="M00854-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
</link>
</ref-setup-item>
</ref>
<eqpconds>
<eqpconds-setup-item>
<condition>Wheels Chocked
</condition>
<itemref>
<link application="frame" linkaction="prompt" linktype="return"
popup="no" xlink:href="O00451-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
</link>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
<harness-indx id="T00001-X-XXX-XXXX-GM39151-S">
<title>Wiring Harness W420
</title>
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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```
<title>W420 Wiring Harness
</title>
<graphic boardno="obj.3" unitmeasure="in">
</graphic>
</figure>
<para esd="no" hcp="no">
</para>
</harness-indx>
<testwithstate id="T00001-X-XXX-XXXX-AT02A0011">
<reasonfortest>
<title>Reason for Test
</title>
<para esd="no" hcp="no">To verify that original symptom still exists.
</para>
</reasonfortest>
<statemanipulation>
<variable description="Engine Compartment Fault" name="AT02A001-F1"
scope="global" value-type="fault">
<initialize>
<fault fault-code="AT02A001-F1" fault-state="possible">
<oktext>Engine compartment blower motor OK.
</oktext>
<badtext>Engine compartment blower motor faulty.
</badtext>
<possibletext>Open circuit in engine compartment blower motor.
</possibletext>
</fault>
</initialize>
</variable>
</statemanipulation>
<statemanipulation>
<variable description="Wiring Harness W420" name="AT02A001-F2"
scope="global" value-type="fault">
<initialize>
<fault fault-code="AT02A001-F2" fault-state="possible">
<oktext>Engine compartment blower motor OK.
</oktext>
<badtext>Engine compartment blower motor faulty.
</badtext>
<possibletext>Wiring harness W420 faulty.
</possibletext>
</fault>
</initialize>
</variable>
</statemanipulation>
<statemanipulation>
<variable description="Question Answers" name="answer" scope="global" value-type="boolean">
</variable>
</statemanipulation>
<statemanipulation>
<variable description="Evaluation Result" name="AT02A00114-EV-182"
scope="global" value-type="integer">
</variable>
</statemanipulation>
<statemanipulation>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<variable description="Evaluation Result" name="AT02A00114-EV-184"
scope="global" value-type="integer">
</variable>
</statemanipulation>
<statemanipulation>
<variable description="Evaluation Result" name="AT02A00114-EV-186"
scope="global" value-type="integer">
</variable>
</statemanipulation>
<statemanipulation>
<variable description="Evaluation Result" name="AT02A00114-EV-188"
scope="global" value-type="integer">
</variable>
</statemanipulation>
<statemanipulation>
<variable description="Evaluation Result" name="AT02A00114-EV-190"
scope="global" value-type="integer">
</variable>
</statemanipulation>
<statemanipulation>
<variable description="Evaluation Result" name="AT02A00114-EV-192"
scope="global" value-type="integer">
</variable>
</statemanipulation>
<statemanipulation>
<variable description="Evaluation Result" name="AT02A00114-EV-194"
scope="global" value-type="integer">
</variable>
</statemanipulation>
<statemanipulation>
<variable description="Evaluation Result" name="AT02A00114-EV-196"
scope="global" value-type="integer">
</variable>
</statemanipulation>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">
<link application="frame" linkaction="prompt" linktype="return"
popup="no" xlink:href="O00354-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Perform Auxiliary Power Unit (APU) start-up.
</prompt>
</link>
</para>
</step1>
<step1 assocfig="T00001-X-XXX-XXXX-AT02A0011-fig1" esd="no" frame="no"
hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check operation of engine compartment fan.
<figure application="both" figtype="normal-page" pane="no" tocentry="1"
id="T00001-X-XXX-XXXX-AT02A0011-fig1">
<title>Engine Compartment Fan
</title>
<graphic boardno="obj.1" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">
<link application="frame" linkaction="prompt" linktype="return"
popup="no" xlink:href="000354-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Perform APU shutdown.
</prompt>
</link>
</para>
</step1>
<interaction>
<dialog popup="no">
<binarymenu answer="yesno">
<prompt>Does engine compartment fan operate when APU is running?
</prompt>
<yesstate>
<statemanipulation>
<variableref name="answer"/>
<expression>
<boolean>
<true/>
</boolean>
</expression>
</statemanipulation>
</yesstate>
<nostate>
<statemanipulation>
<variableref name="answer"/>
<expression>
<boolean>
<false/>
</boolean>
</expression>
</statemanipulation>
</nostate>
</binarymenu>
</dialog>
</interaction>
```

(2) Example XML markup in relationship to Figure 301.

```
<evaluate>
<if>
<expression>
<variableref name="answer"/>
</expression>
<then>
<resultwithstate>
<interaction>
<message button_title="OK" popup="no">
<messageline>Remove wheel chocks.
</messageline>
<aftermessage>
<link application="frame" linkaction="immediate" linktype="return"
popup="no" xlink:href="O00455-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
</link>
```


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Check the source to verify that this is the current version before use.

```
</aftermessage>
</message>
</interaction>
<completed_test>
</completed_test>
</resultwithstate>
</then>
<else>
<resultwithstate>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A0012" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</else>
</if>
</evaluate>
</testwithstate>
```

(3) Example XML markup in relationship to Figure 302.

```
<testwithstate id="T00001-X-XXX-XXXX-AT02A0012">
<reasonfortest>
<title>Reason for Test
</title>
<para esd="no" hcp="no">If waveform type and signal characteristics
do not match, there is an APU failure.
</para>
</reasonfortest>
<step1 assocfig="T00001-X-XXX-XXXX-fig1" esd="no" frame="no" hcp="no"
qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Disconnect P2 connector
<callout assocfig="T00001-X-XXX-XXXX-fig1" label="1"
numref="T00001-X-XXX-XXXX-fig1-ref1"/> of wiring harness W420 from J2 connector
<callout assocfig="T00001-X-XXX-XXXX-fig1" label="2"
numref="T00001-X-XXX-XXXX-fig1-ref2"/> on engine compartment fan
<callout assocfig="T00001-X-XXX-XXXX-fig1" label="3"
numref="T00001-X-XXX-XXXX-fig1-ref3"/> .
<figure application="both" figtype="normal-page" pane="no" tocentry="1"
id="T00001-X-XXX-XXXX-fig1">
<title>Engine Compartment Fan Connections
</title>
<graphic boardno="obj.2" reprodep="4" reprowid="3" unitmeasure="in">
<mapref id="T00001-X-XXX-XXXX-fig1-ref1" label="1" refdes="P2">
<map.circle radius="0.25" unitmeasure="in">
<map.coord unitmeasure="in" x="1" y="1"/>
</map.circle>
</mapref>
<mapref id="T00001-X-XXX-XXXX-fig1-ref2" label="2" refdes="J2">
<map.circle radius="0.25" unitmeasure="in">
<map.coord unitmeasure="in" x="2" y="0.25"/>
</map.circle>
</mapref>
<mapref id="T00001-X-XXX-XXXX-fig1-ref3" label="3">
<map.circle radius="0.25" unitmeasure="in">
<map.coord unitmeasure="in" x="1.5" y="4"/>
</map.circle>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</mapref>
</graphic>
</figure>
</para>
</step1>
```

(4) Example XML markup in relationship to Figure 303.

```
<step1 esd="no" frame="yes" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">
<link application="frame" linkaction="prompt" linktype="return"
popup="no" xlink:href="T00021-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Connect test probes to test equipment.
</prompt>
</link>
</para>
</step1>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">
<link application="frame" linkaction="prompt" linktype="return"
popup="no" xlink:href="T00022-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Connect oscilloscope/spectrum analyzer to test equipment.
</prompt>
</link>
</para>
</step1>
<step1 assocfig="T00001-X-XXX-XXXX-fig1" esd="no" frame="no" hcp="no"
qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Connect oscilloscope/spectrum analyzer
signal reference wire to pin A and test probe to pin B
on P2 connector of wiring harness W420
<link application="frame" linkaction="prompt" linktype="goto"
local="T00001-X-XXX-XXXX-GM39151-S" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="figure">
<pretext> refer to
</pretext>
<prompt>schematic
</prompt>
</link> .
</para>
</step1>
```

(5) Example XML markup in relationship to Figure 304.

```
<step1 esd="no" frame="yes" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">
<link application="frame" linkaction="prompt" linktype="return"
popup="no" xlink:href="O00354-X-XXX-XXXX-" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Perform APU start-up.
</prompt>
</link>
</para>
</step1>
<diagnostic_group action="prompt">
<message button_title="OK">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<title>Perform  Test
</title>
<messageline>Ready to Perform Oscilloscope Test.
</messageline>
<messageline>Configure oscilloscope if required and take readings.
</messageline>
</message>
<diagnostic application="PSW32.EXE" testname="PICO">
<desc>Oscilloscope  Test
</desc>
</diagnostic>
</diagnostic_group>
<interaction>
<dialog popup="no">
<binarymenu answer="yesno">
<prompt>Is waveform a square wave with an average value of 28 V peak,
cycle period length of 55 milliseconds (ms) and frequency of 1.8 kHz?
</prompt>
<yesstate>
<statemanipulation>
<variableref name="answer"/>
<expression>
<boolean>
<true/>
</boolean>
</expression>
</statemanipulation>
</yesstate>
<nostate>
<statemanipulation>
<variableref name="answer"/>
<expression>
<boolean>
<false/>
</boolean>
</expression>
</statemanipulation>
</nostate>
</binarymenu>
</dialog>
</interaction>
<evaluate>
<if>
<expression>
<variableref name="answer"/>
</expression>
<then>
<resultwithstate>
<statemanipulation>
<variableref name="AT02A001-F1"/>
<fault fault-code="AT020A001-F1" fault-state="ok">
</fault>
</statemanipulation>
<statemanipulation>
<variableref name="AT02A001-F2"/>
<fault fault-code="AT02A001-F2" fault-state="ok">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</fault>
</statemanipulation>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A0014" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</then>
```

(6) Example XML markup in relationship to Figure 306.

```
<else>
<resultwithstate>
<fault fault-code="AT02A001-F1" fault-state="bad">
</fault>
<interaction>
<message button_title="OK" popup="no">
<title>FAULT - AT02A001-F1
</title>
<messageline>Replace blower motor.
</messageline>
<aftermessage>
<link application="frame" linkaction="immediate" linktype="return"
popup="no" xlink:href="M00121-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
</link>
</aftermessage>
</message>
</interaction>
<statemanipulation>
<variableref name="AT02A001-F1"/>
<fault fault-code="ATO2A001-F" fault-state="bad">
</fault>
</statemanipulation>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A00121" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</else>
</if>
</evaluate>
</testwithstate>
```

(7) Example XML markup in relationship to Figure 305.

```
<testwithstate id="T00001-X-XXX-XXXX-AT02A0014">
<reasonfortest>
<title>Reason for Test
</title>
<para esd="no" hcp="no">An open circuit on wires 166, 167 or 168 in
wiring harness W420 could cause circuit failure.
</para>
</reasonfortest>
<statemanipulation>
<variable name="AT02A0014-EV-182" scope="global" value-type="integer">
</variable>
</statemanipulation>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<step1 assocfig="T00001-X-XXX-XXXX-AT02A0014-fig1" esd="no" frame="no"
hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Disconnect P2 connector
<callout assocfig="T00001-X-XXX-XXXX-AT02A0014-fig1" label="4"
numref="T00001-X-XXX-XXXX-fig1-ref4"/> of wiring harness W420 from J3 connector
<callout assocfig="T00001-X-XXX-XXXX-AT02A0014-fig1" label="4"
numref="T00001-X-XXX-XXXX-fig1-ref4"/> on APU control panel
<callout assocfig="T00001-X-XXX-XXXX-AT02A0014-fig1" label="6"
numref="T00001-X-XXX-XXXX-fig1-ref6"/> .
<figure application="both" figtype="normal-page" pane="no" tocentry="1"
id="T00001-X-XXX-XXXX-AT02A0014-fig1">
<title>APU Control Panel Connections
</title>
<graphic boardno="obj.8" unitmeasure="in">
<mapref id="T00001-X-XXX-XXXX-AT02A0014-fig1-ref4" label="4" refdes="P2">
<map.circle radius="0.25" unitmeasure="in">
<map.coord unitmeasure="in" x="1" y="1"/>
</map.circle>
</mapref>
<mapref id="T00001-X-XXX-XXXX-AT02A0014-fig1-ref5" label="5" refdes="J2">
<map.circle radius="0.25" unitmeasure="in">
<map.coord unitmeasure="in" x="2" y="0.25"/>
</map.circle>
</mapref>
<mapref id="T00001-X-XXX-XXXX-AT02A0014-fig1-ref6" label="6">
<map.circle radius="0.25" unitmeasure="in">
<map.coord unitmeasure="in" x="1.5" y="4"/>
</map.circle>
</mapref>
<graphic>
</figure>
</para>
</step1>
```

(8) The remaining Example XML markup.

```
<step1 esd="no" frame="yes" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Connect red and black test probes together and hold.
<figure application="both" figtype="normal-page" pane="no" tocentry="1"
id="T00001-X-XXX-XXXX-AT02A0014-fig2">
<title>Connect Red and Black Test Probes
</title>
<graphic boardno="obj.9" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
<diagnostic_group action="prompt">
<message button_title="OK">
<messageline>Ready to Perform Offset
</messageline>
<messageline>AND
</messageline>
<messageline>Release test probes
</messageline>
</message>
<diagnostic protocol="DODACOFFSET" testname="91">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<desc> Offset
</desc>
<sendparameter>
<string>J4
</string>
</sendparameter>
<sendparameter>
<name>REPEAT
</name>
<string>SINGLE
</string>
</sendparameter>
<receiveparameter>
<variableref name="AT02A00140EV0182"/>
</receiveparameter>
</diagnostic>
</diagnostic_group>
<evaluate>
<if>
<expression>
<expression>
<integer>-255
</integer>
<le/>
<variableref name="AT02A0014-EV-182"/>
</expression>
<and/>
<expression>
<variableref name="AT02A0014-EV-182"/>
<ge/>
<integer>255
</integer>
</expression>
</expression>
<then>
<resultwithstate>
<statemanipulation>
<variableref name="AT02A001-F1"/>
<fault fault-code="AT02A001-F11" fault-state="ok">
</fault>
</statemanipulation>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A0016" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</then>
<else>
<resultwithstate>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A0017" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</else>
</if>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</evaluate>
</testwithstate>
<testwithstate id="T00001-X-XXX-XXXX-AT02A0016">
<reasonfortest>
<title>Reason for Test
</title>
<para esd="no" hcp="no">An open circuit on wire 166 in wiring harness
W420 could cause engine compartment fan failure.
</para>
</reasonfortest>
<statemanipulation>
<variable name="AT02A0016-EV-184" scope="global" value-type="integer">
</variable>
</statemanipulation>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Connect red test probe to pin A at P2
connector of wiring harness W420 and connect black test probe
to pin A at P1 connector of wiring harness W420
<link application="frame" linkaction="prompt" linktype="undefined"
local="T00001-X-XXX-XXXX-GM39151-S" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="figure">
<pretext> refer to
</pretext>
<prompt>schematic
</prompt>
</link> .
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.10" unitmeasure="in">
</graphic>
</figure>
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.11" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
<diagnostic_group action="prompt">
<message button_title="OK">
<messageline>Ready to Perform Test #91
</messageline>
</message>
<diagnostic protocol="DODACTEST" testname="91">
<desc>Test #91
</desc>
<sendparameter>
<string>J4
</string>
</sendparameter>
<sendparameter>
<name>REPEAT
</name>
<string>LOOP
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</string>
</sendparameter>
<receiveparameter>
<variableref name="AT02A0016-EV-184"/>=
</receiveparameter>
</diagnostic>
</diagnostic_group>
<evaluate>
<if>
<expression>
<variableref name="AT02A0016-EV-184"/>
<lt/>
<integer>5
</integer>
</expression>
<then>
<resultwithstate>
<statemanipulation>
<variableref name="AT02A001-F1"/>
<fault fault-code="AT02A001-F1" fault-state="bad">
</fault>
</statemanipulation>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A0018" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</then>
<else>
<resultwithstate>
<interaction>
<message button_title="OK" popup="no">
<messageline>Replace wiring harness W420.
</messageline>
<aftermessage>
<link application="frame" linkaction="immediate" linktype="return"
popup="no" xlink:href="M00131-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
</link>
</aftermessage>
</message>
</interaction>
<statemanipulation>
<variableref name="AT02A001-F2"/>
<fault fault-code="ATO2A001-F2" fault-state="bad">
</fault>
</statemanipulation>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A0018" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</else>
</if>
</evaluate>
</testwithstate>
```


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Check the source to verify that this is the current version before use.

<testwithstate id="T00001-X-XXX-XXXX-AT02A0017">
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Click on Reset button to return to main menu.
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.12" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
<interaction>
<message button_title="RESET" popup="no">
<messageline>Test equipment has failed offset test. Problem could be a faulty test probe or internal test box failure.
</messageline>
<messageline>Notify your supervisor.
</messageline>
</message>
</interaction>
<completed_test>
</completed_test>
</testwithstate>
<testwithstate id="T00001-X-XXX-XXXX-AT02A0018">
<reasonfortest>
<title>REASON FOR TEST
</title>
<para esd="no" hcp="no">An open circuit on wire 167 in wiring harness W420 could cause engine compartment fan failure.
</para>
</reasonfortest>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Connect red test probe to pin B at P2 connector of wiring harness W420 and connect black test probe to pin B at P1 connector of wiring harness W420
<link application="frame" linkaction="prompt" linktype="goto" local="T00001-X-XXX-XXXX-GM39151-S" popup="no" xlink:type="simple"xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="figure">
<pretext> refer to
</pretext>
<prompt>schematic
</prompt>
</link>.
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.13" unitmeasure="in">
</graphic>
</figure>
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.14" unitmeasure="in">
</graphic>
</figure>
</para>

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```
</step1>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Click on Perform Test # 91 button.
</para>
</step1>
<diagnostic_group action="prompt">
<message button_title="OK">
<messageline>Ready to Perform Test #91
</messageline>
</message>
<diagnostic protocol="DODACTEST" testname="91">
<desc>DODACTEST
</desc>
<sendparameter>
<name>INPUT
</name>
<string>J4
</string>
</sendparameter>
<sendparameter>
<name>REPEAT
</name>
<string>LOOP
</string>
</sendparameter>
<receiveparameter>
<variableref name="AT02A0018-AV-186"/>
</receiveparameter>
</diagnostic>
</diagnostic_group>
<evaluate>
<if>
<expression>
<variableref name="AT002A0018-EV-186"/>
<lt/>
<integer>5
</integer>
</expression>
<then>
<resultwithstate>
<statemanipulation>
<variableref name="AT02A001-F1"/>
<fault fault-code="AT02A001-F1" fault-state="bad">
</fault>
</statemanipulation>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A00110" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</then>
<else>
<resultwithstate>
<interaction>
<message button_title="OK" popup="no">
<messageline>Replace wiring harness W420.
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</messageline>
<aftermessage>
<link application="frame" linkaction="prompt" linktype="return"
popup="no" xlink:href="M00131-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<returnlink>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A00121" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</returnlink>
</link>
</aftermessage>
</message>
</interaction>
</resultwithstate>
</else>
</if>
</evaluate>
</testwithstate>
<testwithstate id="T00001-X-XXX-XXXX-AT02A00110">
<reasonfortest>
<title>REASON FOR TEST
</title>
<para esd="no" hcp="no">An open circuit on wire 168 in wiring harness
W420 could cause engine compartment fan failure.
</para>
</reasonfortest>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Connect red test probe to pin C at P2
connector of wiring harness W420 and connect black test probe
to pin C at P1 connector of wiring harness W420
<link application="frame" linkaction="prompt" linktype="goto"
local="T00001-X-XXX-XXXX-GM39151-S" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="figure">
<pretext> refer to
</pretext>
<prompt>schematic
</prompt>
</link>
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.15" unitmeasure="in">
</graphic>
</figure>
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.16" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Click on Perform Test # 91 button
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</para>
</step1>
<diagnostic_group action="prompt">
<message button_title="OK">
<messageline>Ready to Perform Test #91
</messageline>
</message>
<diagnostic protocol="DODACTEST" testname="91">
<desc>DODACTEST
</desc>
<sendparameter>
<name>INPUT
</name>
<string>J4
</string>
</sendparameter>
<sendparameter>
<name>REPEAT
</name>
<string>LOOP
</string>
</sendparameter>
<receiveparameter>
<variableref name="AT02A0018-AV-188"/>
</receiveparameter>
</diagnostic>
</diagnostic_group>
<evaluate>
<if>
<expression>
<variableref name="AT02A0018-AV-188"/>
<lt/>
<integer>5
</integer>
</expression>
<then>
<resultwithstate>
<statemanipulation>
<variableref name="AT02A001-F2"/>
<fault fault-code="AT02A001-F2" fault-state="ok">
</fault>
</statemanipulation>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A00112" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</then>
<else>
<resultwithstate>
<interaction>
<message button_title="OK" popup="no">
<messageline>Replace wiring harness W420.
</messageline>
<aftermessage>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<link application="frame" linkaction="prompt" linktype="return"
popup="no" xlink:href="M00131-X-XXX-XXXX" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
<returnlink>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A00121" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</returnlink>
</link>
</aftermessage>
</message>
</interaction>
</resultwithstate>
</else>
</if>
</evaluate>
</testwithstate>
<testwithstate id="T00001-X-XXX-XXXX-AT02A00112">
<reasonfortest>
<title>REASON FOR TEST
</title>
<para esd="no" hcp="no">A short circuit on wire 166 in wiring harness
W420 could cause engine compartment fan failure.
</para>
</reasonfortest>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Connect red and black test probes together and hold.
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.17" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Click on Perform Offset button and release test probes.
</para>
</step1>
<diagnostic_group action="prompt">
<message button_title="TEST">
<messageline>Ready to Perform Offset
</messageline>
</message>
<diagnostic protocol="DODACOFFSET" testname="91">
<desc>Perform Offset
</desc>
<sendparameter>
<name>INPUT
</name>
<string>J4
</string>
</sendparameter>
<sendparameter>
<name>REPEAT
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</name>
<string>SINGLE
</string>
</sendparameter>
<receiveparameter>
<variableref name="AT02A00112-EV-190"/>
</receiveparameter>
</diagnostic>
</diagnostic_group>
<evaluate>
<if>
<expression>
<expression>
<integer>-255
</integer>
<lt/>
<variableref name="AT02A00112-EV-190"/>
</expression>
<and/>
<expression>
<variableref name="AT02A00112-EV-190"/>
<gt/>
<integer>255
</integer>
</expression>
</expression>
<then>
<resultwithstate>
<interaction>
<message button_title="OK" popup="no">
<messageline>Fault Corrected
</messageline>
</message>
</interaction>
<completed_test>
</completed_test>
</resultwithstate>
</then>
<else>
<resultwithstate>
<statemanipulation>
<variableref name="AT02A001-F2"/>
<fault fault-code="AT02A001-F2" fault-state="bad">
</fault>
</statemanipulation>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A00115" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</else>
</if>
</evaluate>
</testwithstate>
<testwithstate id="T00001-X-XXX-XXXX-AT02A00115">
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<para esd="no" hcp="no">Click on Reset button to return to main menu.
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.19" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
<interaction>
<message button_title="RESET" popup="no">
<messageline>Test equipment has failed offset test. Problem could be a faulty test probe or internal test box failure. Notify your supervisor.
</messageline>
<aftermessage>
<link application="frame" linkaction="immediate" linktype="goto" popup="no" xlink:href="X-XXX-XXXX#COVER" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="document">
</link>
</aftermessage>
</message>
</interaction>
</testwithstate>
<testwithstate id="T00001-X-XXX-XXXX-AT02A00121">
<reasonfortest>
<title>REASON FOR TEST
</title>
<para esd="no" hcp="no">To verify that original problem is solved.
</para>
</reasonfortest>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">
<link application="frame" linkaction="prompt" linktype="return" popup="no" xlink:href="M00354-X-XXX-XXXX" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Perform APU start-up.
</prompt>
</link>
</para>
</step1>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check operation of engine compartment fan.
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.22" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">
<link application="frame" linkaction="prompt" linktype="return" popup="no" xlink:href="M00355-X-XXX-XXXX" xlink:type="simple" xmlns:xlink="http://www.w3.org/1999/xlink">
<prompt>Perform APU shutdown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</prompt>
</link>
</para>
</step1>
<interaction>
<dialog  popup="no">
<binarymenu  answer="yesno">
<prompt>Is original problem solved?
</prompt>
<yesstate>
<statemanipulation>
<variableref  name="answer"/>
<expression>
<boolean>
<true/>
</boolean>
</expression>
</statemanipulation>
</yesstate>
<nostate>
<statemanipulation>
<variableref  name="answer"/>
<expression>
<boolean>
<false/>
</boolean>
</expression>
</statemanipulation>
</nostate>
</binarymenu>
</dialog>
</interaction>
<evaluate>
<if>
<expression>
<variableref  name="answer"/>
</expression>
<then>
<resultwithstate>
<interaction>
<message  button_title="OK"  popup="no">
<messageline>Remove  wheel  chocks.
</messageline>
<aftermessage>
<link  application="frame"  linkaction="prompt"  linktype="return"
popup="no"  xlink:href="O00131-X-XXX-XXXX"  xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink">
</link>
</aftermessage>
</message>
</interaction>
<completed_test>
</completed_test>
</resultwithstate>
</then>
<else>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<resultwithstate>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A00123" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</resultwithstate>
</else>
</if>
</evaluate>
</testwithstate>
<testwithstate id="T00001-X-XXX-XXXX-AT02A00123">
<step1 esd="no" frame="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Click on Restart button to troubleshoot symptom.
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>
</title>
<graphic boardno="obj.24" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
<interaction>
<message button_title="RESTART" popup="no">
<messageline>Failure condition has not been fixed.
</messageline>
<messageline>Troubleshoot symptom until failure is fixed.
</messageline>
<messageline>If problem persists, notify your supervisor.
</messageline>
<aftermessage>
<link application="frame" linkaction="immediate" linktype="goto"
local="T00001-X-XXX-XXXX-AT02A0011" popup="no" xlink:type="simple"
xmlns:xlink="http://www.w3.org/1999/xlink" xreftype="test">
</link>
</aftermessage>
</message>
</interaction>
</testwithstate>
</diagnosticwp>
```

b. Frame-based TM stylesheet output example for ***<diagnosticwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

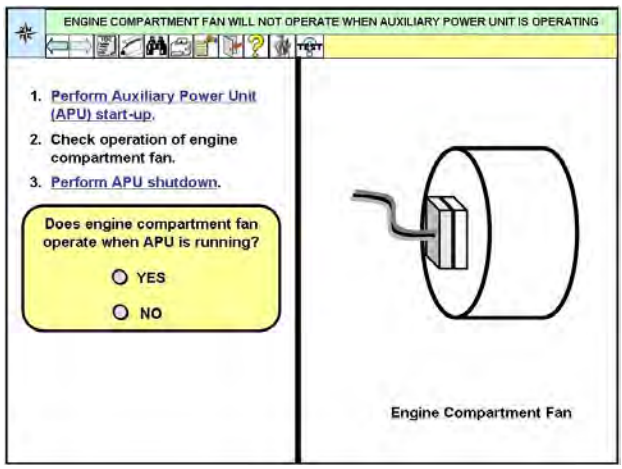

FIGURE 300. Diagnostic work package frame-based – Initial frame.

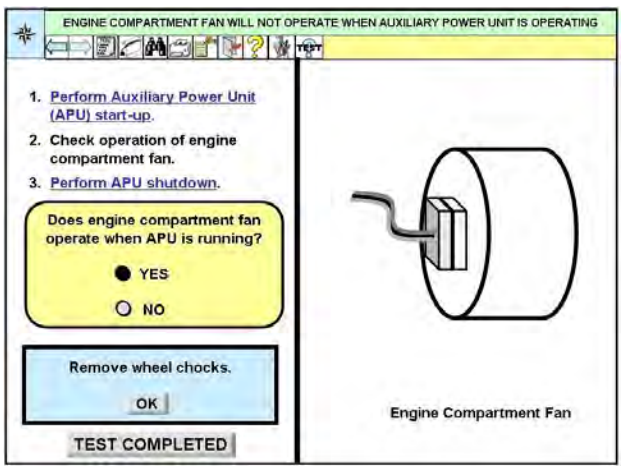

FIGURE 301. Diagnostic work package frame-based – YES response.

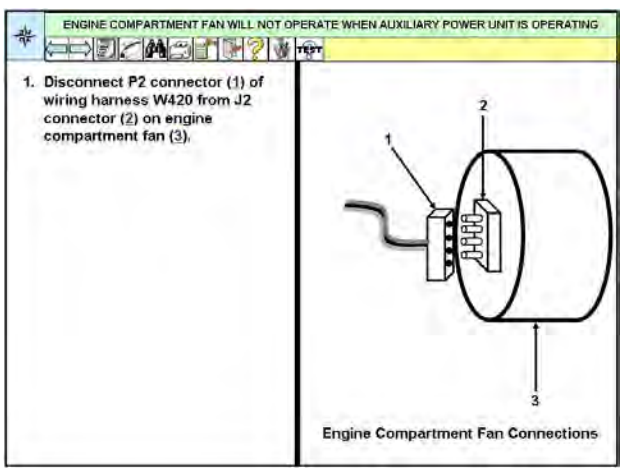

FIGURE 302. Diagnostic work package frame-based – NO response.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

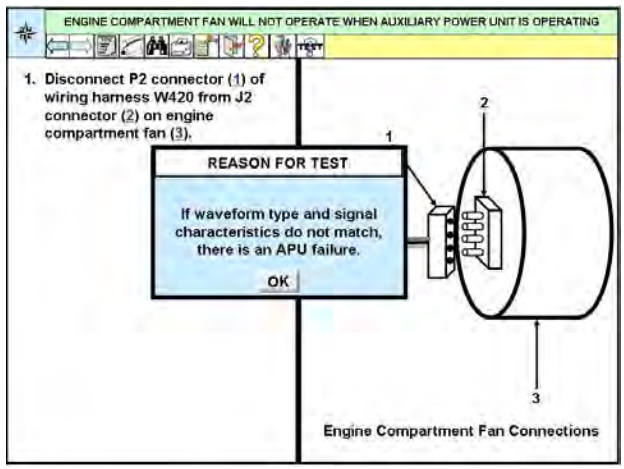

FIGURE 302. Diagnostic work package frame-based – NO response - Continued.

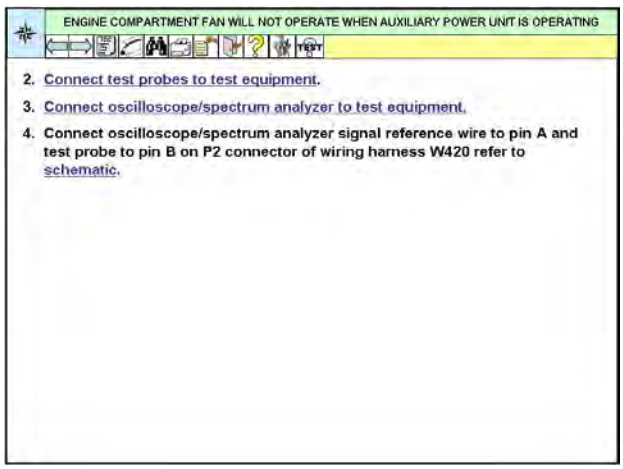

FIGURE 303. Diagnostic work package frame-based – NO response next frame.

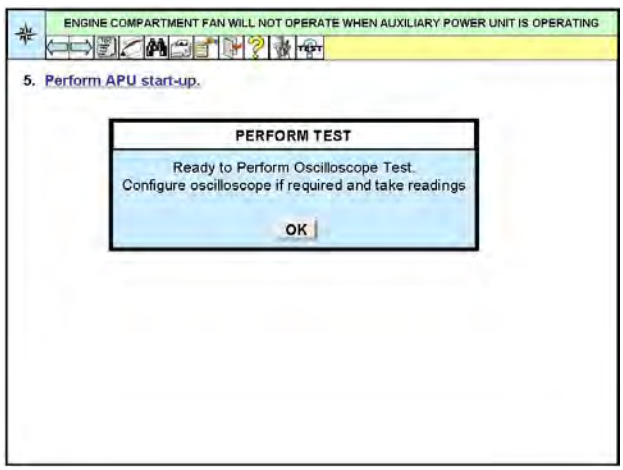

FIGURE 304. Diagnostic work package frame-based – external test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

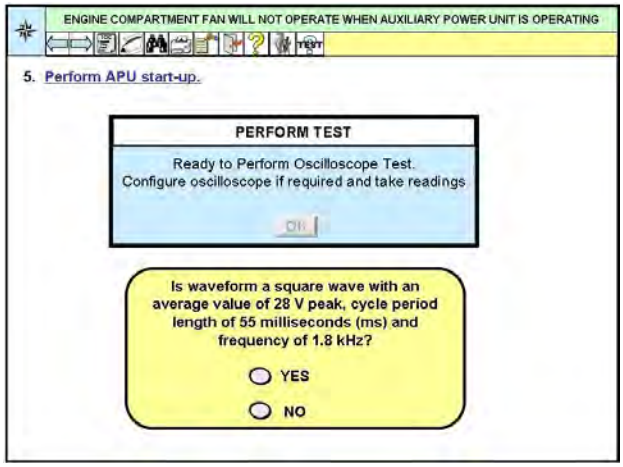

FIGURE 304. Diagnostic work package frame-based – external test - Continued.

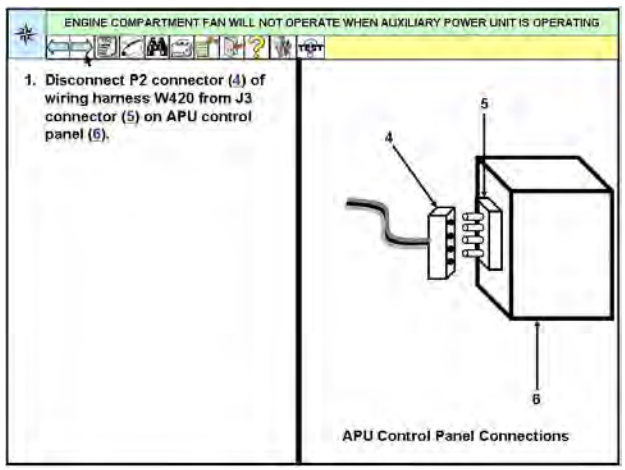

FIGURE 305. Diagnostic work package frame-based — YES response.

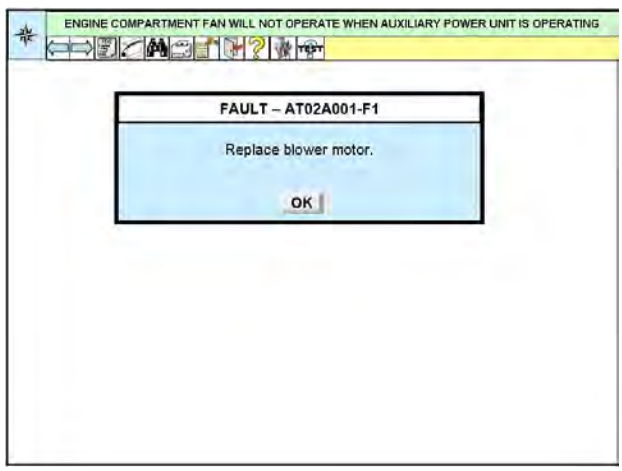

FIGURE 306. Diagnostic work package frame-based — NO response.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

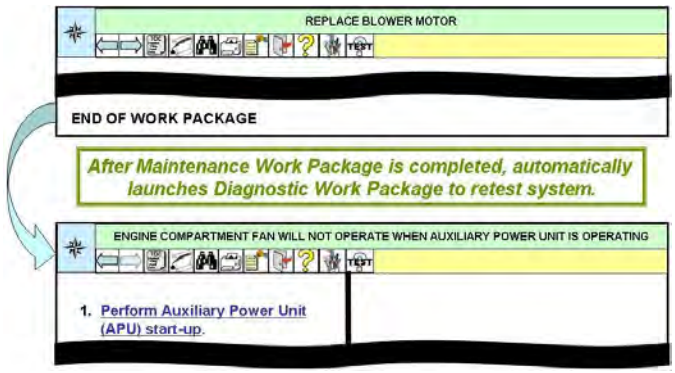

FIGURE 306. Diagnostic work package frame-based — NO response - Continued.

36.12 Preshop analysis work package ***<pshopanalwp>***). The preshop analysis work package is prepared when inspection or testing is more effective in determining useful life of a system, subsystem, or component than a disassembly.

a. The components of ***<pshopanalwp>*** are:

(1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
(2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
(3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
(4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(7) Scope ***<scope>*** (required) is the purpose and coverage of the preshop analysis.
(8) Preparation Procedures ***<proc>*** (required – one or more) (see 33.2) contains the following procedural information of Unpacking and special handling; Checking attached documents; External inspection; and Cleaning and preservation. Each preparation procedure uses a separate ***<proc>***element.
(9) Preshop analysis procedures ***<pshopanal>*** (required) contains how to perform the analysis as either
  (a) Narrative procedure ***<proc>*** (required) (see 33.2).
  (b) Checklist ***<chklist>*** (required) that contains
    1 A blank cover sheet ***<coverpage>*** (required) to detail information about the equipment and identify the person performing the checklist.
    2 Introduction information ***<intro>*** (optional) (see 49.1.4.15) to provide a brief explanation of its use.
    3 Table of test and inspection ***<pshopanal.tab>***/***<table>*** (required) (see 42) contains as a minimum an entry for each test and inspection procedure with inspection point (the item or area to be inspected), condition, action, remarks, and identification of the personnel performing the inspection. Each component in the table would be a column in the table.

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b. DTD fragment for ***<pshopanalwp>***:

```
<!ELEMENT pshopanalwp (wp.metadata?, wpidinfo, wpinfo, (warning*, caution*,
    note*), scope, proc+, pshopanal)>
<!ATTLIST pshopanalwp
   airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
<!ELEMENT pshopanal (proc | chklist)>
<!ATTLIST pshopanal
    assocfig          IDREFS                           #IMPLIED
    changeref         IDREFS                           #IMPLIED
    comment           CDATA                            #IMPLIED
    delchlvl          CDATA                            #IMPLIED
    id                ID                               #IMPLIED
    idref             IDREFS                           #IMPLIED
    inschlvl          CDATA                            #IMPLIED
    security          (uc | fouo |c | s | ts)          #IMPLIED
    skilltrk          CDATA                            #IMPLIED
>
<!ELEMENT chklist (coverpage, intro?, pshopchk.tab)>
<!ATTLIST chklist
    assocfig          IDREFS                           #IMPLIED
    changeref         IDREFS                           #IMPLIED
    comment           CDATA                            #IMPLIED
    delchlvl          CDATA                            #IMPLIED
    id                ID                               #IMPLIED
    idref             IDREFS                           #IMPLIED
    inschlvl          CDATA                            #IMPLIED
    security          (uc | fouo |c | s | ts)          #IMPLIED
    skilltrk          CDATA                            #IMPLIED
>
<!ELEMENT pshopchk.tab (table)>
```

c. DTD fragment is graphically depicted for Preshop Analysis Work Package ***<pshopanalwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

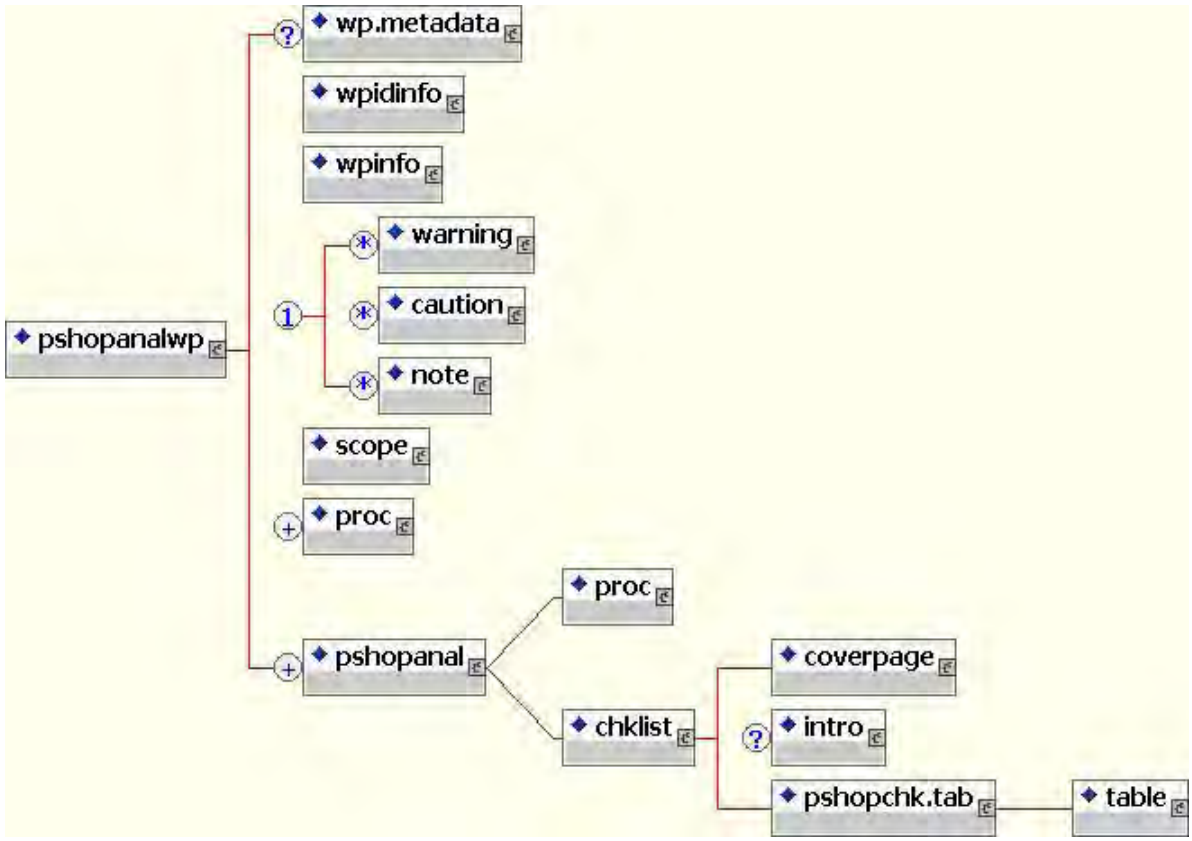

FIGURE 307. Preshop analysis work package DTD hierarchy <pshopanalwp>..

d. Common attributes for ***<pshopanalwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

36.12.1 XML document instance fragment and output for ***<pshopanalwp>***. The XML instance and its stylesheet output for a ***<pshopanalwp>*** is provided below.

a. Example of an XML document instance fragment for ***<pshopanalwp>***:

```
<pshopanalwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="M0003-X-XXXX-XXX" wpseq="0003">
<wpidinfo>
<maintlvl level="depot"/>
<title>Preshop Analysis Procedures
</title>
</wpidinfo>
<wpinfo>
<title>Not Applicable
</title>
<null insert="none"/>
</wpinfo>
<scope>
<title>Scope
</title>
<para>The purpose of the preshop analysis operations is to determine,
at the highest assembly level possible, the work required to
return the Electrical Equipment RL/RIU Rack to a condition
specified under the Scope of this DMWR found in
<xref wpid="G0001-X-XXXX-XXX" posttext=", General Information"/>. If inspection at
the highest assembly level is precluded by missing, damaged, or defective
components, inspection will proceed at the next lower level. The preshop
analysis checklist will be used to record the results of the analysis
and any required maintenance. The preshop analysis checklists are to be
reproduced locally, as required, for recording preshop analysis checks.
</para>
</scope>
<proc>
<title>Unpacking and Special Handling
</title>
<para>Remove the end item, assembly, subassembly, or component from
the container, using care not to damage or change the configuration
of the item. The control measures are those defined in
<xref wpid="G0001-X-XXXX-XXX" posttext=", General Information"/>.
</para>
</proc>
<proc>
<title>Checking Attached Documents
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</title>
*<para>*Check all tags and forms attached to the item to determine reason for removal from service or other discrepancies. Do not remove tags until pertinent information has been placed on appropriate work documents.
</para>
</proc>
<proc>
*<title>*External Inspection
</title>
*<para>*Make an external inspection to determine the completeness of the item, noting any damaged or missing parts. Record the results of the external inspection, using the preshop analysis checklist outlined in this WP.
</para>
</proc>
<proc>
*<title>*Cleaning and Preservation
</title>
*<para>*If cleaning is required to perform the preshop analysis, clean only as necessary to make the analysis. Cleaning will be performed in accordance with
<xref wpid="M0005-X-XXXX-XXX"/> and
<xref wpid="M0007-X-XXXX-XXX"/>.
</para>
</proc>
<proc>
*<title>*Preshop Analysis Test
</title>
*<para>*The test that may be required to perform preshop analysis is listed in
<xref tableid="M0003-X-XXXX-XXX-TABLE1"/>.
</para>
</proc>
<pshopanal>
<chklist>
<coverpage>
<partno>
</partno>
<serialno>
</serialno>
<nsn>
<fsc>
</fsc>
<niin>
</niin>
</nsn>
<modreq>
</modreq>
<reason>
</reason>
<secitem>
</secitem>
<revtag>
</revtag>
<revform>
</revform>
<name>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</name>
<sig>
</sig>
<date>
</date>
</coverpage>
<intro>
<para0>
<title>Preshop Analysis Checklist
</title>
<para>The purpose of each column in the preshop analy-
sis checklist is as follows:
<randlist bullet="yes">
<item>Inspection Point. Indicates the inspection and test necessary
to determine required overhaul maintenance tasks.
</item>
<item>Reference. Indicates the paragraph within the body of this
DMWR or the document where the inspection/test is located.
</item>
<item>Condition. Provides space for maintenance personnel to
list the results of the inspections/tests.
</item>
<item>Action. Provides space for maintenance personnel to list
the maintenance tasks required for overhaul.
</item>
<item>Person Performing Analysis (Sign and Date). Provides
space for maintenance personnel to sign and date upon
completion of each inspection/test.
</item>
</randlist>
</para>
</para0>
</intro>
<pshopchk.tab>
<table id="M0003-X-XXXX-XXX-TABLE1">
<title>Test and Inspection - Electrical Equipment RL/RIU
Rack 11465907 or 11478033
</title>
<tgroup cols="5">
<colspec colname="col1" colwidth="1*"/>
<colspec colname="col2" colwidth="0.66*"/>
<colspec colname="col3" colwidth="1.33*"/>
<colspec colname="col4" colwidth="1.33*"/>
<colspec colname="col5" colwidth="1*"/>
<thead>
<row>
<entry align="center" valign="bottom">INSPECTION AREA
</entry>
<entry align="center" valign="bottom">REFERENCE
</entry>
<entry align="center" valign="bottom">CONDITION
</entry>
<entry align="center" valign="bottom">ACTION
</entry>
<entry align="center" valign="bottom">PERSON PERFORMING ANALYSIS (SIGN AND DATE)
</entry>
```


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Check the source to verify that this is the current version before use.

```
</row>
</thead>
<tbody>
<row>
<entry>Modifications
</entry>
<entry>
<xref  wpid="M00015-X-XXXX-XXX"/>
</entry>
<entry>
</entry>
<entry>
</entry>
<entry>
</entry>
</row>
<row>
<entry>Review of Records and Data
</entry>
<entry>
<xref  wpid="M0003-X-XXXX-XXX"/>
<brk/>
<xref  wpid="M0005-X-XXXX-XXX"/>
<brk/>
<xref  wpid="M0007-X-XXXX-XXX"/>
</entry>
<entry>
</entry>
<entry>
</entry>
<entry>
</entry>
</row>
<row>
<entry>Corrosion
</entry>
<entry>
<xref  wpid="M0003-X-XXXX-XXX"/>
<brk/>
<xref  wpid="M0005-X-XXXX-XXX"/>
<brk/>
<xref  wpid="M0007-X-XXXX-XXX"/>
</entry>
<entry>
</entry>
<entry>
</entry>
<entry>
</entry>
</row>
<row>
<entry>Damaged Connector
</entry>
<entry>
<xref  wpid="M0003-X-XXXX-XXX"/>
<brk/>
```

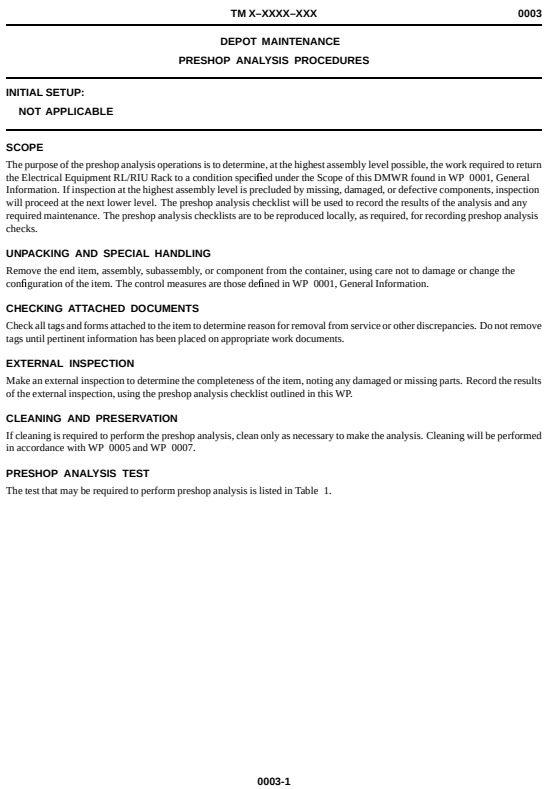

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<xref wpid="M0005-X-XXXX-XXX"/>
<brk/>
<xref wpid="M0007-X-XXXX-XXX"/>
</entry>
</entry>
<entry>
</entry>
<entry>
</entry>
<entry>
</entry>
</row>
<row>
<entry>Damaged or Missing Hardware
</entry>
<entry>
<xref wpid="M0003-X-XXXX-XXX"/>
<brk/>
<xref wpid="M0005-X-XXXX-XXX"/>
<brk/>
<xref wpid="M0007-X-XXXX-XXX"/>
</entry>
<entry>
</entry>
<entry>
</entry>
<entry>
</entry>
</row>
<row>
<entry>Damaged Wiring
</entry>
<entry>
<xref wpid="M0003-X-XXXX-XXX"/>
<brk/>
<xref wpid="M0005-X-XXXX-XXX"/>
<brk/>
<xref wpid="M0007-X-XXXX-XXX"/>
</entry>
<entry>
</entry>
<entry>
</entry>
<entry>
</entry>
</row>
<row>
<entry>Testing/Performance Check
</entry>
<entry>
<xref wpid="M0003-X-XXXX-XXX"/>
<brk/>
<xref wpid="M0005-X-XXXX-XXX"/>
<brk/>
<xref wpid="M0007-X-XXXX-XXX"/>
</entry>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<entry>
</entry>
<entry>
</entry>
<entry>
</entry>
</row>
</tbody>
</tgroup>
</table>
</pshopchk.tab>
</chklist>
</pshopanal>
</pshopanalwp>

b. Page-based TM stylesheet output example for ***<pshopanalwp>***:



TM X–XXXX–XXX 0003

DEPOT MAINTENANCE
PRESHOP ANALYSIS PROCEDURES

INITIAL SETUP:
NOT APPLICABLE

SCOPE

The purpose of the preshop analysis operations is to determine, at the highest assembly level possible, the work required to return the Electrical Equipment RL/RIU Rack to a condition specified under the Scope of this DMWR found in WP 0001, General Information. If inspection at the highest assembly level is precluded by missing, damaged, or defective components, inspection will proceed at the next lower level. The preshop analysis checklist will be used to record the results of the analysis and any required maintenance. The preshop analysis checklists are to be reproduced locally, as required, for recording preshop analysis checks.

UNPACKING AND SPECIAL HANDLING

Remove the end item, assembly, subassembly, or component from the container, using care not to damage or change the configuration of the item. The control measures are those defined in WP 0001, General Information.

CHECKING ATTACHED DOCUMENTS

Check all tags and forms attached to the item to determine reason for removal from service or other discrepancies. Do not remove tags until pertinent information has been placed on appropriate work documents.

EXTERNAL INSPECTION

Make an external inspection to determine the completeness of the item, noting any damaged or missing parts. Record the results of the external inspection, using the preshop analysis checklist outlined in this WP.

CLEANING AND PRESERVATION

If cleaning is required to perform the preshop analysis, clean only as necessary to make the analysis. Cleaning will be performed in accordance with WP 0005 and WP 0007.

PRESHOP ANALYSIS TEST

The test that may be required to perform preshop analysis is listed in Table 1.

0003-1

FIGURE 308. Example of a page-based TM stylesheet output for <pshopanalwp>.

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TM X–XXXX–XXX 0003

PRESHOP ANALYSIS FOR P/N _____

Serial No. _____

NSN _____

MWOs Required _____

Reason(s) for Overhaul/Repair _____

_____

_____

_____

Unpacking Secondary Items Required? _____

Reviewed Tags? _____

Reviewed Forms? _____

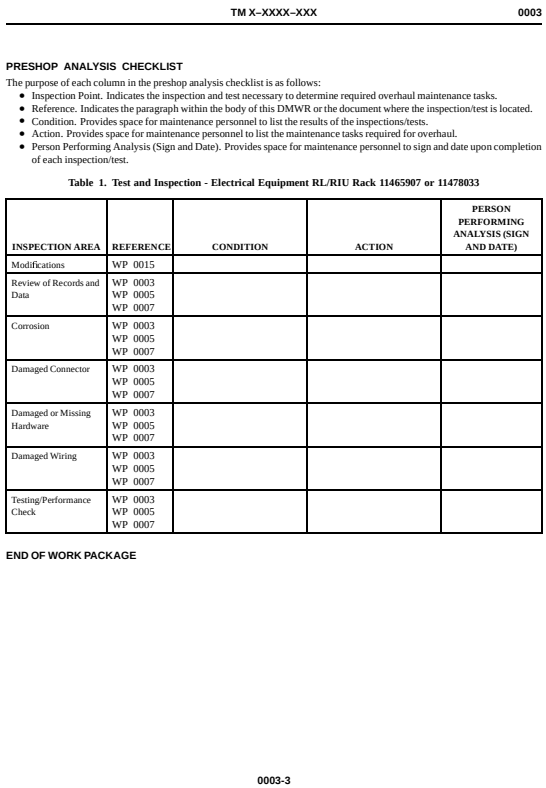
Name (please print) _____

Signature _____ Date _____

0003-2

FIGURE 308. Example of a page-based TM stylesheet output for <pshopanalwp> - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.



TM X–XXXX–XXX 0003

PRESHOP ANALYSIS CHECKLIST

The purpose of each column in the preshop analysis checklist is as follows:

- Inspection Point. Indicates the inspection and test necessary to determine required overhaul maintenance tasks.
- Reference. Indicates the paragraph within the body of this DMWR or the document where the inspection/test is located.
- Condition. Provides space for maintenance personnel to list the results of the inspections/tests.
- Action. Provides space for maintenance personnel to list the maintenance tasks required for overhaul.
- Person Performing Analysis (Sign and Date). Provides space for maintenance personnel to sign and date upon completion of each inspection/test.

Table 1. Test and Inspection - Electrical Equipment RL/RIU Rack 11465907 or 11478033

INSPECTION AREA	REFERENCE	CONDITION	ACTION	PERSON PERFORMING ANALYSIS (SIGN AND DATE)
Modifications	WP 0015			
Review of Records and Data	WP 0003 WP 0005 WP 0007			
Corrosion	WP 0003 WP 0005 WP 0007			
Damaged Connector	WP 0003 WP 0005 WP 0007			
Damaged or Missing Hardware	WP 0003 WP 0005 WP 0007			
Damaged Wiring	WP 0003 WP 0005 WP 0007			
Testing/Performance Check	WP 0003 WP 0005 WP 0007			

END OF WORK PACKAGE

0003-3

FIGURE 308. Example of a page-based TM stylesheet output for <pshopanalwp> - Continued.

36.13 Component checklist work package ***<compchklistwp>***. The element is used in support of preshop analysis procedures. Generally the element ***<compchklist>*** contains no data, but prescribes the fields the stylesheet is generate.

a. The components of ***<compchklistwp>*** are:

(1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
(2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
(3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
(4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(7) Introduction information ***<intro>*** (optional) (see 49.1.4.15) to provide a brief explanation of its use.
(8) Component checklist ***<compchklist>*** (required) contains the depot level checklist for a component.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

b. DTD fragments are:

```
<!ELEMENT compchklistwp (wp.metadata?, wpidinfo, wpinfo, intro,
     compchklist)>
<!ATTLIST compchklistwp
   airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>

<!ELEMENT compchklistwp (name, serialno?, daterec?, recfrom?, compname?,
nsn?, (partno, cageno)*, qty?, qtyrec?, damage?)>
<!ATTLIST compchklistwp
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. DTD fragment is graphically depicted for Component Checklist Work Package ***<compchklistwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

FIGURE 309. Component checklist work package DTD hierarchy <compchklistwp>.

d. Common attributes for ***<compchklistwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

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(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

36.13.1 XML document instance fragment and output for ***<compchklistwp>***. The XML instance and its stylesheet output for a ***<compchklistwp>*** is provided below.

a. Example of an XML document instance fragment for ***<compchklistwp>***:

```
<compchklistwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="t00002-X-XXXX-XXX" wpseq="0080">
<wpidinfo>
<maintlvl level="depot"/>
<title>Component Checklist
</title>
</wpidinfo>
<wpinfo>
<title>Not Applicable
</title>
<null insert="none"/>
</wpinfo>
<intro frame="no">
<para0 esd="no" hcp="no">
<title>Scope
</title>
<para esd="no" hcp="no">This work package includes a list which is to
be copied for each item received for a preshop analysis. After
copying one list for each item, the information required must be
completed on the checklist prior to the preshop analysis.
</para>
</para0>
</intro>
<compchklist>
<name>
</name>
<serialno>
</serialno>
```


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<daterec>
</daterec>
<recfrom>
</recfrom>
<compname>
</compname>
<nsn>
<fsc>
</fsc>
<niin>
</niin>
</nsn>
<qty>
</qty>
<qtyrec>
</qtyrec>
<damage>
</damage>
</compchklist>
</compchklistwp>

b. Page-based TM stylesheet output example for ***<compchklistwp>***:

TM X–XXXX–XXX 0080

DEPOT MAINTENANCE
COMPONENT CHECKLIST

INITIAL SETUP:
NOT APPLICABLE

SCOPE
This work package includes a list which is to be copied for each item received for a preshop analysis. After copying one list for each item, the information required must be completed on the checklist prior to the preshop analysis.

COMPONENT CHECKLIST

Name/nomenclature of the equipment/item _____

Serial number _____

Date received _____

Received from (identify unit) _____

Component name _____

NSN _____

Part number _____

Quantity required _____

Quantity received _____

Visual damage found _____

END OF WORK PACKAGE

0080-1

FIGURE 310. Example of a page-based TM stylesheet output for <compchklistwp>.

36.14 Aviation troubleshooting introduction work package ***<tsintrowp>***. The work package describes the "How to Use" this manual for Aviation Troubleshooting TM (-T). The work package describes the

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testing and troubleshooting processes, general methods used to perform testing and troubleshooting, general flow of troubleshooting processes, and any peculiar electrical subsystems and electronic equipment troubleshooting information. When a troubleshooting index (refer to 36.6) is used, a description and explanation of the index is included.

a. The components of ***<tsintrowp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (5) Describe the "How to Use" this manual for Aviation Troubleshooting TM, select a method.
      (a) Titled paragraph ***<para0>***/***<para0–alt>*** (required – one or more) (see 49.1.1.8).
      (b) Paragraph ***<para>*** (required) (see 49.1.1.5).
      (c) How to Use ***<howtouse>*** (required) (see 31.3.1.1.2).

b. DTD fragment is:

```
<!ELEMENT tsintrowp (wp.metadata?, wpidinfo, geninfo?, ((para0 |
   para0-alt+)+ | para | howtouse))>
<!ATTLIST tsintrowp
    airforce         (yes | no )               "no"
    army             (yes | no )               "no"
    assocfig         IDREFS                     #IMPLIED
    changeref        IDREFS                     #IMPLIED
    comment          CDATA                      #IMPLIED
    crewmember       CDATA                      #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                      #IMPLIED
    deletewp         (yes | no )               "no"
    fgc              CDATA                      #IMPLIED
    frame            (yes | no )               "yes"
    idref            IDREFS                     #IMPLIED
    inschlvl         CDATA                      #IMPLIED
    insertwp         CDATA                      #IMPLIED
    lsa-id           CDATA                      #IMPLIED
    marines          (yes | no )               "no"
    navy             (yes | no )               "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                      #IMPLIED
    tocentry         (2 | 3 | 4 | 5)           "2"
    wpno             ID                         #REQUIRED
    wpseq            CDATA                      #IMPLIED
>
```

c. DTD fragment is graphically depicted for Aviation troubleshooting introduction work package ***<tsintrowp>***:

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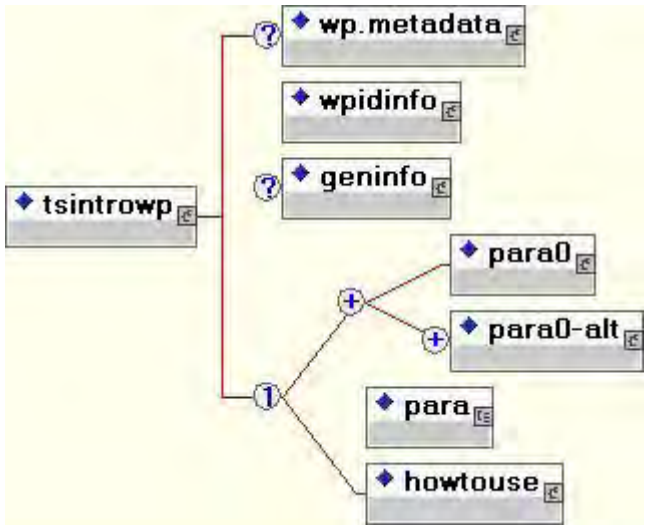

FIGURE 311. Aviation Troubleshooting Introduction Work Package DTD hierarchy <tsintrowp>.

d. Common attributes for ***<tsintrowp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).

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(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

36.15 Aviation troubleshooting technical description work package (***<techdescwp>***. The troubleshooting technical description work package is used only for Aviation Troubleshooting TM (-T). The work package provides for three components that are derived from equipment description work package (see 34.1.2), control/indicators work packages (see 35.2), and theory of operation work package (see 34.1.3).

a. The components of ***<techdescwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (8) At lease one of the technical description procedures (equipment description ***<descproc>***, control/indicators ***<ctrlindproc>***, and/or theory of operation ***<thryproc>***) is needed.
b. DTD fragment is graphically depicted for Aviation Troubleshooting Technical Description Work Package ***<techdescwp>***:

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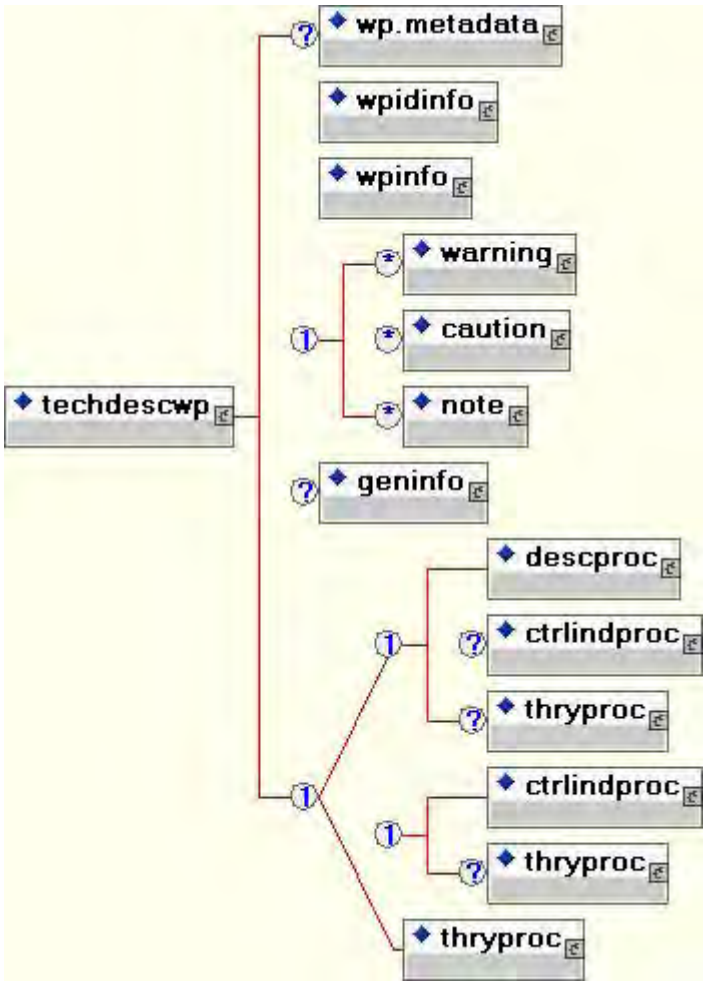

FIGURE 312. Aviation troubleshooting technical description work package DTD hierarchy <techdescwp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment is:

```
<!ELEMENT techdescwp (wp.metadata?, wpidinfo, wpinfo, (warning*, caution*,
     note*), geninfo?, ((descproc, ctrlindproc?, thryproc?) |
     (ctrlindproc, thryproc?) | thryproc))>
<!ATTLIST techdescwp
   airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes for ***<techdescwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37 MAINTENANCE INSTRUCTIONS CHAPTER *<MIM>*.

37.1 Maintenance instructions ***<mim>***. Maintenance instruction chapters are prepared and subdivided into individual work packages that provide maintenance information to enable a technician to receive, process, inspect, clean, service, test and repair the weapon system/equipment and associated weapons replacement assemblies/shop replacement assemblies (WRAs/SRAs) to an acceptable performance standard. The maintenance instruction chapter has twelve (12) maintenance instruction chapter category types. Each category provides the precise structural functionality that is defined in MIL–STD–40051–1/–2. Multiple chapters can be assigned to a maintenance category depending on the type and complexity of the weapon system/equipment. An individual chapter, however, can only be assigned to one maintenance instruction category. The XML elements designate the components to be presented within the individual category, chapter, or work package. Some elements within a chapter or work package are unique to the respective chapter or work package while other elements are common to one or more categories. Each chapter and work package is better defined and identified by the addition of element attributes which are contained in the DTD fragment.

a. The components of a maintenance instruction chapter ***<mim>*** are:
(1) Title page ***<titlepg>*** (required) (see 49.1.1.1).
(2) One of the following maintenance categories, in accordance with MIL–STD–40051–1/–2, is required:
(a) Preventive maintenance checks and services (PMCS) category ***<pmcscategory>*** (see 37.2.1).
(b) General maintenance with PMCS category ***<maintenancepmcscategory>*** (see 37.2.2).
(c) General maintenance category ***<maintenancecategory>*** (see 37.2.3).
(d) Depot maintenance category ***<depotcategory>*** (see 37.2.4).
(e) Aircraft maintenance category ***<aviationcategory>*** (see 37.2.6).
(f) Ammunition maintenance category ***<ammunitioncategory>*** (see 37.2.9).
(g) Auxiliary equipment maintenance category ***<auxiliarycategory>*** (see 37.2.5).
(h) Preventive maintenance services (PMS) category ***<pmscategory>*** (see 37.2.7).
(i) Phased maintenance inspections (PMI) category ***<checklistcategory>*** (see 37.2.8).
(j) Test and inspection category ***<testinspectioncategory>*** (see 37.2.10).
(k) Ammunition marking maintenance category ***<ammomarkingcategory>*** (see 37.2.11).
(l) Shipment/movement and storage maintenance ***<shipmentmovementstoragecategory>*** (see 37.2.12).

b. DTD fragment is graphically depicted for ***<mim>***:

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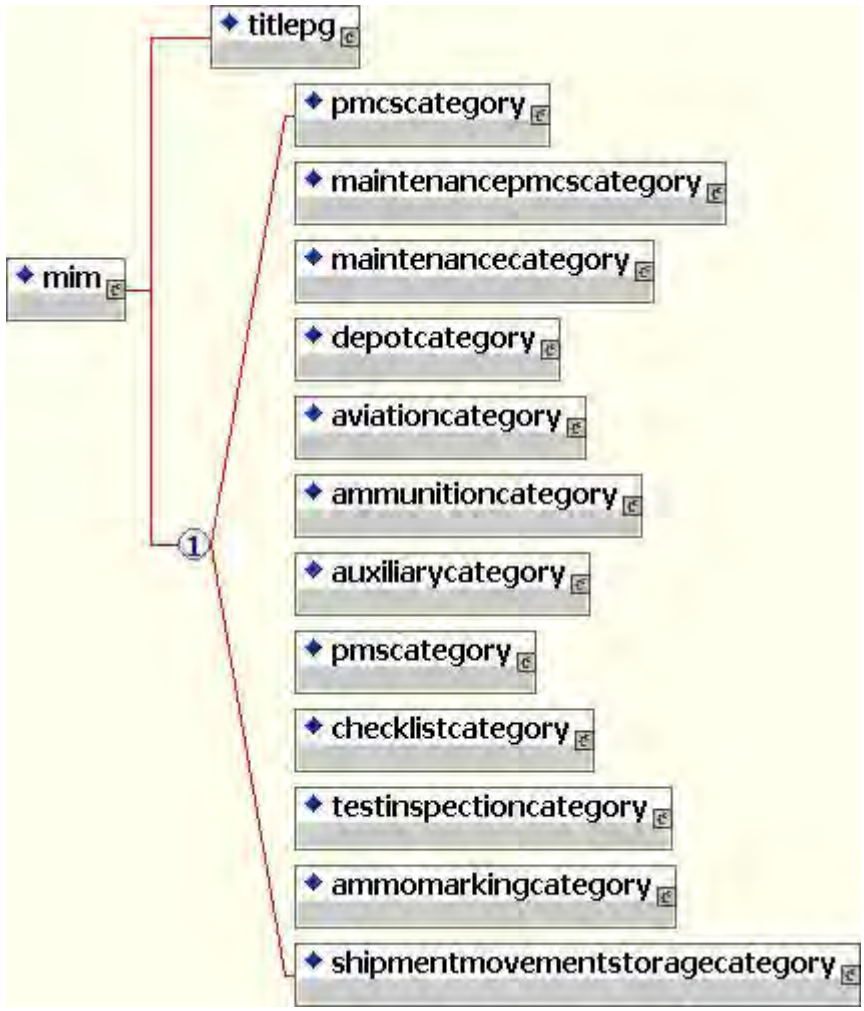

FIGURE 313. Maintenance instructions chapter DTD hierarchy <mim>.

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c. DTD fragment:

```
<!ELEMENT mim (titlepg, (pmcscategory | maintenancepmcscategory |
   maintenancecategory | depotcategory | aviationcategory |
   ammunitioncategory | auxiliarycategory | pmscategory |
   checklistcategory | testinspectioncategory |
   ammomarkingcategory | shipmentmovementstoragecategory))>
<!ATTLIST mim
    assocfig  IDREFS                    #IMPLIED
    changeref IDREFS                    #IMPLIED
    chap-toc  (yes | no )               "yes"
    chngno    CDATA                     #REQUIRED
    comment   CDATA                     #IMPLIED
    delchlvl  CDATA                     #IMPLIED
    frame     (yes | no )               "yes"
    idref     IDREFS                    #IMPLIED
    inschlvl  CDATA                     #IMPLIED
    revno     CDATA                     #REQUIRED
    security  (uc | fouo | c | s | ts )  #IMPLIED
    skilltrk  CDATA                     #IMPLIED
    tocentry  (0 | 1 | 2)               "1"
>
```

d. Common attributes for ***<mim>***.

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **chap-toc** – Create a chapter work package table of contents (optional). Select either "**yes**" or "**no**" with a default value "**yes**".
(4) **chngno** – Change number (required) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(8) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(9) **inschlvl** – Insert change level (optional) (see 49.2.9).
(10) **revno** – Revision number (required) (see 49.2.9).
(11) **security** – Security classification (optional) (see 49.2.13).
(12) **skilltrk** – Skill level (optional) (see 49.2.2).
(13) **tocentry** – In the TM table of contents, the attribute indicates the indenture level. The possible selections are "**0**" (do not include), "**1**" (the highest level or 1st indenture level), or "**2**" (the second indentrue level) (default value is "**1**").

37.2 Maintenance categories. Depending on the type and complexity of the weapon system/equipment, the IETM or TM contains one or more of the following maintenance category chapters and work packages (within the selected categories). In accordance with MIL–STD–40051–1/–2, the content matrices prescribes the optional and/or required maintenance categories and work packages for each technical manual type. The acquiring activity has the responsibility in selecting the maintenance category and the work packages within each category.

37.2.1 Preventive maintenance checks and services (pmcs) category ***<pmcscategory>***. PMCS is based upon the principles of Reliability Centered Maintenance logic. PMCS includes the necessary periodic lubrication instructions, and applicable scheduled corrosion inspections required to maintain operational equipment in good operating order. The PMCS category applies only to weapon systems, equipment, and applicable components, but does not apply to aircraft TMs, DMWRs, and NMWRs. If the complexity of the weapon system/equipment is extensive, multiple PMCS categories can be developed for each maintenance

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level(s). Additionally with a PMCS category, multiple PMCS can be developed for various conditions (i.e., before use, after use, after 5,000 miles, etc.)

a. The work packages for ***<pmcscategory>*** are:
   (1) PMCS introduction work package ***<pmcsintrowp>*** (required) (see 37.5).
   (2) PMCS work package ***<pmcswp>*** (required – one or more) (see 37.6).

b. DTD fragment is graphically depicted for PMCS Category ***<pmcscategory>***:

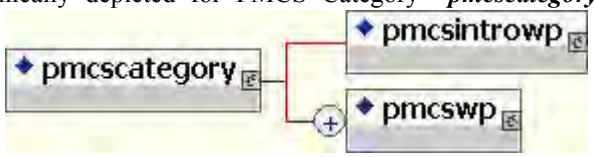

FIGURE 314. PMCS category DTD hierarchy <pmcscategory>.

c. DTD fragment for ***<pmcscategory>***:

```
<!ELEMENT pmcscategory (pmcsintrowp, pmcswp+)>
```

37.2.2 General maintenance with PMCS category ***<maintenancepmcscategory>***. The general maintenance with PMCS category contains the necessary work packages with servicing, PMCS, and assembly instructions required to place a piece of equipment or weapon system into service. This category applies only to weapon systems, equipment, and applicable components, but does not apply to aircraft TMs, DMWRs, and NMWRs. Depending on the complexity of the system, multiple general maintenance categories may be developed to separate information into major end items and/or maintenance levels.

a. The work packages for ***<maintenancepmcscategory>*** are:
   (1) Service upon receipt work package ***<surwp>*** (optional – zero or more) (see 37.3).
   (2) Equipment/user fitting instruction work package ***<perseqpwp>*** (optional – zero or more) (see 37.4).
   (3) PMCS introduction work package ***<pmcsintrowp>*** (required) (see 37.5).
   (4) PMCS work package(s) ***<pmcswp>*** (required – one or more) (see 37.6).
   (5) The following work packages occur in no specific order and require at least one to be developed:
      (a) General maintenance work package ***<gen.maintwp>*** (see 37.8).
      (b) Specific maintenance work package ***<maintwp>*** (see 37.7).
      (c) Lubrication instructions work package ***<lubewp>*** (see 37.9).
   (6) Illustrated list of manufactured items work package ***<manuwp>*** (optional) (see 37.14).
   (7) Torque limits work package ***<torquewp>*** (optional) (see 37.15).
   (8) Wiring diagrams work package ***<wiringwp>*** (optional – zero or more) (see 37.16).

b. DTD fragment is graphically depicted for Maintenance with PMCS Category ***<maintenancepmcscategory>***:

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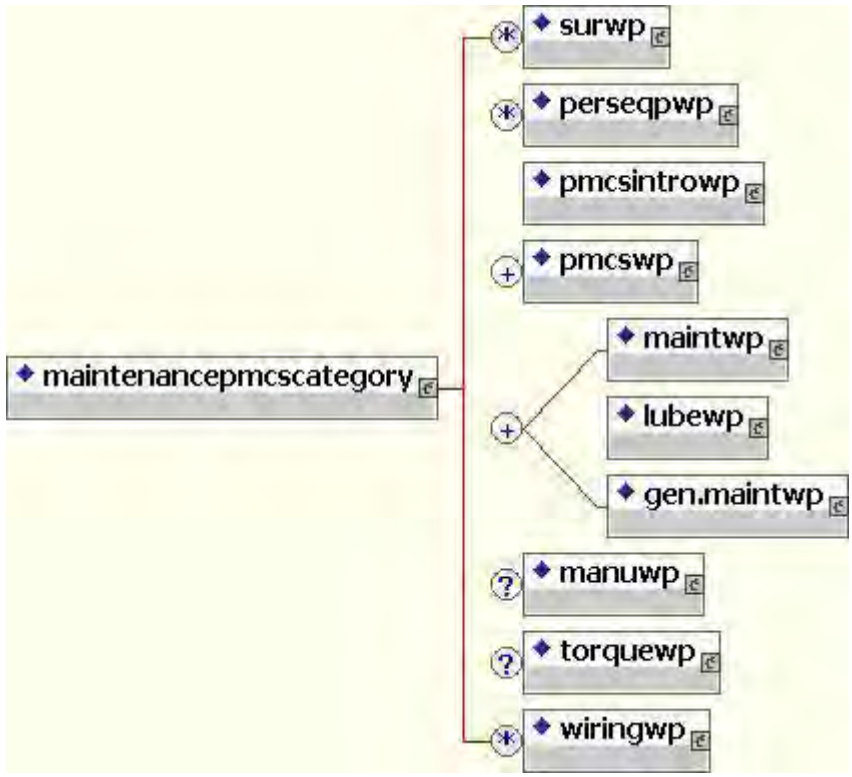

FIGURE 315. Maintenance with PMCS category DTD hierarchy <maintenancepmcscategory>.

c. DTD fragment for ***<maintenancepmcscategory>***:

```
<!ELEMENT maintenancepmcscategory (surwp*, perseqpwp*, pmcsintrowp,
   pmcswp+, (gen.maintwp | maintwp | lubewp)+, manuwp?, torquewp?,
   wiringwp*)>
```

37.2.3 General maintenance category ***<maintenancecategory>***. The general maintenance category contains weapon system/equipment maintenance repair task work packages sufficient to ensure repair of the weapon system/equipment. Maintenance repair tasks are identified in the applicable Maintenance Allocation Chart (MAC). The developer ensures either the PMCS category or general maintenance with PMCS category is developed in accordance with MIL–STD–40051–1/–2. This category applies only to weapon systems, equipment, and applicable components, but does not apply to aircraft TMs, DMWRs, and NMWRs. Depending on the complexity of the system multiple general maintenance categories may be developed to separate information into major end items and/or maintenance levels.

a. The work packages for ***<maintenancecategory>*** are:
   (1) Service upon receipt work package ***<surwp>*** (optional – zero or more) (see 37.3).
   (2) Equipment/user fitting instruction work package ***<perseqpwp>*** (optional – zero or more) (see 37.4).
   (3) The following work packages occur in no specific order and require at least one to be developed:
      (a) General maintenance work package ***<gen.maintwp>*** (see 37.8).
      (b) Specific maintenance work package ***<maintwp>*** (see 37.7).
      (c) Lubrication instructions work package ***<lubewp>*** (see 37.9).
   (4) Illustrated list of manufactured items work package ***<manuwp>*** (optional) (see 37.14).
   (5) Torque limits work package ***<torquewp>*** (optional) (see 37.15).
   (6) Wiring diagrams work package ***<wiringwp>*** (optional – zero or more) (see 37.16).

b. DTD fragment is graphically depicted for General Maintenance Category ***<maintenancecategory>***:

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Check the source to verify that this is the current version before use.

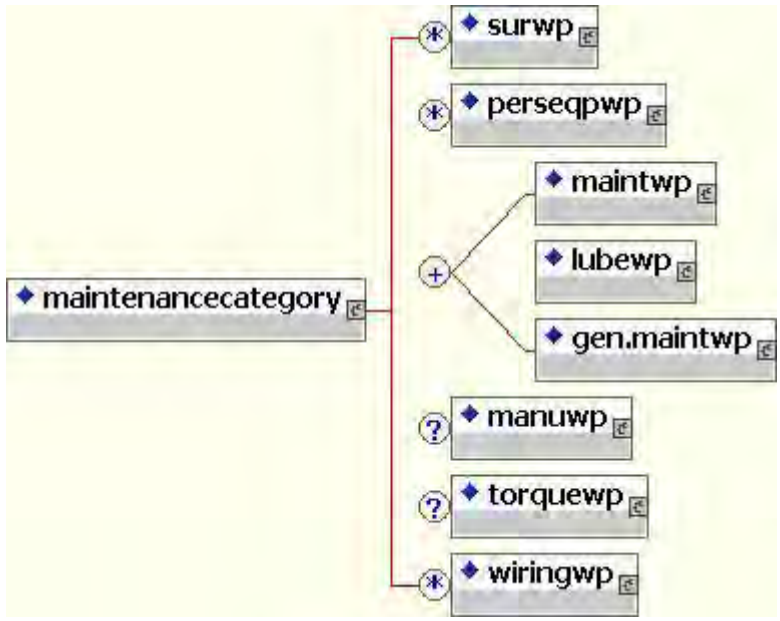

FIGURE 316. General maintenance category DTD hierarchy <maintenancecategory>.

c. DTD fragment for ***<maintenancecategory>***:

```
<!ELEMENT maintenancecategory (surwp*, perseqpwp*, (gen.maintwp | maintwp
   | lubewp)+, manuwp?, torquewp?, wiringwp*)>
```

37.2.4 Depot maintenance category ***<depotcategory>***. The depot maintenance category contains weapon system/equipment maintenance repair task work packages sufficient to ensure complete repair of the weapon system/equipment. Depot maintenance repair tasks are identified in the applicable Maintenance Allocation Chart (MAC). Depending on the complexity of the system multiple general maintenance categories may be developed to separate information into major end items and/or maintenance levels.

a. The work packages for ***<depotcategory>*** are:
   (1) The following work packages occur in no specific order and require at least one to be developed:
      (a) General maintenance work package ***<gen.maintwp>*** (see 37.8).
      (b) Specific maintenance work package ***<maintwp>*** (see 37.7).
      (c) Lubrication instructions work package ***<lubewp>*** (see 37.9).
   (2) Facilities work package ***<facilwp>*** (optional) (see 37.10).
   (3) Overhaul inspection procedures (OIP) work package ***<oipwp>*** (optional – zero or more) (see 37.11).
   (4) Depot mobilization requirements work package ***<mobilwp>*** (optional) (see 37.12).
   (5) Quality assurance (QA) requirements work package ***<qawp>*** (required) (see 37.13).
   (6) Illustrated list of manufactured parts work package ***<manuwp>*** (optional) (see 37.14).
   (7) Torque limits work package ***<torquewp>*** (optional) (see 37.15).
   (8) When aircraft depot specific work packages are developed, the following work packages are available (optional).
      (a) Aircraft inventory master guide work package ***<inventorywp>*** (optional) (see 37.18).
      (b) Storage of aircraft work package ***<storagewp>*** (optional – zero or more) (see 37.19).
      (c) Weighing and loading work package ***<wtloadwp>*** (required) (see 37.20).
   (9) Wiring diagrams work package ***<wiringwp>*** (optional – zero or more) (see 37.16).

b. DTD fragment is graphically depicted for Depot Maintenance Category ***<depotcategory>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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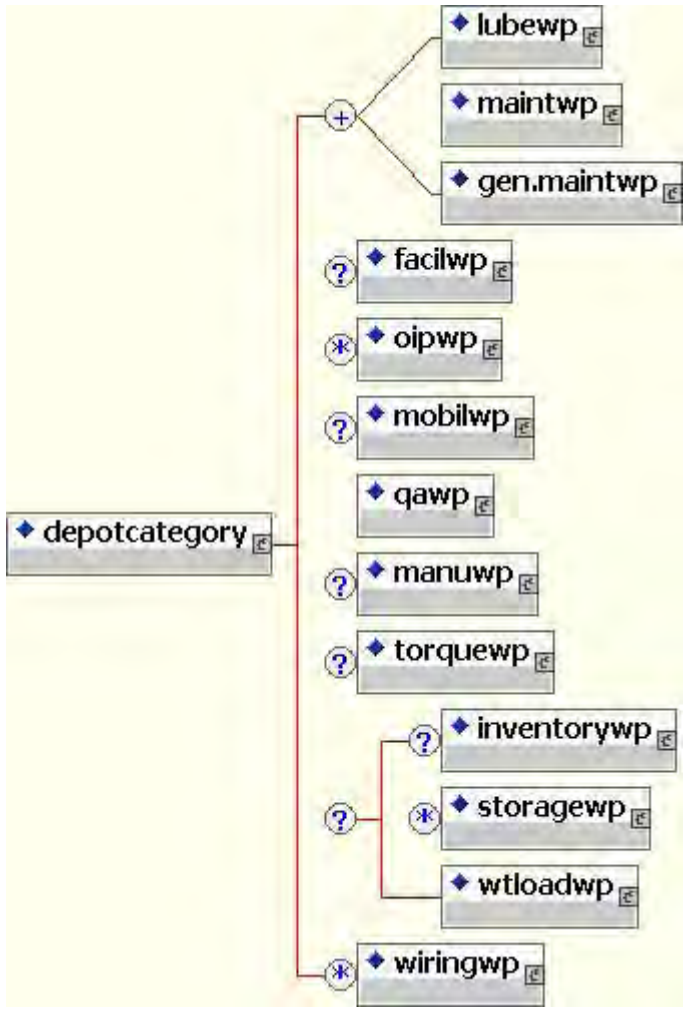

FIGURE 317. Depot maintenance category DTD hierarchy <depotcategory>.

c. DTD fragment for ***<depotcategory>***:

```
<!ELEMENT depotcategory (surwp*, perseqpwp* , (lubewp | maintwp |
   gen.maintwp)+ , facilwp? , oipwp* , mobilwp? , qawp , manuwp? ,
   torquewp? , (inventorywp? , storagewp* , wtloadwp)? , wiringwp*)>
```

37.2.5 Auxiliary equipment maintenance category ***<auxiliarycategory>***. The auxiliary equipment maintenance category is used to prepare maintenance instructions when auxiliary equipment maintenance TMs or maintenance requirement cards have not been procured.

a. The work packages for ***<auxiliarycategory>*** are:
   (1) Auxiliary equipment maintenance work package(s) ***<auxeqpwp>*** (required – one or more) (see 37.21).
   (2) Illustrated list of manufactured parts work package ***<manuwp>*** (optional) (see 37.14).
   (3) Torque limits work package ***<torquewp>*** (optional) (see 37.15).
   (4) Wiring diagrams work package ***<wiringwp>*** (optional – zero or more) (see 37.16).

b. DTD fragment is graphically depicted for Auxiliary Equipment Maintenance Category ***<auxiliarycategory>***:

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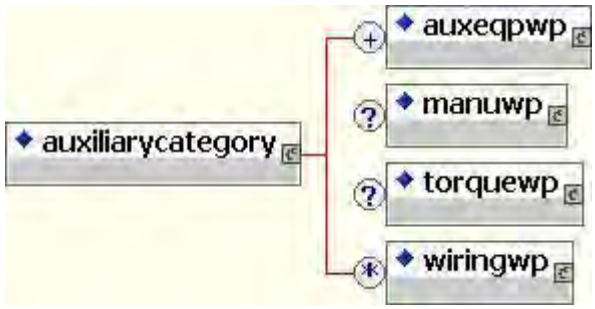

FIGURE 318. Auxiliary equipment maintenance category DTD hierarchy <auxiliarycategory>.

c. DTD fragment for ***<auxiliarycategory>***:

```
<!ELEMENT auxiliarycategory (auxeqpwp+, manuwp?, torquewp?, wiringwp*)>
```

37.2.6 Aircraft maintenance category ***<aviationcategory>***. The aircraft maintenance category contains aircraft weapon system/equipment maintenance repair task work packages sufficient to ensure repair of the aircraft weapon system/equipment and return the aircraft to an operational status. Aviation maintenance repair tasks are identified in the applicable aircraft Maintenance Allocation Chart (MAC).

a. The work packages for ***<aviationcategory>*** are:
   (1) Service upon receipt work package ***<surwp>*** (optional – zero or more) (see 37.3).
   (2) Equipment/user fitting instruction work package ***<perseqpwp>*** (optional – zero or more) (see 37.4).
   (3) The following work packages occur in no specific order and require at least one to be developed:
      (a) Preventive maintenance inspections work package ***<pmiwp>*** (see 37.17).
      (b) Lubrication instructions work package ***<lubewp>*** (see 37.9).
      (c) Specific maintenance work package ***<maintwp>*** (see 37.7).
      (d) General maintenance work package ***<gen.maintwp>*** (see 37.8).
   (4) Illustrated list of manufactured parts work package ***<manuwp>*** (optional) (see 37.14).
   (5) Torque limits work package ***<torquewp>*** (optional) (see 37.15).
   (6) Aircraft inventory master guide work package ***<inventorywp>*** (optional) (see 37.18).
   (7) Storage of aircraft work package ***<storagewp>*** (optional – zero or more) (see 37.19).
   (8) Weighing and loading work package ***<wtloadwp>*** (required) (see 37.20).
   (9) Wiring diagrams work package ***<wiringwp>*** (optional – zero or more) (see 37.16).

b. DTD fragment is graphically depicted for Aircraft Maintenance Category ***<aviationcategory>***:

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Check the source to verify that this is the current version before use.

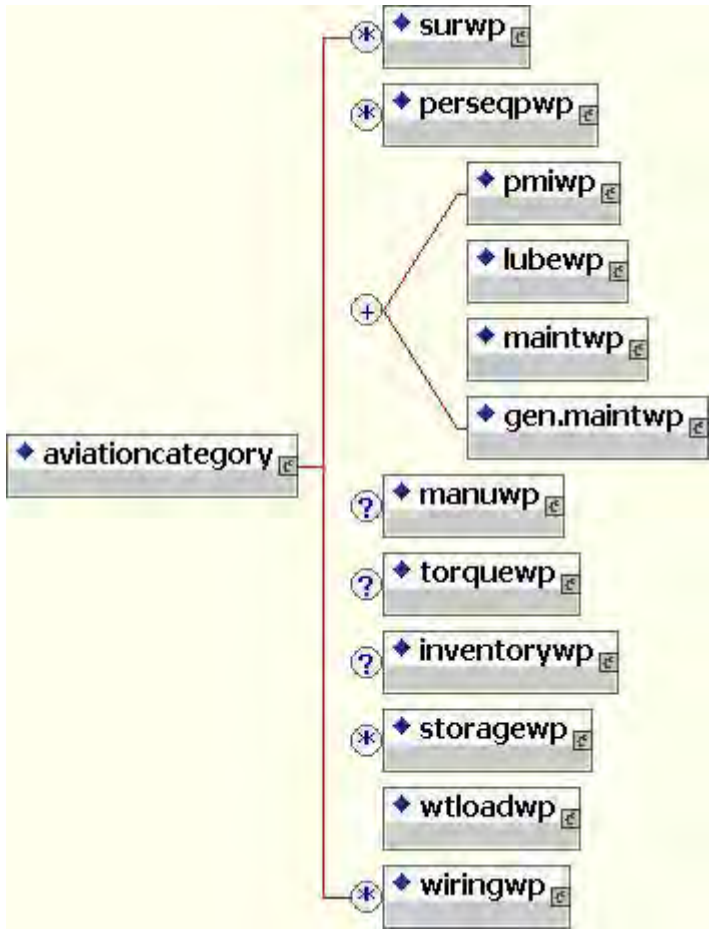

FIGURE 319. Aircraft maintenance category <aviationcategory> DTD hierarchy.

c. DTD fragment for ***<aviationcategory>***:

```
<!ELEMENT aviationcategory (surwp* , perseqpwp* , (pmiwp | lubewp |
   maintwp | gen.maintwp)+ , manuwp? , torquewp? , inventorywp? ,
   storagewp* , wtloadwp , wiringwp*)>
```

37.2.7 Preventive maintenance service (PMS)/preventive maintenance daily (PMD) category ***<pmscategory>***. The PMS/PMD category contains the requirements for aircraft PMS/PMD Inspection work packages.

a. The work package for ***<pmscategory>*** is PMS/PMD inspection work package ***<pms-inspecwp>*** (required – one or more) (see 37.25).

b. DTD fragment is graphically depicted for PMS/PMDCategory ***<pmscategory>***:

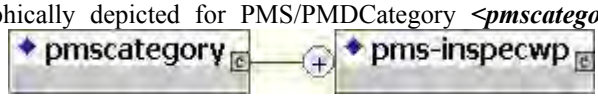

FIGURE 320. PMS/PMD category DTD Hierarchy <pmscategory>.

c. DTD fragment for ***<pmscategory>***:

```
<!ELEMENT pmscategory (pms-inspecwp+)>
```

37.2.8 Phased maintenance inspections (PMI) checklist category ***<checklistcategory>***. The PMI category is used to develop an aircraft phased inspection TM. Each work package ensures given aircraft

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components/sections are inspected on a specific (hourly/time) schedule. When a phased inspection cycle is complete, the cycle starts again thereby ensuring the aircraft is maintained in optimum flying condition.

a. The work package for ***<checklistcategory>*** is phased maintenance inspection work package(s) ***<pmi-cklistwp>*** (required – one or more) (see 37.26).

b. DTD fragment is graphically depicted for PMI Category ***<checklistcategory>***:

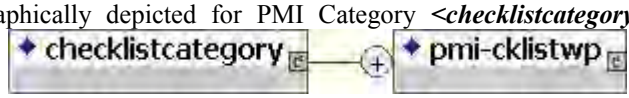

FIGURE 321. PMI category DTD hierarchy <checklistcategory>.

c. DTD fragment for ***<checklistcategory>***:

```
<!ELEMENT checklistcategory (pmi-cklistwp+)>
```

37.2.9 Ammunition maintenance category ***<ammunitioncategory>***. The ammunition maintenance category contains work packages sufficient to ensure proper ammunition care and marking. Additionally, any foreign ammunition identification and description is provided.

a. The work packages for ***<ammunitioncategory>*** are (required – one or more):
   (1) Ammunition maintenance work package ***<ammowp>*** (see 37.22).
   (2) Ammunition marking information work package ***<ammo.markingwp>*** (see 37.23).
   (3) Foreign ammunition (NATO) work package ***<natowp>*** (see 37.24).

b. DTD fragment is graphically depicted for Ammunition Maintenance Category ***<ammunitioncategory>***:

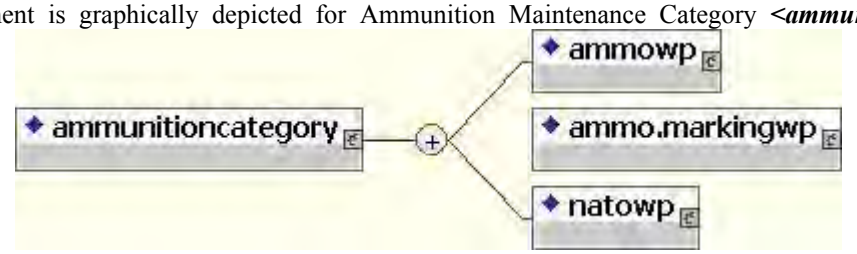

FIGURE 322. Ammunition maintenance category DTD hierarchy <ammunitioncategory>.

c. DTD fragment for ***<ammunitioncategory>***:

```
<!ELEMENT ammunitioncategory (ammowp | ammo.markingwp | natowp)+>
```

37.2.10 Test and inspection maintenance category ***<testinspectioncategory>***. The test and inspection maintenance category is applicable to conventional and chemical ammunition TMs only and contains the requirements for ammunition testing and inspection. The category is intended to use only the test and inspection ***<test-inspect>*** maintenance task within ***<maintwp>***. Using any other maintenance task would be a misuse of the category.

a. The work package for ***<testinspectioncategory>*** is maintenance work package ***<maintwp>*** (required – one or more) (see 37.7).

b. DTD fragment is graphically depicted for test and inspection category ***<testinspectioncategory>***:

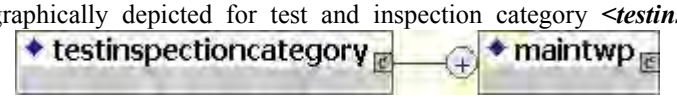

FIGURE 323. Test and inspection maintenance category <testinspectioncategory>.

c. DTD fragment for ***<testinspectioncategory>***:

```
<!ELEMENT testinspectioncategory (maintwp+)>
```

37.2.11 Ammunition marking maintenance category ***<ammomarkingcategory>***. The ammunition marking maintenance category is applicable to conventional and chemical ammunition TMs only and contains the requirements for ammunition identifying, marking, and handling.

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a. The work package for ***<ammomarkingcategory>*** is ammunition marking information work package ***<ammo.markingwp>*** (required – one or more) (see 37.23).
b. DTD fragment is graphically depicted for Ammunition Marking Maintenance Information Category ***<ammomarkingcategory>***:

FIGURE 324. Ammunition marking maintenance category DTD hierarchy <ammomarkingcategory>.

c. DTD fragment for ***<ammomarkingcategory>***:

```
<!ELEMENT ammomarkingcategory (ammo.markingwp+)>
```

37.2.12 Shipment, movement, and storage maintenance category ***<shipmentmovementstoragecategory>***. The shipment, movement, and storage maintenance category is applicable to conventional and chemical ammunition TMs only and contains the requirements for shipping, moving and storage of ammunition. The category is intended to use only the preparation for storage or shipment ***<pss>*** maintenance task within ***<maintwp>***. Using any other maintenance task would be a misuse of the category.

a. The work package for ***<shipmentmovementstoragecategory>*** is maintenance work package ***<maintwp>*** (required – one or more) (see 37.7).
b. DTD fragment is graphically depicted for Shipment, Movement, and Storage Maintenance Category ***<shipmentmovementstoragecategory>***:

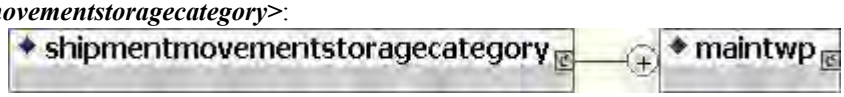

FIGURE 325. Shipment, movement, and storage maintenance category DTD Hierarchy <shipmentmovementstoragecategory>.

c. DTD fragment for ***<shipmentmovementstoragecategory>***:

```
<!ELEMENT shipmentmovementstoragecategory (maintwp+)>
```

37.3 Service upon receipt work package ***<surwp>***. The Service Upon Receipt work package contains information to ensure the equipment is adequately inspected, serviced and operationally tested before it is placed into service. Depending upon the complexity of the equipment being placed into service, the work package can contain one or more service upon receipt tasks.

a. The components of ***<surwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (8) Service upon receipt task ***<surtsk>*** (required – one or more) (see 37.3.1).
b. DTD graphically depicted for ***<surwp>***:

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Check the source to verify that this is the current version before use.

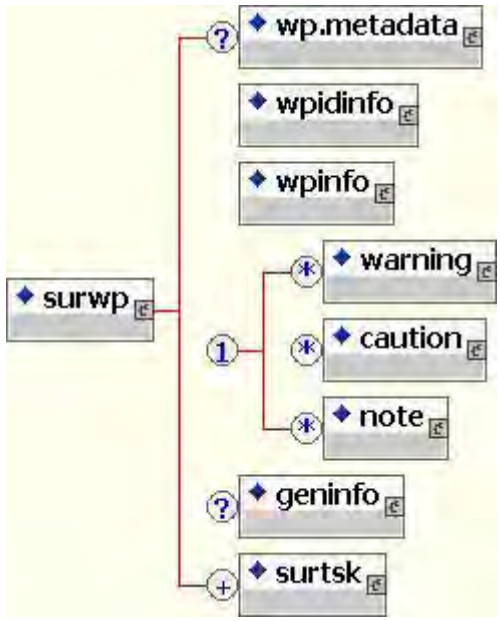

FIGURE 326. Service upon receipt work package DTD hierarchy <surwp>.

c. DTD fragment for ***<surwp>***:

```
<!ELEMENT surwp (wp.metadata?, wpidinfo, wpinfo, (warning* , caution* ,
   note*) , geninfo?, surtsk+)>
<!ATTLIST surwp
   airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes for ***<surwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

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(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.3.1 Service upon receipt task ***<surtsk>***. There are eleven (11) primary service upon receipt tasks. Each task may be repeated in the ***<surwp>***, as necessary, to ensure system components are correctly prepared to be placed into service.

a. The components are:
(1) One of the following service upon receipt task is required:
(a) Siting ***<siting>*** are the instructions when equipment is being placed into service requires that a site be prepared (see 33.1 for task development).
(b) Shelter requirements ***<shltr>*** are the instructions when the equipment is being placed into service requires a permanent or semi–permanent housing (see 33.1 for task development).
(c) Service upon receipt of material ***<surmat>*** (see 37.3.1.1).
(d) Installation ***<install>*** are the instructions for properly installing the equipment, including the use of tools, making the necessary connections, lubricating, calibrating, and adjusting to the equipment (see 33.1 for task development).
(e) Preliminary servicing of equipment ***<preserv>*** are the instructions for all lubrication required on newly installed equipment (see 33.1 for task development).
(f) Preliminary checks and adjustment of equipment ***<prechkadj>*** are the instructions for all checks and adjustments to be made on newly installed equipment (see 33.1 for task development).

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(g) Preliminary calibration of equipment ***<precal>*** are the instructions for all calibration to be made on newly installed equipment (see 33.1 for task development).
(h) Circuit alignment ***<calign>*** (see 37.3.1.2).
(i) Ammunition marking ***<ammo.markings>*** are the instructions for marking ammunition and ammunition containers (see 33.1 for task development).
(j) Classification of ammunition defects ***<ammo.defect>*** are the instruction for performing visual inspection of ammunition/containers (pallets, boxes, etc.) and including the classification and disposition of defective ammunition/containers (see 33.1 for task development).
(k) Ammunition handling ***<ammo.handling>*** (see 37.7.1.3).
(l) Activate ammunition. ***<arm>*** are the instructions for the ammunition activation for preparatory to detonation (see 33.1 for task development).
(m) Other service upon receipt ***<other.surtsk>*** is for task instructions not covered by previously service upon receipt task types (see 33.1 for task development).

(2) Follow-on maintenance ***<followon.maint>*** (optional) are instructions that are accomplished following the completion of a service upon receipt task to clean up or undo actions performed during the task (see 33.1 for task development).

b. DTD graphically depicted for ***<surtsk>***:

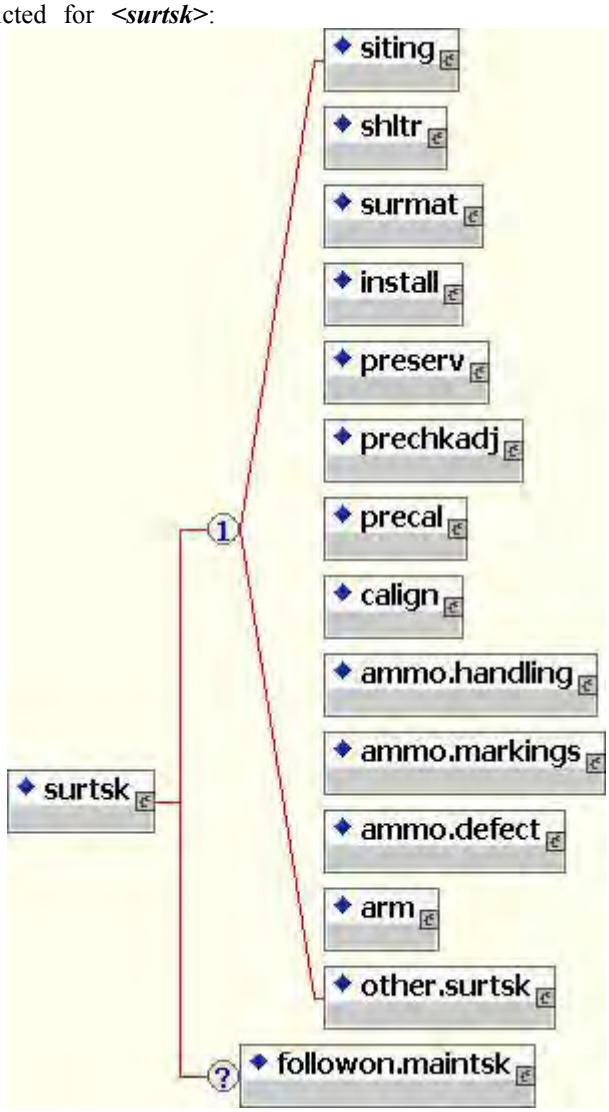

FIGURE 327. Service upon receipt task <surtsk> DTD hierarchy. .

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<surtsk>***:

```
<!ELEMENT surtsk ((siting | shltr | surmat | install | preserv |
   prechkadj | precal | calign | ammo.handling | ammo.markings |
   arm | other.surtsk), followon.maintenance?)>
```

37.3.1.1 Service upon receipt of material task ***<surmat>***. The service upon receipt of material task contains three sub–tasks: unpacking, checking unpacked, and processing unpacked equipment. Each sub–task is a complete task package and can be used independently.

a. The components for ***<surmat>*** are:

(1) When the service upon receipt of material task has an overall task information (optional) the components are:

(a) Task title ***<title>*** (required) (see 49.1.1.3).

(b) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).

(c) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).

(d) Note – ***<note>*** (optional – zero or more) (see 41.1.3).

(e) General information ***<geninfo>*** (optional) (see 49.1.4.12).

(2) At least of the following sub–tasks are required:

(a) Unpacking equipment ***<unpack>*** are instructions to protect received equipment and reusable containers from inadvertent destruction, for packaging unserviceable components, and returning empty reusable containers to the supply system (see 33.1 for task development).

(b) Checking unpacked equipment ***<chkeqp>*** (see 37.3.1.1.1).

(c) Processing unpacked equipment ***<processeqp>*** are instructions for processing the unpacked equipment (e.g., removing excess lubricant from a new rifle) (see 33.1 for task development).

b. DTD graphically depicted for ***<surmat>***:

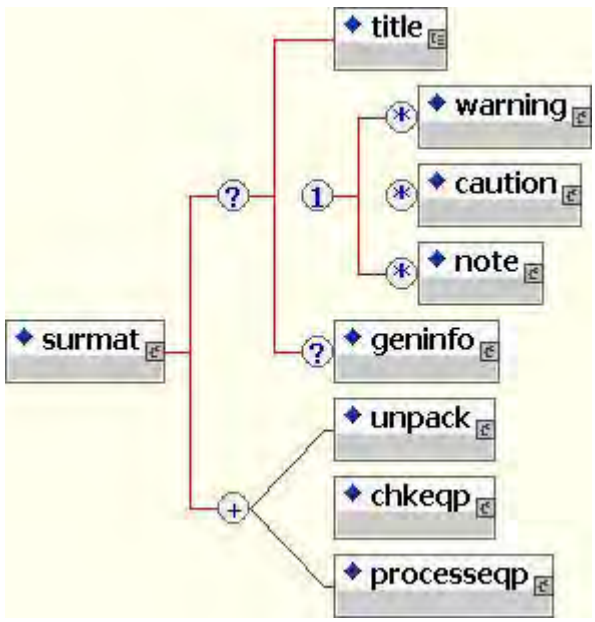

FIGURE 328. Service upon receipt of material task <surmat> DTD hierarchy. .

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c. DTD fragment :

```
<!ELEMENT surmat (title, (warning* | caution* | note*), geninfo?)?),
   (unpack | chkeqp | processeqp+)>
<!ATTLIST surmat
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    date-time-stamp   (date | time | date-time)       #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    esd               (yes | no)                      "no"
    frame             (yes | no)                      "yes"
    hcp               (yes | no)                      "no"
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 3 | 4 | 5)                 "0"
>
```

d. Common attributes:

(1) **applicable** – Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(8) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(9) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(10) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(11) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(12) **inschlvl** – Insert change level (optional) (see 49.2.9).
(13) **security** – Security classification (optional) (see 49.2.13).
(14) **skilltrk** – Skill level (optional) (see 49.2.2).
(15) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**0**").

37.3.1.1.1 Checking unpacked equipment ***<chkeqp>***. The checking unpacked equipment task contains instructions for a condition check of the shipment (including pallets, containers, boxes, and legibility of markings). Two standard informations are available to ensure packing materials were sufficient to protect the equipment during shipment (see 37.3.1.1.2) and the equipment was received in good mechanical condition (see 37.3.1.1.3).

a. The components for ***<chkeqp>*** are:

(1) Task title ***<title>*** (required) (see 49.1.1.3).
(2) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(3) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(4) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(5) General information ***<geninfo>*** (optional) (see 49.1.4.12).
(6) Illustration ***<figure>*** (optional – zero or more) are included to ensure understanding of the task (see 44.1.1).

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(7) Three (3) methods are provided for developing the checking unpacked equipment task. Select one of the following methods:

(a) Method 1 has additional information paragraphs (with any necessary alerts) followed by instructions or descriptive titled paragraphs. The components of the method are:

1 Paragraph ***<para>*** (see 49.1.1.5) and/or paragraph with alerts ***<specpara>*** (see 49.1.1.6) (required – one or more).

2 Select either instructions (optional – zero or more) (see 37.3.1.1.1.a.(7).(b) for method 2) OR descriptive paragraphs (optional – zero or more) (see 37.3.1.1.1.a.(7).(c) for method 3).

(b) Method 2 has instructions. The components of the method (required – one or more) are:

1 Instructions ***<proc>*** (required) (see 33.2).

2 Packaging Material Inspection Criteria standard information ***<crit.insp.tab>*** (optional – zero or more) (see 37.3.1.1.2).

3 Equipment Components Inspection standard information ***<pecul.insp.tab>*** (optional – zero or more) (see 37.3.1.1.3).

(c) Method 3 has descriptive titled paragraphs. The components of the method (required – one or more) are:

1 Descriptive paragraph ***<para0>*** (required) (see 49.1.1.8)/conditional descriptive paragraph ***<para0–alt>*** (required – one or more) (see 48.2.1).

2 Packaging Material Inspection Criteria standard information ***<crit.insp.tab>*** (optional – zero or more) (see 37.3.1.1.2).

3 Equipment Components Inspection standard information ***<pecul.insp.tab>*** (optional – zero or more) (see 37.3.1.1.3).

b. DTD graphically depicted for ***<chkeqp>***:

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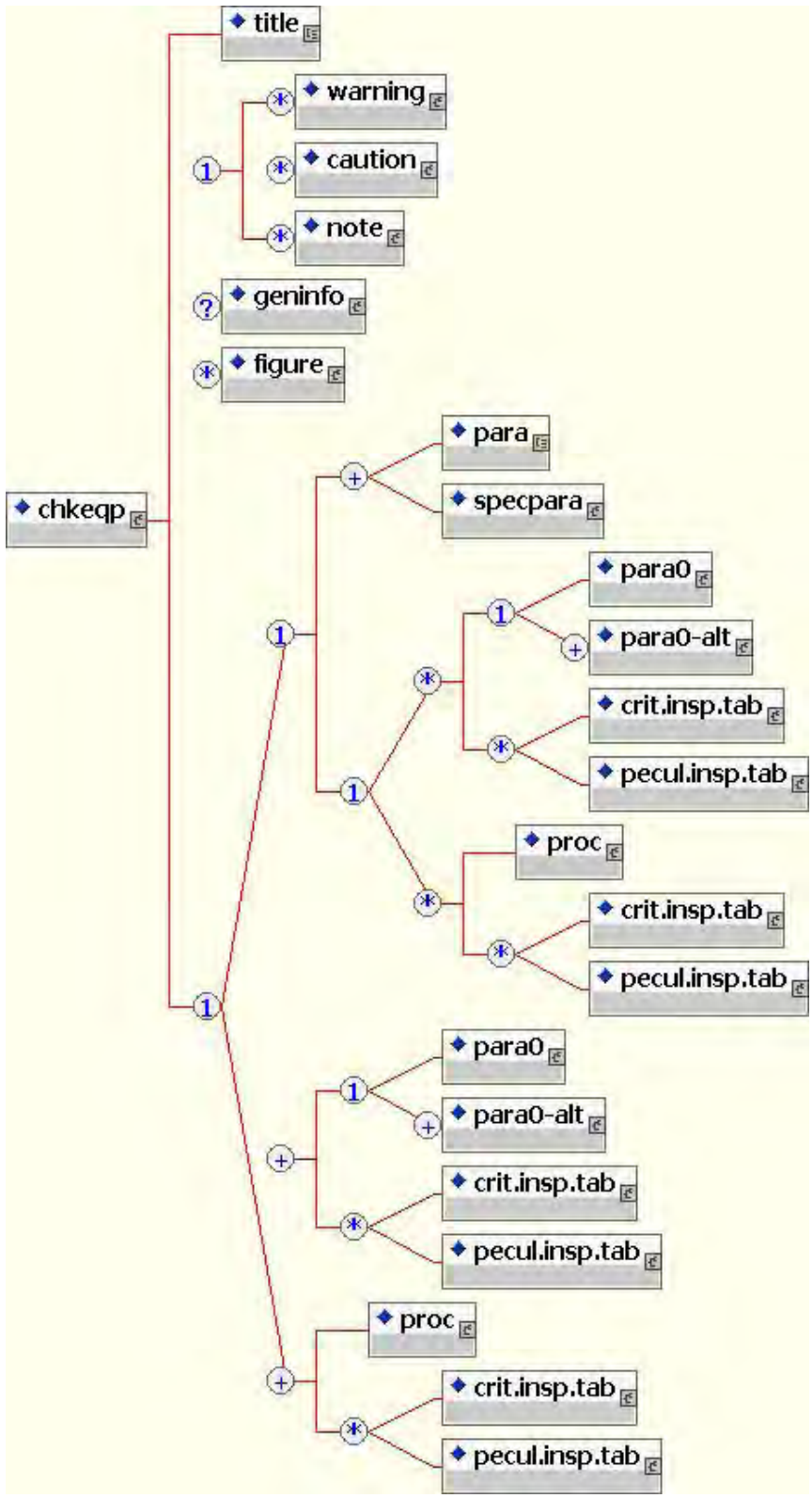

FIGURE 329. Checking unpacked equipment task DTD hierarchy <chkeqp>.

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c. DTD fragment:

```
<!ELEMENT chkeqp (title, (warning*, caution*, note*), geninfo?, figure*,
   (((para | specpara)+, (((para0 | para0-alt+), (crit.insp.tab |
   pecul.insp.tab)*)* | (proc, (crit.insp.tab | pecul.insp.tab)*)*)) |
   ((para0 | para0-alt+), (crit.insp.tab | pecul.insp.tab)*)+ |
   (proc, (crit.insp.tab | pecul.insp.tab)*)+))>
<!ATTLIST chkeqp
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    date-time-stamp   (date | time | date-time)       #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    esd               (yes | no)                      "no"
    frame             (yes | no)                      "yes"
    hcp               (yes | no)                      "no"
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 3 | 4 | 5)                 "0"
>
```

d. Common attributes for ***<chkeqp>*** are:

(1) **applicable** – Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(8) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(9) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(10) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(11) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(12) **inschlvl** – Insert change level (optional) (see 49.2.9).
(13) **security** – Security classification (optional) (see 49.2.13).
(14) **skilltrk** – Skill level (optional) (see 49.2.2).
(15) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**0**").

37.3.1.1.2 Packaging material inspection criteria standard information ***<crit.insp.tab>***. The ***<crit.insp.tab>*** element provides the location of the packaging material item to be inspected prior to unpacking the equipment and establishes acceptable, repairable, and nonrepairable standards for each item inspected. The element is similar to a "table" in a structural table.

a. The components for ***<crit.insp.tab>*** are:

(1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
(2) Standard table header ***<std.table header>*** (required) (see 37.3.1.1.2.1).
(3) Item ***<crit.insp-group>*** (required – one or more) (see 37.3.1.1.2.2)

b. DTD graphically depicted for ***<crit.insp.tab>***:

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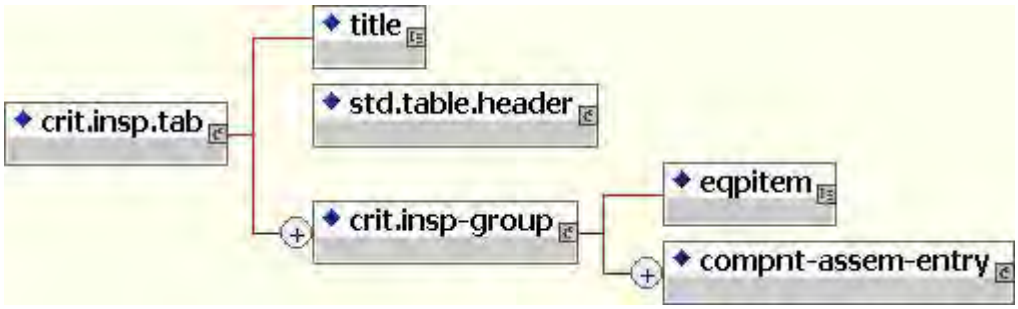

FIGURE 330. Packaging material inspection criteria standard information DTD hierarchy <crit.insp.tab>.

c. DTD fragment for ***<crit.insp.tab>***:

```
<!ELEMENT crit.insp.tab (title, std.table.header, crit.insp-group+)>
<!ATTLIST crit.insp.tab
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1"
>
```

d. Common attributes for ***<crit.insp.tab>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

37.3.1.1.2.1 Packaging material inspection criteria standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table VIII.

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TABLE VIII. Packaging material inspection criteria standard information header *<std.table.header>* boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Packaging material inspection criteriaStandard Information Header	***&surwp.chkequ.crit.insp.tab.std-header;***	Not applicable.

37.3.1.1.2.2 Packaging material criteria inspection grouping ***<crit.insp-group>***. The element groups the major packaging material item, which divides into the individual packaging material items with the associated inspection criteria.

a. The components for ***<crit.insp-group>*** are:
   (1) Packaging material item ***<eqpitem>*** (required) (see 37.3.1.1.2.3). The element is similar to a "row" in a structural table. Additionally, the element is similar to a “cell” in a structural table that spans across all columns.
   (2) Packaging material component and criteria inspection entry ***<compnt-assem-entry>*** (required – one or more) (see 37.3.1.1.2.4). The element is similar to a "row" in a structural table.

b. DTD fragment for ***<crit.insp-group>***:

```
<!ELEMENT crit.insp-group (eqpitem, compnt-assem-entry+)>
<!ATTLIST crit.insp-group
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<crit.insp-group>*** are:
   (1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
   (2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (4) **comment** – Change information (optional) (see 49.2.9).
   (5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (8) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (9) **security** – Security classification (optional) (see 49.2.13).
   (10) **skilltrk** – Skill level (optional) (see 49.2.2).

37.3.1.1.2.3 Packaging material item or component ***<eqpitem>***. The element is used for both the packaging material item name and the packaging material component name. The item name is a major packaging material item, in which each major item has components with associated inspection criteria.

a. The components for ***<eqpitem>*** are:
   (1) Parsable characters or type text. – #PCDATA
   (2) Format text – ***<emphasis>*** (see 49.1.3.1).
   (3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
   (4) Superscript – < ***supscrpt>*** (see 49.1.3.5).
   (5) Cross reference – ***<xref>*** (see 46.2.2).

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(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).

b. DTD fragment for ***<eqpitem>***:

```
<!ELEMENT eqpitem (#PCDATA | emphasis | subscrpt | supscrpt | xref |
   extref | link | help.info | indxref | term | term.def | callout |
   ftnote | ftnref)*>
<!ATTLIST eqpitem
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<eqpitem>*** are:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

37.3.1.1.2.4 Packaging material component and inspection criteria entry ***<compnt-assem-entry>***. The element defines for each component the item name and inspection criteria for acceptable, repairable, and nonrepairable.

a. The components for ***<compnt-assem-entry>*** are:
(1) Equipment item ***<eqpitem>*** (required) (see 37.3.1.1.2.3). The element is similar to a "cell" in a structural table and is entered in column one.
(2) Inspection criteria entry ***<accpt-rpbl-nonrpbl-entry>*** (required – one or more). When multiple inspection criteria entries are used, beyond the first occurrence the element is similar to a "row" in a structural table and the first cell is left blank.
(a) Acceptable criterion ***<accept>*** (required) (see 37.3.1.1.2.4.1). The element is similar to a "cell" in a structural table and is entered in column two.
(b) Repairable criterion ***<repairable>*** (required) (see 37.3.1.1.2.4.1). The element is similar to a "cell" in a structural table and is entered in column three.
(c) Nonrepairable criterion ***<nonrepairable>*** (required) (see 37.3.1.1.2.4.1). The element is similar to a "cell" in a structural table and is entered in column four.

b. DTD graphically depicted for ***<compnt-assem-entry>***:

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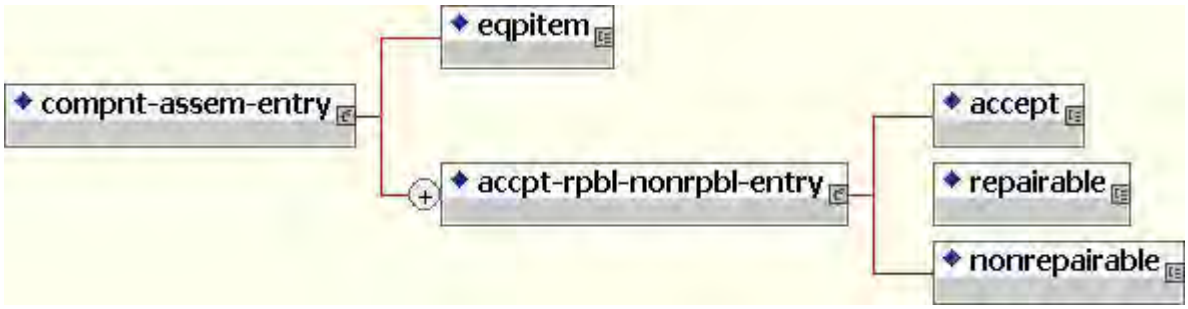

FIGURE 331. Packaging material inspection criteria component/assemble entry DTD hierarchy <compnt-assem-entry>..

c. DTD fragment for ***<compnt-assem-entry>*** and ***<accpt-rpbl-nonrpbl-entry>*** are:

```
<!ELEMENT compnt-assembly-entry (eqpitem, accpt-rpbl-nonrpbl-entry+)>
<!ATTLIST compnt-assembly-entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
<!ELEMENT accpt-rpbl-nonrpbl-entry (accept, repairable, nonrepairable)>
<!ATTLIST accpt-rpbl-nonrpbl-entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<compnt-assem-entry>*** and ***<accpt-rpbl-nonrpbl-entry>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

37.3.1.1.2.4.1 Acceptable criterion ***<accept>***.

a. The components for ***<accept>*** are:

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(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).

b. DTD fragment for ***<accept>***:

```
<!ELEMENT accept (#PCDATA | emphasis | subscrpt | supscrpt | xref |
   extref | link | help.info | indxref | term | term.def | callout |
   ftnote | ftnref)*>
<!ATTLIST accept
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<accept>*** are:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

37.3.1.1.2.4.2 Repairable criterion ***<repairable>***.

a. The components for ***<repairable>*** are:
(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).

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(13) Footnote – < ***ftnote***> (see 45.1.1).
(14) Footnote reference – <***ftnref***> (see 45.1.2).
(15) Control/Indicator – <***ctrlind***> (see 49.1.4.3).
(16) Control/Indicator value – <***ctrlind-val***> (see 49.1.4.4).
(17) DoD ammunition code – <***dodac***> (see 49.1.4.5).
(18) Lubricant value – <***lubricant***> (see 49.1.4.16).
(19) Graphic symbol – <***symbol***> (see 44.3.1).
(20) Torque value – <***torque***> (see 49.1.4.24).
(21) Voltage value – < ***voltage***> (see 49.1.4.25).
(22) Null text – <***null*** > (see 49.1.3.2).
(23) Changed text marker – <***change***> (see 49.1.3.7).

b. DTD fragment for <***repairable***>:

```
<!ELEMENT repairable (#PCDATA | emphasis | subscrpt | supscrpt | xref |
   extref | link | help.info | indxref | term | term.def | callout |
   ftnote | ftnref | ctrlind | ctrlind-val | dodac | lubricant |
   symbol | torque | voltage | null | change)*>
<!ATTLIST repairable
   changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

c. Common attributes for <***repairable***> are:
(1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(2) **comment** – Change information (optional) (see 49.2.9).
(3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(4) **inschlvl** – Insert change level (optional) (see 49.2.9).

37.3.1.1.2.4.3 Nonrepairable criterion <***nonrepairable***>.

a. The components for <***nonrepairable***> are:
(1) Parsable characters or type text. – #PCDATA
(2) Format text – <***emphasis***> (see 49.1.3.1).
(3) Subscript – <***subscrpt***> (see 49.1.3.4).
(4) Superscript – < ***supscrpt***> (see 49.1.3.5).
(5) Cross reference – <***xref***> (see 46.2.2).
(6) External reference – <***extref***> (see 46.2.1).
(7) Enhanced linking – <***link***> (see 46.2.3).
(8) IETM help information – <***help.info***> (see 48.3.2.4).
(9) Index reference – < ***indxref***> (see 31.4.2.2.3).
(10) Term – <***term***> (see 49.1.2.3.1.1).
(11) Term and definition – <***term.def***> (see 49.1.2.3.1).
(12) Figure callout reference – <***callout***> (see 46.2.4.1).
(13) Footnote – < ***ftnote***> (see 45.1.1).
(14) Footnote reference – <***ftnref***> (see 45.1.2).
(15) Control/Indicator – <***ctrlind***> (see 49.1.4.3).
(16) Control/Indicator value – <***ctrlind-val***> (see 49.1.4.4).
(17) DoD ammunition code – <***dodac***> (see 49.1.4.5).
(18) Lubricant value – <***lubricant***> (see 49.1.4.16).
(19) Graphic symbol – <***symbol***> (see 44.3.1).
(20) Torque value – <***torque***> (see 49.1.4.24).
(21) Voltage value – < ***voltage***> (see 49.1.4.25).
(22) Null text – <***null*** > (see 49.1.3.2).
(23) Changed text marker – <***change***> (see 49.1.3.7).

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b. DTD fragment for ***<nonrepairable>***:

```
<!ELEMENT nonrepairable (#PCDATA | emphasis | subscrpt | supscrpt | xref |
   extref | link | help.info | indxref | term | term.def | callout |
   ftnote | ftnref | ctrlind | ctrlind-val | dodac | lubricant | symbol |
   torque | voltage | null | change)*>
<!ATTLIST nonrepairable
   changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

c. Common attributes for ***<nonrepairable>*** are:

(1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(2) **comment** – Change information (optional) (see 49.2.9).
(3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(4) **inschlvl** – Insert change level (optional) (see 49.2.9).

37.3.1.1.3 Equipment components inspection standard information ***<pecul.insp.tab>***. The element lists each item ***<eqpitem>*** of a component by location ***<location>*** that requires inspection. For each item listed an inspection action step(s) ***<step1>*** is provided and, when needed, a reference is made to the associated work package. Each component item may need multiple steps and subordinate steps to describe the inspection requirements. Each first level step can provide a separate reference to a work package for further details on inspection or action to be taken.

a. The components for ***<pecul.insp.tab>*** are:

(1) Standard information title ***<title>*** (required) (see 49.1.1.3).
(2) Standard table header ***<std.table header>*** (required) (see 37.3.1.1.3.1).
(3) Equipment component information ***<pecul.insp-group>*** (required – one or more). The element is similar to a "row" in a structural table. The components for ***<pecul.insp-group>*** are:
(a) Item is located in equipment component name ***<compassem>*** (required) (see 40.4.1.2.2). The element is similar to a “cell” in a structural table and is entered in column one.
(b) Equipment component item inspection entry ***<pecul.insp-entry>*** (required – one or more) (see 37.3.1.1.3.2). The element is similar to a "row" in a structural table.

b. DTD graphically depicted for ***<pecul.insp.tab>***:

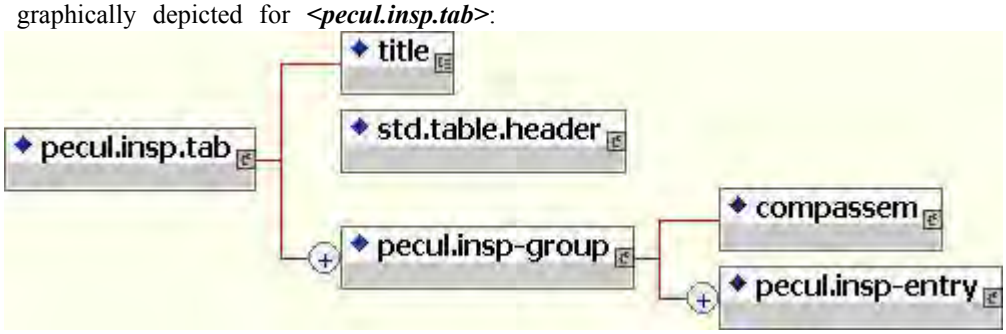

FIGURE 332. Equipment components inspection a standard information DTD hierarchy <pecul.insp.tab>.

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c. DTD fragment for ***<pecul.insp.tab>***:

```
<!ELEMENT pecul.insp.tab (title, std.table.header, pecul.insp-group+)>
<!ATTLIST pecul.insp.tab
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1"
>

<!ELEMENT pecul.insp-group (compassem, pecul.insp-entry+)>
<!ATTLIST pecul.insp-group
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<pecul.insp.tab>*** and ***<pecul.insp-group>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

37.3.1.1.3.1 Equipment components inspection standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table VIII.

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TABLE IX. Equipment components inspection standard information header *<std.table.header>* boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Equipment components inspection Standard Information Header	***&surwp.chkequ.pecul.insp.tab.std-header;***	Not applicable.

37.3.1.1.3.2 Equipment component item inspection entry ***<pecul.insp-entry>***. The element contains the component item and inspection action requirements located for the specified equipment component.

a. The components for ***<pecul.insp-entry>*** are:

(1) Equipment component item name ***<eqpitem>*** (required) (see 37.3.1.1.2.3). The element is similar to a "cell" in a structural table and is entered in column two.

(2) Inspection action ***<pecul.step-entry>*** (required – one or more) is comprised of:

(a) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).

(b) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).

(c) Inspection step level 1 ***<step1>*** (see 33.3) and/or conditional inspection step level 1 ***<step1–alt>*** (see 48.2.1) (required).

(d) Remarks ***<remarks>*** (required) (see 37.3.1.1.3.3). The element is similar to a "cell" in a structural table and is entered in column four.

b. DTD graphically depicted for ***<pecul.insp-entry>***:

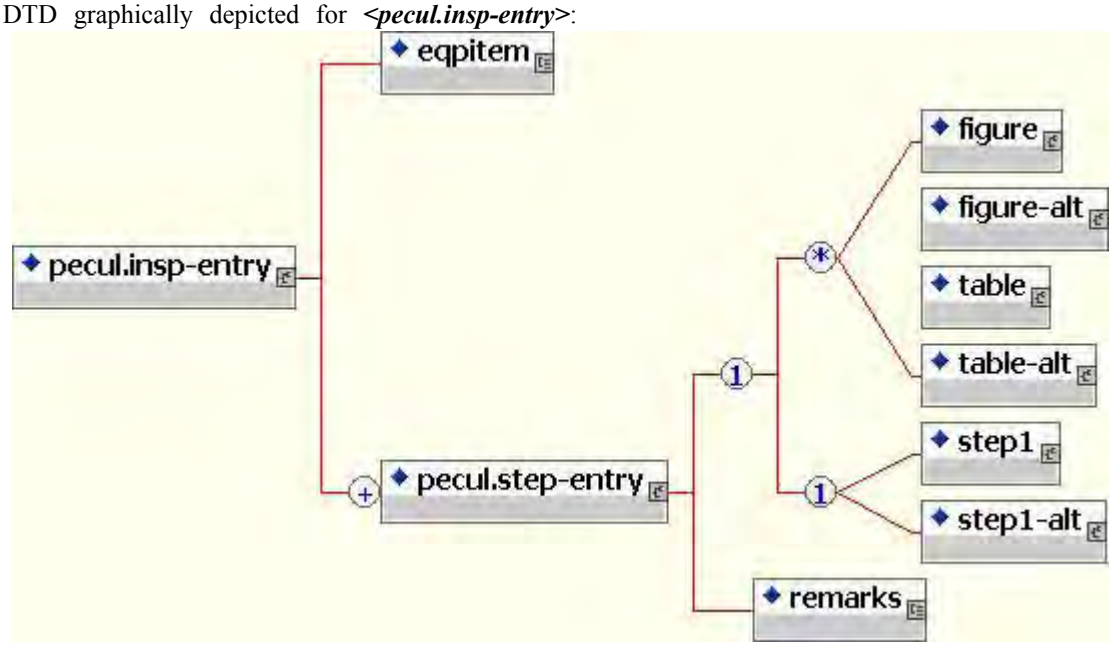

FIGURE 333. Equipment component item inspection entry DTD hierarchy <pecul.insp-entry>.

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c. DTD fragment for ***<pecul.insp-entry>***:

```
<!ELEMENT pecul.insp-entry (eqpitem, pecul.step-entry+)>
<!ATTLIST pecul.insp-entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>

<!ELEMENT pecul.step-entry ((figure | figure-alt | table | table-alt)*,
   (step1 | step1-alt), remarks)>
<!ATTLIST pecul.step-entry
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<pecul.insp-entry>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

37.3.1.1.3.3 Remarks ***<remarks>***. The element contains remarks information pertinent to the inspection action step, usually this is a reference to further detail instructions on the inspection. When no additional instructions are needed for the inspection use the ***<null insert="dash"/>*** indicating no further work packages are needed for the step.

a. The components of Remarks ***<remarks>*** are:

(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).

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(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).

b. DTD fragment for ***<remarks>***:

```
<!ELEMENT remarks (#PCDATA | emphasis | subscrpt | supscrpt | xref |
   extref | link | help.info | indxref | term | term.def | callout |
   ftnote | ftnref | ctrlind | ctrlind-val | dodac | lubricant |
   symbol | torque | voltage | null | change)*>
<!ATTLIST remarks
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Attribute for ***<remarks>*** are:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

37.3.1.2 Circuit alignment ***<calign>***. The element is the instructions for circuit alignment procedures. The circuit alignment task contains three sub–tasks: external connections; switch settings, patch panel connections, and internal control settings; and alignment Each sub–task is a complete task package and can be used independently.

a. The components for ***<calign>*** are:
(1) Task title ***<title>*** (required) (see 49.1.1.3).
(2) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(3) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(4) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(5) General information ***<geninfo>*** (optional) (see 49.1.4.12).
(6) The circuit alignment task requires at least one of following tasks to be used.

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(a) External connections ***<extconn>*** are the instructions for the connections to external lines required for each installation option and conform to the requirements for installing wiring and cabling interconnections (see 33.1 for task development).

(b) Switch settings, patch panel connections, and internal control settings ***<setconn>*** are the instructions for all switch settings, patch panel connections, and internal control settings required for each installation option and mode of operation (see 33.1 for task development).

(c) Alignment procedure ***<align>*** are the instructions for all alignment procedures, including any variations required for different installation options and modes of operation. (see 33.1 for task development).

b. DTD graphically depicted for ***<calign>***:

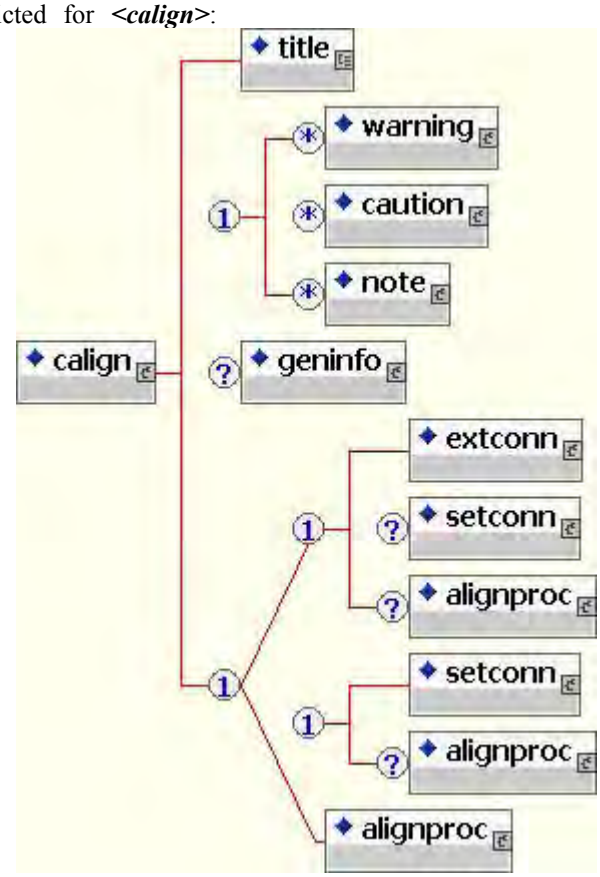

FIGURE 334. Circuit alignment task DTD hierarchy <calign>.

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c. DTD fragment for ***<calign>***:

```
<!ELEMENT calign (title, (warning*, caution*, note*),
   geninfo?, ((extconn, setconn?, alignproc?) |
   (setconn, alignproc?) | alignproc))>
<!ATTLIST calign
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    date-time-stamp   (date | time | date-time)       #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    esd               (yes | no)                      "no"
    frame             (yes | no)                      "yes"
    hcp               (yes | no)                      "no"
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 3 | 4 | 5)                 "0"
>
```

d. Common attributes for ***<calign>*** are:

(1) **applicable** – Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(8) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(9) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(10) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(11) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(12) **inschlvl** – Insert change level (optional) (see 49.2.9).
(13) **security** – Security classification (optional) (see 49.2.13).
(14) **skilltrk** – Skill level (optional) (see 49.2.2).
(15) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**0**").

37.3.2 Service upon receipt work package general entity boilerplates. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed (see Table X).

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TABLE X. Boilerplate Entities for *<chkeqp>*.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Service upon receipt checking unpacked equipment mandatory steps	***&surwp.surmat.chkeqp.inspect;***	Not applicable

37.3.3 XML document instance fragment and output for ***<surwp>***. The XML instance and its stylesheet output for a ***<surwp>*** is provided below.

a. Example of an XML document instance fragment for ***<surwp>***:

```
<surwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="M00001-X-XXX-XXX" wpseq="0109">
<wpidinfo>
<maintlvl level="field"/>
<title>Service Upon Receipt
</title>
</wpidinfo>
<wpinfo>
<tools>
<tools-setup-item>
<name>Measuring Tape
</name>
<itemref>
<xref wpid=”S0006–X-XXX-XXX” itemid="docno="S0006–X-XXX-XXX-item3"/>
</itemref>
</tools-setup-item>
</tools>
<ref>
<ref-setup-item>
<xref wpid="M00125-X-XXX-XX"/>
</ref-setup-item>
<ref-setup-item>
<xref wpid="M00128-X-XXX-XX"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="SF 361" posttext=", Transportation Discrepancy Report"/>
</ref-setup-item>
</ref>
</wpinfo>
<surtsk>
<surmat esd="no" frame="yes" hcp="no" tocentry="0">
<chkeqp esd="no" frame="yes" hcp="no" tocentry="0">
<title>Checking Unpacked Equipment
</title>
<proc>
<title>
</title> %surwp.surmat.chkeqp.inspect;
</proc>
<crit.insp.tab tocentry="1">
<title>Inspection Criteria for Packaging.
</title> %surwp.chkequ.crit.insp.tab.std-header;
<crit.insp-group>
<eqpitem>Wooden Boxes and Crates
```


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Check the source to verify that this is the current version before use.

</eqpitem>
<compnt-assem-entry>
<eqpitem>Hardware
</eqpitem>
<accpt-rpbl-nonrpbl-entry>
<accept>Operative and tight.
</accept>
<repairable>Inoperative or loose.
</repairable>
<nonrepairable>None.
</nonrepairable>
</accpt-rpbl-nonrpbl-entry>
<accpt-rpbl-nonrpbl-entry>
<accept>Nails, screws, and fasteners
</accept>
<repairable>Nails, screws, and fasteners.
</repairable>
<nonrepairable>None.
</nonrepairable>
</accpt-rpbl-nonrpbl-entry>
</compnt-assem-entry>
<compnt-assem-entry>
<eqpitem>Ends
</eqpitem>
<accpt-rpbl-nonrpbl-entry>
<accept>Free from damage.
</accept>
<repairable>Broken or missing cleats and handles.
</repairable>
<nonrepairable>Damage that requires disassembly of box.
</nonrepairable>
</accpt-rpbl-nonrpbl-entry>
</compnt-assem-entry>
<compnt-assem-entry>
<eqpitem>Wood
</eqpitem>
<accpt-rpbl-nonrpbl-entry>
<accept>Splits less than 3 inches long, no closer than 1 inch to edge of board or adjoining split. The board must be secured by at least one nail on each side of the split when it extends to the end of the board.
</accept>
<repairable>Splits more than 3 inches but no closer than 1 inch to edge of board or adjoining split, or **%frac12;**-inch wide. That can be repaired by use of corrugated fasteners.
</repairable>
<nonrepairable>Splits closer than 1 inch to edge of board or adjoining split or over **%frac12;**-inch wide.
</nonrepairable>
</accpt-rpbl-nonrpbl-entry>
</compnt-assem-entry>
</crit.insp-group>
<crit.insp-group>
<eqpitem>Fiber Containers
</eqpitem>
<compnt-assem-entry>
<eqpitem>Metal Ends

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```
</eqpitem>
<accpt-rpbl-nonrpbl-entry>
<accept>Minor rust, cracks, indentations, or splits that would not
impair water proofing or serviceability of container.
</accept>
<repairable>None.
</repairable>
<nonrepairable>Perforations, excessive rust, or ends which are
crushed or not securely crimped to body.
</nonrepairable>
</accpt-rpbl-nonrpbl-entry>
</compnt-assem-entry>
<compnt-assem-entry>
<eqpitem>Body and Cap
</eqpitem>
<accpt-rpbl-nonrpbl-entry>
<accept>No leaks, cuts, or gouges.
</accept>
<repairable>Cuts, tears, or gouges not closer than 1 inch
to closure, less than %frac12; square inch in area, and
unpenetrated layers that can be spot painted.
</repairable>
<nonrepairable>Cuts, tears, or gouges closer than 1 inch to closure, more
than %frac12; square inch in area, or through all impregnated layers
</nonrepairable>
</accpt-rpbl-nonrpbl-entry>
</compnt-assem-entry>
</crit.insp-group>
</crit.insp.tab>
<pecul.insp.tab  tocentry="1">
<title>M29 and M30 Control Surfaces and Containers.
</title> %surwp.chkequ.pecul.insp.tab.std-header;
<pecul.insp-group>
<compassem>
<name>Container
</name>
</compassem>
<pecul.insp-entry>
<eqpitem>Components
</eqpitem>
<pecul.step-entry>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Inspect for rust, fungus, paint
damage, and deformation.
</para>
</step1>
<remarks>
<xref  wpid="M00125-X-XXX-XX"/>
</remarks>
</pecul.step-entry>
<pecul.step-entry>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Reject container if damage prevents
it from functioning properly.
</para>
</step1>
```


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Check the source to verify that this is the current version before use.

<remarks>
<null insert="dash"/>
</remarks>
</pecul.step-entry>
</pecul.insp-entry>
</pecul.insp-group>
<pecul.insp-group>
<compassem>
<name>M29
</name>
</compassem>
<pecul.insp-entry>
<eqpitem>Control Surfaces
</eqpitem>
<pecul.step-entry>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect for dents and scratches on post, trailing edge phenolic, skin, and closure plate.
</para>
</step1>
<remarks>
<xref wpid="M00128-X-XXX-XX"/>
</remarks>
</pecul.step-entry>
<pecul.step-entry>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Reject control surface:
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If post dents or scratches exceed 0.002 in. (0.051 mm).
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If trailing edge phenolic dents exceed 0.040 in. (10.160 mm).
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If skin dents exceed 0.030 in. (7.620 mm) within 2 in. (50.800 mm) of post.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If closure plate dents exceed 0.030 in. (7.620 mm) within 2 in. (50.800 mm) of post.
</para>
</step2>
</step1>
<remarks>
<null insert="dash"/>
</remarks>
</pecul.step-entry>
</pecul.insp-entry>
</pecul.insp-group>
<pecul.insp-group>

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Check the source to verify that this is the current version before use.

```
<compassem>
<name>M30
</name>
</compassem>
<pecul.insp-entry>
<eqpitem>Control  Surfaces
</eqpitem>
<pecul.step-entry>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Inspect for dents and scratches on post and skin.
</para>
</step1>
<remarks>
<null  insert="dash"/>
</remarks>
</pecul.step-entry>
<pecul.step-entry>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Skin dents or scratches up to 0.050 in.
(12.700 mm) are allowable, but should be blended.
</para>
</step1>
<remarks>
<null  insert="dash"/>
</remarks>
</pecul.step-entry>
<pecul.step-entry>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Reject control surface if post dents or
scratches exceed 0.002 in.   (0.051 mm).
</para>
</step1>
<remarks>
<null  insert="dash"/>
</remarks>
</pecul.step-entry>
</pecul.insp-entry>
</pecul.insp-group>
</pecul.insp.tab>
</chkeqp>
</surmat>
</surtsk>
</surwp>
```

b. Page-based TM stylesheet output example for **<*surwp*>**:

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FIELD MAINTENANCE
SERVICE UPON RECEIPT

INITIAL SETUP:

Tools and Special Tools
Measuring Tape (WP 0240, Item 3)

References
WP 0125

References (cont.)
WP 0128
SF 361, Transportation Discrepancy Report

Checking Unpacked Equipment

1. Inspect the equipment for damage incurred during shipment. If the equipment has been damaged, report the damage on SF 361, Transportation Discrepancy Report.
2. Check the equipment against the packing slip to see if the shipment is complete. Report all discrepancies in accordance with applicable service instructions (e.g., for Army instructions, see DA PAM 738-8).
3. Check to see whether the equipment has been modified.

Table 1. Inspection Criteria for Packaging.

COMPONENT	ACCEPTABLE	REPARABLE	NONREPARABLE
Wooden Boxes and Crates			
Hardware	Operative and tight.	Inoperative or loose.	None.
	Nails, screws, and fasteners	Nails, screws, and fasteners.	None.
Ends	Free from damage.	Broken or missing cleats and handles.	Damage that requires disassembly of box.
Wood	Splits less than 3 inches long, no closer than 1 inch to edge of board or adjoining split. The board must be secured by at least one nail on each side of the split when it extends to the end of the board.	Splits more than 3 inches but no closer than 1 inch to edge of board or adjoining split, or ½-inch wide. That can be repaired by use of corrugated fasteners.	Splits closer than 1 inch to edge of board or adjoining split or over ½-inch wide.
Fiber Containers			
Metal Ends	Minor rust, cracks, indentations, or splits that would not impair water proofing or serviceability of container.	None.	Perforations, excessive rust, or ends which are crushed or not securely crimped to body.
Body and Cap	No leaks, cuts, or gouges.	Cuts, tears, or gouges not closer than 1 inch to closure, less than ½ square inch in area, and unpenetrated layers that can be spot painted.	Cuts, tears, or gouges closer than 1 inch to closure, more than ½ square inch in area, or through all impregnated layers

0109-1

FIGURE 335. Example of a page-based TM stylesheet output for <surwp>.

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TM X–XXXX–XXX 0109

CHECKING UNPACKED EQUIPMENT– CONTINUED

Table 2. M29 and M30 Control Surfaces and Containers.

LOCATION	ITEM	ACTION	REMARKS
Container	Components	1. Inspect for rust, fungus, paint damage, and deformation.	WP 0125
		2. Reject container if damage prevents it from functioning properly.	–
M29	Control Surfaces	1. Inspect for dents and scratches on post, trailing edge phenolic, skin, and closure plate.	WP 0128
		2. Reject control surface:	–
		a. If post dents or scratches exceed 0.002 in. (0.051 mm).	
		b. If trailing edge phenolic dents exceed 0.040 in. (10.160 mm).	
		c. If skin dents exceed 0.030 in. (7.620 mm) within 2 in. (50.800 mm) of post.	
		d. If closure plate dents exceed 0.030 in. (7.620 mm) within 2 in. (50.800 mm) of post.	
M30	Control Surfaces	1. Inspect for dents and scratches on post and skin.	–
		2. Skin dents or scratches up to 0.050 in. (12.700 mm) are allowable, but should be blended.	–
		3. Reject control surface if post dents or scratches exceed 0.002 in. (0.051 mm).	–

END OF TASK

END OF WORK PACKAGE

0109-2

FIGURE 335. Example of a page-based TM stylesheet output for <surwp> - Continued.

37.4 Equipment/user fitting instructions work package ***<perseqpwp>***. Equipment/user fitting instructions work package contains instructions for personal use equipment. The information is entered as descriptive text or as procedural steps.

a. The components of ***<perseqpwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Single procedure
      (a) Procedural steps title ***<title>*** (required) (see 49.1.1.3).
      (b) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
      (c) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
      (d) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
      (e) General information ***<geninfo>*** (optional) (see 49.1.4.12).
      (f) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
      (g) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
      (h) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1-alt>*** (see 48.2.1) (required – one or more).
   (5) Multiple procedure ***<proc>*** (required – two or more) (see 33.2).
   (6) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<perseqpwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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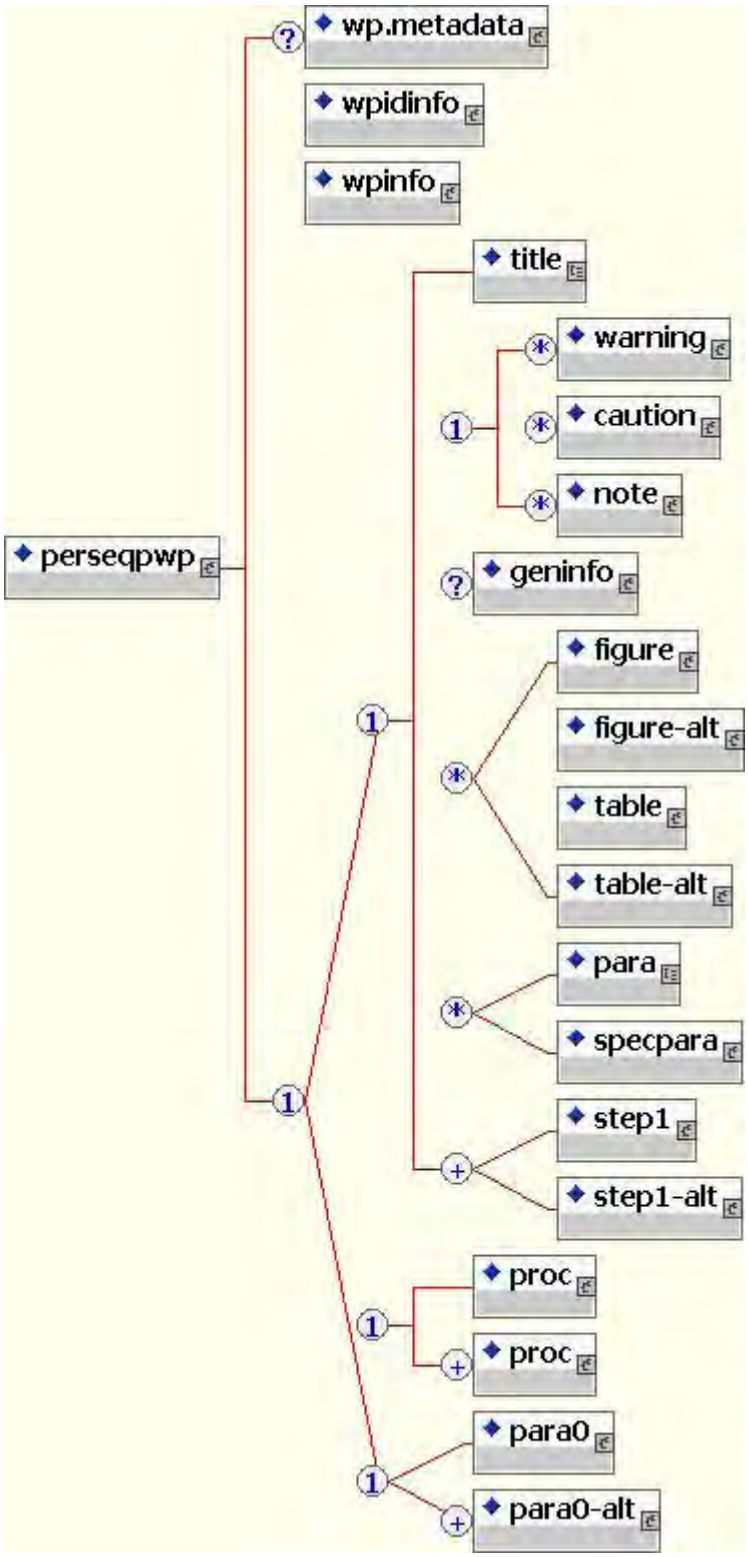

FIGURE 336. Equipment/user fitting instructions work package DTD hierarchy <perseqpwp>.

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c. DTD fragment for ***<perseqpwp>***:

```
<!ELEMENT perseqpwp (wp.metadata?, wpidinfo, wpinfo, ((title, (warning*,
   caution*, note*), geninfo?, (figure | figure-alt | table |
   table-alt)*, (para | specpara)*, (step1 | step1-alt)+) |
   (proc, proc+) | (para0 | para0-alt+)))>
<!ATTLIST perseqpwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attributes for ***<perseqwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.4.1 XML document instance fragment and output for ***<perseqpwp>***. The XML instance and its stylesheet output for a ***<perseqpwp>*** is provided below.

a. Example of an XML document instance fragment for ***<perseqpwp>***:

```
<perseqpwp airforce="no" army="no" deletewp="no" frame="yes"
marines="no" navy="no" tocentry="2" wpno="M00890-X-XXXX-XXX" wpseq="1324">
<wpidinfo>
<maintlvl level="field"/>
<title>DONNING AND FITTING THE SRU-37/P
</title>
</wpidinfo>
<wpinfo>
<title>N/A
</title>
<null insert="none"/>
</wpinfo>
<geninfo>
<title>GENERAL
</title>
<para esd="no" hcp="no">The container assembly is designed to be worn on the
user's chest, upper back, or lower back.  If the SARVIP is to be worn, it
shall be donned first.  The SRU-37/P shall then be donned over the SARVIP.
</para>
</geninfo>
<proc esd="no" frame="yes" hcp="no" tocentry="0">
<title>CHEST MOUNTING OPTION
</title>
<warning haz-abbrev="no">
<title>FLIGHT CONTROL RESTRICTION
</title>
<trim.para esd="no" hcp="no">When the SRU-37/P is worn on the
chest, certain size individuals in certain helicopters may
experience aft cyclic control restriction.
</trim.para>
</warning>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Position the SRU-37/P across the chest and place
both shoulder straps over the shoulders and back.  The right shoulder
strap with male buckle shall be passed under the left armpit and
securely inserted into the female buckle of the left cross strap.
The left shoulder strap with male buckle shall be passed under the
```


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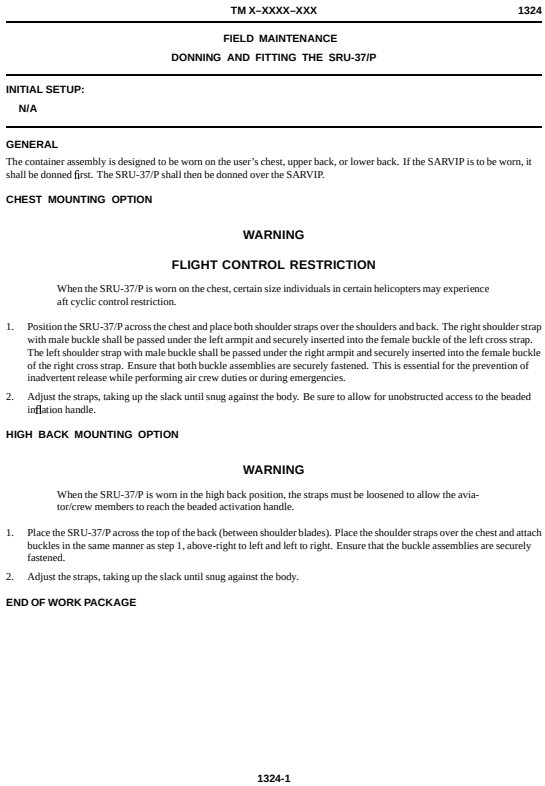
```
right armpit and securely inserted into the female buckle of the
right cross strap.  Ensure that both buckle assemblies are securely
fastened.  This is essential for the prevention of inadvertent release
while performing air crew duties or during emergencies.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Adjust the straps, taking up the slack
until snug against the body.  Be sure to allow for unobstructed
access to the beaded inflation handle.
</para>
</step1>
</proc>
<proc esd="no" frame="yes" hcp="no" tocentry="0">
<title>HIGH BACK MOUNTING OPTION
</title>
<warning haz-abbrev="no">
<trim.para esd="no" hcp="no">When the SRU-37/P is worn in the high back
position, the straps must be loosened to allow the aviator/crew
members to reach the beaded activation handle.
</trim.para>
</warning>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Place the SRU-37/P across the top of the back
(between shoulder blades).  Place the shoulder straps over the chest and
attach buckles in the same manner as step 1, above-right to left and left
to right.  Ensure that the buckle assemblies are securely fastened.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Adjust the straps, taking up the
slack until snug against the body.
</para>
</step1>
</proc>
</perseqpwp>
```

b. Page-based TM stylesheet output example for ***<persepqwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.



TM X–XXXX–XXX 1324

FIELD MAINTENANCE

DONNING AND FITTING THE SRU-37/P

INITIAL SETUP:

N/A

GENERAL

The container assembly is designed to be worn on the user's chest, upper back, or lower back. If the SARVIP is to be worn, it shall be donned first. The SRU-37/P shall then be donned over the SARVIP.

CHEST MOUNTING OPTION

WARNING

FLIGHT CONTROL RESTRICTION

When the SRU-37/P is worn on the chest, certain size individuals in certain helicopters may experience aft cyclic control restriction.

1. Position the SRU-37/P across the chest and place both shoulder straps over the shoulders and back. The right shoulder strap with male buckle shall be passed under the left armpit and securely inserted into the female buckle of the left cross strap. The left shoulder strap with male buckle shall be passed under the right armpit and securely inserted into the female buckle of the right cross strap. Ensure that both buckle assemblies are securely fastened. This is essential for the prevention of inadvertent release while performing air crew duties or during emergencies.
2. Adjust the straps, taking up the slack until snug against the body. Be sure to allow for unobstructed access to the beaded inflation handle.

HIGH BACK MOUNTING OPTION

WARNING

When the SRU-37/P is worn in the high back position, the straps must be loosened to allow the aviator/crew members to reach the beaded activation handle.

1. Place the SRU-37/P across the top of the back (between shoulder blades). Place the shoulder straps over the chest and attach buckles in the same manner as step 1, above-right to left and left to right. Ensure that the buckle assemblies are securely fastened.
2. Adjust the straps, taking up the slack until snug against the body.

END OF WORK PACKAGE

1324-1

FIGURE 337. Example of a page-based TM stylesheet output for <perseqpwp>.

37.5 Preventive maintenance checks and services (PMCS) introduction work package ***<pmcsintrowp>***. The function of the PMCS introduction work package is to explain the purpose and use of the PMCS data such as a description and explanation of each PMCS table column or entry, an explanation of how to document PMCS deficiencies and shortcomings using DA Form 2404, a statement concerning Corrosion Prevention and Control (CPC) with references to the applicable CPC instructions and indicating adherence to the CPC instructions, an identification of the lubricants used in the PMCS, the standard military symbols for the lubricants, and the interval identifiers and interval definitions (e.g., D: daily, W: weekly, MI: miles operated), an oil filter standard statement, an Army Oil Analysis Program (AOAP) sampling interval statement, an AOAP not avaialble/non-enrolled standard statement, when an applicable component is not enrolled in the AOAP, or oil analysis support is not available, a warranty hardtime standard statement, if the equipment is under warranty and is not enrolled in the AOAP, and/or Leakage criteria when the equipment contains fluids such as lubrication oil or hydraulic fluid. Many of these subjects may have standard statements in boilerplate entries (see 37.5.2).

a. The components of ***<pmcsintrowp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Introduction ***<intro>*** (required) (see 49.1.4.15).
   (4) Descriptive introduction titled paragraphs (required – one or more).
      (a) Titled first level paragraph ***<para0>*** (see 49.1.1.8).
      (b) Conditional titled first level paragraph ***<para0–alt>*** (see 48.2.1).
   (5) Fluid leakage statement ***<fluid.leakage>*** (optional) (see 37.5.1).
b. DTD graphically depicted for ***<pmcsintrowp>***:

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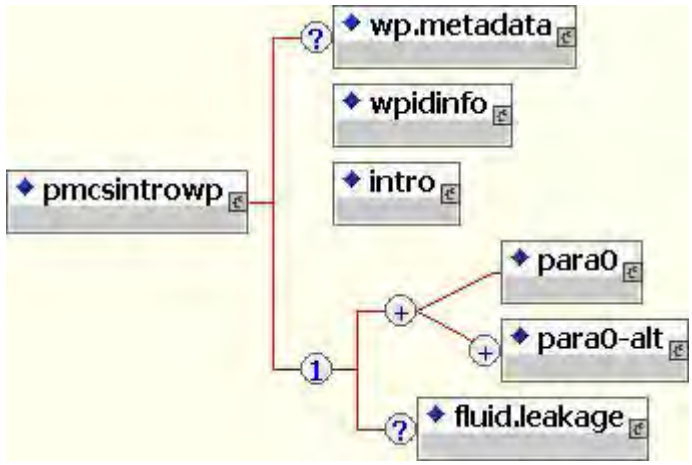

FIGURE 338. PMCS introduction work package DTD hierarchy <pmcsintrowp>.

c. DTD fragment for ***<pmcsintrowp>***:

```
<!ELEMENT pmcsintrowp (wp.metadata?, wpidinfo, intro, ((para0 |
   para0-alt)+, fluid.leakage?))>
<!ATTLIST pmcsintrowp
   airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time)  #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts )  #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes for ***<pmcsintrowp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

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(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.5.1 Fluid leakage statement ***<fluid.leakage>***. In accordance with MIL–STD–40051–1/–2 warnings or cautions can not appear in narrative text. The standard statement in accordance with MIL–STD–40051–1/–2 requires a caution in describing fluid leakage. The DTD includes the ***<fluid.leakage>*** element to handle this exception. Additionally, the standard statement is contained in boilerplate entities (see 37.5.2).

37.5.2 PMCS introduction standard statement boilerplate entries. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed (see Table XI).

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TABLE XI. Boilerplate Entities for *<pmcsintrowp>*.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
AOAP Sampling Interval Statement	***&pmcsintrowp.intro.aoap;***	— —
• Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
• Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***
• AOAP Timeframe – Page-base TMs Only		***<![%page-base;[<!ENTITY pmcsintrowp.intro.aoap.timeframe "***REPLACE WITH APPLICABLE HOUR/MILEAGE TIMEFRAME***"]]>***
• AOAP Reference – Page-base TMs Only • DA PAM 738-751 • Replace IGNORE to INCLUDE • Replace INCLUDE to IGNORE • DA PAM 738–8 • No change is required		***<![%page-base;[<!ENTITY pmcsintrowp.intro.aoap.prescribed '<![***IGNORE***[<extref docno="DA PAM 738-751" posttext=", Functional Users Manual for the Army Maintenance Management Systems - Aviation (TAMMS-A)"/>]]>*** ***<![***INCLUDE***[<extref docno="DA PAM 738-8" posttext=", The Army Maintenance Management System (TAMMS) Users Manual)"/>]]'>]]>***
AOAP Not Available/Non-Enrolled Statement	***&pmcsintrowp.intro.aoap-na;***	— —
• Component or Equipment Name		***<!ENTITY pmcsintrowp.intro.component-equipment "***REPLACE WITH NAME OF COMPONENT/EQUIPMENT***">***
Fluid Leakage Statement	***&pmcsintrowp.intro.fluid-leakage;***	— —
• Component or Equipment Name		***<!ENTITY pmcsintrowp.intro.component-equipment "***REPLACE WITH NAME OF COMPONENT/EQUIPMENT***">***
Conditions to Change Oil Filter Statement	***&pmcsintrowp.intro.oilfilter;***	Not Applicable
Warranty Statement	***&pmcsintrowp.intro.warranty;***	Not Applicable

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37.5.3 XML document instance fragment and output for ***<pmcsintrowp>***. The XML instance and its stylesheet output for ***<pmcsintrowp>*** is provided below.

a. Example of an XML document instance fragment for ***<pmcsintrowp>***:

```
<pmcsintrowp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="M00062-X-XXX-XXX" wpseq="0062">
<wpidinfo>
<maintlvl level="operator"/>
<title>NBCRS FOX M93A1 PMCS PROCEDURES INTRODUCTION
</title>
</wpidinfo>
<intro frame="no">
<para0 esd="no" hcp="no">
<title>INTRODUCTION
</title>
<para esd="no" hcp="no">Preventive Maintenance Checks and Services (PMCS)
are performed to keep the vehicle in operating condition. The checks
are used to find, correct, or report problems. Crew members are to
do the PMCS jobs as shown in the PMCS table. PMCS are done every
day the vehicle is operated, using the PMCS table. Pay attention
to WARNING and CAUTION statements. A WARNING means someone could
be hurt. A CAUTION means equipment could be damaged.
<randlist bullet="yes">
<item>Before you begin operating vehicle equipment, do Before PMCS.
</item>
<item>During operation, do During PMCS.
</item>
<item>After operation, do After PMCS.
</item>
<item>Once a week do Weekly PMCS. If vehicle has not been operated
in a week, also do Before PMCS at the same time.
</item>
<item>Do Monthly PMCS once a month. If vehicle has not been operated
in a month, also do After PMCS at the same time.
</item>
<item>If you are operating the vehicle for the first time, do your
Weekly and Monthly PMCS the first time you do your Before PCMS.
</item>
<item>If you find something wrong when performing PMCS, fix it if you
can, using troubleshooting procedures and/or maintenance procedures.
</item>
<item>The right-hand column of the PMCS table lists conditions that make
the vehicle not fully mission capable. Write up items not-fixed on
<extref docno="DA Form 2404"/> for unit maintenance. For further
information on how to use this form
<extref pretext=", see " docno="DA PAM 738-750"/>.
</item>
<item>If tools required to perform PMCS are not listed in
<xref wpid="M00361-X-XXX-XXX"/>, notify unit maintenance.
</item>
</randlist>
</para>
</para0>
</intro>
<para0 esd="no" hcp="no">
<title>INSPECTION
</title>
```


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<para esd="no" hcp="no">Look for signs of a problem or trouble. Senses help here. You can feel, smell, hear, or see many problems. Be alert when on the vehicle.
</para>
<para esd="no" hcp="no">Inspect to see if items are in good condition. Are they correctly assembled, stowed, secured, excessively worn, leaking, corroded, or properly lubricated? Correct any problems found or notify unit maintenance.
</para>
<para esd="no" hcp="no">There are some common items to check all over the vehicle. These include the following:
<seqlist>
<item>Bolts, clamps, nuts, and screws: Continuously check for looseness. Look for chipped paint, bear metal, rust, or corrosion around bolt and screw heads and nuts. Tighten them when you find them loose. If tools are not available, notify unit maintenance.
</item>
<item>Welds: Many items on the vehicle are welded. To check these welds, look for chipped paint, rust, corrosion, or gaps. When these conditions exist, notify unit maintenance on
<extref docno="DA Form 2404"/>.
</item>
<item>Electrical wires, connectors, and harnesses: Tighten loose connectors. Look for cracked or broken insulation, bare wires, and broken connectors. If any are found, notify unit maintenance.
</item>
<item>Hoses and fluid lines: Look for wear, damage and leaks, and make sure clamps and fittings are tight. Wet spots mean a leak. A stain by a fitting or connector can also mean a leak. When you find a leak, notify unit maintenance.
</item>
</seqlist>
</para>
</para0>
<para0 esd="no" hcp="no">
<title>LUBRICATION SERVICE INTERVALS - NORMAL CONDITIONS
</title>
<para esd="no" hcp="no">For safer, more trouble free operations, make sure that your vehicle is serviced when it needs it. For the proper lubrication and service intervals
<extref docno="LO 9-6665-376-12" pretext=", see "/>.
</para>
</para0>
<para0 esd="no" hcp="no">
<title>LUBRICATION SERVICE INTERVALS - UNUSUAL CONDITIONS
</title>
<para esd="no" hcp="no">Your vehicle will require extra service and care when you operate under unusual conditions. High or low temperatures, long periods of hard use, or continued use in sand, water, mud, or snow will break down the lubricant, requiring you to add or change lubricant more often.
</para>
</para0>
<para0 esd="no" hcp="no">
<title>CLEANING AND LUBRICATION
</title>

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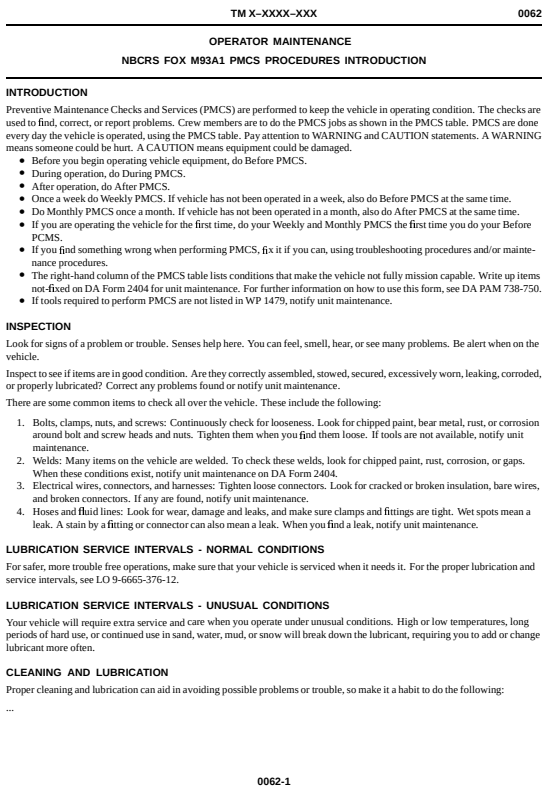
```
<para esd="no" hcp="no">Proper cleaning and lubrication can aid in avoiding
possible problems or trouble, so make it a habit to do the following:
</para>
<para>. . .
</para>
</para0>  &pmcsintrowp.intro.fluid.leakage;
</pmcsintrowp>
```

b. Page-based TM stylesheet output example for ***<pmcsintrowp>***:



TM X–XXXX–XXX 0062

OPERATOR MAINTENANCE

NBCRS FOX M93A1 PMCS PROCEDURES INTRODUCTION

INTRODUCTION

Preventive Maintenance Checks and Services (PMCS) are performed to keep the vehicle in operating condition. The checks are used to find, correct, or report problems. Crew members are to do the PMCS jobs as shown in the PMCS table. PMCS are done every day the vehicle is operated, using the PMCS table. Pay attention to WARNING and CAUTION statements. A WARNING means someone could be hurt. A CAUTION means equipment could be damaged.

- Before you begin operating vehicle equipment, do Before PMCS.
- During operation, do During PMCS.
- After operation, do After PMCS.
- Once a week do Weekly PMCS. If vehicle has not been operated in a week, also do Before PMCS at the same time.
- Do Monthly PMCS once a month. If vehicle has not been operated in a month, also do After PMCS at the same time.
- If you are operating the vehicle for the first time, do your Weekly and Monthly PMCS the first time you do your Before PCMS.
- If you find something wrong when performing PMCS, fix it if you can, using troubleshooting procedures and/or maintenance procedures.
- The right-hand column of the PMCS table lists conditions that make the vehicle not fully mission capable. Write up items not-fixed on DA Form 2404 for unit maintenance. For further information on how to use this form, see DA PAM 738-750.
- If tools required to perform PMCS are not listed in WP 1479, notify unit maintenance.

INSPECTION

Look for signs of a problem or trouble. Senses help here. You can feel, smell, hear, or see many problems. Be alert when on the vehicle.

Inspect to see if items are in good condition. Are they correctly assembled, stowed, secured, excessively worn, leaking, corroded, or properly lubricated? Correct any problems found or notify unit maintenance.

There are some common items to check all over the vehicle. These include the following:

1. Bolts, clamps, nuts, and screws: Continuously check for looseness. Look for chipped paint, bear metal, rust, or corrosion around bolt and screw heads and nuts. Tighten them when you find them loose. If tools are not available, notify unit maintenance.
2. Welds: Many items on the vehicle are welded. To check these welds, look for chipped paint, rust, corrosion, or gaps. When these conditions exist, notify unit maintenance on DA Form 2404.
3. Electrical wires, connectors, and harnesses: Tighten loose connectors. Look for cracked or broken insulation, bare wires, and broken connectors. If any are found, notify unit maintenance.
4. Hoses and fluid lines: Look for wear, damage and leaks, and make sure clamps and fittings are tight. Wet spots mean a leak. A stain by a fitting or connector can also mean a leak. When you find a leak, notify unit maintenance.

LUBRICATION SERVICE INTERVALS - NORMAL CONDITIONS

For safer, more trouble free operations, make sure that your vehicle is serviced when it needs it. For the proper lubrication and service intervals, see LO 9-6665-376-12.

LUBRICATION SERVICE INTERVALS - UNUSUAL CONDITIONS

Your vehicle will require extra service and care when you operate under unusual conditions. High or low temperatures, long periods of hard use, or continued use in sand, water, mud, or snow will break down the lubricant, requiring you to add or change lubricant more often.

CLEANING AND LUBRICATION

Proper cleaning and lubrication can aid in avoiding possible problems or trouble, so make it a habit to do the following:

...

0062-1

FIGURE 339. Example of a page-based TM stylesheet output for <pmcsintrowp>.

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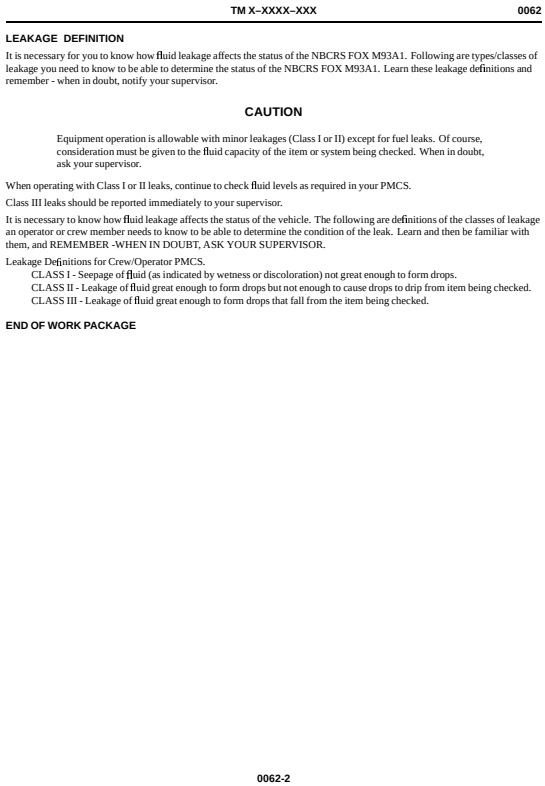

TM X–XXXX–XXX 0062

LEAKAGE DEFINITION

It is necessary for you to know how fluid leakage affects the status of the NBCRS FOX M93A1. Following are types/classes of leakage you need to know to be able to determine the status of the NBCRS FOX M93A1. Learn these leakage definitions and remember - when in doubt, notify your supervisor.

CAUTION

Equipment operation is allowable with minor leakages (Class I or II) except for fuel leaks. Of course, consideration must be given to the fluid capacity of the item or system being checked. When in doubt, ask your supervisor.

When operating with Class I or II leaks, continue to check fluid levels as required in your PMCS.

Class III leaks should be reported immediately to your supervisor.

It is necessary to know how fluid leakage affects the status of the vehicle. The following are definitions of the classes of leakage an operator or crew member needs to know to be able to determine the condition of the leak. Learn and then be familiar with them, and REMEMBER -WHEN IN DOUBT, ASK YOUR SUPERVISOR.

Leakage Definitions for Crew/Operator PMCS.

CLASS I - Seepage of fluid (as indicated by wetness or discoloration) not great enough to form drops.
CLASS II - Leakage of fluid great enough to form drops but not enough to cause drops to drip from item being checked.
CLASS III - Leakage of fluid great enough to form drops that fall from the item being checked.

END OF WORK PACKAGE

0062-2

FIGURE 339. Example of a page-based TM stylesheet output for <pmcsintrowp> - Continued.

37.6 Preventive maintenance checks and services (PMCS) work package ***<pmcswp>***. The PMCS work package contains the data required to perform an equipment PMCS. The PMCS work package includes periodic lubrication instructions, applicable scheduled corrosion inspections, and equipment checks to ensure the equipment is maintained in good working order. Extensive lubrication instructions for a repair action is included in a separate lubrication work package ***<lubewp>*** (see 37.9).

a. The components of ***<pmcswp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) PMCS standard information***<pmcstable>*** (required – one or more) (see 37.6.1).
   (5) Mandatory replacement parts ***<mrplpart>*** (required) (see 37.6.2).

b. DTD graphically depicted for ***<pmcswp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

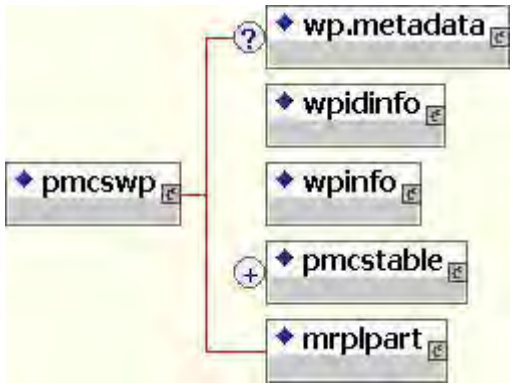

FIGURE 340. PMCS work package DTD hierarchy <pmcswp>.

c. DTD fragment for ***<pmcswp>***:

```
<!ELEMENT pmcswp (wp.metadata?, wpidinfo, wpinfo, pmcstable+, mrplpart)>
<!ATTLIST pmcswp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attributes for ***<pmcswp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

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(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.6.1 PMCS standard information ***<pmcstable>***. The PMCS standard information identifies the detailed requirements and intervals for conducting preventive maintenance checks and services.

a. The components of ***<pmcstable>*** are:
(1) Standard information title ***<title>*** (required) (see 49.1.1.3). The title generally contains the equipment and/or interval to be inspected.
(2) PMCS intervals ***<pmcs-intervals>*** (optional) (see 37.6.1.1).
(3) Standard table header ***<std.table.header>*** (required) (see 37.6.4).
(4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(7) PMCS entry ***<pmcs-entry>*** (required – one or more) (see 37.6.1.2).

b. DTD graphically depicted for ***<pmcstable>***:

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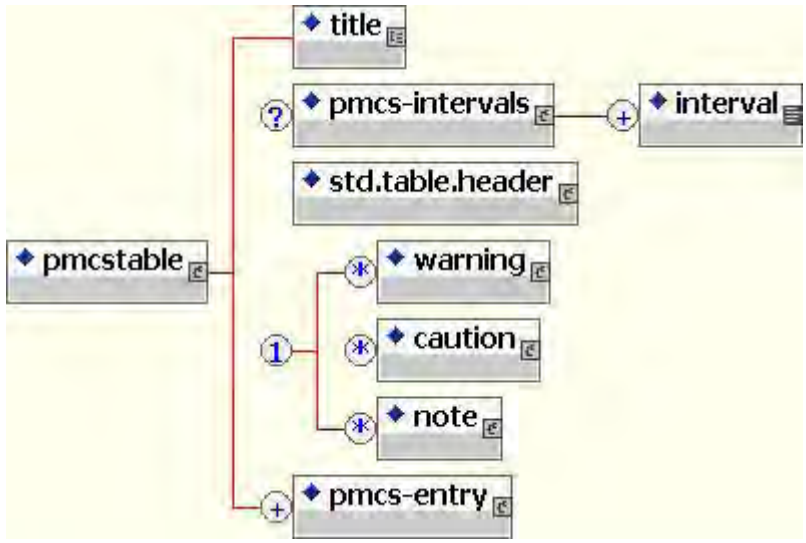

FIGURE 341. PMCS standard information DTD hierarchy <pmcstable>.

c. DTD fragment for ***<pmcstable>***:

```
<!ELEMENT pmcstable (title, pmcs-intervals?, std.table.header, (warning*,
    caution*, note*), pmcs-entry+)>
<ATTLIST pmcstable
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1"
   use-manhours       (yes | no)                       "no"
>
```

d. Unique attributes for ***<pmcstable>*** is **use manhours** – Is man-hours entry used? (default value is ”no”) Identifies whether the PMCS table requires man-hours entries (used for lubrication procedures).

e. Common attributes for ***<pmcstable>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is “**0**” do not include, “**3**” the 3rd level, “**4**” the 4th level, or “**5**” the 5th level (default value is “**1**”).

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37.6.1.1 PMCS interval definition ***<pmcs-intervals>***. The ***<pmcs-intervals>*** is used to define the various PMCS intervals used during this inspection. Defining the intervals provides the information for possible filtering the checks or services for only the requested PMCS intervals to be performed. Each interval used in the PMCS is defined in the element ***<interval>*** (see 37.6.1.2.a.(2)) and should match the ***<interval>*** defined in the element ***<pmcs-entry>***.

37.6.1.2 PMCS entry ***<pmcs-entry>***. The element contains the information to conduct the check or service for an inspection item. The element is similar to a "row" in a structural table.

a. The components of ***<pmcs-entry>*** are:
   (1) Item number ***<itemno>*** (required) – Identifies the number assigned to the item to be checked or serviced and is located under the **ITEM NO.** heading. The element is similar to a "cell" in a structural table and is entered in column one.
   (2) Inspection interval ***<interval>*** (required – one or more) – Identifies when each check is to be performed (i.e., "before", "during", "after", "weekly"), and is located under the **INTERVAL** heading. The element is similar to a "cell" in a structural table and is entered in column two. If filtering the intervals, the interval data should exactly match the interval data within the element ***<pmcs-intervals>*** (see 37.6.1.1).
   (3) Man-hours ***<manhours>*** (optional) – Identifies the man-hours required to perform the lubrication and is located under the **MAN-HOUR** heading. The element is similar to a "cell" in a structural table and is entered in column third.
   (4) Item to be checked or serviced ***<checked>*** (required) – Identifies the item to be checked or serviced, and is located under the **ITEM TO BE CHECKED OR SERVICED** heading. The element is similar to a "cell" in a structural table and is entered in column third or fourth (if Man-hours is requirement).
   (5) PMCS procedure***<pmcsproc>*** (required – one or more) (see 37.6.1.3).

b. DTD fragment is graphically depicted.

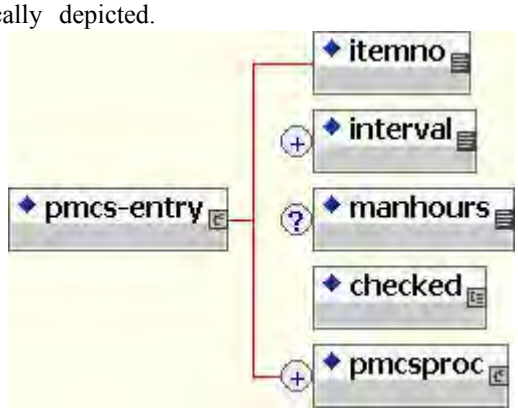

FIGURE 342. PMCS entry DTD hierarchy <pmcs-entry>..

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c. DTD fragments for ***<pmcs-entry>***:

```
<!ELEMENT pmcs-entry (itemno, interval+, manhours?, checked, pmcsproc+)>
<!ATTLIST pmcs-entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<pmcs-entry>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

37.6.1.3 PMCS procedure and equipment not available ***<pmcsproc>***. The element contains each check or service step (located under the **PROCEDURE** heading) with the associated equipment not available criteria (if any) (located under the **EQUIPMENT NOT READY/AVAILABLE IF**). The PMCS step contains a brief step-by-step description of each check or service to be performed. Included within the steps is any information required to accomplish the checks or services, including lubrication, tolerances, adjustment limits, and instrument gage readings and any illustration(s) necessary to identify the step location. Refer to MIL–STD–40051–1/–2, Appendix E, for a list of procedures when the equipment being serviced has a scheduled periodic/scheduled lubrication requirement. Reference, as applicable, the standard statement for lubrication authorization (use boilerplate entities ***&pmcswp.pmcstable.pmcsproc.arctic-oper.auth;*** or ***&pmcswp.pmcstable.pmcsproc.arctic-oper.not-auth;*** see 37.6.3). If a PMCS step has a criteria that renders the equipment not ready and/or available, the equipment condition (e.g., out-of-limits, malfunctions, etc.) is identified with the procedural step (see 37.6.1.4).

a. The components of ***<pmcsproc>*** are:

(1) Procedure title ***<title>*** (optional) (see 49.1.1.3).
(2) PMCS procedural step(s) requires either a
(a) Unnumbered step (single) ***<pmcspara>*** (required) (see 37.6.1.4).
(b) Numbered steps (multiple) ***<pmcsstep1>*** (required – one or more) (see 37.6.1.4).

b. DTD fragment is graphically depicted.

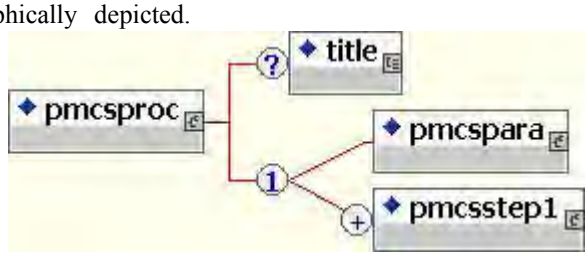

FIGURE 343. PMCS procedure DTD hierarchy <pmcsproc>..

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c. DTD fragments for ***<pmcsproc>***:

```
<!ELEMENT pmcsproc (title?, (pmcspara | pmcsstep1+))>
<!ATTLIST pmcs-entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
   esd               (yes | no)                      "no"
    hcp               (yes | no)                       "no"
   crewmember   CDATA    #IMPLIED
>
```

d. Unique attribute for ***<pmcsproc>*** is **crewmember** – Crew member(s) only (optional). Identifies the crew member(s) that performs the check or service procedural steps. The information displayed before the first step with similar formatting as a PMCS procedural title.

e. Common attributes for ***<pmcsproc>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(12) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).

37.6.1.4 PMCS procedural step with associated equipment not available/ready criteria. Each PMCS procedural step provides the check or service action to perform and if the action has a condition when the equipment would not be available or ready is stated with the step. The PMCS procedural step is located under the **PROCEDURE** heading and the equipment not ready/available criteria is located under the **EQUIPMENT NOT READY/AVAILABLE IF** heading and aligned with the associated step. In a style sheet, each ***<pmcsstep1>*** (after the initial step) is considered a new row with empty cells for previous columns. Creating a new row has two purposes, first allowing necessary page breaks for long procedures and the second maintains the step to equipment not ready/available criteria association.

a. The components of ***<pmcspara>*** are:

(1) Requires either a
(a) PMCS procedural step narrative with alerts ***<specpara>*** (required) (see 49.1.1.6). The element is similar to a "cell" in a structural table and is entered in column four or five (if Man-hours is requirement).
(b) PMCS procedural step narrative ***<para>*** (required) (see 49.1.1.5). The element is similar to a "cell" in a structural table and is entered in column four or five (if Man-hours is requirement).
(2) Equipment not ready/available if criteria ***<eqpnotavail>*** (optional). The element is similar to a "cell" in a structural table and is entered in column five or six (if Man-hours is requirement).

b. The components of ***<pmcsstep1>***, ***<pmcsstep2>***, ***<pmcsstep3>***, and ***<pmcsstep4>*** are:

(1) Requires either a

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(a) PMCS procedural step narrative with alerts ***<specpara>*** (required) (see 49.1.1.6). The element is similar to a "cell" in a structural table and is entered in column four or five (if Man-hours is requirement).

(b) PMCS procedural step narrative ***<para>*** (required) (see 49.1.1.5). The element is similar to a "cell" in a structural table and is entered in column four or five (if Man-hours is requirement).

(2) Equipment not ready/available if criteria ***<eqpnotavail>*** (optional). The element is similar to a "cell" in a structural table and is entered in column five or six (if Man-hours is requirement).

(3) Subordinate procedural steps (optional) ***<pmcsstep2>***, ***<pmcsstep3>***, and ***<pmcsstep4>*** are model similarly to ***<pmcsstep1>***.

c. DTD fragment is graphically depicted.

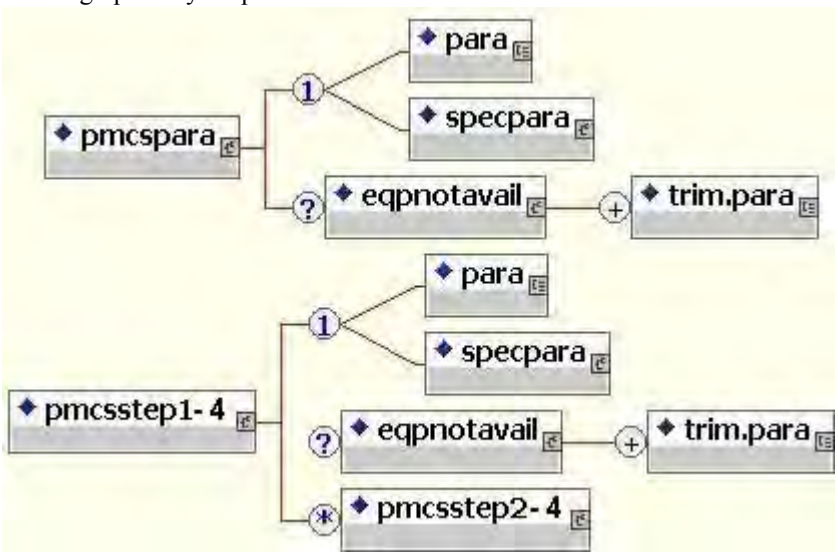

FIGURE 344. PMCS step DTD hierarchy <pmcspara> and <pmcsstep#>.

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d. DTD fragments for ***<pmcspara>***, ***<pmcsstep1>***, ***<pmcsstep2>***, ***<pmcsstep3>***, and ***<pmcsstep4>*** are:

```
<!ELEMENT pmcspara ((para | specpara), eqpnotavail?)>
<!ATTLIST pmcspara
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
<!ELEMENT pmcsstep1 ((para | specpara), eqpnotavail?, pmcsstep2*)>
<!ELEMENT pmcsstep2 ((para | specpara), eqpnotavail?, pmcsstep3*)>
<!ELEMENT pmcsstep3 ((para | specpara), eqpnotavail?, pmcsstep4*)>
<!ELEMENT pmcsstep4 ((para | specpara), eqpnotavail?)>
<!ATTLIST pmcsstep1, pmcsstep2, pmcsstep3, and pmcsstep4 >
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
   esd               (yes | no)                      "no"
    hcp               (yes | no)                       "no"
    qa                (yes | no)                       "no"
   crewmember   CDATA    #IMPLIED
>
```

e. Unique attribute for ***<pmcsstep1>***, ***<pmcsstep2>***, ***<pmcsstep3>***, and ***<pmcsstep4>*** is **crewmember** – Crew member(s) only (optional). Identifies the crew member(s) that performs the check or service with the procedural step. The information displayed after the step number and before the step narrative. Generally, the information is formatted in bold.

f. Common attributes for ***<pmcspara>***, ***<pmcsstep1>***, ***<pmcsstep2>***, ***<pmcsstep3>***, and ***<pmcsstep4>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **esd** – Electrostatic discharge requirement (default value is “no”) (see 49.2.5).
(12) **hcp** – Nuclear hardness requirement (default value is “no”) (see 49.2.5).
(13) **qa** Quality assurance (default value is “no”) (see 49.2.10).

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37.6.2 Mandatory replacement parts ***<mrplpart>***. The mandatory replacement parts ***<mrplpart>*** list contains all items that are replaced during a PMCS. When no mandatory replace parts exists this is stated as a standard statement from MIL–STD–40051–1/–2 (use ***&pmcswp.pmcstable.mrplpart.no-mrp;***).

a. The components of ***<mrplpart>*** are:
   (1) Title ***<title>*** (required).
   (2) Paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
   (3) Mandatory replacement parts list ***<mrpl>*** (optional) (see 40.9.1).

b. DTD fragment for ***<mrplpart>***:

```
<!ELEMENT mrplpart (title, para+, mrpl?)>
<!ATTLIST mrplpart
    assocfig        IDREFS                          #IMPLIED
    changeref       IDREFS                          #IMPLIED
    comment         CDATA                           #IMPLIED
    delchlvl        CDATA                           #IMPLIED
    id              ID                              #IMPLIED
    idref           IDREFS                          #IMPLIED
    inschlvl        CDATA                           #IMPLIED
    security        (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk        CDATA                           #IMPLIED
>
```

c. Common attributes for ***<mrplpart>*** are:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

37.6.3 PMCS work package standard statement boilerplates. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed Table XII.

TABLE XII. Boilerplate Entities for ***<pmcswp>***.

Description	**Boilerplate Entity**	**Selectable Entity to Set Editable Entity to Set**
PMCS lubricant is authorized in arctic operation note	***&pmcswp.pmcstable.pmcsproc.arctic-oper.auth;***	Not applicable
PMCS lubrication is not authorized in arctic operation note	***&pmcswp.pmcstable.pmcsproc.arctic-oper.not-auth;***	Not applicable
PMCS with no mandatory replacement parts statement.	***&pmcswp.pmcstable.mrplpart.no-mrp;***	Not applicable

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37.6.4 PMCS and mandatory replacement parts standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XIII.

TABLE XIII. PMCS standard information header ***<std.table.header>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set
PMCS without Man-hours Standard Information Header	***&pmcswp.pmcstable.std-header;***	Not applicable.
PMCS with Man-hoursStandard Information Header	***&pmcswp.pmcstable.std-header.manhours;***	Not applicable.
Mandatory Replacement Parts Standard Information Header	***&mrplwp.mrpllist.std-header;***	— —
Frame-based		***<!ENTITY % frame-base "INCLUDE"> <!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE"> <!ENTITY % page-base "INCLUDE">***

37.6.5 XML document instance fragment and output for ***<pmcswp>***. The XML instance and its stylesheet output for a ***<pmcswp>*** is provided below.

a. Example of an XML document instance fragment for ***<pmcswp>***:

```
<pmcswp wpno="M0689-X-XXXX-XXX" tocentry="2" frame="yes" army="no"
airforce="no" navy="no" marines="no" wpseq="0063" deletewp="no">
<wpidinfo>
<maintlvl level="operator"/>
<title>NBCRS FOX M93A1
<brk/>Preventive Maintenance checks and Services
</title>
</wpidinfo>
<wpinfo>
<title>N/A
```


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```
</title>
<null/>
</wpinfo>
<pmcstable tocentry="1" use-manhours="no">
<title>Preventive Maintenance checks and Services for NBCRS FOX M93A1.
</title> &pmcswp.pmcstable.std-header;
<note acknowledge="no">
<trim.para hcp="no" esd="no">Driver, commander and crew will inspect for
damaged or missing items while performing checks in walk around sequence.
</trim.para>
</note>
<pmcs-entry>
<itemno>1
</itemno>
<interval>Before
</interval>
<checked>Vehicle Exterior
</checked>
<pmcsproc hcp="no" esd="no" crewmember="COMMANDER">
<pmcsstep1 hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Check for damaged and missing items. Check
lighting fixtures. Make sure all stowed items (pioneer equipment,
etc.) are secured to vehicle for travel.
</para>
</pmcsstep1>
<pmcsstep1 hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Secure stowed items. Replace missing items. Report
major damage of lighting fixtures. Report minor damage after operations.
</para>
<eqpnotavail>
<trim.para hcp="no" esd="no">Any damage or missing items
that will prevent operation.
</trim.para>
</eqpnotavail>
</pmcsstep1>
</pmcsproc>
</pmcs-entry>
<pmcs-entry>
<itemno>2
</itemno>
<interval>Before
</interval>
<checked>Left Side Exercise
</checked>
<pmcsproc hcp="no" esd="no" crewmember="DRIVER">
<pmcspara>
<specpara hcp="no" esd="no">
<note acknowledge="no">
<trim.para hcp="no" esd="no">If leakage is detected, further investigation
is needed to determine the location and cause of the leak.
</trim.para>
</note>
<para hcp="no" esd="no">Check underneath vehicle for evi-
dence of fuel, or coolant.
</para>
</specpara>
```


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<eqpnotavail>
<trim.para hcp="no" esd="no">Any class III leak of oil, fuel leakage.
</trim.para>
</eqpnotavail>
</pmcspara>
</pmcsproc>
</pmcs-entry>
<pmcs-entry>
<itemno>3
</itemno>
<interval>Before
</interval>
<checked>Left Side Tires
</checked>
<pmcsproc hcp="no" esd="no" crewmember="DRIVER">
<pmcspara>
<specpara hcp="no" esd="no">
<warning haz-abbrev="no">
<trim.para hcp="no" esd="no">Operating a vehicle with an improperly inflated tire or with a questionable defect may lead to premature tire failure and cause equipment damage, injury, or death to personnel.
</trim.para>
</warning>
<para hcp="no" esd="no">Check for missing or damaged tires and wheels. Visually check for proper tire inflation.
</para>
</specpara>
<eqpnotavail>
<trim.para hcp="no" esd="no">Missing or damaged tires or wheels. Flat or deflated tires.
</trim.para>
</eqpnotavail>
</pmcspara>
</pmcsproc>
</pmcs-entry>
<pmcs-entry>
<itemno>7
</itemno>
<interval>Before
</interval>
<checked>Reciprocator
</checked>
<pmcsproc hcp="no" esd="no" crewmember="CREW MEMBER 1">
<pmcsstep1 hcp="no" esd="no" qa="no">
<specpara hcp="no" esd="no">
<warning haz-abbrev="no">
<trim.para hcp="no" esd="no">Do not remove cap
<callout label="37" assocfig="M0689-X-XXXX-XXX-fig3"/> when engine is hot.
</trim.para>
</warning>
<note acknowledge="no">
<trim.para hcp="no" esd="no">The reciprocator is charged with a pressure of 550 psi (38.5 kg/cm2) in the M109 and M109A1. With temperature of 100° F or over, 600 psi (42.0 kg/cm2) in the M109A1 is 2 recommended.
</trim.para>
</note>

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```
<para hcp="no" esd="no">Check pins
<callout label="35assocfig="M0689-X-XXXX-XXX-fig3""/> before firing. Pins must
not extend over 3/4 inch (19.1mm); 1/8 inch (3.2mm) is correct. If pins
<callout assocfig="M0689-X-XXXX-XXX-fig3" label="35"/> do extend
beyond 3/4 inch, remove cap
<callout assocfig="M0689-X-XXXX-XXX-fig3" label="37"/> from valve
<callout assocfig="M0689-X-XXXX-XXX-fig3" label="36"/> and hydraulic fluid
<xref wpid="M0746-X-XXXX-XXX" itemno="5" pretext="(" posttext=")"/>
until pins extend 1/8 inch.
<figure application="both" figtype="normal-page" tocentry="1"
pane="no" id="M0689-X-XXXX-XXX-fig3">
<title>Reciprocator
</title>
<graphic boardno="PMCSRECIPROCATOR" unitmeasure="in" hscale="50">
</graphic>
</figure>
</para>
</specpara>
<eqpnotavail>
<trim.para hcp="no" esd="no">Pins
<callout label="35"
assocfig="M0689-X-XXXX-XXX-fig3"/> extended over 3/4 inch
</trim.para>
</eqpnotavail>
</pmcsstep1>
</pmcsproc>
</pmcs-entry> ...
</pmcstable>
<mrplpart>
<title>MANDATORY REPLACEMENT PARTS
</title> &pmcswp.pmcstable.mrplpart.no-mrp;
</mrplpart>
</pmcswp>
```

b. Page-based TM stylesheet output example for ***<pmcswp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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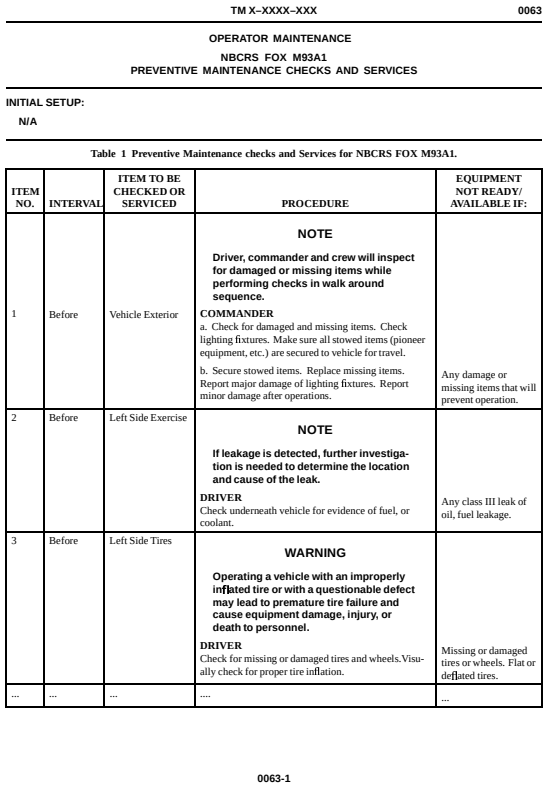

TM X–XXXX–XXX 0063

OPERATOR MAINTENANCE

NBCRS FOX M93A1
PREVENTIVE MAINTENANCE CHECKS AND SERVICES

INITIAL SETUP:

N/A

Table 1 Preventive Maintenance checks and Services for NBCRS FOX M93A1.

ITEM NO.	INTERVAL	ITEM TO BE CHECKED OR SERVICED	PROCEDURE	EQUIPMENT NOT READY/ AVAILABLE IF:
1	Before	Vehicle Exterior	NOTE **Driver, commander and crew will inspect for damaged or missing items while performing checks in walk around sequence.** **COMMANDER** a. Check for damaged and missing items. Check lighting fixtures. Make sure all stowed items (pioneer equipment, etc.) are secured to vehicle for travel. b. Secure stowed items. Replace missing items. Report major damage of lighting fixtures. Report minor damage after operations.	Any damage or missing items that will prevent operation.
2	Before	Left Side Exercise	NOTE **If leakage is detected, further investigation is needed to determine the location and cause of the leak.** **DRIVER** Check underneath vehicle for evidence of fuel, or coolant.	Any class III leak of oil, fuel leakage.
3	Before	Left Side Tires	WARNING **Operating a vehicle with an improperly inflated tire or with a questionable defect may lead to premature tire failure and cause equipment damage, injury, or death to personnel.** **DRIVER** Check for missing or damaged tires and wheels. Visually check for proper tire inflation.	Missing or damaged tires or wheels. Flat or deflated tires.
...	...	...	....	...

0063-1

FIGURE 345. Example of a page-based TM stylesheet output for <pmcswp>.

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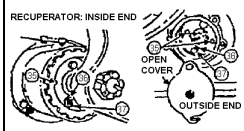
Table 1 Preventive Maintenance checks and Services for NBCRS FOX M93A1.–Continue

ITEM NO.	INTERVAL	ITEM TO BE CHECKED OR SERVICED	PROCEDURE	EQUIPMENT NOT READY/ AVAILABLE IF:
7	Before	Reciprocator	**WARNING** **Do not remove cap(Figure 3 , Item 37) when engine is hot.** **NOTE** **The reciprocator is charged with a pressure of 550 psi (38.5 kg/cm2) in the M109 and M109A1. With temperature of 100° F or over, 600 o psi (42.0 kg/cm2) in the M109A1 is 2 recommended.** **CREW MEMBER 1** a. Check pins (Figure 3 , Item 35)) before firing. Pins must not extend over 3/4 inch (19.1mm); 1/8 inch (3.2mm) is correct. If pins (Figure 3 , Item 35)) do extend beyond 3/4 inch, remove cap (Figure 3 , Item 37)) from valve (Figure 3 , Item 36) and hydraulic fluid (WP 0069, Item 5) until pins extend 1/8 inch.  Figure 3. Reciprocator	Pins(Figure 3 , Item 35) extended over 3/4 inch

MANDATORY REPLACEMENT PARTS

There are no replacement parts required for these PMCS procedures.

END OF WORK PACKAGE

FIGURE 345. Example of a page-based TM stylesheet output for <pmcswp> - Continued.

37.7 Maintenance work packages ***<maintwp>***. Maintenance work packages contains the maintenance task(s) required to maintain or repair the equipment. Maintenance information is divided by functionality in to individual maintenance work packages ***<maintwp>***. Maintenance work packages are not required for aircraft PM and PMS manuals.

a. The components of ***<maintwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (8) Maintenance task ***<maintsk>*** (required – one or more) (see 37.7.1).

b. DTD graphically depicted for ***<maintwp>***:

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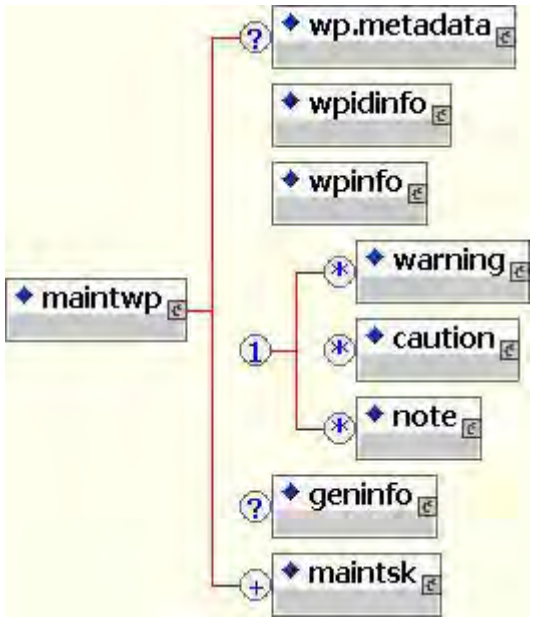

FIGURE 346. Maintenance work packages DTD hierarchy <maintwp>.

c. DTD fragment for ***<maintwp>***:

```
<!ELEMENT maintwp (wp.metadata?, wpidinfo, wpinfo, (warning*, caution*,
   note*), geninfo?, maintsk+)>
<!ATTLIST maintwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attributes for ***<maintwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

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(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.7.1 Maintenance task ***<maintsk>***. The maintenance task element consists of complete start-to-finish maintenance procedural steps in a logical sequence of occurrence. The maintenance task title ***<title>*** is identical to the Functional Group Code (FGC) title used in the applicable Maintenance Allocation Chart (MAC) and Repair Parts and Special Tools List (RPSTL). Each maintenance task ***<maintsk>*** requires defines a specific task function, followed by an optional follow-on or close-up maintenance task ***<followon.maintsk>***, as defined in MIL–STD–40051–1/–2. Most specific tasks have similar XML markup requirements and are explained there in33.1. A few specific tasks, that have unique task requirements, are expanded in the preceding subparagraphs.

a. The components of ***<maintsk>*** are:
(1) Requires one of the following specific task types:
(a) Aviation assembly and preparation for use task ***<prepforuse>*** are the instructions for unpacking, assembly, and installation (see 33.1 for task development).
(b) Servicing task ***<service>*** are the instructions for replenishment of fuel; oil; hydraulic or other fluids; oxygen, nitrogen, other gases; and tire pressure, plus any other such items and materials (except for lubricants) required for complete servicing of the equipment (see 33.1 for task development).

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(c) Ground handling task(s) ***<groundtsk>*** are the descriptions, instructions, and necessary cautions and warnings for equipment, including any information needed in extreme cold, heat, humidity, dust, or other unusual or extreme conditions. Requires one of the following specific task types:
1 Towing task ***<tow>*** (see 33.1 for task development).
2 Jacking task ***<jack>*** (see 33.1 for task development).
3 Parking task ***<park>*** (see 33.1 for task development).
4 Mooring task ***<moor>*** (see 33.1 for task development).
5 Covering task ***<cover>*** (see 33.1 for task development).
6 Hoisting task ***<hoist>*** (see 33.1 for task development).
7 Sling loading task ***<sling>*** (see 33.1 for task development).
8 External power task ***<extpwr>*** (see 33.1 for task development).

(d) Inspection of installed items task ***<inspinstitm>*** are the instructions for inspection of components, assemblies, or parts installed on the equipment (see 33.1 for task development).

(e) Removal task ***<remove>*** is the instructions for the logical removal sequence prescribed by the FGC (see 33.1 for task development).

(f) Disassembly task ***<disassem>*** are the instructions for disassembly of components, assemblies, or subassemblies to the extent specified by the MAC and SMR coded items (see 33.1 for task development).

(g) Cleaning task ***<clean>*** are the instructions, methods, special equipment, and materials that is required to clean the components, assemblies, or subassemblies (see 33.1 for task development).

(h) Inspection-acceptance and rejection criteria task ***<acptrejinsp>*** are the inspection requirements to include acceptance and rejection information sufficient to determine that new, repaired, and used components, assemblies and subassemblies conform to wear limits, fits, and tolerances established (see 33.1 for task development).

(i) Nondestructive testing inspection (NDTI) ***<ndti>*** are the testing inspection instruction that does destroy the components, assemblies, or subassemblies (see 33.1 for task development).

(j) Repair or replacement task ***<repair-rplc>*** are the instructions to restore an item to a completely serviceable or fully mission capable status (see 33.1 for task development).

(k) Alignment task ***<align>*** are the instructions to adjust specified variable elements of an item to bring about optimum or desired performance (see 33.1 for task development).

(l) Painting task ***<paint>*** are the instructions for required painting, refinishing, and marking of assembled components, assemblies, subassemblies, or end item (see 33.1 for task development).

(m) Lubrication task ***<lube>*** are the lubrication instructions, CPC procedures, and general lubrication instructions not specified elsewhere in the manual (see 33.1 for task development).

(n) Assembly task ***<assem>*** are the instructions for assembling items disassembled or removed that make up the components, assemblies, or subassemblies (see 33.1 for task development).

(o) Test and inspection task ***<test-inspect>*** are the instructions for testing and inspection during or after assembly to ensure proper assembly of the item (see 37.7.1.1).

(p) Installation task ***<install>*** are the instructions for installation of the item (see 33.1 for task development).

(q) Adjustment task ***<adjust>*** are the adjustment instructions that may be required before operating the part, system, or end item (see 33.1 for task development).

(r) Calibration task ***<calibration>*** are the equipment that requires calibration after assembly or installation is indicated, and reference is made to the publication containing the applicable calibration instructions (see 33.1 for task development).

(s) Radio interference suppression (RIS) task ***<ris>*** are the instructions, replacement and testing of these RIS components (see 33.1 for task development).

(t) Placing in service task ***<pis>*** are the instructions for actions not previously noted that may be required for an assembly, component, or end item, such as removal of an item from storage, preparation for installation on an end item, final servicing checks, calibration, leak checks, charging, pressurizing, and operational checks (see 33.1 for task development).

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(u) Testing task ***<test-pass>*** are the instructions to test the performance of components, assemblies, and subassemblies prior to installation in the end item (see 33.1 for task development).

(v) Preservation, packaging and marking (PPM) task ***<ppm>*** are the instructions for packaging information and any special or unique preservation, packaging, or marking instructions that apply to the equipment (see 33.1 for task development).

(w) Overhaul and retirement schedule task ***<orsch>*** are the instructions for units of operating equipment that are to be overhauled or retired at the period specified are listed in this task (see 37.7.1.2).

(x) Preparation for storage or shipment task ***<pss>*** are the instructions for preparation before storage or shipment of components, assemblies, and subassemblies (see 33.1 for task development).

(y) Classification of ammunition defects task ***<ammo.defect>*** are the instructions for performing visual inspection of ammunition/containers (pallets, boxes, etc.) and include classification and disposition of defective ammunition/containers (see 33.1 for task development).

(z) Handling ammunition task ***<ammo.handling>*** are the instructions for handling packing and unpacking ammunition (see 37.7.1.3).

(aa) Ammunition markings task ***<ammo.markings>*** are the instructions for marking ammunition and ammunition containers (see 33.1 for task development).

(ab) Ammunition activation task ***<arm>*** are the instructions for activation of ammunition, mines, etc (see 33.1 for task development).

(ac) Addition maintenance task ***<other.maintsk>*** are the instructions for specific maintenance tasks not previously covered in above types (see 33.1 for task development).

(2) Follow-on maintenance task ***<followon.maintsk>*** (optional) are the instructions for a maintenance condition which is accomplished sometime following the completion of a specific task to clean up or undo actions performed during the task (see 33.1 for task development).

b. DTD graphically depicted for ***<maintsk>***:

FIGURE 347. Maintenance task DTD hierarchy <maintsk>.

c. DTD graphically depicted for ***<groundtsk>***

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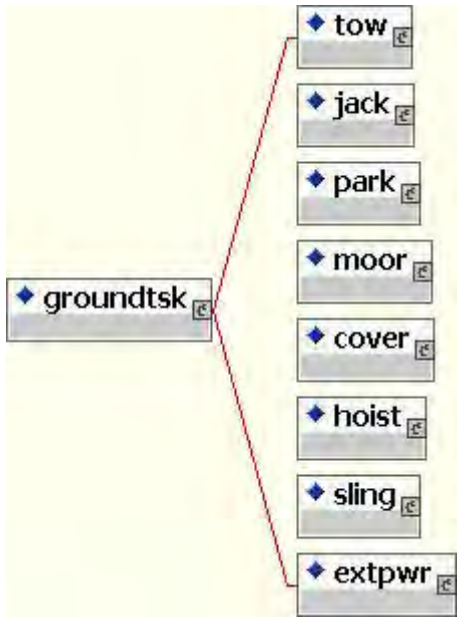

FIGURE 348. Maintenance ground handling task DTD hierarchy <groundtsk>.

d. DTD fragment for ***<maintsk>*** and ***<groundtsk>***:

```
<!ELEMENT maintwp ((prepforuse | service | groundtsk | inspinstitm |
   remove | disassem | clean | acptrejinsp | ndti | repair-rplc |
   align | paint | lube | assem | test-inspect | install | adjust |
   calibration | ris | pis | test-pass | ppm | orsch | pss |
   ammo.defect | ammo.handling | ammo.markings | arm | other.maintsk),
   followon.maintsk?)>

<!ELEMENT groundtsk (tow | jack | park | moor | cover | hoist |
   sling | extpwr)>
```

37.7.1.1 Test and inspection task ***<test-inspect>***. The element is the instructions for testing and inspection during or after assembly to ensure proper assembly of the item.

a. The components of ***<test-inspect>*** are:
   (1) Task title ***<title>*** (required) (see 49.1.1.3).
   (2) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (3) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (4) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (5) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (6) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (7) Procedural steps ***<proc>*** (required – one or more) (see 33.2). Following each procedure the inspection standard information are available.
      (a) Classification of material defects standard information ***<defect.tab>*** (optional – zero or more) (see 37.7.1.1.1).
      (b) Packaging material inspection criteria standard information ***<crit.insp.tab>*** (optional – zero or more) (see 37.3.1.1.2)

b. DTD graphically depicted for ***<test-inspect>***:

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Check the source to verify that this is the current version before use.

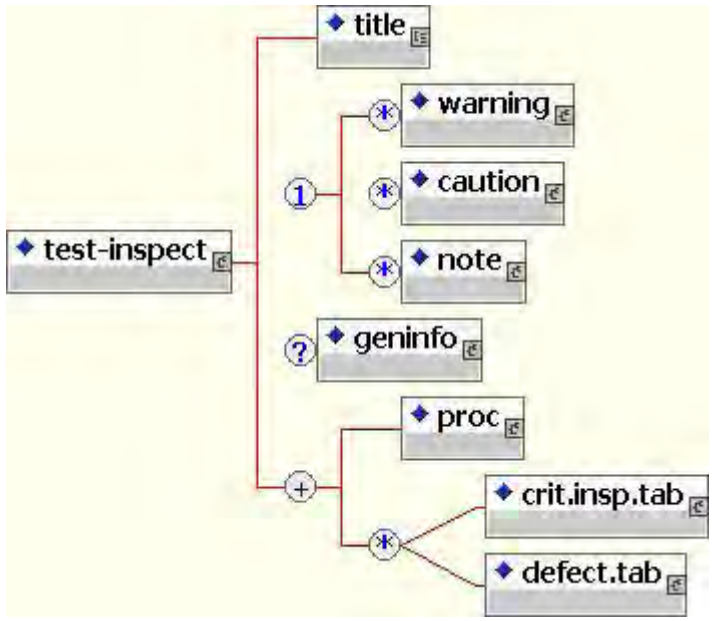

FIGURE 349. Test and inspection task DTD hierarchy <test-inspect>.

c. DTD fragment for ***<test-inspect>***:

```
<!ELEMENT test-inspect (title, (warning*, caution*, note*), geninfo?,
   (proc, (crit.insp.tab | defect.tab)*)+)>
<!ATTLIST test-inspect
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    date-time-stamp   (date | time | date-time)       #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    esd               (yes | no)                      "no"
    frame             (yes | no)                      "yes"
    hcp               (yes | no)                      "no"
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 3 | 4 | 5)                 "0"
>
```

d. Common attributes for ***<test-inspect>*** are:

(1) **applicable** – Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(8) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

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(9) **hcp** – Nuclear hardness requirement (default value is “no”) (see 49.2.5).
(10) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(11) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(12) **inschlvl** – Insert change level (optional) (see 49.2.9).
(13) **security** – Security classification (optional) (see 49.2.13).
(14) **skilltrk** – Skill level (optional) (see 49.2.2).
(15) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is “**0**” do not include, “**3**” the 3rd level, “**4**” the 4th level, or “**5**” the 5th level (default value is “**0**”).

37.7.1.1.1 Classification of material defects standard information ***<defect.tab>***. The classification of material defects table contains instructions for inspection methods or techniques used to detect defective components or end items being processed.

a. The components of ***<defect.tab>*** are:
(1) Standard information title ***<title>*** (required) (see 49.1.1.3).
(2) Standard table header ***<std.table header>*** (required) (see 37.7.1.1.2).
(3) Classification of material defects row ***<defect-row>***. The element is similar to a "row" in a structural table. The component of ***<defect-row>*** are:
(a) Defect category type ***<defecttype>*** (required) is either minor, major, or critical, which is selected by the attribute **type**. The element is similar to a "row" in a structural table. The attribute value is similar to a “cell” in a structural table and is entered in column one.
(b) Defective material grouping ***<defect-group>*** (required – one or more). The element is similar to a "row" in a structural table. The element components are:
1 Defect attributable for each component ***<condition>*** (required) (see 49.1.4.14). The element is similar to a “cell” in a structural table and is entered in column one.
2 Corrective action is either work package reference ***<link>*** (see 46.2.3)/***<xref>*** (see 46.2.2) (required) or single instruction ***<actionreq>*** (required). Generally this is reference to another maintenance or operations checkout work package. The element is similar to a “cell” in a structural table and is entered in column two.
3 Inspection method after corrective action ***<insp-method>*** (required). The inspection method is narrative text such as “Visual”. The element is similar to a “cell” in a structural table and is entered in column three.
4 Acceptable quality level ***<acceptqual>***(required). Provides a short formatted text statement to determine if the defect has been fixed to acceptable level. The element is similar to a “cell” in a structural table and is entered in column four.

b. DTD graphically depicted for ***<defect.tab>***

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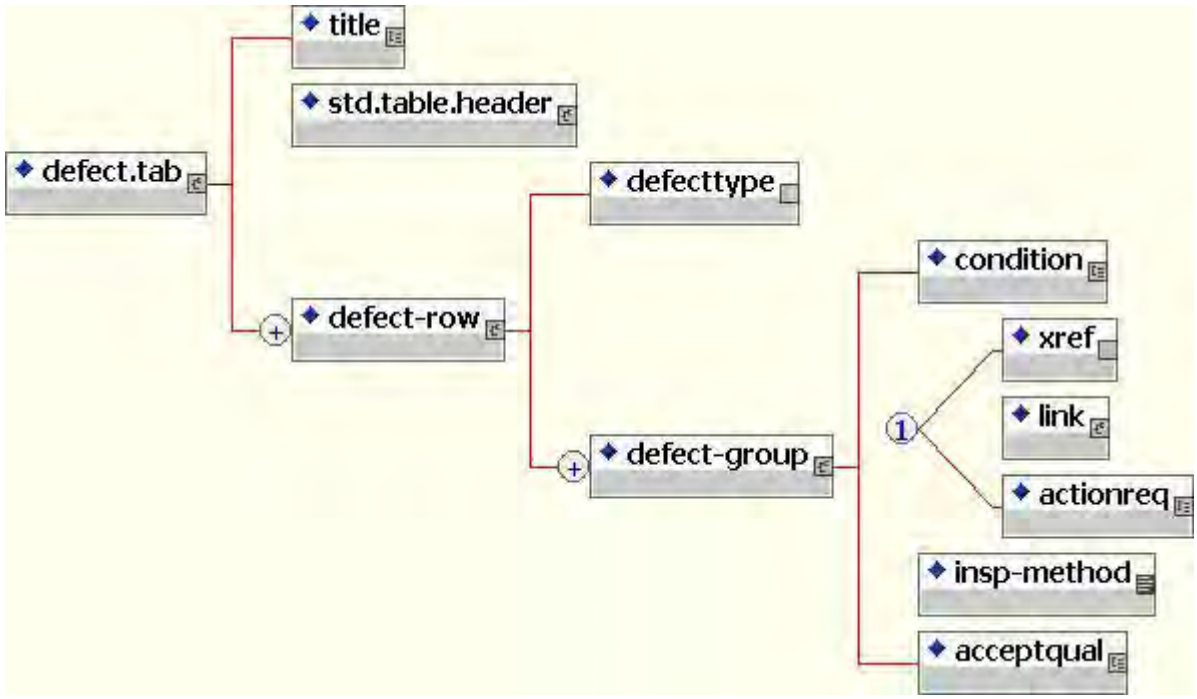

FIGURE 350. Classification of material defects standard information DTD hierarchy <defect.tab>.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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c. DTD fragment for ***<defect.tab>***:

```
<!ELEMENT defect.tab (title, std.table.header, defect-row+>
<!ATTLIST defect.tab
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1">

<!ELEMENT defect-row (defecttype, defect-group+)>
<!ATTLIST defect-row
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>

<!ELEMENT defecttype EMPTY>
<!ATTLIST defecttype
    type              (minor | major | critical)      #REQUIRED>

<!ELEMENT defect-group (condition, (link | xref | actionreq),
   insp-method, acceptqual)>
<!ATTLIST defect-group
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>

<!ELEMENT insp-method (#PCDATA)>

<!ELEMENT acceptqual (#PCDATA | emphasis | subscrpt | supscrpt |
xref | extref | link | help.info | indxref | term | term.def |
callout | ftnote | ftnref | ctrlind | ctrlind-val | dodac |
lubricant | symbol | torque | voltage | null | change)>
```


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Check the source to verify that this is the current version before use.

d. Unique attribute for ***<defecttype>*** is **type** – Defect category type (required) and select one of the following from the list "**minor**", "**major**", or "**critical**".
e. Common attributes for ***<defect.tab>***, ***<defect-row>***, and ***<defect-group>*** are:
(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

37.7.1.1.2 Classification of material defects standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XIV.

TABLE XIV. Classification of material defects standard information header ***<std.table.header>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Classification of material defects Standard Information Header	***&maintwp.test-inspect.defect.tab.tab.std-header;***	Not applicable.

37.7.1.2 Overhaul and retirement schedule task ***<orsch>***. The element is the instructions for units of operating equipment that are to be overhauled or retired at the period specified are listed in this task.
a. The components of ***<orsch>*** are:
(1) Task title ***<title>*** (required) (see 49.1.1.3).
(2) Introduction ***<intro>*** (required) (see 49.1.4.15 and 37.7.2 for standard statement entity).
(3) Overhaul and retirement standard information ***<orsch.tab>*** (required) (see 37.7.1.2.1).
b. DTD graphically depicted for ***<orsch>***:

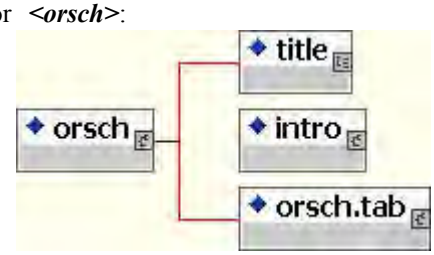

FIGURE 351. Overhaul and retirement schedule task DTD hierarchy <orsch>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<orsch>***:

```
<!ELEMENT orsch (title, intro, orsch.tab)>
<!ATTLIST orsch
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    date-time-stamp   (date | time | date-time)       #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    esd               (yes | no)                      "no"
    frame             (yes | no)                      "yes"
    hcp               (yes | no)                      "no"
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 3 | 4 | 5)                 "0"
>
```

d. Common attributes for ***<orsch>*** are:

(1) **applicable** – Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(8) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(9) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(10) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(11) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(12) **inschlvl** – Insert change level (optional) (see 49.2.9).
(13) **security** – Security classification (optional) (see 49.2.13).
(14) **skilltrk** – Skill level (optional) (see 49.2.2).
(15) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**0**").

37.7.1.2.1 Overhaul and retirement schedule table ***<orsch.tab>***. The overhaul and retirement schedule table identifies units of operating equipment that are to be overhauled or retired at the specified period.

a. The components of ***<orsch.tab>*** are:

(1) Standard information title ***<title>*** (required) (see 49.1.1.3).
(2) Standard table header ***<std.table.header>*** (see 42.3.1).
(3) Overhaul and retirement schedule entry ***<orsch-entry>*** (required – one or more) and the components are:
(a) Part nomenclature ***<name>*** (required) (see 49.1.4.19).
(b) Part overhaul and retirement schedule interval entry***<orsch.interval.entry>*** (see 37.7.1.2.2).

b. DTD graphically depicted for ***<orsch.tab>***:

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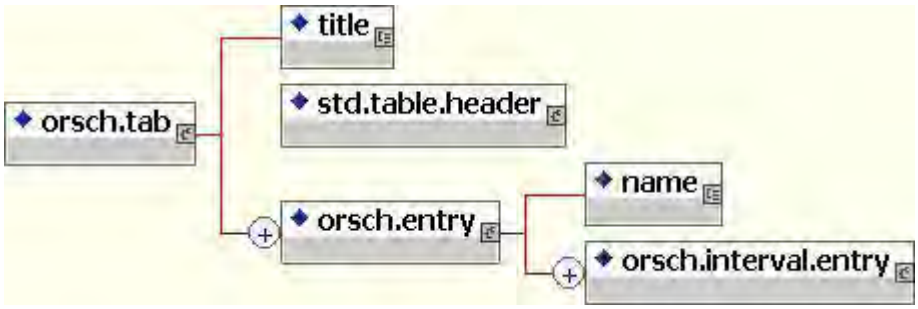

FIGURE 352. Overhaul and retirement schedule standard information DTD hierarchy <orsch.tab>.

c. DTD fragment for ***<orsch.tab>*** and ***<orsch.entry>***:

```
<!ELEMENT orsch.tab (title, std.table.header, orsch.entry+)>
<!ATTLIST orsch.tab
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1"
>

<!ELEMENT orsch.entry (name, orsch.interval.entry+)>
<!ATTLIST orsch.entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<orsch.tab>*** and ***<orsch.entry>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

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(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

37.7.1.2.2 Overhaul and retirement schedule interval entry ***<orsch.interval.entry>***. The element identifies the detailed overhaul and/or retirement hours and requirements for a specified part number.

a. The components of ***<orsch.interval.entry>*** are:
   (1) Part number ***<partno>*** (required) (see 49.1.4.21).
   (2) Commercial and government entity code (CAGEC) ***<cageno>*** (required) (see 49.1.4.2).
   (3) Overhaul interval ***<overhaul.interval>*** (optional) consisting of the following components:
      (a) Overhaul interval hours ***<interval.hours>*** (required) contains the maximum operating time allowed on the part before being overhauled.
      (b) Overhaul interval notes ***<interval.notes>*** (optional) contains any additional information required on the part's overhaul interval.
   (4) Retirement interval ***<retirement.interval>*** (required) consisting of the following components:
      (a) Retirement interval hours ***<interval.hours>*** (required) contains the maximum operating time allowed on the part before being retired.
      (b) Retirement interval notes ***<interval.notes>*** (optional) contains any additional information required on the part's retired interval.

b. DTD graphically depicted for ***<orsch.interval.entry>***:

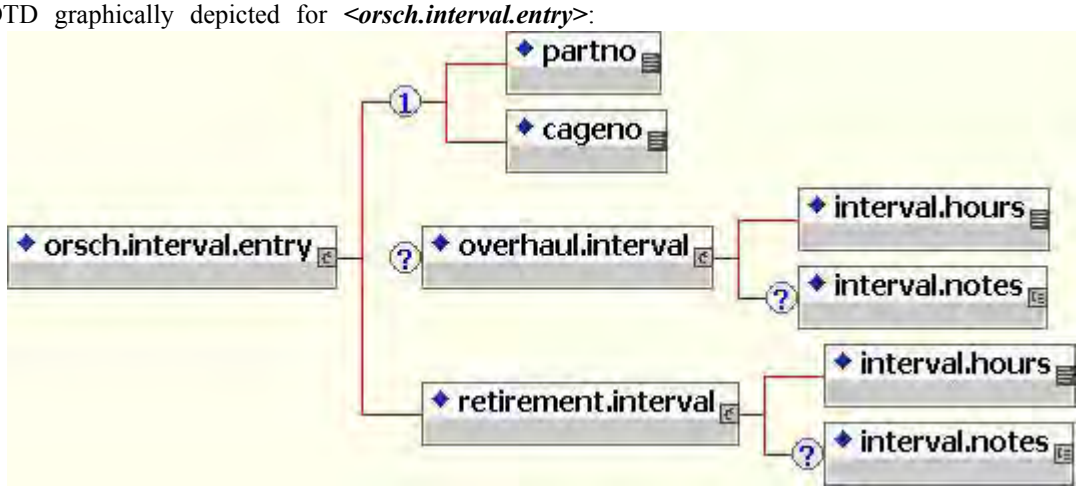

FIGURE 353. Part overhaul and retirement interval entry DTD hierarchy <orsch.interval.entry>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<orsch.interval.entry>***, ***<overhaul.interval>***, and ***<retirement.interval>***:

```
<!ELEMENT orsch.interval.entry (partno, cageno, overhaul.interval?,
   retirement.interval)>
<!ATTLIST orsch.interval.entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>

<!ELEMENT overhaul.interval (interval.hours, interval.notes?)>

<!ELEMENT retirement.interval (interval.hours, interval.notes?)>
```

d. Common attributes for ***<orsch.interval.entry>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

37.7.1.3 Ammunition handling ***<ammo.handling>***. Ammunition handling task contains all procedures required for care and handling of ammunition, including hazard distances, storage, special requirements, prevention of deterioration due to rough handling, exposure to adverse weather condition, or other hazards. Visual inspection criteria is prepared to determine item serviceability.

a. The components of ***<ammo.handling>*** are:

(1) Task title ***<title>*** (required) (see 49.1.1.3)
(2) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(3) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(4) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(5) General information ***<geninfo>*** (optional) (see 49.1.4.12).
(6) At least one of the following tasks are required, in no specific order:
(a) Ammunition unpacking ***<ammo.unpacking>*** (see 33.1 for task development) contains information regarding the unpacking of ammunition, such as any special sequence of action necessary to protect the ammunition, if a special design reusable container is involved for either the end item or components which are authorized for replacement, instructions to report or reenter the empty container through supply channels, and man-hours/total man-hours required for unpacking the equipment.
(b) Ammunition packing ***<ammo.packing>*** (see 33.1 for task development) contains information regarding the packing of ammunition, such as any special sequence of action necessary to protect the ammunition, instructions on how to package defective ammunition, and man-hour/total man-hours required for packing the ammunition.

b. DTD graphically depicted for ***<ammo.handling>***

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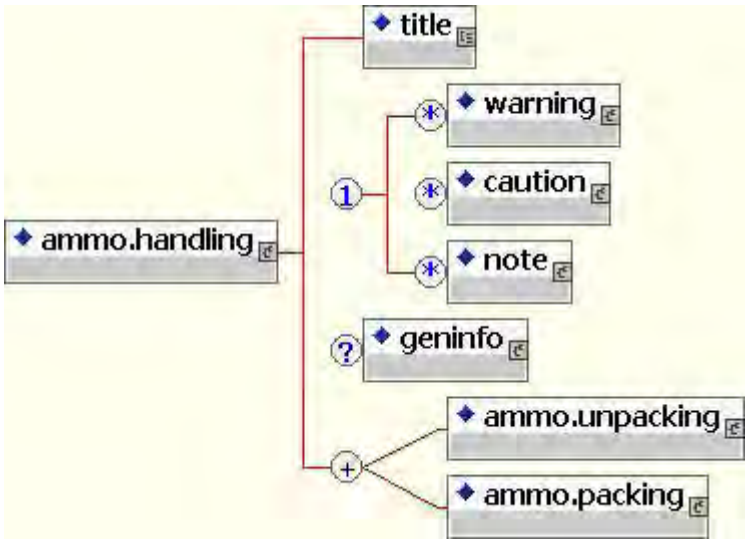

FIGURE 354. Ammunition handling task DTD hierarchy <ammo.handling>.

c. DTD fragment for ***<ammo.handling>***:

```
<!ELEMENT ammo.handling (title, (warning* | caution* |
  note*), geninfo?, (ammo.unpacking | ammo.packing)+)>
<!ATTLIST ammo.handling
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    date-time-stamp   (date | time | date-time)       #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    esd               (yes | no)                      "no"
    frame             (yes | no)                      "yes"
    hcp               (yes | no)                      "no"
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 3 | 4 | 5)                 "0"
>
```

d. Common attributes for ***<ammo.handling>***, ***<ammo.unpacking>***, and ***<ammo.packing>*** are:

(1) **applicable** – Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(8) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(9) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(10) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

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(11) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(12) **inschlvl** – Insert change level (optional) (see 49.2.9).
(13) **security** – Security classification (optional) (see 49.2.13).
(14) **skilltrk** – Skill level (optional) (see 49.2.2).
(15) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**0**").

37.7.2 Maintenance work packages general entity boilerplates. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listedTable XV.

TABLE XV. Boilerplate Entities for ***<maintwp>***.

Description	**Boilerplate Entity**	**Selectable Entity to Set Editable Entity to Set**
Additional information on-data plate statement	***&maintwp.test-pass.data-plate;***	Not applicable
Overhaul and retirement schedule statement	***&maintwp.orsch;***	Not applicable
Packaging information for preservation, packaging and marking (PPM) statement	***&maintwp.ppm.packaging;***	Not applicable
Inspection and test of conventional and chemical ammunition or components containing radioactive materials statements	***&maintwp.test-inspect.disposition-1;***	Not applicable
	&maintwp.test-inspect.disposition-2;	
	&maintwp.test-inspect.disposition-3;	

37.7.3 XML document instance fragment and output for ***<maintwp>***. The maintenance list work package XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

```
<maintwp wpno="M03451–X-XXXX-XXX" tocentry="2" frame="yes" army="no" airforce="no"
navy="no" marines="no" wpseq="0029" deletewp="no" security="uc">
<wpidinfo>
<maintlvl level="sustain"/>
<title>24–VOLT CONNECTOR RECEPTACLE REPAIR
<brk/>ASSEMBLY, REPAIR, REASSEMBLY
</title>
</wpidinfo>
<wpinfo>
<tools>
<tools-setup-item>
<name>Pliers, diagonal cutting
</name>
<itemref>
<xref wpid="S1904-X-XXXX-XXX" itemno="3" pretext=" (" posttext=")"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Screwdrive, flat-tip 3/16-inch
```


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Check the source to verify that this is the current version before use.

```
</name>
<itemref>
<xref wpid="S1904-X-XXXX-XXX" itemno="43" pretext=" (" posttext=")"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Soldering iron, gun type
</name>
<itemref>
<xref wpid="S1904-X-XXXX-XXX" itemno="48" pretext=" (" posttext=")"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Stripper, wire, hand
</name>
<itemref>
<xref wpid="S1905-X-XXXX-XXX" itemno="55" pretext=" (" posttext=")"/>
</itemref>
</tools-setup-item>
</tools>
<mtrlpart>
<mtrlpart-setup-item>
<name>Alcohol, denatured
</name>
<itemref>
<xref wpid="S1903-X-XXXX-XXX" itemno="2" pretext=" (" posttext=")"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Cotton, acid swabbing
</name>
<itemref>
<xref wpid="S1903-X-XXXX-XXX" itemno="4" pretext=" (" posttext=")"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Flux, rosin
</name>
<itemref>
<xref wpid="S1903-X-XXXX-XXX" itemno="8" pretext=" (" posttext=")"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Solder, non-acid
</name>
<itemref>
<xref wpid="S1903-X-XXXX-XXX" itemno="17" pretext=" (" posttext=")"/>
</itemref>
</mtrlpart-setup-item>
</mtrlpart>
<persnreq>
<persnreq-setup-item>
<name>One
</name>
</persnreq-setup-item>
</persnreq>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<ref>
<ref-setup-item>
<xref  wpid="M0382-X-XXXX-XXX"/>
</ref-setup-item>
<ref-setup-item>
<xref  wpid="M0379-X-XXXX-XXX"/>
</ref-setup-item>
</ref>
<eqpconds>
<eqpconds-setup-item>
<condition>24-volt connector receptacle removed
</condition>
<itemref>
<xref wpid="M0213-X-XXXX-XXX" pretext=" (" posttext=")"/>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
<maintsk>
<disassem hcp="no" esd="no" frame="yes" tocentry="0">
<title>Disassembly
</title>
<note acknowledge="no">
<trim.para hcp="no" esd="no">Tag wires to aid in installation
<xref wpid="M0382-X-XXXX-XXX" pretext="(" posttext=")"/>.  If circuit
marker bands are missing or not readable, replace
<xref wpid="M0213-X-XXXX-XXX"
pretext="(" posttext=")"/>.
</trim.para>
</note>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Unscrew and pull back bushing retaining nut
<callout assocfig="m00001-fig1" label="1"/> from shell
<callout assocfig="m00001-fig1" label="4"/>.
</para>
</step1>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Using screwdriver, pry off shell
<callout assocfig="m00001-fig1" label="4"/> from busing
<callout assocfig="m00001-fig1" label="2"/>.
</para>
</step1>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Using pliers, pull out 12 inserts
<callout assocfig="m00001-fig1" label="3"/> from bushing
<callout assocfig="m00001-fig1" label="2"/>
<figure application="both" figtype="normal-page" tocentry="0"
pane="no" id="m00001-fig1">
<title>24-Volt Connector Receptacle
</title>
<graphic boardno="FrontCover" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
</disassem>
```


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```
</maintsk>
<maintsk>
<repair-rplc hcp="no" esd="no" frame="yes" tocentry="0">
<title>Repair
</title>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Using soldering iron, heat insert
<callout assocfig="m00001-fig2" label="5"/> and pull from wires
<callout assocfig="m00001-fig2" label="3"/>.
</para>
</step1>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Pull out 12 wires
<callout assocfig="m00001-fig2" label="3"/>.
</para>
</step1>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Position wires in bushing
<callout assocfig="m00001-fig2" label="2"/> according to chart
<callout assocfig="m00001-fig2" label="1"/>.
</para>
</step1>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Push wires all the way through holes in bushing
<callout assocfig="m00001-fig2" label="2"/>.
</para>
</step1>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Using wire stripper, strip insulation back
<supscrpt>1
</supscrpt>/
<subscrpt>4
</subscrpt>-inch.
</para>
</step1>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Slip an insert
<callout assocfig="m00001-fig2" label="5"/> over each wire, and solder
using non-acid solder and soldering iron.
<figure application="both" figtype="normal-page" tocentry="0"
pane="no" id="m00001-fig2">
<title>24-Volt Connector Receptacle Wiring Diagram
</title>
<graphic boardno="FrontCover" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
</repair-rplc>
</maintsk>
<maintsk>
<assem hcp="no" esd="no" frame="yes" tocentry="0">
<title>Reassembly
</title>
<note acknowledge="no">
```


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```
<trim.para hcp="no" esd="no">Make sure wire ends are clean before
soldering.  If necessary, clean with cleaning solvent and stiff
fiber brush.  Solder must be non-acid type; use rosin flux.  Wires
and soldering iron must be pre-tinned for good connection and
maximum transfer of heat.  After soldering, clean all solder
joints with an acid swabbing brush and alcohol.
</trim.para>
</note>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Push wires
<callout assocfig="m00001-fig2" label="3"/> and inserts
<callout assocfig="m00001-fig2" label="5"/> into bushing
<callout assocfig="m00001-fig2" label="2"/> until inserts
<callout assocfig="m00001-fig2" label="5"/> are seated.
</para>
</step1>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Put bushing
<callout assocfig="m00001-fig2" label="2"/> in place within shell
<callout assocfig="m00001-fig2" label="6"/>.
</para>
</step1>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Screw on bushing retaining nut
<callout assocfig="m00001-fig2" label="4"/> to shell
<callout assocfig="m00001-fig2" label="7"/>.  Tighten busing retaining nut
<callout assocfig="m00001-fig2" label="4"/>.
</para>
</step1>
<step1 skilllevel="novice_expert" hcp="no" esd="no" qa="no">
<para hcp="no" esd="no">Install 24-volt connector receptacle
<xref wpid="M0379-X-XXXX-XXX" pretext=" (" posttext=")"/>.
</para>
</step1>
</assem>
</maintsk>
</maintwp>
```

37.8 General maintenance work packages ***<gen.maintwp>***. This work package contains common, general, or standard maintenance procedures (e.g., specific torque wrench usage, lockwire procedures, "O" ring seal installation, external power connections, etc.) applicable to other maintenance work packages contained within the TM that require the general maintenance procedures to complete the tasks.

a. The components of ***<gen.maintwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (8) Single procedure
      (a) Procedural steps title ***<title>*** (required) (see 49.1.1.3).
      (b) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
      (c) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
      (d) Note – ***<note>*** (optional – zero or more) (see 41.1.3).

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(e) General information ***<geninfo>*** (optional) (see 49.1.4.12).
(f) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(g) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(h) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1-alt>*** (see 48.2.1) (required – one or more).

(9) Multiple procedure ***<proc>*** (required – two or more) (see 33.2).
(10) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<gen.maintwp>***:

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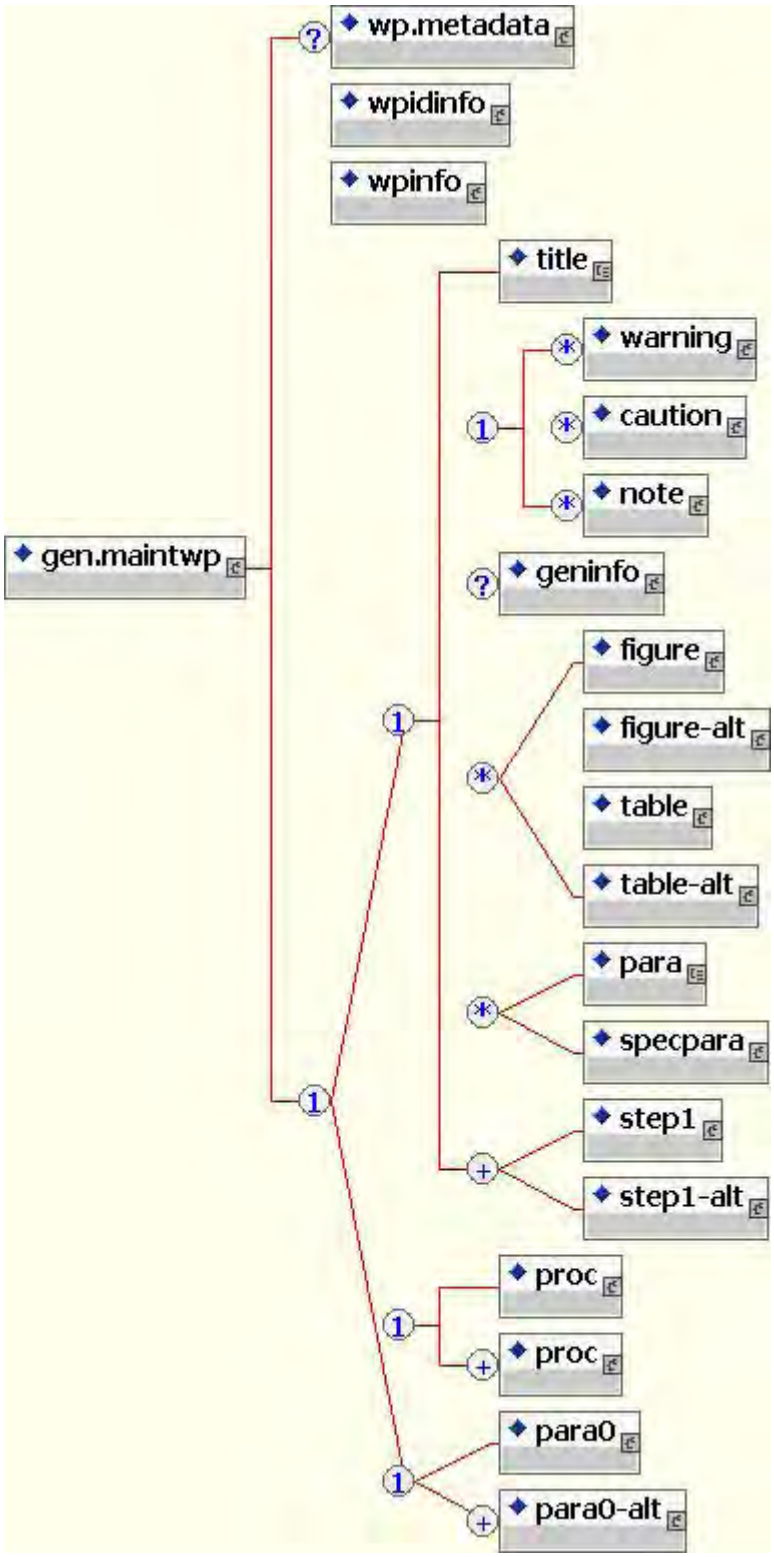

FIGURE 355. Genral maintenance work packages DTD hierarchy <gen.maintwp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<gen.maintwp>***:

```
<!ELEMENT gen.maintwp (wp.metadata?, wpidinfo, wpinfo, ((title,
   (warning*, caution*, note*), geninfo?, (figure | figure-alt |
   table | table-alt)*, (para | specpara)*, (step1 | step1-alt)+) |
   (proc, proc+) | (para0 | para0-alt+)))>
<!ATTLIST gen.maintwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attributes for ***<gen.maintwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.9 Lubrication instructions work package ***<lubewp>***. The lubrication instructions work package consists of lubrication schedules and lubrication charts. Lubrication schedules provide all applications and procedures, lubricants, and lubrication points to completely lubricate the equipment. Lubrication charts consist of a main drawing prepared as a three-dimensional diagram, and enlarged/detailed views that are necessary to identify items which otherwise would be obscured. Lubrication charts contain the lubrication requirements for all parts of the equipment requiring periodic lubrication (other than those lubricated by the main engine oil system, indicating type of lubricant) method of application, and frequency.

a. The components of ***<lubewp>*** are:
(1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
(2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
(3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
(4) Single procedure
(a) Procedural steps title ***<title>*** (required) (see 49.1.1.3).
(b) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(c) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(d) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(e) General information ***<geninfo>*** (optional) (see 49.1.4.12).
(f) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(g) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(h) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1-alt>*** (see 48.2.1) (required – one or more).
(5) Multiple procedure ***<proc>*** (required – two or more) (see 33.2).
(6) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<lubewp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

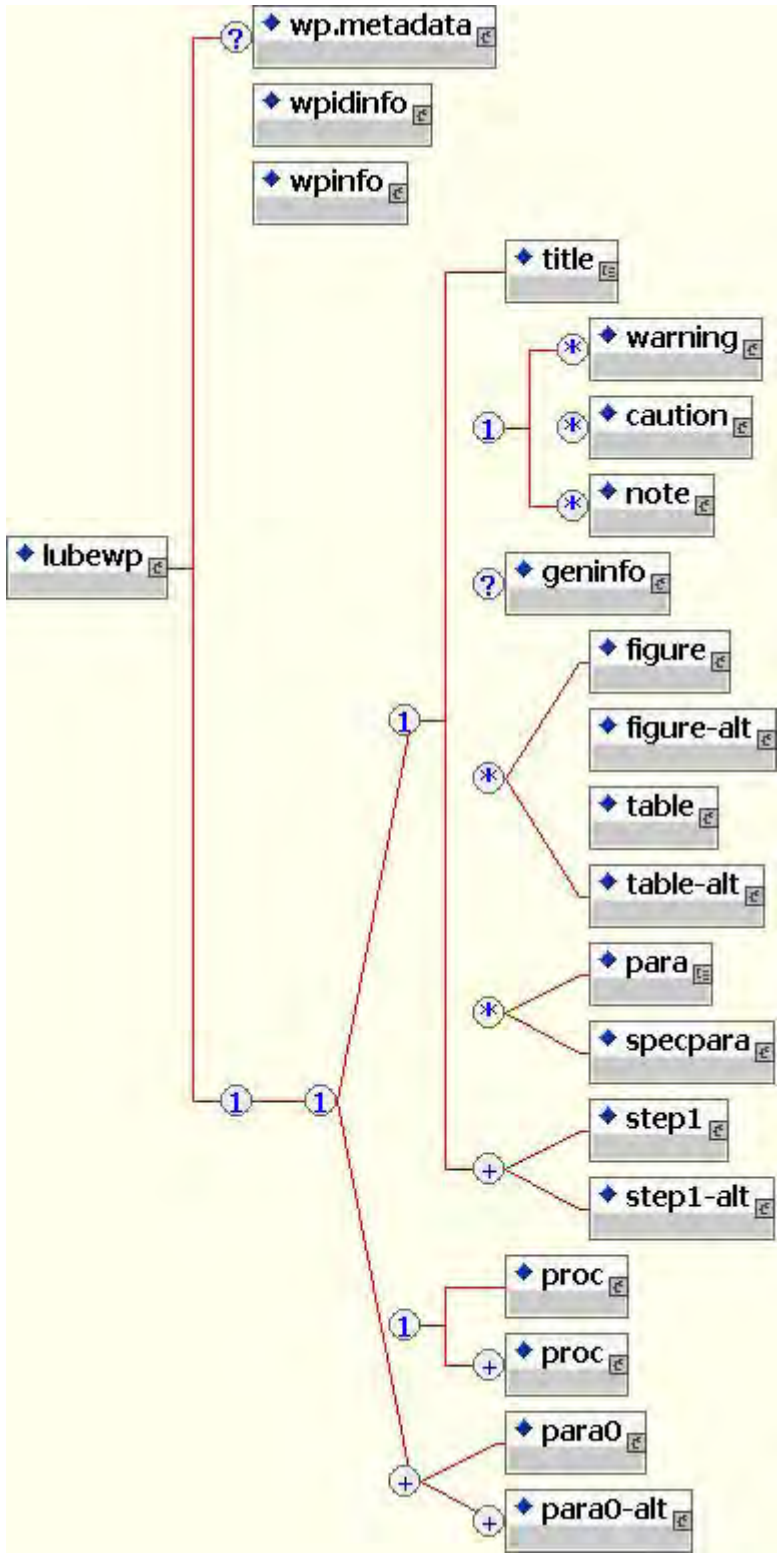

FIGURE 356. Lubrication instructions work package DTD hierarchy <lubewp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<lubewp>***:

```
<!ELEMENT lubewp (wp.metadata?, wpidinfo, wpinfo, ((title,
   (warning*, caution*, note*), geninfo?, (figure | figure-alt |
   table | table-alt)*, (para | specpara)*, (step1 | step1-alt)+) |
   (proc, proc+) | (para0 | para0-alt+)))>
<!ATTLIST lubewp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attributes for ***<lubewp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.9.1 XML document instance fragment and output for ***<lubewp>***. The XML instance and its stylesheet output for a ***<lubewp>*** is provided below.

a. Example of an XML document instance fragment for ***<lubewp>***:

```
<lubewp wpno="M00452-X-XXX-X-XXX" tocentry="2" frame="yes" army="no"
airforce="no" navy="no" marines="no" wpseq="0007" deletewp="no" security="uc">
<wpidinfo>
<maintlvl level="sustain"/>
<title>Lubrication Instructions
</title>
</wpidinfo>
<wpinfo>
<ref>
<ref-setup-item>
<extref docno="MIL-L-23398"/>
</ref-setup-item>
</ref>
</wpinfo>
<title>Intervals
</title>
<para>The following table lists the lubrication schedule for
the maintenance shelter.  The schedule is applicable for all
temperatures between &ndash;25&deg;F and +125&deg;F.
<table>
<title>LUBRICATION INTERVALS
</title>
<tgroup cols="5">
<colspec colname="col1" colwidth="0.35*"/>
<colspec colname="col2" colwidth="0.35*"/>
<colspec colname="col3" colwidth="0.40*"/>
<colspec colname="col4" colwidth="1.90*"/>
<colspec colname="col5" colwidth="2.00*"/>
<thead>
<row>
<entry colsep="0" nameend="col3" namest="col1">Q &ndash; Quarterly
</entry>
<entry nameend="col5" namest="col4">A &ndash; Annually
</entry>
</row>
<row>
<entry align="center" morerows="1" valign="bottom">ITEM NO.
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</entry>
<entry align="center" nameend="col3" namest="col2" valign="bottom"> INTERVALS
</entry>
<entry align="center" morerows="1" valign="bottom">ITEM TO BE SERVICED
</entry>
<entry align="center" morerows="1" valign="bottom">PROCEDURE
</entry>
</row>
<row>
<entry align="center">Q
</entry>
<entry align="center">A
</entry>
</row>
</thead>
<tbody>
<row>
<entry morerows="2" rowsep="0">1
</entry>
<entry rowsep="0">
</entry>
<entry rowsep="0">
</entry>
<entry rowsep="0">HINGE.  STEP
</entry>
<entry morerows="2" rowsep="0">
<step1 id="M00452-X-XXX-X-XXX-step1">
<para>Remove all foreign matter from joints or surfaces
immediately before applying lubricant.
</para>
</step1>
<step1 id="M00452-X-XXX-X-XXX-step2">
<para>Apply air-drying film lubricant
<extref docno="MIL-L-23398"/>, sparingly to prevent contaminant accumulation.
</para>
</step1>
</entry>
</row>
<row>
<entry rowsep="0">  &bull;
</entry>
<entry rowsep="0">
</entry>
<entry rowsep="0">   Deployed
</entry>
</row>
<row>
<entry align="center" rowsep="0">
</entry>
<entry align="center" rowsep="0">  &bull;
</entry>
<entry rowsep="0">   Stored
</entry>
</row>
<row>
<entry morerows="2" rowsep="0">2
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  rowsep="0">HINGE,  HANDHOLD
</entry>
<entry  morerows="2"  rowsep="0">Same  as
<xref  stepend="M00452-X-XXX-X-XXX-step2"  stepstart="M00452-X-XXX-X-XXX-step1"/>
</entry>
</row>
<row>
<entry  align="center"  rowsep="0">  &bull;
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  rowsep="0">   Deployed
</entry>
</row>
<row>
<entry  align="center"  rowsep="0">
</entry>
<entry  align="center"  rowsep="0">  &bull;
</entry>
<entry  rowsep="0">   Stored
</entry>
</row>
<row>
<entry  morerows="2"  rowsep="0">3
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  rowsep="0">HINGE,  DOOR
</entry>
<entry  morerows="2"  rowsep="0">Same  as
<xref  stepend="M00452-X-XXX-X-XXX-step2"  stepstart="M00452-X-XXX-X-XXX-step1"/>
</entry>
</row>
<row>
<entry  align="center"  rowsep="0">  &bull;
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  rowsep="0">   Deployed
</entry>
</row>
<row>
<entry  align="center"  rowsep="0">
</entry>
<entry  align="center"  rowsep="0">  &bull;
</entry>
<entry  rowsep="0">   Stored
</entry>
</row>
```


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```
<row>
<entry  morerows="2"  rowsep="0">4
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  rowsep="0">HANDLE,  DOOR
</entry>
<entry  morerows="2"  rowsep="0">Same  as
<xref  stepend="M00452-X-XXX-X-XXX-step2"  stepstart="M00452-X-XXX-X-XXX-step1"/>
</entry>
</row>
<row>
<entry  align="center"  rowsep="0">  &bull;
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  rowsep="0">   Deployed
</entry>
</row>
<row>
<entry  align="center"  rowsep="0">
</entry>
<entry  align="center"  rowsep="0">  &bull;
</entry>
<entry  rowsep="0">   Stored
</entry>
</row>
<row>
<entry  morerows="2"  rowsep="0">5
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  rowsep="0">RETAINING  DEVICE,  DOOR
</entry>
<entry  morerows="2"  rowsep="0">Same  as
<xref  stepend="M00452-X-XXX-X-XXX-step2"  stepstart="M00452-X-XXX-X-XXX-step1"/>
</entry>
</row>
<row>
<entry  align="center"  rowsep="0">  &bull;
</entry>
<entry  align="center"  rowsep="0">
</entry>
<entry  rowsep="0">   Deployed
</entry>
</row>
<row>
<entry  align="center"  rowsep="0">
</entry>
<entry  align="center"  rowsep="0">  &bull;
</entry>
<entry  rowsep="0">   Stored
```


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```
</entry>
</row>
<row>
<entry morerows="2" rowsep="1">6
</entry>
<entry align="center" rowsep="0">
</entry>
<entry align="center" rowsep="0">
</entry>
<entry rowsep="0">TRAY ASSEMBLY, CABINET
</entry>
<entry morerows="2" rowsep="1">Same as
<xref stepend="M00452-X-XXX-X-XXX-step2" stepstart="M00452-X-XXX-X-XXX-step1"/>
</entry>
</row>
<row>
<entry align="center" rowsep="0"> &bull;
</entry>
<entry align="center" rowsep="0">
</entry>
<entry rowsep="0">  Deployed
</entry>
</row>
<row>
<entry align="center" rowsep="0">
</entry>
<entry align="center" rowsep="0"> &bull;
</entry>
<entry rowsep="0">  Stored
</entry>
</row>
</tbody>
</tgroup>
</table>
</para>
</lubewp>
```

b. Page-based TM stylesheet output example for ***<lubewp>***:

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TM X–XXXX–XXX 0007

SUSTAINMENT MAINTENANCE
LUBRICATION INSTRUCTIONS

INITIAL SETUP:

References

MIL-L-23398

INTERVALS

The following table lists the lubrication schedule for the maintenance shelter. The schedule is applicable for all temperatures between –25°F and +125°F.

Table 1. LUBRICATION INTERVALS

Q – Quarterly A – Annually

ITEM NO.	INTERVALS Q	INTERVALS A	ITEM TO BE SERVICED	PROCEDURE
1			HINGE. STEP	
	•		Deployed	1. Remove all foreign matter from joints or surfaces immediately before applying lubricant.
		•	Stored	2. Apply air-drying film lubricant MIL-L-23398, sparingly to prevent contaminant accumulation.
2			HINGE, HANDHOLD	Same as Steps 1 – 2
	•		Deployed	
		•	Stored	
3			HINGE, DOOR	Same as Steps 1 – 2
	•		Deployed	
		•	Stored	
4			HANDLE, DOOR	Same as Steps 1 – 2
	•		Deployed	
		•	Stored	
5			RETAINING DEVICE, DOOR	Same as Steps 1 – 2
	•		Deployed	
		•	Stored	
6			TRAY ASSEMBLY, CABINET	Same as Steps 1 – 2
	•		Deployed	
		•	Stored	

END OF WORK PACKAGE

0007-1

FIGURE 357. Example of a page-based TM stylesheet output for <lubewp>.

37.10 Facilities work package ***<facilwp>***. The element contains a description of all facilities, such as test stands, test tracks, clean rooms, shielded rooms, or other facilities that are required to do the maintenance work.

a. The components of ***<facilwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Main level paragraph ***<para0>*** (see 49.1.1.8)/***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<facilwp>***:

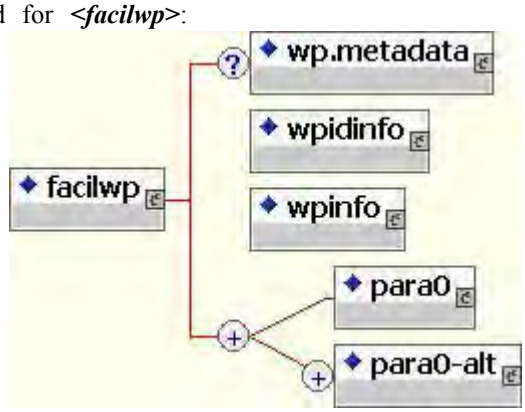

FIGURE 358. Facilities work package DTD hierarchy <facilwp>.

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c. DTD fragment for ***<facilwp>***:

```
<!ELEMENT facilwp (wp.metadata?, wpidinfo, wpinfo, (para0 | para0-alt+)+)>
<!ATTLIST facilwp
    airforce          (yes | no )                 "no"
    army              (yes | no )                 "no"
    assocfig          IDREFS                       #IMPLIED
    changeref         IDREFS                       #IMPLIED
    comment           CDATA                        #IMPLIED
    crewmember        CDATA                        #IMPLIED
    date-time-stamp   (date | time | date-time)  #IMPLIED
    delchlvl          CDATA                        #IMPLIED
    deletewp          (yes | no )                 "no"
    fgc               CDATA                        #IMPLIED
    frame             (yes | no )                 "yes"
    idref             IDREFS                       #IMPLIED
    inschlvl          CDATA                        #IMPLIED
    insertwp          CDATA                        #IMPLIED
    lsa-id            CDATA                        #IMPLIED
    marines           (yes | no )                 "no"
    navy              (yes | no )                 "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                        #IMPLIED
    tocentry          (2 | 3 | 4 | 5)             "2"
    wpno              ID                           #REQUIRED
    wpseq             CDATA                        #IMPLIED
>
```

d. Common attributes for ***<facilwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.10.1 XML document instance fragment and output for ***<facilwp>***. The XML instance and its stylesheet output for a ***<facilwp>*** is provided below.

a. Example of an XML document instance fragment for ***<facilwp>***:

```
<facilwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="M03450–X-XXX-XXX" wpseq="0025">
<wpidinfo>
<maintlvl level="depot"/>
<title>FACILITIES
</title>
</wpidinfo>
<wpinfo>
<title>N/A
</title>
<null insert="none"/>
</wpinfo>
<para0 esd="no" hcp="no">
<title>FACILITIES
</title>
<para esd="no" hcp="no">The Main Fuel Control (MFC) must be overhauled or
maintained in an enclosed and controlled area. The following items are
required and should be installed in the area for safety and the most
expeditious execution of overhaul and maintenance procedures.
<seqlist>
<item>Facilities to perform parts cleaning and protective
finishes application are required as follows:
<seqlist>
<item>Well-ventilated areas for adequate protection when toxic chemical
and flammable vapors are emitted during cleaning and repair operations.
</item>
<item>Emergency washing facilities for personnel who may
accidentally become contaminated or endangered by contact with
toxic or otherwise injurious materials.
</item>
<item>Sinks, containers, spray booths, and manipulating fixtures
to facilities dip, spray, flushing, and air-dry methods of
cleaning and application using: hydrocarbon solvents; corrosion
preventative fingerprint remover (water displacing bath);
chemical conversion materials for aluminum.
</item>
<item>A cold chest with a temperature range of 0&deg; to
-90&deg;F (-17.8&deg; to -67.71&deg;C) is required for
installation of interference fit parts.
</item>
```


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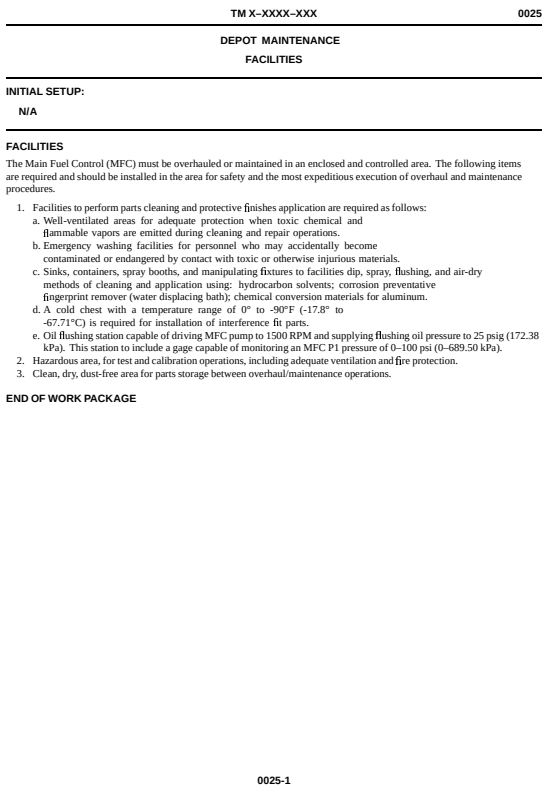
```
<item>Oil flushing station capable of driving MFC pump to 1500
RPM and supplying flushing oil pressure to 25 psig (172.38
kPa).  This station to include a gage capable of monitoring
an MFC P1 pressure of 0-100 psi (0-689.50 kPa).
</item>
</seqlist>
</item>
<item>Hazardous area, for test and calibration operations, including
adequate ventilation and fire protection.
</item>
<item>Clean, dry, dust-free area for parts storage between
overhaul/ maintenance operations.
</item>
</seqlist>
</para>
</para0>
</facilwp>
```

b. Page-based TM stylesheet output example for ***<facilwp>***:



TM X–XXXX–XXX 0025

DEPOT MAINTENANCE
FACILITIES

INITIAL SETUP:

N/A

FACILITIES

The Main Fuel Control (MFC) must be overhauled or maintained in an enclosed and controlled area. The following items are required and should be installed in the area for safety and the most expeditious execution of overhaul and maintenance procedures.

1. Facilities to perform parts cleaning and protective finishes application are required as follows:
   a. Well-ventilated areas for adequate protection when toxic chemical and flammable vapors are emitted during cleaning and repair operations.
   b. Emergency washing facilities for personnel who may accidentally become contaminated or endangered by contact with toxic or otherwise injurious materials.
   c. Sinks, containers, spray booths, and manipulating fixtures to facilities dip, spray, flushing, and air-dry methods of cleaning and application using: hydrocarbon solvents; corrosion preventative fingerprint remover (water displacing bath); chemical conversion materials for aluminum.
   d. A cold chest with a temperature range of 0° to -90°F (-17.8° to -67.71°C) is required for installation of interference fit parts.
   e. Oil flushing station capable of driving MFC pump to 1500 RPM and supplying flushing oil pressure to 25 psig (172.38 kPa). This station to include a gage capable of monitoring an MFC P1 pressure of 0–100 psi (0–689.50 kPa).
2. Hazardous area, for test and calibration operations, including adequate ventilation and fire protection.
3. Clean, dry, dust-free area for parts storage between overhaul/maintenance operations.

END OF WORK PACKAGE

0025-1

FIGURE 359. Example of a page-based TM stylesheet output for <facilwp>.

37.11 Overhaul inspection procedure (OIP) work package ***<oipwp>***. The element is for depots and lists items that have parts with specific characteristics, wear limits, specified performance requirements, or fatigue characteristics or tolerances.

a. The components of ***<oipwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).

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(4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(7) Each part has the following components
(a) General information ***<geninfo>*** (optional) (see 49.1.4.12).
(b) Overhaul inspection procedures table as either defined content specific table ***<oiptab>*** (required) (see 37.11.1) or a standard table ***<table>*** (required) (see section 42).
(c) Associated part illustration ***<figure>*** (optional – zero or more) (see 44.1.1).

b. DTD graphically depicted for ***<oipwp>***:

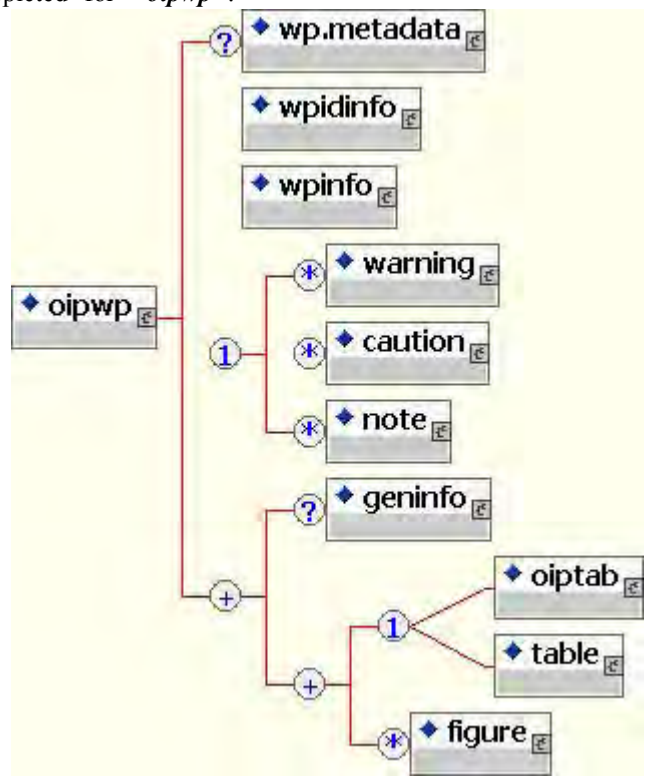

FIGURE 360. Overhaul inspection procedures (OIP) work package DTD hierarchy <oipwp>.

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c. DTD fragment for ***<oipwp>***:

```
<!ELEMENT oipwp (wp.metadata?, wpidinfo, wpinfo, (warning*,
   caution*, note*), (geninfo?, ((oiptab | table), figure*)+)+)>
<!ATTLIST oipwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes for ***<oipwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.11.1 Overhaul inspection procedures table ***<oiptab>***. The overhaul inspection procedures table is to be prepared for each part of the item that requires a critical inspection.

a. The components of ***<oiptab>*** are:
(1) Table title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
(2) Standard table header ***<std.table.header>*** (required) (see 42.3.1).
(3) At least on of the following grouping is required:
(a) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(b) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(c) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(d) Inspection item***<oipitem>*** identifies the detailed requirements. The element is similar to a "row" in a structural table. The components of ***<oipitem>*** are:
1 Quality assurance **qa** attribute (required). The quality assurance attribute identifies procedures which have a major quality assurance effect. The attribute has a declared value of "yes" or "no" (default value is "no"). The attribute is similar to a "cell" in a structural table and is entered in column one.
2 Item number***<itemno>*** (required) identifies the item number assigned to the procedure (see 49.1.4.8). The element is similar to a "cell" in a structural table and is entered in column two.
3 Reference letter ***<callout>*** (optional) identifies the critical inspection characteristics of the parts on the illustrations (see 46.2.4.1). The element is similar to a "cell" in a structural table and is entered in column three.
4 Characteristics description ***<desc>*** (required) identifies the characteristics being inspected (see 49.1.4.17) The element is similar to a "cell" in a structural table and is entered in column four.
5 Inspection method ***<insp-method>*** (required) identifies the inspection method used. The element is similar to a "cell" in a structural table and is entered in column five.
6 Corrective action ***<actionreq>*** (required) required action to be performed to correct the defect. The element is similar to a "cell" in a structural table and is entered in column six.

37.11.2 XML document instance fragment and output for ***<oipwp>***. The XML instance and its stylesheet output for a ***<oipwp>*** is provided below.

a. Example of an XML document instance fragment for ***<oipwp>***:

```
<oipwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="M00189-X-XXX-XX" wpseq="0280">
<wpidinfo>
<maintlvl level="depot"/>
<title>Spur Gear for
<brk/>Overhaul Inspection Procedures
</title>
</wpidinfo>
<wpinfo>
<ref>
```


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<ref-setup-item>
<xref wpid="M00342-X-XXX-XX"/>
</ref-setup-item>
</ref>
</wpinfo>
<oiptab tocentry="1">
<title>Overhaul Inspection Procedures for Spur Gear (Item 5, fig 4).
</title>
<std.table.header cols="6" rowsep="1">
<std.table.row>
<std.table.entry colnum="1" cols.span="0">QA REQ
</std.table.entry>
<std.table.entry colnum="2" cols.span="0">NO.
</std.table.entry>
<std.table.entry colnum="3" cols.span="0">REF LTR
</std.table.entry>
<std.table.entry colnum="4" cols.span="0">CHARACTERISTIC
</std.table.entry>
<std.table.entry colnum="5" cols.span="0">INSP METHOD
</std.table.entry>
<std.table.entry colnum="6" cols.span="0">REQUISITE
</std.table.entry>
</std.table.row>
</std.table.header>
<oipitem qa="no">
<itemno>1
</itemno>
<desc>Serviceability
</desc>
<insp-method>Visual/measure
</insp-method>
<actionreq>Examine for nicks, gouges, burrs, and corrosion, identified below repair damaged areas, 0.020 inch (0.508mm) or less deep, by blending.
</actionreq>
</oipitem>
<oipitem qa="yes">
<itemno>2
</itemno>
<desc>Metal fatigue
</desc>
<insp-method>Magnetic particle inspection
</insp-method>
<actionreq>No fractures or cracks.
</actionreq>
</oipitem>
<oipitem qa="yes">
<itemno>3
</itemno>
<callout assocfig="M00189-X-XXX-XX-fig1" label="A"/>
<desc>Tooth wear
</desc>
<insp-method>Visual
</insp-method>
<actionreq>No pitting, scuffing, scoring, metal flow, or wear steps allowed.
</actionreq>
</oipitem>

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Check the source to verify that this is the current version before use.

```
<oipitem qa="yes">
<itemno>4
</itemno>
<callout assocfig="M00189-X-XXX-XX-fig1" label="B"/>
<desc>Journal wear
</desc>
<insp-method>Measure
</insp-method>
<actionreq>Minimum diameter, 0.9841 inch (24.99mm). Repair (
<xref wpid="M00342-X-XXX-XX"/>).
</actionreq>
</oipitem>
</oiptab>
<figure application="both" figtype="normal-page" id="M00189-X-XXX-XX-fig1"
pane="no" tocentry="1">
<title>Spur Gear.
</title>
<graphic boardno="oip" unitmeasure="in">
</graphic>
</figure>
</oipwp>
```

b. Page-based TM stylesheet output example for ***<oipwp>***:

TM X–XXXX–XXX 0280

DEPOT MAINTENANCE

SPUR GEAR FOR
OVERHAUL INSPECTION PROCEDURES

INITIAL SETUP:

References

WP 0052

Table 1 Overhaul Inspection Procedures for Spur Gear (Item 5, fig 4).

QA REQ	NO.	REF LTR	CHARACTERISTIC	INSP METHOD	REQUISITE
NO	1		Serviceability	Visual/measure	Examine for nicks, gouges, burrs, and corrosion, identified below repair damaged areas, 0.020 inch (0.508mm) or less deep, by blending.
YES	2		Metal fatigue	Magnetic particle inspection	No fractures or cracks.
YES	3	A	Tooth wear	Visual	No pitting, scuffing, scoring, metal flow, or wear steps allowed.
YES	4	B	Journal wear	Measure	Minimum diameter, 0.9841 inch (24.99mm). Repair (WP 0052).

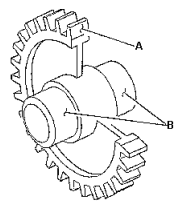

Figure 1. Spur Gear.

END OF WORK PACKAGE

0280-1

FIGURE 361. Example of a page-based TM stylesheet output for <oipwp>.

37.12 Depot mobilization requirements work package ***<mobilwp>***. The element contains the requirements to modify, delete, or add data to the preshop analysis or overhaul procedures during mobilization.

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The introduction paragraph is MIL–STD–40051–1/–2 standard statement and uses the boilerplate entity ***&mobilwp.intro;***.

a. The components of ***<mobilwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) Introduction ***<intro>*** (required) (see 49.1.4.15) includes scope and explanation of mobilization requirements. These statements are specified in MIL–STD–40051–1/–2 and uses the boilerplate entity ***&mobilwp.intro;***.
   (8) Mobilization requirements ***<mobilreq>*** (see 37.12.1).

b. DTD graphically depicted for ***<mobilwp>***:

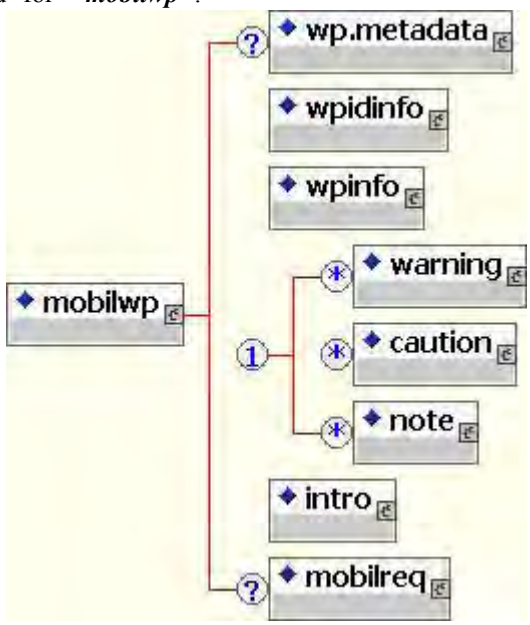

FIGURE 362. Depot mobilization requirements work package DTD hierarchy <mobilwp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<mobilwp>***:

```
<!ELEMENT mobilwp (wp.metadata?, wpidinfo, wpinfo, (warning*, caution*,
   note*), intro, mobilreq?)>
<!ATTLIST mobilwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attributes for ***<mobilewp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.12.1 Depot mobilization requirements ***<mobilreq>***. The element contains a list of actions that are in effect during the depot mobilization and associate the related work packages that are modified. The author either lists the mobilization requirements in tabular list (MIL–STD–40051–1/–2 prescribes the use as standard information ***<mobiltab>***) or in a paragraph references the work package(s) that lists and describes the mobilization requirements.

a. The components of ***<mobilreq>*** are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Paragraph ***<para>*** (required) (see 49.1.1.5).
   (3) Mobilization requirements standard information ***<mobiltab>*** (optional) (see 37.12.2).

b. DTD graphically depicted for ***<mobilreq>***:

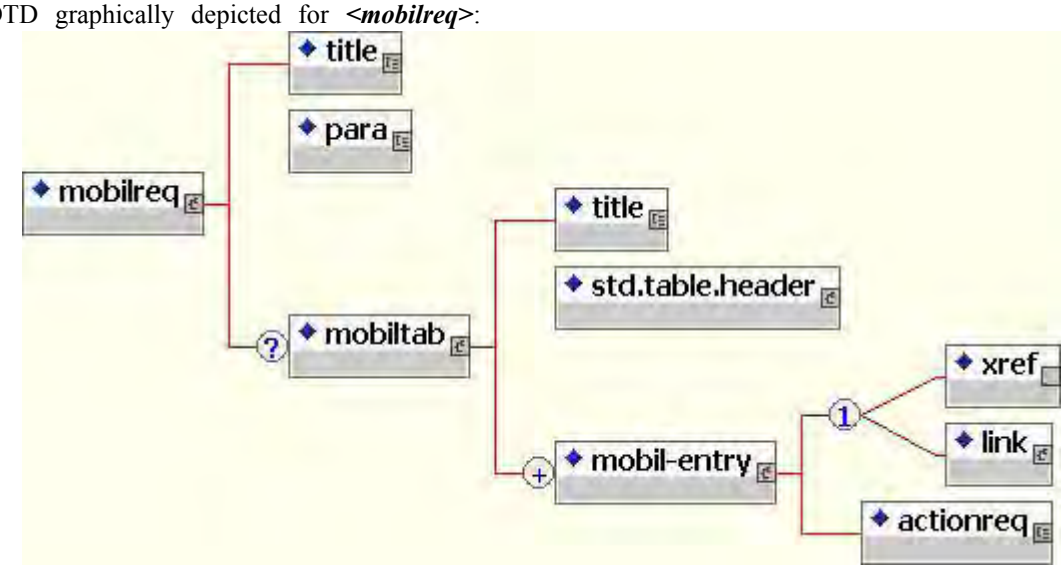

FIGURE 363. Depot mobilization requirements DTD hierarchy <mobilreq>.

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c. DTD fragment for ***<mobilreq>***:

```
<!ELEMENT mobilreq (title, para, mobiltab?)>
<!ATTLIST mobilreq
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<mobilreq>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

37.12.2 Depot mobilization requirements standard information ***<mobiltab>***. Depot mobilization requirements standard information contains the requirements for all analysis and procedures that are modified during mobilization. When multiple mobilization actions occur for the same work package a separate mobilization requirement entry (***<mobil-entry>***) is used.

a. The components of ***<mobiltab>*** are:

(1) Standard information title ***<title>*** (required). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet). (see 49.1.1.3).
(2) Standard information header ***<std.table.header>*** (see 42.3.1).
(3) Mobilization requirement entry ***<mobil-entry>*** (required – one or more). The element is similar to a "row" in a structural table.. The components of ***<mobil-entry>*** are:
(a) Work package cross-reference ***<link>*** (see 46.2.3) or ***<xref>*** (see 46.2.2) (required). The element is similar to a “cell” in a structural table and is entered in column
(b) Action requirement ***<actionreq>*** (required) contains the required action to be performed when mobilization occurs. The element is similar to a “cell” in a structural table and is entered in column

37.12.3 Depot mobilization requirements work package general entity boilerplates. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on “how to use” general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed (see Table XVI)

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TABLE XVI. Boilerplate Entities for *<mobilwp>*.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Depot mobilization requirements work package scope and mobilization requirements	*&mobilwp.intro;*	Not applicable

37.12.4 XML document instance fragment and output for *<mobilwp>*. The XML instance and its stylesheet output for a *<mobilwp>* is provided below.

a. Example of an XML document instance fragment for *<mobilwp>*:

```
<mobilwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="M00150-X-XXX-XX" wpseq="0071">
<wpidinfo>
<maintlvl level="depot"/>
<title>Depot Mobilization
</title>
</wpidinfo>
<wpinfo>
<title>Not Applicable
</title>
<null insert="none"/>
</wpinfo>
<intro frame="no"> &mobilwp.intro;
</intro>
<mobilreq>
<title>Mobilization Requirements
</title>
<para esd="no" hcp="no">
</para>
<mobiltab tocentry="1">
<title>Mobilization Requirements
</title> &mobilwp.mobiltab.tab.std-header;
<mobil-entry>
<xref wpid="M00088-X-XXX-XX"/>
<actionreq>
<xref stepstart="M00088-X-XXX-XX-6" taskid="M00088-X-XXX-XX-TASK"/>. Add
&ldquo;Depending on the urgency of requirements, availability of
materials, and fabrication lead time, provisions of this work package
may be relaxed. When that occurs, any practical method may be used to
inscribe or attach the data to the equipment, i.e., decals.&rdquo;
</actionreq>
</mobil-entry>
<mobil-entry>
<xref wpid="M00090-X-XXX-XX"/>
<actionreq>
<xref stepstart="M00090-X-XXX-XX-3" taskid="M00090-X-XXX-XX-CLEAN"/>.
Add &ldquo;Clean only to the extent necessary to per-
form preshop analysis.&rdquo;
</actionreq>
</mobil-entry>
<mobil-entry>
<xref wpid="M00092-X-XXX-XX"/>
<actionreq>
<xref stepstart="M00092-X-XXX-XX-8" taskid="M00092-X-XXX-XX-TASK"/>. Add
&ldquo;Clean only to the extent necessary to inspect components.&rdquo;
```


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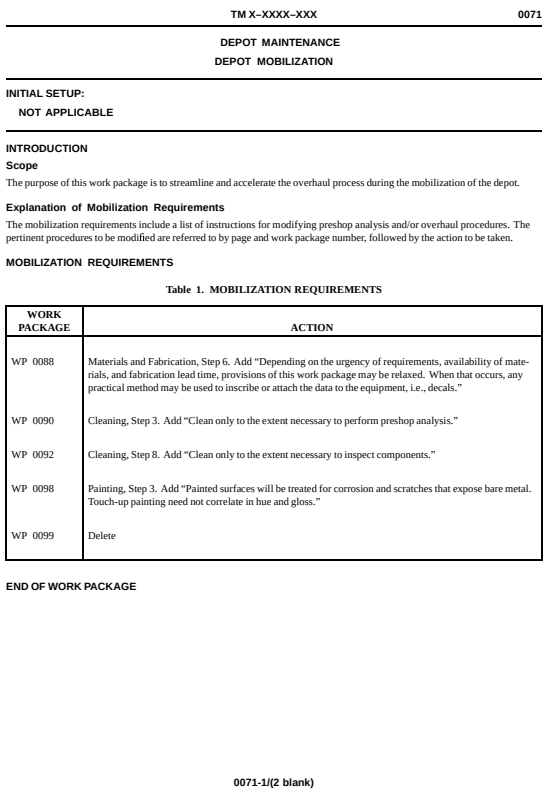
```
</actionreq>
</mobil-entry>
<mobil-entry>
<xref wpid="M00098-X-XXX-XX"/>
<actionreq>
<xref stepstart="M00098-X-XXX-XX-3" taskid="M00098-X-XXX-XX-TASK"/>. Add
&ldquo;Painted surfaces will be treated for corrosion and
scratches that expose bare metal. Touch-up painting need
not correlate in hue and gloss.&rdquo;
</actionreq>
</mobil-entry>
<mobil-entry>
<xref wpid="M00099-X-XXX-XX"/>
<actionreq>Delete
</actionreq>
</mobil-entry>
</mobiltab>
</mobilreq>
</mobilwp>
```

b. Page-based TM stylesheet output example for ***<mobilwp>***:



TM X–XXXX–XXX 0071

DEPOT MAINTENANCE
DEPOT MOBILIZATION

INITIAL SETUP:
NOT APPLICABLE

INTRODUCTION

Scope

The purpose of this work package is to streamline and accelerate the overhaul process during the mobilization of the depot.

Explanation of Mobilization Requirements

The mobilization requirements include a list of instructions for modifying preshop analysis and/or overhaul procedures. The pertinent procedures to be modified are referred to by page and work package number, followed by the action to be taken.

MOBILIZATION REQUIREMENTS

Table 1. MOBILIZATION REQUIREMENTS

WORK PACKAGE	ACTION
WP 0088	Materials and Fabrication, Step 6. Add "Depending on the urgency of requirements, availability of materials, and fabrication lead time, provisions of this work package may be relaxed. When that occurs, any practical method may be used to inscribe or attach the data to the equipment, i.e., decals."
WP 0090	Cleaning, Step 3. Add "Clean only to the extent necessary to perform preshop analysis."
WP 0092	Cleaning, Step 8. Add "Clean only to the extent necessary to inspect components."
WP 0098	Painting, Step 3. Add "Painted surfaces will be treated for corrosion and scratches that expose bare metal. Touch-up painting need not correlate in hue and gloss."
WP 0099	Delete

END OF WORK PACKAGE

0071-1/(2 blank)

FIGURE 364. Example of a page-based TM stylesheet output for <mobilwp>.

37.13 Quality assurance requirements work package ***<qawp>***. The element contains the requirements to prepare a quality assurance program for depot.

a. The components of ***<qawp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).

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(3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
(4) Statement of responsibility ***<responsibility>*** (required) (see 37.13.1).
(5) Definitions ***<definitions>*** (required) (see 37.13.2).
(6) Special requirements for inspection tools and equipment ***<specialreq>*** (optional) (see 37.13.3).
(7) Certification requirements ***<certreq>*** (optional) (see 37.13.4).
(8) Quality program ***<quality_program>*** (optional) (see 37.13.5).
(9) In-process inspections ***<inprocess>*** (required) (see 37.13.6).
(10) Acceptance inspections ***<acceptance>*** (required) (see 37.13.7).
(11) First article inspection ***<first>*** (optional) (see 37.13.8).

b. DTD graphically depicted for ***<qawp>***:

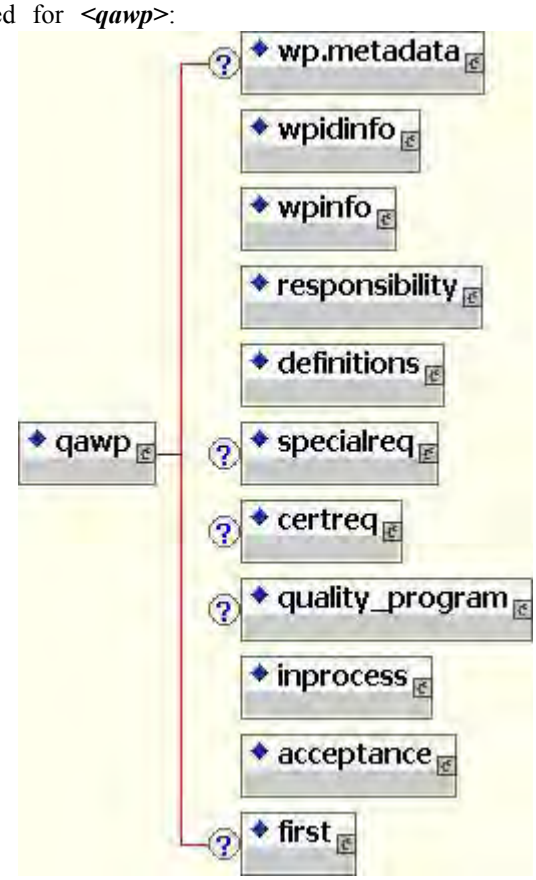

FIGURE 365. Quality assurance requirements work package DTD hierarchy <qawp>.

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c. DTD fragment for ***<qawp>***:

```
<!ELEMENT qawp (wp.metadata?, wpidinfo, wpinfo, responsibility,
   definitions, specialreq?, certreq?, quality_program?, inprocess,
   acceptance, first?)>
<!ATTLIST qawp
    airforce         (yes | no )                 "no"
    army             (yes | no )                 "no"
    assocfig         IDREFS                       #IMPLIED
    changeref        IDREFS                       #IMPLIED
    comment          CDATA                        #IMPLIED
    crewmember       CDATA                        #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                        #IMPLIED
    deletewp         (yes | no )                 "no"
    fgc              CDATA                        #IMPLIED
    frame            (yes | no )                 "yes"
    idref            IDREFS                       #IMPLIED
    inschlvl         CDATA                        #IMPLIED
    insertwp         CDATA                        #IMPLIED
    lsa-id           CDATA                        #IMPLIED
    marines          (yes | no )                 "no"
    navy             (yes | no )                 "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                        #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                           #REQUIRED
    wpseq            CDATA                        #IMPLIED
>
```

d. Common attributes for ***<qawp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.13.1 Statement of responsibility ***<responsibility>***. The element contains the responsibilities of the depot/contractor. Generally only the standard statement is included from MIL–STD–40051–1/–2 and uses the boilerplate entity ***&qawp.responsibility;*** (see 37.13.9).

a. The components of ***<responsibility>*** are:
(1) Title ***<title>*** (required) (see 49.1.1.3).
(2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(4) Select one of the following information types:
(a) Narrative paragraphs
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
(b) Narrative paragraphs with descriptive or narrative titled text
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
(c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<responsibility>***:

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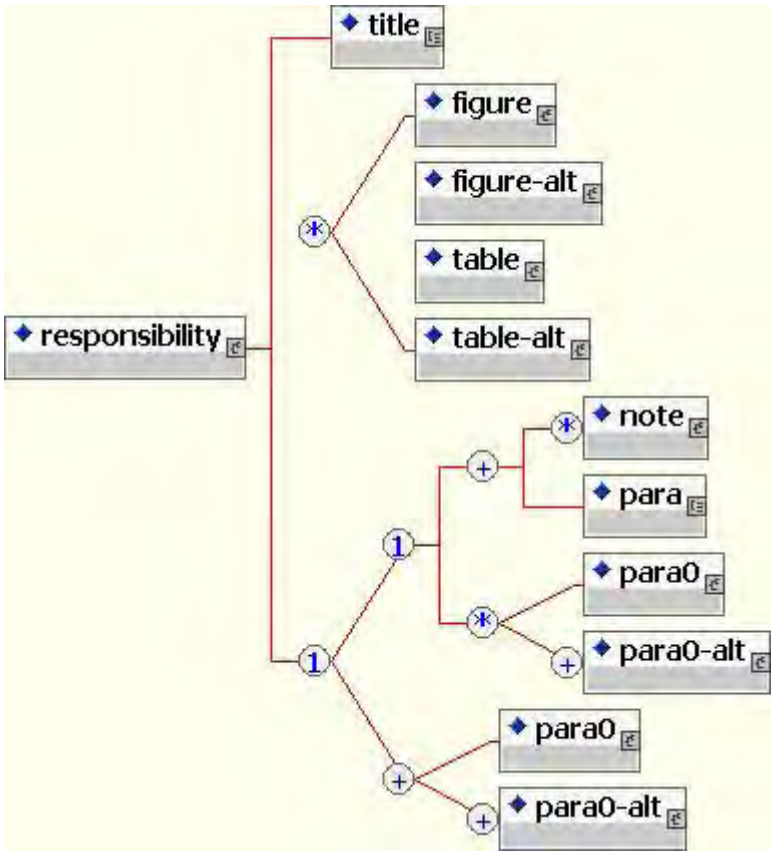

FIGURE 366. Quality assurance requirements work package DTD hierarchy <responsibility>.

c. DTD fragment for ***<responsibility>***:

```
<!ELEMENT responsibility (title, (figure | figure-alt | table |
   table-alt)* (((note*, para)+, (para0 | para0-alt+)*) | (para0 |
   para0-alt+)+))>
<!ATTLIST responsibility
    changeref        IDREFS                  #IMPLIED
    comment          CDATA                   #IMPLIED
    delchlvl         CDATA                   #IMPLIED
    inschlvl         CDATA                   #IMPLIED
>
```

d. Common attributes for ***<responsibility>*** are:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

37.13.2 Definitions ***<definitions>***. The element contains quality assurance terms and definitions that are extensively used in the DMWR/NMWR. If the definitions are listed in another publication, the referenced publications are listed. The definitions may be authored as paragraphs or term/definition listing. Using paragraphs may used to provide an introduction or general information about the definition set or specify the referenced documents for the terms used in the publication. If only a simple term and definition listing is needed use the ***<term.def>*** element.

a. The components of ***<definitions>*** are:

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(1) Definition title ***<title>*** (required) (see 49.1.1.3).
(2) Select a method to author the definitions either by:
(a) Paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
(b) Term and definition ***<term.def>*** (required – one or more) (see 37.13.1).

b. DTD graphically depicted for ***<definitions>***:

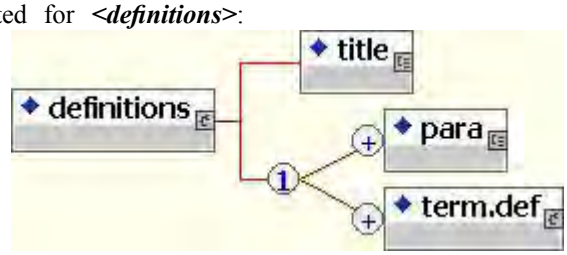

FIGURE 367. Definitions DTD hierarchy <definitions>.

c. DTD fragment for ***<definitions>***:

```
<!ELEMENT definitions (title, (para+ | term.def+))>
<!ATTLIST definitions
    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

d. Common attributes for ***<definitions>*** are:
(1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(2) **comment** – Change information (optional) (see 49.2.9).
(3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(4) **inschlvl** – Insert change level (optional) (see 49.2.9).

a. The components of ***<definitions>*** are:
(1) Required title ***<title>***.
(2) At least one of the following is required (not both):
(a) Paragraph(s) ***<para>***, or
(b) Term and definition(s) ***<term.def>***

37.13.3 Special requirements for inspection tools and equipment ***<specialreq>***. The element contains the special requirements for the maintenance and calibration of tools and test equipment used for quality assurance inspections. The components for ***<specialreq>*** is the same as ***<responsibility>*** (see 37.13.1).

37.13.4 Certification requirements ***<certreq>***. The element contains any certification or licensing requirements for processes, procedures, materials, equipment or personal skills used. When listing the certification or licensing include the appropriate standards, specifications, regulations, or laws that apply. Additionally, reference the text in the DMWR/NMWR where a soldering, welding, magnetic particle inspection certification radioactive substance, or test driver licenses are required. The components for ***<certreq>*** is the same as ***<responsibility>*** (see 37.13.1).

37.13.5 Quality program ***<quality_program>***. The element contains any additional requirements for a quality program. The components for ***<quality_program>*** is the same as ***<responsibility>*** (see 37.13.1).

37.13.6 In-process inspections ***<inprocess>***. The element contains the method used to identify QA inspections. Generally only the standard statement is included from MIL–STD–40051–1/–2 and uses the boilerplate entity ***&qawp.inprocess;*** (see 37.13.9). The components for ***<inprocess>*** is the same as ***<responsibility>*** (see 37.13.1).

37.13.7 Acceptance inspections ***<acceptance>***. The element contains the method used for acceptance inspection. Generally, only the standard statement is included from MIL–STD–40051–1/–2 and uses the

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boilerplate entity ***&qawp.acceptance;*** (see 37.13.9). The components for ***<acceptance>*** is the same as ***<responsibility>*** (see 37.13.1).

37.13.8 First article inspection ***<first>***. The element contains a statement that defines the first article inspection/test and is in accordance with ISO 9000 Series standards or equivalent. The components for ***<first>*** is the same as ***<responsibility>*** (see 37.13.1).

37.13.9 Quality assurance requirements work package requirements work package general entity boilerplates. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed (see Table XVII).

TABLE XVII. Boilerplate Entities for ***<qawp>***.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Responsibilities of the depot/contractor statement	***&qawp.responsibility;***	Not applicable
Method used to identify QA inspections statement	***&qawp.inprocess;***	Not applicable
Method used for acceptance inspection statement	***&qawp.acceptance;***	Not applicable

37.13.10 XML document instance fragment and output for ***<qawp>***. The XML instance and its stylesheet output for a ***<qawp>*** is provided below.

a. Example of an XML document instance fragment for ***<qawp>***:

```
<qawp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="m0567-xx-xxxx-xx" wpseq="0163">
<wpidinfo>
<maintlvl level="depot"/>
<title>QUALITY ASSURANCE REQUIREMENTS
</title>
</wpidinfo>
<wpinfo>
<ref>
<ref-setup-item>
<extref docno="AR 750-43"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="DESCOM-R 702-1"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="MIL-I-45607B"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="MIL-STD-410"/>
</ref-setup-item>
</ref>
</wpinfo>
<responsibility> &qawp.responsibility;
</responsibility>
<definitions>
<title>DEFINITIONS
```


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</title>
<para esd="no" hcp="no">For quality assurance terms and definitions, refer to the glossary in this DMWR.
</para>
</definitions>
<specialreq>
<title>SPECIAL REQUIREMENTS FOR INSPECTION TOOLS AND EQUIPMENT
</title>
<para esd="no" hcp="no">The overhaul facility is responsible for acquisitions, maintenance, calibration, and disposition of all inspection and test equipment. Test equipment to be used by AMC (Army) elements will be acquired in accordance with <extref docno="AR 750-43"/>. All instrumentation and inspection equipment used in compliance with this DMWR shall be calibrated and controlled in accordance with <extref docno="MIL-I-45607B"/> or <extref docno="DESCOM-R-702-1" posttext=", Depot Quality System (Army facility)"/>, with all standards traceable to the National Bureau of Standards. Descriptions of inspecting and measuring equipment are left to the discretion of the over hauling facility to be considered as good shop practice.
</para>
</specialreq>
<certreq>
<title>CERTIFICATION REQUIREMENTS
</title>
<para esd="no" hcp="no">The contractor/depot QA activity shall be responsible for ascertaining and certifying personnel skills, equipment, and materiel meet the requirements of the work to be accomplished, unless otherwise specified in the contract or by PA/CM representative, the contractor/depot QA activity shall provide the PA/CM with statements or other evidence that specifications for such special processes as welding, nondestructive testing, plating, etc., have been complied with. Personnel performing magnetic particle and penetrant tests shall be certified in accordance with <extref docno="MIL-STD-410"/>.
</para>
</certreq>
<inprocess> &qawp.inprocess;
</inprocess>
<acceptance> &qawp.acceptance;
</acceptance>
<first>
<title>FIRST ARTICLE INSPECTION
</title>
<para esd="no" hcp="no">The contractor/depot quality assurance activity shall perform a first article inspection on each of the first overhauled assemblies produced in accordance with the DMWR. After contractor/depot acceptance, the procuring agency's quality assurance representative may perform a separate first overhauled article inspection. The first article inspection shall be conducted as follows: (1) Component Inspection (Chapter 3), (2) Final Inspection (Chapter 4), and (3) Processing for Storage and Shipment (Chapter 6).
</para>
<para0 esd="no" hcp="no">
<title>Submission of Product
</title>

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```
<para esd="no" hcp="no">The inspection of lot size, lot formation, and
presentation of lots shall be as specified by the contracting activity.
</para>
</para0>
<para0 esd="no" hcp="no">
<title>Quality Assurance Component Inspection
</title>
<para esd="no" hcp="no">The component inspection shall be conducted by the
procuring activity's quality assurance representative during production
of first overhauled item to evaluate conformance of materials and
workmanship to drawings and overhaul technical data package.
</para>
</para0>
<para0 esd="no" hcp="no">
<title>Final Acceptance
</title>
<para esd="no" hcp="no">The contractor's/depot's quality assurance activity,
along with the procuring activity's quality assurance representative,
shall conduct the final acceptance inspection in accordance with the
requirements specified herein. During this time, the contractor/depot
quality assurance activity shall have available the written inspection
system plan, procedures, inspection records, and components.
</para>
</para0>
<para0 esd="no" hcp="no">
<title>Processing for Storage and Shipment
</title>
<para esd="no" hcp="no">Inspection of processing for storage and shipment
shall be conducted in accordance with applicable specifications on the
first overhauled item and on one of the next ten processed items under
the procedure established during the first overhauled item.
</para>
</para0>
</first>
</qawp>
```

b. Page-based TM stylesheet output example for ***<qawp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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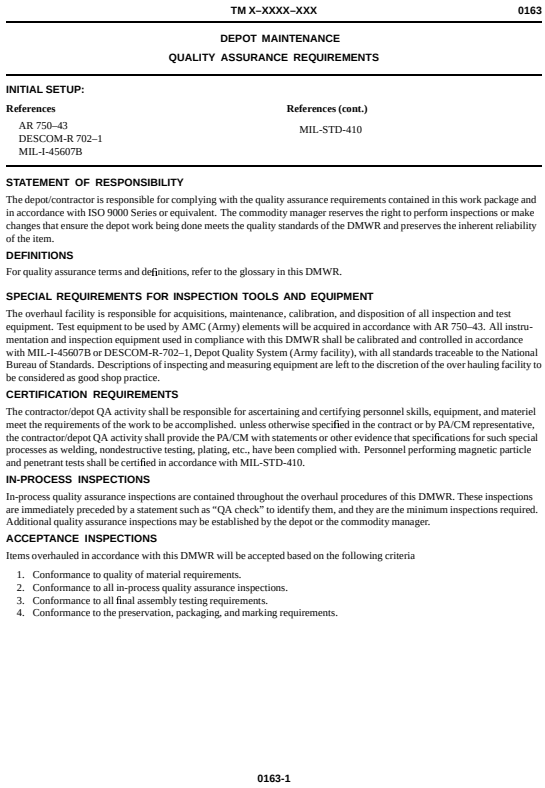

TM X–XXXX–XXX 0163

DEPOT MAINTENANCE
QUALITY ASSURANCE REQUIREMENTS

INITIAL SETUP:

References
AR 750–43
DESCOM-R 702–1
MIL-I-45607B

References (cont.)
MIL-STD-410

STATEMENT OF RESPONSIBILITY

The depot/contractor is responsible for complying with the quality assurance requirements contained in this work package and in accordance with ISO 9000 Series or equivalent. The commodity manager reserves the right to perform inspections or make changes that ensure the depot work being done meets the quality standards of the DMWR and preserves the inherent reliability of the item.

DEFINITIONS

For quality assurance terms and definitions, refer to the glossary in this DMWR.

SPECIAL REQUIREMENTS FOR INSPECTION TOOLS AND EQUIPMENT

The overhaul facility is responsible for acquisitions, maintenance, calibration, and disposition of all inspection and test equipment. Test equipment to be used by AMC (Army) elements will be acquired in accordance with AR 750–43. All instrumentation and inspection equipment used in compliance with this DMWR shall be calibrated and controlled in accordance with MIL-I-45607B or DESCOM-R-702–1, Depot Quality System (Army facility), with all standards traceable to the National Bureau of Standards. Descriptions of inspecting and measuring equipment are left to the discretion of the over hauling facility to be considered as good shop practice.

CERTIFICATION REQUIREMENTS

The contractor/depot QA activity shall be responsible for ascertaining and certifying personnel skills, equipment, and materiel meet the requirements of the work to be accomplished. unless otherwise specified in the contract or by PA/CM representative, the contractor/depot QA activity shall provide the PA/CM with statements or other evidence that specifications for such special processes as welding, nondestructive testing, plating, etc., have been complied with. Personnel performing magnetic particle and penetrant tests shall be certified in accordance with MIL-STD-410.

IN-PROCESS INSPECTIONS

In-process quality assurance inspections are contained throughout the overhaul procedures of this DMWR. These inspections are immediately preceded by a statement such as "QA check" to identify them, and they are the minimum inspections required. Additional quality assurance inspections may be established by the depot or the commodity manager.

ACCEPTANCE INSPECTIONS

Items overhauled in accordance with this DMWR will be accepted based on the following criteria

1. Conformance to quality of material requirements.
2. Conformance to all in-process quality assurance inspections.
3. Conformance to all final assembly testing requirements.
4. Conformance to the preservation, packaging, and marking requirements.

0163-1

FIGURE 368. Example of a page-based TM stylesheet output for <qawp>.

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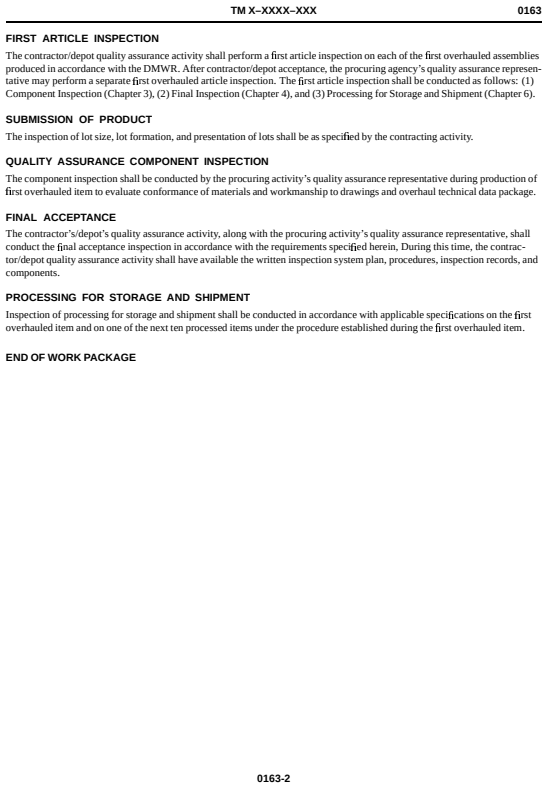

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FIRST ARTICLE INSPECTION

The contractor/depot quality assurance activity shall perform a first article inspection on each of the first overhauled assemblies produced in accordance with the DMWR. After contractor/depot acceptance, the procuring agency's quality assurance representative may perform a separate first overhauled article inspection. The first article inspection shall be conducted as follows: (1) Component Inspection (Chapter 3), (2) Final Inspection (Chapter 4), and (3) Processing for Storage and Shipment (Chapter 6).

SUBMISSION OF PRODUCT

The inspection of lot size, lot formation, and presentation of lots shall be as specified by the contracting activity.

QUALITY ASSURANCE COMPONENT INSPECTION

The component inspection shall be conducted by the procuring activity's quality assurance representative during production of first overhauled item to evaluate conformance of materials and workmanship to drawings and overhaul technical data package.

FINAL ACCEPTANCE

The contractor's/depot's quality assurance activity, along with the procuring activity's quality assurance representative, shall conduct the final acceptance inspection in accordance with the requirements specified herein, During this time, the contractor/depot quality assurance activity shall have available the written inspection system plan, procedures, inspection records, and components.

PROCESSING FOR STORAGE AND SHIPMENT

Inspection of processing for storage and shipment shall be conducted in accordance with applicable specifications on the first overhauled item and on one of the next ten processed items under the procedure established during the first overhauled item.

END OF WORK PACKAGE

0163-2

FIGURE 368. Example of a page-based TM stylesheet output for <qawp> - Continued.

37.14 Illustrated list of manufactured items work package ***<manuwp>***. The illustrated list of manufactured items work package is prepared to identify and include technical information for each item authorized to be manufactured or fabricated by field/AMC or above maintenance personnel (all "MO", "MF", "MH" and "MD" source coded items). When applicable, an external reference may be made to fabrication instructions for tools and equipment.

a. The components of ***<manuwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Introduction ***<intro>*** (required) (see 49.1.4.15 and 37.14.5).
   (5) Manufactured items index ***<manuindx>*** (required) (see 37.14.1).
   (6) Manufactured item ***<manuitem>*** (required) (see 37.14.2).

b. DTD graphically depicted for ***<manuwp>***:

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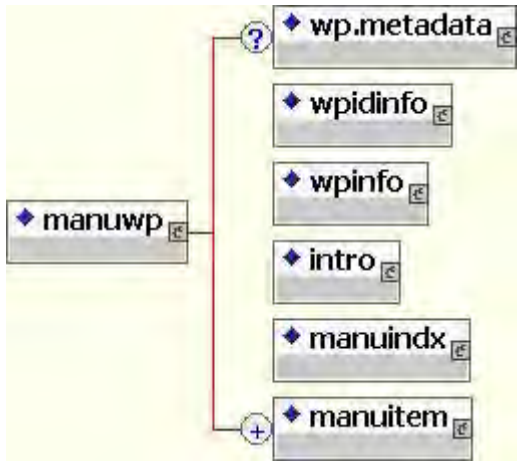

FIGURE 369. Illustrated list of manufactured items work package DTD hierarchy <manuwp>.

c. DTD fragment for ***<manuwp>***:

```
<!ELEMENT manuwp (wp.metadata?, wpidinfo, wpinfo, intro, manuindx,
   manuitem+)>
<!ATTLIST manuwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes for ***<manuwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

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(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.14.1 Manufactured items index ***<manuindx>***. The element lists each manufactured item illustrated. Each manufactured item describes ***<partdesc>*** either the part number ***<partno>***/CAGE code ***<cageno>*** and/or drawing number ***<dwgno>*** in alphabetical order, along with the nomenclature ***<name>***, and an illustration figure number reference ***<figno>*** that contains the manufacturing data.

a. The components of ***<manuindx>*** are:
   (1) Index title ***<title>*** (optional) (see 49.1.1.3).
   (2) Index header ***<std.table.header>*** (optional) (see 42.3.1).
   (3) Manufactured item (required – one or more) components are:
      (a) Manufactured item description ***<partdesc>*** (required) (see 37.14.3).
      (b) Manufactured item figure reference ***<figno>*** (optional) (see 46.2.4.3).

b. DTD graphically depicted for ***<manuindx>***:

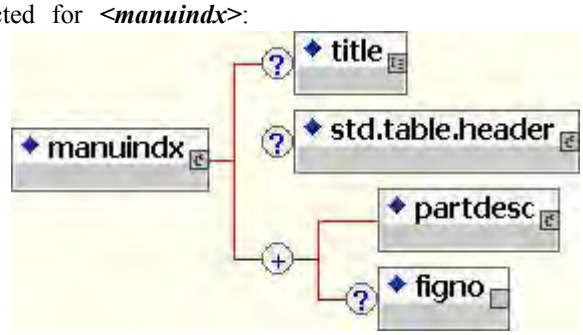

FIGURE 370. Manufactured items index DTD hierarchy <manuindx>.

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c. DTD fragment for ***<manuindx>***:

```
<!ELEMENT manuindx (title?, std.table.header?, (partdesc, figno?)+)>
<!ATTLIST manuindx
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<manuindx>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

37.14.2 Manufactured items ***<manuitem>***. The element contains all the information required to manufacture a part that includes such information as the illustrated manufactured part, instructions, parts list, and/or manufactured part description.

a. The components of ***<manuitem>*** are:

(1) Manufactured item title ***<title>*** (optional) (see 49.1.1.3).
(2) Includes one or more of the following components and in no particular order.
(a) Illustration ***<graphic>*** (see 44.2).
(b) Figure ***<figure>*** (see 44.1.1).
(c) Table ***<table>*** (see 42).
(d) Paragraph ***<para>***.
(e) Paragraph with alerts ***<specpara>*** (see 49.1.1.6).
(f) Instruction step ***<step1>***/***<step1–alt>*** (see 33.3).
(g) Procedure ***<proc>*** (see 33.2).
(h) Material/part list ***<material-list>*** (see 37.14.4).
(i) Manufacture part description ***<partdesc>*** (see 37.14.3).

b. DTD graphically depicted for ***<manuitem>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

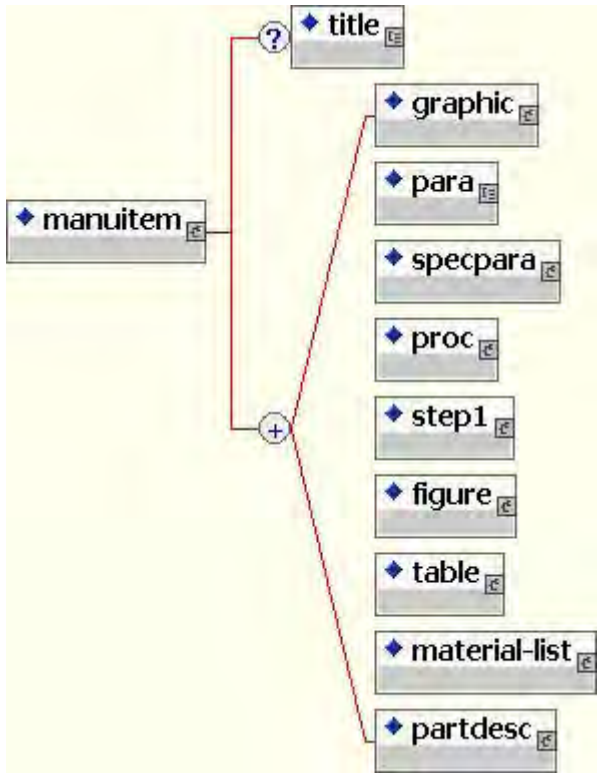

FIGURE 371. Manufactured item DTD hierarchy <manuitem>.

c. DTD fragment for ***<manuitem>***:

```
<!ELEMENT manuitem (title?, (graphic | para | specpara | proc | step1 |
   figure | table | material-list | partdesc)+)>
<!ATTLIST manuitem
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<manuitem>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

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37.14.3 Manufactured part description ***<partdesc>***. The element contains a description of the item to be manufactured, including the part number (with CAGE code) and/or drawing number, along with the nomenclature.

a. The components of ***<partdesc>*** are:
   (1) Part identification either by:
      (a) Part number ***<partno>*** (required) (see 49.1.4.21), CAGE code ***<cageno>*** (required) (see 49.1.4.2), and drawing number ***<dwgno>*** (optional) (see 49.1.4.7).
      (b) Drawing number ***<dwgno>*** (required) (see 49.1.4.7).
   (2) Part nomenclature ***<name>*** (required) (see 49.1.4.19).

b. DTD graphically depicted for ***<partdesc>***:

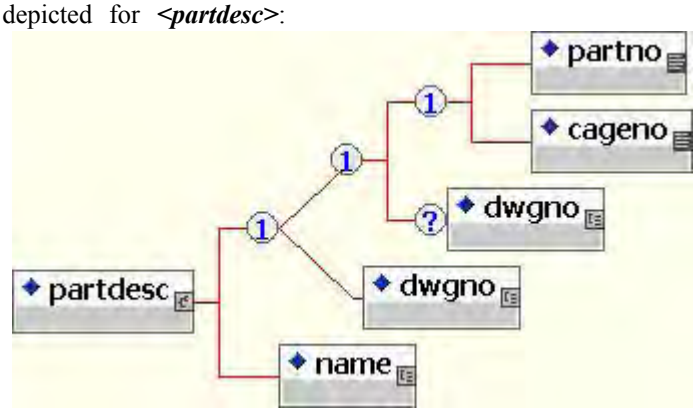

FIGURE 372. Manufactured part description DTD hierarchy <partdesc>.

c. DTD fragment for ***<partdesc>***:

```
<!ELEMENT partdesc ((((partno, cageno), dwgno?) | dwgno), name)>
```

37.14.4 Material list ***<material-list>***. The material list contains the bulk material needed to manufacture the item, and cites the technical characteristics (i.e., standards, specifications, conditions, dimensions, and any other pertinent data). When applicable, references are made to the associated parts information in the RPSTL TM or RPSTL work package (for combined TMs).

a. The components of ***<material-list>*** are:
   (1) Material/part list title ***<title>*** (optional) (see 49.1.1.3).
   (2) Material/part list header ***<std.table.header>*** (optional) (see 42.3.1).
   (3) Each material/part information includes the following components:
      (a) Material/part nomenclature ***<name>*** (required) (see 49.1.4.19).
      (b) Material/part identification is one of following
         1 Part number ***<partno>*** (required), CAGE code ***<cageno>*** (required), and NSN ***<nsn>*** (optional).
         2 Specification or standard ***<extref>*** (required) (see 46.2.1).
         3 Work package ***<link>*** (required) (see 46.2.3).
   (4) Quantity ***<qty>*** (optional) (see 49.1.4.9).
   (5) Material/part information reference ***<itemref>*** (optional) (see 32.5.1.1).

b. DTD graphically depicted for ***<material-list>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

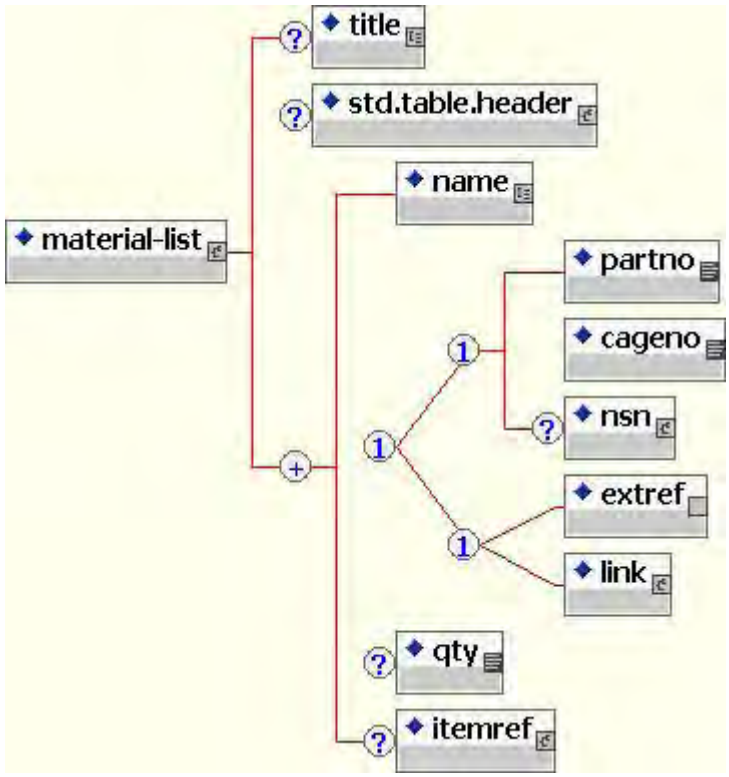

FIGURE 373. Material items list DTD hierarchy <material-list>.

c. DTD fragment for ***<material-list>***:

```
<!ELEMENT material-list (title?, std.table.header?, (name, ((partno,
   cageno, nsn?) | (extref | link)), qty?, itemref?)+))>
<!ATTLIST material-list
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<material-list>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

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37.14.5 Illustrated list of manufactured items work package general entity boilerplates. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed Table XVIII

TABLE XVIII. Boilerplate Entities for *<manuwp>*.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Work package scope, how to use the index of manufactured items, and explanation of the illustrations of manufactured items statement	*&manuwp.intro;*	— —
• Enter the lowest maintenance level allowed to perform this task.	*&manuwp.intro.maint-level;*	*<!ENTITY manuwp.intro.maint-level "REPLACE WITH MAINTENANCE LEVEL">*
• Enter where the RPSTL information is located either in a separate TM (*<extref>*) or within the TM in a parts work package (*<xref>*).	*&manuwp.intro.rpstl;*	*<!ENTITY manuwp.intro.rpstl '<extref docno="REPLACE WITH (WHEN APPLICABLE) A REFERENCE TO THE ASSOCIATED RPSTL TM "/> OR <xref wpid="RPSTL_PART_LIST_WORK_PACKAGE"/>.>*

37.14.6 XML document instance fragment and output for *<manuwp>*. The XML instance and its stylesheet output for a *<manuwp>* is provided below.

a. Example of an XML document instance fragment for *<manuwp>*:

```
<manuwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="m00049-9-2320-365" wpseq="0127">
<wpidinfo>
<maintlvl level="field"/>
<title>Illustrated List Of Manufactured Items
</title>
</wpidinfo>
<wpinfo>
<tools>
<tools-setup-item>
<name>File, Metal
</name>
</tools-setup-item>
<tools-setup-item>
<name>Hacksaw, fine-toothed
</name>
</tools-setup-item>
<tools-setup-item>
<name>Metal cutter
</name>
```


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Check the source to verify that this is the current version before use.

```
</tools-setup-item>
<tools-setup-item>
<name>Welding  torch
</name>
</tools-setup-item>
</tools>
<mtrlpart>
<mtrlpart-setup-item>
<name>Angle  Iron
</name>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Blackout  Shield  Header  Seals
</name>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Bulk  Tubing
</name>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Glue,  GP1TY1,
<extref  docno="MIL-A-46106"/>
</name>
<itemref>
<xref  itemid="s00016-9-2320-365-11"  itemno="11"  wpid="s00016-9-2320-365"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Insulgrip,  CSA  105  C
</name>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>LH  Door  Gasket
</name>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Steel,  ASTM  A569  Sheet,  Hot  Rolled
</name>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Steel,  Flat  Bar
</name>
</mtrlpart-setup-item>
</mtrlpart>
<ref>
<ref-setup-item>
<extref  docno="DOD-STD-00100D(AR)"/>
</ref-setup-item>
<ref-setup-item>
<extref  docno="ANSI  Y14.5M1982"/>
</ref-setup-item>
</ref>
</wpinfo>
<intro  frame="no">  &manuwp.intro;
</intro>
<manuindx  id="m00049-9-2320-365-1">
```

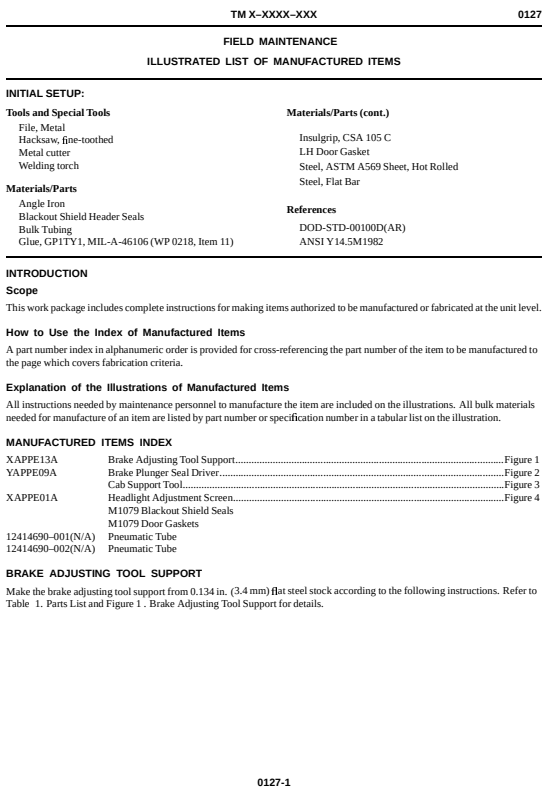

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<title>Manufactured items index
</title>
<std.table.header cols="3" rowsep="1">
<std.table.row>
<std.table.entry colnum="1" cols.span="0" rowsep="0">DRAWING/
</std.table.entry>
<std.table.entry colnum="2" cols.span="0" rowsep="0">
</std.table.entry>
<std.table.entry colnum="3" cols.span="0" rowsep="0">FIGURE
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry colnum="1" cols.span="0" rowsep="1">PART NO.
</std.table.entry>
<std.table.entry colnum="2" cols.span="0" rowsep="1">DESCRIPTION
</std.table.entry>
<std.table.entry colnum="3" cols.span="0" rowsep="1">NO.
</std.table.entry>
</std.table.row>
</std.table.header>
<partdesc>
<dwgno>XAPPE13A
</dwgno>
<name>Brake Adjusting Tool Support
</name>
</partdesc>
<figno idref="m00049-9-2320-365-56"/>
<partdesc>
<dwgno>YAPPE09A
</dwgno>
<name>Brake Plunger Seal Driver
</name>
</partdesc>
<figno idref="m00049-9-2320-365-57"/>
<partdesc>
<dwgno>
</dwgno>
<name>Cab Support Tool
</name>
</partdesc>
<figno idref="m00049-9-2320-365-58a"/>
<partdesc>
<dwgno>XAPPE01A
</dwgno>
<name>Headlight Adjustment Screen
</name>
</partdesc>
<figno idref="m00049-9-2320-365-62"/>
<partdesc>
<dwgno>
</dwgno>
<name>M1079 Blackout Shield Seals
</name>
</partdesc>
<partdesc>
<dwgno>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</dwgno>
<name>M1079  Door  Gaskets
</name>
</partdesc>
<partdesc>
<partno>12414690&ndash;001
</partno>
<cageno>N/A
</cageno>
<name>Pneumatic  Tube
</name>
</partdesc>
<partdesc>
<partno>12414690&ndash;002
</partno>
<cageno>N/A
</cageno>
<name>Pneumatic  Tube
</name>
</partdesc>
</manuindx>  . . .
</manuwp>
```

b. Page-based TM stylesheet output example for ***<manuwp>***:



TM X–XXXX–XXX 0127

FIELD MAINTENANCE

ILLUSTRATED LIST OF MANUFACTURED ITEMS

INITIAL SETUP:

Tools and Special Tools

File, Metal
Hacksaw, fine-toothed
Metal cutter
Welding torch

Materials/Parts

Angle Iron
Blackout Shield Header Seals
Bulk Tubing
Glue, GP1TY1, MIL-A-46106 (WP 0218, Item 11)

Materials/Parts (cont.)

Insulgrip, CSA 105 C
LH Door Gasket
Steel, ASTM A569 Sheet, Hot Rolled
Steel, Flat Bar

References

DOD-STD-00100D(AR)
ANSI Y14.5M1982

INTRODUCTION

Scope

This work package includes complete instructions for making items authorized to be manufactured or fabricated at the unit level.

How to Use the Index of Manufactured Items

A part number index in alphanumeric order is provided for cross-referencing the part number of the item to be manufactured to the page which covers fabrication criteria.

Explanation of the Illustrations of Manufactured Items

All instructions needed by maintenance personnel to manufacture the item are included on the illustrations. All bulk materials needed for manufacture of an item are listed by part number or specification number in a tabular list on the illustration.

MANUFACTURED ITEMS INDEX

XAPPE13A Brake Adjusting Tool Support.....Figure 1
YAPPE09A Brake Plunger Seal Driver.....Figure 2
Cab Support Tool.....Figure 3
XAPPE01A Headlight Adjustment Screen.....Figure 4
M1079 Blackout Shield Seals
M1079 Door Gaskets
12414690–001(N/A) Pneumatic Tube
12414690–002(N/A) Pneumatic Tube

BRAKE ADJUSTING TOOL SUPPORT

Make the brake adjusting tool support from 0.134 in. (3.4 mm) flat steel stock according to the following instructions. Refer to Table 1. Parts List and Figure 1 . Brake Adjusting Tool Support for details.

0127-1

FIGURE 374. Example of a page-based TM stylesheet output for <manuwp>.

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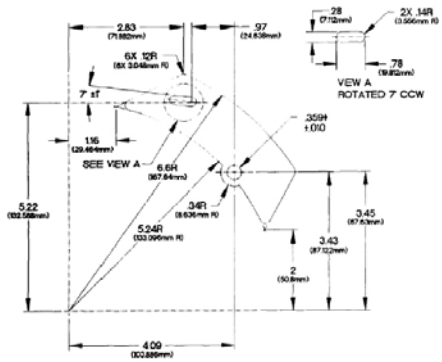

Figure 1. Brake Adjusting Tool Support

1. All dimensions are in inches (millimeters).
2. Cut steel sheet as shown by dimensions on Figure 1. Brake Adjusting Tool Support.
3. De-burr and remove sharp edges.

Table 1. Parts List

Item	Dimension
Steel, ASTM A569 Sheet, Hot Rolled	6.0 in. (152.4 mm) x 6.0 in. (152.4 mm) x 0.134 in. (3.4 cm)

0127-2

FIGURE 374. Example of a page-based TM stylesheet output for <manuwp> - Continued.

37.15 Torque limits work package ***<torquewp>***. The element contains the applicable torque values, data as to bolt grade markings and their proper identification, and specific torque sequencing requirements. Specific instructions such as torque limits for dry and wet fasteners, fastener sizes, thread patterns, etc. is included.

a. The components of ***<torquewp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (4) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (5) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (6) Introduction ***<intro>*** (required) (see 49.1.4.15) contains the scope or how to use the work package.
   (7) Torque limit values ***<torqueval>*** (required) (see 37.15.1).

b. DTD graphically depicted for ***<torquewp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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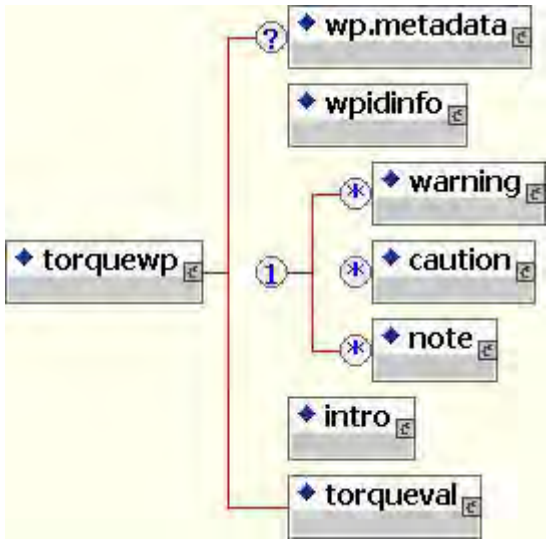

FIGURE 375. Torque limits work package DTD hierarchy <torquewp>.

c. DTD fragment for ***<torquewp>***:

```
<!ELEMENT torquewp  (wp.metadata?, wpidinfo, (warning*, caution*, note*),
intro, torqueval)>
<!ATTLIST torquewp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attributes for ***<torquewp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

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(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.15.1 Torque limit values ***<torqueval>***. The torque limits are generally displayed as tables. If multiple torque limits tables are used, the tables can be separated into multiple main paragraphs (***<para0>***) or multiple tables with different table titles. The groups can be separated by material type (with a table or description for each type) or by size ranges (with a table for common torque material sizes or dimensions). The XML markup example show the torque limits grouped by size range.

a. The components of ***<torqueval>*** are:
(1) Single procedure
(a) Procedural steps title ***<title>*** (required) (see 49.1.1.3).
(b) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(c) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(d) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(e) General information ***<geninfo>*** (optional) (see 49.1.4.12).
(f) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(g) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(h) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1-alt>*** (see 48.2.1) (required – one or more).
(2) Multiple procedure ***<proc>*** (required – two or more) (see 33.2).
(3) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

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b. DTD graphically depicted for ***<torqueval>***:

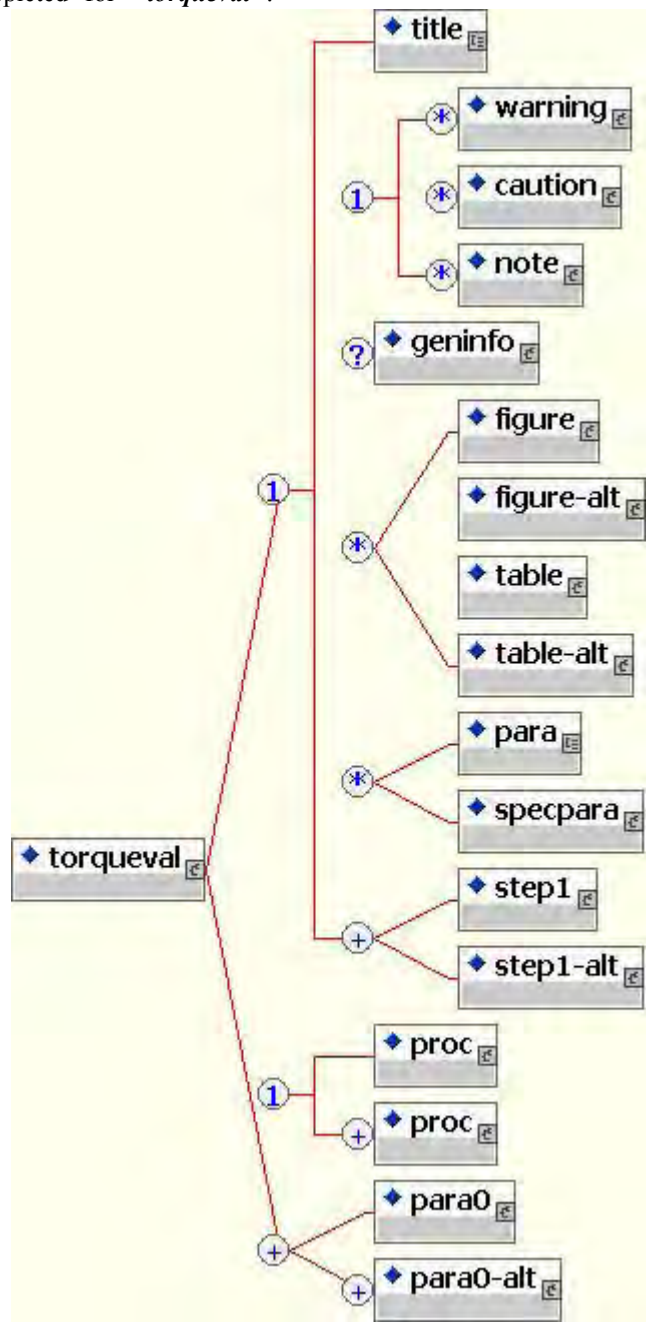

FIGURE 376. Torque limit values DTD hierarchy <torqueval>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<torqueval>***:

```
<!ELEMENT torqueval  ((title, (warning*, caution*,
   note*), geninfo?, (figure | figure-alt | table |
   table-alt)*, (para | specpara)*, (step1 | step1-alt)+) |
   (proc, proc+) | (para0 | para0-alt)+)>
<!ATTLIST torqueval
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<torqueval>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

37.15.2 XML document instance fragment and output for ***<torquewp>***. The XML instance and its stylesheet output for a ***<torquewp>*** is provided below.

a. Example of an XML document instance fragment for ***<torquewp>***:

```
<torquewp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="M1899-X-XXX-XX" wpseq="0321">
<wpidinfo>
<maintlvl level="sustain"/>
<title>TORQUE LIMITS
</title>
</wpidinfo>
<intro frame="no">
<para0>
<title>How To Use Torque Tables
</title>
<para>
<seqlist>
<item>Measure the diameter of the screw you are installing.
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>Measuring Screw.
</title>
<graphic boardno="measurescrew" unitmeasure="in">
</graphic>
</figure>
</item>
```


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Check the source to verify that this is the current version before use.

<item id="M1899-X-XXX-XX-intro2">Count the number of threads per inch or use a pitch grade.
</item>
<item>Under the heading SIZE, look down the left-hand column until you find the diameter of the screw you are installing. (There will usually be two lines beginning with the same size).
</item>
<item>In the second column under SIZE, find the numbers of threads per inch that matches the number of threads you counted in <xref itemid="M1899-X-XXX-XX-intro2"/>. (Not required for metric screws.)
<figure application="both" figtype="normal-page" id="torque-fig2" pane="no" tocentry="1">
<title>Capscrew Head Markings.
</title>
<graphic boardno="capscrew" unitmeasure="in">
</graphic>
</figure>
<note acknowledge="no">
<trim.para>Manufacture’s mark may vary. Standard are all SAE Grade 5 (3-Line). Metric screws are of three grades: 8.8, 10.9, and 12.9 Grades and manufacturer’s marks appear on the screw head.
</trim.para>
</note>
</item>
<item>To Find the grade screw you are installing, match the markings on the head to the correct picture of Capscrew Head Markings in <xref figid="torque-fig2"/> preceding the torque table.
</item>
</seqlist>
</para>
</para0>
</intro>
<torqueval>
<title>Torque Limits
</title>
<para>
<table>
<title>Torque Limits for 7/8 to 3 in
</title>
<tgroup cols="18">
<colspec align="center" colname="col1" colwidth="1.25*"/>
<colspec align="center" colname="col2" colwidth="1.19*"/>
<colspec align="center" colname="col3" colwidth="1.45*"/>
<colspec align="center" colname="col4" colwidth="1.65*"/>
<colspec align="right" colname="col5" colwidth="0.79*"/>
<colspec align="right" colname="col6" colwidth="0.91*"/>
<colspec align="right" colname="col7" colwidth="0.86*"/>
<colspec align="right" colname="col8" colwidth="0.86*"/>
<colspec align="right" colname="col9" colwidth="0.91*"/>
<colspec align="right" colname="col10" colwidth="0.89*"/>
<colspec align="right" colname="col11" colwidth="0.84*"/>
<colspec align="right" colname="col12" colwidth="0.94*"/>
<colspec align="right" colname="col13" colwidth="0.89*"/>
<colspec align="right" colname="col14" colwidth="0.96*"/>
<colspec align="right" colname="col15" colwidth="0.91*"/>
<colspec align="right" colname="col16" colwidth="0.91*"/>

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Check the source to verify that this is the current version before use.

```
<colspec align="right" colname="col17" colwidth="0.94*"/>
<colspec align="right" colname="col18" colwidth="0.91*"/>
<thead>
<row>
<entry morerows="1" valign="middle">Fastener
</entry>
<entry morerows="1" valign="middle">Type
</entry>
<entry morerows="1" valign="middle">Min Tensile Strngn
</entry>
<entry morerows="1" valign="middle">Material
</entry>
<entry align="center" nameend="col18" namest="col5" valign="middle">Body
Size or Outside Diameter of Fastener
</entry>
</row>
<row>
<entry align="center" valign="middle"> &frac78;
</entry>
<entry align="center" valign="middle">1
</entry>
<entry align="center" valign="middle">1&frac18;
</entry>
<entry align="center" valign="middle">1&frac14;
</entry>
<entry align="center" valign="middle">1&frac38;
</entry>
<entry align="center" valign="middle">1&frac12;
</entry>
<entry align="center" valign="middle">1&frac58;
</entry>
<entry align="center" valign="middle">1&frac34;
</entry>
<entry align="center" valign="middle">1&frac78;
</entry>
<entry align="center" valign="middle">2
</entry>
<entry align="center" valign="middle">2&frac14;
</entry>
<entry align="center" valign="middle">2&frac12;
</entry>
<entry align="center" valign="middle">2&frac34;
</entry>
<entry align="center" valign="middle">3
</entry>
</row>
</thead>
<tbody>
<row>
<entry valign="middle">
<graphic boardno="sae0" hscale="75">
</graphic>
</entry>
<entry valign="middle">SAE 0&ndash;1&ndash;2
</entry>
<entry valign="middle">74,000 PSI
```


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```
</entry>
<entry  valign="middle">Low  Carbon  Steel
</entry>
<entry  valign="middle">206
</entry>
<entry  valign="middle">310
</entry>
<entry  valign="middle">450
</entry>
<entry  valign="middle">675
</entry>
<entry  valign="middle">900
</entry>
<entry  valign="middle">1100
</entry>
<entry  valign="middle">1470
</entry>
<entry  valign="middle">1900
</entry>
<entry  valign="middle">2360
</entry>
<entry  valign="middle">2750
</entry>
<entry  valign="middle">3450
</entry>
<entry  valign="middle">4400
</entry>
<entry  valign="middle">7350
</entry>
<entry  valign="middle">9500
</entry>
</row>
<row>
<entry  valign="middle">
<graphic  boardno="sae3"  hscale="75">
</graphic>
</entry>
<entry  valign="middle">SAE  3
</entry>
<entry  valign="middle">100,000  PSI
</entry>
<entry  valign="middle">Medium  Carbon  Steel
</entry>
<entry  valign="middle">372
</entry>
<entry  valign="middle">551
</entry>
<entry  valign="middle">872
</entry>
<entry  valign="middle">1211
</entry>
<entry  valign="middle">1624
</entry>
<entry  valign="middle">1943
</entry>
<entry  valign="middle">2660
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</entry>
<entry  valign="middle">3463
</entry>
<entry  valign="middle">4695
</entry>
<entry  valign="middle">5427
</entry>
<entry  valign="middle">7226
</entry>
<entry  valign="middle">8049
</entry>
<entry  valign="middle">13450
</entry>
<entry  valign="middle">17448
</entry>
</row>  . . .
</tbody>
</tgroup>
</table>
</para>
</torqueval>
</torquewp>
```

b. Page-based TM stylesheet output example for ***<torquewp>***:

TM X–XXXX–XXX 0321

SUSTAINMENT MAINTENANCE
TORQUE LIMITS

HOW TO USE TORQUE TABLES

1. Measure the diameter of the screw you are installing.

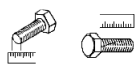

Figure 1. Measuring Screw.

2. Count the number of threads per inch or use a pitch grade.
3. Under the heading SIZE, look down the left-hand column until you find the diameter of the screw you are installing. (There will usually be two lines beginning with the same size).
4. In the second column under SIZE, find the numbers of threads per inch that matches the number of threads you counted in Item 2. (Not required for metric screws.)

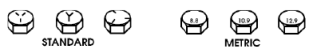

Figure 2. Capscrew Head Markings.

NOTE

Manufacture's mark may vary. Standard are all SAE Grade 5 (3-Line). Metric screws are of three grades: 8.8, 10.9, and 12.9 Grades and manufacturer's marks appear on the screw head.

5. To Find the grade screw you are installing, match the markings on the head to the correct picture of Capscrew Head Markings in Figure 2 preceding the torque table.

TORQUE LIMITS

Table 1. Torque Limits for 7/8 to 3 in

Fastener	Type	Min Tensile Strngn	Material	Body Size or Outside Diameter of Fastener														
				7/8	1	1 1/8	1 1/4	1 3/8	1 1/2	1 5/8	1 3/4	1 7/8	2	2 1/4	2 1/2	2 3/4	3
	SAE 0–1–2	74,000 PSI	Low Carbon Steel	206	310	450	675	900	1100	1470	1900	2360	2750	3450	4400	7350	9500
	SAE 3	100,000 PSI	Medium Carbon Steel	372	551	872	1211	1624	1943	2660	3463	4695	5427	7226	8049	13450	17448

0321-1

FIGURE 377. Example of a page-based TM stylesheet output for <torquewp>.

37.16 Wiring diagrams work package ***<wiringwp>***. The element includes wiring and cable provisions contained in the equipment/end item, including all systems or equipment which can be installed or removed late. Applicable diagrams are explained in relation to the equipment configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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a. The components of ***<wiringwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Wire diagram introduction ***<intro>*** (required) includes the work package scope and a statement(s) explaining that wiring diagrams and essential wiring information are provided for all electrical and electronic systems and circuits. (see 49.1.4.15).
   (5) Wire identification ***<wireid>*** (required) (see 37.16.1).
   (6) Wiring abbreviations ***<abbrev>*** (required) (see 37.16.2).
   (7) Wiring diagrams ***<wiringdiag>*** (required – one or more) (see 37.16.3).

b. DTD graphically depicted for ***<wiringwp>***:

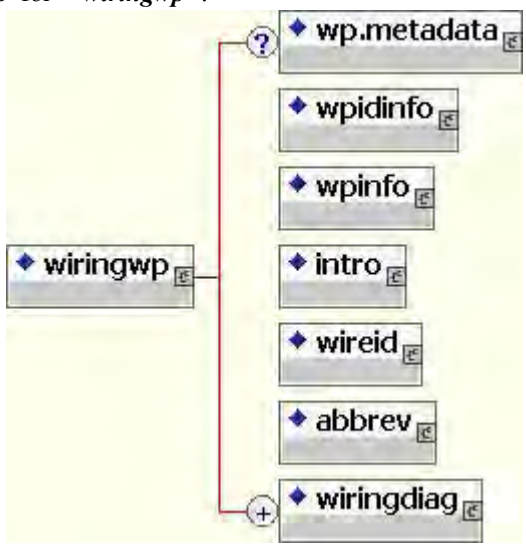

FIGURE 378. Wiring diagrams work package DTD hierarchy <wiringwp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<wiringwp>***:

```
<!ELEMENT wiringwp (wp.metadata?, wpidinfo, wpinfo, intro, wireid,
   abbrev, wiringdiag+)>
<!ATTLIST wiringwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes for ***<wiringwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.16.1 Wire identification ***<wireid>***. The element includes the identification of wires by number, a list of circuit designators, and associated wire identification diagrams.

a. The components of ***<wireid>*** are:
(1) Title ***<title>*** (required) (see 49.1.1.3).
(2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(4) Select one of the following information types:
(a) Narrative paragraphs
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
(b) Narrative paragraphs with descriptive or narrative titled text
1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
(c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<wireid>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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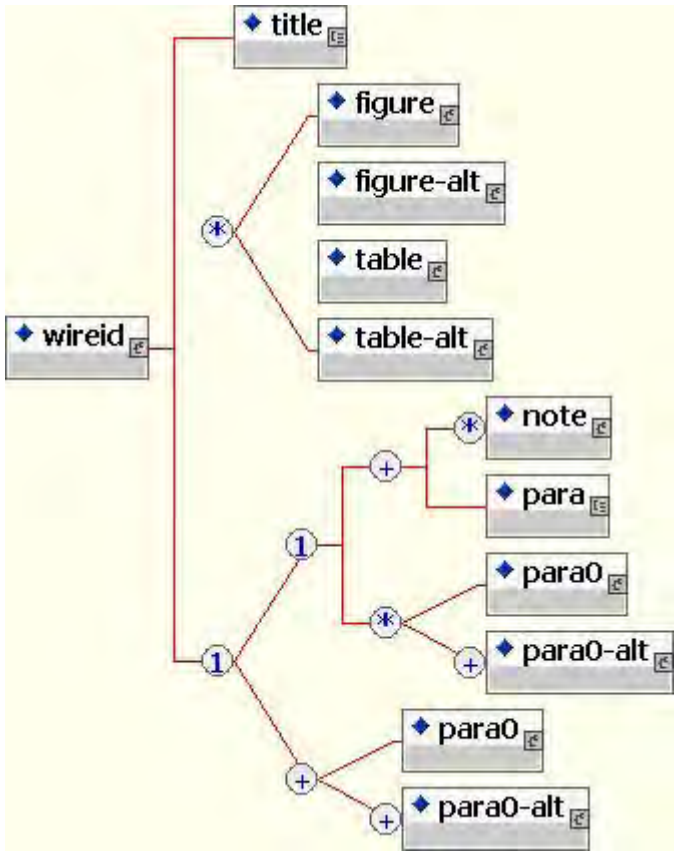

FIGURE 379. Wiring diagram abbreviations DTD hierarchy <wireid>.

c. DTD fragment for ***<wireid>***:

```
<!ELEMENT wireid (title, (figure | figure-alt | table | table-alt)*,
   (((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST wireid
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<wireid>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)

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(9) **security** – Security classification. (optional) (See 49.2.13.)

37.16.2 Abbreviations ***<abbrev>***. The element includes all abbreviations not contained ASME Y14.38. Additionally, text is needed stating that abbreviations are in accordance with ASME Y14.38, except when the abbreviation stands for a marking actually found in the equipment.

a. The components of ***<abbrev>*** are:
   (1) Title ***<title>*** (required) (see 49.1.1.3).
   (2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
   (3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
   (4) Select one of the following information types:
      (a) Narrative paragraphs
         1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
         2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
      (b) Narrative paragraphs with descriptive or narrative titled text
         1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
         2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
         3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
      (c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<abbrev>***:

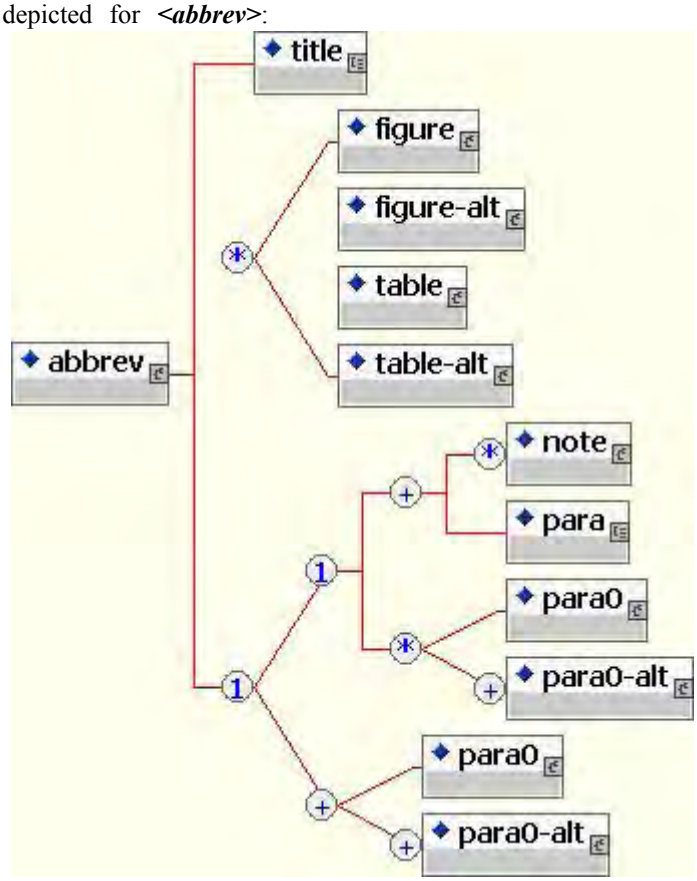

FIGURE 380. Wiring diagram abbreviations DTD hierarchy <abbrev>.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment for ***<abbrev>***:

```
<!ELEMENT abbrev (title, (figure | figure-alt | table | table-alt)*,
   (((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST abbrev
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<abbrev>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

37.16.3 Wiring diagrams ***<wiringdiag>***. The element contains a single electrical or electronic system and circuit with the required number of wiring diagrams.

a. The components are:

(1) Electrical or electronic system and circuit title ***<title>*** (required) (see 49.1.1.3).
(2) Introduction Paragraph ***<trim.para>*** (optional) (see 49.1.1.7).
(3) Wiring diagram ***<figure>*** (required – one or more) (see 44.1.1).

b. DTD graphically depicted for:

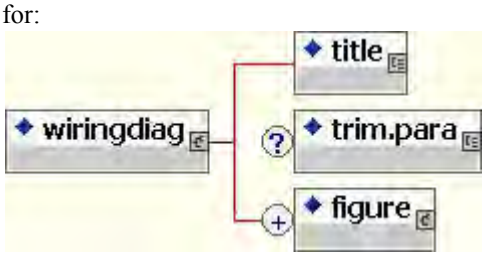

FIGURE 381. Wiring diagrams DTD hierarchy <wiringdiag>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<wiringdiag>***:

```
<!ELEMENT wiringdiag (title, trim.para?, figure+)>
<!ATTLIST wiringdiag
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

37.16.4 XML document instance fragment and output for ***<wiringwp>***. The XML instance and its stylesheet output for a ***<wiringwp>*** is provided below.

a. Example of an XML document instance fragment for ***<wiringwp>***:

<wiringwp airforce="no" army="no" deletewp="no" frame="yes" marines="no" navy="no" tocentry="2" wpno="M0245-X-XXXX-XXX" wpseq="0306">
<wpidinfo>
<maintlvl level="sustain"/>
<title>SHELTER CABLES WIRING DIAGRAMS
</title>
</wpidinfo>
<wpinfo>
<ref>
<ref-setup-item>
<extref docno="ASME-Y14.38"/>
</ref-setup-item>
</ref>
</wpinfo>
<intro frame="no">
<para0 esd="no" hcp="no">
<title>INTRODUCTION
</title>
<subpara1 esd="no" hcp="no">
<title>Scope
</title>
<para esd="no" hcp="no">This work package describes the wiring provisions contained in the equipment, including all systems or equipment which can be installed or removed later (e.g., mission-related systems/ equipment). Wiring diagrams and essential wiring information are provided for all electrical and electronic systems and circuits. All critical wire and cable data has been included.

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Check the source to verify that this is the current version before use.

```
</para>
</subpara1>
</para0>
</intro>
<wireid>
<title>WIRE  IDENTIFICATION
</title>
<para esd="no" hcp="no">All wires have been identified by number.  A tabular
list of circuit designators and wire identification diagrams are included.
<xref posttext=")" pretext="(Refer to " tableid="M0245-X-XXXX-XXX-tab1"/>
<table id="M0245-X-XXXX-XXX-tab1" tocentry="1">
<title>Cable Assembly to Wiring Diagram Cross Reference.
</title>
<tgroup cols="2">
<colspec align="center" colname="col1"/>
<colspec align="center" colname="col2"/>
<thead>
<row>
<entry>Cable  Part  No.
</entry>
<entry>Figure  No.
</entry>
</row>
</thead>
<tbody>
<row>
<entry>A31U116300-11
</entry>
<entry>
<xref figid="M0245-X-XXXX-XXX-fig1"/>
</entry>
</row>
<row>
<entry>A31U116300-13
</entry>
<entry>
<xref figid="M0245-X-XXXX-XXX-fig1"/>
</entry>
</row>
<row>
<entry>A31U116302-5
</entry>
<entry>
<xref figid="M0245-X-XXXX-XXX-fig2"/>
</entry>
</row>
<row>
<entry>A31U116304-3
</entry>
<entry>
<xref figid="M0245-X-XXXX-XXX-fig3"/>
</entry>
</row>
<row>
<entry>A31U116310-3
</entry>
```


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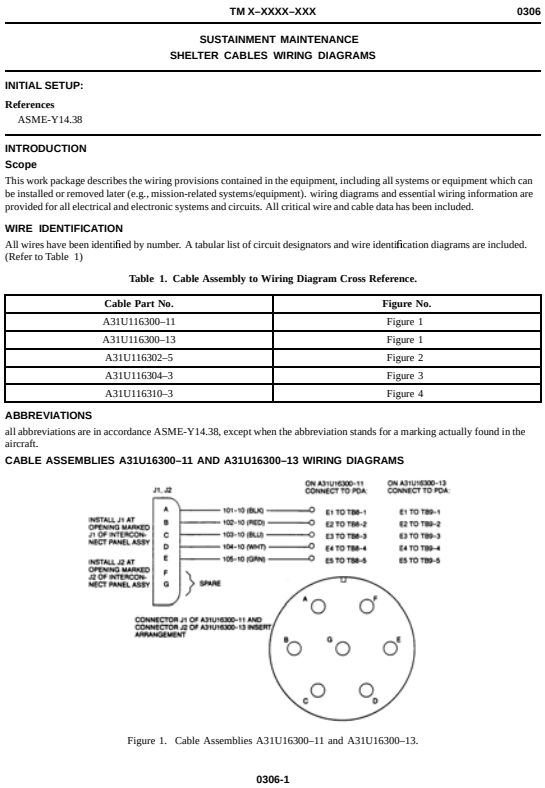
Check the source to verify that this is the current version before use.

```
<entry>
<xref figid="M0245-X-XXXX-XXX-fig4"/>
</entry>
</row>
</tbody>
</tgroup>
</table>
</para>
</wireid>
<abbrev>
<title>ABBREVIATIONS
</title>
<para esd="no" hcp="no">All abbreviations are in accordance
<extref docno="ASME-Y14.38"/>, except when the abbreviation stands
for a marking actually found in the aircraft.
</para>
</abbrev>
<wiringdiag>
<title>Cable Assemblies A31U16300-11 and A31U16300-13 Wiring Diagrams
</title>
<figure application="both" figtype="normal-page" pane="no" tocentry="1"
id="M0245-X-XXXX-XXX-fig1">
<title>Cable Assemblies A31U16300-11 and A31U16300-13.
</title>
<graphic boardno="wiringfig1" hscale="95">
</graphic>
</figure>
</wiringdiag>
<wiringdiag>
<title>Cable Assembly A31U16302-5 Wiring Diagrams
</title>
<figure application="both" figtype="normal-page" pane="no" tocentry="1"
id="M0245-X-XXXX-XXX-fig2">
<title>Cable Assembly A31U16302-5
</title>
<graphic boardno="wiringfig2">
</graphic>
</figure>
</wiringdiag>
<wiringdiag>
<title>Cable Assembly A31U16304-3 WIRING DIAGRAMS
</title>
<figure application="both" figtype="normal-page" pane="no" tocentry="1"
id="M0245-X-XXXX-XXX-fig3">
<title>Cable Assembly A31U16304-3
</title>
<graphic boardno="wiringfig3">
</graphic>
</figure>
</wiringdiag>
<wiringdiag>
<title>Cable Assembly A31U116310-3 Wiring Diagrams
</title>
<figure application="both" figtype="normal-page" pane="no" tocentry="1"
id="M0245-X-XXXX-XXX-fig4">
<title>Cable Assembly A31U116310-3
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</title>
<graphic boardno="wiringfig4">
</graphic>
</figure>
</wiringdiag>
</wiringwp>

b. Page-based TM stylesheet output example for ***<wiringwp>***:



TM X–XXXX–XXX 0306

SUSTAINMENT MAINTENANCE
SHELTER CABLES WIRING DIAGRAMS

INITIAL SETUP:

References
ASME-Y14.38

INTRODUCTION

Scope

This work package describes the wiring provisions contained in the equipment, including all systems or equipment which can be installed or removed later (e.g., mission-related systems/equipment). wiring diagrams and essential wiring information are provided for all electrical and electronic systems and circuits. All critical wire and cable data has been included.

WIRE IDENTIFICATION

All wires have been identified by number. A tabular list of circuit designators and wire identification diagrams are included. (Refer to Table 1)

Table 1. Cable Assembly to Wiring Diagram Cross Reference.

Cable Part No.	Figure No.
A31U116300–11	Figure 1
A31U116300–13	Figure 1
A31U116302–5	Figure 2
A31U116304–3	Figure 3
A31U116310–3	Figure 4

ABBREVIATIONS

all abbreviations are in accordance ASME-Y14.38, except when the abbreviation stands for a marking actually found in the aircraft.

CABLE ASSEMBLIES A31U16300–11 AND A31U16300–13 WIRING DIAGRAMS

Figure 1. Cable Assemblies A31U16300–11 and A31U16300–13.

0306-1

FIGURE 382. Example of a page-based TM stylesheet output for <wiringwp>.

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CABLE ASSEMBLY A31U16302–5 WIRING DIAGRAMS

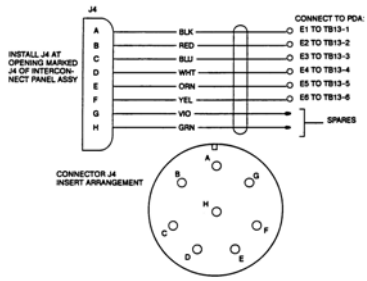

Figure 2. Cable Assembly A31U16302–5

CABLE ASSEMBLY A31U16304–3 WIRING DIAGRAMS

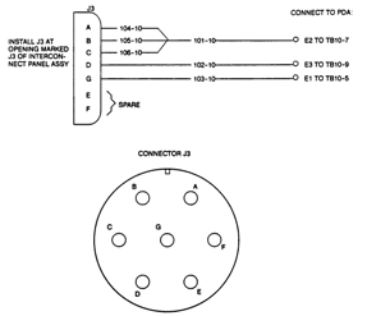

Figure 3. Cable Assembly A31U16304–3

0306-2

FIGURE 382. Example of a page-based TM stylesheet output for <wiringwp> - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

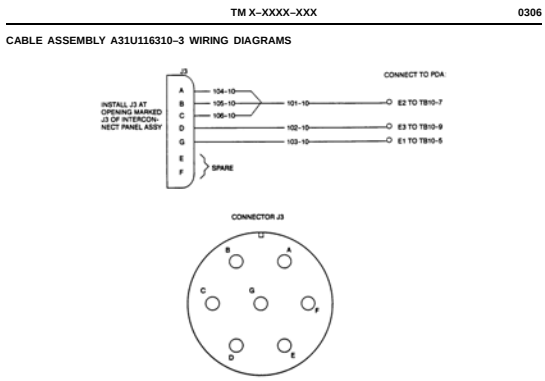

Figure 4. Cable Assembly A31U116310–3

END OF WORK PACKAGE

FIGURE 382. Example of a page-based TM stylesheet output for <wiringwp> - Continued.

37.17 Phased maintenance inspections (PMI) work package ***<pmiwp>***. The element contains the requirements for special inspections, overhaul and retirement schedule, and standards of serviceability applicable to the aircraft. The inspections described in the work package is performed at specified periods when required.

a. The components of ***<pmiwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) General information ***<geninfo>*** (required) (see 49.1.4.12). Standard statements for general information are described in 37.17.2.
   (8) At least one of the following is required, in no specific order:
      (a) Procedure ***<proc>*** (see 33.2).
      (b) Preventive maintenance inspection peculiar inspection table ***<pmi.pecul.tab>*** (see 37.17.1) and preceded with an illustration ***<figure>*** (see 44.1.1) showing the aircraft or accessory inspection sequence by area.

b. DTD graphically depicted for ***<pmiwp>***:

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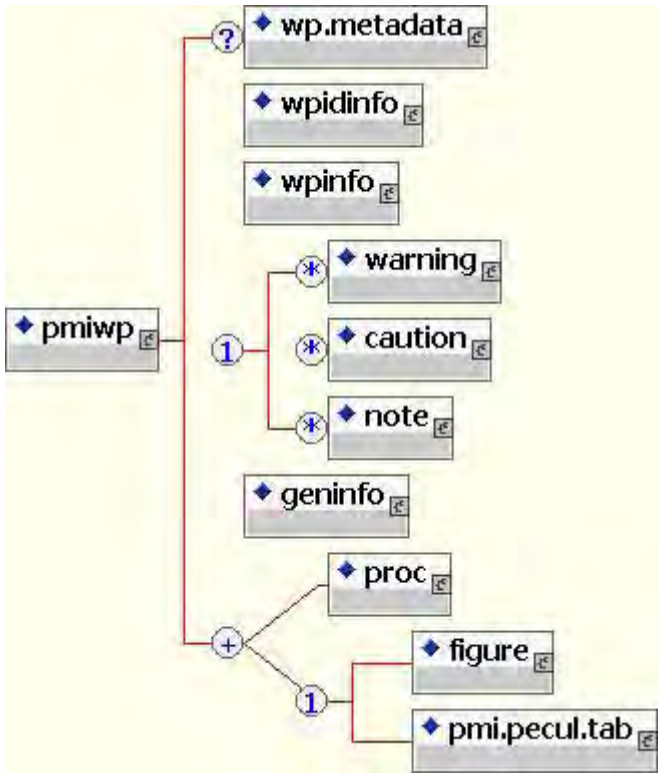

FIGURE 383. Phased maintenance inspections (PMI) work package DTD hierarchy <pmiwp>.

c. DTD fragment for ***<pmiwp>***:

```
<!ELEMENT pmiwp (wp.metadata?, wpidinfo, wpinfo, (warning*, caution*,
   note*), geninfo, (proc | pmi.pecul.tab)+)>
<!ATTLIST pmiwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
```


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Check the source to verify that this is the current version before use.

d. Common attributes for ***<pmiwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.17.1 PMI inspection standard information ***<pmi.pecul.tab>***. The PMI standard information contains components and other items which qualify under the criteria for special inspections, e.g., hard landings, sudden stoppage and overspeed. The inspection data required is aircraft serial or tail number, inspection date, area number, inspection number, inspection interval, component name being inspected, and inspection procedure.

a. The components of ***<pmi.pecul.tab>*** are:

(1) Standard information title ***<title>*** (required) (see 49.1.1.3).
(2) Standard information header ***<std.table.header>*** (optional) (see 42.3.1).
(3) PMI row ***<pmi.pecul-row>*** (required – one or more) and has the following components:
(a) Aircraft serial or tail number ***<serialno>*** (required) (see 47.2.1.2.1).
(b) Inspection date ***<date>*** (required).
(c) PMI inspection entry***<pmi.pecul-entry>*** (required – one or more) and the components are:
1 Area number ***<areano>*** (required) (see 37.17.1.1).
2 Inspection number ***<itemno>*** (required) (see 49.1.4.8).
3 Inspection interval ***<interval>*** (required) (see 37.6.1.2.a.(2)).

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4 Component name being inspected ***<compname>*** (required) (see 37.17.1.2).
5 One of the following is required:
a Inspection procedure ***<proc>*** (required) (see 33.2)
b Inspection steps ***<step1>***/***<step1–alt>*** (required – one or more) (see 33.3).

b. DTD graphically depicted for ***<pmi.pecul.tab>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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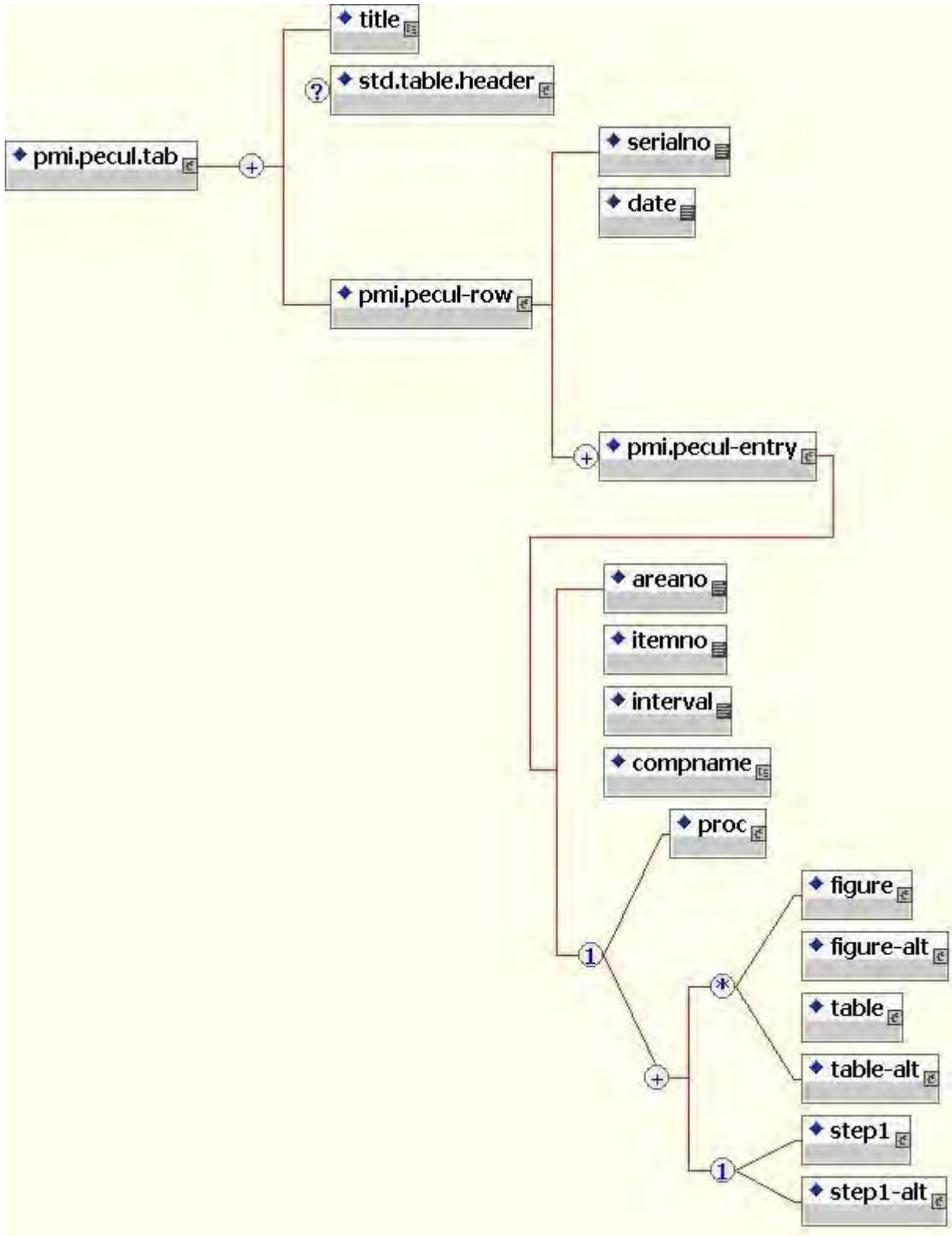

FIGURE 384. PMI inspection standard information DTD hierarchy <pmi.pecul.tab>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<pmi.pecul.tab>***:

```
<!ELEMENT pmi.pecul.tab (title, std.table.header?, pmi.pecul-row)+>
<!ATTLIST pmi.pecul.tab
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1"
>

<!ELEMENT pmi.pecul-row (serialno, date, pmi.pecul-entry+)>
<!ATTLIST pmi.pecul-row
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>

<!ELEMENT pmi.pecul-entry (areano, itemno, interval, compname, (proc |
   ((figure | figure-alt | table | table-alt)*, (step1 | step1-alt))+))>
<!ATTLIST pmi.pecul-entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>

<!ELEMENT compname (#PCDATA | emphasis | subscrpt | supscrpt)*>
```

d. Common attributes for ***<pmi.pecul.tab>***, ***<pmi.pecul-row>***, and ***<pmi.pecul-entry>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).

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(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

37.17.1.1 Area number ***<areano>***. The element ***<areano>*** is used to enter the aviation area number. In page-base, the element is equivalent to an "entry" element in a table and is inserted in the first colum of the PMI inspection standard information table

a. The components of ***<areano>*** are:
(1) Parsable characters or type text. – #PCDATA
b. DTD fragment for ***<pmi.pecul.tab>***:

```
<!ELEMENT areano (#PCDATA)>
```

c. No Attributes.

37.17.1.2 Component name ***<compname>***. The element ***<compname>*** is used to enter the name of component being inspected. In page-base, the element is equivalent to an "entry" element in the ***<pmi.pecul.tab>***.

a. The components of ***<compname>*** are:
(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).
b. DTD graphically depicted for :

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

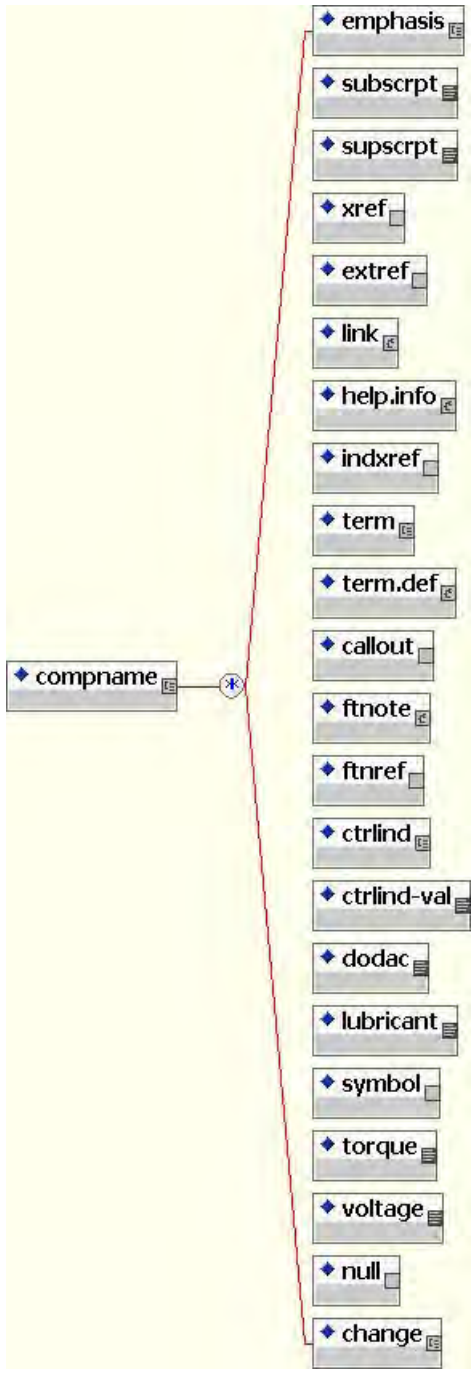

FIGURE 385. Component name <compname>.

c. DTD fragment for ***<compname>***:

```
<!ELEMENT compname (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link | help.info
```

d. No Attributes.

37.17.2 PMI work package general information standard statement general entity. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording.

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MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed Table XIX.

TABLE XIX. Boilerplate Entities for *<pmiwp>*.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
General information and introduction	***&pmiwp.geninfo;***	— —
Avionic equipment TM number		***<!ENTITY pmiwp.geninfo.aircraft "<extref docno='REPLACE WITH APPLICABLE AIRCRAFT INSPECTION CHECKLIST TM'/>">***
Standards of serviceability statement	***&pmiwp.stdserv;***	— —
Special inspection including definition and general information	***&pmiwp.def-geninfo;***	— —

37.17.3 XML document instance fragment and output for ***<pmiwp>***. The XML instance and its stylesheet output for a ***<pmiwp>*** is provided below.

a. Example of an XML document instance fragment for ***<pmiwp>***:

```
<pmiwp airforce="no" army="no" deletewp="no" frame="yes" marines="no" navy="no"
tocentry="2" wpno="m00524-55-1520-241" wpseq="0035">
<wpidinfo>
<maintlvl level="avum-avim"/>
<title>Inspection Requirements
</title>
</wpidinfo>
<wpinfo>
<title>N/A
</title>
<null insert="none"/>
</wpinfo>
<geninfo frame="no"> &pmiwp.geninfo;
<para esd="no" hcp="no">Special inspection frequencies that are
based on flight hours may be accomplished within a plus or
minus ton percent tolerance from the nominal time when such
inspections would ordinarily be duo.
</para>
<para esd="no" hcp="no">Special inspections based on calendar times will
have a tolerance of plus or minus ten percent not to exceed thirty days.
</para>
<para0 esd="no" hcp="no">
<title>Standards of serviceability
</title> &pmiwp.stdserv;
</para0>
<para0 esd="no" hcp="no">
<title>Special Inspection
</title> &pmiwp.def-geninfo;
</para0>
```


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Check the source to verify that this is the current version before use.

</geninfo>
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>Inspection Area Diagram
</title>
<graphic boardno="pmiwp-fig1" unitmeasure="in">
</graphic>
<legend>
<title>Area No.
</title>
<legend.item>
<term>1 Nose Area
</term>
<def>
<para>All surfaces and components in nose compartment and on exterior ahead of crew doors
</para>
</def>
</legend.item>
<legend.item>
<term>2 Cabin and Landing Gear Area
</term>
<def>
<para>All surfaces, components, and equipment inside cabin, and on cabin exterior between forward side of crew doors and aft side of passenger door and cabin overhead. Includes complete landing gear, fuel cell sumps and filler.
</para>
</def>
</legend.item>
<legend.item>
<term>3 Transmission and Pylon Area
</term>
<def>
<para>All surfaces, components, and equipment of the main rotor pylon group, from top of mast to cabin roof. Includes main rotor, mast and rotating controls, transmission with accessorial and mounts, and main drive shaft.
</para>
</def>
</legend.item>
<legend.item>
<term>4 Engine Area
</term>
<def>
<para>All surfaces, components, and equipment associated with engine installation, located above engine work deck and within engine cowling and tailpipe fairing.
</para>
</def>
</legend.item>
<legend.item>
<term>5 Avionics and Aft Fuselage Area
</term>
<def>
<para>All surfaces, components, and equipment in fuselage below engine deck level, between cabin area and tailboom attachment bulkhead.

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Check the source to verify that this is the current version before use.

```
</para>
</def>
</legend.item>
<legend.item>
<term>6  Tailbooms
</term>
<def>
<para>All surfaces, components, and equipment located in the tailboom and
vertical fin structure.  Includes tail rotor, elevator, and control
linkages.  Also all supports, bearing, and shafting mounted on tailboom.
</para>
</def>
</legend.item>
</legend>
</figure>
<pmi.pecul.tab  tocentry="1">
<title>Aircraft  Inspection  Checklist
</title>
<std.table.header  cols="6"  rowsep="1">
<std.table.row>
<std.table.entry  colnum="1"  cols.span="2"  rowsep="1">SERIAL  NO.:
</std.table.entry>
<std.table.entry  colnum="4"  cols.span="2"  rowsep="1">DATE  OF  INSPECTION:
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry  colnum="1"  cols.span="0">AREA  NO.
</std.table.entry>
<std.table.entry  colnum="2"  cols.span="0">INSP.  NO.
</std.table.entry>
<std.table.entry  colnum="3"  cols.span="0">REQUIREMENT  EVERY
</std.table.entry>
<std.table.entry  colnum="4"  cols.span="1">ITEM
</std.table.entry>
<std.table.entry  colnum="6"  cols.span="0">STATUS
</std.table.entry>
</std.table.row>
</std.table.header>
<pmi.pecul-row>
<serialno>
</serialno>
<date  julian="">
</date>
<pmi.pecul-entry>
<areano>All
</areano>
<itemno>1
</itemno>
<interval>AFTER  A  HARD  LANDING
</interval>
<compname>
</compname>
<proc  esd="no"  frame="yes"  hcp="no"  tocentry="0">
<title>
</title>
<geninfo  frame="no">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<para esd="no" hcp="no">Definition: Hard landing is defined as any accident or incident in which ground impact of the helicopter causes severe pitching of main rotor, allowing hard contact of hub with mast, or results in noticeable yielding or cracking of fuselage pylon support structure or landing gear. This definition is confined only to those accidents not involving sudden stoppage of main rotor or tail rotor.
</para>
<para esd="no" hcp="no">Inspections: When a probable hard landing incident has occurred, proceed as follows:
</para>
</geninfo>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check all cowling and doors for proper fit and alignment. Misaligned cowling may indicate a distorted fuselage resulting in major stresses and damage. Remove centermost cover assembly and inspect cap angles for buckling and bending.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Remove all cowling as necessary to perform a complete visual inspection.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect landing gear skid tubes and crosstubes for damage and deflection. Inspect crosstube attachment points for damage or distortion.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect structure with a 10–power magnifying glass around the transmission mounting points. Particular attention should be given to the isolation mount and pylon support mounts. Inspect the transmission support where the spike on the drag pin assembly fits into the stop. This area is directly under the transmission on the top of the cabin.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect upper longerons and adjacent skin near the aft engine firewall (station 167) for loose rivets, cracks, or distortion.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect tail skid tube and mounting for damage. Inspect tailboom internally and externally for cracks, distortion, and loose rivets. Inspect the tailboom attachment points for elongated bolt holes and damaged structure. Inspect external skin of tailboom for cracks in area of attachment of the horizontal stabilizer.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Completely inspect the flight control system from pilot controls to rotor head for bent or damaged tubes, bellcranks, and supports, and for damaged bearings. Particular attention should be given to the pitch links and collective sleeve assembly.

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```
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check for leaks in the hydraulic system and
interference or binding, and for satisfactory operation.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect the transmission and mast
assembly for the following:
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Mast for indentation caused by hard
contact with static stops.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Mast outside diameter (O.D.) for
out-of- round condition as follows:
</para>
<step3 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<specpara esd="no" hcp="no">
<note acknowledge="no">
<trim.para esd="no" hcp="no">Cadium plating on the outside diameter
of mast is 0.0003 to 0.0004 inch.
</trim.para>
</note>
<para esd="no" hcp="no">Using a suitable micrometer, measure the mast
O.D., at a minimum of six places for out-of-round condition; two
opposing measurements under the collar set and above the swashplate,
two opposing measurements midway between the collar set and the
static stop contact area, two opposing measurements approximately
2.000 inches below the static stop contact area on the mast.
</para>
</specpara>
</step3>
<step3 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">If any set of opposing measurements vary
in excess of 0.002 inch, the mast is suspected of bending
and a runout check must be performed.
</para>
</step3>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Structure supporting the transmission
&mdash; mount points for cracks or deformation.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Transmission - drag pin - transmission mount
support on cabin roof for deformation and/or loss of attachment bolt
torque.  Supporting structure for damage and loose or sheared rivets.
</para>
</step2>
</step1>
</proc>
```


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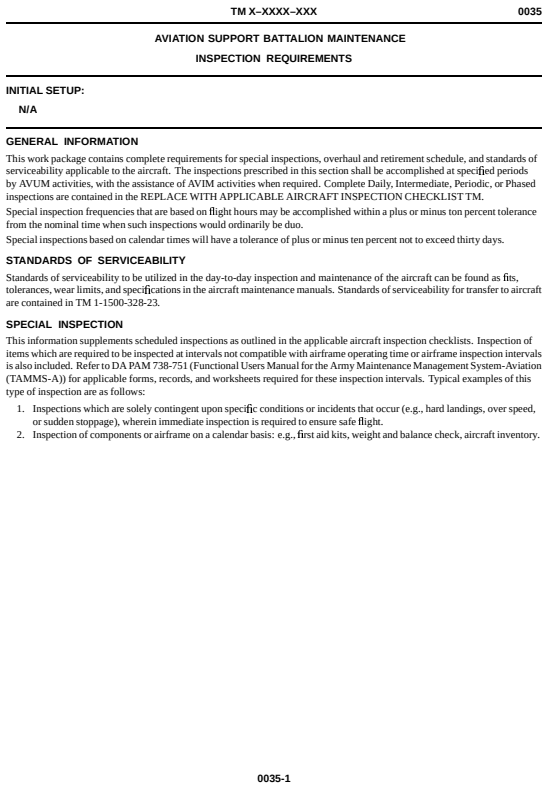
Check the source to verify that this is the current version before use.

```
</pmi.pecul-entry>
<pmi.pecul-entry>
<areano>1,  2,  3
</areano>
<itemno>2
</itemno>
<interval>AFTER  WIRE  STRIKE
</interval>
<compname>
</compname>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Lower WSPS assembly.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect for obvious damage to WSPS.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect attachment area for damage.
</para>
</step2>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Windshield deflector and upper WSPS assembly.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect for obvious damage to WSPS.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect attachment area for damage.
</para>
</step2>
</step1>
</pmi.pecul-entry>
<pmi.pecul-entry>
<areano>1
</areano>
<itemno>3
</itemno>
<interval>AFTER LOWER WSPS GROUND CONTACT
</interval>
<compname>
</compname>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect for obvious damage.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Inspect attachment area for damage.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Remove panel and inspect structure and
directional control tubes and bellcranks for damage.
</para>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</step1>
</pmi.pecul-entry>
</pmi.pecul-row>
</pmi.pecul.tab>
</pmiwp>

b. Page-based TM stylesheet output example for ***<pmiwp>***:



TM X–XXXX–XXX 0035

AVIATION SUPPORT BATTALION MAINTENANCE
INSPECTION REQUIREMENTS

INITIAL SETUP:
N/A

GENERAL INFORMATION

This work package contains complete requirements for special inspections, overhaul and retirement schedule, and standards of serviceability applicable to the aircraft. The inspections prescribed in this section shall be accomplished at specified periods by AVUM activities, with the assistance of AVIM activities when required. Complete Daily, Intermediate, Periodic, or Phased inspections are contained in the REPLACE WITH APPLICABLE AIRCRAFT INSPECTION CHECKLIST TM.

Special inspection frequencies that are based on flight hours may be accomplished within a plus or minus ton percent tolerance from the nominal time when such inspections would ordinarily be duo.

Special inspections based on calendar times will have a tolerance of plus or minus ten percent not to exceed thirty days.

STANDARDS OF SERVICEABILITY

Standards of serviceability to be utilized in the day-to-day inspection and maintenance of the aircraft can be found as fits, tolerances, wear limits, and specifications in the aircraft maintenance manuals. Standards of serviceability for transfer to aircraft are contained in TM 1-1500-328-23.

SPECIAL INSPECTION

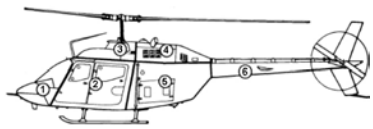
This information supplements scheduled inspections as outlined in the applicable aircraft inspection checklists. Inspection of items which are required to be inspected at intervals not compatible with airframe operating time or airframe inspection intervals is also included. Refer to DA PAM 738-751 (Functional Users Manual for the Army Maintenance Management System-Aviation (TAMMS-A)) for applicable forms, records, and worksheets required for these inspection intervals. Typical examples of this type of inspection are as follows:

1. Inspections which are solely contingent upon specific conditions or incidents that occur (e.g., hard landings, over speed, or sudden stoppage), wherein immediate inspection is required to ensure safe flight.
2. Inspection of components or airframe on a calendar basis: e.g., first aid kits, weight and balance check, aircraft inventory.

0035-1

FIGURE 386. Example of a page-based TM stylesheet output for <pmiwp>.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.



AREA NO.

1 Nose Area	All surfaces and components in nose compartment and on exterior ahead of crew doors
2 Cabin and Landing Gear Area	All surfaces, components, and equipment inside cabin, and on cabin exterior between forward side of crew doors and aft side of passenger door and cabin overhead. Includes complete landing gear, fuel cell sumps and filler.
3 Transmission and Pylon Area	All surfaces, components, and equipment of the main rotor pylon group, from top of mast to cabin roof. Includes main rotor, mast and rotating controls, transmission with accessorial and mounts, and main drive shaft.
4 Engine Area	All surfaces, components, and equipment associated with engine installation, located above engine work deck and within engine cowling and tailpipe fairing.
5 Avionics and Aft Fuselage Area	All surfaces, components, and equipment in fuselage below engine deck level, between cabin area and tailboom attachment bulkhead.
6 Tailbooms	All surfaces, components, and equipment located in the tailboom and vertical fin structure. Includes tail rotor, elevator, and control linkages. Also all supports, bearing, and shafting mounted on tailboom.

Figure 1. Inspection Area Diagram

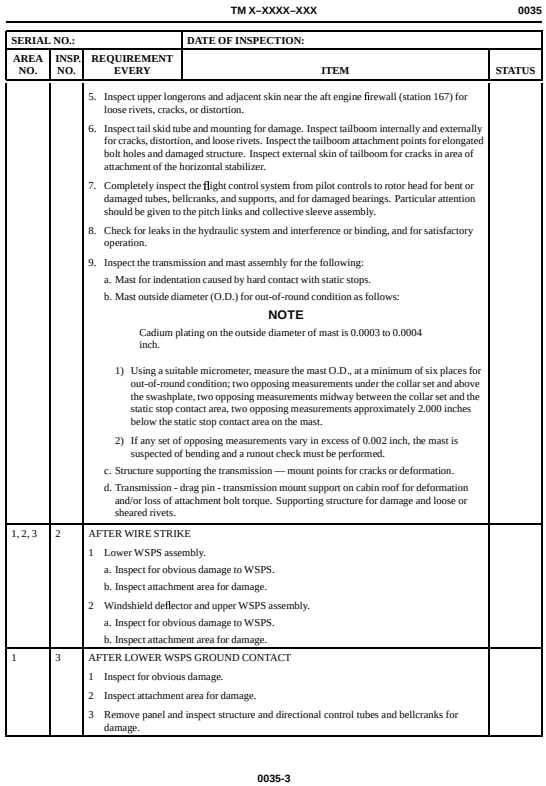
AIRCRAFT INSPECTION CHECKLIST

SERIAL NO.:			DATE OF INSPECTION:	
AREA NO.	**INSP. NO.**	**REQUIREMENT EVERY**	**ITEM**	**STATUS**
All	1	AFTER A HARD LANDING Definition: Hard landing is defined as any accident or incident in which ground impact of the helicopter causes severe pitching of main rotor, allowing hard contact of hub with mast, or results in noticeable yielding or cracking of fuselage pylon support structure or landing gear. This definition is confined only to those accidents not involving sudden stoppage of main rotor or tail rotor. Inspections: When a probable hard landing incident has occurred, proceed as follows: 1. Check all cowling and doors for proper fit and alignment. Misaligned cowling may indicate a distorted fuselage resulting in major stresses and damage. Remove centermost cover assembly and inspect cap angles for buckling and bending. 2. Remove all cowling as necessary to perform a complete visual inspection. 3. Inspect landing gear skid tubes and crosstubes for damage and deflection. Inspect crosstube attachment points for damage or distortion. 4. Inspect structure with a 10–power magnifying glass around the transmission mounting points. Particular attention should be given to the isolation mount and pylon support mounts. Inspect the transmission support where the spike on the drag pin assembly fits into the stop. This area is directly under the transmission on the top of the cabin.	

0035-2

FIGURE 386. Example of a page-based TM stylesheet output for <pmiwp> - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.



TM X–XXXX–XXX 0035

SERIAL NO.:			DATE OF INSPECTION:	
AREA NO.	INSP. NO.	REQUIREMENT EVERY	ITEM	STATUS
		5. Inspect upper longerons and adjacent skin near the aft engine firewall (station 167) for loose rivets, cracks, or distortion. 6. Inspect tail skid tube and mounting for damage. Inspect tailboom internally and externally for cracks, distortion, and loose rivets. Inspect the tailboom attachment points for elongated bolt holes and damaged structure. Inspect external skin of tailboom for cracks in area of attachment of the horizontal stabilizer. 7. Completely inspect the flight control system from pilot controls to rotor head for bent or damaged tubes, bellcranks, and supports, and for damaged bearings. Particular attention should be given to the pitch links and collective sleeve assembly. 8. Check for leaks in the hydraulic system and interference or binding, and for satisfactory operation. 9. Inspect the transmission and mast assembly for the following: a. Mast for indentation caused by hard contact with static stops. b. Mast outside diameter (O.D.) for out-of-round condition as follows: NOTE Cadium plating on the outside diameter of mast is 0.0003 to 0.0004 inch. 1) Using a suitable micrometer, measure the mast O.D., at a minimum of six places for out-of-round condition; two opposing measurements under the collar set and above the swashplate, two opposing measurements midway between the collar set and the static stop contact area, two opposing measurements approximately 2.000 inches below the static stop contact area on the mast. 2) If any set of opposing measurements vary in excess of 0.002 inch, the mast is suspected of bending and a runout check must be performed. c. Structure supporting the transmission — mount points for cracks or deformation. d. Transmission - drag pin - transmission mount support on cabin roof for deformation and/or loss of attachment bolt torque. Supporting structure for damage and loose or sheared rivets.	
1, 2, 3	2	AFTER WIRE STRIKE 1 Lower WSPS assembly. a. Inspect for obvious damage to WSPS. b. Inspect attachment area for damage. 2 Windshield deflector and upper WSPS assembly. a. Inspect for obvious damage to WSPS. b. Inspect attachment area for damage.		
1	3	AFTER LOWER WSPS GROUND CONTACT 1 Inspect for obvious damage. 2 Inspect attachment area for damage. 3 Remove panel and inspect structure and directional control tubes and bellcranks for damage.		

0035-3

FIGURE 386. Example of a page-based TM stylesheet output for <pmiwp> - Continued.

37.18 Aircraft inventory master guide work package ***<inventorywp>***. The aircraft inventory master guide work package contains information on standard inventory procedures to allow determination of inventoriable items of installed and loose equipment that is authorized and required by the aircraft in performance of its mission.

a. The components of ***<inventorywp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) Introduction ***<intro>*** (required) contains a short explanation of the work package scope and purpose that includes information pertaining to necessary steps to ensure the list is accurate, exact, and complete and a reference to DA PAM 738–751 for applicable forms and records (see 49.1.4.15).
   (8) Security ***<security>*** (required) (see 37.18.1).
   (9) Inventoriable items ***<inventoriable>*** (required) (see 37.18.2).
   (10) Periods of inventory ***<prdinv>*** (required) (see 37.18.3).

b. DTD graphically depicted for ***<inventorywp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

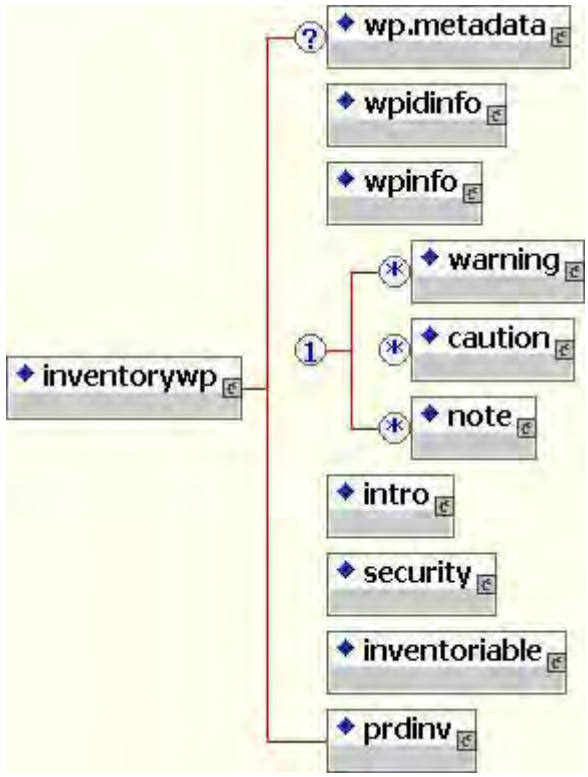


FIGURE 387. Aircraft inventory master guide work package DTD hierarchy <inventorywp> **(Aircraft Only).**

c. DTD fragment for ***<inventorywp>***:

```
<!ELEMENT inventorywp (wp.metadata?, wpidinfo, wpinfo,(warning*,
  caution*, note*), intro, security, inventoriable, prdinv)>
<!ATTLIST inventorywp
    airforce          (yes | no )               "no"
    army              (yes | no )               "no"
    assocfig          IDREFS                     #IMPLIED
    changeref         IDREFS                     #IMPLIED
    comment           CDATA                      #IMPLIED
    crewmember        CDATA                      #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                      #IMPLIED
    deletewp          (yes | no )               "no"
    fgc               CDATA                      #IMPLIED
    frame             (yes | no )               "yes"
    idref             IDREFS                     #IMPLIED
    inschlvl          CDATA                      #IMPLIED
    insertwp          CDATA                      #IMPLIED
    lsa-id            CDATA                      #IMPLIED
    marines           (yes | no )               "no"
    navy              (yes | no )               "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                      #IMPLIED
    tocentry          (2 | 3 | 4 | 5)           "2"
    wpno              ID                         #REQUIRED
    wpseq             CDATA                      #IMPLIED
>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

d. Common attributes for ***<inventorywp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.18.1 Security ***<security>***. The element describes necessary procedures if any aircraft inventory records content is classified, otherwise state no security procedures are needed for this aircraft inventory record.

a. The components of ***<seucrity>*** are:

(1) Title ***<title>*** (required) (see 49.1.1.3).
(2) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(3) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
(4) Select one of the following information types:
  (a) Narrative paragraphs
    1 Note ***<note>*** (optional – zero or more) (see 41.1.3).
    2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
  (b) Narrative paragraphs with descriptive or narrative titled text
    1 Note ***<note>*** (optional – zero or more) (see 41.1.3).

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2 Narrative paragraph ***<para>*** (required – one or more) (see 49.1.1.5).
3 Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (optional – zero or more).
(c) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<seucrity>***:

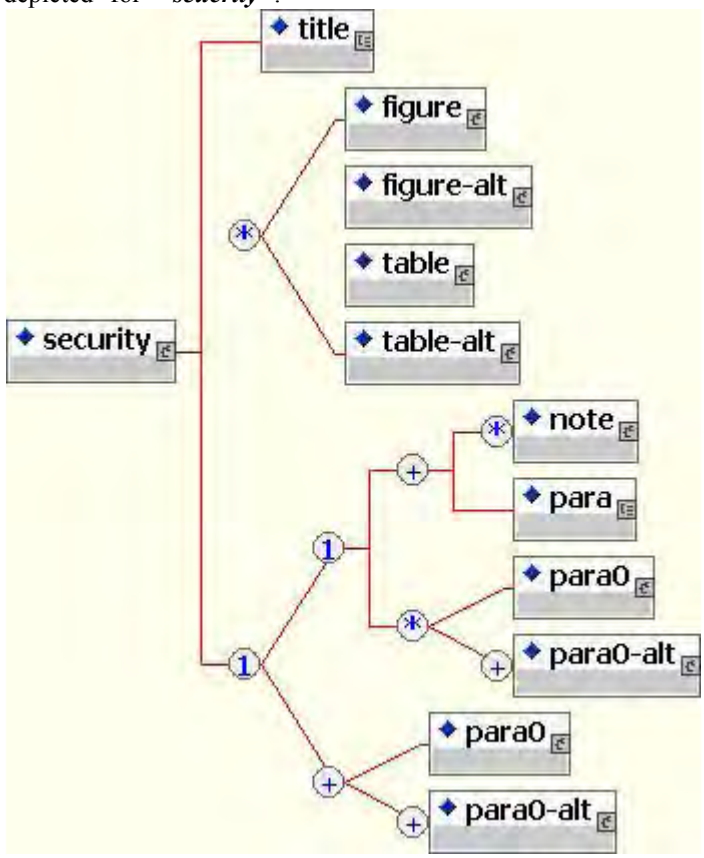

FIGURE 388. Aircraft inventory master guide security requirements DTD hierarchy <security>.

c. DTD fragment for ***<security>***:

```
<!ELEMENT security (title, (figure | figure-alt | table | table-alt)*,
 (((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+))>
<!ATTLIST security
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<security>*** are:
(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).

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(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

37.18.2 Inventoriable items ***<inventoriable>***. The element lists the inventoriable items selection without regard to the agency (governmental or contractual) furnishing the items. Inventoriable items information is also used as source data for DA Form 2408–17. The inventoriable items are generally presented as a table and if applicable an illustration showing a zone or items to be inventoried. The components for ***<inventoriable>*** are the same as ***<security>*** (see 37.18.1).

37.18.3 Periods of inventory ***<prdinv>***. The element contains periods of inventory which are normally performed upon receipt, prior to transfer, upon placing in storage and removal from storage, or every twelve months. MIL–STD–40051–1/–2 provides a state statement for this information and is provided in the boilerplate entity ***&inventorywp.prdinv;*** (see 37.18.4). The components for ***<prdinv>*** are the same as ***<security>*** (see 37.18.1).

37.18.4 Aircraft inventory master guide work package standard statement entity. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed (see Table XX).

TABLE XX. Periods of Inventory Boilerplate Entities.

Description	**Boilerplate Entity**	**Selectable Entity to Set Editable Entity to Set**
Periods of inventory statement	***&inventorywp.prdinv;***	Not applicable

37.18.5 XML document instance fragment and output for ***<inventorywp>***. The XML instance and its stylesheet output for a ***<inventorywp>*** is provided below.

a. Example of an XML document instance fragment for ***<inventorywp>***:

```
<inventorywp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="M0347–X-XXXX-XXX" wpseq="0347">
<wpidinfo>
<maintlvl level="asb"/>
<title>AIRCRAFT INVENTORY MASTER GUIDE
</title>
</wpidinfo>
<wpinfo>
<title>N/A
</title>
<null insert="none"/>
</wpinfo>
<intro frame="no">
```


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<para0 esd="no" hcp="no">
<title>INTRODUCTION
</title>
<subpara1 esd="no" hcp="no">
<title>Scope
</title>
<para esd="no" hcp="no">This work package includes a listing of all inventoriable items of installed and loose equipment authorized and required by the aircraft in performance of its mission. Refer to DA PAM 738-751 for applicable forms and records to use in performing the inventory.
</para>
</subpara1>
</para0>
</intro>
<security>
<title>SECURITY
</title>
<para esd="no" hcp="no">Aircraft inventory records should be unclassified, but any classification of the contents, if necessary, should be in accordance with the existing security regulations.
</para>
</security>
<inventoriable>
<title>INVENTORIABLE ITEMS
</title>
<para esd="no" hcp="no">Modification kits which are reissued or distributed to using organizations for installation, and which are not immediately placed in use, shall be recorded on the aircraft’s <extref docno="DA Form-2408-17" posttext=", Aircraft Inventory Record"/>, and identified as loose equipment until modification is complete. Refer to <xref tableid="M0347–X-XXXX-XXX-tab1"/> for a list of inventoriable items for the aircraft.
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>Aircraft Inventory Section
</title>
<graphic boardno="figure23" unitmeasure="in">
</graphic>
</figure>
<table tocentry="1" id="M0347–X-XXXX-XXX-tab1">
<title>Inventoriable Items.
</title>
<tgroup align="left" char=" " charoff="50" cols="4">
<colspec colname="col1" colwidth="4.00*"/>
<colspec colname="col2" colwidth="3.00*"/>
<colspec colname="col3" colwidth="3.00*"/>
<colspec colname="col4" colwidth="1.00*"/>
<thead valign="bottom">
<row>
<entry>NOMENCLATURE
</entry>
<entry>PART NUMBER
</entry>
<entry>NSN
</entry>
<entry>QTY RQD

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```
</entry>
</row>
</thead>
<tbody  valign="top">
<row>
<entry  nameend="col4"  namest="col1">SECTION A - NOSE ELECTRONICS
</entry>
</row>
<row>
<entry>TADS  Turret
</entry>
<entry>  130760000
</entry>
<entry>1270-01-188-4138
</entry>
<entry  align="center">1
</entry>
</row>
<row>
<entry>PNVS  Turret  and  Faring
</entry>
<entry>130900000
</entry>
<entry>1090-01-169-9415
</entry>
<entry  align="center">1
</entry>
</row>
<row>
<entry>Multiplex  Terminal  -  Type  3
</entry>
<entry>4032298-956
</entry>
<entry>7025-01-210-7768
</entry>
<entry  align="center">1
</entry>
</row>
<row>
<entry>Signal  Data  Converter  -  Eng.
</entry>
<entry>7-311C20019
</entry>
<entry>6629-01-160-3518
</entry>
<entry  align="center">1
</entry>
</row>
<row>
<entry>Radar  Jammer  Transmit  Antenna
</entry>
<entry>SM-A-919685-2
</entry>
<entry>
</entry>
<entry  align="center">1
```


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```
</entry>
</row>
<row>
<entry>Radar Warning Antenna
</entry>
<entry>AS2891/APR-39V
</entry>
<entry>1020-00-024-7608
</entry>
<entry  align="center">2
</entry>
</row>
<row>
<entry>Power Transformer
</entry>
<entry>7-219510089
</entry>
<entry>5950-01-186-8032
</entry>
<entry  align="center">1
</entry>
</row>
<row>
<entry nameend="col4" namest="col1">SECTION B - RIGHT FORWARD AVIONICS BAY
</entry>
</row>
<row>
<entry>Gun Control Box
</entry>
<entry>7-317222500-603
</entry>
<entry>1005-01-211-4165
</entry>
<entry  align="center">1
</entry>
</row>
<row>
<entry>Multiplex Terminal - Type 1
</entry>
<entry>4032297-955
</entry>
<entry>7025-01-211-0130
</entry>
<entry  align="center">1
</entry>
</row>
<row>
<entry>Turret Control Box
</entry>
<entry>7-317222004
</entry>
<entry>5930-01-239-2391
</entry>
<entry  align="center">1
</entry>
</row>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<row>
<entry>PNVS Electronics Box
</entry>
<entry>13080274
</entry>
<entry>4931-01-169-9369
</entry>
<entry align="center">1
</entry>
</row>
<row>
<entry>Display Electronics - IHADSS
</entry>
<entry>7-319430041
</entry>
<entry>1270-01-183-0518
</entry>
<entry align="center">1
</entry>
</row>
<row>
<entry>Sight Electronic Unit - IHADSS
</entry>
<entry>7-319430031
</entry>
<entry>1270-01-183-0519
</entry>
<entry align="center">1
</entry>
</row>
<row>
<entry>Fire Control Computer
</entry>
<entry>7-31900001
</entry>
<entry>1430-01-211-0023
</entry>
<entry align="center">1
</entry>
</row>
<row>
<entry>Remote Electronic Unit
</entry>
<entry>7-317141001
</entry>
<entry>1270-01-187-5778
</entry>
<entry align="center">1
</entry>
</row>
<row>
<entry>Portable Fire Bottle
</entry>
<entry>7-211180002-501
</entry>
<entry>4210-00-555-8837
```


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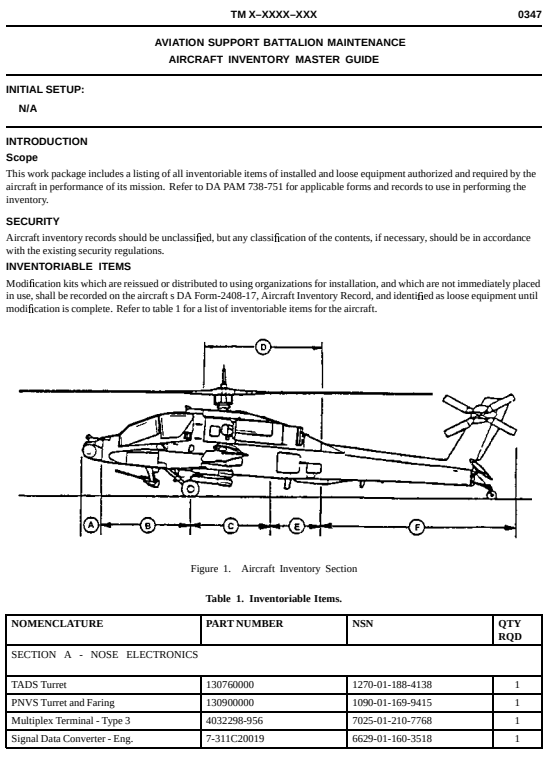
```
</entry>
<entry   align="center">1
</entry>
</row>
<row>
<entry>Electromagnet  Relay
</entry>
<entry>7-211B12033
</entry>
<entry>5945-01-160-5639
</entry>
<entry   align="center">1
</entry>
</row>
</tbody>
</tgroup>
</table>
</para>
</inventoriable>  &inventorywp.prdinv;
</inventorywp>
```

b. Page-based TM stylesheet output example for ***<inventorywp>***:



TM X–XXXX–XXX 0347

AVIATION SUPPORT BATTALION MAINTENANCE
AIRCRAFT INVENTORY MASTER GUIDE

INITIAL SETUP:

N/A

INTRODUCTION

Scope

This work package includes a listing of all inventoriable items of installed and loose equipment authorized and required by the aircraft in performance of its mission. Refer to DA PAM 738-751 for applicable forms and records to use in performing the inventory.

SECURITY

Aircraft inventory records should be unclassified, but any classification of the contents, if necessary, should be in accordance with the existing security regulations.

INVENTORIABLE ITEMS

Modification kits which are reissued or distributed to using organizations for installation, and which are not immediately placed in use, shall be recorded on the aircraft s DA Form-2408-17, Aircraft Inventory Record, and identified as loose equipment until modification is complete. Refer to table 1 for a list of inventoriable items for the aircraft.

Figure 1. Aircraft Inventory Section

Table 1. Inventoriable Items.

NOMENCLATURE	PART NUMBER	NSN	QTY RQD
SECTION A - NOSE ELECTRONICS			
TADS Turret	130760000	1270-01-188-4138	1
PNVS Turret and Faring	130900000	1090-01-169-9415	1
Multiplex Terminal - Type 3	4032298-956	7025-01-210-7768	1
Signal Data Converter - Eng.	7-311C20019	6629-01-160-3518	1

0347-1

FIGURE 389. Example of a page-based TM stylesheet output for <inventorywp>.

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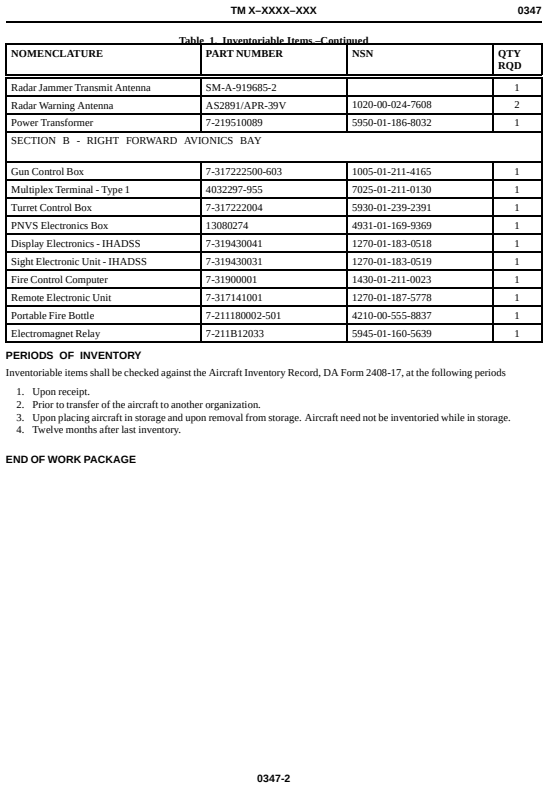

TM X–XXXX–XXX 0347

Table 1. Inventoriable Items –Continued.

NOMENCLATURE	PART NUMBER	NSN	QTY RQD
Radar Jammer Transmit Antenna	SM-A-919685-2		1
Radar Warning Antenna	AS2891/APR-39V	1020-00-024-7608	2
Power Transformer	7-219510089	5950-01-186-8032	1
SECTION B - RIGHT FORWARD AVIONICS BAY			
Gun Control Box	7-317222500-603	1005-01-211-4165	1
Multiplex Terminal - Type 1	4032297-955	7025-01-211-0130	1
Turret Control Box	7-317222004	5930-01-239-2391	1
PNVS Electronics Box	13080274	4931-01-169-9369	1
Display Electronics - IHADSS	7-319430041	1270-01-183-0518	1
Sight Electronic Unit - IHADSS	7-319430031	1270-01-183-0519	1
Fire Control Computer	7-31900001	1430-01-211-0023	1
Remote Electronic Unit	7-317141001	1270-01-187-5778	1
Portable Fire Bottle	7-211180002-501	4210-00-555-8837	1
Electromagnet Relay	7-211B12033	5945-01-160-5639	1

PERIODS OF INVENTORY

Inventoriable items shall be checked against the Aircraft Inventory Record, DA Form 2408-17, at the following periods

1. Upon receipt.
2. Prior to transfer of the aircraft to another organization.
3. Upon placing aircraft in storage and upon removal from storage. Aircraft need not be inventoried while in storage.
4. Twelve months after last inventory.

END OF WORK PACKAGE

0347-2

FIGURE 389. Example of a page-based TM stylesheet output for <inventorywp> - Continued.

37.19 Storage of aircraft work package ***<storagewp>***. The storage of aircraft work package is prepared for each aircraft storage category. The work package includes considerations for selection of the appropriate category and steps to be taken for care of the aircraft during exceptionally wet weather. All essential information for each aircraft storage category includes all procedures for preparing the aircraft for storage and removal from storage, however it excludes information on when or why the aircraft is stored. Each aircraft storage category makes a reference to inspection documents and inspection procedures to be conducted before, during and after storage.

a. The components of ***<storagewp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) Storage of aircraft general information ***<geninfo>*** (required) (see 49.1.4.12) (see 37.19.4 for standard statements requirement).
   (8) Flyable storage requirements ***<flyable>*** (required) (see 37.19.1).
   (9) Short term storage requirements ***<short>*** (required) (see 37.19.2).
   (10) Intermediate storage requirements ***<intermediate>*** (required) (see 37.19.3).

b. DTD graphically depicted for ***<storagewp>***:

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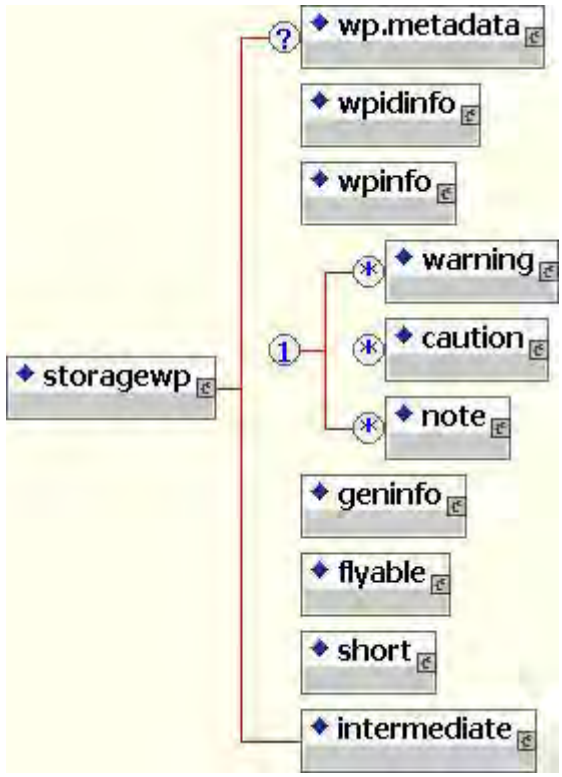

FIGURE 390. Storage of aircraft work package DTD hierarchy <storagewp>.

c. DTD fragment for ***<storagewp>***:

```
<!ELEMENT storagewp (wp.metadata?, wpidinfo, wpinfo, (warning*,
   caution*, note*), geninfo, flyable, short, intermediate)>
<!ATTLIST storagewp
    airforce          (yes | no )               "no"
    army              (yes | no )               "no"
    assocfig          IDREFS                     #IMPLIED
    changeref         IDREFS                     #IMPLIED
    comment           CDATA                      #IMPLIED
    crewmember        CDATA                      #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                      #IMPLIED
    deletewp          (yes | no )               "no"
    fgc               CDATA                      #IMPLIED
    frame             (yes | no )               "yes"
    idref             IDREFS                     #IMPLIED
    inschlvl          CDATA                      #IMPLIED
    insertwp          CDATA                      #IMPLIED
    lsa-id            CDATA                      #IMPLIED
    marines           (yes | no )               "no"
    navy              (yes | no )               "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                      #IMPLIED
    tocentry          (2 | 3 | 4 | 5)           "2"
    wpno              ID                         #REQUIRED
    wpseq             CDATA                      #IMPLIED
>
```


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d. Common attributes for ***<storagewp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.19.1 Flyable storage ***<flyable>***. The element contains the flyable storage and removal procedures for aircraft storage.

a. The components of ***<flyable>*** are:

(1) Single procedure
(a) Procedural steps title ***<title>*** (required) (see 49.1.1.3).
(b) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(c) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(d) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(e) General information ***<geninfo>*** (optional) (see 49.1.4.12).
(f) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
(g) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).

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(h) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1-alt>*** (see 48.2.1) (required – one or more).

(2) Multiple procedure ***<proc>*** (required – two or more) (see 33.2).

(3) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<flyable>***:

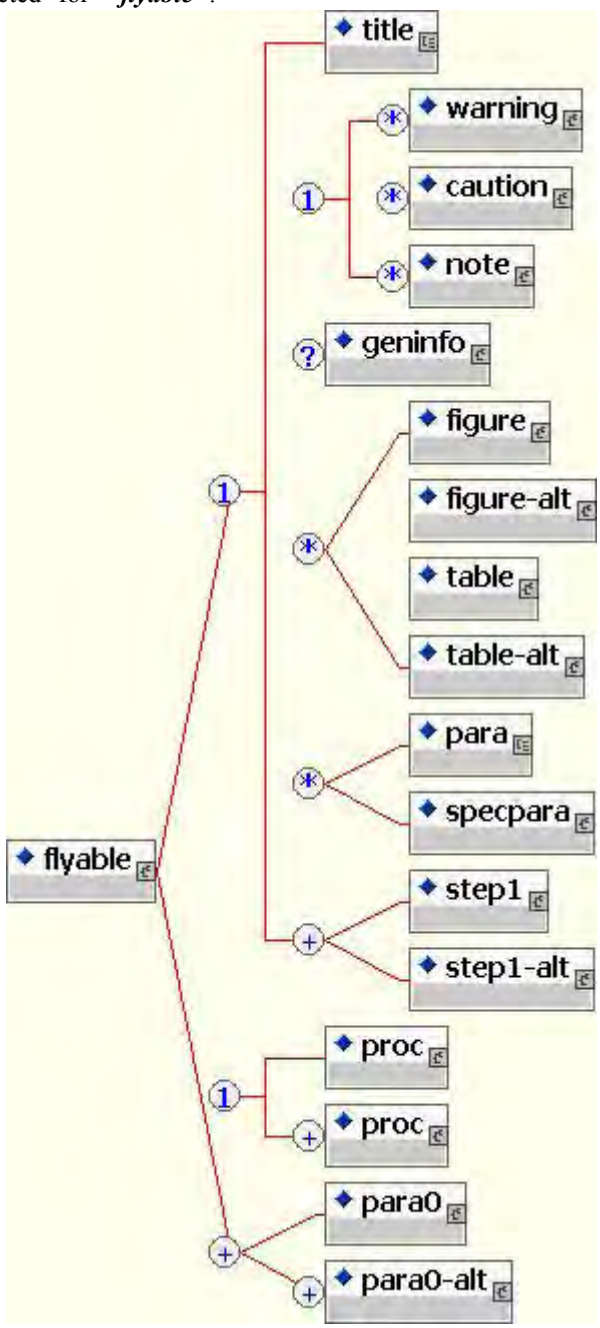

FIGURE 391. Flyable storage DTD hierarchy <flyable>.

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c. DTD fragment for ***<flyable>***:

```
<!ELEMENT flyable ((title, (warning*, caution*,
   note*), geninfo?, (figure | figure-alt | table |
   table-alt)*, (para | specpara)*, (step1 | step1-alt)+) |
   (proc, proc+) | (para0 | para0-alt)+)>
<!ATTLIST flyable
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
   esd               (yes | no)                      "no"
    hcp               (yes | no)                       "no"
>
```

d. Common attributes for ***<flyable>*** are:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(12) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).

37.19.2 Short term storage ***<short>***. The element contains the criteria for short length storage and removal from storage. The components for ***<short>*** is the same as ***<flyable>*** (see 37.19.1).

37.19.3 Intermediate storage ***<intermediate>***. The element contains the criteria for intermediate length storage and removal from storage. The components for ***<intermediate>*** is the same as ***<flyable>*** (see 37.19.1).

37.19.4 Storage of aircraft work package standard statement boilerplates. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed

TABLE XXI. Storage of aircraft work package standard statement boilerplates.

Boilerplate Entity	Description	Element Used In
&storage.geninfo;	Storage of aircraft general information	***<geninfo>***

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37.19.5 XML document instance fragment and output for ***<storagewp>***. The XML instance and its stylesheet output for a ***<storagewp>*** is provided below.

a. Example of an XML document instance fragment for ***<storagewp>***:

```
<storagewp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="M00345–X-XXX-XXXX">
<wpidinfo>
<maintlvl level="asb"/>
<title>STORAGE OF AIRCRAFT
</title>
</wpidinfo>
<wpinfo>
<tools>
<tools-setup-item>
<name>Aircraft mechanic's kit
</name>
<itemref>
<xref wpid="S00342-X-XXX-XXXX" itemid="S00342-X-XXX-XXXX-376"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Adjustable air filtering respirator
</name>
<itemref>
<xref wpid="S00342-X-XXX-XXXX" itemid="S00342-X-XXX-XXXX-262"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Chemical protective gloves
</name>
<itemref>
<xref wpid="S00342-X-XXX-XXXX" itemid="S00342-X-XXX-XXXX-154"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Light duty laboratory apron
</name>
<itemref>
<xref wpid="S00342-X-XXX-XXXX" itemid="S00342-X-XXX-XXXX-27"/>
</itemref>
</tools-setup-item>
</tools>
<mtrlpart>
<mtrlpart-setup-item>
<name>Barrier material
</name>
<itemref>
<xref wpid="S00332-X-XXX-XXXX" itemid="S00332-X-XXX-XXXX-32"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Cloth
</name>
<itemref>
<xref wpid="S00332-X-XXX-XXXX" itemid="S00332-X-XXX-XXXX-52"/>
</itemref>
</mtrlpart-setup-item>
```


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<mtrlpart-setup-item>
<name>Dry cleaning solvent
</name>
<itemref>
<xref wpid="S00332-X-XXX-XXXX" itemid="S00332-X-XXX-XXXX-74"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Lubricating oil
</name>
<itemref>
<xref wpid="S00332-X-XXX-XXXX" itemid="S00332-X-XXX-XXXX-117"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Tape
</name>
<itemref>
<xref wpid="S00332-X-XXX-XXXX" itemid="S00332-X-XXX-XXXX-207"/>
</itemref>
</mtrlpart-setup-item>
</mtrlpart>
<persnreq>
<persnreq-setup-item>
<name>Attack Helicopter Repairer
</name>
<mos>67R
</mos>
<qty>1
</qty>
</persnreq-setup-item>
<persnreq-setup-item>
<name>Attack Helicopter Repairer/Technical Inspector
</name>
<mos>67R3F
</mos>
<qty>1
</qty>
</persnreq-setup-item>
<persnreq-setup-item>
<name>Maintenance Test Pilot
</name>
<mos>152FG
</mos>
<qty>1
</qty>
</persnreq-setup-item>
</persnreq>
<ref>
<ref-setup-item>
<extref docno="TM 1-1500-204-23"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="TM 9-1090-208-23"/>
</ref-setup-item>
<ref-setup-item>

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Check the source to verify that this is the current version before use.

<extref docno="TM 11-6140-203-14-1"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="TM 11-6140-203-14-2"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="TM 55-1500-344-23"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="TM 55-1520-238-10"/>
</ref-setup-item>
</ref>
<eqpconds>
<eqpconds-setup-item>
<condition>Helicopter safed
</condition>
<itemref>
<xref wpid="M00651-X-XXX-XXXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Engine 10–hour/14 day inspection performed
</condition>
<itemref>
<extref docno="TM 55-2840-248-23"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>10–hour/14 day inspection performed
</condition>
<itemref>
<extref docno="TM 1-1520-238-PMS"/>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
<geninfo frame="no">
<para0 esd="no" hcp="no">
<title>GENERAL INFORMATION
</title>
<subpara1 esd="no" hcp="no">
<title>Components Involved of Storage
</title>
<para esd="no" hcp="no">Any component removed for reason of accident shall not be preserved, but is shipped in the same condition it was in after the accident.
</para>
</subpara1>
<subpara1 esd="no" hcp="no">
<title>Categories of Storage
</title>
<para esd="no" hcp="no">
<seqlist>
<item>Flyable storage — no time limit.
</item>
<item>Short term (administrative storage) — 1 to 45 days.

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Check the source to verify that this is the current version before use.

```
</item>
<item>Intermediate storage &mdash; 46 to 180 days.
</item>
</seqlist>
</para>
<note acknowledge="no">
<trim.para esd="no" hcp="no">Refer to
<extref docno="TM 1-1500-204-23"/> for general procedures
for storage of aircraft.
</trim.para>
</note>
<para esd="no" hcp="no">
</para>
</subpara1>
</para0>
</geninfo>
<flyable esd="no" hcp="no">
<title>FLYABLE STORAGE &mdash; NO TIME LIMIT
</title>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Start and operate auxiliary power unit (APU)
<xref wpid="M00981-X-XXX-XXXX" pretext="(" posttext=")"/>.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Allow APU to run for 10 to 15 minutes.
</para>
</step2>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Perform engine ground run-up procedures
<extref docno="TM 55-1520-238-10" pretext="(" posttext=")"/>.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Allow engines to run at idle for 5 minutes.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Shut engines down and allow to cool.
</para>
</step2>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Services fuel system
<xref wpid="M00351-X-XXX-XXXX" pretext="(" posttext=")"/>.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Drain residual water from fuel
cells and service to full.
</para>
</step2>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<specpara esd="no" hcp="no">
<caution>
<trim.para esd="no" hcp="no">If ambient temperature is below &ndash;40&deg;F
(-40&deg;C), remove battery from helicopter
```


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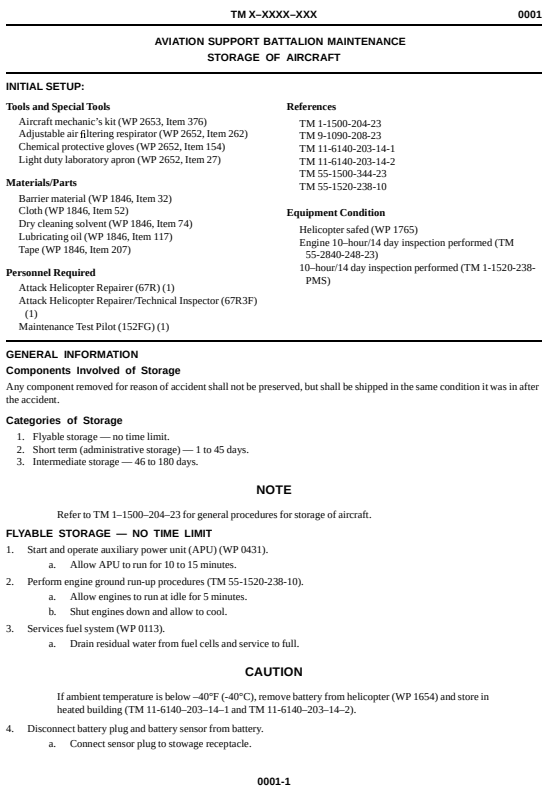
Check the source to verify that this is the current version before use.

```
<xref wpid="M00871–X-XXX-XXXX" pretext="(" posttext=")"/> and
store in heated buliding
<extref docno="TM 11-6140-203-14-1" pretext="(" posttext=" and "/>
<extref docno="TM 11-6140-203-14-2" posttext=")"/>.
</trim.para>
</caution>
<para esd="no" hcp="no">Disconnect battery plug and bat-
tery sensor from battery.
</para>
</specpara>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Connect sensor plug to stowage receptacle.
</para>
</step2>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Clean helicopter
<extref docno="TM 55-1500-344-23" pretext="(" posttext=")"/>.
</para>
</step1>
</flyable> . . .
</storagewp>
```

b. Page-based TM stylesheet output example for ***<storagewp>***:



TM X–XXXX–XXX 0001

AVIATION SUPPORT BATTALION MAINTENANCE
STORAGE OF AIRCRAFT

INITIAL SETUP:

Tools and Special Tools
Aircraft mechanic's kit (WP 2653, Item 376)
Adjustable air filtering respirator (WP 2652, Item 262)
Chemical protective gloves (WP 2652, Item 154)
Light duty laboratory apron (WP 2652, Item 27)

Materials/Parts
Barrier material (WP 1846, Item 32)
Cloth (WP 1846, Item 52)
Dry cleaning solvent (WP 1846, Item 74)
Lubricating oil (WP 1846, Item 117)
Tape (WP 1846, Item 207)

Personnel Required
Attack Helicopter Repairer (67R) (1)
Attack Helicopter Repairer/Technical Inspector (67R3F) (1)
Maintenance Test Pilot (152FG) (1)

References
TM 1-1500-204-23
TM 9-1090-208-23
TM 11-6140-203-14-1
TM 11-6140-203-14-2
TM 55-1500-344-23
TM 55-1520-238-10

Equipment Condition
Helicopter safed (WP 1765)
Engine 10–hour/14 day inspection performed (TM 55-2840-248-23)
10–hour/14 day inspection performed (TM 1-1520-238-PMS)

GENERAL INFORMATION

Components Involved of Storage

Any component removed for reason of accident shall not be preserved, but shall be shipped in the same condition it was in after the accident.

Categories of Storage

1. Flyable storage — no time limit.
2. Short term (administrative storage) — 1 to 45 days.
3. Intermediate storage — 46 to 180 days.

NOTE

Refer to TM 1–1500–204–23 for general procedures for storage of aircraft.

FLYABLE STORAGE — NO TIME LIMIT

1. Start and operate auxiliary power unit (APU) (WP 0431).
   a. Allow APU to run for 10 to 15 minutes.
2. Perform engine ground run-up procedures (TM 55-1520-238-10).
   a. Allow engines to run at idle for 5 minutes.
   b. Shut engines down and allow to cool.
3. Services fuel system (WP 0113).
   a. Drain residual water from fuel cells and service to full.

CAUTION

If ambient temperature is below –40°F (-40°C), remove battery from helicopter (WP 1654) and store in heated building (TM 11-6140–203–14–1 and TM 11-6140–203–14–2).

4. Disconnect battery plug and battery sensor from battery.
   a. Connect sensor plug to stowage receptacle.

0001-1

FIGURE 392. Example of a page-based TM stylesheet output for <storagewp>.

37.20 Weighing and loading work package ***<wtloadwp>***. The element contains description, information, and procedures for aircraft weighing, balancing and loading.

a. The components of ***<wtloadwp>*** are:

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(1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
(2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
(3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
(4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(7) General information ***<geninfo>*** (required) containing the scope of the work package (see 49.1.4.12). The standard statement in accordance with MIL–STD–40051–1/–2 is ***&wtloadwp.geninfo;*** (see 37.20.3).
(8) Weighing form chart ***<formchart>*** (required) (see 37.20.1).
(9) Loading information ***<weightinst>*** (required) (see 37.20.2).

b. DTD graphically depicted for ***<wtloadwp>***:

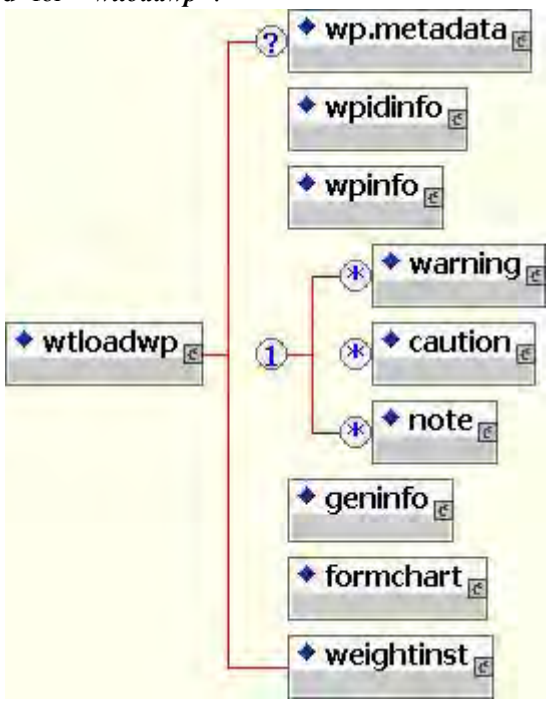

FIGURE 393. Aircraft weighing and loading work package DTD hierarchy <wtloadwp>.

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c. DTD fragment for ***<wtloadwp>***:

```
<!ELEMENT wtloadwp (wp.metadata?, wpidinfo, wpinfo, (warning*, caution*,
   note*), geninfo, formchart, weightinst)>
<!ATTLIST wtloadwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes for ***<wtloadwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.20.1 Weighing form chart ***<formchart>***. The element contains information on the usage of forms and charts within the weighing and loading work package.

a. The components of ***<formchart>*** require at least one or more of the following that are:
(1) Procedure***<proc>*** (see 33.2).
(2) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1).
(3) Illustration***<figure>*** (see 44.1.1).

b. DTD graphically depicted for ***<formchart>***:

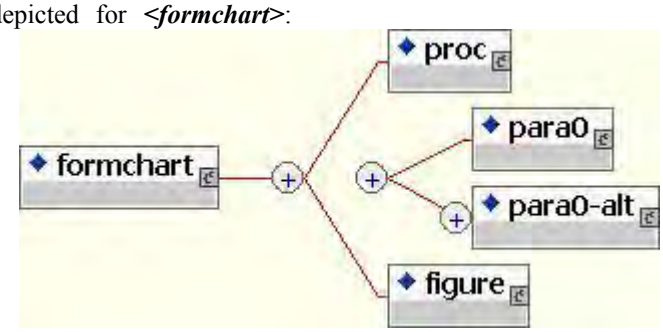

FIGURE 394. Weighing form chart DTD hierarchy <formchart>.

c. DTD fragment for ***<formchart>***:

```
<!ELEMENT formchart (proc | (para0 | para0-alt)+ | figure)+>
<!ATTLIST formchart
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
   esd               (yes | no)                      "no"
    hcp               (yes | no)                       "no"
>
```

d. Common attributes for ***<formchart>*** are:
(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).

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(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(12) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).

37.20.2 Loading information ***<weightinst>***. The element contains the descriptions and instructions for aircraft loading, and computing weight and balance information. Loading information includes weight and balance characteristics, center of gravity limits, weight/balance and loading, and weight and moment tables for load items such as crew, fuel, cargo, and armament.

a. The components of ***<weightinst>*** are:
   (1) Single procedure
      (a) Procedural steps title ***<title>*** (required) (see 49.1.1.3).
      (b) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
      (c) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
      (d) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
      (e) General information ***<geninfo>*** (optional) (see 49.1.4.12).
      (f) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1) (optional – zero or more).
      (g) Table ***<table>*** (see section 42) and/or conditional table ***<table-alt>*** (see 48.2.1) (optional – zero or more).
      (h) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1-alt>*** (see 48.2.1) (required – one or more).
   (2) Multiple procedure ***<proc>*** (required – two or more) (see 33.2).
   (3) Descriptive or narrative titled text ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).
b. DTD graphically depicted for ***<weightinst>***:

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Check the source to verify that this is the current version before use.

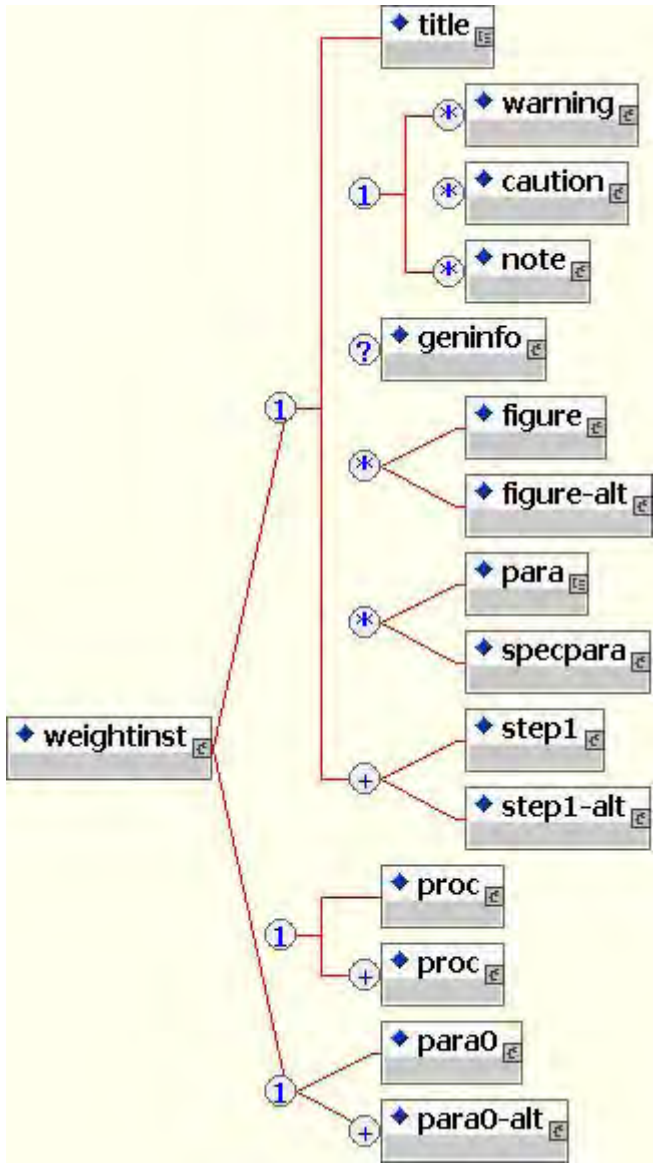

FIGURE 395. Loading Information DTD hierarchy <weightinst>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<weightinst>***:

```
<!ELEMENT weightinst ((title, (warning*, caution*, note*),
   geninfo?, (figure | figure-alt)*, (para | specpara)*,
   (step1 | step1-alt)+) | (proc, proc+) | (para0 | para0-alt+)))>>
<!ATTLIST weightinst
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<weightinst>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

37.20.3 Weighing and loading work package general entity boilerplates. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listedTable XXII.

TABLE XXII. Boilerplate Entities for ***<wtloadwp>***.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Weighing and loading AVIM general information and scope statement	***&wtloadwp.geninfo;***	Not applicable

37.20.4 XML document instance fragment and output for ***<wtloadwp>***. The XML instance and its stylesheet output for a ***<wtloadwp>*** is provided below.

a. Example of an XML document instance fragment for ***<wtloadwp>***:

<wtloadwp airforce="no" army="no" deletewp="no" frame="yes" marines="no" navy="no" tocentry="2" wpno="M00367-X-XXXX-XXX" wpseq="0367">
<wpidinfo>
<maintlvl level=""asb/>
<title>HELICOPTER WEIGHING (AIRCRAFT ON JACKS)
</title>
</wpidinfo>
<wpinfo>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<tools>
<tools-setup-item>
<name>Aircraft mechanic's tool kit
</name>
<itemref>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item376"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Fuselage jack adapter (2)
</name>
<itemref>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item1"/>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item391" pretext="p/o"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>3-ton tripod hydraulic jack
</name>
<itemref>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item184"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>5-ton aircraft landing gear jack
</name>
<itemref>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item183"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Weighing kit
</name>
<itemref>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item196"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Strut locks (2)
</name>
<itemref>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item203"/>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item25" pretext=" p/o"/>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item194" pretext=" and "/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Aircraft maintenance kit (plumb bob leveling kit)
</name>
<itemref>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item207"/>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item25" pretext=" p/o"/>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item194" pretext=" and "/>
</itemref>
</tools-setup-item>
<tools-setup-item>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<name>Tripod aircraft tail stand (2) (extension legs removed)
</name>
<itemref>
<xref wpid="M00893-X-XXXX-XXX" itemid="M00893-X-XXXX-XXX-item344"/>
</itemref>
</tools-setup-item>
</tools>
<mtrlpart>
<mtrlpart-setup-item>
<name>Wood block, 2 in. x 6 in. x 12 in.
</name>
</mtrlpart-setup-item>
</mtrlpart>
<persnreq>
<persnreq-setup-item>
<name>Attack Helicopter Repairer/Technical Inspector
</name>
<mos>67R3F
</mos>
</persnreq-setup-item>
<persnreq-setup-item>
<name>Assistants
</name>
<qty>3
</qty>
</persnreq-setup-item>
</persnreq>
<eqpconds>
<eqpconds-setup-item>
<condition>Helicopter safed
</condition>
<itemref>
<xref wpid="M00057-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Helicopter defueled, SPA method
</condition>
<itemref>
<xref wpid="M00019-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Engine oil system serviced
</condition>
<itemref>
<xref wpid="M00024-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>APU oil system serviced
</condition>
<itemref>
<xref wpid="M00026-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<eqpconds-setup-item>
<condition>Engine nose gearboxes serviced
</condition>
<itemref>
<xref wpid="M00028-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Intermediate gearbox serviced
</condition>
<itemref>
<xref wpid="M00030-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Tail rotor gearbox serviced
</condition>
<itemref>
<xref wpid="M00031-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Main transmission serviced
</condition>
<itemref>
<xref wpid="M00032-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Hydraulic system serviced
</condition>
<itemref>
<xref wpid="M00034-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Access doors L135 and R135 opened
</condition>
<itemref>
<xref wpid="M00102-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Main landing gear shock struts collapsed
</condition>
<itemref>
<xref wpid="M00041-X-XXXX-XXX"/>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
<warning haz-abbrev="no">
<warning.group haz-abbrev="no">
<trim.para esd="no" hcp="no">Helicopter is unstable on jacks. Jack helicopter evenly and carefully on a firm, level, flat surface. Death or serious injury can result if helicopter should fall off jacks.

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Check the source to verify that this is the current version before use.

</trim.para>
</warning.group>
<warning.group haz-abbrev="no">
<trim.para esd="no" hcp="no">When weighing helicopter, ensure that area is roped off and that multiple warning signs that read "HELICOPTER ON JACKS" are prominently displayed. Do not allow unauthorized persons in or around helicopter while it is on jacks.
</trim.para>
</warning.group>
</warning>
<caution>
<trim.para esd="no" hcp="no">Weighing should be done inside a hangar with all doors closed. Weighing the helicopter outdoors is not recommended as wind velocities may affect the accuracy of scale readings.
</trim.para>
</caution>
<geninfo frame="no"> **&wtloadwp.geninfo;**
</geninfo>
<formchart esd="no" hcp="no">
<proc esd="no" frame="yes" hcp="no" tocentry="0">
<title>FORM CHART
</title>
<para>
<figure application="both" figtype="normal-page" pane="no" tocentry="1" id="M00367-X-XXXX-XXX-fig1">
<title>Two view Chart E diagram.
</title>
<graphic boardno="weightwp" unitmeasure="in">
</graphic>
</figure>
</para>
</proc>
</formchart>
<weightinst>
<proc esd="no" frame="yes" hcp="no" tocentry="0">
<title>PRELIMINARY
</title>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">This work package describes helicopter weighing procedures with the use of aircraft jacks and electronic load cells. Helicopter is weighed in a 3.5 degree nose up attitude with the main landing gear locked in a collapsed position.
<xref figid="M00367-X-XXXX-XXX-fig1" pretext="(See “ posttext=" for two view chart diagram.)">
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Helicopter shall be weighed in the BASIC WEIGHT condition. BASIC WEIGHT condition is established with all doors closed, except L135 and R135, seats in the mid position, and includes usable engine and APU oil, trapped fuel, 30mm gun and ammo magazine, and wing pylons.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check for and remove all non-basic weight items such expendable ordnance, clipboards, helmets, personal equipment, tools, etc.
</para>

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```
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Defuel helicopter before weighing.
Helicopter Center of Gravity (CG) cannot be accurately
determined when weighed with fuel on board.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Defuel helicopter on the ground attitude with the
use of a defueling pump.  Drain remainder of fuel from each tank through
sump drains.  Sump drains are found on helicopter bottom centerline
at fuselage stations (F.S.) 172.3 and 255.4.  When the fuel drainage
diminishes to dripping, allow five minutes for balance of fuel to drain.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Pylons are a part of helicopter basic
weight and should be listed on
<extref docno="DD Form 365-1"/> (Chart A) as follows:
<table tocentry="1">
<title>Helicopter basic weight.
</title>
<tgroup align="center" char=" " charoff="50" cols="5">
<colspec colname="col1" colwidth="0.97*"/>
<colspec colname="col2" colwidth="1.59*"/>
<colspec colname="col3" colwidth="0.92*"/>
<colspec colname="col4" colwidth="0.64*"/>
<colspec colname="col5" colwidth="0.90*"/>
<thead>
<row>
<entry valign="bottom">COMPARTMENT
<brk/>AND
<brk/>ITEM NUMBER
</entry>
<entry valign="bottom">ITEMS AND LOCATION
<brk/> (Grouped by compartment)
</entry>
<entry valign="bottom">WEIGHT
<brk/>(lbs)
</entry>
<entry valign="bottom">ARM
<brk/>(in)
</entry>
<entry valign="bottom">MOMENT/100
<brk/>ENTER CONSTANT
<brk/> (in-lbs)
</entry>
</row>
</thead>
<tbody>
<row>
<entry>H
</entry>
<entry>Right Wing (F.S. 185 &ndash; 238)
</entry>
<entry>
```


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Check the source to verify that this is the current version before use.

```
</entry>
<entry>
</entry>
<entry>
</entry>
</row>
<row>
<entry>H
</entry>
<entry>Wing  Pylon  318000001200
</entry>
<entry>58
</entry>
<entry>199.0
</entry>
<entry>115
</entry>
</row>
<row>
<entry>H
</entry>
<entry>Wing  Pylon  31800001
</entry>
<entry>58
</entry>
<entry>199.0
</entry>
<entry>115
</entry>
</row>
<row>
<entry>I
</entry>
<entry>Left  Wing  (F.S.  185  &ndash;  238)
</entry>
<entry>
</entry>
<entry>
</entry>
<entry>
</entry>
</row>
<row>
<entry>I
</entry>
<entry>Wing  Pylon  318000001100
</entry>
<entry>58
</entry>
<entry>199.0
</entry>
<entry>115
</entry>
</row>
<row>
<entry>I
```


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Check the source to verify that this is the current version before use.

```
</entry>
<entry>Wing  Pylon  31800001
</entry>
<entry>58
</entry>
<entry>199.0
</entry>
<entry>115
</entry>
</row>
</tbody>
</tgroup>
</table>
</para>
</step1>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Perform  a  complete  inventory  of  items
installed  on  helicopter.    Use
<extref  pretext="Basic  Weight  Check  List,  "  docno="DD  Form  365-1"/>  found  in  the
helicopter  weight  and  balance  file.  Record  all  discrepancies  from
<extref  docno="DD  Form  365-1"/>  in  Column  I  or  II  (as  appli-
cable)  on  the  reverse  side  of  the
<extref  pretext="Aircraft  Weighing  Record,  "  docno="DD  Form  365-2"/>.
</para>
</step1>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Failure  to  weigh  helicopter  in  the  BASIC  WEIGHT
condition  with  four  pylons  installed  will  cause  an  incorrect  center
of  gravity  calculation  when  converting  from  &ldquo;as  weighed&rdquo;
condition  to  &ldquo;corrected  to  level  attitude&rdquo;  condition.
</para>
</step1>
</proc>
<proc  esd="no"  frame="yes"  hcp="no"  tocentry="0">
<title>HELICOPTER  WEIGHING  PREPARATION
</title>
<step1  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<specpara  esd="no"  hcp="no">
<caution>
<trim.para  esd="no"  hcp="no">Ensure  that  there  are  no  adverse
air  currents  from  fans,  heater  blowers,  etc.,  which  could
affect  the  accuracy  of  scale  readings.
</trim.para>
</caution>
<para  esd="no"  hcp="no">Position  helicopter  in  hangar.
</para>
</specpara>
<step2  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para esd="no" hcp="no">Move helicopter to designated weighing area in hangar
</para>
</step2>
<step2  esd="no"  hcp="no"  qa="no"  skilllevel="novice_expert">
<para  esd="no"  hcp="no">Close  all  hangar  doors  and  rope  off  weighing  area.
</para>
</step2>
</step1>
```


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```
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Position helicopter rotor blades.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Turn main rotor blades clockwise to a position
of 45 degrees relative to helicopter's longitudinal axis.
<figure application="both" figtype="normal-page" pane="no" tocentry="1">
<title>Helicopter longitude axis.
</title>
<graphic boardno="weightwp2" hscale="75" unitmeasure="in">
</graphic>
</figure>
</para>
</step2>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Assemble aircraft weighing kit.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Assemble weighing kit per manufacturer's
instructions supplied with kit.  Use weighing kit.
</para>
</step2>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<specpara esd="no" hcp="no">
<caution>
<trim.para esd="no" hcp="no">Plumb bob bracket and target installation
screws are not removable.  Damage to bracket and target
will result if removal is attempted.
</trim.para>
</caution>
<para esd="no" hcp="no">Install plumb bob bracket
<callout assocfig="M00367-X-XXXX-XXXfig3" label="1"/> and target
<callout assocfig="M00367-X-XXXX-XXXfig3" label="2"/> on helicopter left
side at F.S. 171.0.  Use aircraft maintenance kit.
<figure application="both" figtype="normal-page" id="M00367-X-XXXX-XXXfig3"
pane="no" tocentry="1">
<title>Install plumb bob bracket
</title>
<graphic boardno="weightwp3" hscale="75" unitmeasure="in">
</graphic>
</figure>
</para>
</specpara>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Loosen two upper screws
<callout assocfig="M00367-X-XXXX-XXXfig3" label="3"/>.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Loosen two lower screws
<callout assocfig="M00367-X-XXXX-XXXfig3" label="4"/>.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
```


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```
<para esd="no" hcp="no">Slide bracket
<callout assocfig="M00367-X-XXXX-XXXfig3" label="1"/> down on screws
<callout assocfig="M00367-X-XXXX-XXXfig3" label="3"/>.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Slide target
<callout assocfig="M00367-X-XXXX-XXXfig3" label="2"/> down on screws
<callout assocfig="M00367-X-XXXX-XXXfig3" label="4"/>.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Tighten screws
<callout assocfig="M00367-X-XXXX-XXXfig3" label="3"/> and
<callout assocfig="M00367-X-XXXX-XXXfig3" label="4"/>.
</para>
</step2>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Install and secure plumb bob line
<callout assocfig="M00367-X-XXXX-XXXfig3" label="5"/> with plumb bob
<callout assocfig="M00367-X-XXXX-XXXfig3" label="6"/> in bracket
<callout assocfig="M00367-X-XXXX-XXXfig3" label="1"/>.  Use air-
craft maintenance kit.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Insert line
<callout assocfig="M00367-X-XXXX-XXXfig3" label="5"/> in bracket slot
<callout assocfig="M00367-X-XXXX-XXXfig3" label="7"/>.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Lower plumb bob
<callout assocfig="M00367-X-XXXX-XXXfig3" label="6"/> until just
below edge of target
<callout assocfig="M00367-X-XXXX-XXXfig3" label="2"/>.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Secure line
<callout assocfig="M00367-X-XXXX-XXXfig3" label="5"/> to bracket
<callout assocfig="M00367-X-XXXX-XXXfig3" label="1"/> by wrapping line
<callout assocfig="M00367-X-XXXX-XXXfig3" label="5"/> around rivets
<callout assocfig="M00367-X-XXXX-XXXfig3" label="8"/> and
<callout assocfig="M00367-X-XXXX-XXXfig3" label="9"/>.
</para>
</step2>
</step1>
</proc>
</weightinst>
</wtloadwp>
```

b. Page-based TM stylesheet output example for **<*wtloadwp*>**:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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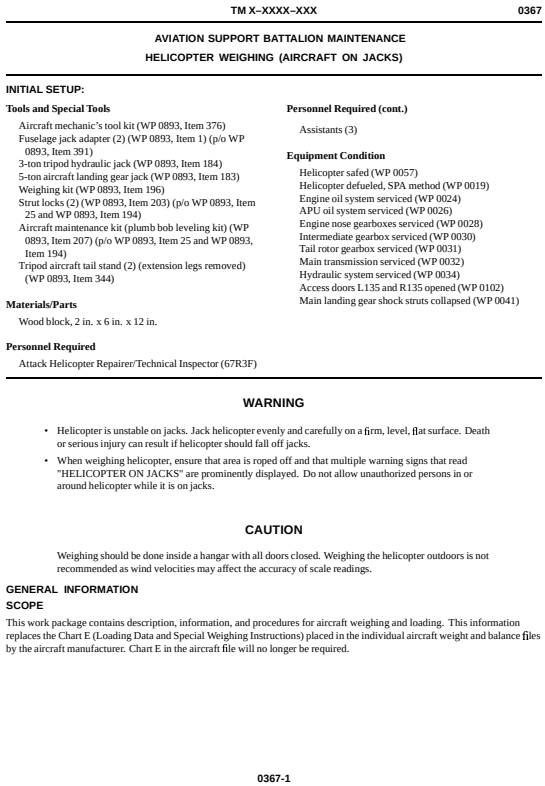

TM X–XXXX–XXX 0367

AVIATION SUPPORT BATTALION MAINTENANCE

HELICOPTER WEIGHING (AIRCRAFT ON JACKS)

INITIAL SETUP:

Tools and Special Tools

Aircraft mechanic's tool kit (WP 0893, Item 376)
Fuselage jack adapter (2) (WP 0893, Item 1) (p/o WP 0893, Item 391)
3-ton tripod hydraulic jack (WP 0893, Item 184)
5-ton aircraft landing gear jack (WP 0893, Item 183)
Weighing kit (WP 0893, Item 196)
Strut locks (2) (WP 0893, Item 203) (p/o WP 0893, Item 25 and WP 0893, Item 194)
Aircraft maintenance kit (plumb bob leveling kit) (WP 0893, Item 207) (p/o WP 0893, Item 25 and WP 0893, Item 194)
Tripod aircraft tail stand (2) (extension legs removed) (WP 0893, Item 344)

Materials/Parts

Wood block, 2 in. x 6 in. x 12 in.

Personnel Required

Attack Helicopter Repairer/Technical Inspector (67R3F)

Personnel Required (cont.)

Assistants (3)

Equipment Condition

Helicopter safed (WP 0057)
Helicopter defueled, SPA method (WP 0019)
Engine oil system serviced (WP 0024)
APU oil system serviced (WP 0026)
Engine nose gearboxes serviced (WP 0028)
Intermediate gearbox serviced (WP 0030)
Tail rotor gearbox serviced (WP 0031)
Main transmission serviced (WP 0032)
Hydraulic system serviced (WP 0034)
Access doors L135 and R135 opened (WP 0102)
Main landing gear shock struts collapsed (WP 0041)

WARNING

- Helicopter is unstable on jacks. Jack helicopter evenly and carefully on a firm, level, flat surface. Death or serious injury can result if helicopter should fall off jacks.
- When weighing helicopter, ensure that area is roped off and that multiple warning signs that read "HELICOPTER ON JACKS" are prominently displayed. Do not allow unauthorized persons in or around helicopter while it is on jacks.

CAUTION

Weighing should be done inside a hangar with all doors closed. Weighing the helicopter outdoors is not recommended as wind velocities may affect the accuracy of scale readings.

GENERAL INFORMATION

SCOPE

This work package contains description, information, and procedures for aircraft weighing and loading. This information replaces the Chart E (Loading Data and Special Weighing Instructions) placed in the individual aircraft weight and balance files by the aircraft manufacturer. Chart E in the aircraft file will no longer be required.

0367-1

FIGURE 396. Example of a page-based TM stylesheet output for <wtloadwp>.

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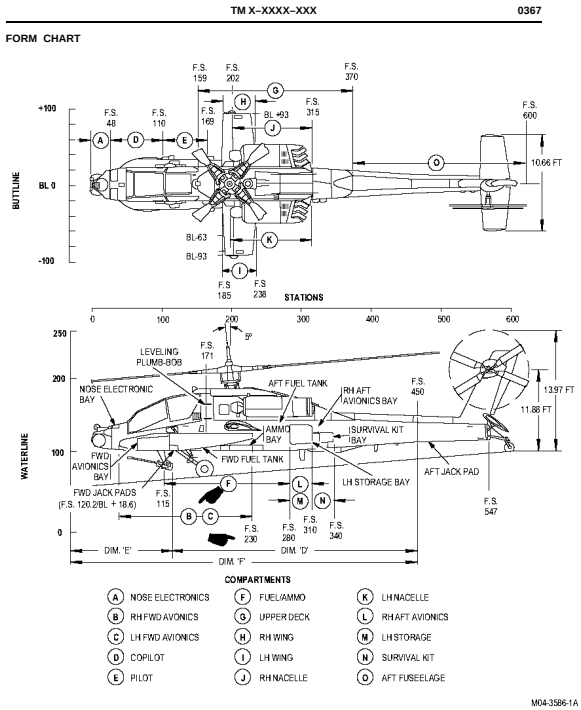

Figure 1. Two view Chart E diagram.

0367-2

FIGURE 396. Example of a page-based TM stylesheet output for <wtloadwp> - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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PRELIMINARY

1. This work package describes helicopter weighing procedures with the use of aircraft jacks and electronic load cells. Helicopter is weighed in a 3.5 degree nose up attitude with the main landing gear locked in a collapsed position. (See Figure 1 for two view chart diagram.)
2. Helicopter shall be weighed in the BASIC WEIGHT condition. BASIC WEIGHT condition is established with all doors closed, except L135 and R135, seats in the mid position, and includes usable engine and APU oil, trapped fuel, 30mm gun and ammo magazine, and wing pylons.
3. Check for and remove all non-basic weight items such expendable ordnance, clipboards, helmets, personal equipment, tools, etc.
4. Defuel helicopter before weighing. Helicopter Center of Gravity (CG) cannot be accurately determined when weighed with fuel on board.
5. Defuel helicopter on the ground attitude with the use of a defueling pump. Drain remainder of fuel from each tank through sump drains. Sump drains are found on helicopter bottom centerline at fuselage stations (F.S.) 172.3 and 255.4. When the fuel drainage diminishes to dripping, allow five minutes for balance of fuel to drain.
6. Pylons are a part of helicopter basic weight and should be listed on DD Form 365-1 (ChartA) as follows:

Table 1. Helicopter basic weight.

COMPARTMENT AND ITEM NUMBER	ITEMS AND LOCATION (Grouped by compartment)	WEIGHT (lbs)	ARM (in)	MOMENT/100 ENTER CONSTANT (in-lbs)
H	Right Wing (F.S. 185 – 238)			
H	Wing Pylon 318000001200	58	199.0	115
H	Wing Pylon 31800001	58	199.0	115
I	Left Wing (F.S. 185 – 238)			
I	Wing Pylon 318000001100	58	199.0	115
I	Wing Pylon 31800001	58	199.0	115

7. Perform a complete inventory of items installed on helicopter. Use Basic Weight Check List, DD Form 365-1 found in the helicopter weight and balance file. Record all discrepancies from DD Form 365-1 in Column I or II (as applicable) on the reverse side of the Aircraft Weighing Record, DD Form 365-2.
8. Failure to weigh helicopter in the BASIC WEIGHT condition with four pylons installed will cause an incorrect center of gravity calculation when converting from "as weighed" condition to "corrected to level attitude" condition.

HELICOPTER WEIGHING PREPARATION

CAUTION

Ensure that there are no adverse air currents from fans, heater blowers, etc., which could affect the accuracy of scale readings.

1. Position helicopter in hangar.
   a. Move helicopter to designated weighing area in hangar
   b. Close all hangar doors and rope off weighing area.
2. Position helicopter rotor blades.
   a. Turn main rotor blades clockwise to a position of 45 degrees relative to helicopter's longitudinal axis.

0367-3

FIGURE 396. Example of a page-based TM stylesheet output for <wtloadwp> - Continued.

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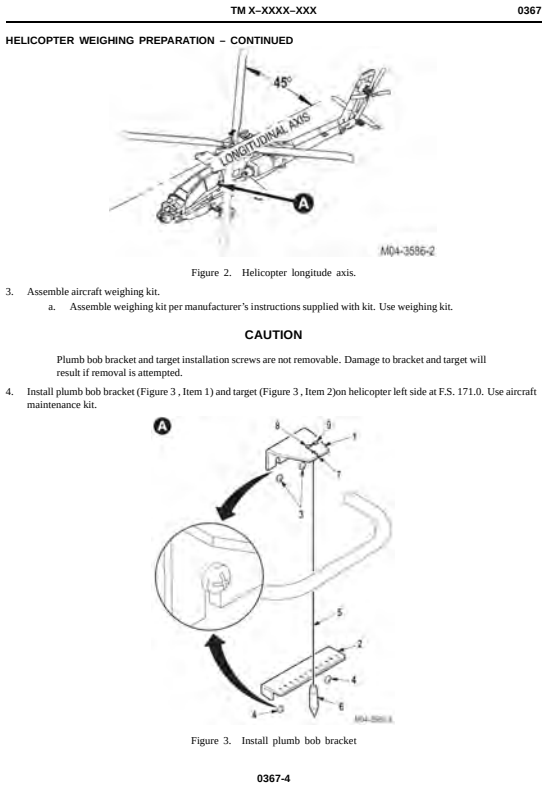
TM X–XXXX–XXX 0367

HELICOPTER WEIGHING PREPARATION – CONTINUED

Figure 2. Helicopter longitude axis.

3. Assemble aircraft weighing kit.
   a. Assemble weighing kit per manufacturer's instructions supplied with kit. Use weighing kit.

CAUTION

Plumb bob bracket and target installation screws are not removable. Damage to bracket and target will result if removal is attempted.

4. Install plumb bob bracket (Figure 3 , Item 1) and target (Figure 3 , Item 2)on helicopter left side at F.S. 171.0. Use aircraft maintenance kit.

Figure 3. Install plumb bob bracket

0367-4

FIGURE 396. Example of a page-based TM stylesheet output for <wtloadwp> - Continued.

37.21 Auxiliary equipment maintenance work package ***<auxeqpwp>***. Auxiliary equipment maintenance instruction work package is prepared when auxiliary equipment (e.g., Modified Tables of Organization and Equipment (MTOE) items, etc.) maintenance TMs are not procured for peculiar equipment furnished by the contractor. When a maintenance task does not fit the information, use a generic procedure.

a. The components of ***<auxeqpwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (8) Use either:
      (a) Maintenance task ***<maintsk>*** (required – one or more) (see 37.7.1)
      (b) Procedure ***<proc>*** (required – one or more) (see 33.2)

b. DTD graphically depicted for ***<auxeqpwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

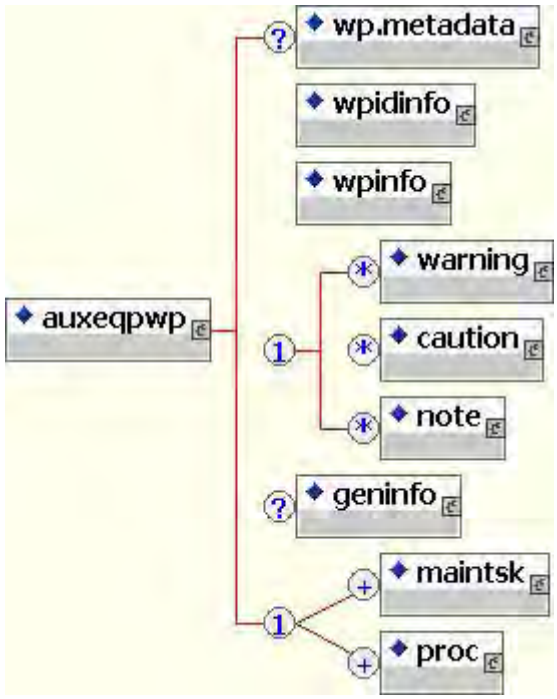

FIGURE 397. Auxiliary equipment maintenance work package DTD hierarchy <auxeqpwp>.

c. DTD fragment for ***<auxeqpwp>***:

```
<!ELEMENT auxeqpwp (wp.metadata?, wpidinfo, wpinfo, (warning*,
    caution*, note*), geninfo?, (maintsk+ | proc+))>
<!ATTLIST auxeqpwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attributes for ***<auxeqwp>*** are :

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.21.1 XML document instance fragment and output for ***<auxeqpwp>***. The XML instance and its stylesheet output for a ***<auxeqpwp>*** is provided below.

a. Example of an XML document instance fragment for ***<auxeqpwp>***:

```
<auxeqpwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no" tocentry="2" wpno="M3192-X-XXXX-XXX" wpseq="0062">
<wpidinfo>
<maintlvl level="field"/>
<title>BLANK FIRING ATTACHMENT (M15A2) MAINTENANCE
</title>
</wpidinfo>
<wpinfo>
<mtrlpart>
<mtrlpart-setup-item>
<name>Cleaner, lubricant, and preservative (CLP)
</name>
<itemref>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<xref itemno="9" wpid="S0034-X-XXXX-XXX"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Coating compound, enamel (RedM16A2)
</name>
<itemref>
<xref itemno="23" wpid="S0034-X-XXXX-XXX"/>
</itemref>
</mtrlpart-setup-item>
</mtrlpart>
</wpinfo>
<warning haz-abbrev="no">
<warning.group haz-abbrev="no">
<trim.para esd="no" hcp="no">Do not keep live ammunition near the work area.
</trim.para>
</warning.group>
<warning.group haz-abbrev="no">
<trim.para esd="no" hcp="no">Only blank cartridge M200 is to be used when the blank firing attachment is attached to the weapon.
</trim.para>
</warning.group>
</warning>
<geninfo frame="no">
<title>GENERAL
</title>
<para esd="no" hcp="no">This work package contains information and instructions to keep auxiliary equipment used with your weapon in good repair.
</para>
</geninfo>
<maintsk>
<remove esd="no" frame="yes" hcp="no" tocentry="0">
<title>REMOVAL
</title>
<caution>
<trim.para esd="no" hcp="no">Do not use tools to tighten the blank firing attachment.
<emphasis emph="bold">USE HANDS ONLY
</emphasis>.
</trim.para>
</caution>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Unscrew slide
<callout assocfig="M3192-X-XXXX-XXX-fig1" label="1"/> to remove from compensator
<callout assocfig="M3192-X-XXXX-XXX-fig1" label="2"/>.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Unhook blank firing attachment
<callout assocfig="M3192-X-XXXX-XXX-fig1" label="3"/> from behind the first groove of compensator
<callout assocfig="M3192-X-XXXX-XXX-fig1" label="2"/>.
<figure application="both" figtype="normal-page" pane="no" id="M3192-X-XXXX-XXX-fig1" tocentry="1">
<title>Unhook blank firing from compensator.
</title>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<graphic boardno="auxwp1" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Screw slide
<callout assocfig="M3192-X-XXXX-XXX-fig2" label="1"/> all the way
in on blank firing attachment
<callout assocfig="M3192-X-XXXX-XXX-fig2" label="3"/>.
<figure application="both" figtype="normal-page" pane="no"
id="M3192-X-XXXX-XXX-fig2" tocentry="1">
<title>Blank firing attachment.
</title>
<graphic boardno="auxwp2" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
</remove>
</maintsk>
<maintsk>
<clean esd="no" frame="yes" hcp="no" tocentry="0">
<title>CLEANING
</title>
<para esd="no" hcp="no">Clean blank firing attachment with
CLP. wipe dry. and coat with CLP.
</para>
</clean>
</maintsk>
<maintsk>
<paint esd="no" frame="yes" hcp="no" tocentry="0">
<title>PAINTING
</title>
<para esd="no" hcp="no">Inspect blank firing attachment for cracks or
distortion. Be sure the parts in the slide are clear and clean. If
blank firing attachment is cracked or distorted, it is unserviceable.
</para>
</paint>
</maintsk>
<maintsk>
<repair-rplc esd="no" frame="yes" hcp="no" tocentry="0">
<title>REPLACEMENT
</title>
<para esd="no" hcp="no">Replace blank firing attachment if unserviceable.
</para>
</repair-rplc>
</maintsk>
<maintsk>
<install esd="no" frame="yes" hcp="no" tocentry="0">
<title>INSTALLATION
</title>
<note acknowledge="no">
<trim.para esd="no" hcp="no">M23 blank firing attachment is stamped "M4
Carbine Only" painted yellow and may be used on the M4 and M4A1 carbines.
M15A2 BFA is painted red and is used on the M16A2 rifle.

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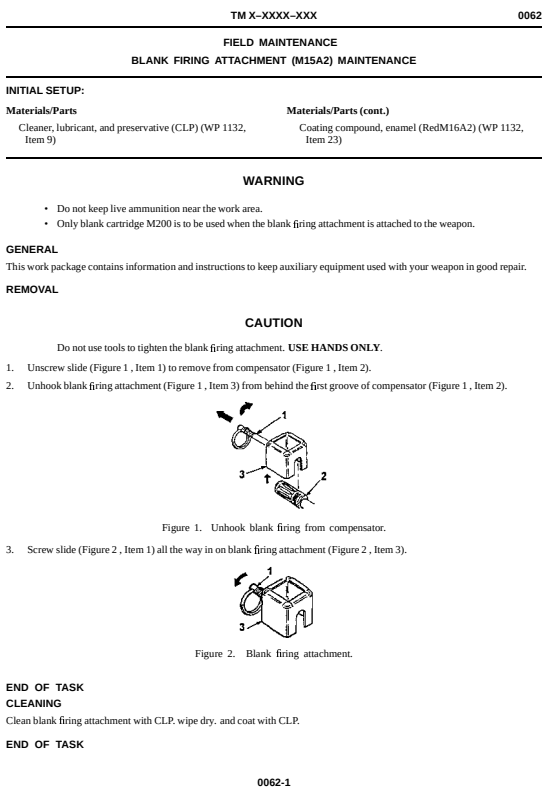
```
</trim.para>
</note>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Unscrew and pull slide
<callout assocfig="M3192-X-XXXX-XXX-fig3" label="1"/> all the way
out on blank firing attachment
<callout assocfig="M3192-X-XXXX-XXX-fig3" label="2"/>.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Hook blank firing attachment
<callout assocfig="M3192-X-XXXX-XXX-fig3" label="2"/> behind the
first groove of the compensator
<callout assocfig="M3192-X-XXXX-XXX-fig3" label="3"/>.
<figure application="both" figtype="normal-page" pane="no"
id="M3192-X-XXXX-XXX-fig3" tocentry="1">
<title>Installation of compensator.
</title>
<graphic boardno="auxwp3" unitmeasure="in">
</graphic>
</figure>
</para>
</step1>
</install>
</maintsk>
</auxeqpwp>
```

b. Page-based TM stylesheet output example for ***<auxeqpwp>***:



TM X–XXXX–XXX 0062

FIELD MAINTENANCE
BLANK FIRING ATTACHMENT (M15A2) MAINTENANCE

INITIAL SETUP:

Materials/Parts
Cleaner, lubricant, and preservative (CLP) (WP 1132, Item 9)

Materials/Parts (cont.)
Coating compound, enamel (RedM16A2) (WP 1132, Item 23)

WARNING

- Do not keep live ammunition near the work area.
- Only blank cartridge M200 is to be used when the blank firing attachment is attached to the weapon.

GENERAL

This work package contains information and instructions to keep auxiliary equipment used with your weapon in good repair.

REMOVAL

CAUTION

Do not use tools to tighten the blank firing attachment. **USE HANDS ONLY**.

1. Unscrew slide (Figure 1 , Item 1) to remove from compensator (Figure 1 , Item 2).
2. Unhook blank firing attachment (Figure 1 , Item 3) from behind the first groove of compensator (Figure 1 , Item 2).

Figure 1. Unhook blank firing from compensator.

3. Screw slide (Figure 2 , Item 1) all the way in on blank firing attachment (Figure 2 , Item 3).

Figure 2. Blank firing attachment.

END OF TASK

CLEANING

Clean blank firing attachment with CLP. wipe dry. and coat with CLP.

END OF TASK

0062-1

FIGURE 398. Example of a page-based TM stylesheet output for <auxeqpwp>.

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TM X–XXXX–XXX 0062

PAINTING

Inspect blank firing attachment for cracks or distortion. Be sure the parts in the slide are clear and clean. If blank firing attachment is cracked or distorted, it is unserviceable.

END OF TASK

REPLACEMENT

Replace blank firing attachment if unserviceable.

END OF TASK

INSTALLATION

NOTE

M23 blank firing attachment is stamped "M4 Carbine Only" painted yellow and may be used on the M4 and M4A1 carbines. M15A2 BFA is painted red and is used on the M16A2 rifle.

1. Unscrew and pull slide (Figure 3 , Item 1) all the way out on blank firing attachment (Figure 3 , Item 2).
2. Hook blank firing attachment (Figure 3 , Item 2) behind the first groove of the compensator (Figure 3 , Item 3).

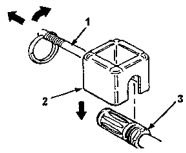

Figure 3. Installation of compensator.

END OF TASK

END OF WORK PACKAGE

0062-2

FIGURE 398. Example of a page-based TM stylesheet output for <auxeqpwp> - Continued.

37.22 Ammunition maintenance work package ***<ammowp>***. The element contains all procedures required for the care and handling of ammunition, information for the disposition of defective ammunition, and the use of authorized cleaning materials and paint.

a. The components of ***<ammowp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (8) One or more of the following is required:
      (a) Ammunition markings ***<ammo.markings>*** (see 33.1 for task development), contains the procedure(s) for marking ammunition and ammunition containers.
      (b) Ammunition defect ***<ammo.defect>*** (see 33.1 for task development), contains the procedure(s) for performing visual inspection of ammunition/containers and includes classification and disposition of defective ammunition/containers.
      (c) Ammunition handling ***<ammo.handling>*** (see 37.7.1.3)
      (d) Service ***<service>*** (see 33.1 for task development), contains the procedure(s) for use of cleaning materials and paint authorized for use in the specified maintenance operations.

b. DTD graphically depicted for ***<ammowp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

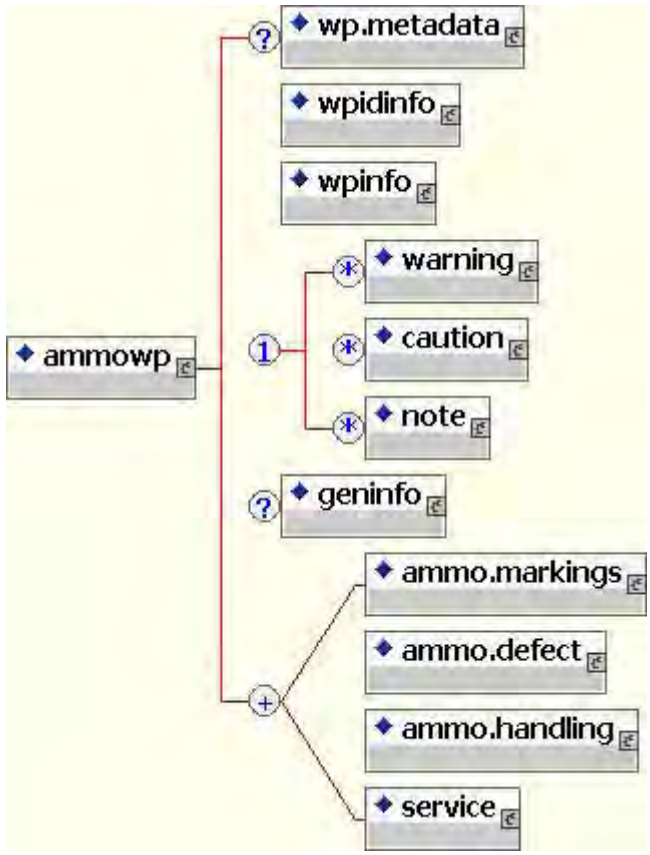

FIGURE 399. Ammunition maintenance work package DTD hierarchy <ammowp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<ammowp>***:

```
<!ELEMENT ammowp (wp.metadata?, wpidinfo, wpinfo, (warning*, caution*,
   note*), geninfo?, (ammo.markings | ammo.defect | ammo.handling |
   service)+)>
<!ATTLIST ammowp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes for ***<ammowp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.22.1 XML document instance fragment and output for ***<ammowp>***. The XML instance and its stylesheet output for a ***<ammowp>*** is provided below.

a. Example of an XML document instance fragment for ***<ammowp>***:

```
<ammowp wpno="M00341-X-XXX-XXX" wpseq="0542">
<wpidinfo>
<maintlvl level="field"/>
<title>Unpacking, Packing, and Replacing Anti-Pilferage Seal, Gasket, and
Packing Materials for Mine Canister Shipping and Storage Container
</title>
</wpidinfo>
<wpinfo>
<tools>
<tools-setup-item>
<name>Cutting Pliers
</name>
<itemref>
<xref wpid="S00121-X-XXX-XXX" itemid="S00121-X-XXX-XXX-item2"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Pliers
</name>
<itemref>
<xref wpid="S00121-X-XXX-XXX" itemid="S00121-X-XXX-XXX-item5"/>
</itemref>
</tools-setup-item>
</tools>
<mtrlpart>
<mtrlpart-setup-item>
<name>Anti-pilferage seal
</name>
<itemref>
<xref wpid="S00119-X-XXX-XXX" itemid="S00121-X-XXX-XXX-item2"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Gasket
</name>
<itemref>
<xref wpid="S00119-X-XXX-XXX" itemid="S00121-X-XXX-XXX-item6"/>
</itemref>
```


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```
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Foam end pad
</name>
<itemref>
<xref wpid="S00119-X-XXX-XXX" itemid="S00121-X-XXX-XXX-item10"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Bottom foam end pad
</name>
<itemref>
<xref wpid="S00119-X-XXX-XXX" itemid="S00121-X-XXX-XXX-item11"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Support cover
</name>
<itemref>
<xref wpid="S00119-X-XXX-XXX" itemid="S00121-X-XXX-XXX-item13"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Support cover
</name>
<itemref>
<xref wpid="S00119-X-XXX-XXX" itemid="S00121-X-XXX-XXX-item14"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Sleeve spacer
</name>
<itemref>
<xref wpid="S00119-X-XXX-XXX" itemid="S00121-X-XXX-XXX-item25"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Strap Assembly
</name>
<itemref>
<xref wpid="S00119-X-XXX-XXX" itemid="S00121-X-XXX-XXX-item27"/>
</itemref>
</mtrlpart-setup-item>
</mtrlpart>
<persnreq>
<persnreq-setup-item>
<name>Person
</name>
<qty>2
</qty>
</persnreq-setup-item>
</persnreq>
</wpinfo>
<geninfo>
<title>General
</title>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<para>This work package describes unpacking and packing procedures for mine canister, shipping and storage container. It also give instructions for replacement of anti-pilferage seal, the gasket, and the packaging materials of the shipping and storage containers.
</para>
</geninfo>
<ammo.handling>
<title>Ammunition Handling
</title>
<ammo.unpacking>
<title>Unpacking Mine Canister Shipping Container
</title>
<warning>
<trim.para>Unloading and loading canisters requires two people. Failure to comply may result in personal injury.
</trim.para>
</warning>
<note>
<trim.para>Canisters may be unloaded or loaded intro free or stacked and palletized shipping containers.
</trim.para>
</note>
<step1>
<para>Using cutting pliers, cut anti-pilferage seal
<callout assocfig="M00341-X-XXX-XXX-fig1" label="1"/>.
</para>
</step1>
<step1>
<para>Remove shipping container cap
<callout assocfig="M00341-X-XXX-XXX-fig1" label="2"/> by lifing locking latch
<callout assocfig="M00341-X-XXX-XXX-fig1" label="3"/> and turning cap to left. Pull cap from container
<callout assocfig="M00341-X-XXX-XXX-fig1" label="4"/> and save for reuse.
</para>
</step1>
<step1>
<para>Remove foam end pad by pulling ends of straps
<callout assocfig="M00341-X-XXX-XXX-fig1" label="5"/> out of container
<callout assocfig="M00341-X-XXX-XXX-fig1" label="4"/> and lifting top foam end pad
<callout assocfig="M00341-X-XXX-XXX-fig1" label="6"/> from con-tainer. Save end pad for reuse.
</para>
</step1>
<step1>
<para>Remove rimmed canister support cover by pulling on straps
<callout assocfig="M00341-X-XXX-XXX-fig1" label="5"/> to lift cover
<callout assocfig="M00341-X-XXX-XXX-fig1" label="7"/> free from container
<callout assocfig="M00341-X-XXX-XXX-fig1" label="4"/>. Save cover for reuse.
</para>
</step1>
<step1>
<para>Remove first canister by pulling on straps
<callout assocfig="M00341-X-XXX-XXX-fig1" label="5"/> until canister breech
<callout assocfig="M00341-X-XXX-XXX-fig1" label="8"/> is exposed from container. Pull canister from container and set first canister aside.
</para>

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```
</step1>
<step1>
<para>Continue pulling on straps
<callout assocfig="M00341-X-XXX-XXX-fig1" label="5"/> to lift foam bumper pad
<callout assocfig="M00341-X-XXX-XXX-fig1" label="9"/> from con-
tainer.  Save pad for reuse.
</para>
</step1>
<step1>
<para>Remove rimless canister support cover by pulling on straps
<callout assocfig="M00341-X-XXX-XXX-fig1" label="5"/> to lift cover
<callout assocfig="M00341-X-XXX-XXX-fig1" label="10"/> free from
container.  Save cover for reuse.
</para>
</step1>
<step1>
<para>Remove second canister from container by pulling on straps
<callout assocfig="M00341-X-XXX-XXX-fig1" label="5"/> until canister breech
<callout assocfig="M00341-X-XXX-XXX-fig1" label="11"/> is exposed from
container.  Pull canister from container and set aside.
</para>
</step1>
<step1>
<para>Remove bottom foam end pad
<callout assocfig="M00341-X-XXX-XXX-fig1" label="12"/>, strap assembly
<callout assocfig="M00341-X-XXX-XXX-fig1" label="5"/>, and fireboard spacer
<callout assocfig="M00341-X-XXX-XXX-fig1" label="13"/>.
</para>
</step1>
<figure id="M00341-X-XXX-XXX-fig1">
<title>Mine Canister Shipping Container
</title>
<graphic boardno="ammowp">
</graphic>
</figure>
</ammo.unpacking>
</ammo.handling>
</ammowp>
```

b. Page-based TM stylesheet output example for ***<ammowp>***:

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TM X–XXXX–XXX 0542

FIELD MAINTENANCE

UNPACKING, PACKING, AND REPLACING ANTI-PILFERAGE SEAL, GASKET, AND PACKING MATERIALS FOR MINE CANISTER SHIPPING AND STORAGE CONTAINER

INITIAL SETUP:

Tools and Special Tools
Cutting Pliers (WP 0723, Item 2)
Pliers (WP 0723, Item 5)

Materials/Parts
Anti-pilferage seal (WP 0721, Item 2)
Gasket (WP 0721, Item 6)
Foam end pad (WP 0721, Item 10)
Bottom foam end pad (WP 0721, Item 11)

Materials/Parts (cont.)
Support cover (WP 0721, Item 13)
Support cover (WP 0721, Item 14)
Sleeve spacer (WP 0721, Item 25)
Strap Assembly (WP 0721, Item 27)

Personnel Required
Person (2)

GENERAL

This work package describes unpacking and packing procedures for mine canister, shipping and storage container. It also give instructions for replacement of anti-pilferage seal, the gasket, and the packaging materials of the shipping and storage containers.

AMMUNITION HANDLING

UNPACKING MINE CANISTER SHIPPING CONTAINER

WARNING

Unloading and loading canisters requires two people. Failure to comply may result in personal injury.

NOTE

Canisters may be unloaded or loaded intro free or stacked and palletized shipping containers.

1. Using cutting pliers, cut anti-pilferage seal (Figure 1 , Item 1).
2. Remove shipping container cap (Figure 1 , Item 2) by lifing locking latch (Figure 1 , Item 3) and turning cap to left. Pull cap from container (Figure 1 , Item 4) and save for reuse.
3. Remove foam end pad by pulling ends of straps (Figure 1 , Item 5) out of container (Figure 1 , Item 4) and lifting top foam end pad (Figure 1 , Item 6) from container. Save end pad for reuse.
4. Remove rimmed canister support cover by pulling on straps (Figure 1 , Item 5) to lift cover (Figure 1 , Item 7) free from container (Figure 1 , Item 4). Save cover for reuse.
5. Remove first canister by pulling on straps (Figure 1 , Item 5) until canister breech (Figure 1 , Item 8) is exposed from container. Pull canister from container and set first canister aside.
6. Continue pulling on straps (Figure 1 , Item 5) to lift foam bumper pad (Figure 1 , Item 9) from container. Save pad for reuse.
7. Remove rimless canister support cover by pulling on straps (Figure 1 , Item 5) to lift cover (Figure 1 , Item 10) free from container. Save cover for reuse.
8. Remove second canister from container by pulling on straps (Figure 1 , Item 5) until canister breech (Figure 1 , Item 11) is exposed from container. Pull canister from container and set aside.
9. Remove bottom foam end pad (Figure 1 , Item 12), strap assembly (Figure 1 , Item 5), and fireboard spacer (Figure 1 , Item 13).

0542-1

FIGURE 400. Example of ammunition work package page-based TM stylesheet output for <ammowp>.

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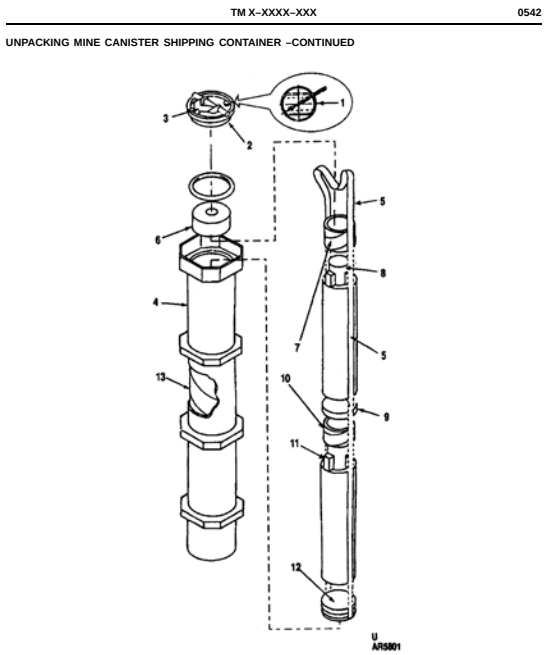

FIGURE 400. Example of ammunition work package page-based TM stylesheet output for <ammowp> - Continued.

37.23 Ammunition marking information work package ***<ammo.markingwp>***. The ammunition marking information work package provides information on ammunition marking ***<ammo.markings>***, classification, identification ***<ammotype>***, care and handling ***<ammo.handling>***, preservation, transportation, authorized rounds, preparation for firing, fuzes, and packing ***<ammo.packing>***.

a. The components of ***<ammo.markingwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (8) One or more of the following is required:
      (a) Ammunition markings ***<ammo.markings>*** (see 33.1 for task development), containing procedures for marking ammunition and ammunition containers.
      (b) Ammunition handling ***<ammo.handling>*** (see 37.7.1.3)
      (c) Ammunition type ***<ammotype>*** (see 37.23.1)

b. DTD graphically depicted for ***<ammo.markingwp>***:

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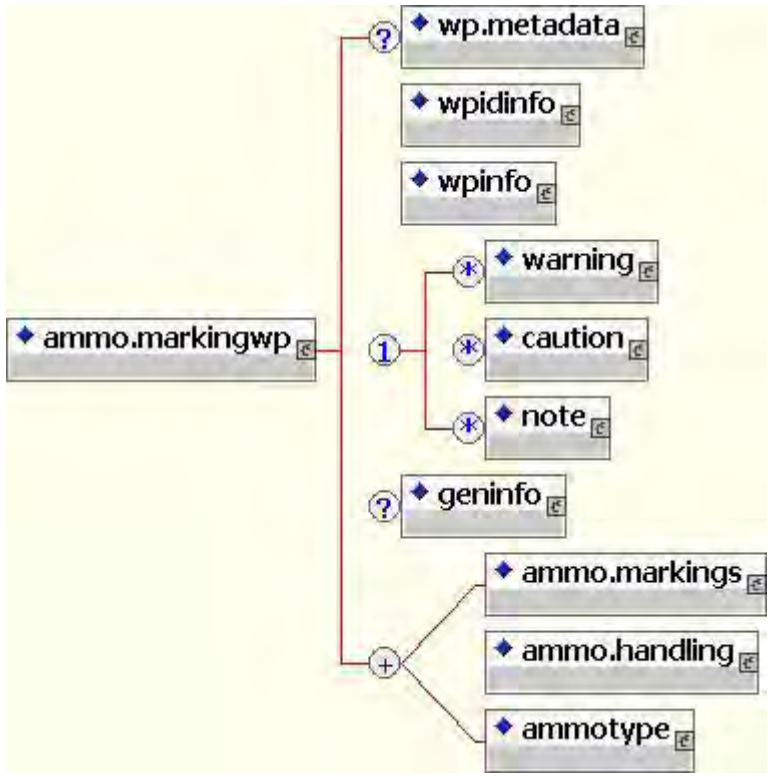

FIGURE 401. Ammunition maintenance work package DTD hierarchy <ammo.markingwp>.

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c. DTD fragment for ***<ammo.markingwp>***:

```
<!ELEMENT ammo.markingwp (wp.metadata?, wpidinfo, wpinfo, (warning*,
   caution*, note*), geninfo?, (ammo.markings | ammo.handling |
   ammotype)+)>
<!ATTLIST ammo.markingwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)           "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attributes for ***<ammo.markingwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.23.1 Ammunition classification and/or identification ***<ammotype>***. The element is used for classifying and/or identifying ammunition in separate task or groups. The description can be implemented using a combination of narrative, tables and illustrations.

a. The components of ***<ammotype>*** are:
   (1) Ammunition name ***<name>*** (required) (see 49.1.4.19)
   (2) Ammunition description ***<para>*** (see 49.1.1.5)/***<specpara>*** (see 49.1.1.6) (required – one or more).
b. DTD graphically depicted for ***<ammo.markingwp>***:

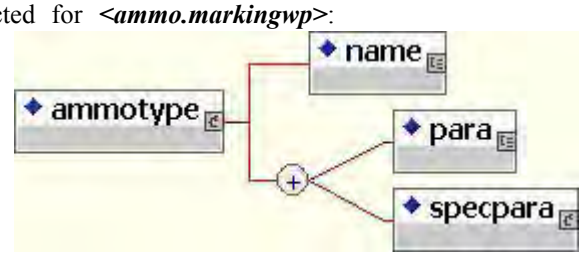

FIGURE 402. Ammunition classification and/or identification DTD hierarchy <ammotype>.

c. DTD fragment for ***<ammotype>***:

```
<!ELEMENT ammotype (name, (para | specpara)+)>
```

37.23.2 XML document instance fragment and output for ***<ammo.markingwp>***. The XML instance and its stylesheet output for a ***<ammo.markingwp>*** is provided below.

a. Example of an XML document instance fragment for ***<ammo.markingwp>***:

<ammo.markingwp airforce="no" army="no" deletewp="no" frame="yes" marines="no" navy="no" tocentry="2" wpno="M00034-X-XXX-XX" wpseq="1839">
<wpidinfo>
<maintlvl level="field"/>
<title>Ammunition Marking Information
</title>
</wpidinfo>
<wpinfo>
<ref-setup-item>
<xref wpid="M00134–X-XXX-XX"/>
</ref-setup-item>
<ref-setup-item>
<extref docno="TM 9-1300-251-20"/>
</ref-setup-item>
</wpinfo>
<ammo.markings esd="no" frame="yes" hcp="no" tocentry="0">

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<title>GENERAL
</title>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Ammunition for the M199 cannon is the separate loading type. The loading of each complete round into the cannon requires three separate operations: loading the fuzed projectile, the propelling charge, and the primer.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">These components are shipped separately; therefore, the cannon crew must know how to store, unpack, inspect, prepare, and load each complete round every time the weapon is fired.
</para>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">The chief of section supervises the loading and the preparation duties performed by cannoneers.
</para>
</step2>
<step2 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">The chief of section must also see that the cannoneers and driver are cross-trained in the specific duties of the care, handling, unpacking, inspection, preparation, and loading of the ammunition components in order to sustain a 24-hour operation or to operate with a reduced crew.
</para>
</step2>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">It is planned that future ammo for all new 155mm weapons will be interchangeable. This will enable projectiles and propelling changes of one NATO nation to be fired from the 155mm weapons of all others. Current items of interchangeability are contained in Chapter 5.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">For maintenance of ammunition,
<extref docno="TM 9-1300-251-20" pretext="see "/>.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<specpara esd="no" hcp="no">
<warning haz-abbrev="no">
<trim.para esd="no" hcp="no">Until safety and reliability testing is completed, the use of ammo other than prescribed in this manual is prohibited.
</trim.para>
</warning>
<warning haz-abbrev="no">
<trim.para esd="no" hcp="no">
<ftnote id="note1" label="1" mark="ctr">
<ftnpara>Do not fire the M650 projectile if the obturating band is missing or broken. If the band is displaced and can be repositioned and remain in the groove, the projectile can be fired.
</ftnpara>

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Check the source to verify that this is the current version before use.

</ftnote> Do not fire the M650 projectile if the obturating band is missing or broken. If the band is displaced and can be repositioned and remain in the groove, the projectile can be fired.
</trim.para>
</warning>
<note acknowledge="no">
<trim.para esd="no" hcp="no">
<ftnote id="note2" label="2" mark="ctr">
<ftnpara>M728 and M732 fired only with "VX" projectile and only in combat emergency.
</ftnpara>
</ftnote> M728 andM732 fired only with "VX" projectile and only in combat emergency.
</trim.para>
<trim.para esd="no" hcp="no">
<ftnote id="note3" label="3" mark="ctr">
<ftnpara>Fuze, MTSQ, M564 is restricted from firing with zone 9 M188A1 propelling charge.
</ftnpara>
</ftnote> Fuze, MTSQ, M564 is restricted from firing with zone 9 M188A1 propelling charge.
</trim.para>
<trim.para esd="no" hcp="no">
<ftnote id="note4" label="4" mark="ctr">
<ftnpara>Authorized, requires removal of supplementary charge.
</ftnpara>
</ftnote> Authorized, requires removal of supplementary charge.
</trim.para>
</note>
<para esd="no" hcp="no">Refer to <xref wpid="M00134–X-XXX-XX"/> for information about the Loose Projectile Restraint System (LPRS). The LPRS is a divider rack for securing loose unfuzed projectiles for transportation in a field artillery companion vehicle.
<table tocentry="1">
<title>Authorized Projectile Fuze Combinations for 8-Inch Howitzer, M110A2 Cannon M201A1.
</title>
<tgroup align="left" char=" " charoff="50" cols="12">
<colspec colname="col1" colwidth="1.80*"/>
<colspec colname="col2" colwidth="0.93*"/>
<colspec colname="col3" colwidth="0.94*"/>
<colspec colname="col4" colwidth="0.93*"/>
<colspec colname="col5" colwidth="0.91*"/>
<colspec colname="col6" colwidth="0.93*"/>
<colspec colname="col7" colwidth="0.94*"/>
<colspec colname="col8" colwidth="0.93*"/>
<colspec colname="col9" colwidth="0.96*"/>
<colspec colname="col10" colwidth="0.93*"/>
<colspec colname="col11" colwidth="0.94*"/>
<colspec colname="col12" colwidth="0.94*"/>
<thead valign="bottom">
<row>
<entry morerows="2" valign="top">TYPE AND MODEL NUMBER OF PROJECTILE
</entry>
<entry nameend="col12" namest="col2" valign="top">FUZE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</entry>
</row>
<row>
<entry  nameend="col3"  namest="col2"  valign="top">PD
</entry>
<entry  nameend="col5"  namest="col4"  valign="top">MT
</entry>
<entry  nameend="col8"  namest="col6"  valign="top">MTSQ
</entry>
<entry  nameend="col10"  namest="col9"  valign="top">PROX(VT)
</entry>
<entry  nameend="col12"  namest="col11"  valign="top">ET
</entry>
</row>
<row>
<entry  valign="middle">M739  SERIES
</entry>
<entry  valign="middle">M1557
</entry>
<entry  valign="middle">M572
</entry>
<entry  valign="middle">M565
</entry>
<entry  valign="middle">M564
</entry>
<entry  valign="middle">M557  SERIES
</entry>
<entry  valign="middle">M582  SERIES
</entry>
<entry  valign="middle">M728
</entry>
<entry  valign="middle">M732
</entry>
<entry  valign="middle">M762
</entry>
<entry  valign="middle">M767
</entry>
</row>
</thead>
<tbody  valign="top">
<row>
<entry  valign="top">Agent  GB,  VX,  M426
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
```


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Check the source to verify that this is the current version before use.

```
</entry>
<entry  align="center"  valign="middle">X
<ftnref  xrefid="note2"/>
</entry>
<entry  align="center"  valign="middle">X
<ftnref  xrefid="note2"/>
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
</row>
<row>
<entry  valign="top">HE,  M106  (Shallow  Cavity)
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
<ftnref  xrefid="note3"/>
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
</row>
<row>
<entry  valign="top">HE,  M106  (Deep  Cavity)
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
<ftnref  xrefid="note3"/>
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</entry>
<entry  align="center"  valign="middle">X
<ftnref  xrefid="note4"/>
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
</row>
<row>
<entry  valign="top">HE,  M404  ICM
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
</row>
<row>
<entry  valign="top">HE,  M509A1  ICM
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
```

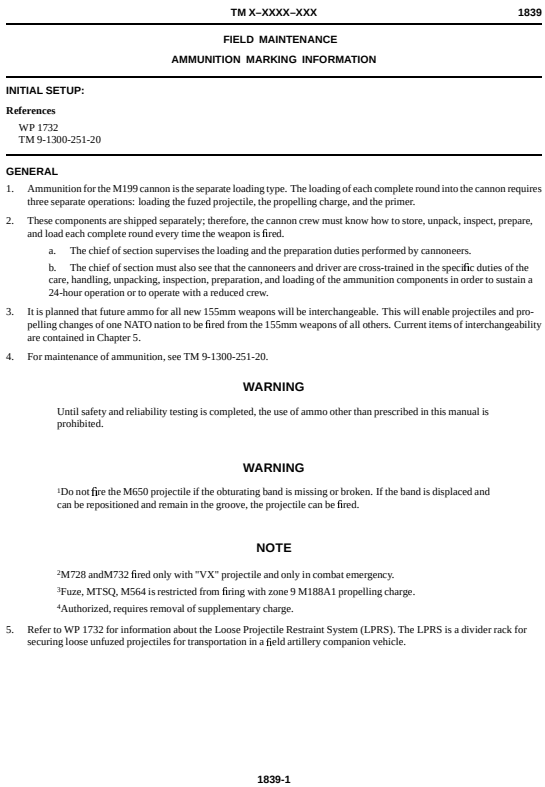

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
</row>
<row>
<entry  valign="top"> HERA, M650 (Rocket Only)
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
</row>
<row>
<entry  valign="top">HERA, M650 (Rocket Only)
<ftnref  xrefid="note1"/>
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
<ftnref  xrefid="note3"/>
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<entry  align="center"  valign="middle">
</entry>
<entry  align="center"  valign="middle">X
</entry>
</row>
</tbody>
</tgroup>
</table>
</para>
</specpara>
</step1>
</ammo.markings>
</ammo.markingwp>
```

b. Page-based TM stylesheet output example for ***<ammo.markingwp>***:



TM X–XXXX–XXX 1839

FIELD MAINTENANCE

AMMUNITION MARKING INFORMATION

INITIAL SETUP:

References

WP 1732
TM 9-1300-251-20

GENERAL

1. Ammunition for the M199 cannon is the separate loading type. The loading of each complete round into the cannon requires three separate operations: loading the fuzed projectile, the propelling charge, and the primer.
2. These components are shipped separately; therefore, the cannon crew must know how to store, unpack, inspect, prepare, and load each complete round every time the weapon is fired.
   a. The chief of section supervises the loading and the preparation duties performed by cannoneers.
   b. The chief of section must also see that the cannoneers and driver are cross-trained in the specific duties of the care, handling, unpacking, inspection, preparation, and loading of the ammunition components in order to sustain a 24-hour operation or to operate with a reduced crew.
3. It is planned that future ammo for all new 155mm weapons will be interchangeable. This will enable projectiles and propelling changes of one NATO nation to be fired from the 155mm weapons of all others. Current items of interchangeability are contained in Chapter 5.
4. For maintenance of ammunition, see TM 9-1300-251-20.

WARNING

Until safety and reliability testing is completed, the use of ammo other than prescribed in this manual is prohibited.

WARNING

[1]Do not fire the M650 projectile if the obturating band is missing or broken. If the band is displaced and can be repositioned and remain in the groove, the projectile can be fired.

NOTE

[2]M728 andM732 fired only with "VX" projectile and only in combat emergency.

[3]Fuze, MTSQ, M564 is restricted from firing with zone 9 M188A1 propelling charge.

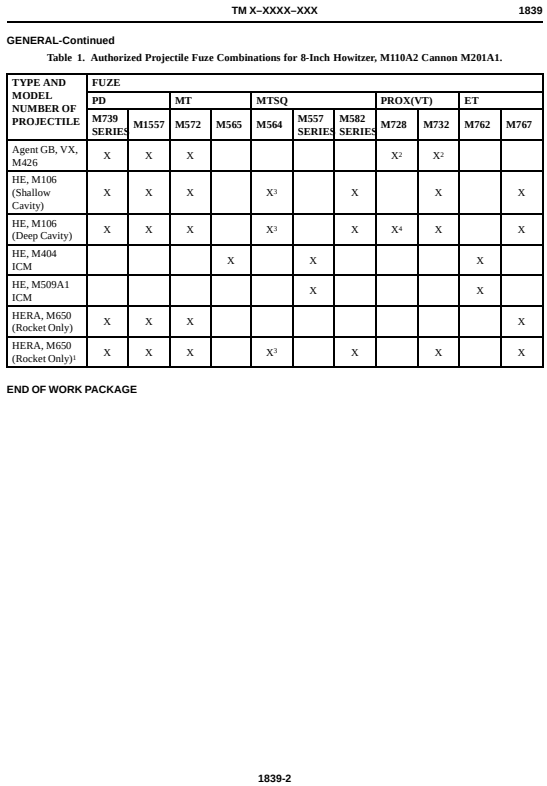
[4]Authorized, requires removal of supplementary charge.

5. Refer to WP 1732 for information about the Loose Projectile Restraint System (LPRS). The LPRS is a divider rack for securing loose unfuzed projectiles for transportation in a field artillery companion vehicle.

1839-1

FIGURE 403. Example of a page-based TM stylesheet output for <ammo.markingwp>.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.



TM X–XXXX–XXX 1839

GENERAL-Continued

Table 1. Authorized Projectile Fuze Combinations for 8-Inch Howitzer, M110A2 Cannon M201A1.

TYPE AND MODEL NUMBER OF PROJECTILE	FUZE										
	PD		MT		MTSQ			PROX(VT)		ET	
	M739 SERIES	M1557	M572	M565	M564	M557 SERIES	M582 SERIES	M728	M732	M762	M767
Agent GB, VX, M426	X	X	X					X[2]	X[2]		
HE, M106 (Shallow Cavity)	X	X	X		X[3]		X		X		X
HE, M106 (Deep Cavity)	X	X	X		X[3]		X	X[4]	X		X
HE, M404 ICM				X		X				X	
HE, M509A1 ICM						X				X	
HERA, M650 (Rocket Only)	X	X	X								X
HERA, M650 (Rocket Only)[1]	X	X	X		X[3]		X		X		X

END OF WORK PACKAGE

1839-2

FIGURE 403. Example of a page-based TM stylesheet output for <ammo.markingwp> - Continued.

37.24 Foreign ammunition (NATO) work package ***<natowp>***. The foreign ammunition (NATO) work package contains the special requirements for foreign ammunition marking, classification, identification, handling, transportation, preparation for firing, and other similar data when applicable.

a. The components of ***<natowp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) General information ***<geninfo>*** (optional) (see 49.1.4.12).
   (8) One or more of the following is required:
      (a) Ammunition markings ***<ammo.markings>*** (see 33.1 for task development), containing procedures for marking ammunition and ammunition containers.
      (b) Ammunition handling ***<ammo.handling>*** (see 37.7.1.3)
      (c) Ammunition type ***<ammotype>*** (see 37.23.1)

b. DTD graphically depicted for ***<natowp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

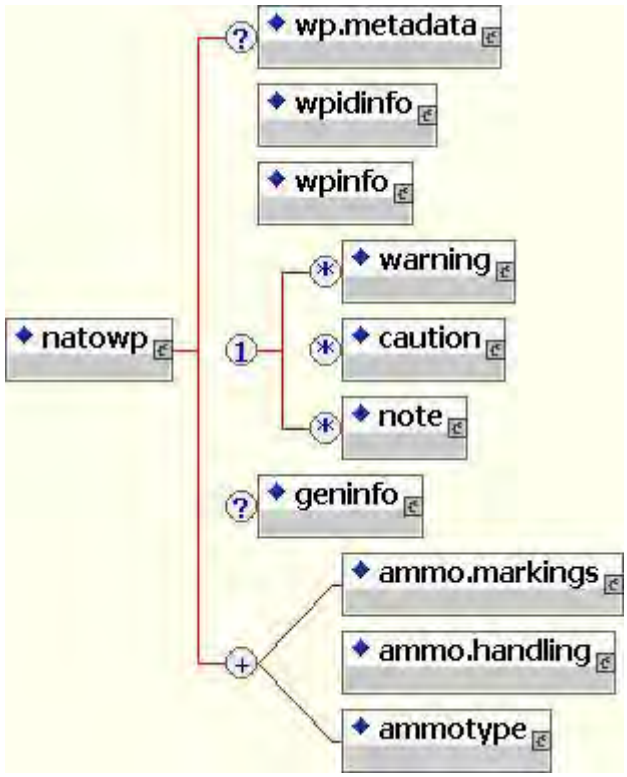


FIGURE 404. Foreign ammunition (NATO) work package DTD hierarchy <natowp>.

c. DTD fragment for ***<natowp>***:

```
<!ELEMENT natowp (wp.metadata?, wpidinfo, wpinfo, (warning*, caution*,
   note*), geninfo?, (ammo.markings | ammo.handling | ammotype)+)>
<!ATTLIST natowp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time)  #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

d. Common attributes for ***<natowp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.24.1 XML document instance fragment and output for ***<natowp>***. The XML instance and its stylesheet output for a ***<natowp>*** is provided below.

a. Example of an XML document instance fragment for ***<natowp>***:

```
<natowp airforce="no" army="no" deletewp="no" frame="yes" marines="no" navy="no"
tocentry="2" wpno="M02543-X-XXXX-XXX" wpseq="0101">
<wpidinfo>
<maintlvl level="field"/>
<title>Foreign Ammunition (NATO)
</title>
</wpidinfo>
<wpinfo>
<title>N/A
</title>
<null insert="none"/>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</wpinfo>
<geninfo frame="no">
<para0 esd="no" hcp="no">
<title>GENERAL
</title>
<para esd="no" hcp="no">Agreements between the United States and NATO allies have established the interoperability of weapon systems and ammunition of the nations. The agreements enable the safe and effective firing of major types of ammunition of the same size from the same compatible size and type weapon of the NATO armies.
</para>
<para esd="no" hcp="no">The following pages cover only authorized German (GE), United Kingdom (UK), Canadian (CA), Netherlands (NL), French (FR), Norwegian (NO), Italian (IT), Danish (DA), Greek (GR), or Belgian (BE) 155mm components. If a munitions item has not yet been authorized, it is because it has not yet been determined to be safe to fire or it has been determined that the munitions item cannot be safely fired from the US weapon system.
</para>
</para0>
</geninfo>
<ammo.markings esd="no" frame="yes" hcp="no" tocentry="0">
<title>M198 Howitzers
</title>
<warning haz-abbrev="no">
<warning.group haz-abbrev="no">
<trim.para esd="no" hcp="no">Only under emergency combat conditions will zone 1 of the M3A1 and DM62 propelling charge be fired from the M199 cannon tube of the M198 howitzer weapon system.
</trim.para>
</warning.group>
<warning.group haz-abbrev="no">
<trim.para esd="no" hcp="no">Do not mix US, GE, UK, CA, NL, FR, NO, IT, OA, GR, or BE components (i.e., projectile, propelling charge, flash reducer, fuze). Fire only components from one nation, except GE primer DM9141. GE must use DM191A1 when firing US, GE, UK, CA, NL, FR, NO, IT, OA, GR, or BE 155mm munitions. Mixing may cause injury to personnel.
</trim.para>
</warning.group>
</warning>
<note acknowledge="no">
<trim.para esd="no" hcp="no">At the conclusion of any training exercise, ammunition drawn from a NATO, nation and not fired should be returned to the troops of the NATO nation from whom it was obtained.
</trim.para>
</note>
<para>
</para>
</ammo.markings>
<ammotype>
<name>AUTHORIZED PROJECTILES — GE
</name>
<para esd="no" hcp="no">The following GE munitions are authorized for use in M198 howitzers:
<deflist>
<term.def>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<term>Projectile
</term>
<def>
<para esd="no" hcp="no">155mm, HE, M107 (TNT-loaded only)
</para>
</def>
</term.def>
<term.def>
<term>Charge, propelling
</term>
<def>
<para esd="no" hcp="no">Green bag, M3A1, zones 1-5
</para>
</def>
</term.def>
<term.def>
<term>Charge, propelling
</term>
<def>
<para esd="no" hcp="no">White bag, M4A2, zones 3-7
</para>
</def>
</term.def>
<term.def>
<term>Fuze
</term>
<def>
<para esd="no" hcp="no">Point-detonating, M557, L85A2
</para>
</def>
</term.def>
<term.def>
<term>Primer
</term>
<def>
<para esd="no" hcp="no">M82
</para>
</def>
</term.def>
</deflist>
</para>
</ammotype>
<ammotype>
<name>AUTHORIZED PROJECTILES — UK
</name>
<para esd="no" hcp="no">The following UK munitions are authorized for use in M198 howitzers:
<deflist>
<term.def>
<term>Projectile
</term>
<def>
<para esd="no" hcp="no">155mm, HE, M107 (TNT-loaded only)
</para>
</def>
</term.def>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<term.def>
<term>Charge, propelling
</term>
<def>
<para esd="no" hcp="no">Green bag, M3A1, zones 1-5
</para>
</def>
</term.def>
<term.def>
<term>Charge, propelling
</term>
<def>
<para esd="no" hcp="no">White bag, M4A2, zones 3-7
</para>
</def>
</term.def>
<term.def>
<term>Fuze
</term>
<def>
<para esd="no" hcp="no">Point-detonating, M557, L85A2
</para>
</def>
</term.def>
<term.def>
<term>Primer
</term>
<def>
<para esd="no" hcp="no">M82
</para>
</def>
</term.def>
</deflist>
</para>
</ammotype>
<ammotype>
<name>AUTHORIZED PROJECTILES — CA
</name>
<specpara esd="no" hcp="no">
<note acknowledge="no">
<trim.para esd="no" hcp="no">M3 and M4A 1 propelling charges do not have flash reducers.
</trim.para>
</note>
<para esd="no" hcp="no">The following CA munitions are authorized for use in M198 howitzers:
<deflist>
<term.def>
<term>Projectile
</term>
<def>
<para esd="no" hcp="no">155mm, HE, M107
</para>
</def>
</term.def>
<term.def>

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Check the source to verify that this is the current version before use.

<term>Charge, propelling
</term>
<def>
<para esd="no" hcp="no">M3, M3A1, M4A1, M4A2
</para>
</def>
</term.def>
<term.def>
<term>Fuze
</term>
<def>
<para esd="no" hcp="no">Point-detonating, M557, M564 proximity, M514A1
</para>
</def>
</term.def>
<term.def>
<term>Primer
</term>
<def>
<para esd="no" hcp="no">M82
</para>
</def>
</term.def>
</deflist>
</para>
</specpara>
</ammotype>
<ammotype>
<name>AUTHORIZED PROJECTILES — NL
</name>
<specpara esd="no" hcp="no">
<note acknowledge="no">
<trim.para esd="no" hcp="no">The M107C1 projectile, M3C1 and M4C3 propelling charges, M557C1 fuze, and M82C1 primer are NL manufacture.
</trim.para>
</note>
<para esd="no" hcp="no">The following NL munitions are authorized for use in M198 howitzers:
<deflist>
<term.def>
<term>Projectile
</term>
<def>
<para esd="no" hcp="no">155mm, HE, M107, M107C1 (TNT-loaded only)
</para>
</def>
</term.def>
<term.def>
<term>Charge, propelling
</term>
<def>
<para esd="no" hcp="no">M3C1, M4C3, M4A1
</para>
</def>
</term.def>
<term.def>

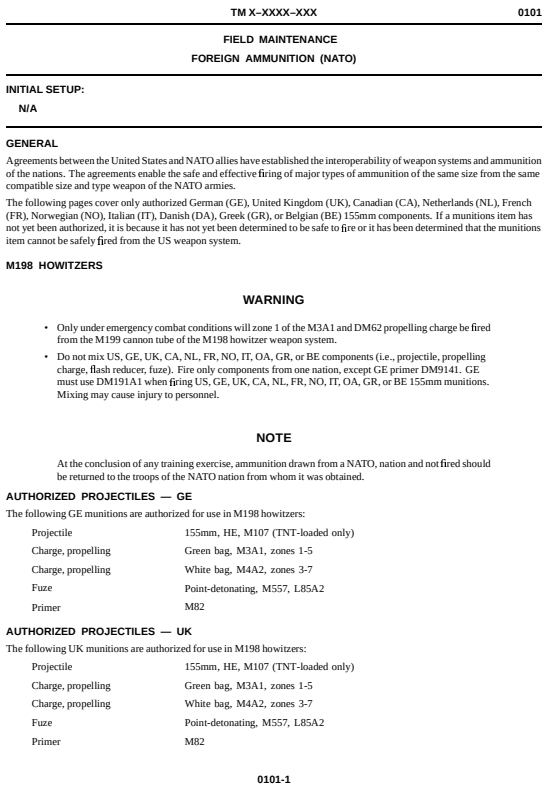
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<term>Fuze
</term>
<def>
<para esd="no" hcp="no">Point-detonating, M557, M557C1
</para>
</def>
</term.def>
<term.def>
<term>Primer
</term>
<def>
<para esd="no" hcp="no">M82, M82C1
</para>
</def>
</term.def>
</deflist>
</para>
</specpara>
</ammotype>
<ammotype>
<name>AUTHORIZED PROJECTILES — FR
</name>
<para esd="no" hcp="no">The following FR munitions are authorized for use in M198 howitzers:
<deflist>
<term.def>
<term>Projectile
</term>
<def>
<para esd="no" hcp="no">155mm, HE, M107
</para>
</def>
</term.def>
<term.def>
<term>Charge, propelling
</term>
<def>
<para esd="no" hcp="no">Green bag, M3, zones 1-5
</para>
</def>
</term.def>
<term.def>
<term>Charge, propelling
</term>
<def>
<para esd="no" hcp="no">White bag, M4A1, zones 5-7
</para>
</def>
</term.def>
<term.def>
<term>Fuze
</term>
<def>
<para esd="no" hcp="no">Point-detonating, M557
</para>
</def>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</term.def>
<term.def>
<term>Primer
</term>
<def>
<para esd="no" hcp="no">Use US M82 only, not MK2A4.
</para>
</def>
</term.def>
</deflist>
</para>
</ammotype>
</natowp>
```

b. Page-based TM stylesheet output example for ***<natowp>***:



TM X–XXXX–XXX 0101

FIELD MAINTENANCE
FOREIGN AMMUNITION (NATO)

INITIAL SETUP:

N/A

GENERAL

Agreements between the United States and NATO allies have established the interoperability of weapon systems and ammunition of the nations. The agreements enable the safe and effective firing of major types of ammunition of the same size from the same compatible size and type weapon of the NATO armies.

The following pages cover only authorized German (GE), United Kingdom (UK), Canadian (CA), Netherlands (NL), French (FR), Norwegian (NO), Italian (IT), Danish (DA), Greek (GR), or Belgian (BE) 155mm components. If a munitions item has not yet been authorized, it is because it has not yet been determined to be safe to fire or it has been determined that the munitions item cannot be safely fired from the US weapon system.

M198 HOWITZERS

WARNING

- Only under emergency combat conditions will zone 1 of the M3A1 and DM62 propelling charge be fired from the M199 cannon tube of the M198 howitzer weapon system.
- Do not mix US, GE, UK, CA, NL, FR, NO, IT, OA, GR, or BE components (i.e., projectile, propelling charge, flash reducer, fuze). Fire only components from one nation, except GE primer DM9141. GE must use DM191A1 when firing US, GE, UK, CA, NL, FR, NO, IT, OA, GR, or BE 155mm munitions. Mixing may cause injury to personnel.

NOTE

At the conclusion of any training exercise, ammunition drawn from a NATO, nation and not fired should be returned to the troops of the NATO nation from whom it was obtained.

AUTHORIZED PROJECTILES — GE

The following GE munitions are authorized for use in M198 howitzers:

Projectile	155mm, HE, M107 (TNT-loaded only)
Charge, propelling	Green bag, M3A1, zones 1-5
Charge, propelling	White bag, M4A2, zones 3-7
Fuze	Point-detonating, M557, L85A2
Primer	M82

AUTHORIZED PROJECTILES — UK

The following UK munitions are authorized for use in M198 howitzers:

Projectile	155mm, HE, M107 (TNT-loaded only)
Charge, propelling	Green bag, M3A1, zones 1-5
Charge, propelling	White bag, M4A2, zones 3-7
Fuze	Point-detonating, M557, L85A2
Primer	M82

0101-1

FIGURE 405. Example of a page-based TM stylesheet output for <natowp>.

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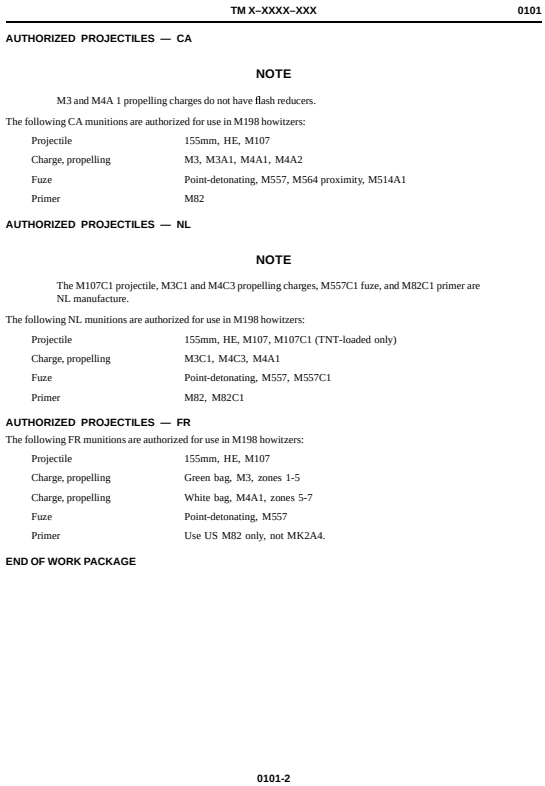
Check the source to verify that this is the current version before use.



TM X–XXXX–XXX 0101

AUTHORIZED PROJECTILES — CA

NOTE

M3 and M4A 1 propelling charges do not have flash reducers.

The following CA munitions are authorized for use in M198 howitzers:

Projectile	155mm, HE, M107
Charge, propelling	M3, M3A1, M4A1, M4A2
Fuze	Point-detonating, M557, M564 proximity, M514A1
Primer	M82

AUTHORIZED PROJECTILES — NL

NOTE

The M107C1 projectile, M3C1 and M4C3 propelling charges, M557C1 fuze, and M82C1 primer are NL manufacture.

The following NL munitions are authorized for use in M198 howitzers:

Projectile	155mm, HE, M107, M107C1 (TNT-loaded only)
Charge, propelling	M3C1, M4C3, M4A1
Fuze	Point-detonating, M557, M557C1
Primer	M82, M82C1

AUTHORIZED PROJECTILES — FR

The following FR munitions are authorized for use in M198 howitzers:

Projectile	155mm, HE, M107
Charge, propelling	Green bag, M3, zones 1-5
Charge, propelling	White bag, M4A1, zones 5-7
Fuze	Point-detonating, M557
Primer	Use US M82 only, not MK2A4.

END OF WORK PACKAGE

0101-2

FIGURE 405. Example of a page-based TM stylesheet output for <natowp> - Continued.

37.25 Preventive maintenance services (PMS)/preventive maintenance daily (PMD) inspection work package ***<pms-inspecwp>***. The element contains each specific inspection interval (i.e,, daily, intermediate, periodic, 10 hour/14 day, 30 hour/42 day, etc.). Within the work package, an inspection checklist has been created for each areas of the aircraft (i.e., nose, fuselage, tail, etc.) and is developed in a table. Generally an illustration(s) is associated with each inspection checklist showing the inspection locations. The “FOD REMINDER” statement is inserted by the stylesheet or preprinted pages.

a. The components of ***<pms-inspecwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) Inspection checklist (required – one or more) components are:
      (a) Illustration ***<figure>*** (optional) (see 44.1.1).
      (b) Inspection checklist ***<table>*** (required) (see 42).

b. DTD graphically depicted for ***<pms-inspecwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

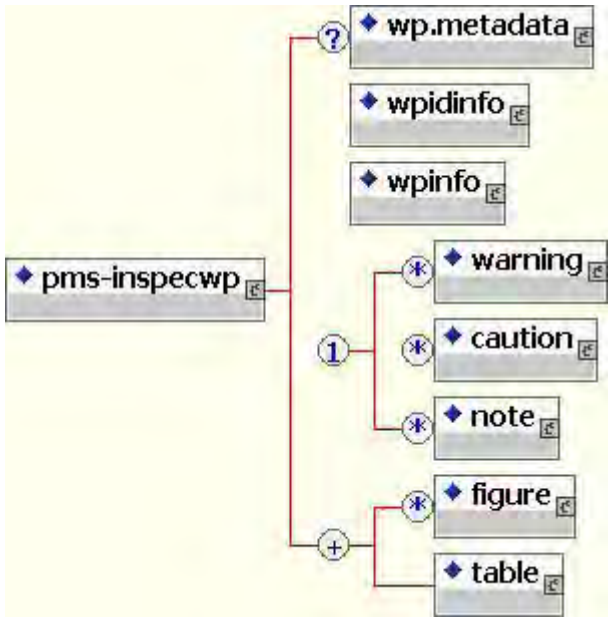


FIGURE 406. Preventive maintenance services inspection work package DTD hierarchy <pms-inspecwp>.

c. DTD fragment for ***<pms-inspecwp>***:

```
<!ELEMENT pms-inspecwp (wp.metadata?, wpidinfo, wpinfo, (warning*,
   caution*, note*), (figure?, table)+)>
<!ATTLIST pms-inspecwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attributes for ***<pms-inspecwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

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(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.25.1 Preventive maintenance services inspection work package general entity boilerplates. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed (see Table XXIII).

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TABLE XXIII. Boilerplate Entities for *<pms-inspecwp>*.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Avionics equipment inspections reference statement	***&pms-inspecwp.perform;***	— —
• Avionic equipment TM number		***<!ENTITY pms-inspecwp.perform.tm "<extref docno='TM 11-(INSERT TM NUMBER)'/>"/>***
Forms and records initial inspection statement	***&pms-inspecwp.checklist;***	Not applicable
Lubrication requirements statement	***%pms-inspecwp.lubrication;***	— —
• Lubrication TM number		***<!ENTITY pms-inspecwp.lubrication.tm "<extref docno='TM 55-(INSERT TM NUMBER)'/>"/>***
Forms and records completion statement	***&pms-inspecwp.last-item;***	Not applicable

37.25.2 XML document instance fragment and output for ***<pms-inspecwp>***. The XML instance and its stylesheet output for a ***<pms-inspecwp>*** is provided below.

a. Example of an XML document instance fragment for ***<pms-inspecwp>***:

```
<pms-inspecwp wpno="M00356-X-XXXX-XXX" wpseq="0039">
<wpidinfo>
<maintlvl level="asb"/>
<title>AH-64A Helicopter 10 Hour/14 day Inspection Checklist
</title>
</wpidinfo>
<wpinfo>
<tools>
<tools-setup-item>
<name>Aircraft Mechanician's Tool Kit
</name>
<itemref>
<xref wpid="S00003-X-XXXX-XXX" itemid="S00003-X-XXXX-XXX-item1"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Tire pressure gage
</name>
<itemref>
<xref wpid="S00003-X-XXXX-XXX" itemid="S00003-X-XXXX-XXX-item7"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>Sample jar (2)
</name>
<itemref>
<xref wpid="S00003-X-XXXX-XXX" itemid="S00003-X-XXXX-XXX-item5"/>
</itemref>
</tools-setup-item>
</tools>
```


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Check the source to verify that this is the current version before use.

<mtrlpart>
<mtrlpart-setup-item>
<name>Rags
</name>
<itemref>
<xref wpid="S00004-X-XXXX-XXX" itemid="S00003-X-XXXX-XXX-item5"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Hydraulic fluid
</name>
<itemref>
<extref docno="MIL-H-5606"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Hydraulic fluid
</name>
<itemref>
<extref docno="MIL-H-83282"/>
</itemref>
</mtrlpart-setup-item>
<mtrlpart-setup-item>
<name>Lubricating oil
</name>
<itemref>
<extref docno="MIL-L-23699"/>
</itemref>
</mtrlpart-setup-item>
</mtrlpart>
<persnreq>
<persnreq-setup-item>
<name>Attack Helicopter Repairer
</name>
<mos>67R
</mos>
<qty>1
</qty>
</persnreq-setup-item>
<persnreq-setup-item>
<name>Assistant
</name>
<qty>1
</qty>
</persnreq-setup-item>
</persnreq>
<eqpconds>
<eqpconds-setup-item>
<condition>Helicopter safed
</condition>
<itemref>
<extref docno="TM 1-1520-238-10"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Weapons systems safed and cleared

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```
</condition>
<itemref>
<extref docno="TM 1-1520-238-10"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Remove protective cover
</condition>
<itemref>
<extref docno="TM 1-1520-238-10"/>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
<figure>
<title>Area Diagram
</title>
<graphic boardno="area">
</graphic>
</figure>
<table frame="none">
<title>Inspection Areas
</title>
<tgroup cols="3" colsep="0" rowsep="0">
<colspec colwidth="0.4*"/>
<colspec colwidth="0.9*"/>
<colspec colwidth="1.7*"/>
<tbody>
<row>
<entry>AREA NO. 1
</entry>
<entry>Fuselage &ndash; Right Side Forward
</entry>
<entry>All surfaces, components, and equipment in forward avionics
bay. Include landing gear and search light.
</entry>
</row>
<row>
<entry>AREA NO. 2
</entry>
<entry>Fuselage &ndash; Right Side Center
</entry>
<entry>All surfaces, components, and equipment aft of crew station to
engine nacelle. Includes transmission deck, right nose gearbox,
engine installation components, wing and stores.
</entry>
</row>
<row>
<entry>AREA NO. 3
</entry>
<entry>Engine Nacelle &ndash; Right
</entry>
<entry>All surfaces, components, and equipment aft of APU exhaust duct.
Includes lower nacelle, aft electronics compartment, and IR suppressor.
</entry>
</row>
```


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<row>
<entry>AREA NO. 4
</entry>
<entry>Fuselage – Right Side Aft
</entry>
<entry>All surfaces, components, and equipment aft of APU exhaust duct and forward of intermediate gearbox. Includes hydraulic ground service panel and aft horizontal tail rotor drive shaft.
</entry>
</row>
<row>
<entry>AREA NO. 5
</entry>
<entry>Tail Section
</entry>
<entry>All surfaces, components, and equipment aft of tailboom area. Includes horizontal stabilator, tail landing gear, intermediate and tail rotor gear boxes, aft vertical tail rotor drive shaft.
</entry>
</row>
<row>
<entry>AREA NO. 6
</entry>
<entry>Fuselage – Left Side Aft
</entry>
<entry>All surfaces, components, and equipment forward of intermediate gearbox and aft of ENCU exhaust duct.
</entry>
</row>
<row>
<entry>AREA NO. 7
</entry>
<entry>Catwalk
</entry>
<entry>All surfaces, components, and equipment in catwalk area. Includes shaft-driven compressor, forward tail rotor drive shaft, fire extinguisher containers, environmental control unit (ENCU), and APU.
</entry>
</row>
<row>
<entry>AREA NO. 8
</entry>
<entry>Main Rotor Mast
</entry>
<entry>All surfaces, components, and equipment in the mast area. Includes main rotor, air data sensor, and upper controls (mixer).
</entry>
</row>
<row>
<entry>AREA NO. 9
</entry>
<entry>Engine Nacelle – Left
</entry>
<entry>All surfaces, components, and equipment forward of ENCU exhaust duct and aft of left nose gearbox. Includes IR suppressor, aft avionics compartment, and lower nacelle.

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```
</entry>
</row>
<row>
<entry>AREA NO. 10
</entry>
<entry>Fuselage &ndash; Left Side Center
</entry>
<entry>All surfaces, components, and equipment forward of engine nacelle
and aft of left forward avionics bay. Includes left transmission deck,
wing and stores, nose gearbox, and engine installation components.
</entry>
</row>
<row>
<entry>AREA NO. 11
</entry>
<entry>Fuselage &ndash; Left Side Forward
</entry>
<entry>All surfaces, components, and equipment in forward
avionics bay. Include landing gear.
</entry>
</row>
<row>
<entry>AREA NO. 12
</entry>
<entry>Nose Section
</entry>
<entry>All surfaces, components, and equipment on or under the helicopter
nose. Includes TADS/PNVS turret and area weapon.
</entry>
</row>
<row>
<entry>AREA NO. 13
</entry>
<entry>Pilot Station
</entry>
<entry>All surfaces, components, and equipment in the pilot
station. Includes windshields, landing gear brake control,
canopy jettison system, lighting and indicator components,
power and flight controls and instruments.
</entry>
</row>
<row>
<entry>AREA NO. 14
</entry>
<entry>CPG Station
</entry>
<entry>All surfaces, components, and equipment in the CPG
station. Includes windshields, landing gear brake control,
canopy jettison system, lighting and indicator components,
power and flight controls and instruments.
</entry>
</row>
</tbody>
</tgroup>
</table>
<figure>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<title>Inspection Area No.  1
</title>
<graphic boardno="area-1">
</graphic>
</figure>
<table>
<title>FUSELAGE &mdash; RIGHT SIDE FORWARD &mdash; POWER OFF
</title>
<tgroup cols="3" rowsep="0" tgroupstyle="0">
<colspec colwidth="0.2*"/>
<colspec align="center" colwidth="0.35*"/>
<colspec colwidth="2.5*"/>
<thead>
<row>
<entry align="center" rowsep="1">Seq.  No.
</entry>
<entry rowsep="1">Location
</entry>
<entry align="center" rowsep="1">Item and Procedure
</entry>
</row>
</thead>
<tbody>
<row>
<entry>1.1
</entry>
<entry> &mdash; &mdash;
</entry>
<entry> &pms-inspecwp.checklist;
</entry>
</row>
<row>
<entry>1.2
</entry>
<entry> &mdash; &mdash;
</entry>
<entry>
<trim.para>Exterior surfaces
</trim.para>
<randlist>
<item>Skin areas for cracks and distortion
</item>
<item>Loose or missing hardware
</item>
<item>Access panels, doors, and fairings for mounting security
</item>
<item>Exposed hydraulic lines for leakage and chafing
</item>
<item>Hydraulic connectors for security
</item>
</randlist>
</entry>
</row>
<row>
<entry>1.3
</entry>
```


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Check the source to verify that this is the current version before use.

```
<entry>1
</entry>
<entry>
<trim.para>Radar Warning Antenna
</trim.para>
<randlist>
<item>Physical damage and mounting security
</item>
</randlist>
</entry>
</row>
<row>
<entry>1.4
</entry>
<entry>2
</entry>
<entry>
<trim.para>Open Avionics Door R90
</trim.para>
<randlist>
<item>Internal panels for cracks and cleanliness
</item>
<item>Loose or missing hardware
</item>
<item>Interior components for physical damage and mounting security
</item>
<item>Electrical connectors for security
</item>
<item>Visible wiring for chafing or damaged insula-
tion and connection security
</item>
<item>Ammo conveyor for damaged or cracked carriers and tracks
</item>
</randlist>
</entry>
</row>
<row>
<entry>1.5
</entry>
<entry>3
</entry>
<entry>
<trim.para>
<emphasis emph="bold">(FSCAP) STATIC PORT
</emphasis>
</trim.para>
<randlist>
<item>
<emphasis emph="bold">(FSCAP) OBSTRUCTIONS, CLEANLINESS, AND DAMAGE
</emphasis>
</item>
</randlist>
</entry>
</row>  ...
</tbody>
</tgroup>
```


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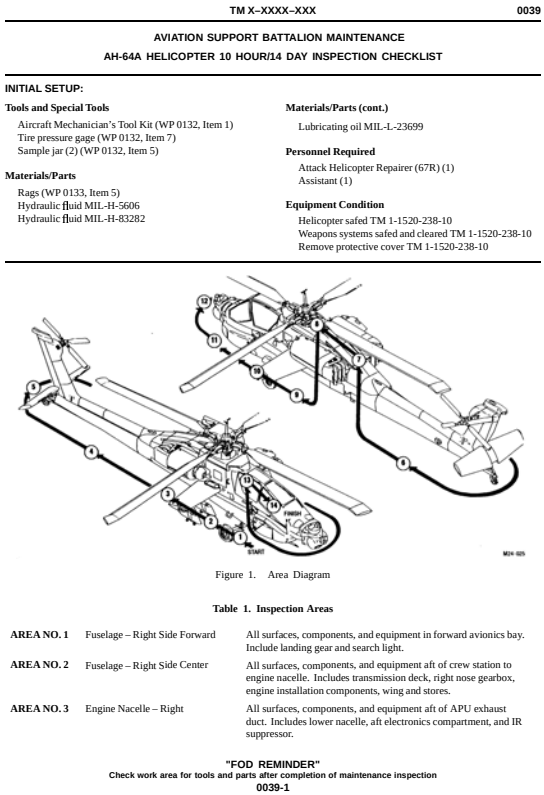
Check the source to verify that this is the current version before use.

</table>
</pms-inspecwp>

b. Page-based TM stylesheet output example for ***<pms-inspecwp>***:



TM X–XXXX–XXX 0039

AVIATION SUPPORT BATTALION MAINTENANCE

AH-64A HELICOPTER 10 HOUR/14 DAY INSPECTION CHECKLIST

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanician's Tool Kit (WP 0132, Item 1)
Tire pressure gage (WP 0132, Item 7)
Sample jar (2) (WP 0132, Item 5)

Materials/Parts

Rags (WP 0133, Item 5)
Hydraulic fluid MIL-H-5606
Hydraulic fluid MIL-H-83282

Materials/Parts (cont.)

Lubricating oil MIL-L-23699

Personnel Required

Attack Helicopter Repairer (67R) (1)
Assistant (1)

Equipment Condition

Helicopter safed TM 1-1520-238-10
Weapons systems safed and cleared TM 1-1520-238-10
Remove protective cover TM 1-1520-238-10

Figure 1. Area Diagram

Table 1. Inspection Areas

AREA NO. 1	Fuselage – Right Side Forward	All surfaces, components, and equipment in forward avionics bay. Include landing gear and search light.
AREA NO. 2	Fuselage – Right Side Center	All surfaces, components, and equipment aft of crew station to engine nacelle. Includes transmission deck, right nose gearbox, engine installation components, wing and stores.
AREA NO. 3	Engine Nacelle – Right	All surfaces, components, and equipment aft of APU exhaust duct. Includes lower nacelle, aft electronics compartment, and IR suppressor.

"FOD REMINDER"
Check work area for tools and parts after completion of maintenance inspection
0039-1

FIGURE 407. Example of a page-based TM stylesheet output for <pms-inspecwp>.

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Table 1. Inspection Areas–Continued

AREA NO. 4	Fuselage – Right Side Aft	All surfaces, components, and equipment aft of APU exhaust duct and forward of intermediate gearbox. Includes hydraulic ground service panel and aft horizontal tail rotor drive shaft.
AREA NO. 5	Tail Section	All surfaces, components, and equipment aft of tailboom area. Includes horizontal stabilator, tail landing gear, intermediate and tail rotor gear boxes, aft vertical tail rotor drive shaft.
AREA NO. 6	Fuselage – Left Side Aft	All surfaces, components, and equipment forward of intermediate gearbox and aft of ENCU exhaust duct.
AREA NO. 7	Catwalk	All surfaces, components, and equipment in catwalk area. Includes shaft-driven compressor, forward tail rotor drive shaft, fire extinguisher containers, environmental control unit (ENCU), and APU.
AREA NO. 8	Main Rotor Mast	All surfaces, components, and equipment in the mast area. Includes main rotor, air data sensor, and upper controls (mixer).
AREA NO. 9	Engine Nacelle – Left	All surfaces, components, and equipment forward of ENCU exhaust duct and aft of left nose gearbox. Includes IR suppressor, aft avionics compartment, and lower nacelle.
AREA NO. 10	Fuselage – Left Side Center	All surfaces, components, and equipment forward of engine nacelle and aft of left forward avionics bay. Includes left transmission deck, wing and stores, nose gearbox, and engine installation components.
AREA NO. 11	Fuselage – Left Side Forward	All surfaces, components, and equipment in forward avionics bay. Include landing gear.
AREA NO. 12	Nose Section	All surfaces, components, and equipment on or under the helicopter nose. Includes TADS/PNVS turret and area weapon.
AREA NO. 13	Pilot Station	All surfaces, components, and equipment in the pilot station. Includes windshields, landing gear brake control, canopy jettison system, lighting and indicator components, power and flight controls and instruments.
AREA NO. 14	CPG Station	All surfaces, components, and equipment in the CPG station. Includes windshields, landing gear brake control, canopy jettison system, lighting and indicator components, power and flight controls and instruments.

"FOD REMINDER"
Check work area for tools and parts after completion of maintenance inspection
0039-2

FIGURE 407. Example of a page-based TM stylesheet output for <pms-inspecwp> - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

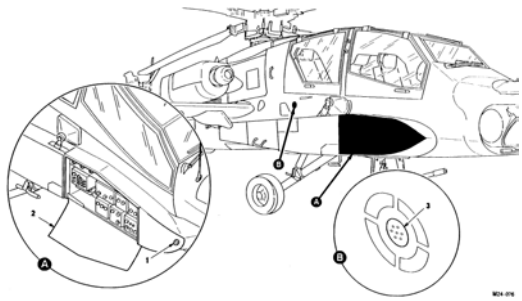

Figure 2. Inspection Area No. 1

Table 2. FUSELAGE — RIGHT SIDE FORWARD — POWER OFF

Seq. No.	Location	Item and Procedure
1.1	— —	Inspect aircraft forms and records for recorded discrepancies (DA PAM 738-750, Functional User's Manual for the Army Maintenance Management System (TAMMS)).
1.2	— —	Exterior surfaces Skin areas for cracks and distortion Loose or missing hardware Access panels, doors, and fairings for mounting security Exposed hydraulic lines for leakage and chafing Hydraulic connectors for security
1.3	1	Radar Warning Antenna Physical damage and mounting security
1.4	2	Open Avionics Door R90 Internal panels for cracks and cleanliness Loose or missing hardware Interior components for physical damage and mounting security Electrical connectors for security Visible wiring for chafing or damaged insulation and connection security Ammo conveyor for damaged or cracked carriers and tracks
1.5	3	(FSCAP) STATIC PORT (FSCAP) OBSTRUCTIONS, CLEANLINESS, AND DAMAGE

"FOD REMINDER"
Check work area for tools and parts after completion of maintenance inspection
0039-3

FIGURE 407. Example of a page-based TM stylesheet output for <pms-inspecwp> - Continued.

37.26 Phased maintenance inspection (PMI) inspection checklist work package ***<pmi-cklistwp>***. The phased maintenance inspection checklist work package contains all of the data required to perform phased maintenance inspections on an aircraft.

a. The components of ***<pmi-cklistwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Work package initial setup ***<wpinfo>*** (required) (see 32.5).
   (4) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
   (5) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
   (6) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
   (7) Inspection area diagram ***<figure>*** (required – one or more) (see 44.1.1). The illustration is for locating the inspection areas and the access doors and panels which require removal at various phased maintenance inspections of the aircraft.
   (8) Note ***<note>*** (required) (see 41.1.3). The standard statement is contained in the entity boilerplate ***&pmi.cklistwp;*** (see 37.26.1 for further boilerplate information).
   (9) The inspection type (optional – zero or more) components are:
      (a) Illustration ***<figure>*** (optional) (see 44.1.1).
      (b) Inspection form ***<tabmat>*** (required) (see 42).

b. DTD graphically depicted for ***<pmi-cklistwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

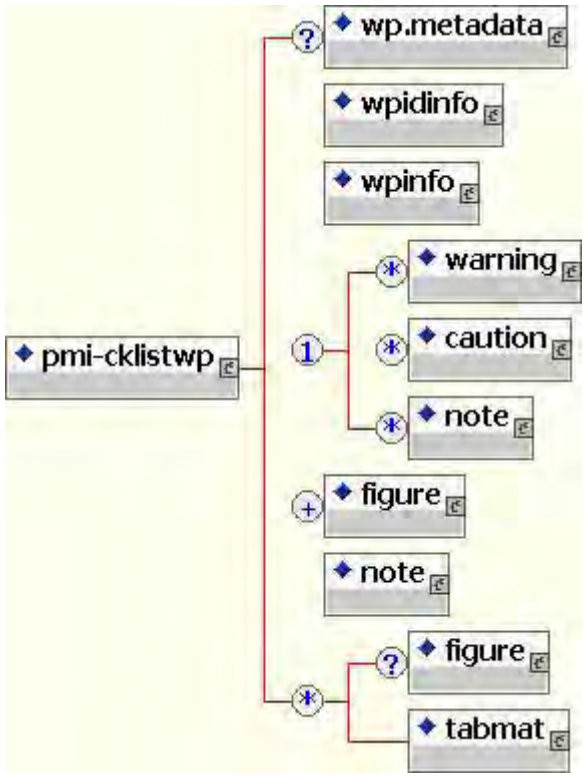

FIGURE 408. Phased maintenance inspection (PMI) checklist work package DTD hierarchy <pmi-cklistwp>.

c. DTD fragment for ***<pmi-cklistwp>***:

```
<!ELEMENT pmi-cklistwp (wp.metadata?, wpidinfo, wpinfo, (warning*,
   caution*, note*), figure+, note, (figure?, tabmat)*)>
<!ATTLIST pmi-cklistwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```


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d. Common attributes for ***<pmi-cklistwp>*** are:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

37.26.1 Phased maintenance inspection checklist work package general entity boilerplates. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listedTable XXIV.

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TABLE XXIV. Boilerplate Entities for *<pmi-cklistwp>*.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Phased maintenance inspection prior to start note	***&pmi-cklistwp.note;***	— —

37.26.2 XML document instance fragment and output for ***<pmi-cklistwp>***. The XML instance and its stylesheet output for a ***<pmi-cklistwp>*** is provided below.

a. Example of an XML document instance fragment for ***<pmi-cklistwp>***:

```
<pmi-cklistwp wpno="m00200-9-1305-201" wpseq="0003">
<wpidinfo>
<maintlvl level="asb"/>
<title>AH-64A Helicopter Phased Maintenance Inspection Checklist
</title>
</wpidinfo>
<wpinfo>
<title>N/A
</title>
<null/>
</wpinfo>
<figure>
<title>Inspection Area Diagram
</title>
<graphic boardno="pmi-cklistwp-fig1">
</graphic>
</figure>
<figure>
<title>Inspection Access Provisions
</title>
<subfig sheet="1" totalsheets="2">
<subtitle>
</subtitle>
<graphic boardno="pmi-cklistwp-fig2">
</graphic>
</subfig>
<subfig sheet="2" totalsheets="2">
<subtitle>
</subtitle>
<graphic boardno="pmi-cklistwp-fig3">
</graphic>
</subfig>
</figure> &pmi-cklistwp.note;
<tabmat>
<tgroup cols="8">
<colspec colname="col1" colwidth="0.55*"/>
<colspec colname="col2" colwidth="0.15*"/>
<colspec colname="col3" colwidth="2.55*"/>
<colspec colname="col4" colwidth="0.45*"/>
<colspec colname="col5" colwidth="2.45*"/>
<colspec colname="col6" colwidth="0.95*"/>
<colspec colname="col7" colwidth="1.35*"/>
<colspec colname="col8" colwidth="0.55*"/>
<thead>
```


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```
<row>
<entry colsep="0" nameend="col3" namest="col1" valign="bottom">PHASE
NO. _____
</entry>
<entry colsep="1" nameend="col8" namest="col4" valign="bottom">PHASED
MAINTENANCE CHECKLIST
</entry>
</row>
<row>
<entry align="center" nameend="col3" namest="col1" rowsep="0"
valign="bottom">Area Name and No.
</entry>
<entry align="center" nameend="col5" namest="col4" rowsep="0"
valign="bottom">Aircraft Serial No.
</entry>
<entry align="center" rowsep="0" valign="bottom">Date
</entry>
<entry align="center" nameend="col8" namest="col7" rowsep="0"
valign="bottom">Total Hrs.    This Area
</entry>
</row>
<row>
<entry align="center" nameend="col3" namest="col1" valign="bottom"> General
</entry>
<entry align="center" nameend="col5" namest="col4" valign="bottom">
</entry>
<entry align="center" valign="bottom">
</entry>
<entry align="center" nameend="col8" namest="col7" valign="bottom">
</entry>
</row>
<row>
<entry align="center" valign="bottom">Inspect Phase Nos.
</entry>
<entry align="center" colsep="1" nameend="col3" namest="col2"
valign="bottom">Inspection Requirements
</entry>
<entry align="center" valign="bottom">Status
</entry>
<entry align="center" valign="bottom">Fault and/or Remarks
</entry>
<entry align="center" nameend="col7" namest="col6" valign="bottom"> Action Taken
</entry>
<entry align="center" valign="bottom">Initial
</entry>
</row>
</thead>
<tbody>
<row>
<entry align="center" morerows="7" rowsep="1">ALL
<brk/>
<emphasis emph="bold">C
</emphasis>
</entry>
<entry colsep="0" morerows="7" rowsep="1">1.
</entry>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<entry colsep="1" morerows="7" rowsep="1">Prior to inspection, check
forms and records for record deficiencies.
</entry>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry align="center" morerows="7" rowsep="1">ALL
<brk/>
<emphasis emph="bold">C
</emphasis>
</entry>
<entry colsep="0" morerows="7" rowsep="1">2.
</entry>
<entry colsep="1" morerows="7" rowsep="1">Fuel tanks will be fully
serviced prior to start of phased inspection.  If maintenance
is to be accomplished which requires defueling, this item may be
deferred until after such maintenance is completed.
</entry>
</entry>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry align="center" morerows="7" rowsep="1">ALL
</entry>
<entry colsep="0" morerows="7" rowsep="1">3.
</entry>
<entry colsep="1" morerows="7" rowsep="1">Clean helicopter per
<extref docno="TM 1-1520-238-23"/>.
</entry>
</entry>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry align="center" morerows="7" rowsep="1">ALL
<brk/>
<emphasis emph="bold">C
</emphasis>
</entry>
<entry colsep="0" morerows="7" rowsep="1">4.
</entry>
<entry colsep="1" morerows="7" rowsep="1">
<emphasis emph="caps">
<emphasis emph="bold">Perform engine run-up and check for
proper engine operation per
<extref docno="TM 55-1520-238-MTF"/>.
</emphasis>
</emphasis>
</entry>
</entry>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
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<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>
<row>
<entry nameend="col7" namest="col6">
</entry>
</row>  . . .
</tbody>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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</tgroup>
</tabmat>
</pmi-cklistwp>

b. Page-based TM stylesheet output example for ***<pmi-cklistwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

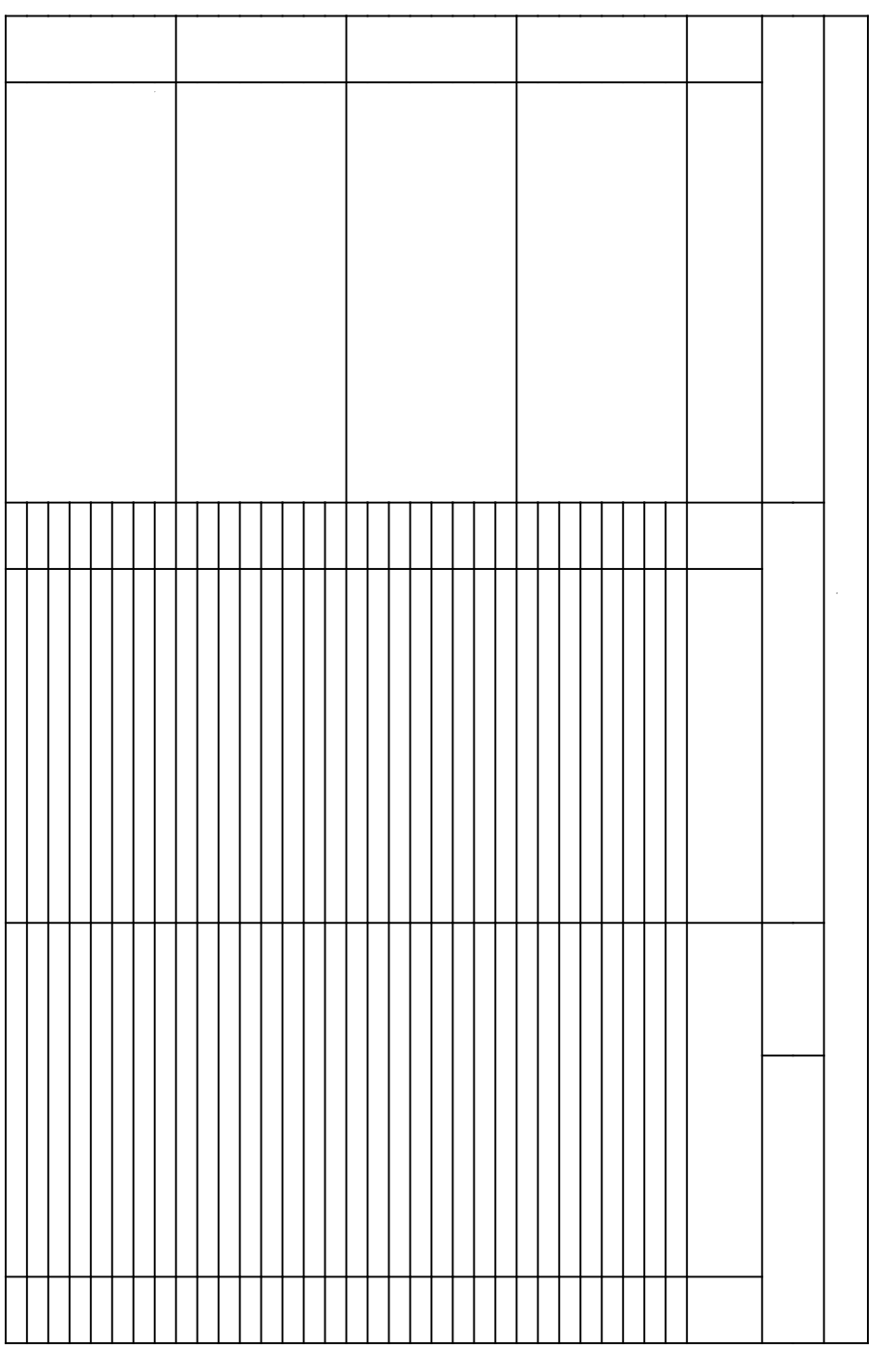

FIGURE 409. Example of a page-based TM stylesheet output for <pmi-cklistwp>.

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Check the source to verify that this is the current version before use.

38 PARTS INFORMATION (RPSTL).

38.1 Overview.

38.1.1 Parts information organization. Parts information, commonly referred to as the RPSTL for Repair Parts and Special Tools List, is a collection of parts information related work packages.

38.1.2 Parts information manual. A separate parts information manual may be prepared. Typically a separate parts manual is prepared:

a. For large or complex weapons systems, or
b. When prescribed by the acquiring activity.

When a separate manual is prepared, it will normally be a combined maintenance manual and it will contain all parts for all levels of maintenance. Depot level repair parts may be included, however depot parts are usually not included in non depot manuals.

38.2 General tagging instructions. Parts information may be tagged for display in either a page based TM or for an IETM. Unlike previous versions MIL-STD-40051-1 and MIL-STD-40051-2 now use the same tags for both types of manuals.

38.3 RPSTL from a data base. The RPSTL may be developed from information contained in a parts or provisioning data base. When a data base is used, all the requirements contained in MIL-STD-40051-1 and MIL-STD-40051-2 are to be met.

38.4 Parts information ***<pim>*** work packages. The ***<pim>*** whether a separate manual or included in maintenance TM contains the following work packages.

a. The components are:
   (1) Introduction work package ***<introwp>*** (required) (see 38.4.1).
   (2) One or more parts list work packages ***<plwp>*** (required – one or more) (see 38.4.2).
   (3) An optional special tools repair parts work package ***<stl-partswp>*** (optional) (see 38.4.3).
   (4) Kit parts work package ***<kitswp>*** (optional) (see 38.4.4).
   (5) Bulk items list work package ***<bulk_itemswp>*** (optional) (see 38.4.5).
   (6) Special tools list work package(s) ***<stlwp>*** (optional) (see 38.4.6).
   (7) National stock number and part number indexes ***<nsnindxwp>*** (optional) (see 38.5.1), ***<pnindxwp>*** (optional) (see 38.5.2). The stock number and part number indexes are required to be used together. If one is used, the other is required by the standard.
   (8) Reference designator index work package ***<refdesindxwp>*** (optional) (see 38.5.3).
b. DTD fragment is graphically depicted.

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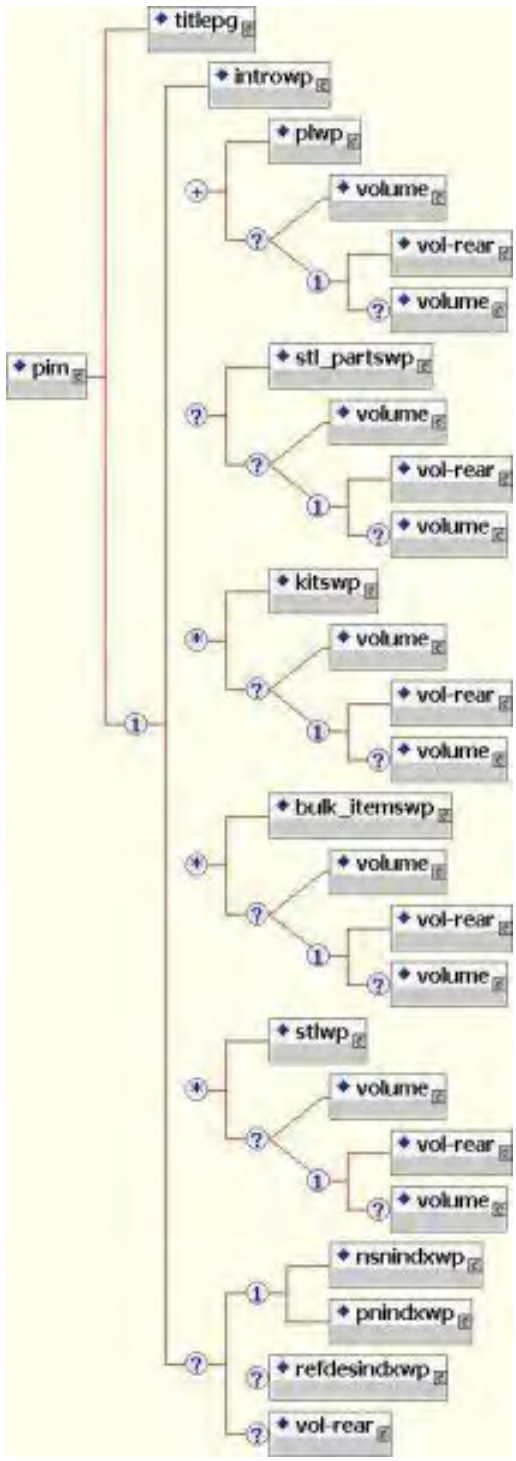

FIGURE 410. Parts information <pim> DTD hierarchy.

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c. The DTD fragment:

```
<!ELEMENT pim (titlepg, (introwp, (plwp, (volume |
(vol-rear, volume?))?)+,
(stl_partswp, (volume | (vol-rear, volume?))?)?, (kitswp, (volume |
(vol-rear, volume?))?)*, (bulk_itemswp, (volume | (vol-rear,
volume?))?)*, (stlwp, (volume | (vol-rear, volume?))?)*, ((nsnindxwp,
pnindxwp), refdesindxwp?, vol-rear?)?))>
<!ATTLIST pim  dmwr-inclus (parts | parts-tools | none) #REQUIRED>
    assocfig  IDREFS                    #IMPLIED
    changeref IDREFS                    #IMPLIED
    chap-toc  (yes | no )               "yes"
    chngno    CDATA                     #REQUIRED
    comment   CDATA                     #IMPLIED
    delchlvl  CDATA                     #IMPLIED
    frame     (yes | no )               "yes"
    idref     IDREFS                    #IMPLIED
    inschlvl  CDATA                     #IMPLIED
    revno     CDATA                     #REQUIRED
    security  (uc | fouo | c | s | ts )  #IMPLIED
    skilltrk  CDATA                     #IMPLIED
    tocentry  (0 | 1 | 2)                "1">
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)
(10) **tocentry** – Table of contents level entry (default value is "**1**") (see 32.3.1.6).

38.4.1 Repair parts introduction work package ***<introwp>***. Both MIL-STD-40051-1 and MIL-STD-40051-2 require an introductory work package. The introduction work package ***<introwp>*** contains information common to all the parts list work packages. This information consists primarily of verbatim text that is provided in MIL-STD-40051-1 or MIL-STD-40051-2. The text to be used is based on the type of TM being acquired. All of the boilerplate text may be entered using text entities. These entities are contained in a series of entity files. The following subparagraphs explain and illustrate how to use these entity files to enter the boilerplate text into the data. Also, the following list is the type of data that is required in a ***<introwp>***.

a. A scope stating the maintenance level or levels and components the parts listings support.
b. A listing of the types of work packages included in the parts listing.
c. A explanation of the columns used in the ***<plwp>*** or ***<stl_partswp>***, ***<kitswp>***, ***<bulk_itemswp>***, and ***<stlwp>*** if they are included. This explanation includes an explanation of the Source, Maintenance, and Recoverability (SMR) codes used in the parts lists.
d. An explanation of the columns used in the ***<nsnindxwp>***, ***<pnindxwp>***, or the ***<refdesindxwp>*** (when included).
e. Any special information that may be needed. This section includes
   (1) Instructions on how to locate repair parts
   (2) An optional listing of useable on codes (UOC) or other methods to identify configuration applicability. This list will be included if there are multiple configurations listed in the parts lists that are identified by a some form of identification code.

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(3) An optional list of related publications. This list will be included if the parts list is contained in a separate manual.

(4) An optional list of abbreviations that contains abbreviations found only in the parts list.

f. Though the majority of the ***<introwp>*** is boilerplate text, the introduction may contain other information at the discretion of the acquiring activity. Other information commonly found in the ***<introwp>*** include a listing of Commercial and Government Entity (CAGE) codes with their associated name and address or an overview graphic showing how the parts listings are broken down.

a. The components are:

(1) Work package metadata element ***<wp.metadata>*** (optional) (see 32.3.2).

(2) Work package identification element ***<wpidinfo>*** (required) (see 32.2).

(3) Index Marker Reference ***<indxref>*** (optional – zero or more). The element is used (see 31.4.2.2.3)

(4) Title ***<title>*** (required) (see 32.4.3).

(5) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).

(6) Narrative titled text first level ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more) if a How to use ***<howtouse>*** is not selected.

(7) Note(s) ***<note>*** (optional – zero or more) (see 41.1.3).

(8) How to use ***<howtouse>***. Element provides the user any special information on how to read and use information and procedures. This tag is optional if the tags ***<para0>*** or ***<para0–alt>*** is used to provide the data for the ***<introwp>***(see 31.3.1.1.2).

b. DTD fragment is graphically depicted.

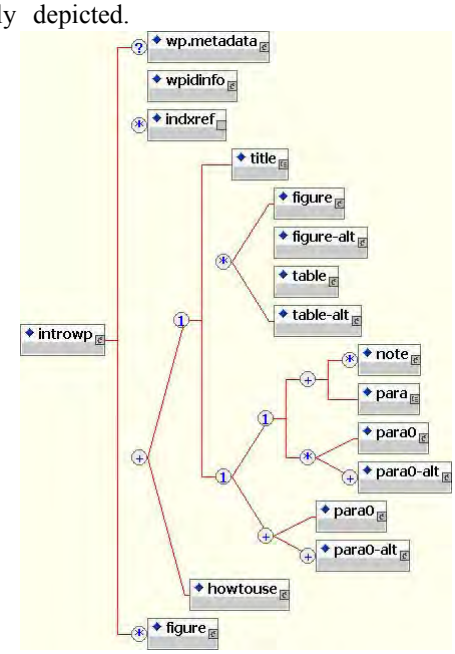

FIGURE 411. Repair parts introduction <introwp> DTD hierarchy..

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c. DTD fragment:

```
<!ELEMENT introwp (wp.metadata?, wpidinfo, indxref*, ((title, (figure | figure-alt)*,
(((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0-alt+)+)) | howtouse)+,
figure*)>.
<!ATTLIST introwp
    airforce          (yes | no )                 "no"
    army              (yes | no )                 "no"
    assocfig          IDREFS                       #IMPLIED
    changeref         IDREFS                       #IMPLIED
    comment           CDATA                        #IMPLIED
    crewmember        CDATA                        #IMPLIED
    date-time-stamp   (date | time | date-time)  #IMPLIED
    delchlvl          CDATA                        #IMPLIED
    deletewp          (yes | no )                 "no"
    fgc               CDATA                        #IMPLIED
    frame             (yes | no )                 "yes"
    idref             IDREFS                       #IMPLIED
    inschlvl          CDATA                        #IMPLIED
    insertwp          CDATA                        #IMPLIED
    lsa-id            CDATA                        #IMPLIED
    marines           (yes | no )                 "no"
    navy              (yes | no )                 "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                        #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                           #REQUIRED
    wpseq             CDATA                        #IMPLIED>
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

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(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

38.4.1.1 Example tagging of a ***<introwp>***.

38.4.1.1.1 Tagged with entites. The preferred method of tagging the boilerplate text required by the standard is with general entities available in the entity files described in 50.6.1. The following is an example of an ***<introwp>*** tagged using the boilerplate entity.

```
<introwp wpno="r00432-1-1111-111">&introwp.intro;</introwp>
```

The first paragraph of the ***<introwp>*** would be resolved to resemble the following:

```
<title>INTRODUCTION
</title>
<para0>
<title>SCOPE
</title>
<para>This RPSTL lists and authorizes spares and repair parts; special
tools; special test, measurement, and diagnostic equipment (TMDE);
and other special support equipment required for performance of
&introwp.intro.lowest-maint;maintenance of the &introwp.intro.item;.
It authorizes the requisitioning, issue, and disposition of
spares, repair parts,and special tools as indicated by the source,
maintenance, and recoverability (SMR) codes.
</para>
</para0>
```

Note that within the ***<para0>***, there are two other entities. The ***&introwp.intro.lowest-maint;*** in this example resolves to the term:

```
DIRECT
```

and the ***&introwp.intro.item;*** would be set to the name of the item or system the RPSTL is dealing with.

38.4.1.1.2 Tagging without entities. Though it is strongly discouraged to tag the ***<introwp>*** without using the available entities. There are cases where the entities may not provide all the information required. In cases such as this, the boilerplate entities should be used to the maximum extent possible. Then any additional information should be added after the entities. The following example uses the basic ***<introwp>*** entity shown in 38.4.1.1.1 and then adds an example list of CAGE codes and an overview figure.

```
<introwp wpno=&rdquo;r00432-1-1111-111&rdquo;>&amp;introwp.intro;
<para0>
<title>Contractor and government entity codes
</title>
<para>The following is a list of the Contractor and Government
Entity (CAGE)Codes used in this parts list.
<deflist>
<term.def>
<term>96412
```


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```
</term>
<def>
<para>
<proponent>
<name>Department of Defense
<address>
<city>Washington
<state>DC
<zip>20301
<figure id="g00432-1-1111-111-1">
<title>Example system overview
<graphic boardno="overview-ex" hscale="80" vscale="80"/>
</para>
</def>
</deflist>
</para>
```

38.4.2 Parts list work package <*plwp*>. The parts list work package(s) <*plwp*> identifies all parts that may be replaced or repaired at the maintenance level(s) for the TM. At least one <*plwp*> is required to be provided. Depending on the complexity of the system, all parts may be listed in a single <*plwp*> or the system may be broken down by functional group codes into multiple <*plwp*>s. Care should be taken when planning parts list layout. Though the <*plwp*> may contain an entire weapons systems parts list, for more complex systems, use of multiple <*plwp*>s is highly recommended. 32.1.1 provides additional information on determining when to prepare a single large or several smaller work packages. An example of a <*plwp*> and all its child elements is provided in 38.4.2.2.

a. The components are.
   (1) Work package metadata element <*wp.metadata*> (optional) (see 32.3.2).
   (2) Work package identification element <*wpidinfo*> (required) (see 32.2).
   (3) Parts item category elements <*pi.category*> (required – one or more).

b. DTD fragment is graphically depicted:

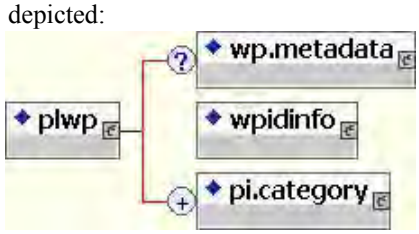

FIGURE 412. Parts list work package <plwp> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT plwp (wp.metadata?, wpidinfo, pi.category+)>
<!ATTLIST plwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED>
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

38.4.2.1 Parts information category ***<pi.category>***. The ***<pi.category>*** is the mechanism that allows the ***<plwp>*** to list either a single functional grouping of parts or the entire system. Normally each ***<pi.category>*** will contain a single functional grouping (see MIL-PRF-49506 for definition). As will be explained in 38.4.2.1.3 , a ***<pi.category>*** is also an option in the ***<pi.item>*** to allow for sub functional groups.

a. The components are:
(1) Illustration ***<figure>*** (required) 44.1.1.
(2) Functional group ***<fncgrp>*** (required) 38.4.2.1.2.
(3) One required standard table header ***<std.table.header>*** (required) 42.3.1.
(4) One or more individual parts ***<pi.item>*** (required – one or more) 38.4.2.1.3

b. DTD fragment is graphically depicted.

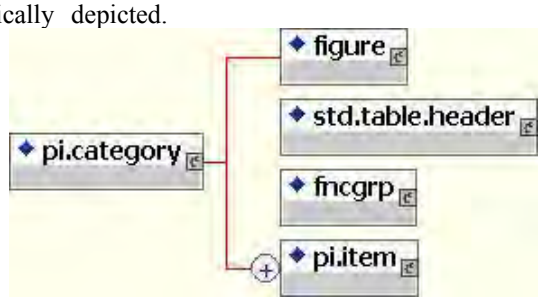

FIGURE 413. Parts information category <pi.category> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT pi.category (figure, std.table.header, fncgrp, pi.item+)>
<!ATTLIST pi.category
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

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(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

38.4.2.1.1 Figure ***<figure>***. The figure is used to provide a visual depiction of the part or parts of the component. The figure is prepared as prescribed in MIL-STD-40051-1 or MIL-STD-40051-2. See 44.1.1

38.4.2.1.2 Functional group ***<fncgrp>***. The function group information is a required entry for each ***<pi.category>***.

a. The components are:
   (1) Function code ***<fnccode>*** (required) (see 38.4.2.1.2.1).
   (2) Function title ***<fnctitle>*** (required) (see 38.4.2.1.2.2).
   (3) A group consisting of zero or more of the following:
      (a) One required part number ***<partno>*** element.
      (b) Zero or one useable on code ***<uoc>*** element.
      (c) Zero or more source, maintainability and recoverability code ***<smr>*** elements.

b. The functional group ***<fncgrp>*** content model is graphically shown below:

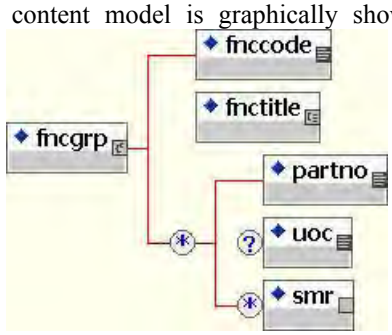

FIGURE 414. Functional group <fncgrp> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT fncgrp (fnccode, fnctitle, (partno, uoc?, smr*)*)>
<!ATTLIST fncgrp
    assocfig        IDREFS                          #IMPLIED
    changeref       IDREFS                          #IMPLIED
    comment         CDATA                           #IMPLIED
    delchlvl        CDATA                           #IMPLIED
    id              ID                              #IMPLIED
    idref           IDREFS                          #IMPLIED
    inschlvl        CDATA                           #IMPLIED
    security        (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk        CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

38.4.2.1.2.1 Function code ***<fnccode>*** element. The ***<fnccode>*** element is a character data element that is used to enter the applicable function code. The author needs to be aware of the how the style sheet for their particular system functions. Some style sheets can generate the lead in text of the word “GROUP”.

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Other systems will require the author to enter this text manually. If the data is entered directly and the style sheet can support the generation, then the word "GROUP" will appear twice.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data. (required)

b. DTD fragment is graphically depicted.

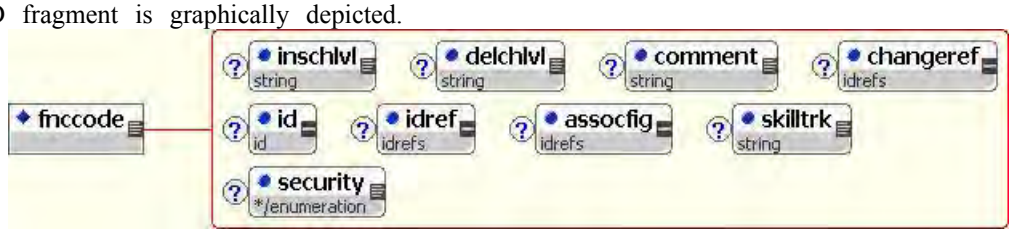

FIGURE 415. Function code <fnccode> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT fnccode (#PCDATA)>
<!ATTLIST fnccode
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

38.4.2.1.2.2 Functional title ***<fnctitle>***. The ***<fnctitle>*** element is an element that allows both character data and certain minor elements such as ***<subscrpt>*** or ***<xref>***. The functional title should be taken from the list of functional codes provided or approved by the acquiring activity.

a. The components are:
   (1) Parsable data (#PCDATA) (optional – zero or more). The element provides the text for the data. (see 9.3.2)
   (2) Emphasis ***<emphasis>*** (optional – zero or more). The element is used to emphasize the text of the entry. (see 49.1.3.1)
   (3) Subscript ***<subscrpt>*** (optional – zero or more). The element is used to format the text as subscript. (see 49.1.3.4)
   (4) Superscript ***<supscrpt>*** (optional – zero or more). The element is used to format the text as superscript. (see 49.1.3.5)
   (5) Cross Reference ***<xref>*** (optional – zero or more). The element is used to reference the work package sequence number, figure, table, step(s), etc. (see 46.2.2)
   (6) External Reference ***<extref>*** (optional – zero or more). The element is used to reference to external document information.(see 46.2.1)

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(7) Linking ***<link>*** (optional – zero or more). The element provides a capability to reference internal or external targets.(see 46.2.3)
(8) Content Sensitive Help ***<help.info>*** (optional – zero or more). The element is used for "help information" about the technical data used in frame-based manuals.(see 48.3.2.4)
(9) Index Marker Reference ***<indxref>*** (optional – zero or more). The element is used (see 31.4.2.2.3)
(10) Term ***<term>*** (optional – zero or more). The element provides a word, phrase, acronym, symbol, or abbreviation in a definition and legend list.(see 49.1.2.3.1.1)
(11) Item Definition ***<term.def>*** (optional – zero or more). The element provides a wrapper for a term and corresponding definition. (see 49.1.2.3.1)
(12) Figure Callout ***<callout>*** (optional – zero or more). The element is used for a figure reference and callout reference, number, letter, or symbol appearing in the figure.(see 46.2.4.1)
(13) Footnote ***<ftnote>*** (optional – zero or more). The element contains the footnote information which is referenced in the document with footnote reference.(see 45.1.1)
(14) Footnote Reference ***<ftnref>*** (optional – zero or more). The element is the reference in the text to the footnote. (see 45.1.2)
(15) Department of Defense Ammunition Code ***<dodac>*** (optional – zero or more). The element provides identification of an ammunition type.(see 49.1.4.5)
(16) Lubricant ***<lubricant>*** (optional – zero or more). The element provides identification of a lubricant.(see 49.1.4.16)
(17) Symbol ***<symbol>*** (optional – zero or more). The element provides a graphic symbol not found in standard ISO character sets that is inserted as a graphic in text.(see 44.3.1)
(18) Torque Value or Limit ***<torque>*** (optional – zero or more). The element provides a torque value or limit embedded in the text or table entry.(see 49.1.4.24)
(19) Voltage Measurement ***<voltage>*** (optional – zero or more). The element provides the identification of a critical voltage measurement.(see 49.1.4.25)
(20) Null ***<null>*** (optional – zero or more). The element specifies that it contains no content and marking is identified by it's attribute.(see 49.1.3.2)
(21) Text Change Information ***<change>*** (optional – zero or more). The element provides the scope and type of changed information that is indicated by being enclosed within start and end tag. (see 49.1.3.7)

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

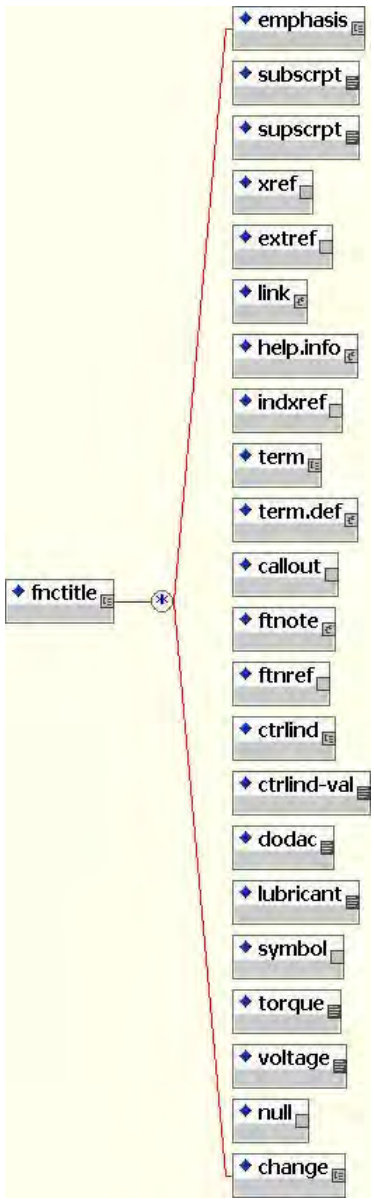

FIGURE 416. Functional group title <fnctitle> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT fnctitle (#PCDATA | emphasis | subscrpt | supscrptxref |
extref | link | help.info | indxref | term | term.def | callout | ftnote |
ftnref ctrlind | ctrlind-val | dodac | lubricant | symbol | torque |
voltage | null | change)*>
<!ATTLIST fnctitle
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

38.4.2.1.2.3 Optional parts information group. The ***<fncgrp>*** may contain an optional and repeatable group consisting of the elements ***<partno>***,***<uoc>***, and ***<smr>***. This information is not normally displayed in a page based TM, but may be used for filtering or display in an IETM. The usage of each element is described below:

a. There is a single ***<partno>*** required each time the group is used. The ***<partno>*** is discussed in 38.4.2.1.3.2.3.
b. Zero or one usable on code ***<uoc>*** which is discussed in 38.4.2.1.3.1.3.
c. Zero or more source, maintainability and recoverability ***<smr>*** code elements. The multiple SMR codes maybe used to for multi service or multiple command differences in provisioning and repairing the identified part. The multiple SMR codes are primarily for differences in service maintenance and provisioning. The ***<smr>*** is discussed in 38.4.2.1.3.2.1

38.4.2.1.3 Parts information ***<pi.item>***. The parts information ***<pi.item>*** element is the heart of the parts list work package ***<plwp>***. This element contains all the pertinent information for each part. The ***<pi.item>*** allows for significant tagging options to support for the fairly limited page based display or a robust IETM.

a. The components are:

(1) Next higher assembly ***<nha_item>*** (optional).
(2) Call out ***<callout>*** (optional).
(3) Usable on-code (UOC) ***<uoc>*** (optional).
(4) Basis of issue ***<boi>*** (optional).
(5) Usable effective serial number ***<usbefserno>*** (optional).
(6) Quantity ***<qty>*** (required) .
(7) Select elements from either Group A or Group B (one selection is (required)):
(a) Group A
1 Source, maintenace, and recoverability code ***<smr>*** (required – one or more).
2 National Stock Number ***<nsn>*** (optional).
3 Part number ***<partno>*** (required).
4 Commercial and Government Entity Code ***<cageno>*** (required).

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5 Name ***<name>*** (required) .
6 Item description ***<desc>*** (required).
7 Total quantity required ***<qty_per_end_item>***.
8 Unit of issue ***<ui>*** (optional).
9 External reference ***<extref>*** (optional).
10 Link ***<link>*** (optional).
11 Reference designator ***<refdes>*** (optional).
12 Maintenance ***<maintenance>*** (optional).

(b) Group B

1 Common part information reference ***<common_part_ref>*** (required).

(8) Part item ***<pi.item>*** (optional).
(9) Kit item ***<kit.itempi.category>*** (optional).
(10) Part category ***<pi.category>*** (optional).
(11) Next lower functional group reference ***<part.breakdown.ref>*** (optional).

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

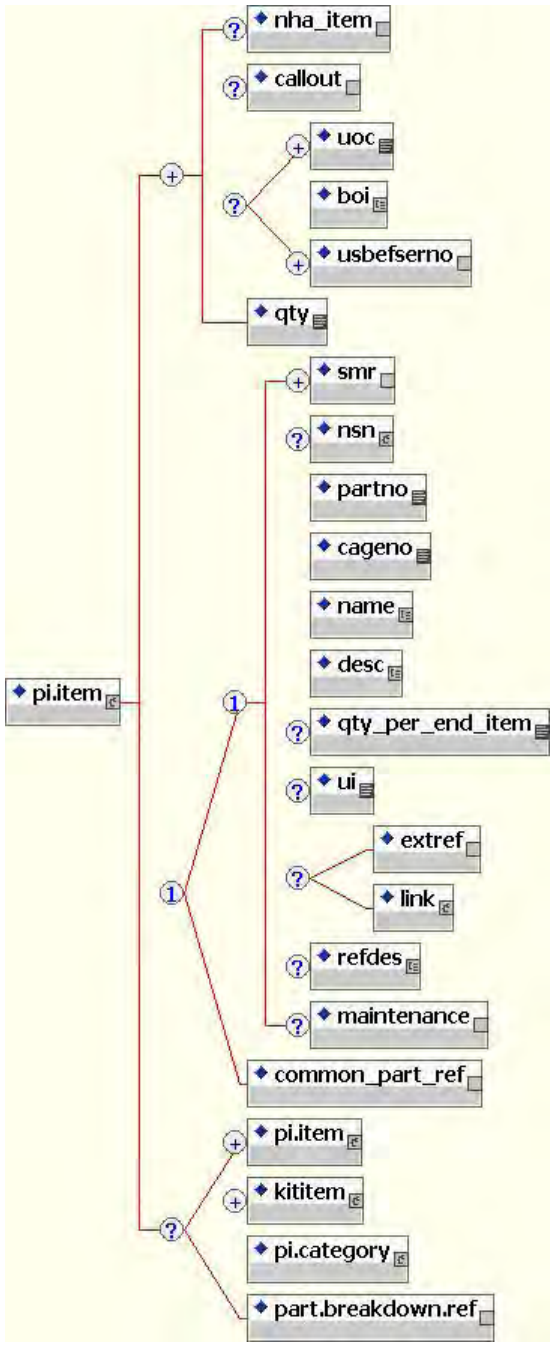

FIGURE 417. Parts item <pi.item> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT pi.item ((nha_item?, callout?, (uoc | boi | usbefserno)?, qty)+,
((smr+, nsn?, partno, cageno, name, desc, qty_per_end_item?, ui?, (extref | link)?,
refdes?, maintenance?) | common_part_ref), (pi.item+ | kititem+ |
pi.category | part.breakdown.ref)?)>
<!ATTLIST pi.item  id      ID      #IMPLIED
           indent CDATA  '0'
           type    (part | exp | coei | bii | aal | tool | special.tool)  'part'
           hci       (yes | no)  'no'
           esd       (yes | no)  'no'
           fscap     (yes | no)  'no'
           mrp       (yes | no)  'no'
           csi       (yes | no)  'no'
           tereq     (yes | no)  'no'>
```

d. Unique attributes:

(1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

(2) **indent** – (optional) when with the ***<pi.item>*** shows breakdown levels through a series of dots. The value of the **indent** should correspond to the desired level of indenture of the part. For page based parts lists, the indent dot appears at the start of the description ***<desc>***. The value of the 'indent' attribute should be a number between 0 and 5. The value of 5 is the historical limit for indenture. Style sheets should be developed that account for at least five dots. Values above 5 may not display properly in all systems. The default value is '0' which provides no dot.

(3) **type** – (optional) attribute is a list that may be used to identify where a part would typically be used. Each item in the list is described in the following list:

(a) The 'part' attribute is the default and indicates the identified part is typically a repair part contained in the ***<plwp>*** (see 38.4.2).

(b) The 'exp' attribute would be used to indicate the part is an expendable or durable item that would be listed in the ***<explistwp>*** (see 40.7).

(c) The 'coei' attribute identifies those parts that are part of the component of end item ***<coei>*** (see 40.5.1.2).

(d) The 'bii' attribute identifies those parts that are primarily basic issue items and found in the ***<bii>*** (see 40.5.1.3).

(e) The 'tool' attribute identifies common tools normally found in the ***<toolidwp>*** (see 40.8).

(f) The 'special.tool' attribute identifies special tools that would be listed in the ***<stlwp>*** (see 38.4.6)

(4) **hci** – Hardness critical item(default value is “no”). The **hci** attribute is a toggle, on or off, used to identify any components falling under the hardness critical item requirements.

(5) **esd** – Electrostatic discharge requirement (default value is “no”) (see 49.2.5).

(6) **fscap** – Flight safety critical aircraft part (default value is “off”). The **fscap** attribute is a toggle, on or off, used to identify aviation parts that are identified as flight safety critical.

(7) **mrp** – Mandatory replacement part (default value is “off”). The **mrp** attribute is a toggle, on or off, used to identify parts that are required to be replaced during the during a maintenance action or inspection. Mandatory replacement parts are listed in the initial setup for the task where they are replaced and in the ***<mrplwp>*** (see 40.9).

(8) The 'csi' attribute is a toggle, on or off, with a default of “off” used to identify critical safety items. Critical safety items are listed in the initial setup of the work package where they are used and in the ***<csi>*** part of the ***<csi.fscap.wp>*** (see 40.10)

(9) **tereq** – Test equipment required (default value is “off”). The **tereq** attribute is a toggle, on or off, used to identify a part as required test equipment. Required test equipment is listed in the ***<tereqtab>*** in the ***<mac>*** (see 40.4).

38.4.2.1.3.1 Unique part information. The following paragraphs provide details on part information that makes the part unique for that usage. Information such as next higher assemblies, index numbers and quantities required.

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38.4.2.1.3.1.1 Next higher assembly ***<nha_item>***. The next higher assembly ***<nha_item>*** is one of the potentially unique parts elements. It is used to identify any specifically listed next higher assembly for the current part. The ***<nha_item>*** is an EMPTY element with a single attribute 'nha.ref' that is an IDREF. The 'nha.ref' points to the ID of the ***<pi.item>*** for the next higher assembly.

```
<nha_item nha.ref=”a1234”/>
```

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT nha_item EMPTY>
<!ATTLIST nha_item   nha.ref IDREF #REQUIRED>
```

c. Unique attributes:
   (1) **nha.ref** – Next higher assembly reference. This is an IDREF.

38.4.2.1.3.1.2 Item callout ***<callout>***. The ***<callout>*** is the second of the unique part elements. The ***<callout>*** is an optional element that may be used to provide the index number relating the current part to the illustration. The ***<callout>*** would be used if there is no automatic integrated method of displaying a the item number on a figure in the flowing text.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment is graphically depicted.

FIGURE 418. Callout <callout> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT callout EMPTY>
<!ATTLIST callout   id ID #IMPLIED
  numref IDREF #IMPLIED
  partref IDREF #IMPLIED
  assocfig IDREF #REQUIRED
  label CDATA #IMPLIED>
```

d. Unique attributes:
   (1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (2) **numref** – Figure numerical reference, IDREF. References the numeric identifier of the callout.
   (3) **partref** – Part number reference, IDREF. References the part identifier to which the callout is being associated.
   (4) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (5) **label** – Label A CDATA attribute that allows the author or system to enter various information about the ***<callout>***.

38.4.2.1.3.1.3 Usable on code ***<uoc>***. The ***<uoc>*** element is part of group that aids in identifying the configuration usage of the part. The other elements that may be used are the ***<boi>*** (see 38.4.2.1.3.1.4) or the ***<usbefserno>*** (see 38.4.2.1.3.1.5). Typically only one of these elements is used, but all may be.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment is graphically depicted.

```
<!ELEMENT uoc (#PCDATA)>
```

c. No attributes.

38.4.2.1.3.1.4 Basis of issue ***<boi>***. The ***<boi>*** is the quantity of an item (special tool) authorized for the end item density spread or for the unit level specified.

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a. The components are:
   (1) Parsable characters or type text. – #PCDATA
   (2) Format text – ***<emphasis>*** (see 49.1.3.1).
   (3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
   (4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
   (5) Cross reference – ***<xref>*** (see 46.2.2).
   (6) External reference – ***<extref>*** (see 46.2.1).
   (7) Enhanced linking – ***<link>*** (see 46.2.3).
   (8) IETM help information – ***<help.info>*** (see 48.3.2.4).
   (9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
   (10) Term – ***<term>*** (see 49.1.2.3.1.1).
   (11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
   (12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
   (13) Footnote – ***< ftnote>*** (see 45.1.1).
   (14) Footnote reference – ***<ftnref>*** (see 45.1.2).
   (15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
   (16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
   (17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
   (18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
   (19) Graphic symbol – ***<symbol>*** (see 44.3.1).
   (20) Torque value – ***<torque>*** (see 49.1.4.24).
   (21) Voltage value – ***< voltage>*** (see 49.1.4.25).
   (22) Null text – ***<null >*** (see 49.1.3.2).
   (23) Changed text marker – ***<change>*** (see 49.1.3.7).
b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

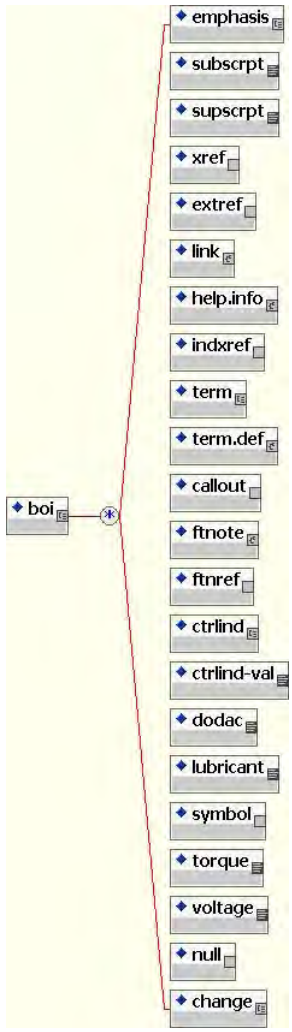

FIGURE 419. Basis of issue <boi> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT boi (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link | help.info |
indxref | term | term.def | callout | ftnote | ftnref | ctrlind | ctrlind-val |
dodac | lubricant | symbol | torque | voltage | null | change)*>
<!ATTLIST boi  inschlvl CDATA  #IMPLIED
      delchlvl CDATA  #IMPLIED
      label  CDATA  #IMPLIED
      texttype CDATA  #IMPLIED
      itemid  NMTOKEN #IMPLIED
      config  CDATA  #IMPLIED
      skilltrk CDATA  #IMPLIED
       idref    IDREFS  #IMPLIED
      assocfig IDREFS  #IMPLIED >
```

d. Common attributes :

(1) **inschlvl** – Insert change level (optional) (see 49.2.9).

(2) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(3) **label** – Label. (optional) A CDATA attribute that allows the author or system to enter various information about the ***<boi>***.

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Check the source to verify that this is the current version before use.

(4) **texttype** – Text type. (optional) This is a CDATA attribute with as yet unspecified usage. Until such time as the desired utilization is provided, recommend that this attribute not be used. Any values entered here may not provide the expected or desired output.
(5) **config** – Configuration (optional). Specifies the equipment configurations to which element applies.
(6) **skilltrk** – Skill level (optional) (see 49.2.2).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

38.4.2.1.3.1.5 Usable effective serial number ***<usbefserno>***. The element ***<usbefserno>*** contains the information needed to identify a range of usable effective serial numbers when part numbers of spare/repair parts are not the same for all serial numbered equipment of the same model. The starting serial number is entered using the attribute 'beginserno' and the ending serial number is entered using the attribute 'endserno'.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment is graphically depicted.

FIGURE 420. Usable effective serial number <usbefserno> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT usbefserno EMPTY>
<!ATTLIST usbefserno   beginserno CDATA #REQUIRED
  endserno CDATA #IMPLIED>
```

d. Unique attributes :
   (1) **beginserno** – Single or beginning serial number (required). This attribute is used to show either a single serial number break in sequence, such as “before serial number 100” or it may be used as the start of a serial number sequence when used with the 'endserno' attribute. E.g., “use with serial numbers 100–110”. This attribute is REQUIRED.
   (2) **endserno** – Ending serial number (optional). This attribute works with the 'beginserno' value to establish a range of allowable serial (or other identifiable sequential listings such as tail number or vehicle ID) numbers.

38.4.2.1.3.1.6 Quantity required ***<qty>***. The last item unique element is the quantity required/used ***<qty>***. This is not a total quantity of the part. Total quantity is a Marine Corps requirement and is covered in 38.4.2.1.3.2.7.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment is graphically depicted.

FIGURE 421. Quantity required <qty> DTD hierarchy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT qty (#PCDATA)>
<!ATTLIST qty   inschlvl CDATA  #IMPLIED
      delchlvl CDATA  #IMPLIED
      label  CDATA  #IMPLIED
      texttype CDATA  #IMPLIED
      itemid  NMTOKEN #IMPLIED
      config  CDATA  #IMPLIED
      skilltrk CDATA  #IMPLIED
       idref    IDREFS  #IMPLIED
      assocfig IDREFS  #IMPLIED>
```

d. Common attributes:

(1) **inschlvl** – Insert change level (optional) (see 49.2.9).
(2) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(3) **label** – Label. (optional) A CDATA attribute that allows the author or system to enter various information about the ***<boi>***.
(4) **texttype** – Text type. (optional) This is a CDATA attribute with as yet unspecified usage. Until such time as the desired utilization is provided, recommend that this attribute not be used. Any values entered here may not provide the expected or desired output.
(5) **config** – Configuration (optional). Specifies the equipment configurations to which element applies.
(6) **skilltrk** – Skill level (optional) (see 49.2.2).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

38.4.2.1.3.2 Common part information. Common parts information is information that is used consistently with a specific part, regardless of where or how it is used. Common information includes such things as nomenclature, part number, or description. The common parts elements are described in the following paragraphs.

38.4.2.1.3.2.1 Source, maintenace, and recoverability code ***<smr>***. The source, maintenance, and recoverability code ***<smr>*** is normally a five character code that indicates the supply source, allowable maintenance level and authorized disposal maintenance level. For the Marine Corps, an optional sixth character may be used. The allowable values for each segment of the SMR code may be found in AR 700-82 which is a joint service regulation. For the Marine Corps, this is MCO 4400.120. The common parts information allows for multiple entries of the ***<smr>*** code. This allows for different support capability on a single part based on service, i.e., Army or Marine Corps, or different major command maintenance concepts. When multiple SMR codes are listed, the description ***<desc>*** should indicate who is authorized to use each code.

a. The components are:

(1) The element is EMPTY and all pertinent information is entered through its attributes.

b. DTD fragment is graphically depicted.

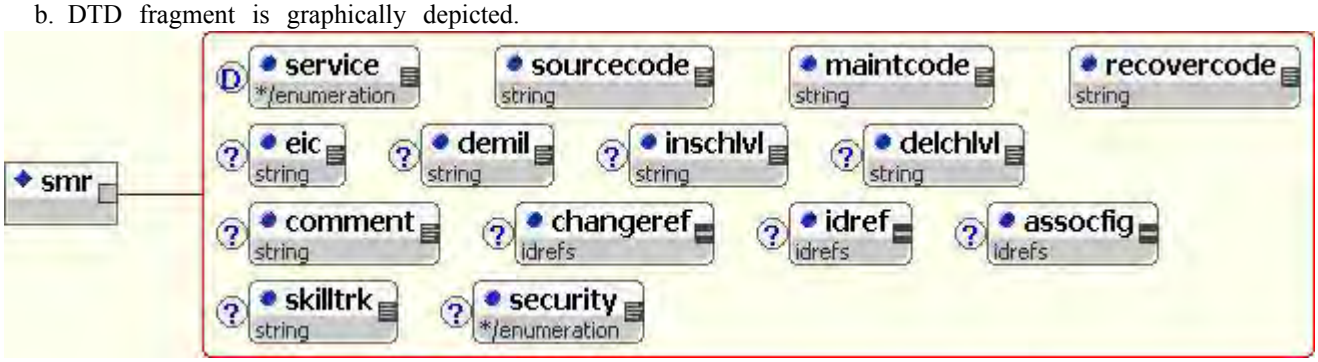

FIGURE 422. Source, maintenace, and recoverability code <smr> DTD hierarchy..

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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c. DTD fragment:

```
<!ELEMENT smr EMPTY>
<!ATTLIST smr   service (army | navy | AF | marine | CG) "army"
      sourcecode CDATA #REQUIRED
      maintcode CDATA #REQUIRED
      recovercode CDATA #REQUIRED
      eic CDATA #IMPLIED
      demil CDATA #IMPLIED
      inschlvl CDATA  #IMPLIED
      delchlvl CDATA  #IMPLIED
      label  CDATA  #IMPLIED
      texttype CDATA  #IMPLIED
      itemid  NMTOKEN #IMPLIED
      config  CDATA  #IMPLIED
      skilltrk CDATA  #IMPLIED
       idref    IDREFS  #IMPLIED
      assocfig IDREFS  #IMPLIED>
```

d. Unique attributes:

(1) **service** – A list of the military services with a default value of “army”.

(2) **sourcecode** – This is a (required)attribute. The **sourcecode** is a two character value that identifies the source of supply where the part may be procured. These are the first two characters in the SMR code.

(3) **maintcode** – This is a is a (required) two character code that identifies the maintenance level that is allowed to perform maintenance on the identified part. These are the third and fourth characters in the SMR code.

(4) **recoverability** – This is a code is a (required) single character code that identifies the maintenance level that is allowed to dispose of the identified part. This is the fifth character in the SMR code.

(5) **demil** – This is an (optional) sixth character. This attribute may be used by the Army when directed by the acquiring activity to identify demilitarization information. An explanation of these codes is contained in AR 700-82. The sixth character may also be used by the Marine Corps. Specific codes for the Marine Corps will be provided by the acquiring activity.

e. Common attributes:

(1) **label** – Label. (optional) A CDATA attribute that allows the author or system to enter various information about the ***<smr>***.

(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(4) **comment** – Change information (optional) (see 49.2.9).

(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(6) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(7) **inschlvl** – Insert change level (optional) (see 49.2.9).

(8) **security** – Security classification (optional) (see 49.2.13).

(9) **skilltrk** – Skill level (optional) (see 49.2.2).

38.4.2.1.3.2.2 National stock number ***<nsn>***. The National Stock Number ***<nsn>*** is an optional element in the parts list. When available, the ***<nsn>*** should be entered. The ***<nsn>*** element is broken into two sub elements, the federal stock class ***<fsc>*** and the national item identification number ***<niin>***. The ***<fsc>*** is the first four digits of the ***<nsn>***. The ***<niin>*** comprises the remainder, the last nine characters, of the ***<nsn>***. When displayed, the ***<nsn>*** is normally separated by dashes. Depending on the style sheet and where the ***<nsn>*** is used, the dashes may be entered by the author, or they may be generated by the style sheet. Authors should verify how the stock number will be displayed and if the acquiring activity desires the use of dashes.

a. The components are:

(1) Federal stock class ***<fsc>*** (required) (see 38.4.2.1.3.2.2.1).

(2) National item identification number ***<niin>*** (required) (see 38.4.2.1.3.2.2.2).

b. DTD fragment is graphically depicted.

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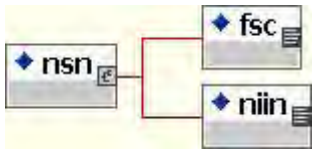

FIGURE 423. National stock number <nsn> DTD hierarchy.

c. DTD fragment:
```
<!ELEMENT nsn (fsc, niin)>
```
d. No attributes.

38.4.2.1.3.2.2.1 Federal stock class ***<fsc>***. The ***<fsc>*** element is used in the ***<nsn>*** to identify the first four digits of the stock number.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:
```
<!ELEMENT fsc (#PCDATA)>
```
c. No attributes.

38.4.2.1.3.2.2.2 National item identification number ***<niin>***. The ***<niin>*** consists of the remaining digits of the ***<nsn>***. The ***<niin>*** are the numbers used to search and sort stock number assets.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:
```
<!ELEMENT niin (#PCDATA)>
```
c. No attributes.

38.4.2.1.3.2.3 Part number ***<partno>***. The part number ***<partno>*** is one of four elements that are required for the common parts information. The ***<partno>*** identifies the manufacturers identifier for the part. When the ***<partno>*** is used in conjunction with the ***<cageno>*** (see 38.4.2.1.3.2.4), a unique identifier for a part is obtained.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment is graphically depicted.

FIGURE 424. Part number <partno> DTD hierarchy.

c. DTD fragment:
```
<!ELEMENT partno (#PCDATA)>
<!ATTLIST partno   idref    IDREFS  #IMPLIED
           assocfig IDREFS  #IMPLIED>
```
d. Unique attributes:
   (1) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

38.4.2.1.3.2.4 Commercial and government entity code ***<cageno>***. The commercial and government entity code ***<cageno>*** is the second of the four required entries for the common part information. The ***<cageno>*** is a five character entry that information on the part's manufacturer, such as name and address.

a. The DTD representation of the ***<cageno>*** content model is shown below:
```
<!ELEMENT cageno (#PCDATA)>
```
b. There are no attributes for the ***<cageno>*** element.

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38.4.2.1.3.2.5 Item nomenclature *<name>*. The item nomenclature *<name>* is the formal designator for the part. A part may have an informal name. The informal name is normally not found in the parts information list.

a. The components are:

b. DTD fragment is graphically depicted.

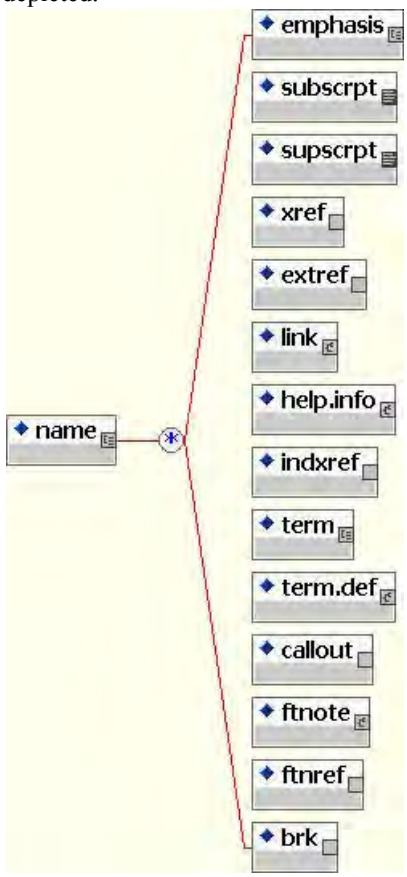

FIGURE 425. Item nomenclature <name> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT name (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link
| help.info | indxref | term | term.def | callout | ftnote | ftnref | brk)*>
<!ATTLIST name
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).

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(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

38.4.2.1.3.2.6 Item description ***<desc>***. The description ***<desc>*** is a block of text that provides some textual representation of the current part. Any special material, markings, dimensions, handling, or other relevant information may be included in the description. While there is no set limit for the description length, descriptions taken from data base systems may have a field length limit imposed through the data base structure.

a. The components are:
(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).

b. DTD fragment is graphically depicted.

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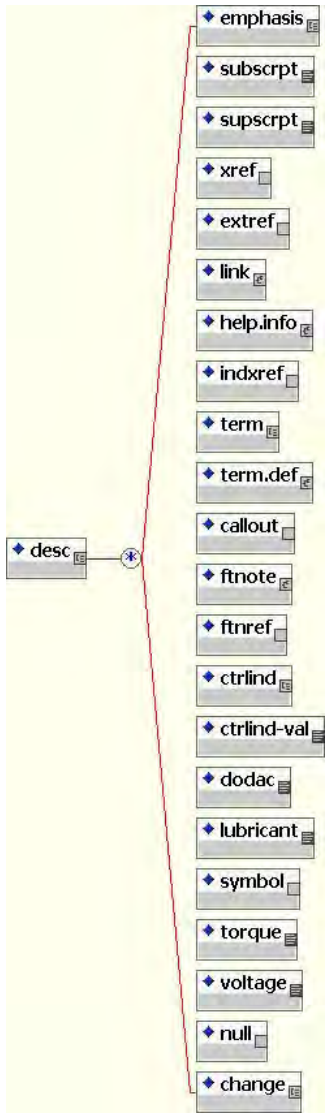

FIGURE 426. Item description <desc> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT desc (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link |
help.info | indxref | term | term.def | callout | ftnote | ftnref |
ctrlind | ctrlind-val | dodac | lubricant | symbol | torque | voltage |
null | change)*>
<!ATTLIST desc
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

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(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

38.4.2.1.3.2.7 Total quantity required ***<qty_per_end_item>***. The ***<qty_per_end_item>*** is a Marine Corps requirement that adds an extra column to the parts list. This column indicates the total quantity of part used throughout the weapons system. The ***<qty_per_end_item>*** is a character only element. It may be populated by direct entry by the author, or by the system if the system can perform simple arithmetic.

a. The components are:
(1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT qty_per_end_item (#PCDATA)>
<!ATTLIST qty_per_end_item
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

c. Common attributes:
(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

38.4.2.1.3.2.8 Unit of issue ***<ui>***. The unit of issue ***<ui>*** is a character field that identifies to the user normal quantity a part is issued. The unit of issue is defined in MIL-PRF-49506.

a. The components are:
(1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT ui (#PCDATA)>
<!ATTLIST ui  um CDATA  #IMPLIED >
```

c. Unique attributes:
(1) **um** – Unit of measure (optional). The difference between a unit of issue and unit of measure tends to be semantic in many respects. The unit of issue tends to be a specific unit such as 'each', 'box', or 'gallon' while a unit of measure is a value required from an abstract unit of issue. For example a unit of issue of a roll (RL) may contain 100 feet of the material. The unit of measure would be 100 feet. Or a unit of measure may specify the type of roll to be ordered if there are rolls of various lengths (i.e., 100 and 500 foot). A more precise explanation of units of issue and units of measure may be found in DOD 4100.39M, Federal Logistics System Information.

38.4.2.1.3.2.9 Linking elements ***<extref>*** or ***<link>***. The ***<pi.item>*** allows for two methods of identifying links to other documents or references outside the current document.

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a. Link ***<link>*** (optional – zero or more)(see 48.3.4.4.4).
b. External reference ***<extref>*** (optional – zero or more) (see 46.2.1).

38.4.2.1.3.2.10 Reference designator ***<refdes>***. The reference designator ***<refdes>*** is an optional element, used primarily with electronic components. The ***<refdes>*** is an alpha-numeric code that when used provides a top down breakdown of the major component. Development of reference designators is based on the requirements in ANSI Y 32.16 additional information on reference designators may also be found in AMC-P 700-25 available through the LOGSA web site.

a. The components are:
   (1) Parsable data (#PCDATA) (optional – zero or more). The element provides the text for the data. (see 9.3.2)
   (2) Emphasis ***<emphasis>*** (optional – zero or more). The element is used to emphasize the text of the entry. (see 49.1.3.1)
   (3) Subscript ***<subscrpt>*** (optional – zero or more). The element is used to format the text as subscript. (see 49.1.3.4)
   (4) Superscript ***<supscrpt>*** (optional – zero or more). The element is used to format the text as superscript. (see 49.1.3.5)
   (5) Cross Reference ***<xref>*** (optional – zero or more). The element is used to reference the work package sequence number, figure, table, step(s), etc. (see 46.2.2)
   (6) External Reference ***<extref>*** (optional – zero or more). The element is used to reference to external document information.(see 46.2.1)
   (7) Linking ***<link>*** (optional – zero or more). The element provides a capability to reference internal or external targets.(see 46.2.3)
b. DTD fragment is graphically depicted.

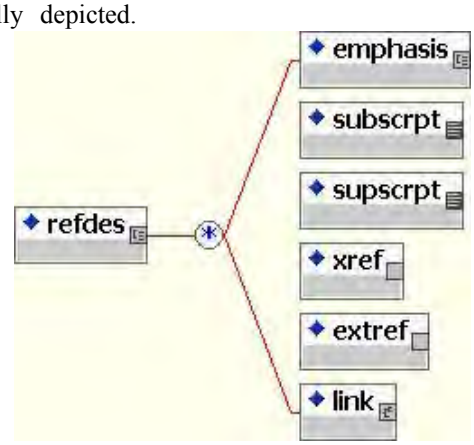

FIGURE 427. Reference designator <refdes> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT refdes (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link)*>
<!ATTLIST refdes   nsn CDATA #IMPLIED
  eic CDATA #IMPLIED

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Unique attributes:
   (1) **nsn** – National stock number (optional). This allows any NSN's associated with the reference designator to be included.
   (2) **eic** – End item code (optional). This allows the entry of a specific end item code with the reference designator.

e. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

38.4.2.1.3.2.11 Common part information reference ***<common_part_ref>***. The ***<common_part_ref>*** element is an optional element that allows the author to reference parts information that is common between two or more ***<pi.item>*** elements. An example would be a screw that is used in several locations. Each time the screw is used, it may have a separate index number if used in the current ***<pi.category>*** or will have a separate figure and index number if used in another ***<pi.category>***. By using the ***<common_part_ref>***'s 'idref' attribute with the 'id' of the initial use of the screw, the complete part information may be entered once and reused without reentry elsewhere.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.

b. DTD fragment:

```
<!ELEMENT common_part_ref EMPTY>
<ATTLIST common_part_ref idref IDREF #REQUIRED>
```

c. Unique attributes:
   (1) **idref** – Reference identifier(s) (required) (see 49.2.6).

d. An example of how this would be tagged is shown below. The first tag example is the initial use of the part with a complete ***<pi.item>***. The second example is the second use of the common part, where the ***<smr>***, ***<nsn>***, ***<partno>***, ***<cageno>***, ***<name>***, and ***<desc>*** information would be displayed using the initially entered data. Initial parts information entry:

<pi.item>
<callout assocfig="a1234567" label="3"/>
<qty>5
</qty>
<smr maintcode="oo" recovercode="d" sourcecode="pa"/>

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<nsn>
<fsc>6610
</fsc>
<niin>010001234
</niin>
</nsn>
<partno>a1234567
</partno>
<cageno>11111
</cageno>
<name>repair kit
</name>
<desc>The description, useableon, or other special information about the part and the quantity requiredfor this useage.
</desc>
</pi.item>
Second use of part:
<pi.item>
<callout assocfig="a1234567" label="4"/>
<qty>6
</qty>
<common_part_ref idref="a1234567-3"/>
</pi.item>

e. When produced and formatted, the above examples would appear as shown in Figure 428

(1) ITEM NO.	(2) SMR CODE	(3) NSN	(4) CAGEC	(5) PART NUMBER	(6) DESCRIPTION AND USABLE ON CODE(UOC)	(7) QTY
					GROUP 0001	
					FIGURE 1 FIGURE 1 TITLE	
3	OO	6610010001234	11111	A1234567	REPAIR KIT .....5 The description, useable on, or other special information about the part and the quantity required for this useage.	
4	OO	6610010001234	11111	a1234567	REPAIR KIT .....6 The description, useable on, or other special information about the part and the quantity required for this useage.	
					END OF FIGURE	

FIGURE 428. Formatted common parts reference example.

38.4.2.2 Tagging the ***<plwp>***. The following subparagraphs provide examples on how to tag a ***<plwp>*** for both page based and framed based TMs. The examples for frame based present only one possible approach to tagging. The page based example, which has a more fixed presentation, should be followed.

The examples are for a basic parts list item. More complex tagging for parts with kit parts, indents or common information is described and illustrated in 38.4.2.3.

38.4.2.2.1 Page based tagging. As stated in 38.4.2.1.3, the order parts information is entered does not follow the page based presentation order.

<plwp wpno="P-00025-9-2350-361">
<wpidinfo>

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```
<maintlvl level="dirsup"/>
<title>plwp title
</title>
</wpidinfo>
<pi.category>
<figure id="a1234567">
<title>figtitle
</title>
<graphic boardno="plwp1" hscale="60" vscale="60">
</graphic>
</figure>
<std.table.header cols="7">
<std.table.row>
<std.table.entry colnum="1">(1)
</std.table.entry>
<std.table.entrycolnum="2">(2)
</std.table.entry>
<std.table.entry colnum="3">(3)
</std.table.entry>
<std.table.entry colnum="4">(4)
</std.table.entry>
<std.table.entrycolnum="5">(5)
</std.table.entry>
<std.table.entry colnum="6">(6)
</std.table.entry>
<std.table.entry colnum="7">(7)
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry colnum="1">ITEM
</std.table.entry>
<std.table.entry colnum="2">SMR
</std.table.entry>
<std.table.entrycolnum="3">
</std.table.entry>
<std.table.entry colnum="4">
</std.table.entry>
<std.table.entry colnum="5">PART
</std.table.entry>
<std.table.entrycolnum="6">DESCRIPTION AND USABLE ON
</std.table.entry>
<std.table.entrycolnum="7">
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry colnum="1">
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry colnum="2">NO.
</std.table.entry>
<std.table.entry colnum="3">CODE
</std.table.entry>
<std.table.entrycolnum="4">NSN
</std.table.entry>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<std.table.entry colnum="">CAGEC
</std.table.entry>
<std.table.entry colnum="5">NUMBER
</std.table.entry>
<std.table.entrycolnum="6">CODE (UOC)
</std.table.entry>
<std.table.entrycolnum="7">QTY
</std.table.entry>
</std.table.row>
</std.table.header>
<fncgrp>
<fnccode>0100
</fnccode>
<fnctitle>engine assembly
</fnctitle>
</fncgrp>
<pi.item>
<callout assocfig="a1234567" label="1"/>
<uoc>all
</uoc>
<qty>1
</qty>
<smr maintcode="oo" recovercode="d" sourcecode="pa"/>
<nsn>
<fsc>6610
</fsc>
<niin>010001234
</niin>
</nsn>
<partno>a1234567
</partno>
<cageno>11111
</cageno>
<name>Item name
</name>
<desc>Description of the item thatis named.
</desc>
</pi.item>
<pi.item>
<callout assocfig="a1234567" label="2"/>
<uoc>some
</uoc>
<qty>10
</qty>
<smr maintcode="oo"recovercode="d" sourcecode="pa"/>
<nsn>
<fsc>6610
</fsc>
<niin>010001234
</niin>
</nsn>
<partno>a1234567
</partno>
<cageno>11111
</cageno>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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```
<name>another  part
</name>
<desc>The description, useable on, or other special informationabout
the part and the quantity required for this useage.
</desc>
</pi.item>
<pi.item>
<callout  assocfig="a1234567"  label="3"/>
<qty>5
</qty>
<smr  maintcode="oo"  recovercode="d"  sourcecode="pa"/>
<nsn>
<fsc>6610
</fsc>
<niin>010001234
</niin>
</nsn>
<partno>a1234567
</partno>
<cageno>11111
</cageno>
<name>another  part
</name>
<desc>The description, useableon, or other special information about
the part and the quantity requiredfor this useage.
</desc>
</pi.item>
</pi.category>
</plwp>
```

Figure 429 illustrates the formatted output.

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DIRECT SUPPORT MAINTENANCE

PLWP TITLEINTRODUCTION

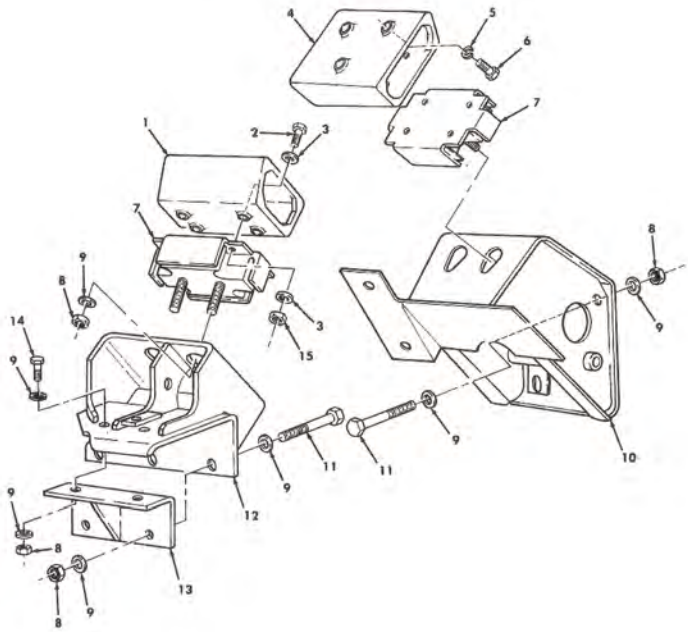

Figure 2 fig title

(1) ITEM NO.	(2) SMR CODE	(3) NSN	(4) CAGEC	(5) PART NUMBER	(6) DESCRIPTION AND USABLE ON CODE(UOC)	(7) QTY
					GROUP 0100 ENGINE ASSEMBLY	
					FIGURE 2 FIG TITLE	
1	PAOOE	6610010001234	11111	A1234567	ITEM NAME.....1 Description of the item that is named. UOC: ALL	

0000 00-8

FIGURE 429. Page based <plwp> example output.

38.4.2.3 Parts items ***<pi.item>*** with kit parts, indents, or part reference info. The ***<pi.item>*** allows for more complex support of parts information. The ***<pi.item>*** supports including other ***<pi.item>***s, a single ***<pi.category>***, parts that are included in a repair kit (***<kititem>***), or a reference to another part. Each option is discussed in the following sub paragraphs.

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38.4.2.3.1 Nested parts items. A parts list may contain subordinate parts. These are tagged with their own ***<pi.item>***. The nested ***<pi.item>***s may be items such as attaching parts or sub components. For items that may have complex component break downs, the nested ***<pi.category>*** should be used IAW 38.4.2.3.2.

Each additional ***<pi.item>*** is tagged as described in 38.4.2.1.3.

38.4.2.3.2 Nested parts category. Normally a parts category would be used independently. It is recommended that the independent approach be used. If it is necessary to list a series of higher level parts together on one illustration/list, the ***<pi.item>*** for the each upper level part may contain a ***<pi.category>*** containing the breakdown for the higher level component. It is this nesting that allows a complete repair parts list to be contained in a single ***<plwp>***.

When a ***<pi.category>*** is included in a ***<pi.item>***, it will be tagged as described in 38.4.2.1.

38.4.2.3.3 Kit parts with part items . MIL-STD-40051-1 and MIL-STD-40051-2 provide two methods of entering parts information for parts contained in repair kits. The first method is to include the kit parts ***<kititem>*** with the repair parts listing, the ***<pi.item>*** and is described in the next paragraphs. The second method is to enter all kit parts in their own work package. The second method is described in 38.4.4.

38.4.2.3.3.1 Kit items ***<kititem>***. The ***<kititem>*** element is used to identify kit parts when they are listed in the ***<pi.item>***. When used, the ***<kititem>***(s) will appear after the last repair part ***<pi.item>*** in the category.

a. The components are:
   (1) Item nomenclature ***<name>*** see 38.4.2.1.3.2.5.
   (2) Quantity required ***<qty>*** see 38.4.2.1.3.1.6.
   (3) Item callout ***<callout>*** see 38.4.2.1.3.1.2.

b. DTD fragment is graphically depicted.

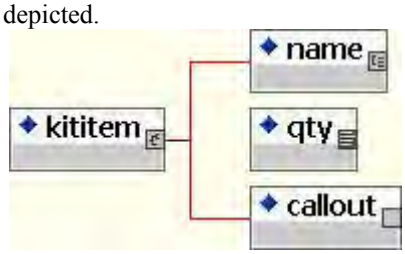

FIGURE 430. Kit items <kititem> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT kititem (name, qty, callout)>
<!ATTLIST kititem
    assocfig        IDREFS                          #IMPLIED
    changeref       IDREFS                          #IMPLIED
    comment         CDATA                           #IMPLIED
    delchlvl        CDATA                           #IMPLIED
    id              ID                              #IMPLIED
    idref           IDREFS                          #IMPLIED
    inschlvl        CDATA                           #IMPLIED
    security        (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk        CDATA                           #IMPLIED>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

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(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

38.4.2.3.3.2 Tagging the ***<kititem>***. The ***<kititem>*** is tagged as part of a parent ***<pi.item>*** that identifies the kit. The following elements in the ***<pi.item>*** are used when tagging the ***<kititem>***.

a. The ***<callout>*** may be used, but if it is used, it is required by MIL-STD-40051-1 or MIL-STD-40051-2 to have the callout values be letters.
b. The ***<qty>*** tag may be used. If the quantity to be used may vary, the letter "V" may be used in place of a specific quantity value.
c. The ***<smr>*** tag is required and will indicate the SMR code of the kit.
d. The ***<nsn>*** may be entered if the kit has an NSN assigned to it.
e. The ***<partno>*** and ***<cageno>*** elements are required to be entered.

```
<pi.item>
<qty>V
</qty>
<smr maintcode="oz" recovercode="z" sourcecode="PA"/>
<partno>k-123456
</partno>
<cageno>11111
</cageno>
<name>repair kit example
</name>
<desc>
</desc>
<kititem>
<name>first repair kit part
</name>
<qty>1
</qty>
<callout assocfig="a1234567" label="5" partref="part-ex-5"/>
</kititem>
<kititem>
<name>second repair kit part
</name>
<qty>1
</qty>
<callout assocfig="a1234567" label="5" partref="part-ex-5"/>
</kititem>
<kititem>
<name>third repair kit part
</name>
<qty>1
</qty>
<callout assocfig="a1234567" label="5" partref="part-ex-5"/>
</kititem>
</pi.item>
```

For a page based TM, the above tagging would appear as shown in Figure 431.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

(1) ITEM NO.	(2) SMR CODE	(3) NSN	(4) CAGEC	(5) PART NUMBER	(6) DESCRIPTION AND USABLE ON CODE(UOC)	(7) QTY
4	PAOOD	6610010001234	11111	A1234567	ANOTHER PART ..... 10 The description, useable on, or other special information about the part and the quantity required for this use age. UOC: SOME	
5	PAOOD	6610010001234	11111	A1234567	REPAIR KIT ..... 5 The description, useable on, or other special information about the part and the quantity required for this use age.	
	PAOZZ		11111	K-123456	REPAIR KIT EXAMPLE ..... V FIRST REPAIR KIT PART (1) 2-5 SECOND REPAIR KIT PART (1) 2-5 THIRD REPAIR KIT PART (1) 2-5 END OF FIGURE	

FIGURE 431. Example <kititem> output.

38.4.2.3.4 Reference to other parts lists ***<part.breakdown.ref>***. The last element that may be used in the ***<pi.item>*** is the ***<part.breakdown.ref>***. The ***<part.breakdown.ref>*** may be used to point to another ***<plwp>***, ***<pi.category>***, or ***<pi.item>***.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT part.breakdown.ref EMPTY>
<!ATTLIST part.breakdown.ref   idref IDREF #REQUIRED>
```

c. Unique attribute:
   (1) **idref** – Reference identifier(s) (required) (see 49.2.6).

38.4.3 Special tools repair parts ***<stl-partswp>***. The special tools repair parts work package is an optional work package. The purpose of the ***<stl-partswp>*** is to provide a listing of any repair parts needed for special tools listed in the special tools work package 38.4.6. There are three criteria in MIL-STD-40051-1 and MIL-STD-40051-2 that are required to be met in order for a ***<stl-partswp>*** to be prepared. These criteria are:

a. The RPSTL identifies special tools in the special tools list work package.
b. The special tool has repair parts that may be replaced at any maintenance level covered in the TM.
c. The special tool does not have repair instructions and parts listed in another technical manual for the special tool.

If a ***<stl-partswp>*** is prepared, it is done in the same manner as a ***<plwp>*** (see 38.4.2 and subordinate paragraphs) except that the functional group code is “SPECIAL TOOLS (REPAIR PARTS)”.

a. The components are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Parts information category ***<pi.category>*** (required – one or more) (see 38.4.2.1).
b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

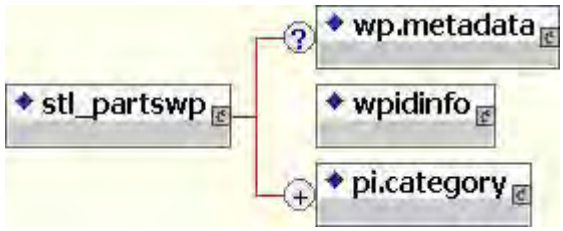

FIGURE 432. Special tools repair parts <stl-partswp> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT stl_partswp (wp.metadata?, wpidinfo, pi.category+)>
<!ATTLIST stl_partswp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

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(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

38.4.4 Repair parts kits work package ***<kitswp>***. As stated in 38.4.2.3.3, there are two methods to enter repair parts that are contained in kits. The following paragraphs describe the second methodMIL-STD-40051-1 and MIL-STD-40051-2, using a separate work package. When the ***<kitswp>*** is used, all repair parts contained in kits will be listed in the work package. Mixing of Option 1 and Option 2 is not allowed by the standard.

a. The components are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Parts information category ***<pi.category>*** (required – one or more) (see 38.4.2.1).
b. DTD fragment is graphically depicted.

FIGURE 433. Repair parts kits work package <kitswp> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT kitswp (wp.metadata?, wpidinfo, pi.category+)>
<!ATTLIST kitswp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).

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(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

38.4.4.1 Tagging the ***<kitwp>***. The following example markup illustrates the recommended method to tag a ***<kitswp>***.

```
<kitswp>
<wpidinfo>
<maintlvl level="field"/>
<title>separate kits work package
</title>
</wpidinfo>
<pi.category>
<figure id="kitswpf-1">
<title>Kits
</title>
</figure>
<std.table.header cols="7">
<std.table.row>
<std.table.entrycolnum="1">(1)
</std.table.entry>
<std.table.entry colnum="2">(2)
</std.table.entry>
<std.table.entry colnum="3">(3)
</std.table.entry>
<std.table.entrycolnum="4">(4)
</std.table.entry>
<std.table.entry colnum="5">(5)
</std.table.entry>
<std.table.entry colnum="6">(6)
</std.table.entry>
<std.table.entrycolnum="7">(7)
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry colnum="1">ITEM
</std.table.entry>
<std.table.entry colnum="2">SMR
</std.table.entry>
<std.table.entrycolnum="3">
</std.table.entry>
<std.table.entry colnum="4">
</std.table.entry>
<std.table.entry colnum="5">PART
</std.table.entry>
<std.table.entry colnum="6">DESCRIPTION AND USABLE ON
</std.table.entry>
<std.table.entry colnum="7">
</std.table.entry>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</std.table.row>
<std.table.row>
<std.table.entry  colnum="1">
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry  colnum="2">NO.
</std.table.entry>
<std.table.entry  colnum="3">CODE
</std.table.entry>
<std.table.entrycolnum="4">NSN
</std.table.entry>
<std.table.entry  colnum="">CAGEC
</std.table.entry>
<std.table.entry  colnum="5">NUMBER
</std.table.entry>
<std.table.entrycolnum="6">CODE  (UOC)
</std.table.entry>
<std.table.entrycolnum="7">QTY
</std.table.entry>
</std.table.row>
</std.table.header>
<fncgrp>
<fnccode>94
</fnccode>
<fnctitle>Repair  kits
</fnctitle>
</fncgrp>
<pi.item>
<uoc>all
</uoc>
<qty>1
</qty>
<smr  maintcode="z"recovercode="dd"  sourcecode="pa"/>
<partno>b1234
</partno>
<cageno>11111
</cageno>
<name>repair  kit  for  xyz
</name>
<desc>
</desc>
<kititem>
<name>first  kit  part
</name>
<qty>2
</qty>
<callout  assocfig="kitswpf-1"  label="1"/>
</kititem>
<kititem>
<name>second  kit  part
</name>
<qty>5
</qty>
<callout  assocfig="kitswpf-1"  label="2"/>
```


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```
</kititem>
<kititem>
<name>third  kit  part
</name>
<qty>5
</qty>
<calloutassocfig="kitswpf-1"  label="3"/>
</kititem>
</pi.item>
<pi.item   id="kit-ex-123">
<uoc>all
</uoc>
<qty>1
</qty>
<smr  maintcode="oo"  recovercode="z"  sourcecode="pa"/>
<nsn>
<fsc>4920
</fsc>
<niin>019897654
</niin>
</nsn>
<partno>C1234
</partno>
<cageno>11111
</cageno>
<name>repairkit  for  abc
</name>
<desc>
</desc>
<kititem>
<name>firstkit  part  for  abc
</name>
<qty>2
</qty>
<callout  assocfig="a1234567"label="4"/>
</kititem>
<kititem>
<name>second  kit  partfor  abc
</name>
<qty>5
</qty>
<callout  assocfig="kitswpf-1"label="5"/>
</kititem>
<kititem>
<name>third  kit  part  for  abc
</name>
<qty>5
</qty>
<callout  assocfig="kitswpf-1"label="6"/>
</kititem>
</pi.item>
</pi.category>
</kitswp>
```


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38.4.5 Bulk items work package ***<bulk_itemswp>***. The bulk items work package ***<bulk_itemswp>*** is an optional component of the overall parts list. Both MIL-STD-40051-1 and MIL-STD-40051-2 state this work package is required if there are bulk items listed in any of the other parts lists such as a ***<plwp>*** or ***<kitswp>***.

The ***<bulk_itemswp>*** content model differs slightly from the other parts listing work packages. The bulk items work package is a single functional group and does not use a figure to illustrate any of the bulk items listed.

a. The components are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Standard Table Header ***<std.table.header>*** (required). The element provides a heading for bulk material list (see 42.3.1).
   (4) Functional group ***<fncgrp>*** (required) see 38.4.2.1.2.
   (5) Parts information ***<pi.item>*** (required – one or more)see 38.4.2.1.3.

b. DTD fragment is graphically depicted.

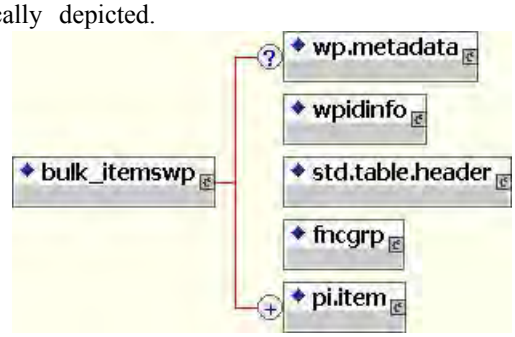

FIGURE 434. Bulk items work package <bulk_itemswp> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT bulk_itemswp (wp.metadata?, wpidinfo, std.table.header, fncgrp, pi.item+)>
<!ATTLIST bulk_itemswp
    airforce         (yes | no )               "no"
    army             (yes | no )               "no"
    assocfig         IDREFS                     #IMPLIED
    changeref        IDREFS                     #IMPLIED
    comment          CDATA                      #IMPLIED
    crewmember       CDATA                      #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                      #IMPLIED
    deletewp         (yes | no )               "no"
    fgc              CDATA                      #IMPLIED
    frame            (yes | no )               "yes"
    idref            IDREFS                     #IMPLIED
    inschlvl         CDATA                      #IMPLIED
    insertwp         CDATA                      #IMPLIED
    lsa-id           CDATA                      #IMPLIED
    marines          (yes | no )               "no"
    navy             (yes | no )               "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                      #IMPLIED
    tocentry         (2 | 3 | 4 | 5)           "2"
    wpno             ID                         #REQUIRED
    wpseq            CDATA                      #IMPLIED>
```

d. Common attributes:

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(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

38.4.5.1 Tagging exceptions for the bulk items work package . When tagging the ***<bulk_itemswp>*** using the ***<pi.item>***, the user should not use the following tags:

a. ***<pi.category>*** – There are no sub categories for the bulk items and the ***<pi.category>*** is not needed.
b. ***<kititem>*** – Though a kit may include a requirement to use a bulk item and may include some quantity of the item in the kit. The ***<bulk_itemswp>*** does not list kits or their items as such.
c. ***<pi.item>*** – There is no need to list any bulk item breakdowns.

38.4.5.2 Tagging instructions for the bulk items work package . When tagging the ***<bulk_itemswp>***, the ***<fncgrp>*** should be tagged using the following. The ***<fncode>*** should be "BULK MATERIAL" and the ***<fnctitle>*** should indicate "FIG BULK" MIL-STD-40051-1 or MIL-STD-40051-2 for complete details.

38.4.6 Special tools work package ***<stlwp>***. The special tools work package ***<stlwp>*** is the last of the parts listing work packages. It is optional and is only required byMIL-STD-40051-1 or MIL-STD-40051-2 when there are any "special tools, special TMDE, and other special support equipment" required to support maintenance of the system.

a. The components are:
(1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).

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(2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
(3) Parts information category ***<pi.category>*** (required – one or more) (see 38.4.2.1).

b. DTD fragment is graphically depicted.

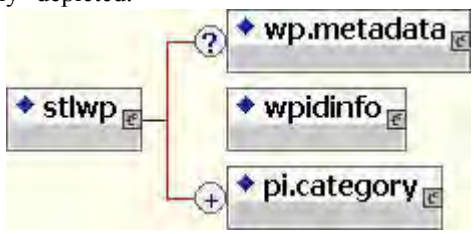

FIGURE 435. Special tools work package <stlwp> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT stlwp (wp.metadata?, wpidinfo, pi.category+)>
<!ATTLIST stlwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
```

d. Common attributes:

(1) Parsable data (#PCDATA) (optional – zero or more). The element provides the text for the data. (see 9.3.2)
(2) Emphasis ***<emphasis>*** (optional – zero or more). The element is used to emphasize the text of the entry. (see 49.1.3.1)
(3) Subscript ***<subscrpt>*** (optional – zero or more). The element is used to format the text as subscript. (see 49.1.3.4)
(4) Superscript ***<supscrpt>*** (optional – zero or more). The element is used to format the text as superscript. (see 49.1.3.5)
(5) Cross Reference ***<xref>*** (optional – zero or more). The element is used to reference the work package sequence number, figure, table, step(s), etc. (see 46.2.2)
(6) External Reference ***<extref>*** (optional – zero or more). The element is used to reference to external document information.(see 46.2.1)
(7) Linking ***<link>*** (optional – zero or more). The element provides a capability to reference internal or external targets.(see 46.2.3)

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(8) Content Sensitive Help ***<help.info>*** (optional – zero or more). The element is used for "help information" about the technical data used in frame-based manuals.(see 48.3.2.4)
(9) Index Marker Reference ***<indxref>*** (optional – zero or more). The element is used (see 31.4.2.2.3)
(10) Term ***<term>*** (optional – zero or more). The element provides a word, phrase, acronym, symbol, or abbreviation in a definition and legend list.(see 49.1.2.3.1.1)
(11) Item Definition ***<term.def>*** (optional – zero or more). The element provides a wrapper for a term and corresponding definition. (see 49.1.2.3.1)
(12) Figure Callout ***<callout>*** (optional – zero or more). The element is used for a figure reference and callout reference, number, letter, or symbol appearing in the figure.(see 46.2.4.1)
(13) Footnote ***<ftnote>*** (optional – zero or more). The element contains the footnote information which is referenced in the document with footnote reference.(see 45.1.1)
(14) Footnote Reference ***<ftnref>*** (optional – zero or more). The element is the reference in the text to the footnote. (see 45.1.2)
(15) Department of Defense Ammunition Code ***<dodac>*** (optional – zero or more). The element provides identification of an ammunition type.(see 49.1.4.5)
(16) Lubricant ***<lubricant>*** (optional – zero or more). The element provides identification of a lubricant.(see 49.1.4.16)
(17) Symbol ***<symbol>*** (optional – zero or more). The element provides a graphic symbol not found in standard ISO character sets that is inserted as a graphic in text.(see 44.3.1)
(18) Torque Value or Limit ***<torque>*** (optional – zero or more). The element provides a torque value or limit embedded in the text or table entry.(see 49.1.4.24)
(19) Voltage Measurement ***<voltage>*** (optional – zero or more). The element provides the identification of a critical voltage measurement.(see 49.1.4.25)
(20) Null ***<null>*** (optional – zero or more). The element specifies that it contains no content and marking is identified by it's attribute.(see 49.1.3.2)
(21) Text Change Information ***<change>*** (optional – zero or more). The element provides the scope and type of changed information that is indicated by being enclosed within start and end tag. (see 49.1.3.7)
(22) Internet Address ***<internet>*** (optional – zero or more). The element provides the reference to the internet by email or home page (see 49.1.4.1.7)
(23) Proponent ***<proponent>*** (optional – zero or more). The element identifies the proponent name and address.(see 49.1.4.22)
(24) Telephone Number ***<phone>*** (optional – zero or more). The element provides the phone number for the connection type and either voice or fax phone number.(see 49.1.4.1.6)
(25) Instruction Plates and Decals ***<instructplt>*** (optional – zero or more). The element is used to specify relevant decals and instruction plates that are located on the equipment.(see 49.1.4.13)

38.4.6.1 Tagging exceptions for a special tools work package. The ***<stlwp>*** is tagged in the same manner as the ***<plwp>*** see 38.4.2 with the following excepctions:
a. The ***<fnccode>*** is "SPECIAL TOOLS"
b. There should be no sub categories so the ***<pi.category>*** within a ***<pi.item>*** is not needed.

38.5 Parts index work packages. There are three optional index work packages allowed in a parts manual. They are:
a. The National Stock Number Index 38.5.1.
b. The Part Number Index 38.5.2.
c. The Reference Designator Index 38.5.3.

Tagging of each of these indexes is the same and will be shown only for the ***<nsnindxwp>***.

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OPERATOR, UNIT, DIRECT AND GENERAL SUPPORT MAINTENANCE

NATIONAL STOCK NUMBER INDEX

STOCK NUMBER	FIG.	ITEM
5305-00-054-5637	21	
5305-00-054-6652	20	
5305-00-054-6654	14	
5305-00-068-5405	20	
5305-00-068-5414	14	
	20	
5305-00-958-6517	20	
	20	
5305-00-988-7602	20	
	22	
5305-00-988-7613	22	
5310-00-531-9514	14	
	20	
5310-00-929-6395	14	
	20	
5310-00-933-8118	14	
5310-00-934-9748	14	
5310-00-934-9761	14	
	20	g11-6625-3613-14P-0058-198
5310-01-138-4232	15	
5310-01-141-6672	14	g11-6625-3613-14P-0052-123
5930-00-241-3547	20	
5935-00-001-8655	8	
5935-00-171-0362	4	
5935-00-205-9383	10	
5935-00-344-1952	9	
5935-00-356-7770	11	
5935-00-394-0564	5	
5935-00-417-4583	7	
5935-00-430-4044	4	
5935-01-024-4027	6	
5940-01-225-6499	15	
5999-00-243-6500	6	
5999-00-473-3551	4	
	5	
	6	
	7	
	8	
	9	
	10	
	11	
	12	
5999-00-478-4402	5	
	8	
	9	
	10	
	11	
5999-01-106-7076	4	

END OF WORK PACKAGE

FIGURE 436. Example national stock number index work package <nsnindxwp>.

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38.5.1 National stock number index work package ***<nsnindxwp>***. The National Stock Number index ***<nsnindx>*** is a required part of the parts listing data. The ***<nsnindxwp>*** lists each stock number listed in each ***<plwp>***, ***<stl-partswp>***, ***<kitswp>***, ***<bulk_itemswp>***, or ***<stlwp>***. The list is in stock number sequence, sorted by the NIIN, or the last nine characters of the NSN. Each stock number only needs to be shown once except as stated for page based TMs. The exception for page based TMs is the entire stock number will be shown after any page break. The National Stock Number index work package does not require any initial setup ***<wpinfo>***.

a. The components are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) National Stock Number index ***<nsnindx>*** (required) (see 38.5.1.1).

b. DTD fragment is graphically depicted.

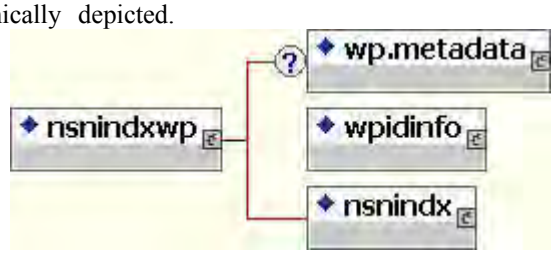

FIGURE 437. National stock number index work package <nsnindxwp> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT nsnindxwp (wp.metadata?, wpidinfo, nsnindx)>
<!ATTLIST nsnindxwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED>
```

d. Common attributes:
   (1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
   (2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
   (3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

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(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

38.5.1.1 National stock number index ***<nsnindx>***. The ***<nsnindx>*** is actual content tag for the ***<nsnindxwp>***. Each index contains a standard table header ***<std.table.header>*** and one or more ***<nsnindxrow>*** elements.

a. The components are:
   (1) National Stock Number index row ***<nsnindxrow>*** (required – one or more) (see 38.5.1.1.1).
   (2) Standard table header ***<std.table.header>*** (see 42.3.1).
b. DTD fragment is graphically depicted.

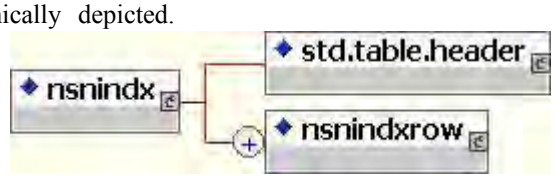

FIGURE 438. National stock number index <nsnindx> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT nsnindx (std.table.header, nsnindxrow+)>
<!ATTLIST nsnindx
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

38.5.1.1.1 National stock number index row ***<nsnindxrow>***. The ***<nsnindxrow>*** allows the entry of each ***<nsn>*** and its associated ***<callout>*** information. The ***<nsn>*** only needs to be entered once. A ***<callout>*** is required for each figure and index reference to the stock number.

a. The components are:

(1) National Stock Number index row ***<nsn>*** (required) (see 38.4.2.1.3.2.2).
(2) Callout ***<callout>*** (required – one or more) (see 46.2.4.1).

b. DTD fragment is graphically depicted.

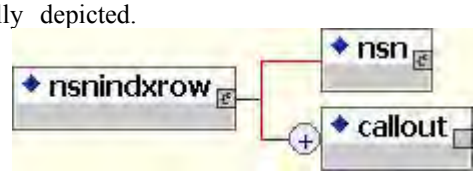

FIGURE 439. National stock number index row <nsnindxrow> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT nsnindxrow (nsn, callout+)>
<!ATTLIST nsnindxrow
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).

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(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

38.5.1.2 Tagging the ***<nsnindxwp>***. The ***<nsnindxwp>*** is made up of standard table header ***<std.table.header>*** one or more ***<nsnindxrow>*** elements. The standard table header lists the header for each column. Each row contains a single ***<nsn>*** and one or more ***<callout>*** elements. The ***<callout>*** is used to provide the reference or link to the figure and index where the NSN is listed. Partial tagging for the example in Figure 436 is shown below.

```
<nsnindxwp wpno="WP0023" wpseq="0023 00">
<wpidinfo>
<maintlvl level="field">
<title>
<indxref ref1="National Stock Number Index">
</title>
</wpidinfo>
<nsnindx>
<std.table.header cols="3">
<std.table.row>
<std.table.entry colnum="1">STOCK NUMBER
</std.table.entry>
<std.table.entry colnum="2">FIGURE
</std.table.entry>
<std.table.entrycolnum="3">ITEM
</std.table.entry>
</std.table.row>
</std.table.header>
<nsnindxrow>
<nsn>7025-01-474-3789
</nsn>
<callout assocfig="D-1" id="n2a">
</nsnindxrow>
<nsnindxrow>
<nsn>7025-01-474-3791
</nsn>
<callout assocfig="D-1" id="n4">
</nsnindxrow>
<nsnindxrow>
<nsn>7025-01-474-3792
</nsn>
<callout assocfig="D-1" id="n4a">
<callout assocfig="D-2" id="n5a1">
<callout assocfig="D-4" id="a21">
</nsnindxrow>
<nsnindxrow>
<nsn>7021-01-474-3793
</nsn>
<callout assocfig="D-1" id="n1a">
</nsnindxrow>
<nsnindxrow>
<nsn>7025-01-474-5753
</nsn>
<callout assocfig="D-1" id="n2">
</nsnindxrow>
```


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```
<nsnindxrow>
<nsn>7021-01-475-0217
</nsn>
<callout assocfig="D-1" id="n1">
</nsnindxrow>
<nsnindxrow>
<nsn>7025-01-475-0229
</nsn>
<callout assocfig="D-1" id="n3">
</nsnindxrow>
<nsnindxrow>
<nsn>7025-01-475-0280
</nsn>
<callout assocfig="D-1" id="n3b">
</nsnindxrow>
<nsnindxrow>
<nsn>7025-01-475-0282
</nsn>
<callout assocfig="D-1" id="n3a">
</nsnindxrow>
<nsnindxrow>
<nsn>7520-01-484-1219
</nsn>
<callout assocfig="D-1" id="n5">
</nsnindxrow>
</nsnindx>
</nsnindxwp>
```

38.5.2 Part number index work package ***<pnindxwp>***. The part number index, like the NSN index, is a required segment of the part listing. The ***<pnindxwp>*** lists each individual part contained in each ***<plwp>***, ***<stl-partswp>***, ***<kitswp>***. ***<bulk_itemswp>***, or ***<stlwp>***. The ***<pnindxwp>*** is sorted in part number sequence and provides the same sequence of information as found in the National Stock Number index (see 38.5.1).

a. The components are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Part number index ***<pnindx>*** (required) (see 38.5.2.1).

b. DTD fragment is graphically depicted.

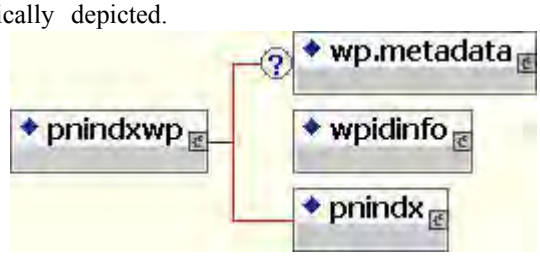

FIGURE 440. Part number index work package <pnindxwp> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT pnindxwp (wp.metadata?, wpidinfo, nsnindx)>
<!ATTLIST pnindxwp
    airforce         (yes | no )               "no"
    army             (yes | no )               "no"
    assocfig         IDREFS                     #IMPLIED
    changeref        IDREFS                     #IMPLIED
    comment          CDATA                      #IMPLIED
    crewmember       CDATA                      #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                      #IMPLIED
    deletewp         (yes | no )               "no"
    fgc              CDATA                      #IMPLIED
    frame            (yes | no )               "yes"
    idref            IDREFS                     #IMPLIED
    inschlvl         CDATA                      #IMPLIED
    insertwp         CDATA                      #IMPLIED
    lsa-id           CDATA                      #IMPLIED
    marines          (yes | no )               "no"
    navy             (yes | no )               "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                      #IMPLIED
    tocentry         (2 | 3 | 4 | 5)           "2"
    wpno             ID                         #REQUIRED
    wpseq            CDATA                      #IMPLIED>
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

38.5.2.1 Part number index ***<pnindx>***. The ***<pnindx>*** is actual content tag for the ***<pnindxwp>***. Each index contains a standard table header ***<std.table.header>*** (see 42.3.1) and one or more ***<pnindxrow>*** (see 38.5.2.1.1) elements.

a. The components are:
(1) Standard table header ***<std.table.header>*** (see 42.3.1).
(2) Parts index row ***<pnindxrow>*** (required – one or more) (see 38.5.2.1.1).

b. DTD fragment is graphically depicted.

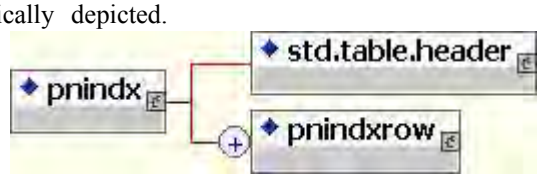

FIGURE 441. Part number index <pnindx> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT pnindx (std.table.header, pnindxrow+)>
<!ATTLIST pnindx
    assocfig        IDREFS                      #IMPLIED
    changeref       IDREFS                      #IMPLIED
    comment         CDATA                       #IMPLIED
    delchlvl        CDATA                       #IMPLIED
    idref           IDREFS                      #IMPLIED
    inschlvl        CDATA                       #IMPLIED
    security        (uc | fouo |c | s | ts)     #IMPLIED
    skilltrk        CDATA                       #IMPLIED>
```

d. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

38.5.2.1.1 Part number index row ***<pnindxrow>*** content model. The ***<pnindxrow>*** allows the entry of each ***<partno>*** and its associated ***<callout>*** information. The ***<partno>*** only needs to be entered once. A ***<callout>*** is required for each figure and index reference to the stock number.

a. The components are:
(1) Part number ***<partno>*** (required) (see 38.4.2.1.3.2.3).

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(2) Commercial and government entity code ***<cageno>*** (required) (see 38.4.2.1.3.2.4)
(3) Callout ***<callout>*** (required – one or more) (see 46.2.4.1).

b. DTD fragment is graphically depicted.

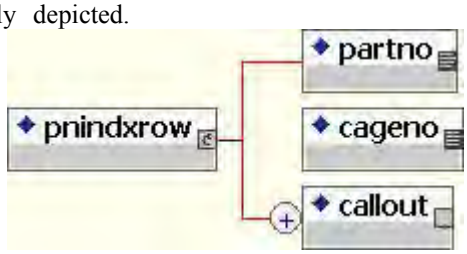

FIGURE 442. Part number index row <pnindxrow> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT pnindxrow (partno, cageno, callout+)>
<!ATTLIST pnindxrow
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

38.5.2.2 Part number index work package ***<pnindxwp>*** tagging. The ***<pnindxwp>*** is tagged in the same manner as the ***<nsnindxwp>***.

```
<pnindxwp wpno="WP0024" wpseq="0024 00">
<wpidinfo>
<maintlvl level="field">
<title>
<indxref ref1="Part Number Index">
</title>
</wpidinfo>
<pnindx>
<std.table.header cols="3">
<std.table.row>
<std.table.entry colnum="1">PART NUMBER
</std.table.entry>
<std.table.entry colnum="2">FIGURE
</std.table.entry>
<std.table.entrycolnum="3">ITEM
</std.table.entry>
</std.table.row>
</std.table.header>
```


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Check the source to verify that this is the current version before use.

<pnindxrow>
<partno>0410-06558-0000
</partno>
<callout assocfig="D-2" id="p6a">
</pnindxrow>
<pnindxrow>
<partno>59755-1
</partno>
<callout assocfig="D-2" id="p6">
</pnindxrow>
<pnindxrow>
<partno>598-48-1
</partno>
<callout assocfig="D-1" id="p5">
</pnindxrow>
<pnindxrow>
<partno>881291-1, –3
</partno>
<callout assocfig="D-1"id="p1">
</pnindxrow>
<pnindxrow>
<partno>881292-1, –2
</partno>
<callout assocfig="D-1" id="p1a">
</pnindxrow>
<pnindxrow>
<partno>881293-1, –2
</partno>
<callout assocfig="D-1"id="p3">
<callout assocfig="D-1" id="p10">
<callout assocfig="D-3"id="5">
<callout assocfig="D-6" id="3">
</pnindxrow>
<pnindxrow>
<partno>881294-1, –2
</partno>
<callout assocfig="D-1"id="p3a">
</pnindxrow>
<pnindxrow>
<partno>881295-1, –2,–3
</partno>
<callout assocfig="D-1" id="p4">
</pnindxrow>
<pnindxrow>
<partno>881296-1
</partno>
<callout assocfig="D-1" id="p2">
</pnindxrow>
<pnindxrow>
<partno>881297-1, –2
</partno>
<callout assocfig="D-1"id="p2a">
</pnindxrow>
<pnindxrow>

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```
<partno>881298-1
</partno>
<callout assocfig="D-1" id="p4a">
</pnindxrow>
<pnindxrow>
<partno>881299-1,  &ndash;2
</partno>
<callout assocfig="D-1"id="p3b">
</pnindxrow>
</pnindx>
</pnindxwp>
```

38.5.3 Reference designator index work package ***<refdesindxwp>***. The reference designator index is an optional segment of the part listing. A reference designator is typically used with electrical or electronic systems or components. The reference designator provides a unique identifier and a hierarchical top to bottom breakdown for the system or component. As such, a reference designator will reference a single item in the parts listing. If the same part is used multiple times on different components of the system each individual use will have its own reference designator. The ***<refdesindxwp>*** lists each individual reference designator contained in each ***<plwp>***, ***<stl-partswp>***, ***<kitswp>***. ***<bulk_itemswp>***, or ***<stlwp>***. The ***<refdesindxwp>*** is sorted in alpha–numeric sequence and contains the same information as found in the National Stock Number index (see 38.5.1).

a. The components are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Index reference marker ***<indxref>*** (optional – zero or more) (see 31.4.2.2.3)
   (4) Reference designator index ***<refdesindx>*** (see 38.5.3.1).

b. DTD fragment is graphically depicted.

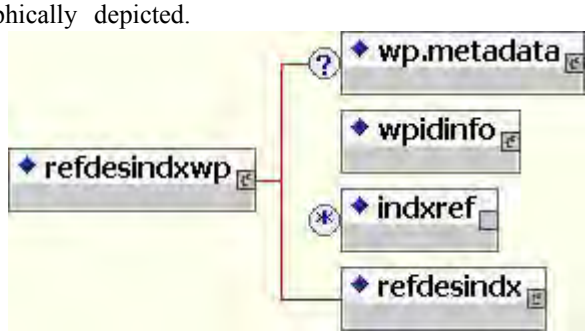

FIGURE 443. Reference designator index work package <refdesindxwp> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT refdesindxwp (wp.metadata?, wpidinfo, indxref*, refdesindx)>
<!ATTLIST refdesindxwp
    airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED>
```

d. Common attributes:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

38.5.3.1 Reference designator index ***<refdesindx>***. The ***<refdesindx>*** is the wrapper element for the listing of all reference designators used in the parts list.

a. The components are:
(1) Standard table header ***<std.table.header>*** (see 42.3.1).
(2) Reference designator index row ***<refdesindxrow>*** (required – one or more) (see 38.5.3.1.1).

b. DTD fragment is graphically depicted.

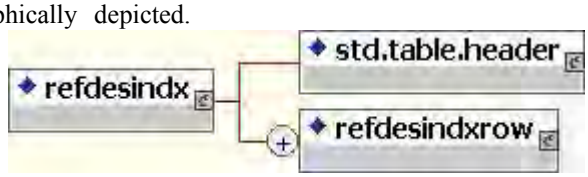

FIGURE 444. Reference designator index <refdesindx> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT refdesindx (std.table.header, refdesindxrow+)>
<!ATTLIST refdesindx
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

38.5.3.1.1 Reference designator index row ***<refdesindxrow>***. The ***<refdesindxrow>*** is similar in nature to the ***<nsnindxrow>*** or ***<pnindxrow>*** with one major exception. In the NSN or Part Number indexes, there may be multiple uses of an NSN or Part Number. In the reference designator, each ***<refdes>*** is unique and will only be listed once.

a. The components are:

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(1) Reference designator ***<refdes>*** (required) (see 38.4.2.1.3.2.10).
(2) Callout ***<callout>*** (required – one or more) (see 46.2.4.1).

b. DTD fragment is graphically depicted.

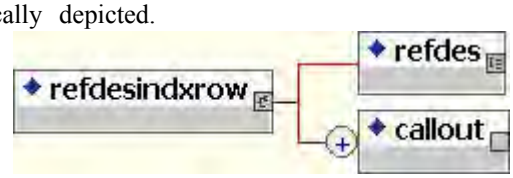

FIGURE 445. Reference designator index row <refdesindxrow> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT refdesindxrow (refdes, callout+)>
<!ATTLIST refdesindxrow
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

38.5.3.2 Example of a reference designator index work package. An example of a page based ***<refdesindxwp>*** is contained in MIL-STD-40051-2. Programs may develop their own display of the ***<refdesindxwp>*** for IETMs.

a. Example of an XML document instance fragment for ***<refdesindxwp>***:

<refdesindxwp wpno="WP0026" wpseq="0026 00">
<wpidinfo>
<maintlvl level="field"/>
<title>
<indxref ref1="Reference Desginator Index">
</title>
</wpidinfo>
<refdesindx>
<std.table.header cols="3">
<std.table.row>
<std.table.entry colnum="1">REF DES
</std.table.entry>
<std.table.entrycolnum="2">FIGURE
</std.table.entry>
<std.table.entry colnum="3">ITEM
</std.table.entry>

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</std.table.row>
</std.table.header>
<refdesindxrow>
*<refdes>*S1
</refdes>
<calloutassocfig="f1" label="15"/>
</refdesindxrow>
<refdesindxrow>
*<refdes>*W2
</refdes>
<calloutassocfig="f1" label="3"/>
</refdesindxrow>
<refdesindxrow>
*<refdes>*2AT1
</refdes>
<calloutassocfig="f2" label="309"/>
</refdesindxrow>
<refdesindxrow>
*<refdes>*2AT10
</refdes>
<calloutassocfig="f2" label="552"/>
</refdesindxrow>
<refdesindxrow>
*<refdes>*2AT11
</refdes>
<calloutassocfig="f2" label="699"/>
</refdesindxrow>
<refdesindxrow>
*<refdes>*2AT12
</refdes>
<calloutassocfig="f2" label="699"/>
</refdesindxrow>
<refdesindxrow>
*<refdes>*2AT13
</refdes>
<calloutassocfig="f2" label="479"/>
</refdesindxrow>
<refdesindxrow>
*<refdes>*2AT14
</refdes>
<calloutassocfig="f2" label="479"/>
</refdesindxrow>
<refdesindxrow>
*<refdes>*2AT2
</refdes>
<calloutassocfig="f2" label="309"/>
</refdesindxrow>
<refdesindxrow>
*<refdes>*2AT3
</refdes>
<calloutassocfig="f2" label="558"/>
</refdesindxrow>
<refdesindxrow>
*<refdes>*2AT4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<//refdes>
<calloutassocfig="f2" label="564"/>
</refdesindxrow>
<refdesindxrow>
<refdes>2AT5
<//refdes>
<calloutassocfig="f2" label="705"/>
</refdesindxrow>
<refdesindxrow>
<refdes>2AT5
<//refdes>
<calloutassocfig="f2" label="479"/>
</refdesindxrow>
<refdesindxrow>
<refdes>2AT6
<//refdes>
<calloutassocfig="f2" label="494"/>
</refdesindxrow>
<refdesindxrow>
<refdes>2AT7
<//refdes>
<calloutassocfig="f2" label="675"/>
</refdesindxrow>
<refdesindxrow>
<refdes>2AT8
<//refdes>
<calloutassocfig="f2" label="624"/>
</refdesindxrow>
<refdesindxrow>
<refdes>2AT9
<//refdes>
<calloutassocfig="f2" label="552"/>
</refdesindxrow>
<refdesindxrow>
<refdes>2A1
<//refdes>
<calloutassocfig="f2" label="489"/>
</refdesindxrow>
</refdesindx>
</refdesindxwp>

39 DESTRUCTION OF ARMY MATERIAL CHAPTER .

39.1 Destruction of Army material ***<dim>***. At the direction of the acquiring activity, destruction procedures may be included as a chapter in a weapons system TM. The requirements contained in 31.9 apply except as noted below.

a. The components are:
   (1) Chapter Title Page ***<titlepg>*** (required). The element provides the title page preceding an information chapter (see 49.1.1.1).
   (2) Destruction general information work package ***<destruct-ginfowp>*** (required) (see 31.9.2).
   (3) Destruction procedures work package ***<destruct-materialwp>*** (required – one or more) (see 31.9.3).

b. DTD fragment is graphically depicted.

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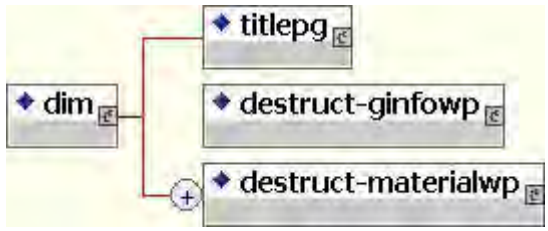

FIGURE 446. Destruction of Army material chapter <dim> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT dim (titlepg, destruct-ginfowp, destruct-materialwp+)>
<!ATTLIST dim
    assocfig  IDREFS                    #IMPLIED
    changeref IDREFS                    #IMPLIED
    chap-toc  (yes | no )              "yes"
    chngno    CDATA                     #REQUIRED
    comment   CDATA                     #IMPLIED
    delchlvl  CDATA                     #IMPLIED
    frame     (yes | no )               "yes"
    idref     IDREFS                    #IMPLIED
    inschlvl  CDATA                     #IMPLIED
    revno     CDATA                     #REQUIRED
    security  (uc | fouo | c | s | ts )  #IMPLIED
    skilltrk  CDATA                     #IMPLIED
    tocentry  (0 | 1 | 2)               "1"
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **chap-toc** – Create a chapter work package table of contents (optional). Select either "**yes**" or "**no**" with a default value "**yes**".
(4) **chngno** – Change number (required) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(8) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(9) **inschlvl** – Insert change level (optional) (see 49.2.9).
(10) **revno** – Revision number (required) (see 49.2.9).
(11) **security** – Security classification (optional) (see 49.2.13).
(12) **skilltrk** – Skill level (optional) (see 49.2.2).
(13) **tocentry** – In the TM table of contents, the attribute indicates the indenture level. The possible selections are "**0**" (do not include), "**1**" (the highest level or 1st indenture level), or "**2**" (the second indentrue level) (default value is "**1**").

39.1.1 Destruction general information work package ***<destruct-ginfowp>***. The ***<destruct-ginfowp>*** is discussed in 31.9.2.

39.1.2 Destruction procedures work package ***<destruct-materialwp>***. The ***<destruct-materialwp>*** is discussed in 31.9.3.

40 SUPPORTING INFORMATION CHAPTER .

40.1 Supporting information ***<sim>***. Supporting information is to be prepared for weapon systems, major equipment, components and applicable support and interface equipment as work packages and contained in a

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Supporting Information Chapter ***<sim>***. Supporting information requirements are included for the preparation of technical data that supplements the specific operation and maintenance information contained in the TM. This supplemental information includes reference data, general maintenance and parts information and associated illustrations to provide the user in preparing supporting information work packages for a TM. Below are the components of a Supporting Information Chapter.

a. The components are:
   (1) Chapter title page ***<titlepg>*** (required) (see 49.1.1.1).
   (2) References Work Package ***<refwp>*** (required) (see 40.2).
   (3) MAC introduction work package ***<macintrowp>*** (optional) (see 40.3) followed by a MAC work package***<macwp>*** (see 40.4).
   (4) A complex TM may require select one or more of the following individual supporting work packages
      (a) Components of End Item (COEI) and Basic Issue Items (BII) Lists work package ***<coeibiiwp>*** (see 40.5).
      (b) Additional Authorization List (AAL) work package ***<aalwp>*** (see 40.6).
      (c) Expendable and durable items list work package ***<explistwp>*** (see 40.7).
      (d) Tool identification list work package ***<toolidwp>*** (see 40.8).
      (e) Mandatory replacement parts list work package ***<mrplwp>*** (see 40.9).
      (f) Critical Safety Items and Flight Safety Critical Aircraft Parts work package ***<csi.fscap.wp>*** (see 40.10).
   (5) A less complex TM may include a small amount of information from one or more of the work packages listed above. The different supporting information chunks are grouped into a support items work package ***<supitemwp>*** (see 40.11).
   (6) Optional additional generic work package(s) ***<genwp>*** (see 40.12).

b. DTD fragment is graphically depicted.

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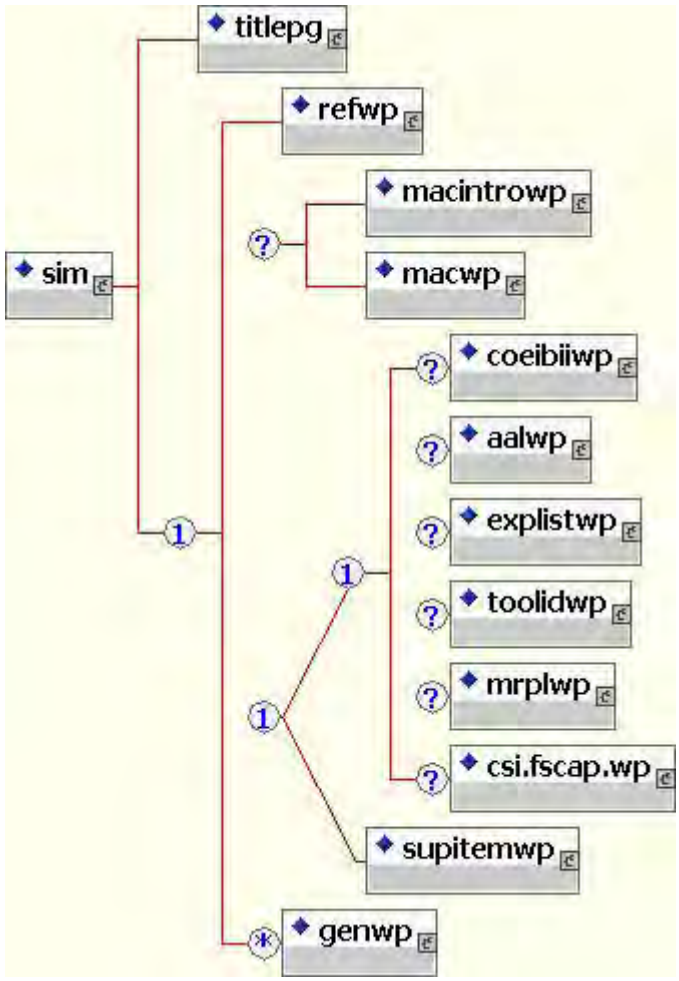

FIGURE 447. SIM chapter <sim> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT sim (titlepg, (refwp, (macintrowp, macwp)?, ((coeibiiwp?, aalwp?,
   explistwp?, toolidwp?, mrplwp?, csi.fscap.wp?) | supitemwp), genwp*))>
<!ATTLIST sim
    assocfig  IDREFS                    #IMPLIED
    changeref IDREFS                    #IMPLIED
    chap-toc  (yes | no )              "yes"
    chngno    CDATA                     #REQUIRED
    comment   CDATA                     #IMPLIED
    delchlvl  CDATA                     #IMPLIED
    frame     (yes | no )               "yes"
    idref     IDREFS                    #IMPLIED
    inschlvl  CDATA                     #IMPLIED
    revno     CDATA                     #REQUIRED
    security  (uc | fouo | c | s | ts )  #IMPLIED
    skilltrk  CDATA                     #IMPLIED
    tocentry  (0 | 1 | 2)               "1"
>
```

d. Unique attributes:

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(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **chap-toc** – Create a chapter work package table of contents (optional). Select either "**yes**" or "**no**" with a default value "**yes**".
(4) **chngno** – Change number (required) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(7) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(8) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(9) **inschlvl** – Insert change level (optional) (see 49.2.9).
(10) **revno** – Revision number (required) (see 49.2.9).
(11) **security** – Security classification (optional) (see 49.2.13).
(12) **skilltrk** – Skill level (optional) (see 49.2.2).
(13) **tocentry** – In the TM table of contents, the attribute indicates the indenture level. The possible selections are "**0**" (do not include), "**1**" (the highest level or 1st indenture level), or "**2**" (the second indentrue level) (default value is "**1**").

e. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.2 References work package ***<refwp>***. The References work package is to be prepared and list all publications referenced in the TM and required by the user to operate and/or maintain the equipment. The publication or form. if known, should be referenced with a link directly or to the web site where it can be obtained.

a. The components are:
(1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
(2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
(3) Work package scope ***<scope>*** (required) (see 49.1.4.23). The scope provides a short introduction of the publication categories listed.
(4) Publication list by categories ***<publist>*** (required – one or more) (see 40.2.1).

b. DTD fragment is graphically depicted.

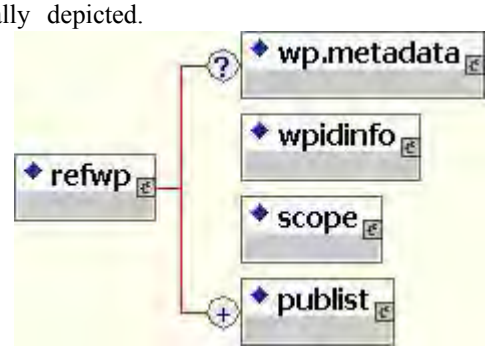

FIGURE 448. Reference work package DTD hierarchy <refwp>..

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c. DTD fragment:

```
<!ELEMENT refwp (wp.metadata?, wpidinfo, scope, publist+)>
<!ATTLIST refwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attribute:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.2.1 Publication list ***<publist>***. Publication list is used for listing all publications, forms, and similar data referenced in the TM that are required to operate or maintain the equipment. This list is identified by category name ***<title>***, any applicable introduction/general information ***<trim.para>***, and associated publications ***<pubident>*** (listed by publication number in alphanumerical order).

a. The components of ***<publist>*** are:
(1) Publication Category Title ***<title>*** (required) (see 49.1.1.3).
(2) Introductory Information Using Reduce Paragraph ***<trim.para>*** (optional – zero or more) (see 49.1.1.7).
(3) Publication Identification Number And Title ***<pubident>*** (required – one or more) (see 40.2.1.1).

b. DTD fragment:

```
<!ELEMENT publist (title, trim.para*, pubident+)>
<!ATTLIST publist
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.2.1.1 Publication identification ***<pubident>***. The publication identification number is the technical manual, Army Regulation number, or other identification publication number and publication title.

a. The components of ***<pubident>*** are:
(1) Publication number identified (required) through either
(a) An External Reference ***<extref>*** (see 46.2.1) or ***<link>*** (see 46.2.3).
(b) A Textual With No Reference ***<name>*** (see 49.1.4.19).
(2) Publication Title ***<title>*** (required) (see 49.1.1.3).

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b. DTD fragment:

```
<!ELEMENT pubident ((name | extref | link), title)>
<!ATTLIST pubident
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.2.2 XML document instance fragment and output for ***<refwp>***.

a. Example of an XML document instance fragment for ***<refwp>***:

```
<refwp wpno="s00001-X-XXXX-XXX" tocentry="2" frame="no" army="no"
airforce="no" navy="no" marines="no" wpseq="0438" deletewp="no">
<wpidinfo>
<maintlvl level="field"/>
<title>References
</title>
</wpidinfo>
<scope frame="no">
<title>Scope
</title>
<para hcp="no" esd="no">This work package lists all field
manuals, forms, technical manuals and miscellaneous pub-
lications referenced in this manual.
</para>
</scope>
<publist>
<title>FIELD MANUALS
</title>
<pubident>
<link application="both" xlink:href="http://www.army.mil/usapa/doctrine/Active_FM.html"
xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>FM 3-3
</prompt>
</link>
<title>NBC Decontamination Avoidance
</title>
</pubident>
<pubident>
```


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Check the source to verify that this is the current version before use.

<link application="both" xlink:href="http://www.army.mil/usapa/doctrine/Active_FM.html" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>FM 3-19
</prompt>
</link>
<title>NBC Reconnaissance
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.army.mil/usapa/doctrine/Active_FM.html" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>FM 9-207
</prompt>
</link>
<title>Operation and Maintenance of Ordnance Material in Cold Weather
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.army.mil/usapa/doctrine/Active_FM.html" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>FM 20-22
</prompt>
</link>
<title>Vehicle Recovery Operations
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.army.mil/usapa/doctrine/Active_FM.html" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>FM 21-11
</prompt>
</link>
<title>First Aid for Soldiers
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.army.mil/usapa/doctrine/Active_FM.html" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>FM 31-70
</prompt>
</link>
<title>Basic Cold Weather Manual
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.army.mil/usapa/doctrine/Active_FM.html" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>FM 31-71
</prompt>
</link>
<title>Northern Operations
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.army.mil/usapa/doctrine/Active_FM.html" xlink:type="simple" linktype="return" linkaction="prompt">

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<prompt>FM 90-3
</prompt>
</link>
<title>Desert Operation
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.army.mil/usapa/doctrine/Active_FM.html"
xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>FM 90-6
</prompt>
</link>
<title>Mountain Operations
</title>
</pubident>
</publist>
<publist>
<title>FORMS
</title>
<pubident>
<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp"
xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>DA Form 2028
</prompt>
</link>
<title>Recommended Changes to Publications and Blank Forms
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp"
xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>DA Form 2062
</prompt>
</link>
<title>Hand Receipt
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp"
xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>DA Form 2404
</prompt>
</link>
<title>Equipment Inspection and Maintenance Worksheet
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp"
xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>DA Form 2408
</prompt>
</link>
<title>Equipment Log Assembly (Records)
</title>
</pubident>
<pubident>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>DA Form 2408-4
</prompt>
</link>
<title>Weapon Record Data
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>DA Form 2408-9
</prompt>
</link>
<title>Equipment Control Record
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>DA Form 2408-14
</prompt>
</link>
<title>Uncorrected Fault Record
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>DA Form 2408-20
</prompt>
</link>
<title>Oil Analysis Log
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>DD 518
</prompt>
</link>
<title>Accident Identification Card
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp" xlink:type="simple" linktype="return" linkaction="prompt">
<prompt>SF 91
</prompt>
</link>
<title>Motor Vehicle Accident Report
</title>
</pubident>
<pubident>
<link application="both" xlink:href="http://www.apd.army.mil/USAPA_PUB_search_F.asp" xlink:type="simple" linktype="return" linkaction="prompt">

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<prompt>SF  368
</prompt>
</link>
<title>Product  Quality  Deficiency  Report
</title>
</pubident>
</publist>
<publist>
<title>TECHNICAL  MANUALS
</title>
<pubident>
<link  application="both"  xlink:href="http://logsa.army.mil/pubs.htm"
xlink:type="simple"  linktype="return"  linkaction="prompt">
<prompt>TM  9-1300-200
</prompt>
</link>
<title>Ammunition,  General
</title>
</pubident>
<pubident>
<link  application="both"  xlink:href="http://logsa.army.mil/pubs.htm"
xlink:type="simple"  linktype="return"  linkaction="prompt">
<prompt>TM  11-5695-286-14
</prompt>
</link>
<title>Hand  Set  Microphone
</title>
</pubident>
<pubident>
<link  application="both"  xlink:href="http://logsa.army.mil/pubs.htm"
xlink:type="simple"  linktype="return"  linkaction="prompt">
<prompt>TM  750-244-6
</prompt>
</link>
<title>Destruction  of  TACOM  Equipment
</title>
</pubident>
</publist>
</refwp>
```

b. Page-based TM stylesheet output example for ***<refwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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FIELD MAINTENANCE

REFERENCES

SCOPE

This work package lists all field manuals, forms, technical manuals and miscellaneous publications referenced in this manual.

FIELD MANUALS

FM 3-3	NBC Decontamination Avoidance
FM 3-19	NBC Reconnaissance
FM 9-207	Operation and Maintenance of Ordnance Material in Cold Weather
FM 20-22	Vehicle Recovery Operations
FM 21-11	First Aid for Soldiers
FM 31-70	Basic Cold Weather Manual
FM 31-71	Northern Operations
FM 90-3	Desert Operation
FM 90-6	Mountain Operations

FORMS

DA Form 2028	Recommended Changes to Publications and Blank Forms
DA Form 2062	Hand Receipt
DA Form 2404	Equipment Inspection and Maintenance Worksheet
DA Form 2408	Equipment Log Assembly (Records)
DA Form 2408-4	Weapon Record Data
DA Form 2408-9	Equipment Control Record
DA Form 2408-14	Uncorrected Fault Record
DA Form 2408-20	Oil Analysis Log
DD 518	Accident Identification Card
SF 91	Motor Vehicle Accident Report
SF 368	Product Quality Deficiency Report

TECHNICAL MANUALS

TM 9-1300-200	Ammunition, General
TM 11-5695-286-14	Hand Set Microphone
TM 750-244-6	Destruction of TACOM Equipment

END OF WORK PACKAGE

FIGURE 449. Example of page-based TM stylesheet output for <refwp>.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. Frame-based TM or IETM stylesheet output example for ***<refwp>***:

REFERENCES

Maintenance: FIELD

SCOPE

This work package lists all field manuals, forms, technical manuals and miscellaneous publications referenced in this manual.

FIELD MANUALS

FM 3-3	NBC Decontamination Avoidance
FM 3-19	NBC Reconnaissance
FM 9–207	Operation and Maintenance of Ordnance Material in Cold Weather
FM 20–22	Vehicle Recovery Operations
FM 21–11	First Aid for Soldiers

FIGURE 450. Example of frame-based TM stylesheet output for <refwp>.

40.3 Maintenance allocation chart (MAC) introduction work package ***<macintrowp>***. The MAC introduction work package contains the textual introduction for standard (five-level), aviation (three–level), or two-level for MAC Work Package. The MAC introduction is specified in MIL–STD–40051–1/–2 to the specific wording.

a. The components are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Introduction text ***<intro>*** (required) (see 40.3.1 for standard text markup guidance and 49.1.4.15).

b. DTD fragment is graphically depicted.

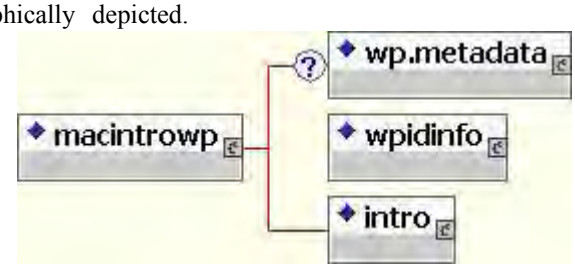

FIGURE 451. MAC introduction work package DTD hierarchy <macintrowp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT macintrowp (wp.metadata?, wpidinfo, intro)>
<!ATTLIST macintrowp
   airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time) #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attribute:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.3.1 MAC introduction work package standard text. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed Table XXV.

TABLE XXV. MAC introduction *<intro>* boilerplate entities.

Description	**Boilerplate Entity**	**Selectable Entity to Set**
Standard MAC Introduction		
Frame-basedFive-Level (LEGACY ONLY)	***&macintrowp.intro-std;***	***<!ENTITY % frame-base "INCLUDE"> <!ENTITY % page-base "IGNORE"> <!ENTITY % mac.2-level "IGNORE"> <!ENTITY % mac.3-level "IGNORE"> <!ENTITY % mac.5-level "INCLUDE"> <!ENTITY % mac.nonav-level "INCLUDE"> <!ENTITY % mac.av-level "IGNORE">***
Page-basedFive-Level (LEGACY ONLY)		***<!ENTITY % frame-base "IGNORE"> <!ENTITY % page-base "INCLUDE"> <!ENTITY % mac.2-level "IGNORE"> <!ENTITY % mac.3-level "IGNORE"> <!ENTITY % mac.5-level "INCLUDE"> <!ENTITY % mac.nonav-level "INCLUDE"> <!ENTITY % mac.av-level "IGNORE">***
Frame-basedTwo-Level		***<!ENTITY % frame-base "INCLUDE"> <!ENTITY % page-base "IGNORE"> <!ENTITY % mac.2-level "INCLUDE"> <!ENTITY % mac.3-level "IGNORE"> <!ENTITY % mac.5-level "IGNORE"> <!ENTITY % mac.nonav-level "INCLUDE"> <!ENTITY % mac.av-level "IGNORE">***
Page-basedTwo-Level		***<!ENTITY % frame-base "IGNORE"> <!ENTITY % page-base "INCLUDE">***

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Check the source to verify that this is the current version before use.

Table XXV MAC introduction *<intro>* boilerplate entities – Continued.

Description	Boilerplate Entity	Selectable Entity to Set
		<!ENTITY % mac.2-level "INCLUDE"> *<!ENTITY % mac.3-level "IGNORE">* *<!ENTITY % mac.5-level "IGNORE">* *<!ENTITY % mac.nonav-level "INCLUDE">* *<!ENTITY % mac.av-level "IGNORE">*
Aviation MAC Introduction		
Frame-basedTwo-Level	*&macintrowp.intro-av;*	*<!ENTITY % frame-base "INCLUDE">* *<!ENTITY % page-base "IGNORE">* *<!ENTITY % mac.2-level "INCLUDE">* *<!ENTITY % mac.3-level "IGNORE">* *<!ENTITY % mac.nonav-level "IGNORE">* *<!ENTITY % mac.av-level "INCLUDE">*
Page-basedTwo-Level		*<!ENTITY % frame-base "IGNORE">* *<!ENTITY % page-base "INCLUDE">* *<!ENTITY % mac.2-level "INCLUDE">* *<!ENTITY % mac.3-level "IGNORE">* *<!ENTITY % mac.5-level "IGNORE">* *<!ENTITY % mac.nonav-level "IGNORE">* *<!ENTITY % mac.av-level "INCLUDE">*
Frame-basedThree-Level (LEGACY ONLY)	*&macintrowp.intro-av;*	*<!ENTITY % frame-base "INCLUDE">* *<!ENTITY % page-base "IGNORE">* *<!ENTITY % mac.2-level "IGNORE">* *<!ENTITY % mac.3-level "INCLUDE">* *<!ENTITY % mac.nonav-level "IGNORE">* *<!ENTITY % mac.av-level "INCLUDE">*
Page-basedThree-Level (LEGACY ONLY)		*<!ENTITY % frame-base "IGNORE">* *<!ENTITY % page-base "INCLUDE">* *<!ENTITY % mac.2-level "IGNORE">* *<!ENTITY % mac.3-level "INCLUDE">* *<!ENTITY % mac.5-level "IGNORE">* *<!ENTITY % mac.nonav-level "IGNORE">* *<!ENTITY % mac.av-level "INCLUDE">*

40.3.2 XML document instance fragment ***<macintrowp>***. The XML instances for the three ***<macintrowp>*** are provided below. The page-based stylesheet output samples can be located in MIL–STD–40051–2.

a. Example of the ***<macintrowp>*** XML document instance fragment for a **Legacy** Five–Level MAC Introduction work package using the MAC boilerplate entity ***&mac.intro-std;*** and setting the selectable as shown in Table XXV.

<macintrowp wpno="s0002–X-XXXX-XXXX" tocentry="2" frame="yes"
army="no" airforce="no" navy="no" marines="no" wpseq="0437" deletewp="no">
<wpidinfo>
<maintlvl level="dirsup"/>
<title>`Maintenance Allocation Chart (MAC) Introduction`

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Check the source to verify that this is the current version before use.

```
</title>
</wpidinfo>
<intro frame="no"> &mac.intro-std;
</intro>
</macintrowp>
```

b. Example of the **<*macintrowp*>** XML document instance fragment for a Two–Level MAC Introduction work package using the MAC boilerplate entity ***&mac.intro-std;*** and setting the selectable as shown in Table XXV.

```
<macintrowp wpno="s0002–X-XXXX-XXXX" tocentry="2" frame="yes" army="no"
airforce="no" navy="no" marines="no" wpseq="0437" deletewp="no">
<wpidinfo>
<maintlvl level="sustain"/>
<title>Maintenance Allocation Chart (MAC)
</title>
</wpidinfo>
<intro frame="yes"> &mac.intro-std;
</intro>
</macintrowp>
```

c. Example of an XML document instance fragment for a **Legacy** Aviation Three–Level MAC Introduction work package using the MAC boilerplate entity ***&mac.intro-av;*** in the instance of the **<*macintrowp*>**:

```
<macintrowp wpno="s0002–X-XXXX-XXXX" tocentry="2" frame="no"
army="no" airforce="no" navy="no" marines="no" wpseq="0437" deletewp="no">
<wpidinfo>
<maintlvl level="avum-avim"/>
<title>Maintenance Allocation Chart (MAC)
</title>
</wpidinfo>
<intro frame="no"> &mac.intro-av;
</intro>
</macintrowp>
```

40.4 Maintenance allocation chart (MAC) work package**<*macwp*>**. The MAC work package is to be prepared in Functional Group Code (FGC) sequence to consolidate and identify those groups on the list which involve identified maintenance functions. The MAC is prepared according to the approved source data provided by the acquiring activity.

a. The components of MAC work package **<*macwp*>** are:
   (1) Work package metadata (information about the work package) **<*wp.metadata*>** (optional) (see 32.3.2).
   (2) Work package identification information **<*wpidinfo*>** (required) (see 32.2).
   (3) A Maintenance Allocation Chart (MAC) **<*mac*>** (required) (see 40.4.1) or an Aviation MAC **<*avmac*>** (required) (see 40.4.2).
   (4) Tools and test equipment requirements table **<*tereqtab*>** (required) (see 40.4.3).
   (5) Remarks table **<*remarktab*>** (required) (see 40.4.4).

b. DTD fragment is graphically depicted.

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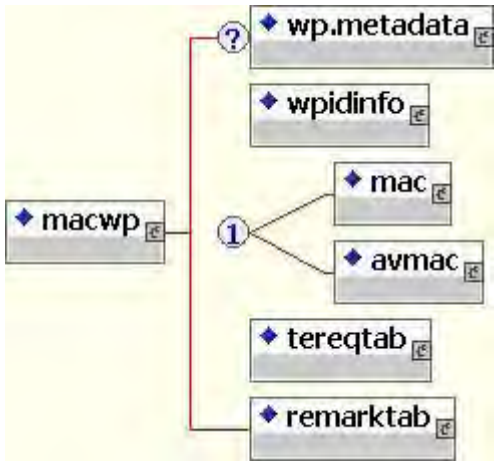

FIGURE 452. MAC work package DTD hierarchy <macwp>..

c. DTD fragment:

```
<!ELEMENT macwp (wp.metadata?, wpidinfo, (mac | avmac), tereqtab, remarktab)>
<!ATTLIST macwp
   airforce          (yes | no )                "no"
    army              (yes | no )                "no"
    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    crewmember        CDATA                       #IMPLIED
    date-time-stamp   (date | time | date-time)  #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    deletewp          (yes | no )                "no"
    fgc               CDATA                       #IMPLIED
    frame             (yes | no )                "yes"
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    insertwp          CDATA                       #IMPLIED
    lsa-id            CDATA                       #IMPLIED
    marines           (yes | no )                "no"
    navy              (yes | no )                "no"
    security          (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk          CDATA                       #IMPLIED
    tocentry          (2 | 3 | 4 | 5)            "2"
    wpno              ID                          #REQUIRED
    wpseq             CDATA                       #IMPLIED
>
```

d. Common attribute:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

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Check the source to verify that this is the current version before use.

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.4.1 Standard maintenance allocation chart (MAC) ***<mac>***. A five- or two-level MAC lists the maintenance functions, levels and times assigned to each item. The MAC can have a two or five-level maintenance concept. This element is equivalent to a table element in a structural table.

a. The components are:
(1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
(2) Standard table header ***<std.table.header>*** (required) (see 40.4.1.1).
(3) Select either the element ***<mac-group>*** for a Legacy five-level MAC or the element ***<mac-group-2lvl>*** for a two-level MAC.
(a) MAC Functional Group Codes (FGC) information five-level ***<mac-group>*** (required – one or more) (see 40.4.1.2)
(b) MAC Functional Group Codes (FGC) information two-level MAC ***<mac-group-2lvl>*** (required – one or more) (see 40.4.1.3).

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b. DTD fragment:

```
<!ELEMENT mac (title, std.table.header, (mac-group+ | mac-group-2lvl+))>
<!ATTLIST mac
    maintconcept      (2 | 5 )                          #REQUIRED
    assocfig          IDREFS                             #IMPLIED
    changeref         IDREFS                             #IMPLIED
    comment           CDATA                              #IMPLIED
    delchlvl          CDATA                              #IMPLIED
    id                ID                                 #IMPLIED
    idref             IDREFS                             #IMPLIED
    inschlvl          CDATA                              #IMPLIED
    security          (uc | fouo |c | s | ts)          #IMPLIED
    skilltrk          CDATA                              #IMPLIED
>
```

c. Unique attributes:
   (1) **maintconcept** – Is the MAC using two or five-level (standard) maintenance concept. It effects the standard information headings.

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

40.4.1.1 MAC standard information headers ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XXVI and Table XXVII.

TABLE XXVI. MAC (five-level) standard information header (LEGACY ONLY) ***<std.table.header>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set
Standard MAC (five-level) (LEGACY ONLY)Standard Information Header	***&macwp.mac.std-header;***	***<!ENTITY % mac.2-level "IGNORE">*** ***<!ENTITY % mac.5-level "INCLUDE">*** ***<!ENTITY % mac.nonav-level "INCLUDE">*** ***<!ENTITY % mac.av-level "IGNORE">***
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***

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TABLE XXVII. MAC (two-level) standard information header *<std.table.header>* boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set
Standard MAC (two-level)Standard Information Header	*&macwp.mac.std-header;*	*<!ENTITY % mac.2-level "INCLUDE">* *<!ENTITY % mac.5-level "IGNORE">* *<!ENTITY % mac.nonav-level "INCLUDE">* *<!ENTITY % mac.av-level "IGNORE">*
Frame-based		*<!ENTITY % frame-base "INCLUDE">* *<!ENTITY % page-base "IGNORE">*
Page-based		*<!ENTITY % frame-base "IGNORE">* *<!ENTITY % page-base "INCLUDE">*

40.4.1.2 MAC functional group code (FGC) information ***<mac-group>***. The element groups the functional group number component(s)/assemble(s) information. The element is similar to a "row" in a structural table.

40.4.1.2.1 Functional group number ***<groupno>***. The element ***<groupno>*** contains the functional group code (FGC) for the maintenance action(s). The element is similar to a “cell” in a structural table and is entered in column one of the MAC table.

a. The components are:
   (1) Parsable Characters - The element contains narrative text #PCDATA parsable character data.
b. DTD fragment

```
<!ELEMENT groupno (#PCDATA)>
<!ATTLIST groupno
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:
d. **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
e. **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
f. **comment** – Reason for change information. (optional) (See 49.2.9.)
g. **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
h. **id** – Unique identifier. (optional) (See 49.2.6.)
i. **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
j. **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
k. **skilltrk** – Training skill level. (optional) (See 49.2.2.)
l. **security** – Security classification. (optional) (See 49.2.13.)

40.4.1.2.2 Component/assembly item ***<compassem>***. The element ***<compassem>*** specifies the component, assembly, subassembly, or module name and, if applicable, the type designator for which maintenance is authorized.

a. The components are:
   (1) Name ***<name>***(required) (see 49.1.4.19)

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(2) Type Designator ***<typedes>*** (optional) (see 49.1.4.19)

b. DTD fragment

```
<!ELEMENT compassem (name, typedes?)>
<!ATTLIST compassem
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:

d. **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

e. **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

f. **comment** – Reason for change information. (optional) (See 49.2.9.)

g. **delchlvl** – Deletion change level. (optional) (See 49.2.9.)

h. **id** – Unique identifier. (optional) (See 49.2.6.)

i. **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

j. **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

k. **skilltrk** – Training skill level. (optional) (See 49.2.2.)

l. **security** – Security classification. (optional) (See 49.2.13.)

40.4.1.2.2.1 Type designator ***<typedes>***. The element ***<typedes>*** provides the equipment type designation.

a. The components are:

(1) The element contains narrative text #PCDATA parsable character data. (see 49.1.4.19)

b. DTD fragment

```
<!ELEMENT typedes (#PCDATA)>
```

c. No Attributes.

40.4.1.2.3 Maintenance action qualification ***<qualify>***. Maintenance action qualification ***<qualify>*** (required – one or more) identifies the maintenance function, estimate hours to perform by maintenance level, required tools and equipment reference(s), and remark reference(s). The element is similar to a "row" in a structural table.

a. The components are:

(1) Maintenance function ***<maintfunc>*** (required). The element is similar to a “cell” in a structural table and is entered in column three of the MAC table. (see 40.4.1.2.3.1)

(2) Maintenance level classification ***<maintclass>*** (required). The element is similar to a “cell” in a structural table and is entered in column 4 of the MAC table. (see 40.4.1.2.3.2)

(3) Tools and test equipment reference(s) ***<terefs>*** (optional). The element is similar to a “cell” in a structural table and is entered in column five of the MAC table.

(4) Remark reference(s) ***<remarkrefs>*** (optional). The element is similar to a “cell” in a structural table and is entered in column six of the MAC table.

b. DTD fragment is graphically depicted.

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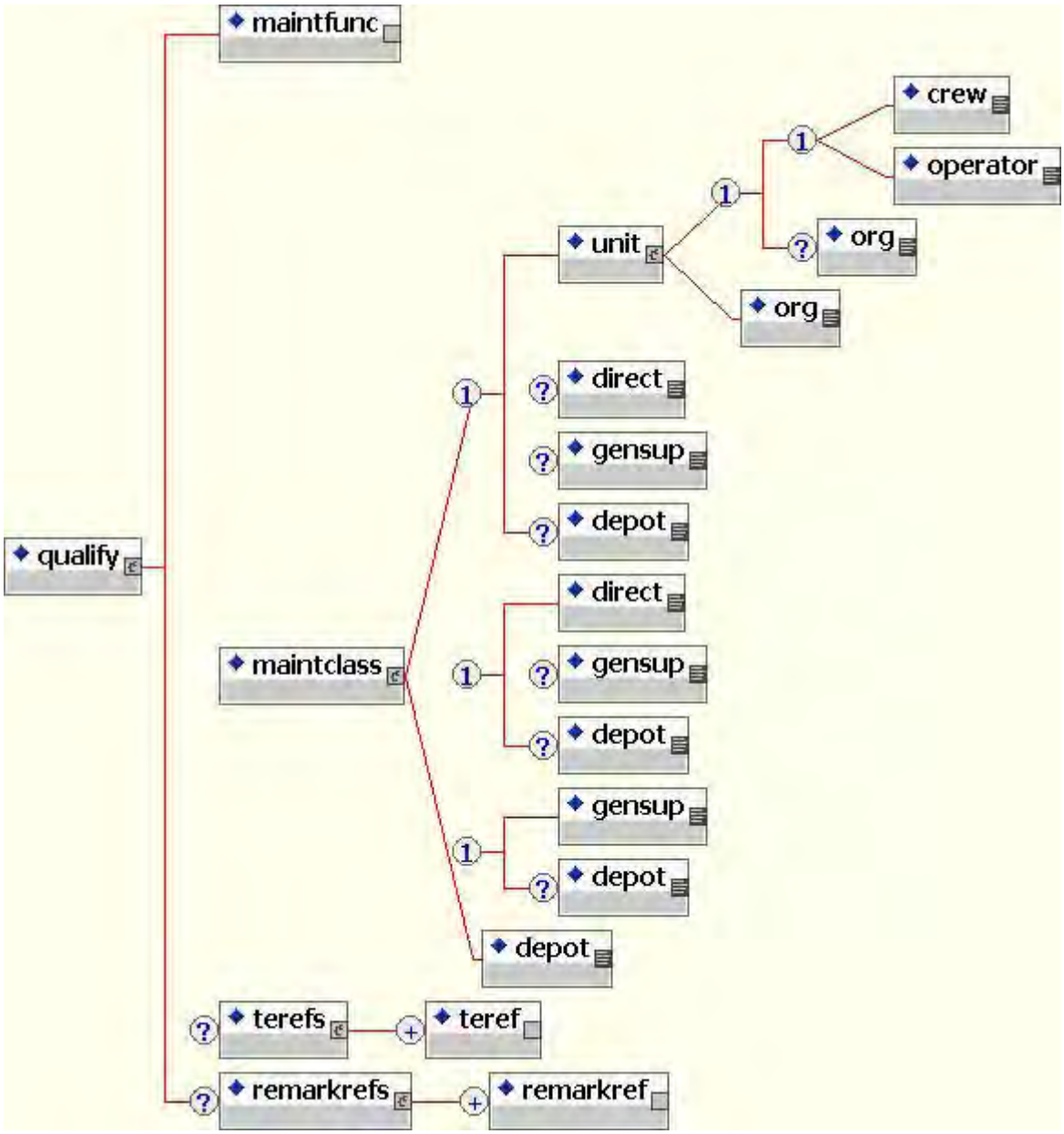

FIGURE 453. Maintenance action qualification DTD hierarchy <qualify>.

c. DTD fragment:

```
<!ELEMENT qualify (maintfunc , maintclass , terefs? , remarkrefs?)>
```

40.4.1.2.3.1 Maintenance function ***<maintfunc>***. The element ***<maintfunc>*** provides the maintenance function (as listed from the MAC introduction) to be performed on the component/assembly item. The maintenance function element is EMPTY, but the "**func**" attribute is used to enter the maintenance function type from the list below. The element is similar to a "cell" in a structural table and is entered in column three.

a. The component is EMPTY, but the "**func**" attribute is used to enter the maintenance function type from the following list.

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(1) **inspect** – Inspect is to determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (e.g., by sight, sound, or feel). This includes scheduled inspection and gagings and evaluation of cannon tubes.
(2) **test** – Test is to verify serviceability by measuring the mechanical, pneumatic, hydraulic, or electrical characteristics of an item and comparing those characteristics with prescribed standards on a scheduled basis, i.e., load testing of lift devices and hydrostatic testing of pressure hoses.
(3) **service** – Service is the operations required periodically to keep an item in proper operating condition; e.g., to clean (includes decontaminate, when required), to preserve, to drain, to paint, or to replenish fuel, lubricants, chemical fluids, or gases. This includes scheduled exercising and purging of recoil mechanisms.
(4) **adjust** – Adjust to maintain or regulate, within prescribed limits, by bringing into proper position, or by setting the operating characteristics to specified parameters.
(5) **align** – Align is to adjust specified variable elements of an item to bring about optimum or desired performance.
(6) **calib** – Calibrate is to determine and cause corrections to be made or to be adjusted on instruments of test, measuring, and diagnostic equipment used in precision measurement. Consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.
(7) **remove** – Remove/instal is to remove and install the same item when required to perform service or other maintenance functions. Install may be the act of emplacing, seating, or fixing into position a spare, repair part, or module (component or assembly) in a manner to allow the proper functioning of an equipment or system.
(8) **replace** – Replace is to remove an unserviceable item and install a serviceable counterpart in its place. "Replace" is authorized by the MAC and assigned maintenance level is shown as the third position code of the Source, Maintenance and Recoverability (SMR) code.
(9) **repair** – Repair is the application of maintenance services, including fault location/troubleshooting, removal/installation, disassembly/assembly procedures, and maintenance actions to identify troubles and restore serviceability to an item by correcting specific damage, fault, malfunction, or failure in a part, subassembly, module (component or assembly), end item, or system.
(10) **overhaul** – Overhaul is the maintenance effort (service/action) prescribed to restore an item to a completely serviceable/operational condition as required by maintenance standards in appropriate technical publications. Overhaul is normally the highest degree of maintenance performed by the Army. Overhaul does not normally return an item to like new condition.
(11) **rebuild** – Rebuild consists of those services/actions necessary for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing standards. Rebuild is the highest degree of material maintenance applied to Army equipment. The rebuild operation includes the act of returning to zero those age measurements (e.g., hours/miles) considered in classifying Army equipment/components.
(12) **none** – None is no maintenance functions to be performed, but is necessary for the FGC hierarchy.

b. Unique Attributes.
(1) **func** (required) References the remark code from the MAC remark table.

40.4.1.2.3.2 Maintenance level classification ***<maintclass>*** . The element ***<maintclass>*** provides the estimated time (in tenth of hours) or if unknown use a dash "–" to perform the maintenance function by maintenance level. The author enters one or more for the appropriate maintenance level(s) estimate time, as listed below. The ***<maintclass>*** is used in **Legacy** five-level MAC tables and.three-level Aviation MAC tables.

a. The components are
(1) Unit ***<unit>*** that is divided into
(a) Crew member ***<crew>*** (optional). The element is similar to a "cell" in a structural table and is entered in column four.

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(b) Operator ***<operator>*** (optional). The element is similar to a "cell" in a structural table and is entered in column five.

(c) Unit organization ***<org>*** (optional). The element is similar to a "cell" in a structural table and is entered in column six.

(2) Direct support ***<direct>*** (optional). The element is similar to a "cell" in a structural table and is entered in column seven.

(3) General support ***<gensup>*** (optional). The element is similar to a "cell" in a structural table and is entered in column eight.

(4) Depot ***<depot>*** (optional). The element is similar to a "cell" in a structural table and is entered in column nine.

b. DTD fragment:

```
<!ELEMENT maintclass ((unit , direct? , gensup? , depot?) |
(direct , gensup? , depot?) | (gensup , depot?) | depot)>
```

c. No Attributes.

40.4.1.2.3.3 Tools and test equipment reference(s) ***<terefs>*** . The element provides the reference to the required tools and/or test equipments for the current maintenance task found in the tools and test equipment table. Each tool or test equipment reference is specified through the element ***<teref>*** and references the target identifier with the attribute "**refs**".

a. The components are

(1) Tools and Equipment Reference ***<teref>***(required – one or more). (see 40.4.1.2.3.3.1)

b. DTD fragment for

```
<!ELEMENT terefs (teref+)>
```

c. No Attributes.

40.4.1.2.3.3.1 Tools and equipment reference ***<teref>***. The element provides the reference to the required tools and/or test equipments for the current maintenance task found in the tools and test equipment table. Each tool or test equipment reference is specified through the element ***<teref>*** and references the target identifier with the attribute "**refs**".

a. The component is EMPTY, but the references the target identifier with the attribute "**refs**".

b. Unique Attributes.

(1) **refs** – A reference to the tools and equipment list. (required)

40.4.1.2.3.4 Remark reference(s) ***<remarkrefs>***. The element remarkrefs provides the reference to remarks concerning the maintenance task found in the remarks table. Each remark reference is specified through the element ***<remarkref>***.

a. The components are

(1) Remark Reference ***<remarkref>*** (required – one or more). (see 40.4.1.2.3.4.1)

b. DTD fragment for

```
<!ELEMENT remarkrefs (remarkref+)>
```

c. No Attributes.

40.4.1.2.3.4.1 Remark reference ***<remarkref>***. The element ***<remarkref>*** is specified through the element ***<remarkref>*** and references the target identifier with the attribute "**refs**".

a. The component is EMPTY, but the references the target identifier with the attribute "**refs**".

b. Unique Attributes.

(1) **refs** – A reference to the tools and equipment list. (required)

40.4.1.3 2 level maintenance – maintenance allocation chart ***<mac-group-2lvl>***. The element ***<mac-group-2lvl>*** lists each the functional group number component(s)/assemble(s) information. In page-base, the element is equivalent to a "row" element in a table.

a. The components are:

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(1) Functional group number ***<groupno>*** (required). (see 40.4.1.2.1)
(2) Select either the component/assembly and qualify or select multiple components/assemblies.
(a) Component/assembly item ***<compassem>*** (required). The element is similar to a “cell” in a structural table and is entered in column two. (see 40.4.1.2.2)
(b) Qualifier – 2 Level Maintenance ***<qualify-2lvl>*** (required – one or more) identifies a MAC component qualification entry that contains maintenance function, maintenance level(s) work time, any tools and/or test equipments, and any remarks. In page-base, the element is equivalent to a "row" element in a table. (see 40.4.1.3.1).

(3) Component/Assemble Grouping – 2 Level Maintenance ***<compassemgroup-2lvl>*** (required – one or more). (see 40.4.1.3.2)

b. DTD graphically depicted for ***<mac-group-2lvl>***:

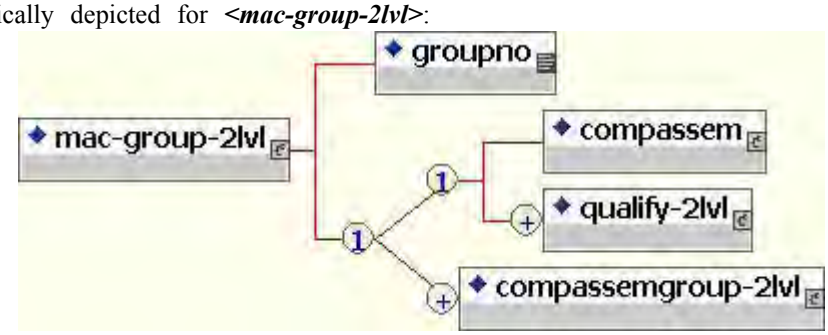

FIGURE 454. MAC functional group <mac-group-2lvl> DTD hierarchy.

c. DTD fragment for:

```
<!ELEMENT mac-group-2lvl (groupno, ((compassem, qualify-2lvl+) | compassemgroup-2lvl+))>
<!ATTLIST mac-group-2lvl

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<mac-group>***:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.4.1.3.1 Qualifier – 2 level maintenance ***<qualify-2lvl>***. The element ***<qualify-2lvl>*** identifies a MAC component qualification entry that contains maintenance function, maintenance level(s) work time, any tools and/or test equipments, and any remarks. In page-base, the element is equivalent to a "row" element in a table.

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a. The components are:
   (1) Maintenance function ***<maintfunc>*** (required). The element is similar to a "cell" in a structural table and is entered in column three of the MAC table. (see 40.4.1.2.3.1)
   (2) Maintenance level classification ***<maintclass-2lvl>*** (required). The element is similar to a "cell" in a structural table and is entered in column 4 of the MAC table. (see 40.4.1.3.1.1)
   (3) Tools and test equipment reference(s) ***<terefs>*** (optional). The element is similar to a "cell" in a structural table and is entered in column five of the MAC table. (see 40.4.1.2.3.3)
   (4) Remark reference(s) ***<remarkrefs>*** (optional). The element is similar to a "cell" in a structural table and is entered in column six of the MAC table. (see 40.4.1.2.3.4)

b. DTD graphically depicted for:

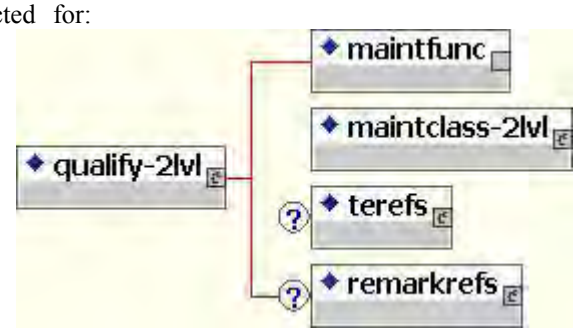

FIGURE 455. Maintenance action qualification <qualify-2lvl> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT qualify-2lvl (maintfunc, maintclass-2lvl, terefs?, remarkrefs?)>
```

40.4.1.3.1.1 Maintenance level classification ***<maintclass-2lvl>***. The element ***<maintclass-2lvl>*** The contains the authorized maintenance level and the time required to perform the task. The time required to complete the task is entered in the appropriate level of maintenance (field and sustainment). The ***<maintclass-2lvl>*** is used in two-level MAC tables and Aviation MAC tables.

a. The components are
   (1) Requires one of the maintenance levels listed below:
      (a) (Field) Operator or Crew maintenance ***<c>*** (optional). The element is similar to a "cell" in a structural table and is entered in column four.
      (b) (Field) Unit maintenance ***<o>*** (optional). The element is similar to a "cell" in a structural table and is entered in column five.
      (c) (Field) Direct Support maintenance ***<f>*** (optional). The element is similar to a "cell" in a structural table and is entered in column six.
      (d) (Sustainment) General Support maintenance ***<h>*** (optional). The element is similar to a "cell" in a structural table and is entered in column six.
      (e) (Sustainment) Depot maintenance ***<d>*** (optional). The element is similar to a "cell" in a structural table and is entered in column seven.

b. DTD fragment:

```
<!ELEMENT maintclass-2lvl ((c, o?, f?, h?, d?) | (o, f?, h?,
 d?) | (f, h?, d?) | (h, d?) | d)>
```

c. No Attributes.

40.4.1.3.2 Component/assemble grouping – 2 level maintenance ***<compassemgroup-2lvl>***. The element ***<compassemgroup-2lvl>*** if it is a list is subdivided into parts, for example by subassemblies, the category element is used. After the category element is entered, the category title and the specific entries are entered.

a. The components are:

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(1) Component/assembly item ***<compassem>*** (required). The element is similar to a “cell” in a structural table and is entered in column two. (see 40.4.1.2.2)
(2) Qualifier – 2 Level Maintenance ***<qualify-2lvl>*** (required – one or more) identifies a MAC component qualification entry that contains maintenance function, maintenance level(s) work time, any tools and/or test equipments, and any remarks. In page-base, the element is equivalent to a "row" element in a table. (see 40.4.1.3.1).

b. DTD fragment

```
<!ELEMENT compassemgroup-2lvl (compassem, qualify-2lvl+)>


>
```

c. No attributes:

40.4.2 Aviation MAC ***<avmac>***. An aviation Maintenance Allocation Chart lists the maintenance functions, levels and times assigned to each item. The Aviation MAC supports either the 2- level aviation MAC or the Legacy 3-level aviation MAC. This element is equivalent to a table element in a structural table.

a. The components of ***<avmac>*** are:
(1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
(2) Standard table header ***<std.table.header>*** (required) (see 40.4.2.1).
(3) Select either the Aviation MAC Functional Group or Functional Grouping for 2 Level Maintenance
(a) Aviation MAC Functional Group Codes (FGC) information (Legacy 3–level aviation MAC) ***<avmac-group>*** (required – one or more) (see 40.4.2.2).
(b) Functional Grouping for 2 Level Maintenance***<avmac-group-2lvl>*** (required – one or more) (see 40.4.2.2).

b. DTD fragment:

```
<!ELEMENT avmac (title, std.table.header, (avmac-group+ | avmac-group-2lvl+))>
<!ATTLIST avmac    maintconcept (2 | 3) #REQUIRED

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<avmac>*** are:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.4.2.1 Aviation MAC standard information headers ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information

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specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XXVIII.

TABLE XXVIII. Aviation MAC standard information header ***<std.table.header>*** boilerplate entities.

Description	**Boilerplate Entity**	**Selectable Entity to Set**
Aviation MAC Standard Information Header	***&macwp.avmac.std-header;***	***<!ENTITY % mac.2-level "IGNORE"> <!ENTITY % mac.3-level "IGNORE"> <!ENTITY % mac.5-level "IGNORE"> <!ENTITY % mac.nonav-level "IGNORE"> <!ENTITY % mac.av-level "INCLUDE">***
Frame-based		***<!ENTITY % frame-base "INCLUDE"> <!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE"> <!ENTITY % page-base "INCLUDE">***

40.4.2.2 Aviation MAC functional group code (FGC) information ***<avmac-group>***. The element groups the functional group number component(s)/assemble(s) information. The element is similar to a "row" in a structural table.

a. The components of ***<avmac-group>*** are:
   (1) Functional group number ***<groupno>*** (required). The element contains the functional group code (FGC) for the maintenance action(s). The element is similar to a "cell" in a structural table and is entered in column one.
   (2) A single component/assembly OR
      (a) Component/assembly item ***<compassem>*** (required). The element specifies the component, assembly, subassembly, or module name ***<name>*** (required) and, if applicable, the type designator ***<typedes>*** (optional) for which maintenance is authorized. The element is similar to a "cell" in a structural table and is entered in column two.
      (b) Aviation maintenance action qualifier ***<avqualify>*** (required – one or more) identifies the maintenance function, maintenance levels, required tools and test equipment reference(s), and remark reference(s) (see 40.4.2.2.1).

   (3) Multiple aviation components/assemblies ***<avcompassem-group>*** (required – one or more) with each component/assembly containing:
      (a) Component/assembly item ***<compassem>*** (required). The element specifies the component, assembly, subassembly, or module name ***<name>*** (required) and. if applicable, the type designator ***<typedes>*** (optional) for which maintenance is authorized. The element is similar to a "cell" in a structural table and is entered in column two.
      (b) Aviation maintenance action qualifier ***<avqualify>*** (required – one or more) identifies the maintenance function, maintenance levels, required tools and test equipment reference(s), and remark reference(s) (see 40.4.2.2.1).
b. DTD graphically depicted for ***<avmac-group>***:

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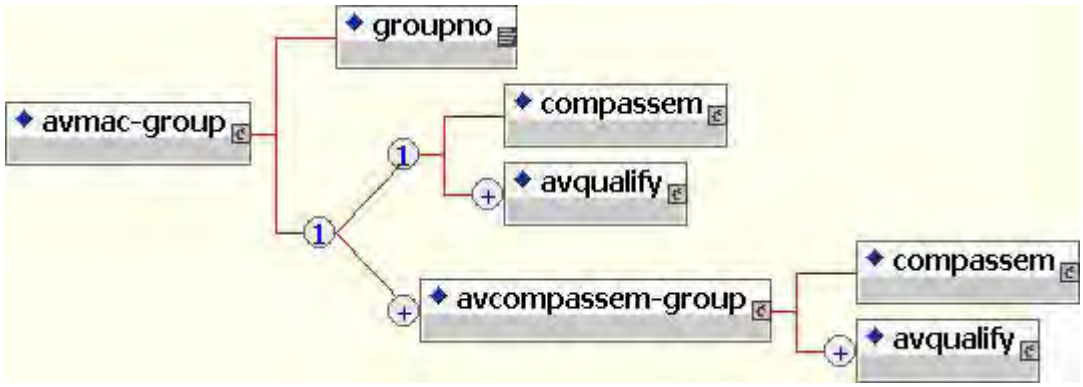

FIGURE 456. Aviation MAC functional group DTD hierarchy <avmac-group>.

c. DTD fragment:

```
<!ELEMENT avmac-group (groupno, ((compassem, avqualify+) | avcompassem-group+))>
<!ATTLIST avmac-group
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<avmac-group>***:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.4.2.2.1 Aviation maintenance action qualification ***<avqualify>***. Aviation maintenance action qualification ***<avqualify>*** (required – one or more) identifies the maintenance function, estimate hours to perform by maintenance level, required tools and equipment reference(s), and remark reference(s). The element is similar to a "row" in a structural table.

a. The components of ***<avqualify>*** are:

(1) Maintenance function ***<maintfunc>*** (required) is the maintenance function (as listed from the MAC introduction) to be performed on the component/assembly item The maintenance function element is EMPTY, but the "**func**" attribute is used to enter the maintenance function type from the list below. The element is similar to a "cell" in a structural table and is entered in column three.

(a) **inspect** – Inspect is to determine the serviceability of an item by comparing its physical, mechanical, and/or electrical characteristics with established standards through examination (e.g., by sight, sound, or feel). This includes scheduled inspection and gagings and evaluation of cannon tubes.

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(b) **test** – Test is to verify serviceability by measuring the mechanical, pneumatic, hydraulic, or electrical characteristics of an item and comparing those characteristics with prescribed standards on a scheduled basis, i.e., load testing of lift devices and hydrostatic testing of pressure hoses.
(c) **service** – Service is the operations required periodically to keep an item in proper operating condition; e.g., to clean (includes decontaminate, when required), to preserve, to drain, to paint, or to replenish fuel, lubricants, chemical fluids, or gases. This includes scheduled exercising and purging of recoil mechanisms.
(d) **adjust** – Adjust to maintain or regulate, within prescribed limits, by bringing into proper position, or by setting the operating characteristics to specified parameters.
(e) **align** – Align is to adjust specified variable elements of an item to bring about optimum or desired performance.
(f) **calib** – Calibrate is to determine and cause corrections to be made or to be adjusted on instruments of test, measuring, and diagnostic equipment used in precision measurement. Consists of comparisons of two instruments, one of which is a certified standard of known accuracy, to detect and adjust any discrepancy in the accuracy of the instrument being compared.
(g) **remove** – Remove/instal is to remove and install the same item when required to perform service or other maintenance functions. Install may be the act of emplacing, seating, or fixing into position a spare, repair part, or module (component or assembly) in a manner to allow the proper functioning of an equipment or system.
(h) **replace** – Replace is to remove an unserviceable item and install a serviceable counterpart in its place. "Replace" is authorized by the MAC and assigned maintenance level is shown as the third position code of the Source, Maintenance and Recoverability (SMR) code.
(i) **repair** – Repair is the application of maintenance services, including fault location/troubleshooting, removal/installation, disassembly/assembly procedures, and maintenance actions to identify troubles and restore serviceability to an item by correcting specific damage, fault, malfunction, or failure in a part, subassembly, module (component or assembly), end item, or system.
(j) **overhaul** – Overhaul is the maintenance effort (service/action) prescribed to restore an item to a completely serviceable/operational condition as required by maintenance standards in appropriate technical publications. Overhaul is normally the highest degree of maintenance performed by the Army. Overhaul does not normally return an item to like new condition.
(k) **rebuild** – Rebuild consists of those services/actions necessary for the restoration of unserviceable equipment to a like new condition in accordance with original manufacturing standards. Rebuild is the highest degree of material maintenance applied to Army equipment. The rebuild operation includes the act of returning to zero those age measurements (e.g., hours/miles) considered in classifying Army equipment/components.
(l) **none** – None is no maintenance functions to be performed, but is necessary for the FGC hierarchy.

(2) Aviation Maintenance level classification ***<avmaintclass>*** (required) provides the estimated time (in tenth of hours) or if unknown use a dash "–" to perform the maintenance function by maintenance level. The author enters one or more for the appropriate maintenance level(s) estimate time, as listed below.
(a) AVUM ***<avum>*** (optional). The element is similar to a "cell" in a structural table and is entered in column four.
(b) AVIM ***<avim>*** (optional). The element is similar to a "cell" in a structural table and is entered in column five.
(c) Depot ***<depot>*** (optional). The element is similar to a "cell" in a structural table and is entered in column six.

(3) Tools and test equipment reference(s) ***<terefs>*** (optional) provides the reference to the required tools and/or test equipments for the current maintenance task found in the tools and test equipment table. Each tool or test equipment reference is specified through the element ***<teref>***

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and references the target identifier with the attribute "**refs**". The element is similar to a "cell" in a structural table and is entered in column seven.

(4) Remark reference(s) ***<remarkrefs>*** (optional) provides the reference to remarks concerning the maintenance task found in the remarks table. Each remark reference is specified through the element ***<remarkref>*** and references the target identifier with the attribute "**refs**". The element is similar to a "cell" in a structural table and is entered in column eight.

b. DTD graphically depicted for ***<avqualify>***:

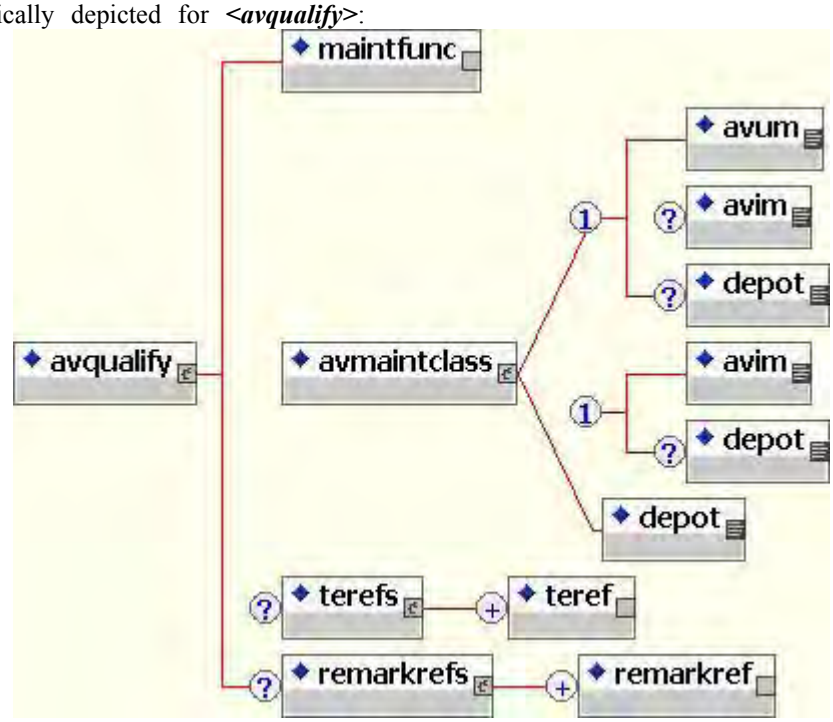

FIGURE 457. Aviation maintenance action qualification DTD hierarchy <avqualify>.

c. DTD fragment for ***<avqualify>***, ***<maintfunc>***, ***<avmaintclass>***, ***<terefs>***, ***<teref>***, ***<remarkrefs>***, and ***<remarkref>*** are:

```
<!ELEMENT avqualify (maintfunc, avmaintclass, terefs?, remarkrefs?)>
<!ELEMENT maintfunc EMPTY>
<!ATTLIST maintfunc
    func  (inspect |  test | service | adjust | align | calib | remove | replace |
         repair | overhaul | rebuild | none )  #REQUIRED>
<!ELEMENT avmaintclass ((avum, avim?, depot?) | (avum, depot?) | depot)>
<!ELEMENT terefs (teref+)>
<!ELEMENT teref EMPTY>
<!ATTLIST teref
     refs IDREF  #REQUIRED >
<!ELEMENT remarkrefs (remarkref+)>
<!ELEMENT remarkref EMPTY>
<!ATTLIST remarkref
    refs IDREF  #REQUIRED >
```

40.4.3 Tools and test equipment requirements standard information ***<tereqtab>***. The element is used to list all the tools and test equipment, both special and common, required to maintain the equipment. The tool and test equipment references from the MAC targets are located within the element.

a. The components of Tools and test equipment requirements standard information ***<tereqtab>*** are:

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(1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
(2) Standard table header ***<std.table.header>*** (required) (see 40.4.3.1).
(3) Tools and test equipment item ***<teref-group>*** (required – one or more) (see 40.4.3.2).

b. DTD graphically depicted for ***<tereqtab>***:

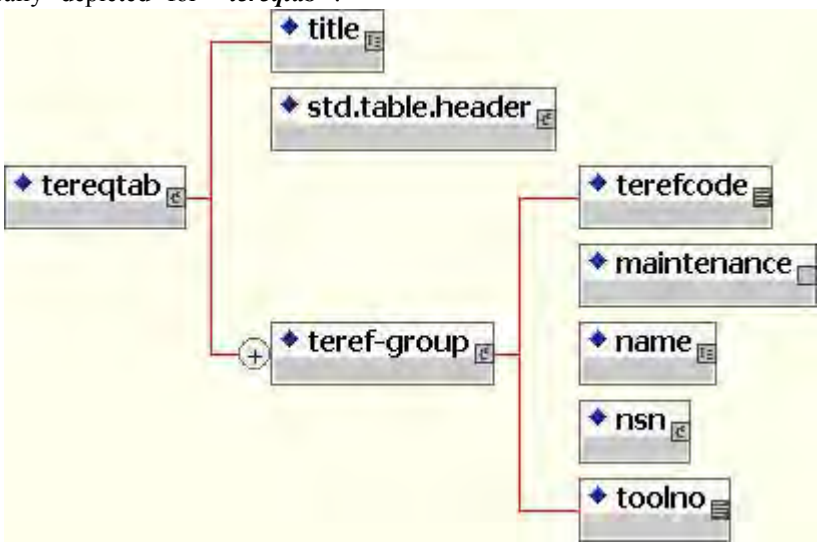

FIGURE 458. Tools and test equipment requirements table DTD hierarchy <tereqtab>.

c. DTD fragment:

```
<!ELEMENT tereqtab (title, std.table.header, teref-group+)>
<!ATTLIST tereqtab
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<tereqtab>***:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.4.3.1 Tools and test equipment requirements standard information headers ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table

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headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in ***&macwp.tereqtab.std-header;***.

40.4.3.2 Tools and test equipment item ***<teref-group>***. The element contains the information to identify and procure the tool or test equipment needed for the maintenance function. The element is similar to a "row" in a structural table.

a. The components of Tools and test equipment item ***<teref-group>*** are:
   (1) Reference code ***<terefcode>*** (required) (see 40.4.3.2.1). The element is similar to a "cell" in a structural table and is entered in column one.
   (2) Lowest allowable maintenance level for usage ***<maintenance>*** (required) (see 40.4.3.2.2). The element is similar to a "cell" in a structural table and is entered in column two.
   (3) Tool name ***<name>*** (required) (see 49.1.4.19). The element is similar to a "cell" in a structural table and is entered in column three.
   (4) National Stock Number (NSN) ***<nsn>*** (required) (see 49.1.4.20). The element is similar to a "cell" in a structural table and is entered in column four.
   (5) Tool number ***<toolno>*** (required) (see 40.4.3.2.3). The element is similar to a "cell" in a structural table and is entered in column five.
b. DTD fragment:

```
<!ELEMENT teref-group (terefcode, maintenance, name, nsn, toolno)>
```

c. No attributes for ***<teref-group>***.

40.4.3.2.1 Reference code ***<terefcode>***. The element contains a tools and equipment reference code. The attribute "**id**" is the target for the MAC element ***<teref>*** and attribute "**refs**". When the MAC references the tools and test equipment item, the text entered is the reference code used by the MAC.

a. The components of Tools and equipment reference code ***<terefcode>*** is typed text (Parsable Characters).
b. DTD fragment:

```
<!ELEMENT terefcode (#PCDATA)>
<!ATTLIST terefcode
    id                  ID                                  #REQUIRED>
```

c. Attribute for ***<terefcode>*** is
   (1) **id** – The unique identifier or target reference in the MAC for the element ***<teref refs=" ">***.

40.4.3.2.2 Lower maintenance level ***<maintenance>***. The element is used to identify the lowest level of maintenance authorized that requires the listed item. The lowest level of maintenance is enter through the required attribute "**lvl**". The stylesheet will use and capitalize the **lvl** attribute value.

a. The components of Lower Maintenance Level ***<maintenance>*** is EMPTY and all text is obtained from the attribute.
b. DTD fragment:

```
<!ELEMENT maintenance EMPTY>
<!ATTLIST maintenance
    lvl                 (c | o | f | h | d )              #REQUIRED >
```

c. Attribute for ***<maintenance>*** is
   (1) **lvl** – Specifies the lowest maintenance level(s) code allowed. The two-level and five-level MAC contains the following maintenance level codes:
      (a) "C" – Crew or operator.
      (b) "O" – Organizational (unit).
      (c) "F" – Direct support (DS).
      (d) "H" – General support (GS).
      (e) "D" – Depot
      The aviation MAC contains the following maintenance level codes:
      (f) "O" – AVUM.

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(g) “F” – AVIM.
(h) “D” – Depot

40.4.3.2.3 Tool number ***<toolno>***. The element contains the manufacturer’s part number, model number, or type number.

a. The components are
   (1) Parsable Characters - The element contains narrative text #PCDATA parsable character data.
b. DTD fragment for ***<toolno>***:

```
<!ELEMENT toolno (#PCDATA)>
<!ATTLIST toolno
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

40.4.4 Remark standard information ***<remarktab>***. The element provides pertinent remarks to the maintenance functions as referenced in the MAC.

a. The components of Remark standard information ***<remarktab>*** are:
   (1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
   (2) Standard table header ***<std.table.header>*** (required) (see 40.4.4.1).
   (3) Remark information ***<remark-group>*** (required – one or more) (see 40.4.4.2).
b. DTD graphically depicted for ***<remarktab>***:

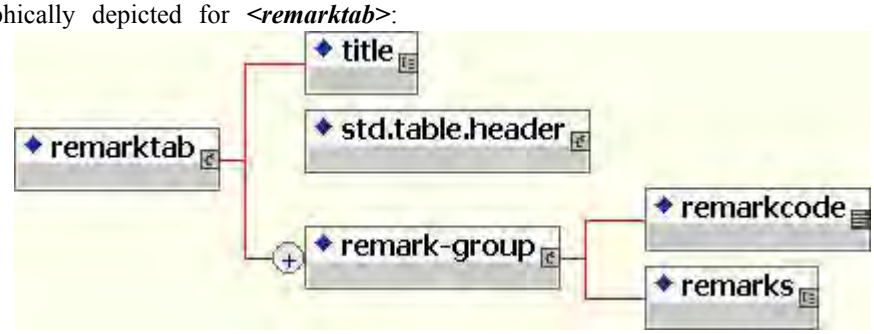

FIGURE 459. Tools and test equipment requirements table <remarktab> DTD hierarchy. .

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c. DTD fragment:

```
<!ELEMENT remarktab (title, std.table.header, remark-group+)>
<!ATTLIST remarktab
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<remarktab>***:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

40.4.4.1 Remarks standard information headers ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in ***&macwp.remarktab.std-header;***.

40.4.4.2 Remark information ***<remark-group>***. The element contains the remark text and reference code. The element is similar to a "row" in a structural table.

a. The components of Remark information ***<remark-group>*** are:
   (1) Reference code ***<remarkcode>*** (required) (see 40.4.4.2.1). The element is similar to a "cell" in a structural table and is entered in column one.
   (2) Remarks ***<remarks>*** (required) (see 40.4.4.2.2). The element is similar to a "cell" in a structural table and is entered in column two.

b. DTD fragment for ***<remark-group>***:

```
<!ELEMENT remark-group (remarkcode, remarks)>
```

c. No attributes for ***<remark-group>***.

40.4.4.2.1 Reference code ***<remarkcode>***. The element contains a remarks reference code. The attribute "**id**" is the target for the MAC element ***<remarkref>*** and attribute "**refs**". When the MAC references the remarks, the text entered is the reference code used by the MAC.

a. The components of Tools and equipment reference code ***<remarkcode>*** is typed text (Parsable Characters).

b. DTD fragment for ***<remarkcode>***:

```
<!ELEMENT remarkcode (#PCDATA)>
<!ATTLIST remarkcode
    id                ID                              #REQUIRED>
```

c. Attribute for ***<remarkcode>*** is

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(1) **id** – The unique identifier or target reference in the MAC for the element ***<remarkref refs=" ">***.

40.4.4.2.2 Remarks ***<remarks>***. The element contains remarks information pertinent to the maintenance function performed as indicated in the MAC.

a. The components of Remarks ***<remarks>*** are:
   (1) Parsable characters or type text. – #PCDATA
   (2) Format text – ***<emphasis>*** (see 49.1.3.1).
   (3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
   (4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
   (5) Cross reference – ***<xref>*** (see 46.2.2).
   (6) External reference – ***<extref>*** (see 46.2.1).
   (7) Enhanced linking – ***<link>*** (see 46.2.3).
   (8) IETM help information – ***<help.info>*** (see 48.3.2.4).
   (9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
   (10) Term – ***<term>*** (see 49.1.2.3.1.1).
   (11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
   (12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
   (13) Footnote – ***< ftnote>*** (see 45.1.1).
   (14) Footnote reference – ***<ftnref>*** (see 45.1.2).
   (15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
   (16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
   (17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
   (18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
   (19) Graphic symbol – ***<symbol>*** (see 44.3.1).
   (20) Torque value – ***<torque>*** (see 49.1.4.24).
   (21) Voltage value – ***< voltage>*** (see 49.1.4.25).
   (22) Null text – ***<null >*** (see 49.1.3.2).
   (23) Changed text marker – ***<change>*** (see 49.1.3.7).

b. DTD fragment for ***<remarks>***:

```
<!ELEMENT remarks (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link |
   help.info | indxref | term | term.def | callout | ftnote | ftnref | ctrlind |
   ctrlind-val | dodac | lubricant | symbol | torque | voltage | null | change)*>
<!ATTLIST remarks
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Attribute for ***<remarks>*** is
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)

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(9) **security** – Security classification. (optional) (See 49.2.13.)

40.4.5 XML document instance fragment and output for a five-level MAC ***<macwp>***. The MAC work package (five-level) XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % mac.2-level "IGNORE">
<!ENTITY % mac.5-level "INCLUDE">
<!ENTITY % mac.nonav-level "INCLUDE">
<!ENTITY % mac.av-level "IGNORE">
```

b. Example of an XML document instance fragment for a five-level MAC ***<mac>***:

```
<macwp wpno="s00003-X-XXXX-XXX" tocentry="2" frame="yes" army="no" airforce="no"
navy="no" marines="no"
wpseq="0440" deletewp="no">
<wpidinfo>
<maintlvl level="unitlvl">
<title>TSEC/ST-34 Maintenance Allocation chart (MAC)
</title>
</wpidinfo>

<!– MAC Five-Level standard information –>
<mac maintconcept="5">
<title>MAC for TSEC/ST-34.
</title> &macwp.mac.std-header;
<mac-group>
<groupno>00
</groupno>
<compassem>
<name>TSEC/ST-34
</name>
</compassem>
<qualify>
<maintfunc func="inspect"/>
<maintclass>
<unit>
<crew>0.1
</crew>
</unit>
</maintclass>
<remarkrefs>
<remarkref refs="remark-a"/>
</remarkrefs>
</qualify>
<qualify>
<maintfunc func="service"/>
<maintclass>
<unit>
<crew>0.2
</crew>
</unit>
</maintclass>
<remarkrefs>
```


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Check the source to verify that this is the current version before use.

```
<remarkref  refs="remark-b"/>
</remarkrefs>
</qualify>
<qualify>
<maintfunc  func="replace"/>
<maintclass>
<unit>
<crew>0.4
</crew>
</unit>
</maintclass>
<terefs>
<teref  refs="tool1"/>
</terefs>
<remarkrefs>
<remarkref  refs="remark-c"/>
<remarkref  refs="remark-d"/>
</remarkrefs>
</qualify>
<qualify>
<maintfunc  func="test"/>
<maintclass>
<unit>
<crew>0.3
</crew>
</unit>
</maintclass>
<remarkrefs>
<remarkref  refs="remark-e">
</remarkrefs>
</qualify>
<qualify>
<maintfunc  func="repair"/>
<maintclass>
<direct>1.5
</direct>
</maintclass>
<terefs>
<teref  refs="tool1">
<teref  refs="tool2">
</terefs>
<remarkrefs>
<remarkref  refs="remark-f">
</remarkrefs>
</qualify>
<qualify>
<maintfunc  func="repair"/>
<maintclass>
<gensup>2.0
</gensup>
</maintclass>
<terefs>
<teref  refs="tool1"/>
<teref  refs="tool2"/>
<teref  refs="tool3"/>
<teref  refs="tool4"/>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<teref refs="tool5"/>
</terefs>
<remarkrefs>
<remarkref refs="remark-g"/>
<remarkref refs="remark-h"/>
</remarkrefs>
</qualify>
<qualify>
<maintfunc func="repair"/>
<maintclass>
<depot>2.0
</depot>
</maintclass>
<terefs>
<teref refs="tool1"/>
<teref refs="tool2"/>
<teref refs="tool3"/>
<teref refs="tool4"/>
<teref refs="tool5"/>
<teref refs="tool6"/>
<teref refs="tool7"/>
<teref refs="tool8"/>
</terefs>
<remarkrefs>
<remarkref refs="remark-i">
</remarkrefs>
</qualify>
<qualify>
<maintfunc func="overhaul"/>
<maintclass>
<depot>16.0
</depot>
</maintclass>
<terefs>
<teref refs="tool1"/>
<teref refs="tool2"/>
<teref refs="tool3"/>
<teref refs="tool4"/>
<teref refs="tool5"/>
<teref refs="tool6"/>
<teref refs="tool7"/>
<teref refs="tool8"/>
</terefs>
<remarkrefs>
<remarkref refs="remark-j">
</remarkrefs>
</qualify>
</mac-group>
<mac-group>
<groupno>01
</groupno>
<compassem>
<name>POWER UNIT, STP-34
</name>
</compassem>
<qualify>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<maintfunc func="inspect"/>
<maintclass>
<unit>
<crew>1.8
</crew>
</unit>
</maintclass>
<remarkrefs>
<remarkref refs="remark-a"/>
</remarkrefs>
</qualify>
<qualify>
<maintfunc func="test"/>
<maintclass>
<unit>
<crew>0.3
</crew>
</unit>
</maintclass>
<remarkrefs>
<remarkref refs="remark-e"/>
</remarkrefs>
</qualify>
<qualify>
<maintfunc func="repair"/>
<maintclass>
<direct>1.8
</direct>
</maintclass>
<terefs>
<teref refs="tool1"/>
<teref refs="tool2"/>
</terefs>
<remarkrefs>
<remarkref refs="remark-f"/>
</remarkrefs>
</qualify>
<qualify>
<maintfunc func="repair"/>
<maintclass>
<gensup>2.0
</gensup>
</maintclass>
<terefs>
<teref refs="tool1"/>
<teref refs="tool2"/>
<teref refs="tool3"/>
<teref refs="tool4"/>
<teref refs="tool5"/>
</terefs>
<remarkrefs>
<remarkref refs="remark-g"/>
<remarkref refs="remark-j"/>
</remarkrefs>
</qualify>
<qualify>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<maintfunc  func="repair"/>
<maintclass>
<depot>2.0
</depot>
</maintclass>
<terefs>
<teref  refs="tool1"/>
<teref  refs="tool2"/>
<teref  refs="tool3"/>
<terefrefs="tool4"/>
<teref  refs="tool5"/>
<teref  refs="tool6"/>
<teref  refs="tool7"/>
<teref  refs="tool8"/>
</terefs>
<remarkrefs>
<remarkref  refs="remark-g"/>
<remarkref  refs="remark-j"/>
</remarkrefs>
</qualify>
</mac-group>
</mac>

<!– Tools and test equipment standard information –>
<tereqtab>
<title>Tools and Test Equipment for TSEC/ST-34.
</title>  &macwp.tereqtab.std-header;
<teref-group>
<terefcode  id="tool1">1
</terefcode>
<maintenance  lvl="h"/>
<name>Automatic test system ST-51
</name>
<nsn>
<fsc>5810
</fsc>
<niin>00-089-4599
</niin>
</nsn>
<toolno>TSEC/ST-51
</toolno>
</teref-group>
<teref-group>
<terefcode  id="tool2">2
</terefcode>
<maintenance  lvl="f"/>
<name>  Multimeter,  digital
</name>
<nsn>
<fsc>6625
</fsc>
<niin>01-139-2512
</niin>
</nsn>
<toolno>AN/PSM-45
</toolno>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</teref-group>
<teref-group>
<terefcode id="tool3">3
</terefcode>
<maintenance lvl="d"/>
<name>Multimeter, digital
</name>
<nsn>
<fsc>6625
</fsc>
<niin>01-145-2430
</niin>
</nsn>
<toolno>AN/USM 486
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool4">4
</terefcode>
<maintenance lvl="h"/>
<name>Oscilliscope
</name>
<nsn>
<fsc>6625
</fsc>
<niin>01-187-7847
</niin>
</nsn>
<toolno>AN/USM 488
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool5">5
</terefcode>
<maintenance lvl="d"/>
<name>Power supply (0-35 VDC 2.4A)
</name>
<nsn>
<fsc>6130
</fsc>
<niin>00-006-5224
</niin>
</nsn>
<toolno>HP 6434B86
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool6">6
</terefcode>
<maintenance lvl="d"/>
<name>Power supply tester
</name>
<nsn>
<fsc>NOT
</fsc>
<niin>APPLICABLE
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</niin>
</nsn>
<toolno>ON502427
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool7">7
</terefcode>
<maintenance lvl="h"/>
<name>Repari and soldering center (page)
</name>
<nsn>
<fsc>4940
</fsc>
<niin>01-031-4541
</niin>
</nsn>
<toolno>PRC-350C/equip
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool8">8
</terefcode>
<maintenance lvl="f">
<name>Tool, kit, electronic equipment
</name>
<nsn>
<fsc>5180
</fsc>
<niin>00-610-8177
</niin>
</nsn>
<toolno>TK-105/6
</toolno>
</teref-group>
</tereqtab>

<!– Remarks standard information –>
<remarktab>
<title>Remarks for TSEC/ST-34.
</title> &macwp.remarktab.std-header;
<remark-group>
<remarkcode id="remark-a">A
</remarkcode>
<remarks>External.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-b">B
</remarkcode>
<remarks>Preventive maintenance checks and services (PMCS)
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-c">C
</remarkcode>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<remarks>Replace rack installed unit, 0.4 hrs.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-d">D
</remarkcode>
<remarks>Bench top use only, 0.1 hrs.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-e">E
</remarkcode>
<remarks>Self-test.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-f">F
</remarkcode>
<remarks>Repair by PMA and authorized component replacement only.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-g">G
</remarkcode>
<remarks>Complete unitand subassembly repair (except STP-34
switching assembly and E-EBO/1).
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-h">H
</remarkcode>
<remarks>Complete unitand subassembly repair.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-i">I
</remarkcode>
<remarks>In compliancewith TSEC/ST-34 CIDOS.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-j">J
</remarkcode>
<remarks>Function performed by specialized repair activity (SRA.(Theater
COMSEC Logistics Support Center-Europe or Lexington-Blue Grass Army Depot)
</remarks>
</remark-group>
</remarktab>
</macwp>
```

40.4.6 XML document instance fragment and output for a two-level MAC ***<macwp>***. The MAC work package (two-level) XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % mac.2-level "INCLUDE">
<!ENTITY % mac.5-level "IGNORE">
<!ENTITY % mac.nonav-level "INCLUDE">
<!ENTITY % mac.av-level "IGNORE">
```

b. Example of an XML document instance fragment for a two-level MAC *<mac>*:

```
<macwp wpno="s00004-X-XXXX-XXX" tocentry="2" frame="yes" army="no" airforce="no"
navy="no" marines="no"
wpseq="0440" deletewp="no">
<wpidinfo>
<maintlvl level="unitlvl">
<title>TSEC/ST-34 Maintenance Allocation chart (MAC)
</title>
</wpidinfo>

<!– MAC Two-Level standard information –>
<mac maintconcept="2"> &macwp.mac.std-header;
<title>MAC for TSEC/ST-34.
</title>
<mac-group>
<groupno>00
</groupno>
<compassem>
<name>POWER UNIT, STP-34
</name>
</compassem>
<qualify>
<maintfunc func="inspect"/>
<maintclass>
<unit>
<crew>1.8
</crew>
</unit>
</maintclass>
<remarkrefs>
<remarkref refs="remark-a">
</remarkrefs>
</qualify>
<qualify>
<maintfunc func="test"/>
<maintclass>
<unit>
<crew>0.3
</crew>
</unit>
</maintclass>
<remarkrefs>
<remarkref refs="remark-b">
</remarkrefs>
</qualify>
<qualify>
<maintfunc func="repair"/>
<maintclass>
<direct>1.8
</direct >
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</maintclass>
<terefs>
<teref  refs="tool1"/>
<teref  refs="tool2"/>
</terefs>
<remarkrefs>
<remarkref  refs="remark-c">
</remarkrefs>
</qualify>
<qualify>
<maintfunc  func="repair"/>
<maintclass>
<gensup>2.0
</gensup  >
</maintclass>
<terefs>
<teref  refs="tool1"/>
<teref  refs="tool2"/>
<teref  refs="tool3"/>
<teref  refs="tool4"/>
<teref  refs="tool5"/>
</terefs>
<remarkrefs>
<remarkref  refs="remark-d"/>
<remarkref  refs="remark-e"/>
</remarkrefs>
</qualify>
<qualify>
<maintfunc  func="repair"/>
<maintclass>
<depot>2.0
</depot  >
</maintclass>
<terefs>
<teref  refs="tool1"/>
<teref  refs="tool2"/>
<teref  refs="tool3"/>
<teref  refs="tool4"/>
</terefs>
<remarkrefs>
<remarkref  refs="remark-d"/>
<remarkref  refs="remark-e"/>
</remarkrefs>
</qualify>
</mac-group>
</mac>

<!– Tools and test equipment standard information –>
<tereqtab>
<title>Tools and Test Equipment for TSEC/ST-34.
</title>  &macwp.tereqtab.std-header;
<teref-group>
<terefcode  id="tool1">1
</terefcode>
<maintenance  lvl="h"/>
<name>Automatic test system ST-51
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</name>
<nsn>
<fsc>5810
</fsc>
<niin>00-089-4599
</niin>
</nsn>
<toolno>TSEC/ST-51
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool2">2
</terefcode>
<maintenance lvl="f"/>
<name> Multimeter, digital
</name>
<nsn>
<fsc>6625
</fsc>
<niin>01-139-2512
</niin>
</nsn>
<toolno>AN/PSM-45
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool3">3
</terefcode>
<maintenance lvl="d"/>
<name>Multimeter, digital
</name>
<nsn>
<fsc>6625
</fsc>
<niin>01-145-2430
</niin>
</nsn>
<toolno>AN/USM 486
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool4">4
</terefcode>
<maintenance lvl="h"/>
<name>Oscilliscope
</name>
<nsn>
<fsc>6625
</fsc>
<niin>01-187-7847
</niin>
</nsn>
<toolno>AN/USM 488
</toolno>
</teref-group>
<teref-group>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<terefcode id="tool5">5
</terefcode>
<maintenance lvl="d"/>
<name>Power supply (0-35 VDC 2.4A)
</name>
<nsn>
<fsc>6130
</fsc>
<niin>00-006-5224
</niin>
</nsn>
<toolno>HP 6434B86
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool6">6
</terefcode>
<maintenance lvl="d"/>
<name>Power supply tester
</name>
<nsn>
<fsc>NOT
</fsc>
<niin>APPLICABLE
</niin>
</nsn>
<toolno>ON502427
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool7">7
</terefcode>
<maintenance lvl="h"/>
<name>Repari and soldering center (page)
</name>
<nsn>
<fsc>4940
</fsc>
<niin>01-031-4541
</niin>
</nsn>
<toolno>PRC-350C/equip
</toolno>
</teref-group>
<teref-group>
<terefcode id="tool8">8
</terefcode>
<maintenance lvl="f">
<name>Tool, kit, electronic equipment
</name>
<nsn>
<fsc>5180
</fsc>
<niin>00-610-8177
</niin>
</nsn>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<toolno>TK-105/6
</toolno>
</teref-group>
</tereqtab>

<!– Remarks standard information –>
<remarktab>
<title>Remarks for TSEC/ST-34.
</title> &macwp.remarktab.std-header;
<remark-group>
<remarkcode id="remark-a">A
</remarkcode>
<remarks>External.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-b">B
</remarkcode>
<remarks>Preventive maintenance checks and services (PMCS)
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-c">C
</remarkcode>
<remarks>Replace rack installed unit, 0.4 hrs.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-d">D
</remarkcode>
<remarks>Bench top use only, 0.1 hrs.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-e">E
</remarkcode>
<remarks>Self-test.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-f">F
</remarkcode>
<remarks>Repair by PMA and authorized component replacement only.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-g">G
</remarkcode>
<remarks>Complete unitand subassembly repair (except STP-34
switching assembly and E-EBO/1).
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-h">H
</remarkcode>
<remarks>Complete unitand subassembly repair.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-i">I
</remarkcode>
<remarks>In compliancewith TSEC/ST-34 CIDOS.
</remarks>
</remark-group>
<remark-group>
<remarkcode id="remark-j">J
</remarkcode>
<remarks>Function performed by specialized repair activity (SRA.(Theater
COMSEC Logistics Support Center-Europe or Lexington-Blue Grass Army Depot)
</remarks>
</remark-group>
</remarktab>
</macwp>
```

40.4.7 XML document instance fragment and output for an aviation (three-level) MAC **<*macwp*>**. The aviation (three-level) MAC work package XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % mac.2-level "IGNORE">
<!ENTITY % mac.5-level "IGNORE">
<!ENTITY % mac.nonav-level "IGNORE">
<!ENTITY % mac.av-level "INCLUDE">
```

b. Example of an XML document instance fragment for an aviation (three-level) MAC **<*avmac*>**:

```
<macwp wpno="s00005-X-XXXX-XXX" tocentry="2" frame="yes" army="no" airforce="no"
navy="no" marines="no" wpseq="0440" deletewp="no">
<wpidinfo>
<maintlvl level="unitlvl">
<title>TSEC/ST-34 Maintenance Allocation Chart (MAC)
</title>
</wpidinfo>

<!– Aviation MAC Three-Level standard information –>
<avmac>
<title>Maintenance Allocation Chart (MAC)
</title > %macwp.avmac.std-header;
<avmac-group>
<groupno>04
</groupno>
<compassem>
<name>POWER PLANT
</name>
</compassem>
<avqualify>
<maintfunc func="none"/>
<avmaintclass>
</avmaintclass>
</avqualify>
</avmac-group>
```


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<avmac-group>
<groupno>0401
</groupno>
<compassem>
<name>ENGINE, TURBINE
</name>
</compassem>
<avqualify>
<maintfunc func="inspect"/>
<avmaintclass>
<avum>--
</avum>
</avmaintclass>
<terefs>
<teref refs="tool1"/>
</terefs>
<remarkrefs>
<remarkref refs="remark-a"/>
</remarkrefs>
</avqualify>
<avqualify>
<maintfunc func="test"/>
<avmaintclass>
<avum>--
</avum>
</avmaintclass>
<remarkrefs>
<remarkref refs="remark-b"/>
</remarkrefs>
</avqualify>
<avqualify>
<maintfunc func="test"/>
<avmaintclass>
<avim>--(3)
</avim>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc func="test"/>
<avmaintclass>
<depot>--
</depot>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc func="service"/>
<avmaintclass>
<avum>0.2
</avum>
</avmaintclass>
<remarkrefs>
<remarkref refs="remark-c"/>
</remarkrefs>
</avqualify>
<avqualify>
<maintfunc func="remove"/>

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<avmaintclass>
<avum>--
</avum>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc func="replace"/>
<avmaintclass>
<avum>--
</avum>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc func="repair"/>
<avmaintclass>
<avum>--
</avum>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc func="repair"/>
<avmaintclass>
<avim>--(4)
</avim>
</avmaintclass>
<remarkrefs>
<remarkref refs="remark-a"/>
</remarkrefs>
</avqualify>
<avqualify>
<maintfunc func="overhaul"/>
<avmaintclass>
<avum>--
</avum>
</avmaintclass>
</avqualify>
</avmac-group>
<avmac-group>
<groupno>040101
</groupno>
<compassem>
<name>EXTERNAL LINES & HOSES
</name>
</compassem>
<avqualify>
<maintfunc func="inspect"/>
<avmaintclass>
<avum>--
</avum>
</avmaintclass>
<terefs>
<teref refs="tool3"/>
</terefs>
<remarkrefs>
<remarkref refs="remark-d"/>
</remarkrefs>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</avqualify>
<avqualify>
<maintfunc  func="test"/>
<avmaintclass>
<avim>--
</avim>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc  func="remove"/>
<avmaintclass>
<avum>--
</avum>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc  func="replace"/>
<avmaintclass>
<avum>--
</avum>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc  func="repair"/>
<avmaintclass>
<avim>--
</avim>
</avmaintclass>
</avqualify>
</avmac-group>
<avmac-group>
<groupno>0402
</groupno>
<compassem>
<name>COMPRESSORSECTION  (COLD  SECTION  MODULE)
</name>
</compassem>
<avqualify>
<maintfunc  func="inspect"/>
<avmaintclass>
<avum>0.1
</avum>
</avmaintclass>
<remarkrefs>
<remarkref  refs="remark-e"/>
</remarkrefs>
</avqualify>
<avqualify>
<maintfunc  func="inspect"/>
<avmaintclass>
<avim>0.2
</avim>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc  func="test"/>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<avmaintclass>
<avim>--
</avim>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc  func="service"/>
<avmaintclass>
<avum>0.2
</avum>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc  func="repair"/>
<avmaintclass>
<avum>0.4
</avum>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc  func="repair"/>
<avmaintclass>
<avim>0.6
</avim>
</avmaintclass>
</avqualify>
<avqualify>
<maintfunc  func="overhaul"/>
<avmaintclass>
<depot>--
</depot>
</avmaintclass>
</avqualify>
</avmac-group>
</avmac>
...
</macwp>
```

40.5 Components of end item (COEI) and basic issue items (BII) lists work package ***<coeibiiwp>***. Components of end item (COEI) and basic issue items (BII) lists work package is to be prepared as an inventory for the equipment to ensure safe and efficient operation. The format of the COEI and BII is based on the number of items and usability. The COEI/BII has two different formats by either Method A (see 40.5.1) or Method B (see 40.5.2).

a. The components of ***<coeibiiwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) COEI/BII introduction ***<intro>*** (required) (see 40.5.1.1 for Method A or 40.5.2.1 for Method B and 49.1.4.15 for element usage).
   (4) Method A (required) using either
      (a) Components of End Item (COEI) standard information ***<coei>*** with all illustrations prior to the standard information (see 40.5.1.2).
      (b) Basic Issue Items (BII) standard information ***<bii>*** with all illustrations prior to the standard information (see 40.5.1.3).
   (5) Method B (required) using either

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(a) Components of End Item (COEI) standard information ***<coei-opt>*** with each illustration include with each item (see 40.5.2.2).

(b) Basic Issue Items (BII) standard information ***<bii-opt>*** with each illustration include with each item (see 40.5.2.3).

b. DTD graphically depicted for ***<coeibiiwp>***:

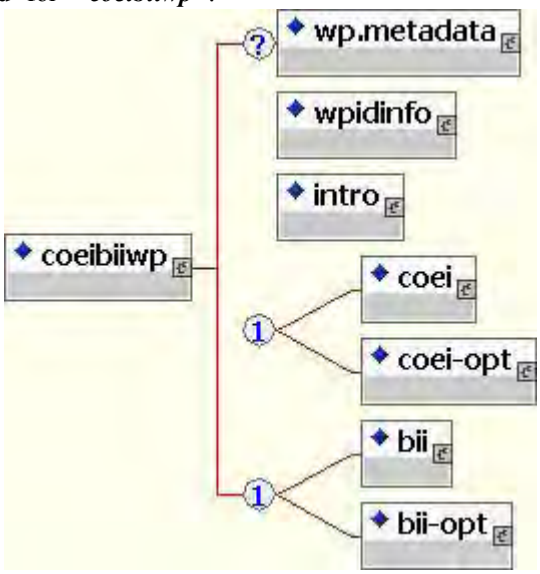

FIGURE 460. COEI and BII work package DTD hierarchy <coeibiiwp>.

c. DTD fragment for ***<coeibiiwp>***:

```
<!ELEMENT coeibiiwp (wp.metadata?, wpidinfo, intro, (coei | coei-opt),
   (bii | bii-opt))>
<!ATTLIST coeibiiwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attribute

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(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.5.1 COEI/BII method A. When there are only a few items the illustrations are placed above the tabular listing (Method A). The difference between Method A and Method B are the following:

Method A has illustrations displaying the end items before the COEI and BII list.
Method A for the COEI and BII list contains the Illustration Number in column one.
Method B displays the illustration with the end item in column two of the COEI and BII list.
Method B for the COEI and BII list contains the Item Number in column one.

40.5.1.1 COEI/BII (method A) introduction ***<intro>*** standard text. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed Table XXXI.

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TABLE XXIX. COEI/BII (Method A) introduction *<intro>* boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
COEI/BII Introduction	***&coeibiiwp.intro;***	***<!ENTITY % coeibiiwp.method-a "INCLUDE">*** ***<!ENTITY % coeibiiwp.method-b "IGNORE">***
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***
End Item Name		***<!ENTITY short.end.item.name "Insert Short End Item Name">***
Exclude UOC		***<!ENTITY % uoc-list "IGNORE">***
Include UOC		***<!ENTITY % uoc-list "INCLUDE">*** ***<!ENTITY intro.uoc-list '<![%uoc-list;[*** ***<deflist>*** ***<title.term.def>*** ***<title>***Code ***</title>*** ***<title>***Used on ***</title>*** ***</title.term.def>*** ***<term.def id="uoc.xxx">*** ***<term>*** *Insert UOC* ***</term>*** ***<def>*** ***<para hcp="no" esd="no">*** *Insert UOC Model Number* ***</para>*** ***</def>*** ***</term.def>*** ***</deflist>]]>'>***

40.5.1.1.1 Editing usable on-code (UOC) list ***&intro.uoc-list;***. When UOC exist for the end items they are listed in the introduction. Including the UOC and model number into introduction boilerplate entity is by editing the common UOC list boiler entity ***&intro.uoc-list;***. This common boilerplate entity is used by the following introductions, to include UOC, which are COEI/BII, AAL, and RPSTL. Once UOC list is defined for the COEI/BII, the same list is used for both AAL and RPSTL. The information entered by replacing the existing ***<term.def>*** element and it's content with a ***<term.def>*** (with ID attribute) for each UOC. Below is an example of ***&intro.uoc-list;*** changed.

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Check the source to verify that this is the current version before use.

```
<!ENTITY  intro.uoc-list  '<![%uoc-list;[
      <deflist>
      <title.term.def>
      <title>Code
      </title>
      <title>Used  on
      </title>
      </title.term.def>
      <term.def  id="uoc.PAA">
      <term>PAA
      </term>
      <def>
      <para  hcp="no"  esd="no">Model  XXX
      </para>
      </def>
      </term.def>
      <term.def  id="uoc.PAB">
      <term>PAB
      </term>
      <def>
      <para  hcp="no"  esd="no">Model  XXXX
      </para>
      </def>
      </term.def>
      <term.def  id="uoc.PAC">
      <term>PAC
      </term>
      <def>
      <para  hcp="no"  esd="no">Model  XXXXX
      </para>
      </def>
      </term.def>
      </deflist>
]]>'>
```

FIGURE 461. Example editing UOC boilerplate entity &intro.uoc-list;.

40.5.1.2 Components of end item (COEI) (method A) ***<coei>***. Components of End Item (COEI) (Method A) contains all COEI illustrated and then list the component for inventory purposes.

a. The components of ***<coei>*** are:
   (1) COEI illustration ***<graphic>*** (required – one or more) (see 44.2). The COEI illustrations may be organized into a single graphic or be comprised of individual graphics. Using a single graphic for all illustration provides greater flexibility in organizing and utilizing white space.
   (2) COEI (Method A) standard information ***<coeitab>*** (required) (see 40.5.1.2.1).

b. DTD graphically depicted for ***<coei>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

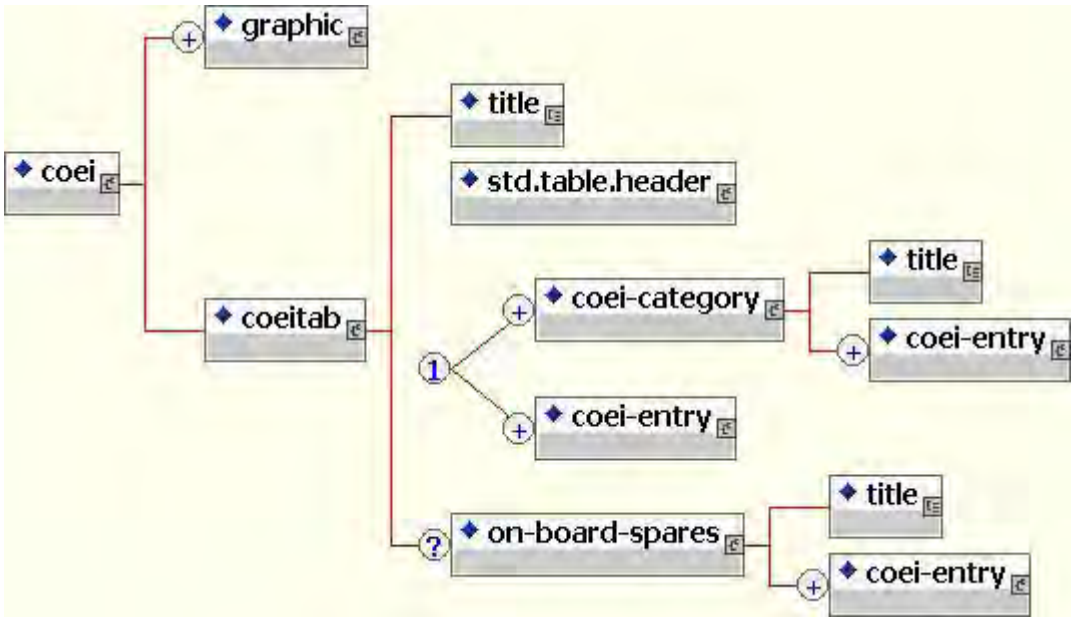

FIGURE 462. COEI (method A) DTD hierarchy <coei>.

c. DTD fragment for ***<coei>***:

```
<!ELEMENT coei (graphic+, coeitab)>
<!ATTLIST coei
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<coei>***:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.1.2.1 COEI (method A) standard information ***<coeitab>***. COEI (Method A) standard information lists all the spare/repair parts that are removed from the major end item and separately packaged or stowed for transportation or movement.

a. The components of ***<coeitab>*** are

(1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
(2) COEI (Method A) standard table header ***<std.table.header>*** (required) (see 40.5.1.2.1.1).

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(3) The COEI is displayed by either
(a) Grouping COEI (Method A) components by categories, use the element ***<coei-category>*** (required – one or more) (see 40.5.1.2.1.2).
(b) Each COEI (Method A) components is listed separately, use the element ***<coei-entry>*** (required – one or more) (see 40.5.1.2.1.3).
(4) On–board spares list ***<on-board-spares>*** (optional) (see 40.5.1.2.1.4).

b. DTD fragment for ***<coeitab>***:

```
<!ELEMENT coeitab (title, std.table.header, (coei-category+ | coei-entry+),
   on-board-spares?)>
<!ATTLIST coeitab
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<coeitab>*** are:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.1.2.1.1 COEI and BII (method A) standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XXX.

TABLE XXX. COEI/BII (Method A) standard information header ***<std.table.header>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set
COEI and BII Method AStandard Information Header	***&coeibiiwp.coeitab-biitab.std-header;***	***<!ENTITY % coeibiiwp.method-a "IGNORE">*** ***<!ENTITY % coeibiiwp.method-b "INCLUDE">***
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***

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Check the source to verify that this is the current version before use.

Table XXX COEI/BII (Method A) standard information header *<std.table.header>* boilerplate entities – Continued.

Description	Boilerplate Entity	Selectable Entity to Set
Page-based		*<!ENTITY % frame-base "IGNORE">* *<!ENTITY % page-base "INCLUDE">*

40.5.1.2.1.2 COEI (method A) category ***<coei-category>***. If a table is subdivided into groups, for example by subassemblies, the category element is used to represent the groups. After the category name is entered, the specific COEI (Method A) entries ***<coei-entry>*** are entered for that category. Usually more than one category is entered in the table.

a. The components of ***<coei-category>*** are:
   (1) Category title ***<title>*** (required) (see 49.1.1.3).
   (2) COEI (Method A) entry(s) ***<coei-entry>*** (required – one or more) (see 40.5.1.2.1.3) for this category.

b. DTD fragment for ***<coei-category>***:

```
<!ELEMENT coei-category (title, coei-entry+)>
<!ATTLIST coei-category
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<coei-category>*** are:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.1.2.1.3 COEI (method A) entry ***<coei-entry>***. The COEI (Method A) entries contain the specific information about the item and references to the associated illustration. The element is similar to a "row" in a structural table.

a. The components of ***<coei-entry>*** are
   (1) Illustration number ***<illno>*** (required) (see 40.5.1.2.1.3.1). The element is similar to a “cell” in a structural table and is entered in column one.
   (2) Usually the COEI entry has a single identification information, but on occasions similar fit and function components are listed under the same illustration number (required – one or more).
      (a) National Stock Number (NSN) ***<nsn>*** (required) identifies the stock number of the item to be used for requisitioning purposes (see 49.1.4.20). The element is similar to a “cell” in a structural table and is entered in column two.
      (b) Item name, description, CAGE code, part number, and usable on-code (UOC) ***<dcpno>*** (required) (see 40.5.1.2.1.3.2).

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1 The elements ***<name>***, ***<desc>***, ***<cageno>***, and ***<partno>*** combined is similar to a "cell" in a structural table and is entered in column three.
   a Name ***<name>*** (required) (see 49.1.4.19).
   b Description ***<desc>*** (optional) (see 49.1.4.17).
   c CAGE code ***<cageno>*** (required – one or more) (see 49.1.4.2).
   d Part number ***<partno>*** (required – one or more) (see 49.1.4.21).
2 Usable on-code (UOC) ***<uoc>*** (required – one or more) (see 38.4.2.1.3.1.3). The element is similar to a "cell" in a structural table and is entered in column four.

(c) Unit of issue ***<ui>*** (required) provides the component size or amount when issued. The element is similar to a "cell" in a structural table and is entered in column four.

(d) Quantity ***<qty>*** (required) (see 49.1.4.9). The element is similar to a "cell" in a structural table and is entered in column five.

b. DTD graphically depicted for ***<coei-entry>***:

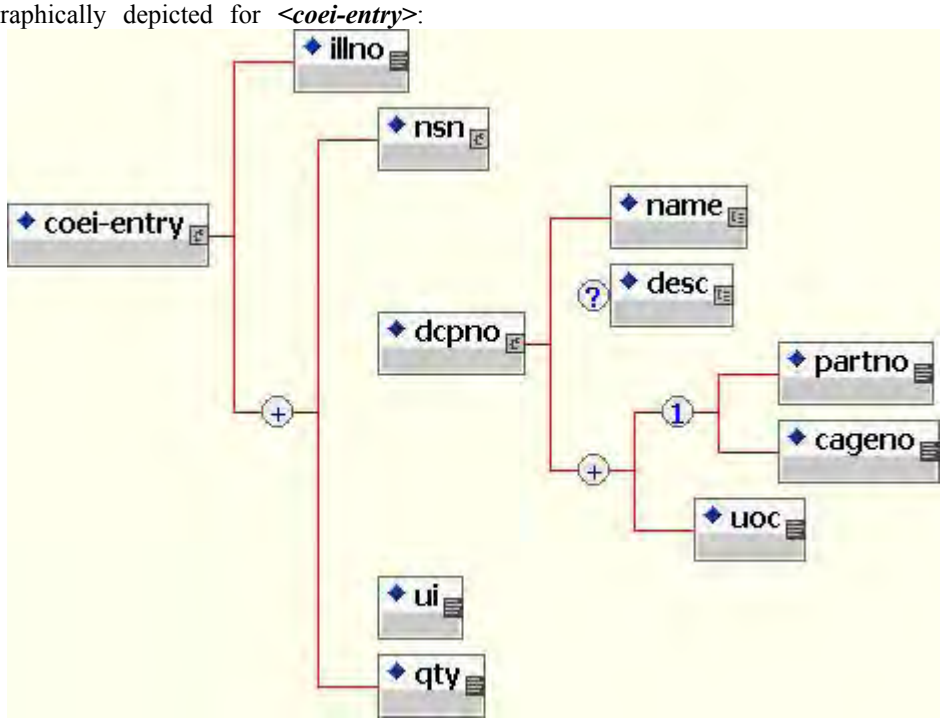

FIGURE 463. COEI (method A) entry DTD hierarchy <coei-entry>.

c. DTD fragment for ***<coei-entry>***:

```
<!ELEMENT coei-entry (illno, (nsn, dcpno, ui, qty)+)>
<!ATTLIST coei-entry
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<coei-entry>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

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(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.1.2.1.3.1 Illustration number ***<illno>***. The element ***<illno>*** illustration call out number relates the illustration to the list. In page-base, the element is equivalent to an "entry" element in a table. The ***<illno>*** is entered in the first column in the BII table and in the COEI table.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment for:

```
<!ELEMENT illno (#PCDATA)>
<!ATTLIST illno

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.1.2.1.3.2 Item name, description, CAGE code, part number, and usable on-code (UOC) ***<dcpno>***. This element ***<dcpno>*** contains the name/nomenclature, additional information or description, CAGE code, part number, and the usable on code (UOC). In page-base, the element is equivalent to an "entry" element in a table. The elements ***<name>***, ***<desc>***, ***<cageno>***, and ***<partno>*** combined is similar to a “cell” in a structural table and is entered in column three in the COMPONENTS OF END ITEM (COEI) AND BASIC ISSUE ITEMS (BII) LISTS and Usable on-code (UOC) ***<uoc>*** is entered in column four. In the Additional Authorization List (AAL) structural table the elements ***<name>***, ***<desc>***, ***<cageno>***, and ***<partno>*** combined is entered in column two and Usable on-code (UOC) ***<uoc>*** is entered in column three.

a. The components are:
   (1) Name ***<name>*** (required) (see 49.1.4.19).
   (2) Description ***<desc>*** (optional) (see 49.1.4.17).
   (3) CAGE code ***<cageno>*** (required – one or more) (see 49.1.4.2).
   (4) Part number ***<partno>*** (required – one or more) (see 49.1.4.21).
   (5) Usable on-code (UOC) ***<uoc>*** (required – one or more) (see 38.4.2.1.3.1.3).

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b. DTD graphically depicted for:

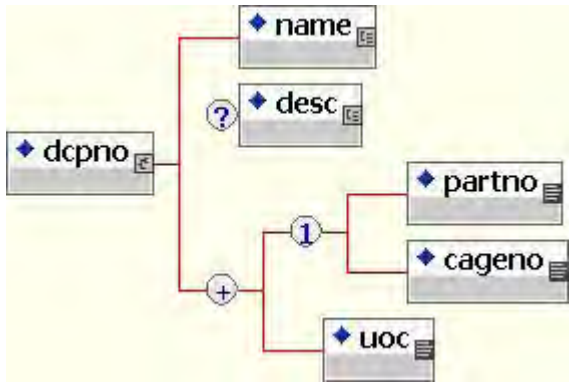

FIGURE 464. Item name, description, CAGE code, part number, and usable on-code (UOC) <dcpno>.

c. DTD fragment for:

```
<!ELEMENT dcpno (name, desc?, ((partno, cageno), uoc+)+)>
<!ATTLIST dcpno
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.1.2.1.4 On–board spares (method A) ***<on-board-spares>***. Any on-board spares, when applicable, are listed under a new heading or category name as "ON-BOARD SPARES" and components are entered in the same format as ***<coei-entry>***.

a. The components of ***<on-board-spares>*** are
(1) On-board spares title ***<title>*** (required) (see 49.1.1.3). MIL–STD–40051–1/–2 states the mandatory title is "ON-BOARD SPARES". The element is similar to a "row" in a structural table. The element is similar to a "cell" in a structural table and is entered in column one.
(2) COEI (Method A) entry(s) ***<coei-entry>*** (required – one or more) (see 40.5.1.2.1.3).

b. DTD fragment for ***<on-board-spares>***:

```
<!ELEMENT on-board-spares (title, coei-entry+)>
```

c. No attributes for ***<on-board-spares>***.

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40.5.1.3 Basic issue items (BII) (method A) ***<bii>***. Basic Issue Items (BII) (Method A) contains all BII illustrated and then list the components required to operate the equipment. The BII items are not part of the end item but are required to operate it.

a. The components of ***<bii>*** are:
   (1) BII illustration ***<graphic>*** (required – one or more) (see 44.2). The BII illustrations may be organized into a single graphic or be comprised of individual graphics. Using a single graphic for all illustration provides greater flexibility in organizing and utilizing white space.
   (2) BII (Method A) standard information ***<biitab>*** (required) (see 40.5.1.3.1).

b. DTD graphically depicted for ***<bii>***:

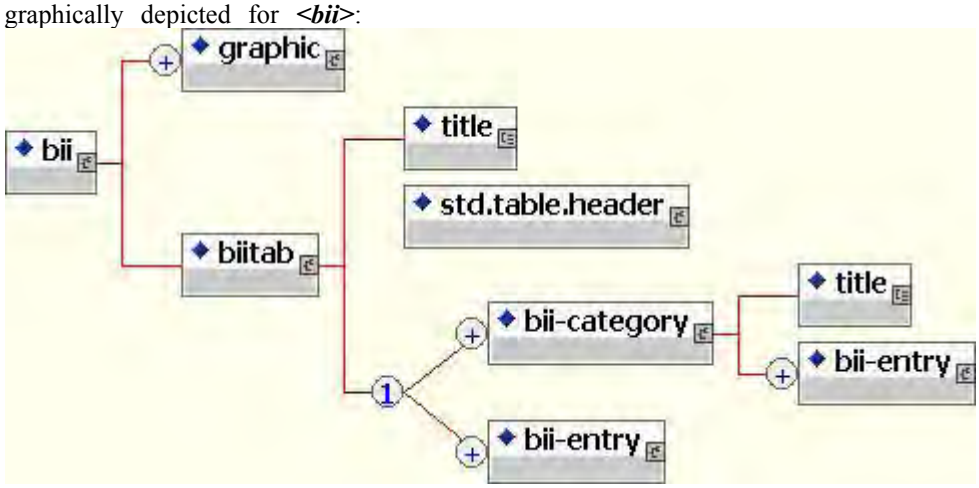

FIGURE 465. BII (method A) DTD hierarchy <bii>.

c. DTD fragment for ***<bii>***:

```
<!ELEMENT bii (graphic+, biitab)>
<!ATTLIST bii
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<bii>***:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.1.3.1 BII (method A) standard information ***<biitab>***. BII (Method A) standard information lists all the components required to operate the equipment.

a. The components of ***<biitab>*** are

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(1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
(2) BII (Method A) standard table header ***<std.table.header>*** (required) (see 40.5.1.2.1.1).
(3) The BII is displayed by either
(a) Grouping BII (Method A) components by categories, use the element ***<bii-category>*** (required – one or more) (see 40.5.1.3.1.1).
(b) Each BII (Method A) components is listed separately, use the element ***<bii-entry>*** (required – one or more) (see 40.5.1.3.1.2).

b. DTD fragment for ***<biitab>***:

```
<!ELEMENT biitab (title, std.table.header, (bii-category+ | bii-entry+))>
<!ATTLIST biitab
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<biitab>*** are:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.1.3.1.1 BII (method A) category ***<bii-category>***. If a table is subdivided into groups, for example by subassemblies, the category element is used to represent the groups. After the category name is entered, the specific BII (Method A) entries ***<bii-entry>*** are entered for that category. Usually more than one category is entered in the table.

a. The components of ***<bii-category>*** are:
(1) Category title ***<title>*** (required) (see 49.1.1.3).
(2) BII (Method A) entry(s) ***<bii-entry>*** (required – one or more) (see 40.5.1.3.1.2) for this category.

b. DTD fragment for ***<bii-category>***:

```
<!ELEMENT bii-category (title, bii-entry+)>
<!ATTLIST bii-category
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```


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c. Common attributes for ***<bii-category>*** are:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.1.3.1.2 BII (method A) entry ***<bii-entry>***. The BII (Method A) entries contain the specific information about the item and references to the associated illustration. The element is similar to a "row" in a structural table.

a. The components of ***<bii-entry>*** are
(1) Illustration number ***<illno>*** (required) (see 40.5.1.2.1.3.1). The element is similar to a "cell" in a structural table and is entered in column one.
(2) Usually the BII entry has a single identification information, but on occasions similar fit and function components are listed under the same illustration number (required – one or more).
(a) National Stock Number (NSN) ***<nsn>*** (required) identifies the stock number of the item to be used for requisitioning purposes (see 49.1.4.20). The element is similar to a "cell" in a structural table and is entered in column two.
(b) Item name, description, CAGE code, part number, and usable on-code (UOC) ***<dcpno>*** (required) (see 40.5.1.2.1.3.2).
1 The elements ***<name>***, ***<desc>***, ***<cageno>***, and ***<partno>*** combined is similar to a "cell" in a structural table and is entered in column three.
a Name ***<name>*** (required) (see 49.1.4.19).
b Description ***<desc>*** (optional) (see 49.1.4.17).
c CAGE code ***<cageno>*** (required – one or more) (see 49.1.4.2).
d Part number ***<partno>*** (required – one or more) (see 49.1.4.21).
2 Usable on-code (UOC) ***<uoc>*** (required – one or more) (see 38.4.2.1.3.1.3). The element is similar to a "cell" in a structural table and is entered in column four.
(c) Unit of issue ***<ui>*** (required) provides the component size or amount when issued. The element is similar to a "cell" in a structural table and is entered in column four.
(d) Quantity ***<qty>*** (required) (see 49.1.4.9). The element is similar to a "cell" in a structural table and is entered in column five.

b. DTD graphically depicted for ***<bii-entry>***:

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Check the source to verify that this is the current version before use.

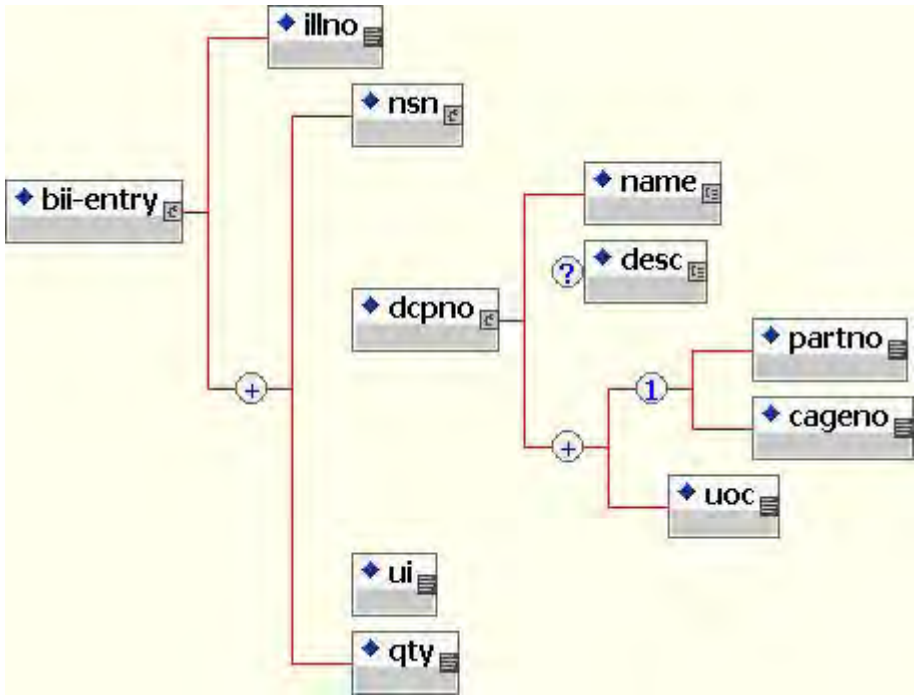

FIGURE 466. BII entry DTD hierarchy <bii-entry>.

c. DTD fragment for ***<bii-entry>***:

```
<!ELEMENT bii-entry (illno, (nsn, dcpno, ui, qty)+)>
<!ATTLIST bii-entry
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<bii-entry>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.1.4 XML document instance fragment and output for COEI/BII work package (method A) standard information ***<coeibiiwp>***. The COEI/BII work package (Method A) XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

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a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % coeibiiwp.method-a "INCLUDE">
<!ENTITY % coeibiiwp.method-b "IGNORE">
<!ENTITY % uoc-list "INCLUDE">
<!ENTITY short.end.item.name "M198 howitzer">
<!ENTITY intro.uoc-list '<![%uoc-list;[
<deflist>
<title.term.def>
<title>Code
</title>
<title>Used on
</title>
</title.term.def>
<term.def id="uoc.PAA">
<term>PAA
</term>
<def>
<para hcp="no" esd="no">Model XXX
</para>
</def>
</term.def>
<term.def id="uoc.PAB">
<term>PAB
</term>
<def>
<para hcp="no" esd="no">Model XXXX
</para>
</def>
</term.def>
<term.def id="uoc.PAC">
<term>PAC
</term>
<def>
<para hcp="no" esd="no">Model XXXXX
</para>
</def>
</term.def>
</deflist>
]]>'>
```

b. Example of an XML document instance fragment for COEI/BII work package (Method A) **<coeibiiwp>**:

```
<coeibiiwp wpno="s00003-X-XXXX-XXX" tocentry="2" frame="yes" army="no"
airforce="no" navy="no" marines="no" wpseq="0441" deletewp="no">
<wpidinfo>
<maintlvl level="unitlvl"/>
<title>M198 HOWITZER
</title>
</wpidinfo> &aal.intro;

<!– START COEI Standard Information –>
<coei>
<graphic boardno="coei" unitmeasure="in">
</graphic>
```


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Check the source to verify that this is the current version before use.

```
<coeitab>
<title>Components of End Item List.
</title> &coeibii.std.table.header;
<coei-entry>
<illno>1
</illno>
<nsn>
<fsc>1005
</fsc>
<niin>00-706-8880
</niin>
</nsn>
<dcpno>
<name>MOUNT, MACHINE GUN 1:cal.    .50
</name>
<desc>(in mount on cupola)
</desc>
<partno>7068880
</partno>
<cageno>19204
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</coei-entry>
<coei-entry>
<illno>2
</illno>
<nsn>
<fsc>1240
</fsc>
<niin>00-344-4643
</niin>
</nsn>
<dcpno>
<name>PERISCOPE:M27 (chief of section)
</name>
<desc>(stowage box cab wall)
</desc>
<partno>7633132
</partno>
<cageno>19200
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</coei-entry>
<coei-entry>
```


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Check the source to verify that this is the current version before use.

<illno>3
</illno>
<nsn>
<fsc>1240
</fsc>
<niin>00-509-2743
</niin>
</nsn>
<dcpno>
<name>PERISCOPE:M45 (driver's)
</name>
<desc>(stowage box driver's compartment)
</desc>
<partno>8213430
</partno>
<cageno>19200
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>3
</qty>
</coei-entry>
<coei-entry>
<illno>4
</illno>
<nsn>
<fsc>1240
</fsc>
<niin>00-864-2930
</niin>
</nsn>
<dcpno>
<name>TELESCOPE, PANORAMIC M117
</name>
<desc>(in mount M145 or telescope box)
</desc>
<partno>7660400
</partno>
<cageno>19200
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</coei-entry>
<coei-entry>
<illno>5
</illno>
<nsn>
<fsc>1240

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</fsc>
<niin>00-491-9676
</niin>
</nsn>
<dcpno>
<name>TELESCOPE, ELBOW: M118CA1
</name>
<desc>(in mount M146)
</desc>
<partno>10559855
</partno>
<cageno>19200
</cageno>
<uoc>PAB
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</coei-entry>
</coeitab>
</coei>
<!– START BII Standard Information –>
<bii>
<graphic boardno="bii" unitmeasure="in">
</graphic>
<biitab>
<title> Basic Issue Items (BII)
</title> &coeibii.std.table.header;
<bii-entry>
<illno>1
</illno>
<nsn>
<fsc>1290
</fsc>
<niin>00-535-7629
</niin>
</nsn>
<dcpno>
<name>LIGHT, AIMING POST; M14
</name>
<partno>7197188
</partno>
<cageno>19200
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>2
</qty>
</bii-entry>
<bii-entry>
<illno>2
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</illno>
<nsn>
<fsc>4930
</fsc>
<niin>00-766-3545
</niin>
</nsn>
<dcpno>
<name>LUBRICATING GUN, HAND: high pressure (in tool bag)
</name>
<partno>102758
</partno>
<cageno>36251
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</bii-entry>
<bii-entry>
<illno>3
</illno>
<nsn>
<fsc>8415
</fsc>
<niin>00-266-8843
</niin>
</nsn>
<dcpno>
<name>MITTENS, CLOTH: (pair) M1942 (in oddment tray)
</name>
<partno>11655982
</partno>
<cageno>19207
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>2
</qty>
</bii-entry>
<bii-entry>
<illno>4
</illno>
<nsn>
<fsc>4930
</fsc>
<niin>00-262-8868
</niin>
</nsn>
<dcpno>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<name>OILER, HAND: steel, pump type, 1pt, spout 9 lg
(in leftcab door stowage box)
</name>
<partno>328
</partno>
<cageno>72798
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</bii-entry>
<bii-entry>
<illno>5
</illno>
<nsn>
<fsc>7240
</fsc>
<niin>00-160-0455
</niin>
</nsn>
<dcpno>
<name>PAIL, UTILITY: 14-qt capability (on vehicle floor)
</name>
<partno>RRP35
</partno>
<cageno>81348
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</bii-entry>
</biitab>
</bii>
</coeibiiwp>
```

40.5.2 COEI/BII method B. When there are numerous items the illustrations may be included within the tabular listing for better usability (Method B). The difference between Method A and Method B are the following:

Method A has illustrations displaying the end items before the COEI and BII list.
Method A for the COEI and BII list contains the Illustration Number in column one.
Method B displays the illustration with the end item in column two of the COEI and BII list.
Method B for the COEI and BII list contains the Item Number in column one.

40.5.2.1 COEI/BII (method B) introduction ***<intro>*** standard text. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD.

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Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed Table XXXI.

TABLE XXXI. COEI/BII (Method B) introduction *<intro>* boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
COEI/BII (Method B)Introduction	***&coeibiiwp.intro;***	***<!ENTITY % coeibiiwp.method-a "IGNORE">*** ***<!ENTITY % coeibiiwp.method-b "INCLUDE">***
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***
End Item Name		***<!ENTITY short.end.item.name "Insert Short End Item Name">***
Exclude UOC		***<!ENTITY % uoc-list "IGNORE">***
Include UOC (see 40.5.1.1.1)		***<!ENTITY % uoc-list "INCLUDE">*** ***<!ENTITY intro.uoc-list '<![%uoc-list;[*** ***<deflist>*** ***<title.term.def>*** ***<title>***Code ***</title>*** ***<title>***Used on ***</title>*** ***</title.term.def>*** ***<term.def id="uoc.xxx">*** ***<term>*** *Insert UOC* ***</term>*** ***<def>*** ***<para hcp="no" esd="no">*** *Insert UOC Model Number* ***</para>*** ***</def>*** ***</term.def>*** ***</deflist>]]>'>***

40.5.2.2 Components of end item (COEI) (method B) ***<coei-opt>***. COEI (Method B) contains a component of end item that lists and illustrates all COEI items for inventory purposes.

a. The components of ***<coei-opt>*** are
   (1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
   (2) COEI (Method B) standard table header ***<std.table.header>*** (required) (see 40.5.2.2.1).
   (3) COEI (Method B) component information ***<coei-opt-entry>*** (required – one or more) (see 40.5.2.2.2).
   (4) On–board spares list (Method B) ***<on-board-spares-opt>*** (optional) (see 40.5.2.2.3).

b. DTD graphically depicted for ***<coei>***:

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Check the source to verify that this is the current version before use.

FIGURE 467. COEI (method B) DTD hierarchy <coei-opt>.

c. DTD fragment for ***<coei-opt>***:

```
<!ELEMENT coei-opt (title, std.table.header, coei-opt-entry+, on-board-spares-opt?)>
<!ATTLIST coei-opt
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<coei-opt>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

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(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.2.2.1 COEI and BII standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XXXII.

TABLE XXXII. COEI/BII (Method B) standard information header ***<std.table.header>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set
COEI and BII Method BStandard Information Header	***&coeibiiwp.coeitab-biitab.std-header;***	***<!ENTITY % coeibiiwp.method-a "INCLUDE">*** ***<!ENTITY % coeibiiwp.method-b "IGNORE">***
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***

40.5.2.2.2 COEI (method B) entry ***<coei-opt-entry>***. The COEI (Method B) entries contain the specific information about the item and component illustration. The element is similar to a "row" in a structural table.

a. The components of ***<coei-opt-entry>*** are:
(1) Item number ***<itemno>*** (required) (see 49.1.4.8). The element is similar to a "cell" in a structural table and is entered in column one.
(2) National Stock Number (NSN) ***<nsn>*** (required) identifies the stock number of the item to be used for requisitioning purposes (see 49.1.4.20). The element is similar to a "cell" in a structural table and is entered in column two.
(3) COEI illustration ***<graphic>*** (required) (see 44.2). The element is similar to a "cell" in a structural table and is entered in column two.
(4) Item name, description, CAGE code, part number, and usable on-code (UOC) ***<dcpno>*** (required) (see 40.5.1.2.1.3.2).
(a) The elements ***<name>***, ***<desc>***, ***<cageno>***, and ***<partno>*** combined is similar to a "cell" in a structural table and is entered in column three.
1 Name ***<name>*** (required) (see 49.1.4.19).
2 Description ***<desc>*** (optional) (see 49.1.4.17).
3 Part number ***<partno>*** (required – one or more) (see 49.1.4.21).
4 CAGE code ***<cageno>*** (required – one or more) (see 49.1.4.2).

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(b) Usable on-code (UOC) ***<uoc>*** (required – one or more) (see 38.4.2.1.3.1.3). The element is similar to a "cell" in a structural table and is entered in column four.

(5) Unit of issue ***<ui>*** (required) provides the component size or amount when issued. The element is similar to a "cell" in a structural table and is entered in column four.

(6) Quantity ***<qty>*** (required) (see 49.1.4.9). The element is similar to a "cell" in a structural table and is entered in column five.

b. DTD fragment for ***<coei-entry>***:

```
<!ELEMENT coei-opt-entry (itemno, nsn, graphic, dcpno, ui, qty)>
<!ATTLIST coei-opt-entry
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<coei-opt-entry>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

(3) **comment** – Reason for change information. (optional) (See 49.2.9.)

(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)

(5) **id** – Unique identifier. (optional) (See 49.2.6.)

(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)

(9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.2.2.3 On–board spares (method B) ***<on-board-spares-opt>***. Any on-board spares, when applicable, are listed under a new heading or category name as "ON-BOARD SPARES" and components are entered in the same format as ***<coei-opt-entry>***.

a. The components of ***<on-board-spares-opt>*** are

(1) On-board spares title ***<title>*** (required) (see 49.1.1.3). MIL–STD–40051–1/–2 states the mandatory title is "ON-BOARD SPARES". The element is similar to a "row" in a structural table. The element is similar to a "cell" in a structural table and is entered in column one.

(2) COEI (Method B) entry(s) ***<coei-opt-entry>*** (required – one or more) (see 40.5.1.2.1.3).

b. DTD fragment for ***<on-board-spares-opt>***:

```
<!ELEMENT on-board-spares-opt (title, coei-opt-entry+)>
```

c. No attributes for ***<on-board-spares-opt>***.

40.5.2.3 Basic of mssue (BII) (method B) ***<bii-opt>***. BII (Method B) contains a component of end item that lists and illustrates all BII items for inventory purposes.

a. The components of ***<bii-opt>*** are

(1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).

(2) BII (Method B) standard table header ***<std.table.header>*** (required) (see 40.5.2.2.1).

(3) BII (Method B) component information ***<bii-opt-entry>*** (required – one or more) (see 40.5.2.3.1).

b. DTD graphically depicted for ***<bii>***:

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Check the source to verify that this is the current version before use.

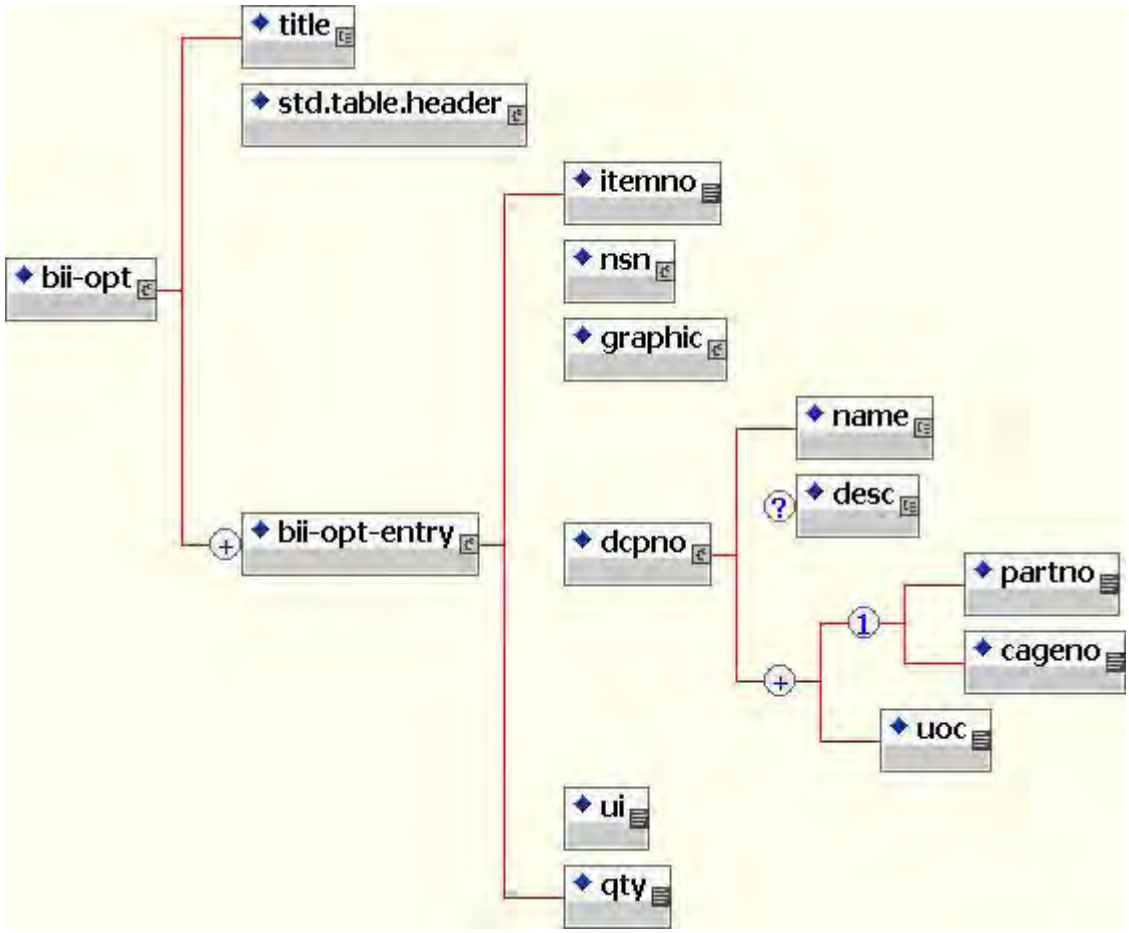

FIGURE 468. BII (method B) DTD hierarchy <bii-opt>.

c. DTD fragment for ***<bii-opt>***:

```
<!ELEMENT bii-opt (title, std.table.header, bii-opt-entry+)>
<!ATTLIST bii-opt
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes for ***<bii-opt>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

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(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.5.2.3.1 BII (method B) entry ***<bii-opt-entry>***. The BII (Method B) entries contain the specific information about the item and component illustration. The element is similar to a "row" in a structural table.

a. The components of ***<bii-opt-entry>*** are:
(1) Item number ***<itemno>*** (required) (see 49.1.4.8). The element is similar to a "cell" in a structural table and is entered in column one.
(2) National Stock Number (NSN) ***<nsn>*** (required) identifies the stock number of the item to be used for requisitioning purposes (see 49.1.4.20). The element is similar to a "cell" in a structural table and is entered in column two.
(3) BII illustration ***<graphic>*** (required) (see 44.2). The element is similar to a "cell" in a structural table and is entered in column two.
(4) Item name, description, CAGE code, part number, and usable on-code (UOC) ***<dcpno>*** (required) (see 40.5.1.2.1.3.2).
(a) The elements ***<name>***, ***<desc>***, ***<cageno>***, and ***<partno>*** combined is similar to a "cell" in a structural table and is entered in column three.
1 Name ***<name>*** (required) (see 49.1.4.19).
2 Description ***<desc>*** (optional) (see 49.1.4.17).
3 Part number ***<partno>*** (required – one or more) (see 49.1.4.21).
4 CAGE code ***<cageno>*** (required – one or more) (see 49.1.4.2).
(b) Usable on-code (UOC) ***<uoc>*** (required – one or more) (see 38.4.2.1.3.1.3). The element is similar to a "cell" in a structural table and is entered in column four.
(5) Unit of issue ***<ui>*** (required) provides the component size or amount when issued. The element is similar to a "cell" in a structural table and is entered in column five.
(6) Quantity ***<qty>*** (required) (see 49.1.4.9). The element is similar to a "cell" in a structural table and is entered in column six.

b. DTD fragment for ***<bii-entry>***:

```
<!ELEMENT bii-opt-entry (itemno, nsn, graphic, dcpno, ui, qty)>
<!ATTLIST bii-opt-entry
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<bii-opt-entry>*** are:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

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40.5.2.4 XML document instance fragment and output for COEI/BII work package (method B) standard information ***<coeibiiwp>***. The COEI/BII work package (Method B) XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % coeibiiwp.method-a "IGNORE">
<!ENTITY % coeibiiwp.method-b "INCLUDE">
<!ENTITY % uoc-list "INCLUDE">
<!ENTITY short.end.item.name "M198 howitzer">
<!ENTITY intro.uoc-list ' %uoc-list;
<deflist>
<title.term.def>
<title>Code
</title>
<title>Used on
</title>
</title.term.def>
<term.def id="uoc.PAA">
<term>PAA
</term>
<def>
<para hcp="no" esd="no">Model XXX
</para>
</def>
</term.def>
<term.def id="uoc.PAB">
<term>PAB
</term>
<def>
<para hcp="no" esd="no">Model XXXX
</para>
</def>
</term.def>
<term.def id="uoc.PAC">
<term>PAC
</term>
<def>
<para hcp="no" esd="no">Model XXXXX
</para>
</def>
</term.def>
</deflist>
'>
```

b. Example of an XML document instance fragment for COEI/BII work package (Method B) ***<coeibiiwp>***:

```
<coeibiiwp wpno="s00003-X-XXXX-XXX" tocentry="2" frame="yes" army="no"
airforce="no" navy="no" marines="no" wpseq="0441" deletewp="no">
<wpidinfo>
<maintlvl level="unitlvl"/>
<title>M198 HOWITZER
</title>
</wpidinfo> &aal.intro;

<!– START COEI Standard Information –>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<coei-opt>
<title>Components of End Item List.
</title> &coeibii.std.table.header;
<coei-entry-opt>
<itemno>1
</itemno>
<nsn>
<fsc>1005
</fsc>
<niin>00-706-8880
</niin>
</nsn>
<graphic boardno="coei1" unitmeasure="in">
<graphic>
<dcpno>
<name>MOUNT, MACHINE GUN 1:cal. .50
</name>
<desc>(in mount on cupola)
</desc>
<partno>7068880
</partno>
<cageno>19204
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</coei-entry-opt>
<coei-entry-opt>
<itemno>2
</itemno>
<nsn>
<fsc>1240
</fsc>
<niin>00-344-4643
</niin>
</nsn>
<graphic boardno="coei2-3" unitmeasure="in">
<graphic>
<dcpno>
<name>PERISCOPE:M27 (chief of section)
</name>
<desc>(stowage box cab wall)
</desc>
<partno>7633132
</partno>
<cageno>19200
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>

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<qty>1
</qty>
</coei-entry-opt>
<coei-entry-opt>
<itemno>3
</itemno>
<nsn>
<fsc>1240
</fsc>
<niin>00-509-2743
</niin>
</nsn>
<graphic boardno="coei2-3" unitmeasure="in">
<graphic>
<dcpno>
<name>PERISCOPE:M45 (driver's)
</name>
<desc>(stowage box driver's compartment)
</desc>
<partno>8213430
</partno>
<cageno>19200
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>3
</qty>
</coei-entry-opt>
<coei-entry-opt>
<itemno>4
</itemno>
<nsn>
<fsc>1240
</fsc>
<niin>00-864-2930
</niin>
</nsn>
<graphic boardno="coei4" unitmeasure="in">
<graphic>
<dcpno>
<name>TELESCOPE, PANORAMIC M117
</name>
<desc>(in mount M145 or telescope box)
</desc>
<partno>7660400
</partno>
<cageno>19200
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>

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```
<qty>1
</qty>
</coei-entry-opt>
<coei-entry-opt>
<itemno>5
</itemno>
<nsn>
<fsc>1240
</fsc>
<niin>00-491-9676
</niin>
</nsn>
<graphic boardno="coei5" unitmeasure="in">
<graphic>
<dcpno>
<name>TELESCOPE, ELBOW: M118CA1
</name>
<desc>(in mount M146)
</desc>
<partno>10559855
</partno>
<cageno>19200
</cageno>
<uoc>PAB
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</coei-entry-opt>
</coeitab>
</coei>
<!– START BII Standard Information –>
<bii-opt>
<title> Basic Issue Items (BII)
</title> &coeibii.std.table.header;
<bii-entry-opt>
<itemno>1
</itemno>
<nsn>
<fsc>1290
</fsc>
<niin>00-535-7629
</niin>
</nsn>
<graphic boardno="bii-1" unitmeasure="in">
</graphic>
<dcpno>
<name>LIGHT, AIMING POST; M14
</name>
<partno>7197188
</partno>
<cageno>19200
</cageno>
<uoc>PAA
```


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```
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>2
</qty>
</bii-entry-opt>
<bii-entry-opt>
<itemno>2
</itemno>
<nsn>
<fsc>4930
</fsc>
<niin>00-766-3545
</niin>
</nsn>
<graphic boardno="bii-2" unitmeasure="in">
</graphic>
<dcpno>
<name>LUBRICATING GUN, HAND: high pressure (in tool bag)
</name>
<partno>102758
</partno>
<cageno>36251
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</bii-entry-opt>
<bii-entry-opt>
<itemno>3
</itemno>
<nsn>
<fsc>8415
</fsc>
<niin>00-266-8843
</niin>
</nsn>
<graphic boardno="bii-3" unitmeasure="in">
</graphic>
<dcpno>
<name>MITTENS, CLOTH: (pair) M1942 (in oddment tray)
</name>
<partno>11655982
</partno>
<cageno>19207
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
```


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Check the source to verify that this is the current version before use.

<qty>2
</qty>
</bii-entry-opt>
<bii-entry-opt>
<itemno>4
</itemno>
<nsn>
<fsc>4930
</fsc>
<niin>00-262-8868
</niin>
</nsn>
<graphic boardno="bii-4" unitmeasure="in">
</graphic>
<dcpno>
<name>OILER, HAND: steel, pump type, 1pt, spout 9 lg
(in leftcab door stowage box)
</name>
<partno>328
</partno>
<cageno>72798
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</bii-entry-opt>
<bii-entry-opt>
<itemno>5
</itemno>
<nsn>
<fsc>7240
</fsc>
<niin>00-160-0455
</niin>
</nsn>
<graphic boardno="bii-5" unitmeasure="in">
</graphic>
<dcpno>
<name>PAIL, UTILITY: 14-qt capability (on vehicle floor)
</name>
<partno>RRP35
</partno>
<cageno>81348
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</bii-entry-opt>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</bii-opt>
</coeibiiwp>

40.6 Additional authorization list (AAL) work package ***<aalwp>***. The AAL work package identifies items that do not have to accompany the end item and do not have to be turned in with the end item. The AAL items are authorized to use by CTA, MTOE, TDA, or JTA. Only used in an Operator's manual.

a. The components of ***<aalwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) AAL introduction ***<intro>*** (required) (see 40.6.1 and 49.1.4.15).
   (4) Additional Authorization List standard information ***<aal>*** (required) (see 40.6.2).

b. DTD graphically depicted for ***<aalwp>***:

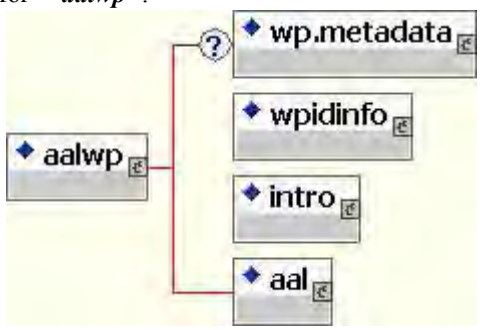

FIGURE 469. Additional authorization list (AAL) work package DTD hierarchy for <aalwp>.

c. DTD fragment for ***<aalwp>***:

```
<!ELEMENT aalwp (wp.metadata?, wpidinfo, intro, aal)>
<!ATTLIST aalwp
   airforce          (yes | no )                "no"
    army             (yes | no )                 "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                 "no"
    security         (uc | fouo | c | s | ts )  #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attribute
   (1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

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(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.6.1 AAL introduction ***<intro>*** standard text. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed Table XXXIII.

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TABLE XXXIII. AAL introduction *<intro>* boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
AAL Introduction	***&aalwp.intro;***	
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***
End Item Name		***<!ENTITY short.end.item.name "Insert Short End Item Name">***
Exclude UOC		***<!ENTITY % uoc-list "IGNORE">***
Include UOC (see 40.5.1.1.1)		***<!ENTITY % uoc-list "INCLUDE">*** ***<!ENTITY intro.uoc-list '<![%uoc-list;[*** ***<deflist>*** ***<title.term.def>*** ***<title>***Code ***</title>*** ***<title>***Used on ***</title>*** ***</title.term.def>*** ***<term.def id="uoc.xxx">*** ***<term>*** *Insert UOC* ***</term>*** ***<def>*** ***<para hcp="no" esd="no">*** *Insert UOC Model Number* ***</para>*** ***</def>*** ***</term.def>*** ***</deflist>]]>'>***

40.6.2 Additional authorization list ***<aal>***. A tabular list containing standard information of all additional authorization items in the additional authorization list. These items are required to operate the equipment but are not classified as COEI or BII items. This element functions as the table element.

a. The components of ***<aal>*** are:
   (1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
   (2) AAL standard table header ***<std.table.header>*** (required) (see 40.6.2.1).
   (3) The AAL is displayed by either
      (a) Grouping components by categories, use the element ***<aal-category>*** (required – one or more) (see 40.6.2.2).
      (b) Each components is listed separately, use the element ***<aal-entry>*** (required – one or more) (see 40.6.2.3).

b. DTD graphically depicted for ***<aal>***:

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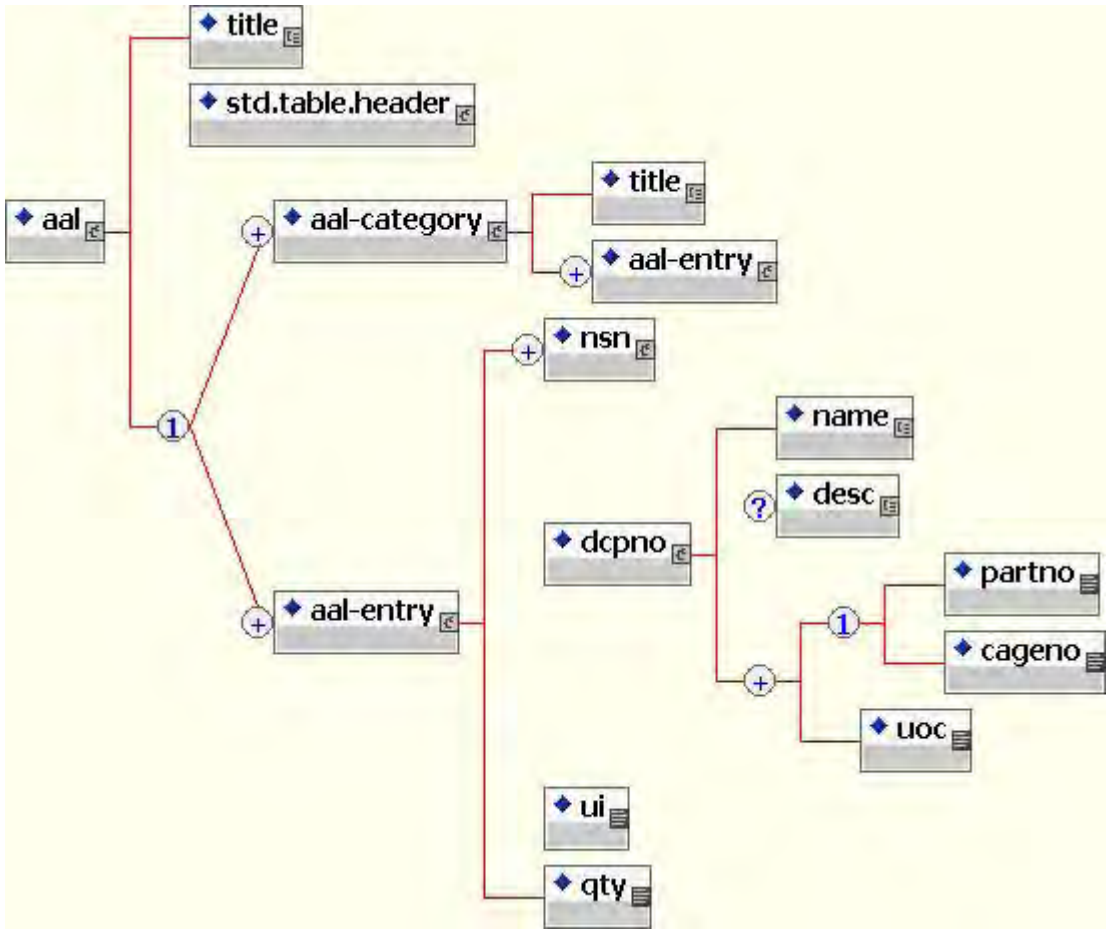

FIGURE 470. Additional authorization list (AAL) standard information DTD hierarchy for <aal>.

c. DTD fragment for ***<aal>***:

```
<!ELEMENT aal (title, std.table.header, (aal-category+ | aal-entry+))>
<!ATTLIST aal
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    frame             (yes | no)                      "no"
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

d. Attributes for ***<aal>***:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **frame** – In frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**no**").

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(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

40.6.2.1 AAL standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XXXIV.

TABLE XXXIV. AAL standard information header ***<std.table.header>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set
AALStandard Information Header	***&aalwp.aal.std-header;***	
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***

40.6.2.2 AAL category ***<aal-category>***. If the standard information is subdivided into groups, for example by subassemblies, the category element is used to represent the groups. After the category name is entered, the specific AAL entries ***<aal-entry>*** are entered for that category. Usually more than one category is entered in the standard information.

a. The components of ***<aal-category>*** are:
   (1) Category title ***<title>*** (required) (see 49.1.1.3).
   (2) AAL entry(s) ***<aal-entry>*** (required – one or more) (see 40.6.2.3) for this category.
b. DTD fragment for ***<aal-category>***:

```
<!ELEMENT aal-category (title, aal-entry+)>
<!ATTLIST aal-category
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    frame             (yes | no)                      "no"
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)         #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<aal-category>*** are:
   (1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

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(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **frame** – In frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**no**").
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

40.6.2.3 AAL entry ***<aal-entry>***. The AAL entries contain the specific information about the item. The element is similar to a "row" in a structural table.

a. The components of ***<aal-entry>*** are
(1) National Stock Number (NSN) ***<nsn>*** (required – one or more) identifies the stock number of the item to be used for requisitioning purposes (see 49.1.4.20). Usually the AAL entry has a single NSN, but on occasions similar fit and function components are listed under the same AAL item. The element is similar to a "cell" in a structural table and is entered in column one.
(2) Item name, description, CAGE code, part number, and usable on-code (UOC) ***<dcpno>*** (required) (see 40.5.1.2.1.3.2).
(a) The elements ***<name>***, ***<desc>***, ***<cageno>***, and ***<partno>*** combined is similar to a "cell" in a structural table and is entered in column two.
1 Name ***<name>*** (required) (see 49.1.4.19).
2 Description ***<desc>*** (optional) (see 49.1.4.17).
3 Part number ***<partno>*** (required – one or more) (see 49.1.4.21).
4 CAGE code ***<cageno>*** (required – one or more) (see 49.1.4.2).
(b) Usable on-code (UOC) ***<uoc>*** (required – one or more) (see 38.4.2.1.3.1.3). The element is similar to a "cell" in a structural table and is entered in column three.
(3) Unit of issue ***<ui>*** (required) provides the component size or amount when issued. The element is similar to a "cell" in a structural table and is entered in column four.
(4) Quantity ***<qty>*** (required) (see 49.1.4.9). The element is similar to a "cell" in a structural table and is entered in column five.

b. DTD fragment for ***<aal-entry>***:

```
<!ELEMENT aal-entry (nsn+, dcpno, ui, qty)>
<!ATTLIST aal-entry
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<aal-entry>*** are:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

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(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.6.3 XML document instance fragment and output for ***<aalwp>***. The AAL XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % uoc-list "IGNORE">
<!ENTITY short.end.item.name "NBCRS FOXM93A1">
```

b. Example of an XML document instance fragment for AAL work package ***<aalwp>***:

```
<aalwp wpno="s00004-X-XXXX-XXX" tocentry="2" frame="yes" army="no" airforce="no"
navy="no" marines="no" wpseq="0442" deletewp="no">
<wpidinfo>
<maintlvl level="unitlvl"/>
<title>NBCRS FOX M93A1 ADDITIONAL AUTHORIZATION LIST (AAL)
</title>
</wpidinfo> &aal.intro;
<aal frame="no">
<title>Additional Authorization List.
</title> &aalwp.aal.std-header;
<aal-entry>
<nsn>
<fsc>6665
</fsc>
<niin>01-105-5623
</niin>
</nsn>
<dcpno>
<name>ALARM, CHEMICAL AGENT
</name>
<partno>8762101
</partno>
<cageno>19200
</cageno>
<uoc>
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
<aal-entry>
<nsn>
<fsc>1240
</fsc>
<niin>01-207-5787
</niin>
</nsn>
<dcpno>
<name>BINOCULARS, MOD, CON M22
</name>
<partno>9370122
```


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</partno>
<cageno> 19200
</cageno>
<uoc>
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
<aal-entry>
<nsn>
<fsc>2590
</fsc>
<niin>01-148-7961
</niin>
</nsn>
<dcpno>
<name>CABLE KIT, SPECIAL PURPOSE
</name>
<partno>223592-2000
</partno>
<cageno>19200
</cageno>
<uoc>
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>2
</qty>
</aal-entry>
<aal-entry>
<nsn>
<fsc>1080
</fsc>
<niin>00-623-7295
</niin>
</nsn>
<dcpno>
<name>CAMOUFLAGE SCREEN WOODLAND/DST POLES
</name>
<partno>11655722
</partno>
<cageno>34623
</cageno>
<uoc>
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
<aal-entry>

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<nsn>
<fsc>1080
</fsc>
<niin>00-103-1246
</niin>
</nsn>
<dcpno>
<name>CAMOUFLAGE SCREEN WOODLAND RAD SCT
</name>
<partno>11655720
</partno>
<cageno>34623
</cageno>
<uoc>
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
<aal-entry>
<nsn>
<fsc>6665
</fsc>
<niin>01-199-4153
</niin>
</nsn>
<dcpno>
<name>CHEMICAL AGENT MONITOR (CAM)
</name>
<partno>11645620
</partno>
<cageno>34623
</cageno>
<uoc>
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
</aal>
</aalwp>

40.7 Expendable and durable items list work package ***<explistwp>***. The expendable and durable items list work package is to provide the user a list of all expendable and durable items called out in the TM text that are necessary to operate and/or maintain the equipment.

a. The components of ***<explistwp>*** are
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Expendable and durable items introduction ***<intro>*** (required) (see 40.7.1 and 49.1.4.15).
   (4) When expendable and durable items exists, the information is entered as standard information ***<explist>***, otherwise a statement is provided stating no expendable and durable item apply.
      (a) Expendable and durable items standard information ***<explist>*** (required) (see 40.7.2).

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(b) Paragraph ***<para>*** stating no expendable and durable item apply (required) (see 49.1.1.5).

b. DTD graphically depicted for ***<explistwp>***:

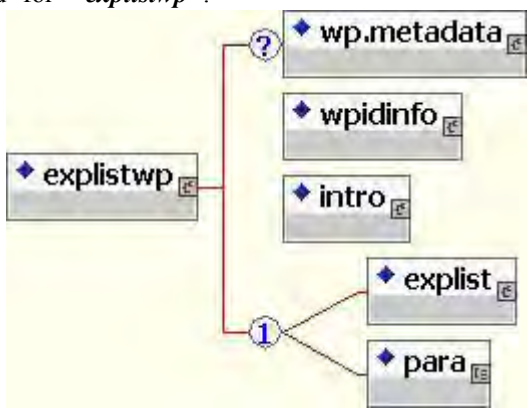

FIGURE 471. Expendable and durable items list work package DTD hierarchy <explistwp>.

c. DTD fragment for ***<explistwp>***:

```
<!ELEMENT explistwp (wp.metadata?, wpidinfo, intro, (explist | para))>
<!ATTLIST explistwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attribute

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

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(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.7.1 Expendable and durable items introduction ***<intro>*** standard text. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed Table XXXV.

TABLE XXXV. Expendable and durable items introduction ***<intro>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Expendable and durable items Introduction	***&explistwp.intro;***	
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***

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Table XXXV Expendable and durable items introduction *<intro>* boilerplate entities – Continued.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
End Item Name		*<!ENTITY short.end.item.name "Insert Short End Item Name">*
Lowest Maintenance Allowed (Remove level that do not apply.)		*<!ENTITY explistwp.intro.level "C=Operator/Crew, O = Unit/AVUM, F = Direct Support/AVIM, H = General Support, D = Depot">*

40.7.2 Expendable and durable items standard information ***<explist>***. The element lists all expendable and durable items called out in the TM text that are necessary to operate and/or maintain the equipment.

a. The components of ***<explist>*** are:
   (1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
   (2) Expendable and durable items standard table header ***<std.table.header>*** (required) (see 40.7.2.1).
   (3) The expendable and durable items is displayed by either
      (a) Grouping components by categories, use the element ***<expdur-category>*** (required – one or more) (see 40.7.2.2).
      (b) Each components is listed separately, use the element ***<expdur-entry>*** (required – one or more) (see 40.7.2.3).

b. DTD graphically depicted for ***<explist>***:

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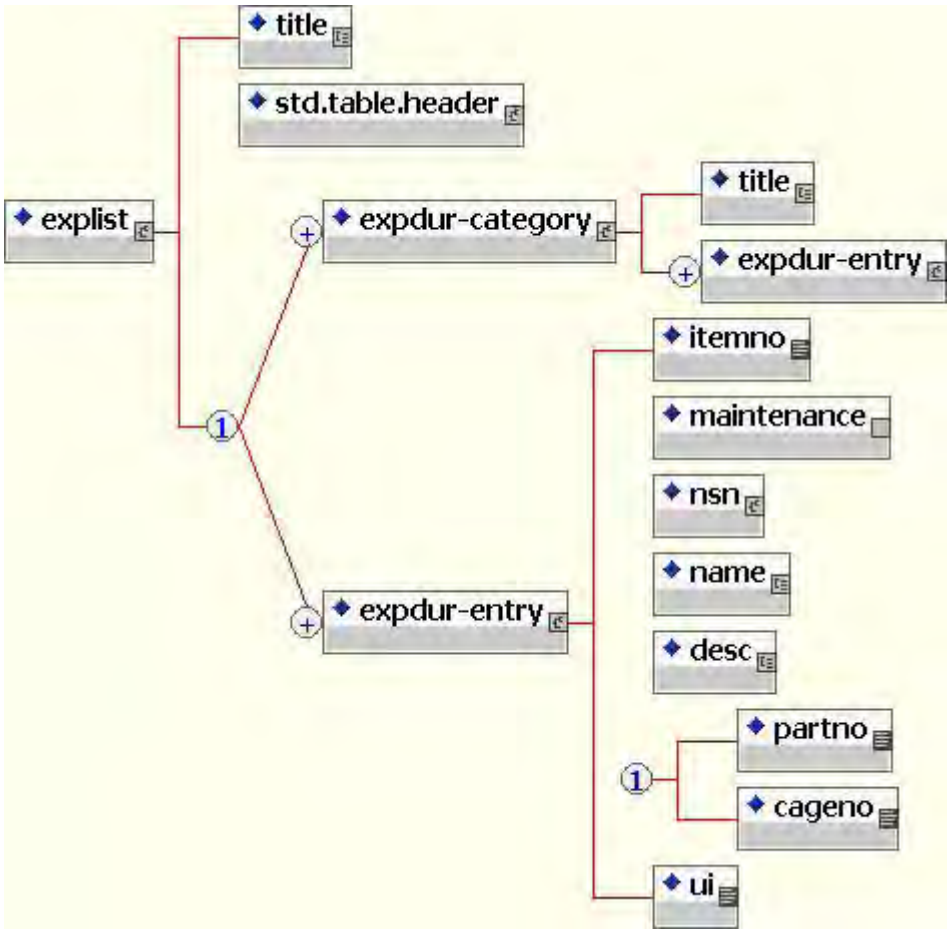

FIGURE 472. Expendable and durable items standard information DTD hierarchy for <explist>.

c. DTD fragment for ***<explist>***:

```
<!ELEMENT explist (title, std.table.header, (expdur-category+ | expdur-entry+))>
<!ATTLIST explist
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<explist>***:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

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(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.7.2.1 Expendable and durable items standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XXXVI.

TABLE XXXVI. Expendable and durable items standard information header ***<std.table.header>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set
Expendable and durable items Standard Information Header	***&explistwp.explist.std-header;***	
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***

40.7.2.2 Expendable and durable items category ***<expdur-category>***. If the standard information is subdivided into groups, for example by subassemblies, the category element is used to represent the groups. After the category name is entered, the specific expendable and durable items entries ***<expdur-entry>*** are entered for that category. Usually more than one category is entered in the standard information.

a. The components of ***<expdur-category>*** are:
   (1) Category title ***<title>*** (required) (see 49.1.1.3).
   (2) Expendable and durable items entry(s) ***<expdur-entry>*** (required – one or more) (see 40.7.2.3) for this category.
b. DTD fragment for ***<expdur-category>***:

```
<!ELEMENT expdur-category (title, expdur-entry+)>
<!ATTLIST expdur-category
    assocfig        IDREFS                          #IMPLIED
    changeref       IDREFS                          #IMPLIED
    comment         CDATA                           #IMPLIED
    delchlvl        CDATA                           #IMPLIED
    id              ID                              #IMPLIED
    idref           IDREFS                          #IMPLIED
    inschlvl        CDATA                           #IMPLIED
    security        (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk        CDATA                           #IMPLIED
>
```

c. Common attributes for ***<expdur-category>*** are:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

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(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.7.2.3 Expendable and durable item entry ***<expdur-entry>***. The expendable and durable item entries contain the specific information about the item. The element is similar to a "row" in a structural table.

a. The components of ***<expdur-entry>*** are
   (1) Item number ***<itemno>*** (required) (see 49.1.4.8). The element is similar to a “cell” in a structural table and is entered in column one.
   (2) Lowest maintenance level ***<maintenance>*** (required) that is allowable to use the listed item (see 40.4.3.2.2). The element is similar to a “cell” in a structural table and is entered in column two.
   (3) National Stock Number (NSN) ***<nsn>*** (required) identifies the stock number of the item to be used for requisitioning purposes (see 49.1.4.20). The element is similar to a “cell” in a structural table and is entered in column three.
   (4) The elements ***<name>***, ***<desc>***, ***<partno>***, and ***<cageno>*** combined is similar to a “cell” in a structural table and is entered in column four.
      (a) Name ***<name>*** (required) (see 49.1.4.19).
      (b) Description ***<desc>*** (required) (see 49.1.4.17).
      (c) Part number ***<partno>*** (required) (see 49.1.4.21).
      (d) CAGE code ***<cageno>*** (required) (see 49.1.4.2).
   (5) Unit of issue ***<ui>*** (required) provides the component size or amount when issued. The element is similar to a “cell” in a structural table and is entered in column five.

b. DTD fragment for ***<expdur-entry>***:

```
<!ELEMENT expdur-entry (itemno, maintenance, nsn, name, desc, (partno, cageno), ui)>
<!ATTLIST expdur-entry
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<expdur-entry>*** are:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

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40.7.3 XML document instance fragment and output for expendable and durable items list work package standard information ***<explistwp>***. The expendable and durable items work package XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY short.end.item.name "NBCRS FOXM93A1">
<!ENTITY explistwp.intro.level "C=Operator/Crew">
```

b. Example of an XML document instance fragment for expendable and durable items list work package ***<explistwp>*** :

```
<explistwp wpno="s00004-X-XXXX-XXX" tocentry="2" frame="yes" army="no"
airforce="no" navy="no" marines="no" wpseq="0059" deletewp="no">
<wpidinfo>
<maintlvl level="unitlvl"/>
<title>Expendable and Durable Items List
</title>
</wpidinfo> &explistwp.intro;
<explist>
<title>Expendable and Durable Items List.
</title> &explistwp.explist.std-header;
<expdur-entry>
<itemno>1
</itemno>
<maintenance lvl="c"/>
<nsn>
<fsc>6810
</fsc>
<niin>00-201-0906
</niin>
</nsn>
<name>Alcohol, denatured, 16 ounce bottle
</name>
<desc>Grade III,
</desc>
<partno>O-E-760
</partno>
<cageno>81348
</cageno>
<ui>BT
</ui>
</expdur-entry>
<expdur-entry>
<itemno>2
</itemno>
<maintenance lvl="c"/>
<nsn>
<fsc>8030
</fsc>
<niin>01-138-1666
</niin>
</nsn>
<name>Antiseize Compound,
</name>
```


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<desc>250-gram tube
</desc>
<partno>MIL-T-5544
</partno>
<cageno>81349
</cageno>
<ui>TU
</ui>
</expdur-entry>
<expdur-entry>
<itemno>3
</itemno>
<maintenance lvl="c"/>
<nsn>
<fsc>6515
</fsc>
<niin>00-059-5235
</niin>
</nsn>
<name>Applicator,
</name>
<desc>disposable,package of 1000
</desc>
<partno>A-A-30016
</partno>
<cageno>58536
</cageno>
<ui>PK
</ui>
</expdur-entry>
<expdur-entry>
<itemno>4
</itemno>
<maintenance lvl="c"/>
<nsn>
<fsc>8020
</fsc>
<niin>00-224-8024
</niin>
</nsn>
<name>Brush, artist,
</name>
<desc>MTL ferrule, round, tapered point, Type I, camel hair
</desc>
<partno>H-B-118
</partno>
<cageno>81348
</cageno>
<ui>EA
</ui>
</expdur-entry>
<expdur-entry>
<itemno>5
</itemno>
<maintenance lvl="c"/>
<nsn>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<fsc>9150
</fsc>
<niin>01-054-6453
</niin>
</nsn>
<name>Cleaner, Lubricant & Preservation (CLP),
</name>
<desc>1-pint bottle with sprayer
</desc>
<partno>MIL-L-63640
</partno>
<cageno>81349
</cageno>
<ui>P
</ui>
</expdur-entry>
</explist>
</explistwp>
```

40.8 Tool identification list work package ***<toolidwp>***. The tool identification list work package is a list of the tools authorized to the levels of maintenance covered in the narrative portion of the TM and referenced by the initial setups.

a. The components of ***<toolidwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Tool identification introduction ***<intro>*** (required) (see 40.8.1 and 49.1.4.15).
   (4) Tool identification list ***<toolidlist>*** (required) (see 40.8.2).

b. DTD graphically depicted for ***<toolidwp>***:

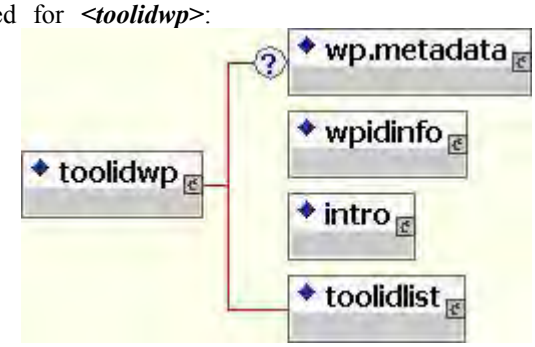

FIGURE 473. Tool identification list work package e DTD hierarchy <toolidwp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<toolidwp>***:

```
<!ELEMENT toolidwp (wp.metadata?, wpidinfo, intro, toolidlist)>
<!ATTLIST toolidwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attribute:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.8.1 Tool identification list introduction ***<intro>*** standard text. MIL–STD–40051–1/–2 contains standard statements that are required for the element and appears in the same wording. MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard statements. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed Table XXXVII.

TABLE XXXVII. Tool identification list introduction ***<intro>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set Editable Entity to Set
Tool identification list Introduction	***&toolidwp.intro;***	
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***
Depot (DMWR/NMWR) Manual		***<!ENTITY % toolidwp.dmwr-nmwr "INCLUDE">*** ***<!ENTITY % toolidwp.common "IGNORE">***
Non-Depot (DMWR/NMWR) Manual		***<!ENTITY % toolidwp.dmwr-nmwr "IGNORE">*** ***<!ENTITY % toolidwp.common "INCLUDE">***
End Item Name		***<!ENTITY short.end.item.name "Insert Short End Item Name">***
Lowest Maintenance Allowed (Remove level that do not apply.)		***<!ENTITY explistwp.intro.level "C=Operator/Crew, O = Unit/AVUM, F = Direct Support/AVIM, H = General Support, D = Depot">***

40.8.2 Tool identification list standard information ***<toolidlist>***. A standard tool identification list includes a tabular listing of all tools required by the initial setup requirements of any procedure in the technical manual.
a. The components of ***<toolidlist>*** are:

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(1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
(2) Tool identification list standard table header ***<std.table.header>*** (required) (see 40.8.2.1).
(3) The tool identification list is displayed by either
  (a) Grouping components by categories, use the element ***<tool-category>*** (required – one or more) (see 40.8.2.2).
  (b) Each components is listed separately, use the element ***<tool-entry>*** (required – one or more) (see 40.8.2.3).

b. DTD graphically depicted for ***<toolidlist>***:

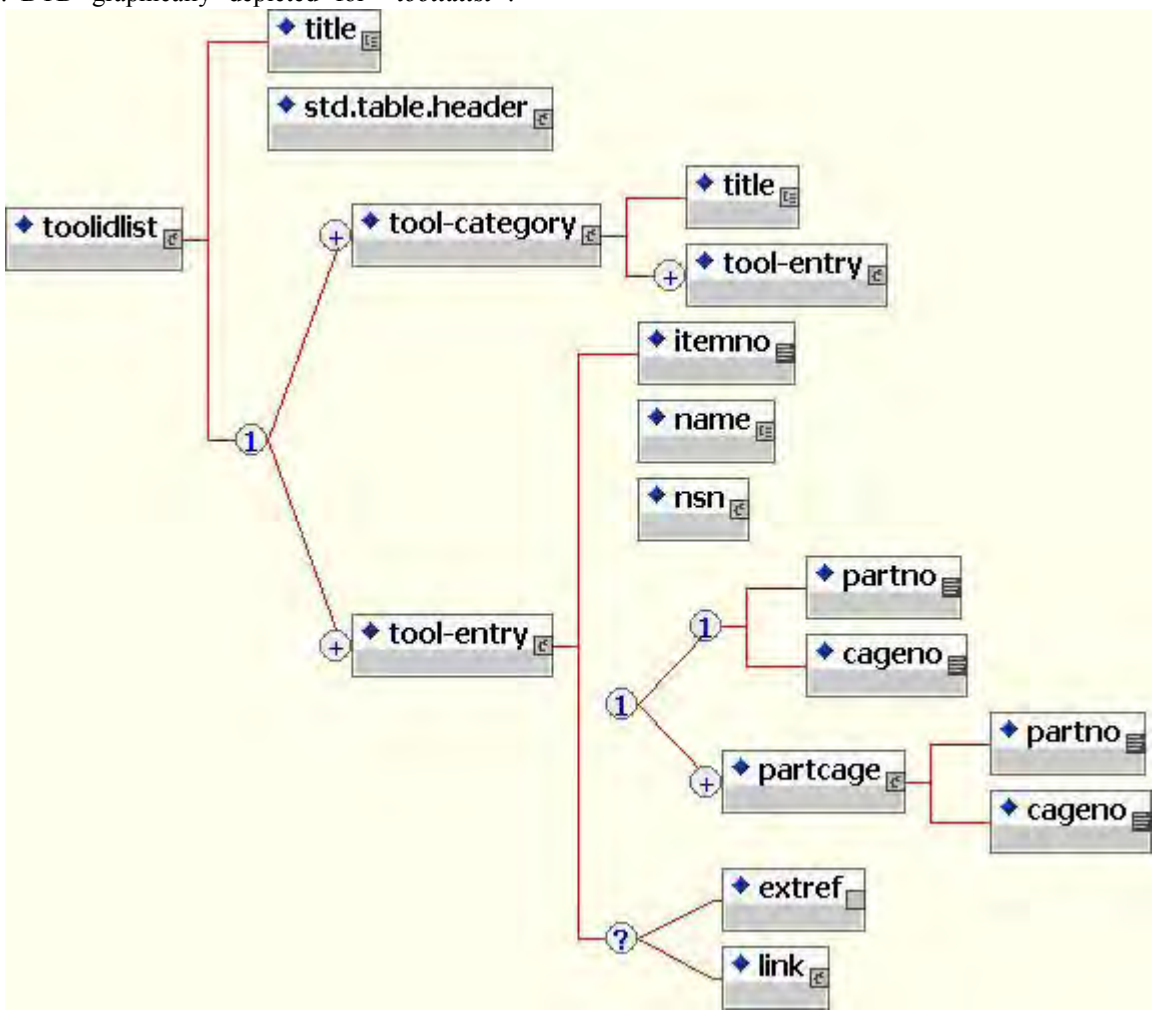

FIGURE 474. Expendable and durable items standard information DTD hierarchy for <toolidlist>.

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c. DTD fragment for ***<toolidlist>***:

```
<!ELEMENT toolidlist (title, std.table.header, (toolid-category+ | toolid-entry+))>
<!ATTLIST toolidlist
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<toolidlist>***:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.8.2.1 Tools identification list standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XXXVIII.

TABLE XXXVIII. Tools identification list standard information header ***<std.table.header>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set
Tools identification listStandard Information Header	***&toolidwp.toolidlist.std-header;***	
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***

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Table XXXVIII Tools identification list standard information header *<std.table.header>* boilerplate entities – Continued.

Description	Boilerplate Entity	Selectable Entity to Set
Depot (DMWR/NMWR) Manual		*<!ENTITY % toolidwp.dmwr-nmwr "INCLUDE">* *<!ENTITY % toolidwp.common "IGNORE">*
Non-Depot (DMWR/NMWR) Manual		*<!ENTITY % toolidwp.dmwr-nmwr "IGNORE">* *<!ENTITY % toolidwp.common "INCLUDE">*

40.8.2.2 Tool identification list category ***<tool-category>***. If the standard information is subdivided into groups, for example by subassemblies, the category element is used to represent the groups. After the category name is entered, the specific tool identification entries ***<tool-entry>*** are entered for that category. Usually more than one category is entered in the standard information.

a. The components are:
   (1) Category title ***<title>*** (required) (see 49.1.1.3).
   (2) Tool identification entry(s) ***<tool-entry>*** (required – one or more) (see 40.8.2.3) for this category.

b. DTD fragment for ***<tool-category>***:

```
<!ELEMENT tool-category (title, tool-entry+)>
<!ATTLIST tool-category
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<tool-category>*** are:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

40.8.2.3 Tool identification list entry ***<tool-entry>***. The tool identification list entries contain the specific information about the item. The element is similar to a "row" in a structural table.

a. The components of ***<tool-entry>*** are
   (1) Item number ***<itemno>*** (required) (see 49.1.4.8). The element is similar to a “cell” in a structural table and is entered in column one.
   (2) Tool name ***<name>*** (required) (see 49.1.4.19). The element is similar to a “cell” in a structural table and is entered in column two.

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(3) National Stock Number (NSN) ***<nsn>*** (required) identifies the stock number of the item to be used for requisitioning purposes (see 49.1.4.20). The element is similar to a "cell" in a structural table and is entered in column three.

(4) Part number and CAGE code is either single ***<partno>*** and ***<cageno>*** or multiple (when the tool has similar form and function) ***<partcage>***. The part number and CAGE code are combined into a "cell" in a structural table and is entered in column four.

(a) Part number ***<partno>*** (required) (see 49.1.4.21).

(b) CAGE code ***<cageno>*** (required) (see 49.1.4.2).

(c) Part number and CAGE code grouping ***<partcage>*** (required).

(5) External link (optional) to detailed information about the tool using either enhanced link ***<link>*** (see 46.2.3) or external reference ***<extref>*** (see 46.2.1). The element is similar to a "cell" in a structural table and is entered in column five.

b. DTD fragment for ***<tool-entry>***:

```
<!ELEMENT tool-entry (itemno, name, nsn, ((partno, cageno) | partcage+),
   (link | extref)?)>
<!ATTLIST tool-entry
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<tool-entry>*** are:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

(3) **comment** – Reason for change information. (optional) (See 49.2.9.)

(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)

(5) **id** – Unique identifier. (optional) (See 49.2.6.)

(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)

(9) **security** – Security classification. (optional) (See 49.2.13.)

40.8.3 XML document instance fragment and output for ***<toolidwp>***. The tool identification list work package XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % toolidwp.dmwr-nmwr "IGNORE">
<!ENTITY % toolidwp.common "INCLUDE">
<!ENTITY short.end.item.name "MICLIC">
```

b. Example of an XML document instance fragment for a tool identification list work package ***<toolidwp>*** :

```
<toolidwp wpno="s00006-X-XXXX-XXX" tocentry="2" frame="yes" army="no"
airforce="no" navy="no" marines="no" wpseq="0444" deletewp="no">
<wpidinfo>
<maintlvl level="unitlvl"/>
<title>MICLIC
```


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```
<brk/>TOOL  IDENTIFICATION  LIST
</title>
</wpidinfo>  &toolidwp.intro;
<toolidlist>
<title>MICLIC  Tool  Identification  List
</title>  &toolidwp.toolidlist.std-header;
<tool-entry>
<itemno>1
</itemno>
<name>Adapter,  socket  wrench,  1/2  inch-3/4  inch
</name>
<nsn>
<fsc>5120
</fsc>
<niin>00-114-5207
</niin>
</nsn>
<partno>11655788-3
</partno>
<cageno>81349
</cageno>
<extref  docno="TM  9-2350-252-20P-1"/>
</tool-entry>
<tool-entry>
<itemno>2
</itemno>
<name>Adapter,  socket  wrench,  3/8  inch-1/2  inch
</name>
<nsn>
<fsc>5120
</fsc>
<niin>00-240-8703
</niin>
</nsn>
<partno>EX503B
</partno>
<cageno>81349
</cageno>
<extref  docno="TM  9-2350-252-20P-1"/>
</tool-entry>
<tool-entry>
<itemno>3
</itemno>
<name>Adapter,  test
</name>
<nsn>
<fsc>4910
</fsc>
<niin>01-138-9334
</niin>
</nsn>
<partno>11629693-1
</partno>
<cageno>81349
</cageno>
<extref  docno="TM  9-1250-252-20P-1"/>
```


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</tool-entry>
<tool-entry>
<itemno>4
</itemno>
<name>Adapter, test
</name>
<nsn>
<fsc>4910
</fsc>
<niin>01-138-9335
</niin>
</nsn>
<partno>11629693-2
</partno>
<cageno>81349
</cageno>
<extref docno="TM 9-2350-252-20P-1"/>
</tool-entry>
<tool-entry>
<itemno>5
</itemno>
<name>Adapter, torque wrench, 1/2 inch drive,1/2 inch
</name>
<nsn>
<fsc>5120
</fsc>
<niin>00-399-1157
</niin>
</nsn>
<partno>2588756
</partno>
<cageno>81349
</cageno>
<extref docno="TM 9-2350-252-20P-1"/>
</tool-entry>
<tool-entry>
<itemno>6
</itemno>
<name>Adapter, torque wrench, 1/2 inch drive,3/4 inch
</name>
<nsn>
<fsc>5120
</fsc>
<niin>00-399-1154
</niin>
</nsn>
<partno>2588757
</partno>
<cageno>81349
</cageno>
<extref docno="TM 9-2350-252-20P-1"/>
</tool-entry>
<tool-entry>
<itemno>7
</itemno>
<name>Adapter, torque wrench, 1/2 inch drive,5/16 inch

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</name>
<nsn>
<fsc>5120
</fsc>
<niin>01-115-1891
</niin>
</nsn>
<partno>12298105-1
</partno>
<cageno>81349
</cageno>
<extref docno="TM 9-2350-252-20P-1"/>
</tool-entry>
<tool-entry>
<itemno>8
</itemno>
<name>Adapter, torque wrench, 1/2 inch drive, 15/16 inch
</name>
<nsn>
<fsc>5120
</fsc>
<niin>00-215-8200
</niin>
</nsn>
<partno>11663358-2
</partno>
<cageno>81349
</cageno>
<extref docno="TM 9-2350-252-20P-1"/>
</tool-entry>
<tool-entry>
<itemno>9
</itemno>
<name>Adjusting tool, belt
</name>
<nsn>
<fsc>4910
</fsc>
<niin>01-128-2670
</niin>
</nsn>
<partno>3375058
</partno>
<cageno>81349
</cageno>
<extref docno="TM 9-2350-252-20P-1"/>
</tool-entry>
<tool-entry>
<itemno>10
</itemno>
<name>Bit, screwdriver, 1/4 inch drive
</name>
<nsn>
<fsc>5120
</fsc>
<niin>00-316-9228

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</niin>
</nsn>
<partno>TMC105A
</partno>
<cageno>81349
</cageno>
<extref docno="TM 9-2350-252-20P-1"/>
</tool-entry>
</toolidlist>
</toolidwp>

40.9 Mandatory replacement parts list (MRPL) work package ***<mrplwp>***. MPRL work package contains the mandatory replacement parts referenced in initial setups and procedural text.

a. The components of ***<mrplwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Introduction ***<intro>*** (required) (see 49.1.4.15).
   (4) Mandatory replacement parts list ***<mrpl>*** (required) (see 40.9.1).

b. DTD graphically depicted for ***<mrplwp>***:

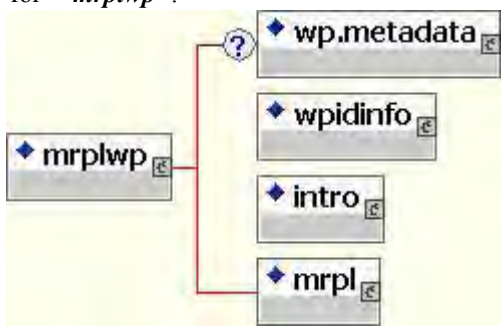

FIGURE 475. Mandatory replacement parts list work package DTD hierarchy <mrplwp>.

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c. DTD fragment for ***<mrplwp>***:

```
<!ELEMENT mrplwp (wp.metadata?, wpidinfo, intro, mrpl)>
<!ATTLIST mrplwp
   airforce          (yes | no )                 "no"
    army             (yes | no )                  "no"
    assocfig         IDREFS                        #IMPLIED
    changeref        IDREFS                        #IMPLIED
    comment          CDATA                         #IMPLIED
    crewmember       CDATA                         #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                         #IMPLIED
    deletewp         (yes | no )                  "no"
    fgc              CDATA                         #IMPLIED
    frame            (yes | no )                  "yes"
    idref            IDREFS                        #IMPLIED
    inschlvl         CDATA                         #IMPLIED
    insertwp         CDATA                         #IMPLIED
    lsa-id           CDATA                         #IMPLIED
    marines          (yes | no )                  "no"
    navy             (yes | no )                  "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                         #IMPLIED
    tocentry         (2 | 3 | 4 | 5)              "2"
    wpno             ID                            #REQUIRED
    wpseq            CDATA                         #IMPLIED
>
```

d. Common attribute

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).

(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).

(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(5) **comment** – Change information (optional) (see 49.2.9).

(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).

(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).

(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).

(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)

(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").

(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(13) **inschlvl** – Insert change level (optional) (see 49.2.9).

(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).

(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.9.1 Mandatory replacement parts list (MRPL) ***<mrpl>***. The element lists the mandatory replacement parts referenced in the work package initial setup.

a. The components of ***<mrpl>*** are:
(1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
(2) MRPL standard table header ***<std.table.header>*** (required) (see 40.9.1.1).
(3) The MRPL is displayed by either
(a) Grouping components by categories, use the element ***<mrpl-category>*** (required – one or more) (see 40.9.1.2).
(b) Each components is listed separately, use the element ***<mrpl-entry>*** (required – one or more) (see 40.9.1.3).

b. DTD graphically depicted for ***<mrpl>***:

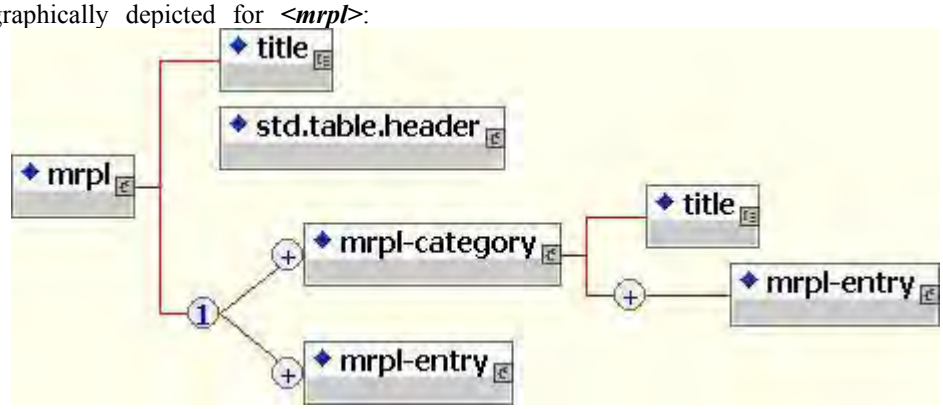

FIGURE 476. Mandatory replacement parts list (MRPL) standard information DTD hierarchy for <mrpl>.

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c. DTD fragment for ***<mrpl>***:

```
<!ELEMENT mrpl (title , std.table.header, (mrpl-category+ | mrpl-entry+))>
<!ATTLIST mrpl
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )       "1"
>
```

d. Attributes for ***<mrpl>***:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

40.9.1.1 MRPL standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XXXIX.

TABLE XXXIX. MRPL standard information header ***<std.table.header>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set
MRPLStandard Information Header	***&mrplwp.mrpllist.std-header;***	
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***

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40.9.1.2 MRPL category ***<mrpl-category>***. If the standard information is subdivided into groups, for example by subassemblies, the category element is used to represent the groups. After the category name is entered, the specific MRPL entries ***<mrpl-entry>*** are entered for that category. Usually more than one category is entered in the standard information.

a. The components of ***<mrpl-category>*** are:
   (1) Category title ***<title>*** (required) (see 49.1.1.3).
   (2) MRPL entry(s) ***<mrpl-entry>*** (required – one or more) (see 40.9.1.3) for this category.
b. DTD fragment for ***<mrpl-category>***:
c. No attributes for ***<mrpl-category>***.
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

40.9.1.3 MRPL entry ***<mrpl-entry>***. The MRPL entries contain the specific information about the item. The element is similar to a "row" in a structural table.

a. The components of ***<mrpl-entry>*** are
   (1) Item number ***<itemno>*** (required) (see 49.1.4.8). The element is similar to a “cell” in a structural table and is entered in column one.
   (2) The part number and CAGE code are combined into a “cell” in a structural table and is entered in column two.
      (a) Part number ***<partno>*** (required) (see 49.1.4.21).
      (b) CAGE code ***<cageno>*** (required) (see 49.1.4.2).
   (3) National Stock Number (NSN) ***<nsn>*** (required) identifies the stock number of the item to be used for requisitioning purposes (see 49.1.4.20). The element is similar to a “cell” in a structural table and is entered in column three.
   (4) Part name ***<name>*** (required) (see 49.1.4.19). The element is similar to a “cell” in a structural table and is entered in column four.
   (5) Quantity ***<qty>*** (required) (see 49.1.4.9). The element is similar to a “cell” in a structural table and is entered in column five.
b. DTD fragment for ***<mrpl-entry>***:

```
<!ELEMENT mrpl-entry (itemno, (partno, cageno) , nsn , name , qty)>
<!ATTLIST mrpl-entry
    applicable        IDREFS                          #IMPLIED
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes for ***<mrpl-entry>*** are:
   (1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
   (2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

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(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

40.9.2 XML document instance fragment and output for ***<mrplwp>***. The MRPL work package XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
```

b. Example of an XML document instance fragment for ***<mrplwp>*** :

```
<mrplwp wpno="s00009-X-XXXX-XXX" tocentry="2" frame="yes" army="no" airforce="no"
navy="no" marines="no" wpseq="0446" deletewp="no">
<wpidinfo>
<maintlvl level="unitlvl"/>
<title>Mandatory Replacement Parts List
</title>
</wpidinfo>
<intro frame="no">
<para0 hcp="no" esd="no">
<title>MANDATORY REPLACEMENT PARTS LIST
</title>
<para hcp="no" esd="no">This work package includes a list of all
mandatory replacement parts referenced in the task initial setups and
procedures.  These are items that must be replaced during maintenance
whether they have failed or not.  This includes items based on
usage intervals such as miles, time, rounds, fired, etc.
</para>
</para0>
</intro>
<mrpl tocentry="1">
<title>Mandatory Replacement Parts List
</title> &mrplwp.mrpllist.std-header;
<mrpl-entry>
<itemno>1
</itemno>
<partno>12286941
</partno>
<cageno>19207
</cageno>
<nsn>
<fsc>2940
</fsc>
<niin>01-086-1605
</niin>
</nsn>
<name>Filter assembly (part of kit, P/N 5705132)
</name>
<qty>1
```


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</qty>
</mrpl-entry>
<mrpl-entry>
<itemno>2
</itemno>
<partno>M83248/1-014
</partno>
<cageno>81349
</cageno>
<nsn>
<fsc>5330
</fsc>
<niin>00-166-0990
</niin>
</nsn>
<name>Preformed packing (item 54 is part of kit, P/N 5705132
</name>
<qty>4
</qty>
</mrpl-entry>
<mrpl-entry>
<itemno>3
</itemno>
<partno>M83248/1-115
</partno>
<cageno>81349
</cageno>
<nsn>
<fsc>5330
</fsc>
<niin>00-166-1066
</niin>
</nsn>
<name>Preformed packing
</name>
<qty>2
</qty>
</mrpl-entry>
<mrpl-entry>
<itemno>4
</itemno>
<partno>M83248/1-904
</partno>
<cageno>81349
</cageno>
<nsn>
<fsc>5330
</fsc>
<niin>00-020-0203
</niin>
</nsn>
<name>Preformed packing
</name>
<qty>5
</qty>
</mrpl-entry>

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Check the source to verify that this is the current version before use.

<mrpl-entry>
<itemno>5
</itemno>
<partno>M83248/1-905
</partno>
<cageno>81349
</cageno>
<nsn>
<fsc>5330
</fsc>
<niin>00-167-5166
</niin>
</nsn>
<name>Preformed packing
</name>
<qty>2
</qty>
</mrpl-entry>
<mrpl-entry>
<itemno>6
</itemno>
<partno>M83248/1-906
</partno>
<cageno>81349
</cageno>
<nsn>
<fsc>5330
</fsc>
<niin>00-020-0186
</niin>
</nsn>
<name>Preformed packing
</name>
<qty>2
</qty>
</mrpl-entry>
<mrpl-entry>
<itemno>7
</itemno>
<partno>M83248/1-908
</partno>
<cageno>81349
</cageno>
<nsn>
<fsc>5330
</fsc>
<niin>00-020-0105
</niin>
</nsn>
<name>Preformed packing
</name>
<qty>3
</qty>
</mrpl-entry>
<mrpl-entry>
<itemno>8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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</itemno>
<partno>M83248/1-910
</partno>
<cageno>81349
</cageno>
<nsn>
<fsc>5330
</fsc>
<niin>00-020-0067
</niin>
</nsn>
<name>Preformed packing
</name>
<qty>3
</qty>
</mrpl-entry>
<mrpl-entry>
<itemno>9
</itemno>
<partno>M83248/1-916
</partno>
<cageno>81349
</cageno>
<nsn>
<fsc>5330
</fsc>
<niin>00-165-4565
</niin>
</nsn>
<name>Preformed packing
</name>
<qty>1
</qty>
</mrpl-entry>
<mrpl-entry>
<itemno>10
</itemno>
<partno>MS35333-39
</partno>
<cageno>96906
</cageno>
<nsn>
<fsc>5310
</fsc>
<niin>00-576-5752
</niin>
</nsn>
<name>Lockwasher
</name>
<qty>4
</qty>
</mrpl-entry>
</mrpl>
</mrplwp>

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Check the source to verify that this is the current version before use.

40.10 Critical safety items (CSI) and flight safety critical aircraft parts (FSCAP) work package ***<csi.fscap.wp>***. The CSI and FSCAP work package contains a listing of all critical safety items and a listing of flight safety critical aircraft parts.

a. The components of ***<csi.fscap.wp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) The work package requires either a CSI, a FSCAP, or both CSI and FSCAP information.
      (a) Critical Safety Items (CSI) ***<csi>*** (see 40.10.1).
      (b) Flight Safety Critical Aircraft Parts (FSCAP) ***<fscap>*** (see 40.10.2)

b. DTD graphically depicted for ***<csi.fscap.wp>***:

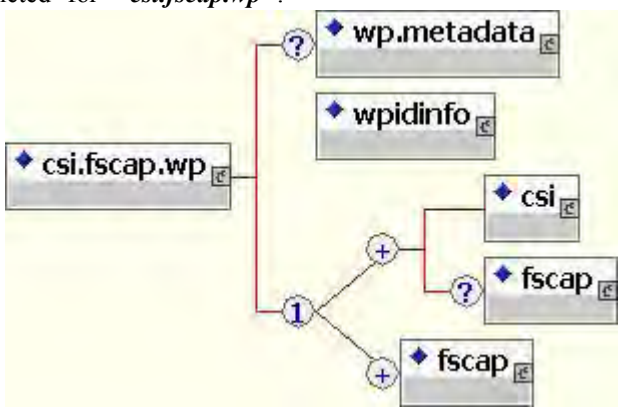

FIGURE 477. CSI and FSCAP work package DTD hierarchy <csi.fscap.wp>.

c. DTD fragment for ***<csi.fscap.wp>***:

```
<!ELEMENT csi.fscap.wp (wp.metadata?, wpidinfo, ((csi, fscap?)+ | fscap+))>
<!ATTLIST csi.fscap.wp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time)  #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
>
```

d. Common attribute

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(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.10.1 Critical safety items (CSI) ***<csi>***. The element identifies the CSI and associated characteristic(s) within overhaul/repair procedures. The location of the critical safety procedures or processes within the depot maintenance work packages is referenced.

a. The components of ***<csi>*** are one of the following:
(1) Uses only a single procedure
(a) Procedural steps title ***<title>*** (required) (see 49.1.1.3).
(b) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(c) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(d) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(e) General information or introduction ***<geninfo>*** (optional) (see 49.1.4.12).
(f) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).
(g) Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1>*** (see 48.2.1) (required – one or more).
(2) Uses multiple procedures

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(a) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(b) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(c) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(d) General information or introduction ***<geninfo>*** (optional) (see 49.1.4.12).
(e) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).
(f) Procedure ***<proc>*** (required – two or more) (see 33.3).

(3) Uses descriptive or narrative text.
(a) Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
(b) Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
(c) Note – ***<note>*** (optional – zero or more) (see 41.1.3).
(d) General information or introduction ***<geninfo>*** (optional) (see 49.1.4.12).
(e) Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).
(f) Narrative titled text first level ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<csi>***:

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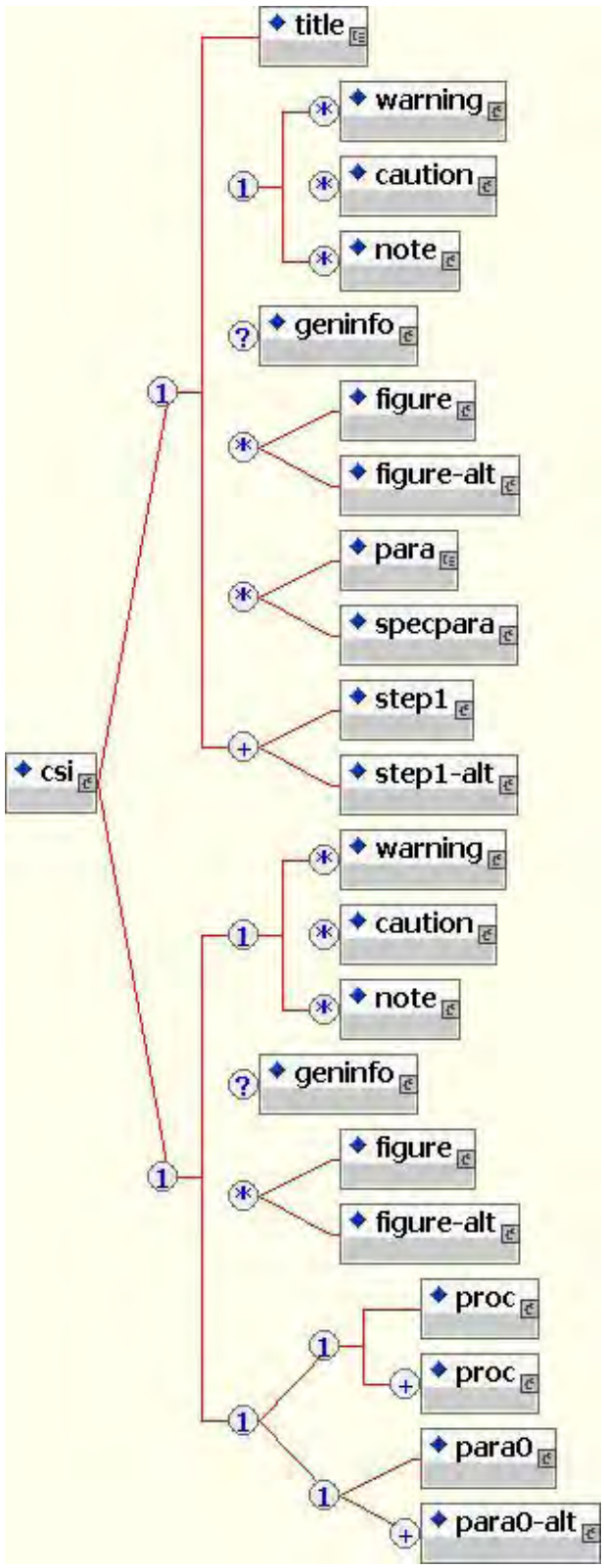

FIGURE 478. Critical safety items (CSI) DTD hierarchy <csi>.

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c. DTD fragment for ***<csi>***:

```
<!ELEMENT csi ((title, (warning*, caution*, note*), geninfo?,
   (figure | figure-alt)*, (para | specpara)*, (step1 | step1-alt)+) |
   ((warning*, caution*, note*), geninfo?, (figure | figure-alt)*,
   ((proc , proc+) | (para0 | para0-alt+))))>
<!ATTLIST csi
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<csi>***:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.10.2 Flight safety critical aircraft parts (FSCAP) ***<fscap>***. This element lists all the flight safety critical aircraft parts and installations identified under the FSCAP program. A critical characteristic is any feature throughout the life cycle of a FSCAP, such as dimension, tolerance, finish, material or assembly, manufacturing process, inspection process, operation, field maintenance requirement, depot overhaul requirement, or other feature that if nonconforming, missing, or degraded, could cause failure or malfunction of a FSCAP.

a. The components of ***<fscap>*** are one of the following:
(1) Select one of the following methods to describe the FSCAP.
(a) Uses only a single procedure
1 Procedural steps title ***<title>*** (required) (see 49.1.1.3).
2 Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
3 Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
4 Note – ***<note>*** (optional – zero or more) (see 41.1.3).
5 General information or introduction ***<geninfo>*** (optional) (see 49.1.4.12).
6 Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).
7 Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1>*** (see 48.2.1) (required – one or more).
(b) Uses multiple procedures
1 Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).
2 Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).
3 Note – ***<note>*** (optional – zero or more) (see 41.1.3).
4 General information or introduction ***<geninfo>*** (optional) (see 49.1.4.12).
5 Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).
6 Procedure ***<proc>*** (required – two or more) (see 33.3).

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(c) Uses descriptive or narrative text.

1 Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).

2 Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).

3 Note – ***<note>*** (optional – zero or more) (see 41.1.3).

4 General information or introduction ***<geninfo>*** (optional) (see 49.1.4.12).

5 Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).

6 Narrative titled text first level ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

(2) FSCAP list ***<fscap.tab>*** (required) (see 40.10.2.1).

b. DTD graphically depicted for ***<fscap>***:

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Check the source to verify that this is the current version before use.

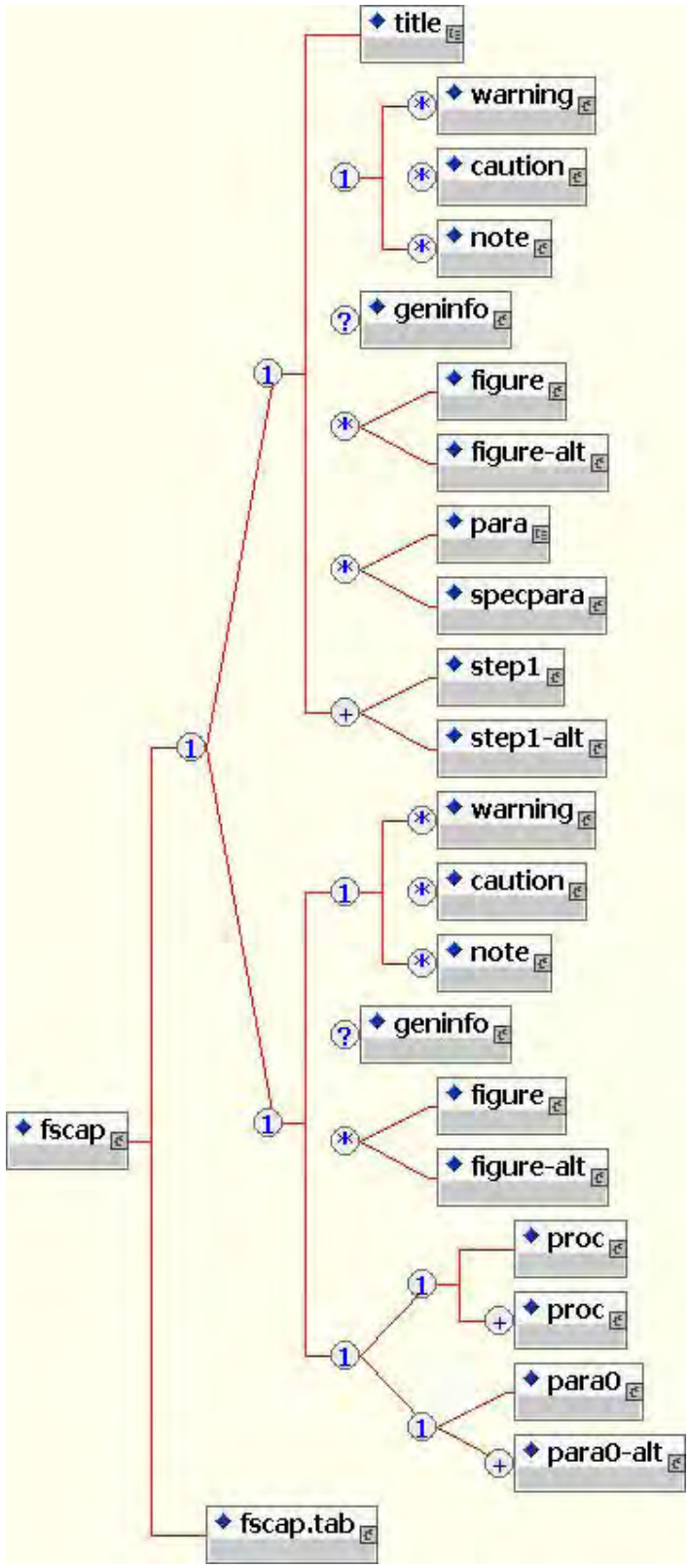

FIGURE 479. Flight safety critical aircraft parts (FSCAP) DTD hierarchy <fscap>.

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c. DTD fragment for ***<fscap>***:

```
<!ELEMENT fscap ((title, (warning*, caution*, note*), geninfo?,
   (figure | figure-alt)*, (para | specpara)*, (step1 | step1-alt)+) |
   ((warning*, caution*, note*), geninfo?, (figure | figure-alt)*,
   ((proc , proc+) | (para0 | para0-alt+))))>
<!ATTLIST fscap
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<fscap>***:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.10.2.1 Flight safety critical aircraft parts table ***<fscap.tab>***. The FSCAP standard information lists the flight safety critical aircraft parts by part number, CAGE Code, nomenclature, and critical characteristic.

a. The components of ***<fscap.tab>*** are:

(1) Standard information title ***<title>*** (required) (see 49.1.1.3). When standard information is represented in tabular format, do not include the table number (this is generated by the stylesheet).
(2) FSCAP standard table header ***<std.table.header>*** (required) (see 40.10.2.1.1).
(3) FSCAP entry(s) ***<fscap-entry>*** (required – one or more) (see 40.10.2.1.2).

b. DTD graphically depicted for ***<fscap.tab>***:

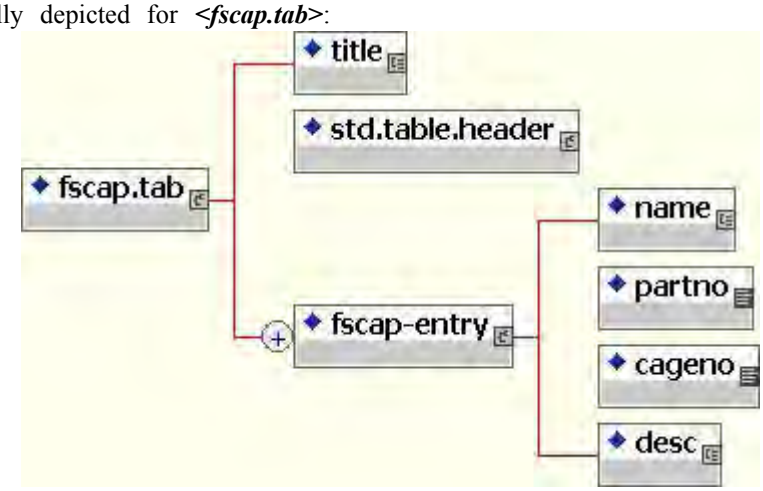

FIGURE 480. FSCAP standard information DTD hierarchy <fscap.tab>.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment for ***<fscap.tab>***:

```
<!ELEMENT fscap.tab (title, std.table.header, fscap-entry+)>
<!ATTLIST fscap.tab
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Attributes for ***<fscap.tab>***:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.10.2.1.1 FSCAP standard information header ***<std.table.header>***. Standard information header provides the author with a means to identify the header information for the various standard information specified throughout MIL–STD–40051–1/–2 (see 42.3.1). MIL–STD–2361 uses boilerplates or XML general entities (see 50) that are a type of replacement text for these standard table headers. Boilerplates are defined in general entities and any variances are defined by selectable (i.e., page-based vs. frame-based) and/or editable (i.e., equipment name) entities contained in the XML DTD. Using boilerplates, authors can reduce text entry time and errors and provide the correct standard statement wording. For information on "how to use" general entity boilerplates, refer to 50.5 and for specific boilerplate usages are listed in Table XL.

TABLE XL. FSCAP standard information header ***<std.table.header>*** boilerplate entities.

Description	Boilerplate Entity	Selectable Entity to Set
FSCAPStandard Information Header	***&csi.fscap.wp.fscaplist.std-header;***	
Frame-based		***<!ENTITY % frame-base "INCLUDE">*** ***<!ENTITY % page-base "IGNORE">***
Page-based		***<!ENTITY % frame-base "IGNORE">*** ***<!ENTITY % page-base "INCLUDE">***

40.10.2.1.2 Flight safety critical aircraft parts entry(s) ***<fscap-entry>***. The FSCAP entries contain the specific information and characteristic about the item. The element is similar to a "row" in a structural table.

a. The components of ***<fscap-entry>*** are: (partno , cageno , name , desc

(1) Item name ***<name>*** (required) (see 49.1.4.19). The element is similar to a "cell" in a structural table and is entered in column one.

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(2) The part number and CAGE code are combined into a "cell" in a structural table and is entered in column two.
(a) Part number ***<partno>*** (required) (see 49.1.4.21).
(b) CAGE code ***<cageno>*** (required) (see 49.1.4.2).
(3) Item characteristic description ***<desc>*** (required) (see 49.1.4.17). The element is similar to a "cell" in a structural table and is entered in column three.

b. DTD fragment for ***<fscap-entry>***:

```
<!ELEMENT fscap-entry (name, partno, cageno, desc)>
<!ATTLIST fscap-entry
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Attributes for ***<fscap-entry>***:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

40.10.2.2 XML document instance fragment and output for ***<fscap.tab>***. The FSCAP standard information XML markup is shown below and the sample style sheet output is shown MIL–STD–40051–1/–2.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
```

b. Example of an XML document instance fragment for ***<fscap.tab>*** :

```
<fscap.tab>
<title>Flight Safety Critical Aircraft Parts (FSCAP)
</title> &csi.fscap.wp.fscaplist.std-header;
<fscap-entry>
<partno>7-211310027-3
</partno>
<cageno>02731
</cageno>
<name>Cluster Gear
</name>
<desc>Process core and surface hardness.
</desc>
</fscap-entry>
<fscap-entry>
<partno>7-113100029-3
</partno>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<cageno>02731
</cageno>
<name>Spur Gear
</name>
<desc>Dimensions and contour of root area.
</desc>
</fscap-entry>
<fscap-entry>
<partno>7-113100121-3
</partno>
<cageno>02731
</cageno>
<name>Nut
</name>
<desc>Process surface hardness.
</desc>
</fscap-entry>
<fscap-entry>
<partno>7-113100141-3
</partno>
<cageno>02731
</cageno>
<name>Spindle
</name>
<desc>Process core hardness.
</desc>
</fscap-entry>
<fscap-entry>
<partno>7-311310016-3
</partno>
<cageno>02731
</cageno>
<name>Carrier Hub
</name>
<desc>Process core and surface hardness.
</desc>
</fscap-entry>
<fscap-entry>
<partno>7-211310035-3
</partno>
<cageno>02731
</cageno>
<name>Gearshaft
</name>
<desc>Process core and surface hardness.
</desc>
</fscap-entry>
<fscap-entry>
<partno>7-211310039-5
</partno>
<cageno>02731
</cageno>
<name>Gearshaft
</name>
<desc>Process core and surface hardness.
</desc>

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</fscap-entry>
<fscap-entry>
<partno>7-311310025-3
</partno>
<cageno>02731
</cageno>
<name>Gear
</name>
<desc>Process core and surface hardness.
</desc>
</fscap-entry>
</fscap.tab>

40.11 Support items work package ***<supitemwp>***. The Support Items Work Package is used when the data contained in the supporting list are minimal and the development of a separate work package is not required. The Support Items Work Package contains the applicable supporting lists.

a. The components of ***<supitemwp>*** are:
   (1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).
   (2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).
   (3) Support items introduction ***<intro>*** (optional) (see 49.1.4.15). A brief introduction may be entered to describe the support items being shown and including entry information for the applicable standard information. The text from the introduction can be copied from the boilerplate entity file (simboil.ent) and used. Note, all selectable entity references (marked sections such as <![%page-base;[...]]>) are removed and include on the words for the type of information being used. Example for a frame-base manual the selectable entity would be

%frame-base;
Entry
%page-base;
Column

   the information in ***%page-base;*** would be removed and the only text remaining is "Entry".
   (4) Components of End Item (COEI) standard information ***<coei>*** (optional) (see 40.5.1.2).
   (5) Basic Issue Items (BII) standard information ***<bii>*** (optional) (see 40.5.1.3).
   (6) Additional Authorization List (AAL) standard information ***<explist>*** (optional) (see 40.6.2).
   (7) Expendable and durable items standard information ***<explist>*** (optional) (see 40.7.2).
   (8) Tool identification standard information ***<toolidlist>*** (optional) (see 40.8.2).
   (9) Mandatory Replacement Parts List (MRPL) standard information ***<mrpl>*** (optional) (see 40.9.1).
   (10) Critical Safety Items (CSI) description ***<csi>*** (optional) (see 40.10.1).
   (11) Flight Safety Critical Aircraft Parts (FSCAP) description ***<fscap>*** (optional) (see 40.10.2).

b. DTD graphically depicted for ***<supitemwp>***:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

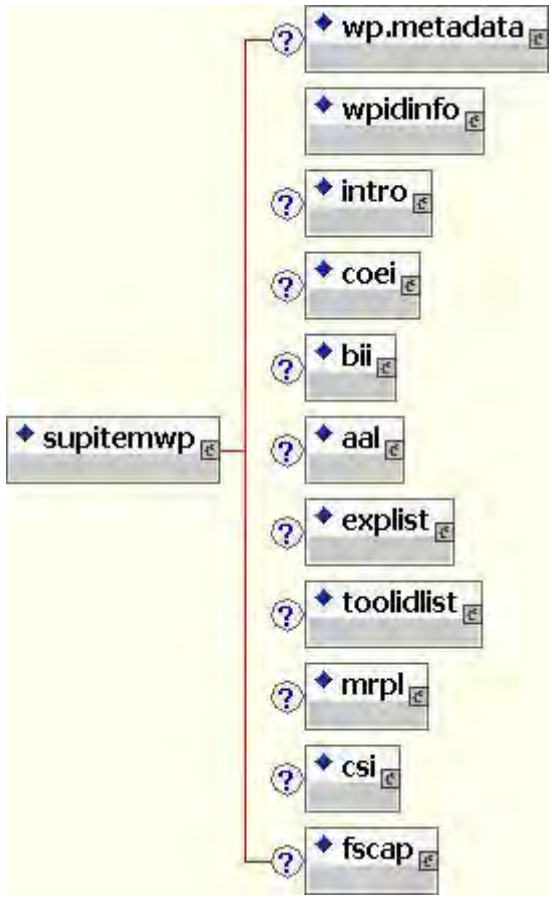

FIGURE 481. Support items work package DTD hierarchy <supitemwp>.

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Check the source to verify that this is the current version before use.

c. DTD fragment for ***<supitemwp>***:

```
<!ELEMENT supitemwp (wp.metadata?, wpidinfo, intro?, coei?, bii?, aal?,
                     explist?, toolidlist?, mrpl?, csi?, fscap?)>
<!ATTLIST supitemwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
```

d. Common attribute

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)

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(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).
(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.11.1 XML document instance fragment and output for ***<supitemwp>***. The XML instance and its stylesheet output for a support items work package ***<supitemwp>*** that consists of a combined introduction ***<intro>***, COEI standard information ***<coei>***, expendable and durable items standard information ***<explist>***, and a MRPL standard information ***<mrpl>***.

a. Example of an XML document instance fragment for selectable and editable entity to generate the correct text in the sample output.

```
<!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE">
<!ENTITY % uoc-list "INCLUDE">
<!ENTITY short.end.item.name "M198 howitzer">
<!ENTITY intro.uoc-list ' %uoc-list;
<deflist>
<title.term.def>
<title>Code
</title>
<title>Used on
</title>
</title.term.def>
<term.def id="uoc.PAA">
<term>PAA
</term>
<def>
<para hcp="no" esd="no">Model XXX
</para>
</def>
</term.def>
<term.def id="uoc.PAB">
<term>PAB
</term>
<def>
<para hcp="no" esd="no">Model XXXX
</para>
</def>
</term.def>
<term.def id="uoc.PAC">
<term>PAC
</term>
<def>
<para hcp="no" esd="no">Model XXXXX
</para>
</def>
</term.def>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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```
</deflist>
'>
```

b. Example of an XML document instance fragment for a Support Items Work Package :

```
<supitemwp wpno="s00010-X-XXXX-XXX" tocentry="2" frame="yes" army="no"
airforce="no" navy="no" marines="no" wpseq="0446" deletewp="no">
<wpidinfo>
<maintlvl level="unitlvl"/>
<title>Support items work package
</title>
</wpidinfo>
<!– Combined introduction into the work package. Uses parts from the
general entities &coeibiiwp.intro; and &expdurwp.intro;.–>
<intro>
<para0 hcp="no" esd="no">
<title>INTRODUCTION
</title>
<subpara1 hcp="no" esd="no">
<title>Scope
</title>
<para hcp="no" esd="no">This work package lists Components of End Item
(COEI), expendable and durable items, and Mandatory Replacement
Parts lists for the &short.end.item.name; to help you inventory
items for safe and efficient operation of the equipment.
</para>
</subpara1>
<subpara1 hcp="no" esd="no">
<title>General
</title>
<para hcp="no" esd="no">The COEI, expendable and durable items, and
MRPL information is divided into the following lists:
</para>
<subpara2 hcp="no" esd="no">
<title>Components of End Item (COEI).
</title>
<para hcp="no" esd="no">This list is for information purposes only and
is not authority to requisition replacements. These items are part
of the &short.end.item.name;. As part of the end item, these items
must be with the end item whenever it is issued or transferred between
property accounts. Items of COEI are removed and separately packaged
for transportation or shipment only when necessary. Illustrations
are furnished to help you find and identify the items.
</para>
</subpara2>
<subpara2 hcp="no" esd="no">
<title>Expendable and durable items.
</title>
<para hcp="no" esd="no">These items are authorized to you by
<extref docno="CTA 50-970" posttext=", Expendable/Durable Items (Except
Medical, Class V Repair Parts, and Heraldic Items)"/>,
<extref docno="CTA 50-909" posttext=", Field and Garrison Furnishings and Equipment"/> or
<extref docno="CTA 8-100" posttext=", Army Medical Department Expendable/Durable Items"/>.
</para>
</subpara2>
<subpara2 hcp="no" esd="no">
<title>Mandatory Replacement Parts List (MRPL).
</title>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<para hcp="no" esd="no">These items are all mandatory replacement parts referenced in the task initial setups and procedures. These are items that must be replaced during maintenance whether they have failed or not. This includes items based on usage intervals such as miles, time, rounds, fired, etc.
</para>
</subpara2>
</subpara1>
<subpara1 hcp="no" esd="no">
<title>Explanation of Columns in the COEI List
</title>
<para hcp="no" esd="no">Column (1) Illus Number. Gives you the number of the item illustrated.
</para>
<para hcp="no" esd="no">Column (2) National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.
</para>
<para hcp="no" esd="no">Column (3) Description, Part Number/(CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The stowage location of COEI and BII is also included in this entry. The last line below the description is the part number and the Commercial and Government Entity Code (CAGEC) (in parentheses).
</para>
<para hcp="no" esd="no">Column (4) Usable on Code. When applicable, gives you a code if the item you need is not the same for different models of equipment. **&intro.uoc;**
</para>
<para hcp="no" esd="no">Column (5) U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number shown in Column (2).
</para>
<para hcp="no" esd="no">Column (6) Qty Rqr. Indicates the quantity required.
</para>
</subpara1>
<subpara1 hcp="no" esd="no">
<title>Explanation of Columns in the Expendable/Durable Items List.
</title>
<para hcp="no" esd="no">Column (1) Item No. This number is assigned to the entry in the list and is referenced in the narrative instructions to identify the item (e.g., Use brake fluid (WP 0098, item 5)).
</para>
<para hcp="no" esd="no">Column (2) Level. This entry identifies the lowest level of maintenance that requires the listed item: **&explistwp.intro.level;**
</para>
<para hcp="no" esd="no">Column (3) National Stock Number (NSN). This is the NSN assigned to the item which you can use to requisition it.
</para>
<para hcp="no" esd="no">Column (4) Item Name, Description, Part Number/(CAGEC). This column provides the other information you need to identify the item. The last line below the description is the part number and the Commercial and Government Entity Code (CAGEC) (in parentheses).
</para>
<para hcp="no" esd="no">Column (5) U/I. Unit of Issue (U/I) code shows the physical measurement or count of an item, such as gallon, dozen, gross, etc.

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```
</para>
</subpara1>
</para0>
</intro>
<!– COEI standard information–>
<coei>
<graphic boardno="coei" unitmeasure="in">
</graphic>
<coeitab>
<title>Components of End Item List.
</title> &coeibiiwp.coeitab-biitab.std-header;
<coei-entry>
<illno>1
</illno>
<nsn>
<fsc>1005
</fsc>
<niin>00-706-8880
</niin>
</nsn>
<dcpno>
<name>MOUNT, MACHINE GUN 1:cal. .50
</name>
<desc>(in mount on cupola)
</desc>
<partno>7068880
</partno>
<cageno>19204
</cageno>
<uoc>PAA
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</coei-entry>
<coei-entry>
<illno>2
</illno>
<nsn>
<fsc>1240
</fsc>
<niin>00-344-4643
</niin>
</nsn>
<dcpno>
<name>PERISCOPE:M27 (chief of section)
</name>
<desc>(stowage box cab wall)
</desc>
<partno>7633132
</partno>
<cageno>19200
</cageno>
<uoc>PAA
```


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```
</uoc>
</dcpno>
<ui>EA
</ui>
<qty>1
</qty>
</coei-entry>
</coeitab>
</coei>
<!– Expendable and durable standard information–>
<explist>
<title>Expendable and Durable Items List.
</title> &explistwp.explist.std-header;
<expdur-entry>
<itemno>1
</itemno>
<maintenance lvl="c"/>
<nsn>
<fsc>6810
</fsc>
<niin>00-201-0906
</niin>
</nsn>
<name>Alcohol, denatured,
</name>
<desc>Grade III, 16 ounce bottle
</desc>
<partno>O-E-760
</partno>
<cageno>81348
</cageno>
<ui>BT
</ui>
</expdur-entry>
<expdur-entry>
<itemno>2
</itemno>
<maintenance lvl="c"/>
<nsn>
<fsc>8030
</fsc>
<niin>01-138-1666
</niin>
</nsn>
<name>Antiseize Compound,
</name>
<desc>250-gram tube
</desc>
<partno>MIL-T-5544
</partno>
<cageno>81349
</cageno>
<ui>TU
</ui>
</expdur-entry>
</explist>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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```
<!– MRPL standard information–>
<mrpl tocentry="1">
<title>Mandatory Replacement Parts List
</title> &mrplwp.mrpllist.std-header;
<mrpl-entry>
<itemno>1
</itemno>
<partno>12286941
</partno>
<cageno>19207
</cageno>
<nsn>
<fsc>2940
</fsc>
<niin>01-086-1605
</niin>
</nsn>
<name>Filter assembly (part of kit, P/N 5705132)
</name>
<qty>1
</qty>
</mrpl-entry>
<mrpl-entry>
<itemno>2
</itemno>
<partno>M83248/1-014
</partno>
<cageno>81349
</cageno>
<nsn>
<fsc>5330
</fsc>
<niin>00-166-0990
</niin>
</nsn>
<name>Preformed packing (item 54 is part of kit, P/N 5705132
</name>
<qty>4
</qty>
</mrpl-entry>
</mrpl>
</supitemwp>
```

c. Example of a page-based TM stylesheet output for Support Items Work Package

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FIELD MAINTENANCE

SUPPORT ITEMS WORK PACKAGE

INTRODUCTION

Scope

This work package lists Components of End Item (COEI), expendable and durable items, and Mandatory Replacement Parts lists for the M198 howitzer to help you inventory items for safe and efficient operation of the equipment.

General

The COEI, expendable and durable items, and MRPL information are divided into the following lists:

Components of End Item

This list is for information purposes only and is not authority to requisition replacements. These items are part of the M198 howitzer. As part of the end item, these items must be with the end item whenever it is issued or transferred between property accounts. Items of COEI are removed and separately packaged for transportation or shipment only when necessary. Illustrations are furnished to help you find and identify the items.

Expendable and Durable items

These items are authorized to you by CTA 50-970, Expendable/Durable Items (Except Medical, Class V Repair Parts, and Heraldic Items), CTA 50-909, Field and Garrison Furnishings and Equipment or CTA 8–100, Army Medical Department Expendable/Durable Items.

Mandatory Replacement Parts List (MRPL)

These items are all mandatory replacement parts referenced in the task initial setups and procedures. These are items that must be replaced during maintenance whether they have failed or not. This includes items based on usage intervals such as miles, time, rounds, fired, etc.

Explanation of Columns in the COEI List

Column (1) Illus Number. Gives you the number of the item illustrated.

Column (2) National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.

Column (3) Description, Part Number/(CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The stowage location of COEI and BII is also included in this entry. The last line below the description is the part number and the Commercial and Government Entity Code (CAGEC) (in parentheses).

Column (4) Usable on Code. When applicable, gives you a code if the item you need is not the same for different models of equipment.

Code	Used on
PAA	Model XXX
PAB	Model XXX
PAC	Model XXXX

Column (5) U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number shown in Column (2).

Column (6) Qty Rqr. Indicates the quantity required.

Explanation of Columns in the Expendable/Durable Items List.

Column (1) Item No. This number is assigned to the entry in the list and is referenced in the narrative instructions to identify the item (e.g., Use brake fluid (WP 0098, item 5)).

Column (2) Level. This entry identifies the lowest level of maintenance that requires the listed item: include as applicable: applicable: C = Operator/Crew, O = Unit/AMC, F = Direct Support/ASB, H = General Support/TASMG, D = Depot.

Column (3) National Stock Number (NSN). This is the NSN assigned to the item which you can use to requisition it.

FIGURE 482. Example of a page-based TM stylesheet output for <supitemwp> page 1.

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Column (4) Item Name, Description, Part Number/(CAGEC). This column provides the other information you need to identify the item. The last line below the description is the part number and the Commercial and Government Entity Code (CAGEC) (in parentheses).

Column (5) U/I. Unit of Issue (U/I) code shows the physical measurement or count of an item, such as gallon, dozen, gross, etc.

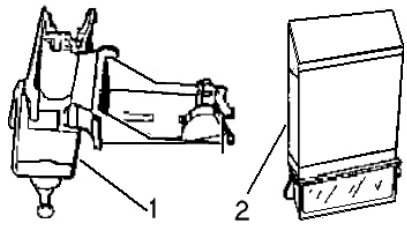

Table 1. Components of End Item List.

(1) Item Number	(2) National Stock Number (NSN)	(3) Description, Part Number/(CAGEC)	(4) Usable On Code	(5) U/I	(6) Qty Rqr.
1	1005-00-706-8880	MOUNT, MACHINE GUN 1:cal. .50 (in mount on cupola) 7068880(19204)	PAA	EA	1
2	1240-00-344-4643	PERISCOPE:M27 (chief of section) (stowage box cab wall) 7633132(19200)	PAA	EA	1

Table 2. Expendable and Durable Items List.

(1) Item No	(2) Level	(3) National Stock Number (NSN)	(4) Item Name, Description, Part Number/(CAGEC)	(5) U/I
1	C	6810-00-201-0906	Alcohol, denatured, Grade III, 16 ounce bottle O-E-760 (81348)	BT
2	C	8030-01-138-1666	Antiseize Compound, 250-gram tube MIL-T-5544 (81349)	TU

Table 3. Mandatory Replacement Parts List

(1) Item No.	(2) Part Number/ (CAGEC)	(3) National Stock Number (NSN)	(4) Nomenclature	(5) Qty
1	12286941 (19207)	2940-01-086-1605	Filter assembly (part of kit, P/N 5705132)	1
2	M83248/1-014 (81349)	5330-00-166-0990	Preformed packing (item 54 is part of kit, P/N 5705132	4

END OF WORK PACKAGE

FIGURE 483. Example of a page-based TM stylesheet output for <supitemwp> page 2.

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40.12 Additional generic work package(s) ***<genwp>***. The additional generic work package is used when a manual contains a work package that does not fit any of the content-specific work packages. There may be more than one additional generic work package contained in the supporting information chapter, and all would occur at the end of the information chapter.

a. The components of ***<genwp>*** are:

(1) Work package metadata (information about the work package) ***<wp.metadata>*** (optional) (see 32.3.2).

(2) Work package identification information ***<wpidinfo>*** (required) (see 32.2).

(3) Select one or more of the following methods to describe the additional generic information.

(a) Uses only a single procedure

1 Procedural steps title ***<title>*** (required) (see 49.1.1.3).

2 Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).

3 Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).

4 Note – ***<note>*** (optional – zero or more) (see 41.1.3).

5 General information or introduction ***<geninfo>*** (optional) (see 49.1.4.12).

6 Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).

7 Procedural step level 1 ***<step1>*** (see 33.3) and/or conditional procedural step level 1 ***<step1>*** (see 48.2.1) (required – one or more).

(b) Uses multiple procedures

1 Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).

2 Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).

3 Note – ***<note>*** (optional – zero or more) (see 41.1.3).

4 General information or introduction ***<geninfo>*** (optional) (see 49.1.4.12).

5 Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).

6 Procedure ***<proc>*** (required – two or more) (see 33.3).

(c) Uses descriptive or narrative text.

1 Warning – ***<warning>*** (optional – zero or more) (see 41.1.1).

2 Caution – ***<caution>*** (optional – zero or more) (see 41.1.2).

3 Note – ***<note>*** (optional – zero or more) (see 41.1.3).

4 General information or introduction ***<geninfo>*** (optional) (see 49.1.4.12).

5 Illustration ***<figure>*** (see 44.1.1) and/or conditional illustration ***<figure-alt>*** (see 48.2.1).(optional – zero or more).

6 Narrative titled text first level ***<para0>*** (see 49.1.1.8) and/or conditional narrative titled text first level ***<para0–alt>*** (see 48.2.1) (required – one or more).

b. DTD graphically depicted for ***<genwp>***:

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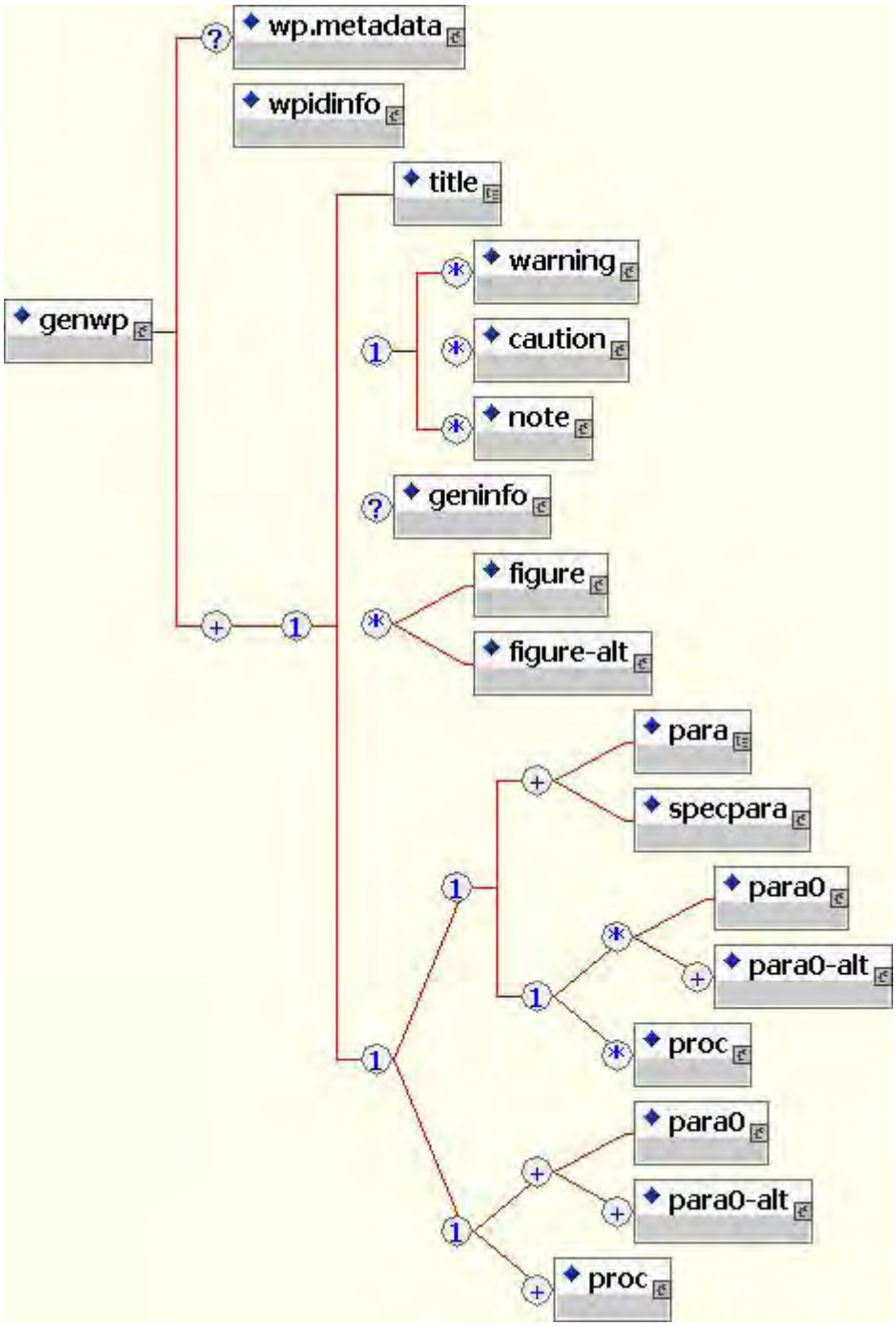

FIGURE 484. Additional generic work package DTD hierarchy <genwp>.

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c. DTD fragment for ***<genwp>***:

```
<!ELEMENT genwp (wp.metadata?, wpidinfo, (((title, (warning*, caution*, note*), geninfo?, (
<!ATTLIST genwp
   airforce          (yes | no )                "no"
    army             (yes | no )                "no"
    assocfig         IDREFS                      #IMPLIED
    changeref        IDREFS                      #IMPLIED
    comment          CDATA                       #IMPLIED
    crewmember       CDATA                       #IMPLIED
    date-time-stamp  (date | time | date-time) #IMPLIED
    delchlvl         CDATA                       #IMPLIED
    deletewp         (yes | no )                "no"
    fgc              CDATA                       #IMPLIED
    frame            (yes | no )                "yes"
    idref            IDREFS                      #IMPLIED
    inschlvl         CDATA                       #IMPLIED
    insertwp         CDATA                       #IMPLIED
    lsa-id           CDATA                       #IMPLIED
    marines          (yes | no )                "no"
    navy             (yes | no )                "no"
    security         (uc | fouo | c | s | ts ) #IMPLIED
    skilltrk         CDATA                       #IMPLIED
    tocentry         (2 | 3 | 4 | 5)            "2"
    wpno             ID                          #REQUIRED
    wpseq            CDATA                       #IMPLIED
```

d. Common attribute:

(1) **airforce** – Indicates that the work package pertains only to the U.S. Air Force (default value is is "no") (see 32.3.1.5).
(2) **army** – Indicates that the work package pertains only to the U.S. Army (default value is is "no") (see 32.3.1.5).
(3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(4) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(5) **comment** – Change information (optional) (see 49.2.9).
(6) **crewmember** – Indicates the information that pertains to a specific crewmember (optional) (see 32.3.1.1).
(7) **date-time-stamp** – To indicate a date–time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp (optional) (see 31.11).
(8) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(9) **deletewp** – Specifies the work package has been deleted. (default value is "no") (see 49.2.9).
(10) **fgc** – Specifies the functional group code that applies to the subject of the element (optional) (see 32.3)
(11) **frame** – In a frame-based TM or IETM, does the element start a new screen or frame of data (optional). Select either "**yes**" or "**no**" (default value is "**yes**").
(12) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(13) **inschlvl** – Insert change level (optional) (see 49.2.9).
(14) **insertwp** – Specifies the new work package change sequence number (point work package) since last the TM revision. Note: After a revision is completed, the attribute is reset to '**0**' and is resequenced in the correct order. (optional) (see 49.2.9).
(15) **lsa-id** – Logistic support analysis identification. Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM. (optional) (see 32.3)
(16) **marines** – Indicates that the work package pertains only to the U.S. Marine Corps (default value is is "no") (see 32.3.1.5).

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(17) **navy** – Indicates that the work package pertains only to the U.S. Navy (default value is is "no") (see 32.3.1.5).
(18) **security** – Security classification (optional) (see 49.2.13).
(19) **skilltrk** – Skill level (optional) (see 49.2.2).
(20) **tocentry** – Table of contents level entry (default value is "**2**") (see 32.3.1.6).
(21) **wpno** – Work package unique identification number (required) (see 32.2.1.1). MIL–STD–40051–1/–2 specifies the identification number format.
(22) **wpseq** – Work package sequence number (optional) (see 32.2.2). MIL–STD–40051–1/–2 specifies the four digit sequential number of the work package for that TM.

40.12.1 XML document instance fragment and output for ***<genwp>***. The XML instance and its stylesheet output for a support items work package ***<genwp>*** that consists of a combined introduction ***<intro>***, COEI standard information ***<coei>***, expendable and durable items standard information ***<explist>***, and a MRPL standard information ***<mrpl>***.

a. Example of an XML document instance fragment for a Support Items Work Package :

```
<genwp airforce="no" army="no" deletewp="no" frame="yes" marines="no" navy="no"
tocentry="2" wpno="S00131-X-XXXX-XXX" wpseq="0420">
<wpidinfo>
<maintlvl level="unitlvl"/>
<title>CARTRIDGE AND PROPELLANT ACTUATED DEVICES
</title>
</wpidinfo>
<proc esd="no" frame="yes" hcp="no" tocentry="0">
<title>CARTRIDGE AND PROPELLANT ACTUATED DEVICES
</title>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">The following list of item are Class V ammunition
items and should be managed in accordance with
<xref wpid="M0143-X-XXX-XXX"/>,
<extref docno="DA PAM 710-2-1"/>.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Shelf life/Service Life (Retirement
Interval) will be in accordance with
<extref docno="TB 9-1300-385" posttext=", MUNITIONS RESTRICTED OR SUSPENDED"/>.
Changes to Shelf and Service life appear in Ammunition Supplemental
Notices between quarterly revisions. This Manual is updated
biannually, request for this document will be referred to
<proponent>
<name>Commander, AMCCOM, ATTN: AMSMC-QAS
</name>
<address>
<city>Rock Island
</city>
<state>IL
</state>
<zip>61299-6000
</zip>
</address>
</proponent>.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Shelf life. The total period of time, beginning
the date of manufacture/cure/assembly that an item may remain in
```


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```
the combined wholesale (including manufacture) and retail storage
system and still remain suitable for issue to, and use by end
user (shelf life is not to be confused with service life which
is a measurement of anticipated total in-use time).
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Parts can be removed during the inspection nearest
the time before removal is due.  Exceptions to this are in
<extref docno="TM 1-1500-328-23"/>.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">The computed item replacement, installation
date, and lot number shall be recorded on
<extref docno="DA Form 2408-18"/> for each installed item.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Be sure to fill out all forms,
records, and worksheets required by
<extref docno="DA PAM 738-751"/> when replacing parts.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Unserviceable item shall be tagged with NSN,
installation date, removal date, reason for removal, lot number,
helicopter type/model and serial number, and aviation unit designation,
and returned to the supporting ammunition supply activity in
the container to transport the replacement item.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">The listing has four headings:
<randlist bullet="yes">
<item>NOMENCLATURE — Name of the part.
</item>
<item>PART NUMBER — Number assigned to the part.  Check
<extref docno="TM 1-1520-238-23P"/> for part numbers of replacement parts.
</item>
<item>NSN — National Stock Number assigned to the part.  Reference
<extref docno="TM 1-1520-238-23P"/>.
</item>
<item>DODAC — Department of Defense Ammunition Code.  Indicates
type of Ammunition and Packaging.  Ref
<extref docno=" SB708-3" pretext="DOD Ammo Codes HDBK, "/>.
</item>
</randlist>
<table frame="all" tocentry="1">
<title>Cartridge and Propellant Actuated Devices
</title>
<tgroup align="left" char=" " charoff="50" cols="4" colsep="1" rowsep="0">
<colspec align="left" colname="c1" colnum="1" colwidth="2.25*"/>
<colspec align="left" colname="c2" colnum="2" colwidth="3.5*"/>
<colspec align="center" colname="c3" colnum="3" colwidth="1.5*"/>
<colspec align="center" colname="c4" colnum="4" colwidth="1.25*"/>
```


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```
<thead  valign="bottom">
<row>
<entry  align="center"  colname="c1"rowsep="1">Nomenclature
</entry>
<entry  align="center"  colname="c2"  rowsep="1">Part  Number
</entry>
<entry  colname="c3"  colsep="1"  rowsep="1">NSN
</entry>
<entry  colname="c4"  colsep="1"  rowsep="1">DODAC
</entry>
</row>
</thead>
<tbody  valign="top">
<row>
<entry  colname="c1">Initiator,  Cartridge MT06  (3 per)
</entry>
<entry  colname="c2">7-311112017-39 51207-3 51207-7 6260964
</entry>
<entry  colname="c3">1377-01-269-6496
</entry>
<entry  colname="c4">1377-MT06
</entry>
</row>
<row>
<entry  colname="c1">Severance  Device  (CA)  MS90,  Cord,  Det
</entry>
<entry  colname="c2">7-311112017-31  51135-1  6260906-1
</entry>
<entry  colname="c3">1377-01-186-9899
</entry>
<entry  colname="c4">1377-MS90
</entry>
</row>
<row>
<entry  colname="c1">Replaced by Thin Line Explosive SP03, Cord, Det
</entry>
<entry  colname="c2">7-311112017-49
</entry>
<entry  colname="c3">1377-01-356-7842
</entry>
<entry  colname="c4">1377-SP03
</entry>
</row>
<row>
<entry  colname="c1">Severance  Device  (CA)  MS91,  Cord,  Det
</entry>
<entry  colname="c2">7-311112017-33  51135-2  6260906-2
</entry>
<entry  colname="c3">1377-01-186-9900
</entry>
<entry  colname="c4">1377-MS91
</entry>
</row>
<row>
<entry  colname="c1">Severance  Device  (CA)  MS93,  Cord,  Det
</entry>
```


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```
<entry colname="c2">7-311112017-37 51135-4 6260906-4
</entry>
<entry colname="c3">1377-01-186-9902
</entry>
<entry colname="c4">1377-MS93
</entry>
</row>
<row>
<entry colname="c1">Replaced by Thin Line Explosive SP02, Cord, Det
</entry>
<entry colname="c2">7-311112017-47
</entry>
<entry colname="c3">1377-01-356-7841
</entry>
<entry colname="c4">1377-SP02
</entry>
</row>
<row>
<entry colname="c1">Severance Device (CA) MS89, Cord, Det
</entry>
<entry colname="c2">7-311112017-43 51135-5 6260906-5
</entry>
<entry colname="c3">1377-01-170-5260
</entry>
<entry colname="c4">1377-MS89
</entry>
</row>
<row>
<entry colname="c1">Severance Device (CA) MS92, Cord, Det
</entry>
<entry colname="c2">7-311112017-35 51135-7 6260906-7
</entry>
<entry colname="c3">1377-01-186-9901
</entry>
<entry colname="c4">1377-MS92
</entry>
</row>
<row>
<entry colname="c1">Severance Device (FWD) MS94, Cord, Det
</entry>
<entry colname="c2">7-311112017-5 51388-1 6260965-1
</entry>
<entry colname="c3">1377-01-184-6112
</entry>
<entry colname="c4">1377-MS94
</entry>
</row>
</tbody>
</tgroup>
</table>
</para>
</step1>
</proc>
</genwp>
```

b. Example of a page-based TM stylesheet output for Additional Generic Work Package

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

FIGURE 485. Example of a page-based TM stylesheet output for <genwp>.

41 ALERTS.

41.1 Warnings, cautions, and notes. Warnings, cautions, and notes identify information of sufficient importance or hazardousness. Definitions of and requirements for warnings, cautions, and notes are specified in MIL-STD-40051-1 and MIL-STD-40051-2. Warnings, cautions, and notes are tagged using the elements ***<warning>*** (see 41.1.1), ***<caution>*** (see 41.1.2), and ***<note>*** (see 41.1.3) respectively. These content models are similar with the differences being as follows:

a. Warnings may include signal words. as specified in the ***<signalword>*** element.
b. For IETMs, the author can specify in the **acknowledge** attribute if a note is acknowledged or not. Warnings and cautions do not have this attribute since they are always acknowledged.
c. Warnings and cautions may include icons (as specified in the ***<icon-set>*** element).

41.1.1 Warnings ***<warning>***. As specified in MIL-STD-40051-1 and MIL-STD-40051-2, a warning contains an operation, procedure, or statement that if not performed properly may result in personal injury or death. Warnings may be grouped and may include signal words or icons.

a. The components are:
   (1) Grouped warnings ***<warning.group>*** (required – two or more for grouped warnings, otherwise not used). Provides narrative for multiple warnings grouped into a single warning item (see 41.1.1.1).
   (2) Title ***<title>*** (optional – zero or one). Provides the capability for a warning to have a title (see 49.1.1.3).
   (3) Icon set ***<icon-set>*** (required – one or more for warnings that include hazard icons, otherwise not used). Identifies a series of hazard icons used as a unit to mark warnings (see 44.3.3).
   (4) Hazard signal word ***<signalword>*** (optional – zero or one). This element contains the signal word relating the warning hazard icons (see 41.1.1.3).
   (5) Reduced paragraph ***<trim.para>*** (either a reduced paragraph or a sequential list is required – one or more). This element (see 49.1.1.7) has the same usage as the paragraph element, but with a reduced content model. The warning text is contained in either the reduced paragraph or in a sequential list. These elements are included in the ***<warning>*** element content model either directly or as elements in ***<warning.group>***.
   (6) Sequential list ***<seqlist>*** (either a sequential list or a reduced paragraph is required – one or more). An ordered list; the sequence of items is denominated by numbers or letters (see 49.1.2.1). The warning text is contained in either the sequential list or in a reduced paragraph. These elements are included in the ***<warning>*** element content model either directly or as an element in ***<warning.group>***.
b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

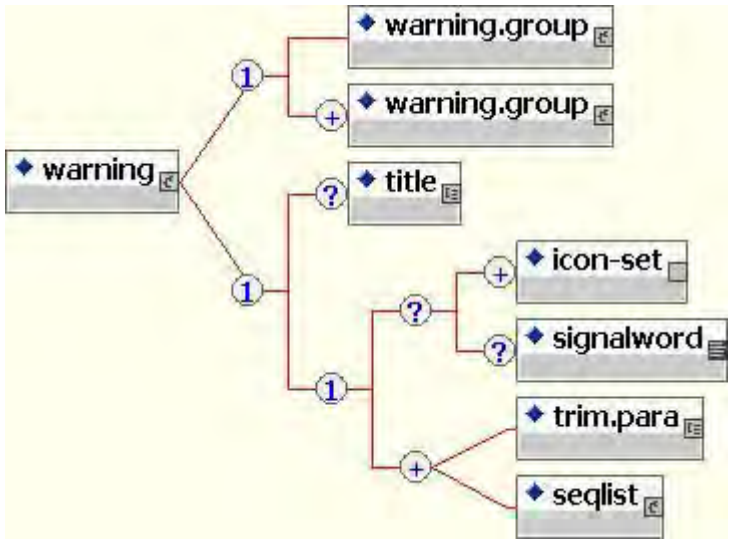

FIGURE 486. Warning <warning> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT warning ((warning.group , warning.group+) | (title?,
     ((icon-set+ , signalword?)? , (trim.para | seqlist)+)))>
<!ATTLIST warning
     haz-abbrev        (yes | no)  'no'
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Unique attributes:

(1) **haz-abbrev** – Determines if an abbreviated warning by hazard icon(s) and signal word only is to be used. A "yes" value indicates that an abbreviated warning by hazard icon(s) and signal word only is to be used. The default is "no".

e. Common Attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

41.1.1.1 Grouped warnings ***<warning.group>***. Grouped warnings provide narrative for multiple warnings grouped into a single warning item. The ***<warning.group>*** element's content model includes the same elements that are used in ***<warning>*** (i.e., ***<title>***, ***<icon-set>***, ***<signalword>***, ***<trim.para>*** and

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<seqlist>). The attributes are also the same. The difference is that each warning is tagged in a separate ***<warning.group>*** element (see 41.1.10 for an example).

a. DTD fragment is graphically depicted.

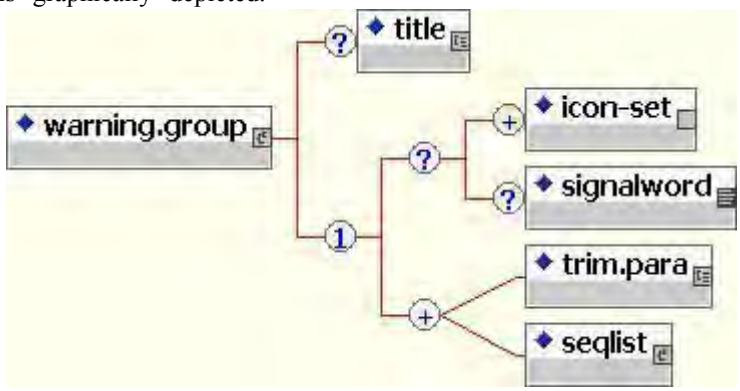

FIGURE 487. Grouped warnings <warning.group> DTD hierarchy.

b. DTD fragment:

```
<!ELEMENT warning.group (title?, ((icon-set+, signalword?)?,
 (trim.para | seqlist)+))>

<!ATTLIST warning.group
     haz-abbrev         (yes | no)   'no'
     assocfig           IDREFS                          #IMPLIED
     changeref          IDREFS                          #IMPLIED
     comment            CDATA                           #IMPLIED
     delchlvl           CDATA                           #IMPLIED
     id                 ID                              #IMPLIED
     idref              IDREFS                          #IMPLIED
     inschlvl           CDATA                           #IMPLIED
     security           (uc | fouo | c | s | ts)        #IMPLIED
     skilltrk           CDATA                           #IMPLIED
```

c. Unique attributes:

(1) **haz-abbrev** — Determines if an abbreviated warning by hazard icon(s) and signal word only is to be used. A "yes" value indicates that an abbreviated warning by hazard icon(s) and signal word only is to be used. The default is "no".

d. Common Attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

41.1.1.2 Icon set ***<icon-set>***. The ***<icon-set>*** element identifies a series of hazard icons used as a unit to mark warnings. The ***<icon-set>*** is explained in 44.3.3. An example of a warning with multiple icons is given in 41.1.5.

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Check the source to verify that this is the current version before use.

41.1.1.3 Hazard signal word ***<signalword>***. The ***<signalword>*** element contains the signal word relating the warning hazard icons. It is only used with an icon set and it's use with the set is not required. The content model is #PCDATA. An example of a warning with multiple icons and a signal word is given in 41.1.6.

a. DTD fragment is graphically depicted.

FIGURE 488. Hazard signal word <signalword> DTD hierarchy.

b. DTD fragment:

```
<!ELEMENT signalword (#PCDATA)>
```

41.1.2 Cautions ***<caution>***. A caution identifies risk of damage to the equipment. The content model for a caution is similar to that of a warning. Like warnings, cautions may be grouped and may include icons. The one major difference is that a caution does not use signal words.

a. The components are:

(1) Grouped Cautions ***<caution.group>*** (required (two or more) for grouped cautions, otherwise not used). Provides narrative for multiple cautions grouped into a single caution item (see 41.1.2.1).

(2) Icon Set ***<icon-set>*** (required (one or more) for cautions that include icons, otherwise not used). Identifies the equipment damage caution icons used to mark cautions (see 41.1.2.2).

(3) Reduced Paragraph ***<trim.para>*** (either a reduced paragraph or a sequential list is required – one or more). This element has the same usage as the paragraph element, but with a reduced content model. The reduced paragraph or a sequential list is included in the ***<caution>*** element content model either directly or as an element in ***<caution.group>***

(4) Sequential List ***<seqlist>*** (either a sequential list or a reduced paragraph is required – one or more). An ordered list; the sequence of items is denominated by numbers or letters. The sequential list or a reduced paragraph is included in the ***<caution>*** element content model either directly or as an element in ***<caution.group>***.

b. DTD fragment is graphically depicted.

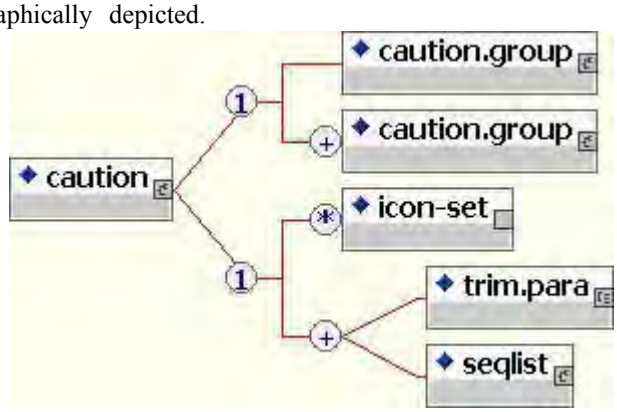

FIGURE 489. Caution <caution> DTD hierarchy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT caution ((caution.group, caution.group+) | (icon-set* ,
                  (trim.para | seqlist)+))>

<!ATTLIST caution
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

41.1.2.1 Grouped cautions ***<caution.group>***. Grouped cautions provide narrative for multiple cautions grouped into a single caution item. The ***<caution.group>*** element's content model includes the same elements that are used in ***<caution>*** (i.e., ***<icon-set>***, ***<trim.para>*** and ***<seqlist>***). The attributes are also the same. The difference is that each caution is tagged in a separate ***<caution.group>*** element. See 41.1.10 for an example of a grouped warning. The tagging for a grouped caution would be similar.

a. DTD fragment is graphically depicted.

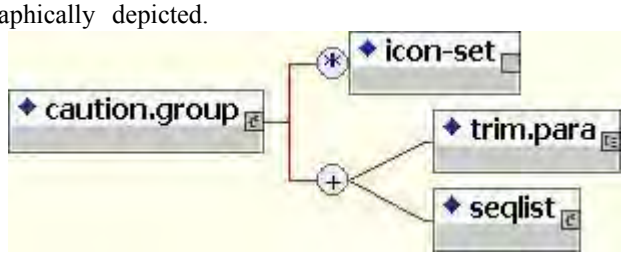

FIGURE 490. Grouped cautions <caution.group> DTD hierarchy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

b. DTD fragment:

```
<!ELEMENT caution.group (icon-set*, (trim.para | seqlist)+)>

<!ATTLIST caution.group
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)        #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

c. Common Attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

41.1.2.2 Icon set ***<icon-set>***. The ***<icon-set>*** element identifies a series of equipment damage icons used as a unit to mark cautions. The content model is empty. An example of a warning with multiple icons is given in 41.1.5. The tagging for a caution would be similar.

a. DTD fragment:

```
<!ELEMENT icon-set EMPTY>

<!ATTLIST icon-set
     alt               CDATA                           #IMPLIED
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     boardno           ENTITY                          #REQUIRED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     hscale            CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     reprowid          CDATA                           #IMPLIED
     reprodep          CDATA                           #IMPLIED
     scalefit          (yes | no )                     #IMPLIED
     security          (uc | fouo | c | s | ts )       #IMPLIED
     skilltrk          CDATA                           #IMPLIED
     unitmeasure       (mm | cm | px | in | pt | pi )  'in'
     vscale      CDATA                                 #IMPLIED>
```

b. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

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(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)
(10) **alt** – Additional text to assist in identifying the graphic (optional) (see 44.2.d.).
(11) **boardno** – External graphic entity name (required) (see 44.2.d.).
(12) **graphicunit** – Graphic unit of measurement (default value is "**in**") (see 44.2.d.).
(13) **hscale** – Horizontal scaling factor (optional) (see 44.2.d.).
(14) **reprowid** – Reproduction area width (optional) (see 44.2.d.).
(15) **reprodep** – Reproduction area depth (optional) (see 44.2.d.).
(16) **scalefit** – Scale the graphic to fit the reproduction area (optional) (see 44.2.d.).
(17) **vscale** – Vertical scaling factor (optional) (see 44.2.d.).

41.1.3 Notes ***<note>***). A note highlights essential procedures, conditions, or statements or conveys important instructional data to the user. Notes are allowed in tasks, procedures, steps, and non-procedural information. Notes may be acknowledged using the attribute **acknowledge**.

a. The components are:
(1) Grouped Notes ***<note.group>*** (required (two or more) for grouped notes, otherwise not used). Provides narrative for multiple notes grouped into a single note item (see 41.1.3.1).
(2) Reduced Paragraph ***<trim.para>*** (either a reduced paragraph or a sequential list is required – one or more). This element has the same usage as the paragraph element, but with a reduced content model. The reduced paragraph or a sequential list is included in the ***<note>*** element content model either directly or as an element in ***<note.group>***
(3) Sequential List ***<seqlist>*** (either a sequential list or a reduced paragraph is required – one or more). An ordered list; the sequence of items is denominated by numbers or letters. The sequential list or a reduced paragraph is included in the ***<note>*** element content model either directly or as an element in ***<note.group>***.

b. DTD fragment is graphically depicted.

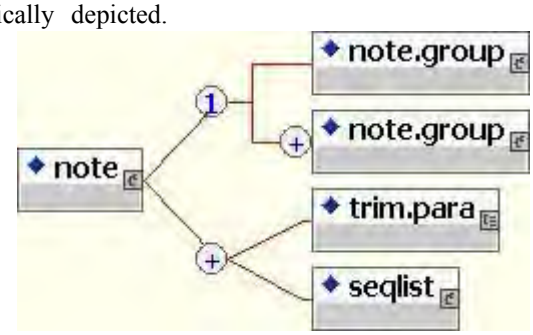

FIGURE 491. Note <note> DTD hierarchy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment:

```
!ELEMENT note ((note.group , note.group+) | (trim.para | seqlist)+)>

<!ATTLIST note
     acknowledge       (yes | no)  'no'
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)       #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

d. Unique attributes:

(1) **acknowledge** – Determines if the note is to be acknowledged. A “yes” value indicates that an acknowledgement will be required. A “no” value indicates that the user will not be presented with an acknowledgement.t The default is “no”.

e. Common Attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

41.1.3.1 Grouped notes ***<note.group>***. Grouped notes provide narrative for multiple notes grouped into a single note item. The ***<note.group>*** element's content model includes the same elements that are used in ***<note>*** (i.e., ***<trim.para>*** and ***<seqlist>***). The attributes are also the same. The difference is that each note is tagged in a separate ***<note.group>*** element. See 41.1.10 for an example of a grouped warning. The tagging for a grouped note would be similar.

a. DTD fragment is graphically depicted.

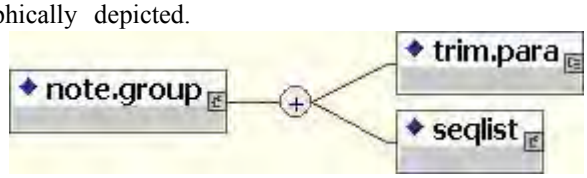

FIGURE 492. Grouped notes <note.group> DTD hierarchy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

b. DTD fragment:

```
<!ELEMENT note.group (trim.para | seqlist)+)>

<!ATTLIST note.group
     acknowledge (yes | no)  'no'
     assocfig          IDREFS                          #IMPLIED
     changeref         IDREFS                          #IMPLIED
     comment           CDATA                           #IMPLIED
     delchlvl          CDATA                           #IMPLIED
     id                ID                              #IMPLIED
     idref             IDREFS                          #IMPLIED
     inschlvl          CDATA                           #IMPLIED
     security          (uc | fouo | c | s | ts)        #IMPLIED
     skilltrk          CDATA                           #IMPLIED>
```

c. Unique attributes:

(1) **acknowledge** — Determines if the note is to be acknowledged. A "yes" value indicates that an acknowledgement will be required. A "no" value indicates that the user will not be presented with an acknowledgement.t The default is "no".

d. Common Attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

41.1.4 Example – Warnings, cautions and notes at the task level. To tag a warning, caution or note at the task level, insert the appropriate ***<warning>***, ***<caution>*** and ***<note>*** elements after the ***<title>*** element for the task. The example shown below depicts a page-based output for a warning at the task level. The warning appears on the page immediately after the task title and before the step information.

a. XML document instance fragment.

```
<opusualwp airforce="no" army="no" deletewp="no" frame="no" marines="no" navy="no"
tocentry="2" wpno="OXXXXX-X-XXXX-XXX" wpseq="0003">
<wpidinfo>
<maintlvl level="operator"/>
<title>START ENGINE WITH OUTSIDE POWER SOURCE
</title>
</wpidinfo>
<wpinfo>
<tools>
<tools-setup-item>
<name>Slave cable
</name>
<itemref>
<xref itemid="S00035-X-XXX-XXX-0048" wpid="S00035-X-XXX-XXX"/>
</itemref>
</tools-setup-item>
<tools-setup-item>
<name>24 volt power source or other vehicle
</name>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<itemref>
<xref itemid="S00035-X-XXX-XXX-0044" wpid="S00035-X-XXXX-XXX"/>
</itemref>
</tools-setup-item>
</tools>
<persnreq>
<persnreq-setup-item>
<name>Driver
</name>
<qty>2
</qty>
</persnreq-setup-item>
</persnreq>
<ref>
<ref-setup-item>
<extrefdocno="TM X-XXX-XXX-XX"/>
</ref-setup-item>
</ref>
<eqpconds>
<eqpconds-setup-item>
<condition>Vehicle unable to start under own power
</condition>
<itemref>
<xref wpid="OYYYY-Y-YYYY-YYY"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Operational vehicle engine stopped
</condition>
<itemref>
<xref posttext=")" pretext="(" wpid="OYYYY-Y-YYYY-YYY"/>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
<opertsk>
<oper esd="no" frame="yes" hcp="no"tocentry="0">
<title>START ENGINE
</title>
<warning haz-abbrev="no">
<trim.para esd="no" hcp="no">Using ether to start engine can result
in engine explosion. Personnel can be injured or killed.
Never use ether to assist starting an engine.
</trim.para>
</warning>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<paraesd="no" hcp="no">Check that master power switch is OFF.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check that engine accessory switch is OFF.
</para>
</step1>
<step1 esd="no"hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no"hcp="no">Check that turret power switch is OFF.
</para>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check that fire suppression switchis in manual.
</para>
</step1>
<step1esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check that starter cutout override switch is OFF.
</para>
</step1>
<step1 esd="no"hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no"hcp="no">Check that fwd and rear bilge pumps switches are OFF.
</para>
</step1>
<step1 esd="no" hcp="no"qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Checkthat smoke screen generator switch is OFF.
</para>
</step1>  . . .
</oper>
</opertsk>
</opusualwp>
```

b. Stylesheet output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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0003

OPERATOR MAINTENANCE
START ENGINE WITH OUTSIDE POWER SOURCE
OPERATION UNDER USUAL CONDITIONS

INITIAL SETUP

Tools and Special Tools

Slave Cable (WP 0024, Item 4)
24 volt power source or other vehicle (WP 0024, Item 6)

Personnel Required

Driver (2)

References

TM X-XXX-XXX-XX

Equipment Condition

Vehicle unable to start under own power (WP 0021)
Operational vehicle engine stopped (WP 0021)

START ENGINE

WARNING

Using ether to start engine can result in engine explosion. Personnel can be injured or killed. Never use ether to assist starting an engine.

1. Check that master power switch is OFF.
2. Check that engine accessory switch is OFF.
3. Check that turret power switch is OFF.
4. Check that fire suppression switch is in manual.
5. Check that starter cutout override switch is OFF.
6. Check that fwd and rear bilge pumps switches are OFF.
7. Check that smoke screen generator switch is OFF.

0003-1

FIGURE 493. Task warning example.

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41.1.5 Example – multiple icons. The ***<icon-set>*** element identifies the icon to be displayed with the warning or caution. More than one ***<icon-set>*** can be identified The example shown below depicts a warning with multiple icons.

a. XML document instance fragment.

```
<warning haz-abbrev="no">
<icon-set boardno="icon-chemical" unitmeasure="in"/>
<icon-set boardno="icon-vapor" unitmeasure="in"/>
<icon-set boardno="icon-eye-protection" unitmeasure="in"/>
<trim.para esd="no" hcp="no">Fluorinated compound OT20 is an irritant to the
eyes and skin. Use safety glasses and latex gloves or barrier cream. Keep
sparks, flames and heat away. Keep grease off skin, eyes, and clothes.
</trim.para>
</warning>
```

b. Stylesheet output.

FIGURE 494. Multiple icons example.

41.1.6 Example – abbreviated hazardous material warning. The XML source and its stylesheet output depicting a warning with multiple icons and a signal word are shown below. The ***<trim.para>*** element is required for all warnings. However. it is not needed for warnings that use a signal word. Enter the ***<trim.para>*** element with an empty value as shown in this example.

a. XML document instance fragment.

```
<warning haz-abbrev="yes">
<icon-set boardno="icon-chemical" unitmeasure="in"/>
<icon-set boardno="icon-vapor" unitmeasure="in"/>
<icon-set boardno="icon-fire" unitmeasure="in"/>
<icon-set boardno="icon-eye-protection" unitmeasure="in"/>
<signalword>ISOPROPYL ALCOHOL, TT-I-735
</signalword>
<trim.para esd="no" hcp="no">
</trim.para>
</warning>
```

b. Stylesheet output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

FIGURE 495. Abbreviated hazardous material warning example.

41.1.7 Example – warnings, cautions and notes for a step. A warning, caution or note that applies to a single step is tagged within the ***<specpara>*** for that step. For example, the XML source and its stylesheet output depicting a warning that applies to the first step in a maintenance task are shown below. The ***<warning>*** element is contained within the ***<specpara>*** in the ***<step1>*** element for the first step. The stylesheet output shows the warning as it would be displayed for an IETM. The step data is not displayed because the user has not yet acknowledged the warning.

a. XML document instance fragment.

```
<maintsk>
<remove esd="no" frame="yes" hcp="no" tocentry="0">
<title>REMOVAL OF EMERGENCY LAMPHOLDER
</title>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<specpara esd="no" hcp="no">
<warning haz-abbrev="no">
<trim.para esd="no" hcp="no">28 Vdc is present at emergency light. To
prevent personal injury, ensure BLACKOUT LIGHTS circuit breaker
is set to OFF before removing lampholder.
</trim.para>
</warning>
<para esd="no" hcp="no">At POWER CONTROL panel, set BLACKOUT
LIGHTS circuit breaker to OFF.
</para>
</specpara>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Remove emergency light shield by removing
four screws, lockwashers, and washers.
</para>
</step1> . . .
</remove>
</maintsk>
```

b. Stylesheet output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

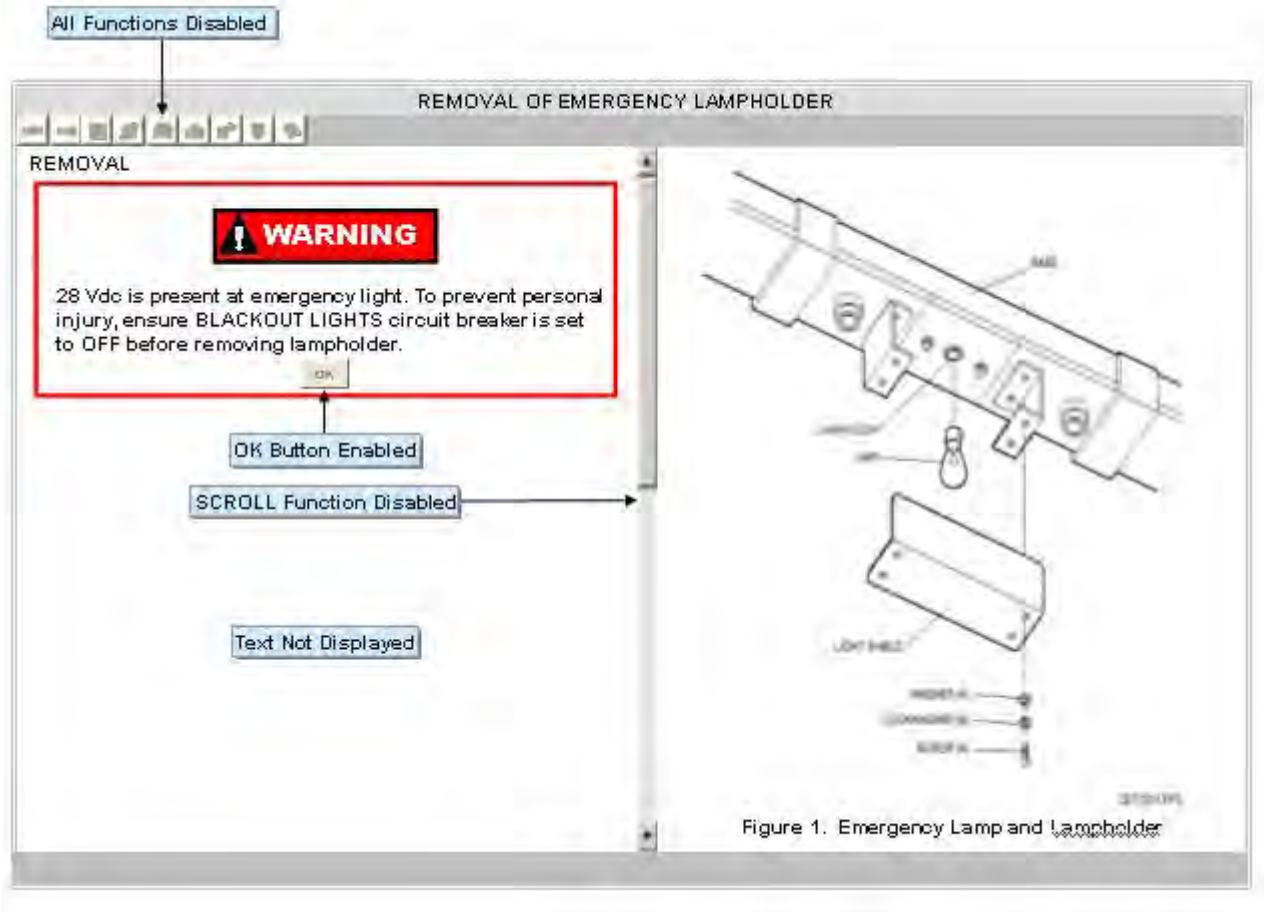

FIGURE 496. Step warning example prior to acknowledgement of warning alert.

41.1.8 Example – warnings, cautions and notes for multiple steps. To tag warnings, cautions or notes that apply to multiple steps, start by tagging them in the first step that they apply to. In that step, set the **id** attribute, which is of type ID, for each warning, caution and note that applies to other steps in that task. This identifier is then referenced in the other step(s) in the appropriate attribute (i.e., **warningref**, **cautionref**, or **noteref**). In cases where a step has several warnings or cautions or notes that pertain to other steps, multiple references can be made since **warningref**, **cautionref**, and **noteref** are of type IDREFS. The XML source and its stylesheet output depicting a warning that applies to the first three steps and a caution that applies to steps four and five are shown below. The ***<warning>*** element is contained in the first step. The **id** attribute in that ***<warning>*** element is set to a unique value. In the second and third steps, the **warningref** attribute is set to the same unique value. Similarly, the ***<caution>*** element is contained in the fourth step. The **id** attribute in that ***<caution>*** element is set to a unique value. In the fifth step, the **cautionref** attribute is set to the same unique value. Figure 497 displays the IETM screen after the warning has been acknowledged by the user and before the caution has been acknowledged. The caution does not prevent the first three steps from being displayed since it does not apply to those steps. The persistent alert icon for the warning (shown in the bottom left portion of the screen) remains as long as steps one, two or three are displayed.

a. XML document instance fragment.

```
<maintsk>
<remove esd="no" frame="yes" hcp="no" tocentry="0">
<title>REMOVAL OF EMERGENCY LAMPHOLDER
</title>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<specpara esd="no" hcp="no">
<warning haz-abbrev="no"id="MXXXXX-X-XXX-XXXX-0046">
<trim.para esd="no"hcp="no">This is an example of a warning that
appliesto the first three steps in a maintenance task.
</trim.para>
</warning>
<para esd="no" hcp="no">AtPOWER CONTROL panel, set BLACKOUT
LIGHTS circuit breaker to OFF.
</para>
</specpara>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert"
warningref="MXXXXX-X-XXX-XXXX-0046">
<para esd="no"hcp="no">Remove emergency light shield by removing
fourscrews, lockwashers, and washers.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert"
warningref="MXXXXX-X-XXX-XXXX-0046">
<para esd="no"hcp="no">Remove lamp by pushing it up, twisting,
and pulling it down from the lamp holder.
</para>
</step1>
<step1 esd="no" hcp="no" id="MXXXXX-X-XXX-XXXX-0047"
qa="no" skilllevel="novice_expert">
<specpara esd="no"hcp="no">
<caution>
<trim.para esd="no"hcp="no">Be careful not to damage wires when
removing emergency light raceway cover.
</trim.para>
</caution>
<para esd="no" hcp="no">Remove emergency light raceway cover by
grasping it firmly and pulling straight down.
</para>
</specpara>
</step1>
<step1 cautionref="MXXXXX-X-XXX-XXXX-0047"esd="no" hcp="no"
qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Remove lamp by pushing it up, twisting,
and pulling it down from the lampholder.
</para>
</step1> ...
</remove>
</maintsk>
```

b. Stylesheet output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

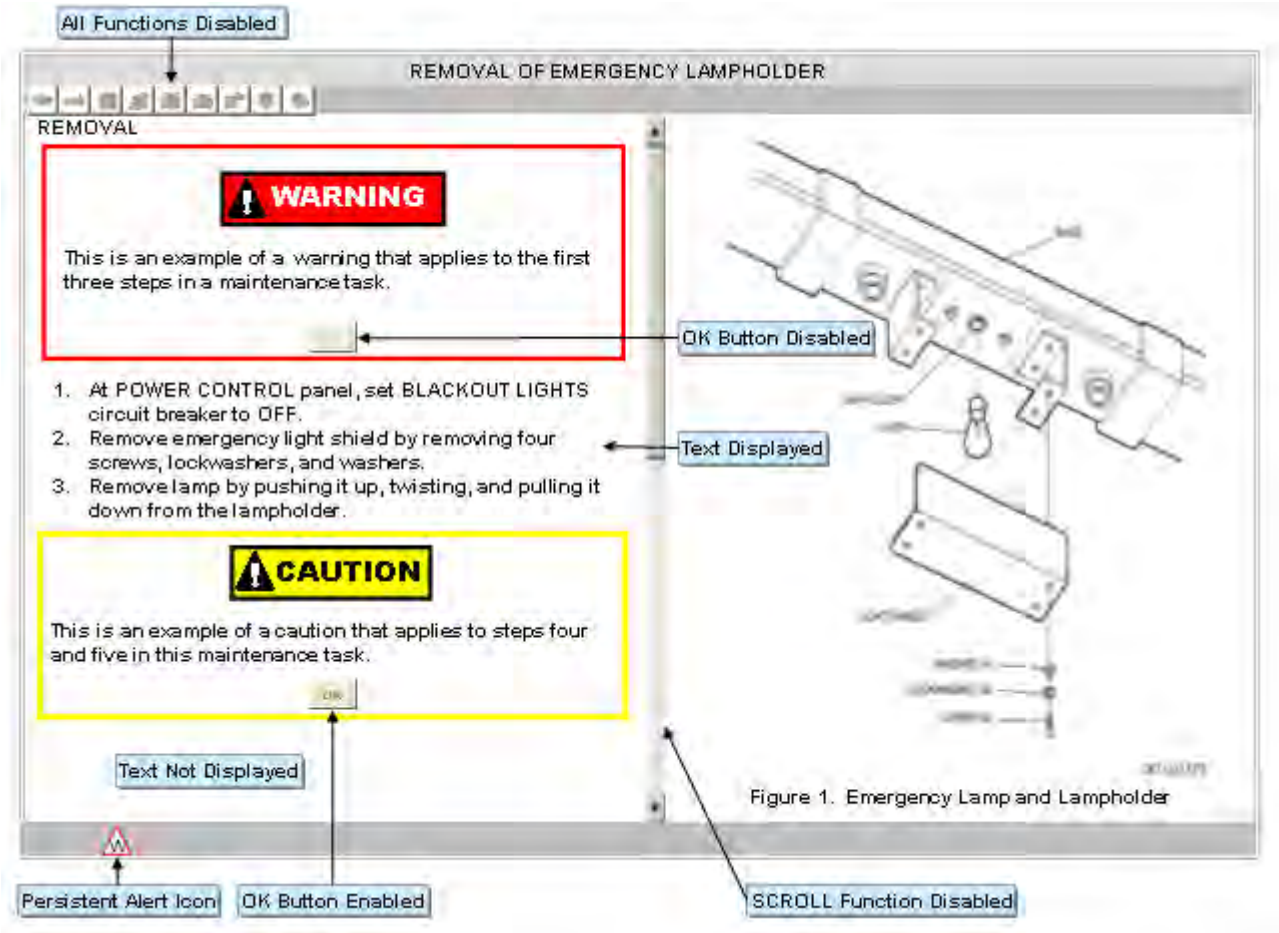

FIGURE 497. Multiple step warning and caution example after acknowledgement of the warning alert.

41.1.9 Example – text entities for warnings, cautions and notes. In a TM/IETM, the same warning, caution or note may appear in several tasks, procedures or steps. Repeating the text for the warning, caution or note can result in errors. Creating text entities for warnings, cautions and notes avoid such errors by letting the text be entered only once. The XML source depicting a text entity for a warning are shown below. This is the same example as shown in 41.1.7. The only different is that the text for the warning as entered in the ***<trim.para>*** element has be replaced with the text entity named "WarningEmergencyLight". The markup to create this text entity is shown at the beginning of the XML document instance fragment.

a. XML document instance fragment.

```
<!ENTITY WarningEmergencyLight "28 Vdc is present at emergency light.
To prevent personal injury, ensure BLACKOUT LIGHTS circuit breaker is
set to OFF before removing lampholder.">. . .
<maintsk>
<remove esd="no" frame="yes" hcp="no" tocentry="0">
<title>REMOVAL OF EMERGENCY LAMPHOLDER
</title>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<specpara esd="no" hcp="no">
<warning haz-abbrev="no">
<trim.para esd="no" hcp="no">&amp;WarningEmergencyLight;
</trim.para>
</warning>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<para esd="no" hcp="no">At POWER CONTROL panel, set BLACKOUT
LIGHTS circuit breaker to OFF.
</para>
</specpara>
</step1>
<step1 esd="no"hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no"hcp="no">Remove emergency light shield by removing
fourscrews, lockwashers, and washers
</para>
</step1> ...
</remove>
```

41.1.10 Example – group warnings, cautions and notes. Warnings, cautions and notes on unrelated topics that pertain to the same task, procedure or step(s) may be grouped under one heading. The XML source and its stylesheet output depicting a group of three warnings for a step are shown below. The tagging for a group of cautions or notes would be similar.

a. XML document instance fragment.

```
<maintsk>
<remove esd="no" frame="yes" hcp="no" tocentry="0">
<title>REMOVAL OF EMERGENCY LAMPHOLDER
</title>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<specpara esd="no" hcp="no">
<warning haz-abbrev="no">
<warning.group haz-abbrev="no">
<trim.para esd="no" hcp="no">This is the first warning in the warning group.
</trim.para>
</warning.group>
<warning.grouphaz-abbrev="no">
<trim.para esd="no" hcp="no">Thisis the second warning in the warning group.
</trim.para>
</warning.group>
<warning.group haz-abbrev="no">
<trim.para esd="no" hcp="no">This is the third warning in the warning group
</trim.para>
</warning.group>
</warning>
<para esd="no" hcp="no">At POWER CONTROL panel, set BLACKOUT
LIGHTS circuit breaker to OFF.
</para>
</specpara>
</step1>
<step1 esd="no"hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no"hcp="no">Remove emergency light shield by removing
fourscrews, lockwashers, and washers
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Remove lamp by pushing it up, twisting,
and pulling it down from the lampholder
</para>
</step1> ...
</remove>
</maintsk>
```

b. Stylesheet output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

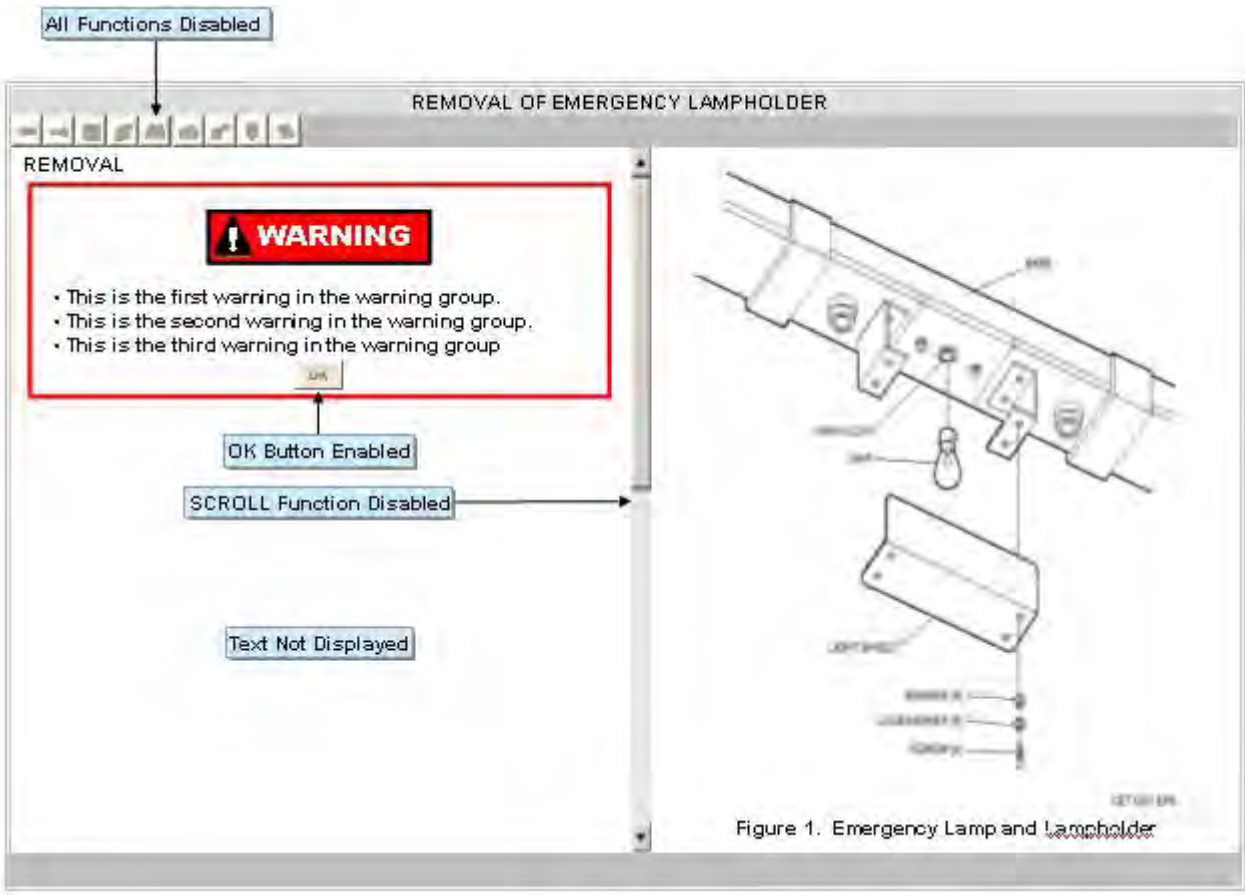

FIGURE 498. Group warning example prior to acknowledgement of warning alert.

41.1.11 Example – multiple warnings, cautions and notes at the task level. To tag multiple warnings, cautions and notes at the task level, insert the appropriate ***<warning>***, ***<caution>*** and ***<note>*** elements after the ***<title>*** element for the task. The warnings appear first and the cautions second. Notes appear after the warnings and cautions. The example shown below depicts a page-based output for a warning, a caution and a note at the task level. The warning, caution and note appear on the page immediately after the task title and before the step information. The **acknowledge** attribute is set to "no" in the ***<note>*** element meaning that there is no requirement to acknowledge the note.

a. XML document instance fragment.

```
<opusualwp airforce="no" army="no" deletewp="no" frame="no" marines="no"
navy="no" tocentry="2" wpno="OXXXXX-X-XXXX-XXX" wpseq="0003">
<wpidinfo>
<maintlvl level="operator"/>
<title>START ENGINE WITH OUTSIDEPOWER SOURCE
</title>
</wpidinfo>
<wpinfo>
<tools>
<tools-setup-item>
<name>Slave cable
</name>
<itemref>
<xref itemid="S00035-X-XXX-XXX-0048" wpid="S00035-X-XXX-XXX"/>
</itemref>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</tools-setup-item>
<tools-setup-item>
<name>24 volt power source or other vehicle
</name>
<itemref>
<xref itemid="S00035-X-XXX-XXX-0044"wpid="S00035-X-XXXX-XXX"/>
</itemref>
</tools-setup-item>
</tools>
<persnreq>
<persnreq-setup-item>
<name>Driver
</name>
<qty>2
</qty>
</persnreq-setup-item>
</persnreq>
<ref>
<ref-setup-item>
<extrefdocno="TM  X-XXX-XXX-XX"/>
</ref-setup-item>
</ref>
<eqpconds>
<eqpconds-setup-item>
<condition>Vehicleunable to start under own power
</condition>
<itemref>
<xref wpid="OYYYY-Y-YYYY-YYY"/>
</itemref>
</eqpconds-setup-item>
<eqpconds-setup-item>
<condition>Operational vehicle engine stopped
</condition>
<itemref>
<xref posttext=")" pretext="(" wpid="OYYYY-Y-YYYY-YYY"/>
</itemref>
</eqpconds-setup-item>
</eqpconds>
</wpinfo>
<opertsk>
<oper esd="no" frame="yes" hcp="no"tocentry="0">
<title>START ENGINE
</title>
<warninghaz-abbrev="no">
<trim.para esd="no" hcp="no">Using etherto start engine can result
in engine explosion.  Personnel canbe injured or killed.
Never use ether to assist starting anengine.
</trim.para>
</warning>
<caution>
<trim.para esd="no" hcp="no">Battery or electrical damagecan
occur if electrical switches are left on.  Turn off
allelectrical switches in both vehicles.
</trim.para>
</caution>
<note acknowledge="no">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<trim.paraesd="no" hcp="no">Steps 1 &mdash; 7 should be done in
both operational and disabled vehicles.
</trim.para>
</note>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check that master power switch is OFF.
</para>
</step1>
<step1 esd="no"hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no"hcp="no">Check that engine accessory switch is OFF.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check that turret power switch is OFF.
</para>
</step1>
<step1 esd="no"hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no"hcp="no">Check that fire suppression switch is in manual.
</para>
</step1>
<step1 esd="no" hcp="no"qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check that starter cutout override switch is OFF.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check that fwd and rear bilge pumps switches are OFF.
</para>
</step1>
<step1 esd="no" hcp="no" qa="no" skilllevel="novice_expert">
<para esd="no" hcp="no">Check that smoke screen generator switch is OFF.
</para>
</step1>  . . .
</oper>
</opertsk>
</opusualwp>
```

b. Stylesheet output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

0003

OPERATOR MAINTENANCE
START ENGINE WITH OUTSIDE POWER SOURCE
OPERATION UNDER USUAL CONDITIONS

INITIAL SETUP

Tools and Special Tools

Slave Cable (WP 0024, Item 4)
24 volt power source or other vehicle (WP 0024, Item 6)

Personnel Required

Driver (2)

References

TM X-XXX-XXX-XX

Equipment Condition

Vehicle unable to start under own power (WP 0021)
Operational vehicle engine stopped (WP 0021)

START ENGINE

WARNING

Using ether to start engine can result in engine explosion. Personnel can be injured or killed. Never use ether to assist starting an engine.

CAUTION

Battery or electrical damage can occur if electrical switches are left on. Turn off all electrical switches in both vehicles.

NOTE

Steps 1 - 7 should be done in both operational and disabled vehicles.

1. Check that master power switch is OFF.
2. Check that engine accessory switch is OFF.
3. Check that turret power switch is OFF.
4. Check that fire suppression switch is in manual.
5. Check that starter cutout override switch is OFF.
6. Check that fwd and rear bilge pumps switches are OFF.
7. Check that smoke screen generator switch is OFF.

0003-1

FIGURE 499. Multiple warning, caution and note example.

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Check the source to verify that this is the current version before use.

42 TABLES.

A table is used when information can be presented more clearly than in text. There are three types of table structures allowed for use with MIL–STD–2361. The three types are Continuous Acquisition Life-Cycle Support (CALS), standard information tables and table material. CALS tables are generic structural tables. Standard information tables (see 42.3) are MIL–STD–40051–1/–2 defined standard information tables, in which columns and rows are inferred from content-specific tags. Table material ***<tabmat>*** (see 42.2) allows data to be formatted structurally as a table but not stated as a table.

42.1 Continuous acquisition life-cycle support (CALS) table. A generic table in MIL–STD–2361 follows the Continuous Acquisition Life-Cycle Support (CALS) table model. It uses structural tags that identify data by its place in the hierarchy of the table and by how the material is formatted in the table.

42.1.1 Table elements ***<table>***. The element ***<table>*** uses the CALS table model. Tables are to be numbered consecutively within each work package in the order of their reference starting with Arabic number 1. All tables have a table title and number. A table can have one table group ***<tgroup>*** or more then one. The table groups can have different numbers of columns but they have the same width as the table. When a table has more then one table group, there is no space between the table groups in the table. The attributes of a table provide the display structure of the table. The attribute value of **frame** control the line rulings of the outer frame of the table.

a. The components are:
   (1) Precondition ***<precond>*** (optional). The element indicates if the parent element is presented according the results from the Boolean expression. (see 42.1.1.1).
   (2) Title ***<title>*** (required). The element identifies the purpose of the table and distinguish that table from others in the TM. (see 49.1.1.3)
   (3) Table Group ***<tgroup>*** (required – one or more). The element identifies a new portion of the table. (see 42.1.1.2)

b. DTD graphically depicted:

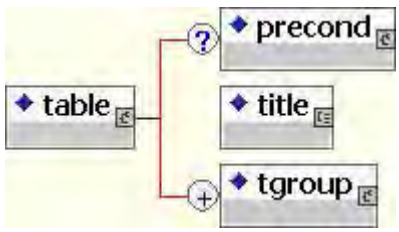

FIGURE 500. CALS table <table> DTD hierarchy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT table (precond?, title, tgroup+)>
<!ATTLIST table
    tabstyle          NMTOKEN                          #IMPLIED
    frame             (top | bottom | topbot
                       | all | sides | none )          #IMPLIED
    colsep            (0|1)                            #IMPLIED
    rowsep            (0|1)                            #IMPLIED
    orient            (port | land )                    #IMPLIED
    label              CDATA                            #IMPLIED
    applicable        IDREFS                           #IMPLIED
    assocfig          IDREFS                           #IMPLIED
    changeref         IDREFS                           #IMPLIED
    comment           CDATA                            #IMPLIED
    delchlvl          CDATA                            #IMPLIED
    id                ID                               #IMPLIED
    idref             IDREFS                           #IMPLIED
    inschlvl          CDATA                            #IMPLIED
    security          (uc | fouo |c | s | ts)          #IMPLIED
    skilltrk          CDATA                             #IMPLIED
    tocentry          (0 | 1 | 2 | 3 | 4 | 5 )        "1"
```

d. Unique attributes:

(1) **tabstyle** - Specified an unique table style defined in the style sheet.
(2) **frame** - Describes position of outer rulings of the table.
(3) **colsep** - Default for all items in this table. If non-zero, display the internal column rulings to the right of each item. If zero, do not display column rulings. Ignored for the last column, where the frame setting applies.
(4) **rowsep** - Default for all items in this table. If non-zero, display the internal row rulings below each item. If zero, do not display row rulings. Ignored for the last row, where the frame setting applies.
(5) **orient** - Orientation of the entire table.
  (a) "PORT" - The table writing direction, along rows, is the same as marginal text.
  (b) "LAND" - The table writing direction is 90° counterclockwise to marginal text.
(6) **label** - Stores the table number for style sheets that can not auto-generate numbers.

e. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **tocentry** – In the TM table of contents indicate the indenture level. The possible selection is "**0**" do not include, "**3**" the 3rd level, "**4**" the 4th level, or "**5**" the 5th level (default value is "**1**").

42.1.1.1 Precondition element ***<precond>***. Precondition element indicates if the parent element is presented according the results from the Boolean expression. The Boolean expression is evaluated by the frame-base viewer's Logic Engine (requires state (variable) information). If the Boolean expression evaluates to "True", then present or process content following the element, otherwise the remaining content is skipped.

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Precondition element contains the element expression ***<expression>*** that defines an equation or expression to evaluate either to a state (variable) information or precondition. Precondition element is used in a frame-based TM and usually with a state table.

a. The components are:
   (1) Expression ***<expression>*** (required). The element defines an equation or expression to evaluate either to a state (variable) information or precondition (see 48.2.2).

b. DTD fragment:

```
<!ELEMENT precond (expression)>
<!ATTLIST precond
    id      ID      #IMPLIED>
```

c. Common attributes:
   (1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

42.1.1.2 Table group ***<tgroup>***. Table group ***<tgroup>*** identifies a new portion of a table. A table group is a set of contiguous rows within a table such as a heading or body of the table group. One table group can contain different number of columns and rows in a table group compare to another table group in the table but each table group has the width as the other which is the same width as the table. The element ***<tgroup>*** also provides the capability to provide a data structural table with or without line rulings or a title.

a. The components are:
   (1) Column specification (optional – zero or more) ***<colspec>*** Column specification provides the details of the column and when new column's specifications is provided, it replaces a previous one in a table group. (see 42.1.1.2.1)
   (2) Spanned column specification (optional – zero or more) ***<spanspec>***. The element identifies a horizontal span of columns and provides a spanning name. (see 42.1.1.2.2)
   (3) Table heading (optional) ***<thead>***. The element provides the heading of the table group columns. (see 42.1.1.2.3)
   (4) Table body (required) ***<tbody>***. The element identifies the body of a table group. (see 42.1.1.2.4).

b. DTD graphically depicted:

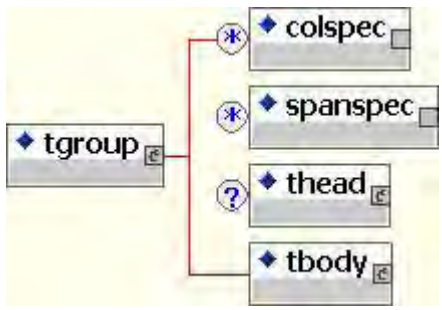

FIGURE 501. CALS table <tgroup> DTD hierarchy.

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c. DTD fragment.

```
<!ELEMENT tgroup (colspec*, spanspec*, thead?, tbody)>
<!ATTLIST tgroup
    cols         CDATA                #REQUIRED
    tgroupstyle  NMTOKEN              #IMPLIED
    colsep       (0|1)                #IMPLIED
    rowsep       (0|1)                #IMPLIED
                  align               (left | right | center |
                                       justify | char )   'left'
   charoff       CDATA                '50'
   char          CDATA                ' ' >
```

d. Unique attributes:

(1) **cols** - Column number in the table (required).
(2) **tgroupstyle** - A unique table group style defined in the style sheet (optional).
(3) **colsep** - Default for all items in this table group. If non-zero, display the internal column rulings to the right of each item. If zero, do not display column rulings. Ignored for the last column, where the frame setting applies. If no value is entered, inherited from table (optional).
(4) **rowsep** - Default for all items in this table group. If non-zero, display the internal row rulings below each item. If zero, do not display row rulings. Ignored for the last row, where the frame setting applies. If no value is entered, inherited from table.
(5) **align** - Text horizontal position within the column. If no value is entered the default value is flush left alignment.
  (a) “LEFT” - Alignment is flush left.
  (b) “RIGHT” - Alignment is flush right.
  (c) “CENTER” - Alignment is centered.
  (d) “JUSTIFY” - Alignment is right/left justified.
  (e) “CHAR” - Alignment is on the left most of the character specified in attribute “CHAR” and position by attribute “CHAROFF”.
(6) **charoff** - For attribute **align** with a value “CHAR”, percent of the current width to the left of the alignment character. If no value is entered the default value is 50%.
(7) **char** - For attribute **align** with a value “CHAR”, the value is aligned on the first character occurrence. If no value is entered the default value is a blank character.

42.1.1.2.1 Column specification ***<colspec>***. Column specification provides the details of the column in a table group. Each column of the table group has it's own ***<colspec>*** and the characteristics default is inherited from either the table group or table heading. When column numbers **colname** are used, the order number of the columns implicitly starts at 1. The attribute **colname**is provided when used in any ***<spanspec>*** and ***<entry>***. A ***<colspec>*** contained in a table head should be complete for all columns. It overrides those on the containing table group ***<tgroup>*** and applies to just the table head ***<thead>***. If the table head doesn't specify it's ***<colspec>*** then the table head inherit the ***<colspec>***(s) of the containing table group ***<tgroup>*** (or the prior table group is used). The ***<colspec>***(s) from the containing table group apply to table body ***<tbody>***. If the ***<colspec>*** element is not used, the table columns size will be equally divided based on the number of columns specified in the **cols** attribute in the ***<tgroup>***. If a ***<colspec>*** is not provided for every column, the table application will set the parameters for columns without a ***<colspec>***. For a four column table that is 6.5 inches wide with only one ***<colspec>*** with a **colwidth** (see (6) below) of .5 inches, the remaining three columns would have a 'colwidth' of 2 inches each. 6.5 minus the .5 inch in the single ***<colspec>*** equals 6 inches. This is divided by the three remaining columns.

a. The components are:

(1) Column specification ***<colspec>*** is EMPTY and the pertaining information is entered through its attributes.

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b. DTD fragment for column specification ***<colspec>***:

```
<!ELEMENT colspec EMPTY>
<!ATTLIST colspec
    colnum       CDATA                    #IMPLIED
    colname      NMTOKEN                  #IMPLIED
    align        (left | right |
                  center | justify |
                  char)                   "center"
    charoff      CDATA                    #IMPLIED
    char         CDATA                    #IMPLIED
    colwidth     CDATA                    #IMPLIED
    colsep       (0|1)                    #IMPLIED
    rowsep       (0|1)                    #IMPLIED>
```

c. Unique attributes:

(1) **colnum** - Identifies the column number, counting from 1 at left of the table.

(2) **colname** - Specifies name of column, used to specify the position in a row, or the start or end of a horizontal span of columns.

(3) **align** - Text horizontal position within the column. If no value is entered the default value is flush left alignment.

(a) “LEFT” - Alignment is flush left.

(b) “RIGHT” - Alignment is flush right.

(c) “CENTER” - Alignment is centered.

(d) “JUSTIFY” - Alignment is right/left justified.

(e) “CHAR” - Alignment is on the left most of the character specified in attribute “CHAR” and position by attribute “CHAROFF”.

(4) **charoff** - For attribute **align** with a value “CHAR”, percent of the current width to the left of the alignment character. If no value is entered the default value is 50%.

(5) **char** - For attribute **align** with a value “CHAR”, the value is aligned on the first character occurrence. If no value is entered the default value is a blank character.

(6) **colwidth** - Enter either a proportional measure of the form number*, i.e., "5*" for 5 times the proportion, or "*" (="1*"); or, fixed measure, i.e., 2pt for 2 points, 3pi for 3 picas, 4in for 4 inches. Coefficients are positive integers.

(7) **colsep** - Default for all items in this table group. If non-zero, display the internal column rulings to the right of each item. If zero, do not display column rulings. Ignored for the last column, where the frame setting applies. If no value is entered, inherited from table.

(8) **rowsep** - Default for all items in this table group. If non-zero, display the internal row rulings below each item. If zero, do not display row rulings. Ignored for the last row, where the frame setting applies. If no value is entered, the value is inherited from table.

42.1.1.2.1.1 Examples of how column specification is used. Two examples of the use of the attributes of column specification and how the values of the attributes affect the columns. The difference between the examples is the value of the attribute **align**. When the attribute **align** contains the value of "char", the percent of the current width to the left of the alignment character is provided through the attribute **charoff**.

a. An example of the column specification element where the information is entered through its attributes to define the column. The attributes describe that the **colnum** is the "1" column of the table, its **colname** is "col1", the **align** of the data is "center", the column width **colwidth**is proportional (*) of ".5*in", column separator **colsep** value "1" means display the internal column rulings to the right of entry and the number "1" in **rowsep** means display the internal horizontal row ruling below each entry unless overridden by the corresponding attributes in a child element such as a cell entry ***<entry>***.

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<colspec colnum="1" colname="col1" align="center" colwidth=".5*in" colsep="1" rowsep="1"/>

b. An example of the column specification element when the attribute **align** has the value of "char". The attributes describe that the **colnum** is the "second" column of the table, its **colname** is "col2", with **charoff** 25% of the **colwidth** of fixed column width of "1.5in" to be set aside to align any text to the left of the first capital letter "C" encountered in the cells in that column. The column will have both **colsep** column and **rowsep** row separators turned on unless overridden by the corresponding attributes in a child element such as the body, or a cell.

<colspec colnum="2" colname="col2" align="C" charoff="25" char="C" colwidth="1.5in" colsep="1" rowsep="1"/>

42.1.1.2.2 Spanned column specification ***<spanspec>*** . Spanned column specification defines a horizontal span of columns and associated attributes that can subsequently be referenced by its **spanname** for repeated use in ***<entry>***(s) in different ***<row>***(s). A ***<spanspec>*** is defined in its context ***<tgroup>***. (The ***<spanspec>*** from the containing ***<tgroup>*** apply to ***<tbody>***.) Spanned column specification uses column names from ***<colspec>*** defined in that context. It gets default values from ***<colspec>*** which provides the **namest** and **nameend** to identify the first and last columns. The spanning is from left column to right column order. The attribute colunm name **colname** is used rather than **colnum** in identifying ***<spanspec>***. The attribute **colname** of the element column specification ***<colspec>*** is when there is a revision the names are independent which may change the number of inserted/deleted columns. When such redefinition occurs, any spanning should be by **namest** and **nameend**. The preferred method that provides better control of spanning columns is to define the spanning in the entry ***<entry>*** of the first column using **namest** and **nameend**.

a. The components are:
   (1) Column specification ***<spanspec>*** is EMPTY and the pertaining information is entered through its attributes.

b. DTD fragment for column specification ***<spanspec>***:

```
<!ELEMENT spanspec EMPTY >
<!ATTLIST spanspec
    namest     CDATA                  #REQUIRED
    nameend    CDATA                  #REQUIRED
    spanname   CDATA                  #REQUIRED
    align      (left | right |
                center | justify |
                char)                  "center"
    charoff    CDATA                 #IMPLIED
    char       CDATA                 #IMPLIED
    colsep     (0 | 1)               #IMPLIED
    rowsep     (0 | 1)               #IMPLIED>
```

c. Unique attributes:
   (1) **namest** - Specifies name of the left most column of span; can be used in combination with "nameend" as an alternative to "spanname." Names are identified in ***<colspec>*** of the current ***<tgroup>***.
   (2) **nameend** - Specifies name of the right most column of span. Names are identified in ***<colspec>*** of the current ***<tgroup>***.
   (3) **spanname** - Specifies name of the spanned columns, used to specify the position in a row.
   (4) **align** - Text horizontal position within the column. If no value is entered the default value is center alignment.
      (a) “LEFT” - Alignment is flush left.
      (b) “RIGHT” - Alignment is flush right.
      (c) “CENTER” - Alignment is centered.

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(d) “JUSTIFY” - Alignment is right/left justified.
(e) “CHAR” - Alignment is on the left most of the character specified in attribute “CHAR” and position by attribute “CHAROFF”.

(5) **charoff** - For attribute **align** with a value “CHAR”, percent of the current width to the left of the alignment character. If no value is entered the default value is 50%.
(6) **char** - For attribute **align** with a value “CHAR”, the value is aligned on the first character occurrence. If no value is entered the default value is a blank character.
(7) **colsep** - Default for all items in this table group. If non-zero, display the internal column rulings to the right of each item. If zero, do not display column rulings. Ignored for the last column, where the frame setting applies. If no value is entered, inherited from table.
(8) **rowsep** - Default for all items in this table group. If non-zero, display the internal row rulings below each item. If zero, do not display row rulings. Ignored for the last row, where the frame setting applies. If no value is entered, the value is inherited from table.

42.1.1.2.2.1 Example of how a spanned column specification is used. An example of a spanned column specification where the information is entered through its attributes to define the spanning of the columns. The attributes describe that the column spanning starts **namest** with the column named "col1" and ends **nameend** with the column named "col3", the name for the spanned column **spanname** is "span1", the alignment of the data is "left", and the number "1" in row separator means display the internal horizontal row ruling below its entry.

<spanspec namest="col1" nameend="col3" spanname="span1" align="left" rowsep="1"/>

42.1.1.2.3 Table head ***<thead>*** . Table head identifies the heading in a table and is a child element of table group ***<tgroup>***. Headings are displayed at the top of the table and again at the top of any continuation after a physical break between rows in a table body ***<tbody>***. Any occurrence of column specification ***<colspec>*** to redefine a column redefines the full set with its own name space and using the same total width as the enclosing ***<tgroup>***. (An implicit default ***<colspec>*** with a width of "1.0in" is assumed for any ***<colspec>*** not explicitly redefined.) These redefined ***<colspec>***(s) apply only to the contained entry ***<entry>***(s). References to **spanname** in such ***<entry>***(s) are not supported; if spanning is required, use the attributes **namest** and **nameend** of the element span specification ***<spanspec>***. If no ***<colspec>*** occurs in this ***<thead>***, the ***<colspec>*** and ***<spanspec>*** sets of the containing ***<tgroup>*** apply.

a. The components are:
(1) Column specification ***<colspec>*** (optional – zero or more). Column specification provides the details of the column and when new column's specifications is provided, it replaces a previous one in a table heading. (see 42.1.1.2.1)
(2) Row ***<row>*** (required – one or more). The element provides a horizontal collection of cells that identifies information in a table group. (see 42.1.1.2.3.1)

b. DTD fragment for table head ***<thead>***:

```
<!ELEMENT thead (colspec*, row+)>
<!ATTLIST thead
    valign       (top | middle | bottom )     "bottom" >
```

c. Unique attribute:
(1) **valign** - Specifies the vertical alignment of content within the entry.

42.1.1.2.3.1 Row ***<row>***. Row is a horizontal collection of cells that identifies information in a table group ***<tgroup>*** of a table. The width of a row is the width of the table. The depth of a row is the depth of the deepest cell. Default values come from the table ***<table>***, table group***<tgroup>*** and column specification ***<colspec>***s attribute list values for like-named attributes.

a. The components are:
(1) Entry(s) ***<entry>*** (required – one or more). (see 42.1.1.2.3.1.1)

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b. DTD fragment for column specification ***<row>***:

```
<!ELEMENT row (entry)+>
<!ATTLIST row
    rowsep            (0|1)      #IMPLIED
    applicable        IDREFS     #IMPLIED

    assocfig          IDREFS                              #IMPLIED
    changeref         IDREFS                              #IMPLIED
    comment           CDATA                               #IMPLIED
    delchlvl          CDATA                               #IMPLIED
    idref             IDREFS                              #IMPLIED
    inschlvl          CDATA                               #IMPLIED
    security          (uc | fouo |c | s | ts)           #IMPLIED
    skilltrk          CDATA                               #IMPLIED>
```

c. Unique attributes:
   (1) **applicable** Applicablity reference (optional) (see 49.2.1.b.(1)).
   (2) **rowsep** - Default for all items in this table group. If non-zero, display the internal row rulings below each item. If zero, do not display row rulings. Ignored for the last row, where the frame setting applies. If no value is entered, inherited from table.

d. Common attributes:
   (1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (3) **comment** – Change information (optional) (see 49.2.9).
   (4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (6) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (7) **security** – Security classification (optional) (see 49.2.13).
   (8) **skilltrk** – Skill level (optional) (see 49.2.2).

42.1.1.2.3.1.1 Entry ***<entry>***. Entry in a table identifies as an entry in a row of a ***<thead>*** and table body ***<tbody>***. Data entered in the cells of an XML table is entered as content of the entry element. The intersection of row location and column location as specified through the entry element locate a table cell, including spanning cells. Default values come from the column specification ***<colspec>*** in table group ***<tgroup>*** or in table head ***<thead>***, and in table body ***<tbody>*** row attribute list values for like-named attributes. A spanning entry not specified by a ***<spanspec>*** gets its defaults from its starting column. Preferred method of spanning columns horizontally is to used ***<entry>*** attributes name start **namest** and name end **nameend**. To straddle specific rows use ***<entry>***'s attribute **morerows**.

a. The components of a table entry ***<entry>*** consist of either zero or more of the following:
   (1) Parsable data (#PCDATA) (optional – zero or more). The element provides the text for the TM number. (see 9.3.2)
   (2) Emphasis ***<emphasis>*** (optional – zero or more). The element is used to emphasize the text of the entry. (see 49.1.3.1)
   (3) Subscript ***<subscrpt>*** (optional – zero or more). The element is used to format the text as subscript. (see 49.1.3.4)
   (4) Superscript ***<supscrpt>*** (optional – zero or more). The element is used to format the text as superscript. (see 49.1.3.5)
   (5) Cross Reference ***<xref>*** (optional – zero or more). The element is used to reference the work package sequence number, figure, table, step(s), etc. (see 46.2.2)
   (6) External Reference ***<extref>*** (optional – zero or more). The element is used to reference to external document information. (see 46.2.1)

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(7) Linking ***<link>*** (optional – zero or more). The element provides a capability to reference internal or external targets. (see 46.2.3)
(8) Content Sensitive Help ***<help.info>*** (optional – zero or more). The element is used for "help information" about the technical data used in frame-based manuals. (see 48.3.2.4)
(9) Controls and Indicators Entry ***<ctrlind>*** (optional – zero or more). The element provides the name of the control or indicator. (see 49.1.4.3)
(10) Reading On ***<ctrlind-val>*** (optional – zero or more). – The element provides a reading from a control or indicator. (see 49.1.4.4)
(11) Department of Defense Ammunition Code ***<dodac>*** (optional – zero or more). The element provides identification of an ammunition type.(see 49.1.4.5)
(12) Lubricant ***<lubricant>*** (optional – zero or more). The element provides identification of a lubricant.(see 49.1.4.16)
(13) Symbol ***<symbol>*** (optional – zero or more). The element provides a graphic symbol not found in standard ISO character sets that is inserted as a graphic in text.(see 44.3.1)
(14) Torque Value or Limit ***<torque>*** (optional – zero or more). The element provides a torque value or limit embedded in the text or table entry. (see 49.1.4.24)
(15) Voltage Measurement ***<voltage>*** (optional – zero or more). The element provides the identification of a critical voltage measurement. (see 49.1.4.25)
(16) Null ***<null>*** (optional – zero or more). The element specifies that it contains no content and marking is identified by it's attribute. (see 49.1.3.2)
(17) Text Change Information ***<change>*** (optional – zero or more). The element provides the scope and type of changed information that is indicated by being enclosed within start and end tag. (see 49.1.3.7)
(18) Reduced Paragraph ***<trim.para>*** (optional – zero or more). The element provides the same usage as a paragraph element, but contains a reduce content model. (see 49.1.1.7)
(19) Graphic ***<graphic>*** (optional – zero or more). The element provides a graphic, which is contained in an external entity. (see 44.2)
(20) First level step in a procedure ***<step1>*** (optional – zero or more). The element provides the first level step. (see 33.3).
(21) Conditional First Level Step***<step1-alt>*** (optional – zero or more) The element provides multiple element options depending on the result from the precondition state (variable) information. (see 48.2.1).
(22) Figure ***<figure>*** (optional – zero or more). The element provides a graphic illustration, multi-sheet illustrations, graphic chart, or text illustrations. (see 44.1.1)
(23) Procedure ***<proc>*** (optional – zero or more) (see 33.1PROCEDURES/TASKS for details on tagging.) (see 33.2)
(24) Notes ***<note>*** (optional – zero or more). The element provides highlights to essential procedures, conditions, or statements or conveys important instructional data to the user. (see 41.1.3)
(25) Line Break ***<brk>*** (optional – zero or more). Causes a line break in the narrative. (see 49.1.3.3)

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b. DTD fragment for column specification ***<entry>***:

```
<!ELEMENT entry (#PCDATA | emphasis | subscrpt | supscrpt | xref |
 extref | link | help.info | indxref | term | term.def | callout |
 ftnote | ftnref | ctrlind | ctrlind-val | dodac | lubricant | symbol |
 torque | voltage | null | change| seqlist | randlist | deflist |
 trim.para | graphic | step1 |step1-alt | figure | proc | note | brk)*>
<!ATTLIST entry
    colname     NMTOKEN                                  #IMPLIED
    spanname    CDATA                                    #IMPLIED
    namest      NMTOKEN                                  #IMPLIED
    nameend     NMTOKEN                                  #IMPLIED
    morerows    CDATA                                    '0'
    colsep      (0|1)                                    #IMPLIED
    rowsep      (0|1)                                    #IMPLIED
    rotate      (yes | no)                               'no'
    valign      (top | bottom
                 | middle                                'top'
   align       (left | right | center |
                justify | char )                         #IMPLIED
    charoff     CDATA                                    #IMPLIED
    char        CDATA                                    #IMPLIED

    assocfig          IDREFS                             #IMPLIED
    changeref         IDREFS                             #IMPLIED
    comment           CDATA                              #IMPLIED
    delchlvl          CDATA                              #IMPLIED
    idref             IDREFS                             #IMPLIED
    inschlvl          CDATA                              #IMPLIED
    security          (uc | fouo |c | s | ts)           #IMPLIED
    skilltrk          CDATA                              #IMPLIED
```

c. Unique attributes:

(1) **colname** - Specifies name of column. Omit if spanname is present.

(2) **spanname** - Specifies name of the spanned columns, used to specify the position in a row.

(3) **namest** - Specifies name of the left most column of span; can be used in combination with "nameend" as an alternative to "spanname." Names are identified in ***<colspec>***, of the current table group ***<tgroup>***.

(4) **nameend** - Specifies name of the right most column of span. Names are identified in column specification ***<colspec>***, of the current table group ***<tgroup>***.

(5) **morerows** - Specifies number of additional rows covered by a vertical straddle.

(6) **colsep** - If one, display the internal column vertical ruling to the right of each item; if zero, do not display it. Ignored for the last column, where the frame setting applies.

(7) **rowsep** - Default for all items in this table group. If non-zero, display the internal row rulings below each item. If zero, do not display row rulings. Ignored for the last row, where the frame setting applies. If no value is entered, inherited from table.

(8) **rotate** - Rotations are not additive to those specified in style sheet. Content is either in the orientation of the table (value is zero) or 90 degrees counter clockwise to table orientation (value is one).

(9) **valign** - Specifies the vertical alignment of content within the entry.

(10) **align** - Text horizontal position within the column. If no value is entered the default value is flush left alignment.

(a) “LEFT” - Alignment is flush left.

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(b) “RIGHT” - Alignment is flush right.
(c) “CENTER” - Alignment is centered.
(d) “JUSTIFY” - Alignment is right/left justified.
(e) “CHAR” - Alignment is on the left most of the character specified in attribute “CHAR” and position by attribute “CHAROFF”.

(11) **charoff** - For attribute **ALIGN** with a value “CHAR”, percent of the current width to the left of the alignment character. If no value is entered the default value is 50%.
(12) **char** - For attribute **align** with a value “CHAR”, the value is aligned on the first character occurrence. If no value is entered the default value is a blank character.

d. Common attributes:
(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

42.1.1.2.3.2 Example of a table heading. An example of the XML for spanning a table heading in a table. Its attribute vertical alignment **valign** specifies alignment of content in the middle of the header. The table header is using the element column specification ***<colspec>*** to define the columns of the table head in the table. The first ***<colspec>*** attributes describe that the column is the "1" column of the table, its name is "col1", the alignment of the data is center, the width of the column is "1.0" inch wide, the number "1" in column separator means display the internal column rulings to the right of entry and the number "1" in row separator means display the internal horizontal row ruling below each entry. The table headings are: first column "ITEM", second column "NAME" and third column "DESCRIPTION".

```
<thead  valign="middle">
<colspec  colnum="1"  colname="col1"  align="center"  colwidth="1.0in"/>
<colspec  colnum="2"  colname="col2"  align="center"  colwidth="2.0in"/>
<colspec  colnum="3"  colname="col3"  align="center"  colwidth="3.0in"/>
<row>
<entry>ITEM
</entry>
<entry>NAME
</entry>
<entry>DESCRIPTION
</entry>
<row>
</thead>
```

42.1.1.2.4 Table body ***<tbody>***. The body of a table group contains one table body ***<tbody>***. If there are multiple table groups in a table then each table group consist of one table body. The columns specification ***<colspec>*** of the enclosing table group ***<tgroup>*** is the default specification for the enclosed table body ***<tbody>***. Table group's attribute vertical alignment **valign** specifies alignment of content of the header. The default for **valign** is "top" unless specified.

a. The components are:
(1) Row ***<row>*** (required – one or more). The element provides a horizontal collection of cells that identifies information in a table group. (see 42.1.1.2.3.1)

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b. DTD fragment for table head ***<tbody>***:

```
<!ELEMENT tbody (row)+>
<!ATTLIST tbody
    valign  (top | middle | bottom )  'top' >
```

c. Unique attribute:

(1) **VALIGN** - Specifies the vertical alignment of content within the entry.

42.2 Tabular material <tabmat>. Tabular material ***<tabmat>*** allows data to be formatted structurally as a table but not identify as a table. The element is used as an outer wrapper for table group to distinguish it as tabular material rather then a table. In some instances the data needs a table format but not regarded as a table. Tabular material does not contain a title and does not contain line rulings. It does not get listed in the table TOC with the tables.

a. The components are:

(1) Table group***<tgroup>*** (required). The element provides the data structure of a table.(see 42.1.1.2)

b. DTD fragment:

```
<!ELEMENT tabmat (tgroup)>
```

c. No attributes.

42.2.1 Example of UOC code tabular material. The following example is of a tabular material XML instance to display the UOC code in the “SPECIAL INFORMATION” section of an Introduction Work Package ***<introwp>*** in a RPSTL. Use the element table material ***<tabmat>*** to format the data for the UOC code. The element table material contains the element table group ***<tgroup>*** that contains the attributes columns **cols="2"** and alignment **align="left"**. The attribute columns state that the tabular material has two columns and the alignment of the text in the entry of the columns is aligned left. Table group contains a table header ***<thead>*** that contains the attribute vertical alignment **valign="bottom"**. Using a table header will bold the text and will align the text close to the bottom of the cell. The data in the heading is underlined. To accomplish the continuous underline use the attribute **rowsep="1"** with a value of one in the table header row. The value "1" puts in a line ruling under the header. The table header in the first column is aligned left which defaults from the attribute **align="left"** set in table group. The second column is centered over the data in the tabular material. Set the attribute in the second entry of the table header row for center alignment **align="center"** so the data in just that entry is aligned center and overrides the alignment that is stated in table group ***<tgroup>***. Table group requires a table body ***<tbody>*** that contains the data rows***<row>***. A row contains two entry(s) which is stated in the attribute **cols="2"** in ***<tgroup>***.

a.

```
<tabmat>
<tgroup  cols="2"  align="left">
<thead  valign="bottom">
<row  rowsep="1">
<entry>Code
</entry>
<entry  align="center">Used  On
</entry>
</row>
</thead>
<tbody>
<row>
<entry>PAA
</entry>
<entry>Model  M114
```


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```
</entry>
</row>
<row>
<entry>PAB
</entry>
<entry>Model  M114A
</entry>
</row>
<row>
<entry>PAC
</entry>
<entry>Model  M114B
</entry>
</row>
</tbody>
</tgroup>
</tabmat>
```

b. The formatted output of how the tabular material would display the UOC code.

Code	Used On
PAA	Model M114
PAB	Model M114A
PAC	Model M114B

FIGURE 502. An example of a tabular material to display the UOC code.

42.3 Standard information tables. MIL-STD-40051-2 contains tables that consist of defined standard information and are designated as "Standard Information Tables". Some table content is mandatory while other content is optional. MIL–STD–2361 provides a unique set of content tags to define the specific data in the standard information tables. The stylesheet output display of a standard information table would display the same as if it was created in a CALS table but it consist of contented data. MIL-STD-40051-2 standard tables have no deviations to the number of columns and the titles in the column headings. A table heading for a MIL-STD-40051-2 standard information table is specified in the element ***<std.table.header>*** of the standard table.

42.3.1 Standard table header ***<std.table.header>***. A table header for a standard table is specified in the element ***<std.table.header>*** in its standard table. A standard table contains one or more standard table rows. MIL–STD–2361 contains boilerplates for MIL–STD–40051–1/–2 standard table headers. See 50 for further information on boilerplates and the entities for the standard table headers.

a. The component are:

(1) Standard table row(s) ***<std.table.row>*** (required – one or more). The element provides a standard information row heading. (see 42.3.1.1

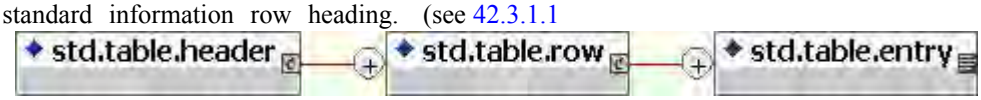

FIGURE 503. Standard table header <std.table.header> DTD hierarchy.

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b. DTD fragment for standard information table header ***<std.table.header>***:

```
<!ELEMENT std.table.header (std.table.row+)>

<!ATTLIST std.table.header
    cols     CDATA          #REQUIRED
    rowsep   CDATA          '1'>
```

c. Unique attributes:
   (1) **cols** - Number of columns in the heading. (required).
   (2) **rowsep** - Row separator. When value is one, the internal horizontal row ruling below each item is displayed. If zero, do not display it.

d. An example of a standard table header XML instance for an Expendable and Durable Items List table.

```
<std.table.header  cols="5"  rowsep="1">
<std.table.row>
<std.table.entry  colnum="1">(1)
</std.table.entry>
<std.table.entry  colnum="2">(2)
</std.table.entry>
<std.table.entry  colnum="3">(3)
</std.table.entry>
<std.table.entry  colnum="4">(4)
</std.table.entry>
<std.table.entry  colnum="5">(5)
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry  colnum="1">Item  No
</std.table.entry>
<std.table.entry  colnum="2">Level
</std.table.entry>
<std.table.entry  colnum="3">National  Stock  Number  (NSN)
</std.table.entry>
<std.table.entry  colnum="4">Item Name, Description, Part Number/(CAGEC)
</std.table.entry>
<std.table.entry  colnum="5">U/I
</std.table.entry>
</std.table.row>
</std.table.header>
```

e.

Table 1. Expendable and Durable Items List

(1)	(2)	(3)	(4)	(5)
Item		National Stock		
No.	Level	Number (NSN)	Item name, Description, Part Number/(CAGEC)	U/I

FIGURE 504. An example of a stylesheet formatted output of a standard table header for the Expendable and Durable Items List table.

42.3.1.1 Standard table row ***<std.table.row>***. The element provides the ability to enter a standard information row heading. In page-base, the element is equivalent to a "row" element in a table.

a. The components are:

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(1) Standard table entry(s) ***<std.table.entry>*** (required – one or more). The element provides a standard information cell entry heading. (see 42.3.1.1.1

b. DTD fragment for standard table row ***<std.table.row>***:

```
<!ELEMENT std.table.row (std.table.entry+)>
```

c. No attributes.

42.3.1.1.1 Standard table entry ***<std.table.entry>***. The element provides the ability to enter a standard information cell entry heading. In page-base, the element is equivalent to an "entry" element in a table. When a column is to be spanned, use the attribute column span **cols.span**.

a. The component of a ***<std.table.entry>*** is:

(1) Parsable data (#PCDATA) (required). The element provides the text for the TM number. (see 9.3.2)

b. DTD fragment for standard table entry ***<std.table.entry>***:

```
<!ELEMENT std.table.entry (#PCDATA)>

<!ATTLIST std.table.entry
    cols.span   CDATA      #IMPLIED
    colnum      CDATA      #REQUIRED
    rowsep      CDATA      #IMPLIED >
```

c. Unique attributes:

(1) **cols.span** - Number of column the cell horizontally spans. To span a column enter the number of columns to be spanned. (optional)

(2) **cols** - Number of columns in the heading. (required)

(3) **rowsep** - Row separator. When value is one, the internal horizontal row ruling below each item is displayed. If zero, do not display it. (optional)

42.3.2 Content tags in a standard table. A standard table contains content tags that have names indicating their data content. Content tags embody the functional requirements contained in the respective functional requirements standards and specifications as found in the standard tables of MIL-STD-40051. Content tags have names indicating their data content, such as Standard Mobilization Requirement table ***<mobiltab>*** containing the content tag mobilization requirements entry tag ***<mobil-entry>***. The ***<mobil-entry>*** is a content tag to wrap the data in the row(s) of the Mobilization Requirement table. Mobilization requirements entry tag ***<mobil-entry>*** is the equivalent to a row tag ***<row>*** in a CALS table. By having its own unique set of content tags the table information can be chuck into data content so it can facilitate information access, sharing, reuse, management, control and change.

a. The following table shows the tag of a CALS table equivalent to the tag of a Standard table.

TABLE XLI. CALS table tag equivalent to a standard mobilization requirement table tag.

CALS TABLE	STANDARD TABLE
<table>	<mobiltab>
<title>	<title>
<tgroup>	
<colspec>	
<thead>	<std.table.header>
<row>	<std.table.row>
<entry>	<std.table.entry>
<row>	<mobil-entry>

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Table XLI CALS table tag equivalent to a standard mobilization requirement table tag – Continued.

<entry>	<xref>
<entry>	<actionreq>

42.3.3 Examples of standard information tables. Below are some examples of standard information tables. The tables contain standard information headings which can be accessed as boilerplates through MIL–STD–2361 DTD. For further examples of standard information tables see Table which reference where they are located in MIL-HDBK-2361.

a. An example of a standard table XML instance for Mobilization requirement table. The boilerplate heading parameter entity ***%mobilwp.mobiltab.tab.std-header;*** for a Depot mobilization requirements table is provided with MIL–STD–2361 DTD.

```
<mobiltab id="m00524-55-1520-250-2">
<std.table.header cols="2" rowsep="1">
<std.table.row>
<std.table.entry colnum="1">WORK PACKAGE
</std.table.entry>
<std.table.entry colnum="2">ACTION
</std.table.entry>
</std.table.row>
</std.table.header>
<mobil-entry>
<xref pagelocid="m00524-55-1520-250-1" wpid"m00524-55-1520-250-1"/>
<actionreq>Add “Depending on the urgency of requirements, availability of
materials, and fabrication lead time, provisions of this work package
may be relaxed. When that occurs, any practical method may be used
to inscribe or attach the data to the equipment, i.e., decals.”
</actionreq>
</mobil-entry>
<mobil-entry>
<xref pagelocid="m00524-55-1520-250-1" wpid"m00524-55-1520-250-1"/>
<actionreq>Cleaning, Step 3. Add “Clean only to the extent
necessary to perform preshop analysis.”
</actionreq>
</mobil-entry>
</mobiltab>
```

b.

Table 1. MOBILIZATION REQUIREMENTS

WORK PACKAGE	ACTION
WP 0088	Add “Depending on the urgency of requirements, availability of materials, and fabrication lead time, provisions of this work package may be relaxed. When that occurs, any practical method may be used to inscribe or attach the data to the equipment, i.e., decals.”
WP 0090	Cleaning, Step 3. Add “Clean only to the extent necessary to perform preshop analysis.”

FIGURE 505. An example of a stylesheet formatted output for the Mobilization standard information table.

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c. An example of a standard table XML instance for a PMCS table containing the "Man-Hour" column. The boilerplate heading parameter entity ***%pmcswp.pmcstable.std-header ;*** for a PMCS table containing the "Man-Hour" column is provided with MIL–STD–2361 DTD.

```
<pmcstable id="m00048-9-2320-365-46">
<title>Preventive Maintenance Checks and Services
</title>
<std.table.header cols="6" rowsep="1">
<std.table.row>
<std.table.entry colnum="1">
</std.table.entry>
<std.table.entry colnum="2">
</std.table.entry>
<std.table.entry>
</std.table.entry>
<std.table.entry colnum="3">ITEM TO BE
</std.table.entry>
<std.table.entry colnum="4">
</std.table.entry>
<std.table.entry colnum="5">
</std.table.entry>
<std.table.entry colnum="6">EQUIPMENT
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry colnum="1">ITEM
</std.table.entry>
<std.table.entry colnum="2">
</std.table.entry>
<std.table.entry colnum="3">MAN-
</std.table.entry>
<std.table.entry colnum="4">CHECKED OR
</std.table.entry>
<std.table.entry colnum="5">
</std.table.entry>
<std.table.entry colnum="6">NOT READY/
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry colnum="1">NO.
</std.table.entry>
<std.table.entry colnum="2">INTERVAL
</std.table.entry>
<std.table.entry colnum="3">HOURS
</std.table.entry>
<std.table.entry colnum="4">SERVICED
</std.table.entry>
<std.table.entry colnum="5">PROCEDURE
</std.table.entry>
<std.table.entry colnum="6">AVAILABLE IF:
</std.table.entry>
</std.table.row>
</std.table.header>
<pmcs-entry id="m00048-9-2320-365-48">
<itemno>1
</itemno>
```


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<interval>Semiannual
</interval>
<manhours>
</manhours>
<checked>Preservice Checks
</checked>
<pmcsproc>
<pmcsstep1id="m00048-9-2320-365-97">
<specpara>
<para>Notice if starter engages smoothly and turns engine at normal cranking speed.
</para>
</specpara>
<eqpnotavail>
<trim.para>Starter inoperative or makes excessive grinding sound.
</trim.para>
</eqpnotavail>
</pmcsstep1>
</pmcsproc>
</pmcs-entry>
<pmcs-entry id="m00048-9-2320-365-49">
<itemno>2
</itemno>
<interval>Semiannual
</interval>
<manhours>
</manhours>
<checked>WHEELS, HUBS, AND CTIS
</checked>
<pmcsproc>
<pmcsstep1 id="m00048-9-2320-365-107">
<specpara>
<warning>
<trim.para>Completely deflate tires before removing from axles only if there is obvious damage to wheel components. Removing damaged tires from axles without deflating tires may cause wheel components to separate. Failure to comply may result in serious injury or death to personnel.
</trim.para>
</warning>
<para>Check wheels for obvious cracks around lug holes. If cracks are found, repair wheel
<xref wpid="m00226-9-2320-365"/>.
</para>
</specpara>
<eqpnotavail>
<trim.para>Cracks are found around lug holes.
</trim.para>
</eqpnotavail>
</pmcsstep1>
<pmcsstep1 id="m00048-9-2320-365-108">
<para>Replace any loose or damaged wheel studs (
<xref wpid="m00227-9-2320-365"/>).
</para>
<eqpnotavail>
<trim.para>More than one lugnut or wheel stud is damaged or missing.
</trim.para>

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Check the source to verify that this is the current version before use.

```
</eqpnotavail>
</pmcsstep1>
<pmcsstep1  id="m00048-9-2320-365-109">
<para>Check  for  oil  leaks.
</para>
<eqpnotavail>
<trim.para>Class  III  leak  is  evident.
</trim.para>
</eqpnotavail>
</pmcsstep1>
</pmcs-entry>
</pmcstable>
```

Table 1. Preventive Maintenance Checks and Services

ITEM NO.	INTERVAL	MAN-HOUR	ITEM TO BE CHECKED OR SERVICED	PROCEDURE	EQUIPMENT NOT READY IF APPLICABLE:
1	Semiannual		Preservice Checks	Notice if starter engages smoothly and turns engine at normal cranking speed.	Starter inoperative or makes excessive grinding sound.
2	Semiannual		WHEELS, HUBS, AND CTIS	a. Check wheels for obvious cracks around lug holes. If cracks are found, repair wheel WP 0045.	Cracks are found around lug holes.
				b. Replace any loose or damaged wheel studs .	More than one lug nut or wheel stud is damaged or missing.
				c. Check for oil leaks.	Class III leak is evident.

FIGURE 506. An example of a stylesheet formatted output for the PMCS standard information table.

42.3.4 List of standard information tables. The following is a list of the MIL–STD–2361 defined standard information tables in accordance with MIL-STD-40051-2. The table headings for these standard tables can be accessed through boilerplates contained in MIL–STD–2361 DTD (see 50). The list of tables includes the name of the standard information table, a reference where it is explained in the MIL–STD–2361 Hand Book and the name of the work package that contains it.

TABLE XLII. MIL–STD–2361 Standard Information Tables.

Standard Table	Work Package
Controls and Indicators ***<ctrlindtab>*** (35.2.1)	Controls and Indicators work package ***<ctrlindwp>***
Criteria Inspection for Packaging ***<crit.insp.tab>***. (37.3.1.1.2)	Service Upon Receipt Work Package ***<surwp>***
Checking Unpacked Equipment Components ***<pecul.insp.tab>*** (37.3.1.1.3)	Service Upon Receipt Work Package ***<surwp>***
Preventive Maintenance Checks and Services (PMCS) ***<pmcstable>*** (37.6.1)	Preventive Maintenance Checks and Services (PMCS) ***<pmcswp>***

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Table XLII MIL–STD–2361 Standard Information Tables – Continued.

Standard Table	Work Package
Classification of Material Defects ***<defect.tab>*** (37.7.1.1.1)	Maintenance Work Packages ***<maintwp>***
Overhaul and Retirement Schedule ***<orsch.tab>*** (37.7.1.2.1)	Maintenance Work Packages ***<maintwp>***
Depot Mobilization Requirements ***<mobiltab>*** (37.12.2)	Depot Mobilization Requirements Work Package ***<mobilwp>***
Overhaul Inspection Procedures ***<oiptab>*** (37.12.2)	Overhaul Inspection Procedure – Work Package ***<oipwp>***
Repair Parts List ***<pi.category>*** (38.4.2.1)	Repair Parts List Work Package ***<plwp>***
Special Tools List ***<pi.category>*** (38.4.2.1)	Parts Information and Repair Parts and Special Tools List ***<stl_partswp>***
NSN Index ***<nsnindx>*** (38.5.1.1)	National Stock Number (NSN) index work package ***<nsnindxwp>***
Part Number Index ***<pnindx>*** (38.5.2.1)	Part Number Index Work Package ***<pnindxwp>***
Reference Designator Index ***<refdesindx>*** (38.5.3.1)	Reference Designator Index Work Package ***<refdesindxwp>***
Standard Maintenance Allocation Chart (MAC) ***<mac>*** (40.4.1)	Standard Maintenance Allocation Chart (MAC) ***<macwp>***
Aviation Maintenance Allocation Chart (AMAC) ***<avmac>*** (40.4.2)	Standard Maintenance Allocation Chart (MAC) ***<macwp>***
Two-Level Maintenance Allocation Chart (MAC) ***<mac>*** (40.4.1)	Standard Maintenance Allocation Chart (MAC) ***<macwp>***
Tools and Test Equipment Requirements for MAC/AMAC ***<tereqtab>*** (40.4.3)	Standard Maintenance Allocation Chart (MAC) ***<macwp>***
Remarks (MAC/AMAC) ***<remarktab>*** (40.4.4)	Standard Maintenance Allocation Chart (MAC) ***<macwp>***
Expendable and Durable Items List ***<explist>*** (40.7.2)	Expendable And Durable Items List Work Package ***<explistwp>***
Mandatory Replacement Parts List ***<mrpl>*** (40.9.1)	Mandatory Replacement Parts Work Package ***<mrplwp>***

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Table XLII MIL–STD–2361 Standard Information Tables – Continued.

Standard Table	Work Package
Component of End Items (COEI) and Basic Issue Items (BII) ***<coeitab>*** (40.5.1.2.1)	Components of end item (COEI) and basic issue items (BII) lists work package (operator only) ***<coeibiiwp>***
Additional Authorization List (AAL) ***<aal>*** (40.6.2)	AAL Work Package ***<aalwp>***
Tools Identification List ***<toolidlist>*** (40.8.2)	Tool identification list work package ***<toolidwp>***
Flight Safety Critical Aircraft Parts (FSCAP) ***<fscap>*** (40.10.2)	Critical safety items (CSI) and flight safety critical aircraft parts (FSCAP) work package ***<csi.fscap.wp>***

43 STANDARD INFORMATION.

Standard information is specified data that does not have deviation to the content requirements including the use of standard headings in tables. The data specified in standard statements and standard tables is considered standard information. Standard information contains content tags that have names indicating their data content. Content tags embody the functional requirements contained in the respective functional requirements standards and specifications. Content tags have names indicating their data content, such as Standard Mobilization Requirement table ***<mobiltab>*** containing the content tag mobilization requirements entry tag ***<mobil-entry>***. The ***<mobil-entry>*** is a content tag to wrap the data in the row(s) of the Mobilization Requirement table. Boilerplates can be used to insert the verbatim text that standard information requires (see 50).

43.1 Presentation. The presentation format is either "page oriented" or "frame-oriented". A page-based and frame-based presentation have the same content coverage. However, the display and presentation of the information is different. Different stylesheets are used for the presentation of page-based or frame-based tables. Standard information formats, hotspot capabilities, linking to other applications (i.e., ordering parts) are determined by the acquiring activity and the IETM developer.

43.1.1 Standard information formats. Standard information can be formatted in the display device as the following possibilities listed below. For further information see MIL-HDBK-1222.

a. Form. It is used to enter and submit additional information or the ability to review the data (i.e., parts ordering).
b. Page-based table. It is used to present the information that does not require or need any special interactive, but to provide reference information.
c. Frame-based tables. It is used to present information in form better for frame view (i.e., the ability to reduce categories not needed).
d. List. It is used to list just enough information (i.e., item name, part number, NSN, etc.) to identify the data to provide a detailed view of the selected item in a separate data pane.
e. Checklist. It is used to record the item was checked and completed. Additionally, the item may denote possible issues of the checked item (i.e., red highlighted item as “Not ready”, yellow highlighted item as “Comment”, button for comments, etc.). Each item may be presented completely as a table (refer to b or c) or list (refer to d). Using the list method, the next check item’s detailed information could be automatically shown after marking the checked item task is completed.

43.1.2 Page oriented standard information. MIL-STD-40051-2 contains defined standard information statements and tables that are designated as “Standard Information Tables”. Standard information tables have no deviations to the number of columns and the titles in the column headings. Some table content is mandatory while other content is optional. MIL–STD–2361 provides a unique set of content tags to define

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the specific data in the standard tables. The stylesheet output display of a standard table is same as if the standard table was tagged as a CALS table. Below are some examples of page-based presentation on standard information statements and tables. For more examples and information see MIL-HDBK-1222

43.1.2.1 Example of a page-based standard information statement export control warning. . The Export Control Warning, ***<export>*** that is used in a TM front page, contains verbatim text that can be also inserted using a boilerplate ***&notices.export;***. The standard information that is specific of ***<export>*** contains a defined title ***<title>*** and specific text ***<text>***.

a. XML document instance fragment.

<export>
<title>Warning –
<title>
<text>This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C., Sec 2751, et. seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et. seq. Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.
</text>
</export>

b. Stylesheet Output.

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TM 3–6665–339–10

TECHNICAL MANUAL

OPERATOR'S MANUAL
FOR

NUCLEAR-BIOLOGICAL-CHEMICAL RECONNAISSANCE SYSTEM (NBCRS) FOX M93A1 (NSN 6665-01–372–1303) (EIC Y60)

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HEADQUARTERS, DEPARTMENT OF THE ARMY

1 JUNE 1996

FIGURE 507. Example of a page-based standard information statement Export Control Warning.

43.1.2.2 Example of a page-based standard information national stock number index table. The standard information that is specific of this table ***<nsnindx>*** are the titles for the standard table heading ***<std.table.header>*** and the data that is requested. The specific data is the NSN number ***<nsn>*** and figure number ***<callout>*** and is wrapped with the tag – National Stock Number index entry or row***<nsnindxrow>***. The stylesheet formats the output of how the standard information tables are displayed.

a. XML document instance fragment.An example of an XML document instance fragment containing a page-based standard information National Stock Number Index table is found in 38.5.1.1.
b. Stylesheet Output.

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UNIT MAINTENANCE
NATIONAL STOCK NUMBER INDEX

STOCK NUMBER	FIG. ITEM
5365-00-003-6807	4
5935-00-005-2826	1
5315-00-012-0123	1
5310-00-016-7216	12
5340-00-021-3495	20
	15
5310-00-027-7247	2
5310-00-030-0580	21
3110-00-034-5257	5
	5
	9
	8
	5
	10
	5
	3
5305-00-038-9048	27
5310-00-045-3296	25
	20
5310-00-045-4007	5
5305-00-052-6456	2
5310-00-054-0041	3
5305-00-054-5637	9
5305-00-054-5638	11
5305-00-054-5647	15
	25
5305-00-054-5648	1
	7
5305-00-054-5649	8
	13
	3
	15
	8
	3
5305-00-054-5650	3
5305-00-054-5651	5
	5
	1
5305-00-054-5652	1
5305-00-054-5653	6
5305-00-054-6653	17
5305-00-054-6654	23
	8
5305-00-054-6655	4
	21
5305-00-054-6657	18
5305-00-054-6666	29
5305-00-054-6669	10
5305-00-054-9263	4
5305-00-056-9961	3
	16
	35
5310-00-057-0573	8
	2
	2
5310-00-058-1823	1
	4
5315-00-058-6062	18
5305-00-059-3657	14
5305-00-059-3658	3
5305-00-059-3661	26
5310-00-061-7326	19
5305-00-066-7327	37
5305-00-066-7369	1
5365-00-067-3836	17
5305-00-068-0543	9
5365-00-068-8011	2
	2
5975-00-074-2072	12
	33
5970-00-074-8780	15
5320-00-076-4071	19
5360-00-079-1713	2
5305-00-079-5835	22
	28

END OF WORK PACKAGE

0049-1

FIGURE 508. Example of a page-based standard information National Stock Number Index table.

43.1.2.3 Example of a page-based standard information two-level MAC table. The standard information that is specific of this table is the titles for the standard table heading ***<std.table.header>*** and the data that

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is requested. The specific data is the data that is wrapped with the tag MAC grouping ***<mac-group>***. The standard data contained in this grouping for a two-level MAC is group number ***<groupno>***, component/assembly ***<compassem>***, and the qualifier. The qualifier tag ***<qualify>*** is a wrapper for the specific data maintenance function ***<maintfunc>***, maintenance classification level ***<maintclass>***, tools and equipment references ***<terefs>*** and remarks code ***<remarks>***. The ***<maintclass>***element contains the table entrys: unit ***<unit>***, direct***<direct>***, general support ***<gensup>*** and DMWR/NMWR support ***<depot>***. The stylesheet formats the output of how the standard information tables are displayed.

a. XML document instance fragment.An example of an XML document instance fragment containing a page-based standard information two-level MAC table is found in 40.4.1.
b. Stylesheet Output.

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TM X-XXX-XXXX-XX 0440

UNIT MAINTENANCE

TSEC/ST-34
MAINTENANCE ALLOCATION CHART (MAC)

Table 1. MAC for TSEC/ST-34.

(1) GROUP NUMBER	(2) COMPONENT/ ASSEMBLY	(3) MAINTENANCE FUNCTION	(4) MAINTENANCE LEVEL					(5) TOOLS AND EQUIPMENT REF CODE	(6) REMARKS CODE
			FIELD			SUSTAINMENT			
			UNIT		DIRECT SUPPORT	GENERAL SUPPORT	DEPOT		
			C	O	F	H	D		
00	TSEC/ST-34	Inspect	0.1						A
		Service	0.2						B
		Replace	0.4					1	C, D
		Test	0.3						E
		Repair			1.5			1, 2	F
		Repair				2.0		1, 2, 3, 4, 5	G, H
		Repair					2.0	1, 2, 3, 4, 5, 6, 7, 8	I
		Overhaul					16.0	1, 2, 3, 4, 5, 6, 7, 8	J
01	POWER UNIT, STP-34	Inspect	0.1						A
		Test	0.3						E
		Repair			1.8			1, 2	F
		Repair				2.0		1, 2, 3, 4, 5	G, J
		Repair					2.0	1, 2, 3, 4, 5, 6, 7, 8	H
0101	PRINT CIRCUIT BOARDS, STP-34								
010101	E-EB0/1	Inspect			0.1				A
		Test			0.5			1, 2	A
		Test					1.0	1, 2, 3, 6, 7, 8	I
		Replace			0.5			1	
		Repair					2.0	1, 2, 3, 4, 6, 7, 8	G
010102	SWITCHING ASSEMBLY	Inspect	0.1		0.1			1	A
		Replace	0.5		0.5			1	
		Test					1.0	1, 2, 3, 6, 7, 8	
		Repair					2.0	1, 2, 3, 4, 6, 7, 8	H
02	LOGIC UNIT, STB-34	Inspect	–						A
		Test	–						E
		Repair	–		1.0			1, 2	F
		Repair				2.0		1, 2, 3, 4, 5	

0440-1

FIGURE 509. Example of a page-based standard information two-level MAC table.

43.1.3 Frame oriented standard information. MIL-STD-40051-1 contains defined standard information but the presentation of it in a frame-based may vary. The standard information can be presented (i.e. text, table, form, etc.) as prescribed by the acquiring activity. The stylists provide how the presentation of the data is viewed. See below for an example of frame-based presentation of standard information. For more examples and information see MIL-HDBK-1222. Some features that can be found in various presentations are:

a. Tables are linked to the appropriate text and displayed when they do not already appear in a pane on the user's viewer.

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b. Scroll bars – the vertical and horizontal scroll bar is used to move through the table when it exceeds the length or width of the data pane.
c. If a table is scrollable, the table has "sticky" column headers.
d. Only the referencing text within a table cell is to be a hotspot.
e. Reference to a table cell or row scrolls the table directly to the referenced cell or row.
f. Reference to a table not in-line uses a single click of a text hotspot or an icon hotspot and displays the object in a separate panning/zooming pane.
g. When in-line table(s) are large or numerous, an icon hotspot may be used in place of the object to speed up the display.

43.1.3.1 Example of a frame-based presentation on Controls and Indicators standard information ***<ctrlindtab>***. An example of a Controls and Indicators standard information ***<ctrlindtab>*** is displayed in three panes in a inner shell of a frame. The standard information that is specific of this table are the titles for the standard table heading ***<std.table.header>*** and the data that is requested. The standard title headings are Key, Control/Indicator, and Function. The data is wrapped with the tag control/indicator row ***<ctrlindrow>***. The control/indicator row wraps the data key ***<key>***, control/indicator ***<ctrlind>***, note ***<note>*** and function ***<function>***. The keys and controls/indicator are linked to the separate panes of data containing the functions of the control/indicator and the illustration. When a key and its controls/indicator of the table is selected it displays the function in a separate pane. The illustration is displayed in a third pane containing callouts ***<callout>***. The callouts of the illustration can contain hotspot capabilities to the keys of the Controls and Indicators standard information providing versatile linkage of the data. The stylesheet formats the output of how the standard information tables are displayed.

a. XML document instance fragment.An example of an XML document instance fragment containing a frame-based presentation on Controls and Indicators standard information is found in 35.2.1.
b. Stylesheet Output.

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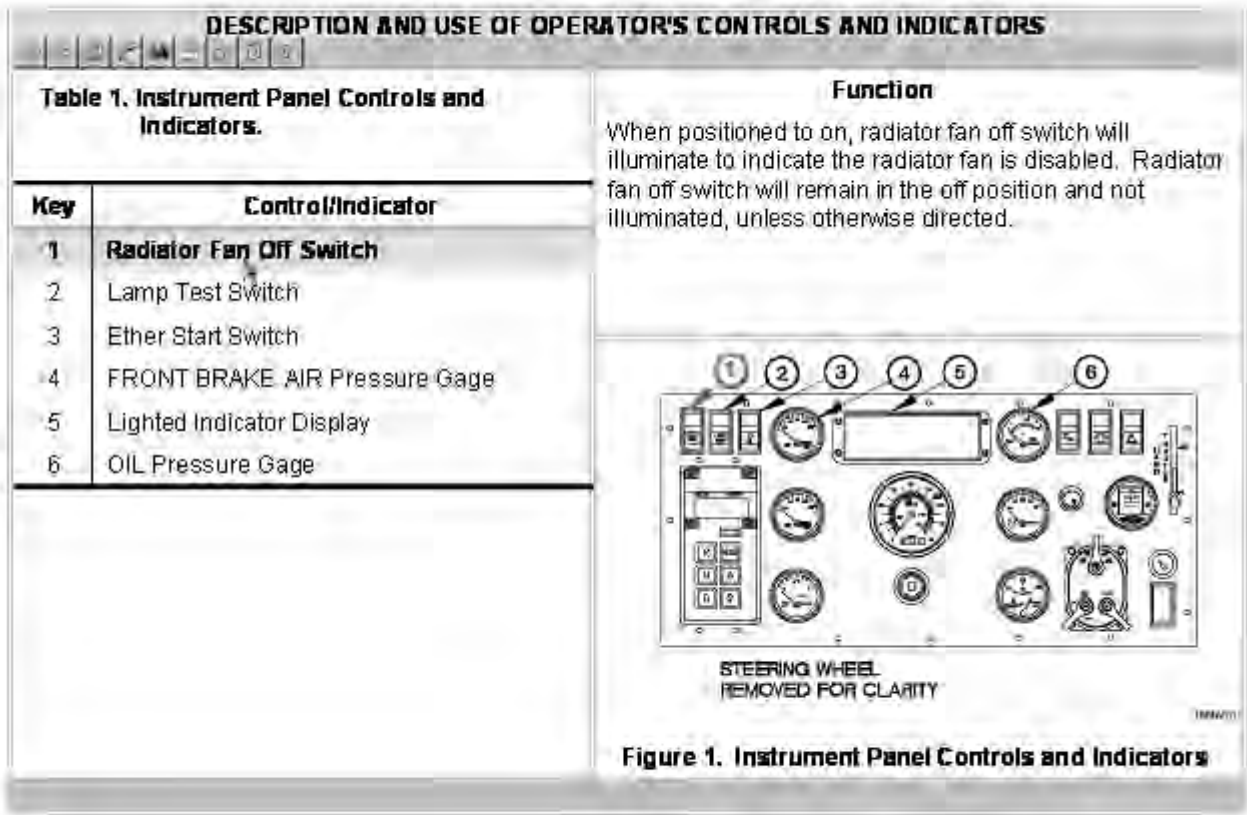

FIGURE 510. Example of a frame-based presentation on controls and indicators standard information. .

44 ILLUSTRATION, GRAPHIC, AND MULTIMEDIA.

44.1 Illustration. Illustration is a general term used for all graphic presentations as per MIL-HDBK-1222. An illustration is defined in two methods as a figure or inline graphic. A figure has the requirements for figure number, title, placement outside the narrative text (usually below the narrative), and text is not wrapped around the figure (i.e., text located to the left or right side of the figure, or within the same cell as a figure within a table) in accordance to MIL–STD–40051–1/–2. Where as an inline graphic is a small graphical object that is untitled and flows within the running narrative text, such as a symbol, icon, or equation.

44.1.1 Figure. The ***<figure>*** element is generally used to display a graphic to provide additional descriptive and location to the associated narrative text. The ***<figure>*** element contains one of the following illustration types – single-sheet, multi-sheet, tabular, or narrative.

a. The components are:
   (1) An optional precondition ***<precond>*** (see 42.1.1.1) that indicates if the figure is to be presented.
   (2) A required figure title (excluding the figure number, which is generated by the style sheet) ***<title>*** (see 49.1.1.3). As noted, the figure title in the source document is inputted before the

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illustration type. However, after the style sheet has completed composition, the title information (with the generated figure number) is placed after the illustration type.

(3) Includes one of the following illustration types:

(a) Single-sheet illustration includes a illustration ***<graphic>*** (see 44.2) followed by an optional legend ***<legend>*** (see 44.4).

(b) Multiple-sheet illustration ***<subfig>*** (see 44.1.3).

(c) Tabular illustration ***<table>*** (see 42) or conditional tabular illustration ***<table-alt>*** (see 48.2.1).

(d) Narrative illustration ***<verbatim>*** (see 49.1.3.6).

b. DTD fragment is graphically depicted.

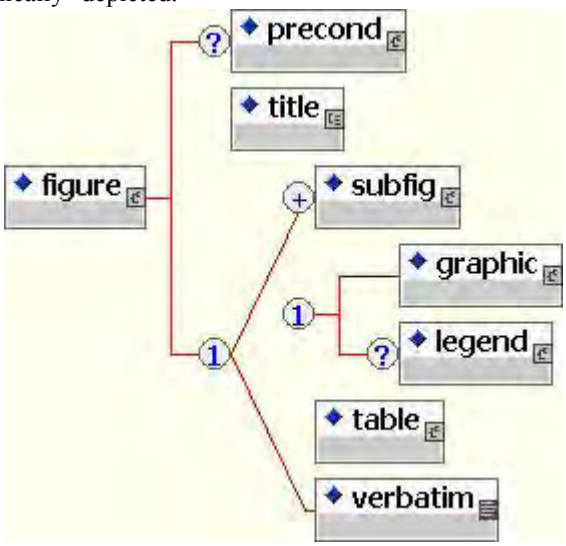

FIGURE 511. Figure DTD hierarchy <figure>.

c. DTD fragment:

```
<!ELEMENT figure (precond? , title , (subfig+ | (graphic , legend?) |
                                        (table | table-alt+) | verbatim))>
<!ATTLIST figure
 application  (page | frame | both )      "both"
  assocfig     IDREFS                       #IMPLIED
 changeref    IDREFS                       #IMPLIED
 comment      CDATA                        #IMPLIED
 delchlvl     CDATA                        #IMPLIED
 figtype      (normal-page | fo-rear )    "normal-page"
  fo-size      (25x11 | 35x11 | 45x11 )    #IMPLIED
 id           ID                           #IMPLIED
 idref        IDREFS                       #IMPLIED
 inschlvl     CDATA                        #IMPLIED
 pane         (yes | no)                   "no"
  security     (uc | fouo | c | s | ts )   #IMPLIED
 skilltrk     CDATA                        #IMPLIED
 tocentry     (0 | 1 | 2 | 3 | 4 | 5 )    "1"
>
```

d. Unique attributes:

(1) **application** – Figure presentation method (optional) (default value is "**both**") defines the figure presentation method as for page-base only ("**page**"), frame-base only ("**frame**"), or for either method ("**both**").

(2) **figtype** – Figure size type (Page-base only) (optional) (default value is "**normal-page**"). Frame–base ignores this attribute. In page-base, the attribute defines if the figure is a oversized

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(foldout) figure and presented at the end of the manual, in the foldout section, ("**fo-rear**") or normal sized figure and presented with the work package ("**normal-page**").

(3) **fo-size** – Foldout size. Defines the paper size (in inches) for a oversized (foldout) figure when the **figtype="fo-rear"** is set. The available foldout sizes (includes the apron) are:
  (a) "25x11" – Foldout size is 25" by 11".
  (b) "35x11" – Foldout size is 35" by 11".
  (c) "45x11" – Foldout size is 45" by 11".

(4) **pane** – New pane (Frame-base only) (optional) (default value is "**no**"). Page–base ignores this attribute. In frame-base, the attribute defines if the figure is shown in separate pane ("**yes**") or in the same pane ("**no**"). The attribute value may be ignored, if the acquiring activity has specified the display engine determines the figure placement.

(5) **tocentry** – TOC level (optional) (default value is "**1**"). Defines the table of content (TOC) placement level. When the attribute value is "0" no TOC entry is entered, otherwise the number specifies the indenture level. The attribute value may be ignore if specified and allow the composition to determine the indenture level.

e. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

44.1.1.1 Allowable figures for page-based TMs and IETMs. The types of graphics that should be used when preparing page-based TMs and IETMs include line drawings, illustrations (tools and test equipment, special tools and test equipment, etc), photographs, engineering drawings, diagrams, charts and graphs. In accordance with MIL-HDBK-1222, the use of multiple-sheet and foldouts illustrations should be limited to page-based TMs only.

44.1.2 Single-sheet illustration type. The single-sheet illustration is the option most common used. The elements used for single-sheet illustration are any precondition or applicability ***<precond>***, a illustration title ***<title>*** (without the figure number), the external graphic ***<graphic>***, and an optional legend ***<legend>***.

44.1.3 Multiple-sheet illustration type ***<subfig>***. The multiple-sheet illustration is used when a graphic requires two or more pages to represent the illustration.

a. The components are:

(1) An optional addendum to the figure title ***<subtitle>***. The supplement text allows additional information about each illustration sheet, the words "(continued)", or any additional words to append to the existing figure title. The subtitle element is not used to insert the "Sheet x of y" numbering, this handled by the attributes "**sheet**" and "**totalsheets**".
(2) A required illustration ***<graphic>***. (see 44.2)
(3) An optional legend ***<legend>***. (see 44.4)

b. DTD fragment is graphically depicted.

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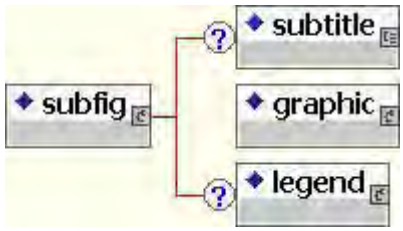

FIGURE 512. Multiple-sheet illustration DTD hierarchy <subfig>.

c. DTD fragment:

```
<!ELEMENT subfig (subtitle? , graphic , legend?)>
<!ATTLIST subfig
 changeref   IDREFS  #IMPLIED
 comment     CDATA   #IMPLIED
 delchlvl    CDATA   #IMPLIED
 id          ID      #IMPLIED
 idrefs      IDREFS  #IMPLIED
 inschlvl    CDATA   #IMPLIED
 sheet       CDATA   #IMPLIED
 totalsheets CDATA   #IMPLIED
>
```

d. Unique attributes for ***<subfig>***:

(1) **sheet** - Used to enter the sheet number of a multi sheet figure. This will override any system generated counters.

(2) **totalsheets** - Provides the value for the total number of sheets in a multi sheet figure. This will override any system generated counters.

e. Common attributes:

(1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(2) **comment** – Change information (optional) (see 49.2.9).

(3) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(4) **inschlvl** – Insert change level (optional) (see 49.2.9).

(5) **id** – Unique identifier. (optional) (See 49.2.6.)

(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

f. Example multiple sheet illustration ***<subfig>*** is shown below.

<figure application="both" figtype="normal-page" pane="no">
<title>`Electrical Power Distribution System`
</title>
<subfig sheet="1" totalsheets="2">
<subtitle>`–Diagram`
</subtitle>
<graphic boardno="g1340482-1" unitmeasure="in">
</graphic>
</subfig>
<subfig sheet="2" totalsheets="2">
<subtitle>`–Components`
</subtitle>
<graphic boardno="g1340482-2"unitmeasure="in">
</graphic>
</subfig>
</figure>

The example would generate the following two figure sheet titles.

(1) Figure 1. Electrical Power Distribution System – Diagrams (Sheet 1 of 2).

(2) Figure 1. Electrical Power Distribution System – Components (Sheet 2 of 2).

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44.1.4 Tabular illustration type. The tabular illustration is used to illustrate a table example and identify the tabular information as a figure (illustration) and not as a table. The elements used for tabular illustration are a illustration title ***<title>*** (without the figure number) and a table (either a table (see 42) or conditional tables (see 47.3)).

44.1.5 Narrative illustration type. The narrative illustration is used for direct typing of information for an illustrate. Unlike normal XML, the ENTER key (carriage return, line break, etc.), space and tab characters are maintained. The composed text uses a fixed spaced font (the space between characters are same amount). The option is used when typing the information is simpler to illustrate or allowing the author the flexibility to quickly update the illustrated data. The elements used for narrative illustration are a illustration title ***<title>*** (without the figure number) and a verbatim text graphic ***<verbatim>***.

44.2 External graphic ***<graphic>***. An external graphic is non-XML data that uses various binary graphical formats (i.e., Computer Graphic Metadata (CGM), Portable Network Graphic (PNG), Bit Map Picture (BMP), etc.). The external graphic file and characteristics (format the placement and size) are defined in the attributes. Additional functionality (for IETMs usage only) is mapping links on the graphic illustration (see 44.2.4).

a. The components are:
   (1) Optional graphical links (hot spots) ***<mapref>*** (see 44.2.4).
b. DTD fragment is graphically depicted.

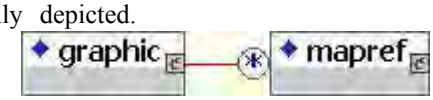

FIGURE 513. Graphic <graphic> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT graphic (mapref*)>
<!ATTLIST graphic
  alt           CDATA                                #IMPLIED
 assocfig      IDREFS                               #IMPLIED
 boardno       ENTITY                               #REQUIRED
 changeref     IDREFS                               #IMPLIED
 comment       CDATA                                #IMPLIED
 delchlvl      CDATA                                #IMPLIED
 graphsty      NMTOKEN                              #IMPLIED
 hplace        (left | right | center | none )   #IMPLIED
 hscale        CDATA                                #IMPLIED
 idref         IDREFS                               #IMPLIED
 inschlvl      CDATA                                #IMPLIED
 reprodep      CDATA                                #IMPLIED
 reprowid      CDATA                                #IMPLIED
 scalefit      (yes | no )                         #IMPLIED
 security      (uc | fouo | c | s | ts )          #IMPLIED
 skilltrk      CDATA                                #IMPLIED
 unitmeasure   (mm | cm | px | in | pt | pi )    "in"
  vscale        CDATA                                  #IMPLIED
>
```

d. Unique attributes:
   (1) **alt** – The attribute contains a description or additional information to be displayed when a mouse-over in frame-based/IETM application is used.
   (2) **boardno** – The required attribute specifies the entity name that points to the external graphic file. See 44.2.1 for how to define the graphic entity name.
   (3) **graphsty** – The attribute specifies a predefined graphic style identifier used in a style sheet.
   (4) **hplace** – The attribute specifies, if any, the horizontal position of the graphic. The values are left, right, center, or none (system specific rules are used). If no specific value is applied the system specific rules are used.

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(5) **hscale** – The attribute specifies the horizontal scaling factor for scaling a graphic, expressed in the form of percentages. Exclude the percent sign (%) when entering an "**hscale**" attribute value (i.e., attribute value of 10 is 10%). See 44.2.3 on further information and consideration concerning scaling.
(6) **reprodep** – The attribute specifies the reproduction area depth of the graphic. Enter a numeric value for the fixed depth, excluding the unit of measure. See 44.2.3 on further information and consideration concerning scaling.
(7) **reprowid** – The attribute specifies the reproduction area width of the graphic. Enter a numeric value for the fixed width, excluding the unit of measure. See 44.2.3 on further information and consideration concerning scaling.
(8) **scalefit** – The attribute specifies if the graphic will be scaled to fit the size of the reproduction area. Declared values are " **yes**" or "**no**". The graphic reproduction area or size is determined by the two attributes "**reprowid**" and "**reprodep**". See 44.2.3 on further information and consideration concerning scaling.
(9) **unitmeasure** – The attribute specifies the unit of measure for the value entered in "**reprowid** " or " **reprodep**" attributes. Declared values for this attribute include "mm" (millimeter), "cm" (centimeter), "px" (pixel), "in" (inch), "pt" (point), or "pi" (pica); default value is "in".
(10) **vscale** – The attribute specifies the vertical scaling factor for scaling a graphic, expressed in the form of percentages. Exclude the percent sign (%) when entering a " **vscale** " attribute value (i.e., attribute value of 10 is 10%). See 44.2.3 on further information and consideration concerning scaling.

e. Common attributes:
(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

44.2.1 Graphic entity name. The graphic entity name is an unique identifier name that points to the actual external graphic file. Since a single graphic may be reused numerous times throughout the publication, the graphic is defined once as an external ENTITY declaration found in the document type declaration at the beginning of the XML source file. The ENTITY declaration identifies entity name (name used in the XML source file), the file name, and graphic type (NOTATION described in 44.2.2). The entity name and the file name does not always have to be the same. The graphic file name could be denoted as an identifier (i.e., X1435930), but the entity name could be a common term (i.e., Bradley-side-view). The graphic entity file definition and usage example is shown Figure 514.

```
<!ENTITY Bradley-side-view SYSTEM "X1435930" NDATA GIF>

...

<graphic boardno="Bradley-side-view"><graphic>
```

FIGURE 514. Graphic entity definition example.

44.2.2 Graphic notations. "Notations identify, by name, the format of unparsed entities, elements with a notation attribute, or specific processing instructions" in accordance with W3C Extensible Markup Language (XML) 1.0. Graphic notations are used to identify the format of the graphic content (non-XML data). Notation declarations define the named notation and provide an external identifier to an accompanying application capable of processing the illustrations in the defined format. A notation's external identifier,

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may either be a PUBLIC or SYSTEM identifier, can be used to reference a processing application. The prescribed notations are declared in the DTD. Figure 515 show examples of notation declarations.

```
<!NOTATION IGES PUBLIC "-//USA-DOD//NOTATION (ASME/ANSI Y14.26M-1987)
                        Initial Graphics Exchange Specification//EN">
                    <!NOTATION  GIF  SYSTEM  "GIF">
```

FIGURE 515. Graphic notation definition example.

The DTD contains the following predefined notation types:

a. Graphic Notations
   (1) “BMP” – Microsoft Windows Bitmap in accordance with ISBN 0-7923-9432-1::Graphic Notation
   (2) “CGM-CHAR” – Computer Graphic Metafile with character encoding in accordance with ISO 8632/2
   (3) “CGM-BINARY” – Computer Graphic Metafile with binary encoding in accordance with ISO 8632/3
   (4) “CGM-CLEAR” – Computer Graphic Metafile with clear text encoding in accordance with ISO 8632/4
   (5) “DITROFF” – Device Independent UNIX Phototypesetter (TROFF)
   (6) “DVI” – TeX Device Independent
   (7) “EPS” – Encapsulated Postscript in accordance with ISBN 0-201-18127-4::Adobe
   (8) “EQN” – Equation UNIX Phototypesetter (TROFF)
   (9) “FAX” – CCITT Group 4 Facsimile Type 1 Untiled Raster
   (10) “GIF” – Graphic Interchange Format
   (11) “GIF87a” – Graphics Interchange Format 87a
   (12) “GIF89a” – Graphics Interchange Format 89a
   (13) “JPG” – Joint Photographic Experts Group
   (14) “JPEG” – Joint Photographic Experts Group
   (15) “IGES” – Initial Graphics Exchange Specification in accordance with ASME/ANSI Y14.26M-1987
   (16) “PCX” – ZSoft PCX bitmap in accordance with ISBN 0-7923-9432-1
   (17) “PDF” – Postscript Document Format
   (18) “PIC” – Picture UNIX Phototypesetter (TROFF)
   (19) “PNG” – Portable Network Graphic in accordance with W3C TR/REC-png
   (20) “PS” – Postscript
   (21) “SGML” – Standard Generalized Markup Language in accordance with ISO 8879:1986
   (22) “TBL” – Table UNIX Phototypesetter (TROFF)
   (23) “TEX” – The TeXbook in accordance with ISBN 0-201-13448-9::Knuth
   (24) “TIFF” – Tagged Image File Format bitmap
   (25) “WMF” – Microsoft Windows Metafile in accordance with ISBN 0-7923-9432-1::Graphic Notation
   (26) “WPG” – WordPerfect Graphic format
   (27) “linespecific” – ASCII text with Linebreak

b. Audio Notations
   (1) “AIF” – Audio Interchange File Format (AIFF)
   (2) “AIFC” – Audio Interchange File Format (AIFF)
   (3) “AIFF” – Audio Interchange File Format (AIFF)
   (4) “AU” – uLaw/AU Audio File
   (5) “CDA” – CD Audio Track
   (6) “CDDA” – Audio Interchange File Format (AIFF)
   (7) “KAR” – Musical Instrument Digital Interface (MIDI)
   (8) “MID” – Musical Instrument Digital Interface (MIDI)
   (9) “MIDI” – Musical Instrument Digital Interface (MIDI)
   (10) “RMI” – Musical Instrument Digital Interface (MIDI)
   (11) “SMF” – Musical Instrument Digital Interface (MIDI)

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(12) “SND” – uLaw/AU Audio File
(13) “ULW” – uLaw/AU Audio File
(14) “WAV” – Waveform Audio File
(15) “WAX” – Windows Media Audio
(16) “WMA” – Windows Media Audio

c. MP3 Notations
(1) “M3U” – MPEG Audio Stream, Layer IIIPlaylist File
(2) “M3URL” – MPEG Audio Stream, Layer III Playlist File
(3) “MP3” – MPEG Audio Stream, Layer III
(4) “SWA” – MPEG Audio Stream, Layer III

d. Video Notation
(1) “ASR” – Microsoft Automap Route
(2) “ASX” – Windows Media Video
(3) “AVI” – Audio Video Interleaved
(4) “DIF” – Digital Video
(5) “DV” – Digital Video
(6) “FLI” – AutoDesk Animator
(7) “MOV” – QuickTime Movie
(8) “QT” – Quicktime Movie
(9) “SMI” – Synchronized Multimedia Integration Language (SMIL)
(10) “SML” – Synchronized Multimedia Integration Language (SMIL)
(11) “VFW” – Video for Windows
(12) “WM” – Windows Media Video
(13) “WMP” – Windows Media Video
(14) “WMV” – Windows Media Video
(15) “WMX” – Windows Media Video
(16) “WVX” – Windows Media Video

e. Motion Picture Notation
(1) “MLV” – MPEG
(2) “MP2” – MPEG-2
(3) “MP2V” – MPEG-2
(4) “MP4” – MPEG-4
(5) “MPA” – MPEG Audio
(6) “MPE” – MPEG
(7) “MPEG” – MPEG
(8) “MPG” – MPEG-1
(9) “MPG4” – MPEG-4
(10) “RTS” – Streaming Movie
(11) “RTSP” – Streaming Movie
(12) “SDP” – Streaming Movie
(13) “SWF” – Macromedia Flash Movie

44.2.3 Scaling graphic. When graphic scaling is desired two methods are used, scaling by percentage and scaling by size. When the percentage attributes (**hscale** or **vscale**) are non-zero values, the scaling by percentage method overrides the scaling by size method.

a. Percentage.Scaling by percentage increases or decreases the graphic size by a specific amount, both vertically and horizontally. Many composition systems recommend (common practice) entering a value in either the **hscale** or the **vscale** attribute – not both. When only a single scaling by percentage attribute value is entered, the composition system will proportionately scale the other direction with the same percentage amount. If different values are entered in both attributes – the image is skewed. When the attribute value is 100, the graphic is actual size. If the graphics needs

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to be 10% greater in size, the attribute(s) value would be 110. If the graphics needs to be 10% lesser in size, the attribute(s) value would be 90.

b. Size.Scaling by size (reproduction area) scales the graphic proportionally to fit the prescribed graphical size. If the graphic size is greater than the reproduction area and the **scalefit** attribute is set to **no**, the graphic will be cropped (in frame-based) or should produce an overflow error (in page-based), otherwise the graphic is scaled proportionally to the specified size.

44.2.4 Graphic hot spot reference ***<mapref>***. The element ***<mapref>*** provides the information to overlay hot spot reference on a graphic ***<graphic>***. Graphic hot spot ***<mapref>*** element defines an area, within a graphic, that enables a link within a graphic to related graphic, related text, or to invoke an object (i.e. multimedia, programs, applications, scripts etc.). The graphic hot spot is designed for use in frame-based/IETM applications. A hot spot reference can display additional content (i.e. acronym, tool tip, etc.) when the mouse is located over the area (mouse-over), when applicable. The hot spot reference has three overlay shapes that are a circle ***<map.circle>***, a rectangle (including a square) ***<map.rectangle>***, and a polygon ***<map.polygon>***.

44.2.4.1 Map coordinate ***<map.coord>***. The map coordinates define the shape by the x (horizontal) and y (vertical) axis coordinates from the upper left corner. An important requirement is knowing the graphic dimension (height and width). Without the graphic dimensions, mapping references are not possible. Additionally, if the frame-based viewer allows graphic resizing, the map reference coordinates***<mapref>*** are resized the same percentage. Knowing the graphical dimensions, any item in the graphic can be outlined with the mapped shapes. The coordinate position attributes "**x**" and "**y**" contain the values and defines a unit of measurement (attribute "**unitmeasure**") as millimeter (mm), centimeter (cm), pixel (px), inch (in), pica (pi), or point (pt).

a. The component of ***<mapref>*** is one of the following overlay shape types:
   (1) Overlay circle shape ***<map.circle>*** (see 44.2.4.2).
   (2) Overlay rectangle or square shape ***<map.rectangle>*** (see 44.2.4.3).
   (3) Overlay user defined shape ***<map.polygon>*** (see 44.2.4.4).

b. DTD fragment is graphically depicted.

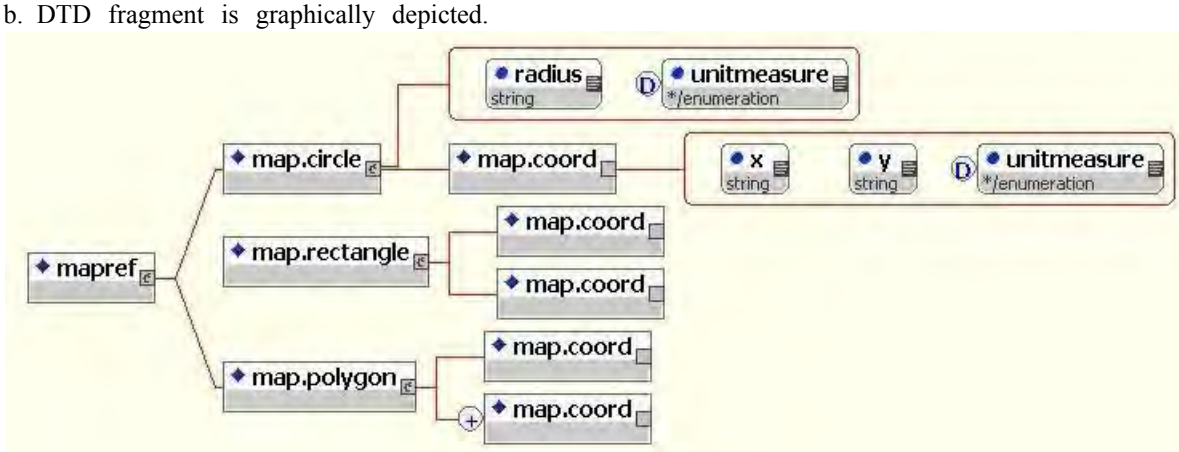

FIGURE 516. Graphic DTD hierarchy <mapref>.

c. DTD fragment:

```
<!ELEMENT mapref (map.circle | map.rectangle | map.polygon)>
<!ATTLIST mapref
 id      ID     #REQUIRED
 label   CDATA  #IMPLIED
 refdes  CDATA  #IMPLIED
 partref IDREF  #IMPLIED
>
```

d. Unique attributes:

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(1) **id** – Unique identifier. (required) The unique identifier is used to link to the graphical hot spot (i.e. ***<callout>*** element).
(2) **label** – Hot spot label. (optional) Provides additional information about the referenced graphical hot spot (i.e., part number, description, NSN).
(3) **partref** – Part or link reference. (optional) A reference to parts information unique identification value assigned to the parts information entry ***<pi.item>***. The reference can also reference a ***<link>*** to start an external application.
(4) **refdes** — Reference designator. (optional) The reference designator illustration index number assigned to applicable repair part or special tool.

44.2.4.2 Map circle hot spot ***<map.circle>***. The element ***<map.circle>*** overlays a circle using the map coordinate ***<map.coord>*** for the circle center and uses the attribute "**radius**" to determine the circle size. An example is placing a 12 point wide circle whose center is 1 inch from the top and 1 inch from left. The XML fragment is

a. **radius** – Circle radius. (required) Sets the radius of the circle based on the coordinates assigned by ***<map.coord>***
b. **unitmeasure** – Unit of measure (optional) (default value is "**in**"). Assigns the unit of measure for the circle's radius. Unit of measure values include "mm" (millimeter), "cm" (centimeter), "px" (pixel), "in" (inch), "pt" (point), or "pi" (pica).

```
<map.circle radius="6" unitmeasure="pt">
<map.coord x="1" y="1" unitmeasure="in">
</map.circle>
```

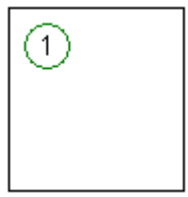

FIGURE 517. Map circle example.

44.2.4.3 Map rectangle hot spot ***<map.rectangle>***. The element ***<map.rectangle>*** overlays a rectangle using two map coordinates (***<map.coord>***) for the upper left corner and lower right corner.

a. Example placing the 12 points by 12 points square whose upper left corner is 72 points from the top and 72 points from left. The two coordinates would be 72 by 72 points for upper left and 84 by 84 points for the lower right. The XML fragment is

```
<map.rectangle>
<map.coord x="72" y="72"unitmeasure="pt"/>
<map.coord x="84"y="84" unitmeasure="pt"/>
</map.rectangle>
```

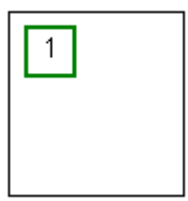

FIGURE 518. Map square example.

b. Example placing the 12 by 16 points rectangle whose upper left corner is 150 points from the top and 400 points from left. The two coordinates would be 150 by 400 points for upper left and 162 by 416 points for the lower right. The XML fragment is

```
<map.rectangle>
<map.coord x="150" y="400"unitmeasure="pt"/>
```


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<map.coord x="162"y="416" unitmeasure="pt"/>
</map.rectangle>

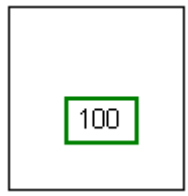

FIGURE 519. Map rectangle example.

44.2.4.4 Map polygon hot spot ***<map.polygon>***. The element ***<map.polygon>*** overlays a shape defined by the map coordinates ***<map.coord>*** by connecting from one map coordinate to the next in the order entered. The last coordinate and the first coordinate will make the final polygon side. A triangle, pentagon, or other shapes can be formed to basically outline a part or item in a graphic. Note the overlay does not need to exactly fit over the image.

a. Example placing a triangle over an object with 12 point sides starting the top angle 24 points from the top and 24 points from the left. The three coordinates would be 24 by 24 points, 30 by 44.4 points, and 18 by 44.4 points. The XML fragment is

<map.polygon>
<map.coord x="24" y="24"unitmeasure="pt"/>
<map.coord x="30"y="44.4" unitmeasure="pt"/>
<map.coordx="18" y="44.4" unitmeasure="pt"/>
</map.polygon>

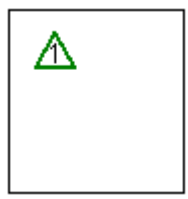

FIGURE 520. Map polygon triangle example.

b. Example placing the 12 points by 12 points square whose upper left corner is 72 points from the top and 72 points from left. The four coordinates would be 72 by 72 points, 84 by 72 points, 84 by 84 points, and 72 by 84 points. The polygon can generate a square, but with much more effort than the rectangle model. The XML fragment is

<map.polygon>
<map.coord x="72" y="72"unitmeasure="pt"/>
<map.coord x="84"y="72" unitmeasure="pt"/>
<map.coordx="84" y="84" unitmeasure="pt"/>
<map.coord x="72" y="84" unitmeasure="pt"/>
</map.polygon>

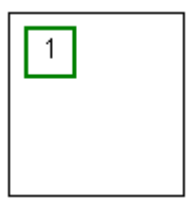

FIGURE 521. Map polygon square example.

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44.2.4.5 Map coordinates ***<map.coord>***. One or more mapping coordinates ***<map.coord>*** element(s) is/are required for each graphic hot spot (***<map.circle>***, ***<map.rectangle>*** or ***<map.polygon>***), depending on the selected shape. The **x** and **y** mapping coordinates are measured from the upper left-hand corner (0,0) of the graphic.

a. The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT map.coord EMPTY>
<!ATTLIST map.coord
      unitmeasure (mm | cm | px | in | pt | pi )    "in"
      x           CDATA                              #REQUIRED
      y           CDATA                              #REQUIRED
>
```

c. Unique attributes:
   (1) **x** – X-axis (horizontal) coordinate. (required) Enter a positive numeric value, omitting the unit of measure.
   (2) **y** – Y-axis (vertical) coordinate. (required) Enter a positive numeric value, omitting the unit of measure.
   (3) **unitmeasure** – Unit of measurement (optional) (default value is "**in**"). Assigns the unit of measure for the (x,y) coordinate from the upper left-hand corner (0,0) of the graphic. Unit of measure values include "mm" (millimeter), "cm" (centimeter), "px" (pixel), "in" (inch), "pt" (point), or "pi" (pica).

44.2.4.6 Creating graphic hot spots in the source data. The following is an example for creating and linking a graphic hot spot.

a. Within the ***<graphic>*** element, insert a graphical hot spot ***<mapref>*** element(s). Enter one for each area identified to be hot spotted.
b. The ***<mapref>*** element requires an overlay hot spot shape. The example is using a rectangle shape ***<map.rectangle>*** element.
c. The map.rectangle
d. One or more mapping coordinates element(s), ***<map.coord>***, is/are required for each illustration hot spot information element (***<map.circle>***, ***<map.rectangle>*** or ***<map.polygon>***), depending on the selected shape. The X,Y mapping coordinates are measured from the upper left-hand corner (0,0) of the graphic. ***<map.coord>*** attributes include:
   (1) "x" – required, x-axis (horizontal) coordinate. Enter a positive numeric value, omitting the unit of measure.
   (2) "y" – required, y-axis (vertical) coordinate. Enter a positive numeric value, omitting the unit of measure.
   (3) "unitmeasure" -- assigns the unit of measure for the (x,y) coordinate from the upper left-hand corner (0,0) of the graphic. Unit of measure values include "mm" (millimeter), "cm" (centimeter), "px" (pixel), "in" (inch), "pt" (point), or "pi" (pica); default value is "in".
e. XML source code example for creating a rectangle hot spot in a graphic that links to the part's information (***<pi.item>***) in the Repair Parts List work package:

<figure id="m00100–9–1305–201–fig23">
<title>Gun Mount Support
</title>
<graphic boardno="m00100–9–1305–201_fig23">
<mapref id="x123" label="87" partref="p00392-9–1305–201_87">
<map.rectangle>
<map.coord x="2.10" y="1.66"graphicunit="in"/>
<map.coord x="2.29"y="1.83" graphicunit="in"/>
</map.rectangle>
</mapref>
</graphic>

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</figure>
XML segment for ***<pi.item>*** in the Repair Parts List work package:
<pi.item id="p00392–9–1305–201_87"indent="0" type="part" fscap="no" mrp="no" csi="no" tereq="no">
<callout assocfig="p00392-9-1305-201-fig6" label="87"/>
<smr service="army" sourcecode="PA" maintcode="OZ" recovercode="Z"/>
<nsn>
<fsc>5310
</fsc>
<niin>00-515-7449
</niin>
</nsn>
<partno>AN960C416L
</partno>
<cageno>88044
</cageno>
<desc>BRACKET
</desc>
<qty>2
</qty>
</pi.item>

44.3 Graphic inline with text. A graphic inline with text is a small graphic set in the same line as the text, such as a symbol (***<symbol>***), warning and caution icon(s) (***<icon-set>***), equation (***<inlinegraphic>***), etc.

44.3.1 Symbol ***<symbol>***. The ***<symbol>*** element is used for a graphic symbol not found in standard ISO character sets that is inserted as a graphic inline with the text.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT symbol EMPTY>
<!ATTLIST symbol
  alt          CDATA                          #IMPLIED
 assocfig     IDREFS                         #IMPLIED
 boardno      ENTITY                         #REQUIRED
 changeref    IDREFS                         #IMPLIED
 comment      CDATA                          #IMPLIED
 delchlvl     CDATA                          #IMPLIED
 hscale       CDATA                          #IMPLIED
 id           ID                             #IMPLIED
 idref        IDREFS                         #IMPLIED
 inschlvl     CDATA                          #IMPLIED
 reprodep     CDATA                          #IMPLIED
 reprowid     CDATA                          #IMPLIED
 scalefit     (yes | no )                   #IMPLIED
 security     (uc | fouo | c | s | ts )     #IMPLIED
 skilltrk     CDATA                          #IMPLIED
 unitmeasure (mm | cm | px | in | pt | pi )  "in"
  vscale       CDATA                           #IMPLIED
>
```

c. Unique attributes:
   (1) **alt** – Additional text to assist in identifying the graphic (optional) (see 44.2.d.).
   (2) **boardno** – External graphic entity name (required) (see 44.2.d.).
   (3) **graphicunit** – Graphic unit of measurement (default value is "**in**") (see 44.2.d.).
   (4) **hscale** – Horizontal scaling factor (optional) (see 44.2.d.).

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(5) **reprowid** – Reproduction area width (optional) (see 44.2.d.).
(6) **reprodep** – Reproduction area depth (optional) (see 44.2.d.).
(7) **scalefit** – Scale the graphic to fit the reproduction area (optional) (see 44.2.d.).
(8) **vscale** – Vertical scaling factor (optional) (see 44.2.d.).

d. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

44.3.2 Inline graphic ***<inlinegraphic>***. The ***<inlinegraphic>*** element identifies a graphic that is maintained with the text (i.e., equation).

a. The components are:
(1) The element is EMPTY and all pertinent information is entered through its attributes.

b. DTD fragment:

```
<!ELEMENT inlinegraphic EMPTY>
<!ATTLIST inlinegraphic
  alt          CDATA                              #IMPLIED
 assocfig     IDREFS                             #IMPLIED
 boardno      ENTITY                             #REQUIRED
 changeref    IDREFS                             #IMPLIED
 comment      CDATA                              #IMPLIED
 delchlvl     CDATA                              #IMPLIED
 hscale       CDATA                              #IMPLIED
 id           ID                                 #IMPLIED
 idref        IDREFS                             #IMPLIED
 inschlvl     CDATA                              #IMPLIED
 reprodep     CDATA                              #IMPLIED
 reprowid     CDATA                              #IMPLIED
 scalefit     (yes | no )                       #IMPLIED
 security     (uc | fouo | c | s | ts )        #IMPLIED
 skilltrk     CDATA                               #IMPLIED
 unitmeasure (mm | cm | px | in | pt | pi )  "in"
  vscale       CDATA                               #IMPLIED
>
```

c. Unique attributes:
(1) **alt** – Additional text to assist in identifying the graphic (optional) (see 44.2.d.).
(2) **boardno** – External graphic entity name (required) (see 44.2.d.).
(3) **graphicunit** – Graphic unit of measurement (default value is "**in**") (see 44.2.d.).
(4) **hscale** – Horizontal scaling factor (optional) (see 44.2.d.).
(5) **reprowid** – Reproduction area width (optional) (see 44.2.d.).
(6) **reprodep** – Reproduction area depth (optional) (see 44.2.d.).
(7) **scalefit** – Scale the graphic to fit the reproduction area (optional) (see 44.2.d.).
(8) **vscale** – Vertical scaling factor (optional) (see 44.2.d.).

d. Common attributes:
(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)

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(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

44.3.3 Icon set ***<icon-set>***. The element ***<icon-set>*** identifies a series of hazard icons used as a unit to mark warnings or cautions; stored as a single graphic entity.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT icon-set EMPTY>
<!ATTLIST icon-set
  alt          CDATA                              #IMPLIED
 assocfig     IDREFS                             #IMPLIED
 boardno      ENTITY                             #REQUIRED
 changeref    IDREFS                             #IMPLIED
 comment      CDATA                              #IMPLIED
 delchlvl     CDATA                              #IMPLIED
 hscale       CDATA                              #IMPLIED
 id           ID                                 #IMPLIED
 idref        IDREFS                             #IMPLIED
 inschlvl     CDATA                              #IMPLIED
 reprodep     CDATA                              #IMPLIED
 reprowid     CDATA                              #IMPLIED
 scalefit     (yes | no )                       #IMPLIED
 security     (uc | fouo | c | s | ts )        #IMPLIED
 skilltrk     CDATA                               #IMPLIED
 unitmeasure (mm | cm | px | in | pt | pi ) "in"
  vscale       CDATA                               #IMPLIED
>
```

c. Unique attributes:
   (1) **alt** – Additional text to assist in identifying the graphic (optional) (see 44.2.d.).
   (2) **boardno** – External graphic entity name (required) (see 44.2.d.).
   (3) **graphicunit** – Graphic unit of measurement (default value is "**in**") (see 44.2.d.).
   (4) **hscale** – Horizontal scaling factor (optional) (see 44.2.d.).
   (5) **reprowid** – Reproduction area width (optional) (see 44.2.d.).
   (6) **reprodep** – Reproduction area depth (optional) (see 44.2.d.).
   (7) **scalefit** – Scale the graphic to fit the reproduction area (optional) (see 44.2.d.).
   (8) **vscale** – Vertical scaling factor (optional) (see 44.2.d.).
d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

44.4 Legend ***<legend>***. The ***<legend>*** element identifies the legend occurring as part of a figure or within a table entry, which is similar to a glossary. The legend identifies a callout reference or keyword term and provides definition with a possible link reference to further detail.

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a. The components are:
   (1) A required a title ***<title>*** (see 49.1.1.3). Usually the title is "LEGEND", but the author may enter different text.
   (2) Optional legend column heading ***<std.table.header>*** (see 42.3.1). Each ***<std.table.header>*** element identifies one row in the legend heading.
   (3) One or more legend item or row ***<legend.item>***. Each legend item or row contains the following columns:
      (a) A term ***<term>*** (see 49.1.2.3.1.1) or callout reference (see 46.2.4.1) being identified.
      (b) The term or callout reference definition or description ***<def>***.(see 49.1.2.3.1.2)
      (c) An optional link to further details on the term or callout reference.

b. Legend DTD graphical hierarchy

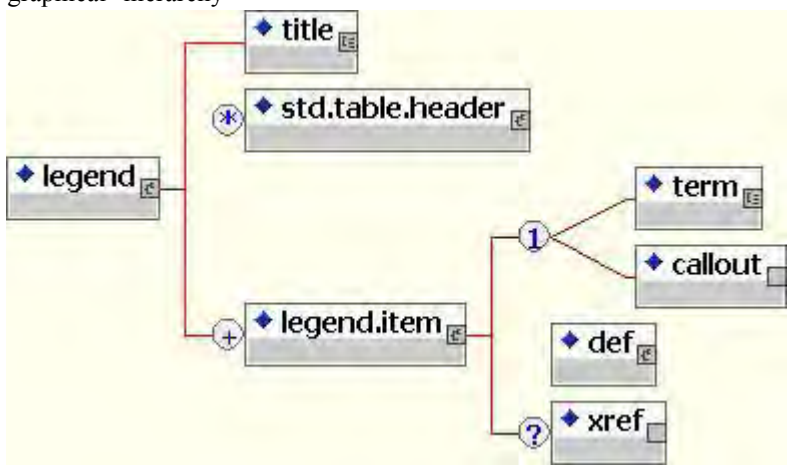

FIGURE 522. Legend <legend> and <legend.item> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT legend (title, std.table.header*, legend.item+)>
<!ATTLIST legend
    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>


<!ELEMENT legend.item ((term | callout) , def , xref?)>
<!ATTLIST legend.item
    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

d. Common attributes for ***<legend>*** and ***<legend.item>***:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

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44.5 Authentication page ***<authent>***. The authentication page is identifying the TM has approved by the U.S. Army. The authentication page is an external graphic obtained through the Army Publishing Directorate (APD) after authentication has been completed.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT authent EMPTY>
<!ATTLIST authent
  alt         CDATA                          #IMPLIED
 boardno     ENTITY                         #REQUIRED
 hscale      CDATA                          #IMPLIED
 reprodep    CDATA                          #IMPLIED
 reprowid    CDATA                          #IMPLIED
 scalefit    (yes | no )                   #IMPLIED
 unitmeasure (mm | cm | px | in | pt | pi )   "in"
  vscale      CDATA                           #IMPLIED
>
```

c. Attributes for ***<authent>***:
   (1) **alt** – Additional text to assist in identifying the graphic (optional) (see 44.2.d.).
   (2) **boardno** – External graphic entity name (required) (see 44.2.d.).
   (3) **graphicunit** – Graphic unit of measurement (default value is "**in**") (see 44.2.d.).
   (4) **hscale** – Horizontal scaling factor (optional) (see 44.2.d.).
   (5) **reprowid** – Reproduction area width (optional) (see 44.2.d.).
   (6) **reprodep** – Reproduction area depth (optional) (see 44.2.d.).
   (7) **scalefit** – Scale the graphic to fit the reproduction area (optional) (see 44.2.d.).
   (8) **vscale** – Vertical scaling factor (optional) (see 44.2.d.).

44.6 Back cover ***<back>***. The back cover is generally a graphic, when applicable, for a metric conversion table covering applicable units.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT back EMPTY>
<!ATTLIST back
    alt              CDATA                          #IMPLIED
    boardno          ENTITY                         #REQUIRED
    hscale           CDATA                          #IMPLIED
    reprodep         CDATA                          #IMPLIED
    reprowid         CDATA                          #IMPLIED
    scalefit         (yes | no)                    #IMPLIED
    unitmeasure      (mm | cm | px | in | pt | pi)    "in"
    vscale           CDATA                           #IMPLIED
>
```

c. Attributes for ***<back>***:
   (1) **alt** – Additional text to assist in identifying the graphic (optional) (see 44.2.d.).
   (2) **boardno** – External graphic entity name (required) (see 44.2.d.).
   (3) **graphicunit** – Graphic unit of measurement (default value is "**in**") (see 44.2.d.).
   (4) **hscale** – Horizontal scaling factor (optional) (see 44.2.d.).
   (5) **reprowid** – Reproduction area width (optional) (see 44.2.d.).
   (6) **reprodep** – Reproduction area depth (optional) (see 44.2.d.).
   (7) **scalefit** – Scale the graphic to fit the reproduction area (optional) (see 44.2.d.).
   (8) **vscale** – Vertical scaling factor (optional) (see 44.2.d.).

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44.7 Multimedia functions. Multimedia refers to the integration of text, graphics, audio, video and/or animation for the enhancement of communication. Multimedia techniques should only be used in an IETM when it results in enhancing the presentation of the information or makes the procedures more effective. Multimedia is used to supplement the narrative and never be used as the primary means of presenting information, as specified in MIL-STD-40051-1. The starting and control keys are defined helper application and may be controlled by parameters being sent through the ***<link>*** element attribute **parameter** (see 46.2.3). Multimedia usage is discussed and approved by the acquiring activity before inclusion in an IETM. Multimedia standards and techniques are defined by the acquiring activity and in accordance with AR 25-30. XML source code example for invoking a multimedia application with a designated video clip:

```
<link xlink:href="c:\video_files\TM_9-2350-294-10-2\t00039–9–2350–294_v36.wmv" application="
frame" xreftype="multimedia" xmlns:xlink="http://www.w3.org/1999/xlink"xlink:type="simple"
linkaction="prompt" linkobject=”WMV”linktype="return" popup="no">
<symbol boardno=”video-icon” unitmeasure=”in”/>
</link>
```

45 FOOTNOTES.

45.1 Footnote elements. Footnotes are used essentially for reference, explanation, comments, or other information. The main elements that provide a footnote in a work package or in a table are footnote ***<ftnote>*** and footnote reference ***<ftnref>***.

45.1.1 Narrative footnote ***<ftnote>***. The element ***<ftnote>*** is used for the body of a footnote in the document. A footnote may contain one or more paragraphs ***<ftnpara>***. The footnote does not appear until the element ***<ftnref>*** is entered. The ***<ftnote>*** does not have to be inserted in a specific spot in a document. It is recommended that it is inserted in the same work package as the footnote reference ***<ftnref>*** for reuse of work packages. The stylesheet will control where the footnote information is displayed.

a. The component of narrative footnote ***<ftnote>*** :
   (1) Paragraph Footnote ***<ftnpara>*** (required – one or more). (see 45.1.1.1)

b. DTD graphically depicted for :

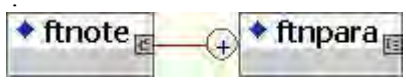

FIGURE 523. Narrative footnote <ftnote> DTD hierarchy for.

c. DTD fragment for ***<ftnote>***:

```
<!ELEMENT ftnote (ftnpara+)>
<!ATTLIST ftnote  id    ID     #REQUIRED
                  mark  (ctr | mark )  'ctr'
                  label CDATA  #IMPLIED >
```

d. Attributes for ***<ftnote>***:
   (1) **ID** - Specifies the footnote identifier.
   (2) **MARK** - Used to specify the footnote prefix marking. When no value is entered the default value is “CTR”.
      (a) “CTR” - The footnote prefix is numbered.
      (b) “MARK” - The footnote prefix is symbol defined in the GPO Manual of Style.
      (c) **LABEL** - Used to specify the number or symbol assigned to the footnote and overrides autogeneration of the number or symbol by the composition system.

45.1.1.1 Paragraph footnote ***<ftnpara>***. The element ***<ftnpara>*** provides data text that contains the footnote information.

a. The components are:

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(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***<indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Ordered (numbered) list – ***<seqlist>*** (see 49.1.2.1).
(13) Unordered list – ***<randlist>*** (see 49.1.2.2).
(14) Definition list – ***<deflist>*** (see 49.1.2.3).
(15) Internet address (e–mail or homepage) – ***<internet>*** (see 49.1.4.1.7).
(16) Proponent or organization address – ***<proponent>*** (see 49.1.4.22).
(17) Telephone number – ***<phone>*** (see 49.1.4.1.6).

b. DTD graphically depicted for :

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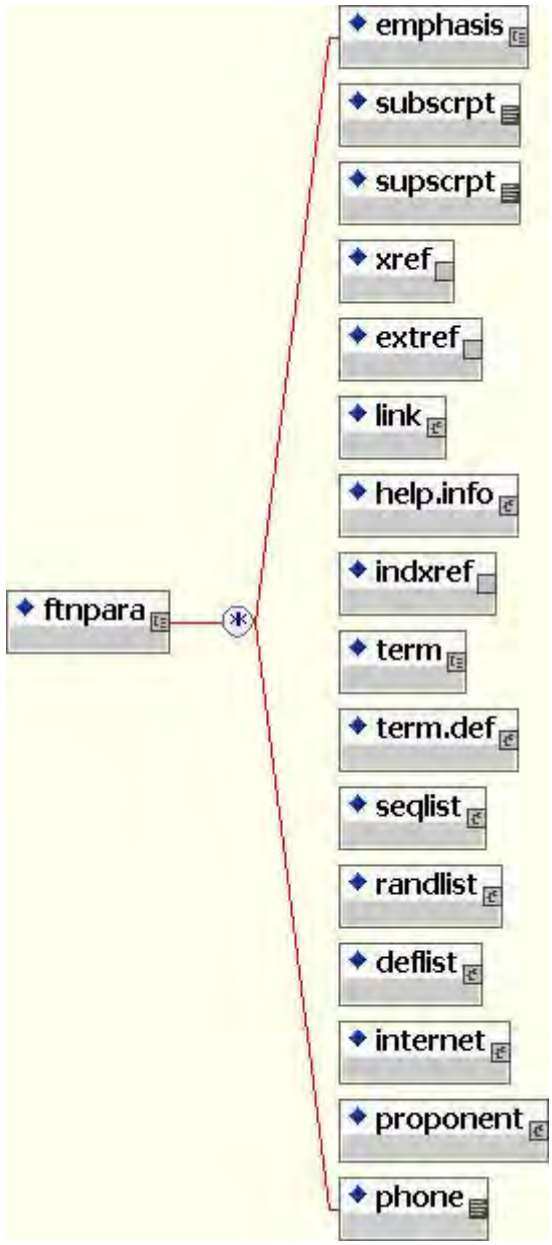

FIGURE 524. Narrative footnote <ftnpara> DTD hierarchy.

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c. DTD fragment for:

```
<!ELEMENT ftnpara  (emphasis | subscrpt | supscrpt|
 xref | extref | link | help.info |
 indxref | term | term.def | internet | proponent | phone | instructplt)*>
<!ATTLIST ftnpara

    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
```

d. Common Attributes.

(1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(2) **comment** – Change information (optional) (see 49.2.9).
(3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(4) **inschlvl** – Insert change level (optional) (see 49.2.9).

45.1.2 Footnote reference. The element ***<ftnref>*** is used to reference a footnote appearing in the footnote page area. The element is EMPTY and all pertinent information is entered through its attributes. The footnote reference is linked to the footnote by identifying it by the use of the required cross-reference identifier's attribute **xrefid**. The footnote reference mark is controlled through the attributes **MARK** or **LABEL** of the element narrative footnote***<ftnote>***.

a. DTD fragment for Footnote Reference ***<ftnref>***:

```
<!ELEMENT ftnref  EMPTY>
<!ATTLIST ftnref
          xrefid  IDREF   #REQUIRED>
```

b. Unique Attributes.

• **XREFID** - Reference to the footnote identifier. The composition system will generate the footnote text at the bottom of the referenced page.

45.2 Displaying footnotes. Footnotes are displayed in a page-based differently compared to how they are displayed in a frame-based document.

45.2.1 Paged-based footnotes. Footnotes are displayed at the bottom of the page of the footnote reference. To indicate the footnote an enumeration or manually-assigned mark is placed at the footnote reference and also at the beginning of the footnote paragraph. Arabic numerals should be used to number the footnotes consecutively. When numbered footnotes may lead to ambiguity such as a chemical formula, other markings such as superscript letters, daggers, and other symbols may be used.

45.2.1.1 Footnotes in tables. When footnotes are used in tables, they are numbered for each table as they are displayed in the table. Footnotes are placed below the closing line of the applicable table unless the table is continued. If the table is continued on other pages, place all footnotes at the bottom of the page on which they are referenced. The footnotes are in consecutive order as they are displayed in the table. Refer to MIL-HDBK-1222 for more information on using footnotes in a table. See the following example of a footnote in a tableTable 525.

a. Example of an XML document instance fragment for footnote ***<ftnote>*** and footnote reference ***<ftnref>*** in a Preventive Maintenance Checks and Services (PMCS) table ***<pmcstable>***.

<pmcstable id="m00048-9-2320-365-46">
<title>Preventive Maintenance Checks and Services
</title>
<std.table.header cols="6" rowsep="1">
<std.table.row>
<std.table.entry colnum="0">
</std.table.entry>

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<std.table.entry cols.span="0" rowsep="0">
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">ITEM TO BE
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">EQUIPMENT
</std.table.row>
<std.table.row>
<std.table.entry cols.span="0" rowsep="0">ITEM
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">MAN-
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">CHECKED OR
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">NOT READY/
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry cols.span="0" rowsep="0">NO.
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">INTERVAL
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">HOURS
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">SERVICED
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">PROCEDURE
</std.table.entry>
<std.table.entry cols.span="0" rowsep="0">AVAILABLE IF:
</std.table.entry>
</std.table.row>
</std.table.header>
<pmcs-entry id="m00048-9-2320-365-47">
<itemno>1
</itemno>
<interval>Before
</interval>
<manhours>.2
</manhours>
<checked>Aural Target Detection
</checked>
<pmcsproc>
<pmcsstep1>
<para>Operate the Antenna Control switch to FWD and REV to direct the antennaat a test target
<ftnref xrefid="testtarget"/>
<ftnote id="testtarget">

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<ftnpara> Forsetting up a test target at a known range/azimuth, refer to procedures inWP 0006 00.
</ftnpara>
</ftnote> of known range.
</para>
<eqpnotavail>
<trim.para>Test target is not heard in the headset as antenna sweeps pastthe target.
</trim.para>
</eqpnotavail>
</pmcsstep1>
<pmcsstep1>
<para>Set PWR switch to ON and Scan Method to MANUAL.
</para>
</pmcsstep1>
</pmcsproc>
</pmcs-entry>
<pmcs-entry id="m00048-9-2320-365-47">
<itemno>2
</itemno>
<interval>Before
</interval>
<manhours>.1
</manhours>
<checked>LEVEL indicator
</checked>
<pmcsproc>
<pmcsstep1>
<para>Check to see that bubble is in center of LEVEL indicator after adjustingleveling wheels as necessary.
</para>
<eqpnotavail>
<trim.para>Leveling bubble will not center in the indicator.
</trim.para>
</eqpnotavail>
</pmcsstep1>
</pmcsproc>
</pmcs-entry>
</pmcstable>
</pmcstable>

b.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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Table I. Preventive Maintenance Checks and Services

ITEM NO	INTERVAL	MAN-HOUR	ITEM TO BE CHECKED OR SERVICED	PROCEDURE	EQUIPMENT: NOT READY AVAILABLE IF:
1	Before	.2	Aural Target Detection	1. Operate the Antenna Control switch to FWD and REV to direct the antenna at a test target [1] of known range. 2. Check to see that bubble is in center of **LEVEL** indicator after adjusting leveling wheels as necessary.	Test target is not heard in the headset as antenna sweeps past the target.
2	Before	.1	LEVEL indicator	Check to see that bubble is in center of **LEVEL** indicator after adjusting leveling wheels as necessary.	Leveling bubble will not center in the indicator.

[1]Test target is not heard in the headset as antenna sweeps past the target.

FIGURE 525. Example of how a footnote is displayed in a page-based document when used in a table.

45.2.2 Framed-based footnotes. XML that is used in a page-based document for a footnote can be the same XML in a frame-based document. The display of the footnote is controlled by the stylesheets. The method to reference footnote data in a frame-based document is through the use of hotspotting or the mouse-over technique. If the mouse-over technique is used, a new window displays the note, etc. Refer to MIL-HDBK-1222 for more information on using hotspotting and mouse-over technique.

45.2.2.1 Footnotes in a table. When referencing in a table the hyperlink from a table cell is the reference text only, not the entire cell. Multiple hotspots within a cell are individually accessible.

a. The example below is how the same footnote that is shown in Figure 525 would display in a frame-based document using the mouse-over technique.

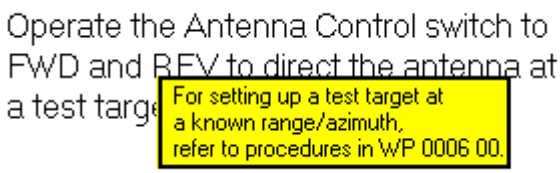

FIGURE 526. Example of the display of a frame-base footnote in a table using the mouse-over technique.

46 LINKING.

Linking or referencing provides a means to connect to associated information. Linking is commonly used in HTML with hot spots to other Web pages that are denoted usually by blue underlined text. In a paper environment the links provide the paragraph number, page number, work package sequence number, etc.

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Linking is created by the target information having a unique identifier and the linking information having a pointer or identification reference to the unique identifier. This section will discuss the various link methods and what is presented for page-based and frame-based presentation.

46.1 Identifier naming convention. A major concern in linking is creating a unique identifier. MIL–STD–40051–1/–2 defines the method for work package identifier creation through the use of: information type (maintenance, troubleshooting, etc), a unique 5 digit number (per TM number and information type), and the TM number where the work package was first created (without the maintenance level, further breakout (i.e., "&P") or volume number). The TM number, after the initial creation, has no meaning to the information. It is simply a method to eliminate identification duplication (i.e. work package identifier is "M00341–1–2345–678"). Work packages are not the only information referenced, but also objects such as figures, tables, steps, etc. Again this information needs to be unique not only in the current TM document, but in other information that references and reuses the information. Since a work package has a unique identifier associated to the information, a good practice is to prefix each identifier in the work package with the corresponding work package identifier (i.e. work package identifier is "M00341–1–2345–678" and a figure could be identified with meaning as "M00341–1–2345–678–fig1" or with unique identifier sequence within the work package as "M00341–1–2345–678–00234"). Since the work package identifier is unique the identifiers within the work package would maintain its uniqueness.

46.2 Linking element. The elements for referencing information are

a. External reference ***<extref>*** (See 46.2.1)
b. Internal reference ***<xref>*** (See 46.2.2)
c. Enhanced external and internal reference ***<link>*** (See 46.2.3)
d. Figures
    (1) Figure callout reference ***<callout>*** (See 46.2.4.1)
    (2) Figure callout mapping and part reference ***<mapref>*** (See 44.2.4)
    (3) Figure Number ***<figno>*** (See 46.2.4.3)
e. Footnote reference ***<ftnref>*** (see 45.1.2)
f. Troubleshooting - Logic Method (see 36.9.1).
    (1) Test block origin ***<origin>***
    (2) Test block ***<testblock>***
    (3) Test end block ***<endblock>***
    (4) Test block reference ***<branchref>***
g. MAC (see 40.4.1)
    (1) Test Equipment Reference ***<teref>***
    (2) Remarks Reference ***<remarkref>***
h. WP Sequence Number Reference ***<wpno>*** (See 46.2.5)

46.2.1 External reference ***<extref>***. The element indicates a reference to another TM, an information chapter outside the XML document, a work package outside the XML document, an external document or other information (not residing in the XML document instance). Note that the attributes for this element contain the content to be displayed and are not XML IDREFs since the references are external to the document instance

a. The components of ***<extref>*** is EMPTY and all pertinent information is entered through its attributes.

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b. DTD fragment for ***<extref>***:

```
<!ELEMENT extref EMPTY>
<!ATTLIST extref
     docno     CDATA                        #IMPLIED
     figid     CDATA                        #IMPLIED
     partid    CDATA                        #IMPLIED
     posttext  CDATA                        #IMPLIED
     pretext   CDATA                        #IMPLIED
     revno     CDATA                        #IMPLIED
     security  (uc | fouo | c | s | ts ) #IMPLIED
     tableid   CDATA                        #IMPLIED
     taskid    CDATA                        #IMPLIED
     wpid      CDATA                        #IMPLIED
>
```

c. Unique attributes for ***<extref>*** are:

(1) **docno** – Document number (optional) is used to the determine the document number to be published and possibly used in a lookup table to establish a hot reference to the information.

(2) **figid** – Figure reference identifier (optional) is used to link directly in the external document to the specific figure identifier value.

(3) **partid** – Part reference identifier (optional) is used to link directly in the external document to the specific part identifier value.

(4) **revno** – Document revision number (optional)" is used to define a certain revision of the referenced document (e.g. MIL-STD 40051B, where the **docno="MIL-STD 40051"** and **revno="B"**.

(5) **tableid** – Table reference identifier (optional) is used to link directly in the external document to the specific table identifier value.

(6) **taskid** – Task reference identifier (optional) is used to link directly in the external document to the specific task identifier value.

(7) **wpid** – Work package reference identifier (optional) is used to link directly in the external document to the specific work package identifier value.

d. Common attributes for ***<extref>*** are:

(1) **posttext** – Post text (optional) is used to provide additional text after the hot reference, but not part of the hot reference (e.g. ")").

(2) **pretext** – Pretext (optional) is used to provide additional text before the hot reference, but not part of the hot reference (e.g. "(see ").

(3) **security** – Security classification (optional) (see 49.2.13).

e. Page-based display.The only information displayed is the attribute values of "pretext", "docno", and "posttext". An example is a link to TM 9-2350-XXX-20.

(1) XML fragment is

<extref pretext="(See " docno="TM 9-2350-XXX-20" posttext=")" wpid="M00054-9-2350-XXX"/>

(2) Prints as "(See TM 9-2350-XXX-20)".

f. Frame-based display.The information to display can vary depending on the desired presentation. The same markup may be presented as "(See TM 9-2350-XXX-20)" or "(See TM 9-2350-XXX-20 referenced WP)". If the lookup table or catalog contains information about the referenced work package the presentation may be presented as "(See TM 9-2350-XXX-20 referenced WP Replace Gun Mount)".

46.2.2 Internal reference ***<xref>***. The element ***<xref>*** is used to specify an internal cross reference to other information in the TM. The ***<xref>*** has an EMPTY content model and the attributes are used to specify what is being referenced. The composition system specifies how the ID referenced value(s) are to be resolved. The element can have multiple ID reference attributes that references different work package components (e.g., work package, task, figure) to create a single reference. An example is a reference to a work package and procedural steps. The element would use three attributes "**wpid**", "**stepstart**" and "**stepend**" to identify the reference location and the composition system would generate "WP 0004 Steps 3-5".

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a. The components of ***<xref>*** is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment for ***<xref>***:

```
<!ELEMENT xref EMPTY>
<!ATTLIST xref
      callout    CDATA                        #IMPLIED
      figid      IDREF                        #IMPLIED
      itemid     IDREF                        #IMPLIED
      itemno     CDATA                        #IMPLIED
      posttext   CDATA                        #IMPLIED
      pretext    CDATA                        #IMPLIED
      security   (uc | fouo | c | s | ts ) #IMPLIED
      stepend    IDREF                        #IMPLIED
      stepstart  IDREF                        #IMPLIED
      tableid    IDREF                        #IMPLIED
      taskid     IDREF                        #IMPLIED
      termdefid  IDREF                        #IMPLIED
      tslocid    IDREF                        #IMPLIED
      wpid       IDREF                        #IMPLIED
>
```

c. Identification reference attributes for ***<xref>***.These attributes displayed values are determined and generated during composition. Having the resolution done during composition allows information to be moved or shifted, and when composed the current reference information is shown. An example is a step in a procedure. The reference is currently pointing at the second step "Step 2" with "M00345-X-XXX-XXX-0354" as the ID. A new step is added to the procedure before the second step. Since the ID did not change and the reference did not change, the reference would generate "Step 3". Identifiecation reference attributes for ***<xref>*** are:
   (1) **figid** – Figure identification reference (optional) is targeted to the figure attribute "**id**". Page-based and frame-based would generate "Figure " and associated figure number.
   (2) **itemid** – Item number identification reference (optional) is targeted to the standard information (i.e., PMCS, AAL, etc) item entry (i.e., ***<pmcs-entry>***, ***<aal-entry>***) attribute "**id**". Page-based and frame-based would generate "Item " and associated number.
   (3) **stepend** – Ending step identification reference (optional) is used only for last step in a range and is targeted to the step attribute "**id**". Page-based and frame-based would use the two attributes to generate the step range as "Steps x-y".
   (4) **stepstart** – Starting step identification reference (optional) is used when only a single step or the first step in a range and is targeted to the step attribute "**id**". Page-based and frame-based would generate "Step " and the associated step number for a single step.
   (5) **tableid** – Table identification reference (optional) is targeted to the table attribute "**id**". Page-based and frame-based would generate "Table " and associated number.
   (6) **termdefid** – Term definition identification reference (optional) is targeted to a glossary or list of acronyms definition (element ***<term.def>***) attribute "**id**". Page-based would not be identified. Frame-based would use a standard system generated reference such as "(Definition)" or highlight for a mouse-over to display the definition
   (7) **tslocid** – Troubleshooting procedure identification reference (optional) is targeted to a procedure (i.e., ***<opcheckproc>***,***<tsproc>***, ***<opcheck-tsproc>***) attribute "**id**".Reference to a troubleshooting procedure object locator identifier defined in the troubleshooting WP ***<tswp>***. The page-based will generate the literal "page " and the page number and, if identified, a logic test or end block identifier. The frame-based will generate, if identified, a logic test or end block identifier or a standard system generated reference or icon such as "Troubleshoot Link" or "Link".
   (8) **wpid** – Work package identification reference (optional) is targeted to the work package attribute "**wpno**". Page-based would generate the "WP " and associated WP sequence number.

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Frame-based would generate either the WP title or a standard system generated reference or icon to a work package such as "WP Link"

d. Typed or hard coded reference information **itemno** attribute.Some references are not readily available to obtain the generated text such as the callout number in an illustration. In this situation the reference is entered in the **itemno** attribute with the remaining associate ID references (i.e. Work package, figure, etc). An example using a cross reference with hard coded item reference.
   (1) XML fragment is

<xref wpid="M00034–X-XXX-XXX" figid="M00034–X-XXX-XXX-0034" callout="2"/>
   (2) Displays as "WP 0039, Figure 1, Item 2"

e. Common attributes for ***<xref>*** are:
   (1) **posttext** – Post text (optional) is used to provide additional text after the hot reference, but not part of the hot reference (e.g. ")").
   (2) **pretext** – Pretext (optional) is used to provide additional text before the hot reference, but not part of the hot reference (e.g. "(see ").
   (3) **security** – Security classification (optional) (see 49.2.13).

46.2.2.1 Multiple Identification references order. The ***<xref>*** element allows multiple ID reference attributes for building referenced information that are separated by ", ". The information is shown in the following order:

a. **wpid** and **termdefid**
b. **taskid**
c. **stepstart**, **stepend**, and **tslocid**
d. **tableid**
e. **itemid** and **itemno**
f. **figid**
g. **callout**

Several examples are shown below with the page-base output.

a. Example showing reference to the 45th work package and its "Replace Gun Mount" task.
   (1) XML fragment is
<xref wpid="M00034-X-XXX-XXX" taskid="M00034–X-XXX-XXX-0001" pretext="(See " posttext=")"/>
   (2) Prints as "(See WP 0045, Replace Gun Mount)"

b. Example showing reference to the 45th work package and the step range from 3 through 6.
   (1) XML fragment is
<xref wpid="M00034-X-XXX-XXX" stepstart="M00034-X-XXX-XXX-0041" stepend="M00034-X-XXX-XXX-0044" pretext="(See " posttext=")"/>
   (2) Prints as "(See WP 0045, Steps 3–6)"

c. Example showing reference to the 45th work package, the third step that contains the first figure in the work package.
   (1) XML fragment is
<xref wpid="M00034–X-XXX-XXX" stepstart="M00034–X-XXX-XXX-0041" figid="M00034–X-XXX-XXX-0234" pretext="(See " posttext=")"/>
   (2) Prints as "(See WP 0045, Step 3, Figure 1)"

d. Example showing reference to the 45th work package and the 34th item number in the expendable and durable list.
   (1) XML fragment is
<xref wpid="S00034–X-XXX-XXX" itemid="S00034–X-XXX-XXX-0041"/>
   (2) Prints as "WP 0045, Item 34"

e. Example showing reference to the 45th work package, the 3rd work package table, and the 2nd work package figure contained within table.
   (1) XML fragment is

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<xref wpid="M00034–X-XXX-XXX" figid="M00034–X-XXX-XXX-0244" tableid="M00034–X-XXX-XXX-0241" pretext="(See " posttext=")"/>

(2) Prints as "(See WP 0045, Table 3, Figure 2)"

46.2.3 Enhanced linking ***<link>***. The element ***<link>*** is used to specify internal cross references (other information in the TM), external cross references (other TMs, publications, documents, multimedia objects [video, sound, animation, etc.], or applications [diagnostic software, calculator, etc.]). The element describes the actions the frame-based viewer performs before and after the linking process. The element defines when it is used for frame-based only, page-based only, or for both. The ***<link>*** element is based on the World Wide Web Consortium (W3C) XML Link (XLink) Language Recommendation.

a. The components of ***<link>*** are:

(1) Popup title bar text ***<title>*** (optional) (see 37.3.1).

(2) Hot reference prompting (optional) has the following components.

(a) Pretext ***<pretext>*** (optional) is used to provide additional text before the hot reference (e.g. "(see "). This additional text is not part of the hot reference.

(b) Hot reference text is either an icon ***<symbol>*** (required) (see 44.3.1), a marker indicating the style sheet generates reference text ***<ref.generate>*** (required), or entered prompt text ***<prompt>*** (required) (see 46.2.3.1).

(c) Post text ***<posttext>*** (optional) is used to provide additional text after the hot reference, but not part of the hot reference (e.g. ")").

(3) Post linking ***<return.link>*** (optional) (see 46.2.3.2).

b. DTD graphically depicted for ***<link>***:

FIGURE 527. Enhanced linking DTD hierarchy <link>.

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c. DTD fragment for ***<link>***:

```
<!ELEMENT link (title? , (pretext? , (symbol | ref.generate | prompt) ,
         posttext?)?, returnlink? )>
<!ATTLIST link
     alt          CDATA                    #IMPLIED
     applicable   IDREFS                   #IMPLIED
     application  (page | frame | both )   "frame"
     assocfig     IDREFS                   #IMPLIED
     changeref    IDREFS                   #IMPLIED
     comment      CDATA                    #IMPLIED
     delchlvl     CDATA                    #IMPLIED
     id           ID                       #IMPLIED
     idref        IDREFS                   #IMPLIED
     inschlvl     CDATA                    #IMPLIED
     linkaction   (prompt | immediate )    "prompt"
     linktype     (goto | return)          "goto"
     local        IDREF                    #IMPLIED
     localrange   IDREF                    #IMPLIED
     parameter    CDATA                    #IMPLIED
     popup        (yes | no)               "no"
     security     (uc | fouo |c | s | ts) #IMPLIED
     skilltrk     CDATA                    #IMPLIED
     xlink:href   CDATA                    #IMPLIED
     xlink:type   (simple | complex )     #FIXED "simple"
     xmlns:xlink  CDATA                    #FIXED
"http://www.w3.org/1999/xlink"
     xreftype     (tm | sc | wp | test | task |
                  step1 | paragraph | figure |
                  table | part | document |
                  setup.item | frame.help |
                  program | multimedia |
                  hotspot | pageno)       #IMPLIED
>
```

d. Unique attributes for ***<link>*** are:

(1) **alt** – Alternate link text (optional) is a brief description of the link's purpose and shown when a mouse over is performed.

(2) **application** – Link prompt base (default value is ”**frame**”) determines the link presentation when a difference between the page-based and frame-based display is required. The **application** attribute defines the link prompt if for frame-based only “**frame**”, page-based only “**page**” or the same for both page-based and frame-based “**both**”.

(3) **linkaction** – Linking action (default value is ”**prompt**”) determines if the no prompt link is executed “**immediate**” or the user is prompted before executing the link “**prompt**” (see 46.2.3.1).

(4) **linktype** – Link return action (default value is ”**goto**”) indicates after completing the linking action does control return back to link position (attribute value is “**return**”) or continues with the linked area and does not return control back to the link position (attribute value is “**goto**”) (see 46.2.3.2).

(5) **local** – Internal reference (optional) indicates an internal (local to the current TM) reference to a target ID. When the attribute is used with **localrange**, the value is the starting reference. In XML, if an attribute ID reference value does not match another element's ID attribute value the document is not a valid XML document.

(6) **localrange** – Internal reference ending range (optional) indicates an internal (local to the current TM) reference to a target ID and is ending reference for a range. In XML, if an attribute ID reference value does not match another element's ID attribute value the document is not a valid XML document.

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(7) **parameter** – Parameter(s) for an external application (optional) provides a delimited list for additional information to execute the application correctly.
(8) **popup** – Popup indicator (default value is no) indicates to the viewer if the link should be shown as a separate window ("**yes**") or inline with the text ("**no**").
(9) **xlink:href** – External reference (optional) is reused from the W3C Xlink Recommendation to find a remote resource (or resource fragment) outside the current TM. The value of the "**xlink:href**" attribute is a Uniform Resource Identifier (URI) reference, as defined in [IETF RFC 2396]. The passed URI reference is directed through a URI resolver to specified information. The URI is generally a Web address, but, as in HTML, an internal reference can be specified through the use a "#" following the address. The information following the "#" is the unique identifier contained in the referenced document (see 46.1 for identifier naming convention). When the option is used the URI resolver will direct the user to the precise location in the referenced document. A good practice in naming the work package is to use the identifier for the work package (i.e. work package identifier is "M00341–1–2345–678" the file name would be "M00341–1–2345–678.xml"). Using this convention would provide a common method to reference work packages in the current or external TMs.
(10) **xlink:type** – XLink reference method (Fixed as **simple**) indicates the support of standard referencing. Current the standard does not support **complex** XLink referencing.
(11) **xreftype** – Target type (optional) defines the type of reference data and indicate possible style sheet generated text, as one of the following types:
(a) tm – Technical manual (TM)
(b) sc – Supply catalog (SC)
(c) wp – Work package (WP)
(d) test – Test procedure
(e) task – Task
(f) step1 – Main step
(g) paragraph – Titled paragraph
(h) figure – Figure
(i) table – Table
(j) part – Part information
(k) document – External document or publication
(l) setup.item – Initial setup item
(m) frame.help – Content specific help
(n) program – External application
(o) multimedia – Multimedia file
(p) hotspot – Hotspot anchor (i.e., graphic mapping)
(q) pageno – Supply current page number

e. Common attributes for ***<link>*** are:
(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

46.2.3.1 Link prompt. The ***<link>*** element, in a frame-based viewer, has the ability to define how the user sees the information and to specify if the reference is immediately linked to the information or if it is prompted before being linked to the information. The ***<link>*** element, in a page-based media, is limited to

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how referenced text is entered or created. The following paragraphs are mainly for frame-based viewers, but also describe how page-based systems would create referenced information generated text.

46.2.3.1.1 Prompted reference. The ***<link>*** element allows the author to determine if the user is prompted before going to the reference or if the user immediately goes to the reference. The action is determined through the attribute **linkaction** that when set to "immediate" will instantaneous jump to the referenced information and when set to "prompt" will display a hot spot text or symbol for the referenced information.

46.2.3.1.1.1 Prompted link. The **linkaction="prompt"** will display a hot spot link before executing the link. The two prompt types are narrative hot spot (uses the elements ***<prompt>*** (fixed text) or ***<ref.generate>*** (generated from the referenced)) or graphical hot spot (uses the element ***<symbol>***). Before (***<pretext>***) and after (***<posttext>***) the prompted data optional text is displayed, but not hot spotted.

46.2.3.1.1.2 Generated text from reference. A powerful XML feature is linking to other information and generating the referenced text information (through a stylesheet). The power in this capability is that information is constantly be added, deleted, moved or rearranged, therefore causing the locator text to change (i.e. "Figure 2" is changed to "Figure 3" when new figure is added before the figure). Using generated locator text each time the information is presented or composed the locators are current with information. The stylesheet determines how the locator is shown by saving the locator text associated with the identifier attribute value. The ***<link>*** element defines when the generated locator text is used by usingl the element ***<ref.generate>***. The format for the generated locator text depends on the ID references used. The following items are the various formats:

a. Work package (i.e. ***<maintwp>***) — "WP ####" (page-based) or "Title" (frame-based)
b. Task (i.e. ***<replace>***) — "Title"
c. Test (i.e. ***<testwithstate>***) — "Title"
d. Procedure (i.e. ***<proc>***) — "Title"
e. Step (i.e. ***<step1>***) — "Step #"
f. Titled paragraph (i.e. ***<para0>***) — "Title"
g. Figure (i.e. ***<figure>***) — "Figure #"
h. Figure callout (i.e. ***<mapref>***) — "Figure #, Item #"
i. Table (i.e. ***<table>***) — "Table #"
j. Supporting information list item (i.e. ***<mrpl-entry>*** in MRPL) — "WP ###, Item #" (page-based) or "Title, Item #" (frame-based)

If the referenced object is outside the scope (in another document, work package or task) of the current location the parent components information is used with the referenced text. An example in page-based is a reference to a figure in another work package. The locator text would generate the work package sequence number and reference figure number (i.e. WP 0034, Figure 3). The rules for the parent component information is the following:

a. Task in another work package — Work package locator information (i.e. WP 0045, Replacement)
b. Test in another work package — Work package locator information (i.e. WP 0059, Engine Compartment Fan Will Not Operate When Auxiliary Power Unit Is Operating)
c. Procedure in another
    (1) Work package — Work package and task locator information (i.e. WP 0245, Replacement, Right Bilge Pump)
    (2) Task but same work package — Task locator information (i.e. Replacement, Right Bilge Pump)
d. Step in another
    (1) Work package — Work package, task, and procedure (if applicable) locator information (i.e. WP 0134, Replacement, Step 3)
    (2) Task but the same work package — Task and procedure (if applicable) locator information (i.e. Replacement, Right Bilge Pump, Step 3)
    (3) Procedure but the same task — Procedure locator information (i.e. Left Bilge Pump, Step 4)
e. Titled paragraph in another

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(1) Work package — Work package locator information (i.e. WP 0003, HMMWV Equipment Description)
(2) Titled paragraph but same work package — All titled paragraphs parents' titles (i.e. Electrical System Theory, Splicing Wires where 1st level paragraph title is "Electrical System Theory" and 2nd level paragraph title is "Splicing Wires")

f. Figure in another work package — Work package locator information (i.e. WP 0045, Figure 3)
g. Figure callout in another work package — Work package locator information (i.e. WP 0004, Figure 3, Item 5)
h. Table in another work package — Work package locator information (i.e. WP 0056, Table 2)

46.2.3.1.1.3 Prompt examples. The following are examples using prompted links.

a. Example showing non-generated text (***<prompt>***).
(1) XML example is:

```
<link xmlns:xlink CDATA #FIXED "http://www.w3.org/1999/xlink" linkaction="prompt"
application="both" local="G00003–X–XXX–XXX" xlink:type="simple"
linktype="goto" xreftype="wp" popup="no">
<pretext>(See
</pretext>
<prompt>maintenance work package
</prompt>
<posttext>)
</posttext>
</link>
```

(2) Generated example frame-based is: "(See maintenance work package)" and page-based is "(See maintenance work package)".

b. Example showing generated text (***<ref.generate>***).
(1) XML example is:

```
<link xmlns:xlink CDATA #FIXED "http://www.w3.org/1999/xlink" linkaction="prompt"
application="both" local="G00003-X-XXX-XXX" xlink:type="simple"
linktype="goto" xreftype="wp" popup="no">
<pretext>(See
</pretext>
<ref.generate/>
<posttext>)
</posttext>
</link>
```

(2) Generated frame-based example is (assuming the referenced work package title is Replace Right Bilge Pump): "(See Replace Right Bilge Pump)" and page-based example is (assuming the reference work package sequence number is "0433") "(See WP 0433)".

c. Example using graphical (***<symbol>***). The graphical prompt is generally used only in frame-based viewers.
(1) XML example is:

```
<link xmlns:xlink CDATA #FIXED "http://www.w3.org/1999/xlink" linkaction="prompt"
application="frame" local="G00003-X-XXX-XXX" xlink:type="simple"
linktype="goto" xreftype="wp" popup="no">
<symbol boardno="gotolink" unitmeasure="in"/>
</link>
```

(2) Generated frame-based example is: " "

46.2.3.1.2 Immediate reference. The **linkaction="immediate"** will jump directly to the referenced information without prompting the user. The "immediate" value is primarily used after evaluating a test result to jump to the next test procedure or corrective action in the diagnostic work package. When **linkaction="immediate"**

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is set, the elements ***<title>***, ***<pretext>***, ***<symbol>***, ***<ref.generate>***, ***<prompt>***, and ***<posttext>*** are ignored, since no information is prompted before jumping to the link. An immediate link XML example:

```
<link xmlns:xlink CDATA #FIXED "http://www.w3.org/1999/xlink" linkaction="immediate"
application="frame" local="G00003-X-XXX-XXX" xlink:type="simple" linktype="goto"
xreftype="wp" popup="no"> . . .
</link>
```

46.2.3.2 Link return action. After linking to the referenced information, the system needs to determine whether to return to the link origin or remain with the linked information after completing the referenced task. The **linktype** attribute is set to "goto" (indicates does not return to the original link) or "return" (indicates returns to the original link). Cases where "goto" would be implemented are next test procedure, follow-on maintenance task, etc. Cases where "return" would be implemented are initial setup part item, equipment condition work packages, additional operation or maintenance work package, etc. After returning from a link, an optional follow-on link ***<returnlink>*** is provided to redirect the user to next information object. An example of follow-on link is after a test evaluation indicates a corrective action and is immediately linked to the information. After the task is completed, the user needs to confirm the corrected fault has been corrected. A follow-on link (prompted or immediate) conducts a retest of the malfunction (determines if the corrections fixed the malfunction and no additional malfunctions exist). The follow-on link example shows an immediately jump to the corrective maintenance work package and after returning will be prompted to retest (determine if the malfunction was fixed).

```
<link xmlns:xlink CDATA #FIXED "http://www.w3.org/1999/xlink" linkaction="immediate"
application="frame" local="M00003-X-XXX-XXX" xlink:type="simple"
linktype="return" xreftype="wp" popup="no">
<returnlink>
<link xmlns:xlink CDATA #FIXED "http://www.w3.org/1999/xlink" linkaction="prompt"
application="frame" local="T00003-X-XXX-XXX" xlink:type="simple"
linktype="goto" xreftype="wp" popup="no">
<prompt>Retest corrective fault
</prompt>
</link>
</returnlink>
</link>
```

46.2.3.3 Launch an application. The ***<link>*** element provides the capability to launch an application or program through the **xlink:href** attribute. The application file location is entered in the **xlink:href** (i.e. launch 1558 test software **xlink:href="1553.dll"**). If any parameters are needed for the application the **parameter** attribute would contain the information (i.e.1553 reset parameters **parameter="RESET RECALIBRATE"**). The application location should be neutral to the directory structure, therefore the application should be in the current directory as the XML source or identified through an URI. An application using a prompted link is described in 46.2.3.1.

46.2.3.4 Launch multimedia. The ***<link>*** element provides the capability to launch multimedia files through the **xlink:href** attribute. Similar to the launch application, the multimedia file is entered in **xlink:href** (i.e. launch HMMWV training video **xlink:href="HMMWV_training.avi"**). The multimedia file location should be neutral to the directory structure, therefore the multimedia file should be in the current directory as the XML source or identified through an URI. The multimedia file is identified with the multimedia rendering application by either the file header information or file extension (as in example "avi"). During the loading of the XML source files, multimedia files, and frame-based viewer, the installation procedure needs to verify the appropriate multimedia rendering applications are available.

46.2.4 Figure linking. Associated figure link elements are figure callout reference ***<callout>***, figure hot spot reference ***<mapref>***, and manufactured items figure number reference ***<figno>***.

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46.2.4.1 Figure callout reference ***<callout>***. The element ***<callout>*** appears in text to reference a callout number, letter, or symbol appearing in a figure. The element is EMPTY and all pertinent information is entered through its attributes. The callout presentation format is "(Figure #, Item #)".

```
<!ELEMENT callout EMPTY>
<!ATTLIST callout
   id       ID     #IMPLIED
   numref   IDREF  #IMPLIED
   partref  IDREF  #IMPLIED
   assocfig IDREF  #REQUIRED
   label    CDATA  #IMPLIED
>
```

46.2.4.1.1 Attribute definition. The ***<callout>*** element has only one required attribute "**assocfig**" that references the figure ***<figure>***. The figure callout number may or may not be linkable in the illustration, thus two methods are used to identify the callout label. The first method is an ID reference "**numref**" using either the Computer Graphic Metadata (CGM) attachment XML reference data file or the element ***<mapref>*** (See 44.2.4) that maps a coordinate(s) for a callout label hot spot within the graphic. The information within the reference provides the callout label. In frame-based system the CGM XML reference data file (or ***<mapref>*** when selected) the link is the callout hot spot in the graphic. The second method is when the composition system has no capability to resolve a callout label reference. The attribute "**label**" is used to enter or hard code the callout label. The final attribute "**partref**" links the associated parts information to the callout label item. The part reference permits the ability to display a parts ordering form.

46.2.4.1.2 Callout example. The following provides several examples using ***<callout>***.

a. Example without a linked figure callout.
   (1) XML fragment is

   <callout assocfig="M00024–X-XXX-XXX-0034" label="2"/>

   (2) Prints as "(Figure 2, Item 2)"

b. Example with a linked callout in conjunction with a mapping reference element ***<mapref>***.
   (1) XML ***<callout>*** fragment

   <callout assocfig="M00024–X-XXX-XXX-0034" numref="M00024–X-XXX-XXX-0348"/>

   (2) XML ***<mapref>*** fragment

   <mapref id=numref="M00024–X-XXX-XXX-0348" label="2">
   <map.circle radius="6" unitmeasure="pt">
   <map.coord x="72" y="72" unitmeasure="pt"/>
   </map.circle>
   </mapref>

   (3) Generates in page-based composition as "(Figure 2, Item 2)".
   (4) Displays in frame-based as a reference "(Figure 2, Item 2)" to the mapped hot spot circle coordinates in the graphic.

46.2.4.2 Figure hot spot reference ***<mapref>***. The element ***<mapref>*** and its child elements are discussed in 44.2.4.

46.2.4.3 Manufactured item figure number reference ***<figno>***. The element ***<figno>*** appears in text to only reference a figure number. The element is EMPTY and all pertinent information is entered through its attribute **idref**. The figure reference presentation format is "Figure #".

46.2.5 Work package sequence number reference ***<wpno>***. The element ***<wpno>*** generates the work package number that is referenced from attribute "**wpref**". The attribute references or links to the work package attribute "**wpno**" (contained in each work package XML element). Page-based systems will display the "WP " and work package sequence number (i.e. WP 0034) and frame-based systems will display the work package title as a hot spot (i.e. Equipment Replacement).

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47 FILTERING.

47.1 Introduction. Filtering is a process that narrows the displayed data to show only a specific and desired sub-set of data. As an example, the complete technical data for an aircraft can be filtered to only display to the user the data that applies to a requested tail-number. MIL-STD-40051-1 describes several types of filtering. including by model series, by modification, by Skill/Maintenance level and by unique identification code. Tags that support filtering are discussed in the following paragraphs

47.2 Effectivity. Filtering based upon differences in components is performed by examining the effectivity of the system. Effectivity is the act or process of identifying weapon systems or end-items and their hardware and software system and subsystems by their associated usable on code (UOC), serial number, model number, part number/CAGEC, NSN, end item code (EIC), software version or modification work order (MWO). Effectivity is included to signify that certain configurations or modifications apply to a given weapon system/equipment. These configurations or modifications may require different maintenance procedures or operating instructions. The maintenance procedures or operating instructions that are applicable to a specific configuration or modification can be associated with that specific configuration or modification through the use of a unique identifier representing effectivity. This identifier is created and saved in the **id** attribute of the ***<applic>*** element. The ***<applic>*** element content model contains an effectivity entry for a given configuration or modification. Elements such as steps will have an **applicable** attribute. The author can enter the unique effectivity identifier(s) into this attribute to signify that the element applies only to a specific configuration(s) or modification(s). When the manual is used for a specific weapon system/equipment, only those elements with an effectivity for that specific weapon system/equipment will be processed and their content displayed where appropriate to the user. Note that when the **applicable** attribute is not set in a given element, the element content applies to all cases. When an element contains different content depending upon the effectivity, and there may be several configurations or modifications, several instances of the element can be created using the alternative conditions capability (see 47.3). In that case the first alternative condition to evaluate to true is the one that is used.

47.2.1 System effectivity ***<applic>***. The System effectivity ***<applic>*** element provides the qualifications to identify the effective system by NSN, part number, unique ID, etc. Each instance of ***<applic>*** has a unique identifier that is created and saved in the **id** attribute of the ***<applic>*** element. This unique identifier is for internal use only and is not seen by the IETM user. It can be created in a manner similar to the work package identification number. (For information on work package identification number structure, refer to MIL-STD-40051-1). The effectivity number can start with an “a”, followed by the work package identification number five digit unique identifier, and the TM number without maintenance level. The ***<applic>*** element content model contains an effectivity entry for a given configuration or modification. This entry can be for a single item, a set of items or a range of items as identified by the ***<single>*** element, the ***<set>*** element and the ***<range>*** element. The optional ***<not>*** element can be used to define an effectivity entry that includes everything except with is identified in the ***<single>***, ***<set>*** element or ***<range>*** elements. The ***<applic>*** element is discussed in 31.2.1

47.2.1.1 Negative ***<not>***. The optional ***<not>*** identifies an effectivity entry that includes everything except what has been identified in the ***<applic>*** element (i.e., in the ***<single>***, ***<set>*** or ***<range>*** elements). The element is EMPTY.

47.2.1.2 Effectivity type set ***<set>***. The ***<set>*** element defines an effectivity filter group using the same information type. The type of information is serial number, Modification Work Order, Usable On Code, Model Number, Software Version, and unique system identifier. The content model is “serialno+ | mwo+ | uoc+ | modelno+ | software_version+ | eqp_id+”.

47.2.1.2.1 Serial number ***<serialno>***. Effectivity can be based upon serial numbers. Differences associated with equipment models or units of the same model that would affect operator or maintenance actions may be related explicitly to serial number ranges. The user would reference the content by serial number or by a range of serial numbers. The content model for the ***<serialno>*** element is #PCDATA.

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47.2.1.3 Range ***<range>***. The ***<range>*** element specifies a range of information for system effectivity. An example for a serial number range would be 00001 – 00100. The type of information is serial number, Modification Work Order, Usable On Code, Model Number, Software Version, and unique system identifier. The content model is "(serialno, serialno) | (mwo, mwo) | (uoc, uoc) | (modelno, modelno) | (software_version, software_version) | (eqp_id, eqp_id) ".

47.2.1.4 Single ***<single>***. The ***<single>*** element specifies a single system effectivity information type. The information types are serial number, Modification Work Order, Usable On Code, Model Number, Software Version, and unique system identifier. The content model is "single (serialno | mwo | uoc | modelno | software_version | eqp_id".

47.2.2 Technical manual system effectivity list ***<applic_ref_list>***. The***<applic_ref_list>*** element contains a listing of all possible system effectivity configurations. When a work package has a system effectivity restriction, the work package references or links to the system effectivity list. The system effectivity ***<applic>*** elements are contained in this list. The IETM software will identify which of the system effectivity ***<applic>*** elements in this list apply to the specific system that the user is working on. This identification may be automatic or it may be obtained by prompting the user at login.

a. The components are:
   (1) System effectivity ***<applic>*** (required – one or more). The System effectivity ***<applic>*** element provides the qualifications to identify the effective system by NSN, part number, unique ID, etc. (see 31.2.1).

b. DTD fragment is graphically depicted.

FIGURE 528. Technical manual system effectivity list <applic_ref_list> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT applic_ref_list (applic+)>
```

d. No attributes:

47.2.3 Effectivity attribute **applicable**. The effectivity attribute **applicable** appears with many elements to identify those configurations that are applicable to the given element. The IETM software will identify which of the system effectivity ***<applic>*** elements in the system effectivity list (***<applic_ref_list>***) apply to the specific system that the user is working on. The effectivity attribute **applicable**, if it exists for an element, will have its referenced id(s) (IDREFS) compared to those selected from the system effectivity list to see if there is at least one match. If there is the element is processed. If there is no match, the element is skipped. The attribute list for an element that contains the **applicable** attribute would include the following:

```
<!ATTLIST ...   applicable IDREFS  #IMPLIED >
```

47.2.4 Effectivity example. The XML source and its stylesheet output (see Figure 529) depicting an operator level maintenance procedure containing a step that is applicable to a specific model is shown below. In the figure, step 3 is shown because the vehicle being worked on is model M2A3. This step would not be displayed if another model was being worked on. The ***<applic_ref_list>*** element shown below contains a listing of all possible system effectivity configurations. The effectivity attribute **applicable** in the ***<step1>*** element that states to "Raise right squad bench seat" identifies that step as pertaining only to model "M2A3".

a. Example of an XML document instance fragment for effectivity.

```
...
<applic_ref_list>
<applic  abbrevcode="M2A2"  id="a0001-X-XXXX-XXX">
<name>Fighting  Vehicle,  Infantry
</name>
<nsn>
```


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<fsc>2350
</fsc>
<niin>01-248-7619
</niin>
</nsn>
<single>
<modelno>M2A2
</modelno>
</single>
</applic>
<applic abbrevcode="M3A2" id="a0002-X-XXXX-XXX">
<name>Fighting Vehicle, Cavalry
</name>
<nsn>
<fsc>2350
</fsc>
<niin>01-248-7620
</niin>
</nsn>
<single>
<modelno>M3A2
</modelno>
</single>
</applic>
<applic abbrevcode="M2A3" id="a0003-X-XXXX-XXX">
<name>Fighting Vehicle, Infantry
</name>
<nsn>
<fsc>2350
</fsc>
<niin>01-436-0005
</niin>
</nsn>
<single>
<modelno>M2A3
</modelno>
</single>
</applic>
<applic abbrevcode="M3A3" id="a0004-X-XXXX-XXX">
<name>Fighting Vehicle, Cavalry
</name>
<nsn>
<fsc>2350
</fsc>
<niin>01-436-0007
</niin>
</nsn>
<single>
<modelno>M3A3
</modelno>
</single>
</applic>
</applic_ref_list> . . .
<maintsk>
<service>
<title>Right front bilge pump

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```
</title>
<step1>
<para>Open power unit access door
<xref posttext=")" pretext="("wpid="M00009-X-XXX-XXXX"/>.
</para>
</step1>
<step1>
<para>Open front hull drain plug.
</para>
<step2>
<para>Open drain plug access door.
</para>
</step2>
<step2>
<para>Remove lock plate from stem.
</para>
</step2>
<step2>
<para>Turn handwheel to open front hull drain plug.
</para>
</step2>
</step1>
<step1 applicable="a0003-X-XXXX-XXX">
<para>Raise right squad bench seat
<xref posttext=")" pretext="("wpid="M00015-X-XXX-XXXX"/>
</para>
</step1>
<step1>
<para>Open rear hull drain plug
</para>
<step2>
<para>Pull up and turn fastener
</para>
</step2>
<step2>
<para>Remove floor plate
</para>
</step2>
<step2>
<para>Remove lock plate from stem
</para>
</step2>
<step2>
<para>Turn handwheelto open rear hull drain plug
</para>
</step2>
</step1>. . .
</service>
</maintsk>
```

b. Stylesheet output.

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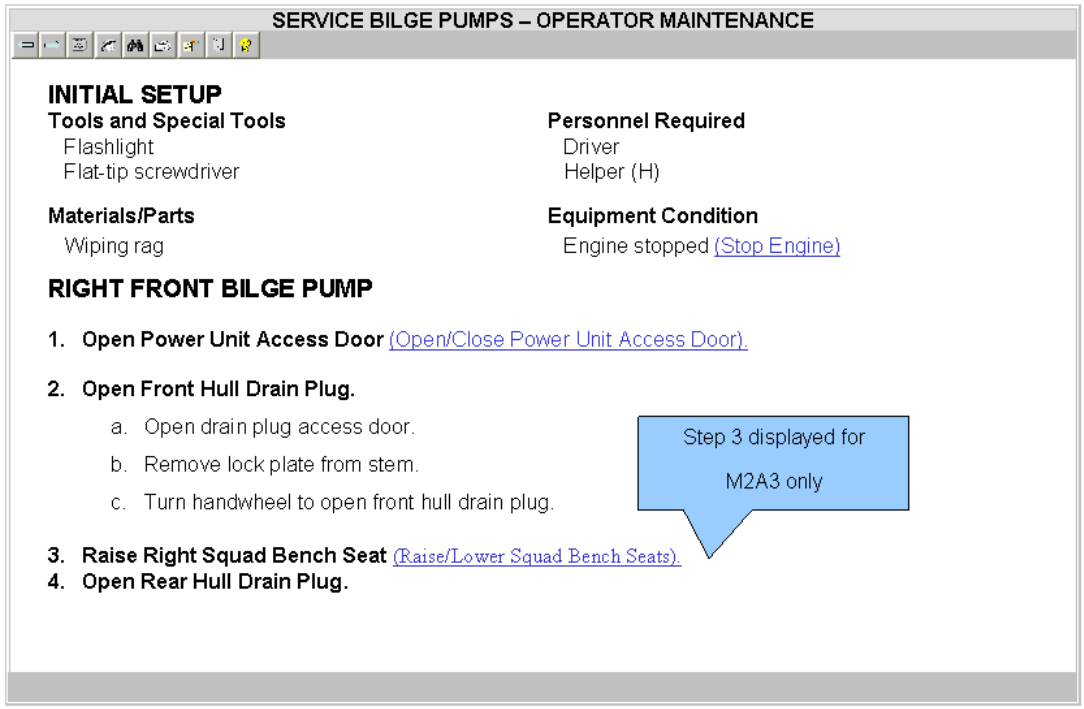

FIGURE 529. Effectivity example.

47.3 Alternative conditions. Alternative conditions enable the user to be presented with information that has been filtered, either to reflect the effectivity of the system and/or to reflect the current system state (see 48.2.1). Filtering with alternative conditions permits the display and processing of different content for an element based upon it's effectivity. For example, a system might have several models and the first step in a procedure might be different for each model. The element ***<step1-alt>*** in the procedure (***<proc>***) element content model would contain a ***<step1>*** element for each model's first step. The effectivity attribute **applicable** in each of those ***<step1>*** elements would be set to the appropriate referenced id(s) in the system effectivity list (***<applic_ref_list>***). If there are no preconditions (***<precond>***) associated with these steps, the first step having the appropriate effectivity will be selected and the remaining steps in the ***<step1-alt>*** element will be ignored. If there are preconditions (see 42.1.1.1), the first step with both a valid effectivity and a "True" evaluation of it's precondition will be the step that is selected. If the above conditions are not satisfied the step would be skipped.

47.3.1 Alternative conditions example. The effectivity example above (see 47.2.4) could have been tagged using alternative conditions. In that case, the three models that do not require the third step to "raise the right squad bench seat" would be grouped as one alternative. This would be particularly useful if those models required that the third step identified an activity other than "raise the right squad bench seat". If the fourth step were again the same for all models, no alternative conditions would be required for that step. The XML source depicting this case is shown below. The ***<applic_ref_list>*** element shown below contains a listing of all possible system effectivity configurations. The effectivity attribute **applicable** in the ***<step1>*** element that states to "Raise right squad bench seat" identifies that step as pertaining only to model

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"M2A3". The effectivity attribute **applicable** in the ***<step1>*** element that states (for the purpose of this example) to "Do something that pertains to the other three steps", identifies that step as pertaining to the other three models. Both of these steps are included in the content model for ***<step1-alt>***.

a. XML document instance fragment.

```
. . .
<applic_ref_list>
<applic abbrevcode="M2A2" id="a0001-X-XXXX-XXX">
<name>Fighting Vehicle, Infantry
</name>
<nsn>
<fsc>2350
</fsc>
<niin>01-248-7619
</niin>
</nsn>
<single>
<modelno>M2A2
</modelno>
</single>
</applic>
<applic abbrevcode="M3A2" id="a0002-X-XXXX-XXX">
<name>Fighting Vehicle, Cavalry
</name>
<nsn>
<fsc>2350
</fsc>
<niin>01-248-7620
</niin>
</nsn>
<single>
<modelno>M3A2
</modelno>
</single>
</applic>
<applic abbrevcode="M2A3" id="a0003-X-XXXX-XXX">
<name>Fighting Vehicle, Infantry
</name>
<nsn>
<fsc>2350
</fsc>
<niin>01-436-0005
</niin>
</nsn>
<single>
<modelno>M2A3
</modelno>
</single>
</applic>
<applic abbrevcode="M3A3" id="a0004-X-XXXX-XXX">
<name>Fighting Vehicle, Cavalry
</name>
<nsn>
<fsc>2350
</fsc>
<niin>01-436-0007
</niin>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
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```
</nsn>
<single>
<modelno>M3A3
</modelno>
</single>
</applic>
</applic_ref_list>  ...
<maintsk>
<service>
<title>Right front bilge pump
</title>
<step1>
<para>Open powerunit access door
<xref posttext=")" pretext="("wpid="M00009-X-XXX-XXXX"/>.
</para>
</step1>
<step1>
<para>Open front hull drain plug.
</para>
<step2>
<para>Open drain plug access door.
</para>
</step2>
<step2>
<para>Remove lock plate from stem.
</para>
</step2>
<step2>
<para>Turn handwheel to open front hull drain plug.
</para>
</step2>
</step1>
<step1-alt>
<step1 applicable="a0001-X-XXXX-XXXa0002-X-XXXX-XXX a0004-X-XXXX-XXX">
<para>Do something that pertains to the other three steps
</para>
</step1>
<step1 applicable="a0003-X-XXXX-XXX">
<para>Raise right squad bench seat
<xref posttext=")" pretext="("wpid="M00015-X-XXX-XXXX"/>
</para>
</step1>
</step1-alt>
<step1>
<para>Open rear hull drain plug
</para>
<step2>
<para>Pull up and turn fastener
</para>
</step2>
<step2>
<para>Remove floor plate
</para>
</step2>
<step2>
<para>Remove lock plate from stem
```


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```
</para>
</step2>
<step2>
<para>Turn handwheel to open rear hull drain plug
</para>
</step2>
</step1>. . .
</service>
</maintsk>
```

47.4 Current system state information. The management of dynamic and static state information is a required capability to achieve intelligent, interactive data display (see 48.1). This is a form of filtering since it narrows the displayed data to show only a specific and desired subset of data. The logic engine, in determining what path to take through the IETM maintains and evaluates this state information (see 48.2). This functionality is fundamental to testing and troubleshooting sequences where the next test is often based on the result of the current test or input from the Materiel interface (dynamic state information). It also allows for the presentation of information to the user to be customized to the Materiel configuration or any system state.

47.5 Service specific information. Filtering can be applied to joint service TMs/IETMs. Each work package includes the attributes **army**, **airforce**, **navy**, **marines** (see 32.3). A "yes value assigned to at least one of these attributes identifies the work package as being applicable only to those specified services. Filtering can also be applied to joint service TMs/IETMs at below the work package level. For example, a table containing fuel indicator correction factors might be only applicable to the Army in a joint service TM. Without filtering the table might be displayed to a Navy user as "Table 3: Fuel Indicator Correction Factors (Army Only)" With filtering the TM could be displayed to that user without table 3. This can be done by setting a state variables and evaluating an expression in the ***<precondition>*** element for that ***<table>*** element (see 48.2).

47.6 Skill level. Filtering at the step level can include the display of steps for a novice only in addition to steps that are for both novice and expert users. All step information is present in the TM. The display of the steps that are intended for novices only can be filtered out when an expert uses the system. The **skilllevel** attribute in the step elements (i.e., ***<step2>*** through ***<step5>***) can be set to either "novice" or "novice_expert". The skill level for a ***<step1>*** element is given the mandatory value of "novice_expert" to indicate that first level steps are always to be displayed for both novices and experts alike (see 33.3).

47.7 Work package level. Each work package has elements and attributes that can serve as inputs into a filtering processing. For example, the ***<wpidinfo>*** element (see 32.2) includes the ***<maintlvl>*** element which specifies the lowest authorized maintenance level to perform the work package. Access to work packages can be restricted to only users who are qualified to perform work at or above that maintenance level. Another example is that each work package has attributes that identified service specific information that can be filtered (see 47.5). Refer to the section on work packages (see 32) to see a complete list of work package elements and attributes.

48 UNIQUE IETM FUNCTIONALITY.

48.1 State tables. The management of dynamic and static state information is a required capability to achieve intelligent, interactive data display. The capability of the system to collect and maintain the system state is achieved through the use of state variables and expressions that use these variables. The system maintains a list of variables and their associated values. This list is called a state table (refer to MIL-STD-40051-1 for state table requirements including state table input, limits and options). The system obtains values for variables through user dialogs and variable assignments. Variables are added as they are encountered in the element ***<variable>***.

48.1.1 Variable declaration ***<variable>***. All variables are declared in a variable declaration element ***<variable>***. The ***<initialize>*** element in the ***<variable>*** element content model assigns a value to the variable. If the ***<initialize>*** element is not used, a null value is set when the a variable is declared. The ***<variable>*** element is only used by the ***<statemanipulation>*** element (see 48.2.3).

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a. The components are:
   (1) Initialize ***<initialize>*** (optional – zero or one) (see 48.1.1.1).
   (2) Dialog Box ***<dialog>*** or ***<dialog-alt>*** (optional – zero or one) (see 48.1.1.2).
   (3) Maintenance Data Collection ***<mdc>*** (optional – zero or more) (see 48.1.1.3).
b. DTD fragment is graphically depicted.

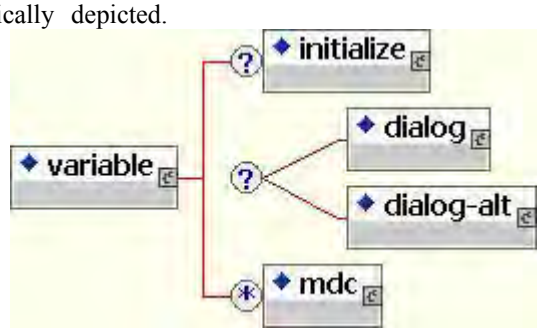

FIGURE 530. Variable declaration <variable> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT variable (initialize?,(dialog | dialog-alt)?, mdc*)>
<!ATTLIST variable  name  CDATA  #REQUIRED
     description  CDATA  #IMPLIED
     config  (yes | no) #IMPLIED
     value-type  (boolean | string | real | integer | fault) #REQUIRED
     precision  CDATA  #IMPLIED
     scope  (global | local)  'global' >
```

d. Unique attributes:
   (1) **name** – The required variable name to be used by the logic engine. Variable names are case insensitive and should not exceed a length of 64 characters.
   (2) **description** – The variable description.
   (3) **config** – Identifies if this variable pertains to a configuration item.
   (4) **value-type** – The required variable declaration type. Valid values are Boolean, string, real, integer, and fault.
   (5) **precision** – The level of precision of real type data.
   (6) **scope** – The state information variable data is kept within a local scope (i.e. it's only valid within the work package) or global (i.e., it's effective for the entire IETM). The default is 'global'.

48.1.1.1 Initialize ***<initialize>***. Initializes a variable to a value. The value is set using either the ***<expression>*** or the ***<fault>*** element in the ***<initialize>*** element content model.

a. The components are:
   (1) Expression ***<expression>*** (optional – zero or one) (see 48.2.2).
   (2) Fault ***<fault>*** (optional – zero or one) (see 48.1.3.5).
b. DTD fragment is graphically depicted.

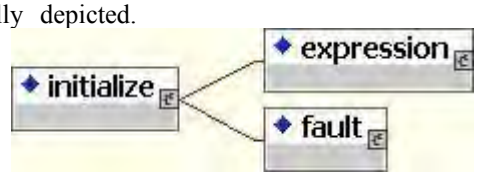

FIGURE 531. Initialize <initialize> DTD hierarchy..

c. DTD fragment:

```
<!ELEMENT initialize (expression | fault)>
```


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48.1.1.2 Dialog box ***<dialog>*** or ***<dialog-alt>***. Defines either a non-conditional or conditional dialog. When a variable is referenced (see 48.1.2), an error condition occurs if a value for that variable does not exist. The author has the option to override this error condition by creating a dialog box (see 48.3) that would be displayed to the user when such an error occurs. This dialog box is specified in the content model for the ***<variable>*** element. The dialog box will include a variable reference that will assign a value to that variable. The author is responsible for correctly creating the dialog box (to include the use of the correct variable name in the variable reference). This dialog box is only processed by the system if the variable is being referenced and a value was not previously assigned.

48.1.1.3 Maintenance data collection ***<mdc>***. Specifies the state variable information value to be used for maintenance data collection (MDC) (i.e. MIL-STD-3008). The state variable information value is stored in the specified MDC variable name and is used in the maintenance records (see 31.11).

48.1.1.4 Variable declaration example. The XML source for several variable declarations is provided below. Each variable has a mandatory name and value type (see 48.1.3) as specified in the **name** and **value-type** attributes. The default value of "global" is set in the **scope** attribute. Several variables have descriptions as entered in the optional **description** attribute. As can be seen, descriptions can be helpful in understanding what the variable represents.

<variable name="Vehicle RPM" scope="global"value-type="integer">
</variable> . . .
<variable name="Channel1field1" scope="global"value-type="string">
</variable> . . .
<variable name="ReadingWithinUpperBound"scope="global" value-type="boolean">
</variable> . . .
<variabledescription="Engine Compartment Fault" name="AT02A001-F1"
scope="global"value-type="fault">
</variable> . . .
<variable description="WiringHarness W420" name="AT02A001-F2" scope="global"
value-type="fault">
</variable>. . .
<variable description="Evaluation Result" name="AT02A00114-EV-182"
scope="global" value-type="integer">
</variable> . . .
<variabledescription="Radius of Map Circle" name="mapcircle"
scope="global"value-type="real">
</variable> . . .
<variable name="Oil filter"scope="local" value-type="fault">
</variable>

48.1.2 Variable reference ***<variableref>***. The element ***<variableref>*** is used to either assign/reassign a value to a variable or to access the variable's value. The variable should have been previously declared in a variable declaration (see 48.1.1). The value can be assigned by the TM author in the source data, set by results from integrated tests, and/or input by the user. The variables are released or lost at the end of a maintenance session, unless specifically requested to save and resume the state table variable values. Evaluating the state table variables is accomplished through the Logic Engine (see 48.2). The content model for ***<variableref>*** is EMPTY. The name of the variable is identified in the **name** attribute.

a. DTD fragment:

```
<!ELEMENT variableref EMPTY>
<!ATTLIST variableref  name CDATA  #REQUIRED >
```

b. Unique attributes:

(1) **name** – The required variable name to be used by the logic engine. Variable names are case insensitive and should not exceed a length of 64 characters.

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48.1.2.1 Variable reference example. The XML source for several variable references is provided below. These references are to the same variable as shown in the example of variable declarations (see 48.1.1.4). Each variable reference has a mandatory name as specified in the **name** attribute.

<variableref name="Vehicle RPM"/> . . .
<variablerefname="Channel1field1"/> . . .
<variableref name="ReadingWithinUpperBound"/> . . .
<variableref name="AT02A001-F1"/> . . .
<variableref name="AT02A001-F2"/> . . .
<variableref name="AT02A00114-EV-182"/> . . .
<variableref name="mapcircle"/> . . .
<variableref name="Oil filter"/>

48.1.3 Values. Data that is used in expressions and assigned to variables areof an explicit data type. The available data types are identified and described in MIL-HDBK-1222. Valid values are Boolean, string, real, integer, and fault. Examples of variables being assigned an explicit data type are given in 48.1.1.4. Value type errors are captured by checking the characters against the defined value type. For instance, the integer type would fail if the value had alpha characters, such as "Yes".

48.1.3.1 Boolean ***<boolean>***. The value for a Boolean data type is true or false. This value is assigned by selecting the appropriate element in the content model for ***<boolean>***.

a. The components of ***<boolean>*** are:
   (1) True ***<true>*** (either this element or element ***<false>*** is required) (see 48.1.3.1.1).
   (2) False ***<false>*** (either this element or element ***<true>*** is required) (see 48.1.3.1.2).
b. DTD fragment is graphically depicted.

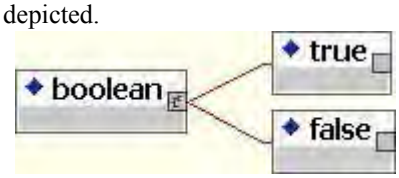

FIGURE 532. Boolean <boolean> DTD hierarchy. .

c. DTD fragment:

```
<!ELEMENT boolean (true | false)>
<!ELEMENT true EMPTY>
<!ELEMENT false EMPTY>
```

d. No attributes.

48.1.3.1.1 True ***<true>***. Assigns the Boolean value to "True". The content model is EMPTY.

48.1.3.1.2 False ***<false>***. Assigns the Boolean value to "False". The content model is EMPTY.

48.1.3.2 String ***<string>***. The value for string data is a set of ASCII characters. The content model is #PCDATA.

a. DTD fragment is graphically depicted.

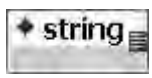

FIGURE 533. String <string> DTD hierarchy. .

b. DTD fragment:

```
<!ELEMENT string (#PCDATA)>
```

c. No attributes.

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Check the source to verify that this is the current version before use.

48.1.3.3 Real ***<real>***. Real data consists of both whole numbers and decimals. The content model is #PCDATA.

a. DTD fragment is graphically depicted.

FIGURE 534. Real <real> DTD hierarchy.

b. DTD fragment:

```
<!ELEMENT real (#PCDATA)>
```

c. No attributes.

48.1.3.4 Integer ***<integer>***. Integer data consists of whole numbers only. The content model is #PCDATA.

a. DTD fragment is graphically depicted.

FIGURE 535. Integer <integer> DTD hierarchy.

b. DTD fragment:

```
<!ELEMENT integer (#PCDATA)>
```

c. No attributes.

48.1.3.5 Fault ***<fault>***. The ***<fault>*** element identifies fault conditions.

a. The components of ***<fault>*** are:
   (1) Fault OK text ***<oktext>*** (optional – zero or one) (see 48.1.3.5.1).
   (2) Fault bad condition text ***<badtext>*** (optional – zero or one) (see 48.1.3.5.2).
   (3) Fault possible text ***<possibletext>*** optional – zero or one) (see 48.1.3.5.3).

b. DTD fragment is graphically depicted.

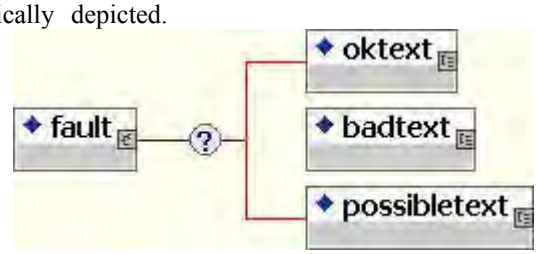

FIGURE 536. Fault <fault> DTD hierarchy.

c. DTD fragment:

```
<<!ELEMENT fault (oktext , badtext , possibletext)?>
<!ATTLIST fault  fault-code  CDATA  #REQUIRED
     fault-ref   IDREF  #IMPLIED
     fault-state  (ok | bad | possible )  #REQUIRED
     weight      CDATA  #IMPLIED
     test        CDATA  #IMPLIED
     dependency  CDATA  #IMPLIED >
```

d. Unique attributes:
   (1) **fault-code** – Identifies the associated fault code.
   (2) **fault-ref** – Provides a link to a reference to fault information.

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(3) **fault-state** – The current state of the test fault code. Enter OK, possible, or bad.
(4) **weight** – Initial weight factor to determine the fault probability.
(5) **test** – Specifies the resolution test procedure, which isolates the fault.
(6) **dependency** – Dependencies that may effect the fault probability.

48.1.3.5.1 Fault OK text ***<oktext>***. This element contains the text for when the fault condition is not the problem.

48.1.3.5.2 Fault bad condition text ***<badtext>***. This element contains the text for when the fault condition is bad.

48.1.3.5.3 Fault possible text ***<possibletext>***. This element contains the text for when the fault condition is possible.

48.2 Logic engine. The logic engine is a software component that determines what path to take through the IETM. This software may be the function of either the IETM presentation application or some form of Maintenance Information System that includes the IETM presentation capability. The functionality that the Logic Engine provides is fundamental to testing and troubleshooting sequences where the next test is often based on the result of the current test or input from the Materiel interface (dynamic state information). It also allows for the presentation of information to the user to be customized to the Materiel configuration or any system state. To do this the Logic Engine will traverse through the work packages, evaluate alternative conditions, maintain state information, display dialogs and evaluate expressions. This section discusses alternative conditions, the manipulation of state information, and the evaluation of expressions. The creation and referencing of state information is addressed in 48.1. Dialog boxes are discussed in 48.3.

48.2.1 Alternative conditions. The logic engine provides the capability for the user to be presented with information that has been context filtered to reflect the current system state. The author can define system conditions which determine when alternative procedures, statements, or steps are to be followed. This is indicated through a precondition expression (see 48.2.1.1). This expression is included in the content model of the elements that represent a set of alternatives. The naming convention for these elements is ***<ELEMENT-alt>***. The content model for these type of elements provides for multiple occurrences of the element. For example, the element ***<dialog>*** creates a dialog box (see 48.3.2). The ***<precond>*** element is included in the content model for dialog. If the author wants to specify the use of alternative dialog boxes based on the evaluation of preconditions, the element ***<dialog-alt>*** would be used. The content model for ***<dialog-alt>*** is as follows:

```
<!ELEMENT dialog-alt (dialog+)>
```

A precondition would be set within each dialog element that is included as an alternative. The set of alternative conditions should produce only one or possibly no valid alternative conditions.

48.2.1.1 Precondition ***<precond>***. The element ***<precond>*** consists of a single Boolean expression (see 48.2.2). The Boolean expression is evaluated by the Logic Engine (requires state (variable) information). If the Boolean expression evaluates to "True", then the data in the element following the expression is presented; otherwise, it is not. An example of a precondition is if test equipment is already connected and a state variable has been set to as connected. The user has no requirement to have the procedure re-displayed, since the test equipment has previously been connected.

a. The components of ***<precond>*** are:
   (1) Expression ***<expression>*** (required) (see 48.2.1.1.1).
b. DTD fragment is graphically depicted.

FIGURE 537. Precondition <precond> DTD hierarchy. .

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c. DTD fragment:

```
<!ELEMENT precond (expression)>
<!ATTLIST precond  id ID  #IMPLIED >
```

d. Common attributes:
   (1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

48.2.1.1.1 Expression ***<expression>***. The element ***<expression>*** (see 48.2.2) in this context, defines an equation or expression to evaluate to the precondition (requires boolean results).

48.2.2 Expression evaluation. The capability to branch, loop, and filter data is made possible by expression evaluation. Expressions are defined in the content of the ***<expression>*** element (see 48.2.2.1). Expressions may include several data types and a rich set of operations are supported for each data type. An expression contains one of four operation forms: a binary operation between two expressions, a unary operation on a single expression, a state table variable, or a constant value. Expressions are authored by combining variables (or other expressions) and values with binary or unary operators. In their simplest form, an expression may be just a variable or a value.

a. Binary Operation. Binary operations are defined as an operation between two expressions. MIL-HDBK-1222 lists, for each operation, the forms that the operation can take, the return value data type, and provides a description. For example:
   (1) Binary Operation: ***<and>***
   (2) Form: Boolean ***<and>*** Boolean
   (3) Return Value: Boolean
   (4) Description: 'True' if both Booleans are 'true', otherwise 'False'.

b. Unary Operation. Unary operations are defined as an operation on a single expression. MIL-HDBK-1222 lists the unary operations the Logic Engine needs to support and for each operation identifies the forms that the operation can take, the returned value data type, and provides a description. For example:
   (1) Unary Operation: ***<factorial>***
   (2) Form: Boolean ***<factorial>*** Integer
   (3) Return Value: Integer
   (4) Description: Returns the product of all integers up to and including the given integer.

   The form information defines the return value types for the expression on each side of the operator.

c. State table variable. If the expression contains only a ***<variableref>*** element, the value of the expression is obtained by looking up the current value of the variable in the state information.

d. Value. Values within an expression represent a constant. A value has an explicit data type through the value type declaration (see 48.1.3). If the expression contains only a value, the value is returned when the expression is evaluated.

48.2.2.1 Expressions ***<expression>***. The content model for the ***<expression>*** element contains numerous elements. These elements are grouped in this section for the purpose of simplifying the discussion of how to tag expressions. For example, there are 19 binary operators. These operators will be addressed as one component.

a. The components are:
   (1) Expression ***<expression>*** (optional) (see 48.2.2.1.1).
   (2) Variable reference ***<variableref>*** (optional) (see 48.2.2.1.2).
   (3) Value content model components (optional) (see 48.2.2.1.3).
   (4) NIL value ***<nil>*** (optional) (see 48.2.2.1.4).
   (5) Unary operator content model components (optional) (see 48.2.2.1.5).
   (6) Binary operator content model components (optional) (see 48.2.2.1.6).

b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

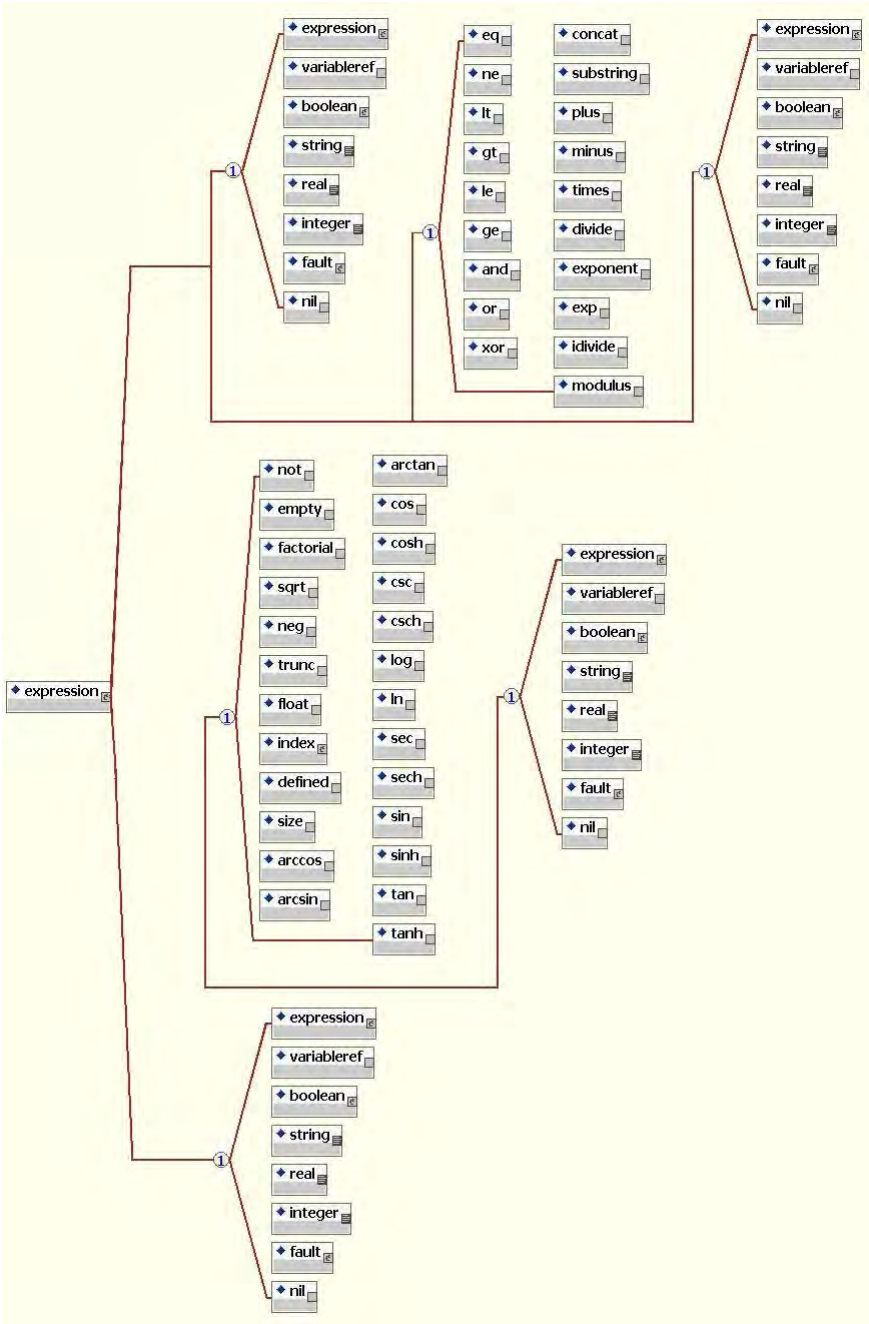

FIGURE 538. Expression <expression> DTD hierarchy. .

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT expression (((expression | variableref | boolean | string | real |
     integer | fault | nil), ((eq | ne | lt | gt | le | ge | and | or | xor |
     concat | substring | plus | minus | times | divide | exponent | exp |
     idivide | modulus), (expression | variableref | boolean | string | real |
     integer | fault | nil))?) | ((not | empty | factorial | sqrt | neg | trunc |
     float | index | defined | size | arccos | arcsin | arctan | cos | cosh |
     csc | csch | log | ln | sec | sech | sin | sinh | tan | tanh), (expression |
     variableref | boolean | string | real | integer | fault | nil)))>
<!ATTLIST expression  id          ID     #IMPLIED
     print-form CDATA  #IMPLIED >
```

d. Unique attributes:

(1) **print-form** – The expression format to be displayed. This attribute can contain descriptive text concerning how to print/display the expression.

e. Common attributes:

(1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

48.2.2.1.1 Nesting of expressions. Provides the capability to nest expressions. The ***<expression>*** element can be nested so that the nested expression can return a value to the parent expression. This returned value can be output as the result of the parent expression or it can used as an operand (i.e., a value to be input to the operator) in a binary or unary operation. An example is provided in 48.2.2.6.

48.2.2.1.2 Variable reference ***<variableref>***. The ***<variableref>*** element provides the capability to access a variable's value (see 48.1.2). This value can be returned as the result of the expression or it can used as an operand (i.e., a value to be input to the operator) in a binary or unary operation. See 48.2.2.3 through 48.2.2.6 for examples.

48.2.2.1.3 Value content model components. The ***<expression>*** element can contain an explicit value that is returned upon evaluation of the expression or it can used as an operand (i.e., a value to be input to the operator) in a binary or unary operation. This value can be of type boolean, string, real, integer, or fault (see 48.1.3). Examples are provided in 48.2.2.2.

48.2.2.1.4 NIL value ***<nil>***. The element when assigned to a state (variable) information will clear any value and causes state (variable) information to be empty. The ***<expression>*** element can contain a NIL value that is returned upon evaluation of the expression or it can used as an operand (i.e., a value to be input to the operator) in an appropriate binary or unary operation. An example is provided in 48.2.2.2.

48.2.2.1.5 Unary operator content model components. A unary operator specifies the type of operation (e.g., factorial, square root, etc.) to be performed on a single expression. The ***<expression>*** element that performs a unary operation will contain a unary operator element (e.g., ***<facorial>***, ***<sqrt>***, etc.) The content model for each unary operator is EMPTY. MIL-HDBK-1222 lists the unary operations the Logic Engine needs to support. Examples are provided in 48.2.2.4.

48.2.2.1.6 Binary operator content model components. A binary operator specifies the type of operation (e.g., less than, greater than, etc.) to be performed between two expressions. The ***<expression>*** element that performs a binary operation will contain a binary operator element (e.g., ***<lt>***, ***<gt>***, etc.) The content model for each binary operator is EMPTY. MIL-HDBK-1222 lists the binary operations the Logic Engine needs to support. Examples are provided in 48.2.2.5.

48.2.2.2 Expressions – value examples. The XML source for several expressions that simply return a constant value is provided below. For example, the expression with the ***<integer>*** tag would return an integer value of "8". The last example would return a null value.

<expression>
<string>CONNECTED
</string>
</expression> . . .

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```
<expression>
<real>1.25
</real>
</expression>   . . .
<expression>
<boolean>
<true/>
</boolean>
</expression>. . .
<expression>
<integer>8
</integer>
</expression>. . .
<expression>
<nil/>
</expression>
```

48.2.2.3 Expressions – variable examples. The XML source for several expressions that return the value of a variable is provided below. The value type being returned would be that of the variable being referenced.

```
<expression>
<variableref name="Vehicle RPM"/>
</expression>. . .
<expression>
<variableref name="Channel1field1"/>
</expression>. . .
<expression>
<variableref name="ReadingWithinUpperbound"/>
</expression>
```

48.2.2.4 Expressions – unary operation examples. The XML source for several expressions having a unary operation is provided below. In these examples the variable named "ReadingWithinUpperBound" is of type Boolean and the variable named "mapcircle" is of type real. The last example depicts a unary operation of an integer constant. The ***<neg>*** operation on the integer value "23" would return a "-23".

```
<expression>
<not/>
<variableref name="ReadingWithinUpperbound"/>
</expression>   . . .
<expression>
<empty/>
<variableref name="ReadingWithinUpperbound"/>
</expression>   . . .
<expression>
<sqrt/>
<variableref name="mapcircle"/>
</expression>   . . .
<expression>
<ln/>
<variableref name="mapcircle"/>
</expression>   . . .
<expression>
<neg/>
<integer>23
</integer>
</expression>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

48.2.2.5 Expressions — binary operation examples. The XML source for two expressions having a binary operation is provided below. In the first example, the variables named "ReadingWithinUpperBound" and "ReadingWithinLowerBound" are of type Boolean. The ***<and>*** operator returns a true value if the two variables are true. A false value is returned if either (or both) variable is false. In the second example, the variable named "mapcircle" is of type real. The Boolean value of true is returned if the value of "mapcircle" is greater than 1.0. The Boolean value of false is returned if the value of "mapcircle" is less than or equal to 1.0.

```
<expression>
<variableref name="ReadingWithinUpperbound"/>
<and/>
<variableref name="ReadingWithinLowerbound"/>
</expression>. . .
<expression>
<variableref name="mapcircle"/>
<gt/>
<real>1.0
</real>
</expression>
```

48.2.2.6 Expressions – complex example. The XML source for a more complex expression is provided below. In this example the logic engine (by evaluating the ***<defined>*** unary operation) checks for the existence of a variable named "ReadingWithinUpperBound". The Boolean result of this operation is ANDed with the result of comparing (using the ***<eq>*** binary operation) the Boolean value of the variable named "ReadingWithinLowerBound" to the Boolean value of true.

```
<expression>
<expression>
<defined/>
<variableref name="ReadingwithinUpperBound"/>
</expression>
<and/>
<expression>
<variableref name="ReadingWithinLowerBound"/>
<eq/>
<boolean>
<true/>
</boolean>
</expression>
</expression>
```

48.2.3 State information manipulation ***<statemanipulation>***. The element ***<statemanipulation>*** performs three state information functions. The first function will initialize a variable by defining the name, type, scope, etc. The second function is to initialize a variable and assign it a value through an expression or fault condition. The third function is to assign a value or stamp the date-time to a previously defined variable.

a. The components are:
   (1) Precondition ***<precond>*** (optional – zero or one) (see 48.2.3.1).
   (2) Variable ***<variable>*** (optional – zero or one) (see 48.2.3.2).
   (3) Variable Reference ***<variableref>*** (optional – zero or one) (see 48.2.3.3).
   (4) Fault ***<fault>*** (optional – zero or one) (see 48.2.3.4).
   (5) Date-Time Stamp ***<date-time_stamp>*** (optional – zero or one) (see 48.2.3.5).
   (6) Expression ***<expression>*** (optional – zero or one) (see 48.2.3.6).
b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

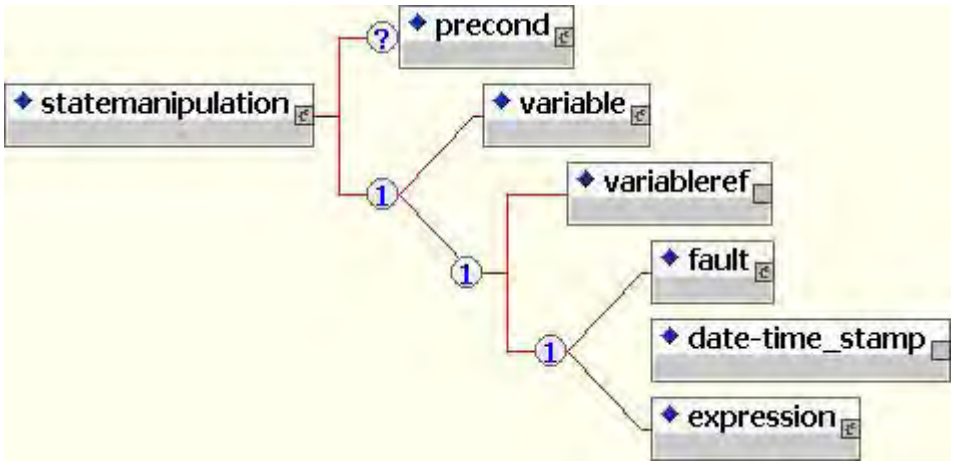

FIGURE 539. State information manipulation <statemanipulation> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT statemanipulation (precond?, (variable | (variableref, (fault |
     date-time_stamp | expression))))>
<!ATTLIST statemanipulation
     applicable IDREFS  #IMPLIED
     inschlvl  CDATA   #IMPLIED
     delchlvl  CDATA   #IMPLIED
     comment   CDATA   #IMPLIED
     changeref IDREFS  #IMPLIED
     id        ID      #IMPLIED
     idref     IDREFS  #IMPLIED
     assocfig IDREFS  #IMPLIED
     skilltrk    CDATA  #IMPLIED
     security  (uc | fouo | c | s | ts )  #IMPLIED >
```

d. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

48.2.3.1 Precondition ***<precond>***. The optional element ***<precond>*** (see 48.2.1.1) contains a Boolean expression. The Boolean expression is evaluated by the Logic Engine. If the Boolean expression evaluates to "True", then the state information manipulation function is performed, otherwise the remaining content in ***<statemanipulation>*** is skipped.

48.2.3.2 Variable ***<variable>***. The variable declaration element ***<variable>*** (see 48.1.1) identifies the variable to be initialized. When an optional ***<fault>*** or ***<expression>*** element is included in the ***<initialize>*** element in the ***<variable>*** element content model, the fault or expression value is assigned to the variable. Otherwise, the variable is assigned a null value.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

48.2.3.3 Variable Reference ***<variableref>***. The ***<variableref>*** element identifies the existing variable to be either assigned a value or have a date-time stamp be recorded. The assignment of a value is done through the ***<fault>*** (see 48.1.3) or ***<expression>*** (see 48.2.2) elements. The element ***<date-time_stamp>*** (see below) records the date-time.

48.2.3.4 Fault ***<fault>***. The element ***<fault>*** is an explicit data type (see 48.1.3.5) that provides text for the fault condition (i.e., OK, Possible Fault, and Known (Bad) Fault). An attribute in ***<fault>*** contains the associated fault code.

48.2.3.5 Date-Time Stamp ***<date-time_stamp>***. The ***<date-time_stamp>*** element is used to record or stamp the date and/or time. This information supports maintenance data collection by recording when a variable was initialized/assigned a value. The DTD fragment for ***<date-time_stamp>*** is as follows.

```
<!ELEMENT date-time_stamp EMPTY>
<!ATTLIST date-time_stamp  date-time  (date | time | date-time )  'date-time' >
```

As can be seen, the content model is EMPTY. The **date-time** attribute determines if the date. time or both are to be recorded. The default is both.

48.2.3.6 Expression ***<expression>***. The ***<expression>*** element (see 48.2.2) evaluates to a value. This value is assigned to the variable being referenced.

48.2.3.7 State information manipulation examples. The XML source for several state information manipulation examples is provided below. These examples show the use of variables and variable references.

```
<statemanipulation>
<variable name="Oilfilter" value-type="fault" scope="local">
<initialize>
<expression>
<fault fault-code="XXX-YYY-ZZZ" fault-state="possible">
<oktext>Oil filter OK.
</oktext>
<badtext>Oil filter faulty.
</badtext>
<possibletext>Oil filter may be faulty.
</possibletext>
</fault>
</expression>
</initialize>
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Vehicle RPM" scope="global" value-type="integer">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel1field1" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="ReadingWithinUpperBound" scope="global" value-type="boolean">
</variable>
</statemanipulation>
<statemanipulation>
<variablerefname="StateInformationVariable1"/>
<expression>
<string>Selected
</string>
</expression>
```


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```
</statemanipulation>
<statemanipulation>
<variableref name="StateInformationVariable2"/>
<expression>
<string>Selected
</string>
</expression>
</statemanipulation>
<statemanipulation>
<variableref name="StateInformationVariable3"/>
<expression>
<string>Selected
</string>
</expression>
</statemanipulation>
<statemanipulation>
<variableref name="Oil filter"/>
<fault fault-code="045-011-005" fault-state="ok">
</fault>
</statemanipulation>
```

48.3 Dialog boxes. Dialog boxes enable the user to communicate with the IETM, thus they are used to obtain information from the user and to provide feedback to the user. There are five kinds of dialog boxes: message, fill-in, menu, multiple-choice, and composite. The element ***<dialog>*** can be used to create each of these dialog box types. Common dialog components (including ***<dialog>***) and each dialog box type are addressed below in separate paragraphs. For each dialog box type, the content model will be discussed first followed by examples showing how to tag the dialog box and the resulting stylesheet output. Note that the binary menu dialog box discussed below is a special case of a menu dialog box.

Dialog boxes can perform common calculations and obtain user acknowledgements. They can display/update state information for systems that provide this capability. Dialog boxes and their components can be filtered. A dynamic enabling feature is also provided. This feature allows for specific menus to be enabled or disabled (i.e., grayed out) depending on menu selections being entered by the user.

Dialog Boxes include several capabilities that are not required by MIL–STD–40051–1/–2. For example, the 2361 DTD permits the author to specify if a dialog box should be displayed as a separate popup or inline with the text. This is done by setting a value in the popup attribute for those elements. This setting can be overridden since it is not a style and format requirement. This may be done for a number of reasons as determined by the procuring activity. One reason could be that the style sheet being used cannot create a separate popup.

48.3.1 Common dialog components. The following elements and attributes are general dialog components used in the dialog objects.

48.3.1.1 Enable selection ***<enable>***. The element ***<enable>*** contains a Boolean expression to determine if the dialog object is active (enabled) or not active (disabled). The Boolean expression is evaluated by the Logic Engine (requires state variable information) each time a dialog data field is entered or selected. If the Boolean expression evaluates to "True", the dialog object is enabled, otherwise the remaining content in the dialog object is disabled (grayed out).

a. The components are:
   (1) Expression ***<expression>*** (required) (see 48.3.1.1.1).
b. DTD fragment is graphically depicted.

FIGURE 540. Enable selection <enable> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT enable (expression)>
```

d. No attributes:

48.3.1.1.1 Expression ***<expression>***. The element ***<expression>*** defines an equation or expression that is evaluated (see 48.2.2)

48.3.1.2 Prompt ***<prompt>***. The element ***<prompt>*** displays text for a response input from the user. Enter text using the ***<emphasis>***, ***<subscript>***, and ***<supscrpt>*** tags as appropriate to generate the displayed text/formatted text. The element ***<variableref>*** displays the current state variable value in the narrative. This utility can be used to display mathematical results, intrusive testing outcomes, etc.

a. The components are:
   (1) Emphasis ***<emphasis>*** (optional – zero or more) (see 49.1.3.1).
   (2) Subscript ***<subscrpt>*** (optional – zero or more) (see 49.1.3.4).
   (3) Superscript ***<supscrpt>*** (optional – zero or more) (see 49.1.3.5).
   (4) Variable reference ***<variableref>*** (optional – zero or more) (see 48.1.2).

b. DTD fragment is graphically depicted.

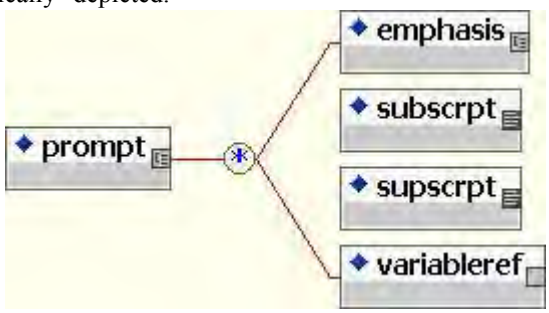

FIGURE 541. Prompt <prompt> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT prompt (#PCDATA | emphasis | subscrpt | supscrpt | variableref)*>
<!ATTLIST prompt  id        ID  #IMPLIED
     position (left | right | top | bottom )  #IMPLIED >
```

d. Unique attributes:
   (1) **position**. – Identifies where the prompt is displayed in relationship to the associated data field (e.g., a fill-in selection).

e. Common attributes:
   (1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

48.3.2 Dialog object ***<dialog>***. The element ***<dialog>*** creates any of the five dialog box types. As described below, this is done by selecting one of more of the elements that represent these types within the ***<dialog>*** content model.

a. The components are:
   (1) Preconditions ***<precond>*** (optional – zero or one) (see 48.3.2.1).
   (2) Title ***<title>*** (optional – zero or one) (see 48.3.2.2).
   (3) Dialog Box Type (required – one or more) (see 48.3.2.3).
   (4) Help information ***<help>*** (optional – zero or one) (see 48.3.2.4).
   (5) State manipulation values (optional – zero or more) (see 48.3.2.5).

b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

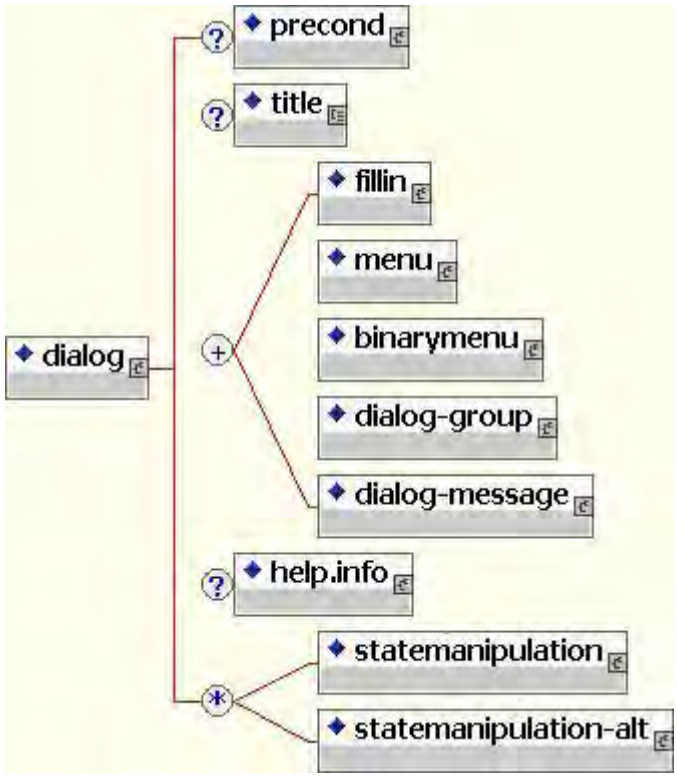

FIGURE 542. Dialog box <dialog> DTD hierarchy. .

c. DTD fragment:

```
<!ELEMENT dialog (precond?, title?, (fillin | menu | binarymenu |
     dialog-group | dialog-message)+, help.info?, (statemanipulation |
     statemanipulation-alt)*)>
<!ATTLIST dialog  popup  (yes | no) #IMPLIED;
     applicable IDREFS  #IMPLIED
     inschlvl  CDATA   #IMPLIED
     delchlvl  CDATA   #IMPLIED
     comment   CDATA   #IMPLIED
     changeref IDREFS  #IMPLIED
     id        ID      #IMPLIED
     idref     IDREFS  #IMPLIED
     assocfig IDREFS  #IMPLIED
     skilltrk    CDATA  #IMPLIED
     security  (uc | fouo | c | s | ts )  #IMPLIED
     cancel_button  (yes | no)  'yes'
     reset_button   (yes | no) #IMPLIED>
```

d. Unique attributes:

(1) **popup** – Identifies how the message box is to be displayed. A “yes” displays the message box as a separate popup window. A “no” displays the dialog box inline with the text.

(2) **cancel_button** – Determines if the dialog box will have a cancel button. A “yes” indicates that the dialog box will display a cancel button, a “no” indicates that it will not. The default is “yes”.

(3) **reset_button** – Determines if the dialog box will have a reset button. A “yes” indicates that the dialog box will display a reset button, a “no” indicates that it will not.

e. Common attributes:

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Check the source to verify that this is the current version before use.

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

48.3.2.1 Precondition ***<precond>***. The optional element ***<precond>*** (see 48.2.1.1) contains a Boolean expression. The Boolean expression is evaluated by the Logic Engine. If the Boolean expression evaluates to "True", then the dialog box is presented to the user, otherwise the remaining content in ***<dialog>*** is skipped.

48.3.2.2 Title ***<title>***. The element ***<title>*** (see 49.1.1.3) contains the optional dialog box title. Not all title element content model components are appropriate for use in tagging a dialog box title. Enter text using the ***<emphasis>***, ***<subscript>***, and ***<supscrpt>*** tags as required to generate the displayed text/formatted text.

48.3.2.3 Dialog box type. The element ***<dialog>*** contains the required choice of fill-in ***<fillin>***, menu ***<menu>***, binary menu ***<binarymenu>***, group dialog ***<dialog-group>***, and/or message ***<dialog-message>*** entry(s). The DTD fragment for this part of ***<dialog>*** is:

```
(fillin | menu | binarymenu | dialog-group | dialog-message)+
```

These elements provide the capability to create the 5 dialog box types. They are addressed below in separate paragraphs for each type.

48.3.2.4 Help information ***<help.info>***. The element ***<help.info>*** provides help information about the data in the dialog box. The text may describe the data that is expected to be entered, where and why the information is needed, etc. This element is optional. When data for ***<help.info>*** is entered, a "HELP" push button is displayed in the dialog box.

48.3.2.5 State manipulation values. The elements ***<statemanipulation>*** (see 48.2.3) and ***<statemanipulation-alt>*** (see 48.2.1) set the state variable information for the dialog box. Several state variables can be set. The state variable is modified when the dialog box is acknowledged by the user.

48.3.3 Fill-in dialog box ***<fillin>***. The element ***<fillin>*** in ***<dialog>*** creates a fill-in-the-blank type of dialog box. This provides the capability to communicate to the IETM by prompting the user to enter text that can then be processed.

a. The components are:
(1) Precondition ***<precond>*** (optional – zero or one) (see 48.3.3.1).
(2) Enable ***<enable>*** (optional – zero or one) (see 48.3.3.2).
(3) Prompt ***<prompt>*** (optional – zero or one) (see 48.3.3.3).
(4) Variable reference ***<variableref>*** (required) (see 48.3.3.4).
(5) Default ***<default>*** (optional – zero or one) (see 48.3.3.5).
(6) Numeric range ***<numrange>*** (optional – zero or one) (see 48.3.3.6).
(7) Validate ***<validate>*** (optional – zero or one) (see 48.3.3.7).

b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

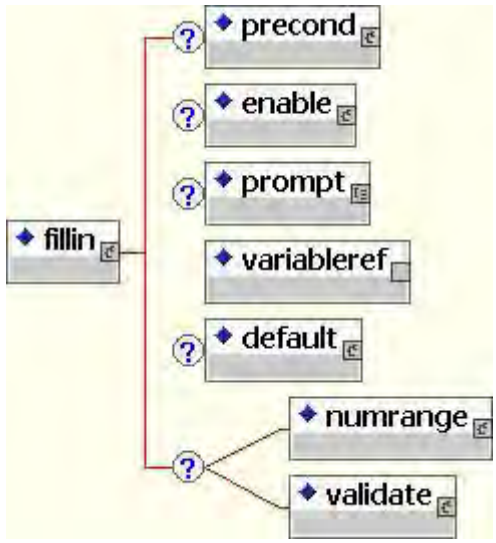

FIGURE 543. Fill-in dialog box <fillin> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT fillin (precond?, enable?, prompt?, variableref, default?,
     (numrange | validate)?)>
<!ATTLIST fillin
     applicable IDREFS  #IMPLIED
     inschlvl  CDATA   #IMPLIED
     delchlvl  CDATA   #IMPLIED
     comment   CDATA   #IMPLIED
     changeref IDREFS  #IMPLIED
     id        ID      #IMPLIED
     idref     IDREFS  #IMPLIED
     assocfig IDREFS  #IMPLIED
     skilltrk    CDATA  #IMPLIED
     security  (uc | fouo | c | s | ts )  #IMPLIED
     mandatory   (yes | no)  'yes'
     fieldsize CDATA  '20' >
```

d. Unique attributes:

(1) **mandatory** – Identifies if the fill-in text is required before the dialog can be completed. The default is "yes" (True). NOTE: An entered space satisfies the mandatory condition.

(2) **fieldsize** – Identifies the fill-in data field width in number of characters. Data that is longer than the displayed field is entered and can be scrolled by using the left and right cursor keys or mouse. The default value is a "20" character width data field.

e. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).

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Check the source to verify that this is the current version before use.

(10) **skilltrk** – Skill level (optional) (see 49.2.2).

48.3.3.1 Precondition ***<precond>***. The optional element ***<precond>*** (see 48.2.1.1) contains a Boolean expression. The Boolean expression is evaluated by the Logic Engine. If the Boolean expression evaluates to "True", then the fill-in is presented to the user, otherwise the remaining content in ***<fillin>*** is skipped.

48.3.3.2 Enable ***<enable>***. The optional element ***<enable>*** determines if the fill-in is active or not active (see 48.3.1.1).

48.3.3.3 Prompt ***<prompt>***. The element ***<prompt>*** contains the prompt text for inputting a response from the user into the associated data field (see 48.3.1.2).

48.3.3.4 Variable reference ***<variableref>***. The element ***<variableref>*** (see 48.1.2) is used to store the user's state information that is input into the fill-in dialog box data field.

48.3.3.5 Default ***<default>***. The element ***<default>*** assigns an optional default value to the fill-in dialog box data field.

a. The components are:
   (1) Expression ***<expression>*** (optional – zero or one) (see 48.3.3.5.1).
   (2) Text ***<text>*** (optional – zero or one) (see 48.3.3.5.2).
   (3) Variable reference ***<variableref>*** (optional – zero or one) (see 48.3.3.5.3).

b. DTD fragment is graphically depicted.

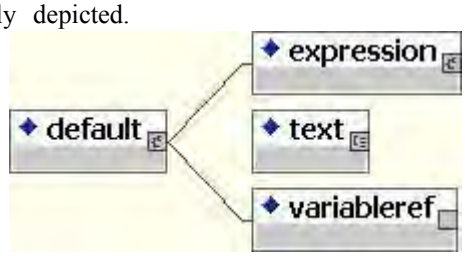

FIGURE 544. Default <default> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT default (expression | text | variableref)>
```

48.3.3.5.1 Expression ***<expression>***. The ***<expression>*** element (see 48.2.2) evaluates to a value. This value is assigned as the default for the fill-in dialog box entry.

48.3.3.5.2 Text ***<text>***. The element ***<text>*** contains textual data that is used as a default for the fill-in dialog box data field. The content model is "(#PCDATA | emphasis | subscrpt | supscrpt)*".

48.3.3.5.3 Variable reference ***<variableref>***. The element ***<variableref>*** (see 48.1.2) is used to retrieve state information that is used as a default for the fill-in dialog box data field.

48.3.3.6 Numeric range ***<numrange>***. The optional element ***<numrange>*** identifies numerical range constraints for a fill-in dialog box data field.

a. The components are:
   (1) Low bound ***<low-bound>*** (optional – zero or one) (see 48.3.3.6.1).
   (2) Double bound ***<double-bound>*** (optional – zero or one) (see 48.3.3.6.2).
   (3) High bound ***<high-bound>*** (optional – zero or one) (see 48.3.3.6.3).

b. DTD fragment is graphically depicted.

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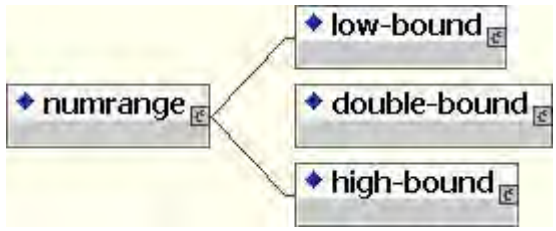

FIGURE 545. Numeric range <numrange> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT numrange (low-bound | double-bound | high-bound)>
```

48.3.3.6.1 Low bound ***<low-bound>***. The element ***<low-bound>*** identifies the minimum acceptable input value (integer or real) for the fill-in dialog box data field. Use this element when there is a minimum range constraint but no maximum.

a. The components are:
   (1) Integer value ***<integer>*** (optional – zero or one) (see 48.1.3.4).
   (2) Real value ***<real>*** (optional – zero or one) (see 48.1.3.3).
   (3) Variable reference ***<variableref>*** (optional – zero or one). (see 48.1.2).
   (4) Expression ***<expression>*** (optional – zero or one). (see 48.2.2).

b. DTD fragment is graphically depicted.

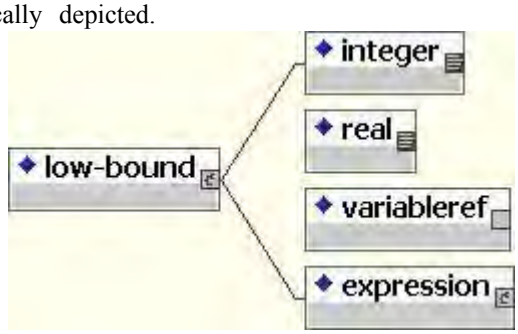

FIGURE 546. Low bound <low-bound> DTD hierarchy..

c. DTD fragment:

```
<!ELEMENT low-bound (integer | real | variableref | expression)>
```

48.3.3.6.2 Double bound ***<double-bound>***. The element ***<double-bound>*** sets the fill-in dialog box data field entry's lowest and highest entered numerical (integer or real) values. Use this element when there is both a minimum and a maximum range constraint.

a. The components are:
   (1) Low bound ***<low-bound>*** (required) (see 48.3.3.6.1).
   (2) High bound ***<high-bound>*** (required) (see 48.3.3.6.3).

b. DTD fragment is graphically depicted.

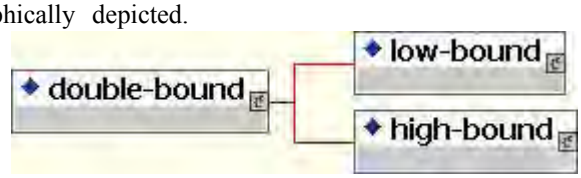

FIGURE 547. Double bound <double-bound> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT double-bound (low-bound, high-bound)>
```

48.3.3.6.3 High bound ***<high-bound>***. The element ***<high-bound>*** identifies the maximum acceptable input value (integer or real) for the fill-in dialog box data field. Use this element when there is a maximum range constraint but no minimum.

a. The components are:
   (1) Integer value ***<integer>*** (optional – zero or one) (see 48.1.3.4).
   (2) Real value ***<real>*** (optional – zero or one) (see 48.1.3.3).
   (3) Variable reference ***<variableref>*** (optional – zero or one). (see 48.1.2).
   (4) Expression ***<expression>*** (optional – zero or one). (see 48.2.2).

b. DTD fragment is graphically depicted.

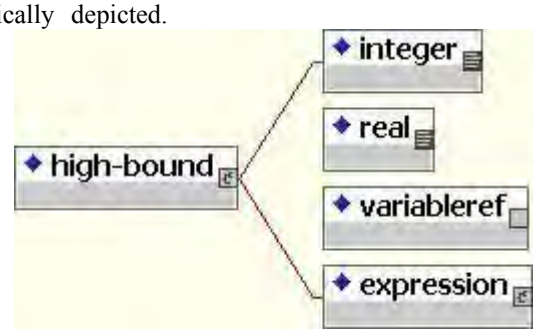

FIGURE 548. High bound <high-bound> DTD hierarchy..

c. DTD fragment:

```
<!ELEMENT high-bound (integer | real | variableref | expression)>
```

48.3.3.7 Validate ***<validate>***. The element ***<validate>*** is used to verify numerical or non-numerical text. The ***<validate>*** element contains the ***<expression>*** element, which is used to evaluate the entered text to a Boolean result.

a. The components are:
   (1) Expression ***<expression>*** (required) (see 48.2.2).

b. DTD fragment is graphically depicted.

FIGURE 549. Validate <validate> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT validate (expression)>
<!ATTLIST validate  errormsg CDATA  #REQUIRED >
```

d. Unique attributes:
   (1) **errormsg** – Indicates the text to be displayed for an invalid input.

48.3.3.8 Fill-in dialog box – multiple entries example. The XML source and its stylesheet output (see Figure 550) for a fill-in dialog box with multiple data fields are provided below. This example depicts a popup dialog box with a single line title and four lines of four data fields. The ***<dialog-group>*** element (see 48.3.8) is used four times to create each line. Each ***<dialog-group>*** element contains four ***<fillin>*** elements. The ***<fillin>*** elements contain a prompt and a data field. The prompt appears to the left of each data field because of the setting in the **position** attribute in the ***<prompt>*** element. The first prompt on each line contains the text and the three other prompts contain a space. The **fieldsize** attributes in the ***<fillin>*** elements are set to 3 indicating that this is the width of each data field. No ***<fillin>*** element contains the

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default element, thus all data fields are initialized as blanks. The value in the **mandatory** attribute in the ***<fillin>*** elements is set to the default of "yes" indicating that an entry in each data field is required. The user will enter state (variable) information into the sixteen data fields. This data will be saved in the specified state variables that are identified by attribute **name** in the element ***<variableref>***>. The element ***<statemanipulation>*** is used to create these variables. The "OK" push button is not enabled because an entry should be made in each of the 16 data fields. The "CANCEL" push button appears since the **cancel_button** attribute in ***<dialog>*** is set to 'yes'.

a. XML document instance fragment.

```
<statemanipulation>
<variable name="Channel1field1" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel1field2" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel1field3" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel1field4" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel2field1" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel2field2" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel2field3" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel2field4" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel3field1" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel3field2" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel3field3" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel3field4" scope="global" value-type="string">
</variable>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</statemanipulation>
<statemanipulation>
<variable name="Channel4field1"scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel4field2" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel4field3" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="Channel4field4" scope="global" value-type="string">
</variable>
</statemanipulation>
<dialog cancel_button="yes" popup="yes">
<title>Enter Title here
</title>
<dialog-group separator="no">
<fillin fieldsize="3" mandatory="yes">
<promptposition="left">Channel 1:
</prompt>
<variablerefname="Channel1field1"/>
</fillin>
<fillin fieldsize="3"mandatory="yes">
<prompt position="left">
</prompt>
<variableref name="Channel1field2"/>
</fillin>
<fillin fieldsize="3" mandatory="yes">
<promptposition="left">
</prompt>
<variableref name="Channel1field3"/>
</fillin>
<fillin fieldsize="3" mandatory="yes">
<prompt position="left">
</prompt>
<variablerefname="Channel1field4"/>
</fillin>
</dialog-group>
<dialog-group separator="no">
<fillin fieldsize="3"mandatory="yes">
<prompt position="left">Channel 2:
</prompt>
<variableref name="Channel2field1"/>
</fillin>
<fillin fieldsize="3" mandatory="yes">
<promptposition="left">
</prompt>
<variableref name="Channel2field2"/>
</fillin>
<fillin fieldsize="3" mandatory="yes">
<prompt position="left">
</prompt>
```


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Check the source to verify that this is the current version before use.

```
<variablerefname="Channel2field3"/>
</fillin>
<fillin fieldsize="3"mandatory="yes">
<prompt position="left">
</prompt>
<variableref name="Channel2field4"/>
</fillin>
</dialog-group>
<dialog-group separator="no">
<fillin fieldsize="3" mandatory="yes">
<prompt position="left">Channel3:
</prompt>
<variableref name="Channel3field1"/>
</fillin>
<fillin fieldsize="3" mandatory="yes">
<prompt position="left">
</prompt>
<variablerefname="Channel3field2"/>
</fillin>
<fillin fieldsize="3"mandatory="yes">
<prompt position="left">
</prompt>
<variableref name="Channel3field3"/>
</fillin>
<fillin fieldsize="3" mandatory="yes">
<promptposition="left">
</prompt>
<variableref name="Channel3field4"/>
</fillin>
</dialog-group>
<dialog-groupseparator="no">
<fillin fieldsize="3" mandatory="yes">
<prompt position="left">Channel 4:
</prompt>
<variableref name="Channel4field1"/>
</fillin>
<fillin fieldsize="3" mandatory="yes">
<promptposition="left">
</prompt>
<variableref name="Channel4field2"/>
</fillin>
<fillin fieldsize="3" mandatory="yes">
<prompt position="left">
</prompt>
<variablerefname="Channel4field3"/>
</fillin>
<fillin fieldsize="3"mandatory="yes">
<prompt position="left">
</prompt>
<variableref name="Channel4field4"/>
</fillin>
</dialog-group>
</dialog>
```

b. Stylesheet output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

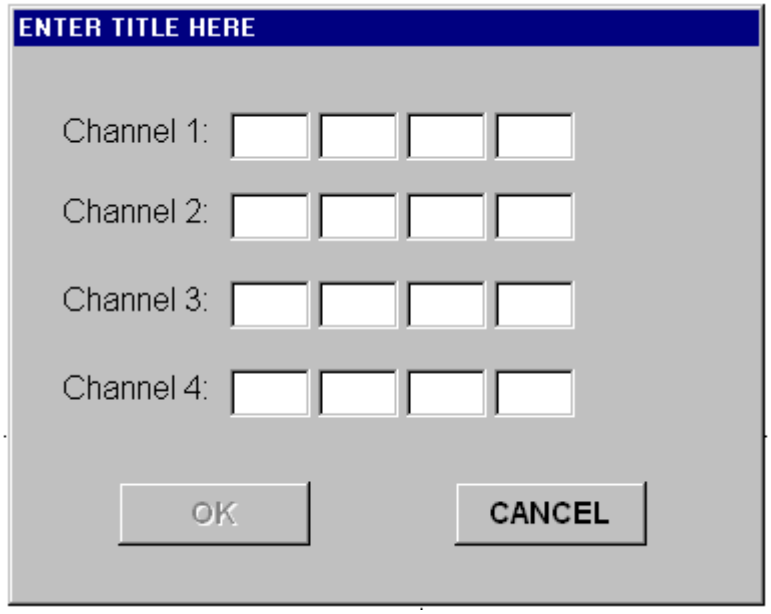

FIGURE 550. Fill-in dialog box – multiple entries example.

48.3.3.9 Fill-in dialog box – number range and assigned default from an expression example. The XML source and its stylesheet output (see Figure 551) for a fill-in dialog box using a number range and an assigned default value from an expression are provided below. This example depicts a popup dialog box with a single line title and one fill-in data field. The ***<fillin>*** element contains a prompt and a data field. The prompt appears to the left of each data field because of the setting in the **position** attribute in the ***<prompt>*** element. The fill-in data will be saved in the variable named "Vehicle RPM" as identified by attribute **name** in the ***<variableref>*** element. The element ***<statemanipulation>***> has been previously used to create this variable. The author defined default value in element ***<default>*** is set to "5" and the allowable range of integer values is set to "0" as the minimum value and "8" as the maximum value. The value of the **mandatory** attribute in the ***<fillin>*** element is set to the default of "yes" indicating that an entry in the data field is required. The **fieldsize** attribute in the ***<fillin>*** element is set to 5 indicating that this is the width of the data field. The "OK" push button is enabled because an entry was made (i.e., the default value of 5) in the data field. The "CANCEL" push button appears since the **cancel_button** attribute in ***<dialog>*** is set to 'yes'.

a. XML document instance fragment.

```
<statemanipulation>
<variable name="Vehicle RPM" scope="global" value-type="integer">
</variable>
</statemanipulation>
<dialog cancel_button="yes" popup="yes">
<fillin fieldsize="5" mandatory="yes">
<prompt position="left">USERRESPONSE
</prompt>
<variableref name="Vehicle RPM"/>
<default>
<expression>
<integer>5
</integer>
</expression>
</default>
<numrange>
<double-bound>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

<low-bound>
<integer>0
</integer>
</low-bound>
<high-bound>
<integer>8
</integer>
</high-bound>
</double-bound>
</numrange>
</fillin>
</dialog>

b. Stylesheet output.

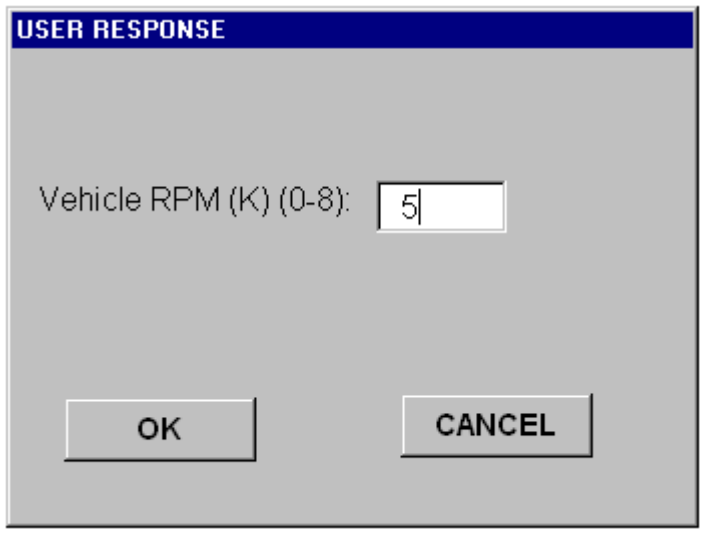

FIGURE 551. Fill-in dialog box – number range and assigned default from an expression example.

48.3.3.10 Fill-in dialog box – validate and assign default from a variable reference example. The XML source and its stylesheet output (see Figure 551) for a fill-in dialog box that validates user input and displays a default from a variable reference are provided below. This example depicts a popup dialog box with a single line title and one fill-in data field. The ***<fillin>*** element contains the prompt and a data field. The prompt as entered in the (***<prompt>***) element appears to the left of each data field because of the setting in the **position** attribute. The fill-in data will be saved in the variable named "Vehicle RPM" as identified by attribute **name** in the ***<variableref>*** element. The element ***<statemanipulation>***> has been previously used to create this variable. The ***<default>*** element in ***<fillin>*** contains the current value of the referenced variable which is "7". The ***<validate>*** element restrict entries to be between "0" as the minimum value and "8" as the maximum value. The value of the **mandatory** attribute in the ***<fillin>*** element is set to the default of "yes" indicating that an entry in the data field is required. The **fieldsize** attribute in the ***<fillin>*** element is set to 5 indicating that this is the width of the data field. The "OK" push button is enabled because an entry was made (i.e., the default value of 7) in the data field. The "CANCEL" push button appears since the **cancel_button** attribute in ***<dialog>*** is set to 'yes'.

a. XML document instance fragment.

<statemanipulation>
<variable name="Vehicle RPM" scope="global"value-type="integer">
<initialize>
<expression>
<integer>7

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
</integer>
</expression>
</initialize>
</variable>
</statemanipulation>
<dialog cancel_button="yes" popup="yes">
<fillin fieldsize="5" mandatory="yes">
<prompt position="left">USER RESPONSE
</prompt>
<variableref name="Vehicle RPM">
<default>
<variableref name="Vehicle RPM">
</default>
<validate errormsg="Value must be integer 0 - 8">
<expression>
<expression>
<integer>0
</integer>
<le>
<variablerefname="Vehicle RPM">
</expression>
<and>
<expression>
<variableref name="VehicleRPM">
<le>
<integer>8
</integer>
</expression>
</expression>
</validate>
</fillin>
</dialog>
```

b. Stylesheet output.

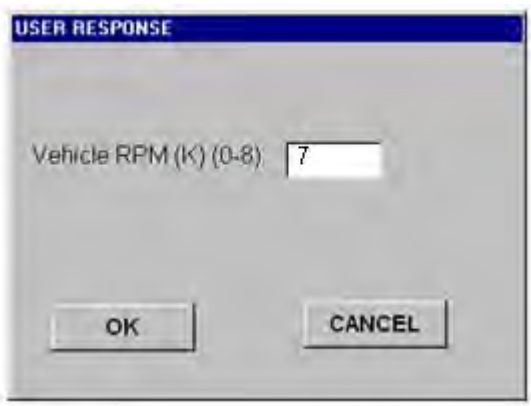

FIGURE 552. Fill-in dialog box – validate and assign default from a variable reference example.

48.3.4 Menu dialog box ***<menu>***. The menu dialog box provides the capability to communicate to the IETM by prompting the user to select one item from a list. The element ***<menu>*** within the element ***<dialog>*** (see 48.3.2 provides this capability in addition to providing a multiple choice capability (see 48.3.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

Note that the element ***<binarymenu>>*** in ***<dialog>*** can also create a menu dialog box in certain cases where there are only two choices (e.g., yes or no). The binary menu dialog box can be simpler to tag (see 48.3.5).

a. The components are:
   (1) Precondition ***<precond>*** (optional – zero or one) (see 48.3.4.1).
   (2) Enable ***<enable>*** (optional – zero or one) (see 48.3.4.2).
   (3) Prompt ***<prompt>*** (required) (see 48.3.4.3).
   (4) Choice ***<choice>*** (required one or more) (see 48.3.4.4).

b. DTD fragment is graphically depicted.

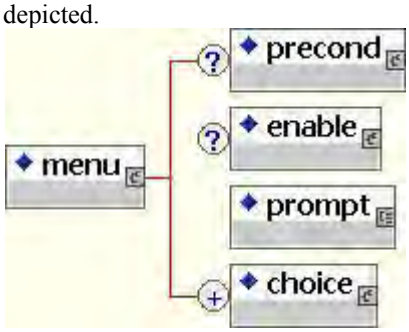

FIGURE 553. Menu <menu> DTD hierarchy. .

c. DTD fragment:

```
<!ELEMENT menu (precond?, enable?, prompt, choice+)>
<!ATTLIST menu
     applicable IDREFS  #IMPLIED
     inschlvl  CDATA   #IMPLIED
     delchlvl  CDATA   #IMPLIED
     comment   CDATA   #IMPLIED
     changeref IDREFS  #IMPLIED
     id        ID      #IMPLIED
     idref     IDREFS  #IMPLIED
     assocfig IDREFS  #IMPLIED
     skilltrk    CDATA  #IMPLIED
     security  (uc | fouo | c | s | ts )  #IMPLIED
     select  (single | multiple )  'single'
     type    (radio | button | pulldown )  #REQUIRED
     mandatory   (yes | no) 'yes'
     flow    (list | inline )  'list'
>
```

d. Unique attributes:
   (1) **select** – A required attribute that specifies if a single selection or multiple selections are permitted. The default is single. For a menu dialog box the select attribute value should be “single”.
   (2) **type** – Defines how the choice items are displayed. The options are as a radio, a button or a pull down menu. For a menu dialog box the type should be set to “radio”.
   (3) **mandatory** – Identifies if a menu choice is required before the dialog can be completed. The default is "yes" (True). For a menu dialog box **mandatory** should be set to “yes”. NOTE: An entered space satisfies the mandatory condition.
   (4) **flow** – Identifies if the selection values are to be displayed vertically (i.e., in a list) or horizontally (i.e. inline). The default is “list”. Either option is acceptable for a menu dialog box.

e. Common attributes:
   (1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
   (2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

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(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

48.3.4.1 Precondition ***<precond>***. The optional element ***<precond>*** (see 48.2.1.1) contains a Boolean expression. The Boolean expression is evaluated by the Logic Engine. If the Boolean expression evaluates to "True", then the menu is presented to the user, otherwise the remaining content in ***<menu>*** is skipped.

48.3.4.2 Enable ***<enable>***. The optional element ***<enable>*** determines if the menu is active or not active (see 48.3.1.1).

48.3.4.3 Prompt ***<prompt>***. The element ***<prompt>*** contains the optional prompt text for the menu (see 48.3.1.2). Any valid value for the **position** attribute is acceptable.

48.3.4.4 Choice ***<choice>***. The element ***<choice>*** is used in an IETM to display a menu choice prompt and to set state (variable) information. Element ***<choice>*** may contain an associated action as either an external application (the element ***<link>***) or a secondary dialog (the element ***<dialog>***). The associated action may be used to provide additional information (sound, image, execute diagnostic test, etc.) to assist with the dialog selection. The action is executed either by selecting the menu choice or clicking a push button (the element ***<button>***) next to the suggested answer. After the action is completed, control returns to the dialog.

a. The components are:
(1) Precondition ***<precond>*** (optional – zero or one) (see 48.3.4.4.1).
(2) Enable ***<enable>*** (optional – zero or one) (see 48.3.4.4.2).
(3) Text ***<text>*** (required) (see 48.3.4.4.3).
(4) Link ***<link>*** (optional – zero or one) (see 48.3.4.4.4).
(5) Dialog ***<dialog>*** (optional – zero or one) (see 48.3.4.4.5).
(6) Button ***<button>*** (optional – zero or more) (see 48.3.4.4.6).
(7) State manipulation values (required – one or more) (see 48.3.4.4.7).

b. DTD fragment is graphically depicted.

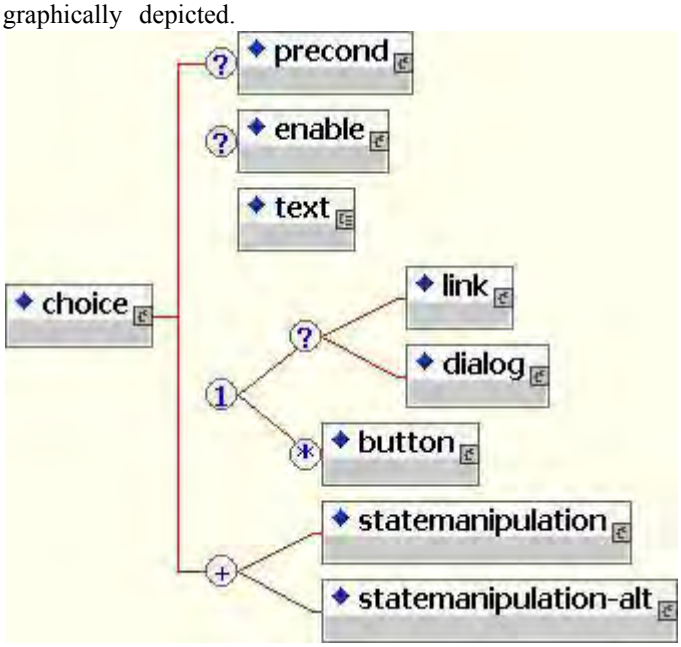

FIGURE 554. Choice <choice> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT choice (precond?, enable?, text, ((link | dialog)? | button*),
     (statemanipulation | statemanipulation-alt)+)>
<!ATTLIST choice
     applicable IDREFS  #IMPLIED
     inschlvl  CDATA   #IMPLIED
     delchlvl  CDATA   #IMPLIED
     comment   CDATA   #IMPLIED
     changeref IDREFS  #IMPLIED
     id        ID      #IMPLIED
     idref     IDREFS  #IMPLIED
     assocfig IDREFS  #IMPLIED
     skilltrk    CDATA  #IMPLIED
     security  (uc | fouo | c | s | ts )  #IMPLIED
     default (yes | no) #IMPLIED  >
```

d. Unique attributes:

(1) **default** – Determines if this selection is the default selection. If **default** is set to "yes" then this selection is the default and it's radio button is shown as being selected when the menu is first displayed. This selection is not the default if the value for **default** is "no" or if it is not specified. If more than one selection has it's default set to "yes" then the first selection in the menu list with a "yes" value for it's **default** attribute is selected as the default.

e. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

48.3.4.4.1 Precondition ***<precond>***. The optional element ***<precond>*** (see 48.2.1.1) contains a Boolean expression. The Boolean expression is evaluated by the Logic Engine. If the Boolean expression evaluates to "True", then the menu choice is presented to the user, otherwise the choice is not displayed.

48.3.4.4.2 Enable ***<enable>***. The optional element ***<enable>*** determines if the menu choice is active or not active (see 48.3.1.1)

48.3.4.4.3 Text ***<text>***. The element ***<text>*** contains textual data to be displayed as a prompt for the menu choice. The content model is "(#PCDATA | emphasis | subscrpt | supscrpt)*".

48.3.4.4.4 Link ***<link>***. Identifies a link (see 46.2.3) to an external application that is associated with the menu selection. After the action is completed, control returns to the dialog.

48.3.4.4.5 Dialog ***<dialog>***. Identifies a secondary dialog (see 48.3.2) that is associated with the menu selection. After the action is completed, control returns to the dialog.

48.3.4.4.6 Button ***<button>***. Presents a push button that can be used (in lieu of selecting the menu choice) to execute an action (to a link or to present a secondary dialog) that is associated with the menu selection.

a. The components are:

(1) Prompt ***<prompt>*** (required). The element ***<prompt>*** contains the prompt text to be displayed for the button (see 48.3.1.2).
(2) Link ***<link>*** (optional – zero or one) (see 48.3.4.4.4).
(3) Dialog ***<dialog>*** (optional – zero or one) (see 48.3.4.4.5).

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Check the source to verify that this is the current version before use.

b. DTD fragment is graphically depicted.

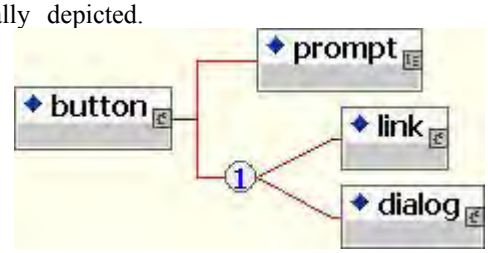

FIGURE 555. Button <button> DTD hierarchy. .

c. DTD fragment:

```
<!ELEMENT button (prompt, (link | dialog))>
```

d. No attributes:

48.3.4.4.7 State manipulation values. The elements ***<statemanipulation>*** (see 48.2.3) and ***<statemanipulation-alt>*** (see 48.2.1) set the state variable information for the menu selection. Several state variables can be set. The state variable is modified when the menu choice is made and the dialog box is acknowledged by the user.

48.3.4.5 Menu dialog box example. The XML source and its stylesheet output (see Figure 556) for a menu dialog box are provided below. This example depicts a popup dialog box with a single line title and three menu selections. Each selection initializes the same variable, but assigns it a different value. The **select** attribute is set by default to “single” indicating that only one choice can be made. The **mandatory** attribute is set by default to “yes” indicating that an entry is required. The first menu selection item is shown as being selected since the value of the **default** attribute for the choice was set to “yes”. An enable condition is set for the third menu selection item. This choice is “grayed” out since the ***<enable>*** element for that menu selection item evaluated to false. The menu prompt appears in bold by using the emphasis tag. The “OK” push button is enabled because a menu selection was made (i.e., the first selection is defaulted to). The “CANCEL” push button appears since the **cancel_button** attribute in ***<dialog>*** is set to 'yes'.

a. XML document instance fragment.

```
<statemanipulation>
<variable name="connected_testXYZ"scope="global" value-type="string">
<initialize>
<expression>
<string>CONNECTED
</string>
</expression>
</initialize>
</variable>
</statemanipulation>
<dialog cancel_button="yes" popup="yes">
<title>Enter TITLE Here
</title>
<menu flow="list"mandatory="yes" select="single" type="radio">
<prompt position="left">
<emphasis emph="bold">Select from the following:
</emphasis>
</prompt>
<choice default="yes">
<text>Selection1
</text>
<statemanipulation>
<variablename="StateInformationVariable" scope="global" value-type="integer">
```


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Check the source to verify that this is the current version before use.

```
<initialize>
<expression>
<integer>1
</integer>
<>
</expression>
</initialize>
</variable>
</statemanipulation>
</choice>
<choicedefault="no">
<text>Selection  2
</text>
<statemanipulation>
<variable  name="StateInformationVariable"scope="global"  value-type="integer">
<initialize>
<expression>
<integer>2
</integer>
<>
</expression>
</initialize>
</variable>
</statemanipulation>
</choice>
<choicedefault="no">
<enable>
<expression>
<variableref   name="connected_testXYZ"/>
<eq/>
<string>DISCONNECTED
</string>
</expression>
</enable>
<text>Selection  3
</text>
<statemanipulation>
<variable  name="StateInformationVariable"  scope="global"  value-type="integer">
<initialize>
<expression>
<integer>4
</integer>
<>
</expression>
</initialize>
</variable>
</statemanipulation>
</choice>
</menu>
</dialog>
```

b. Stylesheet output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

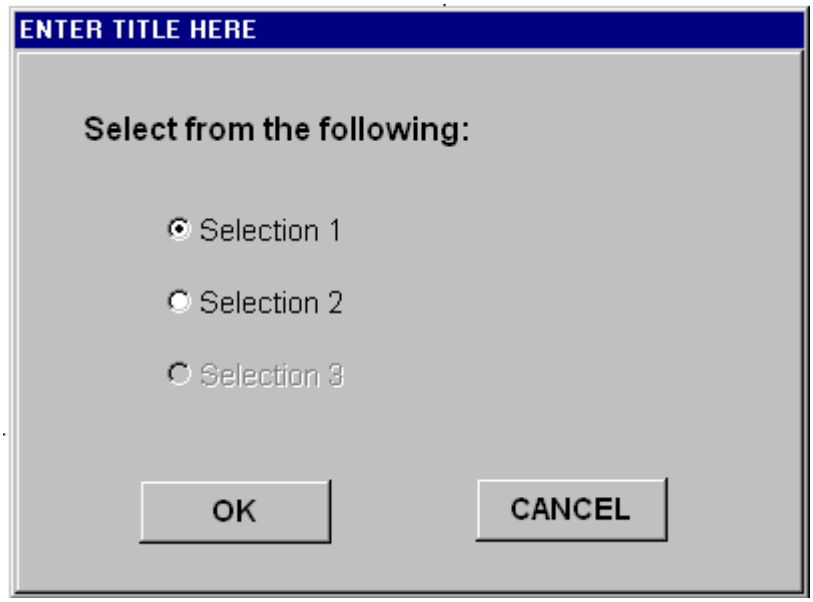

FIGURE 556. Menu dialog box example.

48.3.5 Binary menu dialog box ***<binarymenu>***. The element ***<binarymenu>*** within the element ***<dialog>*** (see 48.3.2 creates a special type of menu dialog box referred to as a binary menu dialog box. This type of dialog box provides the capability to communicate to the IETM by prompting the user to select one item from a two item list (i.e., yes or no, true or false, pass or fail).

a. The components are:
   (1) Precondition ***<precond>*** (optional – zero or one) (see 48.3.5.1).
   (2) Enable ***<enable>*** (optional – zero or one) (see 48.3.5.2).
   (3) Prompt ***<prompt>*** (required) (see 48.3.5.3).
   (4) Yes with state ***<yesstate>*** (required) (see 48.3.5.4).
   (5) No with state ***<nostate>*** (required) (see 48.3.5.5).

b. DTD fragment is graphically depicted.

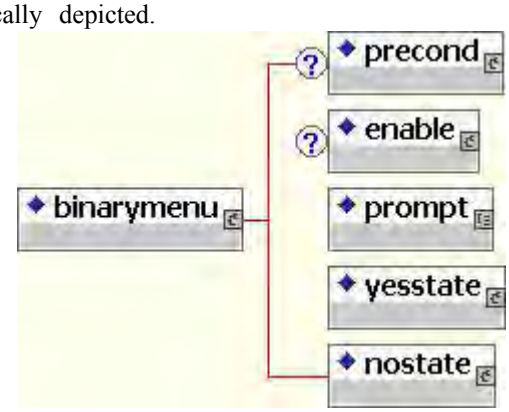

FIGURE 557. Binary menu <binarymenu> DTD hierarchy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT binarymenu (precond? , enable?,  prompt , yesstate , nostate)>
<!ATTLIST binarymenu
     applicable IDREFS  #IMPLIED
     inschlvl  CDATA   #IMPLIED
     delchlvl  CDATA   #IMPLIED
     comment   CDATA   #IMPLIED
     changeref IDREFS  #IMPLIED
     id        ID      #IMPLIED
     idref     IDREFS  #IMPLIED
     assocfig IDREFS  #IMPLIED
     skilltrk    CDATA  #IMPLIED
     security  (uc | fouo | c | s | ts )  #IMPLIED
     answer   (yesno | truefalse | passfail )  'yesno'
     default (yes | no) #IMPLIED
     type (radio | button | pulldown) 'radio'
     select (single) 'single'
     flow    (list | inline )  'list'
     mandatory  (yes | no)  'yes'>
```

d. Unique attributes:

(1) **answer** – Provides the type of answer to display on the screen. There are three types; “yesno”, “truefalse”, and “passfail”. The default is “yesno”.

(2) **default** – Selects one of the radio buttons as a default. If "yes" then the positive radio button is the default, if "no" then the negative radio button is the default, and if not specified, neither radio button is the default.

(3) **type** – Defines how the choice items are displayed. The options are as a radio, a button or a pull down menu). For a binary menu dialog box the type should be set to “radio”.

(4) **select** – A required attribute in both ***<menu>*** and ***<binary>*** that specifies if a single selection or multiple selections are permitted. The default is single which is the only value that is permitted for a binary menu dialog box.

(5) **flow** – Identifies if the selection values are to be displayed vertically (i.e., in a list) or horizontally (i.e. inline). The default is “list”. Either option is acceptable for a binary menu dialog box.

(6) **mandatory** – Identifies if a menu choice is required before the dialog can be completed. The default is "yes" (True). For a binary menu dialog box **mandatory** should be set to “yes”. NOTE: An entered space satisfies the mandatory condition.

e. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).

(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(4) **comment** – Change information (optional) (see 49.2.9).

(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).

(8) **inschlvl** – Insert change level (optional) (see 49.2.9).

(9) **security** – Security classification (optional) (see 49.2.13).

(10) **skilltrk** – Skill level (optional) (see 49.2.2).

48.3.5.1 Precondition ***<precond>***. The optional element ***<precond>*** (see 48.2.1.1) contains a Boolean expression. The Boolean expression is evaluated by the Logic Engine. If the Boolean expression evaluates to "True", then the binary menu is presented to the user, otherwise the remaining content in ***<binarymenu>*** is skipped.

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48.3.5.2 Enable ***<enable>***. The optional element ***<enable>*** determines if the binary menu is active or not active (see 48.3.1.1).

48.3.5.3 Prompt ***<prompt>***. The element ***<prompt>*** contains the prompt text to be displayed for the binary menu (see 48.3.1.2). Any valid value for the **position** attribute is acceptable.

48.3.5.4 Yes with state ***<yesstate>***. The element ***<yesstate>*** is used in an IETM to set state (variable) information. When the positive menu item is chosen in the binary menu, state information (single or multiple) is set.

a. The components are:
   (1) State manipulation values (required – one or more) (see 48.3.5.4.1).
b. DTD fragment is graphically depicted.

FIGURE 558. Yes with state <yesstate> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT yesstate (statemanipulation | statemanipulation-alt)+>
<!ATTLIST yesstate  default  (yes | no)  'no'   >
```

d. Unique attributes:
   (1) **default** – Determines if this selection is the default selection. If **default** is set to "yes" then this selection is the default and it's radio button is shown as being selected when the binary menu is first displayed. This selection is not the default if the value for **default** is “no".

48.3.5.4.1 State manipulation values. The elements ***<statemanipulation>*** (see 48.2.3) and ***<statemanipulation-alt>*** (see 48.2.1) set the state variable information for the binary menu selection. Several state variables can be set. The state variable is modified when the binary menu choice is made and the dialog box is acknowledged by the user.

48.3.5.5 No with state ***<nostate>***. The element ***<nostate>*** is used in an IETM to set state (variable) information. When the negative menu item is chosen in the binary menu, state information (single or multiple) is set.

a. The components are:
   (1) State manipulation values (required – one or more) (see 48.3.5.5.1).
b. DTD fragment is graphically depicted.

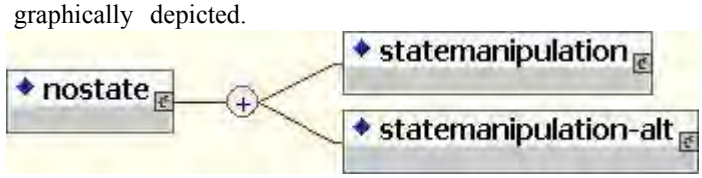

FIGURE 559. No with state <nostate> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT nostate (statemanipulation | statemanipulation-alt)+>
<!ATTLIST nostate  default  (yes | no)  'no'   >
```

d. Unique attributes:

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(1) **default** – Determines if this selection is the default selection. If **default** is set to "yes" then this selection is the default and it's radio button is shown as being selected when the binary menu is first displayed. This selection is not the default if the value for **default** is “no".

48.3.5.5.1 State manipulation values. The elements ***<statemanipulation>*** (see 48.2.3) and ***<statemanipulation-alt>*** (see 48.2.1) set the state variable information for the binary menu selection. Several state variables can be set. The state variable is modified when the binary menu choice is made and the dialog box is acknowledged by the user.

48.3.5.6 Binary menu dialog box example. The XML source and its stylesheet output (see Figure 560) for a binary menu dialog box are provided below. This example depicts a popup dialog box with a single line title and two menu selections (i.e., true or false). A variable is initialized and set to either true or false depending on the user’s selection. Only one choice can be made. The **mandatory** attribute is set by default to “yes” indicating that an entry is required. The first menu selection item is shown as being selected since the value of the **default** attribute for that choice was set to “yes”. The “OK” push button is enabled because a menu selection was made (i.e., the “True” selection is defaulted to). The “CANCEL” push button appears since the **cancel_button** attribute in ***<dialog>*** is set to 'yes'.

a. XML document instance fragment.

```
<dialog ok_caption="OK" popup="yes">
<title>ENTERTITLE HERE
</title>
<binarymenu answer="truefalse" default="yes"flow="list" mandatory="yes"
select="single" type="radio">
<prompt position="left">Is the reading less than 5?
</prompt>
<yesstate default="no">
<statemanipulation>
<variable name="ReadingWithinUpperBound" scope="global" value-type="boolean">
<initialize>
<expression>
<boolean>
<true/>
</boolean>
</expression>
</initialize>
</variable>
</statemanipulation>
</yesstate>
<nostatedefault="no">
<statemanipulation>
<variablename="ReadingWithinUpperBound" scope="global" value-type="boolean">
<initialize>
<expression>
<boolean>
<false/>
</boolean>
</expression>
</initialize>
</variable>
</statemanipulation>
</nostate>
</binarymenu>
</dialog>
```

b. Stylesheet output.

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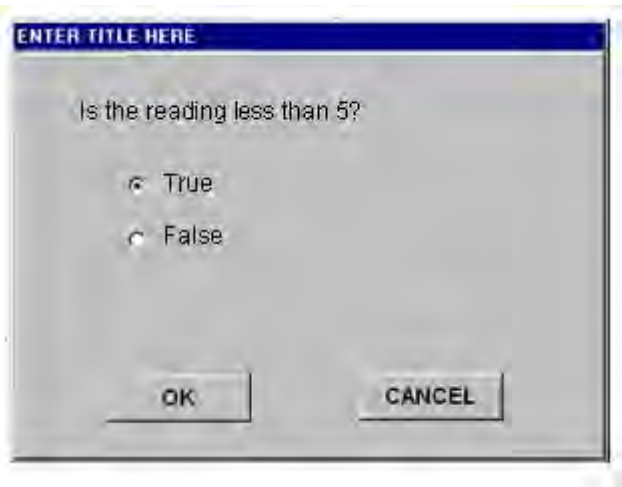

FIGURE 560. Binary menu dialog box example.

48.3.6 Multiple–choice dialog box ***<menu>***. The multiple–choice dialog box provides the capability to communicate to the IETM by prompting the user to select at least one item from a list. The element ***<menu>*** in ***<dialog>*** provides this capability in addition to providing a single choice capability (see 48.3.4). The only tagging difference between a menu dialog box and a multiple–choice dialog box is the setting of the **select** attribute for the element ***<menu>***. The **select** attribute is a required attribute that specifies if a single selection or multiple selections are permitted. The default is single. Set the select attribute to multiple for a multiple–choice dialog box. Otherwise, follow the tagging guidance provided in the Menu dialog box paragraph (see 48.3.4).

48.3.6.1 Multiple-choice dialog box example. The XML source and its stylesheet output (see Figure 561) for a multiple-choice dialog box are provided below. This example depicts a popup dialog box with a single line title and three menu selections. Each choice sets a previously initialized variable to “Selected”. The element ***<statemanipulation>*** has been previously used to create these variables. The user has to select at least one choice. The **select** attributes set to “multiple” indicating that more than one choice can be made. The **mandatory** attribute is set by default to “yes” indicating that an entry is required. Two of the choices are show as being checked because their **default** attribute was set to 'yes'. The “OK” push button is enabled because at least one choice was selected (i.e. the defaults). The menu prompt appears in bold by using the emphasis tag. The “CANCEL” push button appears since the **cancel_button** attribute in ***<dialog>*** is set to 'yes'.

a. XML document instance fragment.

```
<statemanipulation>
<variable name="StateInformationVariable1"scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="StateInformationVariable2" scope="global" value-type="string">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="StateInformationVariable3" scope="global" value-type="string">
</variable>
</statemanipulation>
<dialog cancel_button="yes" popup="yes">
<title>Enter TITLE Here
</title>
<menu flow="list"mandatory="yes" select="multiple" type="radio">
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

```
<prompt position="left">
<emphasis emph="bold">Select from the following:
</emphasis>
</prompt>
<choice default="yes">
<text>Selection1
</text>
<statemanipulation>
<variableref name="StateInformationVariable1"/>
<expression>
<string>Selected
</string>
</expression>
</statemanipulation>
</choice>
<choicedefault="no">
<text>Selection 2
</text>
<statemanipulation>
<variableref name="StateInformationVariable2"/>
<expression>
<string>Selected
</string>
</expression>
</statemanipulation>
</choice>
<choicedefault="yes">
<text>Selection 3
</text>
<statemanipulation>
<variableref name="StateInformationVariable3"/>
<expression>
<string>Selected
</string>
</expression>
</statemanipulation>
</choice>
</menu>
</dialog>
```

b. Stylesheet output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

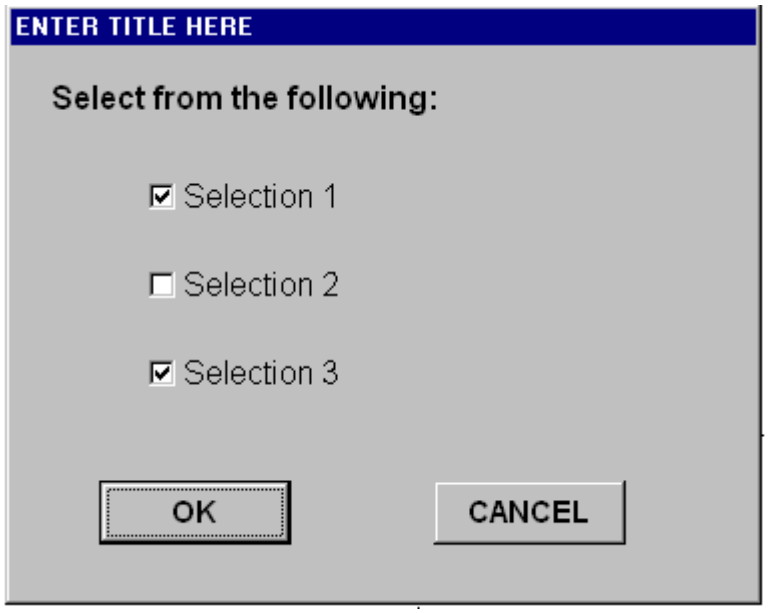

FIGURE 561. Multiple-choice dialog box example.

48.3.7 Message dialog box ***<dialog-message>***. A message dialog box provides information from the IETM to user and does not require any user response, except to continue. The message dialog box can display results from a user requested calculation, intrusive test results, final corrective action, etc. A message dialog box is used when it is deemed necessary that the user acknowledge information such as essential procedures, conditions, statements or important instructional data. The element ***<dialog-message>*** in ***<dialog>*** creates a menu dialog box. For legacy applications, the element ***<message>*** can be used instead.

a. The components are:
   (1) Precondition ***<precond>*** (optional zero or one) (see 48.3.7.1).
   (2) Message text ***<messageline>*** (required) (see 48.3.7.2).

b. DTD fragment is graphically depicted.

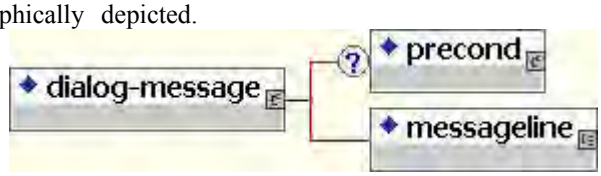

FIGURE 562. Message dialog box <dialog-message> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT dialog-message (precond? , messageline)>
<!ATTLIST dialog-message
     applicable IDREFS  #IMPLIED
     inschlvl  CDATA   #IMPLIED
     delchlvl  CDATA   #IMPLIED
     comment   CDATA   #IMPLIED
     changeref IDREFS  #IMPLIED
     id        ID      #IMPLIED
     idref     IDREFS  #IMPLIED
     assocfig IDREFS  #IMPLIED
     skilltrk    CDATA  #IMPLIED
     security  (uc | fouo | c | s | ts )  #IMPLIED >
```


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Check the source to verify that this is the current version before use.

d. Common attributes:
(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

48.3.7.1 Precondition ***<precond>***. The optional element ***<precond>*** (see 48.2.1.1) contains a Boolean expression. The Boolean expression is evaluated by the Logic Engine. If the Boolean expression evaluates to "True", then the message is presented to the user, otherwise the remaining content in ***<dialog-message>*** is skipped.

48.3.7.2 Message text ***<messageline>***. The element ***<messageline>*** presents textual information in the message dialog box. State information may also be displayed.

a. The components are:
(1) Emphasis ***<emphasis>*** (optional – zero or more). Provides for an emphasis capability (see 49.1.3.1).
(2) Subscript ***<subscrpt>*** (optional – zero or more). Provides for a subscript capability (see 49.1.3.4).
(3) Superscript ***<supscrpt>*** (optional – zero or more). Provides for superscript capability (see 49.1.3.5).
(4) State information may be displayed by using the element ***<variableref>*** (optional – zero or more) (see 48.1.2).

b. DTD fragment is graphically depicted.

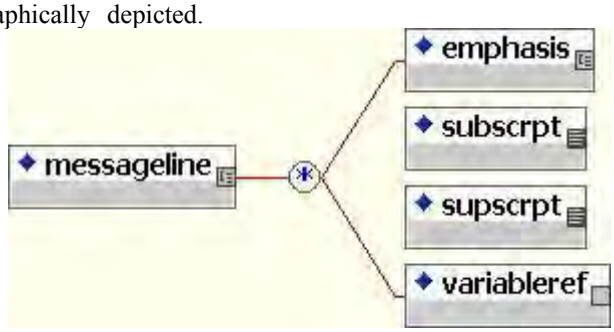

FIGURE 563. Message text <messageline> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT messageline (#PCDATA | emphasis | subscrpt | supscrpt | variableref)*>
```

48.3.7.3 Message dialog box – simple example. The XML source and its stylesheet output (see Figure 564) for a simple example of a message dialog box are provided below. This example depicts a popup message dialog box with a single line title and unformatted message text.

a. XML document instance fragment.

```
<dialog cancel_button="no" popup="yes">
<title>EnterTITLE Here
</title>
<dialog-message>
<messageline>Thetask is completed.
</messageline>
</dialog-message>
</dialog>
```


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Check the source to verify that this is the current version before use.

b. Stylesheet output.

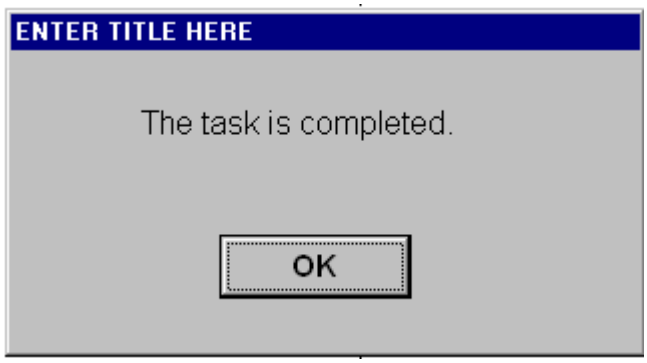

FIGURE 564. Message dialog box – simple example.

48.3.7.4 Message dialog box – complex example. The XML source and its stylesheet output (see Figure 565) for a more complex example of a message dialog box are provided below. This example depicts a popup message dialog box with a title bar text, three formatted message lines (i.e., emphasis is used), display of state information and help text. In this example, the element ***<statemanipulation>*** defines the state information variable and sets its value. The element ***<help.info>*** in ***<dialog>*** creates the help button and accompanying help text.

a. XML document instance fragment.

```
<statemanipulation>
<variable name="connected_testXYZ"scope="global" value-type="string">
<initialize>
<expression>
<string>CONNECTED
</string>
</expression>
</initialize>
</variable>
</statemanipulation>
<dialog cancel_button="yes" popup="yes">
<title>Enter TITLE Here
</title>
<dialog-message>
<messageline>The task is completed.  The
<emphasiscolor="blue" emph="bold">ABC
</emphasis>test equipment is
<variableref name="connected_testXYZ"/>
</messageline>
</dialog-message>
<help.info>
<para esd="no"hcp="no">This message presents the user with the state
of the test equipment (i.e., CONNECTED or DISCONNECTED)
</para>
</help.info>
</dialog>
```

b. Stylesheet output.

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Check the source to verify that this is the current version before use.

FIGURE 565. Message dialog box – complex example.

48.3.8 Composite menu dialog box. A composite dialog box contains any combination of message, fill-in, menu and multiple-choice dialog entries. This section will discuss how combinations of these entries are displayed in a dialog box. This includes a discussion of the element ***<dialog-group>*** which can be used to vary the presentation of these entries in the dialog box.

48.3.8.1 Layout of entries. The elements ***<dialog-message>***, ***<fillin>***, ***<menu>***, and ***<binarymenu>*** elements are entered in the ***<dialog>*** element either directly or as contained in the group dialog element ***<dialog-group>***. This can be seen in the element declarations for ***<dialog>*** and ***<dialog-group>***.

```
<!ELEMENT dialog (precond?, title?, (fillin | menu | binarymenu | dialog-group |
dialog-message)+, help.info?, (statemanipulation | statemanipulation-alt)*)>
<!ELEMENT dialog-group (precond?, enable?, title?, (fillin | menu | binarymenu |
dialog-message | dialog-group)+)>
```

The selection of which method to use is based upon the desired presentation of these fill-in, menu, binary menu, multiple-choice and message entries in the dialog box. This includes deciding upon the placement of the entries from top to bottom and left to right. The element ***<dialog-group>*** provides the capability to display entries horizontally (i.e., inline). The ***<dialog-group>*** element also provides the capability to add a group title. If ***<dialog-group>*** is not used, the entries will be displayed in the dialog box vertically (i.e., as a list). In either case, the order of appearance is determined by the sequence in which the entries are made in the document instance. Note that the presentation of data fields within an entry is still determined by the values that are set for that entry. For example, the data fields in a menu dialog box entry would be displayed as a list if it's flow attribute is set to "list" even if the entry is included in ***<dialog-group>***.

48.3.8.2 Group Dialog Content Model ***<dialog-group>***.

a. The components are:
   (1) Precondition ***<precond>*** (optional – zero or one) (see 48.3.8.2.1).
   (2) Enable ***<enable>*** (optional – zero or one) (see 48.3.8.2.2).
   (3) Title ***<title>*** (optional – zero or one) (see 48.3.8.2.3).
   (4) Dialog Box Type (required – one or more) (see 48.3.8.2.4).
b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

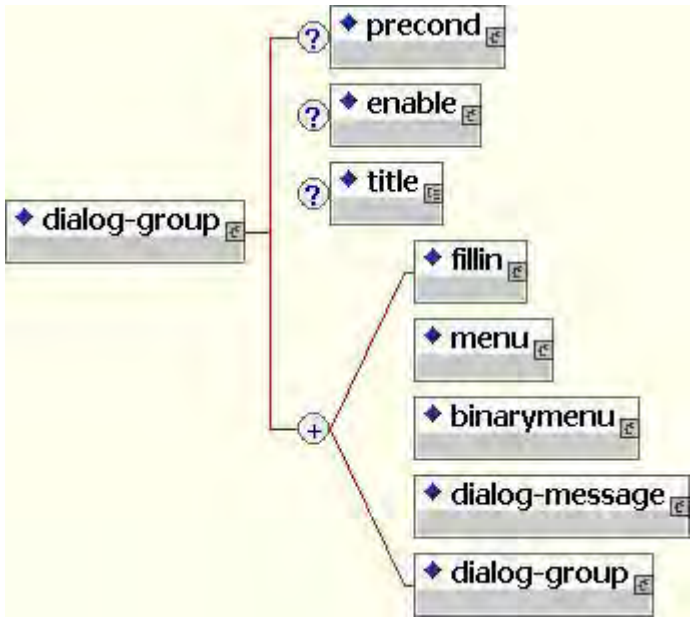

FIGURE 566. Group dialog <dialog-group> DTD hierarchy. .

c. DTD fragment:

```
<!ELEMENT dialog-group (precond?, enable?, title?, (fillin | menu |
     binarymenu | dialog-message | dialog-group)+)>
<!ATTLIST dialog-group
     applicable IDREFS  #IMPLIED
     inschlvl  CDATA   #IMPLIED
     delchlvl  CDATA   #IMPLIED
     comment   CDATA   #IMPLIED
     changeref IDREFS  #IMPLIED
     id        ID      #IMPLIED
     idref     IDREFS  #IMPLIED
     assocfig IDREFS  #IMPLIED
     skilltrk    CDATA  #IMPLIED
     security  (uc | fouo | c | s | ts )  #IMPLIED
     separator (yes | no) 'no' >
```

d. Unique attributes:

(1) **separator** – Indicates if the dialog group is divided by a separator or boxing around dialog objects. Default is no.

e. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

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48.3.8.2.1 Precondition *<precond>*. The optional element ***<precond>*** (see 48.2.1.1) contains a Boolean expression. The Boolean expression is evaluated by the Logic Engine. If the Boolean expression evaluates to "True", then the group dialog box is presented to the user, otherwise the remaining content in ***<dialog-group>*** is skipped.

48.3.8.2.2 Enable *<enable>*. The optional element ***<enable>*** determines if the group dialog is active or not active (see 48.3.1.1).

48.3.8.2.3 Title *<title>*. The element ***<title>*** (see 49.1.1.3) contains the optional group dialog title. This title is displayed above the displayed entries in the group. Not all title element content model components are appropriate for use in tagging a group dialog box title. Enter text using the ***<emphasis>***, ***<subscript>***, and ***<supscrpt>*** tags as required to generate the displayed text/formatted text.

48.3.8.2.4 Dialog box type. The element ***<dialog-group>*** contains the required choice of fill-in ***<fillin>***, menu ***<menu>***, binary menu ***<binarymenu>***, message ***<dialog-message>*** and/or group dialog ***<dialog-group>*** entry(s). These elements provide the capability to create the 5 dialog box types. Please refer to 48.3.2.3 for more information on each type.

48.3.8.3 Composite dialog box example. The XML source and its stylesheet output (see Figure 567) for a composite dialog box are provided below. This example depicts a popup dialog box with a title bar text and 3 types of entries (fill-in, multiple choice and menu). The multiple choice and menu entries appear horizontally because they are in a group. The heading above them is the title for the group. The “OK” push button is not enabled because at least one choice is made in the multiple choice selections (i.e., a check appears in at least one of the squared boxes). The element ***<help.info>*** in ***<dialog>*** creates the help button and accompanying help text. The “CANCEL” push button appears since the **cancel_button** attribute in ***<dialog>*** is set to 'yes'.

a. XML document instance fragment.

```
<statemanipulation>
<variable name="Technician ID" value-type="string" scope="global">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="StateInformationVariable1" value-type="string" scope="global">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="StateInformationVariable2" value-type="string" scope="global">
</variable>
</statemanipulation>
<statemanipulation>
<variablename="StateInformationVariable3" value-type="string" scope="global">
</variable>
</statemanipulation>
<dialog cancel_button="yes" popup="yes"reset_button="no">
<title>USER RESPONSE
</title>
<fillinfieldsize="20" mandatory="yes">
<prompt position="bottom">
<emphasis emph="bold">ENTER TECHNICIAN ID
</emphasis>
</prompt>
<variableref name="Technician ID"/>
<default>
<text>Data Entry Field
</text>
</default>
</fillin>
<dialog-group separator="no">
```


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Check the source to verify that this is the current version before use.

```
<title>
<emphasis emph="bold">MULTIPLE CHOICE: OPTIONS?
</emphasis>
</title>
<menu flow="list"mandatory="yes" select="multiple" type="radio">
<prompt>
</prompt>
<choice>
<text>Option 1
</text>
<statemanipulation>
<variableref name="StateInformationVariable1"/>
<expression>
<string>Selected
</string>
</expression>
</statemanipulation>
</choice>
<choice>
<text>Option2
</text>
<statemanipulation>
<variableref name="StateInformationVariable2"/>
<expression>
<string>Selected
</string>
</expression>
</statemanipulation>
</choice>
<choice>
<text>Option3
</text>
<statemanipulation>
<variableref name="StateInformationVariable3"/>
<expression>
<string>Selected
</string>
</expression>
</statemanipulation>
</choice>
</menu>
<menu flow="list"mandatory="yes" select="single" type="radio">
<prompt>
</prompt>
<choice>
<text>YES
</text>
<statemanipulation>
<variable name="StateInformationVariable" value-type="integer" scope="global">
<initialize>
<expression>
<integer>1
</integer>
</expression>
</initialize>
</variable>
```


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```
</statemanipulation>
</choice>
<choice>
<text>NO
</text>
<statemanipulation>
<variable name="StateInformationVariable" value-type="integer" scope="global">
<initialize>
<expression>
<integer>0
</integer>
</expression>
</initialize>
</variable>
</statemanipulation>
</choice>
<choice>
<text>MAYBE
</text>
<statemanipulation>
<variable name="StateInformationVariable" value-type="integer" scope="global">
<initialize>
<expression>
<integer>2
</integer>
</expression>
</initialize>
</variable>
</statemanipulation>
</choice>
</menu>
</dialog-group>
<help.info>
<para hcp="no" esd="no">This information is presented when
the HELP push button is selected.
</para>
</help.info>
</dialog>
```

b. Stylesheet output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

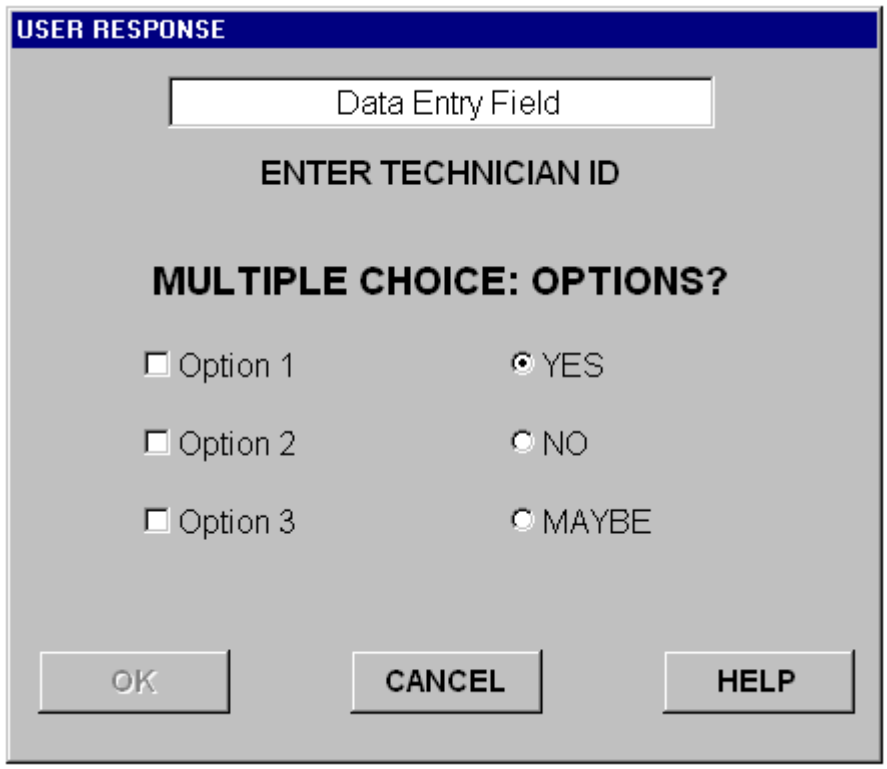

FIGURE 567. Composite dialog box example.

48.4 Message. The element ***<message>*** presents to the user system information, state (variable) information values, or results from evaluations. After acknowledgement, additional action may be specified. This element is primarily intended for legacy applications. The message dialog box ***<dialog-message>*** (see 48.3.7) should be used for new applications.

a. The components are:
   (1) Title ***<title>*** (optional – zero or one) (see 48.4.1).
   (2) Message text ***<messageline>*** (required – one or more) (see 48.4.2).
   (3) Help information ***<help>*** (optional – zero or one) (see 48.4.3).
   (4) Link after message ***<aftermessage>*** (optional – zero or one) (see 48.4.4).

b. DTD fragment is graphically depicted.

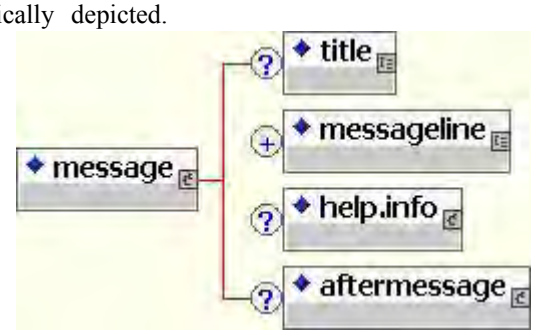

FIGURE 568. Message <message> DTD hierarchy. .

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT message (title?, messageline+, help.info?, aftermessage?)>
<!ATTLIST message   id ID #IMPLIED
     popup (yes | no) #IMPLIED
     button_title CDATA "OK">
```

d. Unique attributes:
   (1) **popup** – Identifies how the message box is to be displayed. A “yes” displays the message box as a separate popup window. A “no” displays the dialog box inline with the text.
   (2) **button_title** – Text for the acknowledgement button. The default is “OK”.

e. Common attributes:
   (1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

48.4.1 Title ***<title>***. The element ***<title>*** (see 49.1.1.3) contains the optional message title. Not all title element content model components are appropriate for use in tagging a message title. Enter text using the ***<emphasis>***, ***<subscript>***, and ***<supscrpt>*** tags as required to generate the displayed text/formatted text.

48.4.2 Message text ***<messageline>***. The element ***<messageline>*** presents textual information in the message box. State information may also be displayed.

a. The components are:
   (1) Emphasis ***<emphasis>*** (optional – zero or more). Provides for an emphasis capability (see 49.1.3.1).
   (2) Subscript ***<subscrpt>*** (optional – zero or more). Provides for a subscript capability (see 49.1.3.4).
   (3) Superscript ***<supscrpt>*** (optional – zero or more). Provides for superscript capability (see 49.1.3.5).
   (4) State information may be displayed by using the element ***<variableref>*** (optional – zero or more) (see 48.1.2).

b. DTD fragment is graphically depicted.

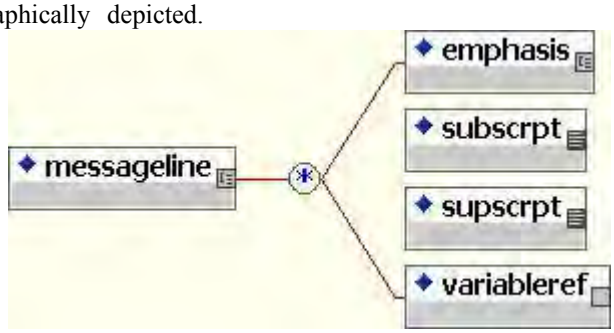

FIGURE 569. Message text <messageline> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT messageline (#PCDATA | emphasis | subscrpt | supscrpt | variableref)*>
```

48.4.3 Help information ***<help.info>***. The element ***<help.info>*** provides help information about the data in the message box. The text may describe the data that is expected to be entered, where and why the information is needed, etc. This element is optional. When data for ***<help.info>*** is entered, a "HELP" push button is displayed in the message box.

48.4.4 Link after message ***<aftermessage>***. This element identifies the specified linkage action to be performed after acknowledging the message box.

a. The components are:
   (1) Linking ***<link>*** (required). The link element provides a capability to reference internal or external targets (see 46.2.3).

b. DTD fragment is graphically depicted.

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FIGURE 570. Link after message <aftermessage> DTD hierarchy. .

c. DTD fragment:

```
<!ELEMENT aftermessage (link)>
```

48.4.5 Message box – simple example. The XML source and its stylesheet output (see Figure 571) for a simple example of a message box are provided below. This example depicts a popup message box with a single line title and unformatted message text.

a. XML document instance fragment.

<interaction>
<message popup="yes">
<title>`Enter TITLEHere`
</title>
<messageline>`The task is completed.`
</messageline>
</message>
</interaction>

b. Stylesheet output

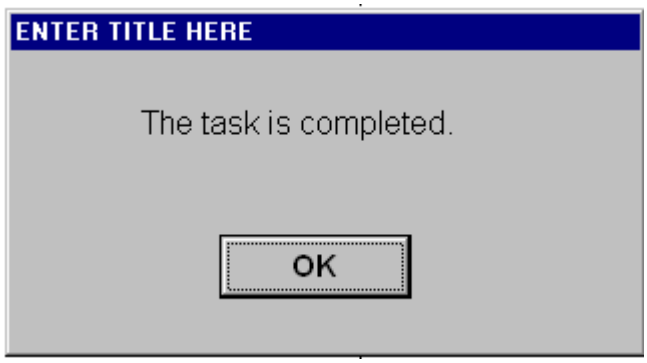

FIGURE 571. Message box – simple example.

49 COMMON STRUCTURE.

49.1 Common elements . The elements are common throughout MIL–STD–2361 and are shared elements used with the various information chapters.

49.1.1 Textual elements.

49.1.1.1 Chapter title page ***<titlepg>***. The chapter title page is required for page-based TMs (refer to MIL-STD-40051-2). As specified in MIL-STD-40051-2, each chapter begins with a chapter title page. A separate chapter title page is not required for pocket size manuals.

a. The components are:
   (1) Name ***<name>*** (required – one or more). The ***<name>*** element defines the equipment or part nomenclature (see 49.1.4.19).(see 31.4.1.4.5.1).
   (2) Part number ***<partno>*** (optional – zero or more). The ***<partno>*** element is the commercial part number (see 49.1.4.21).
   (3) Model number ***<modelno>*** (optional – zero or more). The model number ***<modelno>*** element is the official equipment or equipment piece model number (see 49.1.4.18).

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(4) National stock number ***<nsn>*** (optional – zero or more). The National Stock Number ***<nsn>*** element identifies the item's National Stock Number (NSN) (see 49.1.4.20).

(5) Table of contents ***<contents>*** optional – zero or one). The table of contents ***<contents>*** element provides for an optional chapter level TOC (see 31.3.1.1.8).

b. DTD fragment is graphically depicted.

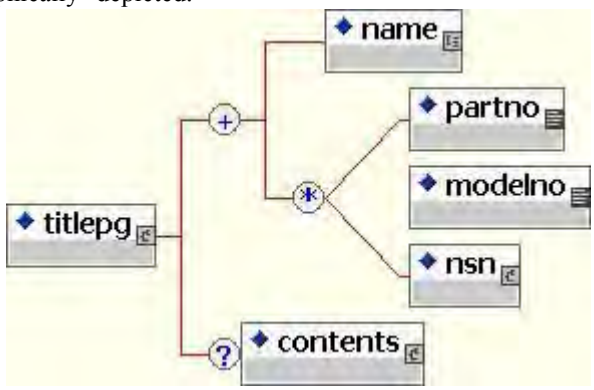

FIGURE 572. Chapter title page <titlepg> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT titlepg ((name, (partno | modelno | nsn)*)+, contents?)>
<!ATTLIST titlepg
    maintlvl (field | sustain | depot | operator | gensup | dirsup |
    unitlvl | avum-avim | inter | tmlvls | asb | amc | tasmg) #REQUIRED

    changeref       IDREFS                  #IMPLIED
    comment         CDATA                   #IMPLIED
    delchlvl        CDATA                   #IMPLIED
    inschlvl        CDATA                   #IMPLIED>
```

d. Unique attributes:

(1) **maintlvl** – Specifies the lowest authorized maintenance level for the chapter. The levels are Fielding (field), Sustainment (sustain), Depot (depot), Operator (operator), General Support (gensup), Direct Support (dirsup), Unit (unitlvl), Aviation Unit Maintenance and Aviation Intermediate Maintenance (avum-avim), Intermediate (inter) (Aviation only), all TM Levels (tmlvls), Aviation Support Battalion (asb), Aviation Maintenance Company (amc), and Theater Aviation Sustainment Maintenance Group (tasmg).

e. Common attributes:

(1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

(2) **comment** – Change information (optional) (see 49.2.9).

(3) **delchlvl** – Deletion change level (optional) (see 49.2.9).

(4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.1.1.1 Chapter title page example. The XML source and its stylesheet output (see Figure 573) depicting a chapter title page is shown below. In this example the publishing system generates the chapter number and displays the TM number. The nomenclature is taken from the ***<name>*** element. The maintenance level is taken from the **maintlvl** attribute.

a. XML document instance fragment.

```
<mim chap-toc="yes" chngno="1" frame="no" revno="0" tocentry="1">
<titlepg maintlvl="unitlvl">
<name>155 MM, M109A6 Howitzer
</name>
```


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

</titlepg> . . .
</mim>

b. Stylesheet output.

TM 3-665-339-10

CHAPTER 6

**UNIT MAINTENANCE INSTRUCTIONS
FOR
155 MM, M109A6 HOWITZER**

FIGURE 573. Chapter title page example.

49.1.1.2 Manual subtitle ***<stitle>***. The element ***<stitle>*** is a subtitle of the TM placed immediately below the prime title to indicate the volume number and contents of every separately bound volume of a TM.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
   (2) Emphasis ***<emphasis>*** (optional – zero or more)(see 49.1.3.1).

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Check the source to verify that this is the current version before use.

(3) Subscript ***<subscrpt>*** (optional – zero or more)(see 49.1.3.4).
(4) Superscript ***<supscrpt>*** (optional – zero or more)(see 49.1.3.5).
(5) Line Break ***<brk>*** (optional – zero or more)(see 49.1.3.3).

b. DTD fragment is graphically depicted.

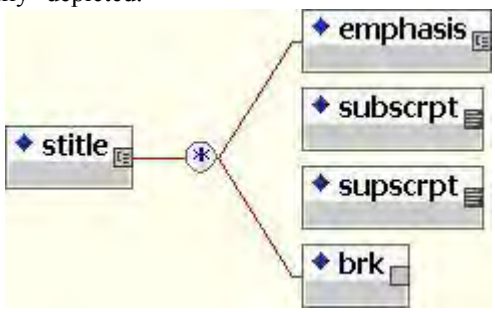

FIGURE 574. Manual subtitle <stitle> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT stitle (#PCDATA | emphasis | subscrpt | supscrpt | brk)*>
<!ATTLIST stitle
     security          (uc | fouo |c | s | ts)          #IMPLIED>

    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
```

d. Common attributes:
(1) **security** – Security classification (optional) (see 49.2.13).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.1.3 Title ***<title>***. The element ***<title>*** is used in multiple contexts within a TM to define the context to be discussed and is presented according to the composition system specifications of a particular context. The element ***<brk>*** provides cosmetic line breaks in titles.

a. The components are:
(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).

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(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).

b. DTD fragment is graphically depicted.

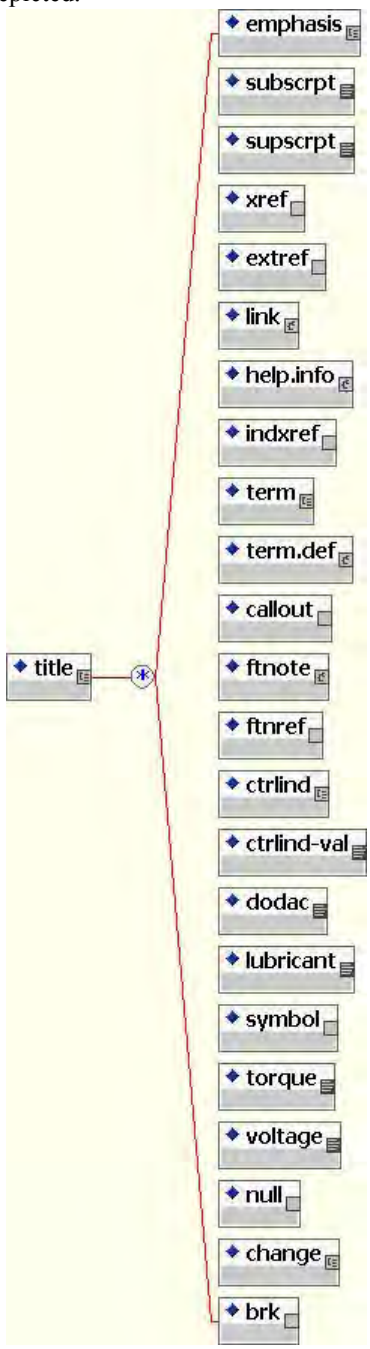

FIGURE 575. Title <title> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT title  (#PCDATA | emphasis | subscrpt
| supscrpt | xref | extref | link | help.info |
indxref | term | term.def | callout | ftnote |
 ftnref | ctrlind | ctrlind-val | dodac |
lubricant | symbol | torque | voltage |
null | change)*>
<!ATTLIST title

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.1.4 Subtitle ***<subtitle>***. The element ***<subtitle>*** is only used in subfigures ***<subfig>*** element and should display the words "Sheet X of Y" where X and Y are numbers for the sheet the user is on out of the total number of sheets in the figure.

a. The components are:

(1) The element contains narrative text #PCDATA parsable character data.
(2) Emphasis ***<emphasis>*** (optional – zero or more)(see 49.1.3.1).
(3) Subscript ***<subscrpt>*** (optional – zero or more)(see 49.1.3.4).
(4) Superscript ***<supscrpt>*** (optional – zero or more)(see 49.1.3.5).

b. DTD fragment is graphically depicted.

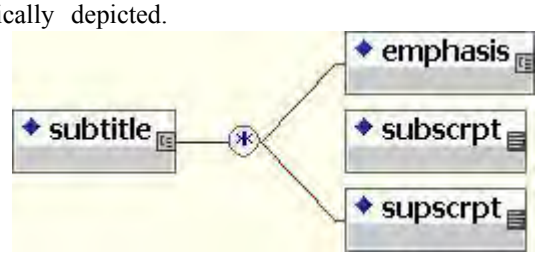

FIGURE 576. Subtitle <subtitle> DTD hierarchy.

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT subtitle  (#PCDATA | emphasis | subscrpt | supscrpt)*>
<!ATTLIST subtitle

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (3) **comment** – Change information (optional) (see 49.2.9).
   (4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (6) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (7) **security** – Security classification (optional) (see 49.2.13).
   (8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.1.5 Paragraph text ***<para>***. A paragraph of text; can also contain embedded inline elements. lists, notes, tables, and figures. The element ***<para>*** is used for paragraphs of text and can also contain embedded inline elements. lists, notes, tables, and figures. referencing and content specific elements which may be contained within either one or more of the parameter entity.

a. The components are:
   (1) Parsable characters or type text. – #PCDATA
   (2) Format text – ***<emphasis>*** (see 49.1.3.1).
   (3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
   (4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
   (5) Cross reference – ***<xref>*** (see 46.2.2).
   (6) External reference – ***<extref>*** (see 46.2.1).
   (7) Enhanced linking – ***<link>*** (see 46.2.3).
   (8) IETM help information – ***<help.info>*** (see 48.3.2.4).
   (9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
   (10) Term – ***<term>*** (see 49.1.2.3.1.1).
   (11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
   (12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
   (13) Footnote – ***< ftnote>*** (see 45.1.1).
   (14) Footnote reference – ***<ftnref>*** (see 45.1.2).
   (15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
   (16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
   (17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
   (18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
   (19) Graphic symbol – ***<symbol>*** (see 44.3.1).
   (20) Torque value – ***<torque>*** (see 49.1.4.24).
   (21) Voltage value – ***< voltage>*** (see 49.1.4.25).
   (22) Null text – ***<null >*** (see 49.1.3.2).
   (23) Changed text marker – ***<change>*** (see 49.1.3.7).
   (24) Internet address (e–mail or homepage) – ***<internet>*** (see 49.1.4.1.7).

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(25) Proponent or organiztion address – ***<proponent>*** (see 49.1.4.22).
(26) Telephone number – ***<phone>*** (see 49.1.4.1.6).
(27) Instruction data plate – ***<instructplt>*** (see 49.1.4.13).
(28) Sequential List ***<seqlist>*** (see 49.1.2.1)
(29) Random List ***<randlist>*** (see 49.1.2.2)
(30) Definition List ***<deflist>*** (see 49.1.2.3)
(31) Figure ***<figure>*** (see 44.1.1)
(32) Alternative figure ***<figure-alt>*** (see 48.2.1)
(33) Inline graphic(s) ***<inlinegraphic>*** (see 44.3.2),
(34) Table ***<table>*** (see 42)
(35) Alternative table ***<table-alt>*** (see 48.2.1)
(36) Verbatim text ***<verbatim>*** (see 49.1.3.6)
(37) User interaction ***<interaction>***(see 49.1.4.10)
(38) State (Variable) Information Manipulation ***<statemanipulation>***(see 48.2.3)
(39) Conditional State (Variable) Information Manipulation ***<statemanipulation-alt>*** (see 48.2.1)
(40) Tabular Material ***<tabmat>*** (see 42.2)

b. DTD fragment is graphically depicted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:02Z
Check the source to verify that this is the current version before use.

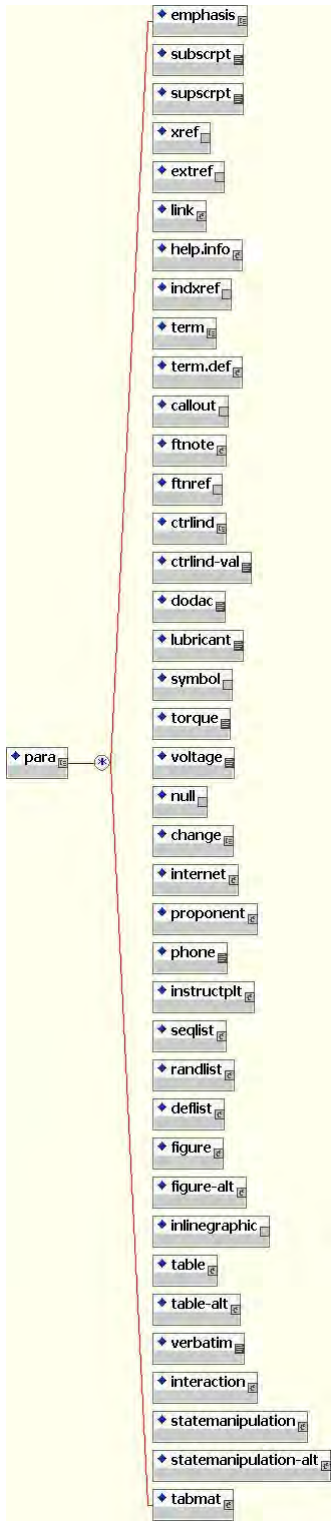

FIGURE 577. Paragraph text <para> DTD hierarchy..

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Check the source to verify that this is the current version before use.

c. DTD fragment:

```
<!ELEMENT para (#PCDATA | emphasis | subscrpt | supscrpt
 | xref | extref | link | help.info | indxref
| term | term.def | callout | ftnote | ftnref
| ctrlind | ctrlind-val | dodac | lubricant
| symbol | torque | voltage | null | change
| internet | proponent | phone | instructpl
| seqlist | randlist | def | figure | figure-alt
| inlinegraphic | table | table-alt | verbatim
| interaction | statemanipulation | statemanipulation-alt
| tabmat)*>
<!ATTLIST para

    esd              (yes | no)                    "no"
    hcp              (yes | no)                     "no"
    assocfig         IDREFS                         #IMPLIED
    changeref        IDREFS                         #IMPLIED
    comment          CDATA                          #IMPLIED
    delchlvl         CDATA                          #IMPLIED
    id               ID                             #IMPLIED
    idref            IDREFS                         #IMPLIED
    inschlvl         CDATA                          #IMPLIED
    security         (uc | fouo |c | s | ts)       #IMPLIED
    skilltrk         CDATA                          #IMPLIED
>
```

d. Common attributes:

(1) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(2) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(3) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(4) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(5) **comment** – Reason for change information. (optional) (See 49.2.9.)
(6) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(7) **id** – Unique identifier. (optional) (See 49.2.6.)
(8) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(9) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(10) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(11) **security** – Security classification. (optional) (See 49.2.13.)

49.1.1.6 Paragraph associated with an alert ***<specpara>***. The element ***<specpara>*** is a paragraph that is specifically associated with warnings, cautions, or notes. The actual narrative data will follow after the warning statement, caution statement, and/or note statement.

a. The components are:

(1) At least one of the following is required:
(a) Warning ***<warning>*** (see 41.1.1).
(b) Caution ***<caution>*** (see 41.1.2).
(c) Note ***<note>*** (see 41.1.3).
(2) Paragraph Text ***<para>*** (required) (see 49.1.1.5)

b. DTD fragment is graphically depicted.

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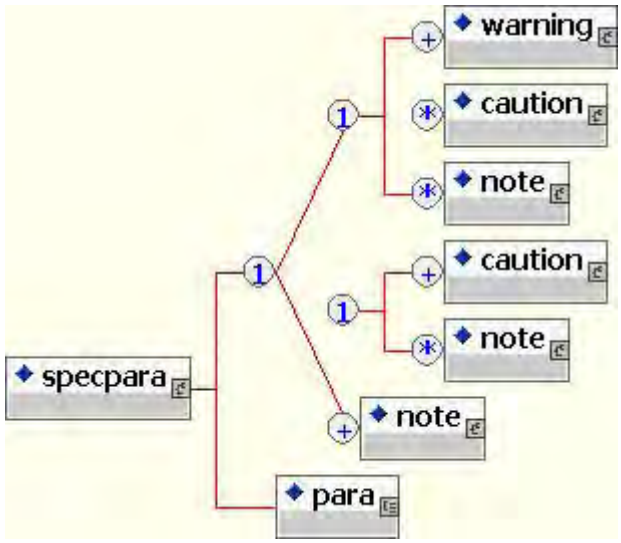

FIGURE 578. Paragraph associated with an Alert <specpara>.

c. DTD fragment:

```
<!ELEMENT specpara (((warning+, caution*, note*) | (caution+, note*) |
note+), para)>
<!ATTLIST specpara

    esd              (yes | no)                      "no"
    hcp              (yes | no)                      "no"
    assocfig         IDREFS                          #IMPLIED
    changeref        IDREFS                          #IMPLIED
    comment          CDATA                           #IMPLIED
    delchlvl         CDATA                           #IMPLIED
    id               ID                              #IMPLIED
    idref            IDREFS                          #IMPLIED
    inschlvl         CDATA                           #IMPLIED
    security         (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk         CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(2) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(3) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(4) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(5) **comment** – Reason for change information. (optional) (See 49.2.9.)
(6) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(7) **id** – Unique identifier. (optional) (See 49.2.6.)
(8) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(9) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(10) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(11) **security** – Security classification. (optional) (See 49.2.13.)

49.1.1.7 Reduced paragraph***<trim.para>***. The element ***<trim.para>*** provides the same usage as paragraph element, but reducing the content model. Several elements (i.e., alerts) require the removal of graphic, figure, table, and list.

a. The components are:

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(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).
(24) Internet address (e–mail or homepage) – ***<internet>*** (see 49.1.4.1.7).
(25) Proponent or organization address – ***<proponent>*** (see 49.1.4.22).
(26) Telephone number – ***<phone>*** (see 49.1.4.1.6).
(27) Instruction data plate – ***<instructplt>*** (see 49.1.4.13).

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

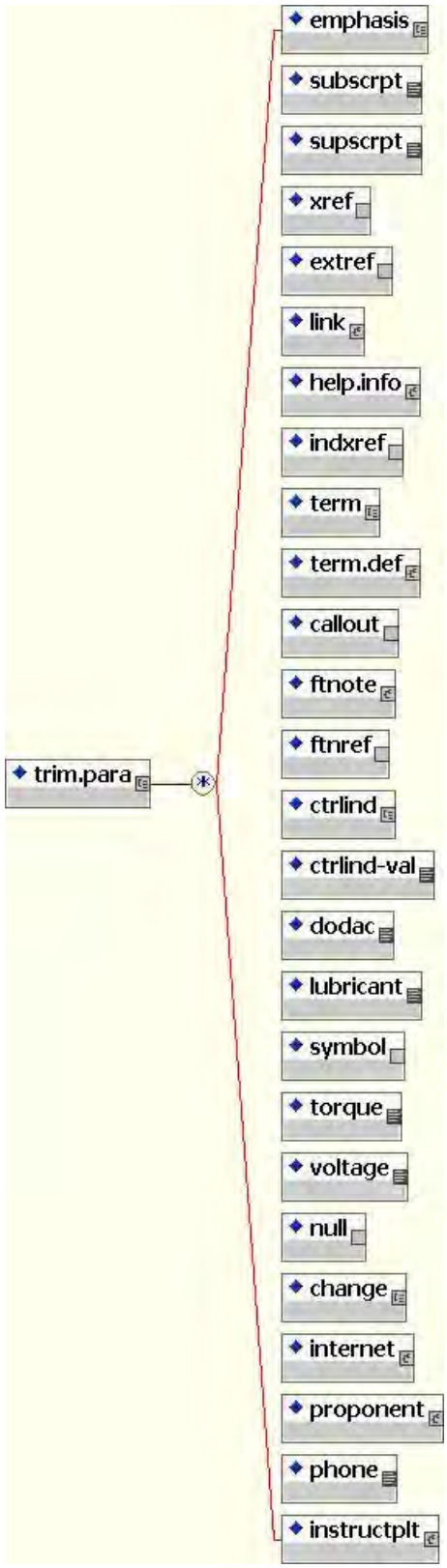

FIGURE 579. Reduced paragraph <trim.para>.

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c. DTD fragment:

```
<!ELEMENT trim.para (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link | help.info
| indxref | term | term.def | callout | ftnote | ftnref | ctrlind | ctrlind-val
| dodac | lubricant | symbol | torque | voltage | null | change | internet
| proponent | phone | instructplt)*>
<!ATTLIST trim.para

    esd              (yes | no)                    "no"
    hcp              (yes | no)                     "no"
    assocfig         IDREFS                         #IMPLIED
    changeref        IDREFS                         #IMPLIED
    comment          CDATA                          #IMPLIED
    delchlvl         CDATA                          #IMPLIED
    id               ID                             #IMPLIED
    idref            IDREFS                         #IMPLIED
    inschlvl         CDATA                          #IMPLIED
    security         (uc | fouo |c | s | ts)       #IMPLIED
    skilltrk         CDATA                          #IMPLIED
>
```

d. Common attributes:

(1) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(2) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).
(3) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(4) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(5) **comment** – Reason for change information. (optional) (See 49.2.9.)
(6) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(7) **id** – Unique identifier. (optional) (See 49.2.6.)
(8) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(9) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(10) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(11) **security** – Security classification. (optional) (See 49.2.13.)

49.1.1.8 Titled paragraph ***<para0>***. The element contains a title and possible paragraph with optional subordinate paragraphs.

a. The components are:

(1) Precondition ***<precond>*** (optional)(see 42.1.1.1)
(2) Title ***<title>*** (required)(see 49.1.1.3).
(3) Select elements at least from either group (a) or group (b):
(a) First Level Subordinate Paragraph ***<subpara1,>*** (see 49.1.1.10) or Alternatet Level Subordinate Paragraph ***<subpara1-alt>***(see 48.2.1)
(b)
1 Note ***<note>***(optional) (see 41.1.3)
2 Paragraph Text ***<para>*** (required – one or more)(see 49.1.1.5)
3 First Level Subordinate Paragraph ***<subpara1,>*** (optional – zero or more) (see 49.1.1.10) or Alternatet Level Subordinate Paragraph ***<subpara1-alt>*** (optional – zero or more)(see 48.2.1)

b. DTD fragment is graphically depicted.

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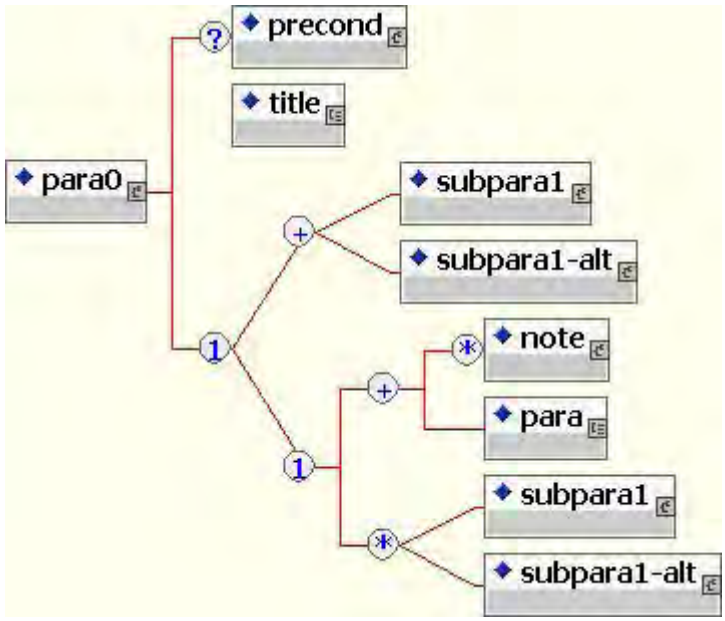

FIGURE 580. Titled paragraph <para0> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT para0 (precond?, title, ((subpara1 | subpara1-alt)+ |
((note*, para)+, (subpara1 | subpara1-alt)*)))>
<!ATTLIST para0
    crewmember          CDATA                #IMPLIED

    applicable         IDREFS                 #IMPLIED
    assocfig           IDREFS                 #IMPLIED
    changeref          IDREFS                 #IMPLIED
    comment            CDATA                  #IMPLIED
    delchlvl           CDATA                  #IMPLIED
    id                 ID                     #IMPLIED
    idref              IDREFS                 #IMPLIED
    inschlvl           CDATA                  #IMPLIED
    security           (uc | fouo |c | s | ts)          #IMPLIED
    skilltrk           CDATA                   #IMPLIED
   esd                (yes | no)              "no"
    hcp                (yes | no)               "no"
>
```

d. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

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(11) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(12) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).

49.1.1.9 Conditional title paragraph ***<para0-alt>***. This element is used to determine if an expression evaluate to true present the paragraph information.

a. The components are:
   (1) Title paragraph(s) ***<para0>*** (required – one or more)(see 49.1.1.8).
b. DTD fragment is graphically depicted.

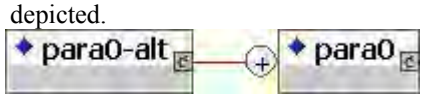

FIGURE 581. Conditional title paragraph <para0-alt> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT para0-alt (para0+)>
<!ATTLIST para0-alt
    id          ID                #IMPLIED>
```

d. Unique attributes:
   (1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).

49.1.1.10 First level subordinate paragraph ***<subpara1>***. This is the first level of subordinate paragraphs.

a. The components are:
   (1) Precondition ***<precond>*** (optional)(see 42.1.1.1) followed by required title ***<title>*** (see 49.1.1.3). and may be followed by either the group (a) or group (b) elements:
      (a) Second Level Subordinate Paragraph ***<subpara2>*** (see 49.1.1.12) or Alternate Second Level Subordinate paragraph ***<subpara2-alt>*** (see 48.2.1).or both the ***<subpara2>*** and ***<subpara2-alt>*** can occur more than once, in no particular order.
      (b) Or optional Note ***<note>*** (see 41.1.3) followed by one or more Paragraph(s) ***<para>*** (see 49.1.1.5), which may be followed by either the ***<subpara>*** or ***<subpara2-alt>*** which can occur more than once, in no particular order.
b. DTD fragment is graphically depicted.

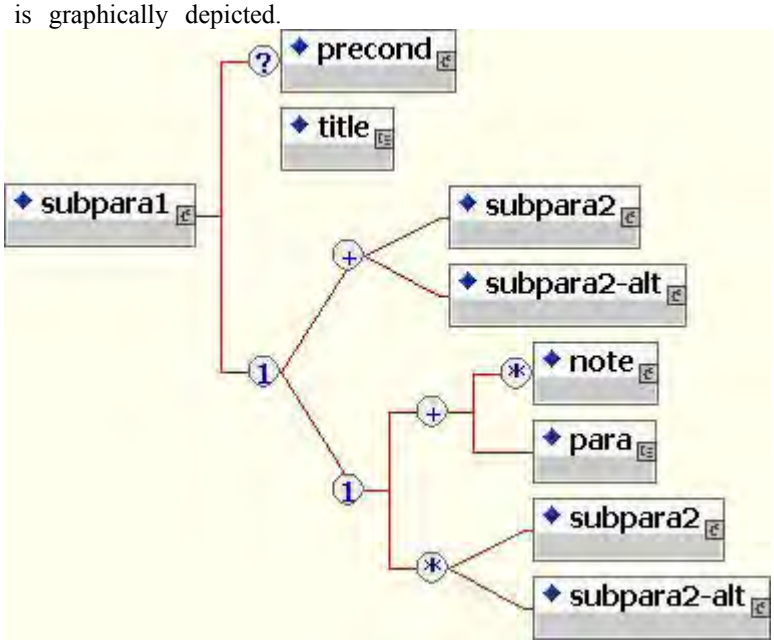

FIGURE 582. First level subordinate paragraph <subpara1> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT subpara1  (precond?, title, ((subpara2 | subpara2-alt)+ |
((note*, para)+,(subpara2 | subpara2-alt)*)))>
<!ATTLIST subpara1
    crewmember          CDATA                   #IMPLIED

    applicable          IDREFS                   #IMPLIED
    assocfig            IDREFS                   #IMPLIED
    changeref           IDREFS                   #IMPLIED
    comment             CDATA                    #IMPLIED
    delchlvl            CDATA                    #IMPLIED
    id                  ID                       #IMPLIED
    idref               IDREFS                   #IMPLIED
    inschlvl            CDATA                    #IMPLIED
    security            (uc | fouo |c | s | ts)  #IMPLIED
    skilltrk            CDATA                    #IMPLIED
   esd                 (yes | no)                "no"
    hcp                 (yes | no)                "no"
>
```

d. Unique attribute:

(1) **crewmember** Crew member that this information pertains to is specified. This will appear in the table prior to the beginning of the procedure.

e. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(12) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).

49.1.1.11 Alternate subpara1 ***<subpara1-alt>***. Alternate subpara1 to support multiple configurations. The attribute **id** provides subpara1-alt to allow linking to alternative state.

a. The components are:

(1) Alternate first level subordinate paragraph ***<subpara1-alt>*** (see 48.2.1).

b. DTD fragment is graphically depicted.

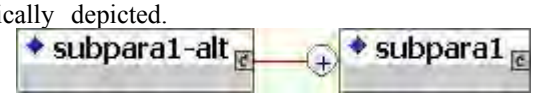

FIGURE 583. Alternate subpara1 <subpara1-alt> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT subpara1-alt  (subpara1+)>
<!ATTLIST subpara1-alt
    id          ID              #IMPLIED>
```

d. Unique attributes:

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(1) **id** - An identifier of the element which is assigned at origination and which remains unchanged as the document is revised or updated even though the automatically assigned enumeration or manually-assigned "labels" change (in some cases many times). The value of the "id" is used when making references to the element from other portions of the document. if no ID is given, none will be maintained and the element can then not be cross-referenced by means of an IDREF on another element or with ***<xref>*** (see 46.2.2).

49.1.1.12 Second level subordinate paragraph ***<subpara2>***. This is the second level of subordinate paragraphs.

a. The components are:

(1) Precondition ***<precond>*** (optional)(see 42.1.1.1) followed by required title ***<title>*** (required)(see 49.1.1.3). and may be followed by either the group (a) or group (b) elements:

(a) Third Level Subordinate Paragraph ***<subpara3>*** (optional – zero or more)(see 49.1.1.14) or Alternate Third Level Subordinate paragraph ***<subpara3-alt>***(optional – zero or more) (see 48.2.1).or both the ***<subpara3>*** and ***<subpara3-alt>*** can occur more than once, in no particular order.

(b) Or Note ***<note>*** (optional – zero or more)(see 41.1.3) followed by one or more Paragraph(s) ***<para>*** (required – one or more)(see 49.1.1.5), which may be followed by either the ***<subpara3>*** (optional – zero or more)or ***<subpara3-alt>*** (optional – zero or more) which can occur more than once, in no particular order.

b. DTD fragment:

```
<!ELEMENT subpara2  (precond?, title, ((subpara3 | subpara3-alt)+ |
((note*, para)+,(subpara3 | subpara3-alt)*)))>
<!ATTLIST subpara2
    crewmember          CDATA                   #IMPLIED

    applicable         IDREFS                    #IMPLIED
    assocfig           IDREFS                    #IMPLIED
    changeref          IDREFS                    #IMPLIED
    comment            CDATA                     #IMPLIED
    delchlvl           CDATA                     #IMPLIED
    id                 ID                        #IMPLIED
    idref              IDREFS                    #IMPLIED
    inschlvl           CDATA                     #IMPLIED
    security           (uc | fouo |c | s | ts)  #IMPLIED
    skilltrk           CDATA                     #IMPLIED
   esd                (yes | no)                "no"
    hcp                (yes | no)                "no"
>
```

c. Unique attribute:

(1) **crewmember** Crew member that this information pertains to is specified. This will appear in the table prior to the beginning of the procedure.

d. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).

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(11) **esd** – Electrostatic discharge requirement (default value is “no”) (see 49.2.5).
(12) **hcp** – Nuclear hardness requirement (default value is “no”) (see 49.2.5).

e.

49.1.1.13 Alternate subpara2 ***<subpara2-alt>***. Alternate subpara2 to support multiple configurations. The attribute **id** provides subpara2-alt to allow linking to alternative state.

a. The components are:
(1) Second Level Subordinate Paragraph ***<subpara2>*** (see 49.1.1.12).

b. DTD fragment:

```
<!ELEMENT subpara2-alt  (subpara2+)>
<!ATTLIST subpara2-alt  id  ID  #IMPLIED>
```

c. Common attributes:
(1) **id** - An identifier of the element which is assigned at origination and which remains unchanged as the document is revised or updated even though the automatically assigned enumeration or manually-assigned "labels" change (in some cases many times). The value of the "id" is used when making references to the element from other portions of the document. if no ID is given, none will be maintained and the element can then not be cross-referenced by means of an IDREF on another element or with ***<xref>*** (see 46.2.2).

49.1.1.14 Third level subordinate paragraph ***<subpara3>***. This is the third level of subordinate paragraphs.

a. The components are:
(1) Precondition ***<precond>*** (optional) (see 42.1.1.1) followed by title ***<title>*** (required) (see 49.1.1.3). and may be followed by either the group (a) or group (b) elements:
(a) Fourth Level Subordinate Paragraph ***<subpara4>*** (required – one or more)(see 49.1.1.16) or Alternate Fourth Level Subordinate paragraph ***<subpara4-alt>*** (required – one or more)(see 48.2.1) or both the ***<subpara4>*** and ***<subpara4-alt>*** can occur more than once, in no particular order.
(b) or Note ***<note>*** (optional – zero or more) (see 41.1.3) followed by one or more Paragraph(s) ***<para>*** (required – one or more) (see 49.1.1.5), which may be followed by either the ***<subpara4>*** (optional – zero or more) or ***<subpara4-alt>*** (optional – zero or more) which can occur more than once, in no particular order.

b. DTD fragment:

```
<!ELEMENT subpara3  (precond?, title, ((subpara4 | subpara4-alt)+ |
((note*, para)+,(subpara4 | subpara4-alt)*)))>
<!ATTLIST subpara3
    crewmember          CDATA                   #IMPLIED

    applicable         IDREFS                          #IMPLIED
    assocfig           IDREFS                          #IMPLIED
    changeref          IDREFS                          #IMPLIED
    comment            CDATA                           #IMPLIED
    delchlvl           CDATA                           #IMPLIED
    id                 ID                              #IMPLIED
    idref              IDREFS                          #IMPLIED
    inschlvl           CDATA                           #IMPLIED
    security           (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk           CDATA                           #IMPLIED
   esd                (yes | no)                      "no"
    hcp                (yes | no)                      "no"
>
```

c. Unique attribute:

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(1) **crewmember** Crew member that this information pertains to is specified. This will appear in the table prior to the beginning of the procedure.

d. Common attributes:
(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **esd** – Electrostatic discharge requirement (default value is "no") (see 49.2.5).
(12) **hcp** – Nuclear hardness requirement (default value is "no") (see 49.2.5).

e.

49.1.1.15 Alternate subpara3 ***<subpara3-alt>***. Alternate subpara3 to support multiple configurations. The attribute **id** provides subpara3-alt to allow linking to alternative state.

a. The components are:
(1) Third Level Subordinate Paragraph ***<subpara3>*** (required – one or more)(see 49.1.1.14).

b. DTD fragment:

```
<!ELEMENT subpara3-alt  (subpara3+)>
<!ATTLIST subpara3-alt
    id          ID              #IMPLIED>
```

c. Unique attributes:
(1) **ID** - An identifier of the element which is assigned at origination and which remains unchanged as the document is revised or updated even though the automatically assigned enumeration or manually-assigned "labels" change (in some cases many times). The value of the "id" is used when making references to the element from other portions of the document. if no ID is given, none will be maintained and the element can then not be cross-referenced by means of an IDREF on another element or with ***<xref>*** (see 46.2.2).

49.1.1.16 Fourth level subordinate paragraph ***<subpara4>***. This is the fourth level of subordinate paragraphs.

a. The components are:
(1) Precondition ***<precond>*** (optional)(see 42.1.1.1).
(2) Title ***<title>*** (required)(see 49.1.1.3).
(3) Note ***<note>*** (optional – zero or more) (see 41.1.3).
(4) Paragraph Text ***<para>*** (required – one or more) (see 49.1.1.5).

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b. DTD fragment:

```
<!ELEMENT subpara4  (precond?, title, (note*, para)+)>
<!ATTLIST subpara4
    crewmember          CDATA                   #IMPLIED

    applicable         IDREFS                          #IMPLIED
    assocfig           IDREFS                          #IMPLIED
    changeref          IDREFS                          #IMPLIED
    comment            CDATA                           #IMPLIED
    delchlvl           CDATA                           #IMPLIED
    id                 ID                              #IMPLIED
    idref              IDREFS                          #IMPLIED
    inschlvl           CDATA                           #IMPLIED
    security           (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk           CDATA                           #IMPLIED
   esd                (yes | no)                      "no"
    hcp                (yes | no)                      "no"
>
```

c. Unique attribute:

(1) **crewmember** Crew member that this information pertains to is specified. This will appear in the table prior to the beginning of the procedure.

d. Common attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).
(2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(3) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(4) **comment** – Change information (optional) (see 49.2.9).
(5) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(6) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
(7) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(8) **inschlvl** – Insert change level (optional) (see 49.2.9).
(9) **security** – Security classification (optional) (see 49.2.13).
(10) **skilltrk** – Skill level (optional) (see 49.2.2).
(11) **esd** – Electrostatic discharge requirement (default value is “no”) (see 49.2.5).
(12) **hcp** – Nuclear hardness requirement (default value is “no”) (see 49.2.5).

49.1.1.17 Alternate subpara4 ***<subpara4-alt>***. Alternate subpara4 to support multiple configurations. The attribute **id** provides subpara4-alt to allow linking to alternative state.

a. The components are:

(1) Fourth Level Subordinate Paragraph ***<subpara4>*** (required – one or more)(see 49.1.1.16).

b. DTD fragment:

```
<!ELEMENT subpara4-alt  (subpara4+)>
<!ATTLIST subpara4-alt
    id          ID              #IMPLIED>
```

c. Unique attributes:

(1) **id** - An identifier of the element which is assigned at origination and which remains unchanged as the document is revised or updated even though the automatically assigned enumeration or manually-assigned "labels" change (in some cases many times). The value of the "id" is used when making references to the element from other portions of the document. if no ID is given, none will be maintained and the element can then not be cross-referenced by means of an IDREF on another element or with ***<xref>*** (see 46.2.2).

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49.1.1.18 Text element *<text>*. The *<text>* element is used in to provide limited character data input in places where a normal mixed content model would not work. An example is where a title may be required, but a full paragraph of text not appropriate.

a. The components are:
   (1) Parsable characters or type text. – #PCDATA
   (2) Format text – ***<emphasis>*** (see 49.1.3.1).
   (3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
   (4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
   (5) Cross reference – ***<xref>*** (see 46.2.2).
   (6) External reference – ***<extref>*** (see 46.2.1).
   (7) Enhanced linking – ***<link>*** (see 46.2.3).

b. DTD fragment is graphically depicted.

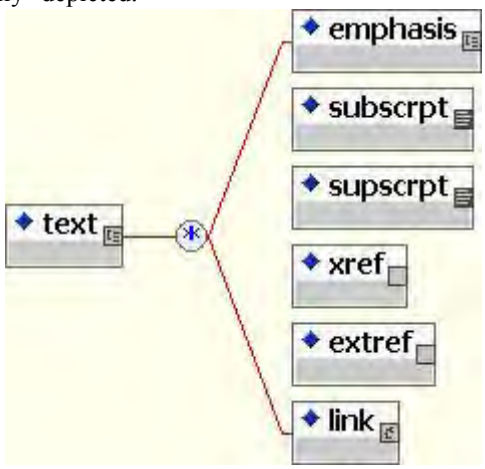

FIGURE 584. <text> content model.

c. DTD fragment:

```
<!ELEMENT text (#PCDATA | emphasis | subscrpt | supscrpt |
xref | extref | link)*>
```

d. No attributes.

49.1.2 List elements.

49.1.2.1 Sequence (ordered) list ***<seqlist>***. The element ***<seqlist>*** is used for a sequential or ordered list. The sequence of items is denominated by numbers or letters. The numbering scheme is the first level list uses numerical counter surrounded by parentheses, the second level list uses lower case alpha characters counter surrounded by parentheses and the third level list uses numerical counter followed by a right parentheses.

a. The components are:
   (1) Title ***<title>*** (optional) (see 49.1.1.3).
   (2) Items ***<item>*** (required – one or more) (see 49.1.2.1.1) are required in a sequence list.

b. DTD fragment is graphically depicted.

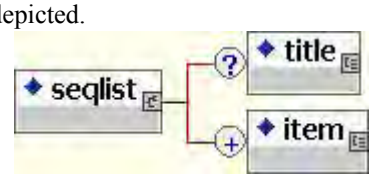

FIGURE 585. Sequence (ordered) list <seqlist> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT seqlist  (title?, item+)>
<!ATTLIST seqlist   >

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

e. Example of an XML instance and the style sheet output for a sequential list ***<seqlist>***.
   (1) Example of an XML instance for a sequential list ***<seqlist>***.

<seqlist>
<item>NOMENCLATURE
</item>
<item>PART NUMBER
</item>
<item>NSN
</item>
<item>DODAC
</item>
</seqlist>

   (2) Style sheet output for a sequential list.

1. NOMENCLATURE
2. PART NUMBER
3. NSN
4. DODAC

FIGURE 586. Example of a sequential list <seqlist>.

49.1.2.1.1 Item ***<item>***. The element ***<item>*** is an entry in a sequential or random list or an equipment item.

a. The components are:
   (1) The following elements are (optional – zero or more).
   (2) Parsable characters or type text. – #PCDATA
   (3) Format text – ***<emphasis>*** (see 49.1.3.1).
   (4) Subscript – ***<subscrpt>*** (see 49.1.3.4).
   (5) Superscript – ***< supscrpt>*** (see 49.1.3.5).
   (6) Cross reference – ***<xref>*** (see 46.2.2).

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(7) External reference – ***<extref>*** (see 46.2.1).
(8) Enhanced linking – ***<link>*** (see 46.2.3).
(9) IETM help information – ***<help.info>*** (see 48.3.2.4).
(10) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(11) Term – ***<term>*** (see 49.1.2.3.1.1).
(12) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(13) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(14) Footnote – ***< ftnote>*** (see 45.1.1).
(15) Footnote reference – ***<ftnref>*** (see 45.1.2).
(16) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(17) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(18) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(19) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(20) Graphic symbol – ***<symbol>*** (see 44.3.1).
(21) Torque value – ***<torque>*** (see 49.1.4.24).
(22) Voltage value – ***< voltage>*** (see 49.1.4.25).
(23) Null text – ***<null >*** (see 49.1.3.2).
(24) Changed text marker – ***<change>*** (see 49.1.3.7).
(25) Note ***<note>*** (see 41.1.3) are required in a sequence list.
(26) Sequence (ordered) list ***<seqlist>*** (see 49.1.2.1) in an item.
(27) Random list ***<randlist>*** (see 49.1.2.2) in an item.
(28) Inline graphic(s) ***<inlinegraphic>*** (see 44.3.2) in an item.
(29) Illustration(s) ***<figure>*** (see 44.1.1) in an item.
(30) Alternative Illustration(s) ***<figure-alt>*** (see 48.2.1) in an item.

b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

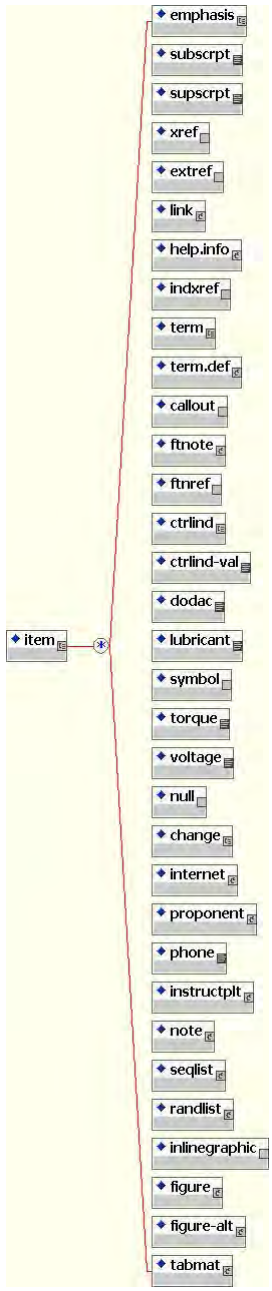

FIGURE 587. Item <item> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT item  (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link | help.info | ir
| figure | figure-alt)*>
<!ATTLIST item
    label          CDATA          #IMPLIED
    applicable     IDREFS         #IMPLIED


    assocfig          IDREFS                              #IMPLIED
    changeref         IDREFS                              #IMPLIED
    comment           CDATA                               #IMPLIED
    delchlvl          CDATA                               #IMPLIED
    id                ID                                  #IMPLIED
    idref             IDREFS                              #IMPLIED
    inschlvl          CDATA                               #IMPLIED
    security          (uc | fouo |c | s | ts)           #IMPLIED
    skilltrk          CDATA                               #IMPLIED
>
```

d. Unique attributes:

(1) **applicable** Applicability reference (optional) (see 49.2.1.b.(1)).

(2) **label** - A label so the authors can enter counter strings rather than have the application do it.

e. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

(3) **comment** – Reason for change information. (optional) (See 49.2.9.)

(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)

(5) **id** – Unique identifier. (optional) (See 49.2.6.)

(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)

(9) **security** – Security classification. (optional) (See 49.2.13.)

49.1.2.2 Random list ***<randlist>***. The element ***<randlist>*** is used for a list of randomly ordered items and are not numbered. The bullet scheme, when specified, is the first level list uses a filled circled and the second level list uses a dash.

a. The components are:

(1) Title ***<title>*** (optional) (see 49.1.1.3).

(2) One or more items ***<item>*** (required – one or more) (see 49.1.2.1.1) are required in a random list.

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b. DTD fragment:

```
<!ELEMENT randlist (title?, item+)>
<!ATTLIST randlist
    bullet          (yes | no)                      'no'
    prefix           CDATA                          #IMPLIED

    assocfig          IDREFS                           #IMPLIED
    changeref         IDREFS                           #IMPLIED
    comment           CDATA                            #IMPLIED
    delchlvl          CDATA                            #IMPLIED
    id                ID                               #IMPLIED
    idref             IDREFS                           #IMPLIED
    inschlvl          CDATA                            #IMPLIED
    security          (uc | fouo |c | s | ts)          #IMPLIED
    skilltrk          CDATA                            #IMPLIED
>
```

c. DTD fragment is graphically depicted.

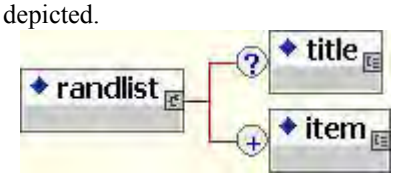

FIGURE 588. Random list <randlist> DTD hierarchy..

d. Unique attributes:

(1) **bullet**- Specifies whether (non-zero number) or not a bullet (0) should precede each item. If no value is entered the default is no bullet displayed.

(2) **prefix** - Specifies a character, word, or symbol (other than a bullet) that should precede each item

e. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)

(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)

(3) **comment** – Reason for change information. (optional) (See 49.2.9.)

(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)

(5) **id** – Unique identifier. (optional) (See 49.2.6.)

(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)

(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)

(9) **security** – Security classification. (optional) (See 49.2.13.)

f. Example of an XML instance and the style sheet output for a random list <randlist>.

(1) Example of an XML instance for random list.

```
<randlist bullet="yes">
<item>NOMENCLATURE – Name of the part.
</item>
<item>PART NUMBER – Number assigned to the part.  Check
<extref docno="TM 1-1520-238-23P"/> for part numbers of replacement parts.
</item>
<item>NSN – National Stock Number assigned to the part.  Reference
<extref docno="TM 1-1520-238-23P"/>.
</item>
<item>DODAC – Department of Defense Ammunition Code.  Indicates
type of Ammunition and Packaging.  Ref
<extref docno=" SB708-3" pretext="DOD Ammo Codes HDBK, "/>.
</item>
```


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</randlist>

(2) Style sheet output for a random list.

- NOMENCLATURE — Name of the part.
- PART NUMBER — Number assigned to the part. Check "TM 1-1520-238-23P", for part numbers of replacement parts.
- NSN — National Stock Number assigned to the part. Reference "TM 1-1520-238-23P".
- DODAC — Department of Defense Ammunition Code. Indicates type of Ammunition and Packaging. Reference DOD Ammo Codes HDBK, "SB708-3"

FIGURE 589. Example of a random list <randlist>.

49.1.2.3 Definition list ***<deflist>***. The element ***<deflist>*** identifies a list of terms and definitions. The term can enclose a word, phrase, abbreviation, or symbol.

a. The components are:
   (1) Title ***<title>*** (optional) (see 49.1.1.3).
   (2) Term and Definition Titles ***<title.term.def>*** (optional – zero or more) of a column contained in a deflist.
   (3) One or more term and definition ***<term.def>*** (required – one or more) (see 49.1.2.3.1) are required in a definition list.

b. DTD fragment is graphically depicted.

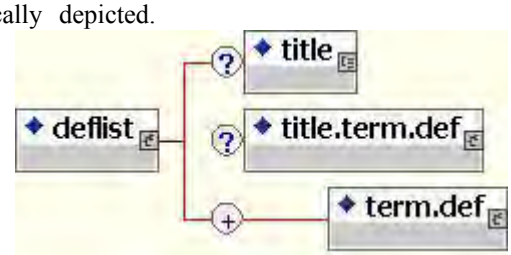

FIGURE 590. Definition list <deflist> DTD hierarchy.

c. DTD fragment

```
<!ELEMENT deflist (title?, title.term.def?, (term.def)+)>
<!ATTLIST deflist

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)

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(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

e. Example of an XML instance and the style sheet output for a definition list ***<deflist>***.

(1) Example of an XML instance for definition list.

```
<deflist>
<title>Definition  List
</title>
<title.term.def>
<title>Name
</title>
<title>Definition  List
</title>
</title.term.def>
<term.def>
<term>NOMENCLATURE
</term>
<def>
<para>Name  of  the  part.
</para>
</def>
</term.def>
<term.def>
<term>PART  NUMBER
</term>
<def>
<para>Number  assigned  to  the  part.    Check
<extref  docno="TM  1-1520-238-23P"/> for part numbers of replacement parts.
</para>
</def>
</term.def>
<term.def>
<term>NSN
</term>
<def>
<para>National Stock Number assigned to the part.  Reference
<extref  docno="TM  1-1520-238-23P"/>.
</para>
</def>
</term.def>
<term.def>
<term>DODAC
</term>
<def>
<para>Department of Defense Ammunition Code.   Indicates type
of Ammunition and Packaging.    Ref
<extref  docno="  SB708-3"  pretext="DOD  Ammo  Codes  HDBK,  "/>.
</para>
</def>
</term.def>
<deflist>
```

(2) Style sheet output for a definition list.

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DEFINITION LIST

Name	Description
NOMENCLATURE	Name of the part.
PART NUMBER	Number assigned to the part. Check "TM 1-1520-238-23P", for part numbers of replacement parts.
NSN.	National Stock Number assigned to the part. Reference "TM 1-1520-238-23P".
DODAC	Department of Defense Ammunition Code. Indicates type of Ammunition and Packaging. Reference DOD Ammo Codes HDBK, "SB708-3".

FIGURE 591. Example of a definition list <deflist>.

49.1.2.3.1 Term definition ***<term.def>***. The element ***<term.def>*** identifies an information row containing a term and a definition. Equivalent to entering "row" in a structural table. The element contains a term and a definition.

a. The components are:
   (1) Term ***<term>*** (required)(see 49.1.2.3.1.1).
   (2) Definition ***<def>*** (required)(see 49.1.2.3.1.2)

b. DTD fragment:

```
<!ELEMENT term.def  (term, def)>
<!ATTLIST term
                id      ID   #REQUIRED>
```

c. The components are:
   (1) Requires a term ***<term>*** (see 49.1.2.3.1.1).
   (2) Requires a definition ***<def>*** (see 49.1.2.3.1.2)

d. Unique attributes:
   (1) **id** - Specifies the identifier of the term definition element.

49.1.2.3.1.1 Term ***<term>***. The element ***<term>*** is a word, phrase, acronym, symbol or abbreviation to be defined in a definition list.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.(see 9.3.2).
   (2) Emphasis ***<emphasis>*** (optional – zero or more) (see 49.1.3.1)
   (3) Subscript ***<subscrpt>*** (optional – zero or more)(see 49.1.3.4)
   (4) Superscript ***<supscrpt>*** (optional – zero or more) (see 49.1.3.5)
   (5) Cross Reference ***<xref>*** (optional – zero or more) (see 46.2.2)
   (6) Linking ***<link>*** (optional – zero or more) (see 46.2.3)

b. DTD fragment:

```
<!ELEMENT term  ("#PCDATA | emphasis | subscrpt | supscrpt | xref | link)*>
<!ATTLIST term  idref  IDREF  #IMPLIED >
```

c. Attributes for ***<term>***:
   (1) **idref** - Reference a definition. (optional)

49.1.2.3.1.2 Definition ***<def>***. The element ***<def>*** is a definition for the term in a definition list. The element contains a general paragraph ***<para>*** (see 49.1.1.5).

a. The components are:

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(1) Paragraph text (required) (see 49.1.1.5)

b. DTD fragment:

```
<!ELEMENT def (para)>
```

49.1.3 Formatting elements.

49.1.3.1 Emphasis ***<emphasis>***. The element ***<emphasis>*** is used for scope and style of emphasized information is indicated by enclosing the material in start and end ***<emphasis>*** tags. Emphasis elements should be used only in situations where the emphasized material is embedded in plain text or where an exception from the usual style of the element specified in the style sheet is needed. Emphasis elements can be nested to specify a combination of styles, such as ***underlined bold italic***. The emphasized text is enclosed in a ***<emphasis>*** start and end element.

a. The components are:
   (1) Parsable characters or type text. – #PCDATA
   (2) Format text – ***<emphasis>*** (see 49.1.3.1).
   (3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
   (4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
   (5) Cross reference – ***<xref>*** (see 46.2.2).
   (6) External reference – ***<extref>*** (see 46.2.1).
   (7) Enhanced linking – ***<link>*** (see 46.2.3).
   (8) IETM help information – ***<help.info>*** (see 48.3.2.4).
   (9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
   (10) Term – ***<term>*** (see 49.1.2.3.1.1).
   (11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
   (12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
   (13) Footnote – ***< ftnote>*** (see 45.1.1).
   (14) Footnote reference – ***<ftnref>*** (see 45.1.2).
   (15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
   (16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
   (17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
   (18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
   (19) Graphic symbol – ***<symbol>*** (see 44.3.1).
   (20) Torque value – ***<torque>*** (see 49.1.4.24).
   (21) Voltage value – ***< voltage>*** (see 49.1.4.25).
   (22) Null text – ***<null >*** (see 49.1.3.2).
   (23) Changed text marker – ***<change>*** (see 49.1.3.7).

b. DTD fragment:

```
<!ELEMENT emphasis  (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link | help.info
| indxref | term | term.def | callout | ftnote | ftnref | ctrlind | ctrlind-val
| dodac | lubricant | symbol | torque | voltage | null | change)*>
<!ATTLIST emphasis
    emph     (caps | bold | italic | bolditalic
              | uline | strikeout | 2line |
              smallcaps | overline)                  #REQUIRED
    color    CDATA                                   #IMPLIED >
```

c. Unique attributes:
   (1) **emph** -
   Chose from the list of possible emphasis characteristics. The list can be expanded through the configuration file. Specifies the type of emphasis to be used.

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(a) “caps” - Specifies the data is capitalized all text.
(b) “bold” - Specifies the data is bold text.
(c) “italic” - Specifies the data is italicized text.
(d) “bolditalic” - Specifies the data is bold and italicized text.
(e) “uline” - Specifies the data is underlined text.
(f) “strikeout” - Specifies the data is strikeout dash through each character.
(g) “2line” - Specifies the data is underlined with two lines.
(h) “smallcaps” - Specifies the data is capitalized using smallcaps.
(i) “overline” - Specifies the data is overlined text.

(2) **color** - Color that is used as a means of emphasis.

49.1.3.2 Null ***<null>***. The element ***<null>*** is used for an element in a table entry which specifically indicates the entry contains no content.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT null  EMPTY>
<!ATTLIST null
    insert  (na | nr | dash | secure | none )  'none' >
```

c. Unique attributes:
   (1) **insert** - Specifies what material is inserted in the table cell containing the null element, i.e., "NA," "NR," a long dash, classified status, or no insertion.

49.1.3.3 Line break ***<brk/>***. The element ***<brk/>*** is used to create a line break in the narrative. May be used optionally in a frame-based manual.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT brk  EMPTY>
```

c. No attributes.

49.1.3.4 Subscript ***<subscrpt>***. The element ***<subscrpt>*** is used to format the text as subscript. The style sheet formats the character or symbol written directly beneath or next to and slightly below a letter or number.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.(see 9.3.2).
b. DTD fragment:

```
<!ELEMENT subscrpt (#PCDATA)>
```

c. No attributes.

49.1.3.5 Superscript ***<supscrpt>***. The element ***<supscrpt>*** is used to format the text as supscript. The style sheet formats the text awritten above and immediately to one side of another.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.(see 9.3.2).
b. DTD fragment:

```
<!ELEMENT subscrpt (#PCDATA)>
```

c. No attributes.

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49.1.3.6 Verbatim ***<verbatim>***. The element ***<verbatim>*** is used for text to be presented verbatim as it is sequenced in the text stream and implies that XML record ends (carriage returns) are to be treated as line endings. Typically, verbatim text is presented in a monospaced font.

a. The component of ***<verbatim>*** is:
   (1) The element contains narrative text #PCDATA parsable character data.(see 9.3.2).
b. DTD fragment:

```
<!ELEMENT verbatim (#PCDATA)>
<!ATTLIST verbatim
    xml:space    (preserve )  #FIXED 'preserve'
```

c. Unique attributes:
   (1) **xml:space** - Specifies XML will perserve space and line break characters.
d. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.3.7 Changed information ***<change>***. The element ***<change>*** is used for the scope and type of changed information indicated by being enclosed in ***<change>*** start and end tag.

a. The components are:
   (1) #PCDATA (see 9.3.2)
   (2) Emphasis ***<emphasis>*** (see 49.1.3.1)
   (3) Subscript ***<subscrpt>*** (see 49.1.3.4)
   (4) Superscript ***<supscrpt>*** (see 49.1.3.5)
   (5) Cross Reference ***<xref>*** (see 46.2.2)
   (6) External Reference ***<extref>*** (see 46.2.1)
   (7) Linking ***<link>*** (see 46.2.3)
   (8) Content Sensitive Help ***<help.info>*** (see 48.3.2.4)
   (9) Index Marker Reference ***<indxref>*** (see 31.4.2.2.3)
   (10) Term ***<term>*** (see 49.1.2.3.1.1)
   (11) Item Definition ***<term.def>*** (see 49.1.2.3.1)
   (12) Figure Call out ***<callout>*** (see 46.2.4.1)
   (13) Footnote ***<ftnote>*** (see 45.1.1)
   (14) Footnote Reference ***<ftnref>*** (see 45.1.2)
   (15) Controls and Indicators Entry ***<ctrlind>*** (see 49.1.4.3)
   (16) Reading On – Controls and Indicators ***<ctrlind-val>*** (see 49.1.4.4)
   (17) Department of Defense Ammunition Code ***<dodac>*** (see 49.1.4.5)
   (18) Lubricant ***<lubricant>*** (see 49.1.4.16)
   (19) Symbol ***<symbol>*** (see 44.3.1)
   (20) Torque Value or Limit ***<torque>*** (see 49.1.4.24)
   (21) Voltage Measurement ***<voltage>*** (see 49.1.4.25)
   (22) Null ***<null>*** (see 49.1.3.2)

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b. DTD fragment:

```
<!ELEMENT change  (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link | help.info |
| voltage | null)*>
<!ATTLIST change

    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

c. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.4 Content specified elements.

49.1.4.1 Address ***<address>***. The element ***<address>*** is used to enter the address.
a. The components are:
   (1) Service Nomenclature ***<servnomen>*** (optional)(see 31.3.1.1.4.3).
   (2) Street Addresses ***<street>*** (optional – zero or more)(see 49.1.4.1.1).
   (3) City ***<city>*** (required)(see 49.1.4.1.2) a.
   (4) State ***<state>*** (required)(see 49.1.4.1.3).
   (5) Zip Code ***<zip>*** (optional)(see 49.1.4.1.4) and country***<country>*** (optional)(see 49.1.4.1.5)
   (6) May have optional ***<phone>*** (optional – zero or more)(see 49.1.4.1.6)
   (7) May have optional service nomenclature ***<internet>*** (optional – zero or more)(see 49.1.4.1.7)
b. DTD fragment is graphically depicted.

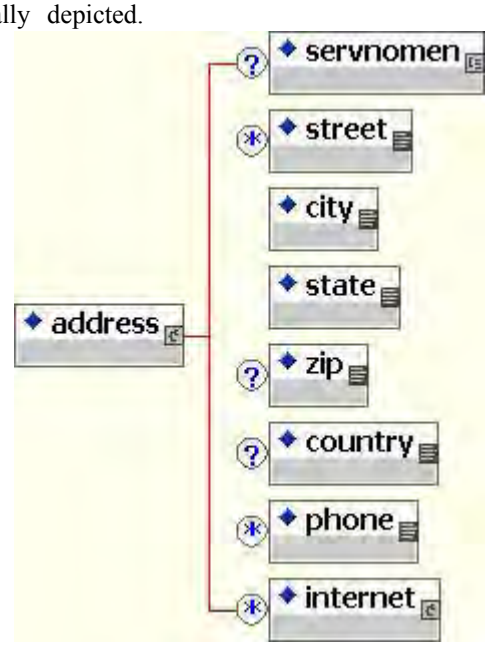

FIGURE 592. Address <address> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT address (servnomen?, street*, city, state, zip?,
 country?, phone*, internet*)>
<!ATTLIST address

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.1.1 Street ***<street>***. The element ***<street>*** is used to enter the name of the street address used in the address block.

a. The components are:

(1) The element contains narrative text #PCDATA parsable character data.

b. DTD fragment:

```
<!ELEMENT street  (#PCDATA)>
<!ATTLIST street

    changeref         IDREFS                    #IMPLIED
    comment           CDATA                     #IMPLIED
    delchlvl          CDATA                     #IMPLIED
    inschlvl          CDATA                     #IMPLIED
>
```

c. Common attributes:

(1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(2) **comment** – Change information (optional) (see 49.2.9).
(3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.4.1.2 City ***<city>***. The element ***<city>*** is used to enter the name of the city used in an address block.

a. The components are:

(1) The element contains narrative text #PCDATA parsable character data.

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b. DTD fragment:

```
<!ELEMENT city  (#PCDATA)>
<!ATTLIST city

    changeref       IDREFS                  #IMPLIED
    comment         CDATA                   #IMPLIED
    delchlvl        CDATA                   #IMPLIED
    inschlvl        CDATA                   #IMPLIED
>
```

c. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.4.1.3 State ***<state>***. The element ***<state>*** is used to enter the name of the state used in address block.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.

b. DTD fragment:

```
<!ELEMENT state  (#PCDATA)>
<!ATTLIST state

    changeref       IDREFS                  #IMPLIED
    comment         CDATA                   #IMPLIED
    delchlvl        CDATA                   #IMPLIED
    inschlvl        CDATA                   #IMPLIED
>
```

c. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.4.1.4 Zip ***<zip>***. The element ***<zip>*** is used to enter the name of the zip used in address block.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.

b. DTD fragment:

```
<!ELEMENT zip  (#PCDATA)>
<!ATTLIST zip

    changeref       IDREFS                  #IMPLIED
    comment         CDATA                   #IMPLIED
    delchlvl        CDATA                   #IMPLIED
    inschlvl        CDATA                   #IMPLIED
>
```

c. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.4.1.5 Country ***<country>***. The element ***<country>*** is used to enter the name of the country used in an address block.

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Check the source to verify that this is the current version before use.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT country  (#PCDATA)>
<!ATTLIST country

    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

c. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.4.1.6 Telephone number ***<phone>***. The element ***<phone>*** is used to enter the telephone number.
a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT phone (#PCDATA)>
<!ATTLIST phone
    type          (dsn | coml | cell | other )   #REQUIRED
    receive       (voice | fax )                  'voice'
    other         CDATA                             #IMPLIED

    changeref         IDREFS                    #IMPLIED
    comment           CDATA                     #IMPLIED
    delchlvl          CDATA                     #IMPLIED
    inschlvl          CDATA                     #IMPLIED
>
```

c. Unique attributes:
   (1) **type** - Specifies the type of the phone.
   (2) **receive** - Specifies if the phone is being used as a voice or a fax
   (3) **other** - Specifies for other data in reference to the phone.
d. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.4.1.7 Internet address***<internet>***. The element ***<internet>*** is used to enter the email address or the address of the homepage.
a. The components are:
   (1) Select either electronic mail address or internet home page address.
      (a) Electronic Mail Address ***<email>*** (required)(see 49.1.4.1.10)
      (b) Internet Home Page Address ***<homepage>*** (see 49.1.4.1.9).

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b. DTD fragment:

```
<!ELEMENT internet  (email | homepage)>
<!ATTLIST internet
    narrative          CDATA                  #IMPLIED
    show.address      (0 | 1)                "1">
```

c. Unique attributes:***<internet>***:
   (1) **narrative** - Narrative to use instead of the email or home page address.
   (2) **show.address** - Show the internet address with narrative attribute text.

49.1.4.1.8 Electronic mail ***<email>***. The element ***<email>*** is used to enter the electronic mail address.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.

b. DTD fragment:

```
<!ELEMENT email  EMPTY>
<!ATTLIST email
    address          CDATA         #REQUIRED

    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

c. Unique attributes:
   (1) **address** - Enter the electronic mail address.

d. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.4.1.9 Internet home page address ***<homepage>***. The element ***<homepage>*** is used to enter the internet home page address.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.

b. DTD fragment:

```
<!ELEMENT homepage  EMPTY>
<!ATTLIST homepage
    protocol     (http | https | ftp )    'http'
    uri           CDATA                    #REQUIRED

    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

c. Unique attributes :
   (1) **protocol** - Type of internet protocol used with the URI address.
   (2) **uri** - Uniform Resource Information (URI) for the Internet home page address.

d. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).

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(4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.4.1.10 Electronic mail ***<email>***. The element ***<email>*** is used to enter the electronic mail address.

a. The components are:
   (1) The element is EMPTY and all pertinent information is entered through its attributes.
b. DTD fragment:

```
<!ELEMENT email EMPTY>
<!ATTLIST email  address     CDATA  #REQUIRED

    changeref        IDREFS                    #IMPLIED
    comment          CDATA                     #IMPLIED
    delchlvl         CDATA                     #IMPLIED
    inschlvl         CDATA                     #IMPLIED
>
```

c. Unique attributes:
   (1) **address** - Enter the electronic mail address.
d. Common attributes:
   (1) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (2) **comment** – Change information (optional) (see 49.2.9).
   (3) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (4) **inschlvl** – Insert change level (optional) (see 49.2.9).

49.1.4.2 Commercial and government entity code (CAGEC) ***<cageno>***. The element ***<cageno>*** is the Commercial and Government Entity Code (CAGEC) and is a five character code assigned to commercial activities that manufacture or supply items used by the Federal Government and to Government activities that control design or are responsible for the development of certain specifications, standards, or drawings which control the design of Government items. CAGE Code assignments are listed in the H4/H8 CAGE Publications.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT cageno  (#PCDATA)>
```

49.1.4.3 Control/indicator ***<ctrlind>***. The element ***<ctrlind>*** control or indicator name is entered. using the **idref** element to link to the control/indicator table for detailed information.

a. The components are:
   (1) Parsable characters or type text. – #PCDATA
   (2) Format text – ***<emphasis>*** (see 49.1.3.1).
   (3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
   (4) Superscript – < ***supscrpt>*** (see 49.1.3.5).
   (5) Cross reference – ***<xref>*** (see 46.2.2).
   (6) External reference – ***<extref>*** (see 46.2.1).
   (7) Enhanced linking – ***<link>*** (see 46.2.3).
   (8) IETM help information – ***<help.info>*** (see 48.3.2.4).
   (9) Index reference – < ***indxref>*** (see 31.4.2.2.3).
   (10) Term – ***<term>*** (see 49.1.2.3.1.1).
   (11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
   (12) Figure call out reference – ***<callout>*** (see 46.2.4.1).
   (13) Footnote – ***<ftnote>*** (see 45.1.1).
   (14) Footnote reference – ***<ftnref>*** (see 45.1.2).
b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

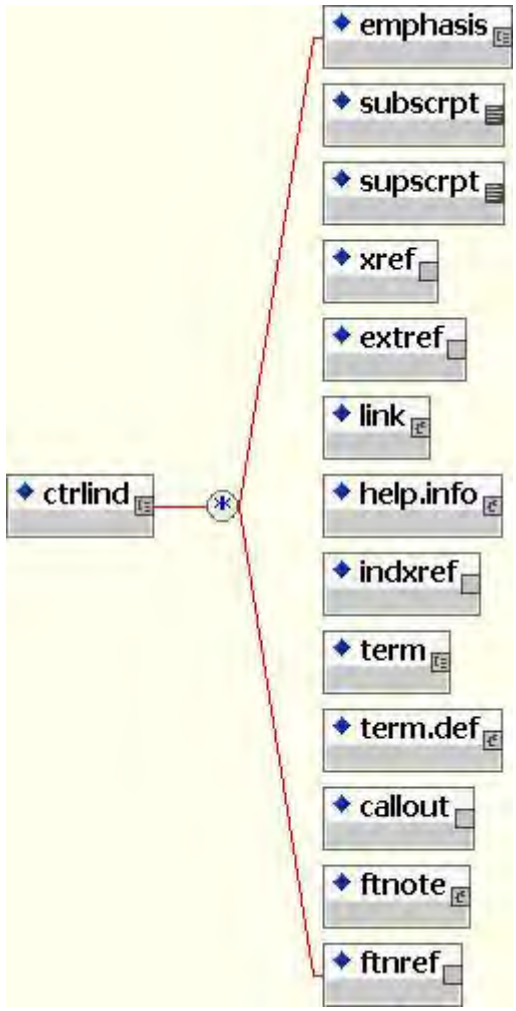

FIGURE 593. Control/indicator <ctrlind> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT ctrlind  (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link | help.info
<!ATTLIST ctrlind

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)

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(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

49.1.4.4 Control/indicator value ***<ctrlind-val>***. The element ***<ctrlind-val>*** identifies the reading on a control or indicator.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data..
b. DTD fragment:

```
<!ELEMENT ctrlind-val  (#PCDATA)>
<!ATTLIST ctrlind-val
    measurement       CDATA       #IMPLIED >

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Unique attributes:
   (1) **meaurement** - Measurement unit for the control or indicator.
d. Common attributes:
   (1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (3) **comment** – Change information (optional) (see 49.2.9).
   (4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (6) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (7) **security** – Security classification (optional) (see 49.2.13).
   (8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.5 Department of defense ammunition code ***<dodac>***. The element ***<dodac>*** is used to identify a type of ammunition. This element is used within the ammunition work package only. The Department of Defense Ammunition Code definitively identifies a type of ammunition. This element is used within the ammunition work package only.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT dodac (#PCDATA)>
```

c. No attributes.

49.1.4.6 Drawing name ***<dwgname>***. The element ***<dwgname>*** is used to identify the title box drawing name of a drawing required to properly perform the tasks within a work package.

a. The components are:
   (1) Parsable characters or type text. – #PCDATA
   (2) Format text – ***<emphasis>*** (see 49.1.3.1).
   (3) Subscript – ***<subscrpt>*** (see 49.1.3.4).

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(4) Superscript – ***<supscrpt>*** (see 49.1.3.5).

b. DTD fragment:

```
<!ELEMENT dwgname  (#PCDATA | emphasis | subscrpt | supscrpt)*>
<!ATTLIST dwgname

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.7 Drawing number ***<dwgno>***. The element ***<dwgno>*** is the drawing number. The ***<dwgno>*** is entered in the first column of the illustrated list of manufactured items.

a. The components are:

(1) The element contains narrative text #PCDATA parsable character data.

b. DTD fragment:

```
<!ELEMENT dwgno  (#PCDATA)>
<!ATTLIST dwgno

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.8 Item number ***<itemno>***. The element ***<itemno>*** provides the item number that is assigned to the entry for reference purpose. In page-base, the element is equivalent to an "entry" element in a table.

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a. The component is:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT itemno (#PCDATA)>
<!ATTLIST itemno
    id          ID          #IMPLIED
    idref       IDREFS      #IMPLIED
    assocfig    IDREFS      #IMPLIE >

>
```

c. Common attributes:
   (1) **id** – Specifies unique identifier (target) to reference (optional) (see 49.2.6).
   (2) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (3) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

49.1.4.9 Quantity ***<qty>***. The element quantity ***<qty>*** indicates the number of personnel or quantity of materials/parts equipment required to perform the procedures in the work package.

a. The component of ***<qty>*** is:
   (1) .The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT qty  (#PCDATA)>
<!ATTLIST qty

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:
   (1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
   (2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
   (3) **comment** – Change information (optional) (see 49.2.9).
   (4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
   (5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (6) **inschlvl** – Insert change level (optional) (see 49.2.9).
   (7) **security** – Security classification (optional) (see 49.2.13).
   (8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.10 User interaction ***<interaction>***. The element ***<interaction>*** provides ability to receive input from dialog prompts (Fill-In or Menu) and transmit results. The main function of the element is to assist with diagnostic. The element also can be used to provide standard mathematical operations during operations. The dialogs are created by technical writer and can not be generated by the IETM user.

a. The components are:
   (1) State (Variable) Information Manipulation ***<statemanipulation>***(see 48.2.3)
   (2) Conditional State (Variable) Information Manipulation ***<statemanipulation-alt>*** (see 48.2.1)
   (3) Dialog Box ***<dialog>*** (see 48.3.2)
   (4) Conditional Dialog Box ***<dialog-alt>*** (see 48.2.1)
   (5) Message ***<message>*** (see 48.4)

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b. DTD fragment is graphically depicted.

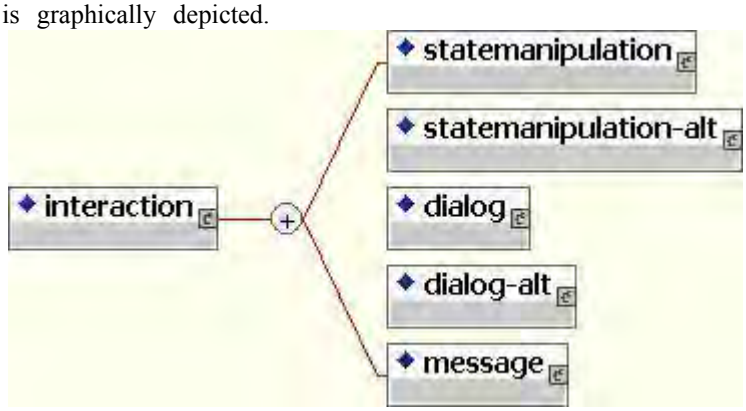

FIGURE 594. User interaction <interaction>.

c. DTD fragment:

```
<!ELEMENT interaction (statemanipulation | statemanipulation-alt
 | dialog | dialog-alt | message)+>
<!ATTLIST interaction

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.11 Reason ***<reason>***. The element reason ***<reason>*** is used to explain the reason for action.

a. The components are:

(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).

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(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).

b. DTD fragment:

```
<!ELEMENT reason  (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link | help.info |
<!ATTLIST reason

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:
(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.12 General or introductory information ***<geninfo>***. The element ***<geninfo>*** contains titled and subtitled paragraphs giving general or introductory information. The element is contained in various work package contexts; such as service upon receipt work package ***<surwp>***, or contained as part of procedural instructions; such as cleaning an aircraft prior to shipping.

a. The components are:
(1) Title ***<title>*** (optional) (see 49.1.1.3)
(2) May be followed by one or more figure(s) ***<figure>***(optional – zero or more) (see 44.1.1).
(3) Contains paragraphs of text that may be grouped into sections or subsections.
(a) Note ***<note>*** (optional – zero or more) (see 41.1.3).
(b) Either General Paragraph ***<para>*** (required – one or more)(see 49.1.1.5) with an optional one or more Primary Level Titled Paragraph(S) ***<para0>*** (see 49.1.1.8) or one or more Primary Level Conditional Titled Paragraph(S) ***<para0–alt>*** (see 49.1.1.3).
(c) Or Primary Level Titled Paragraph(s) ***<para0>*** (see 49.1.1.8) or one or more Primary Level Conditional Titled Paragraph(s) ***<para0–alt>*** (see 49.1.1.3).

b. DTD fragment is graphically depicted.

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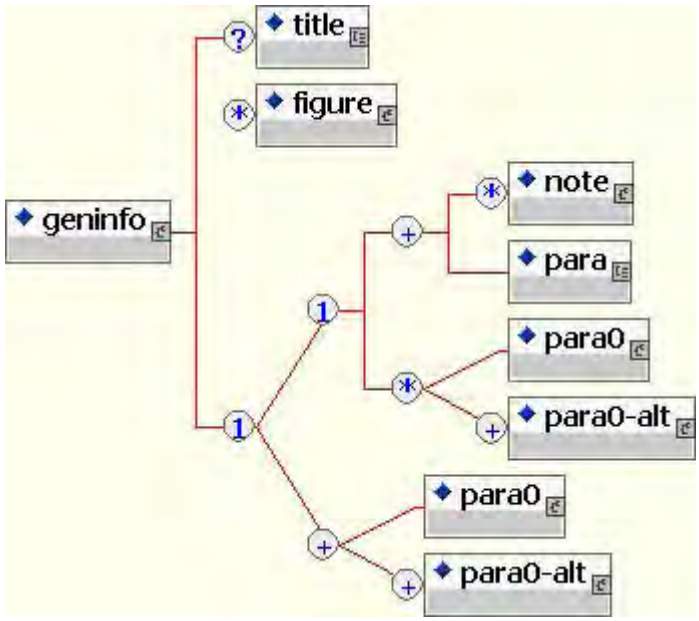

FIGURE 595. General or introductory information <geninfo> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT geninfo (title?, figure*, (((note*, para)+, (para0 | para0-alt+)*) | (para0 | para0
<!ATTLIST geninfo
    frame          (yes | no )          'no'

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Unique attributes:

(1) **frame** Frame indicates to the IETM system that the author's intends a frame break at this element.

e. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

49.1.4.13 Instruction plates and decals ***<instructplt>***. The element ***<instructplt>*** is used to specify the decals and instruction plates that are located on the equipment. This element may include one or more paragraph(s) (***<para>*** see 49.1.1.5) each of which may be preceded by a figure (***<figure>*** see 44.1.1).

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a. The components are:
   (1) Graphic(s) ***<graphic>*** (required – one or more)(see 44.2)
b. DTD fragment:

```
<!ELEMENT instructplt  (graphic+)>
<!ATTLIST instructplt

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Unique attributes:
   (1) **frame** Frame indicates to the IETM system that the author's intends a frame break at this element.
d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

49.1.4.14 System condition statement ***<condition>***. The element ***<condition>*** is used to describe either prerequisite, special environmental or equipment condition statement(s) prior to the work package procedure(s).

a. The components are:
   (1) Parsable characters or type text. – #PCDATA
   (2) Format text – ***<emphasis>*** (see 49.1.3.1).
   (3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
   (4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
   (5) Cross reference – ***<xref>*** (see 46.2.2).
   (6) External reference – ***<extref>*** (see 46.2.1).
   (7) Enhanced linking – ***<link>*** (see 46.2.3).
   (8) IETM help information – ***<help.info>*** (see 48.3.2.4).
   (9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
   (10) Term – ***<term>*** (see 49.1.2.3.1.1).
   (11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
   (12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
   (13) Footnote – ***<ftnote>*** (see 45.1.1).
   (14) Footnote reference – ***<ftnref>*** (see 45.1.2).
b. DTD fragment is graphically depicted.

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Check the source to verify that this is the current version before use.

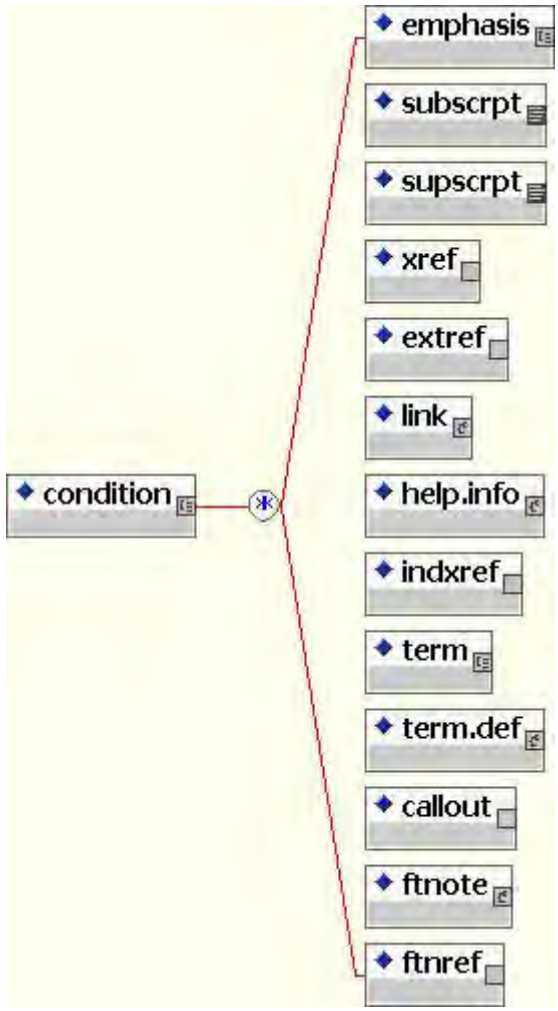

FIGURE 596. System condition statement <condition> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT condition  (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref |
 link | help.info | indxref | term | term.def | callout |
 ftnote | ftnref)>
<!ATTLIST condition

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).

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(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.15 Introductory ***<intro>***. The element ***<intro>*** identifies an introductory section contained in various work packages, which often (but not always) has text that should be entered verbatim as boiler plate text from the governing specification (TMs is MIL-STD-40051-1 or MIL-STD-40051-2).

a. The components are:
   (1) Primary Level Titled Paragraph ***<para0>*** (see 49.1.1.8) or Primary Level Conditional Titled Paragraph***<para0-alt>***(required) (see 48.2.1)
b. DTD fragment is graphically depicted.

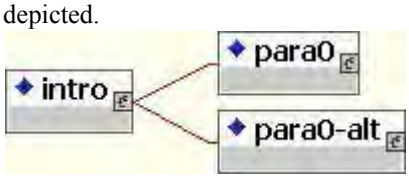

FIGURE 597. Introductory <intro> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT intro  (para0 | para0-alt)>
<!ATTLIST intro
    frame          (yes | no )          'no'
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Unique attributes:
   (1) **frame** Frame indicates to the IETM system that the author's intends a frame break at this element.
e. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

49.1.4.16 Lubricant ***<lubricant>***. The element ***<lubricant>*** identifies a lubricant within text, primarily within a lubrication work package.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.

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b. DTD fragment:

```
<!ELEMENT lubricant  (#PCDATA)>
<!ATTLIST lubricant

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.17 Description ***<desc>***. The element ***<desc>*** provides the item description.

a. The components are:

(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***< ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).
(15) Control/Indicator – ***<ctrlind>*** (see 49.1.4.3).
(16) Control/Indicator value – ***<ctrlind-val>*** (see 49.1.4.4).
(17) DoD ammunition code – ***<dodac>*** (see 49.1.4.5).
(18) Lubricant value – ***<lubricant>*** (see 49.1.4.16).
(19) Graphic symbol – ***<symbol>*** (see 44.3.1).
(20) Torque value – ***<torque>*** (see 49.1.4.24).
(21) Voltage value – ***< voltage>*** (see 49.1.4.25).
(22) Null text – ***<null >*** (see 49.1.3.2).
(23) Changed text marker – ***<change>*** (see 49.1.3.7).

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b. DTD fragment:

```
<!ELEMENT desc  (#PCDATA | emphasis | subscrpt | supscrpt | xref |
 extref | link | help.info | indxref | term | term.def | callout |
ftnote | ftnref | ctrlind | ctrlind-val | dodac | lubricant |
 symbol | torque | voltage | null | change)*>
<!ATTLIST desc

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
```

c. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

49.1.4.18 Model number ***<modelno>***. The element ***<modelno>*** is used to mark any official model number of a piece of equipment embedded in the text stream, work package setup information, and on chapter title pages and front cover of the manual.

a. The components are:

(1) The element contains narrative text #PCDATA parsable character data.

b. DTD fragment:

```
<!ELEMENT modelno  (#PCDATA)>
```

49.1.4.19 Name ***<name>***. The element ***<name>*** is used to identify the official name of a component/assembly.

a. The components are:

(1) Parsable characters or type text. – #PCDATA
(2) Format text – ***<emphasis>*** (see 49.1.3.1).
(3) Subscript – ***<subscrpt>*** (see 49.1.3.4).
(4) Superscript – ***< supscrpt>*** (see 49.1.3.5).
(5) Cross reference – ***<xref>*** (see 46.2.2).
(6) External reference – ***<extref>*** (see 46.2.1).
(7) Enhanced linking – ***<link>*** (see 46.2.3).
(8) IETM help information – ***<help.info>*** (see 48.3.2.4).
(9) Index reference – ***< indxref>*** (see 31.4.2.2.3).
(10) Term – ***<term>*** (see 49.1.2.3.1.1).
(11) Term and definition – ***<term.def>*** (see 49.1.2.3.1).
(12) Figure callout reference – ***<callout>*** (see 46.2.4.1).
(13) Footnote – ***<ftnote>*** (see 45.1.1).
(14) Footnote reference – ***<ftnref>*** (see 45.1.2).

b. DTD fragment is graphically depicted.

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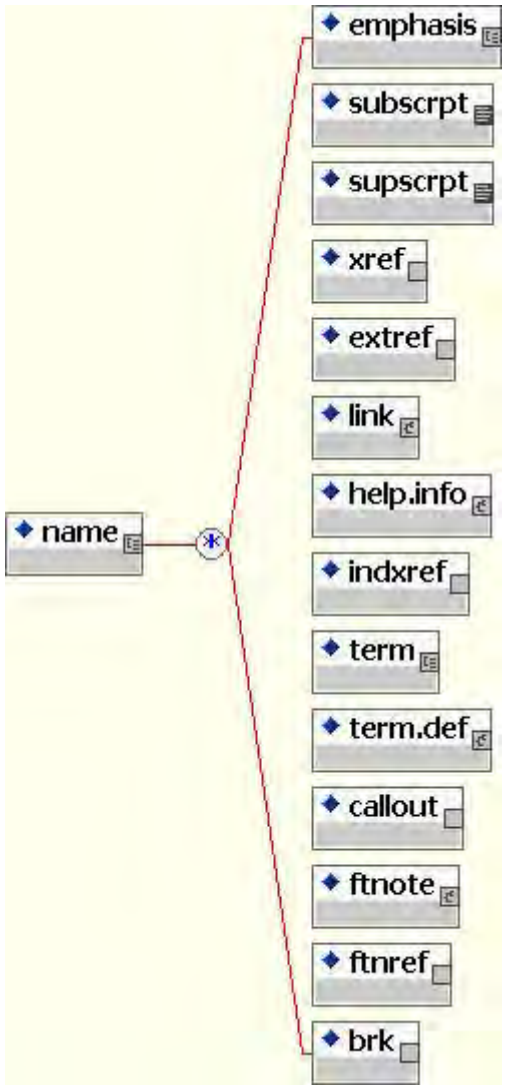

FIGURE 598. Name <name> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT name  (#PCDATA | emphasis | subscrpt | supscrpt | xref | extref | link | help.info | in
<!ATTLIST name

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

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(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.20 National stock number (NSN) ***<nsn>***. The element ***<nsn>*** is the National Stock Number and can be embedded within text stream to further identify the data. The National Stock Number of an item. Used for NSNs embedded in the text stream, on the front cover of the TM and title page of IMs (chapters) as applicable. The NSN ***<nsn>*** is also used as an identifying number in the work package setup information, and as a column in certain standard tables in supporting information such as the tool list.

a. The components are:
   (1) Federal Supply Classification ***<fsc>*** (required) (see 49.1.4.20.1) may occur but neither has to occur; either or both of the tags may occur many times with no specified sequence.
   (2) National Item Identification Number ***<niin>*** (required)(see 49.1.4.20.2).
b. DTD fragment is graphically depicted.

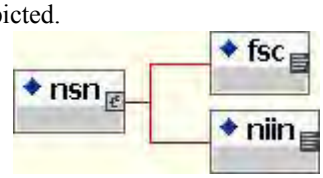

FIGURE 599. National stock number (NSN) <nsn> DTD hierarchy.

c. DTD fragment:

```
<!ELEMENT nsn  (fsc, niin)>
```

d. No attributes.

49.1.4.20.1 Federal supply classification ***<fsc>***. The element ***<fsc>*** Federal Supply Classification is the first four-digits code for the NSN. It is the classification of all items of supply used by the federal government. Each item of supply will be included in one and only one FSC. The FSC is made up of 2 two-digit numeric codes: the federal supply group and the federal supply class. Each Federal Supply Classification (FSC) code is derived from the Federal Supply Groups (FSG). The FSG is the first two digits of the FSC code.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT fsc  (#PCDATA)>
```

c. No attributes.

49.1.4.20.2 National item identification number (NIIN) ***<niin>***. The element ***<niin>*** is the National Item Identification Number (NIIN) and is a nine-digit code that identifies each item of supply. The first two numbers of the NIIN represent the National Codification Bureau Code. This code identifies the country that entered the item into the supply system. The remaining seven digits are sequentially assigned and serve to individually identify each item in the Federal Catalog System (FCS). The element contains narrative text (#PCDATA) parsable character data.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.
b. DTD fragment:

```
<!ELEMENT niin  (#PCDATA)>
```


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c. No attributes.

49.1.4.21 Part number ***<partno>***. The element ***<partno>*** is used to identify a part numbers and can be embedded within text stream to further identify the data.

a. The components are:
   (1) The element contains narrative text #PCDATA parsable character data.

b. DTD fragment:

```
<!ELEMENT partno (#PCDATA)>
<!ATTLIST partno
    idref          IDREFS          #IMPLIED
    assocfig       IDREFS          #IMPLIED>
```

c. Common attributes:
   (1) **idref** – Reference identifier(s) (optional) (see 49.2.6).
   (2) **assocfig** – Associate one or more figures (optional) (see 49.2.6).

49.1.4.22 Proponent ***<proponent>***. The element ***<proponent>*** is used to enter the name (***<name>*** see 49.1.4.19) and address (***<address>*** see 49.1.4.1) of the supporter of the activity.

a. The components are:
   (1) Name ***<name>*** (required) (see 49.1.4.19).
   (2) Address ***<address>*** (required) (see 49.1.4.1).

b. DTD fragment is graphically depicted.

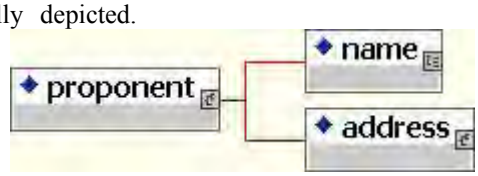

FIGURE 600. Proponent <proponent> DTD hierarchy..

c. DTD fragment:

```
<!ELEMENT proponent (name, address)>
<!ATTLIST proponent  >

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Common attributes:
   (1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
   (2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
   (3) **comment** – Reason for change information. (optional) (See 49.2.9.)
   (4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
   (5) **id** – Unique identifier. (optional) (See 49.2.6.)
   (6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
   (7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
   (8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
   (9) **security** – Security classification. (optional) (See 49.2.13.)

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49.1.4.23 Scope ***<scope>***. The element ***<scope>*** includes a brief statement of what is covered in the TM, information chapter, work package and/or procedure. This includes the type of manual, model numbers and equipment names, purpose of equipment, any special inclusions in the manual and any other pertinent information.

a. The components are:
   (1) Title ***<title>*** (optional) (see 49.1.1.3)
   (2) May be followed by one or more figure(s) ***<figure>***(optional – zero or more) (see 44.1.1).
   (3) Contains paragraphs of text that may be grouped into sections or subsections.
      (a) Note ***<note>*** (optional – zero or more) (see 41.1.3).
      (b) Either General Paragraph ***<para>*** (required – one or more)(see 49.1.1.3) with an optional one or more Primary Level Titled Paragraph(S) ***<para0>*** (see 49.1.1.8) or one or more Primary Level Conditional Titled Paragraph(S) ***<para0–alt>*** (see 49.1.1.3).
      (c) Or Primary Level Titled Paragraph(s) ***<para0>*** (see 49.1.1.8) or one or more Primary Level Conditional Titled Paragraph(s) ***<para0–alt>*** (see 49.1.1.3).

b. DTD fragment is graphically depicted.

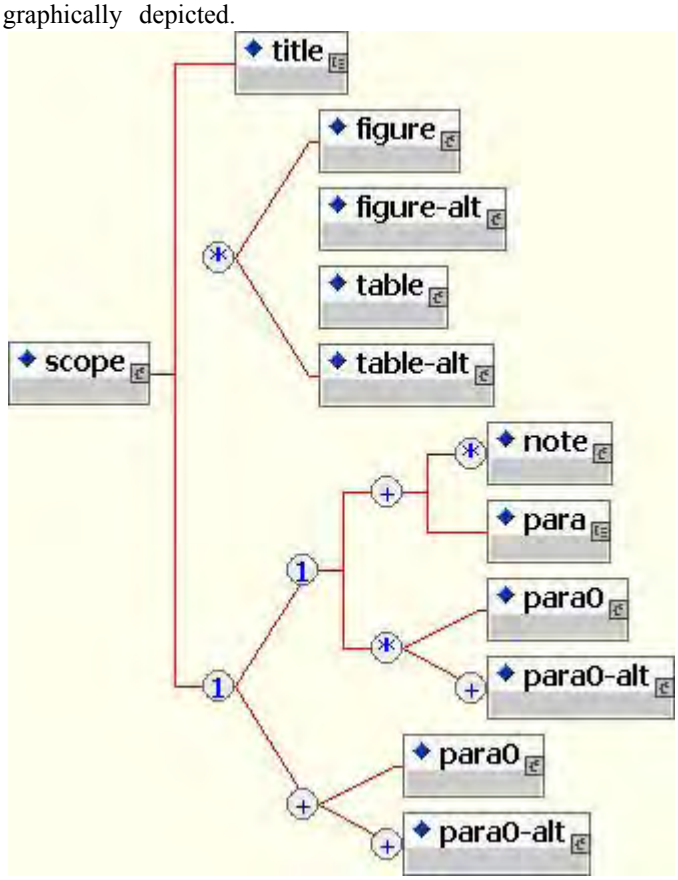

FIGURE 601. Scope <scope> DTD hierarchy.

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c. DTD fragment:

```
<!ELEMENT scope ((title, (figure | figure-alt)*, (((note*, para)+, (para0 | para0-alt+)*) | (
<!ATTLIST scope
    frame          (yes | no )          'no'
    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    id                ID                              #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

d. Unique attributes:

(1) **FRAME** - Indicates to the IETM system the author's intended Frame break.

e. Common attributes:

(1) **assocfig** – Associate one or more figures. (optional) (See 49.2.6.)
(2) **changeref** – Reference one or more change summary entries. (optional) (See 49.2.9.)
(3) **comment** – Reason for change information. (optional) (See 49.2.9.)
(4) **delchlvl** – Deletion change level. (optional) (See 49.2.9.)
(5) **id** – Unique identifier. (optional) (See 49.2.6.)
(6) **idref** – Reference one or more identifiers. (optional) (See 49.2.6.)
(7) **inschlvl** – Insertion change level. (optional) (See 49.2.9.)
(8) **skilltrk** – Training skill level. (optional) (See 49.2.2.)
(9) **security** – Security classification. (optional) (See 49.2.13.)

49.1.4.24 Torque value or limit ***<torque>***. The element ***<torque>*** is used to identify a torque value or limit embedded in the text or table entry.

a. The components are:

(1) The element contains narrative text #PCDATA parsable character data..

b. DTD fragment:

```
<!ELEMENT torque (#PCDATA)>
<!ATTLIST torque
    measurement          CDATA       #IMPLIED

    assocfig          IDREFS                          #IMPLIED
    changeref         IDREFS                          #IMPLIED
    comment           CDATA                           #IMPLIED
    delchlvl          CDATA                           #IMPLIED
    idref             IDREFS                          #IMPLIED
    inschlvl          CDATA                           #IMPLIED
    security          (uc | fouo |c | s | ts)        #IMPLIED
    skilltrk          CDATA                           #IMPLIED
>
```

c. Unique attributes:

(1) **measurement** - Units measuring the torque (lb-in).

d. Common attributes:

(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).

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(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.1.4.25 Voltage ***<voltage>***. The element ***<voltage>*** identifies a critical voltage measurement.
a. DTD fragment:

```
<!ELEMENT voltage (#PCDATA)>
<!ATTLIST voltage
    measurement          CDATA                  #IMPLIED

    assocfig          IDREFS                      #IMPLIED
    changeref         IDREFS                      #IMPLIED
    comment           CDATA                       #IMPLIED
    delchlvl          CDATA                       #IMPLIED
    idref             IDREFS                      #IMPLIED
    inschlvl          CDATA                       #IMPLIED
    security          (uc | fouo |c | s | ts)     #IMPLIED
    skilltrk          CDATA                       #IMPLIED
>
```

b. Unique attributes :
(1) **measurement** - Units measuring the torque (lb-in).
c. Common attributes :
(1) **assocfig** – Associate one or more figures (optional) (see 49.2.6).
(2) **changeref** – Change history or remarks reference (optional) (see 49.2.9).
(3) **comment** – Change information (optional) (see 49.2.9).
(4) **delchlvl** – Deletion change level (optional) (see 49.2.9).
(5) **idref** – Reference identifier(s) (optional) (see 49.2.6).
(6) **inschlvl** – Insert change level (optional) (see 49.2.9).
(7) **security** – Security classification (optional) (see 49.2.13).
(8) **skilltrk** – Skill level (optional) (see 49.2.2).

49.2 Common attributes. The following attributes are common throughout MIL–STD–2361 and are entered in the DTDs using parameter entity references.

49.2.1 Applicability. Defines common elements that may require applicability identification.
a. DTD fragment

```
    applicable        IDREFS        #IMPLIED
```

b. :
(1) **applicable** Reference to the applicable configuration(s) specified in the WP identification information. When using an IETM that can filter information, this information is not presented when not applicable to the current configuration. Presentations that do not filter information, the information will be identified by the assigned associated abbreviation to designate applicability of the information.

49.2.2 Skill level attribute. Designation of the skill level of the user at which the current element of information is aimed. A particular set of values common to all documents has not been created. Currently, the relevant values are set by contract.
a. DTD fragment:

```
skilltrk    CDATA      #IMPLIED
```

49.2.3 Frame break attribute . Identifies where possible frame breaks may occur.

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a. DTD fragment:

```
frame       (yes | no )   'no'
```

49.2.4 Graphic attributes. Attributes in this set supply common graphical properties necessary to present the image.

a. DTD fragment:

```
boardno      ENTITY              #REQUIRED
reprowid      CDATA               #IMPLIED
reprodep      CDATA               #IMPLIED
hscale        CDATA               #IMPLIED
vscale        CDATA               #IMPLIED
scalefit      (yes | no )         #IMPLIED
alt            CDATA               #IMPLIE
unitmeasure   (mm | cm | px
                | in | pt | pi )   'in'
```

b. Attributes :

(1) **boardno** - Board number specifies the name of the entity containing the external graphic file.
(2) **reprowid** - Repro Width specifies the repro area width.
(3) **reprodep** - Repro Depth specifies the repro area depth.
(4) **hscale** - Horizontal scale specifies the horizontal scaling factor. The number 100 is unscaled graphic.
(5) **vscale** - Vertical scaling specifies the vertical scaling factor. The number 100 is unscaled graphic.
(6) **scalefit** - Scale fit specifies the characteristic that allows the graphic to be scaled as needed to fit the size of the reproduction area, when attribute value is non-zero.
(7) **hplace** - Horizontal Placement specifies the horizontal placement in the available repro area. The position is flushed left, flushed right, centered or none.
(8) **alt** - Narrative to identify the graphic.
(9) **unitmeasure** - Unit Measure defines the graphic dimensions unit of measure.

49.2.5 Nuclear hardness and electrostatic discharge markings . The attributes are for general use for any XML element. Marking attributes which specify a task or steps in a procedure relate to establishing nuclear hardness or could damage electrostatic discharge sensitive parts.

a. DTD fragment:

```
hcp     (yes | no )   'no'
esd     (yes | no )   'no'
```

b. Description for:

(1) **hcp** - Marks the task or a step in a procedure relating or contributing to establishing nuclear hardness.
(2) **esd** - Marks a task or a step in a procedure relating to handling or maintenance actions which could damage electrostatic sensitive parts.

49.2.6 References with ID attribute set . Attributes in this set supply identifiers for the current element and references to other element's identifiers.

a. DTD fragment:

```
id        ID       #IMPLIED
idref     IDREFS   #IMPLIED
assocfig IDREFS   #IMPLIED
```

b. Common attributes:

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(1) **id** - An identifier of the element which is assigned at origination and which remains unchanged as the document is revised or updated even though the automatically assigned enumeration or manually-assigned "labels" change (in some cases many times). The value of the "ID" is used when making references to the element from other portions of the document. If no ID is given, none will be maintained and the element can then not be cross-referenced by means of an IDREF.
(2) **idref** - A reference to an identifier(s). The use of this attribute is specified in the composition system as it has no implied or default use.
(3) **assocfig** - A reference to a figure(s) associated with the current element.

49.2.7 Information module resource values . These attributes specify format and content characteristics that apply to the information module as a whole. The attributes are not inherited from one IM to another; however, the attributes at the IM level override the same attributes at the TM level.

a. DTD fragment:

```
chap-toc     (yes | no )        'yes'
revno        CDATA              #REQUIRED
frame        (yes | no )        'yes'
chngno        CDATA               #REQUIRED
tocentry     (0 | 1 | 2)        '1'
```

b. Description for:
(1) **chap-toc** - Specifies whether the chapter includes a table of chapter contents on the chapter title page; the style sheet for the Information Module specifies what contents are extracted to this TOC. A yes value indicates that a TOC should be extracted and printed.
(2) **revno** - Revision number is the IM revision number.
(3) **frame** - Frame indicates to the IETM system the author's intended frame break.
(4) **chngno** - Change number is the number level of the information module.
(5) **tocentry** - Table of Contents Entry defines the indenture level in the TOC. When the level is zero, no entry in the TOC is used.

49.2.8 Maintenance level attribute. Maintenance level attribute values is used for the separation of maintenance activities or functions in the U.S. Army according to the required skills.

a. DTD fragment:

```
maintlevel field | sustain | depot | operator | gensup | dirsup
| unitlvl | avum-avim | inter | tmlvls
```

b. Description for:
(1) "DEPOT" – Applies to depot maintenance level.
(2) "OPERATOR" – Applies to operator maintenance level.
(3) "GENSUP" – Applies to general support (GS) maintenance level.
(4) "DIRSUP" – Applies to direct support (DS) maintenance level.
(5) "UNITLVL" – Applies to unit maintenance level.
(6) "INTER" – Applies to intermediate (DS/GS) maintenance level.
(7) "AVUM-AVIM" – Applies to aviation unit (AVUM) and intermediate (AVIM) maintenance level.
(8) "TMLVLS" – Applies to all maintenance levels.

49.2.9 Change level attributes . Set identifies change history, reason for change, and any additional comment for major level element.

a. DTD fragment:

```
inschlvl          CDATA          #IMPLIED
delchlvl          CDATA          #IMPLIED
comment           CDATA          #IMPLIED
changeref         IDREFS         #IMPLIED
```


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b. Attributes:
   (1) **changeref** – Change Reference is used for the history or remarks reference. (optional)
   (2) **comment** – Comment is used for additional information or comments. (optional)
   (3) **delchlvl** – Deletion change level is used to delete the change level. (optional)
   (4) **inschlvl** – Insert change level is used to insert the change level. (optional)

49.2.10 Quality assurance attribute. Depot and aviation maintenance procedures which have a major quality assurance effect should be identified by the attribute **QA** at the step level.

a. DTD fragment:

```
qa  (yes | no )  'no'
```

b. Description for:
   (1) **qa** - Specifies whether or not the step in the procedure has a major quality assurance effect; a non-zero value indicates that it does.

49.2.11 Referencing attributes. Attributes in this set supply identifiers for the current element and references to other element's identifiers.

a. DTD fragment :

```
idref      IDREFS   #IMPLIED
assocfig   IDREFS   #IMPLIED
```

b. Definitions for:
   (1) **idref** - A reference to an identifier(s). The use of this attribute is specified in the composition system as it has no implied or default use.
   (2) **assocfig** - A reference to a figure(s) associated with the current element.

49.2.12 Link reference type attribute. Define the link element information type being targeted. The link types are Technical Manual (tm), Supply Catalog (sc), Work Package (wp), Test Procedure (test), Task (task), Step Level 1 (step1), Paragraph (paragraph), Figure (figure), Table (table), Part Item (part), External Document (document), Initial Setup Item (setup.item), IETM or Help Frame (frame.help), Application Program (program), Multimedia (multimedia), Hotspot (hotspot), and Page Number (pageno).

a. DTD fragment :

```
tm | sc | wp | test | task | step1 | paragraph |
figure | table | part | document | setup.item |
frame.help | program | multimedia | hotspot
```

49.2.13 Security . The attributes are for general use for any XML element. The attribute defines security classification for the element and is inherent to any children to the element.

a. DTD fragment:

```
security (uc | fouo |c | s | ts)  #IMPLIED
```

b. Definitions for
   (1) **security** - Specifies the security classification of the element. If no value is entered the implied value is unclassified.
      (a) "UC" - Indicates the element is unclassified.
      (b) "FOUO" - Indicates the element is for official use only.
      (c) "C" - Indicates the element is confidential.
      (d) "S" - Indicates the element is secret.
      (e) "TS" - Indicates the element is top secret.

49.2.14 Initial setup attribute set. Defines the required ID for each initial setup item. The concept is to use only the data identified in setup and cross-reference to the initial setup for the value. This is to increase reusability when a work package is shared with multiple systems. When using a work package

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in different weapon system exchange the initial setup data for the weapon system, but maintaining the same identifiers from the previous system.

a. DTD fragment:

```
id          ID      #IMPLIED
applicable  IDREFS  #IMPLIED">
```

b. Definition for:

(1) **id** An identifier of the element which is assigned at origination and which remains unchanged as the document is revised or updated even though the automatically assigned enumeration or manually-assigned "labels" change (in some cases many times). The value of the "id" is used when making references to the element from other portions of the document. if no ID is given, none will be maintained and the element can then not be cross-referenced by means of an IDREF on another element or with ***<xref>***.

(2) **applicable** Reference to the applicable configuration(s) specified in the WP identification information. When using an IETM that can filter information, this information is not presented when not applicable to the current configuration. Presentations that do not filter information, the information will be identified by the assigned associated abbreviation to designate applicability of the information.

49.2.15 Standard information attributes. Defines common attributes associated to a standard table or information.

a. DTD fragment

```
applicable   IDREFS                   #IMPLIED
tocentry    (0 | 1 | 2 | 3 | 4 | 5)  '1'
```

b. Description for:

(1) **applicable** Reference to the applicable configuration(s) specified in the WP identification information. When using an IETM that can filter information, this information is not presented when not applicable to the current configuration. Presentations that do not filter information, the information will be identified by the assigned associated abbreviation to designate applicability of the information.

(2) **tocentry** - Include table information in the table of contents. Zero to five reference levels. Default is at one.

49.2.16 Task attributes. Defines common attributes for all tasks.

a. DTD fragment:

```
frame            (yes | no )                'yes'
tocentry         (0 | 3 | 4 | 5)           '0'
date-time-stamp  (date | time | date-time) #IMPLIED
```

b. Description for:

(1) **frame** - Set a frame break.

(2) **tocentry** - Include table information in the table of contents.

(3) **date-time-stamp** - To indicate a date-time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp.

49.2.17 Tracking attributes. Identifies tracking of document/work package creation, modifications, and versions occurring in more than one document. These attributes may be used with any element type that references this attributes in the document type declaration.

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a. DTD fragment:

```
fgc          CDATA        #IMPLIED
lsa-id       CDATA        #IMPLIED
wpseq        CDATA        #IMPLIED
insertwp     CDATA        #IMPLIED
deletewp     (yes | no )   'no'
```

b. Description for:
   (1) **fgc** - Specifies the functional group code that applies to the subject of the element.
   (2) **changelvl** - Specifies the change level.
   (3) **lsa-id** - Specifies the identification of the subject of the element in logistic support analysis applying to the equipment covered in the TM.
   (4) **insertwp** - Specifies the new work package since last TM revision. The work package is added after the last inserted work package or between two existing work packages in a revision. After a revision is completed, the attribute is reset to ’0’and is resequenced in the correct order
   (5) **deletewp** - Specifies the work package has been deleted. The attribute is used for a place holder for the composition to determine the work package sequence number. The default value is '0'. The IDREF to reference is the deleted work package. The reference is used for a placeholder for output sequence number, until a revision is performed. After a revision is completed the work package is deleted from the assembly.

49.2.18 Work package body attributes . Any of the attributes in the associated attribute set may be used with this element/attribute.

a. DTD fragment :

```
wpno              ID                           #REQUIRED
crewmember        CDATA                        #IMPLIED
tocentry          (2 | 3 | 4 | 5)             '2'
date-time-stamp   (date | time | date-time)    #IMPLIED
frame             (yes | no )                  'yes'
```

b. Description for :
   (1) **wpno** - A unique work package number assigned by the contracting activity at the time the work package is first created. This number remains the same when the work packages reused.
   (2) **crewmember** - The crew member specifically assigned to the work package.
   (3) **tocentry** - Include table information in the table of contents.
   (4) **date-time-stamp** - To indicate a date-time stamp for the task when completed. The author indicates date only, time only or date and time or blank for no time stamp.
   (5) **frame** - Set a frame break.

49.2.19 Work package level . The attributes are for the separation of maintenance activities or functions in the U.S. Army according to the required skills.

a. DTD fragment:

```
level   tm | sc | wp | test | task | step1 | paragraph |
figure | table | part | document | setup.item |
frame.help | program | multimedia | hotspot
     #REQUIRED”>
```

b. Description for:
   (1) **level** - The maintenance level of the work package.
      (a) “DEPOT” – Applies to depot maintenance level.
      (b) “OPERATOR” – Applies to operator maintenance level.
      (c) “GENSUP” – Applies to general support (GS) maintenance level.
      (d) “DIRSUP” – Applies to direct support (DS) maintenance level.

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(e) “UNITLVL” – Applies to unit maintenance level.
(f) “INTER” – Applies to intermediate (DS/GS) maintenance level.
(g) “AVUM-AVIM” – Applies to aviation unit (AVUM) and intermediate (AVIM) maintenance level.
(h) “TMLVLS” – Applies to all maintenance levels.

:

49.2.20 Joint publications for work package resource value attributes. These attributes are used to specify if the work package is for one or more services in a joint publications. The entity consists of the following optional attributes:

a. DTD fragment:

```
army      (yes | no )  'no'
airforce  (yes | no )  'no'
navy      (yes | no )  'no'
marines   (yes | no )  'no'
```

b. Description for:
a. **army** - United States Army
b. **airforce** - United States Airforce
c. **navy** - United States Navy
d. **marines** - United States Marines

49.2.21 Yes or no attribute values. Yes or no attribute values denotes the implied value of the attribute to be either yes or no by logical necessity.

a. DTD fragment :

```
yesorno "(yes | no) #IMPLIED">
```

49.2.22 Yes default value. The “yes” value denotes the value of the attribute to be either yes or no with the attribute default as “yes”.

a. DTD fragmen:

```
yes (yes | no)  'yes'
```

50 BOILERPLATES.

Boilerplates are a type of replacement text known as general entities in MIL–STD–2361. By using boilerplates, authors can reduce text entry time and errors. MIL–STD–2361 uses general entities (see 50.2) as boilerplates to insert standard statements that MIL–STD–40051–1/–2 requires in numerous TMs and appears as the exact wording.

50.1 Boilerplate entities. Boilerplates that are available in MIL–STD–2361 can be a single entity or an entity containing nested entities providing user specific information. Boilerplate text is a simple replacement for parsed text contained in the entity. Boilerplates will parse when inserted correctly with the markup of the TM.

50.2 Single boilerplate entity. A general entity is used for inserting a standard statement. The boilerplate text for this statement is parsed text found between the single or double quotes (**' or "**) and the ending quote. The distribution statement A ***<a.statement>*** shown below is a single general entity.

a. The boilerplate general entity is comprised of the following components.
(1) Starts with a Markup Declaration Open (MDO) “**<!**”
(2) Followed by the XML reserved word “**ENTITY**”
(3) Followed the predefined in MIL–STD–2361 entity name (see 50.6) with at least one space before and after the entity name.

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(4) Followed by the boilerplate text.
(a) The boilerplate text or the replacement text always begins with a double quote (") or singe quote (') and ends with the same quote marking. As note, the quote marking cannot be used within the boilerplate text (except as character entities) (see example in paragraph 50.3). When starting quote marks are used within boilerplate text, the general entity will be invalid XML (will not parse) and the XML application will not recognize the general entity containing the boilerplate text.
(b) The boilerplate can contain just narrative text or the text can be surrounded by the appropriate elements.
(5) Ends with the Markup Declaration Close (MDC) ">".

b. A single general entity declaration is shown in the example below for the distribution statement A.

```
<!ENTITY notices.dist.a.statement "
<a.statement>
<title>DISTRIBUTION STATEMENT A.
</title>
<text>Approved for public release; distribution is unlimited.
</text>
</a.statement>">
```

c. When the general entity is referenced in a document, the general entity name is preceded by the ampersand sign (&) and followed by a semi-colon (;) as shown in the example below.

```
&notices.dist.a.statement;
```

50.3 Nested entities. General entities may be nested. In MIL-STD-2361, this concept is used to allow for boilerplate customization for a specific document and/or local revision. In the example below, the nested entities ***&short.end.item.name;*** and ***&ginfowp.wrntyref-time;*** reflect the short item name and the warranty period as mileage or time frame in the entity warranty information reference statement ***&ginfowp.wrntyref;***. Nested entities may also contain text surrounded by tags. Note that the boilerplate text starts with a single quote (') and ends with a ('). However, within the boilerplate text there are tags such as paragraph ***<para hcp="no" esd="no">*** that contain attributes surrounded with double quote ("). This allows the general entity to be parsed.

a. Shown below is an example of the nested entity boilerplate warranty information ***&ginfowp.wrntyref;*** that has boilerplate text surrounded with tags.

```
<!ENTITY short.end.item.name "M198 Howitzer"> <!ENTITY
ginfowp.wrntyref-time "two years"> <!ENTITY ginfowp.wrntyref '
<title>WARRANTY INFORMATION
</title>
<para>The &short.end.item.name; is warranted for &ginfowp.wrntyref-time;.
The warranty starts on the date found in block 23 of
<extref docno="DA Form 2408-9" posttext=", Equipment Control Record"/>. Report all
defects to your supervisor, who will take appropriate action.
</para>'>
```

b. Shown below is the formatted output for the boilerplate warranty information ***&ginfowp.wrntyref;***. The underline part shown below is the nested entities.**WARRANTY INFORMATION**The M198 Howitzer is warranted for two years. The warranty starts on the date found in block 23 of DA Form 2408–9, Equipment Control Record. Report all defects to your supervisor, who will take appropriate action.

50.4 Inserting a general entity. A boilerplate may be used for inserting a standard statement such as the distribution statement ***<a.statement>*** (see example in 50.2) in a technical manual or used to insert the table header for a standard information table as defined in MIL–STD–40051–1/–2. Some work packages require standard information shown as tables. In the Supporting Information chapter, the Additional Authorization List work package ***<aalwp>*** consists mainly of standard paragraphs, which a boilerplates are used, to insert the text to provide the standard text for the work package. The ***<aalwp>*** also contains the Additional Authorization List (AAL) ***<aal>*** standard information, which can be displayed as a table. When displayed as a table, the AAL standard heading references the general entity ***&aalwp.aal.std-header;*** to populate

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the table header. Below are tag examples of the Additional Authorization List work package ***<aalwp>*** using boilerplates and without using boilerplates.

a. Example of the XML markup for an Additional Authorization List (AAL) work package using the boilerplates ***&aalwp.intro;*** (within ***&aalwp.intro;*** the revisable general entity ***&short.end.item.name;*** needs to be changed) for AAL introduction for ***<aalwp>*** and AAL standard information table header ***&aalwp.aal.std-header;*** for ***<aal>***.

```
<aalwp airforce="no" army="no" deletewp="no" frame="yes" marines="no"
navy="no"tocentry="2" wpno="GXXXXX-X-XXXX-XXX">
<wpidinfo>
<maintlvl level="avum-avim"/>
<title>Additional Authorization List (AAL) work package
</title>
</wpidinfo> &aalwp.intro;
<aal>
<title>Additional authorization title (AAL)
</title> &aalwp.aal.std-header;
<aal-entry>
<nsn>
<fsc>6665
</fsc>
<niin>01-105-5623
</niin>
</nsn>
<dcpno>
<name>ALARM
</name>
<desc>CHEMICAL AGENT
</desc>
<partno>8762101
</partno>
<cageno>19200
</cageno>
</dcpno>
<uoc>
</uoc>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
<aal-entry>
<nsn>
<fsc>1240
</fsc>
<niin>01-207-5787
</niin>
</nsn>
<dcpno>
<name>BINOCULARS
</name>
<desc>MOD,CON M22
</desc>
<partno>9370122
</partno>
<cageno>19200
</cageno>
```


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```
</dcpno>
<uoc>
</uoc>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
<aal-entry>
<nsn>
<fsc>2590
</fsc>
<niin>01-148-7961
</niin>
</nsn>
<dcpno>
<name>CABLE  KIT
</name>
<desc>SPECIAL  PURPOSE
</desc>
<partno>223592-200
</partno>
<cageno>19200
</cageno>
</dcpno>
<uoc>
</uoc>
<ui>EA
</ui>
<qty>2
</qty>
</aal-entry>
</aal>
</aalwp>
```

b. Example of the XML markup for an Additional Authorization List (AAL) work package without the boilerplates ***&aalwp.intro;*** and ***&aalwp.aal.std-header;***.

```
<aalwp  airforce="no"  army="no"  deletewp="no"  frame="yes"  marines="no"
navy="no"  tocentry="2"  wpno="GXXXXX-X-XXXX-XXX">
<wpidinfo>
<maintlvl  level="avum-avim"/>
<title>Additional Authorization List (AAL) work package
</title>
</wpidinfo>
<intro  frame="no">
<para0  hcp="no"  esd="no">
<title>INTRODUCTION
</title>
<subpara1  hcp="no"  esd="no">
<title>Scope
</title>
<para  hcp="no"  esd="no">This work package lists additional items you
are authorized for the support of the NBCRS FOXM93A1.
</para>
</subpara1>
<subpara1  hcp="no"  esd="no">
```


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<title>General
</title>
<para hcp="no" esd="no">This list identifies items that do not have to accompany the NBCRS FOXM93A1. and that do not have to be turned in with it. These items are all authorized to you by CTA, MTOE, TDA, or JTA.
</para>
</subpara1>
<subpara1 hcp="no" esd="no">
<title>Explanation of Columns in the AAL
</title>
<para hcp="no" esd="no">Column (1) National Stock Number (NSN). Identifies the stock number of the item to be used for requisitioning purposes.
</para>
<para hcp="no" esd="no">Column (2) Description, Part Number/(CAGEC). Identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The last line below the description is the part number and the Commercial and Government Entity Code (CAGEC) (in parentheses).
</para>
<para hcp="no" esd="no">Column (3) Usable On Code. When applicable, gives you a code if the item you need is not the same for different models of equipment.
</para>
<para hcp="no" esd="no">Column (4) U/I. Unit of Issue (U/I) indicates the physical measurement or count of the item as issued per the National Stock Number shown in Column (1).
</para>
<para hcp="no" esd="no">Column (5) Qty Recm. Indicates the quantity recommended.
</para>
</subpara1>
</para0>
</intro>
<aal>
<title>Additional authorization title (AAL)
</title>
<std.table.header cols="5" rowsep="1">
<std.table.row>
<std.table.entry colnum="1">(1)
</std.table.entry>
<std.table.entry colnum="2">(2)
</std.table.entry>
<std.table.entry colnum="3">(3)
</std.table.entry>
<std.table.entry colnum="4">(4)
</std.table.entry>
<std.table.entry colnum="5">(5)
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry colnum="1">NATIONAL STOCK
</std.table.entry>
<std.table.entry colnum="2">
</std.table.entry>
<std.table.entry colnum="3">USABLE
</std.table.entry>

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<std.table.entry colnum="4">
</std.table.entry>
<std.table.entry colnum="5">QTY
</std.table.entry>
</std.table.row>
<std.table.row>
<std.table.entry colnum="1">NUMBER (NSN)
</std.table.entry>
<std.table.entry colnum="2">DESCRIPTION, PART NUMBER/(CAGEC)
</std.table.entry>
<std.table.entry colnum="3">ON CODE
</std.table.entry>
<std.table.entry colnum="4">U/I
</std.table.entry>
<std.table.entry colnum="5">RECM
</std.table.entry>
</std.table.row>
</std.table.header>
<aal-entry>
<nsn>
<fsc>6665
</fsc>
<niin>01-105-5623
</niin>
</nsn>
<dcpno>
<name>ALARM
</name>
<desc>CHEMICAL AGENT
</desc>
<partno>8762101
</partno>
<cageno>19200
</cageno>
</dcpno>
<uoc>
</uoc>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
<aal-entry>
<nsn>
<fsc>1240
</fsc>
<niin>01-207-5787
</niin>
</nsn>
<dcpno>
<name>BINOCULARS
</name>
<desc>MOD,CON M22
</desc>
<partno>9370122
</partno>

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```
<cageno>19200
</cageno>
</dcpno>
<uoc>
</uoc>
<ui>EA
</ui>
<qty>1
</qty>
</aal-entry>
<aal-entry>
<nsn>
<fsc>2590
</fsc>
<niin>01-148-7961
</niin>
</nsn>
<dcpno>
<name>CABLE  KIT
</name>
<desc>SPECIAL  PURPOSE
</desc>
<partno>223592-200
</partno>
<cageno>19200
</cageno>
</dcpno>
<uoc>
</uoc>
<ui>EA
</ui>
<qty>2
</qty>
</aal-entry>
</aal>
</aalwp>
```

50.5 MIL-STD-2361 boilerplates. MIL–STD–2361 DTD includes nine (9) boilerplate external entity files. The general entities contain the required standard verbatim text contained in MIL–STD–40051–1/–2 to develop specific work packages in a TM. MIL-STD-2361 DTD provides the user with the predefined boilerplates for developing work packages from the nine (9) boilerplate entities with the content. The Selection (see 50.5.1.2.1) and the Editable (see 50.5.1.2.2) entities are the only two (2) revisable entity files of the nine (9) entity files (see 50.3 about nested entities). The remaining seven (7) boilerplate entities contain boilerplate text that uses entities from the two revisable entity files to provide TM specific information (i.e., short end item name). Some user specific information is known during the TM initial start and some is edited later during the TM development. User specific information is usually a nested entity (see example in 50.3) within the boilerplate statement, which is modified in the Editable entity file. There are two methods (automated and manual) available to edit the revisable entity files, which the XML application requires to make the boilerplates correlated with the TM. An automated application (see example in 50.5.1) is available on the ASRL that prompts the writer for the required user specific information. The manual method (see example in 50.5.1.2) is editing and updates the two revisable entity files with the inputted information (details about each revisable entity is provided as comments in the entity file). The nine entity files are identified below.

a. Revisable entity files.
   (1) Selection. ***<!ENTITY % selection_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 Selection Boilerplate REV 3.16 20070115//EN" "boilerplate/selectboil.ent">***.

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(2) Editable. ***<!ENTITY % editable_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 Editable Boilerplate REV REV 3.16 20070115//EN" "boilerplate/editboil.ent">***.

b. Non-revisable entity files.

(1) Production Front Matter Non-Editable Boilerplates. ***<!ENTITY % prod_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 Production Boilerplate REV 3.16 20070115//EN" "boilerplate/prodboil.ent">***.

(2) General Information Chapter Non-Editable Boilerplates. ***<!ENTITY % gim_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 GIM Boilerplate REV 3.16 20070115//EN" "boilerplate/gimboil.ent">***.

(3) Operator's Procedures Information Chapter Non-Editable Boilerplates. ***<!ENTITY % opim_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 OPIM Boilerplate REV 3.16 20070115//EN" "boilerplate/opimboil.ent">***

(4) Maintenance Information Chapter Non-Editable Boilerplates. ***<!ENTITY % tim_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 TIM Boilerplate REV 3.16 20070115//EN" "boilerplate/timboil.ent">***.

(5) Maintenance Information Chapter Non-Editable Boilerplates. ***<!ENTITY % mim_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 MIM Boilerplate REV 3.16 20070115//EN" "boilerplate/mimboil.ent">***.

(6) Parts Information Chapter Non-Editable Boilerplates. ***<!ENTITY % pim_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 PIM Boilerplate REV 3.16 20070115//EN" "boilerplate/pimboil.ent>***.

(7) Supporting Information Chapter Non-Editable Boilerplates. ***<!ENTITY % sim_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 SIM Boilerplate REV 3.16 20070115//EN" "boilerplate/simboil.ent>***.

50.5.1 Revisable boilerplate entities. The automated application and the manual editing methods are used to revise the boilerplate entities. The two entity files (with entity declaration) are listed below.

a. The Selection entity file is located in the subdirectory “boilerplate” and entity file “selectboil.ent”. The XML entity declaration is ***<!ENTITY % selection_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 Selection Boilerplate REV 3.16 20070115//EN" "boilerplate/selectboil.ent">***.

b. The Editable entity file is located in the subdirectory “boilerplate” and entity file “editboil.ent”. The XML entity declaration is ***<!ENTITY % editable_boilerplate PUBLIC "-//USA-DOD//ENTITIES MIL-STD-2361 Editable Boilerplate REV 3.16 20070115//EN" "boilerplate/editboil.ent">***.

50.5.1.1 Automated boilerplate application. The automated application (downloadable from the ASRL) prompts the writer for the required information to select a TM and edits the data in the Selection and Editable entity files for the equipment specific information needed for the TM. Refer to theTM Boilerplate Application Software Users Guide for assistance on how to use the automated boilerplate application. The TM Boilerplate Application Software Users Guide can be downloaded from the ASRL.

50.5.1.2 Manually editing boilerplates. The procedure to edit the the revisable boilerplate entities requires the Selection (see 50.5.1.2.1) and (see 50.5.1.2.2) Editable entity files to be updated in a word processor that can handle plain text (non-formatted file), which most can achieve. to provide the specific TM (see 50.5.1.2.1). After selecting the TM type to develop, the author then edits the data in the Editable entity file for equipment specific information needed for the TM (see 50.5.1.2.2).

50.5.1.2.1 Edit Selection boilerplate text . Selection boilerplate text provides the means to either turned on (INCLUDE) or off (IGNORE) standard text in the non-revisable entity files, which depends on the TM being developed. Each entity group provides in the documentation comment, selectable choices and instructions on what to change.

a. An example is provided below of selecting the correct entities in the Selection Boilerplate file to develop a page-based TM. The TM is an unclassified, multiple service TM developed for the Army and Marines. The TM includes a RPSTL containing the National Stock Number (NSN), Part Number (PN) and Reference designator indexes. The TM includes a five-level MAC and uses Method A

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for displaying COEI and BII Standard Information. The entities that are used to develop the TM need to contain the value "INCLUDE".

(1) First select if the TM is either a page-based or frame-based TM. If a page-base is required the ***%page-base;*** should display "INCLUDE" and the ***%frame-base;*** should display "IGNORE".

```
<!-- DOCUMENTATION ************************************************
**************************** ** -TITLE- ** Page-Base Or Frame-Base TM
Selection ** ** -INSTRUCTION- ** -IF- Frame-base or IETM ** -THEN- **
-SET- "frame-base" to "INCLUDE" -AND- ** -SET- "page-base" to "IGNORE"
** -IF- Page-base TM ** -SET- "frame-base" to "IGNORE" -AND- ** -SET-
"page-base" to "INCLUDE" ****************************************
********************************** --> <!ENTITY % frame-base "IGNORE">
<!ENTITY % page-base "INCLUDE"> <!-- DOCUMENTATION ********************
************************************************************ ** -TITLE-
** Page-Base Or Frame-Base TM Selection ** ** -INSTRUCTION- ** -IF-
Frame-base or IETM ** -THEN- ** -SET- "frame-base" to "INCLUDE" -AND- **
-SET- "page-base" to "IGNORE" ** -IF- Page-base TM ** -SET- "frame-base"
to "IGNORE" -AND- ** -SET- "page-base" to "INCLUDE" ***************
************************************************************ -->
<!ENTITY % frame-base "IGNORE"> <!ENTITY % page-base "INCLUDE">
```

(2) Select the services to be included in the TM. If a multiple service TM is required for the Army and Marines, set ***%multi-tm;*** to display "INCLUDE", set ***%army-tm;*** to display "INCLUDE", set ***%usmc-tm;*** to display "INCLUDE" and then set the rest of the entities in the multiple service TM part to display "IGNORE". Setting the correct services to be included in the TM provides the user the correct verbatim text to be inserted in the TM for Army and Marine use.

```
<!-- DOCUMENTATION ************************************************
**************************** ** -TITLE- ** Services Included In TM
** ** -USED FOR- ** + Reporting Errors And Recommending Improvements
Statement ** + RPSTL Introduction ** ** -INSTRUCTION- ** -IF- Multiple
Service TM ** -THEN- ** -SET- multi-tm to "INCLUDE" ** -SET- single-tm
to "IGNORE" ** -ELSE- ** -SET- multi-tm to "IGNORE" ** -SET- single-tm
to "INCLUDE" ** TM includes:  ** 1.  Army ** -THEN- ** -SET- army-tm to
"INCLUDE" ** -ELSE- ** -SET- army-tm to "IGNORE" ** 2.  Navy ** -THEN- **
-SET- usn-tm to "INCLUDE" ** -ELSE- ** -SET- usn-tm to "IGNORE" ** 3.  Air
Force ** -THEN- ** -SET- usaf-tm to "INCLUDE" ** -ELSE- ** -SET- usaf-tm
to "IGNORE" ** 4.  Marines ** -THEN- ** -SET- usmc-tm to "INCLUDE" -AND-
** -SET- non-usmc-tm to "IGNORE" ** -ELSE- ** -SET- usmc-tm to "IGNORE"
-AND- ** -SET- non-usmc-tm to "INCLUDE" ** ** -EXAMPLE- ** When Army
and the Navy is included in the TM ** -SET- "multi-tm", "army-tm",
"usn-tm", and "non-usmc-tm" to "INCLUDE" -AND- ** -SET- "single-tm",
"usaf-tm" and "usmc-tm" to "IGNORE".  ** ** -SELECTABLE ENTITY LIST-
** multi-tm ** single-tm ** army-tm ** usn-tm ** usaf-tm ** usmc-tm **
non-usmc-tm ** ********************************************************
************************ --> <!ENTITY % multi-tm "INCLUDE"> <!ENTITY
% single-tm "IGNORE"> <!ENTITY % army-tm "INCLUDE"> <!ENTITY % usn-tm
"IGNORE"> <!ENTITY % usaf-tm "IGNORE"> <!ENTITY % usmc-tm "INCLUDE">
<!ENTITY % non-usmc-tm "IGNORE">
```

(3) Select if the TM is to be unclassified or classified. If the TM is to be unclassified then set ***%class-tm;*** to "IGNORE" then set ***%unclass-tm;*** to "INCLUDE".

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```
<!-- DOCUMENTATION ************************************************
*************************** ** -TITLE- ** Classified TM/IETM ** **
-USED FOR- ** Reporting Errors And Recommending Improvements Statement
** ** -INSTRUCTION- ** -IF- Classified ** -THEN- ** -SET- "unclass-tm"
to "IGNORE" ** -SET- "class-tm" to "INCLUDE" ** -ELSE- -SET- "class-tm"
to "IGNORE" ** -SET- "unclass-tm" to "INCLUDE" ** -SET- "class-tm"
to "IGNORE" ** ** -SELECTABLE ENTITY LIST- ** class-tm ** unclass-tm
*****************************************************************
*********** --> <!ENTITY % class-tm "IGNORE"> <!ENTITY % unclass-tm
"INCLUDE">
```

(4) Select if the TM is either a Preventive Maintenance Services (PMS) or a Phased Maintenance Inspection. (PMI) or it is not one of these types. If the TM is not either a PMS or PMI TM then set ***%pms_or_pmi-tm;*** to "IGNORE"

```
<!-- DOCUMENTATION ************************************************
***************************** ** -TITLE- ** Preventive Maintenance
Services (PMS) or ** Phased Maintenance Inspection (PMI) TM ** **
-INSTRUCTION- ** -IF- PMS or PMI TM ** -THEN- -SET- "pms_or_pmi-tm" to
"INCLUDE" ** -ELSE- -SET- "pms_or_pmi-tm" to "IGNORE" ** ** -SELECTABLE
ENTITY LIST- ** ammo-tm ***************************************
********************************** --> <!ENTITY % pms_or_pmi-tm
"IGNORE">
```

(5) Select if the TM is an Army Conventional And Chemical Ammunition TM. If the TM is not then set ***%ammo-tm;*** to "IGNORE".

```
<!-- DOCUMENTATION ************************************************
***************************** ** -TITLE- ** Army Conventional And
Chemical Ammunition TM ** ** -INSTRUCTION- ** -IF- Conventional and
Chemical Ammunition TM ** -THEN- -SET- "ammo-tm" to "INCLUDE" ** -ELSE-
-SET- "ammo-tm" to "IGNORE" ** ** -SELECTABLE ENTITY LIST- ** ammo-tm
*****************************************************************
********** --> <!ENTITY % ammo-tm "IGNORE">
```

(6) Select if the TM is a separate RPSTL TM or a TM that includes the RPSTL. IF the TM includes a RPSTL and the TM has a number with P then set ***%introwp.RPSTL_manual-tm;*** to "INCLUDE" and set ***%introwp.RPSTL_wp-tm;*** to "IGNORE".

```
<!-- DOCUMENTATION ************************************************
****************************** ** -TITLE- ** Separate RPSTL TM
or TM Includes RPSTL ** ** -INSTRUCTION- ** -IF- No RPSTL in TM
** -THEN- ** -SET- "introwp.RPSTL_manual-tm" to "IGNORE", ** -SET-
"introwp.RPSTL_wp-tm" to "IGNORE" ** -IF- RPSTL TM (TM Number with P)
** -THEN- ** -SET- "introwp.RPSTL_manual-tm" to "INCLUDE", ** -SET-
"introwp.RPSTL_wp-tm" to "IGNORE" ** -IF- TM includes RPSTL (TM Number
with &P) ** -ELSE- ** -SET- "introwp.RPSTL_manual-tm" to "IGNORE" **
-SET- "introwp.RPSTL_wp-tm" to "INCLUDE" ** ** -SELECTABLE ENTITY
LIST- ** RPSTL_manual-tm ** RPSTL_wp-tm ***************************
********************************************* --> <!ENTITY %
RPSTL_manual-tm "INCLUDE"> <!ENTITY % RPSTL_wp-tm "IGNORE">
```

(7) Select if the TM is either a DMWR or NMWR or neither. If the TM is neither then set ***%dmwr;*** to "IGNORE" and set ***%nmwr;*** to "IGNORE".

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```
<!-- DOCUMENTATION ************************************************
**************************** ** -TITLE- ** DMWR or NMWR Manual ** **
-INSTRUCTION- ** -IF- NOT Depot TM ** -THEN- ** -SET- "dmwr" to "IGNORE"
** -SET- "nmwr" to "IGNORE" ** -IF- Depot TM -AND- DMWR ** -THEN- **
-SET- "dmwr" to "INCLUDE" ** -SET- "nmwr" to "IGNORE" ** -IF- Depot TM
-AND- NMWR ** -THEN- ** -SET- "dmwr" to "IGNORE" ** -SET- "nmwr" to
"INCLUDE" ** ** -SELECTABLE ENTITY LIST- ** dmwr ** nmwr ***************
************************************************************** -->
<!ENTITY % dmwr "IGNORE"> <!ENTITY % nmwr "IGNORE">
```

(8) Select if there is any Usable On Code (UOC) to list. If the TM contains any UOC then set the ***%uoc-list;*** to "INCLUDE".

```
<!-- DOCUMENTATION ************************************************
**************************** ** -TITLE- ** Any Usable On Code (UOC)
to List ** ** -INSTRUCTION- ** -IF- TM contains UOC ** -THEN- ** -SET-
"uoc-list" to "INCLUDE" ** -ELSE- ** -SET- "uoc-list" to "IGNORE" **
** -SELECTABLE ENTITY LIST- ** uoc-list ****************************
************************************************ --> <!ENTITY %
uoc-list "INCLUDE">
```

(9) Select if the RPSTL Cross Reference Indexes contain one or more RPSTL Indexes. If the RPSTL includes one or more RPSTL Indexes then set ***%introwp.index;*** to "INCLUDE".

```
<!-- DOCUMENTATION ************************************************
**************************** ** -TITLE- ** RPSTL Cross Reference
Indexes ** ** -INSTRUCTION- ** -IF- RPSTL Includes One or More
RPSTL Indexes ** -THEN- -SET- "introwp.index" to "INCLUDE" ** -ELSE-
-SET- "introwp.index" to "IGNORE" ** ** -SELECTABLE ENTITY LIST- **
introwp.index ************************************************
********************** --> <!ENTITY % introwp.index "INCLUDE">
```

(10) If the RPSTL Cross Reference Indexes ***%introwp.index;*** is set to "INCLUDE" then the Cross Reference Indexes that are to be included in the TM need to be selected. If the RPSTL contains the National Stock Number (NSN), Part Number (PN) and Reference designator index then set the ***%introwp.nsn_pn_refdes-index;*** to "INCLUDE" and the others to "IGNORE".

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```
<!-- DOCUMENTATION ************************************************
**************************** ** -TITLE- ** RPSTL Cross Reference
Index Used ** ** -INSTRUCTION- ** -IF- "introwp.index=INCLUDE" (see
above) ** -THEN- ** -IF- NSN -AND- Part No.  -AND- Ref Des Indices
Used ** -THEN- ** -SET- "introwp.nsn_pn_refdes-index" to "INCLUDE"
** -SET- "introwp.nsn_pn-index" -AND- "introwp.nsn_refdes-index"
-AND- ** "introwp.nsn-index" -AND- "introwp.pn_refdes-index"
-AND- ** "introwp.pn-index" -AND- "introwp.refdes-index" to
"IGNORE" ** -IF- NSN -AND- Part No.   Indices Only Used **
-THEN- ** -SET- "introwp.nsn_pn-index" to "INCLUDE" ** -SET-
"introwp.nsn_pn_refdes-index" -AND- "introwp.nsn_refdes-index" -AND-
** "introwp.nsn-index" -AND- "introwp.pn_refdes-index" -AND- **
"introwp.pn-index" -AND- "introwp.refdes-index" to "IGNORE" ** -IF- NSN
Index Only Used ** -THEN- ** -SET- "introwp.nsn-index" to "INCLUDE" **
-SET- "introwp.nsn_pn_refdes-index" -AND- "introwp.nsn_refdes-index"
-AND- ** "introwp.nsn_pn-index" -AND- "introwp.pn_refdes-index" -AND-
** "introwp.pn-index" -AND- "introwp.refdes-index" to "IGNORE" **
-IF- Part No.  -AND- Ref Des Indices Only Used ** -THEN- ** -SET-
"introwp.pn_refdes-index" to "INCLUDE" ** -SET- "introwp.nsn_pn-index"
-AND- "introwp.nsn_refdes-index" -AND- ** "introwp.nsn-index"
-AND- "introwp.nsn_pn_refdes-index" -AND- ** "introwp.pn-index"
-AND- "introwp.refdes-index" to "IGNORE" ** -IF- Part No.   Index
Only Used ** -THEN- ** -SET- "introwp.pn-index" to "INCLUDE"
** -SET- "introwp.nsn_pn-index" -AND- "introwp.nsn_refdes-index"
-AND- ** "introwp.nsn-index" -AND- "introwp.nsn_pn_refdes-index"
-AND- ** "introwp.pn_refdes-index" -AND- "introwp.refdes-index"
to "IGNORE" ** -IF- Ref Des Index Only Used ** -THEN- ** -SET-
"introwp.refdes-index" to "INCLUDE" ** -SET- "introwp.nsn_pn-index"
-AND- "introwp.nsn_refdes-index" -AND- ** "introwp.nsn-index" -AND-
"introwp.nsn_pn_refdes-index" -AND- ** "introwp.pn_refdes-index" -AND-
"introwp.pn-index" to "IGNORE" ** ** -EXAMPLE- ** If using NSN and
Reference Designator Indexes set ** introwp.nsn_refdes-index "INCLUDE".
** ** -SELECTABLE ENTITY LIST- ** introwp.nsn_pn_refdes-index
(NSN, Part No., Reference Designator) ** introwp.nsn_pn-index
(NSN, Part No.)   ** introwp.nsn_refdes-index (NSN, Reference
Designator) ** introwp.nsn-index (NSN) ** introwp.pn_refdes-index
(Part No., Reference Designator) ** introwp.pn-index (Part No.)   **
introwp.refdes-index (Reference Designator) ************************
************************************************** --> <!ENTITY %
introwp.nsn_pn_refdes-index "INCLUDE"> <!ENTITY % introwp.nsn_pn-index
"IGNORE"> <!ENTITY % introwp.nsn_refdes-index "IGNORE"> <!ENTITY %
introwp.nsn-index "IGNORE"> <!ENTITY % introwp.pn_refdes-index "IGNORE">
<!ENTITY % introwp.pn-index "IGNORE"> <!ENTITY % introwp.refdes-index
"IGNORE">
```

(11) Select the level of the MAC Maintenance Concept. If the MAC is a two–level then set the ***%mac.2-level;*** to "INCLUDE" and the ***%mac.nonav-level;*** to "INCLUDE" and the others to "IGNORE".

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```
<!-- DOCUMENTATION ************************************************
*************************** ** -TITLE- ** MAC Maintenance Concept **
** -INSTRUCTION- ** -IF- MAC is Two-Level Maintenance ** -THEN- ** -SET-
"mac.2-level" to "INCLUDE" ** -SET- "mac.5-level" to "IGNORE" ** -SET-
"mac.nonav-level" to "INCLUDE" ** -SET- "mac.av-level" to "IGNORE" **
-IF- MAC is Five-Level (Legacy) Maintenance ** -SET- "mac.2-level" to
"IGNORE" ** -SET- "mac.5-level" to "INCLUDE" ** -SET- "mac.nonav-level"
to "INCLUDE" ** -SET- "mac.av-level" to "IGNORE" ** -IF- MAC is Aviation
Maintenance ** -SET- "mac.2-level" to "IGNORE" ** -SET- "mac.5-level" to
"IGNORE" ** -SET- "mac.nonav-level" to "IGNORE" ** -SET- "mac.av-level"
to "INCLUDE" ** ** -SELECTABLE ENTITY LIST- ** mac.2-level ** mac.5-level
** mac.nonav-level ** mac.av-level ***********************************
****************************************** --> <!ENTITY % mac.2-level
"INCLUDE"> <!ENTITY % mac.5-level "IGNORE"> <!ENTITY % mac.nonav-level
"INCLUDE"> <!ENTITY % mac.av-level "IGNORE">
```

(12) Select the method to display COEI and BII Standard Information. If Method A is to be used then set ***%coeibiiwp.method-a;*** to "INCLUDE" and ***%coeibiiwp.method-b;*** to "IGNORE".

```
<!-- DOCUMENTATION ************************************************
***************************** ** -TITLE- ** COEI and BII Standard
Information Method Used ** ** The COEI and BII has two methods to display
information Method A and Method B. ** ** ** -INSTRUCTION- ** -IF- Using
METHOD A ** -THEN- ** -SET- "coeibiiwp.method-a" to "INCLUDE" ** -SET-
"coeibiiwp.method-b" to "IGNORE" ** -ELSE- ** -SET- "coeibiiwp.method-a"
to "IGNORE" ** -SET- "coeibiiwp.method-b" to "INCLUDE" ** ** -SELECTABLE
ENTITY LIST- ** coeibiiwp.method-a ** coeibiiwp.method-b **************
************************************************************** -->
<!ENTITY % coeibiiwp.method-a "INCLUDE"> <!ENTITY % coeibiiwp.method-b
"IGNORE">
```

(13) Select the Tool Identification List Maintenance Level. If the maintenance level for the tool identification list is not a DMWR or NMWR then set ***%toolidwp.common;*** to "INCLUDE" and ***%toolidwp.dmwr-nmw;*** to "IGNORE".

```
<!-- DOCUMENTATION ************************************************
****************************** ** -TITLE- ** Tool Identification
List Maintenance Level SELECTION ** ** The Tool Identification has
different narrative for DMWR/NMWR and remaining ** maintenance levels.
** -INSTRUCTION- ** -IF- Depot (DMWR/NMWR) Maintenance Level ** -THEN-
** -SET- "toolidwp.dmwr-nmwr" to "INCLUDE" ** -SET- "toolidwp.common"
to "IGNORE" ** -ELSE- ** -SET- "toolidwp.dmwr-nmwr" to "IGNORE" **
-SET- "toolidwp.common" to "INCLUDE" ** ** -SELECTABLE ENTITY LIST-
** toolidwp.dmwr-nmwr ** toolidwp.common **************************
************************************************* --> <!ENTITY %
toolidwp.dmwr-nmwr "IGNORE"> <!ENTITY % toolidwp.common "INCLUDE">
```

50.5.1.2.2 Editable boilerplate text. Each information chapter has verbatim text, but some require user specific information. Some user specific information is known during the TM start and some will be edited later in the TM development. User specific information is usually a nested entity (see example in 50.3) within the non-revisable boilerplate text entity.

a. Example of a common editable entity ***&short.end.item.name;*** that is used in the boilerplate Reporting Equipment Improvement Recommendations (EIR) statement ***&ginfowp.eir;*** (located in the General Information entity file). Even though the entity ***&short.end.item.name;*** is found in the General Information Entity file, it is only changeable in the Editable entity file. Also, note since the Selection entity file has indicated that the TM is for both Army and Marines the additional required statement for Marines is included in the sample output.

```
<!ENTITY short.end.item.name "INSERT THE SHORT END ITEM NAME">
```

(1) Replace the text in the editable entity ***&short.end.item.name;*** with "M198 Howitzer".

```
<!ENTITY short.end.item.name "M198 Howitzer">
```


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b. Example of the boilerplate Reporting Equipment Improvement Recommendations (EIR) statement ***&ginfowp.eir;*** that would be inserted as an entity in a general information work package with the Editable entity ***&short.end.item.name;*** that was edited using "**M198 Howitzer**".

```
%!ENTITY ginfowp.eir ';
<title>REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR)
</title>
<para hcp="no" esd="no">If your &short.end.item.name; needs improvement,
let us know.  Send us an EIR. You, the user, are the only one who can
tell us what you don't like about your equipment.  Let us know why you
don't like the design or performance.  If you have Internet access, the
easiest and fastest way to report problems or suggestions is to go to
<internet show.address="yes">
<homepage protocol="https" uri="aeps.ria.army.mil/aepspublic.cfm"/> (scroll down and
choose the "Submit Quality Deficiency Report" bar).  The Internet
form lets you choose to submit an Equipment Improvement Recommendation
(EIR), a Product Quality Deficiency Report (PQDR) or a Warranty Claim
Action (WCA). You may also submit your information using a
<extref docno="SF 368" posttext="Product Quality Deficiency Report)"/>.  You
can send your SF 368 via e-mail, regular mail, or facsimile
using the addresses/facsimile numbers specified in
<extref docno="DA PAM 738-750" posttext=",Functional Users Manual for the Army
Maintenance Management System (TAMMS)"/>.  We will send you a reply.
</para>%![%usmc-tm;[;
<para hcp="no" esd="no">For Marine Corps users:  Quality deficiency
reports (QDR) shall be submitted on
<extref docno="SF 368"/> in accordance with
<extref docno="MCO 4855.10"/>.  A reply will be furnished to you.
</para>]]'
```

c. Example of a nested entity within the general entity Reporting Equipment Improvement Recommendations (EIR).

REPORTING EQUIPMENT IMPROVEMENT RECOMMENDATIONS (EIR)

If your M198 Howitzer needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you don't like about your equipment. Let us know why you don't like the design or performance. If you have Internet access, the easiest and fastest way to report problems or suggestions is to go to (scroll down and choose the "Submit Quality Deficiency Report" bar). The Internet form lets you choose to submit an Equipment Improvement Recommendation (EIR), a Product Quality Deficiency Report (PQDR) or a Warranty Claim Action (WCA). You may also submit your information using a SF 368 (Product Quality Deficiency Report). You can send your SF 368 via e-mail, regular mail, or facsimile using the addresses/facsimile numbers specified in DA PAM738-750, Functional Users Manual for the Army Maintenance Management System (TAMMS). We will send you a reply.

For Marine Corps users: Quality deficiency reports (QDR) are to be submitted on SF 368 in accordance with MCO 4855.10. A reply will be furnished to you.

FIGURE 602. Example – a nested entity within a general entity.

50.6 List of boilerplates. The following are lists of entities that are found in MIL–STD–40051–1/–2. The entities are listed in a table under the information chapter they pertain to. The selection list table contains only the entity's name and the entity. The seven other tables provide the entity's name, entity, and where it is referenced in MIL–STD–40051–1/–2. The eight (8) lists consist of the following:

a. Selection list (see 50.6.1)
b. Editable list (see 50.6.2)
c. General information list (see 50.6.4)
d. Operator procedure information list (see 50.6.5)
e. Troubleshooting information list (see 50.6.6)

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f. Maintenance information list (see 50.6.7)
g. Parts information list (see 50.6.8)
h. Supporting information list (see 50.6.9)

50.6.1 Selection entity list. Table XLIII lists the boilerplates that are parameter entities that are found in the Selection entity file listed by entity title and entity name (see 50.5.1.2.1 information on selection boilerplates).

TABLE XLIII. Selection list.

Entity Title	Entity Name
Frame-base TM or IETM	***%frame-base;***
Page-base TM	***%page-base;***
Multiple Service TM	***%multi-tm;***
Single Service TM	***%single-tm;***
U. S. Army TM	***%army-tm;***
U. S. Navy TM	***%usn-tm;***
U. S. Air Force TM	***%usaf-tm;***
U. S. Marines TM	***%usmc-tm;***
Non U. S. Marines TM	***%non-usmc-tm;***
Classified TM/IETM	***%class-tm;***
Unclassified TM/IETM	***%unclass-tm;***
Preventive Maintenance Services (PMS) or Phased Maintenance Inspection (PMI) TM	***%pms_or_pmi-tm;***
Army Conventional and Chemical Ammunition TM	***%ammo-tm;***
TM Number with P	***%RPSTL_manual-tm;***
TM Number with &P	***%RPSTL_wp-tm;***
DMWR Manual	***%dmwr;***
NMWR Manual	***%nmwr;***
Usable On Code (UOC) to List	***%uoc-list;***
RPSTL Cross Reference Indexes	***%introwp.index;***
NSN, Part Number, and Reference Designator Cross Reference Indexes Only Used	***%introwp.nsn_pn_refdes-index;***
NSN and Part Number Cross Reference Indexes Only Used	***%introwp.nsn_pn-index;***
NSN and Reference Designator Cross Reference Indexes Only Used	***%introwp.nsn_refdes-index;***
NSN Cross Reference Indexes Only Used	***%introwp.nsn-index;***
Part Number and Reference Designator Cross Reference Indexes Only Used	***%introwp.pn_refdes-index;***
Part Number Cross Reference Indexes Only Used	***%introwp.pn-index;***
Reference Designator Cross Reference Indexes Only Used	***%introwp.refdes-index;***
MAC 2 Level Maintenance	***%mac.2-level;***
MAC 5 Level Maintenance	***%mac.5-level;***
Non Aviation MAC	***%mac.nonav-level;***
Aviation MAC	***%mac.av-level;***
COEI and BII Standard Information Method A	***%coeibiiwp.method-a;***

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Table XLIII Selection list – Continued.

Entity Title	Entity Name
COEI and BII Standard Information Method B	***%coeibiiwp.method-b;***
Tool Identification List Maintenance DMWR/NMWR	***%toolidwp.dmwr-nmwr;***
Tool Identification List Maintenance	***%toolidwp.commo;***

50.6.2 Editable entity list. Table XLIV lists the boilerplate general entities found in the Editable entity file. The Editable boilerplates contain user specific information (see 50.5.1.2.2 information on Editable boilerplates). The table provides the entity title, name, and where it is referenced in MIL–STD–40051–1/–2. The boilerplates are used to develop TMs in either page-based, frame-based manuals, or both.

TABLE XLIV. Editable boilerplate list.

General Entity	
Title	Name
Short End Item Name	***&short.end.item.name;***
Reason for the distribution restriction	***&notices.dist.determined-reason;***
The date determined for the distribution restriction	***&notices.dist.determined-date;***
The DoD office determining the distribution restriction	***&notices.dist.determined-DoD-office;***
U.S. Army Proponent Activity and Address	***&proponent-address.army;***
U.S. Army Proponent Fax Number	***&proponent-fax.army;***
U.S. Army Proponent Email Address	***&proponent-email.army;***
U.S. Marines Proponent Activity and Address	***&proponent-address.usmc;***
U.S. Marines Proponent Fax	***&proponent-fax.usmc;***
U.S. Marines Proponent Email	***&proponent-email.usmc;***
U.S. Navy Proponent Activity and Address	***&proponent-address.usn;***
U.S. Navy Proponent Fax Number	***&proponent-fax.usn;***
U.S. Navy Proponent Email	***&proponent-email.usn;***
Air Force Proponent Activity and Address	***&proponent-address.usaf;***
U.S. Air Force Proponent Fax Number	***&proponent-fax.usaf;***
U.S. Air Force Proponent Email	***&proponent-email.usaf;***
Reporting Multi-Service Proponent Fax and Email Data (Frame-base Only)	***&proponent-fax;***
Reporting Multi-Service Proponent Fax and Email Data (Frame-base Only)	***&proponent-email;***
"How to Use This Manual" Internal to the TM	***&howtouse.intl-agree.ref;***
"How to Use This Manual" External to the TM	***&howtouse.intl-agree.pg;***
Warranty Period As Mileage Or Time Frame	***&ginfowp.wrntyref-time;***
Quality of Material - Replacement, Repair, or Modification Requirements TM Number	***&ginfowp.qual.mat.info-tm;***
Engineering Change Proposals Address	***&ginfowp.ecp-address;***
Repair Parts WP Reference	***&ginfowp.supdata-partlist.wpref;***
Supporting Data - Repair parts TM Reference	***&ginfowp.supdata-partlist.tmref;***

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Table XLIV Editable boilerplate list – Continued.

General Entity	
Title	Name
PMS Scope and General Information - References and Data Figure Reference	***&pms-ginfowp.fig1;***
PMS Scope and General Information - References and Data Figure Reference	***&pms-ginfowp.fig2;***
PMS Scope and General Information - References and Data Figure Reference	***&pms-ginfowp.fig3;***
PMS Scope and General Information - References and Data (Appropriate aircraft model number)	***&pms-ginfowp.aircraft;***
PMS Scope and General Information - References and Data (Intermediate inspection performed flight hour cycle)	***&pms-ginfowp.fly-hrs.intermediate;***
PMS Scope and General Information - References and Data (Periodic inspection performed flight hour cycle)	***&pms-ginfowp.fly-hrs.periodic;***
PMI General Information (aircraft model)	***&pm-ginfowp.aircraft;***
PMI General Information (flight hour cycle to perform the PM requirements)	***&pm-ginfowp.flight-cycle;***
PMI General Information (flight hour phases)	***&pm-ginfowp.phase-hours;***
PMI General Information (maximum number of phases during the flight cycle)	***&pm-ginfowp.no-phases;***
PMI General Information (latest issue of the aircraft TM number series Maintenance Manuals)	***&pm-ginfowp.tm-maint;***
PMI General Information (aircraft TM number MTF requirements.)	***&pm-ginfowp.tm-mtf;***
PMI General Information (phase hour interval)	***&pm-ginfowp.hour-interval;***
PMI General Information (phase hour cycle)	***&pm-ginfowp.hour-cycle;***
PMI General Information (first phase number example)	***&pm-ginfowp.first;***
PMI General Information (second phase number example)	***&pm-ginfowp.second;***
PMI General Information (last phase number)	***&pm-ginfowp.last-phase;***
PMI General Information (Reference to the WP sequence # or title and figure # that shows the aircraft's inspection areas)	***&pm-ginfowp.fig1;***
PMI General Information (Reference to the required removal locations of access doors and panels inspection areas)	***&pm-ginfowp.fig2;***
Illustration Manufactured Parts Maintenance Level	***&manuwp.intro.maint-level;***
Illustration Manufactured Parts RPSTL Reference	***&manuwp.intro.rpstl;***
PMCS AOAP Hour/Mileage Timeframe	***&pmcsintrowp.intro.aoap.prescribed;***
PMCS AOAP Reference	***&pmcsintrowp.intro.aoap;***
PMCS Introduction Component or Equipment	***&pmcsintrowp.intro.component-equipment;***
PMCS Introduction Not Authorized Arctic Operation Component	***&pmcswp.pmcstable.pmcsproc.arctic-oper.not-auth.component;***

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Table XLIV Editable boilerplate list – Continued.

General Entity	
Title	Name
Maintenance WP Inspection And Test Ammunition Referenced Table	***&maintwp.test-inspect.disposition-2.table-num;***
Aircraft Inspection Checklist TM for PMI WP	***&pmiwp.geninfo.aircraft;***
Referenced Avionics TM for PMS Inspection WP	***&pms-inspecwp.perform.tm;***
Lubrication Requirement Referenced Avionics TM	***&pms-inspecwp.lubrication.tm;***
RPSTL Introduction Lowest Maintenance Level	***&introwp.intro.lowest-maint;***
RPSTL Introduction Equipment Nomenclature	***&introwp.intro.item;***
Maintenance Code "L" Specialized Repair Activity Designator	***&introwp.intro.repair-act-des;***
Referenced Fabrication Instruction TM Number	***&introwp.intro.fab-tm;***
Associated Publications in Separate RPSTL TM	***&introwp.intro.assoc-pubs;***
Reference TM number for Higher Maintenance Level RPSTL Data	***&introwp.intro.higher-maint-tm;***
Uncommon Abbreviations and Definitions in RPSTL	***&introwp.intro.uncommon-abbr;***
UOC List	***&intro.uoc-list;***
Expendable and Durable Maintenance Level Codes	***&explistwp.intro.explain.level;***

50.6.3 Production front matter list. Table XLV lists the boilerplates that are found in the Production Front Matter text entity file. The table provides the entity title, name, and where it is referenced in MIL–STD–40051–1/–2. The boilerplates are used to develop TMs in either page-based, frame-based manuals, or both.

TABLE XLV. Production front matter boilerplate list.

General Entity	
Title	Name
Availability Statement	***&notices.avail;***
Disclosure Notice	***&notices.disclos;***
Distribution Statements A	***&notices.dist.a.statement;***
Distribution Statements B	***&notices.dist.b.statement;***
Distribution Statements C	***&notices.dist.c.statement;***
Distribution Statements D	***&notices.dist.d.statement;***
Distribution Statements E	***&notices.dist.e.statement;***
Distribution Statements F	***&notices.dist.f.statement;***
Distribution Statements X	***&notices.dist.x.statement;***
Export Control Notice Warning	***&notices.export;***
Reporting Errors And Recommending Improvements Statement – Page-Based	
(Army Only), Unclassified/standard, Pocket size TMs, oversize TMs, and TMs with less than eight pages	***&titleblk.reporting.std-pocket.army;***

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Table XLV Production front matter boilerplate list – Continued.

General Entity	
Title	Name
(Marines Only), Unclassified/standard, Pocket size TMs, oversize TMs, and TMs with less than eight pages	***&titleblk.reporting.std-pocket.usmc;***
Multi-Service Each Service's Statement Unclassified/standard, Pocket size TMs, oversize TMs, and TMs with less than eight pages	***&titleblk.reporting.std-pocket.multi-service;***
Multi-service TMs, Unclassified, standard size TMs used for Army, USMC, Navy and/or Air Force.	***&titleblk.reporting.std.multi-service;***
(Army Only) Classified TMs	***&titleblk.reporting.class.army;***
(Marines Only) Classified TMs	***&titleblk.reporting.class.usmc;***
Multi-service TMs, Classified TMs used for Army, USMC, Navy and/or Air Force.	***&titleblk.reporting.class.multi-service;***
Reporting Errors And Recommending Improvements Statement – FRAME-BASED	
Classsifed IETMs	***&reporting.ietm-class;***
(ARMY Only) IETM Army Only Unclassified/Classified IETMs	***&ttitleblk.reporting.ietm.army;***
(USMC Only) IETM Unclassified/Classified IETMs.	***&titleblk.reporting.ietm.usmc;***
Multi-service IETMUnclassified/Classified IETMs used for Army, USMC, Navy and/or Air Force.	***&titleblk.reporting.ietm.multi-service;***
PMS and PMI Title Block With Warning And Note Data	***&titleblk.pm.warning.data;***
International Standardization Agreements Statement	***&howtouse.intl-agree;***

50.6.4 General information list. Table XLVI lists the boilerplates that are found in the General Information Chapter text entity file. The table provides the entity title, name, and where it is referenced in MIL–STD–40051–1/–2. The boilerplates are used to develop TMs in either page-based, frame-based manuals, or both..

TABLE XLVI. General information list.

General Entity	
Title	Name
U. S. Army - Maintenance forms, records and reports (MFRR) statement	***&ginfowp.mfrr-army;***
U.S. Marines - Maintenance forms, records and reports (MFRR) statement	***&ginfowp.mfrr-usmc;***
Multi-Service Maintenance forms, records and reports (MFRR) statement	***&ginfowp.mfrr-multiservice;***
Single-Service Maintenance forms, records and reports (MFRR) statement **	***&ginfowp.mfrr-oneservice;***
Reporting equipment improvement recommendations (EIR) statement	***&ginfowp.eir;***
Hand receipt (HR) manual statement	***&ginfowp.handreceipt;***
Corrosion prevention and control (CPC) statement	***&ginfowp.cpcdata;***

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Table XLVI General information list – Continued.

General Entity	
Title	Name
Warranty information reference statement	***&ginfowp.wrntyref;***
Quality of material information statement	***&ginfowp.qual.mat.info;***
Nuclear Hardness statement	***&ginfowp.hcp;***
Engineering Change Proposals (ECP) statement	***&ginfowp.ecp;***
Modification list statement	***&ginfowp.modification;***
Deviations and Exceptions statement	***&ginfowp.deviation;***
Mobilization Requirements statement	***&ginfowp.mobreq;***
Flight Safety Critical Aircraft Parts Statement	***&ginfowp.fscap;***
Cost Considerations Statement	***&ginfowp.cost;***
Common Tools And Equipment Statement	***&ginfowp.supdata-tools;***
Parts Information Work Package Reference	***&ginfowp.supdata-partlist.wp;***
Parts Information TM (Page ONLY)	***&ginfowp.supdata-partlist.tm;***
Preventive Maintenance Service Scope and General Information (Preventive Maintenance Services Manual only)	***&pms-ginfowp;***
Preventive Maintenance Inspection Scope and General Information (Phased Maintenance Inspection Manual only)	***&pm-ginfowp.geninfo;***

50.6.5 Operator's procedures information list. Table XLVII lists the boilerplates that are found in the Operator's Procedures Information Chapter text entity file. The table provides the entity title, name, and where it is referenced in MIL–STD–40051–1/–2. The boilerplates are used to develop TMs in either page-based, frame-based manuals, or both.

TABLE XLVII. Operator's procedures information list.

General Entity	
Title	Name
Standard Information Header for Controls and Indicators	***&ctrlindwp.ctrlindtab.std-header;***

50.6.6 Troubleshooting information list. Table XLVIII lists the boilerplates that are found in the Troubleshooting Information Chapter text entity file. The table provides the entity title, name, and where it is referenced in MIL–STD–40051–1/–2. The boilerplates are used to develop TMs in either page-based, frame-based manuals, or both.

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TABLE XLVIII. Troubleshooting information list.

General Entity	
Title	Name
Standard Header for Troubleshooting SystemSubsystem Index	***&tsindx.system.header;***
Standard Header for Troubleshooting Symptom/Malfunction Index	***&tsindx.symptom.header;***
Standard Header for Troubleshooting Fault Code/Message Word Index.	***&tsindx.messageword.header;***

50.6.7 Maintenance information list. Table XLIX lists the boilerplates that are found in the Maintenance Information Chapter text entity file. The table provides the entity title, name, and where it is referenced in MIL–STD–40051–1/–2. The boilerplates are used to develop TMs in either page-based, frame-based manuals, or both.

TABLE XLIX. Maintenance information list.

General Entity	
Title	Name
Periods Of Inventory Statement	***&inventorywp.prdinv;***
Overhaul and Retirement Schedule Introduction Statement	***&maintwp.orsch;***
Standard Information Header for Maintenance WP - Overhaul and retirement schedule	***&maintwp.orsch.std-header;***
Maintenance Work Package Inspection And Test Ammunition Statement – Disposition 1	***&maintwp.test-inspect.disposition-1;***
Maintenance WP for Inspection And Test Ammunition Statements – Disposition 2	***&maintwp.test-inspect.disposition-2;***
Maintenance WP for Inspection And Test Ammunition Statements – Disposition 3	***&maintwp.test-inspect.disposition-3;***
Standard Information Header Maintenance WP - Inspection and test ammunition - Material Defect	***&maintwp.test-inspect.defect.tab.std-header;***
Packaging Information for PPM Statement	***&maintwp.ppm.packaging;***
Additional Information On-Data Plate Statement	***&maintwp.test-pass.data-plate;***
Introduction For Illustrated List Of Manufacture	***&manuwp.intro;***
Introduction For Depot Mobilization Requirements	***&mobilwp.intro;***
Standard Information Header for Maintenance WP - Depot mobilization requirements table	***&mobilwp.mobiltab.tab.std-header;***
AOAP Sampling Interval Statement	***&pmcsintrowp.intro.aoap;***
AOAP Not Available/Non-Enrolled Statement	***&pmcsintrowp.intro.aoap-na;***
Fluid Leakage Narrative For PMCS Statement	***&pmcsintrowp.intro.fluid-leakage;***
Conditions To Change Oil Filter Statement	***&pmcsintrowp.intro.oilfilter;***
PMCS Warranty Statement	***&pmcsintrowp.intro.warranty;***
Standard Statement When No MRP Is Included In The PMCS.	***&pmcswp.pmcstable.mrplpart.no-mrp;***
PMCS Introduction Lubricant Is Authorized In Arctic Operation Note	***&pmcswp.pmcstable.pmcsproc.arctic-oper.auth;***

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Table XLIX Maintenance information list – Continued.

General Entity	
Title	Name
PMCS Introduction Lubricant Is Not Authorized In Arctic Operation Note	***&pmcswp.pmcstable.pmcsproc.arctic-oper.not-auth;***
Standard Information Header for PMCS with No Man-Hours Column	***&pmcswp.pmcstable.std-header;***
Standard Information Header for PMCS with Man-Hours Column	***&pmcswp.pmcstable.std-header.manhours;***
PMI Checklist Concerning MTF Note	***&pmi-chklistwp.note;***
PMI General Information Statement	***&pmiwp.def-geninfo;***
PMI General information Statement	***&pmiwp.geninfo;***
PMI Standard Of Serviceability Statement	***&pmiwp.stdserv;***
PMS Inspection WP TM Reference Statement	***&pms-inspecwp.perform;***
PMS Inspection Discrepancy Form Reference Statement.	***&pms-inspecwp.checklist;***
PMS Inspection Last Item To Perform Statement	***&pms-inspecwp.last-item;***
Specific TM Lubrication Requirement Statement	***&pms-inspecwp.lubrication;***
Quality Assurance - Statement of Responsibility	***&qawp.responsibility;***
Quality Assurance - In-process Inspections	***&qawp.inprocess;***
Quality Assurance - Acceptance Inspections	***&qawp.acceptance;***
Storage Aircraft General Information Statement	***&storagewp.geninfo;***
Service Upon Receipt Checking Unpacked Equipment Mandatory Steps	***&surwp.surmat.chkeqp.inspect;***
Standard Information Header for Service Upon Receipt - Checking Unpacked Equipment - Packaging Material	***&surwp.chkequ.crit.insp.tab.std-header;***
Standard Information Header for Service Upon Receipt - Checking Unpacked Equipment Components	***&surwp.chkequ.pecul.insp.tab.std-header;***
Aircraft Weighing And Loading General Information Statement **	***&wtloadwp.geninfo;***

50.6.8 Parts information list. Table L lists the boilerplates that are found in the Parts Information Chapter text entity file. The table provides the entity title, name, and where it is referenced in MIL–STD–40051–1/–2. The boilerplates are used to develop TMs in either page-based, frame-based manuals, or both.

TABLE L. Parts information boilerplates.

General Entity	
Title	Name
RPSTL Introduction	***&introwp.intro;***
RPSTL Index Explanation (for NSN)	***&introwp.index-explain.nsn;***
RPSTL Index Explanation (for PN)	***&introwp.intro.index-explain.pn;***
RPSTL Index Explanation (for Reference Designator Index work package)	***&introwp.intro.index-explain.refdes;***
NSN Index Format Explanation	***&introwp.intro.index-format.nsn;***

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Table L Parts information boilerplates – Continued.

General Entity	
Title	Name
Part Number Index Format Explanation	***&introwp.intro.index-format.pn;***
Reference Designator Index Format Explanation	***&introwp.intro.index-format.refdes;***
Reference Designator How to Locate	***&introwp.intro.locate-repair-parts.refdes;***
Standard Information Header Part List, Special Tools Part List, Kits Part List, Bulk, Items Part List, and Special Tools List.	***&part-list.std-header;***
Standard Information Header for NSN Index	***&nsnindxwp.std-header;***
Standard Information Header for Part Number Index	***&pnindxwp.std-header;***
Standard Information Header for Reference Designator Index	***&refdesindxwp.std-header;***

50.6.9 Supporting information list. Table LI lists the boilerplates that are found in the Supporting Information Chapter text entity file. The table provides the entity title, name, and where it is referenced in MIL–STD–40051–1/–2. The boilerplates are used to develop TMs in either page-based, frame-based manuals, or both.

TABLE LI. Supporting information list.

General Entity	
Title	Name
Standard MAC 2 Level Maintenance Introduction	***&macintrowp.intro-std;***
Standard Information Header for Standard 2 Level Maintenance MAC	***&macwp.mac.std-header;***
Standard MAC 5 Level Maintenance Introduction	***&macintrowp.intro-std;***
Standard Information Header for Standard 5 Level Maintenance MAC	***&macwp.mac.std-header;***
Aviation MAC Introduction	***&macintrowp.intro-av;***
Standard Information Header for Aviation MAC	***&macwp.avmac.std-header;***
Standard Information Header for MAC Tools and Test Equipment	***&macwp.tereqtab.std-header;***
Standard Information Header for MAC Remarks	***&macwp.remarktab.std-header;***
COEI and BII Lists Work Package Introduction	***&coeibiiwp.intro;***
COEI and BII Lists Work Package Supporting Information UOC List	***&intro.uoc;***
COEI and BII Standard Information Header	***&coeibiiwp.coeitab-biitab.std-header;***
AAL Work Package Introduction	***&aalwp.intro;***
AAL Work Package Introduction Supporting Information UOC List	***&intro.uoc;***
AAL Standard Information Header	***&aalwp.aal.std-header;***
Expendable And Durable Items List Work Package Introduction	***&explistwp.explist.intro;***
Expendable and Durable Items List Standard Information Header	***&explistwp.explist.std-header;***

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Table LI Supporting information list – Continued.

General Entity	
Title	Name
Tools Identification List Work Package Introduction	***&toolidwp.intro;***
Tools Identification List Standard Information Header	***&toolidwp.toolidlist.std-header;***

51 NOTES.

51.1 Intended use. U. S. Army publications prepared in Standard Generalized Markup Language (SGML) /Extensible Markup Language (XML) in accordance with the information and guidance contained in this handbook is used for development of Army digital publications.

51.2 Subject term (key word) listing. The following terms are to be used to identify the MIL–STD–2361 document during retrieval searches.

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Administrative Publication Proponent/Editor Interface Process (APPIP)
Army Publishing Directorate (APD)
Army SGML/XML Registry and Library (ASRL)
Army Training Information Architecture (ATIA)
Digital Publications Development (DPD)
Document Type Definition (DTD)
Electronic Delivery
Equipment Publications
Extensible Markup Language (XML)
Formal Public Identifier (FPI)
Formatting Output Specification Instance (FOSI)
Information Reuse
Maintenance Instructions
Operator Instructions
Publishing, Electronic
SGML/XML objects and constructs
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Supporting Information
Training and Doctrine Publications
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Preparing Activity:
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Review Activities:
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